

A PROOF OF THE RIEMANN HYPOTHESIS

JOHN SHIELDS

ABSTRACT. We present a proof strategy for the Riemann Hypothesis based on local jet-normalized blocks of the phase kernel, a value/calibration functional extracted from the leading value-channel deformation, and an N -point odd-moment cancellation mechanism in the transverse channel. The document is written as a living proof draft: every statement is presented in proof form and the notation is frozen globally, so that future revisions can strengthen any remaining technical points without changing the architecture.

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1. INTRODUCTION

The Riemann Hypothesis, originating in Riemann's 1859 memoir [1], asserts that every nontrivial zero of $\zeta(s)$ lies on the critical line $\operatorname{Re} s = \frac{1}{2}$ [2, 3, 4, 5]. The argument developed here is local and geometric. It compares two objects:

- (i) a local jet-whitened block extracted from the zeta/scattering phase on short symmetric windows, and
- (ii) a local off-line toy model corresponding to a split pair away from the critical line.

The original goal was to show that the toy anomaly has a transverse component that survives every local normalization, while the zeta-side transverse channel is annihilated at leading value order and becomes exponentially small after odd-moment cancellation. The present draft should now be read more cautiously: the local analysis has produced a detailed obstruction framework and several conditional rank-two visibility theorems, but the global proof of RH is not closed.

The present proof-state is layered, but the active dependency graph has been narrowed. The upstream proof-level source input is no longer the older finite-row O1/O2/O3 package. It is the canonical Hardy-normalized source survival statement

$$\operatorname{Agg}L^2(q''_{\text{can}}), \quad q''_{\text{can}} = u_H = \partial_t^3 Z_H.$$

The live finite-survival chain is

$$\operatorname{Agg}L^2(q''_{\text{can}}) \implies \text{Graph/O2O3}_{23} \implies \mathfrak{R}_{2Y}\text{-certified } Y12/K9 \text{ carrier} \implies \text{AtomicY12Contradiction}.$$

The remaining global burden is to prove that a hypothetical off-line zero enters this ordinary Hardy-normalized canonical branch, with Fejér–Riesz persistence, raw $2Y$ row/routing certification, and packet-to- $K9$ transfer, all with only polynomial loss. Thus the draft is a conditional local obstruction architecture, not a completed proof of RH.

1.1. Master conditional theorem. The global proof-state is organized around the following conditional theorem. It is the central dependency theorem of the present draft. It does not assert the branch-entry, Fejér–Riesz persistence, raw $2Y$ certificate, or packet-to- $K9$ transfer hypotheses; rather, it records that these are exactly the remaining inputs needed to turn the canonical local obstruction into a high-height exclusion of off-line zeros.

Theorem 1.1 (Master conditional high-height RH). *Assume the following four inputs.*

- (a) **Canonical Hardy branch-entry.** *There are constants $T_0, C, A > 0$ such that every nontrivial zero*

$$\rho = \beta + i\gamma, \quad \beta \neq \frac{1}{2}, \quad |\gamma| \geq T_0,$$

with symmetry packet

$$\mathcal{Z}(\rho) = \{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\},$$

generates, with scale parameter $Q = Q(\gamma) \rightarrow \infty$, an ordinary Hardy-normalized canonical packet family $\mathcal{P}_\rho$. This family is scale-admissible, lies in an ordinary zero-separated Hardy source strip, uses the canonical source

$$q''_{\text{can}} = u_H = \partial_t^3 Z_H(t),$$

and satisfies the canonical source-window and Gate-1/Gate-2 entry bounds with only polynomial loss:

$$W_{\text{branch}}^{\text{can}} \geq Q^{-C} W_{\text{ref}}, \quad \operatorname{Agg}L^2(q''_{\text{can}}) \geq Q^{-A}$$

on a positive polynomial-weight subfamily of packets.

- (b) **Pre-extraction Fejér–Riesz persistence.** *For each active resonant block B in the family $\mathcal{P}_\rho$, the pre-extraction Fejér–Riesz Stack*

$$d\nu_{\text{FR},B} = |\Phi_B(z)|^2 d\theta d\sigma_B, \quad \deg \Phi_B = O(R),$$

is imposed before capped extraction and preserves the selected packet mass up to polynomial loss:

$$\nu_{\text{FR},B}(\mathcal{P}_\rho^{\text{active}}) \geq Q^{-C} \nu_{\text{pre},B}(\mathcal{P}_\rho^{\text{active}}).$$

Moreover no early capped extraction discards a super-polynomial fraction of the active mass. Consequently the active packet family belongs to the bounded-degree Fejér–Riesz charge model needed for Graph/O2O3₂₃, and the graph survival gap satisfies

$$\delta_{\text{graph}} \geq Q^{-A}.$$

- (c) **Raw 2Y row/routing certificate.** For every active packet $p \in \mathcal{P}_p^{\text{active}}$, the raw certificate

$$\mathfrak{R}_{2Y}(p)$$

is verified or imported. In particular, the $C_7/Y12$ detecting rows have polynomial conditioning,

$$L_p^{\text{row}} E_p = I_2, \quad \|L_p^{\text{row}}\|_{\text{op}} \leq Q^C,$$

and every row in the raw table is routed as row-dead, coherent-live, or residual. Only row-dead terms enter the exact killed quotient; every atom-local live Y12 contribution is labelled and retained in the active affine branch.

- (d) **Carrier-corrected packet-to-source-graded K9 transfer.** There is no assumed two-sided comparison between the full analytic packet Hilbert space and the finite corrected K9 jet space. Instead the following four transfer certificates hold.

First, for some $J \geq J_{K9}$, the packet Sobolev trace upper bound holds:

$$\sum_p \mu_p \|\Lambda_p^{\text{gr,full}} v_p\|_{\text{gr}}^2 \leq Q^{C_{\text{tr}}} \sum_p \mu_p W_{\text{ref}}(m_p) \sum_{0 \leq j \leq J} r_p^{2j-1} \int_{K_p^+} |\partial_t^j v_p(t)|^2 \chi_p(t) dt$$

for every analytic packet $v = (v_p)_p$, where

$$\Lambda_p^{\text{gr,full}} = (I - P_p^{\text{full}}) \mathcal{L}_{K9,p}^{\text{gr}} J_{\text{tr},p}.$$

Second, the actual Hardy source admits a carrier decomposition

$$u_H = u_{H,\text{car}} + u_{H,\text{tail}}, \quad u_{H,\text{car}} \in \bigoplus_p \mathcal{V}_{\text{pkt},p}^{\text{car}},$$

with

$$\|u_{H,\text{tail}}\|_{\text{pkt},J}^2 \leq Q^{-B} G_{\text{ref}}$$

and carrier capture

$$\sum_p \mu_p \|\Pi_{\text{eff},p} \Pi_{\text{car},p} u_H\|_{\text{pkt},J,p}^2 \geq Q^{-C_{\text{cap}}} G_{\text{ref}}.$$

Third, on the retained effective carrier, the finite source-graded K9 Gram audit holds:

$$A_p^* G_{\text{jet},p}^{\text{gr}} A_p \geq Q^{-C_{\text{gram}}} G_{\text{pkt},p},$$

where

$$A_p = \begin{bmatrix} \Lambda_p^{\text{gr,full}} \psi_{p,1} & \cdots & \Lambda_p^{\text{gr,full}} \psi_{p,N_p} \end{bmatrix}$$

for a certified basis $\mathcal{B}_p = \{\psi_{p,1}, \dots, \psi_{p,N_p}\}$ of $\mathcal{V}_{\text{pkt},p}^{\text{car}}$.

Fourth, the raw 2Y live/dead routing certificate $\mathfrak{R}_{2Y}(p)$ is verified on these same transferred carrier columns. In particular,

$$L_p^{\text{row}} E_p = I_2, \quad \|L_p^{\text{row}}\|_{\text{op}} \leq Q^C,$$

and every transferred carrier column is classified as row-dead, coherent-live and retained, or residual. No atom-local live Y12 component is quotient-killed.

Consequently,

$$\sum_p \mu_p \|\Lambda_p^{\text{gr,full}} \Pi_{\text{eff},p} \Pi_{\text{car},p} u_H\|_{\text{gr}}^2 \geq Q^{-C_{\text{cap}} - C_{\text{gram}} - C_{2Y}} G_{\text{ref}} - Q^{-B} G_{\text{ref}}.$$

The exponent B is chosen larger than the total downstream loss.

Then there exists $T_* \geq T_0$ such that every nontrivial zero

$$\rho = \beta + i\gamma, \quad |\gamma| \geq T_*,$$

satisfies

$$\beta = \frac{1}{2}.$$

Equivalently, under the four inputs above, RH holds for all sufficiently large heights.

Proof. Suppose, to the contrary, that there is an off-line zero

$$\rho = \beta + i\gamma, \quad \beta \neq \frac{1}{2},$$

with $|\gamma|$ sufficiently large. Let $Q = Q(\gamma)$ be the associated canonical local height/scale parameter.

By canonical Hardy branch-entry, the symmetry packet

$$\{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}$$

produces an ordinary Hardy-normalized canonical packet family $\mathcal{P}_\rho$, with

$$q''_{\text{can}} = u_H = \partial_t^3 Z_H,$$

canonical branch-window domination, and aggregate source survival

$$\text{Agg}L^2(q''_{\text{can}}) \geq Q^{-A}$$

after polynomial loss.

By Fejér–Riesz persistence, the same active packet mass survives inside the pre-extraction bounded-degree charge model. The finite charge anti-lock module therefore supplies

$$\text{Graph/O2O3}_{23}, \quad \delta_{\text{graph}} \geq Q^{-A},$$

again after only polynomial loss.

By the raw certificate $\mathfrak{R}_{2Y}(p)$, the active packets have the required $C_7/Y12$ detecting rows, polynomially conditioned row extraction, and exhaustive labelled live/dead routing. Thus the primitive arithmetic rows are not quotient-killed; all atom-local live $Y12$ contributions enter the retained affine source-graded branch.

Finally, the carrier-corrected transfer package applies to the actual Hardy source. The analytic trace theorem controls tail errors, carrier capture places a polynomial portion of the Hardy source in the finite effective carrier, the finite $K9$ Gram audit converts that carrier mass into the source-graded $K9$ norm, and $\mathfrak{R}_{2Y}(p)$ routes the transferred columns through the retained $C_7/Y12$ affine branch. Thus the active branch satisfies the hypotheses of the canonical source-graded atomic comparison in the same $K9$ -graded norm:

$$\text{Agg}L^2(q''_{\text{can}}) + \text{Graph/O2O3}_{23} + \mathfrak{R}_{2Y} + \text{Transfer}_{K9} \implies \text{AtomicY12Contradiction}.$$

The contradiction is obtained in the same source-graded Hilbert norm, through $\Lambda_p^{\text{gr,full}}$ or $\hat{\Lambda}_{\text{max},p}$. The polynomial losses in the four inputs are absorbed by increasing the constants A and C . Since $Q(\gamma) \rightarrow \infty$ as $|\gamma| \rightarrow \infty$, there is a threshold T_* beyond which the large- Q canonical contradiction applies. Therefore no off-line zero can occur for $|\Im \rho| \geq T_*$. $\square$

Corollary 1.2 (Conditional full RH after a low-height bridge). *Assume the four hypotheses of Theorem 1.1. If, in addition, all nontrivial zeros with*

$$|\Im \rho| < T_*$$

are known to lie on the critical line, then RH follows.

Remark 1.3 (Status of the master theorem). Theorem 1.1 is a dependency theorem, not an assertion that its four inputs have been proved. Its role is to make the remaining proof debt explicit:

$\text{off-line zero} \implies \text{canonical Hardy/Fejér–Riesz}/K9 \text{ branch}$

with polynomial loss, plus the finite row certificate $\mathfrak{R}_{2Y}(p)$. Once those inputs are supplied, the downstream canonical source-graded branch gives the high-height contradiction.

Definition 1.4 (No-loss certificate). Let

$$\mathcal{X} \longrightarrow \mathcal{Y}$$

be a proof-stage transition between packet, source, charge, row, or carrier objects. The transition is called a *no-loss certificate* at exponent C if the relevant active mass, Gram mass, row norm, or source-graded energy satisfies

$$M_Y \geq Q^{-C} M_X - Q^{-B} M_{\text{ref}},$$

where B is chosen larger than the total downstream loss exponent. The transition is called *same-column* if the finite columns used in the source package, carrier Gram audit, and $2Y$ row certificate are the same transferred columns.

A no-loss certificate is never inferred from formal compatibility alone. It is one of the following: an analytic estimate, a packet-routing theorem, a finite Gram/singular-value certificate, a row-table certificate, or an explicitly imported verified audit.

Remark 1.5 (No-loss arrows in the active proof). The active proof chain contains the following no-loss arrows:

RH counterexample	$\longrightarrow$	ordinary Hardy canonical branch,
Hardy source mass	$\longrightarrow$	mass-good ordinary packets,
ordinary packets	$\longrightarrow$	$W_{\text{branch}}^{\text{can}} \geq Q^{-C} W_{\text{ref}},$
raw Whitney packets	$\longrightarrow$	pre-extraction Fejér–Riesz Stack,
FR Stack packets	$\longrightarrow$	Graph/O2O3 ₂₃ ,
Hardy packet source	$\longrightarrow$	finite effective $K9/Y12$ carrier,
finite carrier	$\longrightarrow$	$\mathfrak{R}_{2Y}$ -certified live/dead quotient,
corrected source package	$\longrightarrow$	two-point/four-point finite package.

Each arrow must be backed by a named no-loss certificate in the sense of Definition 1.4. The final source-graded atomic comparison may only be applied after all arrows used by the active branch have been certified with compatible exponents and same-column finite data.

Remark 1.6 (High-height closure is not full low-height closure). Theorem 1.1 is a high-height statement. Even if all its four inputs are proved with effective constants, the resulting threshold

$$T_* = T_*(C, A, B, C_{\text{cap}}, C_{\text{gram}}, C_{2Y}, \dots)$$

leaves a finite low-height verification problem. A complete proof of RH therefore requires either a separate low-height theorem or a certified finite verification below T_* . No asymptotic branch theorem is to be cited as a low-height exclusion unless the effective threshold and the finite verification are explicitly recorded.

Remark 1.7 (RH-entry no-go warning: off-line zeros do not by themselves give Hardy-curvature packets). The canonical Hardy branch-entry input is a genuine analytic theorem. It does not follow from zero symmetry, the functional equation, zero counting, or dyadic Hardy mean-square estimates alone.

Indeed, if an off-line zero

$$\rho = \beta + i\gamma, \quad \eta = \beta - \frac{1}{2} \neq 0,$$

is present, then locally in the critical-line variable the completed function has a factor of the form

$$P_\rho(t) = (t - \gamma)^2 + \eta^2$$

up to the reflected factors and a zero-free background factor G_ρ . Thus

$$Z_H(t) = P_\rho(t) G_\rho(t)$$

locally, while the active source is

$$u_H(t) = \partial_t^3 Z_H(t).$$

The zero controls P_ρ , but

$$(P_\rho G_\rho)''' = P_\rho''' G_\rho + 3P_\rho'' G_\rho' + 3P_\rho' G_\rho'' + P_\rho G_\rho'''$$

contains the zero-free background and may cancel. Therefore the implication

$$\rho \notin \Re s = \frac{1}{2} \implies \int_{p_\rho} |\partial_t^3 Z_H(t)|^2 dt \geq Q^{-A} \mathbf{M}_\rho$$

is not a formal consequence of the local zero quartet. It is the missing zeta-specific localization/non-cancellation theorem in the RH-entry input.

Remark 1.8 (Horizontal displacement bookkeeping). Any entry theorem that resolves the off-line quartet at the horizontal scale

$$\eta = \left| \beta - \frac{1}{2} \right|$$

must record whether the proof constants depend on η^{-1} . Standard zero-distribution input does not supply a lower bound $\eta \geq |\gamma|^{-C}$ for a hypothetical off-line zero. Hence a height-only complexity parameter $Q = Q(\gamma)$ is justified only if the entry proof is genuinely insensitive to arbitrarily small η . Otherwise the branch-entry statement must use

$$Q = Q(\gamma, \eta^{-1}, \kappa_\rho),$$

where κ_ρ records local separation and conditioning. No downstream polynomial-loss estimate may hide an unrecorded super-polynomial dependence on η^{-1} .

Remark 1.9 (Mean-square mass is not localized RH-entry). The Hardy mean-square theorem for $u_H = \partial_t^3 Z_H$ supplies source mass on dyadic or canonical packet families. It does not by itself identify that mass with the packet selected by a given off-line zero. Thus

$$\int_T^{2T} |u_H(t)|^2 dt \gg T(\log T)^c$$

may produce mass-good packets somewhere in the dyadic block, but it does not prove

$$p_\rho \cap \mathcal{P}_{\text{mass-good}} \neq \emptyset$$

with polynomial weight. The RH-entry theorem must contain a separate no-loss localization or packet-overlap statement if the branch is to be triggered by a specified off-line zero.

Classically, RH sits at the intersection of the zero distribution of $\zeta(s)$, the explicit formula relating zeros and primes, and a number of equivalent or partially equivalent positivity, spectral, and functional-analytic reformulations [2, 3].

We introduce a normalized local jet-Gramian associated to the corrected local phase geometry. This object encodes the intrinsic local deformation after removal of the dominant value-channel and background effects, and it is the basic carrier of the odd quotient information used in the argument.

The local framework originally had four main pillars.

1. *Jet normalization*: replace raw point samples by local value/derivative coordinates.
2. *Value/curvature splitting*: separate the leading background-subtracted value channel from the curvature and tail corrections.
3. *Canonical calibration*: extract the correct value scale from the actual leading zeta-side value matrix rather than from an ad hoc scalar functional.
4. *N-point transverse cancellation*: use an odd holomorphic expansion and explicit cancellation coefficients to crush the zeta transverse channel exponentially in the local height parameter.

Remark 1.10 (Audit status of the four pillars). The four pillars above remain part of the local language of the draft, but they should not be read as a complete proof gate. The later audits show that the decisive obstruction is no longer transverse odd-mode detection alone, nor the older finite-row O1/O2/O3 Gate-1 package. The active upstream source statement is the canonical Hardy-normalized aggregate estimate

$$\text{Agg} L^2(q''_{\text{can}}), \quad q''_{\text{can}} = u_H = \partial_t^3 Z_H.$$

The remaining proof debt is global entry and no-loss routing from an off-line zero into the canonical Hardy/Fejér–Riesz/ $K9$ branch, together with the finite raw $2Y$ row/routing certificate. See Remark F.1 for the terminal dependency ledger.

This approach is not aimed at discovering new direct relations among primes. Rather, it studies a local, normalized geometric object attached to the zeta function and argues that an off-line zero would force a forbidden transverse odd mode in that object. In that sense, the method is primarily a zeta-side rigidity argument, with prime-distribution consequences entering only indirectly through the arithmetic content encoded in the zeta phase and, more broadly, through the classical zero–prime linkage expressed by the explicit formula [2, 3, 4, 5]. This local geometric viewpoint is not standard among the main classical families of approaches to RH surveyed in the literature [2, 3].

For orientation, it may be helpful to situate the present framework relative to earlier geometric viewpoints on RH. One such direction is the Speiser–Levinson–Montgomery line, in which the geometry of zeros of $\zeta'(s)$ is used to encode information about zeros of $\zeta(s)$ [6, 7]. Another is the de Branges Hilbert-space program, in which one seeks global kernel/positivity structures attached to entire functions related to ζ [8, 9]. The present approach appears to differ in emphasis: rather than working directly with raw derivative geometry or with a global positivity condition, it constructs a locally normalized transverse invariant after quotienting out the dominant value/gauge directions. This comparison is intended only as heuristic orientation for the reader; no claim is made here that the present framework subsumes those earlier viewpoints. A Mellin-side interpretation of the N -point layer. A second point of orientation concerns the N -point cancellation operator introduced later in the proof. The cancellation coefficients may be viewed as defining a discrete multiplicative distribution whose Mellin transform preserves the constant mode while annihilating the first $N - 1$ odd jets. In this sense, the N -point step is not merely a finite-difference device: it may be read as a local Mellin-side projector acting on the corrected transverse germ. This is formally reminiscent of the Weil/Connes test-function viewpoint, in which one probes the zero set through Mellin/Fourier transforms of multiplicative test functions and explicit- or trace-formula pairings [3, 10, 11]. The present argument does *not* invoke any global trace formula, positivity principle, or spectral realization theorem from that literature; the comparison is included only as heuristic orientation. What is used here is a local, explicit, finite-support projector whose Mellin symbol can be written in closed form and whose odd part starts only at order $2N - 1$.

2. LOCAL KERNEL AND JET NORMALIZATION

2.1. Phase kernel. Let Φ be a real phase on the local spectral variable t , and define the phase kernel

$$(1) \quad K_\Phi(x, y) = \begin{cases} \frac{\sin(\Phi(x) - \Phi(y))}{\pi(x - y)}, & x \neq y, \\ \frac{\Phi'(x)}{\pi}, & x = y. \end{cases}$$

We write

$$q(t) := \Phi'(t).$$

Definition 2.1 (Actual-zeta real phase and canonical Gate–2 normalization). On ordinary smooth actual-zeta packets we use a real completed critical-line phase/scalar

$$\Phi_\zeta.$$

Concretely, Φ_ζ is defined in a real completed gauge, for instance from

$$\Xi(t) = \xi\left(\frac{1}{2} + it\right) \in \mathbb{R}$$

or from an equivalent Hardy-normalized real scalar. It is not a one-sided AFE branch, not P_2 alone, not D_2 alone, and not a proxy zero kernel. If a local real chart is used to define Φ_ζ , every chart denominator is assumed polynomially bounded below on the ordinary packet. If this chart

condition fails, the packet must be treated in another chart or routed to the singular/transition branch. The associated phase derivative is

$$q_\zeta := \mathcal{D}\Phi_\zeta,$$

where $\mathcal{D}$ denotes the normalized packet derivative. In sections where ordinary prime notation is used, this means $q_\zeta = \Phi'_\zeta$ in the local spectral coordinate.

The canonical source curvature is

$$q''_{\text{can}} := \mathcal{D}^2 q_\zeta = \mathcal{D}^3 \Phi_\zeta.$$

Equivalently, in ordinary local prime notation,

$$q''_{\text{can}} = q''_\zeta = \Phi'''_\zeta.$$

This is the full actual-zeta phase-derivative curvature in the same-point Gram normalization. It is not whitened, projectivized, AFE-normalized, or affine-subtracted. Finite zero-side proxies, one-sided AFE truncations, core-only curvatures, and affine-subtracted complement curvatures are diagnostic objects only unless the corresponding error has explicitly been included in q''_{can} or proved perturbative at the stated downstream scale.

When no ambiguity is possible, centered rank-two formulas may write $q, q'', q^{(4)}$ for

$$q_\zeta, \quad q''_{\text{can}}, \quad \mathcal{D}^4 q_\zeta.$$

If a source/core split is being discussed, the affine-subtracted complement curvature is instead denoted

$$q''_{\text{source},R}(m) := (G^{\text{new}})''(m),$$

and should not be identified with q''_{can} unless the entire finite- s row package has explicitly been rebuilt with $q''_{\text{source},R}$ in place of the full actual-zeta curvature.

On ordinary smooth zero-separated packets,

$$q_\zeta, \quad q''_{\text{can}}, \quad \Gamma_G, \quad \Gamma_{G,2}$$

are real-valued. If a critical-line zero lies in the unregularized packet interval, the packet is routed to the singular/transition branch, or else the corresponding regularized phase convention must be used.

Lemma 2.2 (Reality of the completed critical-line scalar). *Let*

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s).$$

Then

$$\Xi(t) := \xi\left(\frac{1}{2} + it\right)$$

is real for all real t .

Proof. The completed function satisfies

$$\xi(s) = \xi(1-s), \quad \overline{\xi(s)} = \xi(\bar{s}).$$

Thus, for $s = \frac{1}{2} + it$,

$$\overline{\xi\left(\frac{1}{2} + it\right)} = \xi\left(\frac{1}{2} - it\right) = \xi\left(1 - \left(\frac{1}{2} + it\right)\right) = \xi\left(\frac{1}{2} + it\right).$$

Therefore $\Xi(t) \in \mathbb{R}$. □

Definition 2.3 (Canonical Gate-2 anisotropy functionals). In the canonical actual-zeta convention of Definition 2.1, define

$$\Gamma_G := q''_{\text{can}} - 6q_\zeta + 2q_\zeta^3,$$

and

$$\Gamma_{G,2} := (1 - q_\zeta^2)q''_{\text{can}} - \frac{1}{6}\mathcal{D}^4 q_\zeta.$$

When the local coordinate is clear, we suppress the ζ -subscript and write q for q_ζ .

Remark 2.4 (Derivative notation in the canonical Gate–2 identities). Throughout the canonical Gate–2 identities, primes denote the same normalized packet derivative $\mathcal{D}$ used in Definition 2.1. Thus

$$q' = \mathcal{D}q_\zeta, \quad q'' = \mathcal{D}^2q_\zeta = q''_{\text{can}}, \quad q^{(4)} = \mathcal{D}^4q_\zeta.$$

In physical coordinates m , one has $\mathcal{D} = \lambda \partial_m$ on ordinary packets. All identities involving $q'', \Gamma_G, \Gamma_{G,2}$, and Γ_G'' are to be read in this normalized derivative convention unless explicitly stated otherwise.

Lemma 2.5 (Differentiated form of the second Gate–2 anisotropy). *With the canonical definitions above,*

$$\Gamma_{G,2} = 2q_\zeta(\mathcal{D}q_\zeta)^2 - \frac{1}{6}\mathcal{D}^2\Gamma_G.$$

Proof. From

$$\Gamma_G = q''_{\text{can}} - 6q_\zeta + 2q_\zeta^3$$

and $q''_{\text{can}} = \mathcal{D}^2q_\zeta$, we compute

$$\mathcal{D}^2\Gamma_G = \mathcal{D}^4q_\zeta - 6q''_{\text{can}} + 12q_\zeta(\mathcal{D}q_\zeta)^2 + 6q_\zeta^2q''_{\text{can}}.$$

Hence

$$\begin{aligned} 2q_\zeta(\mathcal{D}q_\zeta)^2 - \frac{1}{6}\mathcal{D}^2\Gamma_G &= 2q_\zeta(\mathcal{D}q_\zeta)^2 - \frac{1}{6}\mathcal{D}^4q_\zeta + q''_{\text{can}} - 2q_\zeta(\mathcal{D}q_\zeta)^2 - q_\zeta^2q''_{\text{can}} \\ &= (1 - q_\zeta^2)q''_{\text{can}} - \frac{1}{6}\mathcal{D}^4q_\zeta = \Gamma_{G,2}. \end{aligned}$$

□

Lemma 2.6 (Expanded Gate–2 derivative identities). *In the normalized canonical convention,*

$$u = q''_{\text{can}} = q'', \quad g = \Gamma_G = q'' - 6q + 2q^3,$$

and

$$h = \Gamma_{G,2}.$$

Then

$$h = (1 - q^2)q'' - \frac{1}{6}q^{(4)} = (1 - q^2)u - \frac{1}{6}q^{(4)}.$$

Moreover,

$$g' = q''' - 6q' + 6q^2q',$$

and

$$h' = -2qq'q'' + (1 - q^2)q''' - \frac{1}{6}q^{(5)}.$$

Equivalently,

$$h' = -2qq'u + (1 - q^2)u' - \frac{1}{6}q^{(5)}.$$

Proof. The identity for h follows by expanding

$$h = 2q(q')^2 - \frac{1}{6}g''$$

and using

$$g'' = q^{(4)} - 6q'' + 12q(q')^2 + 6q^2q''.$$

Thus

$$\begin{aligned} h &= 2q(q')^2 - \frac{1}{6}q^{(4)} + q'' - 2q(q')^2 - q^2q'' \\ &= (1 - q^2)q'' - \frac{1}{6}q^{(4)}. \end{aligned}$$

The formulas for g' and h' follow by differentiating

$$g = q'' - 6q + 2q^3$$

and

$$h = (1 - q^2)q'' - \frac{1}{6}q^{(4)}.$$

□

Remark 2.7 (Derivative convention and gauge-locking for downstream use). All downstream Gate-2, wall-escape, and D4 row-conversion identities use the same normalized derivative convention

$$\mathcal{D} = \lambda \partial_m.$$

Thus, whenever ordinary prime notation is used in a local packet coordinate in the downstream sections, it is shorthand for $\mathcal{D}$ -differentiation in that normalized local coordinate. In particular,

$$q'' = q''_{\text{can}} = \mathcal{D}^2 q_\zeta, \quad \Gamma_G = q''_{\text{can}} - 6q_\zeta + 2q_\zeta^3,$$

and

$$\Gamma_{G,2} = 2q_\zeta (\mathcal{D} q_\zeta)^2 - \frac{1}{6} \mathcal{D}^2 \Gamma_G.$$

The scalar q_ζ used here is also the scalar that appears in the q_0 -wall and in the D4 source factor. Thus the wall coordinate is not an arbitrary affine primitive of q''_{can} . The affine freedom in solving $\mathcal{D}^2 q = q''_{\text{can}}$ is fixed by the canonical real phase convention above. This gauge-locking is necessary because neither

$$\Gamma_G = q''_{\text{can}} - 6q + 2q^3$$

nor the source factor

$$H_{\text{lead}} = -\frac{q(q''_{\text{can}})^3}{288}$$

is invariant under affine changes $q \mapsto q + am + b$.

For two nearby points around a center T , the natural local point basis is not stable under coalescence. One instead passes to the point-to-jet matrix

$$(2) \quad P_h := \frac{1}{2} \begin{pmatrix} 1 & 1 \\ -1/h & 1/h \end{pmatrix},$$

which turns two point samples into local value and derivative coordinates.

2.2. Jet-limit local blocks. Take two centers $T_1 \neq T_2$. In the point basis, let A_h denote the same-point 2×2 block around a single center and C_h the mixed block between the two centers. Passing to jet coordinates yields finite limits as $h \rightarrow 0$:

$$\tilde{A}_h = P_h A_h P_h^\top \rightarrow J(T), \quad \tilde{C}_h = P_h C_h P_h^\top \rightarrow J_{12}.$$

A direct Taylor expansion gives

$$(3) \quad J(T) = \frac{1}{\pi} \begin{pmatrix} 2q(T) & \frac{1}{2}q'(T) \\ \frac{1}{2}q'(T) & \frac{1}{12}(q''(T) + 2q(T)^3) \end{pmatrix}.$$

For two centers T_1, T_2 , writing

$$\Delta := \Phi(T_1) - \Phi(T_2), \quad s := T_1 - T_2, \quad q_j := q(T_j),$$

one obtains the explicit cross-jet matrix

$$(4) \quad N_{12} = \begin{pmatrix} \frac{2 \sin \Delta}{s} & \frac{\sin \Delta - q_2 s \cos \Delta}{s^2} \\ \frac{q_1 s \cos \Delta - \sin \Delta}{s^2} & \frac{(q_1 + q_2)s \cos \Delta + (q_1 q_2 s^2 - 2) \sin \Delta}{2s^3} \end{pmatrix}.$$

Writing

$$J(T_j) = \frac{1}{\pi} M_j, \quad J_{12} = \frac{1}{\pi} N_{12},$$

the common π^{-1} factor cancels under whitening, and the jet-limit whitened cross block is

$$(5) \quad \Omega_\infty(T_1, T_2) = M_1^{-1/2} N_{12} M_2^{-1/2}.$$

This is the canonical local matrix to compare with the toy anomaly.

3. FINITE- s LOCAL BLOCKS

3.1. Symmetric placement. Fix a midpoint m and a symmetric separation parameter s , and set

$$(6) \quad t_{\pm} = m \pm \frac{s}{2}.$$

At finite s , the same-point jet Gram blocks are naturally two matrices,

$$(7) \quad G_{m,\pm}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_{\pm}) & \frac{1}{2}q'(t_{\pm}) \\ \frac{1}{2}q'(t_{\pm}) & \frac{1}{12}(q''(t_{\pm}) + 2q(t_{\pm})^3) \end{pmatrix}.$$

The mixed block is obtained from (4) by substituting $T_1 = t_-$, $T_2 = t_+$ and $s = t_- - t_+ = -s$:

$$(8) \quad N_m(s) = \frac{1}{\pi} \begin{pmatrix} -\frac{2 \sin \Delta_m(s)}{s} & \frac{\sin \Delta_m(s) + q_+(s) s \cos \Delta_m(s)}{s^2} \\ -\frac{\sin \Delta_m(s) + q_-(s) s \cos \Delta_m(s)}{s^2} & \frac{(q_-(s) + q_+(s)) s \cos \Delta_m(s) + (2 - q_-(s)q_+(s)s^2) \sin \Delta_m(s)}{2s^3} \end{pmatrix}$$

where

$$q_{\pm}(s) := q(t_{\pm}), \quad \Delta_m(s) := \Phi(t_-) - \Phi(t_+).$$

The finite- s whitened local block is therefore

$$(9) \quad \widehat{\Omega}_{\zeta}(s; m) = G_{m,-}(s)^{-1/2} N_m(s) G_{m,+}(s)^{-1/2},$$

followed, when appropriate, by background subtraction and phase-gap subtraction.

3.2. Removable singularity structure. The powers $1/s$, $1/s^2$, $1/s^3$ in (8) are removable at $s = 0$. Indeed, $\Delta_m(s)$ is odd and vanishes at 0, so

$$\Delta_m(s) = O(s), \quad \sin \Delta_m(s) = O(s),$$

which implies that the numerators in (8) vanish to order at least 1, 2, 2, 3 respectively. Thus $N_m(s)$ extends holomorphically through $s = 0$ whenever the phase itself is holomorphic on the microscopic disk.

4. ZETA-SIDE DECOMPOSITION

Write the zeta/scattering phase derivative as

$$(10) \quad q(t) = B_{\zeta}(t) + S(t),$$

where B_{ζ} is the explicit background contribution and S is the background-subtracted zero contribution. The local analysis is carried out after further splitting

$$(11) \quad q(t) = q_{\text{loc}}(t) + g_{\text{sm}}(t) + E_{\text{est}}(t), \quad g_{\text{sm}}(t) := B_{\text{aut}}(t) + T_{\text{far}}(t),$$

where q_{loc} is the designated local model, T_{far} is the omitted-zero tail, and E_{est} is the residual bookkeeping term.

Define the background-subtracted value scale

$$(12) \quad S(m) := q_{\zeta}(m) - B_{\zeta}(m),$$

and the logarithmic control function

$$(13) \quad L(m) := \log\left(3 + |m| + \frac{1}{S(m)}\right).$$

4.1. Curvature estimate. The fundamental zeta-side analytic estimate is the local curvature bound

$$(14) \quad |S''(m)| \ll L(m)S(m)^2.$$

It comes from a single-pair curvature estimate, zero-free-region control, and an optimized near/far decomposition.

5. SMOOTH REMAINDER AFTER AFFINE-JET SUBTRACTION

Fix a short interval $I = [a, b]$ of length $L = b - a$ and midpoint $t_* = (a + b)/2$. Let

$$J_1[g_{\text{sm}}](t) := g_{\text{sm}}(t_*) + g'_{\text{sm}}(t_*)(t - t_*), \quad r(t) := g_{\text{sm}}(t) - J_1[g_{\text{sm}}](t).$$

Assume

$$\|g''_{\text{sm}}\|_{L^\infty(I)} \leq S_2.$$

Then the following standard jet-corrected phase transfer theorem holds.

Theorem 5.1 (Jet-corrected phase transfer). *For every $t \in I$,*

$$|r'(t)| \leq \frac{L}{2} S_2, \quad |r(t)| \leq \frac{L^2}{8} S_2.$$

If Φ and Φ_{jet} are phases normalized by

$$\Phi'(t) = q(t), \quad \Phi'_{\text{jet}}(t) = q_{\text{loc}}(t) + J_1[g_{\text{sm}}](t) + E_{\text{est}}(t), \quad \Phi(t_*) = \Phi_{\text{jet}}(t_*),$$

then

$$\|(\Phi - \Phi_{\text{jet}})'\|_{L^\infty(I)} \leq \frac{L^2}{8} S_2, \quad \|\Phi - \Phi_{\text{jet}}\|_{L^\infty(I)} \leq \frac{L^3}{16} S_2.$$

Proof. Since $r(t_*) = r'(t_*) = 0$ and $r'' = g''_{\text{sm}}$, integrating twice gives the displayed bounds. Integrating once more after identifying $(\Phi - \Phi_{\text{jet}})' = r$ yields the phase estimate. $\square$

6. TAIL CURVATURE BOUND

Let the omitted-zero tail be

$$T_{\text{far}}(t) = \sum_{\rho \notin \mathcal{L}} f_{\beta_\rho, \gamma_\rho}(t),$$

with

$$f_{\beta, \gamma}(t) = \frac{1 - \beta}{(1 - \beta)^2 + (2t - \gamma)^2} + \frac{\beta}{\beta^2 + (2t - \gamma)^2}.$$

For each omitted zero define

$$\Delta_\rho := \text{dist}(2I, \gamma_\rho).$$

Theorem 6.1 (Tail curvature theorem). *One has*

$$\sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \ll \frac{Q}{R^3},$$

where the omitted set is defined by $\Delta_\rho > R$. Consequently,

$$S_2 \leq \|B''_{\text{aut}}\|_{L^\infty(I)} + 48 \sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \ll \|B''_{\text{aut}}\|_{L^\infty(I)} + \frac{Q}{R^3}.$$

In particular, if $R \rightarrow \infty$ with m , then $S_2 = o(Q)$.

Proof. Decompose the omitted zeros into shells

$$\mathcal{S}_k := \{\rho \notin \mathcal{L} : 2^k R < \Delta_\rho \leq 2^{k+1} R\}, \quad k \geq 0.$$

Then

$$\sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \leq \sum_{k \geq 0} \#\mathcal{S}_k (2^k R)^{-4}.$$

Standard zero counting near height m gives $\#\mathcal{S}_k \ll 2^k R Q$, hence

$$\sum_{\rho \notin \mathcal{L}} \Delta_\rho^{-4} \ll \sum_{k \geq 0} (2^k R Q) (2^k R)^{-4} = Q R^{-3} \sum_{k \geq 0} 2^{-3k} \ll \frac{Q}{R^3}.$$

The bound for S_2 follows from the definition. $\square$

7. TOY ANOMALY AND TRANSVERSE OBSTRUCTION

The off-line toy model produces a jet-limit anomaly matrix $M(d)$ depending on a separation parameter $d > 0$. The leading toy deformation has the form

$$(15) \quad \Delta_{\text{toy}}(u; d) = u^2 M(d) + O(u^4),$$

where $u = \beta - \frac{1}{2}$ is the off-line displacement.

Define the transverse functional

$$(16) \quad \Phi_K(X) := x_{12} + x_{21}, \quad X = \begin{pmatrix} x_{11} & x_{12} \\ x_{21} & x_{22} \end{pmatrix}.$$

Then

$$(17) \quad \Phi_K(M(d)) = \frac{2(i(d^2 - 1) - d)}{(d + i)^4}.$$

Hence for every real $d > 0$,

$$\Phi_K(M(d)) \neq 0,$$

and on each compact interval $D \subset (0, \infty)$ there exists $c_K(D) > 0$ such that

$$|\Phi_K(M(d))| \geq c_K(D) \quad (d \in D).$$

8. LEADING VALUE MATRIX AND CANONICAL CALIBRATION

We now make the value-channel derivative A_{val} and its pairing with the toy anomaly matrix $M(d)$ fully explicit.

Work in the symmetric local regime

$$T_1 = m - \frac{\sigma}{2}, \quad T_2 = m + \frac{\sigma}{2},$$

and write

$$B := B_\zeta(m), \quad x := B\sigma.$$

When convenient, we write $A_{\text{val}}(x)$ for $A_{\text{val}}(m, \sigma)$ under the relation $x = B_\zeta(m)\sigma$.

Pure value-channel perturbation. Let the local value-channel parameter be perturbed by

$$q_1 = q_2 = B + \alpha,$$

while all derivative and curvature data are frozen at background level. In the constant-background symmetric approximation, the phase gap is

$$\Delta = \Delta_0 - \alpha\sigma,$$

with baseline value

$$\Delta_0 = -B\sigma = -x.$$

Recall that the jet-limit local block is

$$\Omega_\infty(T_1, T_2) = M_1^{-1/2} N_{12} M_2^{-1/2}.$$

Define the pure value-channel derivative by

$$(18) \quad A_{\text{val}}(m, \sigma) := \left. \frac{\partial}{\partial \alpha} \Omega_\infty(T_1, T_2) \right|_{\alpha=0}.$$

Proposition 8.1 (Explicit formula for the value-channel derivative). *In the symmetric constant-background regime,*

$$A_{\text{val}}(x) = \frac{1}{B} \begin{pmatrix} \frac{x \cos x - \sin x}{x} & \frac{\sqrt{3}(-x^2 \sin x - 2x \cos x + 2 \sin x)}{x^2} \\ \frac{\sqrt{3}(x^2 \sin x + 2x \cos x - 2 \sin x)}{x^2} & \frac{3(x^3 \cos x - 6x \cos x + 3(2 - x^2) \sin x)}{x^3} \end{pmatrix}.$$

Proof. In the constant-background symmetric regime, the same-point jet Gram matrices are diagonal:

$$M_1 = M_2 = M = \begin{pmatrix} 2(B + \alpha) & 0 \\ 0 & (B + \alpha)^3/6 \end{pmatrix}.$$

Hence

$$M^{-1/2} = \begin{pmatrix} (2(B + \alpha))^{-1/2} & 0 \\ 0 & \sqrt{6}(B + \alpha)^{-3/2} \end{pmatrix}.$$

In the same regime, the jet-limit cross block depends only on $(B + \alpha)$, σ , and the phase gap $\Delta = \Delta_0 - \alpha\sigma$. Writing out the explicit jet-limit cross block in this specialization gives

$$N_{12}(\alpha) = \begin{pmatrix} -\frac{2 \sin \Delta}{\sigma} & \frac{\sqrt{3}(\sin \Delta + (B + \alpha)\sigma \cos \Delta)}{\sigma^2} \\ -\frac{\sqrt{3}(\sin \Delta + (B + \alpha)\sigma \cos \Delta)}{\sigma^2} & \frac{3(2 \sin \Delta - 2(B + \alpha)\sigma \cos \Delta - (B + \alpha)^2 \sigma^2 \sin \Delta)}{\sigma^3} \end{pmatrix},$$

up to the common normalization factor already absorbed into the definition of Ω_∞ .

Now differentiate

$$\Omega_\infty(\alpha) = M(\alpha)^{-1/2} N_{12}(\alpha) M(\alpha)^{-1/2}$$

at $\alpha = 0$, using

$$\left. \frac{d}{d\alpha} (B + \alpha)^{-1/2} \right|_{\alpha=0} = -\frac{1}{2} B^{-3/2}, \quad \left. \frac{d}{d\alpha} (B + \alpha)^{-3/2} \right|_{\alpha=0} = -\frac{3}{2} B^{-5/2},$$

and

$$\left. \frac{d}{d\alpha} \sin(\Delta_0 - \alpha\sigma) \right|_{\alpha=0} = -\sigma \cos \Delta_0, \quad \left. \frac{d}{d\alpha} \cos(\Delta_0 - \alpha\sigma) \right|_{\alpha=0} = \sigma \sin \Delta_0.$$

After substituting $\Delta_0 = -x$, $x = B\sigma$, using

$$\sin(-x) = -\sin x, \quad \cos(-x) = \cos x,$$

and simplifying each entry, one obtains

$$A_{\text{val}}(x) = \frac{1}{B} \begin{pmatrix} \frac{x \cos x - \sin x}{x} & \frac{\sqrt{3}(-x^2 \sin x - 2x \cos x + 2 \sin x)}{x^2} \\ \frac{\sqrt{3}(x^2 \sin x + 2x \cos x - 2 \sin x)}{x^2} & \frac{3(x^3 \cos x - 6x \cos x + 3(2 - x^2) \sin x)}{x^3} \end{pmatrix}.$$

□

Corollary 8.2 (Transverse vanishing on the value channel). *One has*

$$\Phi_K(A_{\text{val}}(m, \sigma)) = 0.$$

Proof. By definition,

$$\Phi_K(X) = x_{12} + x_{21}.$$

From Proposition 8.1,

$$A_{\text{val},12}(x) = \frac{\sqrt{3}}{B} \frac{-x^2 \sin x - 2x \cos x + 2 \sin x}{x^2},$$

$$A_{\text{val},21}(x) = \frac{\sqrt{3}}{B} \frac{x^2 \sin x + 2x \cos x - 2 \sin x}{x^2}.$$

These are exact negatives of each other, hence

$$A_{\text{val},12}(x) + A_{\text{val},21}(x) = 0,$$

and therefore

$$\Phi_K(A_{\text{val}}(m, \sigma)) = 0.$$

□

Lemma 8.3 (Small- x expansion). *As $x \rightarrow 0$,*

$$A_{\text{val}}(x) = \frac{1}{B} \begin{pmatrix} -\frac{x^2}{3} + O(x^4) & -\frac{\sqrt{3}x}{3} + O(x^3) \\ \frac{\sqrt{3}x}{3} + O(x^3) & -\frac{3x^2}{5} + O(x^4) \end{pmatrix}.$$

Consequently,

$$\|A_{\text{val}}(x)\|_{\text{F}} \asymp \frac{x}{B} \quad (x \rightarrow 0).$$

Proof. Expand

$$\sin x = x - \frac{x^3}{6} + O(x^5), \quad \cos x = 1 - \frac{x^2}{2} + O(x^4),$$

and substitute into the entries of $A_{\text{val}}(x)$.

For the $(1, 1)$ -entry:

$$\frac{x \cos x - \sin x}{x} = \frac{x(1 - \frac{x^2}{2} + O(x^4)) - (x - \frac{x^3}{6} + O(x^5))}{x} = -\frac{x^2}{3} + O(x^4).$$

For the $(1, 2)$ -entry, expanding the numerator gives

$$-x^2 \sin x - 2x \cos x + 2 \sin x = -x^3 + O(x^5) - 2x(1 - \frac{x^2}{2} + O(x^4)) + 2(x - \frac{x^3}{6} + O(x^5)) = -\frac{x^3}{3} + O(x^5),$$

so

$$\frac{\sqrt{3}(-x^2 \sin x - 2x \cos x + 2 \sin x)}{x^2} = -\frac{\sqrt{3}x}{3} + O(x^3).$$

The $(2, 1)$ -entry is the negative of the $(1, 2)$ -entry, and the $(2, 2)$ -entry similarly simplifies to

$$-\frac{3x^2}{5} + O(x^4).$$

Since the off-diagonal entries are linear in x and dominate the quadratic diagonal terms, the Frobenius norm satisfies

$$\|A_{\text{val}}(x)\|_{\text{F}} \asymp \frac{x}{B}.$$

□

Proposition 8.4 (Pairing with the toy anomaly). *Let $M(d)$ be the toy anomaly matrix. Then, for fixed $d > 0$,*

$$B \langle A_{\text{val}}(x), M(d) \rangle_{\text{F}} = x \frac{2\sqrt{3}d(id+3)}{3(d-i)^4} + x^2 \frac{d(-5d^2+46id+77)}{15(d-i)^5} + O(x^3).$$

In particular, for every compact interval

$$D = [d_-, d_+] \subset (0, \infty),$$

there exists $x_0(D) > 0$ such that for

$$0 < x \leq x_0(D), \quad d \in D,$$

one has

$$|\langle A_{\text{val}}(x), M(d) \rangle_{\text{F}}| \asymp \frac{x}{B}.$$

Proof. Using Lemma 8.3, write

$$A_{\text{val}}(x) = \frac{1}{B} \begin{pmatrix} -\frac{x^2}{3} + O(x^4) & -\frac{\sqrt{3}x}{3} + O(x^3) \\ \frac{\sqrt{3}x}{3} + O(x^3) & -\frac{3x^2}{5} + O(x^4) \end{pmatrix}.$$

Now take the Frobenius pairing with the explicit toy anomaly matrix $M(d)$:

$$\langle A_{\text{val}}(x), M(d) \rangle_{\text{F}} = \sum_{j,k} A_{\text{val},jk}(x) \overline{M_{jk}(d)}.$$

The linear term in x comes entirely from the off-diagonal entries of $A_{\text{val}}(x)$, since the diagonal entries are quadratic. Using the explicit entries of $M(d)$, one finds

$$B \langle A_{\text{val}}(x), M(d) \rangle_{\text{F}} = x \frac{\sqrt{3}}{3} (\overline{M_{21}(d)} - \overline{M_{12}(d)}) + O(x^2).$$

Substituting the explicit formulas for $M_{12}(d)$ and $M_{21}(d)$ and simplifying yields

$$x \frac{2\sqrt{3}d(id+3)}{3(d-i)^4}.$$

Including the quadratic contributions from the diagonal entries and the next terms in the off-diagonal expansion gives

$$B \langle A_{\text{val}}(x), M(d) \rangle_{\text{F}} = x \frac{2\sqrt{3}d(id+3)}{3(d-i)^4} + x^2 \frac{d(-5d^2 + 46id + 77)}{15(d-i)^5} + O(x^3).$$

The coefficient of x is nonzero for every real $d > 0$. Since it depends continuously on d , its modulus is bounded below on every compact interval $D \subset (0, \infty)$. Shrinking $x_0(D)$ if necessary gives

$$|\langle A_{\text{val}}(x), M(d) \rangle_{\text{F}}| \asymp \frac{x}{B}$$

uniformly for $0 < x \leq x_0(D)$ and $d \in D$. $\square$

Proposition 8.5 (Canonical calibration theorem). *For fixed compact $D \subset (0, \infty)$ and $0 < x \leq x_0(D)$, define*

$$\Psi_{x,d}(X) := \frac{\langle A_{\text{val}}(x), X \rangle_{\text{F}}}{\langle A_{\text{val}}(x), M(d) \rangle_{\text{F}}}.$$

Then:

- (1) $\Psi_{x,d}(M(d)) = 1$;
- (2) $\Psi_{x,d}(A_{\text{val}}(x)) \asymp x/B$;
- (3) *if*

$$\Delta_{\zeta} = S(m)A_{\text{val}}(x) + R_{\zeta},$$

then

$$\Psi_{x,d}(\Delta_{\zeta}) = c_{x,d} \frac{x}{B} S(m) + \Psi_{x,d}(R_{\zeta}), \quad c_{x,d} \asymp 1;$$

- (4) *if*

$$\Delta_{\text{toy}}(u; d) = u^2 M(d) + O(u^4),$$

then

$$\Psi_{x,d}(\Delta_{\text{toy}}(u; d)) = u^2 + O(u^4).$$

In particular, whenever

$$\Psi_{x,d}(R_{\zeta}) = o\left(\frac{x}{B} S(m)\right),$$

one obtains

$$(19) \quad u^2 \asymp \frac{x}{B} S(m).$$

Proof. The first statement is immediate from the definition of $\Psi_{x,d}$. By Lemma 8.3 and Proposition 8.4,

$$\Psi_{x,d}(A_{\text{val}}(x)) = \frac{\|A_{\text{val}}(x)\|_{\text{F}}^2}{\langle A_{\text{val}}(x), M(d) \rangle_{\text{F}}} \asymp \frac{(x/B)^2}{x/B} = \frac{x}{B}.$$

The remaining statements follow by linearity. $\square$

Proposition 8.6 (Calibration remainder under a core-stable cutoff choice). *Assume the fixed-core decomposition of Section 9.1 and the core-stable cutoff compatibility of Proposition 9.7. If the cutoff R is chosen so that*

$$R^3 \gg \frac{1}{xQ S(m)},$$

then

$$S_2 = o(xQ^2 S(m)),$$

and consequently

$$\Psi_{x,d}(R_\zeta) = o\left(\frac{x}{B_\zeta(m)} S(m)\right).$$

Proof. By Theorem 6.1,

$$S_2 \ll |B''_{\text{aut}}(m)| + \frac{Q}{R^3}.$$

The background term is negligible, so under the cutoff condition

$$R^3 \gg \frac{1}{xQ S(m)},$$

one has

$$\frac{Q}{R^3} = o(xQ^2 S(m)).$$

Hence

$$S_2 = o(xQ^2 S(m)).$$

By Corollary 9.10, this implies

$$\Psi_{x,d}(R_\zeta) = o\left(\frac{x}{B_\zeta(m)} S(m)\right).$$

The use of such a cutoff does not alter the toy comparison because, by Proposition 9.7, the core $\mathcal{C}$ — and therefore $M(d)$ and A_{val} — is independent of R . $\square$

Remark 8.7 (Status of the calibration remainder estimate). Corollary 9.10 isolates the quantitative bound actually used later:

$$|\Psi_{x,d}(R_\zeta)| \ll_D \frac{S_2}{Q^3}.$$

Thus the calibration remainder step is no longer a deferred analytic-packaging item.

9. CORRECTED FINITE- s HOLOMORPHIC WHITENING

Let $\widehat{\Omega}_\zeta^{\text{corr}}(s; m)$ denote the corrected finite- s local block built from the explicit local model, the affine jet of the smooth remainder, and the jet-corrected phase remainder.

Proposition 9.1 (Microscopic denominator comparability and omitted-smooth holomorphy). *Fix an admissible midpoint m and write*

$$t_\pm = m \pm \frac{s}{2}, \quad Q \asymp \log |m|.$$

For each omitted zero $\rho = \beta_\rho + i\gamma_\rho$, set

$$D_{\rho,\pm}(s) := ((2m - \gamma_\rho) \pm s)^2 + a_\rho^2, \quad a_\rho \in \{\beta_\rho, 1 - \beta_\rho\}.$$

Assume the zero-free region gives

$$a_\rho \geq \frac{\sigma_0}{Q}$$

for some absolute $\sigma_0 > 0$ on the admissible midpoint class. If

$$|s| \leq \frac{\rho_0}{Q}, \quad \rho_0 \leq \frac{\sigma_0}{\sqrt{6}},$$

then

$$\frac{1}{2} \left((2m - \gamma_\rho)^2 + a_\rho^2 \right) \leq |D_{\rho,\pm}(s)| \leq \frac{3}{2} \left((2m - \gamma_\rho)^2 + a_\rho^2 \right).$$

Consequently, for every derivative order $r \leq r_$ needed in the corrected finite- s block entries, the omitted-zero sums defining*

$$T_{\text{far}}(t_\pm), \quad \partial_t T_{\text{far}}(t_\pm), \quad \dots, \quad \partial_t^{r_*} T_{\text{far}}(t_\pm)$$

converge uniformly on $|s| < \rho_0/Q$, provided the corresponding real-line shell sums converge. In particular,

$$g_{\text{sm}}(t_\pm) = B_{\text{aut}}(t_\pm) + T_{\text{far}}(t_\pm)$$

and its required t -derivatives extend holomorphically to

$$|s| < \rho_0/Q.$$

Proof. Set

$$x := 2m - \gamma_\rho, \quad a := a_\rho.$$

Then

$$D_{\rho,\pm}(s) = (x^2 + a^2) \pm 2xs + s^2.$$

Using

$$|2xs + s^2| \leq 2|x||s| + |s|^2 \leq \frac{1}{2}x^2 + 3|s|^2,$$

and the assumptions

$$a \geq \frac{\sigma_0}{Q}, \quad |s| \leq \frac{\rho_0}{Q}, \quad \rho_0 \leq \frac{\sigma_0}{\sqrt{6}},$$

we obtain

$$3|s|^2 \leq \frac{1}{2}a^2.$$

Hence

$$|2xs + s^2| \leq \frac{1}{2}(x^2 + a^2),$$

which gives the stated lower and upper bounds.

Now every differentiated omitted-zero term is a rational combination of powers of $D_{\rho,\pm}(s)^{-1}$ with coefficients depending only on the derivative order. By the above comparability, on $|s| < \rho_0/Q$ these are uniformly comparable to their real-line counterparts. Therefore the same shell decomposition used on the real line yields uniform convergence of the differentiated omitted-zero sums on the complex disk, and hence holomorphy of $T_{\text{far}}(t_\pm)$ and the required t -derivatives there. Adding the holomorphic background term $B_{\text{aut}}(t_\pm)$ gives the corresponding statement for $g_{\text{sm}}(t_\pm)$. $\square$

Proposition 9.2 (Entry-by-entry perturbation estimate for the corrected same-point blocks). *In the corrected finite- s normalization, write*

$$G_{m,\pm}^{\text{corr}}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_\pm) & \frac{1}{2}q'(t_\pm) \\ \frac{1}{2}q'(t_\pm) & \frac{1}{12}(q''(t_\pm) + 2q(t_\pm)^3) \end{pmatrix}, \quad t_\pm = m \pm \frac{s}{2}.$$

Let

$$q(t) = q_0(t) + \delta q(t),$$

where q_0 is the baseline local-model-plus-affine-jet part and δq is the corrected remainder. Define the baseline same-point blocks

$$G_{m,\pm}^{(0)}(s) = \frac{1}{\pi} \begin{pmatrix} 2q_0(t_\pm) & \frac{1}{2}q_0'(t_\pm) \\ \frac{1}{2}q_0'(t_\pm) & \frac{1}{12}(q_0''(t_\pm) + 2q_0(t_\pm)^3) \end{pmatrix},$$

and set

$$R_{m,\pm}(s) := G_{m,\pm}^{\text{corr}}(s) - G_{m,\pm}^{(0)}(s).$$

Assume that on the microscopic window one has

$$|\delta q(t_\pm)| \ll L_I^2 S_2, \quad |\delta q'(t_\pm)| \ll L_I S_2, \quad |\delta q''(t_\pm)| \ll S_2,$$

where L_I is the local window length, and that in the fixed-scaled regime

$$L_I \asymp \frac{x_0}{Q}, \quad |q_0(t_\pm)| \asymp Q.$$

Then

$$\|R_{m,\pm}(s)\| \ll S_2$$

uniformly for $|s| < \rho_0/Q$.

Proof. We estimate the entries of $R_{m,\pm}(s)$ one by one.

(1,1)-entry. Since

$$R_{11} = \frac{1}{\pi} \cdot 2\delta q(t_{\pm}),$$

we have

$$|R_{11}| \ll |\delta q(t_{\pm})| \ll L_I^2 S_2 \ll \frac{S_2}{Q^2} \ll S_2.$$

(1,2)- and (2,1)-entries. Since

$$R_{12} = R_{21} = \frac{1}{\pi} \cdot \frac{1}{2} \delta q'(t_{\pm}),$$

we get

$$|R_{12}| = |R_{21}| \ll |\delta q'(t_{\pm})| \ll L_I S_2 \ll \frac{S_2}{Q} \ll S_2.$$

(2,2)-entry. We have

$$R_{22} = \frac{1}{12\pi} \left(\delta q''(t_{\pm}) + 2 \left((q_0 + \delta q)^3 - q_0^3 \right) (t_{\pm}) \right).$$

Expand the cubic difference:

$$(q_0 + \delta q)^3 - q_0^3 = 3q_0^2 \delta q + 3q_0 (\delta q)^2 + (\delta q)^3.$$

Hence

$$R_{22} = \frac{1}{12\pi} \left(\delta q'' + 6q_0^2 \delta q + 6q_0 (\delta q)^2 + 2(\delta q)^3 \right) (t_{\pm}).$$

Now estimate each term.

First,

$$|\delta q''(t_{\pm})| \ll S_2.$$

Second, using $|q_0(t_{\pm})| \asymp Q$ and $|\delta q(t_{\pm})| \ll L_I^2 S_2 \ll S_2/Q^2$,

$$|q_0(t_{\pm})^2 \delta q(t_{\pm})| \ll Q^2 \cdot \frac{S_2}{Q^2} = S_2.$$

Third,

$$|q_0(t_{\pm})(\delta q(t_{\pm}))^2| \ll Q \left(\frac{S_2}{Q^2} \right)^2 = \frac{S_2^2}{Q^3}.$$

Since from the tail-curvature theorem one already has $S_2 = o(Q)$, for Q sufficiently large this is $O(S_2/Q^2)$ and hence $\ll S_2$.

Finally,

$$|(\delta q(t_{\pm}))^3| \ll \left(\frac{S_2}{Q^2} \right)^3 = \frac{S_2^3}{Q^6} \ll S_2.$$

Therefore

$$|R_{22}| \ll S_2.$$

Collecting the four entrywise bounds, we obtain

$$|R_{11}|, |R_{12}|, |R_{21}|, |R_{22}| \ll S_2.$$

Since all norms on 2×2 matrices are equivalent,

$$\|R_{m,\pm}(s)\| \ll S_2,$$

uniformly on the microscopic disk. This proves the proposition. $\square$

Proposition 9.3 (Entry-by-entry perturbation estimate for the corrected mixed block). *Let*

$$q(t) = q_0(t) + \delta q(t), \quad \Phi(t) = \Phi_0(t) + \delta \Phi(t),$$

where q_0 is the baseline local-model-plus-affine-jet phase derivative and δq is the corrected remainder. Write

$$t_{\pm} = m \pm \frac{s}{2}, \quad \Delta(s) := \Phi(t_-) - \Phi(t_+), \quad \Delta_0(s) := \Phi_0(t_-) - \Phi_0(t_+),$$

and set

$$\delta \Delta(s) := \Delta(s) - \Delta_0(s).$$

Let $N_m^{\text{corr}}(s)$ be the corrected finite- s mixed block built from q, Δ , and let $N_m^{(0)}(s)$ be the corresponding block built from q_0, Δ_0 . Define

$$\tilde{R}_m(s) := N_m^{\text{corr}}(s) - N_m^{(0)}(s).$$

Assume that on the microscopic window one has

$$|\delta q(t_{\pm})| \ll L_I^2 S_2, \quad |\delta q'(t_{\pm})| \ll L_I S_2, \quad |\delta q''(t_{\pm})| \ll S_2,$$

and

$$|\delta \Delta(s)| \ll L_I^3 S_2,$$

with

$$L_I \asymp \frac{x_0}{Q}, \quad |q_0(t_{\pm})| \asymp Q, \quad |s| \asymp L_I.$$

Then

$$\|\tilde{R}_m(s)\| \ll S_2$$

uniformly for $|s| < \rho_0/Q$.

Proof. Write the corrected mixed block entrywise as

$$N_m^{\text{corr}}(s) = \frac{1}{\pi} \begin{pmatrix} -\frac{2 \sin \Delta}{s} & \frac{\sin \Delta + q(t_+)s \cos \Delta}{s^2} \\ -\frac{\sin \Delta + q(t_-)s \cos \Delta}{s^2} & \frac{(q(t_-) + q(t_+))s \cos \Delta + (2 - q(t_-)q(t_+)s^2) \sin \Delta}{2s^3} \end{pmatrix},$$

and analogously for $N_m^{(0)}(s)$ with (q_0, Δ_0) .

We estimate the perturbation entry by entry. Throughout, use

$$\sin(\Delta_0 + \delta \Delta) - \sin \Delta_0 = O(|\delta \Delta|), \quad \cos(\Delta_0 + \delta \Delta) - \cos \Delta_0 = O(|\delta \Delta|).$$

(1,1)-entry.

$$\tilde{R}_{11} = -\frac{2}{\pi s} (\sin \Delta - \sin \Delta_0),$$

hence

$$|\tilde{R}_{11}| \ll \frac{|\delta \Delta|}{|s|} \ll \frac{L_I^3 S_2}{L_I} \ll L_I^2 S_2 \ll S_2.$$

(1,2)-entry. The numerator perturbation is

$$(\sin \Delta - \sin \Delta_0) + (q(t_+) - q_0(t_+))s \cos \Delta + q_0(t_+)s(\cos \Delta - \cos \Delta_0).$$

Therefore

$$|\tilde{R}_{12}| \ll \frac{|\delta \Delta| + |\delta q(t_+)| |s| + |q_0(t_+)| |s| |\delta \Delta|}{|s|^2}.$$

Using

$$|\delta \Delta| \ll L_I^3 S_2, \quad |\delta q(t_+)| \ll L_I^2 S_2, \quad |q_0(t_+)| \asymp Q, \quad |s| \asymp L_I \asymp Q^{-1},$$

we get

$$|\tilde{R}_{12}| \ll \frac{L_I^3 S_2}{L_I^2} + \frac{L_I^3 S_2}{L_I^2} + \frac{Q L_I^4 S_2}{L_I^2} \ll \frac{S_2}{Q} \ll S_2.$$

The same argument gives

$$|\tilde{R}_{21}| \ll S_2.$$

(2,2)-entry. Write the numerator as

$$F(q_-, q_+, \Delta; s) := (q_- + q_+)s \cos \Delta + (2 - q_- q_+ s^2) \sin \Delta.$$

Then

$$\tilde{R}_{22} = \frac{F(q(t_-), q(t_+), \Delta; s) - F(q_0(t_-), q_0(t_+), \Delta_0; s)}{2\pi s^3}.$$

A first-order expansion of F in (q_-, q_+, Δ) gives

$$|F - F_0| \ll |s| |\delta q(t_-)| + |s| |\delta q(t_+)| + |s|^2 Q |\delta q(t_-)| + |s|^2 Q |\delta q(t_+)| + (|s|Q + 1) |\delta \Delta|.$$

Using

$$|\delta q(t_\pm)| \ll L_I^2 S_2, \quad |\delta \Delta| \ll L_I^3 S_2, \quad |s| \asymp L_I \asymp Q^{-1},$$

we obtain

$$|F - F_0| \ll L_I^3 S_2.$$

Hence

$$|\tilde{R}_{22}| \ll \frac{L_I^3 S_2}{|s|^3} \ll S_2.$$

Collecting the four entrywise bounds and using equivalence of norms on 2×2 matrices gives

$$\|\tilde{R}_m(s)\| \ll S_2,$$

uniformly on the microscopic disk. $\square$

Lemma 9.4 (Baseline same-point variation). *Let $G_{m,\pm}^{(0)}(s)$ be the corrected same-point baseline blocks built from the explicit local model together with the affine jet $J_1[g_{\text{sm}}]$. Then, on the microscopic disk $|s| < \rho_0/Q$,*

$$\sup_{|s| < \rho_0/Q} \|G_{m,\pm}^{(0)}(s) - G_{m,\pm}^{(0)}(0)\| \ll \rho_0 Q.$$

Proof. Differentiate the explicit same-point block formula entrywise:

$$G_{m,\pm}^{(0)}(s) = \frac{1}{\pi} \begin{pmatrix} 2q_0(t_\pm) & \frac{1}{2}q_0'(t_\pm) \\ \frac{1}{2}q_0'(t_\pm) & \frac{1}{12}(q_0''(t_\pm) + 2q_0(t_\pm)^3) \end{pmatrix}, \quad t_\pm = m \pm \frac{s}{2}.$$

By the chain rule,

$$\partial_s G_{m,\pm}^{(0)}(s) = \pm \frac{1}{2\pi} \begin{pmatrix} 2q_0'(t_\pm) & \frac{1}{2}q_0''(t_\pm) \\ \frac{1}{2}q_0''(t_\pm) & \frac{1}{12}(q_0'''(t_\pm) + 6q_0(t_\pm)^2 q_0'(t_\pm)) \end{pmatrix}.$$

For the explicit local model plus affine jet, one has the baseline local scales

$$|q_0(t_\pm)| \ll Q, \quad |q_0'(t_\pm)| \ll Q^2, \quad |q_0''(t_\pm)| \ll Q^3, \quad |q_0'''(t_\pm)| \ll Q^4,$$

uniformly on the admissible midpoint class. The cubic combination $q_0'''(t_\pm) + 6q_0(t_\pm)^2 q_0'(t_\pm)$ therefore has size $O(Q^2)$ in the current normalization, and hence

$$\sup_{|s| < \rho_0/Q} \|\partial_s G_{m,\pm}^{(0)}(s)\| \ll Q^2.$$

Integrating along the segment from 0 to s gives

$$\|G_{m,\pm}^{(0)}(s) - G_{m,\pm}^{(0)}(0)\| \leq |s| \sup_{|u| < \rho_0/Q} \|\partial_u G_{m,\pm}^{(0)}(u)\| \ll \frac{\rho_0}{Q} \cdot Q^2 = \rho_0 Q.$$

$\square$

Proposition 9.5 (Corrected finite- s holomorphic whitening). *The corrected same-point and mixed blocks are holomorphic on*

$$|s| < \rho_0/Q.$$

Moreover, after shrinking ρ_0 if necessary, the same-point blocks

$$G_{m,\pm}^{\text{corr}}(s)$$

remain uniformly positive definite on this disk, and the inverse square roots

$$G_{m,\pm}^{\text{corr}}(s)^{-1/2}$$

are holomorphic there. Consequently

$$\widehat{\Omega}_\zeta^{\text{corr}}(s; m) = G_{m,-}^{\text{corr}}(s)^{-1/2} N_m^{\text{corr}}(s) G_{m,+}^{\text{corr}}(s)^{-1/2}$$

is holomorphic on $|s| < \rho_0/Q$.

Proof. Holomorphy of the omitted smooth part and its required t -derivatives follows from Proposition 9.1. The only apparent poles at $s = 0$ in the corrected mixed block come from powers of $1/s$, $1/s^2$, $1/s^3$, and these are removable because $\Delta_m(s)$ is odd in s , $\sin \Delta_m(s) = O(s)$, and the off-diagonal and $(2, 2)$ numerators vanish to orders 2 and 3, respectively. Thus both $G_{m,\pm}^{\text{corr}}(s)$ and $N_m^{\text{corr}}(s)$ are holomorphic on the disk.

By Proposition 9.2 and the tail-curvature theorem,

$$\sup_{|s| < \rho_0/Q} \|R_{m,\pm}(s)\| = o(Q).$$

By Lemma 9.4,

$$\sup_{|s| < \rho_0/Q} \|G_{m,\pm}^{(0)}(s) - G_{m,\pm}^{(0)}(0)\| \ll \rho_0 Q.$$

At $s = 0$, the baseline jet-limit block has spectral gap

$$\lambda_{\min}(G_{m,\pm}^{(0)}(0)) \gg Q$$

on the admissible midpoint class. Therefore

$$\|G_{m,\pm}^{\text{corr}}(s) - G_{m,\pm}^{(0)}(0)\| \leq \|G_{m,\pm}^{(0)}(s) - G_{m,\pm}^{(0)}(0)\| + \|R_{m,\pm}(s)\| \ll \rho_0 Q + o(Q).$$

For ρ_0 sufficiently small and m sufficiently large, this is smaller than half the baseline spectral gap, hence

$$\lambda_{\min}(G_{m,\pm}^{\text{corr}}(s)) \gg Q$$

uniformly on the disk. Holomorphic functional calculus then gives the holomorphic inverse square roots, and the whitened block is holomorphic as a product of holomorphic factors. $\square$

9.1. Fixed anomaly core and cutoff-dependent auxiliary bookkeeping. To separate the anomaly-carrying local model from the cutoff-dependent tail bookkeeping, we fix once and for all a finite local core

$$\mathcal{C}$$

containing the candidate off-line anomaly (for example, the designated off-line pair together with any symmetry-forced companions needed for the local model).

For each cutoff $R \geq R_0$, we then decompose the zero contribution into:

- (1) the fixed core contribution $q_{\text{core}}(t)$, coming from the zeros in $\mathcal{C}$;
- (2) the auxiliary near contribution $q_{\text{aux},R}(t)$, coming from the zeros outside $\mathcal{C}$ but inside the near shell determined by R ;
- (3) the far contribution $T_{\text{far},R}(t)$, coming from the zeros outside the near shell;
- (4) the background contribution $B_{\text{bg}}(t)$;
- (5) the estimation term $E_{\text{est},R}(t)$.

Thus the full phase derivative is decomposed as

$$q(t) = q_{\text{core}}(t) + q_{\text{aux},R}(t) + g_{\text{sm},R}(t) + E_{\text{est},R}(t),$$

where

$$g_{\text{sm},R}(t) := B_{\text{bg}}(t) + T_{\text{far},R}(t).$$

The local toy comparison is attached only to the fixed core $\mathcal{C}$. The auxiliary near part $q_{\text{aux},R}$, the far part $T_{\text{far},R}$, and the estimation term $E_{\text{est},R}$ are absorbed into the corrected zeta-side scalar defined below.

Proposition 9.6 (Corrected local deformation decomposition with fixed core). *Fix the anomaly-carrying core $\mathcal{C}$ as above, and for each cutoff $R \geq R_0$ write*

$$q(t) = q_{\text{core}}(t) + q_{\text{aux},R}(t) + g_{\text{sm},R}(t) + E_{\text{est},R}(t), \quad g_{\text{sm},R}(t) = B_{\text{bg}}(t) + T_{\text{far},R}(t).$$

Let $J_1[g_{\text{sm},R}]$ denote the affine jet of $g_{\text{sm},R}$ at the midpoint, and let $\Phi_{\text{corr},R}$ be the corrected phase obtained from this decomposition.

Using $\Phi_{\text{corr},R}$, define the corrected same-point blocks

$$G_{m,\pm}^{\text{corr},R}(s),$$

the corrected mixed block

$$N_m^{\text{corr},R}(s),$$

and the corrected whitened block

$$\hat{\Omega}_\zeta^{\text{corr},R}(s; m) = G_{m,-}^{\text{corr},R}(s)^{-1/2} N_m^{\text{corr},R}(s) G_{m,+}^{\text{corr},R}(s)^{-1/2}.$$

Let

$$\hat{\Omega}_{\text{bg}}^{\text{corr},R}(s; m)$$

denote the corresponding corrected whitened block at background value-channel parameter $S(m) = 0$, and define the corrected local deformation

$$\Delta_\zeta(m, \sigma) := \hat{\Omega}_\zeta^{\text{corr},R}(s; m) - \hat{\Omega}_{\text{bg}}^{\text{corr},R}(s; m), \quad s = \sigma.$$

Then the corrected local deformation admits the decomposition

$$\boxed{\Delta_\zeta(m, \sigma) = S(m) A_{\text{val}}(m, \sigma) + R_\zeta(m, \sigma),}$$

where $A_{\text{val}}(m, \sigma)$ is the pure value-channel derivative of the jet-limit block associated with the fixed core $\mathcal{C}$, and where

$$\boxed{R_\zeta(m, \sigma) := \Delta_\zeta(m, \sigma) - S(m) A_{\text{val}}(m, \sigma).}$$

Thus, by definition, $R_\zeta(m, \sigma)$ is exactly the non-leading part of the corrected local deformation after the explicit value-channel linear term is removed. In particular, $R_\zeta(m, \sigma)$ is internal to the same full corrected cutoff-dependent object

$$H_{m,R}(s) := \Phi_K(\hat{\Omega}_\zeta^{\text{corr},R}(s; m)),$$

so once the corrected odd holomorphic package is established for $H_{m,R}(s)$, no external tail/error term survives outside the resulting N -point scalar.

Proof. Consider the analytic whitening map

$$\mathcal{W}(G_-, N, G_+) := G_-^{-1/2} N G_+^{-1/2}.$$

By the corrected finite- s holomorphic-whitening theorem, this map is holomorphic on the open set where $G_\pm$ remain uniformly positive definite.

For each cutoff R , the corrected blocks are built from the decomposition

$$q = q_{\text{core}} + q_{\text{aux},R} + g_{\text{sm},R} + E_{\text{est},R}.$$

Thus every auxiliary near contribution from $q_{\text{aux},R}$, every omitted-zero tail contribution from $T_{\text{far},R}$, every affine-jet correction from $J_1[g_{\text{sm},R}]$, and every explicit estimation term $E_{\text{est},R}$ is already absorbed into the corrected phase $\Phi_{\text{corr},R}$, hence into the corrected same-point blocks, the corrected mixed block, the corrected whitened block, and therefore into the corrected transverse scalar

$$H_{m,R}(s) = \Phi_K(\hat{\Omega}_\zeta^{\text{corr},R}(s; m)).$$

Now expand the corrected whitened block around the background value-channel parameter $S(m) = 0$. By analyticity of $\mathcal{W}$,

$$\hat{\Omega}_\zeta^{\text{corr},R}(s; m) = \hat{\Omega}_{\text{bg}}^{\text{corr},R}(s; m) + S(m) D\mathcal{W}_{(\text{bg})}[\dot{G}_-, \dot{N}, \dot{G}_+] + R_\zeta(m, \sigma),$$

where the first-order coefficient

$$D\mathcal{W}_{(\text{bg})}[\dot{G}_-, \dot{N}, \dot{G}_+]$$

is exactly the pure value-channel derivative $A_{\text{val}}(m, \sigma)$ associated with the fixed core $\mathcal{C}$. We therefore define

$$R_\zeta(m, \sigma) := \Delta_\zeta(m, \sigma) - S(m) A_{\text{val}}(m, \sigma),$$

so that $R_\zeta(m, \sigma)$ is, by definition, the full non-leading remainder inside the same corrected cutoff-dependent object.

Subtracting the background block gives

$$\Delta_\zeta(m, \sigma) = S(m) A_{\text{val}}(m, \sigma) + R_\zeta(m, \sigma),$$

as claimed. $\square$

Proposition 9.7 (Core-stable cutoff compatibility). *Fix the anomaly-carrying core $\mathcal{C}$, and for each cutoff $R \geq R_0$ define the corrected scalar*

$$H_{m,R}(s) := \Phi_K(\hat{\Omega}_\zeta^{\text{corr},R}(s; m))$$

from the decomposition

$$q = q_{\text{core}} + q_{\text{aux},R} + g_{\text{sm},R} + E_{\text{est},R}.$$

Then enlarging R changes only the auxiliary near part $q_{\text{aux},R}$ and the far part $T_{\text{far},R}$, while leaving the anomaly-carrying core q_{core} unchanged. Consequently:

- (1) the toy anomaly matrix $M(d)$ and the calibration matrix $A_{\text{val}}(m, \sigma)$ depend only on the fixed core $\mathcal{C}$, not on the cutoff R ;
- (2) all R -dependent auxiliary, tail, jet-correction, and estimation contributions are internal to the full corrected scalar $H_{m,R}(s)$;
- (3) if the corrected odd holomorphic package is proved for $H_{m,R}(s)$, then the N -point odd-moment cancellation theorem applies directly to the full corrected scalar for that cutoff, with no external remainder term.

In particular, the cutoff R may be chosen for quantitative convenience in the curvature/tail estimates without changing the fixed toy comparison attached to the core $\mathcal{C}$.

Proof. Let $R_1 < R_2$. Then the only change in the decomposition

$$q = q_{\text{core}} + q_{\text{aux},R} + g_{\text{sm},R} + E_{\text{est},R}$$

is that some non-core zeros move from the far part into the auxiliary near part:

$$q_{\text{aux},R_2} - q_{\text{aux},R_1} = -(T_{\text{far},R_2} - T_{\text{far},R_1}).$$

The core contribution q_{core} is unchanged by construction.

Thus the toy anomaly matrix $M(d)$, which is attached only to the fixed core, is unchanged, and the value-channel derivative $A_{\text{val}}(m, \sigma)$ associated with that core is unchanged as well. All cutoff dependence is confined to the auxiliary near contribution, the far tail, the affine-jet correction built from $g_{\text{sm},R}$, and the explicit estimation term $E_{\text{est},R}$. These enter only through the corrected phase $\Phi_{\text{corr},R}$, hence only through the full corrected scalar $H_{m,R}(s)$.

Therefore the corrected odd holomorphic package, once established for $H_{m,R}(s)$, applies to the complete cutoff-dependent object. The resulting N -point scalar already contains all auxiliary/tail/error effects for that cutoff, and no external remainder survives. Since the fixed core is unchanged, the toy comparison is stable under enlarging R . This proves the claim. $\square$

Remark 9.8 (Cutoff choice). Proposition 9.7 is a structural statement. It does not assert any particular optimal choice of R ; it only says that once the core $\mathcal{C}$ is fixed, the cutoff may be adjusted in the perturbative bookkeeping without altering the underlying toy comparison.

Proposition 9.9 (Whitened mixed-block transfer with preserved Q^{-3} scale). *Let*

$$\mathcal{W}(G_-, N, G_+) := G_-^{-1/2} N G_+^{-1/2}.$$

Write the corrected finite- s blocks as

$$G_\pm^{\text{corr}}(s) = G_\pm^{(0)}(s) + R_\pm(s), \quad N^{\text{corr}}(s) = N^{(0)}(s) + \tilde{R}(s),$$

and set

$$\widehat{\Omega}_\zeta^{\text{corr}}(s; m) := \mathcal{W}(G_-^{\text{corr}}, N^{\text{corr}}, G_+^{\text{corr}}), \quad \widehat{\Omega}_\zeta^{(0)}(s; m) := \mathcal{W}(G_-^{(0)}, N^{(0)}, G_+^{(0)}).$$

Then

$$\widehat{\Omega}_\zeta^{\text{corr}}(s; m) - \widehat{\Omega}_\zeta^{(0)}(s; m) = \mathcal{T}_-(s; m) + \mathcal{T}_{\text{mix}}(s; m) + \mathcal{T}_+(s; m) + \mathcal{E}_2(s; m),$$

where

$$\begin{aligned} \mathcal{T}_-(s; m) &:= X_-(s) N^{(0)}(s) G_+^{(0)}(s)^{-1/2}, \\ \mathcal{T}_{\text{mix}}(s; m) &:= G_-^{(0)}(s)^{-1/2} \widetilde{R}(s) G_+^{(0)}(s)^{-1/2}, \\ \mathcal{T}_+(s; m) &:= G_-^{(0)}(s)^{-1/2} N^{(0)}(s) X_+(s), \end{aligned}$$

with

$$X_\pm(s) := D(G^{-1/2})_{G_\pm^{(0)}(s)}[R_\pm(s)],$$

and $\mathcal{E}_2$ quadratic in $(R_-, \widetilde{R}, R_+)$.

Moreover,

$$\Phi_K(\mathcal{T}_-(s; m)) \ll \frac{S_2}{Q^3}, \quad \Phi_K(\mathcal{T}_{\text{mix}}(s; m)) \ll \frac{S_2}{Q^3}, \quad \Phi_K(\mathcal{T}_+(s; m)) \ll \frac{S_2}{Q^3},$$

and

$$\Phi_K(\mathcal{E}_2(s; m)) \ll \frac{S_2^2}{Q^3}.$$

Hence

$$\boxed{\Phi_K(\widehat{\Omega}_\zeta^{\text{corr}}(s; m) - \widehat{\Omega}_\zeta^{(0)}(s; m)) \ll \frac{S_2}{Q^3}.}$$

Proof. We first collect the scales used below from results established earlier. The same-point perturbation bound $\|R_\pm(s)\| \ll S_2$ is Proposition 9.2; the mixed-block perturbation bound $\|\widetilde{R}(s)\| \ll S_2$ is Proposition 9.3; and the smallness $S_2 = o(Q)$ is Theorem 6.1. The baseline entry scales

$$\begin{aligned} G_\pm^{(0)}(s) &= \begin{pmatrix} O(Q) & O(1) \\ O(1) & O(Q^3) \end{pmatrix}, \quad G_\pm^{(0)}(s)^{-1/2} = \begin{pmatrix} O(Q^{-1/2}) & O(Q^{-3/2}) \\ O(Q^{-3/2}) & O(Q^{-3/2}) \end{pmatrix}, \\ N^{(0)}(s) &= \begin{pmatrix} O(Q) & O(Q^2) \\ O(Q^2) & O(Q^3) \end{pmatrix} \end{aligned}$$

are read off directly from the explicit same-point formula (7) and the mixed-block formula (8), given the baseline local scales of q_0, q'_0, q''_0 stated in the proof of Lemma 9.4. Uniform positive-definiteness of $G_\pm^{(0)}(s)$ on the microscopic disk is Proposition 9.5.

The whitening map admits the first-order expansion

$$\widehat{\Omega}_\zeta^{\text{corr}} - \widehat{\Omega}_\zeta^{(0)} = D\mathcal{W}_{(0)}[R_-, \widetilde{R}, R_+] + \mathcal{E}_2,$$

where

$$D\mathcal{W}_{(0)}[R_-, \widetilde{R}, R_+] = X_- N^{(0)} G_+^{(0)-1/2} + G_-^{(0)-1/2} \widetilde{R} G_+^{(0)-1/2} + G_-^{(0)-1/2} N^{(0)} X_+.$$

We estimate the three first-order pieces separately.

Step 1: the mixed-block term. By Proposition 9.3 (entrywise form),

$$\widetilde{R}_{11} \ll \frac{S_2}{Q^2}, \quad \widetilde{R}_{12}, \widetilde{R}_{21} \ll \frac{S_2}{Q}, \quad \widetilde{R}_{22} \ll S_2.$$

Combining with the baseline inverse-square-root scales above, the off-diagonal entries of

$$\mathcal{T}_{\text{mix}} := G_-^{(0)-1/2} \widetilde{R} G_+^{(0)-1/2}$$

are all $O(S_2/Q^3)$. Hence

$$\Phi_K(\mathcal{T}_{\text{mix}}) \ll \frac{S_2}{Q^3}.$$

Step 2: the same-point Fréchet derivative. We prove the corresponding estimate for the left same-point term

$$\mathcal{T}_- := X_- N^{(0)} G_+^{(0)-1/2}, \quad X_- := D(G^{-1/2})_{G_-^{(0)}}[R_-].$$

The proof for $\mathcal{T}_+$ is identical.

Write one baseline same-point block in the form

$$G_0 = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad a \asymp Q, \quad b = O(1), \quad c \asymp Q^3,$$

and one perturbation as

$$R = \begin{pmatrix} r_{11} & r_{12} \\ r_{12} & r_{22} \end{pmatrix}, \quad r_{11} \ll \frac{S_2}{Q^2}, \quad r_{12} \ll \frac{S_2}{Q}, \quad r_{22} \ll S_2.$$

Diagonalize

$$G_0 = U \Lambda U^\top, \quad \Lambda = \text{diag}(\lambda_1, \lambda_2),$$

with U orthogonal. Since

$$a \asymp Q, \quad c \asymp Q^3, \quad b = O(1),$$

we have

$$\lambda_1 \asymp Q, \quad \lambda_2 \asymp Q^3.$$

Moreover, if U is a rotation by angle θ , then

$$\tan(2\theta) = \frac{2b}{c-a} = O(Q^{-3}),$$

hence

$$\theta = O(Q^{-3}), \quad U = I + O(Q^{-3}).$$

Set

$$H := U^\top R U.$$

A direct computation gives

$$\begin{aligned} H_{11} &= r_{11} + 2\theta r_{12} + O(\theta^2 r_{22}), \\ H_{12} &= r_{12} + \theta(r_{22} - r_{11}) + O(\theta^2 r_{12}), \\ H_{22} &= r_{22} - 2\theta r_{12} + O(\theta^2 r_{11}). \end{aligned}$$

Using

$$\theta = O(Q^{-3}), \quad r_{11} \ll \frac{S_2}{Q^2}, \quad r_{12} \ll \frac{S_2}{Q}, \quad r_{22} \ll S_2,$$

we obtain

$$H_{11} \ll \frac{S_2}{Q^2}, \quad H_{12} \ll \frac{S_2}{Q}, \quad H_{22} \ll S_2.$$

Now let $f(x) = x^{-1/2}$. For a diagonal matrix Λ , the Fréchet derivative is given entrywise by the Löwner formula

$$Df_\Lambda(H)_{ij} = f^{[1]}(\lambda_i, \lambda_j) H_{ij},$$

where

$$f^{[1]}(u, v) = \begin{cases} \frac{u^{-1/2} - v^{-1/2}}{u - v}, & u \neq v, \\ -\frac{1}{2u^{3/2}}, & u = v. \end{cases}$$

Since

$$\lambda_1 \asymp Q, \quad \lambda_2 \asymp Q^3,$$

we have

$$f^{[1]}(\lambda_1, \lambda_1) \asymp Q^{-3/2}, \quad f^{[1]}(\lambda_2, \lambda_2) \asymp Q^{-9/2}, \quad f^{[1]}(\lambda_1, \lambda_2) \asymp Q^{-7/2}.$$

Therefore, if

$$Y := Df_\Lambda(H),$$

then

$$\begin{aligned} Y_{11} &\ll Q^{-3/2} \cdot \frac{S_2}{Q^2} = \frac{S_2}{Q^{7/2}}, \\ Y_{12} &\ll Q^{-7/2} \cdot \frac{S_2}{Q} = \frac{S_2}{Q^{9/2}}, \\ Y_{22} &\ll Q^{-9/2} \cdot S_2 = \frac{S_2}{Q^{9/2}}. \end{aligned}$$

Since

$$X = U Y U^\top$$

and $U = I + O(Q^{-3})$, these scales persist in the original basis:

$$X_{11} \ll \frac{S_2}{Q^{7/2}}, \quad X_{12}, X_{21} \ll \frac{S_2}{Q^{9/2}}, \quad X_{22} \ll \frac{S_2}{Q^{9/2}}.$$

Step 3: the left same-point term $\mathcal{T}_-$. The baseline mixed-block scales are

$$N^{(0)} = \begin{pmatrix} O(Q) & O(Q^2) \\ O(Q^2) & O(Q^3) \end{pmatrix}, \quad G_+^{(0)-1/2} = \begin{pmatrix} O(Q^{-1/2}) & O(Q^{-3/2}) \\ O(Q^{-3/2}) & O(Q^{-3/2}) \end{pmatrix}.$$

We estimate only the off-diagonal entries, since

$$\Phi_K(Y) = Y_{12} + Y_{21}.$$

For $(\mathcal{T}_-)_{12}$, every summand is $O(S_2/Q^3)$. For example,

$$X_{11} N_{12} (G_+^{-1/2})_{22} \ll \frac{S_2}{Q^{7/2}} \cdot Q^2 \cdot Q^{-3/2} = \frac{S_2}{Q^3},$$

and

$$X_{12} N_{22} (G_+^{-1/2})_{22} \ll \frac{S_2}{Q^{9/2}} \cdot Q^3 \cdot Q^{-3/2} = \frac{S_2}{Q^3}.$$

All other terms are smaller. Thus

$$(\mathcal{T}_-)_{12} \ll \frac{S_2}{Q^3}.$$

The same argument gives

$$(\mathcal{T}_-)_{21} \ll \frac{S_2}{Q^3},$$

and hence

$$\Phi_K(\mathcal{T}_-) \ll \frac{S_2}{Q^3}.$$

By symmetry, the same estimate holds for the right same-point term:

$$\Phi_K(\mathcal{T}_+) \ll \frac{S_2}{Q^3}.$$

Step 4: the quadratic remainder. Every term in $\mathcal{E}_2$ contains at least two perturbation factors from $(R_-, \tilde{R}, R_+)$, and whitening contributes only polynomial Q -weights. Therefore

$$\Phi_K(\mathcal{E}_2) \ll \frac{S_2^2}{Q^3}.$$

By Theorem 6.1, $S_2 = o(Q)$, so this is lower order than S_2/Q^3 .

Combining the estimates for $\mathcal{T}_-$, $\mathcal{T}_{\text{mix}}$, $\mathcal{T}_+$, and $\mathcal{E}_2$ gives

$$\Phi_K(\hat{\Omega}_\zeta^{\text{corr}}(s; m) - \hat{\Omega}_\zeta^{(0)}(s; m)) \ll \frac{S_2}{Q^3},$$

as claimed. □

Corollary 9.10 (Sharpened calibration remainder estimate). *Fix a compact interval*

$$D \subset (0, \infty),$$

and work in the fixed-scaled regime

$$0 < x \leq x_0(D), \quad d \in D, \quad x = B_\zeta(m)\sigma, \quad |\sigma| \asymp Q^{-1}.$$

Write the corrected local deformation in the form

$$\Delta_\zeta(m, \sigma) = S(m)A_{\text{val}}(x) + R_\zeta(m, \sigma).$$

Then

$$|\langle A_{\text{val}}(x), R_\zeta(m, \sigma) \rangle_{\text{F}}| \ll_D \frac{x}{B_\zeta(m)} \frac{S_2}{Q^3},$$

and hence

$$|\Psi_{x,d}(R_\zeta(m, \sigma))| \ll_D \frac{S_2}{Q^3}.$$

In particular, if

$$S_2 = o(xQ^2 S(m)),$$

then

$$\Psi_{x,d}(R_\zeta(m, \sigma)) = o\left(\frac{x}{B_\zeta(m)} S(m)\right).$$

Proof. By Proposition 9.6, after subtraction of the explicit leading value-channel term $S(m)A_{\text{val}}(x)$, the remaining contribution $R_\zeta(m, \sigma)$ is internal to the same corrected transferred package controlled by the whitening perturbation analysis.

We therefore use the entry scales already established in the proof of Proposition 9.9. First, from Proposition 9.3 and the baseline inverse-square-root scales,

$$G_\pm^{(0)}(s)^{-1/2} = \begin{pmatrix} O(Q^{-1/2}) & O(Q^{-3/2}) \\ O(Q^{-3/2}) & O(Q^{-3/2}) \end{pmatrix}, \quad \tilde{R}(s) = \begin{pmatrix} O(S_2/Q^2) & O(S_2/Q) \\ O(S_2/Q) & O(S_2) \end{pmatrix},$$

one gets

$$\mathcal{T}_{\text{mix}}(s; m) = G_-^{(0)}(s)^{-1/2} \tilde{R}(s) G_+^{(0)}(s)^{-1/2} = \begin{pmatrix} O(S_2/Q^3) & O(S_2/Q^3) \\ O(S_2/Q^3) & O(S_2/Q^3) \end{pmatrix}.$$

Next, from the same proof,

$$X_\pm(s) = \begin{pmatrix} O(S_2/Q^{7/2}) & O(S_2/Q^{9/2}) \\ O(S_2/Q^{9/2}) & O(S_2/Q^{9/2}) \end{pmatrix}, \quad N^{(0)}(s) = \begin{pmatrix} O(Q) & O(Q^2) \\ O(Q^2) & O(Q^3) \end{pmatrix},$$

and multiplying once more by $G_\pm^{(0)}(s)^{-1/2}$ gives

$$\mathcal{T}_-(s; m) = \begin{pmatrix} O(S_2/Q^3) & O(S_2/Q^3) \\ O(S_2/Q^3) & O(S_2/Q^3) \end{pmatrix}, \quad \mathcal{T}_+(s; m) = \begin{pmatrix} O(S_2/Q^3) & O(S_2/Q^3) \\ O(S_2/Q^3) & O(S_2/Q^3) \end{pmatrix}.$$

Finally, every entry of the quadratic remainder $\mathcal{E}_2(s; m)$ is

$$O(S_2^2/Q^3),$$

which is lower order because $S_2 = o(Q)$ by Theorem 6.1.

Thus the part of the corrected deformation seen after removal of the leading value-channel term satisfies

$$R_\zeta(m, \sigma) = \begin{pmatrix} O(S_2/Q^3) & O(S_2/Q^3) \\ O(S_2/Q^3) & O(S_2/Q^3) \end{pmatrix}$$

up to lower-order $O(S_2^2/Q^3)$ terms.

Now use the explicit small- x structure of the value-channel derivative: by Lemma 8.3,

$$A_{\text{val}}(x) = \frac{1}{B_\zeta(m)} \begin{pmatrix} O(x^2) & O(x) \\ O(x) & O(x^2) \end{pmatrix}.$$

Therefore

$$|\langle A_{\text{val}}(x), R_\zeta(m, \sigma) \rangle_{\text{F}}| \ll \frac{x}{B_\zeta(m)} \frac{S_2}{Q^3}.$$

By Proposition 8.4,

$$|\langle A_{\text{val}}(x), M(d) \rangle_F| \asymp_D \frac{x}{B_\zeta(m)},$$

so dividing gives

$$|\Psi_{x,d}(R_\zeta(m, \sigma))| \ll_D \frac{S_2}{Q^3}.$$

Finally, since $B_\zeta(m) \asymp Q$,

$$\frac{S_2/Q^3}{(x/B_\zeta(m))S(m)} \asymp \frac{S_2}{xQ^2S(m)} \rightarrow 0$$

under the hypothesis $S_2 = o(xQ^2S(m))$. This proves the final claim. $\square$

10. ODD HOLOMORPHIC TRANSVERSE CHANNEL

Define the corrected transverse scalar

$$(20) \quad H_m(s) := \Phi_K(\widehat{\Omega}_\zeta^{\text{corr}}(s; m)).$$

By symmetry under $s \mapsto -s$, this is odd. By Corollary 8.2, the leading value channel is annihilated by Φ_K , so only the curvature, tail, and corrected finite- s remainder channels contribute.

Proposition 10.1 (Boundary estimate). *On the microscopic boundary*

$$|s| = \rho_0/Q,$$

one has

$$|H_m(s)| \ll_{\rho_0} \frac{L(m)S(m)^2}{Q^3}.$$

Proof. By Proposition 9.6, the corrected local deformation may be written in the form

$$\Delta_\zeta(m, \sigma) = S(m) A_{\text{val}}(m, \sigma) + R_\zeta(m, \sigma),$$

where $R_\zeta(m, \sigma)$ is internal to the full corrected cutoff-dependent scalar

$$H_{m,R}(s) := \Phi_K(\widehat{\Omega}_\zeta^{\text{corr},R}(s; m)).$$

By Proposition 9.7, all cutoff-dependent tail, jet-correction, and estimation contributions are already absorbed into this full corrected scalar, so no external remainder term is carried separately at the boundary stage.

Applying Φ_K to the corrected local decomposition and using Corollary 8.2,

$$\Phi_K(A_{\text{val}}(m, \sigma)) = 0,$$

gives

$$H_{m,R}(s) = \Phi_K(\widehat{\Omega}_\zeta^{\text{corr},R}(s; m)) = \Phi_K(R_\zeta(m, \sigma)).$$

Now compare the full corrected whitened block with its baseline counterpart. By Proposition 9.9,

$$\Phi_K(\widehat{\Omega}_\zeta^{\text{corr},R}(s; m) - \widehat{\Omega}_\zeta^{(0),R}(s; m)) \ll \frac{S_2}{Q^3} \quad (|s| < \rho_0/Q).$$

On the microscopic boundary $|s| = \rho_0/Q$, the tail-curvature theorem and the corrected transfer bounds imply

$$S_2 \ll L(m)S(m)^2$$

in the corrected regime. Therefore

$$|H_{m,R}(s)| \ll \frac{L(m)S(m)^2}{Q^3} \quad (|s| = \rho_0/Q).$$

Since the cutoff bookkeeping has already been absorbed into the full corrected scalar $H_{m,R}(s)$, this is the boundary estimate for the complete corrected object, not merely for a leading part plus an external tail/error remainder. This proves the claim. $\square$

Proposition 10.2 (Odd microscopic expansion). *The function $H_m(s)$ is odd and holomorphic on $|s| < \rho_0/Q$, and therefore admits an expansion*

$$H_m(z/Q^2) = \sum_{r \geq 0} c_{2r+1}(m) \frac{z^{2r+1}}{Q^{2r+4}}, \quad |z| < \rho_0 Q,$$

with

$$|c_{2r+1}(m)| \ll_{\rho_0} L(m) S(m)^2 \rho_0^{-(2r+1)}.$$

Proof. Oddness follows from symmetric placement and the jet-basis swap symmetry:

$$\widehat{\Omega}_\zeta^{\text{corr}}(-s; m) = J_0 \widehat{\Omega}_\zeta^{\text{corr}}(s; m) J_0,$$

with

$$\Phi_K(J_0 X J_0) = -\Phi_K(X).$$

Holomorphy follows from Proposition 9.5, and the coefficient bounds are immediate from Cauchy's estimate applied to Proposition 10.1. $\square$

10.1. Coefficient-level corrected-whitening transfer and exterior odd-jet localization.

Lemma 10.3 (Baseline-scaled matrix gain for corrected whitening). *Let*

$$\mathcal{W}(G_-, N, G_+) := G_-^{-1/2} N G_+^{-1/2}.$$

Assume the baseline same-point and mixed blocks satisfy

$$G_\pm^{(0)}(s) = \begin{pmatrix} O(Q) & O(1) \\ O(1) & O(Q^3) \end{pmatrix}, \quad N^{(0)}(s) = \begin{pmatrix} O(Q) & O(Q^2) \\ O(Q^2) & O(Q^3) \end{pmatrix},$$

uniformly on $|s| < \rho_0/Q$, and that $G_\pm^{(0)}(s)$ remain uniformly positive definite there.

Set

$$S_Q := \begin{pmatrix} Q^{-1/2} & 0 \\ 0 & Q^{-3/2} \end{pmatrix}, \quad \widehat{G}_\pm := S_Q^{-1} G_\pm S_Q^{-1}, \quad \widehat{N} := S_Q^{-1} N S_Q^{-1}.$$

Then

$$\widehat{G}_\pm^{(0)}(s) = \begin{pmatrix} O(1) & O(Q^{-1}) \\ O(Q^{-1}) & O(1) \end{pmatrix}, \quad \widehat{N}^{(0)}(s) = \begin{pmatrix} O(1) & O(1) \\ O(1) & O(1) \end{pmatrix},$$

and

$$\mathcal{W}(G_-, N, G_+) = S_Q \mathcal{W}(\widehat{G}_-, \widehat{N}, \widehat{G}_+) S_Q.$$

In particular, every off-diagonal entry of $\mathcal{W}(G_-, N, G_+)$ carries an explicit matrix-level factor Q^{-2} relative to the corresponding rescaled whitened block. Any additional Q -power appearing in the corrected transverse scalar must therefore come from the scalar normalization used in the definition of that scalar, not from the matrix identity itself.

Proof. The displayed bounds for $\widehat{G}_\pm^{(0)}$ and $\widehat{N}^{(0)}$ are immediate from the definitions and the baseline scales. Since

$$G_\pm = S_Q \widehat{G}_\pm S_Q,$$

functional calculus gives

$$G_\pm^{-1/2} = S_Q^{-1} \widehat{G}_\pm^{-1/2} S_Q^{-1}.$$

Hence

$$\mathcal{W}(G_-, N, G_+) = G_-^{-1/2} N G_+^{-1/2} = S_Q \widehat{G}_-^{-1/2} \widehat{N} \widehat{G}_+^{-1/2} S_Q = S_Q \mathcal{W}(\widehat{G}_-, \widehat{N}, \widehat{G}_+) S_Q.$$

The off-diagonal entries therefore acquire one factor $Q^{-1/2}$ from the upper-left entry of S_Q and one factor $Q^{-3/2}$ from the lower-right entry of S_Q , hence a total matrix-level factor Q^{-2} . $\square$

Proposition 10.4 (Uniform scaled corrected-whitening transfer). *Fix $0 < c_* < \rho_0/8$. For every Q and every radius R satisfying*

$$0 < R \leq c_* Q,$$

set

$$w := \frac{z}{R}, \quad s = \frac{z}{Q^2} = \frac{Rw}{Q^2}.$$

Let

$$\mathfrak{D}_Q(M) := \begin{pmatrix} Q^2 m_{11} & Q m_{12} \\ Q m_{21} & m_{22} \end{pmatrix}, \quad \mathfrak{D}_Q^{-1}(X) = \begin{pmatrix} Q^{-2} x_{11} & Q^{-1} x_{12} \\ Q^{-1} x_{21} & x_{22} \end{pmatrix}.$$

For $R' > 0$, let

$$A_{R'}(M_2) := \left\{ F(z) = \sum_{n \geq 0} F_n z^n : \|F\|_{A_{R'}} := \sum_{n \geq 0} \|F_n\| R'^n < \infty \right\},$$

and write $A_1(M_2)$ for the case $R' = 1$. For scalar holomorphic germs, $A_{R'}(\mathbb{C})$ is defined similarly.

Write the baseline corrected same-point and mixed blocks on the w -disk as

$$G_{\pm;Q,R}^{(0)}(w) := G_{\pm}^{(0)}(Rw/Q^2), \quad N_{Q,R}^{(0)}(w) := N^{(0)}(Rw/Q^2).$$

For a weighted perturbation triple

$$X = (X_-, Y, X_+) \in A_1(M_2)^3,$$

define

$$\delta G_{\pm} := \mathfrak{D}_Q^{-1} X_{\pm}, \quad \delta N := \mathfrak{D}_Q^{-1} Y,$$

and, whenever the corrected scalar perturbation

$$\Phi_K \left(\mathcal{W}(G_{-;Q,R}^{(0)} + \delta G_-, N_{Q,R}^{(0)} + \delta N, G_{+;Q,R}^{(0)} + \delta G_+) - \mathcal{W}(G_{-;Q,R}^{(0)}, N_{Q,R}^{(0)}, G_{+;Q,R}^{(0)}) \right)$$

vanishes at $w = 0$, define

$$\mathcal{T}_{Q,R}(X)(w) := \frac{Q}{Rw} \Phi_K \left(\mathcal{W}(G_{-;Q,R}^{(0)} + \delta G_-, N_{Q,R}^{(0)} + \delta N, G_{+;Q,R}^{(0)} + \delta G_+) - \mathcal{W}(G_{-;Q,R}^{(0)}, N_{Q,R}^{(0)}, G_{+;Q,R}^{(0)}) \right).$$

Then there exist absolute constants

$$\eta > 0, \quad C > 0, \quad B > 0$$

such that for every Q , every $0 < R \leq c_ Q$, and every*

$$X \in A_1(M_2)^3 \quad \text{with} \quad \|X_-\|_{A_1} + \|Y\|_{A_1} + \|X_+\|_{A_1} \leq \eta,$$

the quotient $\mathcal{T}_{Q,R}(X)$ is holomorphic on $|w| < 1$ and admits an absolutely convergent homogeneous expansion

$$\mathcal{T}_{Q,R}(X) = \sum_{p \geq 1} \mathcal{T}_p(X),$$

with

$$\|\mathcal{T}_p(X)\|_{A_1} \leq C B^p (\|X_-\|_{A_1} + \|Y\|_{A_1} + \|X_+\|_{A_1})^p.$$

The constants C, B, η are independent of Q and R .

Moreover, if

$$X_{\pm}(w) = \sum_{m \geq 0} X_{\pm,m} w^m, \quad Y(w) = \sum_{m \geq 0} Y_m w^m,$$

and

$$\mathcal{T}_p(X)(w) = \sum_{n \geq 0} T_{p,n} w^n,$$

then with

$$\mathbf{X}_m := \max(\|X_{-,m}\|, \|Y_m\|, \|X_{+,m}\|),$$

one has the coefficient bound

$$|T_{p,n}| \leq CB^p \sum_{\substack{M \leq n \\ m_1 + \dots + m_p = M \\ m_i \geq 0}} \sum_{m_i \geq 0} \prod_{i=1}^p \mathbf{x}_{m_i}.$$

Proof. On $|w| \leq 2$,

$$\left| \frac{Rw}{Q^2} \right| \leq \frac{2R}{Q^2} \leq \frac{2c_*}{Q} < \frac{\rho_0}{Q},$$

so the baseline blocks

$$G_{\pm;Q,R}^{(0)}(w), \quad N_{Q,R}^{(0)}(w)$$

are holomorphic and uniformly bounded on $|w| \leq 2$, with bounds independent of Q and R . By Lemma 10.3, the same-point baseline blocks remain uniformly positive definite there.

Therefore the corrected-whitening map

$$(\delta G_-, \delta N, \delta G_+) \mapsto \mathcal{W}(G_{-;Q,R}^{(0)} + \delta G_-, N_{Q,R}^{(0)} + \delta N, G_{+;Q,R}^{(0)} + \delta G_+)$$

is analytic on a fixed Banach neighborhood in $A_1(M_2)^3$, uniformly in Q, R . Since Φ_K is linear, the corrected scalar perturbation depends analytically on the input triple. For those structured perturbations for which the corrected scalar perturbation vanishes at $w = 0$, the quotient by w in the definition of $\mathcal{T}_{Q,R}$ is holomorphic. This gives the homogeneous expansion with uniform constants C, B, η .

The coefficient bound follows by testing the homogeneous maps on monomials in $A_1(M_2)$ and using the Wiener-algebra estimate

$$|[w^n]F| \leq \|F\|_{A_1}$$

for scalar coefficients. □

Proposition 10.5 (Pair-radius kernel-class decomposition for one-pair perturbations). *Fix $0 < c_* < \rho_0/8$ and $0 < c \ll 1$. For one exterior zero pair ρ , set*

$$\Delta_\rho := |2m - \gamma_\rho|, \quad R_\rho := c \min(\Delta_\rho, c_* Q).$$

Let $f_\rho(t)$ denote the one-pair phase-derivative contribution, and remove its midpoint affine jet:

$$r_\rho(t) := f_\rho(t) - f_\rho(m) - f'_\rho(m)(t - m).$$

With

$$t_\pm := m \pm \frac{z}{2Q^2},$$

define

$$d_\pm(z) := r_\rho(t_\pm), \quad e_\pm(z) := r'_\rho(t_\pm), \quad g_\pm(z) := r''_\rho(t_\pm),$$

and

$$\eta(z) := \delta \Delta_\rho(z/Q^2) := \int_{t_+}^{t_-} r_\rho(u) du.$$

Then there exist holomorphic kernels $U_\pm, E_\pm, G_\pm, V$ such that

$$d_\pm(z) = \frac{z^2}{Q^4} U_\pm(z), \quad e_\pm(z) = \frac{z}{Q^2} E_\pm(z), \quad g_\pm(z) = G_\pm(z), \quad \eta(z) = \frac{z^3}{Q^6} V(z).$$

For every fixed $H_ > 1$, if $\Delta_\rho \geq H_*$, then*

$$U_\pm, E_\pm, G_\pm, V$$

are holomorphic on $|z| < 2R_\rho$ and satisfy

$$\|U_\pm\|_{A_{R_\rho}} + \|E_\pm\|_{A_{R_\rho}} + \|G_\pm\|_{A_{R_\rho}} + \|V\|_{A_{R_\rho}} \ll \Delta_\rho^{-4}.$$

Let

$$X_\rho := (X_{\rho,-}, Y_\rho, X_{\rho,+})$$

be the weighted one-pair perturbation triple

$$X_{\rho,\pm} := \mathfrak{D}_Q(\delta G_{\rho,\pm}), \quad Y_\rho := \mathfrak{D}_Q(\delta N_\rho),$$

where $\delta G_{\rho,\pm}$ and δN_ρ are the actual one-pair perturbations of the corrected same-point and mixed blocks in the draft.

Then, on $|z| < 2R_\rho$, every entry of every block of $X_\rho(z)$ is linear in

$$U_-, U_+, E_-, E_+, G_-, G_+, V$$

modulo terms containing at least two such pair kernels. In particular,

$$\|X_\rho(R_\rho \cdot)\|_{A_1} \ll \Delta_\rho^{-4}.$$

Proof. The one-pair poles lie at distance $\asymp \Delta_\rho$ in the s -plane, hence at distance $\asymp Q^2 \Delta_\rho$ in the z -plane. Since

$$R_\rho \ll \min(\Delta_\rho, Q),$$

the kernels $U_\pm, E_\pm, G_\pm, V$ are holomorphic on $|z| < 2R_\rho$, and Cauchy gives the displayed A_{R_ρ} -bounds.

Now check the actual corrected formulas.

For the corrected same-point block,

$$G_{m,\pm}^{\text{corr}}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_\pm) & \frac{1}{2}q'(t_\pm) \\ \frac{1}{2}q'(t_\pm) & \frac{1}{12}(q''(t_\pm) + 2q(t_\pm)^3) \end{pmatrix}.$$

Subtracting the baseline and using

$$q = q_0 + d_\pm, \quad q' = q'_0 + e_\pm, \quad q'' = q''_0 + g_\pm,$$

the weighted perturbation entries are:

$$(1, 1) \sim (z/Q)^2 U_\pm, \quad (1, 2), (2, 1) \sim (z/Q) E_\pm,$$

and

$$(2, 2) \sim G_\pm + \left(\frac{q_0}{Q}\right)^2 (z/Q)^2 U_\pm + \left(\frac{q_0}{Q}\right) (z/Q)^4 U_\pm^2 + (z/Q)^6 U_\pm^3.$$

On $|z| \leq 2R_\rho$, the normalized baseline factors

$$\frac{q_0 z}{Q^2}, \quad \frac{q_0^2 z^2}{Q^4}$$

are $O(1)$, so the same-point block is linear in the pair kernels modulo terms with at least two pair kernels.

For the corrected mixed block,

$$N_m^{\text{corr}}(s) = \frac{1}{\pi} \begin{pmatrix} -\frac{2 \sin \Delta}{s} & \frac{\sin \Delta + q(t_+) s \cos \Delta}{s^2} \\ -\frac{\sin \Delta + q(t_-) s \cos \Delta}{s^2} & \frac{(q(t_-) + q(t_+)) s \cos \Delta + (2 - q(t_-) q(t_+) s^2) \sin \Delta}{2s^3} \end{pmatrix}.$$

Write

$$q_\pm = q_{0,\pm} + d_\pm, \quad \Delta = \Delta_0 + \eta.$$

For the $(1, 1)$ -entry, since

$$\eta(z) = \frac{z^3}{Q^6} V(z) = s^3 V(z),$$

$$Q^2 \frac{\sin(\Delta_0 + \eta) - \sin \Delta_0}{s}$$

is linear in V modulo higher pair order.

For the off-diagonal entries, the numerator difference is

$$(\sin(\Delta_0 + \eta) - \sin \Delta_0) + d_\pm s \cos(\Delta_0 + \eta) + q_{0,\pm} s (\cos(\Delta_0 + \eta) - \cos \Delta_0),$$

and after multiplying by Q/s^2 the linear terms are bounded-baseline times $U_\pm$ and V .

For the $(2, 2)$ -entry, define

$$F(q_-, q_+, \Delta; s) := (q_- + q_+) s \cos \Delta + (2 - q_- q_+ s^2) \sin \Delta.$$

Then the perturbation is

$$\frac{F(q_{0,-} + d_-, q_{0,+} + d_+, \Delta_0 + \eta; s) - F(q_{0,-}, q_{0,+}, \Delta_0; s)}{s^3}.$$

Taylor expansion in (d_-, d_+, η) around the baseline gives

$$\partial_{q_-} F_0 d_- + \partial_{q_+} F_0 d_+ + \partial_{\Delta} F_0 \eta$$

plus a remainder quadratic in (d_-, d_+, η) . After dividing by s^3 , the linear part is

$$\tilde{B}_-(z) U_-(z) + \tilde{B}_+(z) U_+(z) + \tilde{B}_{\Delta}(z) V(z),$$

with bounded baseline coefficients $\tilde{B}_-, \tilde{B}_+, \tilde{B}_{\Delta}$, and the remainder carries at least two pair kernels.

So every weighted mixed-block entry is also linear in the pair kernels modulo higher pair order.

Finally, on $|z| \leq 2R_{\rho}$, every factor z/Q is $O(1)$, every normalized baseline factor is $O(1)$, and each pair kernel has $A_{R_{\rho}}$ norm $O(\Delta_{\rho}^{-4})$. Therefore

$$\|X_{\rho}(R_{\rho} \cdot)\|_{A_1} \ll \Delta_{\rho}^{-4}.$$

□

Definition 10.6 (Tagged two-atom pre-whitening source block). Introduce formal source tags τ_1, τ_2 . For a normalized two-atom core $(a_1, h_1; a_2, h_2)$, define the tagged corrected source perturbations by additive substitution into the corrected phase variables:

$$\begin{aligned} d_{\pm}^{(2)} &:= \tau_1 a_1 d_{h_1, \pm} + \tau_2 a_2 d_{h_2, \pm}, & e_{\pm}^{(2)} &:= \tau_1 a_1 e_{h_1, \pm} + \tau_2 a_2 e_{h_2, \pm}, \\ g_{\pm}^{(2)} &:= \tau_1 a_1 g_{h_1, \pm} + \tau_2 a_2 g_{h_2, \pm}, & \eta^{(2)} &:= \tau_1 a_1 \eta_{h_1} + \tau_2 a_2 \eta_{h_2}. \end{aligned}$$

Equivalently,

$$q_{2, \pm} = q_{0, \pm} + d_{\pm}^{(2)}, \quad q'_{2, \pm} = q'_{0, \pm} + e_{\pm}^{(2)}, \quad q''_{2, \pm} = q''_{0, \pm} + g_{\pm}^{(2)}, \quad \Delta_2 = \Delta_0 + \eta^{(2)}.$$

Define

$$\mathcal{B}_2(a_1, h_1; a_2, h_2) := (G_{2,-}^{\text{corr}}, N_2^{\text{corr}}, G_{2,+}^{\text{corr}})$$

by the same corrected same-point and mixed block formulas used above for the one-pair corrected block, with $q_{\pm}, \Delta$ replaced by $q_{2, \pm}, \Delta_2$. Set

$$\mathcal{B}_0 := (G_-^{(0)}, N^{(0)}, G_+^{(0)}).$$

Definition 10.7 (Pair-kernel linear part). Let $\mathcal{K}$ be the filtration on entries of $\mathfrak{D}_Q(\mathcal{B}_2 - \mathcal{B}_0)$ in which each tagged source kernel

$$\tau_i U_{i, \pm}, \quad \tau_i E_{i, \pm}, \quad \tau_i G_{i, \pm}, \quad \tau_i V_i$$

has $\mathcal{K}$ -degree 1, while baseline factors, the amplitudes a_i , Q -scales, and bounded analytic coefficients have degree 0. Let $\text{Lin}_{\mathcal{K}}$ denote projection to $\mathcal{K}$ -degree 1.

Define the one-atom source-linear triple by

$$L_h := \text{Lin}_{\mathcal{K}} \mathfrak{D}_Q(\mathcal{B}_1(h) - \mathcal{B}_0),$$

where $\mathcal{B}_1(h)$ is the one-atom corrected block obtained from the same corrected block formulas.

Lemma 10.8 (Tagged pair-kernel linearity of the source block). *With source tags retained,*

$$\text{Lin}_{\mathcal{K}} \mathfrak{D}_Q(\mathcal{B}_2(a_1, h_1; a_2, h_2) - \mathcal{B}_0) = \tau_1 a_1 L_{h_1} + \tau_2 a_2 L_{h_2}.$$

After this tagged $\mathcal{K}$ -linear projection is taken, applying the augmentation $\tau_1, \tau_2 \mapsto 1$ gives

$$\text{Lin}_{\mathcal{K}} \mathfrak{D}_Q(\mathcal{B}_2 - \mathcal{B}_0) = a_1 L_{h_1} + a_2 L_{h_2}.$$

Proof. For the same-point blocks $G_{2,\pm}^{\text{corr}}$, the $(1,1)$ and off-diagonal entries are linear in $d_{\pm}^{(2)}$ and $e_{\pm}^{(2)}$. The $(2,2)$ entry has the form

$$g_{\pm}^{(2)} + \text{baseline} \cdot d_{\pm}^{(2)} + \text{terms of } \mathcal{K}\text{-degree at least 2},$$

by the same expansion as in the one-pair corrected block calculation.

For the mixed block N_2^{corr} , Taylor expansion in $(d_-^{(2)}, d_+^{(2)}, \eta^{(2)})$ gives a bounded-baseline linear combination of those variables and a remainder of $\mathcal{K}$ -degree at least 2. Since the tagged variables are additive sums of the two source contributions, the $\mathcal{K}$ -linear part is the sum of the two one-atom $\mathcal{K}$ -linear parts, with tags still attached. $\square$

Remark 10.9 (Scope of the tagged source block). The definitions and lemma above are pre-whitening statements. They prove tagged $\mathcal{K}$ -linear additivity of the corrected source block before the nonlinear corrected-whitening transfer

$$\mathcal{W}(G_-, N, G_+) = G_-^{-1/2} N G_+^{-1/2}.$$

They do not by themselves prove transfer-domain vanishing, source-bidegree edge divisibility, quotient-output package coincidence, diagonal merger, the coefficient B_7^f , the quotient $Q_{7,m}^q$, or the projector $\Pi_{1,1}$.

In particular, the tagged pre-whitening block does not automatically induce the homogeneous scalar-grade source tower used in Corollary B.21. That identification is a separate source-model theorem for the actual $\mathfrak{D}_Q$ -scaled corrected-whitening package.

Remark 10.10 (Corrected package map for the two-point and mixed four-point frontiers). The current local frontier is best viewed as a single corrected package problem, not as two independent mixed-geometry problems. The two-point mixed branch and the mixed four-point branch both require the same source-provenance package:

$$\mathcal{B}_2 \longrightarrow C_{\mathcal{W}} \longrightarrow B_7^f \longrightarrow Q_{7,m}^q \longrightarrow \mathcal{H}_7^q.$$

Here $\mathcal{B}_2$ is the tagged two-atom pre-whitening source block of Definition 10.6, $C_{\mathcal{W}}$ is the corrected-whitening matrix cross-effect before Φ_K of Definition C.4, B_7^f is the pre-determinant fixed-sector order-7 coefficient of Definition C.9, and

$$Q_{7,m}^q(C) := [B_7^f(C)] \in \mathfrak{f}/\mathbf{CA}_5^f(m)$$

is the fixed midpoint quotient class. The normalized corrected package representative $C_{\mathcal{W}}^{\text{corr}}$ and its moving fixed-sector representative B_{pkg}^f are defined in Section C.

For the two-point branch, the remaining source input is finite: prove swap symmetry, one-amplitude collapse, and diagonal merger for the actual corrected two-atom package. Once these hold, the quadratic collision factorization

$$\mathcal{I}_2 = a_1 a_2 (h_1 - h_2)^2 \mathcal{J}_2$$

is formal. The role of the order-7 quotient package is to make the corresponding septic quotient defect an actual fixed-target object rather than an uncontrolled two-atom mixed representative.

For the mixed four-point branch, the current reduction says that a genuinely mixed clean branch cannot persist independently in the region

$$\{T_v \neq 0\} \cap \{\mathcal{R} \neq 0\}.$$

Thus every clean mixed branch must exit through one of the scalar walls

$$T_v = 0 \quad \text{or} \quad \mathcal{R} = 0.$$

The package theorem is needed to prevent the corrected source cross-effect from creating a new non-scalar fixed-sector direction in the quotient

$$\mathfrak{f}/\mathbf{CA}_5^f(m).$$

Consequently, the local target is not a scalarized identity after Φ_K . It is the pre- Φ_K , pre-determinant quotient-gauge statement

$$B_7^f(C_W^{\text{corr}}) \in \mathbf{CA}_5^f(m).$$

If one keeps a full corrected package before source-bidegree projection, the equivalent full-package formulation is obtained by applying $(1 - \Pi_{1,1})_{\text{mix}}$ before fixed-sector extraction and quotienting. This implication does not by itself prove that the scalar exits

$$T_v = 0 \quad \text{or} \quad \mathcal{R} = 0$$

are impossible. Its role is narrower: it removes the possibility that non- $(1,1)$ corrected source terms create an independent fixed-sector quotient direction. After that removal, the existing mixed four-point capture theorem reduces every clean mixed branch to the scalar exits, and the two-point residual is pushed into the midpoint quotient edge package.

If this package theorem is proved, the residual two-point mixed obstruction is absorbed into the midpoint quotient edge package, and the mixed four-point source-coupled walls have no independent clean local carrier beyond the scalar exits above.

Remark 10.11 (Conditional role of the corrected package in the mixed branches). The corrected package theorem does not by itself prove RH. Its role is to remove the residual local mixed carrier. If the package theorem closes, then the two-point mixed obstruction reduces to the fixed midpoint quotient edge package, and the mixed four-point branch has no independent clean local carrier outside the scalar exits

$$T_v = 0 \quad \text{or} \quad \mathcal{R} = 0.$$

The remaining global proof still requires the finite-core lower-model transfer or a separate treatment of the near-zero-anomaly branch $\mathcal{N}_k(m, Q)$, hidden-state/ T -horizontal odd-block extraction, and the downstream endgame reductions, including the source-level interaction branch controlled by $J_1 = [s]\Phi_K(\mathcal{I}_{\geq 2})$.

Remark 10.12 (Sensitivity of the mixed four-point branch to Gate B). The mixed four-point branch is sensitive only to the fixed-sector quotient class of a Gate B residual. Let

$$E = eI + fS \in \mathfrak{f}_h.$$

If

$$E \in \mathbf{CA}_5^f(h),$$

then E is invisible in the quotient package. On a good patch, this is equivalently

$$ev_5 - u_5f = 0, \quad A_5^f(h) = u_5I + v_5S.$$

If instead

$$ev_5 - u_5f \neq 0,$$

then E shifts the order-7 quotient scalar:

$$\Delta_7 \mapsto \Delta_7 + (ev_5 - u_5f).$$

Consequently $\lambda = \Delta_7/v_5$ and the downstream scalar package $\mathcal{Z}, \mathcal{R}, T_v$ generally change. Thus a non-gauge Gate B residual would create precisely the independent fixed-sector quotient carrier that the corrected package theorem is designed to remove.

The mixed four-point scalar-exit theorem should therefore be read after Gate B has removed non- $(1,1)$ quotient directions. It is not a substitute for Gate B.

Theorem 10.13 (Mixed four-point dependence descends to the quotient determinant). *Work on a good fixed-sector patch on which*

$$c(h) \neq 0, \quad A_5^f(h) = u_5(h)I + v_5(h)S \neq 0.$$

Let

$$E = eI + fS \in \mathfrak{f}_h$$

be an order-7 fixed-sector residual added to

$$A_7^f = u_7 I + v_7 S,$$

with c, u_5, v_5 fixed. Then the mixed four-point scalar-exit package sees E only through the quotient determinant pairing

$$\omega_h(E, A_5^f(h)) = ev_5 - u_5 f.$$

Equivalently,

$$\Delta_7 = u_7 v_5 - u_5 v_7$$

is transformed by

$$\Delta_7 \mapsto \Delta_7 + \omega_h(E, A_5^f(h)),$$

and every lifted v_5 -patch scalar

$$Q_v, \quad T_v, \quad \mathcal{Z}, \quad \mathcal{R}$$

changes only through this induced change of Δ_7 .

In particular, if

$$E \in \mathbf{C}A_5^f(h),$$

then E is invisible to the mixed four-point geometry. If

$$[E] \neq 0 \quad \text{in} \quad \mathfrak{f}_h/\mathbf{C}A_5^f(h),$$

then the resulting mixed four-point effect is exactly the already-defined quotient-septic obstruction, expressed through Δ_7 . No independent mixed four-point carrier is produced.

Proof. The lifted mixed four-point branch is built, on a v_5 -patch, from

$$x = \frac{v_5}{c}, \quad Y = \frac{u_5}{c}, \quad S = \sigma_v = \frac{\Delta_7}{c v_5},$$

and hence from

$$Q_v = (Y, x, S).$$

The scalar exits $T_v, \mathcal{Z}, \mathcal{R}$ are functions of these data and their x -derivatives. Thus the only possible order-7 input from A_7^f is through Δ_7 .

Replacing A_7^f by $A_7^f + E$, with $E = eI + fS$, changes

$$\Delta_7 = \omega_h(A_7^f, A_5^f)$$

by

$$\omega_h(E, A_5^f) = ev_5 - u_5 f.$$

The same argument on a u_5 -patch uses the parallel quotient coordinate

$$-\frac{\Delta_7}{c u_5}.$$

Hence no raw representative data (u_7, v_7) or $\Phi_K(A_7^f)$ enter as independent mixed four-point inputs. $\square$

Corollary 10.14 (Connection residuals are quotient determinant failures). *For $M \in \{M_A, M_C\}$, set*

$$q_{\text{tot},7}[M] := R_7^{(M)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M], A_i).$$

If

$$q_{\text{tot},7}[M_A] = q_{\text{tot},7}[M_C] = 0,$$

then the residual monomials M_A, M_C create no mixed four-point carrier beyond the scalar exits already present in the lifted branch.

If one of these two scalars is nonzero, the obstruction is exactly a failure of quotient determinant closure:

$$\Delta_7 \mapsto \Delta_7 + q_{\text{tot},7}[M].$$

It is not a separate four-point mechanism.

Remark 10.15 (Three data layers in the mixed four-point audit). The mixed four-point carrier audit separates three logically different layers:

Layer	Data	Role
pre- Φ_K quotient package	$[B_7^f] \in \mathfrak{f}_m / \mathbf{C}A_5^f(m)$	Gate B / UV-026 obstruction ob
Φ_K -visible hidden state	$A_7^f \mapsto A_7^f + \chi A_5^f, \quad \Phi_K(\chi A_5^f) = 2\chi v_5$	visible to transform extraction, quotient
mixed four-point scalar exits	$Q_v, T_v, \mathcal{Z}, \mathcal{R}$ built from c, u_5, v_5, Δ_7	downstream four-point branch geo

Thus quotient invisibility is enough to remove an order-7 corrected source residual from the mixed four-point branch, but it is not by itself enough to solve the package-to-transform hidden-state bridge.

Theorem 10.16 (Mixed four-point factorization through quotient determinant). *Let*

$$A_5^f = u_5 I + v_5 S, \quad A_7^f = u_7 I + v_7 S, \quad c \neq 0,$$

and define

$$\Delta_7 = \omega_h(A_7^f, A_5^f) = u_7 v_5 - u_5 v_7, \quad T = \frac{v_7}{c}.$$

On every good mixed four-point patch, the mixed four-point scalar exits depend only on

$$c, \quad u_5, \quad v_5, \quad \Delta_7.$$

They do not depend on T independently.

Proof. On a $v_5 \neq 0$ patch, the mixed four-point branch is written through the normalized variables

$$x = \frac{v_5}{c}, \quad Y = \frac{u_5}{c}, \quad \sigma_v = \frac{\Delta_7}{c v_5}.$$

All scalar exits are functions of these quantities and their local derivatives. Thus they are determined by

$$c, \quad u_5, \quad v_5, \quad \Delta_7,$$

and do not require u_7 or v_7 separately.

On a $u_5 \neq 0$ patch, one uses the parallel quotient coordinate

$$\sigma_u = -\frac{\Delta_7}{c u_5},$$

and the same factorization holds.

For an order-seven fixed-sector residual

$$E = eI + fS,$$

adding E to A_7^f changes the determinant by

$$\Delta_7 \mapsto \Delta_7 + \omega_h(E, A_5^f) = \Delta_7 + e v_5 - u_5 f.$$

The hidden coordinate changes by

$$T \mapsto T + \frac{f}{c}.$$

Thus the mixed four-point branch sees only the determinant projection $e v_5 - u_5 f$, while the T -coordinate records a separate Φ_K -visible projection. $\square$

Theorem 10.17 (Actual zeta-source mixed four-point bridge). *Assume the corrected Hardy-source package audit, Definition E.138. Then the actual zeta-source correction creates no independent mixed four-point fixed-sector carrier. More precisely,*

$$\left[B_7^f(C_{\mathcal{W}}^{\text{corr}, \zeta}) \right] = 0 \quad \text{in } \mathfrak{f} / \mathbf{C}A_5^f(m),$$

up to $O(Q^{-B})$ residuals. Hence the mixed four-point branch sees only the already-defined quotient determinant channel and exits through the existing scalar walls

$$T_v = 0 \quad \text{or} \quad \mathcal{R} = 0.$$

Proof. The fixed-sector quotient nullity clause of the corrected Hardy-source package audit says exactly that

$$B_7^{\mathfrak{f}}(C_{\mathcal{W}}^{\text{corr}, \zeta}) \in \mathbf{CA}_5^{\mathfrak{f}}(m) + O(Q^{-B}).$$

Passing to the quotient $\mathfrak{f}/\mathbf{CA}_5^{\mathfrak{f}}(m)$ gives the displayed vanishing. The existing mixed four-point quotient-determinant theorem then applies without introducing a new carrier. $\square$

Remark 10.18 (No independent T -dependency in mixed four-point). A pure quotient-line shift

$$E = \chi A_5^{\mathfrak{f}}$$

has

$$\omega_h(E, A_5^{\mathfrak{f}}) = 0,$$

and is therefore invisible to mixed four-point geometry. It changes

$$T$$

by

$$T \longmapsto T + \chi \frac{v_5}{c},$$

and is generally Φ_K -visible on $v_5 \neq 0$. Hence mixed four-point closure is a quotient-determinant problem, whereas two-point transform closure also requires T -horizontal compatibility.

Corollary 10.19 (Single-pair exterior odd-coefficient bound). *Fix $0 < c_* < \rho_0/8$. Let $h_\rho(s)$ be the corrected transverse scalar contributed by one exterior zero pair at absolute distance*

$$\Delta_\rho := |2m - \gamma_\rho|,$$

and write

$$h_\rho(z/Q^2) = \sum_{r \geq 0} a_{2r+1}(\rho) \frac{z^{2r+1}}{Q^{2r+4}}.$$

Then for every fixed $H_* > 1$, if $\Delta_\rho \geq H_*$, one has

$$|a_{2r+1}(\rho)| \ll \frac{1}{Q} \begin{cases} \Delta_\rho^{-(2r+4)}, & \Delta_\rho \leq c_* Q, \\ (c_* Q)^{-2r} \Delta_\rho^{-4}, & \Delta_\rho > c_* Q. \end{cases}$$

Proof. Let

$$R_\rho := c \min(\Delta_\rho, c_* Q).$$

By Proposition 10.5,

$$\|X_\rho(R_\rho \cdot)\|_{A_1} \ll \Delta_\rho^{-4}.$$

For H_* large enough, this lies inside the uniform small ball from Proposition 10.4. Since the one-pair corrected scalar $h_\rho(s)$ is odd in s , the corresponding corrected scalar perturbation vanishes at $w = 0$, so the quotient map $\mathcal{T}_{Q, R_\rho}$ is holomorphic on the one-pair input. Therefore the normalized even germ

$$g_\rho(z) := \frac{Q^3}{z} h_\rho(z/Q^2)$$

satisfies

$$\|g_\rho(R_\rho \cdot)\|_{A_1} \ll \Delta_\rho^{-4}.$$

Hence

$$[z^{2r}] g_\rho(z) \ll R_\rho^{-2r} \Delta_\rho^{-4}.$$

Since

$$g_\rho(z) = \sum_{r \geq 0} \frac{a_{2r+1}(\rho)}{Q^{2r+1}} z^{2r},$$

this gives

$$|a_{2r+1}(\rho)| \ll Q^{2r+1} R_\rho^{-2r} \Delta_\rho^{-4}.$$

Using

$$R_\rho \asymp \min(\Delta_\rho, c_* Q)$$

yields the displayed hybrid bound. $\square$

Lemma 10.20 (Local zero-counting in dyadic shells). *Let*

$$\mathcal{S}_k := \{\rho : 2^k H < \Delta_\rho \leq 2^{k+1} H\}, \quad \Delta_\rho := |2m - \gamma_\rho|, \quad k \geq 0.$$

Then

$$\#\mathcal{S}_k \ll Q(2^k H + 1),$$

where $Q \asymp \log |m|$.

Proof. This is the local zero-counting estimate already used elsewhere in the draft, applied to the interval of ordinates

$$[2m - 2^{k+1} H, 2m + 2^{k+1} H].$$

Its length is $O(2^k H + 1)$, and the local zero density at height m is $O(Q)$. $\square$

Corollary 10.21 (Exterior higher odd-coefficient bound). *Let*

$$H_{\text{out}}(z/Q^2) = \sum_{r \geq 0} a_{2r+1}^{\text{out}}(m, H) \frac{z^{2r+1}}{Q^{2r+4}}$$

be the corrected transverse scalar contributed by all zeros outside absolute radius H from the local center. Then for every $r \geq 0$ and every $H \geq H_$,*

$$|a_{2r+1}^{\text{out}}(m, H)| \ll H^{-(2r+3)} + Q^{-(2r+2)}.$$

Proof. Decompose the exterior zeros into dyadic shells

$$\mathcal{S}_k := \{\rho : 2^k H < \Delta_\rho \leq 2^{k+1} H\}, \quad k \geq 0.$$

By Lemma 10.20,

$$\#\mathcal{S}_k \ll Q(2^k H + 1).$$

For shells with $2^{k+1} H \leq c_* Q$, Corollary 10.19 gives

$$\sum_{\rho \in \mathcal{S}_k} |a_{2r+1}(\rho)| \ll Q(2^k H + 1) \cdot \frac{1}{Q(2^k H)^{2r+4}} \ll (2^k H)^{-(2r+3)}.$$

Summing these shells yields

$$\ll H^{-(2r+3)}.$$

For shells with $2^k H > c_* Q$, Corollary 10.19 gives

$$\sum_{\rho \in \mathcal{S}_k} |a_{2r+1}(\rho)| \ll Q(2^k H + 1) \cdot \frac{1}{Q(c_* Q)^{2r}(2^k H)^4} \ll (c_* Q)^{-2r} (2^k H)^{-3}.$$

Summing the far shells gives

$$\ll Q^{-2r-2}.$$

Adding the near and far ranges proves the claim. $\square$

Corollary 10.22 (Core localization of the extracted odd jet). *Let $N = \lfloor \alpha Q \rfloor$ with $\alpha > 0$, and let $a_{2N-1}^{\leq H}(m)$ denote the coefficient contributed by the zeros with $\Delta_\rho \leq H$. Then for every fixed $H \geq H_*$,*

$$a_{2N-1}(m) = a_{2N-1}^{\leq H}(m) + O(H^{-(2N+1)} + Q^{-2N}).$$

In particular, the first surviving corrected odd coefficient is exponentially localized to any fixed finite core.

Proof. This is exactly Corollary 10.21 with $r = N - 1$. $\square$

11. N -POINT ODD-MOMENT CANCELLATION

For $N \leq \rho_0 Q$, set

$$s_j = \frac{j}{Q^2}, \quad j = 1, \dots, N,$$

and define

$$a_j^{(N)} := (-1)^{j+1} \frac{j \binom{2N}{N+j}}{\binom{2N-2}{N-1}}, \quad 1 \leq j \leq N.$$

Define the corrected N -point transverse scalar

$$(21) \quad \Xi_\zeta^{(N)}(m) := \sum_{j=1}^N a_j^{(N)} H_m(s_j).$$

Lemma 11.1 (Explicit odd-moment-canceling coefficients and ℓ^1 control). *The coefficients $a_j^{(N)}$ satisfy*

$$\sum_{j=1}^N a_j^{(N)} = 1, \quad \sum_{j=1}^N a_j^{(N)} j^{2r+1} = 0 \quad (0 \leq r \leq N-2).$$

Moreover,

$$\sum_{j=1}^N |a_j^{(N)}| = 2N - 1.$$

Consequently, for every integer $k \geq 0$,

$$\left| \sum_{j=1}^N a_j^{(N)} j^k \right| \leq (2N - 1) N^k.$$

Proof. Set

$$D_N := \binom{2N-2}{N-1}.$$

We first prove the normalization. Writing

$$j \binom{2N}{N+j} = N \left(\binom{2N-1}{N+j-1} - \binom{2N-1}{N+j} \right),$$

one gets

$$\sum_{j=1}^N (-1)^{j+1} j \binom{2N}{N+j} = N \sum_{j=1}^N (-1)^{j+1} \left(\binom{2N-1}{N+j-1} - \binom{2N-1}{N+j} \right).$$

Using the partial alternating-binomial identity

$$\sum_{k=0}^m (-1)^k \binom{n}{k} = (-1)^m \binom{n-1}{m},$$

one finds

$$\sum_{k=N}^{2N-1} (-1)^k \binom{2N-1}{k} = (-1)^N \binom{2N-2}{N-1}.$$

Substituting this into the preceding display yields

$$\sum_{j=1}^N (-1)^{j+1} j \binom{2N}{N+j} = \binom{2N-2}{N-1} = D_N,$$

hence

$$\sum_{j=1}^N a_j^{(N)} = 1.$$

Next, for $0 \leq r \leq N - 2$, apply the standard finite-difference identity

$$\sum_{k=0}^{2N} (-1)^k \binom{2N}{k} P(k) = 0$$

to the polynomial

$$P(k) = (k - N)^{2r+2}, \quad \deg P = 2r + 2 < 2N.$$

After the change of variables $j = k - N$, this gives

$$\sum_{j=-N}^N (-1)^j \binom{2N}{N+j} j^{2r+2} = 0.$$

Since $2r + 2$ is even and the $j = 0$ term vanishes,

$$\sum_{j=1}^N (-1)^j \binom{2N}{N+j} j^{2r+2} = 0.$$

Dividing by D_N yields

$$\sum_{j=1}^N a_j^{(N)} j^{2r+1} = 0 \quad (0 \leq r \leq N - 2).$$

Finally, since the signs alternate,

$$\sum_{j=1}^N |a_j^{(N)}| = \frac{1}{D_N} \sum_{j=1}^N j \binom{2N}{N+j}.$$

Using again

$$j \binom{2N}{N+j} = N \left(\binom{2N-1}{N+j-1} - \binom{2N-1}{N+j} \right),$$

this telescopes to

$$\sum_{j=1}^N j \binom{2N}{N+j} = N \binom{2N-1}{N}.$$

Therefore

$$\sum_{j=1}^N |a_j^{(N)}| = N \frac{\binom{2N-1}{N}}{\binom{2N-2}{N-1}} = 2N - 1.$$

The final moment bound is immediate from

$$\left| \sum_{j=1}^N a_j^{(N)} j^k \right| \leq \sum_{j=1}^N |a_j^{(N)}| j^k \leq (2N - 1) N^k. \quad \square$$

Lemma 11.2 (Closed-form Mellin symbol of the discrete N -point projector). *Define*

$$A_N(t) := \sum_{j=1}^N a_j^{(N)} t^j, \quad \hat{\mu}_{N,Q}(z) := A_N(e^{z/Q^2}).$$

Then

$$A_N(t) = \frac{2(2N-1)}{N+1} {}_2F_1(1-N, 2; N+2; t),$$

so in particular

$$\hat{\mu}_{N,Q}(z) = \frac{2(2N-1)}{N+1} e^{z/Q^2} {}_2F_1(1-N, 2; N+2; e^{z/Q^2}).$$

Moreover,

$$A_N(t) - A_N(t^{-1}) = \frac{N(-1)^N}{\binom{2N-2}{N-1}} t^{-N} (1-t)^{2N-1} (1+t),$$

and hence

$$\widehat{\mu}_{N,Q}(z) - \widehat{\mu}_{N,Q}(-z) = \frac{N(-1)^N}{\binom{2N-2}{N-1}} e^{-Nz/Q^2} (1 - e^{z/Q^2})^{2N-1} (1 + e^{z/Q^2}).$$

In particular,

$$\widehat{\mu}_{N,Q}(0) = 1, \quad \partial_z^{2r+1} \widehat{\mu}_{N,Q}(0) = 0 \quad (0 \leq r \leq N-2),$$

and the odd part of $\widehat{\mu}_{N,Q}$ starts only at order $2N-1$.

Proof. Let

$$B_N(t) := \frac{2(2N-1)}{N+1} t {}_2F_1(1-N, 2; N+2; t) = \sum_{j=1}^N b_j^{(N)} t^j.$$

Its coefficients satisfy

$$\frac{b_{j+1}^{(N)}}{b_j^{(N)}} = \frac{(1-N+j-1)(2+j-1)}{(N+2+j-1)j} = \frac{(j-N)(j+1)}{j(N+j+1)}.$$

On the other hand, the coefficients $a_j^{(N)}$ satisfy

$$\frac{a_{j+1}^{(N)}}{a_j^{(N)}} = \frac{(-1)^{j+2}(j+1)\binom{2N}{N+j+1}}{(-1)^{j+1}j\binom{2N}{N+j}} = \frac{(j-N)(j+1)}{j(N+j+1)}.$$

Thus $a_j^{(N)}$ and $b_j^{(N)}$ agree up to an overall constant. Matching the t^1 -coefficient gives

$$b_1^{(N)} = \frac{2(2N-1)}{N+1} = \frac{\binom{2N}{N+1}}{\binom{2N-2}{N-1}} = a_1^{(N)},$$

hence $A_N(t) = B_N(t)$.

For the second identity, define

$$F_N(t) := \sum_{j=-N}^N (-1)^j \binom{2N}{N+j} t^j.$$

By the binomial theorem,

$$F_N(t) = (-1)^N t^{-N} (1-t)^{2N}.$$

Differentiating gives

$$tF'_N(t) = \sum_{j=-N}^N j(-1)^j \binom{2N}{N+j} t^j = -N(-1)^N t^{-N} (1-t)^{2N-1} (1+t).$$

Now

$$A_N(t) - A_N(t^{-1}) = -\frac{1}{\binom{2N-2}{N-1}} \sum_{j=1}^N j(-1)^j \binom{2N}{N+j} (t^j - t^{-j}),$$

while the symmetry $j \mapsto -j$ in the preceding display shows

$$tF'_N(t) = \sum_{j=1}^N j(-1)^j \binom{2N}{N+j} (t^j - t^{-j}).$$

Substituting the closed form for $tF'_N(t)$ yields

$$A_N(t) - A_N(t^{-1}) = \frac{N(-1)^N}{\binom{2N-2}{N-1}} t^{-N} (1-t)^{2N-1} (1+t).$$

Replacing t by e^{z/Q^2} gives the displayed formula for $\widehat{\mu}_{N,Q}(z) - \widehat{\mu}_{N,Q}(-z)$. The normalization $\widehat{\mu}_{N,Q}(0) = 1$ and the low odd-jet vanishing are equivalent to the moment identities in Lemma 11.1. $\square$

Lemma 11.3 (Exact odd-moment generating function). *Define*

$$\mu_k^{(N)} := \sum_{j=1}^N a_j^{(N)} j^k,$$

and, for $|u| < 1/N$,

$$\mathcal{M}_N(u) := \sum_{r \geq 0} \mu_{2r+1}^{(N)} u^{2r+1} = \sum_{j=1}^N \frac{a_j^{(N)} j u}{1 - j^2 u^2}.$$

Then

$$\mathcal{M}_N(u) = (-1)^{N+1} (2N-1)N!(N-1)! \frac{u^{2N-1}}{\prod_{j=1}^N (1 - j^2 u^2)}.$$

In particular, for every real $x > N$,

$$\sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| x^{-(2r+1)} = (2N-1)N!(N-1)! x^2 \frac{\Gamma(x-N)}{\Gamma(x+N+1)}.$$

Proof. The identity

$$\mathcal{M}_N(u) = \sum_{j=1}^N \frac{a_j^{(N)} j u}{1 - j^2 u^2}$$

is immediate from summing the geometric series

$$\frac{j u}{1 - j^2 u^2} = \sum_{r \geq 0} j^{2r+1} u^{2r+1} \quad (|u| < 1/j).$$

Set

$$D_N(u) := \prod_{j=1}^N (1 - j^2 u^2) \mathcal{M}_N(u).$$

Since $\mathcal{M}_N$ is a sum of simple fractions, $D_N(u)$ is an odd polynomial of degree at most $2N-1$. By Lemma 11.1,

$$\mu_{2r+1}^{(N)} = 0 \quad (0 \leq r \leq N-2),$$

so $\mathcal{M}_N(u) = O(u^{2N-1})$ as $u \rightarrow 0$. Hence

$$D_N(u) = C_N u^{2N-1}$$

for some constant C_N .

To determine C_N , compare residues at $u = 1/N$. The residue of $\mathcal{M}_N(u)$ there comes only from the $j = N$ term and equals

$$\operatorname{Res}_{u=1/N} \mathcal{M}_N(u) = -\frac{a_N^{(N)}}{2N}.$$

On the other hand,

$$\operatorname{Res}_{u=1/N} \frac{C_N u^{2N-1}}{\prod_{j=1}^N (1 - j^2 u^2)} = -\frac{C_N}{2N^2 \prod_{k=1}^{N-1} (1 - k^2/N^2)}.$$

Equating these residues gives

$$C_N = a_N^{(N)} N \prod_{k=1}^{N-1} (N^2 - k^2).$$

Now

$$a_N^{(N)} = (-1)^{N+1} \frac{N}{\binom{2N-2}{N-1}},$$

and

$$\prod_{k=1}^{N-1} (N^2 - k^2) = \prod_{k=1}^{N-1} (N - k) \prod_{k=1}^{N-1} (N + k) = (N - 1)! \frac{(2N - 1)!}{N!} = \frac{(2N - 1)!}{N}.$$

Therefore

$$C_N = (-1)^{N+1} \frac{N(2N - 1)!}{\binom{2N-2}{N-1}} = (-1)^{N+1} (2N - 1)N!(N - 1)!.$$

This proves the first formula.

For the tail identity, note that

$$(-1)^{N+1} \mathcal{M}_N(u) = (2N - 1)N!(N - 1)! \frac{u^{2N-1}}{\prod_{j=1}^N (1 - j^2 u^2)}$$

has nonnegative Taylor coefficients for $0 < u < 1/N$. Hence

$$(-1)^{N+1} \mu_{2r+1}^{(N)} \geq 0 \quad (r \geq N - 1),$$

and therefore, for $x > N$,

$$\sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| x^{-(2r+1)} = \left| \mathcal{M}_N \left(\frac{1}{x} \right) \right| = (2N - 1)N!(N - 1)! \frac{x^2}{\prod_{j=1}^N (x^2 - j^2)}.$$

Finally,

$$\prod_{j=1}^N (x^2 - j^2) = \prod_{j=1}^N (x - j) \prod_{j=1}^N (x + j) = \frac{\Gamma(x)}{\Gamma(x - N)} \cdot \frac{\Gamma(x + N + 1)}{\Gamma(x + 1)} = \frac{1}{x} \frac{\Gamma(x + N + 1)}{\Gamma(x - N)}.$$

Substituting this identity yields the claimed formula. $\square$

Remark 11.4 (Structure of the surviving odd tail). Expanding

$$\frac{1}{\prod_{j=1}^N (1 - j^2 u^2)} = \sum_{k \geq 0} h_k(1^2, \dots, N^2) u^{2k},$$

where h_k denotes the complete homogeneous symmetric polynomial, the exact formula of Lemma 11.3 yields

$$\mu_{2N-1+2k}^{(N)} = (-1)^{N+1} (2N - 1)N!(N - 1)! h_k(1^2, \dots, N^2), \quad k \geq 0.$$

Thus the surviving odd moments are not merely bounded: they are explicit and have a fixed sign,

$$(-1)^{N+1} \mu_{2r+1}^{(N)} \geq 0 \quad (r \geq N - 1).$$

This makes the post-cancellation odd tail completely structured rather than mysterious.

Corollary 11.5 (Exact action of the N -point operator on odd germs). *Let*

$$F(s) = \sum_{r \geq 0} b_{2r+1} s^{2r+1}$$

be an odd analytic germ at $s = 0$. Then

$$(\mathcal{P}_{N,Q} F) := \sum_{j=1}^N a_j^{(N)} F\left(\frac{j}{Q^2}\right) = \sum_{r \geq 0} b_{2r+1} \frac{\mu_{2r+1}^{(N)}}{Q^{4r+2}}.$$

Hence, by Remark 11.4,

$$(\mathcal{P}_{N,Q} F) = (-1)^{N+1} (2N - 1)N!(N - 1)! \sum_{k \geq 0} b_{2N-1+2k} h_k(1^2, \dots, N^2) Q^{-(4N+4k-2)}.$$

Thus the N -point operator does not merely annihilate the first $N - 1$ odd jets: on odd germs it acts as an explicit shifted complete-homogeneous transform of the surviving odd Taylor coefficients.

Proposition 11.6 (Exact finite-difference factorization on odd germs). *Let*

$$T_Q := e^{Q^{-2}D}, \quad D := \frac{d}{ds},$$

so that $T_Q F(s) = F(s + Q^{-2})$. Then for every odd analytic germ F at $s = 0$,

$$(\mathcal{P}_{N,Q}F) = \frac{N(-1)^N}{2\binom{2N-2}{N-1}} \left[T_Q^{-N} (1 - T_Q)^{2N-1} (1 + T_Q) F \right]_{s=0}.$$

Equivalently,

$$(\mathcal{P}_{N,Q}F) = \frac{(-1)^{N+1}N}{2\binom{2N-2}{N-1}} \left[T_Q^{-N} (T_Q - 1)^{2N-1} (1 + T_Q) F \right]_{s=0}.$$

In particular, on odd germs the N -point operator factors exactly through the $(2N - 1)$ -st forward difference.

Proof. Let

$$P_{N,Q}(D) := \sum_{j=1}^N a_j^{(N)} e^{(j/Q^2)D} = \sum_{j=1}^N a_j^{(N)} T_Q^j.$$

Then

$$(\mathcal{P}_{N,Q}F) = (P_{N,Q}(D)F)|_{s=0}.$$

By Lemma 11.2, with $t = T_Q$,

$$P_{N,Q}(D) - P_{N,Q}(-D) = \frac{N(-1)^N}{\binom{2N-2}{N-1}} T_Q^{-N} (1 - T_Q)^{2N-1} (1 + T_Q).$$

If F is odd, then

$$(P_{N,Q}(-D)F)|_{s=0} = \sum_{j=1}^N a_j^{(N)} F(-j/Q^2) = - \sum_{j=1}^N a_j^{(N)} F(j/Q^2) = -(\mathcal{P}_{N,Q}F).$$

Hence

$$2(\mathcal{P}_{N,Q}F) = ((P_{N,Q}(D) - P_{N,Q}(-D))F)|_{s=0},$$

which gives the claimed formula. The second form follows from $(1 - T_Q)^{2N-1} = -(T_Q - 1)^{2N-1}$. $\square$

Remark 11.7 (Why the odd tail begins at order $2N - 1$). Proposition 11.6 shows that the disappearance of the first $N - 1$ odd jets is not merely a moment identity. It is built into the operator itself: on odd germs, the N -point projector contains the factor $(T_Q - 1)^{2N-1}$. The surviving odd tail therefore begins at order $2N - 1$ for operator-theoretic reasons, not only by coefficient cancellation.

Lemma 11.8 (Integral form of a forward difference). *For $m \geq 1$, $h > 0$, and an analytic germ f , one has*

$$(e^{hD} - 1)^m f(x) = h^m \int_0^m f^{(m)}(x + hu) \omega_m(u) du,$$

where

$$\omega_m(u) := \frac{1}{(m-1)!} \sum_{r=0}^m (-1)^r \binom{m}{r} (u-r)_+^{m-1}, \quad (u-r)_+ := \max\{u-r, 0\}.$$

The function ω_m is nonnegative, supported on $[0, m]$, and satisfies the symmetry

$$\omega_m(u) = \omega_m(m-u).$$

Proof. The identity

$$(e^{hD} - 1)f(x) = f(x + h) - f(x) = \int_0^h f'(x + t) dt$$

iterates to

$$(e^{hD} - 1)^m f(x) = \int_{[0, h]^m} f^{(m)}(x + t_1 + \cdots + t_m) dt_1 \cdots dt_m.$$

Rescaling $t_i = hu_i$ gives

$$(e^{hD} - 1)^m f(x) = h^m \int_{[0, 1]^m} f^{(m)}(x + h(u_1 + \cdots + u_m)) du_1 \cdots du_m.$$

Pushing forward Lebesgue measure on $[0, 1]^m$ under

$$(u_1, \dots, u_m) \mapsto u_1 + \cdots + u_m$$

yields the density ω_m , namely the classical Irwin–Hall density, which has the stated explicit formula, support, and symmetry. This proves the claim. $\square$

Proposition 11.9 (Positive-kernel representation of the odd projector). *Set*

$$m := 2N - 1, \quad h := Q^{-2},$$

and define, for $0 \leq u \leq N$,

$$\kappa_N(u) := \omega_{2N-1}(N - u) + \omega_{2N-1}(N - 1 - u),$$

where ω_{2N-1} is the density from Lemma 11.8, extended by 0 outside $[0, 2N - 1]$. Then $\kappa_N(u) \geq 0$ on $[0, N]$, and for every odd analytic germ F ,

$$(\mathcal{P}_{N, Q} F) = \frac{N(-1)^{N+1}}{\binom{2N-2}{N-1}} Q^{-(4N-2)} \int_0^N \kappa_N(u) F^{(2N-1)}\left(\frac{u}{Q^2}\right) du.$$

Moreover, $\kappa_N(u) > 0$ for $0 < u < N$.

Proof. Apply Proposition 11.6 and Lemma 11.8 with $m = 2N - 1$ and $h = Q^{-2}$. This gives

$$(\mathcal{P}_{N, Q} F) = \frac{N(-1)^{N+1}}{2\binom{2N-2}{N-1}} h^{2N-1} \int_0^{2N-1} \omega_{2N-1}(v) \left[F^{(2N-1)}((v-N)h) + F^{(2N-1)}((v-N+1)h) \right] dv.$$

Since F is odd, $F^{(2N-1)}$ is even. Using this evenness together with the symmetry

$$\omega_{2N-1}(v) = \omega_{2N-1}(2N - 1 - v),$$

the two terms may be folded onto $[0, N]$, yielding

$$(\mathcal{P}_{N, Q} F) = \frac{N(-1)^{N+1}}{\binom{2N-2}{N-1}} h^{2N-1} \int_0^N [\omega_{2N-1}(N - u) + \omega_{2N-1}(N - 1 - u)] F^{(2N-1)}(uh) du.$$

This is exactly the stated formula. The strict positivity on $0 < u < N$ follows because $N - u \in (0, 2N - 1)$ there, and ω_{2N-1} is strictly positive on the interior of its support. $\square$

Corollary 11.10 (Sign transfer from the first surviving derivative). *Let F be an odd analytic germ. If for some $\sigma \in \{\pm 1\}$,*

$$\sigma F^{(2N-1)}(s) \geq 0 \quad \text{for all } 0 \leq s \leq \frac{N}{Q^2},$$

then

$$\sigma(-1)^{N+1}(\mathcal{P}_{N, Q} F) \geq 0.$$

If moreover $F^{(2N-1)} \not\equiv 0$ on $[0, N/Q^2]$, then the inequality is strict.

Proof. This is immediate from Proposition 11.9, since the kernel κ_N is nonnegative and strictly positive on $(0, N)$. $\square$

Remark 11.11 (Why this is the right downstream target). Proposition 11.9 shows that the N -point layer is not merely an odd-moment cancellation device: on odd germs it is a positive average of the first surviving derivative $F^{(2N-1)}$ over the microscopic interval $[0, N/Q^2]$. Accordingly, any source-level argument that gives a fixed sign, one-sided lower bound, or variation control for

$$H_m^{(2N-1)}(s) \quad \left(0 \leq s \leq \frac{N}{Q^2}\right)$$

would transfer directly to the sign or size of $\Xi_\zeta^{(N)}(m)$. This isolates the next natural analytic target in the N -point route.

Remark 11.12 (Finite-order recurrence of the normalized surviving tail). Define

$$\nu_k := \frac{(-1)^{N+1} \mu_{2N-1+2k}^{(N)}}{(2N-1)N!(N-1)!} = h_k(1^2, \dots, N^2), \quad k \geq 0.$$

Then

$$\sum_{k \geq 0} \nu_k t^k = \frac{1}{\prod_{j=1}^N (1 - j^2 t)} = \frac{1}{1 - e_1 t + e_2 t^2 - \dots + (-1)^N e_N t^N},$$

where $e_r = e_r(1^2, \dots, N^2)$ is the r -th elementary symmetric polynomial in $1^2, \dots, N^2$. Consequently,

$$\nu_k = e_1 \nu_{k-1} - e_2 \nu_{k-2} + \dots + (-1)^{N+1} e_N \nu_{k-N} \quad (k \geq 1),$$

with the conventions $\nu_0 = 1$ and $\nu_k = 0$ for $k < 0$. Thus the surviving odd tail is governed by an exact order- N linear recurrence.

Proposition 11.13 (Partial-fraction form of the surviving odd tail). *Let*

$$G_N(t) := \frac{1}{\prod_{m=1}^N (1 - m^2 t)}.$$

Then G_N has the exact partial-fraction decomposition

$$G_N(t) = \sum_{j=1}^N \frac{B_{N,j}}{1 - j^2 t},$$

where

$$B_{N,j} = \frac{2(-1)^{N-j} j^{2N}}{(N-j)!(N+j)!}.$$

Equivalently,

$$\frac{1}{\prod_{m=1}^N (1 - m^2 u^2)} = \sum_{j=1}^N \frac{2(-1)^{N-j} j^{2N}}{(N-j)!(N+j)!} \frac{1}{1 - j^2 u^2}.$$

Consequently, for every $k \geq 0$,

$$h_k(1^2, \dots, N^2) = \sum_{j=1}^N \frac{2(-1)^{N-j} j^{2N+2k}}{(N-j)!(N+j)!},$$

and hence

$$\mu_{2N-1+2k}^{(N)} = (-1)^{N+1} (2N-1)N!(N-1)! \sum_{j=1}^N \frac{2(-1)^{N-j} j^{2N+2k}}{(N-j)!(N+j)!}.$$

Proof. Since the poles of $G_N(t)$ are simple and located at $t = j^{-2}$ ($1 \leq j \leq N$), one has

$$G_N(t) = \sum_{j=1}^N \frac{B_{N,j}}{1 - j^2 t}$$

for uniquely determined coefficients $B_{N,j}$. Multiplying by $(1 - j^2 t)$ and evaluating at $t = j^{-2}$ gives

$$B_{N,j} = \left. \frac{1}{\prod_{m \neq j} (1 - m^2 t)} \right|_{t=j^{-2}} = \frac{1}{\prod_{m \neq j} (1 - m^2 / j^2)}.$$

Now

$$\prod_{m \neq j} \left(1 - \frac{m^2}{j^2} \right) = j^{-2N+2} \prod_{m \neq j} (j^2 - m^2),$$

and

$$\prod_{m \neq j} (j^2 - m^2) = \prod_{m \neq j} (j - m) \prod_{m \neq j} (j + m).$$

Here

$$\prod_{m \neq j} (j - m) = (j - 1)! (-1)^{N-j} (N - j)!,$$

while

$$\prod_{m \neq j} (j + m) = \frac{(N + j)!}{2j j!}.$$

Therefore

$$\prod_{m \neq j} (j^2 - m^2) = (-1)^{N-j} \frac{(N - j)! (N + j)!}{2j^2},$$

and hence

$$B_{N,j} = \frac{2(-1)^{N-j} j^{2N}}{(N - j)! (N + j)!}.$$

Expanding each geometric series

$$\frac{1}{1 - j^2 u^2} = \sum_{k \geq 0} j^{2k} u^{2k}$$

and comparing coefficients of u^{2k} yields the formula for $h_k(1^2, \dots, N^2)$. The identity for $\mu_{2N-1+2k}^{(N)}$ then follows from Remark 11.4. $\square$

Remark 11.14 (Dominance of the outermost pole for fixed N). For fixed N , the explicit decomposition in Proposition 11.13 shows that the surviving tail is asymptotically dominated by the outermost node $j = N$. Indeed,

$$h_k(1^2, \dots, N^2) = \frac{2N^{2N+2k}}{(2N)!} + O_N((N - 1)^{2k}) \quad (k \rightarrow \infty),$$

and therefore

$$\mu_{2N-1+2k}^{(N)} = (-1)^{N+1} \frac{N^{2N+2k}}{\binom{2N-2}{N-1}} + O_N((N - 1)^{2k}) \quad (k \rightarrow \infty).$$

So for fixed N , the far odd tail is eventually controlled by the single outermost pole $u = \pm 1/N$.

Proposition 11.15 (Kernel form and explicit corrected N -point transverse estimate). *Let*

$$H_m(s) := \Phi_K(\widehat{\Omega}_\zeta^{\text{corr}}(s; m))$$

be the full corrected transverse scalar. For $N = \lfloor \alpha Q \rfloor$ with $0 < \alpha < \rho_0$, define

$$s_j = \frac{j}{Q^2}, \quad \Xi_\zeta^{(N)}(m) := \sum_{j=1}^N a_j^{(N)} H_m(s_j).$$

Then

$$\Xi_\zeta^{(N)}(m) = \frac{1}{2\pi i} \oint_{|\xi|=\rho_0/Q} H_m(\xi) K_{N,Q}(\xi) d\xi,$$

where

$$K_{N,Q}(\xi) := 2 \sum_{j=1}^N a_j^{(N)} \frac{s_j}{\xi^2 - s_j^2} = 2(-1)^{N+1} (2N-1)N!(N-1)! Q^4 \xi \frac{\Gamma(Q^2 \xi - N)}{\Gamma(Q^2 \xi + N + 1)}.$$

Moreover, writing $x := \rho_0 Q$, one has the explicit bound

$$|\Xi_\zeta^{(N)}(m)| \ll_{\rho_0} \rho_0^2 \frac{(2N-1)N!(N-1)!}{Q} \frac{\Gamma(x-N)}{\Gamma(x+N+1)} L(m)S(m)^2.$$

Proof. By Proposition 9.5, the full corrected scalar

$$H_m(s) = \Phi_K(\widehat{\Omega}_\zeta^{\text{corr}}(s; m))$$

is holomorphic on $|s| < \rho_0/Q$. By Proposition 9.7, all cutoff-dependent tail, jet-correction, and estimation contributions are already internal to the full corrected scalar $H_m(s)$; hence no external remainder is carried into $\Xi_\zeta^{(N)}(m)$.

Since H_m is odd, the odd Cauchy formula gives

$$H_m(s) = \frac{1}{2\pi i} \oint_{|\xi|=\rho_0/Q} H_m(\xi) \frac{2s}{\xi^2 - s^2} d\xi.$$

Summing this identity against the coefficients $a_j^{(N)}$ at $s = s_j = j/Q^2$ yields

$$\Xi_\zeta^{(N)}(m) = \frac{1}{2\pi i} \oint_{|\xi|=\rho_0/Q} H_m(\xi) K_{N,Q}(\xi) d\xi$$

with

$$K_{N,Q}(\xi) = 2 \sum_{j=1}^N a_j^{(N)} \frac{s_j}{\xi^2 - s_j^2}.$$

Now

$$\frac{s_j}{\xi^2 - s_j^2} = \frac{1}{\xi} \frac{j/(Q^2 \xi)}{1 - j^2/(Q^4 \xi^2)},$$

so, using Lemma 11.3,

$$K_{N,Q}(\xi) = \frac{2}{\xi} \mathcal{M}_N\left(\frac{1}{Q^2 \xi}\right) = 2(-1)^{N+1} (2N-1)N!(N-1)! \frac{Q^2}{\prod_{j=1}^N (Q^4 \xi^2 - j^2)}.$$

Finally,

$$\prod_{j=1}^N (Q^4 \xi^2 - j^2) = \prod_{j=1}^N (Q^2 \xi - j) \prod_{j=1}^N (Q^2 \xi + j) = \frac{1}{Q^2 \xi} \frac{\Gamma(Q^2 \xi + N + 1)}{\Gamma(Q^2 \xi - N)},$$

which gives

$$K_{N,Q}(\xi) = 2(-1)^{N+1} (2N-1)N!(N-1)! Q^4 \xi \frac{\Gamma(Q^2 \xi - N)}{\Gamma(Q^2 \xi + N + 1)}.$$

For the size bound, Proposition 10.1 gives

$$|H_m(s)| \ll_{\rho_0} \frac{L(m)S(m)^2}{Q^3} \quad (|s| = \rho_0/Q).$$

By Proposition 10.2, one may write

$$H_m(z/Q^2) = \sum_{r \geq 0} c_{2r+1}(m) \frac{z^{2r+1}}{Q^{2r+4}}, \quad |c_{2r+1}(m)| \ll L(m)S(m)^2 \rho_0^{-(2r+1)}.$$

Therefore

$$\Xi_\zeta^{(N)}(m) = \sum_{r \geq 0} c_{2r+1}(m) \frac{\mu_{2r+1}^{(N)}}{Q^{2r+4}}.$$

By Lemma 11.1 and Remark 11.4, this becomes the exact surviving expansion

$$\Xi_{\zeta}^{(N)}(m) = (-1)^{N+1} (2N-1)N!(N-1)! \sum_{k \geq 0} c_{2N-1+2k}(m) h_k(1^2, \dots, N^2) Q^{-(2N+2k+2)}.$$

In particular, the N -point operation isolates the first surviving corrected odd coefficient $c_{2N-1}(m)$ and propagates the higher odd coefficients through an explicit universal positive tail.

For the bound, the coefficient estimate yields

$$|\Xi_{\zeta}^{(N)}(m)| \ll_{\rho_0} \frac{L(m)S(m)^2}{Q^3} \sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| (\rho_0 Q)^{-(2r+1)}.$$

Applying Lemma 11.3 with $x = \rho_0 Q$ gives

$$\sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| (\rho_0 Q)^{-(2r+1)} = (2N-1)N!(N-1)! (\rho_0 Q)^2 \frac{\Gamma(\rho_0 Q - N)}{\Gamma(\rho_0 Q + N + 1)}.$$

Substituting this proves the stated bound. $\square$

Remark 11.16 (Exact surviving-coefficient expansion on the zeta side). With the notation of the proof of Proposition 11.15,

$$\Xi_{\zeta}^{(N)}(m) = \sum_{r \geq 0} c_{2r+1}(m) \frac{\mu_{2r+1}^{(N)}}{Q^{2r+4}} = (-1)^{N+1} (2N-1)N!(N-1)! \sum_{k \geq 0} c_{2N-1+2k}(m) h_k(1^2, \dots, N^2) Q^{-(2N+2k+2)}.$$

Thus the N -point step does not merely remove the first $N-1$ odd jets: it converts the corrected odd Taylor expansion into an explicit universal tail whose leading term is the first surviving corrected odd coefficient $c_{2N-1}(m)$.

Corollary 11.17 (Finite-core localization of the transformed zeta-side scalar). *With the notation of Proposition 11.15 and Corollary 10.21, let*

$$H_{\text{out}}(z/Q^2) = \sum_{r \geq 0} a_{2r+1}^{\text{out}}(m, H) \frac{z^{2r+1}}{Q^{2r+4}}$$

be the corrected transverse scalar contributed by all zero pairs with $\Delta_{\rho} > H$, and define

$$\Xi_{\zeta, \text{out}}^{(N)}(m; H) := \sum_{j=1}^N a_j^{(N)} H_{\text{out}}(s_j), \quad \Xi_{\zeta, \leq H}^{(N)}(m) := \Xi_{\zeta}^{(N)}(m) - \Xi_{\zeta, \text{out}}^{(N)}(m; H).$$

If $HQ > N$, then

$$|\Xi_{\zeta, \text{out}}^{(N)}(m; H)| \ll \frac{(2N-1)N!(N-1)!}{Q} \frac{\Gamma(HQ - N)}{\Gamma(HQ + N + 1)} + (2N-1)N!(N-1)! \frac{\Gamma(Q^2 - N)}{\Gamma(Q^2 + N + 1)}.$$

Consequently,

$$\Xi_{\zeta}^{(N)}(m) = \Xi_{\zeta, \leq H}^{(N)}(m) + O\left(\frac{(2N-1)N!(N-1)!}{Q} \frac{\Gamma(HQ - N)}{\Gamma(HQ + N + 1)}\right) + O\left((2N-1)N!(N-1)! \frac{\Gamma(Q^2 - N)}{\Gamma(Q^2 + N + 1)}\right).$$

Proof. By the exact N -point action on odd germs used in the proof of Proposition 11.15,

$$\Xi_{\zeta, \text{out}}^{(N)}(m; H) = \sum_{r \geq 0} a_{2r+1}^{\text{out}}(m, H) \frac{\mu_{2r+1}^{(N)}}{Q^{2r+4}}.$$

Since $\mu_{2r+1}^{(N)} = 0$ for $0 \leq r \leq N-2$,

$$\Xi_{\zeta, \text{out}}^{(N)}(m; H) = \sum_{r \geq N-1} a_{2r+1}^{\text{out}}(m, H) \frac{\mu_{2r+1}^{(N)}}{Q^{2r+4}}.$$

Taking absolute values and applying Corollary 10.21,

$$|\Xi_{\zeta, \text{out}}^{(N)}(m; H)| \ll \sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| \left(\frac{H^{-(2r+3)}}{Q^{2r+4}} + \frac{Q^{-(2r+2)}}{Q^{2r+4}} \right).$$

Thus

$$|\Xi_{\zeta, \text{out}}^{(N)}(m; H)| \ll \frac{1}{H^2 Q^3} \sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| (HQ)^{-(2r+1)} + \frac{1}{Q^4} \sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| (Q^2)^{-(2r+1)}.$$

Now apply Lemma 11.3 first with $x = HQ$ and then with $x = Q^2$. Since $HQ > N$, both evaluations are valid for all sufficiently large Q , and one obtains

$$\sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| (HQ)^{-(2r+1)} = (2N-1)N!(N-1)!(HQ)^2 \frac{\Gamma(HQ-N)}{\Gamma(HQ+N+1)},$$

and

$$\sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| (Q^2)^{-(2r+1)} = (2N-1)N!(N-1)!Q^4 \frac{\Gamma(Q^2-N)}{\Gamma(Q^2+N+1)}.$$

Substituting these identities proves the displayed bound for $\Xi_{\zeta, \text{out}}^{(N)}(m; H)$, and the decomposition

$$\Xi_{\zeta}^{(N)}(m) = \Xi_{\zeta, \leq H}^{(N)}(m) + \Xi_{\zeta, \text{out}}^{(N)}(m; H)$$

gives the second display. $\square$

Remark 11.18 (Asymptotic finite-core localization at linear depth). Let

$$N = \lfloor \alpha Q \rfloor, \quad 0 < \alpha < \rho_0,$$

and fix $H > \alpha$. Then Corollary 11.17 and Stirling's formula give

$$|\Xi_{\zeta, \text{out}}^{(N)}(m; H)| \ll \exp(-Hf(\alpha/H)Q + O(\log Q)) + \exp(-2\alpha Q \log Q + O(Q)),$$

where

$$f(\beta) = (1+\beta) \log(1+\beta) - 2\beta \log \beta - (1-\beta) \log(1-\beta).$$

In particular,

$$\Xi_{\zeta}^{(N)}(m) = \Xi_{\zeta, \leq H}^{(N)}(m) + O(e^{-Hf(\alpha/H)Q + O(\log Q)}) + O(e^{-2\alpha Q \log Q + O(Q)}).$$

Thus the full transformed zeta-side scalar is determined, up to exponentially small error, by a fixed finite core.

Definition 11.19 (Finite-core N -point scalar). Fix a finite same-scale core $\mathcal{C}$ of zero pairs around the midpoint m , and let

$$H_{\text{core}, m}(s) := \Phi_K(\widehat{\Omega}_{\text{core}}^{\text{corr}}(s; m))$$

denote the corrected transverse scalar contributed by this core, in the same finite- s normalization as $H_m(s)$. For $N \geq 1$, define

$$\Xi_{\text{core}}^{(N)}(m) := \sum_{j=1}^N a_j^{(N)} H_{\text{core}, m}(j/Q^2).$$

If

$$H_{\text{core}, m}(z/Q^2) = \sum_{r \geq 0} c_{\text{core}, 2r+1}(m) \frac{z^{2r+1}}{Q^{2r+4}},$$

then the N -point moment cancellations give

$$\Xi_{\text{core}}^{(N)}(m) = (-1)^{N+1} (2N-1)N!(N-1)! \sum_{k \geq 0} c_{\text{core}, 2N-1+2k}(m) h_k(1^2, \dots, N^2) Q^{-(2N+2k+2)}.$$

Thus the first finite-core coefficient seen by the N -point projector at depth N is

$$c_{\text{core}, 2N-1}(m).$$

Remark 11.20 (Finite-core localization is not a lower law). Corollary 11.17 is an upper/localization statement. It does not supply a lower bound for

$$\Xi_{\text{core}}^{(N)}(m).$$

In particular, it does not rule out cancellation among the finitely many core constituents:

$$c_{\text{core}, 2N-1}(m) = 0$$

or cancellation of several consecutive N -visible odd coefficients. A separate finite-core lower-transfer theorem is required before the N -point contradiction can be applied to genuinely non-pair-like cores.

Remark 11.21 (Exact Laurent tail of the contour kernel). The kernel in Proposition 11.15 has an exact large- ξ Laurent expansion. Indeed, using Lemma 11.3 and Remark 11.4,

$$K_{N,Q}(\xi) = \frac{2}{\xi} \mathcal{M}_N\left(\frac{1}{Q^2\xi}\right) = 2(-1)^{N+1}(2N-1)N!(N-1)!Q^{-4N+2}\xi^{-2N} \sum_{k \geq 0} h_k(1^2, \dots, N^2) Q^{-4k} \xi^{-2k}.$$

So the contour kernel is not merely small: its entire post-cancellation odd tail is explicit, universal, and sign-rigid term-by-term.

Corollary 11.22 (Optimized zeta-side cancellation depth). *Let*

$$N = \lfloor \alpha Q \rfloor, \quad 0 < \alpha < \rho_0, \quad \beta := \frac{\alpha}{\rho_0} \in (0, 1).$$

Then

$$|\Xi_{\zeta}^{(N)}(m)| \ll_{\rho_0, \alpha} \frac{L(m)S(m)^2}{Q} e^{-\delta_*(\alpha, \rho_0)Q},$$

where

$$\delta_*(\alpha, \rho_0) = \rho_0 \left((1 + \beta) \log(1 + \beta) - 2\beta \log \beta - (1 - \beta) \log(1 - \beta) \right).$$

Moreover, $\delta_(\alpha, \rho_0)$ is maximized uniquely at*

$$\beta = \frac{1}{\sqrt{2}}, \quad \text{i.e.} \quad \alpha_{\text{opt}} = \frac{\rho_0}{\sqrt{2}},$$

so the optimal zeta-side cancellation depth is

$$N_{\text{opt}} = \left\lfloor \frac{\rho_0}{\sqrt{2}} Q \right\rfloor,$$

and the corresponding exponential rate is

$$\delta_{\text{opt}} = 2\rho_0 \log(1 + \sqrt{2}).$$

Proof. Set $x := \rho_0 Q$, so that $N = \beta x + O(1)$ with $\beta \in (0, 1)$ fixed. By Proposition 11.15, it suffices to estimate

$$(2N-1)N!(N-1)! \frac{\Gamma(x-N)}{\Gamma(x+N+1)}.$$

A Stirling expansion gives

$$\log \left((2N-1)N!(N-1)! \frac{\Gamma(x-N)}{\Gamma(x+N+1)} \right) = -x f(\beta) + O(\log x),$$

where

$$f(\beta) = (1 + \beta) \log(1 + \beta) - 2\beta \log \beta - (1 - \beta) \log(1 - \beta).$$

Hence

$$(2N-1)N!(N-1)! \frac{\Gamma(x-N)}{\Gamma(x+N+1)} \ll_{\rho_0, \alpha} e^{-x f(\beta)},$$

and therefore

$$|\Xi_{\zeta}^{(N)}(m)| \ll_{\rho_0, \alpha} \frac{L(m)S(m)^2}{Q} e^{-\rho_0 f(\beta)Q}.$$

This proves the first claim.

For the optimization, differentiate:

$$f'(\beta) = \log\left(\frac{1-\beta^2}{\beta^2}\right).$$

Thus $f'(\beta) = 0$ exactly when $\beta = 1/\sqrt{2}$. Since

$$f''(\beta) = -\frac{2\beta}{1-\beta^2} - \frac{2}{\beta} < 0 \quad (0 < \beta < 1),$$

this critical point is the unique maximizer. Finally,

$$f\left(\frac{1}{\sqrt{2}}\right) = 2\log(1+\sqrt{2}),$$

which gives the stated value of δ_{opt} . $\square$

Remark 11.23. Corollary 11.22 optimizes the *zeta-side* cancellation depth. After the exact odd-tail evaluation is imported into Proposition 11.37, the toy side imposes no asymptotic upper restriction on the linear choice $N = \lfloor \alpha Q \rfloor$: for every fixed $\alpha > 0$, the toy microscopic nodes $s_j = j/Q^2$ remain inside the toy analyticity disk for all sufficiently large Q , and the residual odd toy tail is in fact super-exponentially small on the Q -scale. Accordingly one may take the exact zeta-side optimum

$$\alpha = \frac{\rho_0}{\sqrt{2}}.$$

Remark 11.24 (Sharper prefactor in the optimized zeta-side bound). With the notation of Corollary 11.22, set

$$x := \rho_0 Q, \quad N = \beta x + O(1), \quad \beta = \frac{\alpha}{\rho_0} \in (0, 1).$$

A more precise Stirling expansion of the gamma-ratio factor in Proposition 11.15 gives

$$(2N-1)N!(N-1)! \frac{\Gamma(x-N)}{\Gamma(x+N+1)} = \frac{4\pi\beta}{\sqrt{1-\beta^2}} e^{-xf(\beta)} (1+o(1)),$$

where

$$f(\beta) = (1+\beta)\log(1+\beta) - 2\beta\log\beta - (1-\beta)\log(1-\beta).$$

Consequently,

$$|\Xi_\zeta^{(N)}(m)| \ll 2C_\partial(\rho_0)\rho_0 \frac{4\pi\beta}{\sqrt{1-\beta^2}} \frac{L(m)S(m)^2}{Q^2} e^{-\delta_*(\alpha,\rho_0)Q} (1+o(1)).$$

At the zeta-side optimum $\beta = 1/\sqrt{2}$, the prefactor simplifies to 4π , so

$$|\Xi_\zeta^{(N_{\text{opt}})}(m)| \ll 8\pi C_\partial(\rho_0)\rho_0 \frac{L(m)S(m)^2}{Q^2} e^{-2\rho_0\log(1+\sqrt{2})Q} (1+o(1)).$$

This sharper prefactor is not needed for the qualitative contradiction, but it records that, once the boundary constant $C_\partial(\rho_0)$ is made explicit, the zeta-side N -point estimate becomes effective with an explicit leading constant.

Proposition 11.25 (Real profile of the exact N -point kernel). *Let*

$$C_N := (2N-1)N!(N-1)!, \quad \kappa_N(x) := x \frac{\Gamma(x-N)}{\Gamma(x+N+1)} \quad (x > N).$$

Then

$$\kappa_N(x) = \frac{1}{\prod_{j=1}^N (x^2 - j^2)},$$

and for real $x > N$,

$$K_{N,Q}\left(\frac{x}{Q^2}\right) = 2(-1)^{N+1} C_N Q^2 \kappa_N(x).$$

Moreover:

$$(1) \quad \kappa_N(x) > 0 \text{ for all } x > N;$$

- (2) $\kappa'_N(x) < 0$ for all $x > N$;
 (3) for every $x > N$,

$$\sup_{|\xi|=x/Q^2} |K_{N,Q}(\xi)| = 2C_N Q^2 \kappa_N(x) = \left| K_{N,Q}\left(\frac{x}{Q^2}\right) \right|.$$

Proof. From Proposition 11.15,

$$K_{N,Q}(\xi) = 2(-1)^{N+1} C_N Q^4 \xi \frac{\Gamma(Q^2 \xi - N)}{\Gamma(Q^2 \xi + N + 1)}.$$

Substituting $\xi = x/Q^2$ gives

$$K_{N,Q}\left(\frac{x}{Q^2}\right) = 2(-1)^{N+1} C_N Q^2 x \frac{\Gamma(x - N)}{\Gamma(x + N + 1)} = 2(-1)^{N+1} C_N Q^2 \kappa_N(x).$$

Now

$$\Gamma(x + N + 1) = (x + N)(x + N - 1) \cdots x \Gamma(x),$$

and

$$\Gamma(x) = (x - 1)(x - 2) \cdots (x - N) \Gamma(x - N).$$

Hence

$$x \frac{\Gamma(x - N)}{\Gamma(x + N + 1)} = \frac{1}{(x - N) \cdots (x - 1)(x + 1) \cdots (x + N)} = \frac{1}{\prod_{j=1}^N (x^2 - j^2)},$$

which proves the product formula. Positivity is immediate for $x > N$.

For monotonicity, differentiate the logarithm:

$$\frac{d}{dx} \log \kappa_N(x) = - \sum_{j=1}^N \frac{2x}{x^2 - j^2}.$$

Since $x > N$, each denominator is positive, so

$$\frac{d}{dx} \log \kappa_N(x) < 0,$$

hence $\kappa'_N(x) < 0$.

Finally, using the product form of the kernel,

$$K_{N,Q}(\xi) = 2(-1)^{N+1} C_N Q^2 \frac{1}{\prod_{j=1}^N (Q^4 \xi^2 - j^2)},$$

if $|\xi| = x/Q^2$ then

$$|Q^4 \xi^2 - j^2| \geq ||Q^2 \xi|^2 - j^2| = x^2 - j^2 \quad (1 \leq j \leq N),$$

so

$$|K_{N,Q}(\xi)| \leq 2C_N Q^2 \frac{1}{\prod_{j=1}^N (x^2 - j^2)} = 2C_N Q^2 \kappa_N(x).$$

Equality is attained at $\xi = x/Q^2$, proving the final claim. $\square$

Remark 11.26 (No contour loss in the kernel factor). Proposition 11.25 shows that the kernel factor in Proposition 11.15 is controlled exactly by its positive real value on the boundary circle. Thus the boundary estimate used in the zeta-side N -point bound carries no hidden contour-loss from the kernel itself: the worst point is the positive real boundary point

$$\xi = \frac{\rho_0}{Q}.$$

Moreover, since κ_N is strictly decreasing on (N, ∞) , the kernel factor is largest at the smallest admissible boundary radius.

Proposition 11.27 (Positive smoothing preserves the real kernel profile). *Let $\psi \in C_c^\infty(\mathbf{R})$ be even, nonnegative, and normalized by*

$$\int_{\mathbf{R}} \psi(u) du = 1.$$

Set

$$U_\psi := \sup\{|u| : u \in \text{supp } \psi\}.$$

For $x > N + \eta Q^2 U_\psi$, define the smoothed real profile

$$\kappa_{N,\eta}^\psi(x) := \int_{\mathbf{R}} \psi(u) \kappa_N(x + \eta Q^2 u) du,$$

where

$$\kappa_N(x) := x \frac{\Gamma(x - N)}{\Gamma(x + N + 1)} = \frac{1}{\prod_{j=1}^N (x^2 - j^2)}.$$

Then

$$\kappa_{N,\eta}^\psi(x) > 0 \quad \text{and} \quad \frac{d}{dx} \kappa_{N,\eta}^\psi(x) < 0$$

for all $x > N + \eta Q^2 U_\psi$.

Proof. For $x > N + \eta Q^2 U_\psi$ and $u \in \text{supp } \psi$, one has

$$x + \eta Q^2 u > N.$$

By Proposition 11.25,

$$\kappa_N(x + \eta Q^2 u) > 0 \quad \text{and} \quad \kappa'_N(x + \eta Q^2 u) < 0.$$

Since $\psi \geq 0$ and $\int \psi = 1$, averaging preserves positivity:

$$\kappa_{N,\eta}^\psi(x) > 0.$$

Differentiating under the integral sign gives

$$\frac{d}{dx} \kappa_{N,\eta}^\psi(x) = \int_{\mathbf{R}} \psi(u) \kappa'_N(x + \eta Q^2 u) du < 0.$$

□

Remark 11.28 (Smoothed real profile). Proposition 11.27 shows that nonnegative smoothing preserves the qualitative real-axis behavior of the exact kernel: positivity and strict monotonic decay survive averaging. Thus the Mellin-side smoothing introduced later does not destroy the basic real-profile structure of the N -point kernel.

Remark 11.29 (Discrete Mellin projector and test-function viewpoint). Define the additive operator

$$(\mathcal{P}_{N,Q} F) := \sum_{j=1}^N a_j^{(N)} F\left(\frac{j}{Q^2}\right),$$

and the associated multiplicative distribution on $\mathbb{R}_+^\times$,

$$\mu_{N,Q} := \sum_{j=1}^N a_j^{(N)} \delta_{e^{j/Q^2}}.$$

Its Mellin symbol is precisely

$$\widehat{\mu}_{N,Q}(z) = \sum_{j=1}^N a_j^{(N)} e^{zj/Q^2}.$$

Equivalently, if $D := \frac{d}{ds}$, then

$$\mathcal{P}_{N,Q} = P_{N,Q}(D)|_{s=0}, \quad P_{N,Q}(D) := \sum_{j=1}^N a_j^{(N)} e^{(j/Q^2)D}.$$

Thus $\mathcal{P}_{N,Q}$ is a polynomial in the translation semigroup, evaluated at the origin. If

$$F(s) = \sum_{n \geq 0} c_n s^n,$$

then

$$(\mathcal{P}_{N,Q}F) = \sum_{n \geq 0} c_n \frac{\mu_n^{(N)}}{Q^{2n}}, \quad \mu_n^{(N)} := \sum_{j=1}^N a_j^{(N)} j^n.$$

Lemma 11.2 shows that $\widehat{\mu}_{N,Q}(0) = 1$ and that the odd part of $\widehat{\mu}_{N,Q}$ starts only at order $2N - 1$; meanwhile, Corollary 11.5 identifies the exact action of $\mathcal{P}_{N,Q}$ on odd germs as a shifted complete-homogeneous transform of the surviving odd Taylor coefficients.

In this sense, the N -point operator may be viewed as a local discrete Mellin-side projector. This is formally reminiscent of the Weil/Connes test-function viewpoint, where one probes the zero set through Mellin/Fourier transforms of multiplicative test functions and explicit- or trace-formula pairings [3, 10, 11]. The present paper does not use any global trace formula; the comparison is only structural.

A natural smoothed version is obtained by fixing an even $\phi \in C_c^\infty(\mathbb{R})$ with $\widehat{\phi}(0) = 1$ and setting

$$f_{N,Q,\eta}(x) := \sum_{j=1}^N a_j^{(N)} \eta^{-1} \phi\left(\frac{\log x - j/Q^2}{\eta}\right).$$

Its Mellin transform is

$$\widehat{f}_{N,Q,\eta}(z) = \widehat{\phi}(\eta z) \widehat{\mu}_{N,Q}(z).$$

Hence the same low odd-jet annihilation persists after smoothing, while the finite-support projector is replaced by an honest smooth multiplicative test function. This observation is not used elsewhere in the proof, but it places the N -point cancellation layer in a more canonical operator-theoretic setting.

Lemma 11.30 (Smoothed Mellin projector). *Let $\phi \in C_c^\infty(\mathbb{R})$ be even and normalized by*

$$\int_{\mathbb{R}} \phi(u) du = 1.$$

Define its bilateral Laplace transform by

$$\widehat{\phi}(w) := \int_{\mathbb{R}} \phi(u) e^{wu} du.$$

For $\eta > 0$, define the smoothed additive projector

$$(\mathcal{P}_{N,Q,\eta}^\phi F) := \sum_{j=1}^N a_j^{(N)} \int_{\mathbb{R}} \phi(u) F\left(\frac{j}{Q^2} + \eta u\right) du,$$

whenever F is holomorphic on a neighborhood of the integration region.

Then:

(1) *for every exponential mode $F(s) = e^{zs}$,*

$$\mathcal{P}_{N,Q,\eta}^\phi(e^{zs}) = \widehat{\phi}(\eta z) \widehat{\mu}_{N,Q}(z), \quad \widehat{\mu}_{N,Q}(z) = \sum_{j=1}^N a_j^{(N)} e^{zj/Q^2};$$

(2) *the smoothed symbol preserves the exact low odd-jet annihilation:*

$$\widehat{\phi}(0) \widehat{\mu}_{N,Q}(0) = 1, \quad \partial_z^{2r+1}(\widehat{\phi}(\eta z) \widehat{\mu}_{N,Q}(z)) \Big|_{z=0} = 0 \quad (0 \leq r \leq N-2);$$

(3) *if F is holomorphic on a neighborhood Ω containing all points*

$$\frac{j}{Q^2} + \eta u \quad (1 \leq j \leq N, u \in \text{supp } \phi),$$

then

$$|\mathcal{P}_{N,Q,\eta}^\phi F - \mathcal{P}_{N,Q} F| \leq \frac{\eta^2}{2} (2N-1) m_2(\phi) \sup_{\Omega} |F''|,$$

where

$$m_2(\phi) := \int_{\mathbf{R}} |u|^2 |\phi(u)| du.$$

Proof. For $F(s) = e^{zs}$,

$$\mathcal{P}_{N,Q,\eta}^\phi(e^{zs}) = \sum_{j=1}^N a_j^{(N)} e^{zj/Q^2} \int_{\mathbf{R}} \phi(u) e^{\eta zu} du = \widehat{\phi}(\eta z) \widehat{\mu}_{N,Q}(z),$$

which proves the first statement.

Since ϕ is even, $\widehat{\phi}$ is an even entire function. In particular,

$$\widehat{\phi}(0) = \int_{\mathbf{R}} \phi(u) du = 1, \quad \widehat{\phi}^{(2r+1)}(0) = 0 \quad (r \geq 0).$$

Lemma 11.2 gives

$$\widehat{\mu}_{N,Q}(0) = 1, \quad \partial_z^{2r+1} \widehat{\mu}_{N,Q}(0) = 0 \quad (0 \leq r \leq N-2).$$

Applying Leibniz' rule to the product $\widehat{\phi}(\eta z) \widehat{\mu}_{N,Q}(z)$ shows that all odd derivatives up to order $2N-3$ vanish at $z=0$, proving the second statement.

For the quantitative comparison, expand F at j/Q^2 :

$$F\left(\frac{j}{Q^2} + \eta u\right) = F\left(\frac{j}{Q^2}\right) + \eta u F'\left(\frac{j}{Q^2}\right) + \frac{\eta^2 u^2}{2} F''(\xi_{j,u})$$

for some $\xi_{j,u} \in \Omega$. Since ϕ is even,

$$\int_{\mathbf{R}} \phi(u) u du = 0.$$

Also $\int \phi(u) du = 1$. Therefore

$$\int_{\mathbf{R}} \phi(u) F\left(\frac{j}{Q^2} + \eta u\right) du - F\left(\frac{j}{Q^2}\right) = \frac{\eta^2}{2} \int_{\mathbf{R}} \phi(u) u^2 F''(\xi_{j,u}) du.$$

Summing against $a_j^{(N)}$ and using Lemma 11.1,

$$\sum_{j=1}^N |a_j^{(N)}| = 2N-1,$$

gives

$$|\mathcal{P}_{N,Q,\eta}^\phi F - \mathcal{P}_{N,Q} F| \leq \frac{\eta^2}{2} (2N-1) m_2(\phi) \sup_{\Omega} |F''|. \quad \square$$

Remark 11.31 (Smoothing scale). Since $N \asymp Q$, Lemma 11.30 gives

$$\mathcal{P}_{N,Q,\eta}^\phi F = \mathcal{P}_{N,Q} F + O(Q\eta^2 \sup |F''|).$$

Thus any choice $\eta = o(Q^{-1/2})$ makes the smoothing error $o(1)$ while preserving the exact low odd-jet annihilation. This does not enter the present proof, but it shows that the discrete N -point projector admits a genuinely smooth Mellin-side avatar with the same exact cancellation pattern.

Proposition 11.32 (Smoothed comparison for the corrected N -point zeta scalar). *Let $\phi \in C_c^\infty(\mathbf{R})$ be even, normalized by*

$$\int_{\mathbf{R}} \phi(u) du = 1,$$

and set

$$U_\phi := \sup\{|u| : u \in \text{supp } \phi\}, \quad m_2(\phi) := \int_{\mathbf{R}} |u|^2 |\phi(u)| du.$$

Fix $0 < \alpha < \rho_0$, let

$$N = \lfloor \alpha Q \rfloor,$$

and assume

$$0 < \eta \leq \frac{\rho_0 - \alpha}{2U_\phi Q}.$$

Define the smoothed corrected scalar

$$H_{m,\eta}(s) := \int_{\mathbf{R}} \phi(u) H_m(s + \eta u) du,$$

and the corresponding smoothed N -point scalar

$$\Xi_{\zeta,\eta}^{(N)}(m) := \sum_{j=1}^N a_j^{(N)} H_{m,\eta}\left(\frac{j}{Q^2}\right).$$

Then:

(1) $H_{m,\eta}$ is odd and holomorphic on

$$|s| < \frac{\rho_\eta}{Q}, \quad \rho_\eta := \rho_0 - U_\phi \eta Q > \alpha.$$

(2) One has the zeta-side exponential estimate

$$|\Xi_{\zeta,\eta}^{(N)}(m)| \ll_{\phi,\rho_0,\alpha} \frac{L(m)S(m)^2}{Q} e^{-\delta_*(\alpha,\rho_\eta)Q},$$

where

$$\delta_*(\alpha, \rho_\eta) = \rho_\eta \left((1 + \beta_\eta) \log(1 + \beta_\eta) - 2\beta_\eta \log \beta_\eta - (1 - \beta_\eta) \log(1 - \beta_\eta) \right), \quad \beta_\eta := \frac{\alpha}{\rho_\eta}.$$

(3) The smoothed and discrete N -point scalars satisfy

$$|\Xi_{\zeta,\eta}^{(N)}(m) - \Xi_\zeta^{(N)}(m)| \leq \frac{4C_\partial(\rho_0) m_2(\phi)}{(\rho_0 - \alpha)^2} \frac{2N - 1}{Q} \eta^2 L(m) S(m)^2.$$

In particular,

$$|\Xi_{\zeta,\eta}^{(N)}(m) - \Xi_\zeta^{(N)}(m)| \ll_{\phi,\rho_0,\alpha} \eta^2 L(m) S(m)^2.$$

Proof. Since H_m is odd and ϕ is even,

$$H_{m,\eta}(-s) = \int_{\mathbf{R}} \phi(u) H_m(-s + \eta u) du = \int_{\mathbf{R}} \phi(u) H_m(-(s + \eta u)) du = -H_{m,\eta}(s),$$

so $H_{m,\eta}$ is odd. If $|s| < \rho_\eta/Q$ and $u \in \text{supp } \phi$, then

$$|s + \eta u| \leq |s| + \eta U_\phi < \frac{\rho_0 - U_\phi \eta Q}{Q} + \eta U_\phi = \frac{\rho_0}{Q}.$$

Since H_m is holomorphic on $|s| < \rho_0/Q$, it follows that $H_{m,\eta}$ is holomorphic on $|s| < \rho_\eta/Q$. This proves (1).

For $|s| = \rho_\eta/Q$ and $u \in \text{supp } \phi$, the same estimate gives

$$|s + \eta u| \leq \frac{\rho_0}{Q}.$$

Hence Proposition 10.1 implies

$$|H_m(s + \eta u)| \leq C_\partial(\rho_0) \frac{L(m)S(m)^2}{Q^3}.$$

Therefore

$$|H_{m,\eta}(s)| \leq \|\phi\|_{L^1} C_\partial(\rho_0) \frac{L(m)S(m)^2}{Q^3} \quad \left(|s| = \frac{\rho_\eta}{Q} \right).$$

Since $H_{m,\eta}$ is odd and holomorphic on $|s| < \rho_\eta/Q$, the proof of Corollary 11.22 applies verbatim with ρ_0 replaced by ρ_η . This gives

$$|\Xi_{\zeta,\eta}^{(N)}(m)| \ll_{\phi,\rho_0,\alpha} \frac{L(m)S(m)^2}{Q} e^{-\delta_*(\alpha,\rho_\eta)Q},$$

proving (2).

For (3), note that

$$\Xi_{\zeta,\eta}^{(N)}(m) - \Xi_\zeta^{(N)}(m) = (\mathcal{P}_{N,Q,\eta}^\phi - \mathcal{P}_{N,Q})H_m.$$

Apply Lemma 11.30 with $F = H_m$. It remains to bound H_m'' on the relevant region. Since

$$\frac{j}{Q^2} \leq \frac{N}{Q^2} \leq \frac{\alpha}{Q}, \quad \eta U_\phi \leq \frac{\rho_0 - \alpha}{2Q},$$

all points

$$\frac{j}{Q^2} + \eta u \quad (1 \leq j \leq N, u \in \text{supp } \phi)$$

lie in the disk

$$|s| \leq \frac{\alpha}{Q} + \eta U_\phi \leq \frac{\rho_0 + \alpha}{2Q}.$$

Thus their distance to the boundary circle $|s| = \rho_0/Q$ is at least

$$\frac{\rho_0 - \alpha}{2Q}.$$

By Cauchy's derivative estimate and Proposition 10.1,

$$\sup_{\Omega} |H_m''| \leq \frac{2 C_{\partial}(\rho_0) L(m)S(m)^2/Q^3}{((\rho_0 - \alpha)/(2Q))^2} = \frac{8 C_{\partial}(\rho_0)}{(\rho_0 - \alpha)^2} \frac{L(m)S(m)^2}{Q}.$$

Lemma 11.30 therefore yields

$$|\Xi_{\zeta,\eta}^{(N)}(m) - \Xi_\zeta^{(N)}(m)| \leq \frac{\eta^2}{2} (2N - 1) m_2(\phi) \frac{8 C_{\partial}(\rho_0)}{(\rho_0 - \alpha)^2} \frac{L(m)S(m)^2}{Q},$$

which is exactly the stated bound. □

Remark 11.33 (Microscopic smoothing regime). If

$$\eta \ll Q^{-2},$$

then Proposition 11.32 gives

$$\Xi_{\zeta,\eta}^{(N)}(m) = \Xi_\zeta^{(N)}(m) + O\left(\frac{L(m)S(m)^2}{Q^4}\right).$$

At the same time,

$$\rho_\eta = \rho_0 + O(Q^{-1}),$$

so

$$\delta_*(\alpha, \rho_\eta) = \delta_*(\alpha, \rho_0) + O(Q^{-1}).$$

Thus microscopic smoothing preserves the same zeta-side exponential scale while replacing the discrete projector by a genuinely smooth Mellin-side test function.

Lemma 11.34 (Higher-order smoothed Mellin projector). *Fix an integer $M \geq 1$. Let $\phi_M \in C_c^\infty(\mathbf{R})$ be even and satisfy*

$$\int_{\mathbf{R}} \phi_M(u) du = 1, \quad \int_{\mathbf{R}} u^{2\ell} \phi_M(u) du = 0 \quad (1 \leq \ell \leq M - 1).$$

Define its bilateral Laplace transform by

$$\widehat{\phi}_M(w) := \int_{\mathbf{R}} \phi_M(u) e^{wu} du.$$

For $\eta > 0$, define

$$(\mathcal{P}_{N,Q,\eta}^{(M)} F) := \sum_{j=1}^N a_j^{(N)} \int_{\mathbf{R}} \phi_M(u) F\left(\frac{j}{Q^2} + \eta u\right) du,$$

whenever F is holomorphic on a neighborhood of the integration region.

Then:

(1) for every exponential mode $F(s) = e^{zs}$,

$$\mathcal{P}_{N,Q,\eta}^{(M)}(e^{zs}) = \widehat{\phi}_M(\eta z) \widehat{\mu}_{N,Q}(z), \quad \widehat{\mu}_{N,Q}(z) = \sum_{j=1}^N a_j^{(N)} e^{zj/Q^2};$$

(2) one has

$$\widehat{\phi}_M(0) = 1, \quad \widehat{\phi}_M^{(k)}(0) = 0 \quad (1 \leq k \leq 2M-1);$$

(3) the smoothed symbol preserves the exact low odd-jet annihilation:

$$\partial_z^{2r+1}(\widehat{\phi}_M(\eta z) \widehat{\mu}_{N,Q}(z)) \Big|_{z=0} = 0 \quad (0 \leq r \leq N-2);$$

(4) if F is holomorphic on a neighborhood Ω containing all points

$$\frac{j}{Q^2} + \eta u \quad (1 \leq j \leq N, u \in \text{supp } \phi_M),$$

then

$$|\mathcal{P}_{N,Q,\eta}^{(M)} F - \mathcal{P}_{N,Q} F| \leq \frac{\eta^{2M}}{(2M)!} (2N-1) m_{2M}(\phi_M) \sup_{\Omega} |F^{(2M)}|,$$

where

$$m_{2M}(\phi_M) := \int_{\mathbf{R}} |u|^{2M} |\phi_M(u)| du.$$

Proof. For $F(s) = e^{zs}$,

$$\mathcal{P}_{N,Q,\eta}^{(M)}(e^{zs}) = \sum_{j=1}^N a_j^{(N)} e^{zj/Q^2} \int_{\mathbf{R}} \phi_M(u) e^{\eta zu} du = \widehat{\phi}_M(\eta z) \widehat{\mu}_{N,Q}(z),$$

which proves the first statement.

Since ϕ_M is even, all odd moments vanish. By the imposed even-moment conditions, one also has

$$\int_{\mathbf{R}} u^k \phi_M(u) du = 0 \quad (1 \leq k \leq 2M-1).$$

Differentiating $\widehat{\phi}_M$ at 0 gives

$$\widehat{\phi}_M^{(k)}(0) = \int_{\mathbf{R}} u^k \phi_M(u) du,$$

so

$$\widehat{\phi}_M(0) = 1, \quad \widehat{\phi}_M^{(k)}(0) = 0 \quad (1 \leq k \leq 2M-1).$$

This proves the second statement.

By Lemma 11.2,

$$\widehat{\mu}_{N,Q}(0) = 1, \quad \partial_z^{2r+1} \widehat{\mu}_{N,Q}(0) = 0 \quad (0 \leq r \leq N-2).$$

Applying Leibniz' rule to $\widehat{\phi}_M(\eta z) \widehat{\mu}_{N,Q}(z)$ shows that all odd derivatives up to order $2N-3$ vanish at $z=0$, proving the third statement.

For the quantitative comparison, expand F to order $2M-1$ at j/Q^2 :

$$F\left(\frac{j}{Q^2} + \eta u\right) = \sum_{k=0}^{2M-1} \frac{(\eta u)^k}{k!} F^{(k)}\left(\frac{j}{Q^2}\right) + \frac{(\eta u)^{2M}}{(2M)!} F^{(2M)}(\xi_{j,u})$$

for some $\xi_{j,u} \in \Omega$. Integrating against ϕ_M , all terms of order 1 through $2M - 1$ vanish by the moment conditions, while the constant term survives because $\int \phi_M = 1$. Hence

$$\int_{\mathbf{R}} \phi_M(u) F\left(\frac{j}{Q^2} + \eta u\right) du - F\left(\frac{j}{Q^2}\right) = \frac{\eta^{2M}}{(2M)!} \int_{\mathbf{R}} \phi_M(u) u^{2M} F^{(2M)}(\xi_{j,u}) du.$$

Summing against $a_j^{(N)}$ and using Lemma 11.1,

$$\sum_{j=1}^N |a_j^{(N)}| = 2N - 1,$$

gives the claimed bound. $\square$

Proposition 11.35 (Higher-order smoothed comparison for the corrected N -point zeta scalar). *Fix an integer $M \geq 1$, and let $\phi_M \in C_c^\infty(\mathbf{R})$ be even, normalized, and moment-corrected as in Lemma 11.34. Set*

$$U_M := \sup\{|u| : u \in \text{supp } \phi_M\}, \quad m_{2M}(\phi_M) := \int_{\mathbf{R}} |u|^{2M} |\phi_M(u)| du.$$

Fix $0 < \alpha < \rho_0$, let

$$N = \lfloor \alpha Q \rfloor,$$

and assume

$$0 < \eta \leq \frac{\rho_0 - \alpha}{2U_M Q}.$$

Define

$$H_{m,\eta}^{(M)}(s) := \int_{\mathbf{R}} \phi_M(u) H_m(s + \eta u) du,$$

and

$$\Xi_{\zeta,\eta}^{(N,M)}(m) := \sum_{j=1}^N a_j^{(N)} H_{m,\eta}^{(M)}\left(\frac{j}{Q^2}\right).$$

Then:

(1) $H_{m,\eta}^{(M)}$ is odd and holomorphic on

$$|s| < \frac{\rho_{M,\eta}}{Q}, \quad \rho_{M,\eta} := \rho_0 - U_M \eta Q > \alpha;$$

(2) one has

$$|\Xi_{\zeta,\eta}^{(N,M)}(m)| \ll_{\phi_M, \rho_0, \alpha} \frac{L(m)S(m)^2}{Q} e^{-\delta_*(\alpha, \rho_{M,\eta})Q};$$

(3) the smoothed and discrete N -point scalars satisfy

$$|\Xi_{\zeta,\eta}^{(N,M)}(m) - \Xi_{\zeta}^{(N)}(m)| \leq C_{\partial}(\rho_0) (2N - 1) m_{2M}(\phi_M) \left(\frac{2}{\rho_0 - \alpha}\right)^{2M} \eta^{2M} Q^{2M-3} L(m)S(m)^2,$$

and hence

$$|\Xi_{\zeta,\eta}^{(N,M)}(m) - \Xi_{\zeta}^{(N)}(m)| \ll_{\phi_M, \rho_0, \alpha} (\eta Q)^{2M} \frac{L(m)S(m)^2}{Q^2}.$$

Proof. The proof of oddness and holomorphy is identical to that of Proposition 11.32, with ϕ replaced by ϕ_M . Likewise, on $|s| = \rho_{M,\eta}/Q$,

$$|H_{m,\eta}^{(M)}(s)| \leq \|\phi_M\|_{L^1} C_{\partial}(\rho_0) \frac{L(m)S(m)^2}{Q^3}.$$

Therefore the proof of Corollary 11.22 applies verbatim with ρ_0 replaced by $\rho_{M,\eta}$, giving the stated exponential bound.

For the comparison estimate,

$$\Xi_{\zeta,\eta}^{(N,M)}(m) - \Xi_{\zeta}^{(N)}(m) = (\mathcal{P}_{N,Q,\eta}^{(M)} - \mathcal{P}_{N,Q})H_m.$$

Apply Lemma 11.34 with $F = H_m$. Exactly as in the proof of Proposition 11.32, all relevant points lie at distance at least

$$\frac{\rho_0 - \alpha}{2Q}$$

from the boundary circle $|s| = \rho_0/Q$. Hence Cauchy's derivative estimate and Proposition 10.1 give

$$\sup_{\Omega} |H_m^{(2M)}| \leq (2M)! C_{\partial}(\rho_0) \left(\frac{2Q}{\rho_0 - \alpha} \right)^{2M} \frac{L(m)S(m)^2}{Q^3}.$$

Substituting this into Lemma 11.34 yields

$$|\Xi_{\zeta, \eta}^{(N, M)}(m) - \Xi_{\zeta}^{(N)}(m)| \leq C_{\partial}(\rho_0) (2N - 1) m_{2M}(\phi_M) \left(\frac{2}{\rho_0 - \alpha} \right)^{2M} \eta^{2M} Q^{2M-3} L(m)S(m)^2,$$

as claimed. Since $N \asymp Q$, the final big-Oh form follows. $\square$

Remark 11.36 (Saturation at the natural Q^{-1} smoothing scale). Proposition 11.35 shows that higher-order moment correction improves the comparison error to order $2M$, but under the largest admissible smoothing scale

$$\eta \asymp Q^{-1},$$

one still has

$$|\Xi_{\zeta, \eta}^{(N, M)}(m) - \Xi_{\zeta}^{(N)}(m)| \ll \frac{L(m)S(m)^2}{Q^2}.$$

Thus higher-order smoothing does not improve the Q -power at the natural admissible scale; its gain is in canonicity and in the ability to make the smoothed projector arbitrarily close to the discrete one once one works at a smaller scale. For example, if

$$\eta = Q^{-1-\sigma} \quad (\sigma > 0),$$

then

$$|\Xi_{\zeta, \eta}^{(N, M)}(m) - \Xi_{\zeta}^{(N)}(m)| \ll Q^{-2-2M\sigma} L(m)S(m)^2.$$

We isolate the toy microscopic expansion in Appendix A; the only fact used in the main argument is the following N -point preservation statement.

Proposition 11.37 (Toy N -point transverse preservation). *Fix a compact interval*

$$D = [d_-, d_+] \subset (0, \infty),$$

let ρ_{toy} and $u_0(D)$ be as in Lemma A.1, and fix

$$\alpha > 0.$$

For

$$N = \lfloor \alpha Q \rfloor, \quad s_j = \frac{j}{Q^2},$$

define

$$\Xi_{\text{toy}}^{(N)}(u, d) := \sum_{j=1}^N a_j^{(N)} \Phi_K(\Delta_{\text{toy}}(u, d; s_j)).$$

Then, for all sufficiently large Q (depending on D and α), for $d \in D$ and $|u| < u_0(D)$,

$$\Xi_{\text{toy}}^{(N)}(u, d) = u^2 \Phi_K(M(d)) + O(u^4),$$

uniformly in $d \in D$. In particular,

$$|\Xi_{\text{toy}}^{(N)}(u, d)| \asymp u^2$$

uniformly on compact d -ranges.

Proof. By Lemma A.1,

$$\Delta_{\text{toy}}(u, d; s) = u^2 M(d) + u^4 E(u, d; s),$$

where $E(u, d; s)$ is analytic in s on $|s| < \rho_{\text{toy}}$ and uniformly bounded there for $d \in D$ and $|u| < u_0(D)$. Applying the linear functional Φ_K , define

$$R_{u,d}(s) := \Phi_K(E(u, d; s)).$$

Then $R_{u,d}(s)$ is analytic on $|s| < \rho_{\text{toy}}$, and by Cauchy's estimate its Taylor coefficients satisfy

$$R_{u,d}(s) = \sum_{n \geq 0} b_n(u, d) s^n, \quad |b_n(u, d)| \ll \rho_{\text{toy}}^{-n},$$

uniformly for $d \in D$ and $|u| < u_0(D)$.

Therefore

$$\Phi_K(\Delta_{\text{toy}}(u, d; s_j)) = u^2 \Phi_K(M(d)) + u^4 R_{u,d}(s_j),$$

so

$$\Xi_{\text{toy}}^{(N)}(u, d) = u^2 \Phi_K(M(d)) + u^4 \sum_{j=1}^N a_j^{(N)} R_{u,d}(s_j).$$

Expand $R_{u,d}$ at the microscopic nodes $s_j = j/Q^2$:

$$\sum_{j=1}^N a_j^{(N)} R_{u,d}(s_j) = \sum_{n \geq 0} b_n(u, d) \frac{\mu_n^{(N)}}{Q^{2n}}, \quad \mu_n^{(N)} := \sum_{j=1}^N a_j^{(N)} j^n.$$

The constant term contributes

$$b_0(u, d) \mu_0^{(N)} = b_0(u, d),$$

since $\mu_0^{(N)} = \sum_j a_j^{(N)} = 1$ by Lemma 11.1. For the remaining terms, split into even and odd parts.

For the even terms, Lemma 11.1 gives

$$|\mu_{2\ell}^{(N)}| \leq (2N-1)N^{2\ell},$$

so

$$\sum_{\ell \geq 1} |b_{2\ell}(u, d)| \frac{|\mu_{2\ell}^{(N)}|}{Q^{4\ell}} \ll (2N-1) \sum_{\ell \geq 1} \rho_{\text{toy}}^{-2\ell} \frac{N^{2\ell}}{Q^{4\ell}} \ll \frac{1}{Q},$$

because $N \asymp Q$.

For the odd terms, Lemma 11.1 gives

$$\mu_{2r+1}^{(N)} = 0 \quad (0 \leq r \leq N-2),$$

so only the tail $r \geq N-1$ remains. Using the coefficient bound $|b_{2r+1}(u, d)| \ll \rho_{\text{toy}}^{-(2r+1)}$, we obtain

$$\sum_{r \geq N-1} |b_{2r+1}(u, d)| \frac{|\mu_{2r+1}^{(N)}|}{Q^{4r+2}} \ll \sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| (\rho_{\text{toy}} Q^2)^{-(2r+1)}.$$

Now apply Lemma 11.3 with

$$x = \rho_{\text{toy}} Q^2.$$

For all sufficiently large Q , one has $x > N$, and therefore

$$\sum_{r \geq N-1} |\mu_{2r+1}^{(N)}| (\rho_{\text{toy}} Q^2)^{-(2r+1)} = (2N-1)N!(N-1)!(\rho_{\text{toy}} Q^2)^2 \frac{\Gamma(\rho_{\text{toy}} Q^2 - N)}{\Gamma(\rho_{\text{toy}} Q^2 + N + 1)}.$$

Since $N = \lfloor \alpha Q \rfloor$ and $x = \rho_{\text{toy}} Q^2$, Stirling's formula gives

$$(2N-1)N!(N-1)!(\rho_{\text{toy}} Q^2)^2 \frac{\Gamma(\rho_{\text{toy}} Q^2 - N)}{\Gamma(\rho_{\text{toy}} Q^2 + N + 1)} = \exp(-2\alpha Q \log Q + O(Q)).$$

In particular, the residual odd toy tail is super-exponentially small on the Q -scale:

$$\sum_{r \geq N-1} |b_{2r+1}(u, d)| \frac{|\mu_{2r+1}^{(N)}|}{Q^{4r+2}} \ll \exp(-2\alpha Q \log Q + O(Q)).$$

Combining the constant, even, and odd contributions,

$$\sum_{j=1}^N a_j^{(N)} R_{u,d}(s_j) = b_0(u, d) + O(Q^{-1}) + O(e^{-cQ}),$$

and in particular

$$\sum_{j=1}^N a_j^{(N)} R_{u,d}(s_j) = O(1),$$

uniformly for $d \in D$ and $|u| < u_0(D)$. Therefore

$$\Xi_{\text{toy}}^{(N)}(u, d) = u^2 \Phi_K(M(d)) + O(u^4),$$

uniformly in $d \in D$.

Because $\Phi_K(M(d)) \neq 0$ for $d > 0$, and is bounded away from 0 on every compact interval $D \subset (0, \infty)$, the final estimate

$$|\Xi_{\text{toy}}^{(N)}(u, d)| \asymp u^2$$

follows. □

Remark 11.38 (The toy odd tail is negligible beyond all e^{-cQ} -scales). The proof of Proposition 11.37 shows more than the $O(u^4)$ preservation statement used later: after N -point cancellation, the residual odd toy tail is of size

$$\exp(-2\alpha Q \log Q + O(Q)).$$

Thus on the toy side the N -point odd leakage is not merely exponentially small in Q ; it is negligible beyond every e^{-cQ} -scale.

On the toy side, Proposition 11.37 gives

$$(22) \quad \Xi_{\text{toy}}^{(N)}(u, d) = u^2 \Phi_K(M(d)) + O(u^4),$$

so on compact d -ranges,

$$|\Xi_{\text{toy}}^{(N)}(u, d)| \asymp u^2.$$

Using (19),

$$(23) \quad |\Xi_{\text{toy}}^{(N)}(u, d)| \asymp \frac{x_0}{B_\zeta(m)} S(m).$$

Remark 11.39 (Work in progress: calibrated size versus microscopic toy regime). As presently written, the final calibration freezes the comparison scale at a fixed value $x = x_0(D)$, which yields

$$u^2 \asymp \frac{x_0}{B_\zeta(m)} S(m).$$

With only the crude global bounds used later in the contradiction,

$$S(m) \ll Q^2, \quad B_\zeta(m) \asymp Q,$$

this gives at best

$$u^2 \ll x_0 Q,$$

and therefore does not by itself force the calibrated toy parameter u to lie in the microscopic toy regime $|u| < u_0(D)$ required by Proposition 11.37.

A plausible repair is to avoid freezing the scale at $x_0(D)$, and instead to choose the comparison scale adaptively:

$$x = x(m) \in (0, x_0(D)].$$

Since the pairing and calibration statements are naturally formulated for $0 < x \leq x_0(D)$, one expects to choose $x(m)$ so that

$$u^2 \asymp \frac{x(m)}{B_\zeta(m)} S(m)$$

is forced to remain below the toy threshold $u_0(D)^2$. For example, if the implicit upper bound

$$u^2 \leq C \frac{x}{B_\zeta(m)} S(m)$$

is available uniformly, then one may try to take

$$x(m) := \min \left\{ x_0(D), \frac{u_0(D)^2}{2C} \frac{B_\zeta(m)}{S(m)} \right\},$$

which would guarantee

$$u^2 \leq \frac{u_0(D)^2}{2}.$$

To make this repair rigorous, however, the draft must still verify that the entire calibration chain remains valid uniformly in such a variable x -regime. In particular, one still needs to check all of the following:

- (1) The pairing and denominator estimates remain uniform for all $0 < x \leq x_0(D)$, including the regime $x \rightarrow 0$, so that the calibration functional is still well-defined and the main term continues to scale linearly in x .
- (2) The fixed-scaled statements currently written in terms of

$$x = B_\zeta(m)\sigma, \quad |\sigma| \asymp Q^{-1},$$

can be upgraded to a variable/shrinking regime of the form

$$x = B_\zeta(m)\sigma, \quad |\sigma| \leq c Q^{-1},$$

or an equivalent $0 < x \leq x_0(D)$ formulation, without loss of the transfer, whitening, and calibration estimates used later.

- (3) The remainder bound after applying the calibrated functional remains uniform in the variable- x regime; in particular, one needs the normalized error estimate to stay independent of x , or at least sufficiently mild as $x \rightarrow 0$, so that shrinking x to enforce $|u| < u_0(D)$ does not destroy the zeta-side control.
- (4) The cutoff choice required in the calibration remainder estimate can still be made uniformly when $x = x(m)$ is allowed to shrink. Concretely, one must check that the condition

$$R^3 \gg \frac{1}{xQ S(m)}$$

remains compatible with the admissible cutoff regime under the adaptive choice of $x(m)$.

- (5) The final contradiction theorem must be reformulated so that the toy-side main term is compared at the variable scale $x(m)$, rather than at the fixed scale $x_0(D)$, and one must verify that the resulting lower bound still dominates the exponentially small zeta-side bound.

Accordingly, the small- u issue does not presently appear to be a fundamental obstruction, but the current argument should be read as conditional on a variable- x calibration rewrite that has not yet been carried out in the draft.

The following remark records the current status of the toy model in the overall proof strategy.

Remark 11.40 (Work in progress: pair-like versus finite-core endgame). At the current stage of the draft, the toy anomaly model should be understood as a *pair-core first-odd-jet normal form*, not yet as a universal normal form for an arbitrary finite same-scale off-line core.

More precisely, in the pure pair model one has first-odd-jet nondegeneracy:

$$a_1^{\text{toy}}(u, d) = \kappa(d)u^2 + O(u^4), \quad \kappa(d) \neq 0,$$

so the first odd jet already detects the off-line anomaly.

However, the fixed-core analysis developed later in the draft shows that for a general finite same-scale core, cancellation may occur at the first odd order even when a genuine corrected anomaly survives at a higher odd order. Thus the toy model is presently justified as the correct first-odd-jet simplification only under a *pair-like core* hypothesis (or a perturbative enlargement thereof), not yet for an arbitrary finite core.

Accordingly, the argument now has a genuine fork:

- (1) **Pair-like branch.** If the actual same-scale core is perturbatively pair-like in the corrected microscopic sense, then the present toy-model first-jet obstruction should still apply, and the contradiction can be driven by the first odd coefficient.
- (2) **Finite-core branch.** If the actual same-scale core is not perturbatively pair-like, then the correct endgame must be reformulated in terms of a genuine finite-core anomaly model, and the obstruction should be attached not specifically to the first odd jet, but to the *first nonzero odd jet* of the corrected transverse scalar.

Thus the toy model is best viewed, at present, as the first-odd-jet simplification of a pair-like core, rather than as a universal model for all finite cores. This is one of the main places where the present manuscript should still be read as a work in progress.

12. LOCAL PROJECTIVE RIGIDITY OF THE ONE-PAIR AND MULTI-PAIR FAMILIES

Throughout this section, $\mathcal{Q}^{\text{abs}}$ denotes the abstract odd-germ quotient by the value/gauge family (not the height scale $Q \asymp \log |m|$ used elsewhere in the paper). For the first residual step on the cubic-generic region, we will also introduce an explicit finite-order model $\mathcal{R}_h^{(5)}$ through order 5. The latter supplements, but does not globally replace, the ambient abstract quotient $\mathcal{Q}^{\text{abs}}$.

12.1. Cubic leading line and the first residual class for the one-pair family. Let

$$F_h(z) \in \mathcal{Q}^{\text{abs}}[[z^2]]$$

denote the corrected quotient odd germ of the normalized one-pair family, where $\mathcal{Q}^{\text{abs}}$ is the abstract odd-germ quotient by the value/gauge family.

Lemma 12.1 (First-order value-family generator). *Let*

$$\mathcal{W}(G_-, N, G_+) := G_-^{-1/2} N G_+^{-1/2}.$$

Let G_0 and N_0 denote the coincident background same-point and mixed blocks, and write

$$X_0 := G_0^{-1/2}.$$

Let L be the Fréchet derivative of the inverse-square-root map $G \mapsto G^{-1/2}$ at G_0 , so that

$$(G_0 + \varepsilon H)^{-1/2} = X_0 + \varepsilon L(H) + O(\varepsilon^2).$$

If $\dot{G}_{-,1}$, $\dot{N}_1$, and $\dot{G}_{+,1}$ are the first-order odd pure-value variations of the left same-point block, the mixed block, and the right same-point block, respectively, then the induced first-order odd coefficient in the corrected local block is

$$B_1 = L(\dot{G}_{-,1}) N_0 X_0 + X_0 \dot{N}_1 X_0 + X_0 N_0 L(\dot{G}_{+,1}).$$

Equivalently, the first-order family $V_{\text{val}}^{(1)}$ is the image of the linear map

$$(\dot{G}_{-,1}, \dot{N}_1, \dot{G}_{+,1}) \mapsto L(\dot{G}_{-,1}) N_0 X_0 + X_0 \dot{N}_1 X_0 + X_0 N_0 L(\dot{G}_{+,1}).$$

Proof. The corrected local block is obtained by the analytic whitening map

$$\mathcal{W}(G_-, N, G_+) = G_-^{-1/2} N G_+^{-1/2}.$$

Expand the three inputs to first odd order under a pure value deformation:

$$G_-(s) = G_0 + \varepsilon \dot{G}_{-,1} + O(\varepsilon^2), \quad N(s) = N_0 + \varepsilon \dot{N}_1 + O(\varepsilon^2), \quad G_+(s) = G_0 + \varepsilon \dot{G}_{+,1} + O(\varepsilon^2).$$

Then

$$G_{\pm}(s)^{-1/2} = X_0 + \varepsilon L(\dot{G}_{\pm,1}) + O(\varepsilon^2),$$

so multiplying

$$G_-(s)^{-1/2} N(s) G_+(s)^{-1/2}$$

to first order gives

$$X_0 N_0 X_0 + \varepsilon \left(L(\dot{G}_{-,1}) N_0 X_0 + X_0 \dot{N}_1 X_0 + X_0 N_0 L(\dot{G}_{+,1}) \right) + O(\varepsilon^2).$$

The coefficient of ε is exactly the first-order generator B_1 . □

Lemma 12.2 (Whitened first-order reduction). *At the coincident background one has*

$$N_0 = G_0.$$

If

$$\tilde{H} := X_0 H X_0$$

denotes whitening by $X_0 = G_0^{-1/2}$, then the first-order generator of Lemma 12.1 becomes

$$\tilde{B}_1 = \tilde{N}_1 - \frac{1}{2} \tilde{\dot{G}}_{-,1} - \frac{1}{2} \tilde{\dot{G}}_{+,1}.$$

Proof. Since $X_0 G_0 X_0 = I$, whitening conjugates the background to the identity. The Fréchet derivative of $G \mapsto G^{-1/2}$ at I is

$$L_I(H) = -\frac{1}{2}H.$$

Therefore, after whitening the formula of Lemma 12.1 becomes

$$\tilde{B}_1 = L_I(\tilde{\dot{G}}_{-,1}) I I + I \tilde{N}_1 I + I I L_I(\tilde{\dot{G}}_{+,1}) = \tilde{N}_1 - \frac{1}{2} \tilde{\dot{G}}_{-,1} - \frac{1}{2} \tilde{\dot{G}}_{+,1}. \quad \square$$

Lemma 12.3 (First-order same-point span). *The first-order pure-value same-point variations span the full symmetric subspace:*

$$\text{span}\{\tilde{\dot{G}}_{-,1}\} = \text{span}\{\tilde{\dot{G}}_{+,1}\} = \text{Sym}_2(\mathbf{C}).$$

Proof. The corrected same-point block has the form

$$G_{m,\pm}^{\text{corr}}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_{\pm}) & \frac{1}{2}q'(t_{\pm}) \\ \frac{1}{2}q'(t_{\pm}) & \frac{1}{12}(q''(t_{\pm}) + 2q(t_{\pm})^3) \end{pmatrix}.$$

Therefore an infinitesimal pure-value variation at one side gives

$$\delta G_{\pm} = \frac{1}{\pi} \begin{pmatrix} 2\delta q_{\pm} & \frac{1}{2}\delta q'_{\pm} \\ \frac{1}{2}\delta q'_{\pm} & \frac{1}{12}(\delta q''_{\pm} + 6q_{\pm,0}^2 \delta q_{\pm}) \end{pmatrix}.$$

As $\delta q_{\pm}$, $\delta q'_{\pm}$, and $\delta q''_{\pm}$ vary independently, the off-diagonal entry, the (1,1)-entry, and the pure (2,2)-direction may be varied independently. Hence the span is the full space $\text{Sym}_2(\mathbf{C})$. Whitening by X_0 preserves the symmetric subspace and is invertible, so the same conclusion holds for $\tilde{\dot{G}}_{\pm,1}$. □

Lemma 12.4 (First-order mixed skew direction). *Let*

$$\mathfrak{so}_2(\mathbf{C}) := \left\{ \begin{pmatrix} 0 & \xi \\ -\xi & 0 \end{pmatrix} : \xi \in \mathbf{C} \right\}.$$

Then the first-order pure-value mixed variations contain the full skew line:

$$\mathfrak{so}_2(\mathbf{C}) \subset \text{span}\{\tilde{N}_1\}.$$

Equivalently, the first-order mixed family contains a nonzero skew direction.

Proof. Freeze the same-point data and vary only the phase gap:

$$\Delta_m(s) \mapsto \Delta_m(s) + \varepsilon \delta \Delta(s), \quad \delta \Delta(s) = \theta s^3 + O(s^5), \quad \theta \neq 0.$$

Linearizing the mixed block formula in ε , the two off-diagonal entries vary with opposite sign, while the diagonal entries do not contribute at first odd order. More precisely,

$$\delta N_{12}(s) = \frac{\theta}{\pi} s + O(s^3), \quad \delta N_{21}(s) = -\frac{\theta}{\pi} s + O(s^3),$$

up to the overall sign convention for θ . Hence the first odd mixed coefficient of this phase-gap variation is a nonzero element of

$$\mathfrak{so}_2(\mathbf{C}).$$

Whitening by X_0 is invertible, so the whitened first-order mixed family also contains a nonzero skew element. Since $\mathfrak{so}_2(\mathbf{C})$ is one-dimensional, this is equivalent to

$$\mathfrak{so}_2(\mathbf{C}) \subset \text{span}\{\tilde{N}_1\}.$$

□

Proposition 12.5 (Generic triviality of the first quotient order). *The first-order pure-value family fills the full matrix space:*

$$V_{\text{val}}^{(1)} = M_2(\mathbf{C}).$$

Consequently

$$\mathcal{J}_1/V_{\text{val}}^{(1)} = 0,$$

so the generic leading quotient order is at least 3.

Proof. By Lemma 12.2, the first-order generator is

$$\tilde{B}_1 = \tilde{N}_1 - \frac{1}{2}\tilde{G}_{-,1} - \frac{1}{2}\tilde{G}_{+,1}.$$

By Lemma 12.3, the same-point directions $\tilde{G}_{-,1}$ and $\tilde{G}_{+,1}$ span the full symmetric subspace $\text{Sym}_2(\mathbf{C})$. By Lemma 12.4, the mixed directions contain the skew line

$$\mathfrak{so}_2(\mathbf{C}).$$

Therefore the image of the generator contains

$$\text{Sym}_2(\mathbf{C}) \oplus \mathfrak{so}_2(\mathbf{C}) = M_2(\mathbf{C}).$$

Hence

$$V_{\text{val}}^{(1)} = M_2(\mathbf{C}),$$

and so

$$\mathcal{J}_1/V_{\text{val}}^{(1)} = 0.$$

Thus the generic leading quotient order is at least 3. □

Remark 12.6 (Raw versus full value family). The first-order surjectivity statement concerns the full value family as it appears in the three-input derivative

$$D\mathcal{W}_{(\text{bg})}[\dot{G}_-, \dot{N}, \dot{G}_+].$$

This is larger than the raw swap-odd one-pair slice obtained by tying the left and right same-point variations together. In particular, at first order the combined family $V_{\text{val}}^{(1)}$ fills $M_2(\mathbf{C})$, while the odd coefficients of the raw corrected local block are constrained by additional swap/parity structure. Thus the present notation

$$V_{\text{val}}^{(2j+1)}$$

bundles together at least two logically distinct sources of ambiguity: raw pure-value odd jets and additional normalization/basis-gauge directions. The explicit finite-order model introduced later makes this split rigorous through order 5; outside that local model, the ambient notation $V_{\text{val}}^{(2j+1)}$ remains schematic.

Lemma 12.7 (Rigidity of the cubic scalar B). *Let I be a connected parameter interval, and define*

$$B(h) := \frac{q(h)^2 q'(h)}{4} - \frac{q'''(h)}{48}.$$

Then the following are equivalent:

- (1) $B \equiv 0$ on I ;
- (2) $q''' = 12q^2 q'$ on I ;

(3) there exists a constant $C_0 \in \mathbf{C}$ such that

$$q'' = 4q^3 + C_0 \quad \text{on } I.$$

In particular, the hypothesis $B \not\equiv 0$ is equivalent to excluding the special cubic ODE $q'' = 4q^3 + C_0$.

Proof. By definition,

$$48 B = 12q^2 q' - q''.$$

Hence $B \equiv 0$ is equivalent to

$$q''' = (4q^3)'.$$

Integrating on the connected interval I gives

$$q'' = 4q^3 + C_0$$

for some constant $C_0 \in \mathbf{C}$. The converse is immediate by differentiation. $\square$

Proposition 12.8 (Generic cubic leading quotient order). *Let I be a connected one-pair parameter interval, and let $J_3^{\text{pair}}(h)$ denote the cubic jet of the normalized one-pair family.*

Assume:

(1) the cubic value-family satisfies

$$V_{\text{val}}^{(3)} \subset \left\{ \begin{pmatrix} 0 & * \\ * & 0 \end{pmatrix} \right\};$$

(2) the background same-point block is nondegenerate, so that the diagonal entries of $X_0 = G_0^{-1/2}$ satisfy $x_1 \neq x_2$;

(3) the scalar $v(h)$ is not identically zero on I ;

(4) $B(h) \not\equiv 0$ on I , equivalently, by Lemma 12.7, q does not satisfy

$$q'' = 4q^3 + C_0$$

identically on I .

Define the cubic-generic region

$$\mathcal{U}_{\text{cub}} := \{h \in I : v(h)B(h) \neq 0\}.$$

Then $\mathcal{U}_{\text{cub}}$ is open and dense in I , and for every $h \in \mathcal{U}_{\text{cub}}$,

$$[J_3^{\text{pair}}(h)] \neq 0 \quad \text{in } \mathcal{J}_3/V_{\text{val}}^{(3)}.$$

More precisely, the image of $J_3^{\text{pair}}(h)$ in the cubic quotient has nonzero diagonal-traceless component

$$v(h) B(h) (x_1 - x_2) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Consequently,

$$F_h(z) = v_0(h)z^3 + v_1(h)z^5 + \cdots, \quad v_0(h) := [J_3^{\text{pair}}(h)] \neq 0$$

for $h \in \mathcal{U}_{\text{cub}}$. Since the first quotient order is generically trivial, the generic leading quotient order is therefore exactly 3 on $\mathcal{U}_{\text{cub}}$.

Proof. By assumption,

$$\text{diag } J_3^{\text{pair}}(h) = v(h) B(h) (x_1 - x_2) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Because $V_{\text{val}}^{(3)}$ is confined to the off-diagonal sector, it cannot cancel any nonzero diagonal component. Hence for every h with $v(h)B(h)(x_1 - x_2) \neq 0$, the cubic quotient class $[J_3^{\text{pair}}(h)]$ is nonzero.

Since $x_1 \neq x_2$ is fixed and both v and B are not identically zero, the set $\mathcal{U}_{\text{cub}}$ is nonempty; if the dependence on h is real-analytic, its complement is a proper analytic subset, hence $\mathcal{U}_{\text{cub}}$ is open and dense. Finally, Proposition 12.5 shows that the leading quotient order is at least 3, while the present argument shows that it is equal to 3 on $\mathcal{U}_{\text{cub}}$. $\square$

Remark 12.9 (What is genuinely generic here). The cubic-leading statement depends on three logically distinct inputs:

- (1) $x_1 \neq x_2$, a background same-point nondegeneracy condition;
- (2) $v \neq 0$, a one-pair nonfreezing condition;
- (3) exclusion of the special ODE $q'' = 4q^3 + C_0$, equivalently $B \neq 0$.

This separates the structural cubic mechanism from the residual quintic bookkeeping needed later.

Definition 12.10 (Cubic leading line and first residual class). For $h \in \mathcal{U}_{\text{cub}}$, the cubic quotient coefficient

$$v_0(h) = [J_3^{\text{pair}}(h)]$$

is nonzero by Proposition 12.8. Define the cubic leading line by

$$\ell_h := \mathbf{C} v_0(h) \subset \mathcal{Q}^{\text{abs}}.$$

Let $v_1(h) := [J_5^{\text{pair}}(h)]$ denote the quintic quotient coefficient. Its image in the residual quotient

$$\mathcal{Q}^{\text{abs}}/\ell_h$$

is denoted by

$$\bar{v}_1(h) \in \mathcal{Q}^{\text{abs}}/\ell_h.$$

12.2. Normalization gauge at the first residual step. The phrase “value/gauge family” bundles together two logically distinct sources of ambiguity:

- (1) the *pure value family*, recorded coefficientwise by the spaces $V_{\text{val}}^{(2j+1)} \subset \mathcal{J}_{2j+1}$, and
- (2) the *normalization gauge*, consisting of even scalar renormalizations and even jet-basis changes.

For the first residual step it is useful to separate these two.

Definition 12.11 (Normalization gauge). Let $F(z)$ be a matrix-valued odd germ. The normalization gauge acts by

$$F(z) \mapsto a(z) P(z) F(z) P(z)^{-1},$$

where

$$a(z) = 1 + c_2 z^2 + c_4 z^4 + \dots$$

is an even scalar germ with $a(0) = 1$, and

$$P(z) = I + B_2 z^2 + B_4 z^4 + \dots$$

is an even 2×2 matrix germ with $P(0) = I$.

Proposition 12.12 (First-residual transformation law). *Let*

$$F(z) = A_3 z^3 + A_5 z^5 + O(z^7)$$

be an odd matrix germ, and let

$$F^{\text{new}}(z) := a(z) P(z) F(z) P(z)^{-1}$$

with a, P as in Definition 12.11. Then

$$A_3^{\text{new}} = A_3, \quad A_5^{\text{new}} = A_5 + c_2 A_3 + [B_2, A_3].$$

Proof. Expand

$$P(z)^{-1} = I - B_2 z^2 + O(z^4).$$

Then

$$P(z) F(z) P(z)^{-1} = (I + B_2 z^2) (A_3 z^3 + A_5 z^5) (I - B_2 z^2) + O(z^7),$$

so

$$P(z) F(z) P(z)^{-1} = A_3 z^3 + (A_5 + [B_2, A_3]) z^5 + O(z^7).$$

Multiplying by

$$a(z) = 1 + c_2 z^2 + O(z^4)$$

gives

$$F^{\text{new}}(z) = A_3 z^3 + (A_5 + c_2 A_3 + [B_2, A_3]) z^5 + O(z^7),$$

as claimed. $\square$

Corollary 12.13 (Residual effect of normalization gauge). *Let $\ell := \mathbf{C}A_3$. Under the normalization gauge, the quintic coefficient changes by*

$$A_5 \mapsto A_5 + c_2 A_3 + [B_2, A_3].$$

Hence its class in the residual quotient modulo the leading line satisfies

$$[A_5] \mapsto [A_5 + [B_2, A_3]] \quad \text{in} \quad \mathcal{I}_5/\ell.$$

In particular, scalar renormalization is invisible in the first residual quotient.

Proof. This is immediate from Proposition 12.12, since $c_2 A_3 \in \ell$. $\square$

Corollary 12.14 (Trace invariance under normalization gauge). *If A_3 is diagonal-traceless, then every commutator $[B_2, A_3]$ is traceless. Consequently*

$$\text{tr}(A_5^{\text{new}}) = \text{tr}(A_5)$$

for every normalization-gauge transform F^{new} of F .

Proof. The scalar renormalization term $c_2 A_3$ is traceless because A_3 is. The commutator term is traceless because

$$\text{tr}[B_2, A_3] = 0.$$

Hence Proposition 12.12 gives

$$\text{tr}(A_5^{\text{new}}) = \text{tr}(A_5).$$

$\square$

Proposition 12.15 (Second-residual transformation law). *Let*

$$F(z) = A_3 z^3 + A_5 z^5 + A_7 z^7 + O(z^9)$$

be an odd matrix germ, and let

$$F^{\text{new}}(z) := a(z) P(z) F(z) P(z)^{-1}$$

with a, P as in Definition 12.11. Then

$$A_3^{\text{new}} = A_3, \quad A_5^{\text{new}} = A_5 + c_2 A_3 + [B_2, A_3],$$

and the septic coefficient transforms by

$$A_7^{\text{new}} = A_7 + c_2 A_5 + c_4 A_3 + [B_2, A_5] + [B_4, A_3] + c_2 [B_2, A_3] + A_3 B_2^2 - B_2 A_3 B_2.$$

Proof. By Proposition 12.12, the formulas for A_3^{new} and A_5^{new} are already known. It remains to compute the z^7 -coefficient.

Expand

$$P(z)^{-1} = I - B_2 z^2 + (B_2^2 - B_4) z^4 + O(z^6).$$

Then

$$P(z)F(z)P(z)^{-1} = (I + B_2 z^2 + B_4 z^4)(A_3 z^3 + A_5 z^5 + A_7 z^7)(I - B_2 z^2 + (B_2^2 - B_4) z^4) + O(z^9).$$

Collecting the z^7 -terms gives

$$A_7 + [B_2, A_5] + [B_4, A_3] + A_3 B_2^2 - B_2 A_3 B_2.$$

Multiplying by

$$a(z) = 1 + c_2 z^2 + c_4 z^4 + O(z^6)$$

adds the scalar contributions

$$c_2 A_5 + c_4 A_3 + c_2 [B_2, A_3].$$

Hence

$$A_7^{\text{new}} = A_7 + c_2 A_5 + c_4 A_3 + [B_2, A_5] + [B_4, A_3] + c_2 [B_2, A_3] + A_3 B_2^2 - B_2 A_3 B_2,$$

as claimed. $\square$

Lemma 12.16 (Quintic trace witness). *Let $J_{5,K_1}(h)$ denote the K_1 -transport contribution to the quintic jet. At the trace level, the relevant contribution is*

$$\mathrm{tr} J_{5,K_1}(h) = \kappa(h) \mathrm{tr} \left(X_0 J X_4(h) - X_4(h) J X_0 + X_3(h) J X_1(h) - X_1(h) J X_3(h) \right),$$

since

$$\mathrm{tr} (X_2(h) J X_2(h)) = 0$$

for symmetric $X_2(h)$. Therefore

$$\mathrm{tr} J_{5,K_1}(h) = -2\kappa(h) \left((x_1 - x_2)r(h) + \beta(h)(u(h) - w(h)) + v(h)(\gamma(h) - \alpha(h)) \right),$$

where $K_1 = \kappa(h)J$, $v(h)$ is the off-diagonal part of $X_1(h)$, and $\gamma(h) - \alpha(h)$ is the diagonal-traceless part of $X_3(h)$.

Moreover, the linear G_3 -contribution to $\gamma(h) - \alpha(h)$ is

$$(\gamma - \alpha)^{\mathrm{lin}} = \frac{1}{2\pi} \left[x_1^3 \frac{q'''}{24} - x_2^3 \left(\frac{q''''}{576} + \frac{q^2 q'''}{96} + \frac{qq'q''}{16} + \frac{(q')^3}{48} \right) \right].$$

Since

$$v(h) = -\frac{1}{4\pi(s_1 + s_2)} \eta_2(h),$$

the product $v(h)(\gamma(h) - \alpha(h))^{\mathrm{lin}}$ contains the monomial

$$\kappa(h) \eta_2(h) q''''(h)$$

with nonzero coefficient.

Lemma 12.17 (Parity of the whitening expansion at a diagonal base). *Let*

$$G_0 = \mathrm{diag}(\lambda_1, \lambda_2), \quad \lambda_1, \lambda_2 > 0,$$

and let

$$X(H) := (G_0 + H)^{-1/2}$$

for H sufficiently small. Write

$$E := \left\{ \begin{pmatrix} * & 0 \\ 0 & * \end{pmatrix} \right\}, \quad O := \left\{ \begin{pmatrix} 0 & * \\ * & 0 \end{pmatrix} \right\}.$$

Then every multilinear Taylor coefficient of $X(H)$ has definite matrix parity: a coefficient depending on perturbation factors $H_1, \dots, H_n$ lies in E if an even number of the H_j lie in O , and lies in O if an odd number of the H_j lie in O .

In particular, diagonal input gives diagonal output at linear order, and a bilinear coefficient with one diagonal and one off-diagonal input is off-diagonal.

Proof. Use the holomorphic functional calculus

$$(G_0 + H)^{-1/2} = \frac{1}{2\pi i} \oint_{\Gamma} z^{-1/2} (zI - G_0 - H)^{-1} dz,$$

where Γ encloses the spectrum of $G_0 + H$. Expanding the resolvent,

$$(zI - G_0 - H)^{-1} = R(z) + R(z)HR(z) + R(z)HR(z)HR(z) + \dots,$$

with

$$R(z) := (zI - G_0)^{-1},$$

every factor $R(z)$ is diagonal because G_0 is diagonal. Hence the matrix parity of each word

$$R(z)H_{i_1}R(z)H_{i_2} \cdots R(z)H_{i_n}R(z)$$

is exactly the parity of the product $H_{i_1} \cdots H_{i_n}$. Since

$$E \cdot E \subset E, \quad E \cdot O \subset O, \quad O \cdot E \subset O, \quad O \cdot O \subset E,$$

the parity rule follows. Integrating over z preserves the subspaces E and O . $\square$

Lemma 12.18 (No-cancellation of the quintic trace witness). *The monomial*

$$\kappa(h) \eta_2(h) q''''(h)$$

appearing in Lemma 12.16 is not produced by any other quintic contribution to $\text{tr } J_5^{\text{pair}}(h)$. More precisely, on the graded slice corresponding to

$$(K_1, \eta_2, q^{(5)}),$$

the only contributing channels in $\text{tr } J_5^{\text{pair}}(h)$ are

$$-2\kappa(h) (x_1 - x_2)r(h) \quad \text{and} \quad -2\kappa(h) v(h)(\gamma(h) - \alpha(h)),$$

and their total coefficient of $\kappa(h)\eta_2(h)q^{(5)}(h)$ is nonzero. Consequently

$$\text{tr } J_5^{\text{pair}}(h) \not\equiv 0$$

on the generic one-pair region.

Proof. Fix h and isolate the graded slice

$$\kappa(h) \eta_2(h) q^{(5)}(h).$$

We first exclude all non- K_1 contributions. Same-point quintic terms carry no κ -factor, so they cannot contribute. A K_5 -transport term leaves no same-point degree available, and a K_3 -transport term leaves only same-point degree 2, whereas the factor $q^{(5)}(h)$ first appears in same-point degree 3. Hence the target monomial can occur only in the K_1 -transport sector.

Now consider the same-point expansion. By the explicit formula

$$G_{m,\pm}^{\text{corr}}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_{\pm}) & \frac{1}{2}q'(t_{\pm}) \\ \frac{1}{2}q'(t_{\pm}) & \frac{1}{12}(q''(t_{\pm}) + 2q(t_{\pm})^3) \end{pmatrix},$$

the factor $q^{(5)}(h)$ first appears only in the cubic same-point coefficient G_3 , and there only in the diagonal sector, namely through the cubic Taylor term of the $(2, 2)$ -entry coming from $q''(t_{\pm})$. Therefore, by Lemma 12.17, the linear $q^{(5)}$ -contribution to X_3 is diagonal. In particular it contributes to $\gamma - \alpha$, but not to β .

Likewise, the $\eta_2(h)$ -slice of $X_1(h)$ is off-diagonal, because

$$v(h) = -\frac{1}{4\pi(s_1 + s_2)} \eta_2(h).$$

Applying Lemma 12.17 again, the mixed $(\eta_2, q^{(5)})$ -slice of X_4 is off-diagonal. Hence it contributes to r , but not to the diagonal difference $u - w$.

It follows that on the graded slice $(K_1, \eta_2, q^{(5)})$, the term

$$-2\kappa(h) \beta(h)(u(h) - w(h))$$

does not contribute at all. Thus the only surviving K_1 -channels are

$$-2\kappa(h) (x_1 - x_2)r(h) \quad \text{and} \quad -2\kappa(h) v(h)(\gamma(h) - \alpha(h)).$$

To compute their combined coefficient, it suffices by multilinearity to restrict to the two perturbation directions carrying the relevant tags: an off-diagonal first-order perturbation A for the η_2 -slice and a diagonal third-order perturbation D for the $q^{(5)}$ -slice. Accordingly, write

$$G_0 = \text{diag}(a^2, b^2), \quad A = \begin{pmatrix} 0 & u \\ u & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 \\ 0 & d \end{pmatrix},$$

and expand

$$X(\varepsilon, \delta) := (G_0 + \varepsilon A + \delta D)^{-1/2} = X_0 + \varepsilon U + \delta V + \varepsilon \delta W + O(\varepsilon^2, \delta^2).$$

Solving the identity

$$X(\varepsilon, \delta) (G_0 + \varepsilon A + \delta D) X(\varepsilon, \delta) = I$$

order by order gives

$$U_{12} = -\frac{u}{ab(a+b)}, \quad V_{22} = -\frac{d}{2b^3}, \quad W_{12} = \frac{du(a+2b)}{2ab^3(a+b)^2}.$$

Here U_{12} is the η_2 -slice of v , V_{22} is the $q^{(5)}$ -slice of $\gamma - \alpha$, and W_{12} is the mixed $(\eta_2, q^{(5)})$ -slice of r . Therefore the total coefficient of the target slice inside

$$(x_1 - x_2)r + v(\gamma - \alpha)$$

is

$$\left(\frac{1}{a} - \frac{1}{b}\right) W_{12} + U_{12}V_{22} = \frac{du}{a^2b^2(a+b)^2},$$

which is nonzero.

Thus the monomial $\kappa(h)\eta_2(h)q^{(5)}(h)$ occurs in $\text{tr } J_5^{\text{pair}}(h)$ with nonzero total coefficient and is not canceled by any other quintic contribution. Hence

$$\text{tr } J_5^{\text{pair}}(h) \neq 0.$$

□

12.3. Explicit finite-order model of the first residual quotient. For the first residual step it is enough to work with an explicit finite-order model through order 5 on the cubic-generic region. In this range the explicit model is not merely a proxy: Proposition 12.24 canonically identifies it with the order- ≤ 5 truncation of the ambient abstract quotient $\mathcal{Q}^{\text{abs}}$ after passage to the cubic line.

Accordingly, the first residual quintic detector is now rigorous at the finite-order ambient level. What is not addressed here is only a stronger global identification for later uses of $\mathcal{Q}^{\text{abs}}$ beyond order 5 and beyond the first residual layer.

Definition 12.19 (Finite-order truncation of the ambient abstract quotient). Fix $h \in \mathcal{U}_{\text{cub}}$, and let $A_3(h)$ represent the cubic leading line $\ell_h \subset \mathcal{Q}^{\text{abs}}$.

Define the order- ≤ 5 truncation of the ambient abstract quotient at h to be the quotient

$$\mathcal{Q}_h^{\text{abs}(\leq 5)} := \frac{\mathbf{C} A_3(h)z^3 \oplus \mathcal{J}_5 z^5}{(V_{\text{raw}}^{(5)}(h) + \mathcal{N}_h^{(5)})z^5}.$$

Equivalently, $\mathcal{Q}_h^{\text{abs}(\leq 5)}$ is the cubic-quintic odd-germ quotient in which the quintic ambiguity is generated exactly by the raw pure-value quintic slice and the realized normalization-gauge shadows through order 5.

Its first residual quotient is

$$\mathcal{Q}_h^{\text{abs}(\leq 5)} / \ell_h \cong \mathcal{J}_5 / (\mathbf{C} A_3(h) + V_{\text{raw}}^{(5)}(h) + \mathcal{N}_h^{(5)}).$$

Definition 12.20 (Swap involution on cubic-quintic germs). Let

$$J_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

For a truncated odd germ

$$F(z) = A_3 z^3 + A_5 z^5 + O(z^7)$$

define

$$\Sigma(F)(z) := J_0 F(-z) J_0 = -J_0 A_3 J_0 z^3 - J_0 A_5 J_0 z^5 + O(z^7).$$

Thus the induced involution on quintic coefficients is

$$\Sigma_5(A) := -J_0 A J_0.$$

Definition 12.21 (Cubic-normalized swap-symmetric raw corrected germ through order 5). Fix $h \in \mathcal{U}_{\text{cub}}$, and choose a diagonal-traceless matrix

$$A_3(h)$$

representing the cubic leading line ℓ_h .

A cubic-normalized swap-symmetric raw corrected germ through order 5 at h is a truncated odd germ of the form

$$R_h(z) = A_3(h)z^3 + A_5 z^5 + O(z^7)$$

obtained from the corrected one-pair local block by a cubic normalization that preserves the swap symmetry inherited from symmetric placement.

Its raw pure-value quintic slice is the coefficient space

$$V_{\text{raw}}^{(5)}(h)$$

consisting of all quintic coefficients A_5 arising in this way under pure value deformations at the fixed parameter h .

Lemma 12.22 (Raw quintic slice is anti-fixed). *For every $h \in \mathcal{U}_{\text{cub}}$,*

$$V_{\text{raw}}^{(5)}(h) \subset \{A \in M_2(\mathbf{C}) : J_0 A J_0 = -A\}.$$

In particular,

$$\text{tr}(V_{\text{raw}}^{(5)}(h)) = 0.$$

Proof. By Definition 12.21, every cubic-normalized raw corrected representative satisfies

$$R_h(-z) = J_0 R_h(z) J_0.$$

Writing

$$R_h(z) = A_3(h)z^3 + A_5z^5 + O(z^7),$$

coefficient comparison at order z^5 gives

$$J_0 A_5 J_0 = -A_5.$$

Hence every $A_5 \in V_{\text{raw}}^{(5)}(h)$ is anti-fixed. If

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

satisfies $J_0 A J_0 = -A$, then

$$\begin{pmatrix} d & c \\ b & a \end{pmatrix} = \begin{pmatrix} -a & -b \\ -c & -d \end{pmatrix},$$

so $d = -a$, and therefore $\text{tr}(A) = 0$. □

Definition 12.23 (Quintic ambiguity space and explicit first residual quotient). For $h \in \mathcal{U}_{\text{cub}}$, let $V_{\text{raw}}^{(5)}(h)$ be the raw pure-value quintic slice from Definition 12.21.

Define the normalization-gauge quintic shadow space $\mathcal{N}_h^{(5)}$ to be the set of all differences

$$\tilde{A}_5 - A_5$$

arising from two cubic-normalized swap-symmetric raw corrected germs

$$R_h(z) = A_3(h)z^3 + A_5z^5 + O(z^7), \quad \tilde{R}_h(z) = A_3(h)z^3 + \tilde{A}_5z^5 + O(z^7),$$

of the same raw corrected local object through order 5, when those two representatives are related by the allowed normalization gauge.

By Corollary 12.13, every such shadow has the form

$$\tilde{A}_5 - A_5 = c_2 A_3(h) + [B_2, A_3(h)],$$

but $\mathcal{N}_h^{(5)}$ is defined to consist only of the actually realized gauge shadows.

Set

$$V_{\text{amb}}^{(5)}(h) := V_{\text{raw}}^{(5)}(h) + \mathcal{N}_h^{(5)}.$$

Define the explicit first residual quotient by

$$\mathcal{R}_h^{(5)} := \mathcal{J}_5 / (\mathbf{C} A_3(h) + V_{\text{amb}}^{(5)}(h)).$$

Proposition 12.24 (Finite-order comparison theorem). *For each $h \in \mathcal{U}_{\text{cub}}$, the order- ≤ 5 truncation of the ambient abstract quotient and the explicit first residual model agree canonically:*

$$\mathcal{Q}_h^{\text{abs}(\leq 5)} / \ell_h \cong \mathcal{R}_h^{(5)}.$$

Equivalently, after passing to the cubic-generic region and truncating through order 5, the first residual ambiguity in the ambient abstract quotient is exactly the ambiguity built into Definition 12.23.

Proof. By Definition 12.19,

$$\mathcal{Q}_h^{\text{abs}(\leq 5)} = \frac{\mathbf{C} A_3(h) z^3 \oplus \mathcal{J}_5 z^5}{(V_{\text{raw}}^{(5)}(h) + \mathcal{N}_h^{(5)}) z^5}.$$

Passing to the first residual quotient by the cubic line $\ell_h = \mathbf{C} A_3(h)$ gives

$$\mathcal{Q}_h^{\text{abs}(\leq 5)} / \ell_h \cong \mathcal{J}_5 / (\mathbf{C} A_3(h) + V_{\text{raw}}^{(5)}(h) + \mathcal{N}_h^{(5)}).$$

By Definition 12.23,

$$V_{\text{amb}}^{(5)}(h) = V_{\text{raw}}^{(5)}(h) + \mathcal{N}_h^{(5)},$$

so the right-hand side is exactly

$$\mathcal{J}_5 / (\mathbf{C} A_3(h) + V_{\text{amb}}^{(5)}(h)) = \mathcal{R}_h^{(5)}.$$

This is the claimed canonical identification. $\square$

Lemma 12.25 (Gauge shadows are anti-fixed). *For every $h \in \mathcal{U}_{\text{cub}}$,*

$$\mathcal{N}_h^{(5)} \subset \{A \in M_2(\mathbf{C}) : J_0 A J_0 = -A\}.$$

In particular,

$$\text{tr}(\mathcal{N}_h^{(5)}) = 0.$$

Proof. Let

$$R_h(z) = A_3(h) z^3 + A_5 z^5 + O(z^7), \quad \tilde{R}_h(z) = A_3(h) z^3 + \tilde{A}_5 z^5 + O(z^7)$$

be two cubic-normalized swap-symmetric representatives of the same raw corrected local object through order 5, related by the allowed normalization gauge. By Definition 12.21, both satisfy

$$R_h(-z) = J_0 R_h(z) J_0, \quad \tilde{R}_h(-z) = J_0 \tilde{R}_h(z) J_0.$$

Comparing coefficients at order z^5 gives

$$J_0 A_5 J_0 = -A_5, \quad J_0 \tilde{A}_5 J_0 = -\tilde{A}_5.$$

Subtracting,

$$J_0(\tilde{A}_5 - A_5) J_0 = -(\tilde{A}_5 - A_5).$$

Since $\tilde{A}_5 - A_5 \in \mathcal{N}_h^{(5)}$ by Definition 12.23, every gauge shadow is anti-fixed. Tracelessness follows as in Lemma 12.22. $\square$

Proposition 12.26 (Trace-free quintic ambiguity in the explicit finite-order model). *For every $h \in \mathcal{U}_{\text{cub}}$,*

$$V_{\text{amb}}^{(5)}(h) \subset \{A \in M_2(\mathbf{C}) : J_0 A J_0 = -A\}.$$

Equivalently, the involution Σ_5 preserves the quintic ambiguity space and acts trivially on it.

In particular,

$$\text{tr}(V_{\text{amb}}^{(5)}(h)) = 0.$$

Proof. Combine Lemma 12.22 and Lemma 12.25; by Definition 12.23,

$$V_{\text{amb}}^{(5)}(h) = V_{\text{raw}}^{(5)}(h) + \mathcal{N}_h^{(5)}.$$

$\square$

Corollary 12.27 (Trace descends on the explicit first residual quotient). *For every $h \in \mathcal{U}_{\text{cub}}$, trace annihilates*

$$\mathbf{C} A_3(h) + V_{\text{amb}}^{(5)}(h).$$

Hence trace descends to a well-defined linear functional

$$\tau_h : \mathcal{R}_h^{(5)} \rightarrow \mathbf{C}.$$

Proof. By Proposition 12.26, trace vanishes on $V_{\text{amb}}^{(5)}(h)$. It also vanishes on $\mathbf{C} A_3(h)$ because $A_3(h)$ is diagonal-traceless. $\square$

Proposition 12.28 (Generic first residual nonvanishing in the explicit finite-order model). *There exists an open dense subset*

$$\mathcal{U}_{\text{res}} \subset \mathcal{U}_{\text{cub}}$$

such that

$$\bar{v}_1(h) \neq 0 \quad \text{in } \mathcal{R}_h^{(5)}$$

for every $h \in \mathcal{U}_{\text{res}}$.

Hence the one-pair quotient odd germ is not projectively constant on $\mathcal{U}_{\text{res}}$ at the explicit first residual level.

Proof. By Corollary 12.27, trace defines a linear functional

$$\tau_h : \mathcal{R}_h^{(5)} \rightarrow \mathbf{C}.$$

By Lemma 12.18,

$$\text{tr } J_5^{\text{pair}}(h) \neq 0$$

on the cubic-generic region. Hence there is an open dense subset

$$\mathcal{U}_{\text{res}} \subset \mathcal{U}_{\text{cub}}$$

on which

$$\text{tr } J_5^{\text{pair}}(h) \neq 0.$$

For such h ,

$$\tau_h(\bar{v}_1(h)) = \text{tr } J_5^{\text{pair}}(h) \neq 0,$$

so $\bar{v}_1(h) \neq 0$ in $\mathcal{R}_h^{(5)}$. □

Remark 12.29 (Status of the explicit first residual model). The first residual step is now closed at the finite-order ambient level. Indeed, Proposition 12.24 identifies the explicit quotient $\mathcal{R}_h^{(5)}$ canonically with the order- ≤ 5 truncation of the ambient abstract quotient $\mathcal{Q}^{\text{abs}}$ modulo the cubic line on the cubic-generic region.

Thus the remaining issue is no longer a finite-order comparison theorem for the first residual layer. What remains, if one wants a stronger global ambient identification, is a separate theorem showing that later uses of $\mathcal{Q}^{\text{abs}}$ beyond order 5 do not introduce any new ambiguity relevant to the downstream multi-pair and global arguments.

Remark 12.30 (Beyond the first residual trace). The quintic trace is the first scalar residual detector, but it is not the end of the one-pair local structure. The later fixed-sector refinement $A_5^{\mathfrak{f}}, A_7^{\mathfrak{f}}, \Delta_7$ indicates that the residual germ carries further geometric information beyond the first trace channel.

12.4. Fixed-sector refinement at quintic and septic order. The trace witness at quintic order suggests a more structured refinement of the one-pair residual germ than the trace alone records. For bookkeeping purposes, it is convenient to decompose the quintic and septic coefficients into fixed and anti-fixed parts with respect to the involution

$$\iota(A) := J_0 A J_0, \quad J_0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Write

$$\pi_{\mathfrak{f}}(A) := \frac{1}{2}(A + J_0 A J_0), \quad \pi_{\mathfrak{a}}(A) := \frac{1}{2}(A - J_0 A J_0).$$

Then

$$\mathfrak{f} = \text{span}\{I, S\}, \quad \mathfrak{a} = \text{span}\{D, J\},$$

where

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Definition 12.31 (Fixed-sector quintic and septic coefficients). Let

$$F_h(z) = A_3(h)z^3 + A_5(h)z^5 + A_7(h)z^7 + \dots$$

denote the normalized one-pair quotient odd germ on the cubic-generic region. Its fixed-sector quintic and septic coefficients are defined by

$$A_5^{\mathfrak{f}}(h) := \pi_{\mathfrak{f}}(A_5(h)), \quad A_7^{\mathfrak{f}}(h) := \pi_{\mathfrak{f}}(A_7(h)).$$

Write

$$A_5^{\mathfrak{f}}(h) = u_5(h)I + v_5(h)S, \quad A_7^{\mathfrak{f}}(h) = u_7(h)I + v_7(h)S.$$

Definition 12.32 (Second residual determinant). On the cubic-generic region, define

$$\Delta_7(h) := u_7(h)v_5(h) - u_5(h)v_7(h).$$

Remark 12.33 (Interpretation). The quintic fixed-sector vector $A_5^{\mathfrak{f}}(h)$ refines the first residual data beyond the trace detector alone, and the determinant $\Delta_7(h)$ records the next residual quotient at septic order. In the working notes, these form the first two nontrivial layers of a one-pair residual package:

$$c(h), \quad A_5^{\mathfrak{f}}(h), \quad \Delta_7(h).$$

The current draft now records explicit coefficient formulas for the quintic commutator contribution and the septic K_1 -transport contribution in the existing notation $(u, v, w, \alpha, \beta, \gamma, r, \dots)$. The canonical order-7 fixed-sector content is the quotient class

$$[A_7^{\mathfrak{f}}] \in \mathfrak{f}/\mathbf{C}A_5^{\mathfrak{f}},$$

equivalently the second residual determinant

$$\Delta_7 = u_7v_5 - u_5v_7.$$

What remains non-unique is only the raw representative pair (u_7, v_7) , which is defined only up to the one-dimensional septic gauge ambiguity along the quintic fixed-sector line $\mathbf{C}A_5^{\mathfrak{f}}$. Thus the later geometry should be formulated in terms of the canonical quotient-level order-7 data $[A_7^{\mathfrak{f}}]$ and Δ_7 , rather than a uniquely chosen raw pair (u_7, v_7) . At the canonical level, these are the natural variables for the geometric reformulation of the multi-pair problem; raw representatives (u_7, v_7) may be chosen locally by gauge-fixing when convenient, but are not themselves canonical.

Proposition 12.34 (Projected septic gauge law). *Assume the normalization gauge is symmetry-preserving in the sense that*

$$B_2, B_4 \in \mathfrak{f}, \quad A_3 \in \mathfrak{a},$$

with

$$\mathfrak{f} = \text{span}\{I, S\}, \quad \mathfrak{a} = \text{span}\{D, J\}.$$

Then the fixed-sector septic coefficient satisfies

$$(A_7^{\mathfrak{f}})^{\text{new}} = A_7^{\mathfrak{f}} + c_2A_5^{\mathfrak{f}}.$$

Equivalently, the septic fixed-sector ambiguity is one-dimensional and lies along the quintic fixed-sector line.

Proof. By Proposition 12.15,

$$A_7^{\text{new}} = A_7 + c_2A_5 + c_4A_3 + [B_2, A_5] + [B_4, A_3] + c_2[B_2, A_3] + A_3B_2^2 - B_2A_3B_2.$$

Apply the fixed-sector projection $\pi_{\mathfrak{f}}$ term by term.

Since $A_3 \in \mathfrak{a}$, one has

$$\pi_{\mathfrak{f}}(c_4A_3) = 0.$$

Because $B_4 \in \mathfrak{f}$ and $A_3 \in \mathfrak{a}$,

$$[B_4, A_3] \in [\mathfrak{f}, \mathfrak{a}] \subset \mathfrak{a},$$

so

$$\pi_{\mathfrak{f}}[B_4, A_3] = 0.$$

Likewise,

$$[B_2, A_3] \in [\mathfrak{f}, \mathfrak{a}] \subset \mathfrak{a},$$

hence

$$\pi_{\mathfrak{f}}(c_2[B_2, A_3]) = 0.$$

Now decompose

$$A_5 = A_5^{\mathfrak{f}} + A_5^{\mathfrak{a}}.$$

Since $\mathfrak{f} = \text{span}\{I, S\}$ is abelian,

$$[B_2, A_5^{\mathfrak{f}}] = 0.$$

And since $B_2 \in \mathfrak{f}$ and $A_5^{\mathfrak{a}} \in \mathfrak{a}$,

$$[B_2, A_5^{\mathfrak{a}}] \in [\mathfrak{f}, \mathfrak{a}] \subset \mathfrak{a},$$

so

$$\pi_{\mathfrak{f}}[B_2, A_5] = 0.$$

Finally, $B_2 \in \mathfrak{f}$ implies $B_2^2 \in \mathfrak{f}$. Therefore

$$A_3 B_2^2 \in \mathfrak{af} \subset \mathfrak{a}, \quad B_2 A_3 B_2 \in \mathfrak{faf} \subset \mathfrak{a},$$

and hence

$$\pi_{\mathfrak{f}}(A_3 B_2^2 - B_2 A_3 B_2) = 0.$$

Thus the only term surviving fixed projection is

$$c_2 A_5^{\mathfrak{f}},$$

and we conclude

$$(A_7^{\mathfrak{f}})^{\text{new}} = A_7^{\mathfrak{f}} + c_2 A_5^{\mathfrak{f}}. \quad \square$$

Corollary 12.35 (Gauge invariance of the second residual determinant). *Under a symmetry-preserving normalization gauge, the scalar*

$$\Delta_7 = u_7 v_5 - u_5 v_7$$

is invariant.

Proof. By Proposition 12.34,

$$(A_7^{\mathfrak{f}})^{\text{new}} = A_7^{\mathfrak{f}} + c_2 A_5^{\mathfrak{f}}.$$

Writing

$$A_5^{\mathfrak{f}} = u_5 I + v_5 S, \quad A_7^{\mathfrak{f}} = u_7 I + v_7 S,$$

the new septic pair is

$$(u_7^{\text{new}}, v_7^{\text{new}}) = (u_7, v_7) + c_2(u_5, v_5).$$

Therefore

$$\Delta_7^{\text{new}} = (u_7 + c_2 u_5) v_5 - u_5 (v_7 + c_2 v_5) = u_7 v_5 - u_5 v_7 = \Delta_7. \quad \square$$

Definition 12.36 (Septic normalization-shadow sectors). In the notation of Proposition 12.15, define the order-2 and order-4 septic normalization-shadow sectors by

$$A_{7,K_5} := c_2 A_5 + [B_2, A_5],$$

and

$$A_{7,K_3} := c_4 A_3 + [B_4, A_3] + c_2 [B_2, A_3] + A_3 B_2^2 - B_2 A_3 B_2.$$

Then the exact septic transformation law may be written

$$A_7^{\text{new}} = A_7 + A_{7,K_5} + A_{7,K_3}.$$

When the normalization gauge is symmetry-preserving, these are the septic K_5 - and K_3 -transport shadow sectors.

Proposition 12.37 (Projected septic K_5 -shadow law). *Assume the normalization gauge is symmetry-preserving in the sense of Proposition 12.34. Then*

$$A_{7,K_5}^{\mathfrak{f}} = c_2 A_5^{\mathfrak{f}}.$$

In particular,

$$A_{7,K_5}^{\mathfrak{f}} \in \mathbf{C} A_5^{\mathfrak{f}}.$$

Proof. By Definition 12.36,

$$A_{7,K_5} = c_2 A_5 + [B_2, A_5].$$

Apply the fixed-sector projection $\pi_{\mathfrak{f}}$. Decompose

$$A_5 = A_5^{\mathfrak{f}} + A_5^{\mathfrak{a}}.$$

Since $\mathfrak{f} = \text{span}\{I, S\}$ is abelian,

$$[B_2, A_5^{\mathfrak{f}}] = 0.$$

And since $B_2 \in \mathfrak{f}$ and $A_5^{\mathfrak{a}} \in \mathfrak{a}$,

$$[B_2, A_5^{\mathfrak{a}}] \in [\mathfrak{f}, \mathfrak{a}] \subset \mathfrak{a},$$

so its fixed-sector projection vanishes. Therefore

$$A_{7,K_5}^{\mathfrak{f}} = \pi_{\mathfrak{f}}(c_2 A_5 + [B_2, A_5]) = c_2 A_5^{\mathfrak{f}}. \quad \square$$

Proposition 12.38 (Projected septic K_3 -shadow vanishing). *Assume the normalization gauge is symmetry-preserving in the sense of Proposition 12.34. Then*

$$A_{7,K_3}^{\mathfrak{f}} = 0.$$

Proof. By Definition 12.36,

$$A_{7,K_3} = c_4 A_3 + [B_4, A_3] + c_2 [B_2, A_3] + A_3 B_2^2 - B_2 A_3 B_2.$$

Since $A_3 \in \mathfrak{a}$, one has

$$c_4 A_3 \in \mathfrak{a}.$$

Because $B_4 \in \mathfrak{f}$ and $A_3 \in \mathfrak{a}$,

$$[B_4, A_3] \in [\mathfrak{f}, \mathfrak{a}] \subset \mathfrak{a}.$$

Likewise,

$$[B_2, A_3] \in [\mathfrak{f}, \mathfrak{a}] \subset \mathfrak{a},$$

hence

$$c_2 [B_2, A_3] \in \mathfrak{a}.$$

Finally, $B_2 \in \mathfrak{f}$ implies $B_2^2 \in \mathfrak{f}$, so

$$A_3 B_2^2 \in \mathfrak{a}\mathfrak{f} \subset \mathfrak{a}, \quad B_2 A_3 B_2 \in \mathfrak{f}\mathfrak{a}\mathfrak{f} \subset \mathfrak{a}.$$

Thus every term of A_{7,K_3} is anti-fixed, and therefore

$$A_{7,K_3}^{\mathfrak{f}} = \pi_{\mathfrak{f}}(A_{7,K_3}) = 0. \quad \square$$

Remark 12.39 (Interpretation of the septic shadow sectors). Propositions 12.37 and 12.38 show that the lower-order normalization shadows at septic order contribute no new canonical fixed-sector information beyond the already-known quintic line $\mathbf{C}A_5^{\mathfrak{f}}$.

Proposition 12.40 (Explicit fixed-sector formulas at quintic order). *Write*

$$X_1 = \begin{pmatrix} u & v \\ v & w \end{pmatrix}, \quad X_3 = \begin{pmatrix} \alpha & \beta \\ \beta & \gamma \end{pmatrix}.$$

Then the fixed-sector part of the basic quintic commutator channel

$$X_3 J X_1 - X_1 J X_3$$

is

$$\pi_{\mathfrak{f}}(X_3 J X_1 - X_1 J X_3) = (v(\gamma - \alpha) + \beta(u - w))I + (\gamma u - \alpha w)S.$$

Consequently, for the quintic fixed-sector coefficient

$$A_5^{\mathfrak{f}}(h) = u_5(h)I + v_5(h)S,$$

the commutator-type contribution is explicitly

$$u_5^{(\text{comm})} = v(\gamma - \alpha) + \beta(u - w), \quad v_5^{(\text{comm})} = \gamma u - \alpha w.$$

Proof. A direct computation gives

$$X_3 J X_1 - X_1 J X_3 = \begin{pmatrix} -2\alpha v + 2\beta u & -\alpha w + \gamma u \\ -\alpha w + \gamma u & -2\beta w + 2\gamma v \end{pmatrix}.$$

The fixed-sector projection of a symmetric matrix

$$\begin{pmatrix} A & B \\ B & C \end{pmatrix}$$

is

$$\pi_{\mathfrak{f}} \begin{pmatrix} A & B \\ B & C \end{pmatrix} = \frac{A+C}{2} I + B S.$$

Applying this to the displayed matrix gives

$$\pi_{\mathfrak{f}}(X_3 J X_1 - X_1 J X_3) = \frac{(-2\alpha v + 2\beta u) + (-2\beta w + 2\gamma v)}{2} I + (\gamma u - \alpha w) S,$$

which simplifies to

$$(v(\gamma - \alpha) + \beta(u - w)) I + (\gamma u - \alpha w) S.$$

□

Proposition 12.41 (Explicit septic K_1 -transport fixed-sector formulas). *Write*

$$\begin{aligned} X_0 &= \text{diag}(x_1, x_2), & X_1 &= \begin{pmatrix} u & v \\ v & w \end{pmatrix}, & X_2 &= \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \\ X_4 &= \begin{pmatrix} p & r \\ r & s \end{pmatrix}, & X_5 &= \begin{pmatrix} \alpha_5 & \beta_5 \\ \beta_5 & \gamma_5 \end{pmatrix}, & X_6 &= \begin{pmatrix} \mu_6 & \nu_6 \\ \nu_6 & \lambda_6 \end{pmatrix}. \end{aligned}$$

Then the fixed-sector parts of the three septic K_1 -transport channels are

$$\begin{aligned} \pi_{\mathfrak{f}}(X_5 J X_1 - X_1 J X_5) &= (\beta_5(u - w) + v(\gamma_5 - \alpha_5)) I + (\gamma_5 u - \alpha_5 w) S, \\ \pi_{\mathfrak{f}}(X_2 J X_4 - X_4 J X_2) &= (b(p - s) + r(c - a)) I + (cp - as) S, \end{aligned}$$

and

$$\pi_{\mathfrak{f}}(X_0 J X_6 - X_6 J X_0) = (x_2 - x_1) \nu_6 I + (x_2 \mu_6 - x_1 \lambda_6) S.$$

Consequently, if the septic K_1 -transport fixed-sector coefficient is written

$$A_{7,K_1}^{\mathfrak{f}}(h) = u_{7,K_1}(h) I + v_{7,K_1}(h) S,$$

then

$$\begin{aligned} u_{7,K_1} &= \kappa \left((x_2 - x_1) \nu_6 + \beta_5(u - w) + v(\gamma_5 - \alpha_5) + b(p - s) + r(c - a) \right), \\ v_{7,K_1} &= \kappa \left(x_2 \mu_6 - x_1 \lambda_6 + \gamma_5 u - \alpha_5 w + cp - as \right). \end{aligned}$$

Proof. For two symmetric matrices

$$X = \begin{pmatrix} a & b \\ b & c \end{pmatrix}, \quad Y = \begin{pmatrix} p & q \\ q & r \end{pmatrix},$$

a direct calculation gives

$$X J Y - Y J X = \begin{pmatrix} -2aq + 2bp & -ar + cp \\ -ar + cp & -2br + 2cq \end{pmatrix}.$$

Taking fixed-sector projection yields

$$\pi_{\mathfrak{f}}(X J Y - Y J X) = (b(p - r) + q(c - a)) I + (cp - ar) S.$$

Applying this formula to $(X, Y) = (X_5, X_1)$ and $(X, Y) = (X_2, X_4)$ gives the first two displayed identities.

For the X_0/X_6 channel, a direct multiplication gives

$$X_0 J X_6 - X_6 J X_0 = \begin{pmatrix} -2x_1 \nu_6 & -x_1 \lambda_6 + x_2 \mu_6 \\ -x_1 \lambda_6 + x_2 \mu_6 & 2x_2 \nu_6 \end{pmatrix},$$

whose fixed-sector part is

$$(x_2 - x_1)\nu_6 I + (x_2\mu_6 - x_1\lambda_6)S.$$

Summing the three channels gives the stated formulas for u_{7,K_1} and v_{7,K_1} . $\square$

Definition 12.42 (Septic K_1 -residual determinant). If

$$A_{7,K_1}^f(h) = u_{7,K_1}(h) I + v_{7,K_1}(h) S,$$

define the corresponding septic K_1 -residual determinant by

$$\Delta_{7,K_1}(h) := u_{7,K_1}(h)v_5(h) - u_5(h)v_{7,K_1}(h).$$

Corollary 12.43 (Direct $q^{(7)}$ -driven septic slice). *On the direct linear $q^{(7)}$ -driven slice, the septic K_1 -transport fixed-sector contribution is I -directional. More precisely,*

$$v_{7,K_1}^{(q^{(7)})} = 0,$$

and

$$u_{7,K_1}^{(q^{(7)})} = \kappa\left((x_2 - x_1)\nu_6^{(q^{(7)})} + v(\gamma_5 - \alpha_5)^{(q^{(7)})}\right),$$

so in particular

$$\Delta_7^{(q^{(7)})} = u_7^{(q^{(7)})}v_5.$$

Proof. On the direct $q^{(7)}$ -driven slice, the linear X_5 -input is diagonal, hence

$$\beta_5^{(q^{(7)})} = 0.$$

The mixed X_6 -slice is off-diagonal on this grading, so

$$\mu_6^{(q^{(7)})} = \lambda_6^{(q^{(7)})} = 0.$$

Moreover the X_2/X_4 -channel does not lie on the pure $(K_1, \eta_2, q^{(7)})$ -slice. Therefore the S -component vanishes and the displayed formula for $u_{7,K_1}^{(q^{(7)})}$ follows from Proposition 12.41. Since

$$\Delta_7 = u_7v_5 - u_5v_7,$$

the direct $q^{(7)}$ -slice satisfies

$$\Delta_7^{(q^{(7)})} = u_7^{(q^{(7)})}v_5. \quad \square$$

Proposition 12.44 (Noncancellation of the direct septic K_1 witness). *On the graded slice*

$$(K_1, \eta_2, q^{(7)}),$$

the only contributing channels to

$$u_{7,K_1}^{(q^{(7)})}$$

are

$$\kappa(x_2 - x_1)\nu_6^{(q^{(7)})} \quad \text{and} \quad \kappa v(\gamma_5 - \alpha_5)^{(q^{(7)})},$$

and their total coefficient of the monomial

$$\kappa(h)\eta_2(h)q^{(7)}(h)$$

is nonzero. Consequently,

$$u_{7,K_1}^{(q^{(7)})}(h) \neq 0$$

on the generic one-pair region.

Proof. Fix h and isolate the graded slice

$$\kappa(h)\eta_2(h)q^{(7)}(h).$$

By Corollary 12.43, the direct $q^{(7)}$ -driven slice is I -directional and is given by

$$u_{7,K_1}^{(q^{(7)})} = \kappa\left((x_2 - x_1)\nu_6^{(q^{(7)})} + v(\gamma_5 - \alpha_5)^{(q^{(7)})}\right).$$

Thus it remains only to show that the two displayed channels do not cancel.

Now consider the same-point expansion. By the explicit formula

$$G_{m,\pm}^{\text{corr}}(s) = \frac{1}{\pi} \begin{pmatrix} 2q(t_{\pm}) & \frac{1}{2}q'(t_{\pm}) \\ \frac{1}{2}q'(t_{\pm}) & \frac{1}{12}(q''(t_{\pm}) + 2q(t_{\pm})^3) \end{pmatrix},$$

the factor $q^{(7)}(h)$ first appears only in the quintic same-point coefficient G_5 , and there only in the diagonal sector, namely through the quintic Taylor term of the $(2, 2)$ -entry coming from $q''(t_{\pm})$. Therefore, by Lemma 12.17, the linear $q^{(7)}$ -contribution to X_5 is diagonal. In particular it contributes to $\gamma_5 - \alpha_5$, but not to β_5 .

Likewise, the $\eta_2(h)$ -slice of $X_1(h)$ is off-diagonal, because

$$v(h) = -\frac{1}{4\pi(s_1 + s_2)} \eta_2(h).$$

Applying Lemma 12.17 again, the mixed $(\eta_2, q^{(7)})$ -slice of X_6 is off-diagonal. Hence it contributes to ν_6 , but not to μ_6 or λ_6 .

Thus the only contributing channels on the graded slice

$$(K_1, \eta_2, q^{(7)})$$

are exactly

$$\kappa(x_2 - x_1)\nu_6^{(q^{(7)})} \quad \text{and} \quad \kappa v(\gamma_5 - \alpha_5)^{(q^{(7)})}.$$

To compute their combined coefficient, it suffices by multilinearity to restrict to the two perturbation directions carrying the relevant tags: an off-diagonal first-order perturbation A for the η_2 -slice and a diagonal fifth-order perturbation D for the $q^{(7)}$ -slice. Accordingly, write

$$G_0 = \text{diag}(a^2, b^2), \quad A = \begin{pmatrix} 0 & u \\ u & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 \\ 0 & d \end{pmatrix},$$

and expand

$$X(\varepsilon, \delta) := (G_0 + \varepsilon A + \delta D)^{-1/2} = X_0 + \varepsilon U + \delta V + \varepsilon \delta W + O(\varepsilon^2, \delta^2).$$

Solving the identity

$$X(\varepsilon, \delta) (G_0 + \varepsilon A + \delta D) X(\varepsilon, \delta) = I$$

order by order gives

$$U_{12} = -\frac{u}{ab(a+b)}, \quad V_{22} = -\frac{d}{2b^3}, \quad W_{12} = \frac{du(a+2b)}{2ab^3(a+b)^2}.$$

Here U_{12} is the η_2 -slice of v , V_{22} is the $q^{(7)}$ -slice of $\gamma_5 - \alpha_5$, and W_{12} is the mixed $(\eta_2, q^{(7)})$ -slice of ν_6 . Therefore the total coefficient of the target slice inside

$$(x_2 - x_1)\nu_6 + v(\gamma_5 - \alpha_5)$$

is

$$\left(\frac{1}{a} - \frac{1}{b}\right) W_{12} + U_{12}V_{22} = \frac{du}{a^2b^2(a+b)^2},$$

which is nonzero.

Thus the monomial

$$\kappa(h) \eta_2(h) q^{(7)}(h)$$

occurs in $u_{7,K_1}^{(q^{(7)})}(h)$ with nonzero total coefficient, so

$$u_{7,K_1}^{(q^{(7)})}(h) \neq 0.$$

□

Remark 12.45 (Septic K_1 trace target). At the graded level, the septic K_1 -transport channel is the first odd transport sector capable of coupling the new same-point $q^{(7)}$ -slice into the fixed-sector septic residual. In particular, the expected highest new contribution to the septic determinant is carried by the K_1 -channel, in parallel with the earlier quintic trace witness at order 5.

Lemma 12.46 (Parity elimination of the pure same-point septic sector). *Let $t_{\pm} = h \pm s/2$, and let the odd two-point germ be defined as the odd part under the involution*

$$s \mapsto -s \quad (\text{equivalently } t_+ \leftrightarrow t_-).$$

Let $A_{7,\text{pure}}$ denote the pure same-point septic sector, i.e. the septic contribution built only from the same-point blocks and their whitening, with no transport factor coming from the ordered pair (t_+, t_-) .

Then $A_{7,\text{pure}}$ is even in s , hence vanishes in the odd two-point germ. In particular,

$$A_{7,\text{pure}} = 0 \quad \text{in the odd germ}, \quad A_{7,\text{pure}}^f = 0.$$

Consequently, pure same-point septic terms do not contribute to Δ_7 , and in particular do not contribute to the weight-one part $[\Delta_7]^{[1]}$.

Proof. A pure same-point septic term depends only on midpoint jet data at h , i.e. on the same-point expansion at the midpoint, and carries no transport factor coming from the ordered pair (t_+, t_-) . Therefore it is invariant under exchanging the two points:

$$t_+ \leftrightarrow t_-.$$

Since this exchange is exactly the involution $s \mapsto -s$, every pure same-point septic term is an even function of s .

By definition, the odd two-point germ is obtained by projecting to the odd part under $s \mapsto -s$. Hence every even contribution vanishes in that projection. Therefore

$$A_{7,\text{pure}} = 0 \quad \text{in the odd germ.}$$

Applying fixed-sector projection gives

$$A_{7,\text{pure}}^f = 0.$$

Since Δ_7 is extracted from the odd pair germ after fixed-sector projection and quotient normalization, pure same-point septic terms contribute nothing to Δ_7 , and therefore nothing to $[\Delta_7]^{[1]}$. $\square$

Corollary 12.47 (Weight-one septic reduction to the direct K_1 slice). *Let $[\cdot]^{[1]}$ denote the part linear in the first genuinely new same-point $q^{(7)}$ -slice. Then*

$$[\Delta_7]^{[1]} = [\Delta_{7,K_1}]^{[1]}.$$

Equivalently, the highest genuinely new $q^{(7)}$ -coefficient of the septic residual determinant comes only from the direct septic K_1 -transport slice.

Proof. The first genuinely new $q^{(7)}$ -slice enters the same-point expansion at order 5, so

$$X_j^{[1]} = 0 \quad (j < 5).$$

Hence every septic transport sector built only from lower same-point orders is weight zero; in particular, the K_3 - and K_5 -transport sectors do not contribute to $[\Delta_7]^{[1]}$.

By Lemma 12.46, pure same-point septic terms also do not contribute to $[\Delta_7]^{[1]}$, because they vanish already in the odd two-point germ.

Thus the only septic sector that can see the weight-one $q^{(7)}$ -slice is the direct K_1 -transport sector, and therefore

$$[\Delta_7]^{[1]} = [\Delta_{7,K_1}]^{[1]}. \quad \square$$

Proposition 12.48 (Generic nonvanishing of the direct septic slice under nontrivial quintic fixed-sector coefficient). *Let J be a connected open interval contained in the quintic-generic region. Assume*

$$v_5 \not\equiv 0 \quad \text{on } J.$$

Then the direct $q^{(7)}$ -driven slice of the full total septic residual determinant is generically nonzero on J . Equivalently,

$$\Delta_7^{(q^{(7)})} \not\equiv 0 \quad \text{on } J.$$

Proof. By Corollary 12.47,

$$[\Delta_7]^{[1]} = [\Delta_{7,K_1}]^{[1]}.$$

Thus the highest genuinely new direct $q^{(7)}$ -coefficient of the total septic residual determinant is the same as that of the direct K_1 -transport slice.

By Proposition 12.44, the direct septic K_1 witness is generically nonzero on the quintic-generic region. In particular,

$$u_{7,K_1}^{(q^{(7)})} \neq 0 \quad \text{on } J.$$

Moreover, by Corollary 12.43, on the direct $q^{(7)}$ -driven slice one has

$$v_{7,K_1}^{(q^{(7)})} = 0, \quad \Delta_{7,K_1}^{(q^{(7)})} = u_{7,K_1}^{(q^{(7)})} v_5.$$

Since $v_5 \neq 0$ on J , it follows that

$$\Delta_{7,K_1}^{(q^{(7)})} \neq 0 \quad \text{on } J.$$

Therefore

$$\Delta_7^{(q^{(7)})} \neq 0 \quad \text{on } J,$$

as claimed. $\square$

Theorem 12.49 (Quotient-septic closure). *Let the septic odd-germ coefficient be decomposed as*

$$A_7 = A_{7,K_1} + A_{7,K_5} + A_{7,K_3} + A_{7,\text{pure}},$$

where A_{7,K_1} is the direct K_1 -transport sector, A_{7,K_5} and A_{7,K_3} are the septic normalization-shadow sectors from Definition 12.36, and $A_{7,\text{pure}}$ is the pure same-point septic sector from Lemma 12.46.

Then

$$A_7^{\mathfrak{f}} = A_{7,K_1}^{\mathfrak{f}} + A_{7,K_5}^{\mathfrak{f}},$$

with

$$A_{7,K_5}^{\mathfrak{f}} \in \mathbf{C} A_5^{\mathfrak{f}}.$$

Consequently,

$$[A_7^{\mathfrak{f}}] = [A_{7,K_1}^{\mathfrak{f}}] \quad \text{in } \mathfrak{f}/\mathbf{C}A_5^{\mathfrak{f}},$$

and

$$\Delta_7 = \Delta_{7,K_1}.$$

Proof. By Proposition 12.38,

$$A_{7,K_3}^{\mathfrak{f}} = 0.$$

By Lemma 12.46,

$$A_{7,\text{pure}}^{\mathfrak{f}} = 0.$$

Therefore

$$A_7^{\mathfrak{f}} = A_{7,K_1}^{\mathfrak{f}} + A_{7,K_5}^{\mathfrak{f}}.$$

By Proposition 12.37,

$$A_{7,K_5}^{\mathfrak{f}} \in \mathbf{C} A_5^{\mathfrak{f}}.$$

Hence $A_7^{\mathfrak{f}}$ and $A_{7,K_1}^{\mathfrak{f}}$ define the same class in the quotient

$$\mathfrak{f}/\mathbf{C}A_5^{\mathfrak{f}}.$$

Writing

$$A_5^{\mathfrak{f}} = u_5 I + v_5 S, \quad A_{7,K_1}^{\mathfrak{f}} = u_{7,K_1} I + v_{7,K_1} S,$$

this quotient equality means precisely that

$$A_7^{\mathfrak{f}} = A_{7,K_1}^{\mathfrak{f}} + \chi A_5^{\mathfrak{f}}$$

for some scalar χ . Therefore

$$\Delta_7 = (u_{7,K_1} + \chi u_5) v_5 - u_5 (v_{7,K_1} + \chi v_5) = u_{7,K_1} v_5 - u_5 v_{7,K_1} = \Delta_{7,K_1},$$

where Δ_{7,K_1} is as in Definition 12.42. $\square$

Remark 12.50 (Status of the order-7 fixed-sector package). The order-7 fixed-sector analysis is now closed at the canonical quotient level. By Theorem 12.49, the full total septic fixed-sector class satisfies

$$[A_7^f] = [A_{7,K_1}^f] \quad \text{in } \mathfrak{f}/\mathbf{C}A_5^f,$$

and the canonical second residual determinant is

$$\Delta_7 = \Delta_{7,K_1}.$$

What remains non-unique is only the raw representative of A_7^f inside the affine line

$$A_{7,K_1}^f + \mathbf{C}A_5^f,$$

as recorded by Proposition 12.34. Equivalently, the full lower/mixed septic completion of the raw total pair (u_7, v_7) is not canonically determined. This remaining gauge-dependent choice does not affect the quotient class $[A_7^f]$ or the residual determinant Δ_7 used in the later geometric arguments.

Remark 12.51 (Local gauge-fixing for raw septic representatives). Although the raw representative pair (u_7, v_7) is not canonical, local choices may be made by fixing a gauge section along the quintic fixed-sector line. Indeed, Proposition 12.34 gives

$$(u_7^{\text{new}}, v_7^{\text{new}}) = (u_7, v_7) + c_2(u_5, v_5).$$

Hence on any region where $v_5 \neq 0$, one may choose

$$c_2 = -\frac{v_7}{v_5},$$

so that

$$v_7^{\text{new}} = 0, \quad u_7^{\text{new}} = \frac{\Delta_7}{v_5}.$$

Similarly, on any region where $u_5 \neq 0$, one may choose

$$c_2 = -\frac{u_7}{u_5},$$

so that

$$u_7^{\text{new}} = 0, \quad v_7^{\text{new}} = -\frac{\Delta_7}{u_5}.$$

Thus raw septic representatives can be chosen locally by gauge-fixing, but such choices are auxiliary and are not canonical.

12.5. Normalized planar curve and curvature bridge. On the region where $c(h) \neq 0$, define the normalized scalars

$$\rho_1(h) := \frac{v_5(h)}{c(h)}, \quad \rho_2(h) := \frac{\Delta_7(h)}{c(h)},$$

and the associated planar invariant curve

$$\Gamma(h) := (\rho_1(h), \rho_2(h)).$$

Proposition 12.52 (Non-affinity of the second normalized residual scalar). *Let J be a connected open interval contained in the quintic-generic region and in $\{h : c(h) \neq 0\}$. Assume*

$$v_5 \not\equiv 0 \quad \text{on } J.$$

Then there do not exist constants $A, B \in \mathbf{C}$ such that

$$\rho_2(h) \equiv A\rho_1(h) + B$$

on J .

Equivalently,

$$\Delta_7(h) \not\equiv A v_5(h) + B c(h)$$

on J .

Proof. Suppose

$$\Delta_7(h) \equiv A v_5(h) + B c(h)$$

on J . Then the direct $q^{(7)}$ -driven slice of the right-hand side vanishes, because v_5 and c are governed by the cubic/quintic layers and carry no direct septic $q^{(7)}$ -slice.

On the other hand, by Proposition 12.48, the direct $q^{(7)}$ -driven slice of Δ_7 is generically nonzero on J . This is a contradiction. Dividing by $c(h)$ yields

$$\rho_2(h) \neq A \rho_1(h) + B. \quad \square$$

Definition 12.53 (Normalized derivative numerators). Define

$$M(h) := c(h)v_5'(h) - c'(h)v_5(h), \quad N(h) := c(h)\Delta_7'(h) - c'(h)\Delta_7(h).$$

Then

$$\rho_1'(h) = \frac{M(h)}{c(h)^2}, \quad \rho_2'(h) = \frac{N(h)}{c(h)^2}.$$

Proposition 12.54 (Curvature bridge for the normalized planar curve). *Let*

$$K_\Gamma(h) := \rho_1'(h)\rho_2''(h) - \rho_2'(h)\rho_1''(h)$$

denote the affine curvature numerator of the planar curve $\Gamma(h)$. Then

$$K_\Gamma(h) = \frac{M(h)N'(h) - N(h)M'(h)}{c(h)^4}.$$

In particular, if $K_\Gamma \equiv 0$ on a connected open interval J on which

$$c \neq 0, \quad M \neq 0, \quad v_5 \neq 0,$$

then ρ_2 is affine in ρ_1 on J , contradicting Proposition 12.52. Hence on every such interval one has

$$K_\Gamma \neq 0.$$

Proof. By Definition 12.53,

$$\rho_1' = \frac{M}{c^2}, \quad \rho_2' = \frac{N}{c^2}.$$

Differentiating gives

$$\rho_1'' = \frac{M'c - 2Mc'}{c^3}, \quad \rho_2'' = \frac{N'c - 2Nc'}{c^3}.$$

Therefore

$$K_\Gamma = \frac{M}{c^2} \frac{N'c - 2Nc'}{c^3} - \frac{N}{c^2} \frac{M'c - 2Mc'}{c^3} = \frac{MN' - NM'}{c^4}.$$

If $K_\Gamma \equiv 0$ on a connected open set with $M \neq 0$, then

$$0 = MN' - NM' = M^2 \left(\frac{N}{M} \right)',$$

so N/M is constant there:

$$\frac{N}{M} = A.$$

Since

$$\frac{N}{M} = \frac{\rho_2'}{\rho_1'},$$

it follows that

$$\rho_2' = A \rho_1',$$

hence

$$\rho_2 = A \rho_1 + B$$

on that connected interval. If in addition

$$c \neq 0, \quad v_5 \neq 0,$$

there, this contradicts Proposition 12.52. The final claim follows. $\square$

Remark 12.55 (Geometric consequence). Proposition 12.54 is the local-to-global bridge from the order-7 fixed-sector analysis back to the secant/tangent program. On the open set where

$$c(h) \neq 0, \quad M(h) \neq 0, \quad M(h)N'(h) - N(h)M'(h) \neq 0,$$

the normalized planar curve $\Gamma(h)$ decomposes into C^2 branches in the affine chart (ρ_1, ρ_2) , each of which is strictly convex or strictly concave. Thus the remaining global obstruction is not same-branch secancy, but the possibility of a common four-secant line meeting two separated ECT branches twice each.

Definition 12.56 (Projective Wronskian of the strengthened one-pair triple). Define

$$W(h) := \det \begin{pmatrix} c(h) & v_5(h) & \Delta_7(h) \\ c'(h) & v_5'(h) & \Delta_7'(h) \\ c''(h) & v_5''(h) & \Delta_7''(h) \end{pmatrix}.$$

Proposition 12.57 (Projective-Wronskian bridge). *With $M(h)$ and $N(h)$ as in Definition 12.53, one has*

$$M(h)N'(h) - N(h)M'(h) = c(h) W(h).$$

Consequently,

$$K_\Gamma(h) = \frac{W(h)}{c(h)^3}$$

on the region where $c(h) \neq 0$.

Proof. Using

$$M = c v_5' - c' v_5, \quad N = c \Delta_7' - c' \Delta_7,$$

a direct expansion gives

$$MN' - NM' = (c v_5' - c' v_5)(c \Delta_7'' - c'' \Delta_7) - (c \Delta_7' - c' \Delta_7)(c v_5'' - c'' v_5).$$

Expanding and regrouping yields

$$MN' - NM' = c \det \begin{pmatrix} c & v_5 & \Delta_7 \\ c' & v_5' & \Delta_7' \\ c'' & v_5'' & \Delta_7'' \end{pmatrix} = c W.$$

The displayed formula for K_Γ now follows from Proposition 12.54. □

Lemma 12.58 (Exact identities for W and W'). *Let*

$$F(h) := (c(h), v_5(h), \Delta_7(h)), \quad M(h) := c(h)v_5'(h) - c'(h)v_5(h).$$

Then

$$W(h) = \det(F(h), F'(h), F''(h))$$

admits the exact expansion

$$W = M \Delta_7'' + (v_5 c'' - c v_5'') \Delta_7' + (c' v_5'' - v_5' c'') \Delta_7,$$

and

$$W'(h) = \det(F(h), F'(h), F'''(h))$$

admits the exact expansion

$$W' = M \Delta_7''' + (v_5 c''' - c v_5''') \Delta_7' + (c' v_5''' - v_5' c''') \Delta_7.$$

Proof. Expand the determinant $\det(F, F', F'')$ along the third column. This gives

$$W = \det \begin{pmatrix} c & v_5 \\ c' & v_5' \end{pmatrix} \Delta_7'' - \det \begin{pmatrix} c & v_5 \\ c'' & v_5'' \end{pmatrix} \Delta_7' + \det \begin{pmatrix} c' & v_5' \\ c'' & v_5'' \end{pmatrix} \Delta_7,$$

which is exactly the first identity.

For W' , differentiate $W = \det(F, F', F'')$. The terms $\det(F', F', F'')$ and $\det(F, F'', F'')$ vanish by repeated columns, so

$$W' = \det(F, F', F''').$$

Expanding along the third column gives the second formula. □

Proposition 12.59 (Exclusion of W -flat intervals on the $c \neq 0$, $M \neq 0$ region). *Let J be a connected open interval on which*

$$c(h) \neq 0, \quad M(h) \neq 0.$$

Then

$$W \not\equiv 0 \quad \text{on } J.$$

Proof. Suppose for contradiction that

$$W \equiv 0 \quad \text{on } J.$$

By Proposition 12.57,

$$K_\Gamma(h) = \frac{W(h)}{c(h)^3} \equiv 0 \quad \text{on } J.$$

Since

$$M = c v'_5 - c' v_5$$

and $c \neq 0$ on J , the hypothesis $M \neq 0$ on J rules out $v_5 \equiv 0$ on J . Hence $v_5 \not\equiv 0$ on J .

By Proposition 12.54, it follows that

$$\rho_2(h) = A \rho_1(h) + B$$

for some constants $A, B \in \mathbf{C}$, i.e.

$$\Delta_7(h) = A v_5(h) + B c(h) \quad \text{on } J.$$

This contradicts Proposition 12.52. Therefore

$$W \not\equiv 0 \quad \text{on } J. \quad \square$$

Remark 12.60 (Interpretation of the λ -reduction). On the direct highest-new slice one has

$$\Delta_7^{[1]} = \lambda^{[1]} v_5$$

by Corollary 12.43 together with Corollary 12.47. Thus if $W \equiv 0$ on a connected open interval with $c \neq 0$ and $M \neq 0$, then

$$v_5 \not\equiv 0$$

on that interval, because $M = c v'_5 - c' v_5$. The affine dependence

$$\Delta_7 = A v_5 + B c$$

forced by Proposition 12.59 implies

$$\Delta_7^{[1]} = A v_5,$$

hence

$$\lambda^{[1]} \equiv A$$

on that interval. In this sense, the present argument excludes W -flatness on an $M \neq 0$ region without requiring a separate differential equation for $\lambda^{[1]}$; what remains open is not the canonical quotient-level order-7 invariant, but only the lower/mixed septic completion of a raw representative pair (u_7, v_7) , which is irrelevant to the quotient class $[A_7^f]$ and to Δ_7 .

Definition 12.61 (Projective graph and dual tangent-line variables). On the region where $v_5 \neq 0$, define

$$\mu := \frac{c}{v_5}, \quad \lambda := \frac{\Delta_7}{v_5}.$$

On the subregion where $M \neq 0$, define

$$L := v_5 \Delta'_7 - v'_5 \Delta_7, \quad \beta := \frac{d\lambda}{d\mu}, \quad m := \lambda - \mu \beta.$$

Proposition 12.62 (Closed minor system and dual tangent-line curve). *Let*

$$F := (c, v_5, \Delta_7), \quad U := (L, -N, M).$$

Then

$$F \times F' = U, \quad U \times U' = W F.$$

Equivalently,

$$MN' - NM' = cW, \quad ML' - LM' = v_5W, \quad NL' - LN' = \Delta_7W.$$

On the region where $v_5M \neq 0$,

$$\mu' = -\frac{M}{v_5^2}, \quad \beta = -\frac{L}{M}, \quad m = \frac{N}{M},$$

and

$$\beta' = -\frac{v_5W}{M^2}, \quad m' = \frac{cW}{M^2}.$$

Hence the dual tangent-line curve

$$T(h) := (\beta(h), m(h))$$

satisfies

$$T'(h) = \frac{W(h)}{M(h)^2} (-v_5(h), c(h)).$$

Moreover,

$$\det(T'(h), T''(h)) = \frac{W(h)^2}{M(h)^3}.$$

Proof. The identity

$$F \times F' = (L, -N, M)$$

is the direct expansion of the cross product. Differentiating gives

$$U' = F \times F''.$$

Applying the vector identity

$$(A \times B) \times (A \times C) = \det(A, B, C) A$$

with $A = F$, $B = F'$, $C = F''$ yields

$$U \times U' = (F \times F') \times (F \times F'') = \det(F, F', F'') F = W F,$$

which is exactly the three displayed scalar identities.

Next,

$$\mu' = \left(\frac{c}{v_5} \right)' = \frac{c'v_5 - cv_5'}{v_5^2} = -\frac{M}{v_5^2},$$

and

$$\lambda' = \left(\frac{\Delta_7}{v_5} \right)' = \frac{v_5\Delta_7' - v_5'\Delta_7}{v_5^2} = \frac{L}{v_5^2},$$

so

$$\beta = \frac{\lambda'}{\mu'} = -\frac{L}{M}.$$

Also

$$\mu L + M\lambda = \frac{c}{v_5}(v_5\Delta_7' - v_5'\Delta_7) + (cv_5' - c'v_5)\frac{\Delta_7}{v_5} = c\Delta_7' - c'\Delta_7 = N,$$

hence

$$m = \lambda - \mu\beta = \lambda + \mu\frac{L}{M} = \frac{N}{M}.$$

The formulas for β' and m' follow immediately from

$$ML' - LM' = v_5W, \quad MN' - NM' = cW.$$

Finally, write

$$T' = \alpha V, \quad \alpha := \frac{W}{M^2}, \quad V := (-v_5, c).$$

Then

$$\det(T', T'') = \alpha^2 \det(V, V') = \frac{W^2}{M^4} \det \begin{pmatrix} -v_5 & -v'_5 \\ c & c' \end{pmatrix} = \frac{W^2}{M^3},$$

because

$$\det(V, V') = cv'_5 - c'v_5 = M.$$

□

Corollary 12.63 (Graph curvature identities in (μ, λ) -coordinates). *On the region where $v_5 M \neq 0$,*

$$\lambda_{\mu\mu} = \frac{v_5^3}{M^3} W,$$

and

$$\lambda_{\mu\mu\mu} = -\frac{v_5^5}{M^4} \left(W' + 3 \left(\frac{v'_5}{v_5} - \frac{M'}{M} \right) W \right).$$

In particular, at a point where $W = 0$, $M \neq 0$, and $v_5 \neq 0$,

$$\lambda_{\mu\mu\mu} = -\frac{v_5^5}{M^4} W'.$$

Proof. Since

$$\frac{d}{d\mu} = \frac{1}{\mu'} \frac{d}{dh} = -\frac{v_5^2}{M} \frac{d}{dh},$$

the identity

$$\lambda_{\mu\mu} = \frac{v_5^3}{M^3} W$$

follows from

$$\lambda_\mu = \beta, \quad \beta' = -\frac{v_5 W}{M^2}.$$

Differentiating once more gives

$$\lambda_{\mu\mu\mu} = -\frac{v_5^2}{M} \frac{d}{dh} \left(\frac{v_5^3}{M^3} W \right),$$

which expands to the displayed formula. The last statement is the specialisation $W = 0$. □

Proposition 12.64 (Projectivized tangent-angle flow and contact identity). *On any simply connected subinterval of the region*

$$v_5 \neq 0, \quad M \neq 0,$$

choose a local lift Θ of the projectivized tangent direction by

$$(\cos \Theta, \sin \Theta) = \frac{1}{\sqrt{M^2 + L^2}} (M, -L).$$

Then

$$\Theta' = -\frac{v_5 W}{M^2 + L^2}.$$

Moreover, with

$$\mu = \frac{c}{v_5}, \quad \beta = -\frac{L}{M}, \quad m = \frac{N}{M},$$

one has the exact contact identity

$$dm + \mu d\beta = 0.$$

Equivalently, on every interval where β' does not vanish,

$$\frac{dm}{d\beta} = -\mu.$$

In particular, at a generic turning boundary with

$$M(h_b) = 0, \quad N(h_b) \neq 0, \quad v_5(h_b) \neq 0,$$

the projectivized slope direction is regular: one has

$$L(h_b) \neq 0,$$

so the compactified tangent direction extends smoothly across the boundary. Thus the only critical points of the compactified tangent-direction map are the zeros of W .

Proof. Differentiate the angle of the vector $(M, -L)$:

$$\Theta' = \frac{M(-L)' - (-L)M'}{M^2 + L^2} = -\frac{ML' - LM'}{M^2 + L^2}.$$

By Proposition 12.62,

$$ML' - LM' = v_5 W,$$

hence

$$\Theta' = -\frac{v_5 W}{M^2 + L^2}.$$

Next, by Definition 12.61,

$$m = \lambda - \mu\beta.$$

Since $\beta = d\lambda/d\mu$, one has

$$d\lambda = \beta d\mu,$$

and therefore

$$dm = d\lambda - \beta d\mu - \mu d\beta = -\mu d\beta.$$

This is exactly

$$dm + \mu d\beta = 0.$$

Finally, at a generic turning boundary with $v_5(h_b) \neq 0$, one has

$$\mu(h_b) = \frac{c(h_b)}{v_5(h_b)} \neq 0.$$

Using

$$N = \mu L + M\lambda$$

from Proposition 12.62, and evaluating at h_b where $M(h_b) = 0$, gives

$$N(h_b) = \mu(h_b)L(h_b).$$

Since $N(h_b) \neq 0$ and $\mu(h_b) \neq 0$, it follows that

$$L(h_b) \neq 0.$$

Hence

$$M(h_b)^2 + L(h_b)^2 \neq 0,$$

so the projectivized tangent direction is regular there. $\square$

Remark 12.65 (Determinant interpretation on a fixed-sign W -interval). On any connected interval where c , M , and W are nonzero and W keeps one sign, the graph $\lambda = \lambda(\mu)$ is strictly convex or strictly concave. Consequently, for any ordered triple of distinct points on that interval, the determinant

$$D(s, t, u) = \det(F(s), F(t), F(u))$$

has the corresponding fixed Vandermonde sign. Thus the unresolved global issue is not interval-wise sign regularity, but propagation across turning seams and possible nonadjacent recurrence of tangent-slope / tangent-line data.

Proposition 12.66 (Local determinant sign transport across a generic turning boundary). *Let h_b be a generic turning boundary:*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

Then there exists a neighborhood U of h_b and a sign $\sigma_{\text{turn}} \in \{\pm 1\}$ such that for any three distinct points $h_1, h_2, h_3 \in U$,

$$\text{sgn } D(h_1, h_2, h_3) = \sigma_{\text{turn}} \text{sgn}((h_2 - h_1)(h_3 - h_1)(h_3 - h_2)),$$

where

$$D(h_1, h_2, h_3) := \det(F(h_1), F(h_2), F(h_3)), \quad F = (c, v_5, \Delta_7).$$

Equivalently, the triple (c, v_5, Δ_7) is locally discretely sign-regular of order 3 across a generic turning boundary.

Proof. Work in the normalized affine chart

$$x(h) := \frac{v_5(h)}{c(h)}, \quad y(h) := \frac{\Delta_7(h)}{c(h)}.$$

At a generic turning boundary one has

$$x'(h_b) = 0, \quad x''(h_b) \neq 0, \quad y'(h_b) \neq 0.$$

Hence y is a valid local coordinate near h_b , and the normalized curve may be written as

$$x = \chi(y)$$

with

$$\chi'(y_b) = 0, \quad \chi''(y_b) \neq 0.$$

After shrinking U , the sign of χ'' is constant on U .

Now for $h_i \in U$, write

$$x_i := x(h_i), \quad y_i := y(h_i).$$

Since

$$F(h_i) = c(h_i) (1, x_i, y_i),$$

we have

$$D(h_1, h_2, h_3) = c(h_1)c(h_2)c(h_3) \det \begin{pmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{pmatrix}.$$

Using $x_i = \chi(y_i)$, the 3×3 determinant factors as

$$\det \begin{pmatrix} 1 & x_1 & y_1 \\ 1 & x_2 & y_2 \\ 1 & x_3 & y_3 \end{pmatrix} = (y_2 - y_1)(y_3 - y_1)(y_3 - y_2) \chi[y_1, y_2, y_3],$$

where $\chi[y_1, y_2, y_3]$ is the second divided difference of χ . By the scalar divided-difference mean-value theorem,

$$\chi[y_1, y_2, y_3] = \frac{1}{2} \chi''(\eta)$$

for some η between the y_i . Thus $\chi[y_1, y_2, y_3]$ has fixed sign on U .

Also c has fixed sign on a sufficiently small neighborhood of h_b , and since $y'(h_b) \neq 0$, the map $h \mapsto y(h)$ is monotone on U . Hence

$$\operatorname{sgn}((y_2 - y_1)(y_3 - y_1)(y_3 - y_2)) = \operatorname{sgn}(y')^3 \operatorname{sgn}((h_2 - h_1)(h_3 - h_1)(h_3 - h_2)),$$

with $\operatorname{sgn}(y')$ constant on U .

Collecting the fixed signs of c , y' , and χ'' gives the desired constant σ_{turn} . $\square$

Proposition 12.67 (Dual asymptote at a generic turning boundary). *Let h_b be a generic turning boundary:*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

Assume $v_5(h_b) \neq 0$, and write

$$\mu_b := \frac{c(h_b)}{v_5(h_b)}, \quad \lambda_b := \frac{\Delta_7(h_b)}{v_5(h_b)}.$$

Let

$$T(h) = (\beta(h), m(h)) = \left(-\frac{L(h)}{M(h)}, \frac{N(h)}{M(h)} \right)$$

be the dual tangent-line curve on the neighboring $M \neq 0$ branches. Then both one-sided dual arcs have the common asymptote

$$m = -\mu_b \beta + \lambda_b,$$

and, more precisely,

$$m + \mu_b \beta - \lambda_b = \frac{1}{4a\beta} + o\left(\frac{1}{\beta}\right) \quad \text{as } |\beta| \rightarrow \infty,$$

where $a \neq 0$ is the quadratic coefficient in the local turning-point normal form

$$\mu - \mu_b = a(\lambda - \lambda_b)^2 + O((\lambda - \lambda_b)^3).$$

In particular, the two adjacent dual arcs approach the same asymptote from opposite sides. This asymptotic separation is the dual-curve explanation of the local adjacent turning-boundary exclusion theorem proved later.

Proof. At a generic turning boundary one has $\mu'(h_b) = 0$ and $\lambda'(h_b) \neq 0$, so λ is a valid local coordinate and

$$\mu - \mu_b = a(\lambda - \lambda_b)^2 + O((\lambda - \lambda_b)^3), \quad a \neq 0.$$

Differentiating gives

$$\frac{d\mu}{d\lambda} = 2a(\lambda - \lambda_b) + O((\lambda - \lambda_b)^2),$$

hence

$$\beta = \frac{d\lambda}{d\mu} = \frac{1}{2a(\lambda - \lambda_b)} + O(1).$$

Thus $|\beta| \rightarrow \infty$ as $h \rightarrow h_b$ from either side.

Now

$$m = \lambda - \mu\beta,$$

so

$$m + \mu_b \beta - \lambda_b = (\lambda - \lambda_b) - (\mu - \mu_b)\beta.$$

Using

$$\mu - \mu_b = a(\lambda - \lambda_b)^2 + O((\lambda - \lambda_b)^3)$$

and

$$\beta = \frac{1}{2a(\lambda - \lambda_b)} + O(1),$$

we obtain

$$(\mu - \mu_b)\beta = \frac{1}{2}(\lambda - \lambda_b) + O((\lambda - \lambda_b)^2).$$

Therefore

$$m + \mu_b \beta - \lambda_b = \frac{1}{2}(\lambda - \lambda_b) + O((\lambda - \lambda_b)^2).$$

Substituting

$$\lambda - \lambda_b = \frac{1}{2a\beta} + o\left(\frac{1}{\beta}\right)$$

gives

$$m + \mu_b \beta - \lambda_b = \frac{1}{4a\beta} + o\left(\frac{1}{\beta}\right).$$

Since $\beta \rightarrow +\infty$ on one side and $\beta \rightarrow -\infty$ on the other, the correction term changes sign, so the two one-sided dual arcs approach the same asymptote from opposite sides. $\square$

Proposition 12.68 (ECT intervals for the strengthened one-pair triple). *Let I be a connected interval on which*

$$c(h) \neq 0, \quad M(h) \neq 0, \quad W(h) \neq 0.$$

Then the triple

$$(c, v_5, \Delta_7)$$

is an extended complete Chebyshev system on I . Equivalently, every nontrivial linear combination

$$Ac(h) + Bv_5(h) + C\Delta_7(h)$$

has at most two zeros on I , counting multiplicity.

In particular, in the affine chart

$$x = \frac{v_5}{c}, \quad y = \frac{\Delta_7}{c},$$

every affine line meets $\Gamma(I)$ in at most two points, counting tangency with multiplicity.

Proof. Define

$$x(h) := \frac{v_5(h)}{c(h)}, \quad y(h) := \frac{\Delta_7(h)}{c(h)}.$$

Since the one-pair family is real-analytic on the generic region, the functions x and y are C^2 on I .

By Definition 12.53,

$$x'(h) = \frac{M(h)}{c(h)^2}, \quad y'(h) = \frac{N(h)}{c(h)^2}.$$

Because $M(h) \neq 0$ and $c(h) \neq 0$ on I , one has

$$x'(h) \neq 0 \quad (h \in I),$$

so x is strictly monotone on I and may be used as a parameter on $\Gamma(I)$.

Moreover,

$$\frac{dy}{dx} = \frac{y'(h)}{x'(h)} = \frac{N(h)}{M(h)}.$$

Differentiating with respect to x gives

$$\frac{d^2y}{dx^2} = \frac{\frac{d}{dh}(N/M)}{x'(h)} = \frac{(MN' - NM')/M^2}{M/c^2}.$$

Using Proposition 12.57,

$$MN' - NM' = cW,$$

hence

$$\frac{d^2y}{dx^2} = \frac{c(h)^3 W(h)}{M(h)^3}.$$

Since $c(h) \neq 0$, $M(h) \neq 0$, and $W(h) \neq 0$ on I , it follows that

$$\frac{d^2y}{dx^2} \neq 0 \quad (h \in I).$$

Therefore $\Gamma(I)$ is the graph of a strictly convex or strictly concave C^2 function $y = y(x)$.

Now let

$$L(h) := A c(h) + B v_5(h) + C \Delta_7(h)$$

be a nontrivial linear combination. Since $c(h) \neq 0$ on I ,

$$L(h) = 0 \iff A + B x(h) + C y(h) = 0.$$

Thus zeros of L correspond exactly to intersections of $\Gamma(I)$ with the affine line

$$A + B x + C y = 0.$$

If $C = 0$, then the equation is

$$A + B x = 0,$$

which has at most one solution because x is strictly monotone on I . If $C \neq 0$, the equation may be written as

$$y = -\frac{A}{C} - \frac{B}{C}x,$$

and a strictly convex or strictly concave graph meets an affine line in at most two points, counting tangency with multiplicity.

Hence every nontrivial linear combination

$$A c + B v_5 + C \Delta_7$$

has at most two zeros on I , counting multiplicity. This is exactly the ECT property for the triple (c, v_5, Δ_7) . The affine-line statement is the same conclusion in the chart (x, y) . $\square$

Definition 12.69 (ECT interval). An ECT interval is a connected interval

$$I \subset \{h : c(h) \neq 0, M(h) \neq 0, W(h) \neq 0\}.$$

Equivalently, by Proposition 12.68, it is a connected interval on which the triple

$$(c, v_5, \Delta_7)$$

is an extended complete Chebyshev system.

Corollary 12.70 (At most two points from one ECT interval). *Let I be an ECT interval. Then every affine line in the chart*

$$x = \frac{v_5}{c}, \quad y = \frac{\Delta_7}{c}$$

meets $\Gamma(I)$ in at most two points, counting tangency with multiplicity.

In particular, if a finite configuration is cut out by a single affine relation in the normalized chart, then its contribution from any fixed ECT interval has cardinality at most two.

Proof. This is exactly the affine-line formulation of Proposition 12.68. $\square$

Remark 12.71 (Branch geometry on an ECT interval). On an ECT interval I , the quantities c , M , and W have fixed nonzero sign. Hence

$$x = \frac{v_5}{c}$$

is a monotone parameter on I , and in the affine chart

$$\Gamma(I) = \left\{ \left(\frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)} \right) : h \in I \right\}$$

is a single C^2 branch which is strictly convex or strictly concave according to the sign of

$$\frac{c(h)^3 W(h)}{M(h)^3}.$$

The later tangent-separator analysis is therefore an adjacent-branch problem, not a same-branch secancy problem.

Definition 12.72 (Tangent-separator kernel). Let

$$F(h) := (c(h), v_5(h), \Delta_7(h)).$$

For s, t in the generic region, define

$$K(s, t) := \det(F(s), F'(s), F(t)).$$

For fixed s , the zeros of $t \mapsto K(s, t)$ are exactly the points $\Gamma(t)$ lying on the affine tangent line to Γ at $\Gamma(s)$. Equivalently, $K(s, t) = 0$ is the tangent-separator incidence condition for the tangent at s and the point at t .

Lemma 12.73 (Affine form of the tangent-separator kernel). *On the affine chart $c \neq 0$, write*

$$x(h) := \frac{v_5(h)}{c(h)}, \quad y(h) := \frac{\Delta_7(h)}{c(h)}.$$

Then for every s, t with $c(s)c(t) \neq 0$,

$$K(s, t) = c(s)^2 c(t) x'(s) \left((y(t) - m(s)x(t)) - (y(s) - m(s)x(s)) \right),$$

where

$$m(s) = \frac{dy}{dx}(s) = \frac{N(s)}{M(s)}.$$

Equivalently,

$$K(s, t) = c(s)^2 c(t) x'(s) (\phi_{m(s)}(t) - \tau_s),$$

with

$$\phi_{m(s)}(t) := y(t) - m(s)x(t), \quad \tau_s := y(s) - m(s)x(s).$$

Proof. Write

$$F(h) = (c(h), c(h)x(h), c(h)y(h)).$$

Then

$$F'(s) = (c'(s), c'(s)x(s) + c(s)x'(s), c'(s)y(s) + c(s)y'(s)).$$

Subtracting $x(s)$ times the first column from the second and $y(s)$ times the first column from the third gives

$$K(s, t) = \det \begin{pmatrix} c(s) & 0 & 0 \\ c'(s) & c(s)x'(s) & c(s)y'(s) \\ c(t) & c(t)(x(t) - x(s)) & c(t)(y(t) - y(s)) \end{pmatrix}.$$

Expanding along the first row yields

$$K(s, t) = c(s)^2 c(t) (x'(s)(y(t) - y(s)) - y'(s)(x(t) - x(s))).$$

Since

$$m(s) = \frac{y'(s)}{x'(s)},$$

this becomes

$$K(s, t) = c(s)^2 c(t) x'(s) ((y(t) - m(s)x(t)) - (y(s) - m(s)x(s))),$$

as claimed. $\square$

Lemma 12.74 (Basic properties of the tangent-separator kernel). *For every s in the generic region,*

$$K(s, s) = 0, \quad \partial_t K(s, s) = 0, \quad \partial_t^2 K(s, s) = W(s).$$

In particular, on any interval where W has fixed sign, the tangent line at $\Gamma(s)$ is locally supporting.

Proof. The first two identities follow because, at $t = s$, the third column of the determinant equals $F(s)$, and after differentiating once in t it equals $F'(s)$. Differentiating a second time gives

$$\partial_t^2 K(s, s) = \det(F(s), F'(s), F''(s)) = W(s). \quad \square$$

Lemma 12.75 (Compact first-zero reduction on adjacent intervals). *Let $I_1 < I_2$ be adjacent ECT intervals in the affine $x = v_5/c$ ordering. Assume there are compact subintervals*

$$J_1 \subseteq I_1, \quad J_2 \subseteq I_2$$

such that every zero of the tangent-separator kernel

$$K(s, t) = \det(F(s), F'(s), F(t)), \quad F(h) = (c(h), v_5(h), \Delta_7(h)),$$

with $s \in I_2$, $t \in I_1$, lies in $J_2 \times J_1$.

If K has a zero on $J_2 \times J_1$, then there exists a zero

$$(s_*, t_*) \in J_2 \times J_1$$

such that s_ is minimal in the $x = v_5/c$ ordering among all s -coordinates of zeros of K on $J_2 \times J_1$.*

Proof. Because K is continuous, its zero set in the compact set $J_2 \times J_1$ is closed, hence compact. The projection of that zero set to the s -axis is therefore compact as well. Since $x = v_5/c$ is monotone on I_2 , the induced x -ordering is equivalent to the interval ordering on J_2 . Thus the projected zero set has a minimal element s_* , and there exists $t_* \in J_1$ such that (s_*, t_*) is a zero of K . $\square$

Lemma 12.76 (First adjacent zero implies tangency). *In the setting of Lemma 12.75, let*

$$(s_*, t_*) \in J_2 \times J_1$$

be a zero with minimal s -coordinate in the $x = v_5/c$ ordering. Then

$$\partial_t K(s_*, t_*) = 0.$$

Consequently,

$$K(s_*, t_*) = 0, \quad \partial_t K(s_*, t_*) = 0$$

force the tangent line at $\Gamma(s_*)$ to be tangent also at $\Gamma(t_*)$.

Proof. Assume for contradiction that

$$\partial_t K(s_*, t_*) \neq 0.$$

Since

$$(s_*, t_*) \in J_2 \times J_1 \subseteq I_2 \times I_1,$$

the implicit function theorem applies in the open set $I_2 \times I_1$ and gives a local C^1 zero curve

$$t = t(s)$$

through (s_*, t_*) , defined for s in a neighborhood of s_* , such that

$$K(s, t(s)) = 0.$$

For s sufficiently close to s_* with $s < s_*$, one still has

$$(s, t(s)) \in I_2 \times I_1$$

and

$$K(s, t(s)) = 0.$$

By the zero-capture hypothesis in Lemma 12.75, every zero of K with $s \in I_2$ and $t \in I_1$ lies in $J_2 \times J_1$. Hence these nearby zeros also lie in $J_2 \times J_1$, contradicting the minimality of s_* among all zeros of K on $J_2 \times J_1$.

Therefore

$$\partial_t K(s_*, t_*) = 0.$$

Finally, the identities

$$K(s, t) = \det(F(s), F'(s), F(t)), \quad \partial_t K(s, t) = \det(F(s), F'(s), F'(t))$$

show that $K(s_*, t_*) = 0$ places $\Gamma(t_*)$ on the tangent line at $\Gamma(s_*)$, while $\partial_t K(s_*, t_*) = 0$ places the tangent direction at t_* on that same line. Hence the line is tangent to Γ at both $\Gamma(s_*)$ and $\Gamma(t_*)$; that is, it is a common tangent line. $\square$

Proposition 12.77 (Adjacent slope-range separation). *Let $I_1 < I_2$ be adjacent ECT intervals, meeting at a boundary point h_b .*

(1) *If*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0,$$

then the interior tangent-slope ranges of I_1 and I_2 are disjoint.

(2) *If*

$$W(h_b) = 0, \quad M(h_b) \neq 0,$$

then $x = v_5/c$ remains a valid local parameter, but no adjacent tangent separation statement is asserted here. In that case the local affine jet is governed by

$$\left. \frac{d^2 y}{dx^2} \right|_{h_b} = 0, \quad \left. \frac{d^3 y}{dx^3} \right|_{h_b} = \frac{c(h_b)^5}{M(h_b)^4} W'(h_b),$$

so the generic behavior is an ordinary affine inflection unless further special structure forces $W'(h_b) = 0$.

Proof. In case (1), $M(h_b) = 0$ is a simple zero and $N(h_b) \neq 0$. Therefore

$$m(h) = \frac{N(h)}{M(h)}$$

has opposite one-sided blow-up:

$$m(h) \rightarrow \pm\infty$$

as $h \rightarrow h_b^\pm$. Hence the interior tangent-slope ranges on the two sides are disjoint.

Case (2) is the local affine-jet computation recorded in Remark 12.85; it identifies the cubic-inflection coefficient but does not produce a tangent-separation theorem. $\square$

Lemma 12.78 (Endpoint tangent exclusion at a generic turning boundary). *Let $I_1 < I_2$ be adjacent ECT intervals meeting at a boundary point h_b , and assume*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

Then the boundary tangent line at $\Gamma(h_b)$ is vertical and meets neither adjacent interval in its interior.

Proof. The boundary tangent line is

$$x = \rho_1(h_b), \quad \rho_1(h) = \frac{v_5(h)}{c(h)}.$$

Since $M(h_b) = 0$ is a simple zero,

$$\rho_1'(h_b) = 0, \quad \rho_1''(h_b) \neq 0,$$

so ρ_1 has a strict local extremum at h_b . Therefore

$$\rho_1(h) - \rho_1(h_b)$$

has fixed sign on each adjacent interval and vanishes only at the boundary point. Hence the vertical tangent line meets neither adjacent interval in its interior. $\square$

Theorem 12.79 (Conditional adjacent tangent-separator sign regularity across turning boundaries). *Let $I_1 < I_2$ be adjacent ECT intervals, and assume their common boundary point h_b is a generic turning boundary:*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

Assume in addition the zero-capture hypothesis of Lemma 12.75 on the adjacent pair $I_2 \times I_1$. Then

$$K(s, t) \neq 0 \quad (s \in I_2, t \in I_1),$$

where

$$K(s, t) = \det(F(s), F'(s), F(t)), \quad F(h) = (c(h), v_5(h), \Delta_7(h)).$$

Equivalently, tangent lines from the right interval do not meet the adjacent left interval across a generic turning boundary.

Proof. By Lemma 12.78, for a generic turning boundary there is a neighborhood of the boundary pair on which $K(s, t) \neq 0$ for $s \in I_2, t \in I_1$.

Assume for contradiction that some zero $(s_0, t_0) \in I_2 \times I_1$ exists. By the zero-capture hypothesis, there are compact subintervals

$$J_1 \Subset I_1, \quad J_2 \Subset I_2$$

such that every zero of K on $I_2 \times I_1$ lies in $J_2 \times J_1$. Replacing them by smaller compact subintervals still containing the zero set, using the boundary-near zero-free region furnished by Lemma 12.78, we may also assume that $J_2 \times J_1$ is disjoint from that boundary-near region. Then the zero set of K in the compact set $J_2 \times J_1$ is nonempty and closed, so by Lemma 12.75 there exists a zero $(s_*, t_*) \in J_2 \times J_1$ whose s -coordinate is minimal in the $x = v_5/c$ ordering among all zeros in $J_2 \times J_1$.

By Lemma 12.76,

$$K(s_*, t_*) = 0, \quad \partial_t K(s_*, t_*) = 0,$$

so the tangent line at $\Gamma(s_*)$ is also tangent at $\Gamma(t_*)$.

Across a generic turning boundary, Proposition 12.77 shows that the interior tangent-slope ranges of the two adjacent intervals are disjoint. Hence no common interior tangent slope can occur. This contradicts the existence of (s_*, t_*) . Therefore no such zero exists, and

$$K(s, t) \neq 0 \quad (s \in I_2, t \in I_1). \quad \square$$

Remark 12.80 (Status of the turning-boundary adjacent theorem). Theorem 12.79 remains a conditional turning-boundary theorem: the local compact first-zero and first-zero-implies-tangency mechanism is in place, but it still depends on a separate zero-capture statement for the directional tangent-separator kernel $K_{21}(s, t)$ on the full adjacent pair $I_2 \times I_1$.

However, by Theorem 12.96, full zero-capture is no longer needed to exclude actual adjacent pair-pair lines. What remains open here is therefore only the stronger support-ordering / directional-kernel route, not the local adjacent $2 + 2$ exclusion itself.

Lemma 12.81 (Boundary-near exclusion for both directional kernels at a generic turning boundary). *Let $I_1 < I_2$ be adjacent ECT intervals meeting at a generic turning boundary*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

Define

$$K_{21}(s, t) := \det(F(s), F'(s), F(t)), \quad K_{12}(t, s) := \det(F(t), F'(t), F(s)),$$

with

$$F(h) = (c(h), v_5(h), \Delta_7(h)).$$

Then there is a neighborhood of the boundary pair on which both directional kernels are nonzero:

$$K_{21}(s, t) \neq 0, \quad K_{12}(t, s) \neq 0$$

for $s \in I_2, t \in I_1$ sufficiently close to h_b .

Proof. Write

$$x(h) := \frac{v_5(h)}{c(h)}, \quad y(h) := \frac{\Delta_7(h)}{c(h)}.$$

Since

$$y'(h) = \frac{N(h)}{c(h)^2},$$

the hypothesis $N(h_b) \neq 0$ implies that y is a valid local coordinate near h_b . Thus the normalized planar curve may be written locally as a graph

$$x = \chi(y).$$

At h_b ,

$$\frac{dx}{dy} = \frac{M(h)}{N(h)},$$

so

$$\left. \frac{dx}{dy} \right|_{h_b} = 0.$$

Differentiating once more,

$$\frac{d^2x}{dy^2} = \frac{c(h)^2}{N(h)} \cdot \frac{M'(h)N(h) - M(h)N'(h)}{N(h)^2},$$

and since $M(h_b) = 0$,

$$\left. \frac{d^2x}{dy^2} \right|_{h_b} = \frac{c(h_b)^2 M'(h_b)}{N(h_b)^2} \neq 0.$$

Hence, after shrinking to a small neighborhood of h_b , the graph $x = \chi(y)$ is strictly convex or strictly concave.

For such a graph, the tangent line at any point meets the graph only at its point of tangency. Therefore no tangent line from one side can meet the opposite side inside a sufficiently small neighborhood of the boundary. Equivalently, both directional tangent-separator kernels are nonzero near the boundary pair. $\square$

Lemma 12.82 (Mirrored first adjacent zero implies tangency). *Let $J_1 \Subset I_1$, $J_2 \Subset I_2$ be compact interior subintervals across a generic turning boundary, and define*

$$K_{12}(t, s) := \det(F(t), F'(t), F(s)), \quad t \in I_1, s \in I_2.$$

Assume every zero of K_{12} on $I_1 \times I_2$ lies in $J_1 \times J_2$. Let $(t_, s_*) \in J_1 \times J_2$ be a zero whose t -coordinate is minimal in the $x = v_5/c$ ordering among all zeros of K_{12} on $I_1 \times I_2$. Then*

$$\partial_s K_{12}(t_*, s_*) = 0.$$

Proof. Because $M \neq 0$ on the open interval I_1 , the function

$$x(h) = \frac{v_5(h)}{c(h)}$$

is a valid local coordinate on I_1 . Since every zero of K_{12} lies in the compact interior rectangle $J_1 \times J_2$, the chosen point (t_*, s_*) is interior in the ambient domain $I_1 \times I_2$.

Suppose

$$\partial_s K_{12}(t_*, s_*) \neq 0.$$

Then by the implicit function theorem there is a C^1 local zero curve

$$s = s(t)$$

through (t_*, s_*) , defined for t near t_* , such that

$$K_{12}(t, s(t)) = 0.$$

Because t_* is interior in I_1 , there are nearby points with smaller $x = v_5/c$ ordering still in I_1 . For such t , the point $(t, s(t))$ is also a zero of K_{12} , contradicting the minimality of t_* . Therefore

$$\partial_s K_{12}(t_*, s_*) = 0.$$

□

Theorem 12.83 (Conditional symmetric directional-kernel exclusion across turning boundaries). *Let $I_1 < I_2$ be adjacent ECT intervals meeting at a generic turning boundary*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

Assume:

- (1) *the zero-capture hypothesis required in Theorem 12.79 for the kernel K_{21} on $I_2 \times I_1$;*
- (2) *the symmetric zero-capture hypothesis for K_{12} , namely that every zero of*

$$K_{12}(t, s) = \det(F(t), F'(t), F(s))$$

on $I_1 \times I_2$ lies in some compact interior rectangle $J_1 \times J_2 \Subset I_1 \times I_2$.

Then both directional kernels are zero-free:

$$K_{21}(s, t) \neq 0 \quad (s \in I_2, t \in I_1),$$

and

$$K_{12}(t, s) \neq 0 \quad (t \in I_1, s \in I_2).$$

Proof. The first assertion is exactly Theorem 12.79.

For the second assertion, boundary-near exclusion for K_{12} is provided by Lemma 12.81. Assume for contradiction that K_{12} has a zero. By the symmetric zero-capture hypothesis, every zero lies in a compact interior rectangle $J_1 \times J_2$. Choose a zero $(t_*, s_*) \in J_1 \times J_2$ whose t -coordinate is minimal in the $x = v_5/c$ ordering among all zeros of K_{12} on $I_1 \times I_2$.

By Lemma 12.82,

$$\partial_s K_{12}(t_*, s_*) = 0.$$

Together with

$$K_{12}(t_*, s_*) = 0,$$

this implies that the tangent line at $\Gamma(t_*)$ is also tangent at $\Gamma(s_*)$. Hence the two adjacent intervals share a common interior tangent slope, contradicting Proposition 12.77. Therefore K_{12} has no zero. □

Remark 12.84 (Status of the directional-kernel reduction). Lemma 12.91 reduces an adjacent pair-pair line across a generic turning boundary to a compact interior zero of one of the two directional tangent-separator kernels. Theorem 12.83 shows that the remaining turning-boundary issue is no longer orientation selection between K_{21} and K_{12} , but the symmetric zero-capture problem for these two kernels.

Remark 12.85 (The $W = 0$ obstruction and the cubic-inflection coefficient). Let

$$x(h) := \frac{v_5(h)}{c(h)}, \quad y(h) := \frac{\Delta_7(h)}{c(h)}.$$

At a boundary point h_b with

$$W(h_b) = 0, \quad M(h_b) \neq 0,$$

the coordinate x remains a valid local parameter. Since

$$\frac{d^2y}{dx^2} = \frac{c(h)^3 W(h)}{M(h)^3},$$

one has

$$\left. \frac{d^2y}{dx^2} \right|_{h_b} = 0.$$

Differentiating once more with respect to x gives

$$\left. \frac{d^3y}{dx^3} \right|_{h_b} = \frac{c(h_b)^5}{M(h_b)^4} W'(h_b).$$

Hence, after translating the point to the origin and subtracting the tangent line, the normalized curve has local expansion

$$y = \frac{c(h_b)^5 W'(h_b)}{6M(h_b)^4} x^3 + O(x^4).$$

Therefore a generic $W = 0$, $M \neq 0$ boundary is an ordinary affine inflection unless additional special structure forces

$$W'(h_b) = 0.$$

In particular, the tangent-separator route is not presently justified across such boundaries, and the real remaining geometric question is whether the actual triple (c, v_5, Δ_7) imposes any non-generic vanishing of $W'(h_b)$.

Lemma 12.86 (Fixed-slope intercept sets for a strictly convex ECT branch). *Let I be an ECT interval in the affine chart*

$$x = \frac{v_5}{c}, \quad y = \frac{\Delta_7}{c},$$

and suppose $\Gamma(I)$ is a strictly convex C^2 graph

$$\Gamma(I) = \{(x, \psi_I(x)) : x \in J_I\}$$

over an open interval $J_I \subset \mathbf{R}$.

For each slope $m \in \mathbf{R}$, define

$$\mathcal{B}_I(m) := \{\psi_I(x) - mx : x \in J_I\}.$$

Then:

(1) $\mathcal{B}_I(m)$ is an interval.

(2) A line $y = mx + b$ meets $\Gamma(I)$ if and only if

$$b \in \mathcal{B}_I(m).$$

(3) If m lies in the interior tangent-slope range of I , there is a unique $x_I(m) \in J_I$ such that

$$\psi'_I(x_I(m)) = m,$$

and the tangent line of slope m has intercept

$$\tau_I(m) := \psi_I(x_I(m)) - m x_I(m).$$

Moreover,

$$\tau_I(m) = \inf \mathcal{B}_I(m).$$

(4) For such m , a line $y = mx + b$ is 2-secant to $\Gamma(I)$ if and only if

$$b \in \text{int } \mathcal{B}_I(m).$$

Equivalently, the 2-secant intercepts of slope m are exactly the interior points of the intercept interval.

Proof. For fixed m , put

$$\phi_m(x) := \psi_I(x) - mx.$$

Since ϕ_m is continuous on the interval J_I , its image

$$\phi_m(J_I) = \mathcal{B}_I(m)$$

is an interval. This proves (1), and (2) is immediate from

$$(x, \psi_I(x)) \in \{y = mx + b\} \iff \psi_I(x) - mx = b.$$

If m lies in the interior tangent-slope range, strict convexity implies that ψ'_I is strictly monotone, hence there is a unique $x_I(m)$ with $\psi'_I(x_I(m)) = m$. At that point,

$$\phi'_m(x_I(m)) = \psi'_I(x_I(m)) - m = 0,$$

and since

$$\phi''_m(x) = \psi''_I(x) > 0,$$

the point $x_I(m)$ is the unique minimizer of ϕ_m . Thus

$$\tau_I(m) = \phi_m(x_I(m)) = \inf \mathcal{B}_I(m),$$

which proves (3).

Finally, because ϕ_m is strictly convex with a unique minimum, a level b is attained at two points exactly when it lies in the interior of the interval $\phi_m(J_I) = \mathcal{B}_I(m)$. This is precisely the 2-secant condition, proving (4). $\square$

Proposition 12.87 (Support-curve calculus on an ECT branch). *Let I be as in Lemma 12.86, and assume m lies in the interior tangent-slope range of I . Then*

$$x'_I(m) = \frac{1}{\psi''_I(x_I(m))} > 0,$$

$$\tau'_I(m) = -x_I(m),$$

and

$$\tau''_I(m) = -\frac{1}{\psi''_I(x_I(m))} < 0.$$

In particular, the tangent-intercept function $m \mapsto \tau_I(m)$ is strictly concave on the interior tangent-slope range.

Proof. The defining relation

$$\psi'_I(x_I(m)) = m$$

implies, by differentiation in m ,

$$\psi''_I(x_I(m)) x'_I(m) = 1,$$

hence

$$x'_I(m) = \frac{1}{\psi''_I(x_I(m))} > 0.$$

Now

$$\tau_I(m) = \psi_I(x_I(m)) - m x_I(m).$$

Differentiate:

$$\tau'_I(m) = \psi'_I(x_I(m)) x'_I(m) - x_I(m) - m x'_I(m) = m x'_I(m) - x_I(m) - m x'_I(m) = -x_I(m).$$

Differentiating once more gives

$$\tau_I''(m) = -x_I'(m) = -\frac{1}{\psi_I''(x_I(m))} < 0.$$

□

Corollary 12.88 (Single crossing and betweenness of support curves). *Let $I_i < I_j$ be two ordered strict convex ECT branches in the affine chart. On every common interior tangent-slope interval,*

$$(\tau_{I_j} - \tau_{I_i})'(m) = -(x_{I_j}(m) - x_{I_i}(m)) < 0.$$

Hence τ_{I_i} and τ_{I_j} cross at most once; equivalently, the two ordered branches admit at most one common interior tangent slope.

More generally, let

$$I_i < I_j < I_k$$

be three ordered strict convex ECT branches, active on a common interior slope interval. If τ_{I_i} and τ_{I_j} cross at m_{ij} and τ_{I_j} and τ_{I_k} cross at m_{jk} with

$$m_{ij} < m_{jk},$$

then there is a unique

$$m_{ik} \in (m_{ij}, m_{jk})$$

such that

$$\tau_{I_i}(m_{ik}) = \tau_{I_k}(m_{ik}).$$

So the support curves

$$b = \tau_I(m)$$

form a pseudoline arrangement on every common slope window.

Proof. Since the branches are ordered in the $x = v_5/c$ coordinate, one has

$$x_{I_i}(m) < x_{I_j}(m)$$

whenever both tangencies exist. Therefore

$$(\tau_{I_j} - \tau_{I_i})'(m) = -(x_{I_j}(m) - x_{I_i}(m)) < 0,$$

so $\tau_{I_j} - \tau_{I_i}$ is strictly decreasing and can vanish at most once.

For the betweenness statement, set

$$\delta_{ab}(m) := \tau_{I_b}(m) - \tau_{I_a}(m).$$

Each δ_{ab} is strictly decreasing. If

$$\delta_{ij}(m_{ij}) = 0, \quad \delta_{jk}(m_{jk}) = 0, \quad m_{ij} < m_{jk},$$

then

$$\delta_{ik}(m_{ij}) = \delta_{ij}(m_{ij}) + \delta_{jk}(m_{ij}) = \delta_{jk}(m_{ij}) > 0,$$

because δ_{jk} vanishes later at m_{jk} , and similarly

$$\delta_{ik}(m_{jk}) = \delta_{ij}(m_{jk}) + \delta_{jk}(m_{jk}) = \delta_{ij}(m_{jk}) < 0.$$

By strict monotonicity, δ_{ik} has a unique zero in (m_{ij}, m_{jk}) . □

Proposition 12.89 (Ordered-pair overlap windows are one-shot). *Let $I_1 < I_2$ be two ordered strict convex ECT branches in the affine chart, with*

$$\Gamma(I_j) = \{(x, \psi_j(x)) : x \in J_j\}, \quad J_1 < J_2.$$

For $j = 1, 2$, let

$$\mathcal{B}_{I_j}(m) = \{\psi_j(x) - mx : x \in J_j\}, \quad u_j(m) := \sup \mathcal{B}_{I_j}(m),$$

and let $\tau_j(m)$ be the tangent intercept on the interior tangent-slope range.

Then the directional overlap window

$$\Sigma_{2 \rightarrow 1} := \{m : \tau_1(m) \leq \tau_2(m) \leq u_1(m)\}$$

is an interval, possibly empty.

Likewise the pair-pair overlap window

$$\Sigma_{12}^{\text{pp}} := \{m : \text{int } \mathcal{B}_{I_1}(m) \cap \text{int } \mathcal{B}_{I_2}(m) \neq \emptyset\}$$

is an interval, possibly empty.

More precisely:

(1) on every slope subinterval where one fixed endpoint support line

$$m \mapsto y_\xi - m\xi$$

realizes $u_1(m)$, one has

$$\frac{d}{dm}(u_1(m) - \tau_2(m)) = x_{I_2}(m) - \xi > 0;$$

(2) on every slope subinterval where one fixed endpoint support line

$$m \mapsto y_\eta - m\eta$$

realizes $u_2(m)$, one has

$$\frac{d}{dm}(u_2(m) - \tau_1(m)) = x_{I_1}(m) - \eta < 0.$$

In particular, for a fixed ordered pair, directional or pair-pair overlap cannot disappear and later reappear.

Proof. By Corollary 12.88,

$$\tau_2(m) - \tau_1(m)$$

is strictly decreasing on the common interior slope range. Therefore the set

$$\{m : \tau_1(m) \leq \tau_2(m)\}$$

is an interval.

Now fix a slope subinterval on which one endpoint support line

$$m \mapsto y_\xi - m\xi$$

realizes $u_1(m)$. Then

$$u_1(m) = y_\xi - m\xi,$$

so

$$\frac{d}{dm}(u_1(m) - \tau_2(m)) = -\xi - (-x_{I_2}(m)) = x_{I_2}(m) - \xi.$$

Because every point of J_2 lies strictly to the right of every endpoint of J_1 , one has

$$x_{I_2}(m) - \xi > 0.$$

Hence

$$u_1(m) - \tau_2(m)$$

is strictly increasing on that piece. Therefore the set

$$\{m : \tau_2(m) \leq u_1(m)\}$$

is an interval after patching across the finitely many endpoint-switch slopes. Thus

$$\Sigma_{2 \rightarrow 1} = \{m : \tau_1(m) \leq \tau_2(m)\} \cap \{m : \tau_2(m) \leq u_1(m)\}$$

is an interval.

Similarly, on a slope subinterval where one endpoint support line

$$m \mapsto y_\eta - m\eta$$

realizes $u_2(m)$,

$$\frac{d}{dm}(u_2(m) - \tau_1(m)) = -\eta - (-x_{I_1}(m)) = x_{I_1}(m) - \eta < 0,$$

because every point of J_1 lies strictly to the left of every endpoint of J_2 .

Finally, by Lemma 12.86,

$$\text{int } \mathcal{B}_{I_1}(m) \cap \text{int } \mathcal{B}_{I_2}(m) \neq \emptyset$$

is equivalent to

$$\tau_2(m) < u_1(m) \quad \text{and} \quad \tau_1(m) < u_2(m).$$

The first inequality holds on an interval because $u_1 - \tau_2$ is piecewise strictly increasing; the second holds on an interval because $u_2 - \tau_1$ is piecewise strictly decreasing. Their intersection is therefore an interval. $\square$

Lemma 12.90 (Support-overlap truncation across a generic turning boundary). *Let $I_1 < I_2$ be adjacent ECT intervals meeting at a generic turning boundary*

$$c(h_b) \neq 0, \quad M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

For a fixed finite slope m , define

$$\phi_m(h) := \frac{\Delta_7(h) - mv_5(h)}{c(h)}, \quad \mathcal{B}_I(m) := \phi_m(I).$$

Assume

$$b \in \mathcal{B}_{I_1}(m) \cap \mathcal{B}_{I_2}(m).$$

Then there exists a neighborhood U of h_b and compact subintervals

$$J_1 \subseteq I_1 \setminus U, \quad J_2 \subseteq I_2 \setminus U$$

such that

$$b \in \mathcal{B}_{J_1}(m) \cap \mathcal{B}_{J_2}(m).$$

Proof. Differentiate:

$$\phi'_m(h) = \frac{N(h) - mM(h)}{c(h)^2}.$$

At $h = h_b$, since $M(h_b) = 0$,

$$\phi'_m(h_b) = \frac{N(h_b)}{c(h_b)^2} \neq 0.$$

Hence for each fixed finite m , after shrinking to a small neighborhood U of h_b , the function ϕ_m is strictly monotone on each one-sided piece

$$I_1 \cap U, \quad I_2 \cap U.$$

Therefore, for each $i = 1, 2$, the equation

$$\phi_m(h) = b$$

has at most one solution in $I_i \cap U$.

But the assumption

$$b \in \mathcal{B}_{I_i}(m)$$

means exactly that there is at least one solution of $\phi_m(h) = b$ somewhere on I_i . Therefore, after removing the boundary neighborhood U , at least one such solution

$$h_i^\sharp \in I_i \setminus U$$

remains on each side. Choose small compact subintervals

$$J_i \subseteq I_i \setminus U$$

with $h_i^\sharp \in J_i$. Then by continuity,

$$b = \phi_m(h_i^\sharp) \in \phi_m(J_i) = \mathcal{B}_{J_i}(m),$$

for $i = 1, 2$. This proves the claim. $\square$

Lemma 12.91 (Compact adjacent pair-pair implies a directional kernel zero). *Let $I_1 < I_2$ be adjacent ECT intervals meeting at a generic turning boundary, and let*

$$L : y = mx + b$$

be a finite-slope line that is 2-secant to both $\Gamma(I_1)$ and $\Gamma(I_2)$. Then, after removing a sufficiently small boundary neighborhood, there exist compact subintervals

$$J_1 \subseteq I_1, \quad J_2 \subseteq I_2$$

and points $s \in J_2, t \in J_1$ such that one of the two directional kernels vanishes:

$$K_{21}(s, t) := \det(F(s), F'(s), F(t)) = 0,$$

or

$$K_{12}(t, s) := \det(F(t), F'(t), F(s)) = 0,$$

where

$$F(h) = (c(h), v_5(h), \Delta_7(h)).$$

Proof. Because L is 2-secant to each branch, Lemma 12.86 shows that m lies in the interior tangent-slope range of each interval. Hence each branch has a unique interior tangent line of slope m , with tangent intercept

$$\tau_{I_i}(m) = \inf \mathcal{B}_{I_i}(m).$$

At a generic turning boundary,

$$\frac{dy/dh}{dx/dh} = \frac{N(h)}{M(h)}$$

has one-sided blow-up, so no finite-slope tangency can occur arbitrarily close to h_b . Therefore, after shrinking away a boundary neighborhood and applying Lemma 12.90, we may choose compact subintervals

$$J_1 \subseteq I_1, \quad J_2 \subseteq I_2$$

such that:

- (1) $b \in \mathcal{B}_{J_1}(m) \cap \mathcal{B}_{J_2}(m)$;
- (2) the unique slope- m tangency point of each branch lies in the corresponding J_i .

Hence each $\mathcal{B}_{J_i}(m)$ is a compact interval whose lower endpoint is the tangent intercept

$$\tau_i := \tau_{J_i}(m) = \inf \mathcal{B}_{J_i}(m).$$

Now set

$$b_* := \inf(\mathcal{B}_{J_1}(m) \cap \mathcal{B}_{J_2}(m)).$$

Since the intersection of two compact intervals is again a compact interval,

$$b_* = \max(\tau_1, \tau_2).$$

So b_* is equal to one of the two tangent intercepts, and it still lies in the support set of the other branch.

If $b_* = \tau_2$, then the slope- m tangent line to $\Gamma(J_2)$ meets $\Gamma(J_1)$. Thus there are points $s \in J_2, t \in J_1$ such that

$$\det(F(s), F'(s), F(t)) = 0,$$

i.e.

$$K_{21}(s, t) = 0.$$

If $b_* = \tau_1$, then the slope- m tangent line to $\Gamma(J_1)$ meets $\Gamma(J_2)$. Thus there are points $t \in J_1, s \in J_2$ such that

$$\det(F(t), F'(t), F(s)) = 0,$$

i.e.

$$K_{12}(t, s) = 0.$$

This proves the lemma. □

Lemma 12.92 (Strip exclusion for K_{21} against a compact opposite branch). *Let $I_1 < I_2$ be adjacent ECT intervals meeting at a generic turning boundary*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

Let $J_1 \Subset I_1$ be a compact subinterval. Then there exists a one-sided neighborhood $U_2 \subset I_2$ of the shared boundary point h_b such that

$$K_{21}(s, t) := \det(F(s), F'(s), F(t)) \neq 0 \quad (s \in U_2, t \in J_1),$$

where $F(h) = (c(h), v_5(h), \Delta_7(h))$.

Proof. Work in the affine chart

$$x(h) := \frac{v_5(h)}{c(h)}, \quad y(h) := \frac{\Delta_7(h)}{c(h)}.$$

At a generic turning boundary one has

$$x'(h_b) = 0, \quad x''(h_b) \neq 0, \quad y'(h_b) \neq 0.$$

Hence y is a valid local coordinate and the normalized curve may be written near h_b as a graph

$$x = \chi(y)$$

with

$$\chi'(y_b) = 0, \quad \chi''(y_b) \neq 0.$$

Thus x has a strict local extremum at the boundary, while the tangent slope

$$m(h) = \frac{dy/dh}{dx/dh} = \frac{N(h)}{M(h)}$$

blows up one-sidedly as $h \rightarrow h_b$ from either adjacent interval.

Because $J_1 \Subset I_1$ is compact away from h_b , there exists $\delta > 0$ such that

$$|x(t) - x(h_b)| \geq 2\delta \quad (t \in J_1).$$

After shrinking to a small one-sided neighborhood $U_2 \subset I_2$ of h_b , we may also assume

$$|x(s) - x(h_b)| \leq \delta \quad (s \in U_2).$$

Hence

$$|x(t) - x(s)| \geq \delta \quad (s \in U_2, t \in J_1).$$

Fix a preliminary compact neighborhood $U_2^0 \Subset I_2$ of h_b and set

$$B := \sup\{|y(t) - y(s)| : t \in J_1, s \in U_2^0\} < \infty.$$

Since $m(s) \rightarrow \pm\infty$ as $s \rightarrow h_b$ in I_2 , we may shrink further to $U_2 \subset U_2^0$ so that

$$|m(s)| > \frac{2B}{\delta} \quad (s \in U_2).$$

If $K_{21}(s, t) = 0$, then the tangent line at $\Gamma(s)$ passes through $\Gamma(t)$, equivalently

$$y(t) - y(s) = m(s)(x(t) - x(s)).$$

Taking absolute values gives

$$B \geq |y(t) - y(s)| = |m(s)| |x(t) - x(s)| > \frac{2B}{\delta} \cdot \delta = 2B,$$

a contradiction. Therefore $K_{21}(s, t) \neq 0$ on $U_2 \times J_1$. □

Lemma 12.93 (Strip exclusion for K_{12} against a compact opposite branch). *Let $I_1 < I_2$ be adjacent ECT intervals meeting at a generic turning boundary*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

Let $J_2 \Subset I_2$ be a compact subinterval. Then there exists a one-sided neighborhood $U_1 \subset I_1$ of the shared boundary point h_b such that

$$K_{12}(t, s) := \det(F(t), F'(t), F(s)) \neq 0 \quad (t \in U_1, s \in J_2),$$

where $F(h) = (c(h), v_5(h), \Delta_7(h))$.

Proof. This is the left-right mirror of Lemma 12.92. $\square$

Lemma 12.94 (Compact first-contact for K_{21}). *Let $J_1 \in I_1$, $J_2 \in I_2$ be compact subintervals across a generic turning boundary. Assume there is a relatively open collar $V_2 \subset J_2$ at the boundary-facing end of J_2 such that*

$$K_{21}(s, t) \neq 0 \quad (s \in V_2, t \in J_1).$$

Assume also that the zero set of K_{21} in $J_2 \times J_1$ is nonempty. Then there exists a zero $(s_, t_*) \in J_2 \times J_1$ such that:*

- (1) s_* is boundary-nearest in the $x = v_5/c$ ordering among all $s \in J_2$ for which $K_{21}(s, t) = 0$ for some $t \in J_1$;
- (2)

$$\partial_t K_{21}(s_*, t_*) = 0.$$

Proof. Define

$$A_{21} := \{s \in J_2 : \exists t \in J_1 \text{ with } K_{21}(s, t) = 0\}.$$

Since the zero set of K_{21} in $J_2 \times J_1$ is compact and nonempty, the set A_{21} is compact and nonempty. By hypothesis,

$$A_{21} \cap V_2 = \emptyset.$$

Hence A_{21} has a boundary-nearest element s_* in the $x = v_5/c$ ordering on J_2 . Choose $t_* \in J_1$ with

$$K_{21}(s_*, t_*) = 0.$$

Suppose

$$\partial_t K_{21}(s_*, t_*) \neq 0.$$

Then by the implicit function theorem there is a C^1 local zero curve

$$t = t(s)$$

through (s_*, t_*) , defined for s near s_* , such that

$$K_{21}(s, t(s)) = 0.$$

Because $s_* \notin V_2$ and V_2 is an open collar at the boundary-facing end of J_2 , there exist nearby points $s \in J_2$ that are boundary-nearer than s_* in the $x = v_5/c$ ordering. For such s , the point $(s, t(s))$ is also a zero of K_{21} , contradicting the minimality of s_* . Therefore

$$\partial_t K_{21}(s_*, t_*) = 0.$$

$\square$

Lemma 12.95 (Compact first-contact for K_{12}). *Let $J_1 \in I_1$, $J_2 \in I_2$ be compact subintervals across a generic turning boundary. Assume there is a relatively open collar $V_1 \subset J_1$ at the boundary-facing end of J_1 such that*

$$K_{12}(t, s) \neq 0 \quad (t \in V_1, s \in J_2).$$

Assume also that the zero set of K_{12} in $J_1 \times J_2$ is nonempty. Then there exists a zero $(t_, s_*) \in J_1 \times J_2$ such that:*

- (1) t_* is boundary-nearest in the $x = v_5/c$ ordering among all $t \in J_1$ for which $K_{12}(t, s) = 0$ for some $s \in J_2$;
- (2)

$$\partial_s K_{12}(t_*, s_*) = 0.$$

Proof. Define

$$A_{12} := \{t \in J_1 : \exists s \in J_2 \text{ with } K_{12}(t, s) = 0\}.$$

Since the zero set of K_{12} in $J_1 \times J_2$ is compact and nonempty, the set A_{12} is compact and nonempty. By hypothesis,

$$A_{12} \cap V_1 = \emptyset.$$

Hence A_{12} has a boundary-nearest element t_* in the $x = v_5/c$ ordering on J_1 . Choose $s_* \in J_2$ with

$$K_{12}(t_*, s_*) = 0.$$

Suppose

$$\partial_s K_{12}(t_*, s_*) \neq 0.$$

Then by the implicit function theorem there is a C^1 local zero curve

$$s = s(t)$$

through (t_*, s_*) , defined for t near t_* , such that

$$K_{12}(t, s(t)) = 0.$$

Because $t_* \notin V_1$ and V_1 is an open collar at the boundary-facing end of J_1 , there exist nearby points $t \in J_1$ that are boundary-nearer than t_* in the $x = v_5/c$ ordering. For such t , the point $(t, s(t))$ is also a zero of K_{12} , contradicting the minimality of t_* . Therefore

$$\partial_s K_{12}(t_*, s_*) = 0.$$

□

Theorem 12.96 (Adjacent pair-pair exclusion across a generic turning boundary). *Let $I_1 < I_2$ be adjacent ECT intervals meeting at a generic turning boundary*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0.$$

Then no finite-slope affine line in the chart

$$x = \frac{v_5}{c}, \quad y = \frac{\Delta_7}{c}$$

is 2-secant to both $\Gamma(I_1)$ and $\Gamma(I_2)$.

Proof. Assume for contradiction that a finite-slope line

$$L : y = mx + b$$

is 2-secant to both adjacent branches. By Lemma 12.91, after removing a small boundary neighborhood there exist compact subintervals

$$J_1 \subseteq I_1, \quad J_2 \subseteq I_2$$

and a zero of one of the two directional kernels.

First suppose there exist $s_0 \in J_2$, $t_0 \in J_1$ such that

$$K_{21}(s_0, t_0) = 0.$$

By Lemma 12.92, there exists a one-sided neighborhood $U_2 \subset I_2$ of h_b such that

$$K_{21}(s, t) \neq 0 \quad (s \in U_2, t \in J_1).$$

Choose a compact interval $\tilde{J}_2 \subseteq I_2$ containing s_0 and whose boundary-facing collar V_2 lies in U_2 . Then the zero set of K_{21} in $\tilde{J}_2 \times J_1$ is nonempty, and Lemma 12.94 yields a zero

$$(s_*, t_*) \in \tilde{J}_2 \times J_1$$

such that

$$\partial_t K_{21}(s_*, t_*) = 0.$$

Together with $K_{21}(s_*, t_*) = 0$, this means that the tangent line at $\Gamma(s_*)$ is also tangent at $\Gamma(t_*)$. Since $s_* \in I_2$ and $t_* \in I_1$ are interior points, this gives a common interior tangent slope across the adjacent branches, contradicting Proposition 12.77.

Now suppose instead there exist $t_0 \in J_1$, $s_0 \in J_2$ such that

$$K_{12}(t_0, s_0) = 0.$$

By Lemma 12.93, there exists a one-sided neighborhood $U_1 \subset I_1$ of h_b such that

$$K_{12}(t, s) \neq 0 \quad (t \in U_1, s \in J_2).$$

Choose a compact interval $\tilde{J}_1 \subseteq I_1$ containing t_0 and whose boundary-facing collar V_1 lies in U_1 . Then the zero set of K_{12} in $\tilde{J}_1 \times J_2$ is nonempty, and Lemma 12.95 yields a zero

$$(t_*, s_*) \in \tilde{J}_1 \times J_2$$

such that

$$\partial_s K_{12}(t_*, s_*) = 0.$$

Together with $K_{12}(t_*, s_*) = 0$, this means that the tangent line at $\Gamma(t_*)$ is also tangent at $\Gamma(s_*)$. Again this gives a common interior tangent slope across the adjacent branches, contradicting Proposition 12.77.

Both cases are impossible. Therefore no finite-slope line is 2-secant to both adjacent branches across a generic turning boundary. $\square$

Remark 12.97 (What the unconditional adjacent turning theorem does and does not yet replace). Theorem 12.96 gives an unconditional exclusion of finite-slope pair-pair lines across a single generic turning boundary. It is stronger than the older local directional-kernel reduction in that it bypasses full symmetric zero-capture.

However, it does not by itself replace Theorem 12.100. To obtain a full global no-four-secant theorem from it, one would still need a separate propagation statement reducing nonadjacent pair-pair lines to the adjacent turning-boundary case.

Remark 12.98 (Status of the directional-kernel reduction). Lemma 12.91 reduces an adjacent pair-pair line across a generic turning boundary to a compact interior zero of one of the two directional tangent-separator kernels.

There are now two closures of that reduction in the draft:

- (1) Theorem 12.96 gives an unconditional closure for actual adjacent pair-pair lines, via strip exclusion and compact first-contact.
- (2) Theorem 12.83 gives a stronger zero-freeness statement for both directional kernels, but only conditionally on symmetric zero-capture.

Thus orientation selection between K_{21} and K_{12} is no longer the main issue. What remains open only for the stronger tangent-separator / support-ordering route is the symmetric zero-capture problem on the full adjacent rectangle.

Lemma 12.99 (Adjacent support-ordering). *Let $I_1 < I_2$ be adjacent ECT intervals under the hypotheses of Theorem 12.79. Then, for every slope $m \in \mathbf{R}$ such that both fixed-slope intercept sets are nonempty,*

$$\mathcal{B}_{I_1}(m) \cap \mathcal{B}_{I_2}(m) = \emptyset.$$

More precisely, one has one of the two strict alternatives

$$\sup \mathcal{B}_{I_1}(m) < \inf \mathcal{B}_{I_2}(m) \quad \text{or} \quad \sup \mathcal{B}_{I_2}(m) < \inf \mathcal{B}_{I_1}(m),$$

and the same alternative holds for all such m on the adjacent pair.

Proof. Fix m such that both $\mathcal{B}_{I_1}(m)$ and $\mathcal{B}_{I_2}(m)$ are nonempty.

First assume m lies in the interior tangent-slope range of I_2 . Let $s \in I_2$ be the unique point with tangent slope m . Then by Lemma 12.86,

$$\tau_{I_2}(m) = \inf \mathcal{B}_{I_2}(m).$$

By Theorem 12.79,

$$K(s, t) \neq 0 \quad (t \in I_1).$$

By Lemma 12.73,

$$K(s, t) = c(s)^2 c(t) x'(s) (\phi_m(t) - \tau_{I_2}(m)), \quad \phi_m(t) := y(t) - mx(t).$$

Since $c(s)^2 c(t) x'(s)$ has fixed sign on I_1 and $K(s, t)$ is continuous and nowhere zero there, the quantity

$$\phi_m(t) - \tau_{I_2}(m)$$

has constant sign on I_1 . To determine which sign occurs, move s toward the common boundary point from inside I_2 . By Lemma 12.78, the limiting boundary tangent line does not meet I_1 , so

I_1 lies entirely on one side of that tangent line. By continuity, the same side persists for nearby interior tangents. Indeed, if the side changed as the tangency point s moved along I_2 , then by the connectedness of I_2 and continuity of the tangent-line data there would exist an intermediate point $s_* \in I_2$ for which the tangent line at $\Gamma(s_*)$ meets $\Gamma(I_1)$. Equivalently, there would exist $t_* \in I_1$ such that

$$K(s_*, t_*) = 0,$$

contradicting Theorem 12.79. Hence the same side holds for all $s \in I_2$. Therefore either

$$\phi_m(t) < \tau_{I_2}(m) \quad \text{for all } t \in I_1,$$

or

$$\phi_m(t) > \tau_{I_2}(m) \quad \text{for all } t \in I_1.$$

In the first case,

$$\sup \mathcal{B}_{I_1}(m) \leq \tau_{I_2}(m) = \inf \mathcal{B}_{I_2}(m),$$

and since equality would force an intersection with the tangent line, the inequality is strict. The second case is symmetric.

Now suppose m is not in the interior tangent-slope range of I_2 , but both intercept sets are nonempty. Then m lies on one side of the tangent slope range of I_2 , so ϕ_m is strictly monotone on I_2 , and $\mathcal{B}_{I_2}(m)$ is an interval with endpoint attained in the one-sided limit at the adjacent boundary. Passing to that boundary-support line and using Lemma 12.78 again shows that I_1 lies strictly on one side of that boundary-support line. Hence the same strict ordering of intercept sets holds in this case as well.

Thus $\mathcal{B}_{I_1}(m)$ and $\mathcal{B}_{I_2}(m)$ are disjoint for every slope m for which both are nonempty, and the side ordering is constant on the adjacent pair. $\square$

Theorem 12.100 (Partial global no-secant theorem on turning-boundary generic blocks). *Assume every adjacent pair of ECT intervals is separated by a generic turning boundary*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0,$$

and assume moreover that the zero-capture hypothesis required in Theorem 12.79 holds for every adjacent turning-boundary pair. Then no affine line in the chart

$$x = \frac{v_5}{c}, \quad y = \frac{\Delta_7}{c}$$

is 2-secant to two distinct ECT intervals. In particular, there is no global four-secant line for the normalized planar curve $\Gamma(h)$ on such a turning-boundary generic block.

Proof. Under the stated hypotheses, every adjacent boundary is of turning type and Theorem 12.79 applies to each adjacent pair. For such adjacent pairs, Lemma 12.99 applies. Hence adjacent fixed-slope intercept sets are strictly ordered. Iterating this ordering over the $x = v_5/c$ ordering of ECT intervals shows that no affine line can be 2-secant to two distinct ECT intervals. Therefore no global four-secant line exists on the block. $\square$

Corollary 12.101 (Partial exclusion of pair-pair branch configurations). *Under the hypotheses of Theorem 12.100, a bad finite core cannot contain two distinct ECT intervals each contributing a pair of points on a turning-boundary generic block.*

Proof. Each ECT interval contributes at most two points to the zero set of an affine line in the normalized chart, and by Theorem 12.100 no affine line can be 2-secant to two different ECT intervals on such a block. Therefore a pair-pair configuration across two distinct ECT intervals is impossible there. $\square$

Remark 12.102 (Strip picture and clean-block compression). For each ECT interval I in the normalized affine chart

$$x = \frac{v_5}{c}, \quad y = \frac{\Delta_7}{c},$$

it is convenient to package the fixed-slope 2-secant data by the open secant strip

$$\mathcal{S}_I := \{(m, b) : b \in \text{int } \mathcal{B}_I(m)\},$$

where $\mathcal{B}_I(m)$ is the intercept interval from Lemma 12.86. Thus

$$(m, b) \in \mathcal{S}_I$$

if and only if the affine line $y = mx + b$ is 2-secant to $\Gamma(I)$.

Under the hypotheses of Theorem 12.100, these strips are pairwise disjoint across distinct ECT intervals on the block:

$$\mathcal{S}_{I_i} \cap \mathcal{S}_{I_j} = \emptyset \quad (i \neq j).$$

Equivalently, a pair-pair state can use at most one ECT interval from such a turning-boundary generic block. In the later finite-core language, every such block behaves as a single transport chamber rather than as an independent source of $2 + 2$ combinatorics.

Remark 12.103 (Canonical quotient-level closure versus the honest order-7 lane). The direct $q^{(7)}$ -slice of total Δ_7 is now closed at the weight-one level by Proposition 12.48. Together with Theorem 12.49, this closes the canonical quotient-level order-7 package.

However, this does *not* mean that order 7 has disappeared from the later endgame. The honest transverse order-7 datum is not raw septic equality of representatives, but the quotient-septic class

$$[A_7^{\mathfrak{f}}] \in \mathfrak{f}/\mathbf{CA}_5^{\mathfrak{f}}.$$

Once two corrected germs agree through cubic and quintic order and differ only by odd reparametrization plus the already-quotiented lower scalar/value gauge, the remaining septic mismatch is automatically a multiple of $A_5^{\mathfrak{f}}$, so the quotient-septic defect vanishes. What is not yet proved is that cubic/quintic merger alone forces that odd-reparametrization comparability.

Accordingly, the honest order-7 theorem target is not “prove raw septic equality.” It is a provenance-sensitive package statement, for example:

- same reduced image germ at coincidence; or
- collision-functoriality of the corrected block triple on the collision/cancellation locus.

If such a statement holds, the quotient-septic collapse becomes formal afterward. So order 7 is canonically closed, but not yet globally exhausted.

Remark 12.104 (Canonical quotient-septic closure is still locally free). The canonical quotient-level order-7 package is now closed, but that does not yet make the quotient-septic geometry rigid.

Indeed, once the lower one-pair data (c, u_5, v_5) are fixed, the canonical septic residual

$$\Delta_7 = u_7 v_5 - u_5 v_7$$

is still locally free. On a region where $v_5 \neq 0$ one may gauge-fix

$$v_7^{\text{new}} = 0, \quad u_7^{\text{new}} = \frac{\Delta_7}{v_5},$$

while on a region where $u_5 \neq 0$ one may gauge-fix

$$u_7^{\text{new}} = 0, \quad v_7^{\text{new}} = -\frac{\Delta_7}{u_5}.$$

So after the lower one-pair data are fixed, the canonical quotient-septic curve

$$\Gamma(h) = \left(\frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)} \right)$$

is still locally determined by an arbitrary analytic choice of Δ_7 .

Accordingly, no theorem using only arbitrary canonical quotient-septic geometry of (c, v_5, Δ_7) can by itself close the later repeated-source / discriminant branch. Any successful higher-order hybrid must therefore be provenance-sensitive, not merely quotient-septic.

Remark 12.105 (Status and ceiling of the highest-new direct provenance slice). On the region $v_5 \neq 0$, set

$$\mu := \frac{c}{v_5}, \quad \lambda := \frac{\Delta_7}{v_5}, \quad \phi := \lambda^{[1]}.$$

By Corollary 12.43, Corollary 12.47, and Theorem 12.49, the highest-new quotient-level septic scalar is already reduced to the direct $q^{(7)}$ -driven K_1 -slice:

$$\Delta_7^{[1]} = v_5 \phi, \quad \phi = \kappa \left((x_2 - x_1) \nu_6^{(q^{(7)})} + v(\gamma_5 - \alpha_5)^{(q^{(7)})} \right).$$

On the subregion where $M \neq 0$, the direct highest-new dual variables are

$$\beta^{[1]} = \phi_\mu, \quad m^{[1]} = \phi - \mu \phi_\mu,$$

and by Corollary 12.63,

$$\phi_{\mu\mu} = \frac{v_5^3}{M^3} W^{[1]}.$$

Thus the direct highest-new quotient-septic geometry is exactly the ordinary tangent-line geometry of the scalar graph

$$y = \phi(\mu).$$

In particular, any bitangent line or four-secant line for the graph $y = \phi(\mu)$, and any nonadjacent slope re-entry for

$$\beta^{[1]} = \phi_\mu,$$

forces at least two zeros of

$$\phi_{\mu\mu},$$

equivalently of

$$W^{[1]}.$$

So the global direct-slice propagation problem may be viewed as a no-recurrence problem for the scalar graph ϕ .

However, the explicit direct provenance formula only shows that

$$\phi = A q^{(7)} \quad (A := \kappa \eta_2),$$

with A generically nonvanishing on a generic one-pair block. By Proposition 13.8 below, this still leaves ϕ fully universal on each such block. So the direct graph picture is now a precise dictionary, not by itself a live rigidity theorem.

Remark 12.106 (Status of the $W = 0$, $M \neq 0$ side). On the region $c \neq 0$, $M \neq 0$, Proposition 12.59 excludes open-interval flatness $W \equiv 0$. At an isolated boundary

$$W(h_b) = 0, \quad M(h_b) \neq 0,$$

the normalized planar curve in the chart

$$x = \frac{v_5}{c}, \quad y = \frac{\Delta_7}{c}$$

is an ordinary odd-order affine inflection whenever W changes sign, so adjacent tangent-separator sign regularity fails there in general.

However, Theorem 12.107 shows that every finite-order isolated boundary of type $W = 0$, $M \neq 0$ is locally sterile for pair-pair birth. Thus the remaining issue on this side is not local creation of a bad $2 + 2$ line, but global continuation / recurrence of determinant-zero or tangent-slope data across multiple intervals.

The exact identities for W and W' continue to show that this residual obstruction is still encoded through the canonical order-7 package Δ_7 , not by a genuinely new primitive order-9 invariant.

Theorem 12.107 (Finite-order $W = 0$ boundaries are locally sterile for pair-pair birth). *Let h_b satisfy*

$$c(h_b) \neq 0, \quad M(h_b) \neq 0, \quad W(h_b) = 0,$$

and assume that W has finite vanishing order $r \geq 1$ at h_b :

$$W(h) = (h - h_b)^r \widetilde{W}(h), \quad \widetilde{W}(h_b) \neq 0.$$

Define

$$F(h) := (c(h), v_5(h), \Delta_7(h)), \quad D(s, t, u) := \det(F(s), F(t), F(u)).$$

Then, after shrinking to a sufficiently small neighborhood U of h_b , the following holds.

If $s < t < h_b$ lie in U , then:

- (1) if r is even, the function $u \mapsto D(s, t, u)$ has no zero on the opposite side $h_b < u \in U$;
- (2) if r is odd, the function $u \mapsto D(s, t, u)$ has at most one zero on the opposite side $h_b < u \in U$.

The same conclusion holds with left and right exchanged. In particular, no finite-order boundary of type $W = 0$, $M \neq 0$ can support a local pair-pair birth.

Proof. Work in the affine chart

$$x(h) := \frac{v_5(h)}{c(h)}, \quad y(h) := \frac{\Delta_7(h)}{c(h)}.$$

Since

$$x'(h) = \frac{M(h)}{c(h)^2},$$

the hypothesis $M(h_b) \neq 0$ implies that x is a valid local coordinate near h_b . After translating so that $x(h_b) = 0$ and subtracting the tangent line, we may write the normalized curve as

$$y = f(x), \quad f(0) = f'(0) = 0.$$

By Proposition 12.57,

$$\frac{d^2 y}{dx^2} = \frac{c(h)^3 W(h)}{M(h)^3}.$$

Because $x'(h_b) \neq 0$, the finite vanishing order of W in h is the same as its finite vanishing order in the local coordinate x . Therefore

$$f''(x) = \lambda x^r + O(x^{r+1}) \quad (\lambda \neq 0).$$

Fix two same-side points

$$x_1 < x_2 < 0$$

sufficiently close to 0, and define

$$\widehat{D}(x_1, x_2, u) := \det \begin{pmatrix} 1 & x_1 & f(x_1) \\ 1 & x_2 & f(x_2) \\ 1 & u & f(u) \end{pmatrix}.$$

Then

$$\widehat{D}(x_1, x_2, u) = (x_2 - x_1)(u - x_1)(u - x_2) H_{x_1, x_2}(u),$$

where

$$H_{x_1, x_2}(u) := f[x_1, x_2, u]$$

is the second divided difference of f .

If r is even, then f'' has fixed sign near 0. By the Hermite–Genocchi formula for second divided differences, $H_{x_1, x_2}(u)$ has that same fixed sign for $u > 0$ sufficiently small. Hence $\widehat{D}(x_1, x_2, u) \neq 0$ on the opposite side.

If r is odd, then

$$f'''(x) = \lambda r x^{r-1} + O(x^r)$$

has fixed nonzero sign near 0, because $r - 1$ is even. Differentiating with respect to the repeated node gives

$$\partial_u H_{x_1, x_2}(u) = f[x_1, x_2, u, u].$$

By the Hermite–Genocchi formula for repeated-node divided differences, $f[x_1, x_2, u, u]$ has the same fixed nonzero sign for $u > 0$ sufficiently small. Thus $u \mapsto H_{x_1, x_2}(u)$ is strictly monotone on the opposite side, and therefore has at most one zero there.

Finally,

$$D(s, t, u) = c(s)c(t)c(u) \widehat{D}(x(s), x(t), x(u)),$$

and c is nonzero on a small neighborhood of h_b , so the zero set of D is exactly the zero set of $\widehat{D}$. The theorem follows. $\square$

Remark 12.108 (Consequence for the $W = 0$, $M \neq 0$ side). Theorem 12.107 shows that finite-order $W = 0$, $M \neq 0$ boundaries are not local birthplaces of pair-pair configurations. Combined with Lemma 12.109 and Corollary 12.110, this yields a sharper reduction of the one-pair propagation problem.

Namely, after removing local births at generic turning boundaries and at finite-order isolated $W = 0$, $M \neq 0$ boundaries, any surviving nonadjacent pair-pair obstruction must come from a *remote* interior zero of one of the endpoint-gap functions

$$G_{\xi,J}(h) = y_{\xi} - y(h) - m(h)(\xi - x(h)),$$

where (ξ, y_{ξ}) is an endpoint support point and J is an ordered target branch.

Since each $G_{\xi,J}$ is strictly monotone on J , each endpoint/branch pair can contribute at most one such tangency. Thus the remaining obstruction on the $W = 0$ side is no longer local birth at an affine inflection seam, but a finite remote-incidence problem.

Lemma 12.109 (Endpoint-gap monotonicity on an ordered target branch). *Let J be an ECT interval in the affine chart*

$$x = \frac{v_5}{c}, \quad y = \frac{\Delta_7}{c},$$

and let (ξ, y_{ξ}) be a fixed support point with

$$\xi < x(h) \quad (h \in J).$$

Define

$$G_{\xi,J}(h) := y_{\xi} - y(h) - m(h)(\xi - x(h)), \quad m(h) = \frac{N(h)}{M(h)}.$$

Then

$$G'_{\xi,J}(h) = \frac{c(h)W(h)}{M(h)^2} (x(h) - \xi).$$

In particular, $G_{\xi,J}$ is strictly monotone on J .

Equivalently, a fixed support point strictly to the left of J lies on at most one tangent line to $\Gamma(J)$. The symmetric statement holds for a fixed support point strictly to the right of J .

Proof. Using

$$y'(h) = m(h)x'(h),$$

differentiate:

$$G'_{\xi,J}(h) = -y'(h) - m'(h)(\xi - x(h)) + m(h)x'(h) = m'(h)(x(h) - \xi).$$

By Proposition 12.62,

$$m'(h) = \frac{c(h)W(h)}{M(h)^2},$$

so

$$G'_{\xi,J}(h) = \frac{c(h)W(h)}{M(h)^2} (x(h) - \xi).$$

On an ECT interval, c , W , and M have fixed nonzero sign, and by assumption $x(h) - \xi > 0$ on J , so $G'_{\xi,J}$ has fixed nonzero sign. Therefore $G_{\xi,J}$ is strictly monotone. $\square$

Corollary 12.110 (Turning-boundary inertness for endpoint contacts). *Let J be an ECT interval ending at a generic turning boundary h_b with*

$$M(h_b) = 0, \quad M'(h_b) \neq 0, \quad N(h_b) \neq 0, \quad c(h_b) \neq 0.$$

Assume $v_5(h_b) \neq 0$, and let

$$x_b := \frac{v_5(h_b)}{c(h_b)}.$$

If (ξ, y_{ξ}) is a fixed support point with $\xi < x_b$, then

$$G_{\xi,J}(h) \rightarrow \pm\infty \quad \text{as } h \rightarrow h_b \text{ from within } J.$$

In particular, endpoint-support contact cannot be created or destroyed at a generic turning-boundary end of an ordered branch.

Proof. By Proposition 12.62,

$$m(h) = \frac{N(h)}{M(h)} \rightarrow \pm\infty \quad (h \rightarrow h_b),$$

while

$$x(h) \rightarrow x_b, \quad y(h) \rightarrow y_b := \frac{\Delta_7(h_b)}{c(h_b)}.$$

Hence

$$G_{\xi,J}(h) = (y_\xi - y(h)) + m(h)(x(h) - \xi).$$

Since $x(h) - \xi \rightarrow x_b - \xi > 0$, the second term dominates and has fixed sign for h sufficiently close to h_b . Therefore

$$G_{\xi,J}(h) \rightarrow \pm\infty.$$

□

Remark 12.111 (Endpoint purification for bounded strip-overlap components). Let

$$\mathcal{P} := \bigcup_{i < j-1} (\mathcal{S}_{I_i} \cap \mathcal{S}_{I_j})$$

denote the nonadjacent pair-pair strip-overlap set on a compact patch, where the union runs over the ECT intervals in the normalized affine chart.

By Theorem 12.96 and Theorem 12.107, a bounded connected component

$$U \subset \mathcal{P}$$

cannot be created or destroyed at a generic turning boundary, and cannot have a first entrance or last exit at a finite-order boundary of type

$$W = 0, \quad M \neq 0.$$

Moreover, by Lemma 12.109, each fixed endpoint support point can meet a given ordered target branch at most once. Thus any surviving bounded strip-overlap component is controlled by a finite collection of isolated remote endpoint/branch tangencies, rather than by diffuse local birth along a seam.

So the remaining nonadjacent pair-pair problem is best viewed simply as a finite remote-incidence problem.

Remark 12.112 (Audit: residual $W = 0$ pathology). By Proposition 12.59, the $W = 0$, $M \neq 0$ side is no longer obstructed by open-interval flatness $W \equiv 0$ on a component with $c \neq 0$ and $M \neq 0$. By Theorem 12.107, every finite-order isolated $W = 0$ boundary is locally sterile for pair-pair birth. Thus any remaining $W = 0$ issue would have to come either from non-analytic / infinite-order flatness or from a genuinely global continuation phenomenon, not from a generic local inflection birth mechanism.

12.6. From the one-pair residual germ to two-pair and minimal-core rigidity.

Remark 12.113 (Strengthened organizing principle). The original version of this subsection was organized around the cubic leading line and the first residual trace witness. The later working analysis suggests a stronger one-pair package:

$$c(h), \quad A_5^f(h), \quad \Delta_7(h),$$

or equivalently the normalized affine curve

$$h \mapsto \left(\frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)} \right).$$

The intended two-pair and minimal-core rigidity statements should ultimately be reformulated in terms of this strengthened one-pair invariant geometry.

Remark 12.114 (Subsection guide). This subsection is best read as a rigidity program built on the strengthened one-pair local package

$$c(h), \quad A_5^f(h), \quad \Delta_7(h),$$

or equivalently the normalized planar curve

$$h \mapsto \left(\frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)} \right).$$

The projective coincidence conditions remain the organizing principle for the two-pair and minimal-core analysis. The finite-order discussion below now isolates the exact source-level criterion that would force the two-point quadratic interaction factorization through order 7, but the current attack also exhibits two backup lanes: a weaker quotient-diagonal exact exclusion route and, on good overlap patches, a mixed λ -shear / projective-shadow route. Beyond that, the remaining steps are the verification of one of these two-point lanes, the genuine multi-pair interaction bookkeeping, and the extraction into the finite-core endgame.

Remark 12.115 (Interpretation of residual notation in the multi-pair architecture). In the remainder of this section, the symbols

$$F_h(z), \quad \ell_h, \quad \bar{v}_1(h), \quad \mathcal{Q}^{\text{abs}}/\ell_h$$

are used schematically to describe the intended cubic-leading / first-residual architecture of multi-pair and minimal-core configurations.

The fully explicit rigorous first-residual object established earlier is

$$\mathcal{R}_h^{(5)} = \mathcal{J}_5 / (\mathbf{C} A_3(h) + V_{\text{amb}}^{(5)}(h)).$$

Accordingly, whenever a later statement invokes the first residual class in a theorem-strength way, it should be interpreted through $\mathcal{R}_h^{(5)}$. The present two-pair subsection now isolates the exact finite-order source criterion through order 7 together with two weaker backup routes. What remains architectural is the verification of one of these two-point routes for the actual corrected two-atom package, together with its extension to genuine multi-pair and minimal-bad-core interaction terms.

Let $F_h(z)$ denote the corrected quotient odd germ of the normalized one-pair family. By Proposition 12.8, on the cubic-generic region we have

$$F_h(z) = v_0(h)z^3 + v_1(h)z^5 + \cdots, \quad v_0(h) \neq 0,$$

with cubic leading line

$$\ell_h := \mathbf{C} v_0(h) \subset \mathcal{Q}^{\text{abs}}.$$

Under the additional hypotheses of Proposition 12.28, there is an open dense subset $\mathcal{U}_{\text{res}} \subset \mathcal{U}_{\text{cub}}$ on which the explicit first residual class

$$\bar{v}_1(h) \in \mathcal{R}_h^{(5)}$$

is nonzero.

In the strengthened fixed-sector refinement, the one-pair local package is recorded by

$$c(h), \quad A_5^f(h) = u_5(h)I + v_5(h)S, \quad \Delta_7(h),$$

and hence by the normalized planar curve

$$\Gamma(h) := \left(\frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)} \right) = (\rho_1(h), \rho_2(h)).$$

By Proposition 12.54, the affine curvature numerator of Γ is

$$K_\Gamma(h) = \rho_1'(h)\rho_2''(h) - \rho_2'(h)\rho_1''(h) = \frac{M(h)N'(h) - N(h)M'(h)}{c(h)^4},$$

where

$$M(h) = c(h)v_5'(h) - c'(h)v_5(h), \quad N(h) = c(h)\Delta_7'(h) - c'(h)\Delta_7(h).$$

Thus the strengthened one-pair geometry is governed by the curve $\Gamma(h)$ and its secant/tangent behavior on the region where

$$c(h) \neq 0, \quad M(h) \neq 0, \quad M(h)N'(h) - N(h)M'(h) \neq 0.$$

Definition 12.116 (Amplitude-invariant strengthened datum). On the region where

$$c(h) \neq 0,$$

define

$$\widehat{\Psi}(h) := \left(\frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)^2} \right).$$

Lemma 12.117 (Scaling law for the strengthened package). *If the normalized one-pair odd germ is multiplied by a scalar λ , then*

$$c \mapsto \lambda c, \quad A_5^f \mapsto \lambda A_5^f, \quad A_7^f \mapsto \lambda A_7^f,$$

and hence

$$\Delta_7 \mapsto \lambda^2 \Delta_7.$$

Consequently the datum

$$\widehat{\Psi}(h) = \left(\frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)^2} \right)$$

is invariant under scalar rescaling of the full one-pair germ.

Proof. The cubic, quintic, and septic coefficients of the odd germ all scale linearly under multiplication of the germ by λ . Since

$$A_5^f = u_5 I + v_5 S, \quad A_7^f = u_7 I + v_7 S,$$

it follows that

$$u_5 \mapsto \lambda u_5, \quad v_5 \mapsto \lambda v_5, \quad u_7 \mapsto \lambda u_7, \quad v_7 \mapsto \lambda v_7.$$

Therefore

$$\Delta_7 = u_7 v_5 - u_5 v_7$$

scales as

$$\Delta_7 \mapsto \lambda^2 \Delta_7,$$

which gives the claim. □

Remark 12.118 (Two normalizations: geometric versus projective). The affine curve

$$\Gamma(h) = \left(\frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)} \right)$$

is the correct object for the one-pair differential geometry, because its derivatives are governed by the projective Wronskians M , N , and W .

For the multi-pair coincidence problem, however, the natural scalar-invariant datum is $\widehat{\Psi}(h)$, not $\Gamma(h)$. The reason is exactly the scaling law in Lemma 12.117: under scalar rescaling of the full one-pair germ, Δ_7/c is not invariant, whereas Δ_7/c^2 is.

In this formulation, the geometric secant/tangent obstruction is well controlled on ECT intervals and across generic turning-point boundaries. The remaining global issue is no longer a purely local $W = 0$, $M \neq 0$ obstruction, but the global propagation / recurrence problem for the determinant-zero and tangent configurations that survive the local sterility results proved later in the paper.

Definition 12.119 (Two-pair decomposition). For a normalized two-pair core with amplitudes a_1, a_2 and offsets h_1, h_2 , write its quotient odd germ schematically as

$$F_{(a_1, h_1; a_2, h_2)}(z) = a_1 F_{h_1}(z) + a_2 F_{h_2}(z) + \mathcal{I}_{12}(z),$$

where $\mathcal{I}_{12}(z)$ denotes the pair-interaction germ.

In the explicit finite-order corrected source-level model through order 7 constructed below, the cubic scalar, quintic fixed-sector, and septic fixed-sector projections of $\mathcal{I}_{12}$ are quadratic in the amplitudes and divisible by $(h_1 - h_2)^2$.

Proposition 12.120 (Exact strengthened two-pair coincidence through order 7). *Let $(a_1, h_1; a_2, h_2)$ be a normalized two-pair core with*

$$F_{(a_1, h_1; a_2, h_2)}(z) \equiv 0.$$

Assume moreover that, after passage to the explicit corrected finite-order model through order 7, the interaction contribution vanishes at the three relevant levels, namely:

$$a_1 c(h_1) + a_2 c(h_2) = 0,$$

$$a_1 A_5^f(h_1) + a_2 A_5^f(h_2) = 0,$$

and, after the second identity identifies the common quotient line

$$\mathbf{C}A_5^f(h_1) = \mathbf{C}A_5^f(h_2),$$

the second residual classes satisfy

$$a_1 [A_7^f](h_1) + a_2 [A_7^f](h_2) = 0 \quad \text{in} \quad \mathfrak{f}/\mathbf{C}A_5^f(h_1).$$

Then

$$\widehat{\Psi}(h_1) = \widehat{\Psi}(h_2).$$

Equivalently,

$$\frac{u_5(h_1)}{c(h_1)} = \frac{u_5(h_2)}{c(h_2)}, \quad \frac{v_5(h_1)}{c(h_1)} = \frac{v_5(h_2)}{c(h_2)}, \quad \frac{\Delta_7(h_1)}{c(h_1)^2} = \frac{\Delta_7(h_2)}{c(h_2)^2}.$$

Proof. Set

$$\lambda := -\frac{a_2}{a_1}.$$

The cubic identity gives

$$c(h_1) = \lambda c(h_2).$$

The quintic fixed-sector identity gives

$$A_5^f(h_1) = \lambda A_5^f(h_2).$$

Writing

$$A_5^f(h_i) = u_5(h_i)I + v_5(h_i)S,$$

we obtain

$$u_5(h_1) = \lambda u_5(h_2), \quad v_5(h_1) = \lambda v_5(h_2).$$

Hence

$$\frac{u_5(h_1)}{c(h_1)} = \frac{u_5(h_2)}{c(h_2)}, \quad \frac{v_5(h_1)}{c(h_1)} = \frac{v_5(h_2)}{c(h_2)}.$$

Now the second identity implies

$$\mathbf{C}A_5^f(h_1) = \mathbf{C}A_5^f(h_2),$$

so the residual quotient spaces are canonically identified. The order-7 residual identity therefore says

$$[A_7^f](h_1) = \lambda [A_7^f](h_2)$$

in the common quotient. Equivalently, there exists $\mu \in \mathbf{C}$ such that

$$A_7^f(h_1) = \lambda A_7^f(h_2) + \mu A_5^f(h_2).$$

Writing

$$A_7^f(h_i) = u_7(h_i)I + v_7(h_i)S,$$

and using

$$\Delta_7(h) = u_7(h)v_5(h) - u_5(h)v_7(h),$$

we obtain

$$\Delta_7(h_1) = \det(A_7^f(h_1), A_5^f(h_1)) = \det(\lambda A_7^f(h_2) + \mu A_5^f(h_2), \lambda A_5^f(h_2)) = \lambda^2 \Delta_7(h_2).$$

Since also $c(h_1) = \lambda c(h_2)$, it follows that

$$\frac{\Delta_7(h_1)}{c(h_1)^2} = \frac{\Delta_7(h_2)}{c(h_2)^2}.$$

Thus $\widehat{\Psi}(h_1) = \widehat{\Psi}(h_2)$. □

Theorem 12.121 (Quantitative defect-thickened strengthened coincidence). *Let $(a_1, h_1; a_2, h_2)$ be a normalized bad two-pair core, and write*

$$c_i := c(h_i), \quad A_{5,i}^{\mathfrak{f}} := A_5^{\mathfrak{f}}(h_i) = u_{5,i}I + v_{5,i}S, \quad \Delta_i := \Delta_7(h_i).$$

Assume

$$a_1 \neq 0, \quad c_1 c_2 \neq 0.$$

Suppose the cubic, quintic, and septic quotient defects are written as

$$a_1 c_1 + a_2 c_2 + E_{12}^{(3)} = 0,$$

$$a_1 A_{5,1}^{\mathfrak{f}} + a_2 A_{5,2}^{\mathfrak{f}} + E_{12}^{(5)} = 0,$$

and

$$a_1 [A_{7,1}^{\mathfrak{f}}] + a_2 [A_{7,2}^{\mathfrak{f}}] + \overline{E}_{12}^{(7;1)} = 0 \quad \text{in} \quad \mathfrak{f}/\mathbf{C}A_{5,1}^{\mathfrak{f}}.$$

Set

$$\lambda := -\frac{a_2}{a_1}, \quad e := \frac{E_{12}^{(3)}}{a_1}, \quad E := \frac{E_{12}^{(5)}}{a_1} = e_u I + e_v S.$$

Choose any representative $R \in \mathfrak{f}$ of the quotient class

$$-\frac{1}{a_1} \overline{E}_{12}^{(7;1)},$$

so that

$$A_{7,1}^{\mathfrak{f}} = \lambda A_{7,2}^{\mathfrak{f}} + R + \eta A_{5,1}^{\mathfrak{f}}$$

for some $\eta \in \mathbf{C}$. Then the strengthened invariant

$$\widehat{\Psi}(h) = \left(\frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)^2} \right)$$

satisfies the exact identities

$$\begin{aligned} \frac{u_{5,1}}{c_1} - \frac{u_{5,2}}{c_2} &= \frac{u_{5,2}e - c_2 e_u}{c_1 c_2}, \\ \frac{v_{5,1}}{c_1} - \frac{v_{5,2}}{c_2} &= \frac{v_{5,2}e - c_2 e_v}{c_1 c_2}, \end{aligned}$$

and

$$\frac{\Delta_1}{c_1^2} - \frac{\Delta_2}{c_2^2} = \frac{-\lambda c_2^2 \det(A_{7,2}^{\mathfrak{f}}, E) + c_2^2 \det(R, A_{5,1}^{\mathfrak{f}}) + \Delta_2(2\lambda c_2 e - e^2)}{c_1^2 c_2^2}.$$

Proof. The cubic defect identity gives

$$c_1 = \lambda c_2 - e.$$

The quintic defect identity gives

$$A_{5,1}^{\mathfrak{f}} = \lambda A_{5,2}^{\mathfrak{f}} - E,$$

hence

$$u_{5,1} = \lambda u_{5,2} - e_u, \quad v_{5,1} = \lambda v_{5,2} - e_v.$$

Therefore

$$\frac{u_{5,1}}{c_1} - \frac{u_{5,2}}{c_2} = \frac{u_{5,1}c_2 - u_{5,2}c_1}{c_1 c_2} = \frac{(\lambda u_{5,2} - e_u)c_2 - u_{5,2}(\lambda c_2 - e)}{c_1 c_2} = \frac{u_{5,2}e - c_2 e_u}{c_1 c_2},$$

and similarly

$$\frac{v_{5,1}}{c_1} - \frac{v_{5,2}}{c_2} = \frac{v_{5,2}e - c_2 e_v}{c_1 c_2}.$$

For the third coordinate, by the choice of R we have

$$A_{7,1}^f = \lambda A_{7,2}^f + R + \eta A_{5,1}^f$$

for some $\eta \in \mathbf{C}$. Using

$$\Delta_i = \det(A_{7,i}^f, A_{5,i}^f),$$

we obtain

$$\Delta_1 = \det(A_{7,1}^f, A_{5,1}^f) = \lambda \det(A_{7,2}^f, A_{5,1}^f) + \det(R, A_{5,1}^f),$$

since the $\eta A_{5,1}^f$ term drops out. Substituting

$$A_{5,1}^f = \lambda A_{5,2}^f - E$$

gives

$$\det(A_{7,2}^f, A_{5,1}^f) = \lambda \Delta_2 - \det(A_{7,2}^f, E),$$

hence

$$\Delta_1 - \lambda^2 \Delta_2 = -\lambda \det(A_{7,2}^f, E) + \det(R, A_{5,1}^f).$$

Also,

$$c_1 = \lambda c_2 - e$$

implies

$$\lambda^2 c_2^2 - c_1^2 = 2\lambda c_2 e - e^2.$$

Therefore

$$\frac{\Delta_1}{c_1^2} - \frac{\Delta_2}{c_2^2} = \frac{c_2^2 \Delta_1 - c_1^2 \Delta_2}{c_1^2 c_2^2} = \frac{c_2^2 (\Delta_1 - \lambda^2 \Delta_2) + \Delta_2 (\lambda^2 c_2^2 - c_1^2)}{c_1^2 c_2^2},$$

which is exactly the claimed formula.

Finally, the expression

$$\det(R, A_{5,1}^f)$$

is independent of the chosen representative of the quotient defect, since replacing R by $R + \xi A_{5,1}^f$ does not change the determinant. $\square$

Corollary 12.122 (Bad two-pair cores are defect-scale close on good $M \neq 0$ patches). *Let $J \subseteq \{c \neq 0, M \neq 0\}$ be a compact interval. Assume*

$$|a_1| \geq a_* > 0$$

for the normalized bad two-pair core under consideration. Then there exists a constant $C_J > 0$ such that whenever $h_1, h_2 \in J$,

$$|h_1 - h_2| \leq C_J \left(|E_{12}^{(3)}| + \|E_{12}^{(5)}\| \right).$$

In particular, on good $M \neq 0$ patches, bad two-pair cores force the two parameters to be quantitatively close once the cubic/quintic interaction defects are small.

Proof. The second coordinate of $\widehat{\Psi}$ is

$$x(h) := \frac{v_5(h)}{c(h)}, \quad x'(h) = \frac{M(h)}{c(h)^2}.$$

Since $J \subseteq \{c \neq 0, M \neq 0\}$, there exists $\eta_J > 0$ such that

$$|x(h_1) - x(h_2)| \geq \eta_J |h_1 - h_2| \quad (h_1, h_2 \in J).$$

On the other hand, Theorem 12.121 gives the exact identity

$$x(h_1) - x(h_2) = \frac{v_{5,2}e - c_2 e_v}{c_1 c_2},$$

with

$$e = \frac{E_{12}^{(3)}}{a_1}, \quad e_v = S\text{-coefficient of } \frac{E_{12}^{(5)}}{a_1}.$$

Since J is compact and a_1 is bounded away from 0, the right-hand side is bounded by

$$C'_J \left(|E_{12}^{(3)}| + \|E_{12}^{(5)}\| \right)$$

for some constant C'_J . Combining the two estimates yields the claim. $\square$

Proposition 12.123 (No sublinear two-pair defects on good patches). *Let*

$$J \subseteq \{c \neq 0, M \neq 0\}$$

be a compact interval. Then there exists a constant $C_J > 0$ such that every normalized bad two-pair core with $h_1, h_2 \in J$ and $h_1 \neq h_2$ satisfies

$$|E_{12}^{(3)}| + \|E_{12}^{(5)}\| + |\overline{E}_{12}^{(7;1)}| \geq C_J^{-1} |h_1 - h_2|.$$

In particular, there is no sequence of nontrivial normalized bad two-pair cores in J for which

$$|E_{12}^{(3)}| + \|E_{12}^{(5)}\| + |\overline{E}_{12}^{(7;1)}| = o(|h_1 - h_2|).$$

Proof. By Corollary 12.122, there exists a constant $C'_J > 0$ such that

$$|h_1 - h_2| \leq C'_J \left(|E_{12}^{(3)}| + \|E_{12}^{(5)}\| \right)$$

for every normalized bad two-pair core with $h_1, h_2 \in J$. Since

$$|E_{12}^{(3)}| + \|E_{12}^{(5)}\| \leq |E_{12}^{(3)}| + \|E_{12}^{(5)}\| + |\overline{E}_{12}^{(7;1)}|,$$

the same estimate yields

$$|h_1 - h_2| \leq C'_J \left(|E_{12}^{(3)}| + \|E_{12}^{(5)}\| + |\overline{E}_{12}^{(7;1)}| \right).$$

Renaming C'_J as C_J proves the first claim. The final statement is immediate: if a sequence satisfied

$$|E_{12}^{(3)}| + \|E_{12}^{(5)}\| + |\overline{E}_{12}^{(7;1)}| = o(|h_1 - h_2|),$$

then the displayed lower bound would force

$$|h_1 - h_2| = o(|h_1 - h_2|),$$

which is impossible for a nontrivial two-point core with $h_1 \neq h_2$. $\square$

Lemma 12.124 (Minimal finite-order source criterion for quadratic two-point factorization). *Let V be a finite-dimensional complex vector space, and let*

$$\mathfrak{P}_2(a_1, h_1; a_2, h_2) \in V$$

be analytic on a neighborhood of the small two-atom data. Assume that there is a one-atom package

$$F_h \in V$$

such that the following three identities hold:

$$\mathfrak{P}_2(a_1, h_1; a_2, h_2) = \mathfrak{P}_2(a_2, h_2; a_1, h_1),$$

$$\mathfrak{P}_2(0, h_1; a_2, h_2) = a_2 F_{h_2}, \quad \mathfrak{P}_2(a_1, h_1; 0, h_2) = a_1 F_{h_1},$$

and

$$\mathfrak{P}_2(a_1, h; a_2, h) = (a_1 + a_2) F_h.$$

Define the interaction remainder

$$\mathcal{I}_2(a_1, h_1; a_2, h_2) := \mathfrak{P}_2(a_1, h_1; a_2, h_2) - a_1 F_{h_1} - a_2 F_{h_2}.$$

Then there exists an analytic map

$$\mathcal{J}_2 : (\text{small two-atom data}) \rightarrow V$$

such that

$$\boxed{\mathcal{I}_2(a_1, h_1; a_2, h_2) = a_1 a_2 (h_1 - h_2)^2 \mathcal{J}_2(a_1, h_1; a_2, h_2).}$$

Proof. By the one-amplitude collapse identities,

$$\mathcal{I}_2(0, h_1; a_2, h_2) = 0, \quad \mathcal{I}_2(a_1, h_1; 0, h_2) = 0.$$

Hence analyticity implies divisibility by $a_1 a_2$:

$$\mathcal{I}_2(a_1, h_1; a_2, h_2) = a_1 a_2 G(h_1, h_2)$$

for an analytic

$$G : (h_1, h_2) \mapsto V.$$

Swap symmetry gives

$$G(h_1, h_2) = G(h_2, h_1).$$

On the diagonal,

$$\mathcal{I}_2(a_1, h; a_2, h) = \mathfrak{P}_2(a_1, h; a_2, h) - a_1 F_h - a_2 F_h = (a_1 + a_2) F_h - a_1 F_h - a_2 F_h = 0,$$

so

$$G(h, h) = 0.$$

Now set

$$m = \frac{h_1 + h_2}{2}, \quad s = h_2 - h_1, \quad \tilde{G}(m, s) := G\left(m - \frac{s}{2}, m + \frac{s}{2}\right).$$

Since G is symmetric, $\tilde{G}(m, s)$ is even in s , and since $G(h, h) = 0$, one has

$$\tilde{G}(m, 0) = 0.$$

Therefore $\tilde{G}(m, s)$ is divisible by s^2 :

$$\tilde{G}(m, s) = s^2 H(m, s)$$

for analytic H . Returning to (h_1, h_2) gives

$$\mathcal{I}_2(a_1, h_1; a_2, h_2) = a_1 a_2 (h_1 - h_2)^2 \mathcal{J}_2(a_1, h_1; a_2, h_2),$$

as claimed. $\square$

Theorem 12.125 (Actual zeta-source two-point bridge). *Assume the corrected Hardy-source package audit, Definition E.138. Then the actual zeta source satisfies the finite-order source criterion of Lemma 12.124. Consequently the two-point interaction remainder factors as*

$$\mathcal{I}_2^\zeta = a_1 a_2 (h_1 - h_2)^2 \mathcal{J}_2^\zeta + O(Q^{-B}).$$

In particular, on compact good patches and after the K9/Y12 carrier routing, the exact close two-point bad core is excluded up to the declared residual scale.

Proof. One-amplitude collapse gives divisibility by $a_1 a_2$. Swap symmetry makes the remaining two-point defect even under $h_1 \leftrightarrow h_2$. Diagonal merger makes the defect vanish when $h_1 = h_2$. Hence the defect is divisible by $(h_1 - h_2)^2$, giving the displayed factorization. The residual term is the residual allowed in the corrected package audit. The carrier-visible source lower bound from Theorem E.142 prevents the factorized source contribution from being vacuous in the K9/Y12 branch. $\square$

Corollary 12.126 (Conditional quadratic defect estimate and two-point exclusion). *Assume that the actual corrected two-pair finite-order package through order 7 admits a source-level realization*

$$\mathfrak{P}_2(a_1, h_1; a_2, h_2) \in \mathbf{C} \oplus \mathfrak{f} \oplus (\mathfrak{f}/\mathbf{C}A_5^\mathfrak{f})$$

satisfying the three hypotheses of Lemma 12.124, with

$$F_h = (c(h), A_5^\mathfrak{f}(h), [A_7^\mathfrak{f}](h)).$$

Then on compacta,

$$\mathcal{E}^{(3)} = a_1 a_2 O(\delta^2), \quad \mathcal{E}^{(5)} = a_1 a_2 O(\delta^2), \quad \overline{\mathcal{E}}^{(7)} = a_1 a_2 O(\delta^2), \quad \delta := |h_1 - h_2|.$$

If the normalized defects are obtained by dividing by a_1 , then on normalized families with $|a_2|$ bounded one has

$$E_{12}^{(3)} = O(\delta^2), \quad E_{12}^{(5)} = O(\delta^2), \quad \overline{E}_{12}^{(7;1)} = O(\delta^2).$$

Consequently, the two-point bad-core branch is excluded on compact good patches

$$J \subseteq \{c \neq 0, M \neq 0\}.$$

Proof. Apply Lemma 12.124 to the cubic, quintic fixed-sector, and septic quotient outputs, then argue exactly as in the present proof using Theorem 12.121. $\square$

Proposition 12.127 (Conditional exact two-point exclusion from quotient diagonal collapse). *Let*

$$J \subseteq \{c \neq 0\}$$

be a compact interval. Assume there exists an actual corrected two-atom quotient map

$$\mathcal{F}_2(a_1, h_1; a_2, h_2) \in \mathcal{Q}^{\text{abs}}[[z^2]]$$

defined on normalized two-pair data, with the following properties:

- (1) **Continuity.** *The map $\mathcal{F}_2$ is continuous in the normalized two-pair parameters.*
- (2) **Diagonal collapse.** *For every $h \in J$ and every normalized amplitude pair (a_1, a_2) ,*

$$\mathcal{F}_2(a_1, h; a_2, h) = F_h,$$

where F_h is the normalized one-pair quotient odd germ.

- (3) **Precompactness of normalized amplitudes.** *Every sequence of normalized two-pair data with*

$$h_{1,n}, h_{2,n} \in J, \quad |h_{1,n} - h_{2,n}| \rightarrow 0$$

admits a subsequence along which the full normalized parameter tuple converges.

Then there is no sequence of nontrivial normalized bad two-pair cores

$$(a_{1,n}, h_{1,n}; a_{2,n}, h_{2,n})$$

with

$$h_{1,n}, h_{2,n} \in J, \quad 0 < |h_{1,n} - h_{2,n}| \rightarrow 0.$$

Equivalently, under the three assumptions above, the exact close two-point bad-core branch is excluded on compact subsets of $\{c \neq 0\}$.

Proof. Assume for contradiction that there exists a sequence of nontrivial normalized bad two-pair cores

$$(a_{1,n}, h_{1,n}; a_{2,n}, h_{2,n})$$

with

$$h_{1,n}, h_{2,n} \in J, \quad 0 < |h_{1,n} - h_{2,n}| \rightarrow 0.$$

Since each core is bad, its quotient odd germ vanishes identically, so

$$\mathcal{F}_2(a_{1,n}, h_{1,n}; a_{2,n}, h_{2,n}) = 0 \quad \text{for all } n.$$

By the precompactness assumption, after passing to a subsequence we may assume that the full normalized parameter tuple converges:

$$a_{1,n} \rightarrow a_1^*, \quad a_{2,n} \rightarrow a_2^*, \quad h_{1,n} \rightarrow m, \quad h_{2,n} \rightarrow m$$

for some $m \in J$. By continuity of $\mathcal{F}_2$,

$$0 = \lim_{n \rightarrow \infty} \mathcal{F}_2(a_{1,n}, h_{1,n}; a_{2,n}, h_{2,n}) = \mathcal{F}_2(a_1^*, m; a_2^*, m).$$

Applying diagonal collapse gives

$$0 = \mathcal{F}_2(a_1^*, m; a_2^*, m) = F_m.$$

But $m \in J \subseteq \{c \neq 0\}$, so the normalized one-pair quotient germ F_m is nonzero: its cubic leading coefficient is nonzero on $\{c \neq 0\}$. This contradiction proves the claim. $\square$

Remark 12.128 (A weaker exact two-point route). Proposition 12.127 is strictly weaker than the quadratic defect-factorization route of Lemma 12.124. It does not produce

$$\mathcal{I}_2 = a_1 a_2 (h_1 - h_2)^2 \mathcal{J}_2$$

and does not yield the quantitative defect estimates used in Theorem 12.121. Instead, it isolates a smaller alternative input: a genuine corrected two-atom quotient map, continuity in normalized parameters, and exact diagonal collapse to the one-pair quotient class.

Thus there are now two distinct two-point routes in the draft:

- (1) the stronger finite-order source-level route, which would force quadratic defect bounds and hence quantitative defect-scale closeness;
- (2) the weaker quotient-diagonal route, which would already exclude exact close two-point bad cores on compact subsets of $\{c \neq 0\}$.

At present, the draft does not prove either route outright. The first remains a finite-order source-level merger problem; the second remains a continuity and diagonal-collapse problem for the actual corrected two-atom quotient germ.

Remark 12.129 (Exact remaining source-level input on the two-point branch). Lemma 12.124 isolates the exact finite-order source identities needed for the two-point branch. It is enough to verify, for the actual corrected two-pair cubic/quintic/septic package, only:

- (1) swap symmetry;
- (2) one-amplitude collapse;
- (3) diagonal merger

$$\mathfrak{P}_2(a_1, h; a_2, h) = (a_1 + a_2) F_h.$$

Once these identities are known, the quadratic factorization

$$\mathcal{I}_2 = a_1 a_2 (h_1 - h_2)^2 \mathcal{J}_2$$

is formal. So the remaining issue on the two-point branch is not another asymptotic estimate, but verification of this finite-order source-level merger statement for the corrected two-atom package.

Remark 12.130 (Audit: even-amplitude obstruction for the naive source-summed model). The most obvious attempt to prove the source-level merger theorem is to insert a weighted one-atom datum

$$q_{a,h} = a q_h, \quad \Phi_{a,h} = a \Phi_h$$

directly into the corrected same-point and mixed block formulas, whiten, and then extract the cubic/quintic/septic outputs. This route cannot by itself prove the required one-atom linearity

$$\mathfrak{P}_2(a \delta_h) = a F_h.$$

Indeed the explicit corrected block formulas have the form

$$G_{a,h,\pm}^{\text{corr}}(s) = a \mathcal{G}_{h,\pm}(a^2; s), \quad N_{a,h}^{\text{corr}}(s) = a \mathcal{N}_h(a^2; s),$$

so after whitening the overall factor of a cancels:

$$\widehat{\Omega}_{a,h}^{\text{corr}}(s) = \mathcal{G}_{h,-}(a^2; s)^{-1/2} \mathcal{N}_h(a^2; s) \mathcal{G}_{h,+}(a^2; s)^{-1/2}.$$

Hence every fixed finite-order coefficient extracted from the whitened block is an even analytic function of the source weight a . In particular, the naive source-summed corrected block package cannot satisfy analytic one-atom linearity or coincident-atom additivity unless it vanishes identically. So the remaining two-point theorem must be proved using a different lift/transport object, not the raw source-summed whitened coefficients themselves.

Remark 12.131 (Audit: quotient and single-determinant routes are shear-blind). On the mixed two-point branch, any route based only on cubic/quintic source minors, the septic quotient class, or the pointwise patch gauges $v_7 = 0$ and $u_7 = 0$ is exactly blind to the repeated-node shear parameter.

For an exact bad two-pair core, the cubic/quintic defect vector has the form

$$d = \Lambda s_2 - s_1, \quad \Lambda := -\frac{a_2}{a_1}, \quad s_i = (c_i, u_{5,i}, v_{5,i}),$$

so

$$s_2 \wedge d = s_1 \wedge s_2$$

is independent of Λ . Likewise, any pointwise representative extracted only from the septic quotient class and $A_{5,1}^f$ carries no more information than the same determinant scalar already seen by the quotient. Thus the remaining mixed two-point burden is genuinely source-coupled rather than quotient-determinantal.

Proposition 12.132 (Projective compactness of the moving λ -shear defects). *Assume a close exact mixed two-point bad branch on a compact good overlap patch*

$$U \subseteq \{c \neq 0, v_5 \neq 0, M \neq 0\},$$

and write

$$\Lambda_n := -\frac{a_{2,n}}{a_{1,n}}, \quad \delta_n f := f(h_{1,n}) - \Lambda_n f(h_{2,n}), \quad \lambda = \frac{\Delta_7}{v_5}.$$

Then the mixed branch equations are equivalent to the existence of branch-generated moving shears $(\eta_n, \xi_n, \gamma_n)$ such that

$$\delta_n(c) = \eta_n \delta_n(\lambda), \quad \delta_n(v_5) = \xi_n \delta_n(\lambda), \quad \delta_n(u_5) = \gamma_n \delta_n(\lambda).$$

Set

$$d_n := (\delta_n(c), \delta_n(v_5), \delta_n(u_5), \delta_n(\lambda)) \in \mathbf{R}^4.$$

Then, after passing to a subsequence, exactly one of the following holds.

- (1) $d_n = 0$ for all n in the subsequence. Equivalently, along that subsequence one already has exact repeated-source transport simultaneously for

$$c, \quad v_5, \quad u_5, \quad \lambda.$$

- (2) $d_n \neq 0$ for all n in the subsequence, and the normalized defect directions

$$\hat{d}_n := \frac{d_n}{\|d_n\|} \in S^3$$

converge. Equivalently, the projective classes

$$[d_n] = [\delta_n(c) : \delta_n(v_5) : \delta_n(u_5) : \delta_n(\lambda)] \in \mathbf{P}^3$$

admit a convergent subsequence.

Moreover, on any subsequence for which $\delta_n(\lambda) \neq 0$ and the projective limit lies in the affine chart with nonzero λ -coordinate, the affine λ -shears converge:

$$\eta_n = \frac{\delta_n(c)}{\delta_n(\lambda)} \rightarrow \eta_*, \quad \xi_n = \frac{\delta_n(v_5)}{\delta_n(\lambda)} \rightarrow \xi_*, \quad \gamma_n = \frac{\delta_n(u_5)}{\delta_n(\lambda)} \rightarrow \gamma_*.$$

For the limiting affine shear one has only the asymptotic fixed-shear identities

$$\delta_n(c - \eta_* \lambda) = (\eta_n - \eta_*) \delta_n(\lambda),$$

$$\delta_n(v_5 - \xi_* \lambda) = (\xi_n - \xi_*) \delta_n(\lambda),$$

$$\delta_n(u_5 - \gamma_* \lambda) = (\gamma_n - \gamma_*) \delta_n(\lambda).$$

Hence a close exact mixed branch automatically reduces only to a projectively frozen secant-shadow problem. In general it does not reduce to exact repeated-source transport for a single fixed triple

$$(\eta_*, \xi_*, \gamma_*).$$

Proof. The identities

$$\delta_n(c) = \eta_n \delta_n(\lambda), \quad \delta_n(v_5) = \xi_n \delta_n(\lambda), \quad \delta_n(u_5) = \gamma_n \delta_n(\lambda)$$

are just a rewriting of the branch equations. If $d_n = 0$ infinitely often, pass to that subsequence. Otherwise $d_n \neq 0$ for all large n , and compactness of the unit sphere gives a convergent subsequence of $\hat{d}_n$, equivalently of the associated projective classes in $\mathbf{P}^3$.

On the affine chart $\delta_n(\lambda) \neq 0$, the projective coordinates are exactly

$$\eta_n = \frac{\delta_n(c)}{\delta_n(\lambda)}, \quad \xi_n = \frac{\delta_n(v_5)}{\delta_n(\lambda)}, \quad \gamma_n = \frac{\delta_n(u_5)}{\delta_n(\lambda)},$$

so convergence in that chart is precisely convergence of the affine λ -shears. The last three identities are immediate from the definitions. $\square$

Remark 12.133 (Affine freezing is stronger than projective freezing on the mixed two-point branch). Proposition 12.132 shows that the automatic compactness on the mixed two-point branch is projective, not affine. Even on a good overlap patch, the affine λ -shears can diverge if $\delta_n(\lambda)$ vanishes to higher order than $\delta_n(c), \delta_n(v_5), \delta_n(u_5)$. And even when

$$(\eta_n, \xi_n, \gamma_n) \rightarrow (\eta_*, \xi_*, \gamma_*),$$

one gets only asymptotic transport for

$$c - \eta_* \lambda, \quad v_5 - \xi_* \lambda, \quad u_5 - \gamma_* \lambda,$$

not exact transport at each stage of the branch. Therefore the downstream derivative-geometric reduction should be formulated in terms of the projective defect direction

$$[\delta_n(c) : \delta_n(v_5) : \delta_n(u_5) : \delta_n(\lambda)],$$

or equivalently in any affine chart on which the projective limit has nonzero coordinate, rather than as an exact frozen- λ repeated-source branch.

Remark 12.134 (Collision-cancellation chart for close two-point cores). On a compact good patch

$$J \subseteq \{c \neq 0, M \neq 0\},$$

a close exact two-point bad core is naturally studied not in the raw diagonal chart $(h_1, h_2) \rightarrow (h, h)$, but in the cancellation chart

$$m := \frac{h_1 + h_2}{2}, \quad \delta := h_2 - h_1, \quad \lambda := -\frac{a_2}{a_1}, \quad \omega := \frac{\lambda - 1}{\lambda + 1}.$$

Indeed, the cubic equation

$$a_1 c(h_1) + a_2 c(h_2) = 0$$

forces $\lambda \rightarrow 1$ whenever $\delta \rightarrow 0$ on J .

The variable ω is the natural swap-compatible amplitude coordinate: under interchange of the two atoms one has

$$\lambda \mapsto \lambda^{-1}, \quad (m, \omega, \delta) \mapsto (m, -\omega, -\delta).$$

Writing

$$\mathcal{D}_h := (c(h), u_5(h), v_5(h)),$$

and using

$$\lambda = \frac{1 + \omega}{1 - \omega} = 1 + 2\omega + O(\omega^2),$$

one gets the first-order expansion

$$\mathcal{D}_{h_1} - \lambda \mathcal{D}_{h_2} = -2\omega \mathcal{D}_m - \delta \mathcal{D}'_m + O((|\omega| + |\delta|)^2).$$

After polar blow-up

$$(2\omega, \delta) = \rho(\cos \theta, \sin \theta),$$

the leading cancellation datum is therefore the projective source direction

$$\Theta_{m,\theta} := \cos \theta \mathcal{D}_m + \sin \theta \mathcal{D}'_m.$$

So any future cubic/quintic limit theorem for close cancelling two-point cores should be phrased in this blown-up collision-cancellation chart rather than as a Taylor expansion through the singular zero-source point.

Remark 12.135 (Work in progress: parity/projective factorization and the local good-patch frontier). The collision-cancellation chart suggests a substantial weakening of the hypotheses needed for local exclusion on the residual exact fixed-shear lane.

Suppose that an actual corrected defect $E(m, \omega, \delta)$ is analytic in this swap-compatible chart, is swap-even,

$$E(m, -\omega, -\delta) = E(m, \omega, \delta),$$

and vanishes on the collision locus

$$E(m, 0, 0) = 0.$$

Then its Taylor expansion has no linear terms, and with

$$\kappa := \frac{2\omega}{\delta}$$

one can write

$$E(m, \omega, \delta) = \delta^2 \mathcal{H}(m, \kappa, \delta^2)$$

for an analytic germ $\mathcal{H}$.

So the entire first blown-up interaction layer is parity-sterile: the first surviving layer is already quadratic and one-projective-parameter. Along an involutive exact fixed-shear branch this becomes quotient-visible, and on the quartic–sextic rung the quadratic coefficient is expected to be controlled by the single quotient transport scalar Q .

A sharper downstream target is the following conditional edge law for the actual corrected cubic/quintic defects:

$$E_{12}^{(3)} = \delta^2 \mathcal{H}_3(m, \kappa, \delta^2), \quad E_{12}^{(5)} = \delta^2 \mathcal{H}_5(m, \kappa, \delta^2),$$

with edge values

$$\mathcal{H}_3(m, \kappa, 0) = c'(m) - \frac{\kappa}{2}c(m), \quad \mathcal{H}_5(m, \kappa, 0) = (A_5^{\mathfrak{f}})'(m) - \frac{\kappa}{2}A_5^{\mathfrak{f}}(m).$$

On the cubic-killing slope

$$\kappa_*(m) = 2 \frac{c'(m)}{c(m)},$$

this would give the reduced quintic obstruction

$$\mathfrak{R}(m, 0) = \frac{M(m)}{c(m)}, \quad M := c(A_5^{\mathfrak{f}})' - c'A_5^{\mathfrak{f}}.$$

Hence on every compact good patch

$$J \subseteq \{c \neq 0, M \neq 0\},$$

sufficiently close cancelling bad cores would already be excluded.

Beyond that good-patch region, the same edge-law package would force any quartic–sextic survivor into the exceptional locus $M = 0$, where the residual problem should reduce, after Weierstrass preparation, to finitely many prepared branches and then to a one-variable compatibility / resultant / norm test. This is a reduction of the remaining burden, not a closed theorem at present.

Gate A and Gate B.. The corrected-package audit uses two separate gates. We call them Gate A and Gate B to prevent the leading homogeneous cubic audit from being confused with the full raw edge-divisibility theorem.

Remark 12.136 (Roadmap of the Gate A/Gate B correction). The corrected-package audit separates into two layers.

Gate A is the leading homogeneous cubic audit. It is a finite, order-7, leading-source calculation. Under the homogeneous scalar-grade convention, Gate A closes at the aggregate fixed-sector level:

$$F_1^{\text{lead cubic}} = 0.$$

Gate B is the raw edge-divisibility theorem. It concerns the full non-(1, 1) projected raw corrected-whitening cross-effect

$$(1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}$$

on the collision chart

$$h_1 = h - \frac{\delta}{2}, \quad h_2 = h + \frac{\delta}{2}, \quad a_2 = -a_1(1 + \kappa\delta).$$

Gate B is the assertion

$$[z^7][\delta^1] \pi_{fh} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} = 0.$$

The proof of Gate B below uses four inputs:

- (1) Gate A aggregate closure;
- (2) odd total-grade subset-complement cancellation for amplitude terms;
- (3) moving Taylor-frame telescoping for the source-motion terms;
- (4) 2-jet source-frame normalization, which removes the affine lower boundary.

Only after Gate B is proved may one form the normalized edge package

$$-a_1^{-1} \delta^{-2} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}.$$

Definition 12.137 (Gate A: leading homogeneous cubic audit). Work under the homogeneous scalar-grade convention

$$\text{Gr}_a r = \frac{r^{(a+2)}(h)}{(a+2)!} (t-h)^{a+2}.$$

Gate A is the audit of the leading homogeneous cubic non-(1, 1) source sector at ordinary order 7. In the active convention, this is the leading (1, 1, 5)-type cubic sector. The Gate A families are

$$L_1 Y R_1, \quad L_2 Y R_0, \quad L_0 Y R_2, \\ L_2 N_0 R_1, \quad L_1 N_0 R_2,$$

and

$$L_3 N_0 R_0, \quad L_0 N_0 R_3.$$

Gate A asks whether these leading homogeneous non-(1, 1) cubic terms produce a fixed-sector order-7 obstruction.

Definition 12.138 (Gate B: raw edge-divisibility). Gate B is the raw edge-divisibility theorem for the full projected corrected-whitening cross-effect. Let

$$C_{\mathcal{W}}^{\text{raw}} = \mathcal{W}(\mathcal{B}_2) - \mathcal{W}(\mathcal{B}_1(a_1, h_1)) - \mathcal{W}(\mathcal{B}_1(a_2, h_2)) + \mathcal{W}(\mathcal{B}_0),$$

where

$$\mathcal{W}(G_-, N, G_+) = G_-^{-1/2} N G_+^{-1/2}.$$

Apply the source-bidegree projection before fixed-sector extraction, determinant scalarization, or quotienting:

$$(1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}.$$

On the collision chart

$$h_1 = h - \frac{\delta}{2}, \quad h_2 = h + \frac{\delta}{2}, \quad a_2 = -a_1(1 + \kappa\delta),$$

Gate B is the assertion

$$(1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} = O(a_1 \delta^2)$$

through fixed-sector ordinary order 7. Equivalently, the first-collision coefficient

$$F_1 := [z^7][\delta^1] \pi_{f_h}(1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}$$

vanishes.

Remark 12.139 (Why Gate B must be raw). Gate B must be proved before forming the normalized package

$$-a_1^{-1} \delta^{-2} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}.$$

If $F_1 \neq 0$, this normalized expression has a δ^{-1} -pole and the post-edge package is not holomorphic. Therefore one must not use the post-edge quotient statement

$$Q_{7,m}^q(C) = Q_{7,m}^q(\Pi_{1,1}C)$$

as the first proof of edge divisibility. Quotient compatibility is a post-edge statement unless it is reformulated at the raw edge level.

Lemma 12.140 (Order exhaustion for Gate A). *Under the homogeneous scalar-grade convention, every leading homogeneous cubic non-(1,1) contribution to B_7^f is one of the seven families*

$$L_1 Y R_1, \quad L_2 Y R_0, \quad L_0 Y R_2, \quad L_2 N_0 R_1, \quad L_1 N_0 R_2, \quad L_3 N_0 R_0, \quad L_0 N_0 R_3.$$

The three Y-slot families

$$L_1 Y R_1, \quad L_2 Y R_0, \quad L_0 Y R_2$$

are absent at B_7^f . The non-Y families reduce to the leading placements

$$L_2^{\{1,1\}} N_0[0] R_1^{[5]}, \quad L_2^{\{1,5\}} N_0[0] R_1^{[1]},$$

and their right-side mirrors, together with the all-left/all-right placements

$$L_3^{\{1,1,5\}} N_0[0] R_0, \quad L_0 N_0[0] R_3^{\{1,1,5\}}.$$

Proof sketch. The Y-slot absence is the homogeneous scalar-grade order-count statement: under this convention the relevant Y-middle entries do not reach ordinary order 7. The remaining non-Y placements are the only ones whose Fréchet degree and homogeneous source grades can sum to 7 with the baseline middle $N_0[0]$. This is scoped to the homogeneous scalar-grade convention and does not apply to an ordinary matrix-output $M^{[5]}$ convention. $\square$

Lemma 12.141 (Aggregate closure of the leading $L_2 N_0 R_1 / L_1 N_0 R_2$ gate). *Work under the homogeneous scalar-grade convention. Let*

$$N := N_0[0], \quad W := W_{-,0} = G_0^{-1/2},$$

and let Σ_- denote the Sylvester inverse

$$G_0^{1/2} Z + Z G_0^{1/2} = C.$$

The leading same-point source inputs satisfy

$$X_{-,1}^{[1]} = -\frac{1}{48\pi Q^2} (I - D),$$

and

$$X_{-,5}^{[5]} = -\frac{1}{92160\pi Q^{10}} (I - D) = \lambda X_{-,1}^{[1]}, \quad \lambda := \frac{1}{1920Q^8}.$$

Set

$$A := S_{-,1}^{[1]} = \Sigma_-(X_{-,1}^{[1]}), \quad M := W A W,$$

and

$$T_2^{1,1} := \Sigma_-(-2AA), \quad H := 2W A W A W - W T_2^{1,1} W.$$

Then

$$S_{-,5}^{[5]} = \lambda A, \quad T_6^{1,5} = \lambda T_2^{1,1}, \quad H_{-,6}^{1,5} = \lambda H.$$

Consequently the leading fixed-sector tag vectors satisfy

$$C_{112}^{L_2 N_0 R_1} = C_{122}^{L_2 N_0 R_1} = \lambda \pi_{\mathfrak{f}}(HNM),$$

and

$$C_{112}^{L_1 N_0 R_2} = C_{122}^{L_1 N_0 R_2} = -\lambda \pi_{\mathfrak{f}}(MNH).$$

Since H, N, M are symmetric and

$$(HNM)^T = MNH,$$

while $\pi_{\mathfrak{f}}$ is transpose-invariant, one has

$$\pi_{\mathfrak{f}}(HNM) = \pi_{\mathfrak{f}}(MNH).$$

Therefore

$$C_{112}^{L_2 N_0 R_1} + C_{112}^{L_1 N_0 R_2} = 0,$$

and

$$C_{122}^{L_2 N_0 R_1} + C_{122}^{L_1 N_0 R_2} = 0.$$

Thus the leading homogeneous non- Y $L_2 N_0 R_1 / L_1 N_0 R_2$ gate closes at the aggregate fixed-sector level before quotienting.

Lemma 12.142 (Aggregate closure of the leading all-left/all-right cubic gate). *Assume the midpoint parity convention extends to the third-Fréchet coefficient table:*

$$G_{n;a,b,c}^+ = (-1)^n G_{n;a,b,c}^-.$$

Under the homogeneous scalar-grade convention, the first all-left/all-right source multiset is $(1, 1, 5)$, and it contributes at ordinary order 7. Let

$$G := G_{7;1,1,5}^-, \quad W := W_{-,0} = W_{+,0}, \quad N := N_0[0].$$

Then

$$G_{7;1,1,5}^+ = -G.$$

The leading all-left and all-right fixed-sector contributions are

$$C_{1,1,5}^{L_3 N_0 R_0} = \pi_{\mathfrak{f}}(GNW),$$

and

$$C_{1,1,5}^{L_0 N_0 R_3} = -\pi_{\mathfrak{f}}(WNG).$$

Since G, N, W are symmetric,

$$(GNW)^T = WNG.$$

Because $\pi_{\mathfrak{f}}$ is transpose-invariant,

$$\pi_{\mathfrak{f}}(GNW) = \pi_{\mathfrak{f}}(WNG).$$

Hence

$$C_{1,1,5}^{L_3 N_0 R_0} + C_{1,1,5}^{L_0 N_0 R_3} = 0 \quad \text{in } \mathfrak{f}.$$

Thus the leading all-left/all-right homogeneous third-Fréchet gate closes at the aggregate fixed-sector level before quotienting.

Corollary 12.143 (Gate A closure). *Under the homogeneous scalar-grade convention and the stated midpoint-parity conventions,*

$$F_1^{\text{lead cubic}} = 0.$$

Equivalently, the leading homogeneous cubic non- $(1, 1)$ sector contributes no fixed-sector $B_7^{\mathfrak{f}}$ obstruction and no first-collision Gate B obstruction.

Remark 12.144 (Aggregate closure versus family-by-family gauge). The preceding Gate A closures are aggregate fixed-sector cancellations. They do not assert that the individual families

$$L_2 N_0 R_1, \quad L_1 N_0 R_2, \quad L_3 N_0 R_0, \quad L_0 N_0 R_3$$

are separately proportional to $A_5^{\mathfrak{f}}(h)$. The package theorem requires the aggregate corrected-whitening cross-effect, not a family-by-family gauge statement. No determinant scalarization, no Φ_K -scalar invisibility, and no downstream diagonal merger is used in these closures.

Remark 12.145 (Scope of Gate A). Gate A is scoped to the active homogeneous scalar-grade convention. It does not apply to an ordinary matrix-output $M^{[5]}$ convention, nor to a bookkeeping system that treats lower-shadow grades 0, 2, 4 as independent ordinary matrix-output inputs. Such conventions require a separate no-double-counting theorem.

Remark 12.146 (Corrected package theorem after the Gate A/Gate B split). The current corrected-package frontier is no longer the raw seven-family Fréchet shortcut. The proof architecture is layered as follows.

First, Gate A closes the leading homogeneous cubic non- $(1, 1)$ audit at the aggregate fixed-sector level. This uses the homogeneous scalar-grade convention, the absence of the Y -slot families at B_7 , the aggregate $L_2N_0R_1/L_1N_0R_2$ cancellation, and the aggregate $L_3N_0R_0/L_0N_0R_3$ cancellation, the last under the stated third-Fréchet midpoint parity convention.

Second, Gate B proves raw edge-divisibility for the projected raw corrected-whitening cross-effect:

$$[z^7][\delta^1] \pi_{jh} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} = 0.$$

This is the raw theorem needed before the amplitude-normalized package

$$-a_1^{-1} \delta^{-2} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}$$

can be formed.

Only after Gate B is proved does the corrected fixed-target package become available for the quotient-gauge theorem

$$B_7^f(C_{\mathcal{W}}^{\text{corr}}) \in \mathbf{CA}_5^f(m),$$

equivalently

$$Q_{7,m}^q(C_{\mathcal{W}}^{\text{corr}}) = 0.$$

Thus any statement of the form

$$Q_{7,m}^q(C) = Q_{7,m}^q(\Pi_{1,1}C)$$

is post-edge bookkeeping unless it is explicitly reformulated as a raw edge-divisibility statement.

The older raw seven-family finite-coefficient tests remain useful only as diagnostic coefficient tools. They are not the theorem object and should not be used in place of the corrected package representative.

Definition 12.147 (Homogeneous scalar grade). For a residual scalar source r , define

$$\text{Gra}_a r := \frac{r^{(a+2)}(m)}{(a+2)!} (t-m)^{a+2}.$$

Thus grade 1 corresponds to the cubic scalar jet $r^{(3)}(m)$, and grade 5 corresponds to the septic scalar jet $r^{(7)}(m)$.

Lemma 12.148 (Transpose parity of the source-linear mixed input). *For the full source-linear scaled mixed input $M(z)$ before Φ_K , one has*

$$M(-z) = M(z)^T.$$

Consequently the diagonal entries of $M(z)$ are even in z , while the odd ordinary powers occur in the off-diagonal antisymmetric channel. Through ordinary order 7, the coefficient $[z^5]M$ is off-diagonal antisymmetric and is supported by the even source derivatives

$$r^{(2)}(m), \quad r^{(4)}(m), \quad r^{(6)}(m),$$

not by the homogeneous grade-five source $r^{(7)}(m)$.

Remark 12.149 (Verification sketch for the transpose-parity mixed input). Changing z to $-z$ interchanges the two endpoints in the scaled mixed block. The mixed block is therefore transformed by endpoint swap, which is matrix transpose in the left/right basis. Hence the source-linear mixed input satisfies

$$M(-z) = M(z)^T.$$

The diagonal and symmetric off-diagonal channels are even, while the antisymmetric off-diagonal channel is odd. The audited order-7 expansion places $[z^5]M$ in that antisymmetric channel, supported by

$$r^{(2)}(m), \quad r^{(4)}(m), \quad r^{(6)}(m),$$

not by the homogeneous grade-five source $r^{(7)}(m)$.

Lemma 12.150 (*Y-slot families are absent at B_7 under homogeneous grade*). *Under Definition 12.147, the Y-slot cubic families*

$$L_1 Y R_1, \quad L_2 Y R_0, \quad L_0 Y R_2$$

do not contribute to $B_7^\dagger$.

More precisely, the homogeneous grade-five mixed input associated to $\text{Gr}_5 r$ has no ordinary z^5 middle coefficient in the mixed block; its first possible mixed contribution occurs too late to combine with the corresponding first- or second-Fréchet same-point factors at total ordinary order 7. Consequently the three Y-slot families above start at ordinary order at least 8.

Remark 12.151 (Scope of Lemma 12.150). Lemma 12.150 is an order-count absence theorem, not a determinant-proportionality theorem. It is scoped to the homogeneous scalar grade convention. If a future theorem defines $M^{[5]}$ instead as an ordinary matrix-output order-5 projection, or as a parity-corrected non-homogeneous source projection, then the Y-slot families must be reopened and audited separately. Such a convention also requires a no-double-counting theorem against the natural grade 0, 2, 4 same-point shadows.

Remark 12.152 (Status of the homogeneous-grade lemmas after Gate A/B). Lemmas 12.148 and 12.150 are now inputs to the Gate A/Gate B closure mechanism.

Under the homogeneous scalar-grade convention, the Y-slot families are absent at B_7 , the leading non-Y Gate A aggregate closes in $\mathfrak{f}_h$, and Gate B closes through ordinary order 7 after all-subset midpoint parity, moving source-frame telescoping, and affine boundary removal are imposed.

This does not mean that lower-shadow grades 0, 2, 4 are discarded. They may appear in the moving Taylor tower. Interior lower-shadow terms are handled by the telescoping identity of Theorem B.17; only the affine grade -1 boundary is removed by the 2-jet source-frame normalization of Lemma B.18.

Remark 12.153 (Grade-zero sector after the Gate B source-frame audit). The homogeneous scalar-grade convention

$$\text{Gr}_a r = \frac{r^{(a+2)}(m)}{(a+2)!} (t-m)^{a+2}$$

does not discard the grade-zero component

$$\text{Gr}_0 r = \frac{r^{(2)}(m)}{2!} (t-m)^2.$$

Affine removal eliminates only the constant and affine modes. Thus $r^{(2)}(m)$ is a genuine source-frame direction, not baseline data.

The Gate B result treats this direction differently from the old open-status notes. Grade 0 is allowed in the moving homogeneous Taylor tower. Its amplitude contributions are covered by the odd total-grade subset-complement cancellation, including zero-padded labelled multisets. Its source-motion lower boundary is

$$(\partial_h \text{Gr}_0 r)_{\text{lower}} = -r^{(2)}(h)(t-h),$$

which is affine and is removed by the 2-jet source-frame projection $\Pi_{\geq 2, h}$.

Consequently grade 0 is neither an independent leftover Gate B residual nor something silently absorbed into q_0 . It is a live source-frame direction whose first-collision contribution is killed by the Gate B source-frame mechanism.

Remark 12.154 (Residual-source tower before highest-new reduction). The homogeneous scalar-grade convention organizes the local residual source tower, but it does not by itself identify all residual-source coefficients with the highest-new tower $q^{(7)}$. Finite quotient-gauge identities must first be tested in the residual-source variables supplied by the corrected source block. Only a residual that survives this package-level test should be passed to the later same-tower / highest-new reduction.

Thus the statement that the eventual local bridge should run through the single highest-new source $q^{(7)}$ is a downstream package-to-jet bridge claim. It is not a substitute for the pre- Φ_K , pre-determinant fixed-sector quotient-gauge calculation.

Remark 12.155 (Quarantined matrix-output $M^{[5]}$ convention). A nonzero ordinary matrix-output order-5 convention for $M^{[5]}$ is not the active homogeneous scalar-grade convention. If such a convention is later introduced, then the Y -slot families

$$L_1 Y R_1, \quad L_2 Y R_0, \quad L_0 Y R_2$$

must be reopened. That route requires a no-double-counting theorem separating the ordinary matrix-output order-5 contribution from the natural homogeneous scalar grades 0, 2, 4, whose source derivatives also have same-point shadows. The earlier raw-seven-family numerical failure recorded in Remark B.42 rejects the raw matrix-output shortcut; it does not reject the homogeneous scalar-grade gate of Lemma 12.150 or the corrected package representative of Section C.

Remark 12.156 (Historical role of the first non- Y cubic gate). Before the aggregate Gate A closure was isolated, the first non- Y homogeneous cubic gate was the pair

$$L_2 N_0 R_1 \quad \text{and} \quad L_1 N_0 R_2.$$

The finite coefficient inputs for this diagnostic gate were

$$\begin{aligned} [z^2] \mathcal{L}_-^{\{1,1\}}, & \quad [z^6] \mathcal{L}_-^{\{1,5\}}, \\ [z^2] \mathcal{R}_+^{\{1,1\}}, & \quad [z^6] \mathcal{R}_+^{\{1,5\}}, \end{aligned}$$

together with

$$N_0[0], \quad [z^1] \Lambda_{\pm}^{[1]}, \quad [z^5] \Lambda_{\pm}^{[5]}.$$

These data are still useful for auditing the finite coefficient tables.

However, in the current proof they are no longer an open Gate A target. Lemma 12.141 proves that the $L_2 N_0 R_1 / L_1 N_0 R_2$ leading homogeneous gate closes at the aggregate fixed-sector level. Therefore this remark records historical coefficient-tool provenance, not a remaining theorem obligation.

Remark 12.157 (Superseded coefficient test for the first non- Y gate). The earlier coefficient-level audit asked for the four fixed-sector tag vectors

$$C_{112}^{L_2 N_0 R_1}, \quad C_{122}^{L_2 N_0 R_1}, \quad C_{112}^{L_1 N_0 R_2}, \quad C_{122}^{L_1 N_0 R_2}.$$

That was the correct finite test before the aggregate closure mechanism was identified.

In the current draft this test is superseded by Lemma 12.141, which proves the cancellation in $\mathfrak{f}$ directly:

$$C_{112}^{L_2 N_0 R_1} + C_{112}^{L_1 N_0 R_2} = 0, \quad C_{122}^{L_2 N_0 R_1} + C_{122}^{L_1 N_0 R_2} = 0.$$

Thus the old determinant checks against $A_5^{\mathfrak{f}}$ should not be read as remaining obligations for Gate A. They remain only as a useful coordinate-level audit if the homogeneous scalar-grade convention is later changed or the aggregate proof is challenged.

Remark 12.158 (All-left/all-right cubic families: aggregate status). The all-left/all-right families

$$L_3 N_0 R_0, \quad L_0 N_0 R_3$$

are not closed family-by-family. However, under the homogeneous scalar-grade convention and assuming third-Fréchet midpoint parity, their leading $(1, 1, 5)$ aggregate satisfies

$$L_3 N_0 R_0 + L_0 N_0 R_3 = 0 \quad \text{in } \mathfrak{f}.$$

This is stronger than quotient-gauge for the package-level Gate A target. It does not assert that either family is separately proportional to $A_5^\mathfrak{f}$.

If a coefficient table uses ordered triples rather than symmetric multilinear coefficients, the aggregate must include all ordered permutations with the same convention on both left and right sides before this cancellation is invoked.

Remark 12.159 (Current UV chain and status). The current package chain is as follows.

- (1) **UV-022.** Define an invariant source-weight-linear input for the corrected block and obtain exact one-amplitude collapse at the package level.
- (2) **UV-023.** Establish the raw first-collision edge law for the non- $(1, 1)$ projected corrected-whitening cross-effect through ordinary order 7:

$$[z^7][\delta^1] \pi_{\mathfrak{f}_h} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} = 0.$$

In the present draft this is supplied, conditionally, by Corollary B.21. The mechanism is not a $\mathcal{T}_2$ -only truncation theorem, not source-weight linearization, and not post-edge quotient compatibility. It uses Gate A aggregate closure, odd total-grade subset-complement cancellation, grade-preserving source-motion cancellation, moving source-frame telescoping, affine boundary removal, and full-tower differentiation before ordinary-order truncation.

- (3) **UV-024.** Define the pre-determinant fixed-sector coefficient

$$B_7^\mathfrak{f}(C)$$

for the actual corrected-whitening matrix cross-effect before Φ_K , and define

$$Q_{7,m}^q(C) := [B_7^\mathfrak{f}(C)] \in \mathfrak{f}/\mathbf{C}A_5^\mathfrak{f}(m).$$

- (4) **UV-025.** Supply the tagged pre-whitening source block of Definition 10.6. This is now a guarded source-level input, not a quotient theorem.
- (5) **Gate A / UV-026 leading cubic audit.** Under the homogeneous scalar-grade convention, the leading homogeneous cubic non- $(1, 1)$ sector is closed at the aggregate fixed-sector level. The Y -slot families are absent at B_7 , and the non- Y aggregate pairs

$$L_2 N_0 R_1 + L_1 N_0 R_2, \quad L_3 N_0 R_0 + L_0 N_0 R_3$$

vanish in $\mathfrak{f}$, the latter assuming third-Fréchet midpoint parity.

- (6) **Gate B raw edge-divisibility.** The projected raw corrected-whitening cross-effect satisfies

$$[z^7][\delta^1] \pi_{\mathfrak{f}_h} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} = 0$$

through ordinary order 7, provided the all-subset midpoint parity, leading Y -middle parity, moving source-frame telescoping, and affine boundary-removal lemmas are imposed. This supplies the raw edge divisibility needed before amplitude normalization and before the lower package table $Z_j[M]$ is formed.

- (7) **Connection correction subtarget.** After Gate B, compute the transported moving line coefficients

$$A_i = [(h - m)^i] \tau_{h \rightarrow m} A_5^\mathfrak{f}(h), \quad 1 \leq i \leq 7,$$

and the pre-trivialization lower edge coefficients

$$Z_j[M_A], \quad Z_j[M_C], \quad 0 \leq j \leq 6,$$

for

$$M_A = \dot{r}_3 r_2 \dot{r}_4, \quad M_C = \dot{r}_3 \dot{r}_2 r_4.$$

The required cancellation identities are

$$R_7^{(A)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_A], A_i) = 0,$$

and

$$R_7^{(C)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_C], A_i) = 0.$$

These are package-level pre-trivialization connection identities, not raw seven-family identities and not post-trivialization $B_j[M]$ identities.

Shortcut proofs using Φ_K -scalar hiding, determinant scalarization, swap symmetry, one-pair gauge, or the UV-025 source block alone do not prove UV-026. The theorem must be proved at the pre- Φ_K , pre-determinant fixed-sector quotient level of Section C.

In this formulation, UV-026 is a theorem about the normalized corrected package representative. The raw seven-family Fréchet sum is only a diagnostic object; its failure under a matrix-output $M^{[5]}$ convention is recorded in Remark B.42 and does not replace the corrected package theorem.

Remark 12.160 (Superseded Gate B status notes). Earlier WIP notes treated $p \geq 4$ transfer filtration and lower-shadow absorption as open obstructions. That was the correct status before the homogeneous transfer audit. The current proof replaces those open obligations as follows:

- (1) positive-grade amplitude terms of total grade 7 vanish by odd total-grade subset-complement cancellation;
- (2) the grade-preserving part of X_h vanishes by the same cancellation;
- (3) the grade-lowering part of X_h telescopes across adjacent moving Taylor grades;
- (4) the sole lower boundary produced by Gr_0 is affine and is removed by 2-jet source-frame normalization.

Thus the remaining role of the old notes is as guardrails: do not prove Gate B by post-edge quotient compatibility, do not demand family-by-family vanishing, and do not truncate the source tower before differentiating.

Remark 12.161 (The old $d \geq 4$ residual partition list is superseded). Earlier audits listed positive-grade residual partitions such as

$$(1, 1, 1, 4), \quad (1, 1, 2, 3), \quad (1, 2, 2, 2), \\ (1, 1, 1, 1, 3), \quad (1, 1, 1, 2, 2), \quad (1, 1, 1, 1, 1, 2), \quad (1, 1, 1, 1, 1, 1, 1)$$

as live $d \geq 4$ transfer-filtration candidates. Those lists are now superseded by the odd total-grade subset-complement cancellation and the moving source-frame telescoping theorem. They should be used only as historical diagnostics for checking that the aggregate proof has included all labelled placements, not as remaining theorem obligations.

Remark 12.162 (Lower-shadow sectors and zero-padding after Gate B). The old lower-shadow obstruction was not a finite missing table. If grade 0 is treated as an independent source factor, then the source-grade partition inventory is non-finite:

$$(0, 1, 1, 5), \quad (0, 0, 1, 1, 5), \quad (\underbrace{0, \dots, 0}_N, 1, 1, 5), \quad \dots$$

and similarly for

$$(\underbrace{0, \dots, 0}_N, 7), \quad (\underbrace{0, \dots, 0}_N, 1, 2, 4), \quad (\underbrace{0, \dots, 0}_N, 2, 2, 3).$$

The Gate B proof does not close this tower by finite enumeration. It closes it uniformly.

More precisely, lower-shadow grades 0, 2, 4 are not reclassified as baseline data and are not silently discarded. They remain part of the moving homogeneous Taylor source tower. Their amplitude contributions of total homogeneous grade 7 are killed by odd total-grade subset-complement cancellation for arbitrary labelled transfer degree. Their source-motion contributions are handled by the moving source-frame telescoping identity, and the only lower

boundary generated by differentiating Gr_0 is the affine grade -1 mode, which is removed by the 2-jet source-frame normalization.

Thus the correct post-Gate B status is: lower shadows are live source-frame directions, but not live Gate B first-collision residuals. They should not be listed as an infinite family of remaining residual partitions after Corollary B.21.

Remark 12.163 (Counterterm route if Gate B dependencies fail). The present draft proves Gate B under the stated midpoint-parity, source-frame telescoping, and 2-jet normalization hypotheses. If any of those hypotheses fails in the final source model, then one cannot simply subtract the residual after determinant scalarization.

A legitimate counterterm would have to be a canonical pre- Φ_K , pre-determinant, source-bidegree-compatible edge counterterm $C_{\text{diag/edge}}$ determined by the raw edge jets

$$[z^7][\delta^0]\pi_{f_h}(1 - \Pi_{1,1})_{\text{mix}}C_{\mathcal{W}}^{\text{raw}}, \quad [z^7][\delta^1]\pi_{f_h}(1 - \Pi_{1,1})_{\text{mix}}C_{\mathcal{W}}^{\text{raw}}.$$

It must be natural under atom swap, must preserve the $(1 - \Pi_{1,1})_{\text{mix}}$ source-bidegree sector, and must be defined before Φ_K , determinant pairing, and quotienting. Otherwise it is merely subtracting the obstruction by hand.

Remark 12.164 (Relation to the package-to-jet bridge). The corrected package program above is not an alternative to the zeta-side odd-jet extractor. The extractor side is already available; the missing bridge is the package-to-jet theorem identifying the first surviving corrected odd coefficient from finite-core package data. In the fixed-shear corner, the finite quotient-visible transport has already been compressed to the remaining package-level data. Thus any surviving obstruction after the corrected package quotient theorem must be treated as a package-to-transform or hidden-state problem, not as another planar or scalarized local detector.

In particular, Gate A and Gate B are package-construction results. They do not by themselves prove the final finite-core contradiction or the package-to- $\Xi_{\zeta}^{(N)}$ hidden-state theorem. Any residual Φ_K -visible affine-lift or higher odd-block state remains a separate package-to-transform issue.

Remark 12.165 (Open theorem target: finite-core lower transfer). The finite-core branch would be closed by a theorem of the following form.

Finite-core lower-transfer theorem. Let $\mathcal{C}$ be a normalized genuinely finite same-scale core whose corrected package anomaly is nonzero after all quotient and hidden-state identifications relevant to Φ_K . Then there exist an admissible $N = N(m)$, a sign $\sigma_{\mathcal{C}} \in \{\pm 1\}$, and a constant $c_{\mathcal{C}} > 0$, stable on the compact core-parameter patch, such that

$$|\Xi_{\text{core}}^{(N)}(m)| \geq c_{\mathcal{C}} \frac{x}{B_{\zeta}(m)} S(m).$$

The present draft does not prove this theorem. The corrected order-7 package controls the local quotient construction and removes the known two-point/package quotient residuals, but it does not by itself determine the N -visible coefficient

$$c_{\text{core}, 2N-1}(m)$$

for a genuinely non-pair-like finite core, nor does it rule out cancellation among finite-core constituents.

Definition 12.166 (Nonsingular actual-source k -core stratum). A compact parameter patch Θ_k is called a nonsingular actual-source k -core stratum if the following conditions hold.

- (1) Θ_k is contained in a finite-dimensional real-analytic parameter manifold with fixed core cardinality k , fixed source combinatorics, and fixed collision/blow-up chart.
- (2) The corrected finite-core transverse scalar

$$H_{\text{core}, \theta}(z) = \Phi_K \left(\widehat{\Omega}_{\text{core}, \theta}^{\text{corr}}(z/Q^2; m) \right)$$

is real-analytic jointly in (θ, z) on a neighborhood of $\Theta_k \times \{0\}$.

- (3) The whitening, inverse-square-root, and midpoint-trivialization maps used to define $\widehat{\Omega}_{\text{core},\theta}^{\text{corr}}$ are nonsingular and analytic on the patch.
- (4) The chosen coordinate chart does not divide by a vanishing quantity. In particular, if the chart uses c^{-1} , u_5^{-1} , v_5^{-1} , or a determinant coordinate relative to $A_5^{\dagger}$, the corresponding quantity is nonzero on the patch.

Theorem 12.167 (Finite odd-germ determinacy on compact actual-source strata). *Let Θ_k be a compact nonsingular actual-source k -core stratum. Write*

$$H_{\text{core},\theta}(z) = \sum_{r \geq 0} c_{\text{core},2r+1}(\theta) z^{2r+1}.$$

Then there exists an integer $R_{\Theta_k} \geq 0$ such that

$$c_{\text{core},1}(\theta) = c_{\text{core},3}(\theta) = \cdots = c_{\text{core},2R_{\Theta_k}+1}(\theta) = 0$$

implies

$$H_{\text{core},\theta} \equiv 0$$

as an odd germ.

Equivalently, the zero-anomaly locus

$$\mathcal{Z}_k = \{\theta \in \Theta_k : H_{\text{core},\theta} \equiv 0\}$$

is cut out on Θ_k by finitely many odd coefficient equations.

Proof. By joint analyticity of $H_{\text{core},\theta}(z)$, each coefficient

$$c_{\text{core},2r+1}(\theta)$$

is real-analytic on a neighborhood of Θ_k .

Fix $\theta_0 \in \Theta_k$. Let

$$\mathcal{I}_{\theta_0} = (c_{\text{core},1}, c_{\text{core},3}, c_{\text{core},5}, \dots) \subset \mathcal{O}_{\Theta_k, \theta_0}$$

be the ideal generated by the odd coefficient germs in the local analytic ring at θ_0 . The local analytic ring is Noetherian. Hence $\mathcal{I}_{\theta_0}$ is finitely generated. Therefore there are indices $r_1, \dots, r_s$ such that

$$\mathcal{I}_{\theta_0} = (c_{\text{core},2r_1+1}, \dots, c_{\text{core},2r_s+1})$$

as germs at θ_0 .

Set

$$R(\theta_0) = \max(r_1, \dots, r_s).$$

After shrinking to a neighborhood U_{θ_0} , every coefficient $c_{\text{core},2r+1}$ is an analytic linear combination of

$$c_{\text{core},1}, c_{\text{core},3}, \dots, c_{\text{core},2R(\theta_0)+1}.$$

Thus, on U_{θ_0} , vanishing of all odd coefficients up to order $2R(\theta_0) + 1$ forces every odd coefficient to vanish. Hence $H_{\text{core},\theta} \equiv 0$ as an odd germ.

The compactness of Θ_k gives a finite subcover

$$U_{\theta_1}, \dots, U_{\theta_M}.$$

Taking

$$R_{\Theta_k} = \max_{1 \leq j \leq M} R(\theta_j)$$

proves the claim uniformly on Θ_k . □

Remark 12.168 (Strata excluded from one finite-determinacy patch). Theorem 12.167 is stratum-local. The following loci must be separated before applying it.

- (1) *Amplitude-drop strata:*

$$a_j = 0,$$

where a k -core degenerates to a smaller core.

(2) *Collision strata:*

$$h_i = h_j,$$

unless a collision/blow-up coordinate chart has been chosen in which the corrected source construction remains analytic.

(3) *Whitening singularities:* loci where the relevant Gram blocks lose invertibility or the inverse-square-root branches cease to be analytic.

(4) *$c = 0$ strata:* these must be excluded from charts using

$$T = \frac{v_7}{c}$$

or any other c^{-1} -normalization.

(5) $A_5^f = 0$ *strata:* these are rank-jump loci for the quotient line

$$\mathbf{C}A_5^f.$$

(6) $v_5 = 0$ and $u_5 = 0$ *patch boundaries:* these must be separated when using the corresponding v_5 -patch or u_5 -patch coordinates.

(7) *Hidden-state rank jumps:* loci where the augmented package coordinate

$$T = \frac{v_7}{c}$$

or the Φ_K -visible lift ceases to be represented by the chosen analytic chart.

After separating these loci, finite odd-germ determinacy applies on each compact nonsingular analytic stratum.

Corollary 12.169 (Coefficient floor away from the zero-anomaly locus). *Let Θ_k be a compact nonsingular actual-source k -core stratum, and let R_{Θ_k} be the finite-determinacy depth from Theorem 12.167. Define*

$$\mathcal{Z}_k = \{\theta \in \Theta_k : H_{\text{core},\theta} \equiv 0\}.$$

For $\eta > 0$, set

$$\Theta_{k,\eta} = \{\theta \in \Theta_k : \text{dist}(\theta, \mathcal{Z}_k) \geq \eta\}.$$

Then there exists $\gamma_\eta > 0$ such that every $\theta \in \Theta_{k,\eta}$ satisfies

$$\max_{0 \leq r \leq R_{\Theta_k}} |c_{\text{core},2r+1}(\theta)| \geq \gamma_\eta.$$

Proof. The function

$$D(\theta) = \max_{0 \leq r \leq R_{\Theta_k}} |c_{\text{core},2r+1}(\theta)|$$

is continuous on Θ_k . By Theorem 12.167,

$$D(\theta) = 0 \iff \theta \in \mathcal{Z}_k.$$

Thus $D > 0$ on the compact set $\Theta_{k,\eta}$, and hence D has a positive minimum there. $\square$

Remark 12.170 (Finite determinacy is not zeta-scale transfer). Theorem 12.167 is qualitative. It proves only

$$H_{\text{core},\theta} \neq 0 \implies \exists N \leq R_{\Theta_k} + 1 : c_{\text{core},2N-1}(\theta) \neq 0.$$

It does not prove

$$|\Xi_{\text{core}}^{(N)}(m)| \gtrsim \frac{x}{B_\zeta(m)} S(m).$$

Nor does it provide a sign law, a tail-domination estimate for the filtered N -point coefficient, or an explicit bound for R_{Θ_k} depending only on k .

The remaining quantitative problem is to compare the actual zeta core's distance from

$$\mathcal{Z}_k$$

with the calibration scale

$$\frac{x}{B_\zeta(m)} S(m).$$

That is the calibrated separation problem, not a consequence of Noetherian finite determinacy.

Remark 12.171 (Later refinement of calibrated separation). The distance formulation below is an earlier geometric form of the finite-core quantitative gap. The current proof-state uses the sharper sampled-vector formulation in Subsection 16.30. In that formulation the finite-determinacy vector is decomposed as

$$C_{\text{fd}} = P_{\text{fd}} + J_{\text{fd}},$$

and the live hypothesis is the sampled vector lower bound

$$\|P_{\text{fd}}(\theta(m)) + J_{\text{fd}}(\theta(m))\|_{\infty} \geq \Lambda(m, Q).$$

Thus the branch

$$\mathcal{N}_k(m, Q)$$

defined below should be read as a geometric precursor to the sharper interaction-aligned branch

$$\mathcal{N}_k^{\text{align}}(m, Q).$$

Hypothesis 12.172 (Calibrated finite-core separation). Let Θ_k be a compact nonsingular actual-source k -core stratum, with zero-anomaly locus

$$\mathcal{Z}_k = \{\theta : H_{\text{core}, \theta} \equiv 0\}.$$

For actual zeta finite cores $\theta(m)$, either

$$\theta(m) \in \mathcal{Z}_k,$$

or $\theta(m)$ lies in a controlled incidence stratum, or else

$$\text{dist}(\theta(m), \mathcal{Z}_k) \geq D_{\Theta_k}(m, Q),$$

where $D_{\Theta_k}(m, Q)$ is large enough to force

$$\left| c_{\text{core}, 2N-1} + \sum_{j \geq 1} c_{\text{core}, 2N-1+2j} h_j(1^2, \dots, N^2) Q^{-2j} \right| \gtrsim \frac{Q^{2N+2}}{(2N-1)N!(N-1)!} \frac{x}{B_{\zeta}(m)} S(m)$$

for some

$$N \leq R_{\Theta_k} + 1.$$

Definition 12.173 (Near-zero-anomaly finite-core branch). On a compact nonsingular actual-source k -core stratum Θ_k , define

$$\mathcal{N}_k(m, Q) := \{\theta(m) \in \Theta_k : 0 < \text{dist}(\theta(m), \mathcal{Z}_k) < D_{\Theta_k}(m, Q), \quad \theta(m) \notin \mathcal{E}_{\text{geom}}\}.$$

This is the branch where the corrected finite-core anomaly is nonzero but too close to the zero-anomaly locus for the finite-determinacy–Łojasiewicz argument to imply

$$|\Xi_{\text{core}}^{(N)}(m)| \gtrsim \frac{x}{B_{\zeta}(m)} S(m).$$

Remark 12.174 (Status of the near-zero-anomaly branch). The present draft does not prove

$$\mathcal{N}_k(m, Q) = \emptyset.$$

In particular, proximity to $\mathcal{Z}_k$ is not currently shown to imply membership in a controlled incidence stratum such as

$$M = 0, \quad W = 0, \quad K_v = 0, \quad T_v = 0,$$

nor collision, patch-boundary, endpoint-gap incidence, amplitude drop, whitening singularity, $c = 0$, $A_5^{\text{f}} = 0$, or hidden-state rank jump. Thus finite-core closure requires either calibrated anti-tangency from $\mathcal{Z}_k$ or a separate analysis of $\mathcal{N}_k(m, Q)$.

Definition 12.175 (First interaction branch). For a normalized same-scale finite core, write

$$H_{\text{core}}(s) = \Phi_K(F_{\text{core}})(s) = \sum_{r \geq 0} c_{\text{core}, 2r+1} s^{2r+1},$$

and decompose

$$F_{\text{core}} = \sum_{\nu} a_{\nu} F_{h_{\nu}} + \mathcal{I}_{\geq 2}.$$

Define

$$J_1 := [s]\Phi_K(\mathcal{I}_{\geq 2}).$$

The first interaction branch is

$$\mathfrak{Z}_{\text{int}} = \left\{ \sum_{\nu} a_{\nu} K(d_{\nu}) + J_1 = 0 \right\} \setminus \mathcal{E}_{\text{geom}},$$

where $\mathcal{E}_{\text{geom}}$ denotes the union of the controlled geometric, collision, patch-boundary, and endpoint-gap incidence strata.

Remark 12.176 (Interaction rigidity remains a source-level obligation). The current ECT and lifted-curve geometry controls defect-free affine incidence. It does not by itself control the genuinely interacting scalar

$$J_1 = [s]\Phi_K(\mathcal{I}_{\geq 2}).$$

The known structural properties of $\mathcal{I}_{\geq 2}$, such as one-amplitude collapse and collision divisibility, force J_1 to vanish on some boundary strata, but they do not force

$$J_1 \equiv 0,$$

nor do they impose a sign condition or place J_1 in the radical of the controlled incidence ideal.

Thus the finite-core proof needs an additional interaction-rigidity theorem: J_1 must be shown to be zero, sign-controlled, or governed by the controlled incidence equations. Without such a theorem,

$$\mathfrak{Z}_{\text{int}}$$

remains a separate finite-core branch.

Remark 12.177 (Source-level first-interaction rigidity remains open). At two-source order, the first interaction coefficient has the form

$$\mathfrak{j}_{ij}^{(2)} = [s]\Phi_K \left(D\mathcal{W}_0[\mathcal{B}_{ij}^{(2)}] + D^2\mathcal{W}_0[\mathcal{B}_i^{(1)}, \mathcal{B}_j^{(1)}] \right).$$

The direct Hessian contribution and the bilinear whitening contribution split into the independent channels

$$dd, \quad d\eta + \eta d, \quad \eta\eta,$$

together with bilinear whitening products. No current parity, endpoint-symmetry, ECT, or controlled-incidence theorem forces their sum to vanish, gives it a calibrated sign/cone condition, or places it in the radical of the controlled incidence ideal.

Thus

$$J_1 = [s]\Phi_K(\mathcal{I}_{\geq 2})$$

remains an independent finite-core obstruction unless a separate source-level first-interaction rigidity theorem is proved.

Remark 12.178 (Scope after Gate B). Gate B supplies the raw δ^2 -edge divisibility needed to construct the order-7 edge-normalized corrected package. It does not by itself prove the final quotient-gauge theorem, diagonal merger, κ -independence, the connection cancellations for M_A, M_C , or the package-to- $\Xi_{\zeta}^{(N)}$ bridge.

The M_A, M_C connection branch is now closed under the source-grade reflection convention by Theorem C.21. Thus the remaining post-Gate-B local obstruction is not the $Z_j[M_A], Z_j[M_C]$ table. The remaining local actual-source/Bottleneck-C obstruction is the promoted quadratic mixed-block coefficient

$$E_{A,Y}^{\text{BC}}$$

of Definition 12.200. Its quotient test is

$$e_{A,Y}v_5 - u_5f_{A,Y} = 0,$$

and its transform-safe test is

$$f_{A,Y} = 0.$$

Remark 12.179 (Reduced package equality does not imply hidden-state equality). The reduced datum $\widehat{\Psi}$ forgets the raw septic representative in the affine line

$$A_7^f \mapsto A_7^f + \chi A_5^f.$$

This forgotten direction is not automatically invisible to the transverse functional. Writing

$$A_5^f = u_5I + v_5S,$$

one has

$$\Phi_K(A_5^f) = 2v_5.$$

Thus reduced-package equality alone cannot imply agreement modulo $\ker \Phi_K$ through the first surviving odd order, except on the special locus $v_5 = 0$ or after an additional theorem showing cancellation in the actual corrected block. The missing hidden-state theorem is therefore a transform-level state-locality statement modulo $\ker \Phi_K$, not a formal consequence of quotient-septic closure.

Definition 12.180 (Φ_K -visible septic state). On the fixed target fiber

$$\mathfrak{f}_m = \mathbf{C}I \oplus \mathbf{C}S,$$

write

$$A_7^f = u_7I + v_7S.$$

The Φ_K -visible septic state is the class

$$[A_7^f]_\Phi := A_7^f \bmod \ker(\Phi_K|_{\mathfrak{f}_m}),$$

equivalently the scalar

$$\phi_7 := \Phi_K(A_7^f) = 2v_7.$$

If the c -normalized package is being used, the normalized visible septic state is

$$\theta_7 := \frac{\phi_7}{c} = \frac{2v_7}{c}.$$

Proposition 12.181 (Formal non-descent of the Φ_K -visible septic state). *On the generic locus $v_5 \neq 0$, the quotient class*

$$[A_7^f] \in \mathfrak{f}_m / \mathbf{C}A_5^f(m)$$

does not determine the Φ_K -visible septic state

$$[A_7^f]_\Phi \in \mathfrak{f}_m / \ker(\Phi_K|_{\mathfrak{f}_m}).$$

In particular, quotient package equality does not imply equality of the first Φ_K -visible corrected odd coefficient.

Proof. Write

$$A_5^f = u_5I + v_5S, \quad A_7^f = u_7I + v_7S.$$

The quotient identifies

$$A_7^f \sim A_7^f + \chi A_5^f.$$

However

$$\Phi_K(A_7^f + \chi A_5^f) - \Phi_K(A_7^f) = \chi \Phi_K(A_5^f) = 2\chi v_5.$$

Thus, if $v_5 \neq 0$, the forgotten quotient direction is visible to Φ_K .

Equivalently, take the formal model

$$A_5^f = S, \quad A_7^{f,\chi} = \chi S.$$

Then

$$[A_7^{f,\chi}] = 0 \quad \text{in} \quad \mathfrak{f}_m / \mathbf{C}S$$

for every χ , but

$$\Phi_K(A_7^{\mathfrak{f},\chi}) = 2\chi.$$

Hence the reduced quotient package does not determine the Φ_K -visible septic state. $\square$

Remark 12.182 (Package-to-transform bridge after augmentation). The quotient-only package-to-transform bridge is impossible on the generic patch $v_5 \neq 0$, since

$$A_7^{\mathfrak{f}} \sim A_7^{\mathfrak{f}} + \chi A_5^{\mathfrak{f}}$$

preserves the quotient class but changes

$$\Phi_K(A_7^{\mathfrak{f}})$$

by

$$2\chi v_5.$$

The correct theorem object is therefore the transform-safe augmented package

$$\widehat{\Psi}_{\text{red}}^+ = (\widehat{\Psi}_{\text{red}}, T), \quad T = \frac{v_7}{c}.$$

With this augmentation, the first Φ_K -visible septic coefficient is read off as

$$\Phi_K(A_7^{\mathfrak{f}}) = 2cT.$$

The remaining compatibility problem is not quotient-only state-locality; it is the hidden-horizontal diagonal-merger condition

$$\partial_\kappa T(m, \kappa, 0) = 0.$$

Definition 12.183 (Transform-safe augmented reduced package). Work on a good patch with

$$c \neq 0, \quad A_5^{\mathfrak{f}} = u_5 I + v_5 S.$$

Write

$$A_7^{\mathfrak{f}} = u_7 I + v_7 S, \quad \Delta_7 := u_7 v_5 - u_5 v_7.$$

The transform-safe augmented reduced package is

$$\widehat{\Psi}_{\text{red}}^+ := (\widehat{\Psi}_{\text{red}}, T), \quad T := \frac{v_7}{c}.$$

Equivalently, on the v_5 -patch one may write

$$\widehat{\Psi}_{\text{red}}^+ = (c, u_5, v_5, \Delta_7, T).$$

The normalized Φ_K -visible coordinate is

$$\theta_{\Phi,7} := \frac{\Phi_K(A_7^{\mathfrak{f}})}{c} = 2T.$$

Theorem 12.184 (Augmented package-to-transform readout). *Let*

$$\widehat{\Psi}_{\text{red}}^+ = (c, u_5, v_5, \Delta_7, T)$$

be the transform-safe augmented package on a good patch with $c \neq 0$. Then the first Φ_K -visible septic coefficient of the corrected finite-s two-point transform is determined by the augmented package:

$$\Phi_K(A_7^{\mathfrak{f}}) = 2v_7 = 2cT.$$

Equivalently,

$$\frac{1}{c}[s^7]\Phi_K(\widehat{\Omega}^{\text{corr}}(s; m)) = 2T.$$

Thus the two-point transform-visible septic state factors through $\widehat{\Psi}_{\text{red}}^+$, not through $\widehat{\Psi}_{\text{red}}$ alone.

Remark 12.185 (Why the old quotient-only bridge must be abandoned). The reduced quotient package remembers only

$$[A_7^f] \in \mathfrak{f}_m / \mathbf{C}A_5^f(m).$$

It therefore identifies

$$A_7^f \sim A_7^f + \chi A_5^f.$$

But

$$\Phi_K(A_5^f) = 2v_5.$$

Hence, on the generic patch $v_5 \neq 0$, the quotient-forgotten line direction is Φ_K -visible:

$$\Phi_K(A_7^f + \chi A_5^f) - \Phi_K(A_7^f) = 2\chi v_5.$$

Therefore no functional of $\widehat{\Psi}_{\text{red}}$ alone can determine $\Phi_K(A_7^f)$ on the generic patch. The missing coordinate is exactly

$$T = \frac{v_7}{c}.$$

Theorem 12.186 (Sharp no-descent criterion for the Φ_K -visible septic state). *Let*

$$\mathfrak{f}_m = \mathbf{C}I \oplus \mathbf{C}S, \quad A_5^f = u_5I + v_5S.$$

Let

$$q : \mathfrak{f}_m \rightarrow \mathfrak{f}_m / \mathbf{C}A_5^f$$

be the quotient map. There exists a functional

$$\Theta_{\Phi,7}$$

on the quotient such that

$$\Phi_K(A_7^f) = \Theta_{\Phi,7}(q(A_7^f))$$

for all $A_7^f \in \mathfrak{f}_m$ if and only if

$$\Phi_K(A_5^f) = 0.$$

Equivalently, since

$$\Phi_K(A_5^f) = 2v_5,$$

such descent is possible if and only if

$$v_5 = 0.$$

Proof. The quotient identifies

$$A_7^f \sim A_7^f + \chi A_5^f.$$

Therefore Φ_K descends to the quotient if and only if it is constant on these affine fibers, i.e.

$$\Phi_K(A_7^f + \chi A_5^f) = \Phi_K(A_7^f)$$

for all A_7^f and all χ . By linearity this is equivalent to

$$\chi \Phi_K(A_5^f) = 0$$

for all χ , hence to

$$\Phi_K(A_5^f) = 0.$$

Writing

$$A_5^f = u_5I + v_5S$$

gives

$$\Phi_K(A_5^f) = 2v_5.$$

Thus descent holds exactly on the locus $v_5 = 0$. □

Theorem 12.187 (Augmented Bottleneck C condition). *On the good patch $c \neq 0$, diagonal merger for the transform-safe package*

$$\widehat{\Psi}_{\text{red}}^+ = (c, u_5, v_5, \Delta_7, T)$$

requires both

$$\partial_\kappa Q_{7,m}^q(m, \kappa, 0) = 0$$

and

$$\partial_\kappa T(m, \kappa, 0) = 0.$$

The first condition is quotient-horizontal merger. The second is hidden-horizontal merger. On the generic patch $v_5 \neq 0$, the first does not imply the second.

Remark 12.188 (Scalar Bottleneck C condition). Let

$$P_7(a_1, a_2) = [z^7][\delta^1]\tau_{h \rightarrow m}\pi_{f_h}(1 - \Pi_{1,1})_{\text{mix}}C_{\mathcal{W}}^{\text{raw}}.$$

The quotient Bottleneck C condition is

$$\partial_{a_2} P_7(a_1, -a_1) \in \mathbf{CA}_5^f(m).$$

The transform-safe condition is instead

$$\Phi_K(\partial_{a_2} P_7(a_1, -a_1)) = 0.$$

Equivalently, if

$$P_{7,\Phi}(a_1, a_2) := \frac{1}{c}\Phi_K(P_7(a_1, a_2)),$$

then the scalar condition is

$$P_{7,\Phi}(a_1, a_2) \in (a_1 + a_2)^2 \mathbf{C}[a_1, a_2].$$

This scalar condition is independent of quotient-valued double-balanced divisibility on the generic patch $v_5 \neq 0$.

Lemma 12.189 (Hidden-horizontal exceptional divisor condition). *Work on a good patch with*

$$c(m) \neq 0, \quad A_5^f(m) = u_5 I + v_5 S, \quad v_5 \neq 0.$$

Let

$$F_7(a_1, a_2, \delta) = [z^7]\tau_{h \rightarrow m}\pi_{f_h}(1 - \Pi_{1,1})_{\text{mix}}C_{\mathcal{W}}^{\text{raw}} = \delta P_7(a_1, a_2) + O(\delta^2).$$

Then the augmented hidden coordinate $T = v_7/c$ is horizontal on the exceptional divisor if and only if

$$\Phi_K(\partial_{a_2} P_7(a_1, -a_1)) = 0.$$

Equivalently,

$$\partial_\kappa T(m, \kappa, 0) = 0 \quad \Longleftrightarrow \quad \frac{1}{c}\Phi_K(\partial_{a_2} P_7(a_1, -a_1)) = 0.$$

Proof. Along the collision chart

$$a_2 = -a_1(1 + \kappa\delta),$$

one has

$$P_7(a_1, a_2) = P_7(a_1, -a_1) - a_1\kappa\delta\partial_{a_2} P_7(a_1, -a_1) + O(\delta^2).$$

After the standard edge normalization, the exceptional κ -slope is represented by

$$\partial_{a_2} P_7(a_1, -a_1).$$

Applying Φ_K/c gives the slope of the normalized transform-visible coordinate. Since $T = \Phi_K(A_7^f)/(2c)$, the displayed criterion follows. $\square$

Proposition 12.190 (Formal transform-visible line-gauge leakage). *On a good patch with*

$$c \neq 0, \quad A_5^f = u_5 I + v_5 S, \quad v_5 \neq 0,$$

quotient-horizontal Bottleneck C does not imply T-horizontal merger.

Proof. Consider an order-7 fixed-sector edge carrier of the form

$$[\delta^1]A_7^{\mathfrak{f},\chi} = \lambda(m) a_1 a_2 (a_1 + a_2) A_5^{\mathfrak{f}}(m).$$

Since it is pointwise parallel to $A_5^{\mathfrak{f}}(m)$, its quotient class vanishes:

$$[A_7^{\mathfrak{f},\chi}] = 0 \quad \text{in} \quad \mathfrak{f}_m / \mathbb{C} A_5^{\mathfrak{f}}(m).$$

Thus it is invisible to quotient Bottleneck C.

However,

$$a_1 a_2 (a_1 + a_2) \big|_{a_2 = -a_1} = 0,$$

but

$$\partial_{a_2} (a_1 a_2 (a_1 + a_2)) \big|_{a_2 = -a_1} = -a_1^2.$$

Moreover,

$$\Phi_K(A_5^{\mathfrak{f}}) = 2v_5.$$

Therefore the induced hidden-state slope is

$$[\delta^1] \partial_{a_2} T_{\chi}(a_1, -a_1, \delta) = -a_1^2 \lambda(m) \frac{v_5(m)}{c(m)}.$$

On the generic patch $c \neq 0$, $v_5 \neq 0$, this is nonzero unless

$$\lambda(m) = 0.$$

Thus a quotient-invisible line-gauge edge carrier can be transform-visible. $\square$

Corollary 12.191 (First live T -slope carrier). *Let*

$$E_{A,Y}^{\text{BC}} = e_{A,Y} I + f_{A,Y} S$$

be the frozen Bottleneck-C coefficient of the promoted nonlinear mixed-block carrier. The amplitude factor of this carrier is

$$a_1 a_2 (a_1 + a_2),$$

and

$$\partial_{a_2} (a_1 a_2 (a_1 + a_2)) \big|_{a_2 = -a_1} = -a_1^2.$$

Consequently its contribution to the hidden-horizontal slope is

$$\partial_{\kappa} T \big|_{E_{A,Y}^{\text{BC}}} = -a_1^2 \frac{\Phi_K(E_{A,Y}^{\text{BC}})}{2c} = -a_1^2 \frac{f_{A,Y}}{c}.$$

Thus the augmented transform-safe package requires

$$f_{A,Y} = 0.$$

In particular, if

$$E_{A,Y}^{\text{BC}} = \lambda A_5^{\mathfrak{f}},$$

then

$$\partial_{\kappa} T \big|_{E_{A,Y}^{\text{BC}}} = -a_1^2 \lambda \frac{v_5}{c}.$$

On the generic $v_5 \neq 0$ patch, a nonzero line-gauge $E_{A,Y}^{\text{BC}}$ closes the quotient condition but still leaks through T .

Remark 12.192 (Local diagnostic for the hidden septic state). On the quintic-good patch

$$\{c \neq 0, v_5 \neq 0\},$$

one useful local diagnostic enlargement of the reduced package is

$$\tilde{\Psi}_{\min} = (x, Y, S, T) = \left(\frac{v_5}{c}, \frac{u_5}{c}, \frac{\Delta_7}{cv_5}, \frac{v_7}{c} \right).$$

This tuple is useful only as normalized fixed-sector septic data. It is not a canonical global theorem object and it does not give finite determination of the first surviving odd jet. The global target remains the transform-level first Φ_K -visible corrected odd coefficient, modulo $\ker \Phi_K$.

Remark 12.193 (Bottleneck C is exceptional-divisor fiber selection). For Bottleneck C, the obstruction is not the choice of a local septic representative such as $v_7 = 0$, $u_7 = 0$, or $T = v_7/c$. Those are diagnostic data for the hidden-state problem. The C-side object has already quotiented raw septic representatives away. Its remaining obstruction is the exceptional-divisor trace

$$B(m, \kappa) := \tilde{\Psi}_{\text{red}}^{\text{corr}}(m, \kappa, 0).$$

Analyticity, swap-evenness, fixed codomain, scalar normalization, and the central value $B(m, 0) = \hat{\Psi}(m)$ do not force

$$\partial_\kappa B = 0.$$

A valid C proof must supply actual diagonal merger, same-reduced-image, or collision-functoriality for the corrected two-atom package.

Remark 12.194 (Formal C properties do not rule out slope leakage). Formal properties such as analyticity, swap-evenness, fixed codomain, scalar normalization, and the central value at $\kappa = 0$ do not by themselves force the C-side exceptional-divisor trace to be independent of κ . A formal model may have

$$B_3 = Z + \varepsilon\kappa,$$

or, at the quotient-defect level,

$$R \mapsto R + R_\kappa, \quad \det(R_\kappa, A_5^f) = c^2 \varepsilon \kappa.$$

Thus representative-gauge invariance only kills additions parallel to A_5^f . It does not kill determinant-changing quotient-defect data. A C proof must control the actual corrected exceptional-divisor quotient class, not merely the formal shape of the reduced package.

Remark 12.195 (Atom swap does not impose $\kappa \mapsto -\kappa$ on the exceptional divisor). In the collision chart

$$h_1 = m - \frac{\delta}{2}, \quad h_2 = m + \frac{\delta}{2}, \quad a_2 = -a_1(1 + \kappa\delta),$$

interchanging the two labelled atoms gives

$$\delta \mapsto -\delta, \quad a_1 \mapsto -a_1(1 + \kappa\delta), \quad \kappa \mapsto \frac{\kappa}{1 + \kappa\delta}.$$

Hence on the exceptional divisor $\delta = 0$, the slope coordinate is fixed:

$$\kappa \mapsto \kappa.$$

Therefore atom-swap symmetry does not force the exceptional trace

$$B(m, \kappa)$$

to be even in κ , and in particular it does not force

$$\partial_\kappa B(m, \kappa) = 0.$$

Swap symmetry removes the wrong-parity terms in the transverse collision variable; it does not remove quotient-visible dependence on the exceptional slope.

Remark 12.196 (Missing theorem: quotient-valued diagonal merger). Bottleneck C requires a quotient-valued diagonal merger theorem. On every good patch on which

$$c(m) \neq 0, \quad A_5^f(m) \neq 0,$$

the corrected two-atom collision package must factor through the blow-down of the exceptional divisor:

$$(m, \kappa, 0) \mapsto m.$$

Equivalently,

$$B(m, \kappa) = B(m, 0) = \hat{\Psi}(m) \quad \text{for all sufficiently small } \kappa.$$

At the order-7 quotient level this means

$$B_7^f(m, \kappa, 0) - B_7^f(m, 0, 0) \in \mathbf{C}A_5^f(m).$$

This is stronger than choosing a local septic gauge such as $v_7 = 0$ or $u_7 = 0$. Such a gauge selects a representative inside a fixed quotient class; it does not prove that the quotient class itself is independent of the exceptional slope κ .

Definition 12.197 (Quotient-valued two-atom collision functor). On a good patch with

$$A_5^f(m) \neq 0,$$

define the order-7 quotient-valued two-atom collision functor by the ordered composite

$$\begin{aligned} \mathcal{B}_2 &\longmapsto C_{\mathcal{W}}^{\text{raw}} \longmapsto (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} \longmapsto -a_1^{-1} \delta^{-2} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} \\ &\longmapsto \pi_{\mathfrak{f}_h} [z^{\leq 7}] \longmapsto \tau_{h \rightarrow m} \longmapsto [z^7] \longmapsto \mathfrak{f}_m / \mathbf{C} A_5^f(m). \end{aligned}$$

In formulas,

$$\mathcal{C}_7^q(m, \kappa, \delta) := [[z^7] \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} (-a_1^{-1} \delta^{-2} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}})].$$

The exceptional-divisor trace is

$$B(m, \kappa) = \mathcal{C}_7^q(m, \kappa, 0).$$

Thus quotient-valued diagonal merger is precisely the statement that

$$\mathcal{C}_7^q(m, \kappa, 0) = \mathcal{C}_7^q(m, 0, 0),$$

or equivalently that $\mathcal{C}_7^q$ factors on the exceptional divisor through the blow-down map

$$(m, \kappa, 0) \longmapsto m.$$

Proposition 12.198 (Formal quotient-visible exceptional-slope leakage). *The formal properties analyticity, atom-swap compatibility, fixed codomain, scalar normalization, $B(m, 0) = \widehat{\Psi}(m)$ do not imply quotient-valued diagonal merger.*

Proof. Work on a good patch and choose

$$E_m \in \mathfrak{f}_m$$

with

$$\omega_m(E_m, A_5^f(m)) = 1.$$

Thus E_m is nonzero in

$$\mathfrak{f}_m / \mathbf{C} A_5^f(m).$$

In the collision chart

$$h_1 = m - \frac{\delta}{2}, \quad h_2 = m + \frac{\delta}{2}, \quad a_2 = -a_1(1 + \kappa\delta),$$

atom swap acts by

$$\delta \longmapsto -\delta, \quad \kappa \longmapsto \frac{\kappa}{1 + \kappa\delta}.$$

Define

$$H(\kappa, \delta) := \frac{1}{2} \left(\kappa + \frac{\kappa}{1 + \kappa\delta} \right).$$

Then H is analytic near $(\kappa, \delta) = (0, 0)$, satisfies

$$H\left(\frac{\kappa}{1 + \kappa\delta}, -\delta\right) = H(\kappa, \delta),$$

and restricts to

$$H(\kappa, 0) = \kappa$$

on the exceptional divisor.

Let $Z_0(m) \in \mathfrak{f}_m$ be any representative of $\widehat{\Psi}(m)$, and define the formal normalized septic edge representative

$$\mathcal{R}_\varepsilon(m, \kappa, \delta; z) = z^7 (Z_0(m) + \varepsilon H(\kappa, \delta) E_m).$$

Equivalently, before edge normalization one may write the raw model as

$$\mathcal{N}_\varepsilon = -a_1 \delta^2 z^7 (Z_0(m) + \varepsilon H(\kappa, \delta) E_m).$$

This raw model is $O(a_1\delta^2)$, so it is compatible with raw edge-divisibility. It is analytic and atom-swap compatible by construction.

At $\kappa = 0$,

$$B_\varepsilon(m, 0) = [Z_0(m)] = \widehat{\Psi}(m).$$

But on the exceptional divisor,

$$B_\varepsilon(m, \kappa) = [Z_0(m) + \varepsilon\kappa E_m],$$

and hence

$$\partial_\kappa B_\varepsilon(m, \kappa) = \varepsilon[E_m] \neq 0$$

in

$$\mathfrak{f}_m/\mathbf{C}A_5^{\mathfrak{f}}(m).$$

Therefore the listed formal properties do not force quotient-valued diagonal merger. $\square$

Remark 12.199 (Balancing does not impose hidden κ -horizontality). The balancing condition

$$a_2 = -a_1(1 + \kappa\delta)$$

gives

$$\partial_\kappa a_2 = -a_1\delta.$$

This single factor of δ does not force the normalized exceptional slope to vanish after the δ^{-2} edge normalization. A raw term of the form

$$-a_1\delta^2\kappa E_m$$

is still of the allowed Gate-B edge order and survives normalization as

$$\kappa E_m.$$

Equivalently, with $\kappa = 2\omega/\delta$, the leakage term

$$2\varepsilon\omega\delta E_m = \varepsilon\kappa\delta^2 E_m$$

is invisible before edge normalization but visible on the exceptional divisor. Thus balancing kills the zeroth source moment; it does not by itself kill the projective exceptional slope.

Definition 12.200 (Promoted quadratic mixed-block Bottleneck-C coefficient). Let

$$Y_{12}^{(2)} := D^2 N_0[\xi_1, \xi_2]$$

denote the $\mathcal{K}^2$, source-bidegree $(1, 1)$ nonlinear Taylor remainder of the actual corrected mixed block $N^{\text{corr}}(d_-, d_+, \eta)$.

The promoted channel has amplitude factor

$$a_1 a_2 (a_1 + a_2).$$

On the collision chart

$$a_2 = -a_1(1 + \kappa\delta),$$

this amplitude already contributes the first δ -edge:

$$a_1 a_2 (a_1 + a_2) = a_1^3 \kappa \delta + O(\delta^2).$$

Therefore the Bottleneck-C coefficient for this promoted channel is the frozen internal coefficient

$$E_{A,Y}^{\text{BC}} := \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \text{Sym}[z^7] D(G_-^{-1/2})[A_-] Y_{12}^{(2)} R_0.$$

Equivalently,

$$E_{A,Y}^{\text{BC}} = \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \text{Sym} \sum_{\chi \in \{dd, d\eta + \eta d, \eta\eta\}} \sum_{g=4}^7 \sum_{r+b=7-g} H_{r,0}^A Y_{g,0}^\chi W_{+,b}.$$

Write

$$E_{A,Y}^{\text{BC}} = e_{A,Y} I + f_{A,Y} S.$$

The quotient Bottleneck-C obstruction is

$$Q_{A,Y} := e_{A,Y} v_5 - u_5 f_{A,Y}.$$

The transform-safe obstruction is

$$\Phi_K(E_{A,Y}^{\text{BC}}) = 2f_{A,Y}.$$

For comparison, the literal source-position edge

$$E_{A,Y}^{\text{pos}} := \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \text{Sym}[z^7 \delta^1] D(G_-^{-1/2})[A_-] Y_{12}^{(2)} R_0$$

belongs to the next source-motion layer and should not be confused with the first Bottleneck-C amplitude-derivative coefficient.

Definition 12.201 (Active Bottleneck-C source-linear index set). Let

$$\mathcal{A}_{\text{BC}} \subseteq \{2, 3, 4, 5, 6\}$$

denote the set of same-point source-linear Fréchet primitive indices that can enter the promoted mixed-block Bottleneck-C coefficient

$$E_{A,Y}^{\text{BC}}.$$

In the finite Fréchet convention of Section B, the lead top-row obstruction uses

$$a = 2, \quad \Lambda_{2,0}^- = -\frac{1}{24\pi q^3} E_{22},$$

so $2 \in \mathcal{A}_{\text{BC}}$. Older shorthand using $a \in \{1, 5\}$ should be read only as a deprecated table convention and not as the active finite Fréchet index set.

Remark 12.202 (Executable table for $E_{A,Y}^{\text{BC}}$). The frozen Bottleneck-C coefficient is computed from the rows

$$\mathcal{E}_{\chi;g,r,b}^a := \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \text{Sym} \left(\Lambda_{a,r}^- Y_{g,0}^\chi W_{+,b} \right),$$

where

$$\begin{aligned} a &\in \mathcal{A}_{\text{BC}}, & \chi &\in \{dd, d\eta + \eta d, \eta\eta\}, \\ 4 &\leq g \leq 7, & r + b &= 7 - g. \end{aligned}$$

Thus

$$E_{A,Y}^{\text{BC}} = \sum_{\chi \in \{dd, d\eta + \eta d, \eta\eta\}} \sum_{a \in \mathcal{A}_{\text{BC}}} \sum_{\substack{4 \leq g \leq 7 \\ r+b=7-g}} \mathcal{E}_{\chi;g,r,b}^a.$$

The skew row

$$Y_{d\eta - \eta d}^{(2)}$$

is excluded because it is killed by fixed-sector projection.

Remark 12.203 (Status of $E_{A,Y}^{\text{BC}}$). The recurrences for

$$W_{+,0}, W_{+,1}, W_{+,2}, W_{+,3}$$

and

$$\Lambda_{a,0}^-, \Lambda_{a,1}^-, \Lambda_{a,2}^-, \Lambda_{a,3}^-$$

reduce $E_{A,Y}^{\text{BC}}$ to a finite symbolic coefficient table. However, the present draft does not yet supply the scalar midpoint source jets needed to evaluate

$$E_{A,Y}^{\text{BC}} = e_{A,Y} I + f_{A,Y} S.$$

The missing scalar jets are

$$q_0^{(k)}(m), \quad 0 \leq k \leq 9,$$

and

$$(r_i^{[a]})^{(k)}(m), \quad i = 1, 2, \quad a \in \mathcal{A}_{\text{BC}}, \quad 2 \leq k \leq 9.$$

Moreover, the active top row

$$(a, r, g, b) = (2, 0, 7, 0), \quad \chi = \eta\eta_{\text{odd}},$$

recorded in Subsection 16.3, gives a concrete Φ_K -visible survivor unless

$$\nu_{21,7} = 0$$

or unless it is cancelled by another physical top-row contribution. Thus the current local obstruction is sharper than an unevaluated table: the package must prove cancellation of the odd- $\eta\eta$ top row, or add a physical correction cancelling it.

Remark 12.204 (First live promoted nonlinear mixed-block obstruction). The same-point cubic q^3 defect is fixed-sector invisible in same-point-only Fréchet responses by endpoint-swap/transpose symmetry. The first remaining actual-source/Bottleneck-C carrier is the promoted nonlinear mixed-block quadratic remainder

$$Y_{12}^{(2)} = D^2 N_0[\xi_1, \xi_2],$$

coupled to one same-point linear whitening slot.

The amplitude factor of this channel is

$$a_1 a_2 (a_1 + a_2).$$

Since

$$a_1 a_2 (a_1 + a_2) = a_1^3 \kappa \delta + O(\delta^2) \quad \text{on } a_2 = -a_1(1 + \kappa \delta),$$

the first Bottleneck-C coefficient is

$$E_{A,Y}^{\text{BC}},$$

not the internal source-position coefficient $E_{A,Y}^{\text{pos}}$.

Thus quotient-valued Bottleneck C for this channel is exactly

$$Q_{A,Y} = e_{A,Y} v_5 - u_5 f_{A,Y} = 0.$$

The augmented transform-safe package requires the stronger scalar condition

$$f_{A,Y} = 0, \quad \text{equivalently} \quad \Phi_K(E_{A,Y}^{\text{BC}}) = 0.$$

On the generic $v_5 \neq 0$ patch, quotient closure together with T -horizontal closure forces

$$E_{A,Y}^{\text{BC}} = 0.$$

Lemma 12.205 (Promoted mixed-block coefficient table). *In the notation of Definition 12.200, the frozen Bottleneck-C coefficient of the promoted quadratic mixed-block channel is*

$$E_{A,Y}^{\text{BC}} = \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \text{Sym} \sum_{\chi \in \{dd, d\eta + \eta d, \eta\eta\}} \sum_{g=4}^7 \sum_{r+b=7-g} H_{r,0}^A Y_{g,0}^\chi W_{+,b}.$$

Here

$$H_{r,0}^A = [z^r] D(G_-^{-1/2}) [A_-], \quad W_{+,b} = [z^b] G_+^{-1/2},$$

and

$$Y_{g,0}^\chi = [z^g] Y_\chi^{(2)}$$

for

$$\chi \in \{dd, d\eta + \eta d, \eta\eta\}.$$

The skew combination

$$Y_{d\eta - \eta d}^{(2)}$$

is not included, since it is killed by fixed-sector projection.

If

$$E_{A,Y}^{\text{BC}} = e_{A,Y} I + f_{A,Y} S,$$

then the quotient obstruction is

$$e_{A,Y} v_5 - u_5 f_{A,Y},$$

and the transform-safe obstruction is

$$f_{A,Y}.$$

Remark 12.206 (Current status of the promoted mixed-block coefficient table). The finite Fréchet/Sylvester machinery determines the shape of $E_{A,Y}^{\text{BC}}$, but the present draft does not yet tabulate the scalar midpoint jet data needed to decide

$$E_{A,Y}^{\text{BC}} = 0, \quad E_{A,Y}^{\text{BC}} \in \mathcal{CA}_5^f(m), \quad \text{or} \quad (E_{A,Y}^{\text{BC}})^q \neq 0.$$

The exact missing finite table is

$$\left\{ W_{+,0}, W_{+,1}, W_{+,2}, W_{+,3}, \quad \Lambda_{a,0}^-, \Lambda_{a,1}^-, \Lambda_{a,2}^-, \Lambda_{a,3}^-, \right. \\ \left. Y_{g,0}^{dd}, \quad Y_{g,0}^{d\eta+\eta d}, \quad Y_{g,0}^{\eta\eta} \quad (4 \leq g \leq 7) \right\}.$$

Until that table is evaluated, $E_{A,Y}^{\text{BC}}$ remains the first unresolved local actual-source/Bottleneck-C obstruction.

Remark 12.207 (Frozen Bottleneck-C coefficient versus source-position edge). The coefficient $E_{A,Y}^{\text{BC}}$ is the first Bottleneck-C amplitude-ratio carrier. It uses the frozen internal coefficient because the amplitude factor

$$a_1 a_2 (a_1 + a_2)$$

already supplies the first δ -edge on the collision chart

$$a_2 = -a_1(1 + \kappa\delta).$$

The literal source-position edge

$$E_{A,Y}^{\text{pos}} := \tau_{h \rightarrow m} \pi_{fh} \text{Sym}[z^7 \delta^1] D(G_-^{-1/2})[A_-] Y_{12}^{(2)} R_0$$

belongs to the next source-motion layer and should not be used as the first Bottleneck-C amplitude-derivative coefficient.

Remark 12.208 (Shortcut routes excluded for the corrected package theorem). The following routes do not prove the corrected package quotient theorem:

- (1) Φ_K -scalar invisibility or scalar hiding after Φ_K .
- (2) Determinant scalarization used as the definition of the quotient class.
- (3) Swap symmetry or one-pair K_5/K_3 gauge laws without the actual two-atom package object.
- (4) The tagged pre-whitening source block alone.
- (5) The raw seven-family Fréchet sum; tested source-constrained nonconstant baselines leave nonzero determinant residuals.
- (6) Cubic/quintic edge laws plus the abstract source criterion alone.
- (7) Existing one-pair order-7 or centered D_2 formulas promoted directly to a two-atom order-7 Hessian.
- (8) Signed source-weight lift alone; it restores oddness but leaves higher $a^3, a^5, \dots$ amplitude terms.

Every positive route must work before Φ_K , before determinant scalarization, and inside the fixed-target quotient package.

Remark 12.209 (UV-010 boundary: edge law before descent). The midpoint quotient trivialization is available on the good patch

$$A_5^f(m) \neq 0,$$

but this does not construct the order-7 two-atom edge package. The UV-010 target is the normalized fixed-midpoint quotient edge law

$$\overline{E}_{12}^{(7;1)} = \delta^2 \mathcal{H}_7^q(m, \kappa, \delta^2),$$

with

$$[R]_{\text{edge}} = -\mathcal{H}_7^q(m, \kappa, 0).$$

Descent, κ -independence, and diagonal merger are downstream of this edge-law construction. The Gate B edge-divisibility theorem constructs the holomorphic environment needed for

the corrected package representative; it should not be conflated with the further UV-010 task of identifying the exceptional-divisor value $[R]_{\text{edge}}$ with the C-side reduced-package trace. Determinant scalarization is good-patch bookkeeping after the quotient-valued edge object exists; it is not a substitute for the edge package.

Remark 12.210 (What remains in the two-pair rewrite). Theorem 12.121 still provides the correct algebraic bridge from the two-pair core to the strengthened invariant $\widehat{\Psi}$, and Lemma 12.124 still isolates the finite-order source criterion that would close the close two-point branch on compact good patches.

What remains, however, is no longer raw bookkeeping through orders 3, 5, 7 and no longer a search for another local mixed branch taxonomy. The overlap-side geometry is now sharply split:

$$(K_v = 0, S \neq 0)$$

is already mixed low-order,

$$(S = 0, K_v \neq 0)$$

is already absorbed by nonmixed 4-point geometry, and the only genuinely residual mixed corner is

$$(S = 0, K_v = 0).$$

There the branch is quotient-diagonal / involutive, and the remaining burden is best read through the three-front compression of Remark 12.211.

Accordingly, the live downstream two-point programs are now the following.

First, a good-patch collision-cancellation theorem for the *actual* corrected cubic/quintic defects: enough regularity and swap-evenness in the blown-up cancellation chart to justify the quadratic/projective edge-law package.

Second, a quotient-diagonal closure theorem on the exact fixed-shear corner: for example state-locality, transport closure, or a diagonal-collapse result for the actual corrected defect package after quotient-visible descent.

Third, a provenance-sensitive package theorem — for example same reduced image germ at coincidence or collision-functoriality on the collision/cancellation locus — strong enough to force the later quotient-septic collapse formally.

These are conditional downstream programs, not yet closure theorems. What the present draft already shows is that the residual burden is concentrated there, rather than in a missing local branch classification.

Remark 12.211 (Current three-front compression of the remaining burden). The two-point endgame is now substantially narrower than a generic “one more local sign argument” picture.

On the nonmixed/global affine side, the remaining burden is an RH-specific exclusion problem for the isolated remote endpoint/branch incidences encoded by the monotone gap functions $G_{\xi,J}$. On the mixed side, the only genuinely residual overlap contribution is the exact fixed-shear corner

$$\{S = 0\} \cap \{K_v = 0\},$$

already reduced to a quotient-diagonal / involutive transport problem. On the honest higher-order side, the relevant target is no longer raw order-7 equality of representatives, but a provenance-sensitive package statement such as same reduced image germ at coincidence or collision-functoriality on the collision/cancellation locus.

Accordingly, what remains is not a diffuse interaction heuristic. It is a three-front problem: remote endpoint-gap rigidity, exact fixed-shear quotient transport, and honest package-level coincidence/functoriality.

Remark 12.212 (Minimal-core reformulation). The proposition above shows that the correct projective datum for the downstream multi-pair and minimal-core extraction is the amplitude-invariant strengthened map

$$\widehat{\Psi}(h) = \left(\frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)^2} \right),$$

not merely the pair $(\ell_h, \bar{v}_1(h))$.

By contrast, the affine curve

$$\Gamma(h) = \left(\frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)} \right)$$

remains the correct object for the one-pair secant/tangent geometry. The finite-core endgame should therefore be viewed as a two-stage problem: first extract defect-corrected coincidence information for $\widehat{\Psi}$, then convert that information into affine-line incidence constraints for Γ .

Remark 12.213 (Finite-core compression to a remote endpoint/branch incidence problem). The two-stage reformulation above suggests a useful downstream combinatorial compression.

Fix a compact relevant patch J . By Theorem 12.100, every maximal turning-boundary-generic block contributes no pair-pair geometry, so such a block behaves as a single transport chamber. By Theorem 12.107, the isolated finite-order $W = 0$, $M \neq 0$ boundaries are locally sterile for pair-pair birth. Moreover, on the repeated-source side, Proposition 12.299, Corollary 12.315, Proposition 12.317, and Corollary 12.319 show that odd-order $\mathcal{R}$ -boundaries are relay-type rather than arbitrary first-root events.

After these absorptions, any surviving minimal bad core contribution from one-pair incidence must come through the finite remote endpoint/branch tangencies of Lemma 12.109 and Remark 13.5, equivalently through isolated interior zeros of the strictly monotone endpoint-gap functions

$$G_{\xi,J}(h) = y_{\xi} - y(h) - m(h)(\xi - x(h)).$$

On the mixed side, the only residual contribution is the exact fixed-shear corner $\{S = 0\} \cap \{K_v = 0\}$, already treated separately as a quotient-diagonal transport problem.

Thus the remaining global nonadjacent recurrence problem is cleanly a finite remote-incidence problem for the $G_{\xi,J}$, not a diffuse local-birth problem on a graph of generic gates.

Remark 12.214 (Ceiling of the remaining affine/global lane). The derivative-geometric reductions on the nonmixed/global side now have a clean positive content and a clean ceiling.

The positive content is that fixed-slope tangency/incidence has been reduced to a finite remote endpoint/branch incidence problem: support curves are strictly concave, have single-crossing and betweenness, and the endpoint-gap functions

$$G_{\xi,J}(h) = y_{\xi} - y(h) - m(h)(\xi - x(h))$$

are strictly monotone on ordered target branches. Thus each endpoint/branch pair contributes at most one remote tangency.

The ceiling is that this affine/pseudoline structure is universal for analytic plane curves. So no further purely planar, interval-graph, pseudoline, or local convexity argument can by itself finish this lane. Any final contradiction here must use RH-specific source structure constraining the possible zeros of the functions $G_{\xi,J}$.

Definition 12.215 (Minimal bad core). A normalized finite core is called bad if its quotient odd germ vanishes identically. A minimal bad core is a bad normalized finite core with minimal number of constituent pairs.

Theorem 12.216 (Defect-free quintic circuit reduction). *Let*

$$F_{\text{core}}(z) = \sum_{j=1}^k a_j F_{h_j}(z) + \mathcal{I}_{\geq 2}(z)$$

be a normalized finite core, and assume that the multi-pair interaction germ $\mathcal{I}_{\geq 2}(z)$ contributes nothing to the cubic and quintic coefficients. Write

$$A_5^{\dagger}(h) = u_5(h)I + v_5(h)S, \quad V(h) := (c(h), u_5(h), v_5(h)) \in \mathbf{C}^3,$$

and on $\{c \neq 0\}$ define

$$\Xi(h) := \left(\frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)} \right) \in \mathbf{C}^2.$$

If

$$F_{\text{core}}(z) \equiv 0,$$

then

$$\sum_{j=1}^k a_j V(h_j) = 0.$$

Equivalently, with

$$b_j := a_j c(h_j),$$

one has

$$\sum_{j=1}^k b_j = 0, \quad \sum_{j=1}^k b_j \Xi(h_j) = 0.$$

Thus the points $\Xi(h_j)$ are affinely dependent.

Proof. Taking the cubic coefficient in the identity

$$F_{\text{core}}(z) \equiv 0$$

and using the hypothesis that $\mathcal{I}_{\geq 2}$ contributes nothing at cubic order gives

$$\sum_{j=1}^k a_j c(h_j) = 0.$$

Taking the quintic fixed-sector coefficient and using the corresponding defect-free hypothesis at order 5 gives

$$\sum_{j=1}^k a_j A_5^f(h_j) = 0.$$

Since

$$A_5^f(h_j) = u_5(h_j)I + v_5(h_j)S$$

and I, S are linearly independent, this is equivalent to

$$\sum_{j=1}^k a_j u_5(h_j) = 0, \quad \sum_{j=1}^k a_j v_5(h_j) = 0.$$

Together these three identities are exactly

$$\sum_{j=1}^k a_j V(h_j) = 0.$$

If $c(h_j) \neq 0$ for all j , dividing the last two identities by $c(h_j)$ after setting $b_j = a_j c(h_j)$ yields

$$\sum_{j=1}^k b_j = 0, \quad \sum_{j=1}^k b_j \Xi(h_j) = 0,$$

which is the affine dependence statement. $\square$

Corollary 12.217 (Defect-free minimal quintic circuits have size at most four). *In the defect-free order- ≤ 5 model of Theorem 12.216, every minimal bad core has size at most 4. More precisely:*

- (1) if $k = 2$, then $\Xi(h_1) = \Xi(h_2)$;
- (2) if $k = 3$, then the three points $\Xi(h_1), \Xi(h_2), \Xi(h_3)$ are collinear;
- (3) if $k = 4$, then the four points $\Xi(h_1), \dots, \Xi(h_4)$ form an affine circuit in $\mathbf{C}^2$.

Proof. By Theorem 12.216, a minimal defect-free bad core produces a minimal affine dependence among finitely many points of the affine plane $\mathbf{C}^2$. Such a minimal affine dependent set has size at most 4. The cases $k = 2, 3, 4$ are the standard affine configurations in dimension 2: coincidence, collinear triple, or four-point affine circuit. $\square$

Definition 12.218 (Lifted v_5 -patch coordinates). On a connected interval $U \subset \{c \neq 0, v_5 \neq 0\}$, define

$$x(h) := \frac{v_5(h)}{c(h)}, \quad y(h) := \frac{u_5(h)}{c(h)}, \quad \sigma_v(h) := \frac{\Delta_7(h)}{c(h)v_5(h)}.$$

The associated lifted planar and space curves are

$$\Lambda_v(h) := (x(h), \sigma_v(h)) \in \mathbf{C}^2, \quad Q_v(h) := (y(h), x(h), \sigma_v(h)) \in \mathbf{C}^3.$$

Proposition 12.219 (Defect-free lifted affine dependence on a v_5 -patch). *Let*

$$F_{\text{core}}(z) = \sum_{j=1}^k a_j F_{h_j}(z) + \mathcal{I}_{\geq 2}(z)$$

be a normalized finite core, and assume:

- (1) *all parameters h_j lie in one connected interval $U \subset \{c \neq 0, v_5 \neq 0\}$;*
- (2) *the multi-pair interaction germ $\mathcal{I}_{\geq 2}(z)$ contributes nothing to the cubic, quintic, or septic coefficients.*

Then, after choosing on U the local septic gauge $v_7 = 0$, the vanishing identity

$$F_{\text{core}}(z) \equiv 0$$

implies

$$\sum_{j=1}^k b_j = 0, \quad \sum_{j=1}^k b_j y(h_j) = 0, \quad \sum_{j=1}^k b_j x(h_j) = 0, \quad \sum_{j=1}^k b_j \sigma_v(h_j) = 0,$$

where $b_j := a_j c(h_j)$. Equivalently, the points $Q_v(h_j) \in \mathbf{C}^3$ are affinely dependent.

Proof. The cubic and quintic identities are exactly those in Theorem 12.216, so after setting $b_j = a_j c(h_j)$ we obtain

$$\sum_{j=1}^k b_j = 0, \quad \sum_{j=1}^k b_j y(h_j) = 0, \quad \sum_{j=1}^k b_j x(h_j) = 0.$$

On the interval U we may choose the local septic gauge $v_7 = 0$. Then

$$u_7(h) = \frac{\Delta_7(h)}{v_5(h)} = c(h) \sigma_v(h).$$

Since the septic interaction is assumed absent, taking the septic fixed-sector coefficient of $F_{\text{core}} \equiv 0$ yields

$$\sum_{j=1}^k a_j u_7(h_j) = 0.$$

Substituting $u_7(h_j) = c(h_j) \sigma_v(h_j)$ gives

$$\sum_{j=1}^k b_j \sigma_v(h_j) = 0.$$

Hence the points $Q_v(h_j)$ are affinely dependent. □

Corollary 12.220 (Geometric meaning of defect-free lifted affine dependence). *Assume the hypotheses of Proposition 12.219. Then:*

- (1) *if $k = 2$, the two points $Q_v(h_1)$ and $Q_v(h_2)$ coincide;*
- (2) *if $k = 3$, the three points $Q_v(h_1), Q_v(h_2), Q_v(h_3)$ are collinear in $\mathbf{C}^3$;*
- (3) *if $k = 4$, the four points $Q_v(h_1), \dots, Q_v(h_4)$ are coplanar in $\mathbf{C}^3$.*

Proof. By Proposition 12.219, the points $Q_v(h_j)$ are affinely dependent. For $k = 2$, affine dependence of two points is exactly coincidence. For $k = 3$, affine dependence of three points is exactly collinearity. For $k = 4$, affine dependence of four points is exactly coplanarity. □

Corollary 12.221 (Explicit four-point determinant form on a v_5 -patch). *Assume the hypotheses of Proposition 12.219 with $k = 4$. Then the coplanarity of the four lifted points is equivalent to*

$$\det \begin{pmatrix} 1 & y(h_1) & x(h_1) & \sigma_v(h_1) \\ 1 & y(h_2) & x(h_2) & \sigma_v(h_2) \\ 1 & y(h_3) & x(h_3) & \sigma_v(h_3) \\ 1 & y(h_4) & x(h_4) & \sigma_v(h_4) \end{pmatrix} = 0.$$

Equivalently,

$$\det \begin{pmatrix} 1 & \frac{u_5(h_1)}{c(h_1)} & \frac{v_5(h_1)}{c(h_1)} & \frac{\Delta_7(h_1)}{c(h_1)v_5(h_1)} \\ 1 & \frac{u_5(h_2)}{c(h_2)} & \frac{v_5(h_2)}{c(h_2)} & \frac{\Delta_7(h_2)}{c(h_2)v_5(h_2)} \\ 1 & \frac{u_5(h_3)}{c(h_3)} & \frac{v_5(h_3)}{c(h_3)} & \frac{\Delta_7(h_3)}{c(h_3)v_5(h_3)} \\ 1 & \frac{u_5(h_4)}{c(h_4)} & \frac{v_5(h_4)}{c(h_4)} & \frac{\Delta_7(h_4)}{c(h_4)v_5(h_4)} \end{pmatrix} = 0.$$

Multiplying row j by $v_5(h_j)$ gives the canonical determinant identity

$$\det \begin{pmatrix} v_5(h_1) & u_5(h_1) & \frac{v_5(h_1)^2}{c(h_1)} & \frac{\Delta_7(h_1)}{c(h_1)} \\ v_5(h_2) & u_5(h_2) & \frac{v_5(h_2)^2}{c(h_2)} & \frac{\Delta_7(h_2)}{c(h_2)} \\ v_5(h_3) & u_5(h_3) & \frac{v_5(h_3)^2}{c(h_3)} & \frac{\Delta_7(h_3)}{c(h_3)} \\ v_5(h_4) & u_5(h_4) & \frac{v_5(h_4)^2}{c(h_4)} & \frac{\Delta_7(h_4)}{c(h_4)} \end{pmatrix} = 0.$$

Equivalently, multiplying instead row j by $c(h_j)v_5(h_j)$ gives the fully polynomial version

$$\det \begin{pmatrix} c(h_1)v_5(h_1) & u_5(h_1)v_5(h_1) & v_5(h_1)^2 & \Delta_7(h_1) \\ c(h_2)v_5(h_2) & u_5(h_2)v_5(h_2) & v_5(h_2)^2 & \Delta_7(h_2) \\ c(h_3)v_5(h_3) & u_5(h_3)v_5(h_3) & v_5(h_3)^2 & \Delta_7(h_3) \\ c(h_4)v_5(h_4) & u_5(h_4)v_5(h_4) & v_5(h_4)^2 & \Delta_7(h_4) \end{pmatrix} = 0.$$

Proof. For $k = 4$, Corollary 12.220 shows that the four lifted points

$$Q_v(h_j) = (y(h_j), x(h_j), \sigma_v(h_j))$$

are coplanar. This is equivalent to the vanishing of the affine 4×4 determinant with first column 1. The displayed reformulations follow by substituting the definitions of x, y, σ_v and multiplying rows by the indicated nonzero factors. $\square$

Remark 12.222 (Finite-core meaning of the lifted v_5 -patch geometry). The defect-free finite-core reduction now places the lifted v_5 -patch geometry into a sharp global context.

Indeed, by Theorem 12.216, Corollary 12.217, and Proposition 12.219, defect-free bad cores on a good v_5 -patch reduce to affine dependence of the lifted points

$$Q_v(h) = (y(h), x(h), \sigma_v(h)),$$

with effective ceiling

$$k \leq 4.$$

Thus the three low core sizes have the following exact lifted meanings:

$$k = 2 \iff Q_v(h_1) = Q_v(h_2),$$

$$k = 3 \iff Q_v(h_1), Q_v(h_2), Q_v(h_3) \text{ are collinear,}$$

$$k = 4 \iff Q_v(h_1), \dots, Q_v(h_4) \text{ are coplanar.}$$

So the lifted 3-point and 4-point analyses are not auxiliary side computations: they are exactly the $k = 3$ and $k = 4$ geometric layers in the finite-core program. In particular, once the lifted 3-point branch is globally rigid on good v_5 -patches, the remaining finite-core burden shifts

upward to the two-point source-level merger problem and to the mixed/nonmixed lifted 4-point mechanisms.

Remark 12.223 (Parallel u_5 -patch version). There is a completely parallel local story on any connected interval in $\{c \neq 0, u_5 \neq 0\}$, obtained by choosing the local septic gauge $u_7 = 0$ instead of $v_7 = 0$. Then one defines

$$\tau_u(h) := -\frac{\Delta_7(h)}{c(h)u_5(h)}, \quad Q_u(h) := \left(\frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)}, \tau_u(h) \right),$$

and the same defect-free affine-dependence conclusions hold with Q_v replaced by Q_u . Since the present one-pair geometry already uses

$$x(h) = \frac{v_5(h)}{c(h)}$$

as the distinguished ECT coordinate, the $v_5 \neq 0$ patch is the natural first setting and will be used below.

Proposition 12.224 (Centered quintic-patch dichotomy). *Assume the centered diagonal family is a genuine analytic branch in the underlying h -geometry, and that the centered scalar package from Remark 12.379 holds on that branch. Then, after shrinking, there exist one-sided punctured subarcs on which at least one of the following holds:*

(1)

$$c \neq 0, \quad M \neq 0, \quad v_5 \neq 0,$$

so the branch enters a good v_5 -patch;

(2)

$$c \neq 0, \quad u_5 \neq 0,$$

so the parallel u_5 -patch of Remark 12.223 applies.

In particular, the centered diagonal branch enters some quintic patch on one-sided punctured arcs.

Proof. By Corollary 12.383,

$$A_5^f(s) \neq 0,$$

so at least one of u_5 or v_5 is a nonzero analytic germ on the centered branch.

If $v_5 \not\equiv 0$, then its zeros are isolated. Since

$$c(s) = C_0 + C_2 s^2 + \cdots, \quad C_0 \neq 0,$$

we also have $c \neq 0$ near the basepoint. On any one-sided punctured arc where $v_5 \neq 0$, define

$$x = \frac{v_5}{c}.$$

This is a nonconstant analytic germ. If $M \equiv 0$ on such an arc, then

$$x' = \left(\frac{v_5}{c} \right)' = \frac{M}{c^2} \equiv 0,$$

so x would be constant, contradiction. Hence M is not identically zero on that arc, and by analyticity its zeros are isolated. After shrinking again, one obtains

$$c \neq 0, \quad M \neq 0, \quad v_5 \neq 0,$$

i.e. a good v_5 -patch.

If instead $v_5 \equiv 0$, then $u_5 \neq 0$. Its zeros are isolated, so after shrinking to a one-sided punctured arc one has

$$c \neq 0, \quad u_5 \neq 0,$$

and the u_5 -patch of Remark 12.223 applies. □

Corollary 12.225 (Conditional punctured v_5 -patch promotion). *Under the hypotheses of Proposition 12.224, if*

$$v_5 \not\equiv 0$$

on the centered diagonal branch, then after shrinking there exist one-sided punctured subarcs contained in

$$\{c \neq 0, M \neq 0, v_5 \neq 0\}.$$

Hence on such arcs

$$x = \frac{v_5}{c}$$

is automatically a valid local coordinate.

Proof. This is exactly the first branch of the proof of Proposition 12.224. $\square$

Proposition 12.226 (Linear independence of $1, u_5/c, v_5/c$). *Let J be a connected open interval on which*

$$c \neq 0, \quad M \neq 0.$$

Then the three functions

$$1, \quad \frac{u_5}{c}, \quad \frac{v_5}{c}$$

are linearly independent on J .

Proof. Assume there exist constants $A, B, C \in \mathbf{C}$, not all zero, such that

$$A + B \frac{u_5}{c} + C \frac{v_5}{c} \equiv 0 \quad \text{on } J.$$

Multiplying by c gives

$$A c + B u_5 + C v_5 \equiv 0.$$

If $B = 0$, then

$$A c + C v_5 \equiv 0,$$

so v_5/c is constant on J . But

$$\left(\frac{v_5}{c}\right)' = \frac{M}{c^2},$$

and $M \neq 0$ on J . This is impossible.

Hence $B \neq 0$, and after renormalization we may write

$$u_5 = \alpha v_5 + \beta c$$

for constants α, β .

Now isolate the graded slice corresponding to

$$(K_1, \eta_2, q^{(5)}).$$

By Lemmas 12.16 and 12.18, the tagged monomial

$$\kappa(h) \eta_2(h) q^{(5)}(h)$$

occurs with nonzero coefficient in $\text{tr } J_5^{\text{pair}}(h)$. Since

$$A_5^{\text{f}}(h) = u_5(h)I + v_5(h)S$$

and S is traceless, this means exactly that the same tagged monomial occurs in the I -coefficient $u_5(h)$.

It cannot occur in $c(h)$, since c is cubic and carries no quintic $q^{(5)}$ -slice. It also cannot occur in $v_5(h)$. Indeed, on the tagged slice, the only surviving quintic K_1 -transport channels are those already isolated in Lemma 12.18. For the channel

$$X_0 J X_4 - X_4 J X_0,$$

Lemma 12.17 shows that the relevant $(\eta_2, q^{(5)})$ -slice is diagonal, hence contributes only to the I -component. For the commutator channel

$$X_3 J X_1 - X_1 J X_3,$$

Proposition 12.40 gives

$$v_5^{(\text{comm})} = \gamma u - \alpha w.$$

On the tagged slice the η_2 -input in X_1 is purely off-diagonal, so $u = w = 0$, while the $q^{(5)}$ -input in X_3 is diagonal. Hence

$$v_5^{(\text{comm})} = 0$$

on that tagged slice.

Thus the monomial $\kappa\eta_2q^{(5)}$ occurs in u_5 but in neither v_5 nor c , contradicting

$$u_5 = \alpha v_5 + \beta c.$$

This proves the claim. $\square$

Corollary 12.227 (Automatic u_5 -patch on a good $M \neq 0$ interval). *Let J be a connected open interval on which*

$$c \neq 0, \quad M \neq 0.$$

Then $u_5 \not\equiv 0$ on J . Consequently, every connected component of

$$J \setminus u_5^{-1}(0)$$

is a valid local u_5 -patch in the sense of Remark 12.223.

Proof. If $u_5 \equiv 0$ on J , then the three functions

$$1, \quad \frac{u_5}{c}, \quad \frac{v_5}{c}$$

would be linearly dependent, contradicting Proposition 12.226. Thus $u_5 \not\equiv 0$ on J , so its zeros are isolated and each connected component of $J \setminus u_5^{-1}(0)$ lies in $\{c \neq 0, u_5 \neq 0\}$. The parallel u_5 -patch geometry then follows from Remark 12.223. $\square$

Proposition 12.228 (Non-affinity of the lifted septic scalar). *Let J be a connected open interval contained in the quintic-generic region and in*

$$\{h : c(h) \neq 0, v_5(h) \neq 0\}.$$

Then there do not exist constants $A, B \in \mathbf{C}$ such that

$$\sigma_v(h) \equiv A x(h) + B \quad \text{on } J.$$

Equivalently,

$$\Delta_7(h) \not\equiv A v_5(h)^2 + B c(h)v_5(h) \quad \text{on } J.$$

Proof. If

$$\sigma_v(h) \equiv A x(h) + B,$$

then multiplying by $c(h)v_5(h)$ gives

$$\Delta_7(h) \equiv A v_5(h)^2 + B c(h)v_5(h).$$

The right-hand side is built only from the cubic and quintic layers and therefore has no direct septic $q^{(7)}$ -slice. But by Proposition 12.48, the direct $q^{(7)}$ -slice of Δ_7 is generically nonzero on J . This is a contradiction. $\square$

Definition 12.229 (Lifted derivative numerator and curvature numerator). On $\{c \neq 0, v_5 \neq 0\}$, define

$$P_v(h) := c(h)v_5(h)\Delta_7'(h) - \Delta_7(h)(c(h)v_5(h))'.$$

Then

$$\sigma_v'(h) = \frac{P_v(h)}{c(h)^2 v_5(h)^2}.$$

Define also the affine curvature numerator of the lifted planar curve

$$\Lambda_v(h) = (x(h), \sigma_v(h))$$

by

$$K_v(h) := x'(h)\sigma_v''(h) - \sigma_v'(h)x''(h).$$

Proposition 12.230 (Curvature bridge for the lifted v_5 -patch curve). *On $\{c \neq 0, M \neq 0, v_5 \neq 0\}$,*

$$\frac{d\sigma_v}{dx} = \frac{P_v}{M v_5^2},$$

and

$$K_v(h) = \frac{M(h)^2}{c(h)^4} \frac{d}{dh} \left(\frac{P_v(h)}{M(h) v_5(h)^2} \right).$$

In particular, if $K_v \equiv 0$ on a connected open interval $J \subset \{c \neq 0, M \neq 0, v_5 \neq 0\}$, then σ_v is affine in x on J , contradicting Proposition 12.228. Hence

$$K_v \not\equiv 0$$

on every such interval.

Proof. Since

$$x = \frac{v_5}{c}, \quad x' = \frac{M}{c^2}, \quad \sigma'_v = \frac{P_v}{c^2 v_5^2},$$

we obtain

$$\frac{d\sigma_v}{dx} = \frac{\sigma'_v}{x'} = \frac{P_v}{M v_5^2}.$$

Differentiating with respect to h and multiplying by

$$x' = \frac{M}{c^2}$$

gives

$$K_v = x'^2 \frac{d}{dx} \left(\frac{d\sigma_v}{dx} \right) = \frac{M^2}{c^4} \frac{d}{dh} \left(\frac{P_v}{M v_5^2} \right).$$

If $K_v \equiv 0$ on a connected interval with $M \neq 0$, then

$$\frac{P_v}{M v_5^2}$$

is constant there, so $d\sigma_v/dx$ is constant and σ_v is affine in x . This contradicts Proposition 12.228. $\square$

Corollary 12.231 (Local defect-free three-point exclusion on lifted ECT subintervals). *Let $J \subset \{c \neq 0, M \neq 0, v_5 \neq 0\}$ be a connected interval on which K_v has fixed nonzero sign. Then the triple*

$$1, \quad x, \quad \sigma_v$$

is an ECT system on J . In particular, no affine line meets the lifted planar curve

$$\Lambda_v(h) = (x(h), \sigma_v(h))$$

in three points of J . Hence the defect-free three-point branch is excluded on such a subinterval.

Proof. The Wronskian of $1, x, \sigma_v$ is exactly K_v . If K_v has fixed nonzero sign on J , the ECT criterion applies. $\square$

Corollary 12.232 (Lifted order-7 Wronskian is not identically zero). *Let J be a connected open interval contained in the quintic-generic region and in*

$$\{h : c(h) \neq 0, M(h) \neq 0, v_5(h) \neq 0\}.$$

Then

$$T_v(h) := W_h \left(1, \frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h) v_5(h)} \right)$$

is not identically zero on J .

Proof. Write

$$y := \frac{u_5}{c}, \quad x := \frac{v_5}{c}, \quad \sigma_v := \frac{\Delta_7}{c v_5}.$$

If $T_v \equiv 0$ on J , then by analyticity the four functions

$$1, y, x, \sigma_v$$

are locally linearly dependent on J , hence linearly dependent on all of the connected interval J . By Proposition 12.226, the first three are linearly independent, so there exist constants A, B, C such that

$$\sigma_v = A + By + Cx.$$

Multiplying by $c v_5$ gives

$$\Delta_7 = A c v_5 + B u_5 v_5 + C v_5^2.$$

The right-hand side is built entirely from the cubic/quintic layers and therefore carries no direct $q^{(7)}$ -slice. This contradicts Proposition 12.48, which gives

$$\Delta_7^{(q^{(7)})} \neq 0$$

on J . □

Proposition 12.233 (Exact x -coordinate identities for the lifted 4-point invariants). *On a good v_5 -patch, write*

$$x = \frac{v_5}{c}, \quad Y = \frac{u_5}{c}, \quad S = \sigma_v = \frac{\Delta_7}{c v_5}.$$

Then:

(1) *the lifted planar curvature numerator satisfies*

$$K_v = (x')^3 S''(x);$$

(2) *the lifted 4-point Wronskian satisfies*

$$T_v = (x')^6 W_x(1, Y, x, S) = (x')^6 (S'' Y''' - S''' Y'');$$

(3) *equivalently,*

$$T_v = (x')^6 S''^2 \frac{d}{dx} \left(\frac{Y''}{S''} \right) = K_v^2 \frac{d}{dx} \left(\frac{Y''}{S''} \right).$$

In particular, K_v and S'' have the same zero set and the same vanishing order on every good v_5 -patch.

Proof. By definition,

$$K_v = x' \sigma_v'' - \sigma_v' x''.$$

Since $S = \sigma_v$ as a function of the local coordinate x , we have

$$\sigma_v' = S' x', \quad \sigma_v'' = S'' (x')^2 + S' x'',$$

hence

$$K_v = x' (S'' (x')^2 + S' x'') - S' x' x'' = (x')^3 S''.$$

Next,

$$T_v = W_h(1, Y, x, S).$$

Under reparametrization $h \mapsto x(h)$, a 4×4 Wronskian gains the factor $(x')^{0+1+2+3} = (x')^6$, so

$$T_v = (x')^6 W_x(1, Y, x, S).$$

Now

$$W_x(1, Y, x, S) = \det \begin{pmatrix} 1 & Y & x & S \\ 0 & Y' & 1 & S' \\ 0 & Y'' & 0 & S'' \\ 0 & Y''' & 0 & S''' \end{pmatrix} = S'' Y''' - S''' Y'',$$

which proves the second identity.

Finally,

$$S''Y''' - S'''Y'' = S''^2 \frac{d}{dx} \left(\frac{Y''}{S''} \right),$$

and using $K_v = (x')^3 S''$ gives

$$(x')^6 S''^2 = K_v^2.$$

This yields the third identity. $\square$

Proposition 12.234 (Canonical scalar invariants $\mathcal{Z}$ and $\mathcal{R}$ on a nonmixed lifted patch). *On a good v_5 -patch, write*

$$x = \frac{v_5}{c}, \quad Y = \frac{u_5}{c}, \quad S = \sigma_v = \frac{\Delta_7}{c v_5}.$$

On the nonmixed subregion where $K_v \neq 0$, define

$$\mathcal{Z} := \frac{Y''}{S''}, \quad \mathcal{R} := \frac{d\mathcal{Z}}{dx}.$$

Then:

(1)

$$\mathcal{R} = \frac{T_v}{K_v^2};$$

(2) *under the affine gauge*

$$Y \mapsto Y + \alpha + \beta x + \gamma S,$$

one has

$$\mathcal{Z} \mapsto \mathcal{Z} + \gamma, \quad \mathcal{R} \mapsto \mathcal{R}.$$

Hence $\mathcal{R}$ is the canonical scalar invariant of the lifted 4-point branch on every nonmixed patch.

Proof. By Proposition 12.233,

$$T_v = (x')^6 S''^2 \frac{d}{dx} \left(\frac{Y''}{S''} \right) = K_v^2 \frac{d}{dx} \left(\frac{Y''}{S''} \right),$$

so

$$\mathcal{R} = \frac{d}{dx} \left(\frac{Y''}{S''} \right) = \frac{T_v}{K_v^2}.$$

For the gauge transformation, one has

$$(Y + \alpha + \beta x + \gamma S)'' = Y'' + \gamma S'',$$

hence

$$\mathcal{Z} \mapsto \frac{Y'' + \gamma S''}{S''} = \mathcal{Z} + \gamma.$$

Differentiating with respect to x gives $\mathcal{R} \mapsto \mathcal{R}$. $\square$

Remark 12.235 (Below-lifted Wronskian-ratio reformulation). Let

$$x = \frac{v_5}{c}, \quad Y = \frac{u_5}{c}, \quad S = \frac{\Delta_7}{c v_5}, \quad \lambda := \frac{\Delta_7}{v_5}, \quad M := c v'_5 - c' v_5.$$

For any scalar germ f , define

$$J_f(h) := \det \begin{pmatrix} c(h) & v_5(h) & f(h) \\ c'(h) & v'_5(h) & f'(h) \\ c''(h) & v''_5(h) & f''(h) \end{pmatrix}.$$

Then

$$\frac{d^2}{dx^2} \left(\frac{f}{c} \right) = \frac{c^3}{M^3} J_f.$$

Applying this to $f = u_5$ and $f = \lambda$ gives

$$Y'' = \frac{c^3}{M^3} J_{u_5}, \quad S'' = \frac{c^3}{M^3} J_\lambda,$$

hence

$$\mathcal{Z} := \frac{Y''}{S''} = \frac{J_{u_5}}{J_\lambda}.$$

Differentiating in the x -coordinate yields

$$\mathcal{R} := \mathcal{Z}' = \frac{T_v}{K_v^2} = \frac{c^2}{M} \frac{d}{dh} \left(\frac{J_{u_5}}{J_\lambda} \right).$$

Thus the lifted 4-point branch may be viewed equivalently as a parity/divisibility problem for the Wronskian ratio J_{u_5}/J_λ . On the nonmixed side, however, the later source-level no-go results show that the ratio/Wronskian package by itself is not yet restrictive enough: the remaining burden is a genuine source-coupling theorem, not a generic Wronskian-ratio argument.

Proposition 12.236 (Source-level Wronskian for the nonmixed 4-point branch). *Define*

$$\mathcal{W}_{\text{src}} := J_\lambda J'_{u_5} - J_{u_5} J'_\lambda.$$

Then

$$\mathcal{R} = \frac{c^2}{M} \frac{\mathcal{W}_{\text{src}}}{J_\lambda^2}.$$

In particular, on every nonmixed arc on which $J_\lambda \neq 0$, the analytic scalars $\mathcal{R}$ and $\mathcal{W}_{\text{src}}$ differ only by a nonvanishing analytic factor. Hence:

- (1) $\mathcal{R} = 0 \iff \mathcal{W}_{\text{src}} = 0$;
- (2) $\text{ord } \mathcal{R} = \text{ord } \mathcal{W}_{\text{src}}$;
- (3) the local parity classification of isolated zeros of $\mathcal{R}$ transfers verbatim to $\mathcal{W}_{\text{src}}$.

Moreover, under the affine gauge change

$$u_5 \mapsto u_5 + \alpha c + \beta v_5 + \gamma \lambda$$

one has

$$J_{u_5} \mapsto J_{u_5} + \gamma J_\lambda,$$

and therefore

$$\mathcal{W}_{\text{src}} \mapsto \mathcal{W}_{\text{src}}.$$

So $\mathcal{W}_{\text{src}}$ is the canonical source-level gauge-invariant nonmixed scalar.

Proof. Differentiate the ratio

$$\mathcal{Z} = \frac{J_{u_5}}{J_\lambda}$$

from Remark 12.235:

$$\frac{d}{dh} \left(\frac{J_{u_5}}{J_\lambda} \right) = \frac{J_\lambda J'_{u_5} - J_{u_5} J'_\lambda}{J_\lambda^2} = \frac{\mathcal{W}_{\text{src}}}{J_\lambda^2}.$$

Substituting this into

$$\mathcal{R} = \frac{c^2}{M} \frac{d}{dh} \left(\frac{J_{u_5}}{J_\lambda} \right)$$

gives the displayed identity.

On a nonmixed arc, $J_\lambda \neq 0$ and $c^2/M \neq 0$, so $\mathcal{R}$ and $\mathcal{W}_{\text{src}}$ differ by a nonvanishing analytic factor. This proves the zero-set and parity statements.

For gauge invariance, the terms αc and βv_5 do not change J_{u_5} , while $\gamma \lambda$ adds γJ_λ . Therefore

$$(J_{u_5} + \gamma J_\lambda)' = J'_{u_5} + \gamma J'_\lambda,$$

and

$$J_\lambda (J'_{u_5} + \gamma J'_\lambda) - (J_{u_5} + \gamma J_\lambda) J'_\lambda = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda.$$

□

Remark 12.237 (Determinant-level parity on a centered good patch). Assume the centered scalar package from Remark 12.379 and suppose that $x = v_5/c$ is being used as the induced odd collision-side coordinate. Then

$$u_5, \lambda := \frac{\Delta_7}{v_5} \text{ are odd in } s, \quad c \text{ is even in } s, \quad v_5 \text{ is odd in } s.$$

Hence the rows of the defining determinants for J_{u_5} and J_λ have parities

$$(e, o, o), \quad (o, e, e), \quad (e, o, o),$$

so each determinant monomial is odd. Therefore

$$J_{u_5}(s) \text{ and } J_\lambda(s) \text{ are odd in } s.$$

Consequently,

$$\mathcal{Z} = \frac{J_{u_5}}{J_\lambda}$$

is even and

$$\mathcal{R} = \mathcal{Z}'$$

is odd, in agreement with the parity package obtained from the direct Y''/S'' -analysis.

Corollary 12.238 (Centered parity of the source-level Wronskian). *Under the hypotheses of Remark 12.237, the source-level Wronskian*

$$\mathcal{W}_{\text{src}}(s) = J_\lambda(s) J'_{u_5}(s) - J_{u_5}(s) J'_\lambda(s)$$

is odd in the centered variable s .

Consequently, any isolated centered zero of $\mathcal{W}_{\text{src}}$ has odd order if and only if it is a potentially branch-seeding zero for the nonmixed 4-point geometry; even-order centered zeros are locally sterile.

Proof. By Remark 12.237,

$$J_{u_5}(s) \text{ and } J_\lambda(s) \text{ are odd in } s.$$

Hence

$$J'_{u_5}(s) \text{ and } J'_\lambda(s) \text{ are even in } s.$$

Therefore

$$\mathcal{W}_{\text{src}}(s) = J_\lambda(s) J'_{u_5}(s) - J_{u_5}(s) J'_\lambda(s)$$

is odd in s . The final statement follows from Proposition 12.236, which transfers the parity classification from $\mathcal{R}$ to $\mathcal{W}_{\text{src}}$. $\square$

Corollary 12.239 (Mixed low-order exclusion on a centered good v_5 -patch). *Under the hypotheses of Remark 12.237, one has*

$$\mathcal{Z}(x) \text{ even}, \quad \mathcal{R}(x) \text{ odd}, \quad K_v(x) \text{ odd}.$$

Consequently,

$$(r, \text{ord}_0 \mathcal{Z}) \notin \{(1, 1), (1, 3), (2, 2)\}.$$

Equivalently, none of the explicit mixed low-order models

$$(m, n) = (3, 4), (3, 6), (4, 6)$$

can occur on such a centered good patch.

Proof. Since $\mathcal{Z}$ is even, its finite vanishing order must be even, so the cases

$$(r, \text{ord}_0 \mathcal{Z}) = (1, 1), (1, 3)$$

are impossible. Since K_v is odd, its vanishing order $r = \text{ord}_0 K_v$ must be odd, so

$$(r, \text{ord}_0 \mathcal{Z}) = (2, 2)$$

is impossible as well. $\square$

Theorem 12.240 (Local exhaustion of the mixed 4-point branch near the centered base). *Assume the centered scalar package from Remark 12.379, the right-pincer local reduction from Corollary 12.388, and the finite-jet scalar descent from Corollary 12.377. Then every sufficiently small nearby genuinely new mixed 4-point obstruction near the centered diagonal base is excluded.*

Equivalently, near the centered base the mixed 4-point branch is locally exhausted: the clean branch $V_1 \neq 0$ is excluded by the centered good- v_5 -patch parity package, while every late- v_5 branch is either locally sterile on the u_5 -patch or already absorbed into the defect-free planar quintic-circuit regime.

Proof. By Corollary 12.388, every sufficiently small nearby genuinely new mixed 4-point obstruction must lie on the clean branch

$$V_1 \neq 0, \quad \text{equivalently} \quad \text{ord}_s v_5 = 1.$$

On that branch, Corollary 12.377 gives

$$c(s) = C_0 + C_2 s^2 + C_4 s^4 + \cdots.$$

Since $v_5(s)$ is odd with $V_1 \neq 0$, the induced patch coordinate

$$x(s) = \frac{v_5(s)}{c(s)}$$

is odd and has odd inverse. Since $u_5(s)$ is odd and $\Delta_7(s)$ is even, it follows that

$$Y(x) = \frac{u_5}{c}(x) \quad \text{and} \quad S(x) = \frac{\Delta_7}{cv_5}(x)$$

are odd in x . Hence

$$\mathcal{Z}(x) = \frac{Y''}{S''} \text{ is even,} \quad K_v = (x')^3 S''$$

has odd vanishing order, so by Corollary 12.239 none of the explicit mixed low-order obstruction patterns

$$(r, \text{ord}_0 \mathcal{Z}) \in \{(1, 1), (1, 3), (2, 2)\}$$

can occur on the clean branch.

Thus no sufficiently small nearby genuinely new mixed 4-point obstruction remains. The late- v_5 branches were already removed by Corollary 12.388, and the clean branch is excluded as above. $\square$

Remark 12.241 (What the mixed local exhaustion theorem now accomplishes). Theorem 12.240 closes the mixed 4-point route locally near the centered diagonal base in a self-contained way: the clean branch is eliminated by the canonical scalar descent together with the centered good- v_5 -patch parity package, while every late- v_5 branch is already removed by the u_5 -patch right-pincer package as either locally sterile or already-quintic.

Remark 12.242 (What remains after local mixed exhaustion). Theorem 12.240 exhausts the mixed 4-point branch in the centered-base local regime. Any remaining mixed 4-point escape must therefore be genuinely global/nonlocal: it cannot come from a new local mixed septic-jet pattern near the centered diagonal base.

Equivalently, the later mixed/global problem is no longer a local normal-form problem. It is a finite-core / global propagation problem for the normalized line-incidence geometry, with the residual mixed two-point derivative geometry concentrated on the overlap exceptional set and, after the off- $K_v = 0$ handoffs, sharply focused on the exact fixed-shear $K_v = 0$ slice.

Remark 12.243 (How mixed local exhaustion constrains later hybrid routes). Theorem 12.240 is useful beyond the mixed local lane itself. It shows that any later hybrid argument trying to constrain the nonmixed repeated-source / source-coupling branch cannot appeal to a new local mixed 4-point septic jet pattern near the centered diagonal base, because that possibility is already exhausted. Thus any remaining mixed input capable of affecting the nonmixed branch must be genuinely global/nonlocal, or must enter through external canonical/multi-pair structure rather than through a new local mixed obstruction.

Remark 12.244 (A second below-lifted ratio does not add a new obstruction). Let

$$\Theta := \frac{W_{u_5}}{W},$$

where W denotes the corresponding projective-chart Wronskian in the $(c, v_5, \cdot)$ -frame. In the gauge $c \equiv 1$, $v_5 = x$, one has the exact relation

$$\Theta = \mathcal{Z} \frac{S''}{xS'' + 2S'}, \quad (\mathcal{Z} - x\Theta)S'' = 2\Theta S'.$$

So Θ does not provide an independent local obstruction beyond $(\mathcal{Z}, \mathcal{R})$: whenever $2\Theta/(\mathcal{Z} - x\Theta)$ is analytic, one can solve locally for S and then recover Y from $Y'' = \mathcal{Z}S''$. In particular, attacks based only on a generic differential relation between $\mathcal{Z}$ and Θ are too weak to close the 4-point branch.

Corollary 12.245 (Local defect-free four-point exclusion on lifted ECT subintervals). *Let $J \subset \{c \neq 0, M \neq 0, v_5 \neq 0\}$ be a connected interval on which T_v has fixed nonzero sign. Then the quadruple*

$$1, \quad \frac{u_5}{c}, \quad \frac{v_5}{c}, \quad \frac{\Delta_7}{c v_5}$$

is an ECT system on J . Equivalently, no affine plane meets the lifted space curve

$$Q_v(h) = \left(\frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h) v_5(h)} \right)$$

in four points of J . Hence the defect-free four-point branch is excluded on such a subinterval.

Proof. The Wronskian of the displayed quadruple is exactly T_v . If T_v has fixed nonzero sign on J , the ECT criterion applies. $\square$

Lemma 12.246 (Osculating determinant expansion for shrinking affine circuits). *Let $Q : I \rightarrow \mathbf{C}^d$ be real-analytic on an interval I , and define*

$$\Delta_{d+1}(t_0, \dots, t_d) := \det \begin{pmatrix} 1 & Q(t_0) \\ \vdots & \vdots \\ 1 & Q(t_d) \end{pmatrix}.$$

Fix $h_ \in I$, and let*

$$t_j = h_* + \varepsilon \lambda_j \quad (j = 0, \dots, d),$$

where the λ_j are pairwise distinct and $\varepsilon \rightarrow 0$. Then

$$\Delta_{d+1}(t_0, \dots, t_d) = \frac{\varepsilon^{d(d+1)/2}}{\prod_{k=1}^d k!} V(\lambda_0, \dots, \lambda_d) \det(Q'(h_*), Q''(h_*), \dots, Q^{(d)}(h_*)) + O(\varepsilon^{d(d+1)/2+1}),$$

where

$$V(\lambda_0, \dots, \lambda_d) := \prod_{0 \leq i < j \leq d} (\lambda_j - \lambda_i)$$

is the Vandermonde determinant.

In particular, if there is a sequence of exact affine $(d+1)$ -circuits

$$\Delta_{d+1}(t_{0,n}, \dots, t_{d,n}) = 0$$

with pairwise distinct nodes $t_{j,n} \rightarrow h_$, then*

$$\det(Q'(h_*), Q''(h_*), \dots, Q^{(d)}(h_*)) = 0.$$

Proof. Write the Taylor expansion

$$Q(h_* + \varepsilon \lambda) = Q(h_*) + \sum_{m=1}^d \frac{\varepsilon^m \lambda^m}{m!} Q^{(m)}(h_*) + O(\varepsilon^{d+1}),$$

uniformly for λ in bounded sets. Subtract the first row $(1, Q(t_0))$ from each of the remaining rows. The first column then becomes $(1, 0, \dots, 0)^\top$, so Δ_{d+1} reduces to the determinant of the $d \times d$ matrix whose j -th row ($j = 1, \dots, d$) is

$$Q(t_j) - Q(t_0) = \sum_{m=1}^d \frac{\varepsilon^m (\lambda_j^m - \lambda_0^m)}{m!} Q^{(m)}(h_*) + O(\varepsilon^{d+1}).$$

Factor the column vectors $Q'(h_*), \dots, Q^{(d)}(h_*)$. This gives

$$\Delta_{d+1}(t_0, \dots, t_d) = \det(Q'(h_*), \dots, Q^{(d)}(h_*)) \det\left(\frac{\varepsilon^m (\lambda_j^m - \lambda_0^m)}{m!}\right)_{1 \leq j, m \leq d} + O(\varepsilon^{d(d+1)/2+1}).$$

Pulling $\varepsilon^m/m!$ from the m -th column yields the factor

$$\frac{\varepsilon^{1+\dots+d}}{\prod_{k=1}^d k!} = \frac{\varepsilon^{d(d+1)/2}}{\prod_{k=1}^d k!}.$$

The remaining determinant is the standard Vandermonde determinant $V(\lambda_0, \dots, \lambda_d)$. This proves the expansion.

If $\Delta_{d+1}(t_{0,n}, \dots, t_{d,n}) = 0$ for a sequence of shrinking distinct-node circuits, divide by the nonzero Vandermonde factor and let $\varepsilon \rightarrow 0$. The leading term must vanish, giving

$$\det(Q'(h_*), Q''(h_*), \dots, Q^{(d)}(h_*)) = 0. \quad \square$$

Remark 12.247 (Planar curvature and lifted torsion as the first derivative obstructions). Lemma 12.246 shows that shrinking exact affine circuits are controlled, at first nontrivial order, by derivative-side osculating determinants. For a planar curve q , shrinking exact collinear triples force

$$\det(q', q'') = 0,$$

and for a space curve Q , shrinking exact coplanar quadruples force

$$\det(Q', Q'', Q''') = 0.$$

Thus the defect-free lifted 3-point and 4-point branches are not merely incidence problems: they are rooted at curvature- and torsion-type derivative invariants of the corresponding lifted curves.

Proposition 12.248 (Shrinking exact lifted 3-point circuits concentrate into $K_v = 0$). *Let $P \subseteq \{c \neq 0, v_5 \neq 0, M \neq 0\}$ be compact, and use*

$$x = \frac{v_5}{c}, \quad S = \sigma_v = \frac{\Delta_7}{c v_5}.$$

Suppose there is a sequence of distinct triples

$$x_{1,n}, x_{2,n}, x_{3,n} \in x(P), \quad x_{1,n}, x_{2,n}, x_{3,n} \rightarrow x_*,$$

such that

$$\det \begin{pmatrix} 1 & x_{1,n} & S(x_{1,n}) \\ 1 & x_{2,n} & S(x_{2,n}) \\ 1 & x_{3,n} & S(x_{3,n}) \end{pmatrix} = 0 \quad \text{for every } n.$$

Then

$$K_v(x_*) = 0.$$

Proof. Apply Lemma 12.246 with $d = 2$ to the planar lifted curve

$$q(x) := (x, S(x)).$$

The exact collinearity condition in the statement is precisely the vanishing of

$$\det \begin{pmatrix} 1 & x_1 & S(x_1) \\ 1 & x_2 & S(x_2) \\ 1 & x_3 & S(x_3) \end{pmatrix}.$$

Hence shrinking exact lifted 3-point circuits force

$$\det(q'(x_*), q''(x_*)) = 0.$$

Since

$$q'(x) = (1, S'(x)), \quad q''(x) = (0, S''(x)),$$

one gets

$$\det(q'(x_*), q''(x_*)) = S''(x_*).$$

On the x -patch, Proposition 12.233 gives

$$K_v = (x')^3 S'',$$

and $x' \neq 0$ on a good v_5 -patch. Therefore $S''(x_*) = 0$ implies

$$K_v(x_*) = 0. \quad \square$$

Corollary 12.249 (Uniform small-diameter exclusion away from $K_v = 0$). *Let $P_0 \subseteq \{c \neq 0, v_5 \neq 0, M \neq 0\}$ be compact and assume*

$$P_0 \cap \{K_v = 0\} = \emptyset.$$

Then there exists $\rho_3 > 0$ such that no exact lifted 3-point circuit contained in P_0 has x -diameter $< \rho_3$.

Proof. Otherwise there would exist a sequence of exact lifted 3-point circuits in P_0 with diameters tending to 0. By compactness, after passing to a subsequence the three nodes would converge to some $x_* \in x(P_0)$. This contradicts Proposition 12.248. $\square$

Proposition 12.250 (Shrinking exact lifted 4-point circuits concentrate into $T_v = 0$). *Let $P \subseteq \{c \neq 0, v_5 \neq 0, M \neq 0\}$ be compact, and use*

$$x = \frac{v_5}{c}, \quad Y = \frac{u_5}{c}, \quad S = \sigma_v = \frac{\Delta_7}{c v_5}.$$

Suppose there is a sequence of distinct quadruples

$$x_{1,n}, x_{2,n}, x_{3,n}, x_{4,n} \in x(P), \quad x_{1,n}, x_{2,n}, x_{3,n}, x_{4,n} \rightarrow x_*,$$

such that

$$\det \begin{pmatrix} 1 & Y(x_{1,n}) & x_{1,n} & S(x_{1,n}) \\ 1 & Y(x_{2,n}) & x_{2,n} & S(x_{2,n}) \\ 1 & Y(x_{3,n}) & x_{3,n} & S(x_{3,n}) \\ 1 & Y(x_{4,n}) & x_{4,n} & S(x_{4,n}) \end{pmatrix} = 0 \quad \text{for every } n.$$

Then

$$T_v(x_*) = 0.$$

Proof. Apply Lemma 12.246 with $d = 3$ to the lifted space curve

$$Q(x) := (Y(x), x, S(x)).$$

The exact coplanarity condition in the statement is precisely the vanishing of

$$\det \begin{pmatrix} 1 & Y(x_1) & x_1 & S(x_1) \\ 1 & Y(x_2) & x_2 & S(x_2) \\ 1 & Y(x_3) & x_3 & S(x_3) \\ 1 & Y(x_4) & x_4 & S(x_4) \end{pmatrix}.$$

Hence shrinking exact lifted 4-point circuits force

$$\det(Q'(x_*), Q''(x_*), Q'''(x_*)) = 0.$$

But

$$\det(Q'(x), Q''(x), Q'''(x)) = W_x(1, Y, x, S)(x),$$

and Proposition 12.233 gives

$$T_v = (x')^6 W_x(1, Y, x, S).$$

Since $x' \neq 0$ on a good v_5 -patch, it follows that

$$T_v(x_*) = 0. \quad \square$$

Corollary 12.251 (Uniform small-diameter exclusion away from $T_v = 0$). *Let $P_0 \in \{c \neq 0, v_5 \neq 0, M \neq 0\}$ be compact and assume*

$$P_0 \cap \{T_v = 0\} = \emptyset.$$

Then there exists $\rho_4 > 0$ such that no exact lifted 4-point circuit contained in P_0 has x -diameter $< \rho_4$.

Proof. Otherwise there would exist a sequence of exact lifted 4-point circuits in P_0 with diameters tending to 0. By compactness, after passing to a subsequence the four nodes would converge to some $x_* \in x(P_0)$. This contradicts Proposition 12.250. $\square$

Definition 12.252 (Repeated-node tangent-contact determinants on a v_5 -patch). On a $v_5 \neq 0$ patch with $x = v_5/c$, $Y = u_5/c$, and $S = \Delta_7/(cv_5)$, define

$$H_3(a, u) := \det \begin{pmatrix} 1 & a & S(a) \\ 0 & 1 & S'(a) \\ 1 & u & S(u) \end{pmatrix} = S(u) - S(a) - S'(a)(u - a),$$

and

$$H_4(a, u, v) := \det \begin{pmatrix} 1 & Y(a) & a & S(a) \\ 0 & Y'(a) & 1 & S'(a) \\ 1 & Y(u) & u & S(u) \\ 1 & Y(v) & v & S(v) \end{pmatrix}.$$

Thus $H_3(a, u) = 0$ means that $(u, S(u))$ lies on the tangent line to Λ_v at a , while $H_4(a, u, v) = 0$ means that $Q_v(u)$ and $Q_v(v)$ lie in the tangent plane to Q_v at a .

Proposition 12.253 (Repeated-node limits of exact lifted determinants). *For the exact lifted determinants*

$$\Delta_3(a, b, u) := \det \begin{pmatrix} 1 & a & S(a) \\ 1 & b & S(b) \\ 1 & u & S(u) \end{pmatrix},$$

and

$$\Delta_4(a, b, u, v) := \det \begin{pmatrix} 1 & Y(a) & a & S(a) \\ 1 & Y(b) & b & S(b) \\ 1 & Y(u) & u & S(u) \\ 1 & Y(v) & v & S(v) \end{pmatrix},$$

one has

$$\frac{\Delta_3(a, b, u)}{b - a} \rightarrow H_3(a, u) \quad (b \rightarrow a),$$

and

$$\frac{\Delta_4(a, b, u, v)}{b - a} \rightarrow H_4(a, u, v) \quad (b \rightarrow a).$$

Proof. Subtract the first row from the second row in each determinant and divide by $b - a$. The resulting second row is the first difference quotient of the corresponding row entries, and the limit $b \rightarrow a$ gives the derivative row. $\square$

Theorem 12.254 (Local classification of the lifted 3-point branch near a finite-order K_v -zero). *Let $x = 0$ be an isolated finite-order zero of K_v on a good v_5 -patch. After affine normalization of $\Lambda_v = (x, S)$, write*

$$S(x) = x^m + O(x^{m+1}), \quad m = r + 2 \geq 3,$$

where $r = \text{ord}_0 K_v$. Then the repeated-node tangent-contact equation

$$H_3(a, u) = 0$$

with $a < 0 < u$ has the following local classification:

- (1) *if r is even (equivalently m is even), there is no opposite-side tangent-contact branch near $(a, u) = (0, 0)$;*

(2) if r is odd (equivalently m is odd), there exists a unique real-analytic opposite-side branch

$$u = \Upsilon(a), \quad a < 0, \quad |a| \ll 1,$$

with asymptotic expansion

$$\Upsilon(a) = \lambda_m(-a) + O(a^2),$$

where $\lambda_m > 1$ is the unique positive root of

$$P_m(\lambda) := \lambda^m - m\lambda - (m-1).$$

Proof. Set $a = -\alpha$ with $\alpha > 0$, and write $u = \lambda\alpha$ with $\lambda > 0$. For the model germ $S(x) = x^m$, the tangent-contact equation is

$$H_3(-\alpha, \lambda\alpha) = \alpha^m \begin{cases} \lambda^m + m\lambda + (m-1), & m \text{ even}, \\ \lambda^m - m\lambda - (m-1), & m \text{ odd}. \end{cases}$$

If m is even, the displayed factor is strictly positive for every $\lambda > 0$, so no opposite-side branch exists.

If m is odd, define

$$P_m(\lambda) := \lambda^m - m\lambda - (m-1).$$

Then

$$P_m(1) = -2(m-1) < 0, \quad P_m(\lambda) \rightarrow +\infty \text{ as } \lambda \rightarrow +\infty,$$

and

$$P'_m(\lambda) = m(\lambda^{m-1} - 1)$$

is negative on $(0, 1)$ and positive on $(1, \infty)$. Therefore P_m has a unique positive root $\lambda_m > 1$.

For the perturbed germ $S(x) = x^m + O(x^{m+1})$, divide $H_3(-\alpha, \lambda\alpha)$ by α^m . The resulting equation is

$$P_m(\lambda) + O(\alpha) = 0.$$

If m is even, positivity of the model factor persists for α small, so there is still no branch. If m is odd, the root λ_m is simple, so the implicit function theorem yields a unique analytic branch

$$\lambda = \lambda(\alpha) = \lambda_m + O(\alpha),$$

equivalently

$$u = \lambda_m(-a) + O(a^2).$$

This proves the classification. □

Corollary 12.255 (One-sided exclusion for repeated-node tangent contact). *Under the hypotheses of Theorem 12.254, there exists $\varepsilon > 0$ such that*

$$H_3(a, u) \neq 0$$

for every same-side pair with

$$-\varepsilon < a < u < 0 \quad \text{or} \quad 0 < a < u < \varepsilon.$$

Equivalently, every sufficiently small off-diagonal solution of

$$H_3(a, u) = 0$$

is necessarily opposite-side.

Proof. Using

$$H_3(a, u) = S(u) - S(a) - S'(a)(u - a),$$

one has the exact integral identity

$$H_3(a, u) = \int_a^u (u - t) S''(t) dt.$$

Since $x = 0$ is an isolated finite-order zero of $K_v = (x')^3 S''$ on a good v_5 -patch, the analytic function S'' has constant nonzero sign on each punctured one-sided interval

$$(-\varepsilon, 0), \quad (0, \varepsilon)$$

after shrinking. If $a < u$ lie on the same side of 0, then

$$u - t > 0$$

for all $t \in (a, u)$, and $S''(t)$ has fixed nonzero sign on the whole interval. So the integrand has fixed nonzero sign, and therefore

$$H_3(a, u) \neq 0. \quad \square$$

Corollary 12.256 (No infinite-order flatness for K_v on a good v_5 -patch). *On a good v_5 -patch, every zero of K_v has finite order. Equivalently, the lifted curvature $S''(x)$ has no infinite-order zero on such a patch.*

Proof. By Proposition 12.233,

$$K_v = (x')^3 S''(x),$$

and on a good v_5 -patch one has $x' \neq 0$. Thus K_v and S'' have the same zero set and the same vanishing order.

If K_v had an infinite-order zero at some interior point x_* , then so would S'' . Since S'' is analytic in the local coordinate x , this would force

$$S'' \equiv 0$$

on a neighborhood of x_* . Hence S would be affine in x there, contrary to Proposition 12.228. $\square$

Lemma 12.257 (Slope–intercept parametrization for the lifted 3-point branch). *Assume the affine normalization from Theorem 12.254, so that*

$$S(0) = S'(0) = 0, \quad S'' < 0 \text{ on the source side } I_- = (\alpha, 0).$$

Let

$$I_+ = (0, \beta)$$

be the maximal open interval to the right of 0 on which S' is strictly increasing. Equivalently, S'' does not change sign on I_+ , and any zeros of S'' there are isolated even-order zeros.

For

$$m \in J_- := S'(I_-),$$

define the inverse source point

$$a = a(m) := (S'|_{I_-})^{-1}(m),$$

and the source tangent intercept

$$\tau_-(m) := S(a(m)) - m a(m).$$

For

$$m \in J_+ := S'(I_+),$$

define the inverse target tangent point

$$u_+(m) := (S'|_{I_+})^{-1}(m),$$

and the target tangent intercept

$$\tau_+(m) := S(u_+(m)) - m u_+(m).$$

Then

$$\tau'_-(m) = -a(m), \quad \tau'_+(m) = -u_+(m).$$

Hence on

$$J := J_- \cap J_+$$

the intercept gap

$$D(m) := \tau_-(m) - \tau_+(m)$$

satisfies

$$D'(m) = u_+(m) - a(m) > 0.$$

Proof. Differentiate

$$\tau_-(m) = S(a(m)) - m a(m)$$

to obtain

$$\tau'_-(m) = S'(a(m))a'(m) - a(m) - m a'(m) = -a(m),$$

because $S'(a(m)) = m$. The same computation gives

$$\tau'_+(m) = -u_+(m).$$

Subtracting yields

$$D'(m) = u_+(m) - a(m).$$

Since $a(m) < 0 < u_+(m)$, the derivative is strictly positive. $\square$

Theorem 12.258 (Global continuation / nonrecurrence of the odd-order lifted 3-point branch). *Assume the hypotheses and affine normalization of Theorem 12.254, and suppose that $r = \text{ord}_0 K_v$ is odd, so the unique local opposite-side branch exists.*

Let

$$I_- = (\alpha, 0)$$

be the source interval on which $S'' < 0$, and let

$$I_+ = (0, \beta)$$

be the maximal open interval to the right of 0 on which S' is strictly increasing. Set

$$J := S'(I_-) \cap S'(I_+).$$

Then there exists a unique maximal open interval

$$J_{\text{br}} = (0, m_*) \subset J$$

and a unique real-analytic function

$$u : J_{\text{br}} \rightarrow I_+$$

such that

$$H_3(a(m), u(m)) = 0, \quad a(m) := (S'|_{I_-})^{-1}(m), \quad u(m) > 0,$$

and for small $m > 0$ this branch agrees with the local branch from Theorem 12.254.

Moreover:

(1) *the intercept gap*

$$D(m) := \tau_-(m) - \tau_+(m)$$

is strictly positive on all of J , and

$$D'(m) = u_+(m) - a(m) > 0;$$

(2) *along the branch one has*

$$\partial_u H_3(a(m), u(m)) = S'(u(m)) - m \neq 0,$$

so there is no interior common-tangent event;

(3) *the branch is uniquely continued on J_{br} , with*

$$u'(m) = \frac{u(m) - a(m)}{S'(u(m)) - m},$$

and the source coordinate is strictly monotone,

$$a'(m) = \frac{1}{S''(a(m))} < 0;$$

(4) *if $m_* < \sup J$, then*

$$u(m) \longrightarrow \beta \quad (m \uparrow m_*).$$

Hence the branch cannot end in the interior of I_+ ; it can stop only when it reaches the boundary of the source/target slope-overlap block.

In particular, the odd-order lifted 3-point branch has no interior recurrence, no self-return, no interior accumulation, and no infinite-order flatness. Its only remaining endpoint mechanism is boundary exit of the relevant good-patch / monotone-slope block.

Proof. For

$$m \in J,$$

set

$$\Phi(m, u) := S(u) - m u - \tau_-(m).$$

Then

$$\Phi(m, u) = H_3(a(m), u).$$

Step 1: positivity of the intercept gap. By the odd-order local classification theorem, for all sufficiently small $m > 0$ there exists a point $u(m) \in I_+$ such that

$$\Phi(m, u(m)) = 0.$$

Fix such a small m . Since S' is strictly increasing on I_+ , the function

$$u \mapsto \Phi(m, u)$$

has strictly increasing derivative

$$\partial_u \Phi(m, u) = S'(u) - m,$$

hence a unique critical point, namely $u_+(m)$, and that critical point is its unique minimum on I_+ . Therefore

$$\Phi(m, u_+(m)) < 0.$$

But

$$\Phi(m, u_+(m)) = S(u_+(m)) - m u_+(m) - \tau_-(m) = \tau_+(m) - \tau_-(m) = -D(m),$$

so $D(m) > 0$ for all sufficiently small $m > 0$.

By Lemma 12.257,

$$D'(m) = u_+(m) - a(m) > 0$$

on all of J . Hence $D(m) > 0$ for every $m \in J$.

Step 2: no interior common tangency. Suppose

$$\Phi(m, u) = 0$$

for some $m \in J$ and $u \in I_+$. If also

$$\partial_u \Phi(m, u) = S'(u) - m = 0,$$

then by uniqueness of the inverse point on I_+ one has $u = u_+(m)$, and thus

$$0 = \Phi(m, u_+(m)) = -D(m),$$

contrary to $D(m) > 0$. Therefore every branch point satisfies

$$\partial_u \Phi(m, u) \neq 0.$$

Equivalently, there is no interior common-tangent point on the odd-order branch.

Step 3: uniqueness and analytic continuation. For fixed $m \in J$, the derivative

$$u \mapsto \partial_u \Phi(m, u) = S'(u) - m$$

is strictly increasing on I_+ , and

$$\partial_u \Phi(m, 0) = S'(0) - m = -m < 0.$$

Hence $\Phi(m, \cdot)$ first decreases and then increases, so it has at most one zero on I_+ . Thus the local branch from Theorem 12.254 is unique on every connected continuation interval.

Let

$$E := \{m \in J : \exists u \in I_+ \text{ with } \Phi(m, u) = 0\}.$$

By Step 2, every zero is simple in the u -variable, so the implicit function theorem shows that E is open and that the zero depends real-analytically on m . Let

$$J_{\text{br}} = (0, m_*)$$

be the connected component of E containing all sufficiently small positive m . This gives the unique maximal analytic continuation

$$u = u(m), \quad m \in J_{\text{br}}.$$

Differentiating

$$\Phi(m, u(m)) = 0$$

gives

$$(S'(u(m)) - m)u'(m) - u(m) + a(m) = 0,$$

hence

$$u'(m) = \frac{u(m) - a(m)}{S'(u(m)) - m}.$$

Also

$$a'(m) = \frac{1}{S''(a(m))} < 0$$

because $S'' < 0$ on the source interval I_- . So the source coordinate is strictly monotone and cannot return to a previous value.

Step 4: endpoint control. Assume that

$$m_* < \sup J$$

but that $u(m)$ does not approach β as $m \uparrow m_*$. Then there exists a compact subinterval

$$K \Subset I_+$$

and a sequence $m_n \uparrow m_*$ with

$$u(m_n) \in K.$$

After passing to a subsequence,

$$u(m_n) \rightarrow u_* \in K.$$

By continuity,

$$\Phi(m_*, u_*) = 0.$$

Since $m_* < \sup J$, one has $m_* \in J$, so by Step 2,

$$\partial_u \Phi(m_*, u_*) \neq 0.$$

The implicit function theorem therefore extends the branch analytically past m_* , contradicting maximality of J_{br} .

Hence

$$u(m) \rightarrow \beta \quad (m \uparrow m_*).$$

This proves that the branch cannot terminate by an interior fold, interior common tangency, or interior accumulation.

Finally, by Corollary 12.256, every zero of K_v has finite order. Since even-order zeros do not change the sign of S'' , they do not break the strict increase of S' on I_+ . Therefore the endpoint β can only be the ambient boundary of the good patch or the first odd-order zero of K_v to the right. $\square$

Remark 12.259 (Structural package for the global lifted 3-point branch). The correct global parameter on the odd-order lifted 3-point branch is the source tangent slope

$$m = S'(a).$$

In that parameter, the branch is governed by the source/target tangent intercepts

$$\tau_-(m) = S(a(m)) - m a(m), \quad \tau_+(m) = S(u_+(m)) - m u_+(m),$$

and their gap

$$D(m) = \tau_-(m) - \tau_+(m), \quad D'(m) = u_+(m) - a(m) > 0.$$

Thus interior common tangency is exactly the equation

$$D(m) = 0,$$

but this never occurs on the odd-order continuation branch because $D > 0$ there. So the global 3-point problem is not a hidden recurrence problem inside the patch: once born at an odd-order K_v -zero, the branch is a one-parameter analytic graph with monotone source coordinate, and even-order K_v -zeros are invisible to it except as isolated points where S'' vanishes without changing the monotone-slope block.

Remark 12.260 (Target-block form of the intercept-gap monotonicity). The slope–intercept monotonicity from Lemma 12.257 is not special to the first right-hand target block.

Let I be any open interval to the right of the source interval I_- on which S' is monotone and hence invertible. For

$$m \in S'(I_-) \cap S'(I),$$

define

$$u_I(m) := (S'|_I)^{-1}(m), \quad \tau_I(m) := S(u_I(m)) - m u_I(m),$$

and

$$D_I(m) := \tau_-(m) - \tau_I(m).$$

Then exactly the same envelope calculation gives

$$\tau'_I(m) = -u_I(m), \quad D'_I(m) = u_I(m) - a(m) > 0.$$

Thus for any monotone-slope target block, interior common tangency against the fixed source interval I_- is again the scalar equation

$$D_I(m) = 0.$$

So any future continuation across successive odd-order K_v -zeros must be organized blockwise through these monotone intercept-gap functions rather than through a hidden recurrence mechanism inside a block.

Remark 12.261 (Derivative-geometric meaning of the lifted 3-point branch). The lifted 3-point branch is now best viewed as a derivative-geometric propagation theorem for the planar lifted curve

$$\Lambda_v = (x, S).$$

By Lemma 12.246 and Proposition 12.248, shrinking exact lifted 3-point circuits are detected by the osculating determinant and therefore concentrate into the lifted curvature-zero set

$$K_v = (x')^3 S''(x) = 0.$$

Theorem 12.254 then identifies the local repeated-node branch structure at those curvature zeros, and Theorem 12.258 shows that the odd-order branch propagates globally on each monotone-slope target block by the slope–intercept envelope identities

$$\tau'_-(m) = -a(m), \quad \tau'_+(m) = -u_+(m), \quad D'(m) = u_+(m) - a(m) > 0.$$

So the lifted 3-point theory is no longer an ad hoc repeated-node analysis. It is the global propagation of an osculating-contact defect for the curve Λ_v , organized by the derivative geometry of its tangent lines and curvature.

Remark 12.262 (Exact nonmixed reduction in terms of $\mathcal{Z}$ and $\mathcal{R}$). At a nonmixed lifted point

$$K_v(0) \neq 0, \quad T_v(0) = 0,$$

Propositions 12.233 and 12.234 give

$$\mathcal{Z} = \frac{Y''}{S''}, \quad \mathcal{R} = \mathcal{Z}' = \frac{T_v}{K_v^2}.$$

Since $K_v(0) \neq 0$, one has $S''(0) \neq 0$, so after analytic reparametrization and affine target normalization preserving coplanarity one may write the lifted germ as

$$Q_v(\xi) = (\tilde{Y}(\xi), \xi, \xi^2), \quad \tilde{Y}(0) = \tilde{Y}'(0) = \tilde{Y}''(0) = 0.$$

In these normalized coordinates,

$$\mathcal{Z}(\xi) = \frac{1}{2} \tilde{Y}''(\xi), \quad \mathcal{R}(\xi) = \frac{1}{2} \tilde{Y}'''(\xi).$$

Thus the nonmixed lifted 4-point problem is exactly the jet problem of excluding odd-order zeros of $\mathcal{R}$. Equivalently, it is the problem of excluding even leading degrees of $\tilde{Y}$ in the normalized chart.

In particular, the low-order dangerous nonmixed cases are exactly:

$$\begin{aligned} \text{ord}_0 \mathcal{R} = 1 &\iff \tilde{Y}'''(0) = 0, \tilde{Y}^{(4)}(0) \neq 0, \\ \text{ord}_0 \mathcal{R} = 3 &\iff \tilde{Y}^{(3)}(0) = \tilde{Y}^{(4)}(0) = \tilde{Y}^{(5)}(0) = 0, \tilde{Y}^{(6)}(0) \neq 0, \\ \text{ord}_0 \mathcal{R} = 5 &\iff \tilde{Y}^{(3)}(0) = \dots = \tilde{Y}^{(7)}(0) = 0, \tilde{Y}^{(8)}(0) \neq 0. \end{aligned}$$

Equivalently, the dangerous nonmixed low-order list is

$$\text{ord}_0 \mathcal{Z} \in \{2, 4, 6\} \iff \text{ord}_0 \mathcal{R} \in \{1, 3, 5\}.$$

So the nonmixed 4-point endgame has been reduced to a finite jet problem for the canonical invariant $\mathcal{R}$.

Lemma 12.263 (Repeated-node tangent-plane determinant in the normalized nonmixed chart).
In the normalized nonmixed chart of Remark 12.262, define

$$\tilde{H}_4(a, u, v) := \det \begin{pmatrix} 1 & \tilde{Y}(a) & a & a^2 \\ 0 & \tilde{Y}'(a) & 1 & 2a \\ 1 & \tilde{Y}(u) & u & u^2 \\ 1 & \tilde{Y}(v) & v & v^2 \end{pmatrix}.$$

Then

$$\tilde{H}_4(a, u, v) = (u - a)^2(v - a)^2 \left(\tilde{Y}[a, a, v] - \tilde{Y}[a, a, u] \right),$$

where

$$\tilde{Y}[a, a, t] := \frac{\tilde{Y}(t) - \tilde{Y}(a) - \tilde{Y}'(a)(t - a)}{(t - a)^2}$$

is the repeated-node second divided difference. Equivalently, for $u \neq v$,

$$\tilde{H}_4(a, u, v) = (u - a)^2(v - a)^2(v - u) \tilde{Y}[a, a, u, v].$$

Proof. Subtract the first row from the third and fourth rows. Since

$$u^2 - a^2 = (u - a)(u + a), \quad v^2 - a^2 = (v - a)(v + a),$$

this gives

$$\tilde{H}_4(a, u, v) = (u - a)(v - a) \det \begin{pmatrix} 1 & \tilde{Y}(a) & a & a^2 \\ 0 & \tilde{Y}'(a) & 1 & 2a \\ 0 & \frac{\tilde{Y}(u) - \tilde{Y}(a)}{u - a} & 1 & u + a \\ 0 & \frac{\tilde{Y}(v) - \tilde{Y}(a)}{v - a} & 1 & v + a \end{pmatrix}.$$

Expanding along the first column yields

$$\tilde{H}_4(a, u, v) = (u - a)(v - a) \det \begin{pmatrix} \tilde{Y}'(a) & 1 & 2a \\ \frac{\tilde{Y}(u) - \tilde{Y}(a)}{u - a} & 1 & u + a \\ \frac{\tilde{Y}(v) - \tilde{Y}(a)}{v - a} & 1 & v + a \end{pmatrix}.$$

Now subtract the first row from the second and third rows:

$$\tilde{H}_4(a, u, v) = (u - a)(v - a) \det \begin{pmatrix} \tilde{Y}'(a) & 1 & 2a \\ (u - a)\tilde{Y}[a, a, u] & 0 & u - a \\ (v - a)\tilde{Y}[a, a, v] & 0 & v - a \end{pmatrix}.$$

Expanding along the middle column gives

$$\tilde{H}_4(a, u, v) = (u - a)^2(v - a)^2 \left(\tilde{Y}[a, a, v] - \tilde{Y}[a, a, u] \right),$$

which is the first formula. Since

$$\tilde{Y}[a, a, u, v] = \frac{\tilde{Y}[a, a, v] - \tilde{Y}[a, a, u]}{v - u},$$

the second formula follows for $u \neq v$. □

Proposition 12.264 (Exact nonmixed tangent-plane contacts force nearby zeros of $\mathcal{R}$). *In the normalized nonmixed chart of Remark 12.262, assume $a, u, v \in \mathbf{R}$ are distinct and satisfy*

$$\tilde{H}_4(a, u, v) = 0.$$

Then there exists

$$w \in \text{conv}\{a, u, v\}$$

such that

$$\mathcal{R}(w) = 0.$$

Equivalently,

$$\tilde{Y}'''(w) = 0.$$

Proof. By Lemma 12.263, since a, u, v are distinct,

$$0 = \tilde{H}_4(a, u, v) = (u - a)^2(v - a)^2(v - u) \tilde{Y}[a, a, u, v].$$

Hence

$$\tilde{Y}[a, a, u, v] = 0.$$

By the Hermite–Genocchi formula for repeated-node divided differences, there exists

$$w \in \text{conv}\{a, u, v\}$$

such that

$$\tilde{Y}[a, a, u, v] = \frac{\tilde{Y}'''(w)}{3!}.$$

Therefore

$$\tilde{Y}'''(w) = 0.$$

In the normalized nonmixed chart one has

$$\mathcal{R}(\xi) = \frac{1}{2} \tilde{Y}'''(\xi),$$

so $\mathcal{R}(w) = 0$. □

Corollary 12.265 (Sign-definite nonmixed intervals are exactly-contact sterile). *Let J be an open interval in the normalized nonmixed chart such that*

$$\mathcal{R} \text{ has fixed nonzero sign on } J.$$

Then there do not exist distinct points $a, u, v \in J$ such that

$$\tilde{H}_4(a, u, v) = 0.$$

Equivalently, there are no exact nonmixed tangent-plane contacts with all three nodes in J .

Proof. If such a, u, v existed, Proposition 12.264 would produce some

$$w \in \text{conv}\{a, u, v\} \subset J$$

with $\mathcal{R}(w) = 0$, contradicting the sign assumption. □

Remark 12.266 (Derivative-side interpretation of exact nonmixed contacts). Proposition 12.264 strengthens the shrinking-circuit statement Proposition 12.250. It is not merely a limit theorem: every exact nonmixed tangent-plane contact already forces a zero of the canonical derivative invariant inside the convex hull of the three nodes,

$$\mathcal{R} = \frac{1}{2} \tilde{Y}'''.$$

Thus the nonmixed 4-point branch is controlled not only asymptotically by $T_v = 0$, but pointwise by the zero set of a single analytic scalar.

Proposition 12.267 (Interval-persistent nonmixed 4-point degeneracy forces affine planarity). *Let J be a connected open interval in a good v_5 -patch such that*

$$K_v \neq 0 \quad \text{on } J.$$

Assume

$$T_v \equiv 0 \quad \text{on } J.$$

Then on J one has

$$\mathcal{R} \equiv 0, \quad \mathcal{Z} \equiv \gamma$$

for some constant $\gamma \in \mathbf{C}$, and hence

$$Y = \gamma S + ax + b$$

for some constants $a, b \in \mathbf{C}$. Equivalently, the lifted curve

$$x \mapsto (Y(x), x, S(x))$$

lies in an affine plane on J .

In particular, any interval-persistent nonmixed exact lifted 4-point branch collapses to an affine-planar residual rather than remaining genuinely three-dimensional.

Proof. On the nonmixed interval J , Proposition 12.234 gives

$$\mathcal{R} = \frac{T_v}{K_v^2}.$$

Since $T_v \equiv 0$ and $K_v \neq 0$ on J , it follows that

$$\mathcal{R} \equiv 0 \quad \text{on } J.$$

By definition,

$$\mathcal{R} = \frac{d\mathcal{Z}}{dx},$$

so $\mathcal{Z}$ is constant on J :

$$\mathcal{Z} \equiv \gamma.$$

Using

$$\mathcal{Z} = \frac{Y''}{S''},$$

we obtain

$$Y'' = \gamma S''.$$

Integrating twice with respect to x yields

$$Y = \gamma S + ax + b$$

for constants $a, b \in \mathbf{C}$. Therefore the lifted curve satisfies the affine linear equation

$$Y - \gamma S - ax - b = 0$$

throughout J , so it lies in an affine plane. □

Corollary 12.268 (Recurrent exact nonmixed contacts force affine-planar collapse). *Let J be a connected open interval in a good v_5 -patch such that*

$$K_v \neq 0 \quad \text{on } J.$$

Assume that every nonempty open subinterval $I \subset J$ contains an exact nonmixed tangent-plane contact triple. Equivalently, after local nonmixed normalization on I , there exist distinct points $a, u, v \in I$ such that

$$\tilde{H}_4(a, u, v) = 0.$$

Then

$$\mathcal{R} \equiv 0 \quad \text{on } J,$$

hence

$$T_v \equiv 0 \quad \text{on } J,$$

and therefore the lifted curve is affine-planar on J :

$$Y = \gamma S + ax + b$$

for some constants $\gamma, a, b \in \mathbf{C}$.

Proof. Let $I \subset J$ be any nonempty open subinterval. By assumption, I contains an exact nonmixed tangent-plane contact triple. After nonmixed normalization on I , Proposition 12.264 produces a point $w_I \in I$ such that

$$\mathcal{R}(w_I) = 0.$$

Thus every nonempty open subinterval of J contains a zero of $\mathcal{R}$. Since $\mathcal{R}$ is analytic, its zero set cannot have an accumulation point in J unless

$$\mathcal{R} \equiv 0 \quad \text{on } J.$$

Because $K_v \neq 0$ on J and

$$\mathcal{R} = \frac{T_v}{K_v^2},$$

it follows that

$$T_v \equiv 0 \quad \text{on } J.$$

Now apply Proposition 12.267 to conclude that

$$Y = \gamma S + ax + b$$

on J . □

Lemma 12.269 (Derivative formula for the repeated-node nonmixed profile). *In the normalized nonmixed chart, define*

$$D_a(t) := \tilde{Y}[a, a, t] = \frac{\tilde{Y}(t) - \tilde{Y}(a) - \tilde{Y}'(a)(t-a)}{(t-a)^2}, \quad D_a(a) := \frac{1}{2} \tilde{Y}''(a).$$

Then $D_a(t)$ extends real-analytically across $t = a$, and

$$D_a(t) = \int_0^1 (1-s) \tilde{Y}''(a + s(t-a)) ds.$$

Moreover,

$$D'_a(t) = \int_0^1 s(1-s) \tilde{Y}'''(a + s(t-a)) ds = 2 \int_0^1 s(1-s) \mathcal{R}(a + s(t-a)) ds.$$

Proof. Taylor's integral formula gives

$$\tilde{Y}(t) = \tilde{Y}(a) + \tilde{Y}'(a)(t-a) + (t-a)^2 \int_0^1 (1-s) \tilde{Y}''(a + s(t-a)) ds.$$

Dividing by $(t-a)^2$ proves the first formula for $t \neq a$, and the analytic extension across $t = a$ is immediate. Differentiating under the integral sign gives

$$D'_a(t) = \int_0^1 s(1-s) \tilde{Y}'''(a + s(t-a)) ds.$$

Since $\mathcal{R} = \frac{1}{2} \tilde{Y}'''$ in the normalized nonmixed chart, the last formula follows. □

Proposition 12.270 (Parity classification of isolated nonmixed zeros of $\mathcal{R}$). *Let $\xi_0 \in \mathbf{R}$ be an isolated zero of the nonmixed invariant*

$$\mathcal{R}(\xi) = \frac{1}{2} \tilde{Y}'''(\xi),$$

and write

$$\mathcal{R}(\xi) = (\xi - \xi_0)^m \rho(\xi), \quad m \geq 1, \quad \rho(\xi_0) \neq 0.$$

(i) Even-order sterility. *If m is even, then there exists a neighborhood U of ξ_0 such that for every $a \in U$, the function*

$$D_a(t) := \tilde{Y}[a, a, t]$$

is strictly monotone on U . In particular, there do not exist distinct $a, u, v \in U$ such that

$$\tilde{H}_4(a, u, v) = 0.$$

(ii) Odd-order birth. If m is odd, then there exists a neighborhood U of ξ_0 and a real-analytic function $\tau : U \rightarrow U$ such that

$$\tau(\xi_0) = \xi_0, \quad \tau'(\xi_0) = -1, \quad D'_a(\tau(a)) = 0 \quad (a \in U).$$

Moreover,

$$D''_a(\tau(a)) \neq 0 \quad (a \in U \setminus \{\xi_0\}),$$

so for each $a \in U \setminus \{\xi_0\}$, the function D_a has a strict local extremum at $t = \tau(a)$. Consequently, for each such a , there exist distinct points $u_-(a), u_+(a) \in U$ with

$$u_-(a) < \tau(a) < u_+(a), \quad D_a(u_-(a)) = D_a(u_+(a)),$$

and hence

$$\tilde{H}_4(a, u_-(a), u_+(a)) = 0.$$

Proof. By translation in the normalized chart, we may assume $\xi_0 = 0$. Thus

$$\mathcal{R}(\xi) = \xi^m \rho(\xi), \quad \rho(0) \neq 0.$$

For the even-order case, shrink to a neighborhood U on which ρ has fixed nonzero sign. Since m is even, $\xi^m \geq 0$, so $\mathcal{R}$ has fixed sign on U , vanishing only at 0. By Lemma 12.269,

$$D'_a(t) = 2 \int_0^1 s(1-s) \mathcal{R}(a + s(t-a)) ds.$$

For $a \neq t$, the affine segment $a + s(t-a)$ is not identically zero, so the integrand has fixed sign and is not identically zero. Hence $D'_a(t)$ has fixed nonzero sign for all $a \neq t$ in U , and therefore each D_a is strictly monotone on U . Thus D_a is injective on U , and by Lemma 12.263 there are no exact contacts $\tilde{H}_4(a, u, v) = 0$ with distinct $a, u, v \in U$.

Now assume m is odd. Write $a = h$ and $t = \lambda h$. Then Lemma 12.269 gives

$$D'_h(\lambda h) = 2h^m G(h, \lambda),$$

where

$$G(h, \lambda) := \int_0^1 s(1-s)(1+s(\lambda-1))^m \rho(h(1+s(\lambda-1))) ds.$$

This is real-analytic in (h, λ) , and

$$G(0, \lambda) = \rho(0) J_m(\lambda), \quad J_m(\lambda) := \int_0^1 s(1-s)(1+s(\lambda-1))^m ds.$$

Since m is odd,

$$J_m(-1) = \int_0^1 s(1-s)(1-2s)^m ds = 0$$

by the symmetry $s \mapsto 1-s$, while

$$J'_m(-1) = m \int_0^1 s^2(1-s)(1-2s)^{m-1} ds > 0$$

because $m-1$ is even. Thus

$$\partial_\lambda G(0, -1) = \rho(0) J'_m(-1) \neq 0.$$

By the implicit function theorem, there exists a unique real-analytic function $\lambda(h)$ near 0 such that

$$\lambda(0) = -1, \quad G(h, \lambda(h)) = 0.$$

Define

$$\tau(h) := \lambda(h)h.$$

Then

$$\tau(0) = 0, \quad \tau'(0) = \lambda(0) = -1,$$

and

$$D'_h(\tau(h)) = 0.$$

It remains to show that this critical point is a strict local extremum. By Lemma 12.269,

$$D''_a(t) = 2 \int_0^1 s^2(1-s) \mathcal{R}'(a + s(t-a)) ds.$$

Substituting $a = h$, $t = \lambda(h)h$, and using

$$\mathcal{R}'(\xi) = m\xi^{m-1}\rho(\xi) + \xi^m\rho'(\xi),$$

one obtains

$$D''_h(\tau(h)) = 2h^{m-1}\rho(0)J'_m(-1) + O(h^m).$$

Since $\rho(0)J'_m(-1) \neq 0$, it follows that

$$D''_h(\tau(h)) \neq 0$$

for $h \neq 0$ sufficiently small. Therefore $t = \tau(h)$ is a strict local extremum of D_h . Hence there exist distinct points $u_-(h)$, $u_+(h)$ arbitrarily close to $\tau(h)$ such that

$$D_h(u_-(h)) = D_h(u_+(h)).$$

By Lemma 12.263,

$$\tilde{H}_4(h, u_-(h), u_+(h)) = (h - u_-(h))^2(h - u_+(h))^2(D_h(u_+(h)) - D_h(u_-(h))) = 0.$$

This proves the odd-order birth statement. $\square$

Remark 12.271 (Reflection law for odd-order birth). In the odd-order case of Proposition 12.270, the critical-point branch satisfies

$$\tau(a) = 2\xi_0 - a + O((a - \xi_0)^2).$$

Thus the local exact nonmixed contact family born at any odd-order zero of $\mathcal{R}$ has the same first-order reflection geometry across that zero as in the simple-zero case.

Proposition 12.272 (Odd-zero critical branch is uniformly unrooted). *Let ξ_0 be an isolated odd-order zero of*

$$\mathcal{R}(\xi) = \frac{1}{2} \tilde{Y}'''(\xi),$$

and, after translation, write

$$\xi_0 = 0, \quad \mathcal{R}(\xi) = \xi^m \rho(\xi), \quad m \geq 1 \text{ odd}, \quad \rho(0) \neq 0.$$

Let

$$D_a(t) := \tilde{Y}[a, a, t], \quad D_a(a) = \frac{1}{2} \tilde{Y}''(a) = \mathcal{Z}(a).$$

Then there exists $\varepsilon > 0$ and a unique real-analytic function

$$\mu : (-\varepsilon, \varepsilon) \rightarrow \mathbf{R}, \quad \mu(0) = 0,$$

such that for every sufficiently small $h > 0$ the unique nontrivial critical pair

$$a(h) := h(\mu(h) - 1), \quad x(h) := h(\mu(h) + 1)$$

satisfies

$$D'_{a(h)}(x(h)) = 0,$$

and every critical pair

$$a < x, \quad |a| + |x| < \varepsilon, \quad D'_a(x) = 0$$

is of this form.

Moreover,

$$D_{a(h)}(x(h)) - D_{a(h)}(a(h)) = -\frac{\rho(0)}{m+2} h^{m+1} + O(h^{m+2}).$$

Equivalently,

$$D_{a(h)}(x(h)) - D_{a(h)}(a(h)) = -\frac{\rho(0)}{2^{m+1}(m+2)} (x(h) - a(h))^{m+1} + o((x(h) - a(h))^{m+1}).$$

In particular, after shrinking ε if needed, every nontrivial nearby critical pair satisfies

$$D_a(x) \neq D_a(a),$$

indeed

$$\operatorname{sgn}(D_a(x) - D_a(a)) = -\operatorname{sgn} \rho(0).$$

So no sheared repeated-node tangent-contact event can occur in a sufficiently small neighborhood of an isolated odd zero of $\mathcal{R}$.

Proof. Write every nearby pair as

$$a = c - h, \quad x = c + h, \quad h = \frac{x - a}{2} > 0, \quad \mu = \frac{c}{h}.$$

By Lemma 12.269,

$$D'_a(x) = 2 \int_0^1 s(1-s) \mathcal{R}(a + s(x-a)) ds.$$

With the change of variables $u = 2s - 1$, this becomes

$$D'_{c-h}(c+h) = \frac{1}{4} \int_{-1}^1 (1-u^2) \mathcal{R}(c+hu) du = h^m F(h, \mu),$$

where

$$F(h, \mu) := \frac{1}{4} \int_{-1}^1 (1-u^2)(\mu+u)^m \rho(h(\mu+u)) du.$$

This is real-analytic in (h, μ) near $(0, 0)$. At $h = 0$,

$$F(0, \mu) = \frac{\rho(0)}{4} K_m(\mu), \quad K_m(\mu) := \int_{-1}^1 (1-u^2)(\mu+u)^m du.$$

Since m is odd,

$$K_m(0) = 0,$$

and

$$K'_m(\mu) = m \int_{-1}^1 (1-u^2)(\mu+u)^{m-1} du > 0$$

for all real μ , because $m-1$ is even. Hence $\mu = 0$ is the unique real root of K_m , and it is simple. Therefore

$$\partial_\mu F(0, 0) \neq 0,$$

so the implicit function theorem gives a unique real-analytic branch

$$\mu = \mu(h), \quad \mu(0) = 0,$$

parameterizing all nearby critical pairs.

Now write

$$\mathcal{Z}(\xi) = \mathcal{Z}(0) + \xi^{m+1} \eta(\xi), \quad \eta(0) = \frac{\rho(0)}{m+1},$$

since $\mathcal{Z}' = \mathcal{R} = \xi^m \rho(\xi)$. Using

$$D_a(x) - D_a(a) = \frac{1}{2} \int_{-1}^1 (1-u) (\mathcal{Z}(c+hu) - \mathcal{Z}(c-h)) du,$$

we obtain

$$D_{c-h}(c+h) - D_{c-h}(c-h) = h^{m+1} G(h, \mu),$$

where

$$G(h, \mu) := \frac{1}{2} \int_{-1}^1 (1-u) \left((\mu+u)^{m+1} \eta(h(\mu+u)) - (\mu-1)^{m+1} \eta(h(\mu-1)) \right) du.$$

Again G is real-analytic. Evaluating at $(0, 0)$ gives

$$G(0, 0) = \frac{\eta(0)}{2} \int_{-1}^1 (1-u)(u^{m+1} - 1) du = -\frac{\rho(0)}{m+2} \neq 0.$$

Therefore along the critical branch,

$$D_{a(h)}(x(h)) - D_{a(h)}(a(h)) = h^{m+1}G(h, \mu(h)) = -\frac{\rho(0)}{m+2}h^{m+1} + O(h^{m+2}),$$

which has fixed nonzero sign for h sufficiently small. This proves the claim. $\square$

Corollary 12.273 (Sheared repeated-node tangent-contact is impossible near an isolated odd $\mathcal{R}$ -zero). *Let b be an isolated odd-order zero of $\mathcal{R}$ on a genuine opened nonmixed arc. Then there exists a neighborhood U of b such that no pair*

$$a < x, \quad a, x \in U,$$

can satisfy

$$\tilde{Y}_{\mathcal{Z}(a)}''(a) = 0, \quad \tilde{H}_3(a, x) = 0, \quad \partial_x \tilde{H}_3(a, x) = 0.$$

Equivalently, no rooted sheared repeated-node tangent-contact event occurs in a sufficiently small neighborhood of an isolated odd zero of $\mathcal{R}$.

Proof. In the normalized nonmixed chart centered at b , the rooted sheared event is exactly the conjunction

$$D'_a(x) = 0, \quad D_a(x) = D_a(a) = \mathcal{Z}(a).$$

Proposition 12.272 shows that for every sufficiently small nontrivial critical pair one has

$$D_a(x) - D_a(a) \neq 0.$$

So the rooted sheared event is impossible near b . $\square$

Proposition 12.274 (Quadratic-envelope factorization on a nonmixed interval). *Let J be a connected open interval in a good v_5 -patch such that*

$$K_v \neq 0 \quad \text{on } J.$$

Write

$$Q_v(h) = (Y(h), x(h), S(h)), \quad \mathcal{Z} = \frac{Y''}{S''}, \quad \mathcal{R} = \mathcal{Z}' = \frac{T_v}{K_v^2}.$$

For a fixed source point $a \in J$, define

$$A_a(x) := Y(x) - Y(a) - Y'(a)(x - a),$$

$$B_a(x) := S(x) - S(a) - S'(a)(x - a),$$

and

$$\Gamma_a(x) := \frac{A_a(x)}{B_a(x)} \quad (x \neq a).$$

Then:

(i) Tangent-contact denominator. For $x \neq a$,

$$B_a(x) = \int_a^x (x - t) S''(t) dt,$$

so $B_a(x) \neq 0$ on $J \setminus \{a\}$.

(ii) Exact repeated-source factorization. For distinct $a, u, v \in J$,

$$H_4(a, u, v) = -B_a(u) B_a(v) (\Gamma_a(u) - \Gamma_a(v)).$$

(iii) Integral representation. For $x \neq a$,

$$\Gamma_a(x) = \frac{\int_a^x (x - t) \mathcal{Z}(t) S''(t) dt}{\int_a^x (x - t) S''(t) dt}.$$

(iv) Positive-kernel derivative formula. For $x \neq a$,

$$\Gamma'_a(x) = \frac{1}{B_a(x)^2} \int_a^x \Omega_{a,x}(r) \mathcal{R}(r) dr,$$

where

$$\Omega_{a,x}(r) := \int_a^r \int_r^x (t-s) S''(s) S''(t) dt ds.$$

Moreover, after orienting x so that S'' has fixed sign on J ,

$$\Omega_{a,x}(r) > 0 \quad (a < r < x).$$

Proof. Since $K_v = (x')^3 S''$ and $x' \neq 0$ on a good v_5 -patch, the function S'' has fixed nonzero sign on J . Therefore

$$B_a(x) = \int_a^x (x-t) S''(t) dt$$

has fixed nonzero sign for $x \neq a$, proving (i).

For (ii), use the repeated-node determinant

$$H_4(a, u, v) = \det \begin{pmatrix} 1 & Y(a) & a & S(a) \\ 0 & Y'(a) & 1 & S'(a) \\ 1 & Y(u) & u & S(u) \\ 1 & Y(v) & v & S(v) \end{pmatrix}.$$

Subtract from the u -row and v -row the corresponding affine tangent data at a . This gives

$$H_4(a, u, v) = \det \begin{pmatrix} 1 & Y(a) & a & S(a) \\ 0 & Y'(a) & 1 & S'(a) \\ 0 & A_a(u) & 0 & B_a(u) \\ 0 & A_a(v) & 0 & B_a(v) \end{pmatrix} = A_a(u)B_a(v) - A_a(v)B_a(u),$$

hence

$$H_4(a, u, v) = -B_a(u)B_a(v)(\Gamma_a(u) - \Gamma_a(v)).$$

For (iii), Taylor's integral formula gives

$$A_a(x) = \int_a^x (x-t) Y''(t) dt = \int_a^x (x-t) \mathcal{Z}(t) S''(t) dt,$$

while

$$B_a(x) = \int_a^x (x-t) S''(t) dt.$$

Dividing yields the formula for Γ_a .

For (iv), set

$$N_a(x) := \int_a^x (x-t) \mathcal{Z}(t) S''(t) dt, \quad D_a(x) := \int_a^x (x-t) S''(t) dt = B_a(x).$$

Then $\Gamma_a = N_a/D_a$, so

$$\Gamma'_a(x) = \frac{N'_a(x)D_a(x) - N_a(x)D'_a(x)}{D_a(x)^2}.$$

A symmetrization gives

$$N'_a D_a - N_a D'_a = \frac{1}{2} \int_a^x \int_a^x (\mathcal{Z}(t) - \mathcal{Z}(s))(t-s) S''(s) S''(t) ds dt.$$

Since $\mathcal{Z}(t) - \mathcal{Z}(s) = \int_s^t \mathcal{R}(r) dr$, Fubini yields

$$N'_a D_a - N_a D'_a = \int_a^x \Omega_{a,x}(r) \mathcal{R}(r) dr$$

with

$$\Omega_{a,x}(r) = \int_a^r \int_r^x (t-s) S''(s) S''(t) dt ds.$$

Because $t-s > 0$ on the region of integration and S'' has fixed sign on J , one has $\Omega_{a,x}(r) > 0$ after orienting the x -coordinate. This proves (iv). $\square$

Corollary 12.275 (Repeated-source monotonicity on one-sign $\mathcal{R}$ -blocks). *Let $I \subset J$ be a subinterval on which*

$\mathcal{R}$ has fixed nonzero sign.

Then for every $a \in I$, the function $x \mapsto \Gamma_a(x)$ is strictly monotone on each component of $I \setminus \{a\}$. In particular, there do not exist points

$$a < u < v \quad \text{in } I$$

such that

$$H_4(a, u, v) = 0.$$

Proof. This is immediate from Proposition 12.274: the derivative $\Gamma'_a(x)$ is a positive-kernel average of $\mathcal{R}$, so it has the same sign as $\mathcal{R}$ on each one-sign block. The factorization of H_4 then excludes repeated-source exact contacts there. $\square$

Remark 12.276 (Below-lifted source-determinant reformulation of Γ_a). Let

$$\lambda := \frac{\Delta_7}{v_5}, \quad x = \frac{v_5}{c}, \quad Y = \frac{u_5}{c}, \quad S = \frac{\lambda}{c}.$$

For any scalar germ f , define

$$D_f(a, h) := \det \begin{pmatrix} c(a) & v_5(a) & f(a) \\ c'(a) & v'_5(a) & f'(a) \\ c(h) & v_5(h) & f(h) \end{pmatrix}.$$

A direct row reduction gives

$$D_f(a, h) = c(a)^2 c(h) \left(x'(a) (F(h) - F(a)) - F'(a) (x(h) - x(a)) \right), \quad F = \frac{f}{c}.$$

In particular,

$$\Psi_a(h) := \frac{D_{u_5}(a, h)}{D_\lambda(a, h)} = \Gamma_{x(a)}(x(h)).$$

Thus the repeated-source nonmixed 4-point geometry is already encoded by a finite ratio of source determinants.

Likewise, if one sets

$$\mathfrak{D}(a, h) := \det \begin{pmatrix} c(a) & v_5(a) & \lambda(a) & u_5(a) \\ c'(a) & v'_5(a) & \lambda'(a) & u'_5(a) \\ c(h) & v_5(h) & \lambda(h) & u_5(h) \\ c'(h) & v'_5(h) & \lambda'(h) & u'_5(h) \end{pmatrix},$$

then a Plücker identity yields

$$\frac{d}{dh} \Psi_a(h) = \frac{W_h(c, v_5)(a) \mathfrak{D}(a, h)}{D_\lambda(a, h)^2}.$$

So the sign of the quadratic-envelope derivative is governed by one explicit 4×4 source determinant.

Moreover, combining the identity

$$\Psi_a(h) = \Gamma_{x(a)}(x(h))$$

with Proposition 12.274 gives the exact source-level kernel formula

$$\mathfrak{D}(a, h) = c(a)^2 c(h)^2 x'(a) x'(h) \int_{x(a)}^{x(h)} \Omega_{x(a), x(h)}(r) \mathcal{R}(r) dr.$$

After orienting the x -coordinate so that $x' > 0$ on the interval, the sign of $\mathfrak{D}(a, h)$ is therefore exactly the sign of the corresponding positive-kernel average of $\mathcal{R}$. In particular, on every one-sign $\mathcal{R}$ -block the quadruple

$$(c, v_5, \lambda, u_5)$$

is osculatory-ECT in the sense that $\mathfrak{D}(a, h)$ has fixed nonzero sign for $a < h$.

Corollary 12.277 (Source-level Wronskian consequences for the nonmixed branch). *On a good nonmixed v_5 -patch, the following hold.*

- (1) *If there is an exact nonmixed tangent-plane contact, then there exists a source point h_* in the source convex hull of the contact nodes such that*

$$\mathcal{W}_{\text{src}}(h_*) = 0.$$

- (2) *If $\mathcal{W}_{\text{src}}$ has fixed nonzero sign on an open nonmixed interval, then there are no exact nonmixed tangent-plane contacts on that interval.*

- (3) *If every nonempty open subinterval of a nonmixed interval contains an exact nonmixed tangent-plane contact, then*

$$\mathcal{W}_{\text{src}} \equiv 0$$

on that interval, hence

$$\mathcal{R} \equiv 0,$$

and therefore the lifted curve collapses to affine-planar geometry.

Proof. For (1), Proposition 12.264 produces a point in the convex hull of the three normalized nodes where $\mathcal{R} = 0$. Pulling back through the monotone source-to- x coordinate on the nonmixed patch and using Proposition 12.236 gives $\mathcal{W}_{\text{src}} = 0$ at the corresponding source point.

For (2), by Proposition 12.236, $\mathcal{W}_{\text{src}}$ and $\mathcal{R}$ have the same sign on a nonmixed arc. So Corollary 12.275 applies.

For (3), every nonempty open subinterval contains a zero of $\mathcal{W}_{\text{src}}$ by (1), so analyticity forces $\mathcal{W}_{\text{src}} \equiv 0$. Proposition 12.236 then gives $\mathcal{R} \equiv 0$, and Corollary 12.268 gives affine-planar collapse. $\square$

Remark 12.278 (Chebyshev formulation of the nonmixed burden). Corollary 12.277 shows that the nonmixed 4-point branch is controlled source-level by the pair

$$(J_\lambda, J_{u_5}).$$

If this pair were a Chebyshev pair on an opened nonmixed arc, i.e.

$$\mathcal{W}_{\text{src}} \neq 0$$

throughout that arc, then the entire nonmixed 4-point branch would be sterile there. Thus a one-sign theorem for $\mathcal{W}_{\text{src}}$ would still be sufficient. However, the later source-level no-go theorem shows that generic Wronskian algebra alone is not enough to force such a statement. The actual remaining burden is therefore a source-coupling theorem that excludes the analytically possible odd zeros of $\mathcal{W}_{\text{src}}$ on genuine opened nonmixed arcs.

Proposition 12.279 (Lift of the source-level Wronskian to a 4×4 Wronskian). *With*

$$J_\lambda = W(c, v_5, \lambda), \quad J_{u_5} = W(c, v_5, u_5), \quad M = W(c, v_5) = cv'_5 - c'v_5,$$

one has the exact identity

$$\boxed{\mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda = W(J_\lambda, J_{u_5}) = M W(c, v_5, \lambda, u_5).}$$

Thus, on a nonmixed arc where $M \neq 0$, the odd-zero problem for $\mathcal{W}_{\text{src}}$ is equivalent to the odd-zero problem for the source-level 4×4 Wronskian

$$\mathcal{U}_{\text{src}} := W(c, v_5, \lambda, u_5).$$

Proof. By definition,

$$\mathcal{W}_{\text{src}} = W(J_\lambda, J_{u_5}).$$

Now apply the Sylvester–Wronski identity

$$W(W(f_1, f_2, f_3), W(f_1, f_2, f_4)) = W(f_1, f_2) W(f_1, f_2, f_3, f_4)$$

with

$$(f_1, f_2, f_3, f_4) = (c, v_5, \lambda, u_5).$$

This gives

$$W(J_\lambda, J_{u_5}) = W(c, v_5) W(c, v_5, \lambda, u_5) = M W(c, v_5, \lambda, u_5),$$

which is exactly the claimed identity. $\square$

Proposition 12.280 (The current source-level Wronskian package is not yet a contradiction engine for nonmixed odd zeros). *Let I be an interval on which*

$$c \neq 0, \quad M = W(c, v_5) \neq 0, \quad J_\lambda = W(c, v_5, \lambda) \neq 0.$$

Then:

(1) *for every analytic forcing $F : I \rightarrow \mathbf{C}$, there exists an analytic function u_5 on I such that*

$$W(c, v_5, \lambda, u_5) = F;$$

(2) *consequently, for every analytic forcing $G : I \rightarrow \mathbf{C}$, there exists an analytic function u_5 on I such that*

$$\mathcal{W}_{\text{src}} = G;$$

(3) *equivalently, for every analytic forcing $R : I \rightarrow \mathbf{C}$, there exists an analytic function u_5 on I such that*

$$\mathcal{R} = R.$$

In particular, at the current source-level ratio/Wronskian/ 4×4 -determinant level, the zero set, parity, and odd/even vanishing pattern of $\mathcal{W}_{\text{src}}$ and $\mathcal{R}$ are formally unconstrained.

Proof. Expand the 4×4 Wronskian along the u_5 -column. Since the highest derivative of u_5 appearing in $W(c, v_5, \lambda, u_5)$ is u_5''' , with coefficient exactly

$$W(c, v_5, \lambda) = J_\lambda,$$

there exist analytic coefficients a_0, a_1, a_2 on I such that

$$W(c, v_5, \lambda, u_5) = J_\lambda u_5''' + a_2 u_5'' + a_1 u_5' + a_0 u_5.$$

Because $J_\lambda \neq 0$ on I , this is a third-order linear analytic ODE for u_5 with nonvanishing leading coefficient. Standard analytic ODE existence therefore gives an analytic solution u_5 on I for every analytic forcing F . This proves (1).

For (2), given analytic G , set

$$F := \frac{G}{M}.$$

Since $M \neq 0$ on I , F is analytic. By (1), choose u_5 such that

$$W(c, v_5, \lambda, u_5) = F.$$

Then Proposition 12.279 gives

$$\mathcal{W}_{\text{src}} = M W(c, v_5, \lambda, u_5) = G.$$

For (3), given analytic R , set

$$G := \frac{M J_\lambda^2}{c^2} R.$$

Since $c \neq 0$, $M \neq 0$, and $J_\lambda \neq 0$ on I , this is analytic. By (2), choose u_5 such that $\mathcal{W}_{\text{src}} = G$. Then Proposition 12.236 gives

$$\mathcal{R} = \frac{c^2}{M} \frac{\mathcal{W}_{\text{src}}}{J_\lambda^2} = \frac{c^2}{M} \frac{G}{J_\lambda^2} = R. \quad \square$$

Remark 12.281 (Ceiling of the current source-level reduction). Propositions 12.279 and 12.280 show that the current source-level reduction has been pushed to its natural ceiling. The nonmixed 4-point branch is indeed equivalent to an odd-zero problem for the gauge-invariant source-level Wronskian

$$\mathcal{W}_{\text{src}},$$

and even to the 4×4 Wronskian $W(c, v_5, \lambda, u_5)$. But at that level the odd-zero pattern is still formally free.

Accordingly, the missing ingredient can no longer be a generic Wronskian or determinant argument. It must be a genuine *source-coupling theorem*: one that uses additional identities

coming from the actual RH source structure to show that the analytically possible u_5 -freedom exhibited above is not realizable on a genuine opened nonmixed arc.

Remark 12.282 (First-loss rigidification of the quadratic-envelope scalar). Set

$$M(a, x) := B_a(x)^2 \Gamma'_a(x) = A'_a(x) B_a(x) - A_a(x) B'_a(x).$$

Then

$$\partial_x M(a, x) = S''(x) B_a(x) (\mathcal{Z}(x) - \Gamma_a(x)).$$

Moreover, at any point where $M(a, x) = 0$,

$$\partial_a M(a, x) = S''(a) C_a(x) (\mathcal{Z}(a) - \Gamma_a(x)),$$

where

$$C_a(x) := S(a) - S(x) + S'(x)(x - a) = \int_a^x (t - a) S''(t) dt.$$

Thus any formally attained first loss of monotonicity for Γ_a forces

$$\Gamma_a(x) = \mathcal{Z}(a).$$

After the affine shear

$$\tilde{Y}_\gamma := Y - \gamma S, \quad \gamma = \mathcal{Z}(a),$$

this yields

$$\tilde{Y}_\gamma''(a) = 0, \quad \tilde{H}_3(a, x) = 0, \quad \partial_x \tilde{H}_3(a, x) = 0.$$

So a first loss of quadratic-envelope monotonicity can only occur through a sheared repeated-node tangent-contact configuration rooted at a curvature-zero point. In particular, the remaining nonmixed obstruction is diagonal in nature: it is governed by odd zeros of $\mathcal{R}$, not by an independent separated source-target mechanism.

Proposition 12.283 (The current 3-point package is not by itself a contradiction engine for nonmixed first loss). *Fix an analytic function S on an interval I with*

$$S'' \neq 0 \quad \text{on } I.$$

Let $a < x$ be two points of I . Then there exist an analytic function Y on I and a constant $\gamma \in \mathbf{R}$ such that, with

$$\tilde{Y}_\gamma := Y - \gamma S,$$

one has

$$\tilde{Y}_\gamma''(a) = 0, \quad \tilde{H}_3(a, x) = 0, \quad \partial_x \tilde{H}_3(a, x) = 0,$$

where

$$\tilde{H}_3(a, u) := \tilde{Y}_\gamma(u) - \tilde{Y}_\gamma(a) - \tilde{Y}_\gamma'(a)(u - a).$$

In particular, the rigidified repeated-node tangent-contact event from Remark 12.282 is formally compatible with an arbitrary fixed nonmixed planar curve

$$\Lambda_v = (x, S),$$

so the established 3-point package for Λ_v alone cannot exclude that event.

Proof. Define

$$T_{a,x}(t) := (t - a)^3(t - x)^2.$$

Then

$$T_{a,x}(a) = 0, \quad T'_{a,x}(a) = 0, \quad T''_{a,x}(a) = 0, \quad T_{a,x}(x) = 0, \quad T'_{a,x}(x) = 0.$$

Hence

$$T_{a,x}(x) - T_{a,x}(a) - T'_{a,x}(a)(x - a) = 0, \quad T'_{a,x}(x) - T'_{a,x}(a) = 0.$$

Now fix any $\gamma \in \mathbf{R}$ and set

$$Y := \gamma S + T_{a,x}.$$

Then

$$\tilde{Y}_\gamma = Y - \gamma S = T_{a,x},$$

so

$$\tilde{Y}_\gamma''(a) = 0, \quad \tilde{H}_3(a, x) = 0, \quad \partial_x \tilde{H}_3(a, x) = 0.$$

Since S itself has not changed and $S'' \neq 0$ on I , the original 3-point geometry of $\Lambda_v = (x, S)$ on I is exactly the same as before. Thus the sheared repeated-node tangent-contact event is not ruled out by the planar derivative geometry of Λ_v alone. $\square$

Proposition 12.284 (A sheared repeated-node event forces two interior $\mathcal{R}$ -zeros). *Let J be a connected opened nonmixed arc, and let*

$$a < x \quad \text{lie in } J.$$

Set

$$\gamma := \mathcal{Z}(a), \quad \tilde{Y}_\gamma := Y - \gamma S,$$

and let

$$\tilde{H}_3(a, u) := \tilde{Y}_\gamma(u) - \tilde{Y}_\gamma(a) - \tilde{Y}_\gamma'(a)(u - a).$$

Assume

$$\tilde{Y}_\gamma''(a) = 0, \quad \tilde{H}_3(a, x) = 0, \quad \partial_x \tilde{H}_3(a, x) = 0.$$

Then one of the following holds:

- (1) $\mathcal{R} \equiv 0$ on $[a, x]$;
- (2) there exist points

$$a < r_1 < r_2 < x$$

such that

$$\mathcal{Z}(r_1) = \mathcal{Z}(r_2) = \mathcal{Z}(a).$$

Consequently there exist

$$\xi_1 \in (a, r_1), \quad \xi_2 \in (r_1, r_2)$$

with

$$\mathcal{R}(\xi_1) = \mathcal{R}(\xi_2) = 0.$$

In particular, every nontrivial sheared repeated-node event contains at least two interior zeros of $\mathcal{R}$.

Proof. Since

$$\tilde{Y}_\gamma'' = Y'' - \gamma S'' = (\mathcal{Z} - \gamma)S'',$$

the choice $\gamma = \mathcal{Z}(a)$ gives

$$\tilde{Y}_\gamma'' = (\mathcal{Z} - \mathcal{Z}(a))S''.$$

Also,

$$\partial_x \tilde{H}_3(a, x) = \tilde{Y}_\gamma'(x) - \tilde{Y}_\gamma'(a) = \int_a^x \tilde{Y}_\gamma''(t) dt,$$

and

$$\tilde{H}_3(a, x) = \int_a^x (x - t) \tilde{Y}_\gamma''(t) dt.$$

Therefore the hypotheses imply

$$\int_a^x (\mathcal{Z}(t) - \mathcal{Z}(a)) S''(t) dt = 0,$$

and

$$\int_a^x (x - t) (\mathcal{Z}(t) - \mathcal{Z}(a)) S''(t) dt = 0.$$

Choose $\varepsilon \in \{\pm 1\}$ so that

$$\varepsilon S'' > 0 \quad \text{on } J,$$

and define the positive measure

$$d\mu(t) := \varepsilon S''(t) dt, \quad \phi(t) := \mathcal{Z}(t) - \mathcal{Z}(a).$$

Then

$$\int_a^x \phi \, d\mu = 0, \quad \int_a^x (x-t)\phi \, d\mu = 0.$$

Hence also

$$\int_a^x t \phi(t) \, d\mu(t) = 0.$$

If $\phi \equiv 0$ on $[a, x]$, then

$$\mathcal{R} = \phi' \equiv 0 \quad \text{on } [a, x],$$

which is alternative (1).

Assume now that $\phi \not\equiv 0$. Then ϕ cannot have fixed sign on $[a, x]$, because μ is positive and

$$\int_a^x \phi \, d\mu = 0.$$

It also cannot have exactly one sign change. Indeed, if $c \in (a, x)$ were the unique sign-change point, then

$$(t-c)\phi(t) \geq 0 \quad \text{on } [a, x],$$

with strict inequality on a set of positive μ -measure. Therefore

$$\int_a^x (t-c)\phi(t) \, d\mu(t) > 0.$$

But

$$\int_a^x (t-c)\phi(t) \, d\mu(t) = \int_a^x t \phi(t) \, d\mu(t) - c \int_a^x \phi(t) \, d\mu(t) = 0,$$

a contradiction.

So ϕ has at least two sign changes on (a, x) . Since $\phi(a) = 0$ and ϕ is continuous, there exist

$$a < r_1 < r_2 < x$$

such that

$$\phi(r_1) = \phi(r_2) = 0,$$

i.e.

$$\mathcal{Z}(r_1) = \mathcal{Z}(r_2) = \mathcal{Z}(a).$$

Applying Rolle's theorem to $\phi = \mathcal{Z} - \mathcal{Z}(a)$ on $[a, r_1]$ and on $[r_1, r_2]$ gives

$$\mathcal{R}(\xi_1) = 0 \quad \text{for some } \xi_1 \in (a, r_1), \quad \mathcal{R}(\xi_2) = 0 \quad \text{for some } \xi_2 \in (r_1, r_2).$$

This proves (2). □

Corollary 12.285 (Isolated $\mathcal{R}$ -zeros are shear-sterile). *Let $b \in J$ be an isolated zero of $\mathcal{R}$. Then there exists a neighborhood $U \subset J$ of b such that no pair*

$$a < x, \quad a, x \in U,$$

can satisfy

$$\gamma = \mathcal{Z}(a), \quad \tilde{Y}_\gamma''(a) = 0, \quad \tilde{H}_3(a, x) = 0, \quad \partial_x \tilde{H}_3(a, x) = 0.$$

Proof. Choose U so small that b is the only zero of $\mathcal{R}$ in U . If such a pair $a < x$ existed in U , then Proposition 12.284 would produce two interior zeros of $\mathcal{R}$ in $(a, x) \subset U$, a contradiction. □

Corollary 12.286 (Diagonal sheared recurrence forces affine-planar collapse). *Let J be a connected opened nonmixed arc. Suppose there exists a sequence*

$$a_n < x_n, \quad a_n, x_n \in J, \quad a_n, x_n \rightarrow b, \quad x_n - a_n \rightarrow 0,$$

such that, with

$$\gamma_n := \mathcal{Z}(a_n), \quad \tilde{Y}_{\gamma_n} := Y - \gamma_n S,$$

one has

$$\tilde{Y}_{\gamma_n}''(a_n) = 0, \quad \tilde{H}_{3,n}(a_n, x_n) = 0, \quad \partial_x \tilde{H}_{3,n}(a_n, x_n) = 0,$$

where

$$\tilde{H}_{3,n}(a, u) := \tilde{Y}_{\gamma_n}(u) - \tilde{Y}_{\gamma_n}(a) - \tilde{Y}_{\gamma_n}'(a)(u - a).$$

Then

$$\mathcal{R} \equiv 0$$

on a neighborhood of b . Hence on some nontrivial interval containing b one has

$$Y = \gamma S + \alpha x + \beta,$$

so the lifted curve is locally affine-planar there.

Proof. If for some n the first alternative in Proposition 12.284 holds, then $\mathcal{R} \equiv 0$ on $[a_n, x_n]$ and we are done.

Otherwise, for every n the interval (a_n, x_n) contains at least two zeros of $\mathcal{R}$. Since

$$a_n, x_n \rightarrow b, \quad x_n - a_n \rightarrow 0,$$

these zeros accumulate at b . Because $\mathcal{R}$ is analytic, this implies

$$\mathcal{R} \equiv 0$$

on a neighborhood of b . Therefore $\mathcal{Z}$ is locally constant, so

$$Y'' = \gamma S''$$

there, and integrating twice gives

$$Y = \gamma S + \alpha x + \beta$$

on a nontrivial interval around b . □

Proposition 12.287 (Uniform fixed-level nonrecurrence for the sheared 3-point branch). *Let J be a connected opened nonmixed arc, fix a real scalar γ , and set*

$$\tilde{Y}_\gamma := Y - \gamma S = \frac{u_5 - \gamma\lambda}{c}, \quad f_\gamma := u_5 - \gamma\lambda, \quad J_\gamma := J_{f_\gamma} = J_{u_5} - \gamma J_\lambda.$$

Then

$$\tilde{Y}_\gamma'' = \frac{c^3}{M^3} J_\gamma.$$

Assume that $b \in J$ is an isolated finite-order odd zero of J_γ . After affine normalization of the planar curve

$$(x, \tilde{Y}_\gamma)$$

at b , let

$$I_- = (\alpha, 0)$$

be the source interval on which

$$\tilde{Y}_\gamma'' < 0,$$

and let

$$I_+ = (0, \beta)$$

be the maximal open interval to the right of 0 on which

$$\tilde{Y}_\gamma'$$

is strictly increasing. Set

$$\mathcal{J}_\gamma := \tilde{Y}_\gamma'(I_-) \cap \tilde{Y}_\gamma'(I_+).$$

For

$$m \in \mathcal{J}_\gamma,$$

write

$$a_\gamma(m) := (\tilde{Y}_\gamma'|_{I_-})^{-1}(m),$$

and define

$$\tilde{H}_{3,\gamma}(a, u) := \tilde{Y}_\gamma(u) - \tilde{Y}_\gamma(a) - \tilde{Y}_\gamma'(a)(u - a).$$

Then there exists a unique maximal open interval

$$\mathcal{J}_{\gamma, \text{br}} = (0, m_{\gamma,*}) \subset \mathcal{J}_\gamma$$

and a unique real-analytic function

$$u_\gamma : \mathcal{J}_{\gamma, \text{br}} \rightarrow I_+$$

such that

$$\tilde{H}_{3,\gamma}(a_\gamma(m), u_\gamma(m)) = 0 \quad (m \in \mathcal{J}_{\gamma, \text{br}}),$$

and this branch agrees near $m = 0$ with the unique local opposite-side branch born at the odd zero b .

Moreover, along the whole branch one has

$$\partial_u \tilde{H}_{3,\gamma}(a_\gamma(m), u_\gamma(m)) = \tilde{Y}'_\gamma(u_\gamma(m)) - m \neq 0,$$

so there is no interior common-tangent event, no interior recurrence, and no interior accumulation. If

$$m_{\gamma,*} < \sup \mathcal{J}_\gamma,$$

then

$$u_\gamma(m) \longrightarrow \beta \quad (m \uparrow m_{\gamma,*}),$$

so the branch can stop only by boundary exit of the relevant slope-overlap block.

Equivalently, using

$$\tilde{H}_{3,\gamma}(a, u) = A_a(u) - \gamma B_a(u) = B_a(u)(\Gamma_a(u) - \gamma),$$

one has along the branch

$$\Gamma_{a_\gamma(m)}(u_\gamma(m)) = \gamma, \quad \Gamma'_{a_\gamma(m)}(u_\gamma(m)) \neq 0.$$

In particular, for every fixed γ , the distinguished level γ of the repeated-source scalar Γ_a has no interior critical point on the sheared odd-zero branch.

Proof. By Remark 12.235, for any scalar germ f one has

$$\frac{d^2}{dx^2} \left(\frac{f}{c} \right) = \frac{c^3}{M^3} J_f.$$

Applying this with

$$f = f_\gamma = u_5 - \gamma \lambda$$

gives

$$\tilde{Y}_\gamma'' = \frac{c^3}{M^3} J_\gamma.$$

Hence b is an isolated finite-order odd zero of J_γ if and only if it is an isolated finite-order odd zero of the planar curvature

$$\tilde{Y}_\gamma''.$$

Now apply Theorem 12.254 and Theorem 12.258 with the affine-normalized analytic function

$$\tilde{Y}_\gamma$$

playing the role of the planar graphing function S . The proofs of those results use only the planar repeated-node tangent-contact equation for the chosen analytic function, its curvature parity at the base point, and the maximal monotone-slope block on the opposite side. Therefore they apply verbatim to

$$\tilde{H}_{3,\gamma}(a, u) = \tilde{Y}_\gamma(u) - \tilde{Y}_\gamma(a) - \tilde{Y}'_\gamma(a)(u - a),$$

and yield the unique local opposite-side branch, its unique continuation on the maximal slope-overlap block, the nonvanishing of

$$\partial_u \tilde{H}_{3,\gamma},$$

and the boundary-exit alternative. This proves the first part.

For the reformulation in terms of Γ_a , note from the definitions of A_a , B_a , and Γ_a that

$$\tilde{H}_{3,\gamma}(a, u) = (Y(u) - Y(a) - Y'(a)(u - a)) - \gamma(S(u) - S(a) - S'(a)(u - a)) = A_a(u) - \gamma B_a(u).$$

Since

$$A_a(u) = \Gamma_a(u) B_a(u),$$

this becomes

$$\tilde{H}_{3,\gamma}(a, u) = B_a(u)(\Gamma_a(u) - \gamma).$$

By Proposition 12.274,

$$B_a(u) \neq 0 \quad (u \neq a),$$

so along the branch

$$\tilde{H}_{3,\gamma}(a_\gamma(m), u_\gamma(m)) = 0 \iff \Gamma_{a_\gamma(m)}(u_\gamma(m)) = \gamma.$$

Differentiating

$$\tilde{H}_{3,\gamma}(a, u) = B_a(u)(\Gamma_a(u) - \gamma)$$

with respect to u gives

$$\partial_u \tilde{H}_{3,\gamma}(a, u) = B'_a(u)(\Gamma_a(u) - \gamma) + B_a(u)\Gamma'_a(u).$$

On the branch, where

$$\Gamma_a(u) = \gamma,$$

this simplifies to

$$\partial_u \tilde{H}_{3,\gamma}(a, u) = B_a(u)\Gamma'_a(u).$$

Since the first part already gives

$$\partial_u \tilde{H}_{3,\gamma}(a_\gamma(m), u_\gamma(m)) \neq 0$$

and

$$B_{a_\gamma(m)}(u_\gamma(m)) \neq 0,$$

it follows that

$$\Gamma'_{a_\gamma(m)}(u_\gamma(m)) \neq 0$$

along the whole branch. This proves the final statement. $\square$

Remark 12.288 (No-go for the fixed-level chamber-exit route). One might hope to strengthen Proposition 12.287 as follows: if a fixed level γ is chosen near a strict local extremum of $\mathcal{Z}$, so that

$$J_\gamma = (\mathcal{Z} - \gamma)J_\lambda$$

has two nearby simple zeros

$$r_- < r_+,$$

then the corresponding sheared opposite-side 3-point branch should be unable to reach the chamber boundary $u = r_+$ without first encountering an interior common-tangent point

$$\partial_u \tilde{H}_{3,\gamma} = 0.$$

This is false already at the level of the local planar/sheared equations.

Indeed, consider the model functions

$$S_{\text{mod}}(x) := x^2, \quad \mathcal{Z}_{\text{mod}}(x) := 1 - x^2, \quad \mathcal{R}_{\text{mod}}(x) := \mathcal{Z}'_{\text{mod}}(x) = -2x,$$

and fix the level

$$\gamma := 0.$$

Set

$$F_{\text{mod}}(x) := x^2 - \frac{x^4}{6}.$$

Then

$$F''_{\text{mod}}(x) = 2(1 - x^2) = (\mathcal{Z}_{\text{mod}}(x) - \gamma) S''_{\text{mod}}(x),$$

so the two simple zeros of the sheared curvature are

$$r_- = -1, \quad r_+ = 1.$$

Let

$$H_{\text{mod}}(a, u) := F_{\text{mod}}(u) - F_{\text{mod}}(a) - F'_{\text{mod}}(a)(u - a).$$

A direct computation gives

$$H_{\text{mod}}(a, u) = -\frac{(u - a)^2}{6} (3a^2 + 2au + u^2 - 6).$$

Hence the nontrivial opposite-side branch issued from the odd zero $r_- = -1$ is

$$u(a) = -a - \sqrt{6 - 2a^2}, \quad -\frac{5}{3} < a < -1.$$

Along this branch one has

$$-1 < u(a) < 1, \quad u(a) \uparrow 1 \quad \text{as} \quad a \downarrow -\frac{5}{3},$$

so the branch runs inside the chamber and stops by boundary exit at the second curvature zero.

Moreover,

$$\partial_u H_{\text{mod}}(a, u) = F'_{\text{mod}}(u) - F'_{\text{mod}}(a) = -\frac{u-a}{3} (a^2 + au + u^2 - 3).$$

Substituting $u = u(a)$ yields

$$a^2 + au(a) + u(a)^2 - 3 = -a^2 + a\sqrt{6 - 2a^2} + 3 < 0 \quad \left(-\frac{5}{3} < a < -1\right),$$

and therefore

$$\partial_u H_{\text{mod}}(a, u(a)) > 0 \quad \left(-\frac{5}{3} < a < -1\right).$$

So there is *no* interior common-tangent point anywhere on the branch before boundary exit. In fact, if one continues the same algebraic branch past the chamber boundary, the first common tangent occurs only later at

$$(a, u) = (-\sqrt{3}, \sqrt{3}),$$

strictly after the chamber has already been crossed.

Thus fixed-level sheared nonrecurrence cannot be sharpened to a chamber-exit contradiction by planar arguments alone. The remaining obstruction is therefore genuinely source-coupled: any successful odd-zero exclusion must use the source-pinned moving shear

$$f_a = u_5 - \mathcal{Z}(a)\lambda$$

or some equivalent RH-specific coupling, rather than only a frozen level γ and the planar branch geometry of

$$\tilde{Y}_\gamma = Y - \gamma S.$$

Remark 12.289 (Source-level form of the shear transfer). For fixed $a \in J$, set

$$f_a := u_5 - \mathcal{Z}(a)\lambda.$$

Then

$$\tilde{Y}_{\mathcal{Z}(a)} = \frac{f_a}{c}, \quad J_{f_a} = J_{u_5} - \mathcal{Z}(a)J_\lambda = (\mathcal{Z} - \mathcal{Z}(a))J_\lambda,$$

and

$$D_{f_a}(a, h) = D_{u_5}(a, h) - \mathcal{Z}(a) D_\lambda(a, h).$$

Thus the sheared repeated-node conditions are exactly a repeated-node tangent-contact event for the source triple (c, v_5, f_a) :

$$J_{f_a}(a) = 0, \quad D_{f_a}(a, h) = 0, \quad \partial_h D_{f_a}(a, h) = 0.$$

Proposition 12.284 shows that such a source-level event forces either $J_{f_a} \equiv 0$ on the interval or at least two interior zeros of J_{f_a} , equivalently at least two interior zeros of $\mathcal{R}$ and hence of $\mathcal{W}_{\text{src}}$.

Proposition 12.290 (Source-determinant form of the odd-zero unrooted defect). *Let J be a good nonmixed v_5 -patch, and let $b \in J$ be an isolated odd-order zero of $\mathcal{R}$. Use the local x -coordinate on J centered at b , so that*

$$x(b) = 0, \quad \mathcal{R}(x) = x^m \rho(x), \quad m \geq 1 \text{ odd}, \quad \rho(0) \neq 0.$$

For each source point $a \in J$, set

$$\Psi_a(h) := \frac{D_{u_5}(a, h)}{D_\lambda(a, h)}, \quad f_a := u_5 - \mathcal{Z}(x(a))\lambda.$$

Then, after shrinking J around b if necessary, the following hold.

(1) *The critical locus*

$$\partial_h \Psi_a(h) = 0 \quad \text{equivalently} \quad \mathfrak{D}(a, h) = 0$$

is the graph of a unique real-analytic branch

$$h = \sigma(a), \quad \sigma(b) = b, \quad \sigma(a) \neq a \quad (a \neq b).$$

(2) *One has the exact identity*

$$D_{f_a}(a, h) = D_{u_5}(a, h) - \mathcal{Z}(x(a)) D_\lambda(a, h) = D_\lambda(a, h) (\Psi_a(h) - \mathcal{Z}(x(a))).$$

(3) *Along the critical branch,*

$$\Psi_a(\sigma(a)) - \mathcal{Z}(x(a)) = -\frac{\rho(0)}{2^{m+1}(m+2)} (x(\sigma(a)) - x(a))^{m+1} + o(|x(\sigma(a)) - x(a)|^{m+1}).$$

(4) *Consequently,*

$$D_{f_a}(a, \sigma(a)) = -\frac{c(b)^4 J_\lambda(b) \rho(0)}{2^{m+2}(m+2) M(b)^2} (x(\sigma(a)) - x(a))^{m+3} + o(|x(\sigma(a)) - x(a)|^{m+3}).$$

In particular, for all sufficiently small $a \neq b$ one has

$$D_{f_a}(a, \sigma(a)) \neq 0,$$

and indeed

$$\operatorname{sgn} D_{f_a}(a, \sigma(a)) = -\operatorname{sgn}(J_\lambda(b) \rho(0)).$$

So near an isolated odd zero of $\mathcal{R}$, the born repeated-source critical branch

$$\mathfrak{D}(a, h) = 0$$

is uniformly unrooted in source-determinant language:

$$D_{u_5}(a, h) - \mathcal{Z}(x(a)) D_\lambda(a, h) \neq 0.$$

Proof. By Remark 12.276,

$$\Psi_a(h) = \Gamma_{x(a)}(x(h)).$$

Therefore the critical locus

$$\partial_h \Psi_a(h) = 0$$

is exactly the pullback through the source-to- x coordinate of the local critical branch

$$\partial_x \Gamma_{x(a)}(x) = 0.$$

So item (1) is the source-coordinate form of Proposition 12.272.

For item (2), linearity of the determinant in the third column gives

$$D_{f_a}(a, h) = D_{u_5}(a, h) - \mathcal{Z}(x(a)) D_\lambda(a, h).$$

Dividing by $D_\lambda(a, h) \neq 0$ on the nonmixed patch yields

$$\frac{D_{f_a}(a, h)}{D_\lambda(a, h)} = \Psi_a(h) - \mathcal{Z}(x(a)).$$

For item (3), apply Proposition 12.272 to the x -coordinate pair

$$(x(a), x(\sigma(a))).$$

Since

$$\Psi_a(\sigma(a)) = \Gamma_{x(a)}(x(\sigma(a))),$$

that proposition gives exactly

$$\Psi_a(\sigma(a)) - \mathcal{Z}(x(a)) = -\frac{\rho(0)}{2^{m+1}(m+2)} (x(\sigma(a)) - x(a))^{m+1} + o(|x(\sigma(a)) - x(a)|^{m+1}).$$

For item (4), apply the row-reduction identity from Remark 12.276 to

$$f = f_a, \quad F = \frac{f_a}{c} = Y - \mathcal{Z}(x(a))S.$$

This gives

$$D_{f_a}(a, h) = c(a)^2 c(h) x'(a) \tilde{H}_{3, \mathcal{Z}(x(a))}(x(a), x(h)),$$

and since

$$\tilde{H}_{3, \mathcal{Z}(x(a))}(x(a), x(h)) = B_{x(a)}(x(h))(\Psi_a(h) - \mathcal{Z}(x(a))),$$

we obtain

$$D_{f_a}(a, h) = c(a)^2 c(h) x'(a) B_{x(a)}(x(h))(\Psi_a(h) - \mathcal{Z}(x(a))).$$

Along the branch $h = \sigma(a)$, write

$$\delta(a) := x(\sigma(a)) - x(a).$$

Then

$$B_{x(a)}(x(\sigma(a))) = \frac{1}{2} S''(b) \delta(a)^2 + o(|\delta(a)|^2),$$

while

$$x'(b) = \frac{M(b)}{c(b)^2}, \quad S''(b) = \frac{c(b)^3}{M(b)^3} J_\lambda(b)$$

by Remark 12.235. Therefore

$$c(a)^2 c(\sigma(a)) x'(a) B_{x(a)}(x(\sigma(a))) = \frac{c(b)^4 J_\lambda(b)}{2M(b)^2} \delta(a)^2 + o(|\delta(a)|^2).$$

Multiplying by item (3) gives

$$D_{f_a}(a, \sigma(a)) = -\frac{c(b)^4 J_\lambda(b) \rho(0)}{2^{m+2} (m+2) M(b)^2} \delta(a)^{m+3} + o(|\delta(a)|^{m+3}),$$

which is the claimed asymptotic. Since $m+3$ is even, the leading term has fixed nonzero sign for all sufficiently small $a \neq b$. $\square$

Remark 12.291 (Ceiling of the local shear-transfer route). Proposition 12.290 shows that near an isolated odd zero of $\mathcal{R}$, the genuinely born local repeated-source branch is the critical branch

$$\mathfrak{D}(a, h) = 0 \quad \text{equivalently} \quad \partial_h \Psi_a(h) = 0,$$

but its rooted numerator

$$D_{f_a}(a, h) = D_{u_5}(a, h) - \mathcal{Z}(x(a)) D_\lambda(a, h)$$

has a fixed nonzero leading term there.

So the rooted sheared repeated-node tangent-contact condition is not the local source avatar of diagonal odd-zero birth. In particular, no argument using only the local odd-zero model and the repeated-source critical-branch equations can prove an implication of the form

$$\mathfrak{D}(a, h) = 0 \implies D_{f_a}(a, h) = 0$$

near odd zeros.

This is a different obstruction from Remark 12.281: the latter says the generic source-level Wronskian/ 4×4 package is formally too free, while the present remark says that even along the *actual local odd-zero branch*, the rooted numerator fails to vanish with a universal signed defect. Any successful shear-transfer theorem must therefore use genuinely additional RH source coupling beyond the local odd-zero critical-branch analysis.

Proposition 12.292 (Repeated-source first loss is exactly diagonal odd-zero birth). *Let J be a connected opened nonmixed arc, so*

$$K_v \neq 0 \quad \text{on } J.$$

For $a < x$ in J , define

$$A_a(x) := Y(x) - Y(a) - Y'(a)(x - a), \quad B_a(x) := S(x) - S(a) - S'(a)(x - a),$$

$$\Gamma_a(x) := \frac{A_a(x)}{B_a(x)}, \quad M(a, x) := B_a(x)^2 \Gamma'_a(x).$$

Equivalently,

$$M(a, x) = A'_a(x) B_a(x) - A_a(x) B'_a(x).$$

Let

$$\Sigma_M := \{(a, x) \in J^2 : a < x, M(a, x) = 0\}.$$

Then the following hold.

(1) No attained separated first loss. There is no pair $(a_*, x_*) \in \Sigma_M$ such that

$$x_* = \inf\{x : \exists a < x \text{ with } (a, x) \in \Sigma_M\}.$$

(2) Any first-loss sequence is diagonal. If $(a_n, x_n) \in \Sigma_M$ and

$$x_n \downarrow x_* := \inf\{x : \exists a < x \text{ with } (a, x) \in \Sigma_M\},$$

then

$$x_n - a_n \rightarrow 0.$$

(3) Diagonal accumulation forces a zero of $\mathcal{R}$. If $(a_n, x_n) \in \Sigma_M$ and

$$a_n, x_n \rightarrow b, \quad x_n - a_n \rightarrow 0,$$

then

$$\mathcal{R}(b) = 0.$$

(4) Even-order zeros are diagonally sterile. If b is an isolated even-order zero of $\mathcal{R}$, then there exists a neighborhood U of b such that

$$\Sigma_M \cap (U \times U) = \emptyset.$$

(5) Odd-order zeros seed diagonal first loss. If b is an isolated odd-order zero of $\mathcal{R}$, then every neighborhood U of b contains points

$$(a, x) \in \Sigma_M \quad \text{with} \quad a < x, \quad a, x \in U.$$

Accordingly, repeated-source first loss on a nonmixed arc is not an independent obstruction mechanism: it occurs if and only if one diagonally approaches an odd-order zero of $\mathcal{R}$.

Proof. For (1), suppose $(a_*, x_*) \in \Sigma_M$ were an attained separated first-loss pair. Then for fixed a_* one has

$$M(a_*, x) > 0 \quad (a_* < x < x_*),$$

by the definition of x_* . Since $B_{a_*}(x) \neq 0$ on a nonmixed arc, this implies

$$\Gamma'_{a_*}(x) > 0 \quad (a_* < x < x_*),$$

so Γ_{a_*} is strictly increasing on (a_*, x_*) . Therefore

$$\Gamma_{a_*}(x_*) > \lim_{x \downarrow a_*} \Gamma_{a_*}(x) = \mathcal{Z}(a_*).$$

But Remark 12.282 shows that every formally attained first loss satisfies

$$\Gamma_{a_*}(x_*) = \mathcal{Z}(a_*),$$

a contradiction. Thus no attained separated first-loss pair exists.

For (2), suppose $(a_n, x_n) \in \Sigma_M$ and $x_n \downarrow x_*$, but that $x_n - a_n \not\rightarrow 0$. Passing to a subsequence, we may assume

$$a_n \rightarrow a_* < x_*.$$

By continuity of M ,

$$M(a_*, x_*) = 0,$$

so $(a_*, x_*) \in \Sigma_M$, contradicting (1). Hence $x_n - a_n \rightarrow 0$.

For (3), since $B_a(x) \neq 0$ on a nonmixed arc, the condition $(a_n, x_n) \in \Sigma_M$ is equivalent to

$$\Gamma'_{a_n}(x_n) = 0.$$

In the quadratic-envelope notation from Proposition 12.274,

$$\Gamma'_a(x) = \frac{1}{B_a(x)^2} \int_a^x \Omega_{a,x}(r) \mathcal{R}(r) dr.$$

Equivalently, after local nonmixed normalization at b , this becomes the repeated-node profile derivative formula

$$D'_a(x) = 2 \int_0^1 s(1-s) \mathcal{R}(a + s(x-a)) ds.$$

Letting $a_n, x_n \rightarrow b$ with $x_n - a_n \rightarrow 0$, continuity yields

$$0 = \lim_{n \rightarrow \infty} D'_{a_n}(x_n) = 2 \left(\int_0^1 s(1-s) ds \right) \mathcal{R}(b) = \frac{1}{3} \mathcal{R}(b).$$

Hence $\mathcal{R}(b) = 0$.

For (4), if b is an isolated even-order zero of $\mathcal{R}$, then Proposition 12.270 gives a neighborhood U of b on which every repeated-node profile D_a is strictly monotone. Therefore $D'_a(x) \neq 0$ for $a \neq x$ in U , hence $\Gamma'_a(x) \neq 0$ and

$$\Sigma_M \cap (U \times U) = \emptyset.$$

For (5), if b is an isolated odd-order zero of $\mathcal{R}$, then Proposition 12.270 gives, after local normalization, a local repeated-node extremum branch for the profiles D_a . In particular, every neighborhood U of b contains points $a < x$ with

$$D'_a(x) = 0.$$

Since $B_a(x) \neq 0$ on the nonmixed arc, this is equivalent to

$$\Gamma'_a(x) = 0,$$

hence $(a, x) \in \Sigma_M \cap (U \times U)$.

The final statement is the combination of (2), (4), and (5). $\square$

Remark 12.293 (Effect on the rigidified first-loss lane). Proposition 12.292 shows that the repeated-source / first-loss mechanism is not an independent obstruction: no attained separated first loss can occur, every first-loss sequence is forced to the diagonal, and diagonal first loss occurs exactly at isolated odd-order zeros of $\mathcal{R}$.

At the same time, Corollary 12.273 shows that near an isolated odd zero of $\mathcal{R}$, the rooted sheared repeated-node tangent-contact event itself is locally absent; and Corollary 12.286 shows that any genuine diagonal realization of the sheared event already forces the affine-planar collapse case

$$\mathcal{R} \equiv 0.$$

So the sheared bottleneck closes in the collapse direction, not by a purely planar contradiction for (x, S) .

Proposition 12.287 shows moreover that for each fixed shear level γ , the corresponding sheared opposite-side 3-point branch is already globally rigid on its maximal slope-overlap block: it has no interior common-tangent event, no interior recurrence, and no interior accumulation. Thus the remaining issue is not sheared 3-point recurrence at fixed γ , but the source-coupling / level-selection step that would force the distinguished level

$$\gamma = \mathcal{Z}(b)$$

arising from a hypothetical odd zero of $\mathcal{W}_{\text{src}}$ to be hit by the relevant critical-value configuration of Γ_a .

Remark 12.288 shows that this fixed-level route cannot be pushed further by planar chamber-exit arguments alone: even with two nearby roots of $\mathcal{Z} = \gamma$, the sheared opposite-side branch may run to the second curvature zero by boundary exit without any interior common-tangent event. Thus the remaining step must be genuinely source-coupled, involving the source-pinned moving shear

$$f_a = u_5 - \mathcal{Z}(a)\lambda$$

or an equivalent RH-specific coupling, rather than only the frozen-level geometry of

$$\tilde{Y}_\gamma = Y - \gamma S.$$

Remark 12.294 (The repeated-source lane is now closed). Proposition 12.292 shows that the repeated-source / first-loss mechanism does not define an additional nonmixed obstruction beyond the odd-zero problem for $\mathcal{R}$. No attained separated first loss can occur, every first-loss sequence is forced to the diagonal, and diagonal first loss occurs exactly at isolated odd-order zeros of $\mathcal{R}$. Thus the genuinely remaining nonmixed burden is purely the global/source-level control of odd zeros of $\mathcal{R}$, equivalently of the source-level Wronskian $\mathcal{W}_{\text{src}}$.

Proposition 12.295 (Single-sign-change source segment cannot realize rigidified first loss). *Let J be an opened nonmixed arc, and let $a < x$ in J . After local nonmixed normalization on $[a, x]$, write*

$$R_{a,x}(s) := \mathcal{R}(a + s(x - a)), \quad 0 \leq s \leq 1.$$

Then

$$\Gamma'_a(x) = 2 \int_0^1 s(1-s) R_{a,x}(s) ds,$$

and

$$\Gamma_a(x) - \mathcal{Z}(a) = (x - a) \int_0^1 (1-s)^2 R_{a,x}(s) ds.$$

Assume that $R_{a,x}$ has exactly one sign change on $(0, 1)$. If

$$\Gamma'_a(x) = 0,$$

then

$$\text{sgn}(\Gamma_a(x) - \mathcal{Z}(a))$$

is the sign of $R_{a,x}$ on the initial side of the sign change. In particular,

$$\Gamma_a(x) \neq \mathcal{Z}(a).$$

Equivalently, no pair (a, x) for which $\mathcal{R}$ changes sign exactly once on (a, x) can satisfy simultaneously

$$\Gamma'_a(x) = 0, \quad \Gamma_a(x) = \mathcal{Z}(a).$$

Hence any rigidified first-loss segment must cross at least two sign changes of $\mathcal{R}$.

Proof. In the normalized nonmixed chart, $S(t) = t^2$, so $\Gamma_a = D_a$ with

$$D_a(t) = \int_0^1 (1-s) \tilde{Y}''(a + s(t-a)) ds, \quad D_a(a) = \mathcal{Z}(a).$$

Differentiating in t gives

$$D'_a(t) = \int_0^1 s(1-s) \tilde{Y}'''(a + s(t-a)) ds = 2 \int_0^1 s(1-s) \mathcal{R}(a + s(t-a)) ds,$$

which is the first displayed formula at $t = x$. Integrating $\tilde{Y}''(a + s(t-a))$ by parts in s yields

$$D_a(t) - D_a(a) = (t-a) \int_0^1 (1-s)^2 \mathcal{R}(a + s(t-a)) ds,$$

which is the second displayed formula at $t = x$.

Let

$$k_1(s) := s(1-s), \quad k_0(s) := (1-s)^2,$$

so that

$$\frac{k_0(s)}{k_1(s)} = \frac{1-s}{s}$$

is strictly decreasing on $(0, 1)$.

Write $R_{a,x} = \sigma\rho$ with $\rho \geq 0$ and σ changing sign exactly once at some $s_0 \in (0, 1)$. By hypothesis,

$$0 = \int_0^1 k_1(s) \sigma(s) \rho(s) ds.$$

Split

$$I := \int_0^1 k_0(s) \sigma(s) \rho(s) ds = \int_0^{s_0} \frac{k_0}{k_1}(s) k_1(s) \sigma(s) \rho(s) ds + \int_{s_0}^1 \frac{k_0}{k_1}(s) k_1(s) \sigma(s) \rho(s) ds.$$

Let $\epsilon := \operatorname{sgn}(\sigma|_{(0,s_0)})$. Then $k_1 \sigma \rho \geq 0$ with sign ϵ on $(0, s_0)$ and sign $-\epsilon$ on $(s_0, 1)$. Because k_0/k_1 is strictly decreasing, it is bounded below by its value at s_0 on $(0, s_0)$ and bounded above by its value at s_0 on $(s_0, 1)$. Using

$$\int_0^{s_0} k_1 \sigma \rho = - \int_{s_0}^1 k_1 \sigma \rho$$

and strict monotonicity on the support of ρ gives

$$\epsilon I > 0.$$

Hence

$$\Gamma_a(x) - \mathcal{Z}(a) = (x - a) I \neq 0,$$

with sign ϵ (after orienting $x > a$). This proves the claim. $\square$

Corollary 12.296 (Source-level two-sign-change obstruction). *Let $a < h$ lie on an opened nonmixed arc in source coordinates. Assume*

$$\Psi'_a(h) = 0.$$

If the source-level Wronskian

$$\mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda$$

changes sign exactly once on the source interval $[a, h]$, then

$$\Psi_a(h) \neq \mathcal{Z}(a).$$

Equivalently, the simultaneous source-level first-loss conditions

$$\Psi'_a(h) = 0, \quad \Psi_a(h) = \mathcal{Z}(a)$$

force at least two sign changes of $\mathcal{W}_{\text{src}}$ on $[a, h]$.

Proof. By Remark 12.276, $\Psi_a(h) = \Gamma_{x(a)}(x(h))$, and on the nonmixed arc $\mathcal{W}_{\text{src}}$ and $\mathcal{R}$ differ only by a nonvanishing analytic factor (Proposition 12.236). So a single sign change of $\mathcal{W}_{\text{src}}$ on $[a, h]$ pulls back to a single sign change of $\mathcal{R}$ on $[x(a), x(h)]$. Apply Proposition 12.295. $\square$

Proposition 12.297 (Transport equations for a repeated-source critical branch). *Let J be an opened nonmixed arc, and work in a local normalized nonmixed chart on which*

$$S(t) = t^2, \quad \Gamma_a = D_a.$$

Let $U \subset J$ be an interval, and let

$$\tau : U \rightarrow J$$

be a real-analytic function such that

$$\tau(a) > a, \quad D'_a(\tau(a)) = 0 \quad (a \in U).$$

Set

$$V(a) := D_a(\tau(a)).$$

Then:

(1)

$$V'(a) = \frac{2}{\tau(a) - a} (V(a) - \mathcal{Z}(a)).$$

(2)

$$D''_a(\tau(a)) = \frac{2}{(\tau(a) - a)^2} (\mathcal{Z}(\tau(a)) - V(a)).$$

(3) *At any point $a \in U$ where $D''_a(\tau(a)) \neq 0$,*

$$\tau'(a) = \frac{V(a) - \mathcal{Z}(a)}{\mathcal{Z}(\tau(a)) - V(a)}.$$

Proof. Write

$$D_a(t) = \frac{\tilde{Y}(t) - \tilde{Y}(a) - \tilde{Y}'(a)(t-a)}{(t-a)^2}.$$

Differentiating in a gives

$$\partial_a D_a(t) = \frac{2}{t-a} (D_a(t) - \mathcal{Z}(a)).$$

Evaluating at $t = \tau(a)$ and using $D'_a(\tau(a)) = 0$ yields

$$V'(a) = \frac{d}{da} D_a(\tau(a)) = \partial_a D_a(\tau(a)) = \frac{2}{\tau(a) - a} (V(a) - \mathcal{Z}(a)),$$

which proves (1).

Now use the rigidification scalar

$$M(a, t) := B_a(t)^2 \Gamma'_a(t).$$

In the normalized chart,

$$B_a(t) = (t-a)^2, \quad \Gamma_a = D_a, \quad M(a, t) = (t-a)^4 D'_a(t).$$

By Remark 12.282,

$$\partial_t M(a, t) = 2(t-a)^2 (\mathcal{Z}(t) - D_a(t)).$$

At $t = \tau(a)$ we have $D'_a(\tau(a)) = 0$, so

$$\partial_t M(a, \tau(a)) = (\tau(a) - a)^4 D''_a(\tau(a)).$$

Comparing the two formulas gives

$$(\tau(a) - a)^4 D''_a(\tau(a)) = 2(\tau(a) - a)^2 (\mathcal{Z}(\tau(a)) - V(a)),$$

hence (2).

Finally, still by Remark 12.282, at points where $M(a, t) = 0$ one has

$$\partial_a M(a, t) = 2(t-a)^2 (\mathcal{Z}(a) - D_a(t)),$$

in the normalized chart. Since

$$M(a, \tau(a)) = 0,$$

differentiating in a gives

$$0 = \frac{d}{da} M(a, \tau(a)) = \partial_a M(a, \tau(a)) + \partial_t M(a, \tau(a)) \tau'(a).$$

Substituting the formulas above yields

$$0 = 2(\tau(a) - a)^2 (\mathcal{Z}(a) - V(a)) + 2(\tau(a) - a)^2 (\mathcal{Z}(\tau(a)) - V(a)) \tau'(a).$$

If $D''_a(\tau(a)) \neq 0$, then by (2),

$$\mathcal{Z}(\tau(a)) - V(a) \neq 0,$$

so solving for $\tau'(a)$ gives (3). □

Corollary 12.298 (One-sign-change chamber is no-return for the born critical branch). *In the setting of Proposition 12.297, assume moreover that for every $a \in U$ the segment*

$$[a, \tau(a)]$$

contains exactly one sign change of $\mathcal{R}$.

Then for every $a \in U$:

(1)

$$\text{sgn}(V(a) - \mathcal{Z}(a))$$

is the sign of the initial lobe of $\mathcal{R}$ on $[a, \tau(a)]$;

(2)

$$\text{sgn}(\mathcal{Z}(\tau(a)) - V(a))$$

is the sign of the terminal lobe of $\mathcal{R}$ on $[a, \tau(a)]$;

(3)

$$(V(a) - \mathcal{Z}(a))(\mathcal{Z}(\tau(a)) - V(a)) < 0;$$

(4)

$$D_a''(\tau(a)) \neq 0, \quad \tau'(a) < 0;$$

(5)

$$V(a) \notin [\mathcal{Z}(a), \mathcal{Z}(\tau(a))].$$

In particular,

$$V(a) \neq \mathcal{Z}(a),$$

so the branch cannot realize the rigidified first-loss condition anywhere on U .

Proof. The first sign statement is exactly Proposition 12.295 applied to the critical pair $(a, \tau(a))$.

For the second sign statement, write

$$R_{a, \tau(a)}(s) := \mathcal{R}(a + s(\tau(a) - a)).$$

Since

$$\mathcal{Z}(\tau(a)) - \mathcal{Z}(a) = (\tau(a) - a) \int_0^1 R_{a, \tau(a)}(s) ds$$

and

$$V(a) - \mathcal{Z}(a) = (\tau(a) - a) \int_0^1 (1-s)^2 R_{a, \tau(a)}(s) ds,$$

one gets

$$\mathcal{Z}(\tau(a)) - V(a) = (\tau(a) - a) \int_0^1 (2s - s^2) R_{a, \tau(a)}(s) ds.$$

At the critical pair,

$$0 = D_a'(\tau(a)) = 2 \int_0^1 s(1-s) R_{a, \tau(a)}(s) ds,$$

so

$$\int_0^1 (2s - s^2) R_{a, \tau(a)}(s) ds = \int_0^1 s^2 R_{a, \tau(a)}(s) ds.$$

Hence

$$\mathcal{Z}(\tau(a)) - V(a) = (\tau(a) - a) \int_0^1 s^2 R_{a, \tau(a)}(s) ds.$$

Now set

$$k_1(s) := s(1-s), \quad k_2(s) := s^2.$$

Then

$$\frac{k_2(s)}{k_1(s)} = \frac{s}{1-s}$$

is strictly increasing on $(0, 1)$. Repeating the weighted signed-mass argument from the proof of Proposition 12.295 shows that

$$\operatorname{sgn}(\mathcal{Z}(\tau(a)) - V(a))$$

is the sign of the terminal lobe of $\mathcal{R}$.

The two signs are therefore opposite, which proves (3). Then Proposition 12.297(2) gives

$$D_a''(\tau(a)) \neq 0,$$

and Proposition 12.297(3) gives

$$\tau'(a) < 0.$$

Finally, opposite signs of

$$V(a) - \mathcal{Z}(a) \quad \text{and} \quad \mathcal{Z}(\tau(a)) - V(a)$$

mean exactly that $V(a)$ lies strictly outside the endpoint-curvature interval

$$[\mathcal{Z}(a), \mathcal{Z}(\tau(a))].$$

□

Proposition 12.299 (First rooting requires a second odd zero along the born repeated-source branch). *Let b be an isolated odd-order zero of $\mathcal{R}$ on an opened nonmixed arc, and let*

$$t = \tau(a)$$

denote an analytic continuation of the local born repeated-source critical branch, so that

$$\tau(a) > a, \quad D'_a(\tau(a)) = 0.$$

Assume that U is a connected continuation interval on which, for every $a \in U$, the segment

$$[a, \tau(a)]$$

contains exactly one sign change of $\mathcal{R}$.

Then for every $a \in U$ one has

$$D_a(\tau(a)) \neq \mathcal{Z}(a).$$

Equivalently, on the corresponding source branch

$$h = \sigma(a), \quad \mathfrak{D}(a, \sigma(a)) = 0,$$

one has

$$\Psi_a(\sigma(a)) \neq \mathcal{Z}(x(a)),$$

or in determinant form,

$$D_{u_5}(a, \sigma(a)) - \mathcal{Z}(x(a)) D_\lambda(a, \sigma(a)) \neq 0.$$

Consequently, any first parameter at which the analytic continuation of the born repeated-source branch realizes a genuine repeated-source obstruction inside the opened nonmixed arc must occur only after the corresponding segment ceases to be one-sign-change. Internally, away from affine-planar collapse and without leaving the arc, this can happen only when the segment acquires a second odd-order zero of $\mathcal{R}$, equivalently of

$$\mathcal{W}_{\text{src}}.$$

Proof. The first claim is exactly Corollary 12.298(5): on any continuation interval on which each segment

$$[a, \tau(a)]$$

contains exactly one sign change of $\mathcal{R}$, the born critical branch cannot realize the rigidified first-loss condition

$$D_a(\tau(a)) = \mathcal{Z}(a).$$

Via Remark 12.276,

$$\Psi_a(h) = \Gamma_{x(a)}(x(h)),$$

so the same statement in source coordinates is

$$\Psi_a(\sigma(a)) \neq \mathcal{Z}(x(a)).$$

Using Proposition 12.290(2), this is equivalent to

$$D_{u_5}(a, \sigma(a)) - \mathcal{Z}(x(a)) D_\lambda(a, \sigma(a)) \neq 0.$$

For the final statement, as long as the continued branch remains inside the opened nonmixed arc and the corresponding segment remains one-sign-change, the first part shows that rooting cannot occur. Thus any first rooted point must be preceded by failure of the one-sign-change condition.

Now even-order zeros of $\mathcal{R}$ do not change sign, so internally the sign-change count on the segment can change only when an endpoint of the segment meets another odd-order zero of $\mathcal{R}$ (unless the branch exits the arc or enters the affine-planar collapse case). Finally, on a nonmixed arc the functions $\mathcal{R}$ and

$$\mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda$$

differ only by a nonvanishing analytic factor (Proposition 12.236), so odd zeros of the two objects coincide. $\square$

Proposition 12.300 (Discriminant entry into the adjacent-two-odd-zero chamber is not a first rooted point). *Let J be an opened nonmixed arc, and work in a local normalized nonmixed chart on which*

$$S(t) = t^2, \quad \Gamma_a = D_a.$$

Let

$$\tau : (a_-, a_0] \rightarrow J$$

be a real-analytic continuation of the born repeated-source critical branch, so that

$$\tau(a) > a, \quad D'_a(\tau(a)) = 0 \quad (a \in (a_-, a_0]).$$

Assume:

(1) for every $a \in (a_-, a_0)$, the segment

$$[a, \tau(a)]$$

contains exactly one sign change of $\mathcal{R}$;

(2) the target endpoint reaches the next adjacent odd zero at $a = a_0$, i.e.

$$q := \tau(a_0)$$

is an odd-order zero of $\mathcal{R}$, there are no odd zeros of $\mathcal{R}$ in $(\tau(a), q)$ for $a < a_0$ close to a_0 , and the segment

$$[a_0, q]$$

still contains exactly one sign change of $\mathcal{R}$.

Set

$$V(a) := D_a(\tau(a)).$$

Define the sheared discriminant and the RH-selected slice by

$$\mathcal{D} := \{(a, \gamma, h) : a < h, D_{f_\gamma}(a, h) = 0, \partial_h D_{f_\gamma}(a, h) = 0\},$$

$$\mathcal{S}_{\text{RH}} := \{(a, \gamma, h) : \gamma = \mathcal{Z}(a)\}.$$

Then

$$P(a) := (a, V(a), \tau(a)) \in \mathcal{D} \quad (a \in (a_-, a_0]),$$

but at the adjacent-odd-zero entrance point one has

$$V(a_0) - \mathcal{Z}(a_0) \neq 0.$$

More precisely,

$$V(a_0) - \mathcal{Z}(a_0) = (q - a_0) \int_0^1 (1 - s)^2 \mathcal{R}(a_0 + s(q - a_0)) ds,$$

and this quantity has the sign of the initial lobe of $\mathcal{R}$ on $[a_0, q]$.

Consequently,

$$P(a_0) \notin \mathcal{S}_{\text{RH}},$$

and there exists a neighborhood N of $P(a_0)$ in

$$J \times \mathbf{R} \times J$$

such that

$$N \cap \mathcal{D} \cap \mathcal{S}_{\text{RH}} = \emptyset.$$

In particular, a first rooted point cannot be created at the moment when the next adjacent odd zero first enters the active segment. If rooting occurs at all, it must occur only after the branch has moved into a genuine two-sign-change chamber of positive width.

Proof. By Remark 12.276,

$$\Psi_a(h) = \Gamma_{x(a)}(x(h)),$$

and on a nonmixed patch

$$D_{f_\gamma}(a, h) = D_\lambda(a, h)(\Psi_a(h) - \gamma).$$

Since

$$D_\lambda(a, h) \neq 0 \quad (a < h),$$

we have

$$D_{f_\gamma}(a, h) = 0, \quad \partial_h D_{f_\gamma}(a, h) = 0 \iff \gamma = \Psi_a(h), \quad \partial_h \Psi_a(h) = 0.$$

Along the continued born branch,

$$\partial_h \Psi_a(\tau(a)) = 0, \quad \Psi_a(\tau(a)) = V(a),$$

so indeed

$$P(a) = (a, V(a), \tau(a)) \in \mathcal{D}.$$

At $a = a_0$, the integral formula for D_a gives

$$V(a_0) - \mathcal{Z}(a_0) = (q - a_0) \int_0^1 (1 - s)^2 \mathcal{R}(a_0 + s(q - a_0)) ds.$$

Also, since $D'_{a_0}(q) = 0$,

$$0 = D'_{a_0}(q) = 2 \int_0^1 s(1 - s) \mathcal{R}(a_0 + s(q - a_0)) ds.$$

Set

$$R_0(s) := \mathcal{R}(a_0 + s(q - a_0)), \quad k_0(s) := (1 - s)^2, \quad k_1(s) := s(1 - s).$$

Then

$$\frac{k_0(s)}{k_1(s)} = \frac{1 - s}{s}$$

is strictly decreasing on $(0, 1)$. Because R_0 has exactly one sign change on $(0, 1)$, the same weighted signed-mass argument used in Proposition 12.295 implies that

$$\int_0^1 k_0(s) R_0(s) ds$$

has the sign of the initial lobe of R_0 , hence is nonzero. Therefore

$$V(a_0) - \mathcal{Z}(a_0) \neq 0.$$

Thus

$$P(a_0) = (a_0, V(a_0), q) \in \mathcal{D} \quad \text{but} \quad P(a_0) \notin \mathcal{S}_{\text{RH}}.$$

Since both $\mathcal{D}$ along the continued branch and the function

$$(a, \gamma, h) \mapsto \gamma - \mathcal{Z}(a)$$

are continuous, there is a neighborhood N of $P(a_0)$ on which

$$\gamma - \mathcal{Z}(a)$$

keeps the sign of

$$V(a_0) - \mathcal{Z}(a_0).$$

Hence

$$N \cap \mathcal{D} \cap \mathcal{S}_{\text{RH}} = \emptyset.$$

The final statement is immediate: a rooted point is exactly a point of

$$\mathcal{D} \cap \mathcal{S}_{\text{RH}},$$

so no such point can be born at the adjacent-odd-zero entrance. □

Remark 12.301 (What the discriminant analysis now leaves open). Proposition 12.300 shows that a rooted point is not created *at the entry* into the adjacent-two-odd-zero chamber. Corollary 12.315 shows more: on any continuation interval on which the anchor-side value

$$\mathcal{R}(a)$$

keeps a fixed sign, the RH-selected slice

$$\gamma = \mathcal{Z}(a)$$

does not cross the continued discriminant curve at all.

Thus the adjacent-two-odd-zero chamber is not itself the residual obstruction. What remains open is narrower and genuinely global: whether a rooted crossing can occur only after the anchor passes through an odd-order zero of $\mathcal{R}$ (equivalently of $\mathcal{W}_{\text{src}}$), or after branch exit, or after affine-planar collapse. In other words, the remaining discriminant-theoretic bottleneck is a *delayed post-anchor no-crossing problem*: not an entry-point singularity problem, and not crossing at the anchor odd zero itself, but a later crossing on a dangerous post-anchor chamber.

Proposition 12.302 (Rooted-defect transport and stationary first root). *In the setting of Proposition 12.297, define*

$$L(a) := \tau(a) - a, \quad G(a) := V(a) - \mathcal{Z}(a) = D_a(\tau(a)) - \mathcal{Z}(a).$$

Then

$$G'(a) = \frac{2}{L(a)} G(a) - \mathcal{R}(a).$$

Hence, if $a_* \in U$ satisfies

$$G(a_*) = 0,$$

then

$$G'(a_*) = -\mathcal{R}(a_*).$$

If in addition

$$\mathcal{Z}(\tau(a_*)) \neq \mathcal{Z}(a_*),$$

then

$$D''_{a_*}(\tau(a_*)) \neq 0, \quad \tau'(a_*) = 0, \quad \tau''(a_*) = -\frac{\mathcal{R}(a_*)}{\mathcal{Z}(\tau(a_*)) - \mathcal{Z}(a_*)}.$$

Proof. Since

$$G(a) = V(a) - \mathcal{Z}(a),$$

Proposition 12.297(1) gives

$$G'(a) = V'(a) - \mathcal{R}(a) = \frac{2}{\tau(a) - a} (V(a) - \mathcal{Z}(a)) - \mathcal{R}(a) = \frac{2}{L(a)} G(a) - \mathcal{R}(a).$$

Evaluating at a root a_* of G gives

$$G'(a_*) = -\mathcal{R}(a_*).$$

Now assume also that

$$\mathcal{Z}(\tau(a_*)) \neq \mathcal{Z}(a_*).$$

Since $G(a_*) = 0$, this means

$$\mathcal{Z}(\tau(a_*)) \neq V(a_*).$$

So Proposition 12.297(2) yields

$$D''_{a_*}(\tau(a_*)) = \frac{2}{(\tau(a_*) - a_*)^2} (\mathcal{Z}(\tau(a_*)) - V(a_*)) \neq 0.$$

Then Proposition 12.297(3) applies near a_* and gives

$$\tau'(a) = \frac{G(a)}{\mathcal{Z}(\tau(a)) - V(a)}.$$

Evaluating at a_* gives

$$\tau'(a_*) = 0.$$

Differentiate this identity once more. Since $G(a_*) = 0$ and

$$\mathcal{Z}(\tau(a_*)) - V(a_*) = \mathcal{Z}(\tau(a_*)) - \mathcal{Z}(a_*) \neq 0,$$

one gets

$$\tau''(a_*) = \frac{G'(a_*)}{\mathcal{Z}(\tau(a_*)) - V(a_*)} = -\frac{\mathcal{R}(a_*)}{\mathcal{Z}(\tau(a_*)) - \mathcal{Z}(a_*)},$$

as claimed. $\square$

Proposition 12.303 (Nondegenerate rooted point is a fold of the moving-shear equation).
Work in a local normalized nonmixed chart on which

$$S(t) = t^2.$$

Define

$$\mathcal{F}(a, t) := A_a(t) - \mathcal{Z}(a) B_a(t) = (t - a)^2 (D_a(t) - \mathcal{Z}(a)).$$

Let

$$a_* < t_*$$

satisfy

$$\mathcal{F}(a_*, t_*) = 0, \quad \partial_t \mathcal{F}(a_*, t_*) = 0,$$

and assume

$$\mathcal{R}(a_*) \neq 0, \quad \mathcal{Z}(t_*) \neq \mathcal{Z}(a_*).$$

Then

$$\partial_a \mathcal{F}(a_*, t_*) = -(t_* - a_*)^2 \mathcal{R}(a_*),$$

and

$$\partial_{tt} \mathcal{F}(a_*, t_*) = 2(\mathcal{Z}(t_*) - \mathcal{Z}(a_*)).$$

In particular,

$$\partial_a \mathcal{F}(a_*, t_*) \neq 0, \quad \partial_{tt} \mathcal{F}(a_*, t_*) \neq 0.$$

Hence near (a_*, t_*) the zero set

$$\mathcal{F}(a, t) = 0$$

is the graph of a unique real-analytic function

$$a = \alpha(t)$$

with

$$\alpha(t_*) = a_*, \quad \alpha'(t_*) = 0, \quad \alpha''(t_*) = \frac{2(\mathcal{Z}(t_*) - \mathcal{Z}(a_*))}{(t_* - a_*)^2 \mathcal{R}(a_*)}.$$

Equivalently,

$$a - a_* = \frac{\mathcal{Z}(t_*) - \mathcal{Z}(a_*)}{(t_* - a_*)^2 \mathcal{R}(a_*)} (t - t_*)^2 + o((t - t_*)^2).$$

Proof. Since

$$\mathcal{F}(a, t) = (t - a)^2 (D_a(t) - \mathcal{Z}(a)),$$

the conditions

$$\mathcal{F}(a_*, t_*) = 0, \quad \partial_t \mathcal{F}(a_*, t_*) = 0$$

imply

$$D_{a_*}(t_*) = \mathcal{Z}(a_*), \quad D'_{a_*}(t_*) = 0.$$

So (a_*, t_*) is a rooted critical point.

Differentiate $\mathcal{F}$ with respect to a :

$$\partial_a \mathcal{F}(a, t) = -2(t - a)(D_a(t) - \mathcal{Z}(a)) + (t - a)^2 (\partial_a D_a(t) - \mathcal{R}(a)).$$

At the rooted point,

$$D_{a_*}(t_*) - \mathcal{Z}(a_*) = 0,$$

and by Proposition 12.297,

$$\partial_a D_a(t) = \frac{2}{t - a} (D_a(t) - \mathcal{Z}(a)),$$

so

$$\partial_a \mathcal{F}(a_*, t_*) = -(t_* - a_*)^2 \mathcal{R}(a_*).$$

Next,

$$\partial_t \mathcal{F}(a, t) = 2(t - a)(D_a(t) - \mathcal{Z}(a)) + (t - a)^2 D'_a(t),$$

hence

$$\partial_{tt} \mathcal{F}(a, t) = 2(D_a(t) - \mathcal{Z}(a)) + 4(t - a)D'_a(t) + (t - a)^2 D''_a(t).$$

At the rooted critical point this becomes

$$\partial_{tt} \mathcal{F}(a_*, t_*) = (t_* - a_*)^2 D''_{a_*}(t_*).$$

Applying Proposition 12.297(2) and using

$$D_{a_*}(t_*) = \mathcal{Z}(a_*),$$

one gets

$$\partial_{tt} \mathcal{F}(a_*, t_*) = 2(\mathcal{Z}(t_*) - \mathcal{Z}(a_*)).$$

By the two nonvanishing assumptions,

$$\partial_a \mathcal{F}(a_*, t_*) \neq 0.$$

So the implicit function theorem gives a unique real-analytic graph

$$a = \alpha(t)$$

for the zero set of $\mathcal{F}$ near (a_*, t_*) . Since

$$\partial_t \mathcal{F}(a_*, t_*) = 0,$$

one has

$$\alpha'(t_*) = 0.$$

Differentiating

$$\mathcal{F}(\alpha(t), t) = 0$$

twice at $t = t_*$ gives

$$\partial_a \mathcal{F}(a_*, t_*) \alpha''(t_*) + \partial_{tt} \mathcal{F}(a_*, t_*) = 0,$$

hence

$$\alpha''(t_*) = -\frac{\partial_{tt} \mathcal{F}(a_*, t_*)}{\partial_a \mathcal{F}(a_*, t_*)} = \frac{2(\mathcal{Z}(t_*) - \mathcal{Z}(a_*))}{(t_* - a_*)^2 \mathcal{R}(a_*)}.$$

The Taylor expansion of α then yields the stated quadratic form. $\square$

Remark 12.304 (First rooted point: local fold data versus the actual remaining burden). Let b be an isolated odd-order zero of $\mathcal{R}$, and let

$$t = \tau(a)$$

be an analytic continuation of the born repeated-source critical branch. By Proposition 12.299, any first rooted point along this continuation can occur only after the segment has acquired a second odd-order zero of $\mathcal{R}$.

At such a rooted point, if

$$\mathcal{R}(a_*) \neq 0, \quad \mathcal{Z}(\tau(a_*)) \neq \mathcal{Z}(a_*),$$

then Proposition 12.302 shows that the rooted defect

$$G(a) = D_a(\tau(a)) - \mathcal{Z}(a) \quad \text{equivalently} \quad G(a) = \Psi_a(\sigma(a)) - \mathcal{Z}(a)$$

crosses transversely and that the target map is stationary there:

$$\tau'(a_*) = 0.$$

Moreover, Proposition 12.303 shows that the moving-shear equation

$$A_a(x) - \mathcal{Z}(a) B_a(x) = 0 \quad \text{equivalently} \quad D_{f_a}(a, h) = 0, \quad f_a = u_5 - \mathcal{Z}(a)\lambda,$$

has a local fold at such a nondegenerate rooted point.

Thus the transport/fold package determines the *local shape* of any prospective rooted crossing. However, the later route-2 results show that this is not the final obstruction theorem. Proposition 12.300 rules out birth of the first rooted point at entry into the adjacent-two-odd-zero chamber; Proposition 12.314 and Corollary 12.315 exclude rooted crossing on continuation intervals where the anchor-side value keeps fixed sign; Proposition 12.317 rules out rooted crossing at an odd anchor zero itself or immediately after passage; and Proposition 12.322 together with Corollary 12.323 show that even delayed rooted crossing on dangerous post-anchor chambers cannot be excluded from the current planar/route-2 package alone.

Accordingly, this remark should be read only as local fold data. The remaining route-2 burden is no longer a “first-fold exclusion” theorem, but the search for the first genuinely non-universal source-coupled input beyond the normalized planar model $(\mathcal{Z}, \mathcal{R}, D_a)$.

Proposition 12.305 (At most one odd zero implies no genuine repeated-source obstruction). *Let J be an opened nonmixed arc. Assume that $\mathcal{R}$ has at most one odd-order zero on J . Then there is no pair $a < x$ in J such that*

$$\Gamma'_a(x) = 0, \quad \Gamma_a(x) = \mathcal{Z}(a).$$

Equivalently, there is no source-level pair $a < h$ on the corresponding source arc such that

$$\Psi'_a(h) = 0, \quad \Psi_a(h) = \mathcal{Z}(a).$$

Proof. On a nonmixed arc, $\mathcal{W}_{\text{src}}$ differs from $\mathcal{R}$ only by a nonvanishing analytic factor (Proposition 12.236), so odd zeros and sign changes of the two objects coincide.

If $\mathcal{R}$ has at most one odd-order zero on J , then every segment contained in J contains at most one sign change of $\mathcal{R}$ (equivalently of $\mathcal{W}_{\text{src}}$), since even-order zeros do not change sign. Hence Corollary 12.296 excludes any simultaneous source-level solution of

$$\Psi'_a(h) = 0, \quad \Psi_a(h) = \mathcal{Z}(a).$$

Via Remark 12.276, this is equivalent to the planar formulation

$$\Gamma'_a(x) = 0, \quad \Gamma_a(x) = \mathcal{Z}(a).$$

□

Remark 12.306 (Minimal unresolved configuration on this route). Proposition 12.305 rules out every opened nonmixed arc carrying at most one odd zero of $\mathcal{R}$, and Corollary 12.315 rules out rooted crossing on the standard adjacent-two-odd-zero chamber as long as the anchor-side value $\mathcal{R}(a)$ keeps a fixed sign.

The new odd-anchor passage theorem (Proposition 12.317) shows that no rooted crossing can occur at the anchor odd zero itself or immediately after it is crossed, while Corollary 12.319 shows that the subsequent post-anchor chambers alternate dangerous and safe, beginning with a dangerous first post-anchor chamber.

Remark 12.320 shows that the first dangerous chamber is already formally compatible with the local scalar rooted equations and fold data. Accordingly, the first genuinely unresolved configuration is no longer the adjacent-two-odd-zero chamber itself, nor the anchor odd zero passage, but a *delayed* crossing on a dangerous post-anchor chamber. Thus the sharpened source-coupling target is a delayed no-crossing / profile-realizability exclusion theorem for the moving shear

$$f_a = u_5 - \mathcal{Z}(a)\lambda$$

on genuine opened nonmixed arcs.

Remark 12.307 (No-go for a pure level-set / barrier theorem for $\mathcal{Z}$). A natural next guess would be that the remaining adjacent-odd-zero case could be excluded by a theorem depending only on the level geometry of

$$\mathcal{Z}, \quad \mathcal{R} = \mathcal{Z}',$$

together with the first-root transport identities

$$G(a_*) := V(a_*) - \mathcal{Z}(a_*) = 0, \quad G'(a_*) = -\mathcal{R}(a_*), \quad \tau'(a_*) = 0,$$

and the fact that a rooted point forces two interior recrossings of the level

$$\mathcal{Z}(r) = \mathcal{Z}(a_*).$$

At the current level of abstraction this is false: the level-set geometry of $\mathcal{Z}$ alone is still too free.

Indeed, already in the normalized planar determinant model one can realize all of the above with exactly two simple odd zeros of $\mathcal{R}$ and with a nondegenerate rooted critical pair. Take

$$S(t) = t^2, \quad \mathcal{R}(t) = 5t^3 + 5t^2 - 7t + 1, \quad \mathcal{Z}(t) = \int_0^t \mathcal{R}(u) du = t - \frac{7}{2}t^2 + \frac{5}{3}t^3 + \frac{5}{4}t^4.$$

Then $\mathcal{R}$ has exactly two simple zeros in $(0, 1)$, numerically

$$t \approx 0.165726, \quad t \approx 0.660736,$$

so on $[0, 1]$ the sign pattern is $+, -, +$ with exactly two odd sign changes. Now choose an analytic function $\tilde{Y}$ such that

$$\frac{1}{2} \tilde{Y}'' = \mathcal{Z}.$$

For the normalized determinant function $D_a(x)$ associated to $(x, \tilde{Y})$, the rooted critical conditions at $(a, x) = (0, 1)$ become

$$D'_0(1) = 2 \int_0^1 s(1-s) \mathcal{R}(s) ds, \quad D_0(1) - \mathcal{Z}(0) = \int_0^1 (1-s)^2 \mathcal{R}(s) ds.$$

For this $\mathcal{R}$, both moments vanish:

$$\int_0^1 s(1-s) \mathcal{R}(s) ds = 0, \quad \int_0^1 (1-s)^2 \mathcal{R}(s) ds = 0.$$

Hence

$$D'_0(1) = 0, \quad D_0(1) = \mathcal{Z}(0),$$

so $(0, 1)$ is an exact rooted repeated-source critical pair.

Moreover,

$$D''_0(1) = 2 \int_0^1 s^2 \mathcal{R}(s) ds = \frac{5}{6} > 0,$$

so the critical branch is nondegenerate and the implicit-function theorem gives a real-analytic continuation

$$x = \tau(a), \quad \tau(0) = 1, \quad \tau'(0) = 0.$$

Also, with

$$V(a) := D_a(\tau(a)), \quad G(a) := V(a) - \mathcal{Z}(a),$$

one has at the rooted point

$$G(0) = 0, \quad G'(0) = -\mathcal{R}(0) = -1.$$

Finally, the level equation

$$\mathcal{Z}(t) = \mathcal{Z}(0) = 0$$

has two interior solutions,

$$t \approx 0.368001, \quad t \approx 0.851550,$$

so the rooted point comes with exactly the two interior recrossings of the moving level demanded by the source-pinned shear picture.

Thus a theorem using only the scalar level-set topology of $\mathcal{Z}$ together with the first-root identities above cannot suffice to exclude the remaining adjacent-odd-zero configuration. Any successful next step must use additional structure: either the provenance of $(a, \tau(a))$ as the *continued born branch*, or a genuinely source-coupled constraint such as the moving shear

$$f_a = u_5 - \mathcal{Z}(a)\lambda,$$

rather than only the abstract level geometry of $\mathcal{Z}$.

Remark 12.308 (No-go for branch provenance alone). One might try to strengthen Remark 12.307 by using not just the scalar level geometry of $\mathcal{Z}$, but also the *provenance* of the rooted point: namely, that $(a, \tau(a))$ lies on the continued branch born at an earlier odd zero of $\mathcal{R}$. At the planar normalized D_a -level, this is still not enough. There are explicit analytic models in which the born branch from one odd zero continues to a genuine rooted point before leaving the interval bounded by the adjacent odd zeros.

A concrete example is obtained by taking

$$S(t) = t^2, \quad \mathcal{R}(t) := t(t-1)(t^2 - 4t - 2) = t^4 - 5t^3 + 2t^2 + 2t,$$

and

$$\mathcal{Z}(t) := \int_0^t \mathcal{R}(u) du = \frac{1}{5}t^5 - \frac{5}{4}t^4 + \frac{2}{3}t^3 + t^2.$$

Then the odd zeros of $\mathcal{R}$ are

$$2 - \sqrt{6}, \quad 0, \quad 1, \quad 2 + \sqrt{6},$$

so 0 is an odd zero and its adjacent odd zero on the left is $2 - \sqrt{6}$.

Now choose an analytic function $\tilde{Y}$ such that

$$\frac{1}{2} \tilde{Y}'' = \mathcal{Z},$$

and let $D_a(x)$ denote the normalized determinant/divided-difference function for the planar graph $(x, \tilde{Y})$. Near the odd zero 0 there is the usual born critical branch

$$x = \tau(a), \quad a < 0 < x, \quad D'_a(\tau(a)) = 0,$$

with

$$\tau(a) = -a + O(a^2),$$

and the local odd-zero analysis gives

$$G(a) := D_a(\tau(a)) - \mathcal{Z}(a) < 0$$

for $a < 0$ sufficiently close to 0.

However, along this same connected born branch one finds

$$G(-0.5801) < 0, \quad G(-0.5901) > 0,$$

so by continuity there exists

$$a_* \in (-0.5901, -0.5801)$$

such that

$$G(a_*) = 0.$$

Thus the *continued born branch* reaches a rooted point

$$D'_{a_*}(\tau(a_*)) = 0, \quad D_{a_*}(\tau(a_*)) = \mathcal{Z}(a_*).$$

Numerically,

$$a_* \approx -0.5805673462, \quad \tau(a_*) \approx 0.3888336023,$$

and

$$D''_{a_*}(\tau(a_*)) > 0,$$

so this rooted point is nondegenerate; in particular,

$$\tau'(a_*) = 0$$

as predicted by the transport system.

Moreover, the segment

$$[a_*, \tau(a_*)]$$

contains exactly the born odd zero 0 together with the adjacent odd zero $2 - \sqrt{6}$, and no others. So even the strengthened statement

“continued born-branch provenance plus adjacent-odd-zero geometry already excludes rooting” is false in the planar normalized model.

Therefore the remaining exclusion cannot depend only on abstract branch continuation data for

$$D'_a(\tau(a)) = 0$$

together with the scalar geometry of $\mathcal{Z}$. Any successful next step must use genuinely source-coupled structure, such as the actual quotient

$$\Psi_a(h) = \frac{D_{u_5}(a, h)}{D_\lambda(a, h)},$$

the moving shear

$$f_a = u_5 - \mathcal{Z}(a)\lambda,$$

or some equivalent RH-specific relation invisible in the planar D_a model.

Remark 12.309 (No-go for disconjugacy/ECT alone). A further natural idea is to recast the moving-shear family through the third-order operator

$$L[y] := \frac{W(c, v_5, \lambda, y)}{J_\lambda},$$

and try to exclude rooted moving-shear events by proving that L is disconjugate, Chebyshev, or sign-regular on opened nonmixed arcs. At the present level of abstraction this is still not enough.

Indeed, already in the normalized source model

$$c(t) = 1, \quad v_5(t) = t, \quad \lambda(t) = t^2,$$

one has

$$J_f = W(c, v_5, f) = f'', \quad J_\lambda = 2, \quad L[y] = \frac{W(c, v_5, \lambda, y)}{J_\lambda} = \frac{1}{2} y''''.$$

So the homogeneous equation $L[y] = 0$ has basis

$$1, \quad t, \quad t^2,$$

hence is maximally disconjugate / ECT on every interval. Nevertheless, the planar normalized counterexample from Remark 12.308 fits exactly into this framework: for a suitable analytic u_5 and the source-pinned family

$$f_a := u_5 - \mathcal{Z}(a)\lambda,$$

one has

$$L[f_a] = L[u_5], \quad J_{f_a}(a) = 0,$$

and along the continued born branch there is still a rooted point

$$D_{f_a}(a, h) = 0, \quad \partial_h D_{f_a}(a, h) = 0.$$

Thus disconjugacy, ECT, or sign-regularity of the operator L by itself cannot exclude the remaining rooted moving-shear configuration. Any successful next step must use additional source-coupled structure beyond the abstract third-order ODE viewpoint, such as special positivity or monotonicity for the distinguished quotient

$$\Psi_a(h) = \frac{D_{u_5}(a, h)}{D_\lambda(a, h)}$$

or for the particular slice

$$f_a = u_5 - \mathcal{Z}(a)\lambda.$$

Remark 12.310 (Ceiling of the present scalar/discriminant analysis). The preceding results sharply delimit what can and cannot be obtained from the current planar/scalar reformulation.

On the positive-kernel side, one-sign-change chambers are sterile for genuine repeated-source first loss: the born critical branch is simple, has strictly negative slope, and cannot satisfy

$$D_a(\tau(a)) = \mathcal{Z}(a)$$

before the active segment acquires another odd zero of $\mathcal{R}$ (Proposition 12.299). On the discriminant side, the adjacent-odd-zero entrance itself cannot create the first rooted point (Proposition 12.300).

The new anchor-sign and odd-anchor-passage arguments remove a substantial part of the remaining freedom. On any continuation interval on which the anchor-side value

$$\mathcal{R}(a)$$

keeps fixed sign, the RH-selected slice

$$\gamma = \mathcal{Z}(a)$$

cannot cross the continued discriminant curve. When the anchor itself passes through an odd-order zero of $\mathcal{R}$, the integrating-factor defect keeps its incoming sign across that odd anchor zero, so a rooted crossing cannot occur there or immediately after it. The ensuing post-anchor chambers then alternate dangerous and safe, beginning with a dangerous first post-anchor chamber.

However, two no-go facts now show that the present scalar/discriminant package has reached its ceiling. First, Remark 12.320 shows that already on the first dangerous post-anchor chamber, the local normalized rooted equations and fold/transversality data are formally realizable in the planar scalar model. Second, Proposition 12.322 and Corollary 12.323 show that the entire current route-2 source package

$$\Psi_a, \quad \mathfrak{D}, \quad \mathcal{W}_{\text{src}}, \quad D_{f_a}$$

is universal under normalization: every analytic normalized planar model lifts to a genuine analytic source model for this package.

Accordingly, after the fixed-sign and odd-anchor exclusions, the remaining problem is *not* a delayed no-crossing / profile-realizability problem inside the present scalar/discriminant package. It is to find the first genuinely non-universal source-coupled input, i.e. structure not already captured by the normalized planar model $(\mathcal{Z}, \mathcal{R}, D_a)$.

So any further progress on the nonmixed repeated-source branch must use genuinely source-coupled structure beyond the present universal package, possibly through stronger provenance laws, higher canonical/mixed data, or another RH-specific input invisible in the planar normalized model.

Proposition 12.311 (Discriminant reduction for the moving-shear family). *Let J be a good nonmixed v_5 -patch. For $\gamma \in \mathbf{R}$, set*

$$f_\gamma := u_5 - \gamma\lambda.$$

Define the sheared discriminant set

$$\mathcal{D} := \{(a, \gamma, h) \in J \times \mathbf{R} \times J : a < h, D_{f_\gamma}(a, h) = 0, \partial_h D_{f_\gamma}(a, h) = 0\}.$$

Then on the region $a < h$ one has the exact equivalence

$$(a, \gamma, h) \in \mathcal{D} \iff \gamma = \Psi_a(h), \quad \mathfrak{D}(a, h) = 0.$$

Equivalently,

$$\mathcal{D} = \{(a, \Psi_a(h), h) : a < h, \mathfrak{D}(a, h) = 0\}.$$

In particular, on a nonmixed patch the sheared discriminant is not an independent two-dimensional catastrophe surface in (a, γ, h) -space: it is the graph of the repeated-source critical locus

$$\mathfrak{D}(a, h) = 0$$

in the extra γ -direction.

Proof. By Proposition 12.290(2), for every (a, h) with $a < h$ one has

$$D_{f_\gamma}(a, h) = D_\lambda(a, h)(\Psi_a(h) - \gamma).$$

Differentiating in h gives

$$\partial_h D_{f_\gamma}(a, h) = \partial_h D_\lambda(a, h)(\Psi_a(h) - \gamma) + D_\lambda(a, h) \partial_h \Psi_a(h).$$

On a good nonmixed patch,

$$D_\lambda(a, h) \neq 0 \quad (a < h).$$

Therefore

$$D_{f_\gamma}(a, h) = 0 \iff \gamma = \Psi_a(h).$$

At such a point the first term in the formula for $\partial_h D_{f_\gamma}$ vanishes, so

$$\partial_h D_{f_\gamma}(a, h) = 0 \iff D_\lambda(a, h) \partial_h \Psi_a(h) = 0 \iff \partial_h \Psi_a(h) = 0.$$

Finally, by Remark 12.276,

$$\partial_h \Psi_a(h) = \frac{W_h(c, v_5)(a) \mathfrak{D}(a, h)}{D_\lambda(a, h)^2},$$

and the prefactor is nonvanishing on the nonmixed patch. Hence

$$\partial_h \Psi_a(h) = 0 \iff \mathfrak{D}(a, h) = 0.$$

Combining the two equivalences proves the result. $\square$

Proposition 12.312 (Generic rooted discriminant point is a transverse crossing). *In the setting of Proposition 12.311, define the RH-selected slice*

$$\mathcal{S}_{\text{RH}} := \{(a, \gamma, h) \in J \times \mathbf{R} \times J : \gamma = \mathcal{Z}(x(a))\}.$$

Let

$$(a_*, \gamma_*, h_*) \in \mathcal{D} \cap \mathcal{S}_{\text{RH}}$$

and assume

$$\partial_h^2 D_{f_{\gamma_*}}(a_*, h_*) \neq 0, \quad \mathcal{R}(x(a_*)) \neq 0.$$

Then there is a neighborhood of (a_*, γ_*, h_*) in which $\mathcal{D}$ is a unique real-analytic curve

$$a \mapsto (a, \gamma(a), h(a)),$$

with

$$\gamma(a) = \Psi_a(h(a)).$$

Moreover one has

$$\gamma'(a_*) = 0, \quad h'(a_*) = 0,$$

and

$$\left. \frac{d}{da} (\gamma(a) - \mathcal{Z}(x(a))) \right|_{a=a_*} = -\mathcal{R}(x(a_*)) x'(a_*) \neq 0.$$

Hence the discriminant curve $\mathcal{D}$ meets the RH-selected slice $\mathcal{S}_{\text{RH}}$ transversely at (a_*, γ_*, h_*) .

In fact the γ -projection of the discriminant curve has a nondegenerate local extremum there:

$$\gamma''(a_*) = -\frac{2x'(a_*)^2}{x(h_*) - x(a_*)} \mathcal{R}(x(a_*)).$$

Proof. By Proposition 12.311, the discriminant is cut out by

$$\partial_h \Psi_a(h) = 0, \quad \gamma = \Psi_a(h).$$

At the given point,

$$\partial_h^2 D_{f_{\gamma_*}}(a_*, h_*) \neq 0.$$

Since

$$D_{f_\gamma}(a, h) = D_\lambda(a, h)(\Psi_a(h) - \gamma),$$

differentiating twice in h and evaluating at a discriminant point gives

$$\partial_h^2 D_{f_{\gamma_*}}(a_*, h_*) = D_\lambda(a_*, h_*) \partial_h^2 \Psi_{a_*}(h_*).$$

Because $D_\lambda(a_*, h_*) \neq 0$, this implies

$$\partial_h^2 \Psi_{a_*}(h_*) \neq 0.$$

So by the implicit function theorem there is a unique real-analytic function

$$h = h(a)$$

with

$$h(a_*) = h_*, \quad \partial_h \Psi_a(h(a)) = 0.$$

Setting

$$\gamma(a) := \Psi_a(h(a))$$

gives the asserted local parametrization of $\mathcal{D}$.

Now pass to the local normalized nonmixed x -coordinate and write

$$\alpha := x(a), \quad \tau(\alpha) := x(h(a)), \quad V(\alpha) := D_\alpha(\tau(\alpha)).$$

By Remark 12.276,

$$\gamma(a) = \Psi_a(h(a)) = V(\alpha).$$

Since $(a_*, \gamma_*, h_*) \in \mathcal{S}_{\text{RH}}$, one has

$$V(\alpha_*) = \mathcal{Z}(\alpha_*), \quad \alpha_* := x(a_*).$$

By Proposition 12.297, at a rooted noncollapse point one has

$$V'(\alpha_*) = 0, \quad \tau'(\alpha_*) = 0.$$

Therefore

$$\gamma'(a_*) = V'(\alpha_*) x'(a_*) = 0,$$

and

$$x'(h_*) h'(a_*) = \tau'(\alpha_*) x'(a_*) = 0,$$

hence

$$h'(a_*) = 0.$$

For the transverse-crossing derivative, differentiate

$$\gamma(a) - \mathcal{Z}(x(a)) = V(\alpha) - \mathcal{Z}(\alpha)$$

to get

$$\frac{d}{da} (\gamma(a) - \mathcal{Z}(x(a))) = x'(a) (V'(\alpha) - \mathcal{R}(\alpha)).$$

At $a = a_*$, since $V'(\alpha_*) = 0$, this becomes

$$\left. \frac{d}{da} (\gamma(a) - \mathcal{Z}(x(a))) \right|_{a=a_*} = -\mathcal{R}(\alpha_*) x'(a_*),$$

which is nonzero by hypothesis. This proves transversality.

Finally, differentiate

$$V'(\alpha) = \frac{2}{\tau(\alpha) - \alpha} (V(\alpha) - \mathcal{Z}(\alpha))$$

from Proposition 12.297(1). At the rooted point,

$$V(\alpha_*) - \mathcal{Z}(\alpha_*) = 0, \quad \tau'(\alpha_*) = 0,$$

so

$$V''(\alpha_*) = \frac{2}{\tau(\alpha_*) - \alpha_*} (V'(\alpha_*) - \mathcal{R}(\alpha_*)) = -\frac{2}{\tau(\alpha_*) - \alpha_*} \mathcal{R}(\alpha_*).$$

Hence

$$\gamma''(a_*) = V''(\alpha_*) x'(a_*)^2 = -\frac{2 x'(a_*)^2}{x(h_*) - x(a_*)} \mathcal{R}(x(a_*)),$$

as claimed. □

Remark 12.313 (What the discriminant/catastrophe route actually reduces to). The discriminant reformulation sharpens the geometry, but it also shows the limit of purely local catastrophe arguments on this route.

Indeed, by Proposition 12.311, the sheared discriminant on a nonmixed patch is already one-dimensional: it is the graph of the repeated-source critical locus

$$\mathfrak{D}(a, h) = 0$$

in the γ -direction. And by Proposition 12.312, a generic rooted point is not a tangential meeting of sheets, nor a local fold/cusp obstruction for the RH-selected slice

$$\gamma = \mathcal{Z}(x(a)).$$

It is a *transverse crossing* of a smooth discriminant curve by that moving slice, with oriented crossing sign controlled by

$$-\mathcal{R}(x(a_*)).$$

Accordingly, the discriminant/catastrophe route does not reduce the adjacent odd-zero problem to a local singularity classification in (a, γ, h) -space. The remaining issue is global: to exclude the existence of such a transverse crossing along the continued born branch before the segment exits the arc, collapses affine-planarly, or acquires the next odd zero of $\mathcal{W}_{\text{src}}$.

Proposition 12.314 (Anchor-sign preservation excludes first rooting). *Let J be an opened nonmixed arc, and work in a local normalized nonmixed chart on which*

$$S(t) = t^2, \quad \Gamma_a = D_a.$$

Let

$$I = (a_1, a_0] \subset J$$

and let

$$\tau : I \rightarrow J$$

be a real-analytic continuation of the repeated-source critical branch such that

$$\tau(a) > a, \quad D'_a(\tau(a)) = 0 \quad (a \in I).$$

Set

$$V(a) := D_a(\tau(a)), \quad G(a) := V(a) - \mathcal{Z}(a).$$

Assume that there exists $\epsilon \in \{\pm 1\}$ such that

$$\epsilon \mathcal{R}(a) > 0 \quad \text{for every } a \in I,$$

and

$$\epsilon G(a_0) > 0.$$

Then

$$\epsilon G(a) > 0 \quad \text{for every } a \in I.$$

In particular,

$$G(a) \neq 0 \quad (a \in I),$$

so the branch cannot realize the rigidified first-loss condition anywhere on I .

Proof. Assume for contradiction that G vanishes somewhere on I . Since G is continuous, the set

$$Z := \{a \in I : G(a) = 0\}$$

is closed in I and nonempty. Let

$$a_* := \sup Z.$$

Then $a_* < a_0$, and by the choice of a_* one has

$$\epsilon G(a) > 0 \quad \text{for all } a \in (a_*, a_0].$$

Hence the one-sided derivative at a_* satisfies

$$\epsilon G'(a_*) \geq 0.$$

On the other hand, by Proposition 12.302,

$$G'(a) = \frac{2}{\tau(a) - a} G(a) - \mathcal{R}(a).$$

Evaluating at the root $a = a_*$ gives

$$G'(a_*) = -\mathcal{R}(a_*).$$

Since $\epsilon \mathcal{R}(a_*) > 0$ by hypothesis, this yields

$$\epsilon G'(a_*) = -\epsilon \mathcal{R}(a_*) < 0,$$

contradicting

$$\epsilon G'(a_*) \geq 0.$$

Therefore G has no zero on I . Since

$$\epsilon G(a_0) > 0,$$

the sign of G on all of I is ϵ . □

Corollary 12.315 (No crossing on the adjacent-two-odd-zero chamber under anchor-sign preservation). *Let $b < q$ be adjacent odd-order zeros of $\mathcal{R}$ on an opened nonmixed arc J . Work in a local normalized nonmixed chart, and let*

$$I = (a_1, a_0] \subset J, \quad \tau : I \rightarrow J$$

be a real-analytic continuation of the born repeated-source critical branch such that

$$\tau(a) > a, \quad D'_a(\tau(a)) = 0 \quad (a \in I),$$

with

$$\tau(a_0) = q.$$

Assume that for every $a \in I$ with $a < a_0$, the segment

$$[a, \tau(a)]$$

contains exactly two sign changes of $\mathcal{R}$, occurring at the odd zeros b and q , and that

$$\mathcal{R}(a)$$

has fixed sign on I .

Then

$$\operatorname{sgn}(G(a)) = \operatorname{sgn}(\mathcal{R}(a)) \quad (a \in I),$$

hence

$$G(a) \neq 0 \quad (a \in I).$$

Equivalently, the RH-selected slice

$$\gamma = \mathcal{Z}(a)$$

does not cross the continued discriminant curve

$$(a, \gamma, h) = (a, V(a), \tau(a))$$

anywhere on I .

Proof. At the entry point $a = a_0$, one has

$$\tau(a_0) = q,$$

and the segment

$$[a_0, q]$$

contains exactly one sign change of $\mathcal{R}$, namely at b . Therefore Proposition 12.295 applied to the critical pair (a_0, q) gives

$$\operatorname{sgn}(G(a_0)) = \operatorname{sgn}(\mathcal{R}(a_0)).$$

Now apply Proposition 12.314 with

$$\epsilon := \operatorname{sgn}(\mathcal{R}|_I).$$

□

Remark 12.316 (What this really removes from the previous caveat). The preceding corollary shows that the geometric hypothesis “the anchor stays on the original birth-side outer lobe” was stronger than necessary. The proof uses only the intrinsic condition that the anchor-side value

$$\mathcal{R}(a)$$

keep a fixed sign along the continuation interval.

Accordingly, the remaining unresolved possibility on this route is narrower, but it is not yet exhausted. A rooted point can occur only after the anchor itself passes through an odd-order zero of $\mathcal{R}$, or after branch exit, or after affine-planar collapse. Moreover, Proposition 12.317 shows that no rooted crossing can occur at that anchor odd zero itself or immediately after passage, and Corollary 12.319 shows that the subsequent post-anchor chambers alternate dangerous and safe. Thus the true residual obstruction is no longer the raw adjacent-two-odd-zero chamber

itself. But by the universality threshold for the current route-2 package, by the local freedom of arbitrary canonical quotient-septic geometry, and by the full-block universality of the direct provenance slice

$$\phi = A q^{(7)},$$

even the delayed post-anchor scenario cannot be excluded inside those packages alone.

Proposition 12.317 (Odd-anchor passage has no sign reset). *Let J be an opened nonmixed arc, and work in a local normalized nonmixed chart on which*

$$S(t) = t^2, \quad \Gamma_a = D_a.$$

Let

$$q \in J$$

be an isolated odd-order zero of $\mathcal{R}$, and let

$$\tau : (q - \delta, q + \delta) \rightarrow J$$

be a real-analytic continuation of the repeated-source critical branch such that

$$\tau(a) > a, \quad D'_a(\tau(a)) = 0 \quad (a \in (q - \delta, q + \delta)).$$

Set

$$G(a) := D_a(\tau(a)) - \mathcal{Z}(a), \quad L(a) := \tau(a) - a,$$

and define

$$\mathcal{A}'(a) := \frac{2}{L(a)}, \quad \mathcal{G}(a) := e^{-\mathcal{A}(a)} G(a).$$

Assume that there exists $\epsilon \in \{\pm 1\}$ such that

$$\epsilon \mathcal{R}(a) > 0, \quad \epsilon G(a) > 0 \quad \text{for every } a \in (q, q + \delta).$$

Then

$$\epsilon G(q) > 0.$$

In particular, $G(q) \neq 0$. Hence, after possibly shrinking δ ,

$$\epsilon G(a) > 0 \quad \text{for every } a \in (q - \delta, q + \delta).$$

Since q is an odd-order zero of $\mathcal{R}$, the sign of $\mathcal{R}$ flips across q ; therefore on the outgoing side

$$\text{sgn } G(a) = -\text{sgn } \mathcal{R}(a) \quad (a \in (q - \delta, q)).$$

So a rooted crossing cannot occur at the anchor odd zero itself, nor immediately after passage across it.

Proof. By Proposition 12.302,

$$G'(a) = \frac{2}{L(a)} G(a) - \mathcal{R}(a),$$

hence

$$\mathcal{G}'(a) = -e^{-\mathcal{A}(a)} \mathcal{R}(a).$$

Assume for contradiction that $G(q) = 0$. Then $\mathcal{G}(q) = 0$, and for $a \in (q, q + \delta)$,

$$\mathcal{G}(a) = \mathcal{G}(q) + \int_q^a \mathcal{G}'(s) ds = - \int_q^a e^{-\mathcal{A}(s)} \mathcal{R}(s) ds.$$

Since $e^{-\mathcal{A}(s)} > 0$ and $\epsilon \mathcal{R}(s) > 0$ on $(q, q + \delta)$, this gives

$$\epsilon \mathcal{G}(a) < 0 \quad (a \in (q, q + \delta)),$$

hence

$$\epsilon G(a) < 0 \quad (a \in (q, q + \delta)),$$

contradicting the hypothesis. Therefore $G(q) \neq 0$. Since

$$\epsilon G(a) > 0 \quad (a \in (q, q + \delta)),$$

continuity yields

$$\epsilon G(q) \geq 0,$$

and the nonvanishing just proved forces

$$\epsilon G(q) > 0.$$

A second continuity argument then gives

$$\epsilon G(a) > 0 \quad (a \in (q - \delta, q + \delta))$$

after shrinking δ if necessary. Finally, because q is an odd-order zero of $\mathcal{R}$, its sign flips across q , so on $(q - \delta, q)$ one has

$$\operatorname{sgn} G(a) = -\operatorname{sgn} \mathcal{R}(a).$$

In particular G cannot vanish there. $\square$

Proposition 12.318 (Weighted-defect recursion on post-anchor chambers). *Assume the setting of Proposition 12.317. Let*

$$q_1 > q_2 > \cdots > q_N$$

be successive odd-order zeros of $\mathcal{R}$ crossed by the anchor along the continued branch after the adjacent-two-odd-zero chamber, and set

$$I_j := (q_{j+1}, q_j) \quad (1 \leq j \leq N-1).$$

Assume that the continued branch exists real-analytically on

$$\bigcup_{j=1}^{N-1} I_j.$$

For each j , let

$$\epsilon_j := \operatorname{sgn}(\mathcal{R}|_{I_j}) \in \{\pm 1\}, \quad m_j(a) := \int_a^{q_j} e^{-\mathcal{A}(s)} |\mathcal{R}(s)| ds \quad (a \in I_j),$$

and

$$M_j := m_j(q_{j+1}) > 0.$$

Then on each I_j ,

$$\mathcal{G}(a) = \mathcal{G}(q_j) + \epsilon_j m_j(a).$$

Consequently:

(1) *if*

$$\operatorname{sgn} \mathcal{G}(q_j) = \epsilon_j,$$

then $\mathcal{G}$ has no zero on I_j ;

(2) *if*

$$\operatorname{sgn} \mathcal{G}(q_j) = -\epsilon_j,$$

then $\mathcal{G}$ has a zero on I_j if and only if

$$M_j > |\mathcal{G}(q_j)|,$$

and in that case the zero is unique;

(3) *if $\mathcal{G}$ has no zero on I_j , then*

$$\operatorname{sgn} \mathcal{G}(q_{j+1}) = \operatorname{sgn} \mathcal{G}(q_j).$$

Since $\epsilon_{j+1} = -\epsilon_j$, the chamber type

$$\operatorname{sgn} \mathcal{G}(q_j) \quad \text{versus} \quad \epsilon_j$$

alternates from one post-anchor chamber to the next as long as no rooted crossing has yet occurred.

Proof. On I_j one has

$$\mathcal{G}'(a) = -e^{-\mathcal{A}(a)}\mathcal{R}(a) = -\epsilon_j e^{-\mathcal{A}(a)}|\mathcal{R}(a)|.$$

Hence for $a \in I_j$,

$$\mathcal{G}(a) = \mathcal{G}(q_j) + \int_{q_j}^a \mathcal{G}'(s) ds = \mathcal{G}(q_j) + \epsilon_j \int_a^{q_j} e^{-\mathcal{A}(s)}|\mathcal{R}(s)| ds,$$

which is the stated formula.

Now $m_j(a)$ is strictly increasing as a decreases from q_j to q_{j+1} , with

$$m_j(q_j) = 0, \quad m_j(q_{j+1}) = M_j.$$

So if

$$\operatorname{sgn} \mathcal{G}(q_j) = \epsilon_j,$$

then

$$\mathcal{G}(a) = \epsilon_j (|\mathcal{G}(q_j)| + m_j(a))$$

has fixed sign on I_j and no zero. If instead

$$\operatorname{sgn} \mathcal{G}(q_j) = -\epsilon_j,$$

then

$$\mathcal{G}(a) = \epsilon_j (-|\mathcal{G}(q_j)| + m_j(a)),$$

so a zero occurs precisely when $m_j(a) = |\mathcal{G}(q_j)|$, which happens if and only if

$$M_j > |\mathcal{G}(q_j)|.$$

Because m_j is strictly monotone, that zero is unique.

Finally, if no zero occurs on I_j , then $\mathcal{G}$ keeps one sign throughout the interval, so in particular

$$\operatorname{sgn} \mathcal{G}(q_{j+1}) = \operatorname{sgn} \mathcal{G}(q_j).$$

Since the ϵ_j alternate at successive odd zeros, the relative chamber type alternates as claimed. $\square$

Corollary 12.319 (The first post-anchor chamber is dangerous). *Let q_1 be the first odd-order zero of $\mathcal{R}$ crossed by the anchor after the adjacent-two-odd-zero chamber, and let*

$$I_1 := (q_2, q_1)$$

be the first post-anchor chamber. Assume the continued born branch extends real-analytically across q_1 and on I_1 . Then:

- (1) *no rooted crossing occurs at q_1 or immediately after crossing it;*
- (2) *on I_1 one has*

$$\operatorname{sgn} \mathcal{G}(q_1) = -\epsilon_1,$$

so I_1 is the first dangerous post-anchor chamber;

- (3) *as long as no rooted crossing has yet occurred, the post-anchor chambers alternate dangerous/safe/dangerous/safe...*

In particular, the first chamber in which a rooted crossing is even eligible after the anchor-sign-preservation interval is the first post-anchor chamber. Without an additional diagonal-capture theorem, however, this eligibility does not by itself determine the odd-zero count on the active segment.

Proof. The adjacent-two-odd-zero chamber is root-free by Corollary 12.315. At its left endpoint q_1 , the anchor crosses an odd-order zero of $\mathcal{R}$. Proposition 12.317 shows that $\mathcal{G}$ keeps its incoming sign across q_1 , while $\mathcal{R}$ changes sign there. Thus on the outgoing chamber I_1 the signs are opposite:

$$\operatorname{sgn} \mathcal{G}(q_1) = -\epsilon_1.$$

The no-root statement at and immediately after q_1 is also exactly Proposition 12.317. The alternation of chamber types then follows from Proposition 12.318(3). $\square$

Remark 12.320 (No-go for purely local scalar exclusion on the first dangerous post-anchor chamber). At a rooted critical point, after normalizing the active interval to

$$x = a + sL, \quad 0 \leq s \leq 1, \quad L = \tau(a) - a,$$

the profile

$$R_a(s) := \mathcal{R}(a + sL)$$

satisfies the two local scalar moment identities

$$\int_0^1 (1-s)^2 R_a(s) ds = 0, \quad \int_0^1 s(1-s) R_a(s) ds = 0.$$

Once nonlocal-target landing is admitted, the first dangerous post-anchor chamber need not correspond to a fixed odd-zero count or to a rigid four-lobe pattern on the active segment. Even in the simplest three-lobe sign pattern

$$+, -, +$$

(or its global negative), the local scalar rooted equations are already formally realizable.

Indeed, for any $A > 0$, define

$$R_A(s) := \begin{cases} A, & 0 < s < \frac{1}{3}, \\ -\frac{7}{2}A, & \frac{1}{3} < s < \frac{2}{3}, \\ \frac{11}{2}A, & \frac{2}{3} < s < 1. \end{cases}$$

A direct calculation gives

$$\int_0^1 (1-s)^2 R_A(s) ds = 0, \quad \int_0^1 s(1-s) R_A(s) ds = 0,$$

and also

$$\int_0^1 R_A(s) ds = \int_0^1 s R_A(s) ds = \int_0^1 s^2 R_A(s) ds = A.$$

So the local rooted-point equations and the nondegenerate fold data are already formally compatible with a three-lobe profile.

Thus no argument using only the local normalized rooted profile, the two vanishing scalar moments, and the local transversality/fold signs can exclude rooting on the first dangerous post-anchor chamber. Any successful next step must use genuinely source-coupled structure invisible in the planar normalized D_a model.

Remark 12.321 (What route 2 now reduces to). The discriminant analysis first showed that a rooted point is not born *at the entry* into the adjacent-two-odd-zero chamber. The stronger anchor-sign argument showed more: on any continuation interval on which the anchor-side value

$$\mathcal{R}(a)$$

keeps a fixed sign, the RH-selected slice

$$\gamma = \mathcal{Z}(a)$$

cannot cross the discriminant curve at all.

The odd-anchor analysis sharpened this further: when the anchor itself passes through an odd-order zero of $\mathcal{R}$, the integrating-factor defect

$$\mathcal{G}(a) = e^{-\mathcal{A}(a)} (D_a(\tau(a)) - \mathcal{Z}(a))$$

keeps its incoming sign across that odd anchor zero, so a rooted crossing cannot occur there or immediately after passage. The subsequent post-anchor chambers then alternate dangerous and safe, beginning with a dangerous first post-anchor chamber (Corollary 12.319).

However, two no-go facts are now clear. First, Remark 12.320 shows that already on that first dangerous post-anchor chamber, the local normalized rooted equations and fold data are formally realizable in the planar scalar model. Second, the current route-2 source package itself is still universal under the normalized lift

$$c = 1, \quad v_5 = t, \quad \lambda = t^2, \quad u_5'' = 2\mathcal{Z}$$

(Proposition 12.322), so even the source-level objects

$$\Psi_a, \quad \mathfrak{D}, \quad \mathcal{W}_{\text{src}}, \quad D_{f_a}$$

do not yet distinguish the RH branch from an arbitrary analytic normalized model.

So route 2 no longer reduces merely to a delayed no-crossing / profile-realizability problem *within the current package*. It reduces to finding the first genuinely *non-universal* source-coupled input, i.e. an ingredient not already captured by the normalized planar D_a model.

Within the present draft, the first plausible places for such new input lie beyond the current universal quartet, for example in continuum-strength provenance constraints on q over whole one-pair blocks, or in external mixed/canonical structure not already absorbed by the normalized planar model.

Proposition 12.322 (Universality threshold for the current route-2 package). *Let $I \subset \mathbf{R}$ be an interval, let*

$$\mathcal{Z} : I \rightarrow \mathbf{R}$$

be real-analytic, and set

$$\mathcal{R} := \mathcal{Z}'.$$

Define analytic source functions on I by

$$c(t) := 1, \quad v_5(t) := t, \quad \lambda(t) := t^2,$$

and choose u_5 so that

$$u_5''(t) = 2\mathcal{Z}(t).$$

Then:

(1)

$$x = \frac{v_5}{c} = t, \quad S = \frac{\lambda}{c} = t^2, \quad Y = \frac{u_5}{c} = u_5;$$

(2) *for every analytic f ,*

$$J_f = W(c, v_5, f) = W(1, t, f) = f'',$$

hence

$$J_\lambda = 2, \quad J_{u_5} = 2\mathcal{Z}, \quad \mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda = 4\mathcal{R};$$

(3) *the repeated-source ratio is exactly the normalized planar one:*

$$\Psi_a(h) = \Gamma_a(h) = D_a(h);$$

(4) *the critical determinant equation is exactly the normalized planar critical-point equation:*

$$\mathfrak{D}(a, h) = 0 \iff \partial_h \Psi_a(h) = 0 \iff \partial_h D_a(h) = 0;$$

(5) *for the source-pinned moving shear*

$$f_a := u_5 - \mathcal{Z}(a)\lambda,$$

one has

$$J_{f_a} = 2(\mathcal{Z} - \mathcal{Z}(a)),$$

and

$$D_{f_a}(a, h) = D_\lambda(a, h)(\Psi_a(h) - \mathcal{Z}(a)).$$

Thus every analytic normalized planar model $(\mathcal{Z}, \mathcal{R}, D_a)$ lifts to a genuine analytic source model for the entire current route-2 package.

Proof. The identities

$$x = \frac{v_5}{c} = t, \quad S = \frac{\lambda}{c} = t^2, \quad Y = \frac{u_5}{c} = u_5$$

are immediate. Also,

$$J_f = W(c, v_5, f) = W(1, t, f) = f'',$$

so

$$J_\lambda = 2, \quad J_{u_5} = 2\mathcal{Z}.$$

Hence

$$\mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda = 2(2\mathcal{Z}') - 2\mathcal{Z} \cdot 0 = 4\mathcal{R}.$$

Because the source coordinate and the normalized planar coordinate coincide,

$$x(t) = t,$$

the source repeated-source quotient is exactly the planar normalized quadratic envelope:

$$\Psi_a(h) = \Gamma_a(h) = D_a(h).$$

Therefore the source critical determinant equation is exactly the planar critical-point equation

$$\mathfrak{D}(a, h) = 0 \iff \partial_h D_a(h) = 0.$$

Finally,

$$f_a = u_5 - \mathcal{Z}(a)\lambda$$

satisfies

$$J_{f_a} = u_5'' - 2\mathcal{Z}(a) = 2(\mathcal{Z} - \mathcal{Z}(a)),$$

and Proposition 12.290(2) gives

$$D_{f_a}(a, h) = D_\lambda(a, h)(\Psi_a(h) - \mathcal{Z}(a)).$$

So all route-2 source objects reduce exactly to the normalized planar ones. $\square$

Corollary 12.323 (The current route-2 package is universal under normalization). *Any configuration formally realizable by an analytic normalized planar model*

$$(\mathcal{Z}, \mathcal{R}, D_a)$$

is also formally realizable by a genuine analytic source model for the current route-2 package

$$(c, v_5, \lambda, u_5), \quad \Psi_a, \quad \mathfrak{D}, \quad \mathcal{W}_{\text{src}}, \quad D_{f_a}.$$

Accordingly, no contradiction theorem phrased only in terms of the current route-2 package can distinguish the RH branch from an arbitrary analytic normalized planar model.

Proof. This is immediate from Proposition 12.322. $\square$

Remark 12.324 (Explicit source-level countermodel to the current route-2 package). Take

$$\mathcal{R}(t) := t(t-1)(t^2 - 4t - 2)$$

and

$$\mathcal{Z}(t) := \int_0^t \mathcal{R}(u) du = \frac{1}{5}t^5 - \frac{5}{4}t^4 + \frac{2}{3}t^3 + t^2,$$

and realize this normalized planar model by

$$c(t) = 1, \quad v_5(t) = t, \quad \lambda(t) = t^2, \quad u_5''(t) = 2\mathcal{Z}(t).$$

Let

$$q := 2 - \sqrt{6}.$$

In this genuine analytic source model, the unique born repeated-source critical branch from the odd zero 0 continues analytically down to $a = q$, lands there with a nonlocal target

$$\tau(q) \approx 0.3540339256 \neq q,$$

satisfies the anchor-sign safe relation on $(q, 0)$, crosses the odd anchor zero q with no sign reset, and then undergoes a delayed rooted crossing at

$$a_* \approx -0.5805673462, \quad \tau(a_*) \approx 0.3888336023.$$

Thus diagonal capture fails, and the whole delayed-crossing scenario is realized inside a genuine analytic source model for the current route-2 package.

Remark 12.325 (The next live target must be non-universal). By Corollary 12.323, the present lower route-2 package is exhausted: any further progress must use a genuinely non-universal source-coupled input, i.e. an ingredient not already absorbed by the normalized planar model $(\mathcal{Z}, \mathcal{R}, D_a)$.

However, the thread analysis shows that this non-universal input cannot be identified merely with arbitrary canonical quotient-septic geometry, nor even with the explicit direct $q^{(7)}$ -slice by itself. Indeed, Remark 12.104 shows that arbitrary canonical Δ_7 is still locally free once the lower one-pair data are fixed, and Proposition 13.8 together with Corollary 13.9 show that the direct highest-new provenance slice

$$\phi = A q^{(7)}$$

is fully universal on each generic one-pair block, even after any finite family of linear provenance/matching conditions on q .

So any viable continuation must use something stronger: either a *continuum-strength provenance law* for q on a whole block, or an external mixed/canonical theorem that constrains q beyond finite jet data. Within the present draft, the first plausible places for such genuinely non-universal input lie in the mixed / canonical package above arbitrary quotient-septic geometry.

Remark 12.326 (What remains on the nonmixed 4-point side). The preceding results now give a full local derivative-side picture of the nonmixed exact-contact geometry. Exact nonmixed tangent-plane contacts force zeros of $\mathcal{R}$ (Proposition 12.264); intervals on which $\mathcal{R}$ has fixed nonzero sign are exactly-contact sterile (Corollary 12.265); recurrent exact contacts force affine-planar collapse (Corollary 12.268); and isolated zeros of $\mathcal{R}$ are classified by parity (Proposition 12.270).

At source level, the same structure is encoded by the gauge-invariant Wronskian

$$\mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda,$$

which differs from $\mathcal{R}$ only by a nonvanishing analytic factor on a nonmixed arc (Proposition 12.236). Thus exact nonmixed contacts force zeros of $\mathcal{W}_{\text{src}}$, one-sign $\mathcal{W}_{\text{src}}$ -intervals are sterile, and recurrent contacts force $\mathcal{W}_{\text{src}} \equiv 0$ (Corollary 12.277).

Moreover, Proposition 12.279 lifts $\mathcal{W}_{\text{src}}$ to the 4×4 Wronskian

$$W(c, v_5, \lambda, u_5),$$

while Proposition 12.280 shows that, at the current source-level ratio/Wronskian/ 4×4 -determinant level, the odd-zero pattern of $\mathcal{W}_{\text{src}}$ and $\mathcal{R}$ is still formally unconstrained. Thus generic determinant algebra alone is no longer enough to close the nonmixed branch.

The repeated-source / first-loss lane is now also closed. For each fixed source point a , repeated-source exact contacts factor through the quadratic-envelope scalar

$$\Gamma_a(x) = \frac{Y(x) - Y(a) - Y'(a)(x - a)}{S(x) - S(a) - S'(a)(x - a)},$$

equivalently through the source-determinant quotient

$$\Psi_a(h) = \frac{D_{u_5}(a, h)}{D_\lambda(a, h)},$$

with derivative controlled by the osculatory determinant $\mathfrak{D}(a, h)$. The formal first-loss event rigidifies to a sheared repeated-node tangent-contact configuration (Remark 12.282), but this is no longer a live independent bottleneck. Proposition 12.292 shows that no attained separated first loss can occur, every first-loss sequence is forced to the diagonal, and diagonal first loss occurs exactly at isolated odd-order zeros of $\mathcal{R}$. Moreover, Proposition 12.287 shows that for each fixed shear level γ , the corresponding sheared opposite-side 3-point branch is globally rigid on its maximal slope-overlap block; and Proposition 12.272 together with Corollary 12.273 show that near an isolated odd zero of $\mathcal{R}$, the diagonal critical branch is uniformly unrooted, so the rooted sheared repeated-node tangent-contact event itself does not occur there. Thus the sheared lane is closed both in the fixed- γ nonrecurrence direction and in the local isolated-odd-zero direction.

Accordingly, the genuinely live nonmixed 4-point burden is no longer local birth, nor repeated-source first loss, nor generic determinant sign. The positive-kernel route now shows more: on any chamber where the repeated-source segment sees exactly one sign change of $\mathcal{R}$, the born critical branch is simple, has strictly negative slope, and cannot realize the rigidified first-loss condition

$$\Gamma_a(x) = \mathcal{Z}(a).$$

At source level, the same branch is characterized by

$$\mathfrak{D}(a, h) = 0,$$

but Proposition 12.290 shows that its rooted numerator

$$D_{u_5}(a, h) - \mathcal{Z}(x(a)) D_\lambda(a, h)$$

has a fixed signed leading term near an isolated odd zero, so the local branch is uniformly unrooted there. Thus a local branch of

$$\mathfrak{D}(a, h) = 0$$

cannot become a genuine obstruction until the source segment acquires a second odd zero (equivalently a second sign-changing zero) of

$$\mathcal{W}_{\text{src}},$$

or exits the local one-sign-change chamber. By Proposition 12.305, the positive-kernel route already excludes every opened nonmixed arc carrying at most one odd zero of $\mathcal{W}_{\text{src}}$ (equivalently of $\mathcal{R}$).

Moreover, once one follows an analytic continuation of the born branch beyond the local birth chamber, the rooted defect

$$G(a) = D_a(\tau(a)) - \mathcal{Z}(a)$$

satisfies an exact first-order transport equation, and any nondegenerate rooted point is forced to be a stationary point of the target map and a local fold of the moving-shear equation

$$A_a(x) - \mathcal{Z}(a) B_a(x) = 0$$

(Propositions 12.302 and 12.303). The discriminant analysis rules out birth of such a rooted point at entry into the adjacent-two-odd-zero chamber (Proposition 12.300), and the anchor-sign argument rules out any rooted crossing on a continuation interval on which the anchor-side value

$$\mathcal{R}(a)$$

keeps a fixed sign (Proposition 12.314 and Corollary 12.315).

Beyond that, Proposition 12.317 shows that even after the anchor crosses an odd-order zero of $\mathcal{R}$, no rooted crossing can occur at that odd anchor zero itself or immediately after passage, while Corollary 12.319 shows that the ensuing post-anchor chambers alternate dangerous and safe. However, Remark 12.320 shows that the local normalized rooted equations and fold data are already formally compatible with rooting on the first dangerous post-anchor chamber. More strongly, Proposition 12.322 and Corollary 12.323 show that the entire current route-2 source package is universal under normalization, and Remark 12.324 exhibits a genuine analytic source model realizing safe adjacent-two-odd-zero behavior, nonlocal-target landing at the next odd anchor zero, odd-anchor no-reset, and delayed rooted crossing.

So the sharpened remaining nonmixed burden is no longer a delayed no-crossing / profile-realizability theorem *within the current route-2 package*. The current lower package is already exhausted, and the thread analysis shows that arbitrary canonical quotient-septic geometry is still too free, while even the explicit direct provenance slice

$$\phi = A q^{(7)}$$

is fully universal on each generic one-pair block. More strongly, any finite family of linear provenance or matching conditions on q is still too weak to exclude direct-slice recurrence geometry.

Accordingly, any further progress must use a genuinely non-universal continuum-strength provenance law for q , or else an external mixed/canonical input that constrains q on a whole block and is not already absorbed by the normalized planar model $(\mathcal{Z}, \mathcal{R}, D_a)$. By Theorem 12.240, such an external mixed input cannot come from a new sufficiently small local mixed 4-point obstruction near the centered diagonal base; any useful mixed contribution must therefore be genuinely global/nonlocal, or arise from deeper canonical/multi-pair structure.

Remark 12.327 (Compact status summary for the nonmixed 4-point branch). The nonmixed 4-point branch is now reduced to a source-level odd-zero problem. Locally, exact nonmixed contacts force zeros of $\mathcal{R}$, and isolated zeros are classified by parity: even-order zeros are sterile, while odd-order zeros seed local exact contacts. Repeated-source first loss is not a separate mechanism: it occurs if and only if one diagonally approaches an odd-order zero of $\mathcal{R}$. The sheared repeated-node lane is also closed in the collapse direction: any genuine diagonal realization of the rigidified sheared event already forces $\mathcal{R} \equiv 0$ locally, hence affine-planar collapse.

On a nonmixed arc, the same structure is encoded by the gauge-invariant source-level Wronskian

$$\mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda,$$

which differs from $\mathcal{R}$ only by a nonvanishing analytic factor. Repeated-source first loss is not a separate mechanism: it occurs if and only if one diagonally approaches an odd-order zero of $\mathcal{R}$, and the corresponding rooted sheared repeated-node event is locally absent near an isolated odd zero of $\mathcal{R}$. Near such an isolated odd zero, the born repeated-source branch

$$\mathfrak{D}(a, h) = 0$$

is uniformly unrooted at source level:

$$D_{u_5}(a, h) - \mathcal{Z}(x(a)) D_\lambda(a, h) \neq 0$$

with a fixed signed leading term (Proposition 12.290). For each fixed shear level γ , the associated sheared opposite-side 3-point branch is globally rigid on its maximal slope-overlap block. Moreover, the positive-kernel / source-determinant route now excludes every opened nonmixed arc carrying at most one odd zero of $\mathcal{W}_{\text{src}}$: on any chamber where the repeated-source segment sees exactly one sign change of $\mathcal{R}$, the born critical branch is simple, has strictly negative slope, and cannot realize the rigidified first-loss condition. Beyond that chamber, the discriminant analysis rules out birth of a rooted point at entry into the adjacent-two-odd-zero chamber, and the anchor-sign argument rules out any crossing on a continuation interval where the anchor-side value

$$\mathcal{R}(a)$$

keeps a fixed sign.

The odd-anchor passage theorem shows that once the anchor itself crosses an odd-order zero of $\mathcal{R}$, the integrating-factor defect keeps its incoming sign across that odd anchor zero, so no rooted crossing can occur there or immediately after it. The subsequent post-anchor chambers alternate dangerous and safe. However, the local normalized rooted equations and the local fold/transversality data are already formally realizable on the first dangerous post-anchor chamber, and more strongly the entire current route-2 source package is universal under normalization. Even arbitrary canonical quotient-septic geometry remains too free, and the explicit direct provenance slice

$$\phi = A q^{(7)}$$

is fully universal on each generic one-pair block; indeed, even after any finite family of linear provenance/matching conditions on q , direct-slice recurrence geometry is still realizable. Thus the genuinely remaining nonmixed burden is no longer a problem inside the current route-2 or arbitrary quotient-septic package at all. Any further progress must use a genuinely non-universal continuum-strength provenance law for q , or an external mixed/canonical theorem that constrains q on a whole block and is not already absorbed by the normalized planar model. By the centered-base mixed local exhaustion theorem, this external mixed input cannot be a

new local mixed 4-point septic-jet phenomenon near the diagonal base; it must be genuinely global/nonlocal or come from deeper canonical/multi-pair structure.

Remark 12.328 (Fixed-slope interval graphs, support pseudolines, and the remaining nonmixed global burden). There is a useful combinatorial reformulation of the nonadjacent pair-pair geometry on a nonmixed patch.

For fixed slope m , Lemma 12.86 shows that each ECT interval I contributes an intercept interval

$$\mathcal{B}_I(m) = \{\psi_I(x) - mx : x \in J_I\},$$

and that a line $y = mx + b$ is 2-secant to $\Gamma(I)$ exactly when

$$b \in \text{int } \mathcal{B}_I(m).$$

So for fixed m , the same-slope pair-pair incidence graph is an interval graph on the b -axis. In particular, any genuinely minimal surviving nonadjacent recurrence cannot live at one fixed slope; it must be slope-changing.

There is also a sharper support-curve reformulation. By Proposition 12.87, Corollary 12.88, and Proposition 12.89, the support curves

$$b = \tau_I(m)$$

are strictly concave pseudolines on common slope windows: each ordered pair has at most one common interior tangent slope, triple crossings obey the usual betweenness law, and each ordered pair has at most one directional or pair-pair overlap window in slope.

Thus any genuinely minimal surviving nonadjacent recurrence cannot be fixed slope, cannot be pair-repeating, and cannot disappear and later reappear on the same ordered pair.

There is also a useful endpoint-gap reformulation. For every endpoint support point (ξ, y_ξ) and every ordered target branch J , the endpoint-gap function

$$G_{\xi,J}(h) := y_\xi - y(h) - m(h)(\xi - x(h))$$

is strictly monotone on J by Lemma 12.109. So each endpoint/branch pair can contribute at most one remote tangent incidence.

These observations give the sharp current picture of the remaining nonmixed global burden: any surviving bad core must be slope-changing, pair-nonrepeating, and supported by a finite table of remote endpoint/branch incidences.

It is equally important to record the ceiling of this reduction. By Remark 13.4, intrinsic one-pair affine geometry is universal for analytic plane curves. Thus pure planarity, parity, interval-graph combinatorics, fixed-slope support order, or pseudoline combinatorics by itself cannot close the remaining branch. Any final contradiction must therefore use RH-specific source/provenance/canonical structure, not only abstract strip/support geometry.

Accordingly, the natural next global target is not another local normal-form analysis, but an RH-specific sign theorem for the remote endpoint-gap functions $G_{\xi,J}$.

Remark 12.329 (Derivative-geometry meaning of the nonmixed invariant $\mathcal{R}$). On a nonmixed patch,

$$\mathcal{R} = \frac{T_v}{K_v^2} = \frac{d}{dx} \left(\frac{Y''}{S''} \right).$$

Thus $\mathcal{R}$ measures the failure of the lifted space curve

$$x \mapsto (Y(x), x, S(x))$$

to be affine-planar at the derivative level: $\mathcal{R} = 0$ means the ratio Y''/S'' is locally constant, equivalently that the curve has vanishing affine torsion and lies in an affine plane after integrating twice. In this sense, the nonmixed 4-point branch is controlled by a single scalar derivative invariant rather than by raw coplanarity alone.

Remark 12.330 (Exact mixed reduction in terms of $\mathcal{Z}$ and $\mathcal{R}$). At a mixed lifted point

$$K_v(0) = 0, \quad T_v(0) = 0,$$

set

$$r := \text{ord}_0 K_v, \quad s := \text{ord}_0 T_v.$$

By Proposition 12.233,

$$K_v = (x')^3 S'',$$

so on a good v_5 -patch one has

$$S''(x) = \kappa x^r + O(x^{r+1}), \quad \kappa \neq 0.$$

After removing affine terms and integrating twice,

$$S(x) = \frac{\kappa}{(r+1)(r+2)} x^{r+2} + O(x^{r+3}).$$

Thus the mixed exponent m of the monomial model

$$Q_v(x) \sim (x^n, x, x^m)$$

is exactly

$$m = r + 2.$$

Now let n be the first degree of $Y(x)$ not absorbed by $\text{span}\{1, x, S\}$, so that

$$Y(x) = \alpha x^n + O(x^{n+1}), \quad \alpha \neq 0, \quad n > r + 2.$$

Using

$$T_v = (x')^6 (S''Y''' - S'''Y''),$$

one finds

$$s = n + r - 3, \quad n = s - r + 3.$$

Equivalently, on the nonmixed complement of $K_v = 0$,

$$\mathcal{Z} = \frac{Y''}{S''}, \quad \mathcal{R} = \mathcal{Z}' = \frac{T_v}{K_v^2}$$

satisfy the order bookkeeping

$$\text{ord}_0 \mathcal{Z} = n - r - 2 = s - 2r + 1, \quad \text{ord}_0 \mathcal{R} = s - 2r.$$

Thus the dangerous low-order mixed models are finite and explicit:

$$(m, n) = (3, 4), (3, 6), (4, 6),$$

equivalently

$$(r, s) = (1, 2), (1, 4), (2, 5),$$

equivalently

$$(r, \text{ord}_0 \mathcal{Z}) = (1, 1), (1, 3), (2, 2),$$

or

$$(r, \text{ord}_0 \mathcal{R}) = (1, 0), (1, 2), (2, 1).$$

So the mixed lifted 4-point endgame has also been reduced to a finite jet obstruction list, now phrased in terms of the invariant pair

$$(\mathcal{Z}, \mathcal{R}).$$

Remark 12.331 (Exact finite obstruction coefficients for the low-order 4-point branch). The explicit low-order 4-point obstruction list can be reduced to a finite jet package.

Nonmixed side. Suppose $K_v(0) \neq 0$, and write

$$s_k := S^{(k)}(0), \quad y_k := Y^{(k)}(0), \quad s_2 \neq 0.$$

Then the Taylor expansion

$$\mathcal{Z}(x) = z_0 + z_1 x + z_2 x^2 + z_3 x^3 + z_4 x^4 + z_5 x^5 + z_6 x^6 + O(x^7)$$

has canonical even coefficients

$$z_2, \quad z_4, \quad z_6,$$

and the dangerous nonmixed low-order cases are exactly:

$$n = 4 \iff z_2 \neq 0, \quad n = 6 \iff z_2 = 0, z_4 \neq 0, \quad n = 8 \iff z_2 = z_4 = 0, z_6 \neq 0.$$

Explicitly,

$$z_2 = \frac{s_2^2 y_4 - 2s_2 s_3 y_3 + (2s_3^2 - s_2 s_4) y_2}{2 s_2^3},$$

$$z_4 = \frac{s_2^4 y_6 - s_2^3 (4s_3 y_5 + 6s_4 y_4 + 4s_5 y_3 + s_6 y_2) + 2s_2^2 (6s_3^2 y_4 + 12s_3 s_4 y_3 + 4s_3 s_5 y_2 + 3s_4^2 y_2) - 12s_2 s_3^2 (2s_3 y_3 + 3s_4 y_2)}{24 s_2^5}$$

and

$$z_6 = \frac{\mathcal{P}_8(s_2, \dots, s_8; y_2, \dots, y_8)}{720 s_2^7},$$

where $\mathcal{P}_8$ is the explicit degree-8 numerator polynomial obtained by Taylor division of $\mathcal{Z} = Y''/S''$.

Mixed side. Suppose $K_v(0) = T_v(0) = 0$, and let

$$r := \text{ord}_0 K_v.$$

For the two low-order mixed cases $r = 1, 2$, Taylor division of $\mathcal{Z} = Y''/S''$ gives the exact obstruction coefficients

$$\kappa_{1,1}, \quad \kappa_{1,3}, \quad \kappa_{2,2},$$

defined by the expansions

$$\mathcal{Z}(x) = \frac{y_2}{s_3} \frac{1}{x} + \left(\frac{y_3}{s_3} - \frac{s_4 y_2}{2s_3^2} \right) + \kappa_{1,1} x + \kappa_{1,2} x^2 + \kappa_{1,3} x^3 + O(x^4) \quad (r = 1),$$

and

$$\mathcal{Z}(x) = \frac{2y_2}{s_4} \frac{1}{x^2} + \left(\frac{2y_3}{s_4} - \frac{2s_5 y_2}{3s_4^2} \right) \frac{1}{x} + \left(\frac{y_4}{s_4} - \frac{2s_5 y_3}{3s_4^2} - \frac{s_6 y_2}{6s_4^2} + \frac{2s_5^2 y_2}{9s_4^3} \right) + \kappa_{2,1} x + \kappa_{2,2} x^2 + O(x^3) \quad (r = 2).$$

The dangerous mixed low-order list is then exactly:

$$\begin{aligned} (m, n) = (3, 4) &\iff \kappa_{1,1} \neq 0, \\ (m, n) = (3, 6) &\iff \kappa_{1,3} \neq 0, \\ (m, n) = (4, 6) &\iff \kappa_{2,2} \neq 0. \end{aligned}$$

Thus the full currently explicit *low-order* 4-point obstruction list is equivalent to the finite set of coefficients

$$\boxed{z_2, z_4, z_6, \kappa_{1,1}, \kappa_{1,3}, \kappa_{2,2}}.$$

On the mixed side, the three coefficients

$$\kappa_{1,1}, \quad \kappa_{1,3}, \quad \kappa_{2,2}$$

exhaust the presently explicit low-order obstruction list.

On the nonmixed side, however, the parity classification from Proposition 12.270 shows that local exact-contact birth occurs at every isolated odd-order zero of $\mathcal{R}$. Accordingly,

$$z_2, \quad z_4, \quad z_6$$

should now be viewed only as the first three explicit low-order nonmixed obstruction coefficients, not as a complete local nonmixed list.

Remark 12.332 (Wronskian-order reformulation of the mixed low-order list). On a good v_5 -patch one has

$$Y'' = \frac{c^3}{M^3} J_{u_5}, \quad S'' = \frac{c^3}{M^3} J_\lambda, \quad \mathcal{Z} = \frac{J_{u_5}}{J_\lambda}, \quad \lambda = \frac{\Delta_7}{v_5}.$$

Thus, if

$$r := \text{ord}_0 J_\lambda = \text{ord}_0 K_v, \quad q := \text{ord}_0 J_{u_5} = \text{ord}_0 Y'',$$

then

$$m = r + 2, \quad n = q + 2, \quad \text{ord}_0 \mathcal{Z} = q - r.$$

Hence the dangerous mixed models

$$(m, n) = (3, 4), (3, 6), (4, 6)$$

are exactly the Wronskian order-pairs

$$(\text{ord}_0 J_\lambda, \text{ord}_0 J_{u_5}) = (1, 2), (1, 4), (2, 4).$$

If

$$J_\lambda(x) = x^r A(x), \quad J_{u_5}(x) = x^r B(x), \quad A(0) \neq 0,$$

then

$$\mathcal{Z}(x) = \frac{B(x)}{A(x)}.$$

For $r = 1$, the coefficients $\kappa_{1,1}$ and $\kappa_{1,3}$ are exactly the x^1 - and x^3 -coefficients of the regularized ratio B/A ; for $r = 2$, $\kappa_{2,2}$ is exactly the x^2 -coefficient of B/A .

Moreover, since

$$J_c = J_{v_5} = 0,$$

replacing

$$u_5 \mapsto u_5 + a c + b v_5 + \gamma \lambda$$

changes J_{u_5} only by

$$J_{u_5} \mapsto J_{u_5} + \gamma J_\lambda.$$

Thus the nonconstant Taylor coefficients of $\mathcal{Z}$ are canonical. In particular, the currently explicit mixed low-order list is already a finite Wronskian-ratio obstruction list.

Remark 12.333 (Regularized source scalar for the mixed residual quotient). In the notation of Remark 12.332, write locally

$$J_\lambda(x) = x^r A(x), \quad J_{u_5}(x) = x^r B(x), \quad A(0) \neq 0,$$

and set

$$Q(x) := \frac{B(x)}{A(x)} = \frac{J_{u_5}(x)}{J_\lambda(x)}.$$

Then

$$Q'(x) = \frac{J_\lambda J'_{u_5} - J_{u_5} J'_\lambda}{J_\lambda^2} = \frac{\mathcal{W}_{\text{src}}}{J_\lambda^2},$$

where

$$\mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda$$

is the source-level Wronskian from Proposition 12.236.

Moreover, under the affine gauge change

$$u_5 \mapsto u_5 + \alpha c + \beta v_5 + \gamma \lambda$$

one has

$$J_{u_5} \mapsto J_{u_5} + \gamma J_\lambda, \quad Q \mapsto Q + \gamma,$$

so

$$Q'(x) = \frac{\mathcal{W}_{\text{src}}}{J_\lambda^2}$$

is gauge-invariant. Thus, modulo the trivial additive constant in the regularized ratio Q , the mixed residual quotient germ is already controlled by the single scalar

$$\mathcal{D} := Q' = \frac{\mathcal{W}_{\text{src}}}{J_\lambda^2}.$$

On the nonmixed complement of $K_v = 0$, Remark 12.235 and Proposition 12.236 give

$$\mathcal{D} = \frac{M}{c^2} \mathcal{R}.$$

So the exact fixed-shear residual lane is governed by the same regularized source scalar already present on the nonmixed side, rather than by a second primitive pointwise highest-new field.

Proposition 12.334 (Projective secant-shadow reduction on an overlap patch). *Let*

$$U \subseteq \{c \neq 0, u_5 \neq 0, v_5 \neq 0, M \neq 0\}$$

be a good overlap patch. Set

$$\lambda := \frac{\Delta_7}{v_5}, \quad x := \frac{v_5}{c}, \quad Y := \frac{u_5}{c}, \quad S := \frac{\lambda}{c} = \frac{\Delta_7}{c v_5}.$$

Assume there exists a close exact mixed two-point bad branch

$$(h_{1,n}, h_{2,n}, \Lambda_n), \quad h_{1,n} \neq h_{2,n}, \quad h_{1,n}, h_{2,n} \in U, \quad h_{1,n}, h_{2,n} \rightarrow m \in U,$$

and let $(\eta_n, \xi_n, \gamma_n)$ be the branch-generated moving λ -shears, so that

$$\begin{aligned} (c - \eta_n \lambda)(h_{1,n}) &= \Lambda_n(c - \eta_n \lambda)(h_{2,n}), \\ (v_5 - \xi_n \lambda)(h_{1,n}) &= \Lambda_n(v_5 - \xi_n \lambda)(h_{2,n}), \\ (u_5 - \gamma_n \lambda)(h_{1,n}) &= \Lambda_n(u_5 - \gamma_n \lambda)(h_{2,n}). \end{aligned}$$

For each n , set

$$A_n := c - \eta_n \lambda, \quad B_n := v_5 - \xi_n \lambda, \quad F_n := u_5 - \gamma_n \lambda.$$

Then, after discarding finitely many terms, the branch determines the exact projective secant shadow

$$\Sigma(h_{1,n}, h_{2,n}) := [x_1 S_2 - x_2 S_1 : Y_1 S_2 - Y_2 S_1 : S_2 - S_1],$$

where

$$x_i = x(h_{i,n}), \quad Y_i = Y(h_{i,n}), \quad S_i = S(h_{i,n}).$$

More precisely, for all large n one has the exact identity

$$[B_n(h_{1,n}) : F_n(h_{1,n}) : A_n(h_{1,n})] = [B_n(h_{2,n}) : F_n(h_{2,n}) : A_n(h_{2,n})] = \Sigma(h_{1,n}, h_{2,n}).$$

In particular, the secant shadow is independent of the moving shears $(\eta_n, \xi_n, \gamma_n)$ and of Λ_n .

Equivalently, in raw coordinates,

$$\Sigma(h_1, h_2) = [v_{5,1} \lambda_2 - v_{5,2} \lambda_1 : u_{5,1} \lambda_2 - u_{5,2} \lambda_1 : c_1 \lambda_2 - c_2 \lambda_1].$$

As $h_2 \rightarrow h_1 = a$, the secant shadow converges projectively to the tangent shadow

$$\Pi(a) := [x S_x - S : Y S_x - S Y_x : S_x]_{|x=a}.$$

A direct calculation gives

$$\Pi(x) \wedge \partial_x \Pi(x) = S(x) (S_x Y_{xx} - S_{xx} Y_x, S_{xx}, -S_{xx}).$$

Hence Π is projectively immersive wherever

$$S \neq 0 \quad \text{and} \quad S_{xx} \neq 0.$$

Since on the v_5 -patch

$$K_v = (x')^3 S_{xx},$$

any such close exact mixed two-point bad branch can accumulate only on

$$\{S = 0\} \cup \{K_v = 0\}.$$

Proof. Since

$$x' = \left(\frac{v_5}{c}\right)' = \frac{M}{c^2},$$

the hypothesis $M \neq 0$ implies that $x = v_5/c$ is a local coordinate near m . After discarding finitely many terms we may therefore assume

$$x(h_{1,n}) \neq x(h_{2,n}).$$

Fix such an n , and abbreviate

$$\eta = \eta_n, \quad \xi = \xi_n, \quad \gamma = \gamma_n.$$

Write

$$p_i := (x_i, Y_i, 1) \in \mathbf{R}^3, \quad V := (\xi, \gamma, \eta) \in \mathbf{R}^3, \quad i = 1, 2.$$

Since

$$A_n = c(1 - \eta S), \quad B_n = c(x - \xi S), \quad F_n = c(Y - \gamma S),$$

the exact repeated-source equations imply

$$[B_n(h_{1,n}) : F_n(h_{1,n}) : A_n(h_{1,n})] = [p_1 - S_1 V]$$

and

$$[B_n(h_{2,n}) : F_n(h_{2,n}) : A_n(h_{2,n})] = [p_2 - S_2 V].$$

Because these two projective points are equal, there is a common projective value.

If $S_1 = S_2$, then

$$[p_1 - S_1 V] = [p_2 - S_1 V]$$

forces $p_1 = p_2$, hence $x_1 = x_2$, contradicting the choice of local coordinate. Therefore

$$S_1 \neq S_2.$$

Subtracting

$$p_1 - S_1 V = p_2 - S_2 V$$

gives

$$p_1 - p_2 = (S_1 - S_2)V,$$

hence

$$V = \frac{p_1 - p_2}{S_1 - S_2}.$$

Substituting back,

$$p_1 - S_1 V = p_1 - S_1 \frac{p_1 - p_2}{S_1 - S_2} = \frac{S_2 p_1 - S_1 p_2}{S_2 - S_1}.$$

Therefore the common projective value is exactly

$$[S_2 p_1 - S_1 p_2] = [x_1 S_2 - x_2 S_1 : Y_1 S_2 - Y_2 S_1 : S_2 - S_1],$$

which is the asserted secant shadow and is independent of $(\eta_n, \xi_n, \gamma_n)$ and of Λ_n . The raw-coordinate formula is the same identity multiplied through by $c_1 c_2$.

Now divide the homogeneous secant vector by $x_2 - x_1$ and let $h_2 \rightarrow h_1 = a$. The three components converge to

$$x S_x - S, \quad Y S_x - S Y_x, \quad S_x,$$

so the secant shadow converges projectively to

$$\mathbf{\Pi}(a) = [x S_x - S : Y S_x - S Y_x : S_x].$$

Differentiating $\mathbf{\Pi}$ in the local x -coordinate gives

$$\mathbf{\Pi}(x) \wedge \partial_x \mathbf{\Pi}(x) = S(x) (S_x Y_{xx} - S_{xx} Y_x, S_{xx}, -S_{xx}).$$

Therefore $\mathbf{\Pi}$ is projectively immersive wherever $S \neq 0$ and $S_{xx} \neq 0$. On a close bad branch the secant shadows converge to tangent shadows, so accumulation is possible only where immersion fails, i.e. on

$$\{S = 0\} \cup \{S_{xx} = 0\}.$$

Finally, on the v_5 -patch one has

$$K_v = (x')^3 S_{xx},$$

so $\{S_{xx} = 0\} = \{K_v = 0\}$ on U . □

Remark 12.335 (Useful overlap scalar). On the same overlap patch, set

$$\tau_u := -\frac{\Delta_7}{c u_5}, \quad \mathcal{G} := x' S_x + Y' \tau'_u.$$

Then $\mathcal{G}$ admits the exact factorization

$$\mathcal{G} = -\frac{W(u_5, v_5) W(u_5, \lambda)}{c^2 u_5^2}, \quad \lambda = \frac{\Delta_7}{v_5},$$

where $W(f, g) = fg' - f'g$. Equivalently,

$$\mathcal{G} = \frac{u_5 v_5' - u_5' v_5}{c^2} \left(\frac{\Delta_7}{u_5 v_5} \right)'.$$

In particular, $\mathcal{G} \neq 0$ on any connected overlap patch contained in $\{c \neq 0, M \neq 0\}$: the factor $W(u_5, v_5)$ cannot vanish identically there without forcing linear dependence of

$$1, \quad \frac{u_5}{c}, \quad \frac{v_5}{c},$$

and $W(u_5, \lambda)$ cannot vanish identically there without forcing

$$\Delta_7 = \kappa u_5 v_5,$$

which is incompatible with the direct septic slice carried by Δ_7 . This does not yet exclude the mixed two-point branch, but it records the first genuinely nontrivial Y - S overlap coupling scalar exposed by the current attack.

At highest-new weight, this scalar remains in the same differential tower as the direct septic source field:

$$\mathcal{G}^{[1]} = -\frac{W(u_5, v_5)}{c^2 u_5^2} W(u_5, A q^{(7)}), \quad A = \kappa \eta_2.$$

So $\mathcal{G}$ does not produce a second primitive pointwise highest-new generator. It does, however, provide the first useful external-looking rank-2 pointwise coupling on the overlap patch: together,

$$\lambda^{[1]} = A q^{(7)} \quad \text{and} \quad \mathcal{G}^{[1]}$$

recover $(q^{(7)}, (q^{(7)})')$ generically.

Proposition 12.336 (Same-tower closure for the current explicit highest-new overlap/canonical package). *Work on a good overlap patch*

$$U \subseteq \{c \neq 0, u_5 \neq 0, v_5 \neq 0, M \neq 0\}.$$

Let K_{low} denote the differential field generated by the lower controlled fields on U , in particular by

$$c, u_5, v_5, M, A = \kappa \eta_2$$

and their analytic derivatives, and write

$$r := q^{(7)}, \quad x := \frac{v_5}{c}, \quad Y := \frac{u_5}{c}, \quad S := \frac{\Delta_7}{c v_5} = \frac{\lambda}{c}.$$

Then every currently explicit pointwise highest-new field obtained from the quotient-septic package (Δ_7, λ) , the overlap fields (x, Y, S, τ_u) , the dual tangent-line fields $(\mu, \lambda, \beta, m, W)$, and the exact fixed-shear overlap constructions by finitely many algebraic operations, x -derivatives, Wronskians, finite jet extractions, and projectivizations lies in the differential algebra

$$K_{\text{low}} \langle r \rangle.$$

More precisely, the weight-one part of each such field is K_{low} -linear in finitely many x -jets

$$\partial_x^j S^{[1]}, \quad j = 0, \dots, N,$$

for some finite N , and

$$S^{[1]} = \frac{A}{c} r.$$

In particular, the currently explicit pointwise highest-new fields

$$\Delta_7^{[1]}, \lambda^{[1]}, \beta^{[1]}, m^{[1]}, W^{[1]}, S^{[1]}, \mathcal{G}^{[1]}, J_\lambda^{[1]}, \mathcal{W}_{\text{src}}^{[1]},$$

the projected fixed-shear germ at weight one, and the quartic-sextic coefficient $\kappa_{2,2}$ are all same-tower. Hence no second primitive pointwise highest-new generator appears in the current explicit overlap/canonical package. This is a statement about the present explicit pointwise package only, and does not address nonlocal, relational, or downstream transported-state obstructions.

Proof. On the overlap patch, the lower fields x, Y, μ, c, u_5, v_5 , and the coordinate-change factors are already fixed at weight zero. Thus all highest-new weight enters only through

$$\lambda^{[1]} = A r, \quad S^{[1]} = \frac{\lambda^{[1]}}{c} = \frac{A}{c} r.$$

Linearizing any allowed algebraic/differential expression at weight one kills quadratic interactions among highest-new pieces, so the weight-one part is a K_{low} -linear combination of finitely many x -jets of $S^{[1]}$. Since

$$\partial_x = \frac{c^2}{M} \partial_h$$

on U , each $\partial_x^j S^{[1]}$ lies in $K_{\text{low}}\langle r \rangle$. Projectivization only divides by lower nonvanishing factors, and finite jet coefficients such as $\kappa_{2,2}$ are extracted from finitely many such jets. The stated list therefore remains same-tower. $\square$

Corollary 12.337 (Best current external-looking rank-two coupling). *On the same overlap patch,*

$$\begin{pmatrix} \lambda^{[1]} \\ \mathcal{G}^{[1]} \end{pmatrix} = \begin{pmatrix} A & 0 \\ -\frac{W(u_5, v_5)}{c^2 u_5^2} (u_5 A' - u_5' A) & -\frac{W(u_5, v_5)}{c^2 u_5} A \end{pmatrix} \begin{pmatrix} r \\ r' \end{pmatrix}.$$

In particular,

$$\det \begin{pmatrix} A & 0 \\ -\frac{W(u_5, v_5)}{c^2 u_5^2} (u_5 A' - u_5' A) & -\frac{W(u_5, v_5)}{c^2 u_5} A \end{pmatrix} = -\frac{W(u_5, v_5)}{c^2 u_5} A^2.$$

So wherever

$$A \neq 0, \quad u_5 \neq 0, \quad W(u_5, v_5) \neq 0,$$

the pair $(\lambda^{[1]}, \mathcal{G}^{[1]})$ is a genuine rank-two pointwise coupling to (r, r') . It does not produce a new generator, but it is the best current external-looking rank-two coupling on the overlap patch.

Proof. This is the explicit highest-new expansion of Remark 12.335, using

$$\lambda^{[1]} = A r, \quad \mathcal{G}^{[1]} = -\frac{W(u_5, v_5)}{c^2 u_5^2} (u_5 A r' + (u_5 A' - u_5' A) r).$$

$\square$

Remark 12.338 (Work in progress: explicit-pointwise bridge to the local good-patch detector). Proposition 12.336 and Corollary 12.337 show that the present explicit highest-new package has only one currently visible pointwise source, namely the same tower generated by

$$r = q^{(7)},$$

even though the pair

$$(\lambda^{[1]}, \mathcal{G}^{[1]})$$

recovers (r, r') generically on the overlap patch.

This shifts the next bridge question. The issue is no longer whether another explicit pointwise highest-new field appears, but whether this single explicit pointwise direction is already seen by the local good-patch compensator. If it is, then the first same-order local compensator should already be quotient-septic-visible; if it is not, then the entire current explicit pointwise package is invisible to that local detector, and any surviving obstruction must already lie in nonlocal transport, collision-functoriality failure, or other provenance-sensitive two-atom structure.

At present this is a theorem target, not a proved equivalence.

Remark 12.339 (Ceiling of the current pointwise highest-new package). The present explicit highest-new audit has now reached a stable negative conclusion. All currently visible pointwise highest-new fields — including

$$\Delta_7^{[1]}, \lambda^{[1]}, \beta^{[1]}, m^{[1]}, W^{[1]}, S^{[1]}, \mathcal{G}^{[1]}$$

and the quartic–sextic transported coefficient package — remain in the differential algebra generated by the lower data together with

$$r = q^{(7)}.$$

So no second primitive explicit pointwise highest-new generator has emerged.

Combined with Remark 12.103 and Remark 12.130, this means that the honest remaining order-7/two-point burden is not “one more local field” and not the naive source-summed corrected-block route. What remains is package-level and provenance-sensitive: for example a reduced-image-germ theorem or collision-functoriality strong enough to force quotient-septic collapse.

Proposition 12.340 (Exact fixed-shear overlap branches off $K_v = 0$ force a fold at $S = 0$). *Work on the overlap patch of Proposition 12.334, and write*

$$\Phi(x) := \left(\frac{x - \xi_* S}{1 - \eta_* S}, \frac{Y - \gamma_* S}{1 - \eta_* S} \right), \quad S = \frac{\Delta_7}{c v_5}.$$

Assume there is a close exact fixed-shear mixed 2-point branch

$$x_{1,n} \neq x_{2,n}, \quad x_{1,n}, x_{2,n} \rightarrow a, \quad \Phi(x_{1,n}) = \Phi(x_{2,n}) \quad (n \geq 1).$$

If

$$K_v(a) \neq 0,$$

then

$$S(a) = 0, \quad \Phi'(a) = 0, \quad S_x(a) \neq 0.$$

More precisely,

$$\begin{aligned} 0 &= \Phi'_1(a) = 1 - (\xi_* - \eta_* x(a)) S_x(a), \\ 0 &= \Phi'_2(a) = Y_x(a) - (\gamma_* - \eta_* Y(a)) S_x(a). \end{aligned}$$

In particular, setting

$$\beta := \xi_* - \eta_* x(a) = \frac{1}{S_x(a)},$$

and centering the local x -coordinate at a , the first-coordinate fold model

$$U(x) := (x - x(a)) - \beta S(x)$$

satisfies

$$U(a) = 0, \quad U_x(a) = 0, \quad U_{xx}(a) = -\beta S_{xx}(a) \neq 0.$$

Hence, after analytic source reparametrization, the first coordinate of the fixed-shear projected germ is

$$t^2.$$

So any exact fixed-shear mixed branch away from $K_v = 0$ is already forced into the same fold regime that underlies the nonmixed odd-zero analysis. In particular, the remaining off- $K_v = 0$ burden is best understood as an absorption problem into the existing nonmixed odd-zero / affine-planar package, rather than as a search for a new mixed local carrier.

Proof. A fixed-shear branch is, in particular, a close exact mixed two-point bad branch on the same overlap patch. By Proposition 12.334, any accumulation point of such a branch lies in

$$\{S = 0\} \cup \{K_v = 0\}.$$

So if $K_v(a) \neq 0$, then necessarily

$$S(a) = 0.$$

Because

$$\Phi(x_{1,n}) = \Phi(x_{2,n}), \quad x_{1,n} \neq x_{2,n}, \quad x_{1,n}, x_{2,n} \rightarrow a,$$

the mean value theorem applied to each coordinate of Φ gives points

$$\zeta_{1,n}, \zeta_{2,n} \in (x_{1,n}, x_{2,n})$$

such that

$$\Phi'_1(\zeta_{1,n}) = 0, \quad \Phi'_2(\zeta_{2,n}) = 0.$$

Passing to the limit yields

$$\Phi'(a) = 0.$$

Differentiating the two coordinates of Φ gives

$$\begin{aligned}\Phi'_1 &= \frac{1 - (\xi_* - \eta_* x)S_x - \eta_* S}{(1 - \eta_* S)^2}, \\ \Phi'_2 &= \frac{Y_x - (\gamma_* - \eta_* Y)S_x - \eta_* S Y_x}{(1 - \eta_* S)^2}.\end{aligned}$$

Evaluating at $x = a$ and using $S(a) = 0$ yields the displayed identities

$$\begin{aligned}0 &= \Phi'_1(a) = 1 - (\xi_* - \eta_* x(a))S_x(a), \\ 0 &= \Phi'_2(a) = Y_x(a) - (\gamma_* - \eta_* Y(a))S_x(a).\end{aligned}$$

The first identity implies

$$S_x(a) \neq 0 \quad \text{and} \quad \beta := \xi_* - \eta_* x(a) = \frac{1}{S_x(a)}.$$

Now center the local x -coordinate at a and set

$$U(x) := (x - x(a)) - \beta S(x).$$

Then

$$U(a) = 0, \quad U_x(a) = 1 - \beta S_x(a) = 0, \quad U_{xx}(a) = -\beta S_{xx}(a).$$

Since on the v_5 -patch

$$K_v = (x')^3 S_{xx}, \quad x' = \frac{M}{c^2} \neq 0,$$

the hypothesis $K_v(a) \neq 0$ implies

$$S_{xx}(a) \neq 0.$$

Hence

$$U_{xx}(a) \neq 0.$$

By the analytic Morse lemma in one variable, after analytic source reparametrization the first coordinate is t^2 . $\square$

Proposition 12.341 (Off-septic-zero K_v -accumulation is already mixed low-order). *Work on the overlap patch of Proposition 12.334. Assume there exists a close exact mixed two-point bad branch accumulating at a point a such that*

$$K_v(a) = 0, \quad S(a) \neq 0.$$

Then

$$Y_{xx}(a) = 0.$$

Equivalently, on the v_5 -patch one has

$$J_\lambda(a) = 0, \quad J_{u_5}(a) = 0.$$

So any such accumulation already lies in the existing mixed low-order obstruction regime.

Proof. By Proposition 12.334, any close exact mixed two-point bad branch determines the tangent shadow

$$\mathbf{\Pi}(x) = [xS_x - S : YS_x - SY_x : S_x]$$

with projective immersion defect

$$\mathbf{\Pi}(x) \wedge \partial_x \mathbf{\Pi}(x) = S(x)(S_x Y_{xx} - S_{xx} Y_x, S_{xx}, -S_{xx}).$$

At a point a with $K_v(a) = 0$, one has $S_{xx}(a) = 0$ on the v_5 -patch, so this simplifies to

$$\mathbf{\Pi}(a) \wedge \partial_x \mathbf{\Pi}(a) = S(a) Y_{xx}(a) (S_x(a), 0, S(a) - x(a)S_x(a)).$$

If $S(a) \neq 0$ and $Y_{xx}(a) \neq 0$, then the vector on the right is nonzero, so $\mathbf{\Pi}$ is projectively immersive at a , contradicting accumulation of secant shadows there. Hence

$$Y_{xx}(a) = 0.$$

On the v_5 -patch,

$$J_\lambda = \frac{M^3}{c^3} S_{xx}, \quad J_{u_5} = \frac{M^3}{c^3} Y_{xx},$$

so $K_v(a) = 0$ implies $J_\lambda(a) = 0$, and the displayed vanishing of $Y_{xx}(a)$ gives $J_{u_5}(a) = 0$. $\square$

Corollary 12.342 (Simple septic-zero fixed-shear branches off $K_v = 0$ transfer to nonmixed four-point geometry). *Work on the overlap patch of Proposition 12.340. Assume there is a close exact fixed-shear mixed 2-point branch*

$$x_{1,n} \neq x_{2,n}, \quad x_{1,n}, x_{2,n} \rightarrow a, \quad \Phi(x_{1,n}) = \Phi(x_{2,n}),$$

and

$$K_v(a) \neq 0.$$

Then, after shrinking to a sufficiently small neighborhood of a , this lane produces a local interval-persistent nonmixed 4-point family for the lifted curve

$$Q(x) = (x, Y(x), S(x)).$$

Consequently it is absorbed by the existing nonmixed 4-point theory; in particular, it is not a surviving genuinely mixed 2-point derivative-geometric obstruction.

Proof. By Proposition 12.340,

$$S(a) = 0, \quad S_x(a) \neq 0,$$

and the first coordinate of the fixed-shear projected germ is in fold normal form. Thus, after analytic source reparametrization, there is a local parameter t with

$$\Phi(x(t)) = (t^2, \mathcal{V}(t))$$

and repeated nearby values force

$$\mathcal{V}(t) = \mathcal{V}(-t)$$

for all sufficiently small t . Equivalently, in the actual local septic coordinate $s := S$, there are even analytic germs E_x, E_Y such that

$$\frac{x(s) - \xi_* s}{1 - \eta_* s} = E_x(s), \quad \frac{Y(s) - \gamma_* s}{1 - \eta_* s} = E_Y(s), \quad E_x(-s) = E_x(s), \quad E_Y(-s) = E_Y(s).$$

Define

$$Q(s) := (x(s), Y(s), s).$$

If $\eta_* \neq 0$, set

$$P := \left(\frac{\xi_*}{\eta_*}, \frac{\gamma_*}{\eta_*}, \frac{1}{\eta_*} \right).$$

Then the parity formulas imply that $Q(s), Q(-s), P$ are collinear for all small s . If $\eta_* = 0$, then

$$Q(s) - Q(-s) = 2s(\xi_*, \gamma_*, 1),$$

so all symmetric secants are parallel to one fixed direction, which is the same projective statement with a point at infinity. Hence for every small $0 < s_1 < s_2$, the four points

$$Q(-s_2), Q(-s_1), Q(s_1), Q(s_2)$$

are coplanar.

Finally, by Proposition 12.340,

$$x_s(a) = \frac{1}{S_x(a)} \neq 0,$$

so x is monotone in s near a . Thus the four coplanar points above lie on one local x -arc in the nonmixed ordered pattern, yielding a local interval-persistent nonmixed 4-point family. The final claim follows from Proposition 12.267. $\square$

Remark 12.343 (Exact shadow-adapted freezing on the simple septic-zero slice). Work on a good overlap patch and suppose

$$S = \frac{\Delta_7}{c v_5}$$

is a local coordinate near a simple septic-zero point a , so

$$S(a) = 0, \quad S_x(a) \neq 0.$$

Write $s := S$ and let

$$C(s) := (x(s), Y(s)), \quad x = \frac{v_5}{c}, \quad Y = \frac{u_5}{c}.$$

For any distinct nearby secant pair $s_1 \neq s_2$, define

$$\xi(s_1, s_2) := \frac{x(s_1) - x(s_2)}{s_1 - s_2}, \quad \gamma(s_1, s_2) := \frac{Y(s_1) - Y(s_2)}{s_1 - s_2}.$$

Then one has the exact identities

$$x(s_1) - \xi(s_1, s_2) s_1 = x(s_2) - \xi(s_1, s_2) s_2,$$

$$Y(s_1) - \gamma(s_1, s_2) s_1 = Y(s_2) - \gamma(s_1, s_2) s_2.$$

Equivalently, on the simple septic-zero slice every exact secant-shadow pair already carries an exact fixed-shear normalization with

$$\eta = 0, \quad \xi = \xi(s_1, s_2), \quad \gamma = \gamma(s_1, s_2).$$

Thus the apparently projective, non-affinely-frozen septic-zero lane is not a separate carrier mechanism on the simple $S = 0$ slice: in the shadow-adapted chart it freezes exactly and therefore falls under the same exact fixed-shear analysis used in Corollary 12.342.

Remark 12.344 (Status of the mixed λ -shear route). Proposition 12.334 shows that the downstream secant-shadow reduction is already exact and fully projective: a close exact mixed two-point bad branch determines the projective secant shadow

$$\Sigma(h_1, h_2) = [x_1 S_2 - x_2 S_1 : Y_1 S_2 - Y_2 S_1 : S_2 - S_1], \quad S = \frac{\Delta_7}{c v_5},$$

independently of any affine freezing of the branch-generated λ -shears. On the simple septic-zero slice $S = 0$, $S_x \neq 0$, there is moreover a canonical shadow-adapted exact freezing with $\eta = 0$; see Remark 12.343.

The overlap geometry is now sharply split into three pieces.

First, Proposition 12.341 shows that if a close mixed branch accumulates at a point with

$$K_v = 0, \quad S \neq 0,$$

then

$$J_\lambda = J_{u_5} = 0,$$

so this part of the $K_v = 0$ locus is already absorbed by the existing mixed low-order obstruction package.

Second, Corollary 12.342 shows that in the conditional exact fixed-shear overlap lane, the slice

$$\{S = 0\} \cap \{K_v \neq 0\}$$

is not a surviving mixed 2-point carrier lane: it transfers into the established nonmixed 4-point geometry.

Thus the only genuinely residual exact fixed-shear mixed corner is

$$\{S = 0\} \cap \{K_v = 0\}.$$

On that corner, Proposition 12.354 shows that every finite parity-normalized transport jet descends analytically to the quotient variable $u = t^2$. In the quartic–sextic case, Corollary 12.355 shows that every finite quotient-visible transport state is locally an analytic function of Q alone, and Corollary 12.356 rules out any third local reset beyond the involution symmetry. So the residual branch is no longer merely quotient-diagonal / involutive; its finite transport state has local rank 1.

If the actual corrected cubic/quintic defects are state-local in this descended transport state, then Proposition 12.357 already excludes the corner. Under the stronger natural merger law, Corollary 12.358 shows that the interaction remainder dies identically along the involutive branch. Thus any still-live mixed two-point obstruction on

$$\{S = 0\} \cap \{K_v = 0\}$$

must come from genuinely relational or provenance-sensitive two-atom data beyond the descended transport state.

Remark 12.345 (Conditional parity reduction on the exact fixed-shear $K_v = 0$ slice). In the additional conditional exact fixed-shear overlap lane, the residual $K_v = 0$ slice should be read as quotient-diagonal / involutive rather than as a new mixed local branch-classification problem. Indeed, after centering and analytic source reparametrization, the projected germ has the form

$$\Phi(t) = (t^m, G(t)),$$

and exact repeated values force

$$m \text{ even}, \quad G(-t) = G(t).$$

So, after quotienting by analytic functions of the first coordinate, the genuinely mixed residue is an even/even quotient germ

$$(t^m, t^n + \cdots), \quad m, n \text{ even}, \quad n > m.$$

In particular, among the current explicit low-order mixed models

$$(m, n) = (3, 4), (3, 6), (4, 6),$$

only the quartic–sextic case

$$(4, 6)$$

can survive on the exact fixed-shear overlap $K_v = 0$ slice. The subsequent quotient-transport reformulation will show that this is the first rung of an even-support residual ladder

$$(m, n) = (r + 2, r + 2e + 4), \quad r \in 2\mathbf{Z}_{\geq 0}, \quad e = \text{ord}_0 \hat{\mathcal{D}}.$$

Lemma 12.346 (Local exact repeated-value locus on the involutive fixed-shear slice). *Assume the setup of Remark 12.345, so that in a centered analytic source coordinate*

$$\Phi(t) = (t^m, G(t)), \quad m \text{ even}, \quad G(-t) = G(t).$$

After shrinking to a sufficiently small neighborhood of $t = 0$, one has

$$\Phi(t_1) = \Phi(t_2), \quad t_1 \neq t_2 \iff t_2 = -t_1.$$

Equivalently, the local exact repeated-value locus is exactly the involution graph

$$\{(t, -t) : 0 < |t| < \varepsilon\}.$$

In particular, on the exact fixed-shear $K_v = 0$ slice there is no local hidden partner choice beyond the involution $t \leftrightarrow -t$.

Proof. If $\Phi(t_1) = \Phi(t_2)$, then comparing first coordinates gives

$$t_1^m = t_2^m.$$

Since m is even and the source coordinate is real, this implies

$$|t_1| = |t_2|.$$

If $t_1 \neq t_2$, then necessarily $t_2 = -t_1$. Conversely, because m is even and G is even,

$$\Phi(-t) = \Phi(t)$$

for all sufficiently small t . This proves the claim. □

Proposition 12.347 (Quotient transport scalar and residual ladder on the exact fixed-shear $K_v = 0$ slice). *Assume the additional conditional exact fixed-shear overlap lane from Remark 12.345. Let*

$$r := \text{ord}_0 J_\lambda = \text{ord}_0 K_v, \quad q := \text{ord}_0 J_{u_5},$$

and write in the centered involutive source coordinate t

$$J_\lambda(t) = t^r A(t), \quad J_{u_5}(t) = t^r B(t), \quad A(0) \neq 0.$$

Set

$$Q(t) := \frac{B(t)}{A(t)} = \frac{J_{u_5}(t)}{J_\lambda(t)}.$$

Then the following hold.

(1) *Q is even:*

$$Q(-t) = Q(t).$$

Hence there exists a unique analytic germ $\widehat{Q}$ in

$$u := t^2$$

such that

$$Q(t) = \widehat{Q}(t^2).$$

(2) *If*

$$\widehat{\mathcal{D}}(u) := \widehat{Q}'(u),$$

then

$$\frac{\mathcal{W}_{\text{src}}(t)}{J_\lambda(t)^2} = Q'(t) = 2t \widehat{\mathcal{D}}(t^2),$$

and therefore

$$\mathcal{W}_{\text{src}}(t) = 2t^{2r+1} A(t)^2 \widehat{\mathcal{D}}(t^2).$$

(3) *If*

$$e := \text{ord}_0 \widehat{\mathcal{D}} \in \mathbf{Z}_{\geq 0},$$

then

$$q - r = 2e + 2, \quad m = r + 2, \quad n = r + 2e + 4,$$

and

$$\text{ord}_0 \mathcal{W}_{\text{src}} = 2r + 1 + 2e.$$

(4) *In particular,*

$$r \text{ is even, } \quad q - r \in 2\mathbf{Z}_{\geq 2}, \quad \text{ord}_0 \mathcal{W}_{\text{src}} \text{ is odd.}$$

So the exact fixed-shear residual lane lies on an even-support ladder over $K_v = 0$, while the associated source-level Wronskian vanishes to odd order.

(5) *The currently surviving quartic–sextic residual model is exactly the first rung of this ladder:*

$$(m, n) = (4, 6) \iff (r, e) = (2, 0).$$

Proof. By Remark 12.345, the projected germ has the form

$$\Phi(t) = (t^m, G(t)), \quad m \text{ even, } \quad G(-t) = G(t).$$

Thus the quotient-diagonal mixed residue is even/even, so the regularized ratio $Q = B/A$ is even, giving the factorization

$$Q(t) = \widehat{Q}(t^2).$$

Differentiating yields

$$Q'(t) = 2t \widehat{Q}'(t^2) = 2t \widehat{\mathcal{D}}(t^2).$$

On the other hand, Remark 12.333 gives

$$Q'(t) = \frac{\mathcal{W}_{\text{src}}(t)}{J_\lambda(t)^2}.$$

Since

$$J_\lambda(t)^2 = t^{2r} A(t)^2,$$

this gives

$$\mathcal{W}_{\text{src}}(t) = 2t^{2r+1}A(t)^2\widehat{\mathcal{D}}(t^2).$$

If $e = \text{ord}_0 \widehat{\mathcal{D}}$, then

$$\widehat{Q}(u) - \widehat{Q}(0) \sim u^{e+1},$$

so

$$Q(t) - Q(0) \sim t^{2e+2}.$$

Hence

$$q - r = \text{ord}_0(Q(t) - Q(0)) = 2e + 2.$$

Now Remark 12.332 gives

$$m = r + 2, \quad n = q + 2,$$

so

$$n = r + 2e + 4.$$

Also

$$\text{ord}_0 \mathcal{W}_{\text{src}} = 2r + 1 + 2e.$$

Since m is even, $r = m - 2$ is even. The final statements follow immediately. $\square$

Proposition 12.348 (Sign-run compression and the three-run criterion on a quotient-visible exact fixed-shear branch). *Assume the quotient-visible exact fixed-shear reduction, and let*

$$[a, b] \subset \mathbf{R}_u$$

be a connected compact quotient branch on which the descended transport potential

$$\widehat{Q} : [a, b] \rightarrow \mathbf{R}$$

is analytic and nonconstant, with

$$\widehat{\mathcal{D}} = \widehat{Q}'.$$

Let

$$a = s_0 < s_1 < \cdots < s_\ell = b$$

be the sign-change partition of $\widehat{\mathcal{D}}$: on each run

$$R_j := (s_j, s_{j+1}), \quad j = 0, \dots, \ell - 1,$$

the derivative $\widehat{\mathcal{D}}$ has fixed nonzero sign, and the sign alternates from run to run. Define

$$d_j := \widehat{Q}(s_j), \quad \Omega_j := (\min\{d_j, d_{j+1}\}, \max\{d_j, d_{j+1}\}),$$

and the run amplitudes

$$B_j := |d_{j+1} - d_j| = \int_{R_j} |\widehat{\mathcal{D}}(u)| du.$$

Then the following hold.

(1) *For every regular value*

$$\gamma \notin \widehat{Q}(\{\widehat{\mathcal{D}} = 0\}),$$

one has the exact multiplicity formula

$$\#\widehat{Q}^{-1}(\gamma) = \#\{j : \gamma \in \Omega_j\}.$$

(2) *For every*

$$0 \leq j \leq \ell - 3,$$

one has the exact distance-two overlap criterion

$$\Omega_j \cap \Omega_{j+2} \neq \emptyset \iff B_j + B_{j+2} > B_{j+1}.$$

Equivalently,

$$\Omega_j \cap \Omega_{j+2} = \emptyset \iff B_{j+1} \geq B_j + B_{j+2}.$$

(3) *Two consecutive distance-two separations are impossible. More precisely, there is no j such that both*

$$\Omega_j \cap \Omega_{j+2} = \emptyset \quad \text{and} \quad \Omega_{j+1} \cap \Omega_{j+3} = \emptyset.$$

In particular, if the branch has no nonadjacent overlap of regular values, then

$$\ell \leq 3.$$

Proof. On each run R_j , the function $\widehat{Q}$ is strictly monotone, so for a regular value γ it has exactly one preimage on R_j if and only if γ lies in the open image interval Ω_j , and otherwise no preimage there. Summing over runs proves (1).

For (2), fix j and, after replacing $\widehat{Q}$ by $-\widehat{Q}$ if needed, assume that the signs on R_j, R_{j+1}, R_{j+2} are

$$+, -, +.$$

Then

$$d_j < d_{j+1} > d_{j+2} < d_{j+3},$$

with

$$B_j = d_{j+1} - d_j, \quad B_{j+1} = d_{j+1} - d_{j+2}, \quad B_{j+2} = d_{j+3} - d_{j+2}.$$

In this configuration,

$$\Omega_j = (d_j, d_{j+1}), \quad \Omega_{j+2} = (d_{j+2}, d_{j+3}),$$

and since $d_{j+2} < d_{j+1}$ automatically, these open intervals overlap if and only if

$$d_j < d_{j+3}.$$

But

$$d_{j+3} - d_j = (d_{j+1} - d_j) - (d_{j+1} - d_{j+2}) + (d_{j+3} - d_{j+2}) = B_j - B_{j+1} + B_{j+2},$$

so

$$d_j < d_{j+3} \iff B_j + B_{j+2} > B_{j+1}.$$

This proves (2).

For (3), if both consecutive separations held, then by (2) one would have

$$B_{j+1} \geq B_j + B_{j+2}, \quad B_{j+2} \geq B_{j+1} + B_{j+3},$$

which is impossible because all $B_k > 0$. Hence among any two consecutive distance-two comparisons, at least one overlap is forced. If $\ell \geq 4$, such a consecutive comparison exists, so no-nonadjacent-overlap forces $\ell \leq 3$. $\square$

Corollary 12.349 (Three-run classification on a quotient-visible exact fixed-shear branch). *In the setting of Proposition 12.348, assume that*

$$\ell = 3.$$

Then the branch has no nonadjacent overlap of regular values if and only if

$$B_1 \geq B_0 + B_2.$$

If

$$B_1 > B_0 + B_2,$$

the middle run separates the two outer windows by a genuine gap. If

$$B_1 = B_0 + B_2,$$

the outer windows touch only at a critical value, so there is still no regular value with nonadjacent re-entry.

Proof. When $\ell = 3$, the only nonadjacent comparison is (Ω_0, Ω_2) , so the claim is exactly Proposition 12.348(2). $\square$

Remark 12.350 (Touching is sterile in the three-run quotient model). In the equality case

$$B_1 = B_0 + B_2,$$

Corollary 12.349 says that the outer windows meet only at a boundary critical value. Thus the same-value relation between the two outer runs is zero-dimensional downstairs in the quotient variable $u = t^2$: there is no open value interval carrying a same-value branch between the outer runs.

Accordingly, the equality case should be viewed as critical-value touching, not as a genuine open transport overlap. The genuinely live open-overlap three-run regime is

$$B_0 + B_2 > B_1,$$

while

$$B_1 > B_0 + B_2$$

is the separation regime.

Remark 12.351 (Separation is not a transport lane in the three-run quotient model). In the strict separation regime

$$B_1 > B_0 + B_2,$$

the outer windows are disjoint, so there is no outer-overlap interval

$$\Omega_0 \cap \Omega_2$$

and hence no inverse-branch-gap transport problem between the two outer runs. Accordingly, the inverse-branch-gap / Δ / Ξ setup belongs only to the genuine outer-overlap regime

$$B_0 + B_2 > B_1,$$

while the strict separation regime is already a no-reentry regime downstairs in the quotient variable.

Corollary 12.352 (Quartic–sextic quotient injectivity and local no-reentry). *Assume the quartic–sextic residual case*

$$(m, n) = (4, 6).$$

Then

$$Q(t) = Q(0) + \kappa_{2,2}t^2 + O(t^4), \quad \widehat{Q}(u) = Q(0) + \kappa_{2,2}u + O(u^2),$$

so

$$\widehat{\mathcal{D}}(0) = \kappa_{2,2} \neq 0.$$

Equivalently,

$$\kappa_{2,2} = \frac{1}{2} \left(\frac{\mathcal{W}_{\text{src}}}{J_\lambda^2} \right)' (0) = \frac{\mathcal{W}_{\text{src}}^{(5)}(0)}{60 (J_\lambda''(0))^2}.$$

Hence $\widehat{Q}$ is a local analytic diffeomorphism in the quotient variable $u = t^2$. In particular, after quotienting the involution $t \leftrightarrow -t$, there is no local exact fixed-shear re-entry: every nearby fixed-shear level is attained by at most one quotient point.

Proof. In the quartic–sextic case, Proposition 12.347 gives

$$(r, e) = (2, 0).$$

So $\widehat{\mathcal{D}}(0) \neq 0$, which is exactly the displayed coefficient $\kappa_{2,2}$. The explicit formulas follow from Remark 12.332 and the identity

$$Q'(t) = \frac{\mathcal{W}_{\text{src}}}{J_\lambda^2}.$$

Since

$$\widehat{Q}'(0) = \widehat{\mathcal{D}}(0) \neq 0,$$

the analytic inverse function theorem implies that $\widehat{Q}$ is a local diffeomorphism in $u = t^2$. $\square$

Remark 12.353 (The quartic–sextic rung is inert for the lobe-amplitude obstruction). On the first residual rung

$$(m, n) = (4, 6),$$

Proposition 12.347 gives

$$(r, e) = (2, 0),$$

and Corollary 12.352 gives

$$\widehat{\mathcal{D}}(0) = \kappa_{2,2} \neq 0.$$

So the quartic–sextic basepoint is not a sign-change point of $\widehat{\mathcal{D}}$, contributes no compressed node on the quotient branch, and does not participate in any three-run run-amplitude obstruction. The first possible quotient-visible bad core therefore begins only at a higher-rung critical point of $\widehat{\mathcal{D}}$.

Proposition 12.354 (Finite transport-jet quotient descent on the exact fixed-shear corner). *Assume the additional conditional exact fixed-shear overlap lane from Remark 12.345, and work along a local involutive branch*

$$(h_1, h_2) = (h(t), h(-t)).$$

For each $k \geq 0$, define the parity-normalized transport jet

$$\Theta_k(t) := \begin{cases} Q^{(k)}(t), & k \text{ even}, \\ t^{-1}Q^{(k)}(t), & k \text{ odd}, \end{cases}$$

where

$$Q(t) = \frac{J_{u_5}(t)}{J_\lambda(t)} = \widehat{Q}(t^2).$$

Then every Θ_k is even analytic. Equivalently, for each $k \geq 0$ there exists a unique analytic quotient germ $\widetilde{\Theta}_k(u)$ such that

$$\Theta_k(t) = \widetilde{\Theta}_k(t^2).$$

In particular, the canonical descended scalar

$$X(t) := \frac{Q'(t)}{t}$$

is even analytic and satisfies

$$X(t) = 2\widehat{Q}'(t^2) = 2\widehat{\mathcal{D}}(t^2).$$

Proof. Since $Q(t) = \widehat{Q}(t^2)$ is even, $Q^{(k)}$ has the usual parity: $Q^{(k)}$ is even for k even and odd for k odd. Every odd analytic germ is divisible by t , so Θ_k is analytic for odd k . In both cases Θ_k is even, hence factors uniquely through $u = t^2$. The final identity is immediate from

$$Q'(t) = 2t\widehat{Q}'(t^2) = 2t\widehat{\mathcal{D}}(t^2).$$

□

Corollary 12.355 (Rank-one quotient-visible transport state on the quartic–sextic rung). *Assume the quartic–sextic residual case*

$$(m, n) = (4, 6).$$

For every $N \geq 0$, set

$$\mathcal{T}_N(t) := (\Theta_0(t), \dots, \Theta_N(t)).$$

Then there exists an analytic map

$$\mathbf{F}_N$$

defined near $Q(0)$ such that

$$\mathcal{T}_N(t) = \mathbf{F}_N(Q(t)).$$

Equivalently, every finite quotient-visible transport state is locally an analytic function of the single scalar Q alone.

Proof. By Corollary 12.352, $\widehat{Q}$ is a local analytic coordinate in $u = t^2$. Since each $\Theta_k(t) = \widetilde{\Theta}_k(t^2)$, each descended jet $\widetilde{\Theta}_k(u)$ can be re-expressed analytically as a function of $\widehat{Q}(u)$, hence of $Q(t)$. Stacking these functions gives $\mathbf{F}_N$. $\square$

Corollary 12.356 (No finite-order hidden reset on the quartic–sextic residual corner). *Assume the quartic–sextic residual case. Let*

$$\mathcal{T}_N(t) := (\Theta_0(t), \dots, \Theta_N(t))$$

be any finite quotient-visible transport state. Then

$$\mathcal{T}_N(t_1) = \mathcal{T}_N(t_2) \implies t_2 = \pm t_1.$$

In particular, no finite quotient-visible transport state produces a third local reset beyond the involution symmetry $t \leftrightarrow -t$.

Proof. By Corollary 12.355, equality of $\mathcal{T}_N$ implies equality of Q . But

$$Q(t) = \widehat{Q}(t^2),$$

and Corollary 12.352 shows that $\widehat{Q}$ is locally injective in $u = t^2$. Hence $t_1^2 = t_2^2$, so $t_2 = \pm t_1$. $\square$

Proposition 12.357 (Conditional state-local defect closure on the residual exact fixed-shear corner). *Assume the setup of Proposition 12.370. Suppose there exists $N \geq 0$ and analytic functionals*

$$\mathfrak{e}_3, \mathfrak{e}_5$$

defined near $\mathcal{T}_N(0)$ such that along the involutive branch

$$E_{12}^{(3)}(t) = \mathfrak{e}_3(\mathcal{T}_N(t)), \quad E_{12}^{(5)}(t) = \mathfrak{e}_5(\mathcal{T}_N(t)),$$

and

$$\mathfrak{e}_3(\mathcal{T}_N(0)) = \mathfrak{e}_5(\mathcal{T}_N(0)) = 0.$$

Then

$$E_{12}^{(3)}(t) = O(t^2), \quad E_{12}^{(5)}(t) = O(t^2),$$

hence

$$E_{12}^{(3)} = O((h_1 - h_2)^2), \quad E_{12}^{(5)} = O((h_1 - h_2)^2),$$

and no such nontrivial exact residual fixed-shear involutive bad branch can persist on a compact good patch

$$J \in \{c \neq 0, M \neq 0\}.$$

Proof. By Proposition 12.354, each Θ_k is even analytic, so

$$\mathcal{T}_N(t) = \mathcal{T}_N(0) + O(t^2).$$

Analyticity of $\mathfrak{e}_3, \mathfrak{e}_5$ and vanishing at $\mathcal{T}_N(0)$ therefore give

$$E_{12}^{(3)}(t) = O(t^2), \quad E_{12}^{(5)}(t) = O(t^2).$$

Since

$$h_1 - h_2 = h(t) - h(-t) = 2h'(0)t + O(t^3),$$

this becomes

$$E_{12}^{(3)} = O((h_1 - h_2)^2), \quad E_{12}^{(5)} = O((h_1 - h_2)^2).$$

Now apply Corollary 12.122. $\square$

Corollary 12.358 (Stronger conditional package collapse on the involutive branch). *Assume the setup of Proposition 12.370. Suppose that for some $N \geq 0$ the corrected two-pair package along the involutive branch is state-local in the sense that*

$$\mathcal{P}_2(t) = \mathfrak{P}_2(a_1(t), a_2(t); \mathcal{T}_N(t), \mathcal{T}_N(t)),$$

and the corresponding one-pair package is

$$F_{h_1(t)} = F_{h_2(t)} = \mathfrak{F}(\mathcal{T}_N(t)).$$

Assume moreover the natural merger law

$$\mathfrak{P}_2(a_1, a_2; Y, Y) = (a_1 + a_2)\mathfrak{F}(Y).$$

Then the interaction remainder

$$\mathcal{I}_2(t) := \mathcal{P}_2(t) - a_1(t)F_{h_1(t)} - a_2(t)F_{h_2(t)}$$

vanishes identically along the involutive branch:

$$\mathcal{I}_2(t) \equiv 0.$$

Proof. Substitute the displayed state-local forms into the definition of $\mathcal{I}_2(t)$, then apply the merger law with $Y = \mathcal{T}_N(t)$. $\square$

Remark 12.359 (Any remaining residual obstruction is genuinely relational). The preceding results show that on the quartic–sextic residual corner the finite quotient-visible transport state has local rank 1 and no finite-order hidden reset. Therefore, if the exact fixed-shear residual corner is still genuinely live, the obstruction cannot come from:

- a second local transport variable on the quartic–sextic rung;
- a third local reset inside any finite quotient-visible transport state;
- the explicit quotient-visible three-run run-amplitude obstruction on the quartic–sextic rung, since Corollary 12.352 gives $\widehat{\mathcal{D}}(0) \neq 0$ there;
- a two-pair model built only from the descended one-atom transport state, assuming the natural merger law.

So any still-live obstruction on

$$\{S = 0\} \cap \{K_v = 0\}$$

must come from the first genuinely relational or provenance-sensitive two-atom input that survives after quotient-visible transport descent, or else from non-finite-order dependence.

Remark 12.360 (Quotient-cusp interpretation and the no-reset issue). The involutive quotient picture on the residual exact fixed-shear $K_v = 0$ slice is best read as a one-branched quotient singularity rather than as a new branching mixed carrier.

Indeed, in the notation of Remark 12.345, one has

$$\Phi(t) = (t^m, G(t)), \quad m \text{ even}, \quad G(-t) = G(t),$$

so after passing to the quotient parameter

$$u = t^2$$

the projected germ descends to a single analytic quotient branch

$$\bar{\Phi}(u) = (u^{m/2}, \bar{G}(u)).$$

In particular, in the quartic–sextic case

$$(m, n) = (4, 6),$$

the quotient branch is cusp-like:

$$\bar{\Phi}(u) \sim (u^2, u^3 + \dots).$$

Thus any local quantity that genuinely factors through the quotient-diagonal projected germ — equivalently through the quotient-visible fixed-shear transport data — is single-valued in u and has trivial local monodromy at the involutive point. In this sense, the residual exact fixed-shear $K_v = 0$ locus is best viewed as a quotient-cusp splice rather than as a new local source of cycle rank.

At the same time, this does *not* yet by itself prove global no-reentry on the $K_v = 0$ lane: one still has to show that the full transported overlap / critical-value state relevant to the later finite-core problem is indeed quotient-visible. So the live mixed residual question is now a *no-hidden-reset* question at $K_v = 0$, not another local branch classification problem.

Remark 12.361 (What the quotient-transport package does and does not prove). The quotient-transport package now does more than give a single pointwise scalar compression of the residual exact fixed-shear $K_v = 0$ slice.

First, Proposition 12.354 shows that every finite parity-normalized transport jet descends analytically to the quotient variable

$$u = t^2.$$

Second, on the quartic–sextic rung, Corollary 12.355 shows that every finite quotient-visible transport state is locally an analytic function of the single scalar

$$Q(t) = \widehat{Q}(t^2)$$

alone, and Corollary 12.356 rules out any third local reset inside that descended state beyond the involution symmetry $t \leftrightarrow -t$.

What is *not* proved here is that the actual corrected two-pair defect package is already state-local in this descended transport state. If that state-locality held, then Proposition 12.357 would already exclude the residual branch, and under the stronger natural merger law Corollary 12.358 would make the interaction remainder vanish identically.

Accordingly, the present residual mixed burden should now be read as follows: the exact fixed-shear corner is no longer blocked by a missing local branch classification or by hidden finite-order transport dynamics. Any remaining obstruction must come from genuinely relational or provenance-sensitive two-atom structure, or from non-finite-order dependence, not from the descended transport state alone.

Remark 12.362 (No finite local rescue on the exact fixed-shear corner). The current quotient-transport package leaves very little residual finite local freedom on the exact fixed-shear corner.

First, every finite parity-normalized transport jet descends to the quotient variable $u = t^2$. Second, on the quartic–sextic rung the descended transport state has local rank one and no third local reset beyond the involution $t \leftrightarrow -t$. Third, the explicit three-run run-amplitude obstruction is already inert there, because on that rung

$$\widehat{D}(0) = \widehat{Q}'(0) \neq 0.$$

So if the residual exact fixed-shear corner still survives, it cannot be explained by a second local transport variable, by a hidden finite-order reset, or by the explicit lobe-amplitude obstruction on the quartic–sextic rung. Any remaining obstruction must therefore be genuinely nonlocal on the transport side, or package/provenance-sensitive on the two-atom side.

Proposition 12.363 (Local scalar repeated-value classification). *Let f be a real-analytic function on an open neighborhood of $a \in \mathbb{R}$ with $f'(a) = 0$, and let $m \geq 2$ be the order of vanishing of $f - f(a)$ at a , so*

$$f(u) - f(a) = c_m(u - a)^m + O((u - a)^{m+1}), \quad c_m \neq 0.$$

Then there exist a sign $\sigma \in \{+1, -1\}$ and a local real-analytic chart x near a with $x(a) = 0$ and $x'(a) \neq 0$ such that

$$f(u) = f(a) + \sigma x(u)^m.$$

The local repeated-value locus

$$\{(u_1, u_2) : u_1 \neq u_2, f(u_1) = f(u_2)\}$$

near (a, a) is:

- (Even m .) *If m is even, it is the involution graph*

$$u_2 = \iota(u_1) := x^{-1}(-x(u_1)),$$

together with $u_1 = u_2$. The involution ι fixes a and is analytic with $\iota'(a) = -1$; the repeated-value branch is a single local fold half-arc attached to a .

- (Odd m .) *If m is odd, then f is a local analytic homeomorphism near a and the nondiagonal repeated-value locus is empty near (a, a) ; the point a is sterile.*

Proof. By the Malgrange/Morse-type preparation for analytic functions of one variable with an isolated critical point, one may write $f(u) - f(a) = \sigma h(u)^m$ for a local analytic chart h with $h(a) = 0$ and $h'(a) \neq 0$, where $\sigma = \text{sgn}(c_m)$ when m is even and $\sigma = +1$ absorbing the sign into h when m is odd. Setting $x := h$ gives the stated normal form.

For even m , $f(u_1) = f(u_2)$ near (a, a) is equivalent to $x(u_1)^m = x(u_2)^m$, i.e. $x(u_2) = \pm x(u_1)$; the plus sign is the diagonal, the minus sign is $u_2 = x^{-1}(-x(u_1)) = \iota(u_1)$. That ι is involutive with $\iota(a) = a$ and $\iota'(a) = -1$ is immediate from $x'(a) \neq 0$ and $x(a) = 0$.

For odd m , $x(u_1)^m = x(u_2)^m$ forces $x(u_1) = x(u_2)$, hence $u_1 = u_2$ near a since x is a local homeomorphism. $\square$

Corollary 12.364 (Local scalar-bud decomposition on the close exact fixed-shear lane). *Apply Proposition 12.363 to $f = \widehat{Q}$ on the nonconstant quotient branch in the exact fixed-shear corner $\{S = 0\} \cap \{K_v = 0\}$. Let a be a zero of $\widehat{D} = \widehat{Q}'$ of order $m - 1 \geq 1$. Then:*

- *If m is even (i.e. a is an odd-order zero of $\widehat{D}$), the local close exact repeated-value carrier at a is exactly one fold half-arc, namely the involution graph $\{u_2 = \iota(u_1)\}$ of Proposition 12.363.*
- *If m is odd (i.e. a is an even-order zero of $\widehat{D}$), the local close exact repeated-value carrier at a is empty; the basepoint is sterile.*

In particular, on a compact nonconstant quotient branch in the exact fixed-shear corner, the close exact repeated-value carrier is a finite disjoint union of local fold half-arcs, one per odd zero of $\widehat{D}$, with even zeros sterile. No connected nonlocal close carrier remains on that branch.

Proof. By Lemma 12.346, away from zeros of $\widehat{Q}'$ the close exact repeated-value locus is empty, so any close carrier accumulates only at zeros of $\widehat{D}$. At each such zero a the local structure is dictated by Proposition 12.363 applied to $\widehat{Q}$, with $m = \text{ord}_a(\widehat{Q} - \widehat{Q}(a)) = \text{ord}_a \widehat{D} + 1$; the two cases transcribe to odd vs. even order of $\widehat{D}$. Compactness then bounds the number of such basepoints, and the carrier is their finite disjoint union of local fold half-arcs. $\square$

Remark 12.365 (Constant-branch scalarization and the scalar ceiling). There is also a constant-quotient branch. If

$$\widehat{D} \equiv 0,$$

then the quotient identities give

$$Y_{xx} = \gamma_0 S_{xx}, \quad Y = \gamma_0 S + \beta_0 x + \alpha_0.$$

After affine gauge one may reduce to $Y \equiv 0$, and the exact fixed-shear overlap equations then force $\gamma_* = 0$. So the constant branch reduces to the repeated-value problem for the single planar scalar

$$V(t) := U_{\xi_*, \eta_*}(x(t), S(t)), \quad U_{\xi, \eta}(x, S) := \frac{x - \xi S}{1 - \eta S},$$

to which Proposition 12.363 applies pointwise in t . At basepoints with $V'(t_0) \neq 0$, V is a local homeomorphism and close exact repeated-value pairs are excluded; Corollary 12.352 rules out the basepoint case $\widehat{D}(0) \neq 0$, and at basepoints where V has a critical point the local carrier is again a single fold half-arc. Thus the constant exact fixed-shear branch also contributes only local fold half-arcs.

In this form the close exact fixed-shear lane contributes no connected nonlocal close carrier, only local scalar fold half-arcs. Corollary 12.364 is the theorem-level scalar ceiling on that lane: the remaining intrinsic obstruction on the close exact fixed-shear side is now equality of consecutive lobe amplitudes across these local fold half-arcs, not a hidden nonlocal carrier.

Remark 12.366 (Connected-carrier sterility versus scalar multiplicity on the close exact lane). Corollary 12.364 and Remark 12.365 do more than say that close exact repeated values are locally scalar. They also identify the global graph type of the close carrier: after shrinking the close threshold, each connected component is a single local fold bud attached at one odd scalar seed. In particular, each component is one-sided and pendant, never a cycle edge, and there is no connected nonlocal close carrier on the exact fixed-shear lane.

What this closes is the connected-carrier question, not abstract scalar multiplicity. Pure scalar analyticity can still realize multiplicity-4 repeated-value levels, equivalently equal consecutive lobe amplitudes, so no truthful purely scalar theorem can bound the close multiplicity by 2 in full generality. The point is instead that such multiplicity does not organize into a connected global close carrier on the exact fixed-shear lane.

Remark 12.367 (Work in progress: inverse-branch-gap endgame in the overlap regime on the strict three-run lane). Stay in the setting of Proposition 12.348 and Corollary 12.349, and assume the three-run overlap regime

$$B_0 + B_2 > B_1.$$

Then the outer windows overlap on an open value interval

$$\mathcal{I} := \Omega_0 \cap \Omega_2,$$

so there are inverse branches

$$\phi_0 : \mathcal{I} \rightarrow R_0, \quad \phi_2 : \mathcal{I} \rightarrow R_2$$

of $\widehat{Q}$. Define the inverse-branch gap

$$\Delta(q) := \phi_2(q) - \phi_0(q),$$

and the transport-curvature scalar

$$\Xi(u) := \frac{\widehat{Q}''(u)}{(\widehat{Q}'(u))^3}, \quad \Upsilon(q) := \Xi(\phi_0(q)) - \Xi(\phi_2(q)).$$

Then

$$\Delta'(q) = \frac{1}{\widehat{Q}'(\phi_2(q))} - \frac{1}{\widehat{Q}'(\phi_0(q))}, \quad \Delta''(q) = \Upsilon(q).$$

So the strict three-run overlap problem may be studied through the one-variable gap function Δ . Any interior critical point $q_* \in \mathcal{I}$ yields a same-value / same-slope coincidence: with

$$u_0 := \phi_0(q_*), \quad u_2 := \phi_2(q_*),$$

one has

$$\widehat{Q}(u_0) = \widehat{Q}(u_2), \quad \widehat{Q}'(u_0) = \widehat{Q}'(u_2).$$

Thus the overlap obstruction is already a concrete outer-pair witness, not merely an abstract transport mismatch.

It is natural to package the higher mismatch by

$$J_1(u) := \frac{1}{\widehat{Q}'(u)}, \quad J_{n+1}(u) := J_1(u) \partial_u J_n(u).$$

The sharp current expectation is that any genuine overlap witness should force a finite-order even jet mismatch: for some smallest $m \geq 1$,

$$J_2(u_0) = J_2(u_2), \dots, J_{2m-1}(u_0) = J_{2m-1}(u_2), \quad J_{2m}(u_0) \neq J_{2m}(u_2),$$

with sign corresponding to a local minimum of Δ .

There is also a sharper descent fork, now correctly understood as conditional on actual overlap. If Ξ descends to value space across the two outer runs, then $\Upsilon \equiv 0$, so Δ is affine. Any theorem excluding nonconstant affine same-value transport between the two outer runs would then rule out the overlap regime. A stronger rigid alternative, if one can justify the endpoint propagation, would be exact translated replication

$$\Omega_0 = \Omega_2, \quad B_0 = B_2, \quad \phi_2(q) - \phi_0(q) \equiv C > 0, \quad \widehat{Q}(u + C) = \widehat{Q}(u) \quad (u \in R_0).$$

So the genuine three-run *overlap* regime now has the following corrected endgame fork: either Ξ -descent fails on an actual outer-overlap interval, giving the first explicit nonlocal transport obstruction, or Ξ -descent holds and the problem rigidifies to affine, and potentially translated, outer transport. In particular, the inverse-branch-gap setup belongs only to the overlap regime

$$B_0 + B_2 > B_1,$$

whereas

$$B_1 > B_0 + B_2$$

is already the separation regime and carries no such outer-transport lane.

Equivalently, the quantitative overlap scalar is

$$B_0 + B_2 - B_1,$$

and one may write

$$\Delta'(q) = \Delta'(q_0) + \int_{q_0}^q \Upsilon(s) ds,$$

so a transport-mass bound too small to cancel the initial inverse-slope gap would exclude genuine overlap.

This is a sharpened endgame formulation, not yet a closure theorem.

Remark 12.368 (No-go: overlap geometry alone does not force a witness). The corrected strict-overlap transport lane should not be read as if open outer overlap by itself automatically forces an interior same-value / same-slope witness in every endpoint configuration. Outside the stricter endpoint-sign- forcing subcases, the endpoint behavior of

$$W(q) := \Delta'(q) = \frac{1}{\widehat{Q}'(\phi_2(q))} - \frac{1}{\widehat{Q}'(\phi_0(q))}$$

depends on which outer branch is singular or regular at each overlap endpoint. So overlap geometry alone does not determine witness existence in full generality.

Equivalently, outside the strict sign-forcing cases, one needs extra input: endpoint slope comparison, a branch-independent closure law, or genuinely source/package-level structure. Thus the corrected overlap lane has reached a real ceiling for pure overlap geometry.

Remark 12.369 (Conditional rigidity: H -descent forces affine replication). Still in the corrected overlap regime, let

$$H(u) := \frac{\widehat{Q}''(u)}{(\widehat{Q}'(u))^2}, \quad J_1(u) := \frac{1}{\widehat{Q}'(u)}.$$

If H descends to value space across the two outer inverse branches, in the sense that

$$H(\phi_0(q)) = H(\phi_2(q))$$

on the overlap interval, then the reciprocal-slope branches

$$y_i(q) := J_1(\phi_i(q))$$

satisfy the same linear first-order ODE

$$y' = -h(q)y.$$

Hence

$$\frac{y_2(q)}{y_0(q)} = \Lambda$$

is constant, so

$$J_1(\phi_2(q)) = \Lambda J_1(\phi_0(q)), \quad \phi_2'(q) = \Lambda \phi_0'(q),$$

and therefore

$$\phi_2(q) = \Lambda \phi_0(q) + B.$$

So H -descent does not merely give a soft transport coincidence: it forces the outer inverse branches into affine replication. In particular, the translated / affine-copy exceptional branch is a genuinely rigid symmetry branch, not a soft hidden-state loophole.

Proposition 12.370 (Residual involutive defect criterion on the exact fixed-shear corner). *Assume the additional conditional exact fixed-shear overlap lane near a residual accumulation point*

$$a \in \{S = 0\} \cap \{K_v = 0\}.$$

Let a local exact repeated-value branch be parametrized by

$$(h_1, h_2) = (h(t), h(-t)), \quad h(0) = a, \quad h'(0) \neq 0,$$

with h analytic. Let

$$E_{12}^{(3)}(t), \quad E_{12}^{(5)}(t)$$

denote the cubic and quintic normalized interaction defects along this branch.

Assume:

$$E_{12}^{(3)}(t) = E_{12}^{(3)}(-t), \quad E_{12}^{(5)}(t) = E_{12}^{(5)}(-t),$$

and

$$E_{12}^{(3)}(0) = 0, \quad E_{12}^{(5)}(0) = 0.$$

Then

$$E_{12}^{(3)}(t) = O(t^2), \quad E_{12}^{(5)}(t) = O(t^2).$$

Since

$$h_1 - h_2 = h(t) - h(-t) = 2h'(0)t + O(t^3),$$

it follows that

$$E_{12}^{(3)} = O((h_1 - h_2)^2), \quad E_{12}^{(5)} = O((h_1 - h_2)^2).$$

Consequently no such nontrivial exact residual fixed-shear involutive bad branch can persist on a compact good patch

$$J \in \{c \neq 0, M \neq 0\}.$$

Proof. By evenness and vanishing at $t = 0$, the Taylor expansions have no constant or odd-order terms:

$$E_{12}^{(3)}(t) = \alpha_2 t^2 + O(t^4), \quad E_{12}^{(5)}(t) = \beta_2 t^2 + O(t^4).$$

So

$$E_{12}^{(3)}(t) = O(t^2), \quad E_{12}^{(5)}(t) = O(t^2).$$

Since

$$h(t) - h(-t) = 2h'(0)t + O(t^3),$$

we have

$$t = O(h_1 - h_2),$$

and therefore

$$E_{12}^{(3)} = O((h_1 - h_2)^2), \quad E_{12}^{(5)} = O((h_1 - h_2)^2).$$

Now apply Corollary 12.122 on a compact good patch

$$J \in \{c \neq 0, M \neq 0\}.$$

It gives

$$|h_1 - h_2| \leq C_J \left(|E_{12}^{(3)}| + \|E_{12}^{(5)}\| \right) = O((h_1 - h_2)^2).$$

For sufficiently small nonzero $|h_1 - h_2|$ this is impossible. Hence no such nontrivial exact residual fixed-shear involutive bad branch can persist. $\square$

Corollary 12.371 (Package-level reduction on the residual exact fixed-shear corner). *In the setting of Proposition 12.370, it is enough to verify for the actual corrected cubic and quintic two-pair defects that:*

- (1) they are invariant under swapping the two atoms along the involutive parametrization $t \leftrightarrow -t$; and
- (2) they vanish on coincident atoms.

Under these two package-level inputs, the exact residual fixed-shear corner

$$\{S = 0\} \cap \{K_v = 0\}$$

is excluded on compact good overlap patches.

Proof. The two hypotheses are exactly the evenness and diagonal-vanishing inputs of Proposition 12.370. $\square$

Remark 12.372 (The residual exact fixed-shear corner is now a package-level problem). The point of Proposition 12.370 is that it does not require construction of the full corrected two-atom quotient map. Once the residual exact fixed-shear branch is recognized as quotient-diagonal / involutive, the remaining burden is no longer a new local mixed branch classification. It is the package-level collapse of the actual finite-order two-pair defects on coincident atoms, or equivalently a no-hidden-reset statement for whatever transported overlap state survives on the quotient side.

Frame convention for the scalar block. On the cubic-generic one-pair locus we fix the cubic-normalized frame so that the distinguished cubic line is the diagonal-traceless line $\mathbf{CD}$, with

$$D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad J_0 D J_0 = -D.$$

Thus the scalar coefficient extracted below is not an auxiliary choice: it is the canonical coefficient of the cubic line in this fixed cubic-generic frame.

Definition 12.373 (Canonical scalar extraction from the raw one-pair stack). Let

$$R(z) = A_3 z^3 + A_5 z^5 + A_7 z^7 + O(z^9)$$

be a cubic-generic raw one-pair corrected germ in the cubic-normalized frame. Write

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad K = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

We use the Frobenius pairing

$$\langle X, Y \rangle_F := \text{tr}(X^* Y).$$

The scalar channel of the cubic coefficient is defined by projection to the diagonal-traceless line $\mathbf{CD}$:

$$c(R) := \frac{\langle A_3, D \rangle_F}{\langle D, D \rangle_F} = \frac{1}{2} \text{tr}(D A_3).$$

Equivalently, the scalar normalization equations are

$$\pi_I(A_3) = 0, \quad \pi_S(A_3) = 0, \quad \pi_K(A_3) = 0, \quad A_3 = c(R) D,$$

where π_I, π_S, π_K denote the Frobenius projections to the lines $\mathbf{CI}, \mathbf{CS}, \mathbf{CK}$.

Remark 12.374 (Compatibility with the existing gauge laws). By the already-proved gauge law

$$A_3^{\text{new}} = A_3,$$

the cubic coefficient is unchanged by the admissible one-pair normalization gauge. Hence the scalar extraction in Definition 12.373 is gauge independent on the raw one-pair stack.

Lemma 12.375 (Scalar normalization is Σ -invariant). *Let*

$$\Sigma(F)(z) := J_0 F(-z) J_0$$

be the centered reflection involution, and let

$$\Sigma_3(A) := -J_0 A J_0$$

be its induced action on cubic coefficients. Assume

$$J_0 D J_0 = -D.$$

Then the scalar extraction

$$c(R) = \frac{1}{2} \text{tr}(D A_3)$$

from Definition 12.373 is Σ -invariant:

$$c(\Sigma R) = c(R).$$

Proof. If $R(z) = A_3 z^3 + O(z^5)$, then

$$\Sigma(R)(z) = J_0 R(-z) J_0 = (-J_0 A_3 J_0) z^3 + O(z^5) = \Sigma_3(A_3) z^3 + O(z^5).$$

Therefore

$$c(\Sigma R) = \frac{1}{2} \operatorname{tr}(D(-J_0 A_3 J_0)).$$

Using cyclicity of trace,

$$\operatorname{tr}(D(-J_0 A_3 J_0)) = \operatorname{tr}((-J_0 D J_0) A_3).$$

Since $J_0 D J_0 = -D$, we get

$$-J_0 D J_0 = D,$$

hence

$$c(\Sigma R) = \frac{1}{2} \operatorname{tr}(D A_3) = c(R).$$

□

Lemma 12.376 (Uniqueness of the scalar output). *Let R be a cubic-generic raw one-pair corrected germ. Then the scalar $c(R)$ from Definition 12.373 is uniquely determined. In particular, if*

$$A_3 = cD = \tilde{c}D,$$

then $c = \tilde{c}$.

Proof. If

$$(c - \tilde{c})D = 0,$$

then $c - \tilde{c} = 0$ because $D \neq 0$. Equivalently,

$$c = \frac{1}{2} \operatorname{tr}(D A_3)$$

is the unique Frobenius coefficient of A_3 along the line $\mathbf{C}D$. □

Corollary 12.377 (Finite-jet scalar descent on the centered branch). *Let R_s be the centered one-pair branch, and assume*

$$R_{-s} = \Sigma(R_s).$$

Then the scalar coefficient

$$c(s) := c(R_s)$$

is even:

$$c(-s) = c(s).$$

In particular,

$$C_1 = C_3 = C_5 = 0.$$

Proof. By Lemma 12.375,

$$c(R_{-s}) = c(\Sigma R_s) = c(R_s).$$

Thus $c(-s) = c(s)$. The vanishing of the odd Taylor coefficients follows immediately. □

Remark 12.378 (Why the scalar block is the only remaining mixed-local bottleneck). Through quintic order, the quotient-level ambiguity is already confined to the J_0 -anti-fixed traceless sector, while at septic order the canonical datum is the gauge-invariant residual Δ_7 . Thus, on the clean branch $V_1 \neq 0$, the only remaining mixed-local theorem is the scalar descent

$$C_1 = C_3 = C_5 = 0.$$

Once this holds, the finite-jet bridge to the good v_5 -patch coordinate $x = v_5/c$ is formal and the explicit mixed low-order obstruction list

$$(r, \operatorname{ord}_0 \mathcal{Z}) \in \{(1, 1), (1, 3), (2, 2)\}$$

is excluded immediately.

Remark 12.379 (Centered symmetric-placement Taylor package and diagonal-base local exclusion). Let

$$t_{\pm} = h \pm \frac{s}{2},$$

and regard $s \mapsto -s$ as swapping the two microscopic points. For any symmetric 2×2 coefficient matrix, the diagonal half-sums are even in s , the diagonal half-differences are odd in s , and the off-diagonal entries are even in s . Applying this to the explicit formulas already in the draft gives

$$u_5(s), v_5(s) \text{ odd}, \quad u_{7,K_1}(s), v_{7,K_1}(s) \text{ odd},$$

Writing

$$u_5(s) = U_1 s + U_3 s^3 + U_5 s^5 + \cdots, \quad v_5(s) = V_1 s + V_3 s^3 + V_5 s^5 + \cdots,$$

the first coefficients are

$$U_1 = 2(v_0 \delta \alpha_1 + \beta_0 \delta u_1),$$

$$U_3 = 2(v_0 \delta \alpha_3 + v_2 \delta \alpha_1 + \beta_0 \delta u_3 + \beta_2 \delta u_1),$$

$$U_5 = 2(v_0 \delta \alpha_5 + v_2 \delta \alpha_3 + v_4 \delta \alpha_1 + \beta_0 \delta u_5 + \beta_2 \delta u_3 + \beta_4 \delta u_1),$$

and

$$V_1 = 2(\bar{\alpha}_0 \delta u_1 + \delta \alpha_1 \bar{u}_0),$$

$$V_3 = 2(\bar{\alpha}_0 \delta u_3 + \bar{\alpha}_2 \delta u_1 + \delta \alpha_1 \bar{u}_2 + \delta \alpha_3 \bar{u}_0),$$

$$V_5 = 2(\bar{\alpha}_0 \delta u_5 + \bar{\alpha}_2 \delta u_3 + \bar{\alpha}_4 \delta u_1 + \delta \alpha_1 \bar{u}_4 + \delta \alpha_3 \bar{u}_2 + \delta \alpha_5 \bar{u}_0).$$

Likewise, writing

$$u_{7,K_1}(s) = \hat{U}_1 s + \hat{U}_3 s^3 + \cdots, \quad v_{7,K_1}(s) = \hat{V}_1 s + \hat{V}_3 s^3 + \cdots,$$

one gets

$$\hat{U}_1 = 2\kappa(\delta x_1 \nu_{6,0} + \beta_{5,0} \delta u_1 + v_0 \delta \alpha_{5,1} + b_0 \delta p_1 + r_0 \delta a_1),$$

$$\hat{V}_1 = 2\kappa(\delta x_1 \bar{\mu}_{6,0} - \bar{x}_0 \delta \mu_{6,1} + \bar{\alpha}_{5,0} \delta u_1 + \delta \alpha_{5,1} \bar{u}_0 + \bar{a}_0 \delta p_1 + \delta a_1 \bar{p}_0),$$

and

$$\hat{U}_3 = 2\kappa(\delta x_1 \nu_{6,2} + \delta x_3 \nu_{6,0} + \beta_{5,0} \delta u_3 + \beta_{5,2} \delta u_1 + v_0 \delta \alpha_{5,3} + v_2 \delta \alpha_{5,1} + b_0 \delta p_3 + b_2 \delta p_1 + r_0 \delta a_3 + r_2 \delta a_1),$$

$$\hat{V}_3 = 2\kappa(\delta x_1 \bar{\mu}_{6,2} + \delta x_3 \bar{\mu}_{6,0} - \bar{x}_0 \delta \mu_{6,3} - \bar{x}_2 \delta \mu_{6,1} + \bar{\alpha}_{5,0} \delta u_3 + \bar{\alpha}_{5,2} \delta u_1 + \delta \alpha_{5,1} \bar{u}_2 + \delta \alpha_{5,3} \bar{u}_0 + \bar{a}_0 \delta p_3 + \bar{a}_2 \delta p_1 + \delta a_1 \bar{p}_2 + \delta a_3 \bar{p}_0).$$

and therefore

$$\Delta_7(s) = u_{7,K_1}(s) v_5(s) - u_5(s) v_{7,K_1}(s)$$

is even in s , with Taylor expansion

$$\Delta_7(s) = D_2 s^2 + D_4 s^4 + D_6 s^6 + \cdots,$$

where

$$D_2 = \hat{U}_1 V_1 - U_1 \hat{V}_1,$$

$$D_4 = \hat{U}_1 V_3 + \hat{U}_3 V_1 - U_1 \hat{V}_3 - U_3 \hat{V}_1,$$

$$D_6 = \hat{U}_1 V_5 + \hat{U}_3 V_3 + \hat{U}_5 V_1 - U_1 \hat{V}_5 - U_3 \hat{V}_3 - U_5 \hat{V}_1.$$

Assume now that

$$c(s) = C_0 + C_2 s^2 + C_4 s^4 + \cdots, \quad C_0 \neq 0.$$

Then

$$x(s) = \frac{v_5(s)}{c(s)}$$

is an odd series

$$x(s) = X_1 s + X_3 s^3 + X_5 s^5 + \cdots$$

with

$$X_1 = \frac{V_1}{C_0},$$

$$X_3 = \frac{V_3 C_0 - V_1 C_2}{C_0^2},$$

$$X_5 = \frac{V_5 C_0^2 - V_3 C_0 C_2 + V_1 (C_2^2 - C_0 C_4)}{C_0^3}.$$

In particular, if $V_1 \neq 0$, then x is an odd local coordinate with odd inverse

$$s(x) = \sigma_1 x + \sigma_3 x^3 + \sigma_5 x^5 + \cdots,$$

where

$$\sigma_1 = \frac{C_0}{V_1}, \quad \sigma_3 = -\frac{C_0(V_3 C_0 - V_1 C_2)}{V_1^4},$$

and so on.

Similarly,

$$\lambda(s) := \frac{\Delta_7(s)}{v_5(s)}$$

is an odd series

$$\lambda(s) = L_1 s + L_3 s^3 + L_5 s^5 + \cdots$$

with

$$\begin{aligned} L_1 &= \frac{D_2}{V_1}, \\ L_3 &= \frac{D_4 V_1 - D_2 V_3}{V_1^2}, \\ L_5 &= \frac{D_6 V_1^2 - D_4 V_1 V_3 + D_2 (V_3^2 - V_1 V_5)}{V_1^3}. \end{aligned}$$

So, whenever x is used as the induced odd collision-side coordinate, parity in s transfers to parity in x , and one obtains

$$Y(x) = \frac{u_5}{c}(x) \text{ odd}, \quad \lambda(x) = \frac{\Delta_7}{v_5}(x) \text{ odd}, \quad S(x) = \frac{\lambda}{c}(x) \text{ odd}.$$

Consequently,

$$Y''(x) \text{ odd}, \quad S''(x) \text{ odd},$$

hence

$$\mathcal{Z}(x) = \frac{Y''}{S''} \text{ even}, \quad \mathcal{R}(x) = \mathcal{Z}'(x) \text{ odd}, \quad K_v = (x')^3 S'' \text{ odd}.$$

This is the parity package underlying the provisional mixed-branch exclusion discussed below.

If $x = v_5/c$ is used as the induced odd collision-side coordinate, then the centered diagonal germ has the odd/odd form

$$Q(x) \sim (x^n, x, x^m), \quad m, n \text{ odd}, \quad n > m \geq 3.$$

In that regime one obtains the exact factorizations

$$S''(x) = x^{m-2} \Sigma_m[A](x^2), \quad \mathcal{Z}(x) = x^{n-m} H(x^2), \quad \mathcal{R}(x) = x^{n-m-1} J(x^2),$$

with $H(0) \neq 0$ and $J(0) \neq 0$. Hence there exists $\varepsilon > 0$ such that for

$$0 < |x| < \varepsilon$$

one has

$$K_v(x) \neq 0, \quad T_v(x) \neq 0.$$

So, in the diagonal-based local regime where this odd collision-side coordinate description is valid, the centered diagonal base has a punctured neighborhood free of $K_v = 0$ and $T_v = 0$ points. In particular, the currently explicit low-order dangerous patterns do not arise at the symmetric diagonal base itself.

Work in progress. The odd/even transfer from the centered microscopic variable s to a general good-patch coordinate $x = v_5/c$ should still be treated as a live theorem-level bridge outside the diagonal-based local regime. What is no longer open at the same level is total quintic collapse on the centered branch: by Lemma 12.382 and Corollary 12.383, one already has

$$u_5(s) \neq 0, \quad A_5^f(s) \neq 0.$$

So the remaining mixed-side task is not to prove that some quintic channel survives, but to promote this centered quintic nontriviality into an actual punctured v_5 - or u_5 -patch.

Lemma 12.380 (Total centered quintic collapse forces source-level degeneration). *On the centered symmetric-placement branch one has*

$$u_5 = 2(v\delta\alpha + \beta\delta u), \quad v_5 = 2(\bar{\alpha}\delta u + \bar{u}\delta\alpha).$$

If

$$u_5 \equiv 0, \quad v_5 \equiv 0,$$

then

$$\begin{pmatrix} \beta(s) & v(s) \\ \bar{\alpha}(s) & \bar{u}(s) \end{pmatrix} \begin{pmatrix} \delta u(s) \\ \delta\alpha(s) \end{pmatrix} = 0$$

as an identity of analytic germs. Consequently, either

$$\Delta_5(s) := \beta(s)\bar{u}(s) - v(s)\bar{\alpha}(s) \equiv 0,$$

or

$$\delta u(s) \equiv 0, \quad \delta\alpha(s) \equiv 0.$$

Proof. This is immediate from the displayed formulas for u_5 and v_5 . If the 2×2 coefficient matrix has nonzero determinant, then the only solution is $\delta u = \delta\alpha \equiv 0$. If its determinant vanishes identically, then $\Delta_5 \equiv 0$. $\square$

Corollary 12.381 (Total centered quintic collapse forces total canonical transport collapse). *If*

$$u_5 \equiv 0, \quad v_5 \equiv 0$$

on the centered branch, then

$$A_5^\dagger \equiv 0, \quad \Delta_7 \equiv 0.$$

In particular, total centered quintic collapse forces collapse of the canonical quintic/septic transport package on that branch.

Proof. The identity

$$A_5^\dagger = u_5 I + v_5 S$$

gives $A_5^\dagger \equiv 0$ immediately. Likewise

$$\Delta_7 = u_{7,K_1} v_5 - u_5 v_{7,K_1}$$

vanishes identically when $u_5 \equiv v_5 \equiv 0$. $\square$

Lemma 12.382 (Centered tagged quintic witness survives specialization). *On the tagged slice*

$$(K_1, \eta_2, q^{(5)}),$$

the first centered odd coefficient

$$U_1$$

of

$$u_5(s) = U_1 s + U_3 s^3 + U_5 s^5 + \dots$$

has a nonzero $\kappa \eta_2 q^{(5)}$ -coefficient. In particular

$$U_1 \neq 0$$

on that tagged slice, and hence

$$u_5(s) \not\equiv 0$$

as a centered germ.

Proof. The centered coefficient formula is

$$U_1 = 2(v_0 \delta \alpha_1 + \beta_0 \delta u_1).$$

By the tagged-slice analysis already used in the proof of Lemma 12.18, the η_2 -slice of X_1 is off-diagonal, so

$$\delta u_1 = 0,$$

and the $q^{(5)}$ -slice of X_3 is diagonal, so

$$\beta_0 = 0.$$

Hence

$$U_1 = 2v_0 \delta \alpha_1$$

on the tagged slice.

Now use the explicit two-direction computation from the proof of Lemma 12.18. There one writes

$$G_0 = \text{diag}(a^2, b^2), \quad A = \begin{pmatrix} 0 & u \\ u & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & 0 \\ 0 & d \end{pmatrix},$$

and expands

$$X(\varepsilon, \delta) = X_0 + \varepsilon U + \delta V + \varepsilon \delta W + O(\varepsilon^2, \delta^2).$$

The proof gives

$$U_{12} = -\frac{u}{ab(a+b)} \neq 0, \quad V_{22} = -\frac{d}{2b^3} \neq 0.$$

Here U_{12} gives the tagged η_2 -contribution to the off-diagonal channel v_0 , while V_{22} gives the tagged $q^{(5)}$ -contribution to the diagonal-difference channel $\gamma - \alpha = 2\delta \alpha_1$. Thus the tagged slice contributes nontrivially to the product $v_0 \delta \alpha_1$, and hence to $U_1 = 2v_0 \delta \alpha_1$. $\square$

Corollary 12.383 (Centered quintic fixed-sector nontriviality). *On the centered diagonal branch,*

$$A_5^{\mathbf{f}}(s) = u_5(s)I + v_5(s)S$$

is not identically zero. Equivalently,

$$(u_5(s), v_5(s)) \neq (0, 0).$$

Proof. By Lemma 12.382, one has $u_5(s) \neq 0$. Since I and S are linearly independent, the fixed-sector package $A_5^{\mathbf{f}}(s) = u_5(s)I + v_5(s)S$ cannot vanish identically. $\square$

Lemma 12.384 (Valuation form of the septic transport constraint). *Assume on the centered branch that*

$$\lambda = \frac{\Delta_7}{v_5}$$

is analytic odd. Write

$$\text{ord}_s u_5 = 2m + 1, \quad \text{ord}_s v_5 = 2r + 1, \quad r \geq m.$$

Then

$$\text{ord}_s v_{7,K_1} \geq \text{ord}_s v_5 - \text{ord}_s u_5 + 1 = 2(r - m) + 1.$$

Equivalently, if

$$U_1 = \cdots = U_{2m-1} = 0, \quad U_{2m+1} \neq 0,$$

and

$$V_1 = \cdots = V_{2r-1} = 0, \quad V_{2r+1} \neq 0,$$

then

$$\widehat{V}_1 = \widehat{V}_3 = \cdots = \widehat{V}_{2(r-m)-1} = 0,$$

and the first potentially surviving even coefficient of Δ_7 is

$$D_{2r+2} = \widehat{U}_1 V_{2r+1} - U_{2m+1} \widehat{V}_{2(r-m)+1}.$$

Proof. Since λ is analytic odd and

$$\Delta_7 = \lambda v_5,$$

one has

$$\text{ord}_s \Delta_7 \geq \text{ord}_s v_5 + 1 = 2r + 2.$$

Suppose

$$\text{ord}_s v_{7,K_1} = 2d + 1 < 2(r - m) + 1.$$

Then

$$\text{ord}_s(u_5 v_{7,K_1}) = (2m + 1) + (2d + 1) \leq 2r,$$

whereas

$$\text{ord}_s(u_{7,K_1} v_5) \geq 1 + (2r + 1) = 2r + 2$$

because u_{7,K_1} is odd. Therefore the leading term of

$$\Delta_7 = u_{7,K_1} v_5 - u_5 v_{7,K_1}$$

would occur at order at most $2r$, contradicting $\text{ord}_s \Delta_7 \geq 2r + 2$. So

$$\text{ord}_s v_{7,K_1} \geq 2(r - m) + 1.$$

The coefficient statement is the corresponding triangular coefficient form: vanishing of

$$D_2, D_4, \dots, D_{2r}$$

forces

$$\widehat{V}_1, \widehat{V}_3, \dots, \widehat{V}_{2(r-m)-1}$$

to vanish successively, and the first potentially surviving coefficient is

$$D_{2r+2} = \widehat{U}_1 V_{2r+1} - U_{2m+1} \widehat{V}_{2(r-m)+1}.$$

□

Lemma 12.385 (Late- v_5 u_5 -patch affine-normalization transplant). *Assume the centered scalar package from Remark 12.379, and assume*

$$U_1 \neq 0.$$

Then

$$y := \frac{u_5}{c}$$

is an odd local coordinate near the centered base. Suppose in addition that

$$\text{ord}_s v_5 = 2r + 1 \geq 3.$$

On the corresponding u_5 -patch write

$$x(y) := \frac{v_5}{c}, \quad \tau_u(y) := -\frac{\Delta_7}{c u_5}, \quad \tilde{\tau}_u(y) := \tau_u(y) - \tau'_u(0) y.$$

Then:

(1) $x(y)$ is odd and satisfies

$$x(y) = O(y^{2r+1});$$

in particular $\text{ord}_0 x \geq 3$;

(2) $\tilde{\tau}_u(y)$ is odd and either vanishes identically or has odd order at least 3;

(3) if $\tilde{\tau}_u \not\equiv 0$, then after an ambient affine normalization consisting of a coordinate permutation and subtraction of a suitable odd multiple of one higher-order channel from the other, the germ Q_u is carried to the same odd/odd local normal form

$$(y^n, y, y^m), \quad n > m \geq 3, \quad m, n \text{ odd},$$

already analyzed in Remark 12.379.

Consequently, in the subcase $\tilde{\tau}_u \not\equiv 0$, the late- v_5 corner is locally sterile on punctured u_5 -patch arcs: no genuinely new mixed low-order 4-point branch is born there.

Proof. By Lemma 12.382,

$$u_5(s) = U_1 s + O(s^3), \quad c(s) = C_0 + O(s^2), \quad C_0 \neq 0,$$

so

$$y(s) = \frac{u_5(s)}{c(s)}$$

is an odd local coordinate with odd inverse $s = s(y)$.

Since v_5 is odd and c is even in s ,

$$x(y) = \frac{v_5}{c}$$

is odd in y . Because $\text{ord}_s u_5 = 1$ and $\text{ord}_s v_5 = 2r + 1$, one has

$$x(y) = O(y^{2r+1}),$$

so $\text{ord}_0 x \geq 3$.

Likewise Δ_7 is even while c is even and u_5 is odd, hence

$$\tau_u(y) = -\frac{\Delta_7}{c u_5}$$

is odd in y . Therefore

$$\tilde{\tau}_u(y) := \tau_u(y) - \tau'_u(0) y$$

is odd and either identically zero or of odd order at least 3.

Assume now that $\tilde{\tau}_u \neq 0$. Then the normalized germ is

$$Q_u(y) = (y, x(y), \tilde{\tau}_u(y)),$$

with the last two channels odd and of order at least 3. Apply the ambient coordinate permutation

$$(Y, X, Z) \mapsto (X, Y, Z).$$

This preserves affine dependence/coplanarity geometry. If the two higher-order odd channels have the same leading odd order, subtract a suitable odd multiple of one from the other to separate those leading orders. After this affine normalization one reaches a germ of the form

$$(y^n B(y^2), y, y^m A(y^2)), \quad n > m \geq 3, \quad m, n \text{ odd},$$

with $A(0)B(0) \neq 0$, i.e. the same odd/odd local normal form analyzed in Remark 12.379. The same local calculation then gives a punctured neighborhood free of nearby mixed and nonmixed 4-point obstruction points. $\square$

Lemma 12.386 (Planar-lift absorption on the u_5 -patch). *Assume the hypotheses of Lemma 12.385, and suppose*

$$\tilde{\tau}_u \equiv 0.$$

Equivalently, there is a constant $a \in \mathbf{C}$ such that

$$\Delta_7 = -a u_5^2$$

locally on the u_5 -patch. Then after an ambient affine shear one has

$$Q_u \sim (y, x, 0), \quad y = \frac{u_5}{c}, \quad x = \frac{v_5}{c}.$$

In particular, the lifted curve is affine-planar, and the remaining local content is only the planar affine dependence of

$$\Xi(h) := \left(\frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)} \right).$$

Moreover, in the defect-free order- ≤ 5 finite-core setting of Theorem 12.216, this residual corner is absorbed into the already-known quintic-circuit regime: it does not define a genuinely new mixed/septic 4-point branch.

Proof. The identity $\tilde{\tau}_u \equiv 0$ means exactly that

$$\tau_u(y) = a y$$

for some constant a . Since

$$\tau_u = -\frac{\Delta_7}{c u_5}, \quad y = \frac{u_5}{c},$$

this is equivalent to

$$-\frac{\Delta_7}{c u_5} = a \frac{u_5}{c},$$

hence

$$\Delta_7 = -a u_5^2.$$

Thus

$$Q_u = (y, x, ay).$$

Applying the ambient affine shear

$$(Y, X, Z) \mapsto (Y, X, Z - aY)$$

gives

$$Q_u \sim (y, x, 0).$$

So the lifted curve is affine-planar.

Now assume additionally the defect-free order- ≤ 5 finite-core hypotheses from Theorem 12.216. By Remark 12.223, the same defect-free affine-dependence conclusions hold with Q_v replaced by Q_u . Since the third coordinate of Q_u vanishes after the shear, affine dependence of the lifted points is equivalent to affine dependence of the planar points

$$\Xi(h_j) = \left(\frac{u_5(h_j)}{c(h_j)}, \frac{v_5(h_j)}{c(h_j)} \right) \in \mathbf{C}^2.$$

But this is exactly the planar quintic-circuit geometry organized by Theorem 12.216 and Corollary 12.217. Hence this residual corner is already-quintic rather than a genuinely new mixed septic branch. $\square$

Corollary 12.387 (Right-pincer trichotomy near the centered base). *Near the centered diagonal base, the mixed local picture is reduced to the following three cases:*

$$V_1 \neq 0 \implies \text{clean } v_5\text{-patch branch};$$

$$V_1 = 0, \tilde{\tau}_u \neq 0 \implies \text{late-}v_5 \text{ branch locally sterile on } Q_u;$$

$$V_1 = 0, \tilde{\tau}_u \equiv 0 \iff \Delta_7 = -a u_5^2 \implies \text{planar-lift corner, hence already-quintic}.$$

So every late- v_5 branch near the centered base is either locally sterile or already absorbed into the quintic-circuit regime. The only genuinely new mixed local branch left to study there is the clean branch

$$\text{ord}_s v_5 = 1.$$

Proof. Combine Lemma 12.385 and Lemma 12.386, together with the fact that v_5 is odd in the centered variable. $\square$

Corollary 12.388 (Right-pincer local reduction near the centered base). *Assume the centered scalar package from Remark 12.379, and assume also that the centered tagged quintic witness gives*

$$U_1 \neq 0.$$

Then every sufficiently small nearby genuinely new mixed 4-point obstruction point near the centered base must lie on the clean branch

$$\text{ord}_s v_5 = 1,$$

equivalently

$$V_1 \neq 0.$$

In particular, every late- v_5 branch near the centered base is either locally sterile on the u_5 -patch or already absorbed into the defect-free quintic-circuit regime.

Proof. Because v_5 is odd in the centered variable, either

$$\text{ord}_s v_5 = 1 \quad \text{or} \quad \text{ord}_s v_5 \geq 3 \text{ odd.}$$

If $\text{ord}_s v_5 \geq 3$, then Lemma 12.385 and Lemma 12.386 show that the branch is either locally sterile on the u_5 -patch or planar-lift and hence already-quintic. So a sufficiently small nearby genuinely new mixed 4-point obstruction cannot lie on such a late- v_5 branch. Therefore any such genuinely new mixed local branch must lie on the only remaining possibility,

$$\text{ord}_s v_5 = 1. \quad \square$$

Proposition 12.389 (Realization lemma for lifted germs). *Let $\tilde{Y}(\xi)$ and $\tilde{S}(\xi)$ be arbitrary analytic germs at $\xi = 0$. Define*

$$c(\xi) \equiv 1, \quad v_5(\xi) := 1 + \xi, \quad u_5(\xi) := \tilde{Y}(\xi), \quad \Delta_7(\xi) := (1 + \xi)\tilde{S}(\xi).$$

Then on a sufficiently small neighborhood of 0,

$$c \neq 0, \quad v_5 \neq 0, \quad M = c v_5' - c' v_5 = 1 \neq 0,$$

so this is a good v_5 -patch. The normalized lifted coordinates are

$$x = \frac{v_5}{c} = 1 + \xi, \quad Y = \frac{u_5}{c} = \tilde{Y}(\xi), \quad S = \frac{\Delta_7}{c v_5} = \tilde{S}(\xi).$$

After the affine shift $x \mapsto x - 1$, the lifted germ becomes

$$Q_v(\xi) = (\tilde{Y}(\xi), \xi, \tilde{S}(\xi)).$$

Thus every analytic pair of germs $(\tilde{Y}, \tilde{S})$ occurs, after affine normalization, as the lifted pair (Y, S) on some good v_5 -patch.

Proof. The displayed formulas are immediate from the definitions. The hypotheses $c \neq 0$, $v_5 \neq 0$, and $M = 1 \neq 0$ hold on a sufficiently small neighborhood of 0, and the formulas for x, Y, S are direct. $\square$

Corollary 12.390 (No new obstruction for $(\mathcal{Z}, \mathcal{R})$ is available at the current lifted level). *No identity involving only*

$$c, \quad v_5, \quad u_5, \quad \Delta_7$$

and their derivatives can force either of the two dream conclusions:

$$\mathcal{R} \text{ is even in the normalized nonmixed coordinate,}$$

or the mixed-branch exclusion

$$(r, \text{ord}_0 \mathcal{Z}) \notin \{(1, 1), (1, 3), (2, 2)\} \quad \text{at every mixed point.}$$

In particular, the nonmixed models

$$Q_v(\xi) = (\xi^n, \xi, \xi^2), \quad n \geq 4 \text{ even,}$$

are all realizable at the present lifted level; the cases

$$n = 4, 6, 8$$

are simply the first currently explicit low-order examples. The dangerous mixed models

$$Q_v(\xi) = (\xi^n, \xi, \xi^m), \quad (m, n) = (3, 4), (3, 6), (4, 6),$$

are all realizable at the present lifted level.

Consequently, any RH-specific obstruction to the low-order dangerous 4-point patterns must use information not encoded in the current lifted variables

$$(c, v_5, u_5, \Delta_7)$$

alone. It must come from deeper source-level corrected quotient-germ structure, matrix-level identities behind (u_5, Δ_7) , or an additional symmetry not yet visible in the present lifted formulation.

Proof. Apply Proposition 12.389 with

$$(\tilde{Y}, \tilde{S}) = (\xi^n, \xi^2)$$

for the nonmixed cases, and with

$$(\tilde{Y}, \tilde{S}) = (\xi^n, \xi^m)$$

for the mixed cases. These realize exactly the dangerous low-order patterns for $(\mathcal{Z}, \mathcal{R})$ listed above. $\square$

Remark 12.391 (Updated nonmixed status versus mixed local exhaustion). The local nonmixed lifted route is now closed at the derivative-geometric level. Exact nonmixed contacts force zeros of $\mathcal{R}$; one-sign $\mathcal{R}$ -intervals are sterile; recurrent exact contacts force affine-planar collapse; and isolated zeros of $\mathcal{R}$ are classified by parity. In addition, the repeated-source / first-loss mechanism is now fully reduced: no attained separated first loss can occur, every first-loss sequence is forced to the diagonal, and diagonal first loss occurs exactly at odd zeros of $\mathcal{R}$. Thus the nonmixed burden is no longer a local parity theorem or a separate repeated-source obstruction. It is the global/source-level control of which odd zeros of

$$\mathcal{R} = \frac{T_v}{K_v^2}, \quad \text{equivalently} \quad \mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda,$$

can actually arise along opened nonmixed arcs.

On the mixed side, the decisive pair-level object remains

$$A_5^\dagger(s) = u_5(s)I + v_5(s)S.$$

By Corollary 12.383, this package is not identically zero, so after shrinking to one-sided punctured arcs the centered branch enters either a good v_5 -patch or the parallel u_5 -patch. Near the centered base, however, the theorem package now goes further: every late- v_5 branch is already either locally sterile on the automatic u_5 -patch or absorbed into the planar defect-free quintic-circuit regime, so the only genuinely new mixed local branch left there is the clean branch

$$V_1 \neq 0.$$

Accordingly, the mixed-side local problem is no longer a vague codimension-two obstruction, nor merely a patch-opening issue. Near the centered base it has been reduced to the single clean branch $V_1 \neq 0$, and on that branch the remaining theorem-level task is the finite scalar descent

$$c(-s) = c(s) \pmod{s^6}, \quad \text{equivalently} \quad C_1 = C_3 = C_5 = 0,$$

which makes the centered-to-patch finite-jet parity bridge formal.

Remark 12.392 (What the four-point branch now reduces to). The lifted 4-point branch is now organized by the canonical invariant pair

$$\mathcal{Z} = \frac{Y''}{S''} = \frac{J_{u_5}}{J_\lambda}, \quad \mathcal{R} = \mathcal{Z}' = \frac{T_v}{K_v^2}, \quad \lambda = \frac{\Delta_7}{v_5}.$$

On the nonmixed side, after normalization to

$$Q_v(\xi) = (\tilde{Y}(\xi), \xi, \xi^2),$$

the local derivative-side picture is now explicit. Exact nonmixed contacts force zeros of

$$\mathcal{R} = \frac{1}{2} \tilde{Y}''',$$

isolated zeros are classified by parity, one-sign $\mathcal{R}$ -intervals are exactly-contact sterile, and recurrent exact contacts force affine-planar collapse.

The explicit low-order list

$$\text{ord}_0 \mathcal{R} \in \{1, 3, 5\},$$

equivalently the coefficients

$$z_2, \quad z_4, \quad z_6,$$

should therefore now be regarded only as the first explicit low-order nonmixed odd-order cases, not as the full local nonmixed picture.

For fixed source point a , repeated-source exact contacts factor through the quadratic-envelope scalar

$$\Gamma_a(x) = \frac{Y(x) - Y(a) - Y'(a)(x - a)}{S(x) - S(a) - S'(a)(x - a)},$$

equivalently through the below-lifted quotient

$$\Psi_a(h) = \frac{D_{u_5}(a, h)}{D_\lambda(a, h)}.$$

Its derivative is governed by the osculatory 4×4 source determinant $\mathfrak{D}(a, h)$. By Proposition 12.292, this repeated-source / first-loss mechanism is now completely reduced: no attained separated first loss can occur, and any first loss is exactly the diagonal birth at an isolated odd-order zero of $\mathcal{R}$.

Thus the remaining nonmixed burden is no longer local parity, nor a separate first-loss geometry, but purely the global/source-level control of which odd zeros of $\mathcal{W}_{\text{src}}$, equivalently of $\mathcal{R}$, occur on genuine opened arcs.

On the mixed side, the explicit low-order dangerous list is

$$(m, n) = (3, 4), (3, 6), (4, 6),$$

equivalently

$$(r, s) = (1, 2), (1, 4), (2, 5),$$

equivalently

$$(r, \text{ord}_0 \mathcal{Z}) = (1, 1), (1, 3), (2, 2),$$

equivalently

$$(r, \text{ord}_0 \mathcal{R}) = (1, 0), (1, 2), (2, 1).$$

Equivalently, in the below-lifted Wronskian formulation from Remark 12.332, the dangerous mixed models are exactly the order-pairs

$$(\text{ord}_0 J_\lambda, \text{ord}_0 J_{u_5}) = (1, 2), (1, 4), (2, 4).$$

So the mixed low-order branch is already a finite canonical Wronskian-ratio obstruction problem, captured in local mixed x -coordinates by

$$\kappa_{1,1}, \quad \kappa_{1,3}, \quad \kappa_{2,2}.$$

By the realization lemma, no identity involving only

$$(c, v_5, u_5, \Delta_7)$$

and their derivatives can eliminate these mixed patterns in general.

At the pair level, the centered fixed-sector quintic package

$$A_5^{\mathfrak{f}}(s) = u_5(s)I + v_5(s)S$$

is now the natural object controlling whether any quintic patch opens at all. By Corollary 12.383,

$$A_5^{\mathfrak{f}}(s) \not\equiv 0,$$

so the centered branch cannot suffer total quintic collapse.

Near the centered base, however, the mixed-side local picture is now sharper than a mere patch dichotomy. By Corollary 12.388, every sufficiently small nearby genuinely new mixed 4-point obstruction must lie on the clean branch

$$\text{ord}_s v_5 = 1,$$

while every late- v_5 branch is either locally sterile on the automatic u_5 -patch or already absorbed into the defect-free planar quintic-circuit regime. Thus the genuinely new mixed local geometry near the centered base has been reduced to the single clean branch $V_1 \neq 0$.

Accordingly, the remaining mixed-side theorem is no longer another lifted jet computation, nor a pointwise coefficient nonvanishing statement such as $V_1 \neq 0$ in isolation. It is the finite scalar descent on the clean branch, namely

$$c(-s) = c(s) \pmod{s^6}, \quad \text{equivalently} \quad C_1 = C_3 = C_5 = 0,$$

which makes the centered good- v_5 -patch parity bridge formal. After that centered local step, the clean-patch mixed-global problem is no longer open mixed geometry: it is absorbed by the mixed-to-scalar capture theorem (Theorem 12.408), with the source-level viewpoint serving only as the correct language for the lower-core and source-coupled handoff walls.

Remark 12.393 (What the new nonmixed/source-level machinery changes for the mixed side). The new repeated-source/source-determinant viewpoint and the no-go principle excluding purely planar (x, S) -closure do not create a new mixed-local closure theorem. Rather, they clarify the remaining mixed burden.

First, no argument using only planar (x, S) -geometry can close the 4-point branch, so any remaining mixed route must use either the centered scalar descent or genuine coupled Y - S source-level information.

Second, if a below-lifted backup route is needed, the natural mixed object is not an S -only invariant, but a coupled source-minor ratio built from u_5 and

$$\lambda = \frac{\Delta_7}{v_5},$$

i.e. a below-lifted avatar of

$$\mathcal{Z} = \frac{J_{u_5}}{J_\lambda}.$$

Thus the new nonmixed/source-level machinery functions not only as a strategic constraint and as a backup language for the mixed side, but also as the correct external handoff target for the $A = 0$ planar collapse; see Remark 12.394. It does not replace the local mixed exhaustion theorem near the centered base, but it does absorb the nonmixed affine-planar lane that remains after the local mixed reduction.

Remark 12.394 (The $A = 0$ collapse is already the nonmixed $\mathcal{R}$ -lane). Work on a good v_5 -patch and write an exact lifted 4-point relation in the form

$$A Y + B x + C S + D = 0, \quad Y = \frac{u_5}{c}, \quad x = \frac{v_5}{c}, \quad S = \frac{\Delta_7}{c v_5}.$$

If $A = 0$, then the relation reduces to

$$B x + C S + D = 0,$$

so the four nodes lie on an affine line in the planar (x, S) -curve. Thus the positive-dimensional $A = 0$ regime is not a separate mixed branch at all: it is precisely the nonmixed affine-planar exact-contact regime. In particular, the relevant theorem-strength input is already on the nonmixed side: exact nonmixed contacts force zeros of $\mathcal{R}$ (Proposition 12.264), one-sign $\mathcal{R}$ -intervals are sterile (Corollary 12.265), and interval-persistent / recurrent exact contacts force affine-planar collapse (Corollary 12.268). Accordingly, $A = 0$ should be viewed as the external handoff from the mixed side into the already-handled nonmixed $\mathcal{R}$ -package, not as a fourth independent mixed-global lane.

Remark 12.395 (Status of the lifted order-7 geometry). The lifted geometry is now substantially more stratified than a single zero-capture problem.

First, shrinking exact lifted three-point and four-point circuits can accumulate only at

$$K_v = 0 \quad \text{and} \quad T_v = 0,$$

respectively, with uniform small-diameter exclusion away from those zero sets; equivalently, they are rooted at the curvature- and torsion-type osculating determinants of the corresponding lifted planar and space curves (Lemma 12.246, Propositions 12.248 and 12.250).

Second, repeated-node limits are now explicit: three-point collapse limits become tangent-line contacts for the planar lifted curve $\Lambda_v = (x, S)$, and four-point collapse limits become tangent-plane contacts for the lifted space curve $Q_v = (Y, x, S)$.

Third, the three-point side is now blockwise globally controlled on good v_5 -patches. The curvature zero set $K_v = 0$ has no infinite-order flat points; even-order K_v -zeros are sterile; odd-order K_v -zeros carry exactly one local opposite-side tangent-contact branch; and that branch

continues uniquely on the maximal right monotone-slope block, with no interior common-tangent event, no source-side return, and no interior accumulation. Thus the three-point side is no longer a live internal recurrence problem on such patches; only boundary exit of the relevant source/target slope-overlap block can remain, and in the four-point mixed-to-scalar picture this is used only as an external lower-core handoff target rather than as a new mixed carrier lane.

On the mixed four-point side, the right-pincer package now gives an analogous local simplification near the centered base: every late- v_5 branch is either locally sterile on the u_5 -patch or already absorbed into the planar quintic-circuit regime, so the only genuinely new mixed local branch left there is the clean branch $\text{ord}_s v_5 = 1$.

Fourth, the four-point side splits sharply into nonmixed and mixed regimes. On the nonmixed side, finite-order odd T_v -zeros already support genuine local exact branch sheets in the model geometry, so a blanket local sterility theorem is false. On the mixed side, the dangerous low-order obstruction list is finite and explicit:

$$(m, n) = (3, 4), (3, 6), (4, 6),$$

equivalently

$$(r, s) = (1, 2), (1, 4), (2, 5),$$

equivalently

$$(r, \text{ord}_0 \mathcal{Z}) = (1, 1), (1, 3), (2, 2),$$

equivalently

$$(r, \text{ord}_0 \mathcal{R}) = (1, 0), (1, 2), (2, 1).$$

Fifth, below the lifted variables one has the exact Wronskian-ratio reformulation

$$\mathcal{Z} = \frac{J_{u_5}}{J_\lambda}, \quad \mathcal{R} = \frac{c^2}{M} \frac{d}{dh} \left(\frac{J_{u_5}}{J_\lambda} \right), \quad \lambda = \frac{\Delta_7}{v_5},$$

while the projective curvature ratio Θ yields no new independent local obstruction beyond $(\mathcal{Z}, \mathcal{R})$.

Sixth, the centered symmetric-placement Taylor package gives a strong local collision-side parity picture:

$$u_5, v_5, u_{7,K_1}, v_{7,K_1} \text{ odd in } s, \quad \Delta_7 \text{ even in } s,$$

with explicit septic coefficients D_2, D_4, D_6 . In the diagonal-centered odd/odd normal form

$$Q(x) \sim (x^n, x, x^m), \quad m, n \text{ odd}, \quad n > m \geq 3,$$

one gets a punctured neighborhood free of $K_v = 0$ and $T_v = 0$ points. Thus the explicit low-order dangerous 4-point patterns do not arise at the symmetric diagonal base itself.

So the mixed four-point side is no longer waiting for a new local geometry theorem. On the mixed side, the sufficiently small problem is now organized by the mixed-to-scalar capture principle: after the source-coupled and patch-loss reductions, no sufficiently small genuinely mixed branch on a nonplanar good patch can persist entirely inside

$$\{T_v \neq 0\} \cap \{\mathcal{R} \neq 0\},$$

except after entry into already-absorbed planar, lower-core, repeated-node, repeated-source, or quintic-collapse regimes. This is formalized below by the local clean continuation proposition, the positive-scale endpoint corollary, and Theorem 12.408. On the still-live nonmixed side, any genuinely new input would have to be RH-specific source- or matrix-level structure imposing stronger constraints on

$$\mathcal{R} = \frac{T_v}{K_v^2}.$$

Remark 12.396 (Live attack routes). At the present stage, the live routes are now quite sharply separated.

- (1) **Two-point branch: the residual burden is now split more sharply.** The stronger algebraic route is still Lemma 12.124: prove that the actual corrected two-pair cubic/quintic/septic package is analytic on small two-atom data and satisfies swap symmetry, one-amplitude collapse, and diagonal merger

$$\mathfrak{P}_2(a_1, h; a_2, h) = (a_1 + a_2)F_h.$$

The weaker exact route is Proposition 12.127, which would already remove exact close two-point bad cores on compact subsets of $\{c \neq 0\}$ from continuity and diagonal collapse of the actual corrected two-atom quotient germ.

What is now ruled out is the naive source-summed corrected-block route: its extracted one-atom finite-order coefficients are even in the source weight and therefore cannot supply the required linear merger law. On the mixed side, determinant-only and quotient-only arguments are likewise shear-blind. The genuinely new derivative-geometric lane is the exact projective λ -shear / secant-shadow route on a good overlap patch.

By Proposition 12.334, a close exact mixed two-point bad branch can accumulate only on

$$\{S = 0\} \cup \{K_v = 0\}, \quad S = \frac{\Delta_7}{c v_5}.$$

By Proposition 12.341, the part of $K_v = 0$ with $S \neq 0$ is already absorbed by the existing mixed low-order obstruction package. By Corollary 12.342, the conditional exact fixed-shear overlap lane on

$$\{S = 0\} \cap \{K_v \neq 0\}$$

transfers into the established nonmixed 4-point geometry.

So the only genuinely residual exact fixed-shear mixed corner is

$$\{S = 0\} \cap \{K_v = 0\}.$$

There the branch is quotient-diagonal / involutive, the quartic–sextic first rung is already quotient-injective, and the remaining exact fixed-shear burden is now best viewed as split into two sublanes.

The *close* exact fixed-shear lane is now scalar-local at theorem level: by Corollary 12.364, on a compact nonconstant quotient branch the close exact repeated-value carrier is a finite disjoint union of local fold half-arcs, one per odd zero of

$$\widehat{\mathcal{D}} = \widehat{\mathcal{Q}}',$$

with even zeros sterile; by Remark 12.365, the constant-quotient branch reduces, after affine gauge, to repeated values of a single planar scalar and also contributes only local fold half-arcs, with Corollary 12.352 ruling out the $\widehat{\mathcal{D}}(0) \neq 0$ basepoint. So this lane no longer admits a connected nonlocal close carrier; the remaining intrinsic obstruction there is equality of consecutive lobe amplitudes across these local fold half-arcs.

The *three-run overlap* fixed-shear lane should still be read conditionally as an inverse-branch-gap transport problem on an actual outer-overlap interval, namely in the regime

$$B_0 + B_2 > B_1.$$

On such an interval one has outer inverse branches and a gap function

$$\Delta(q) = \phi_2(q) - \phi_0(q).$$

Any interior critical point of Δ then yields a same-value / same-slope witness, and the current endgame fork is: either the transport-curvature scalar fails to descend across the outer runs, giving the first explicit nonlocal transport obstruction, or descent holds and the outer shells rigidify to exact translated replication. What is not yet settled at theorem level is an unconditional activation theorem from open overlap alone.

Thus the remaining two-point work is now concentrated in three very specific targets: finite-order source-level merger, quotient-diagonal continuity/diagonal collapse, and a transport/package theorem on the three-run overlap fixed-shear lane; the scalar-local ceiling on the close exact fixed-shear lane is already captured by Corollary 12.364.

- (2) **Three-point branch (derivative-geometric layer now stabilized).** The lifted three-point side is now the most mature local branch. Shrinking exact lifted 3-point circuits are tied to the osculating determinant and hence to

$$K_v = 0,$$

the local branch structure at finite-order K_v -zeros is completely classified by parity, and the odd-order branch is blockwise globally rigid on good v_5 -patches: infinite-order flatness is excluded, the branch continues uniquely on the maximal right monotone-slope block, and there is no interior common-tangent recurrence or source-side return. Inside the finite-core program this stabilizes the lifted $k = 3$ collinearity layer; any residual issue there is only boundary exit of the relevant source/target slope-overlap block, not an internal three-point propagation mechanism.

- (3) **Four-point branch (mixed side now captured by scalar exits).** On the nonmixed side, the exact-contact and repeated-source / first-loss packages are now organized by the scalar

$$\mathcal{R} = \frac{T_v}{K_v^2},$$

with one-sign intervals sterile and interval-persistent exact contacts forcing affine-planar collapse. On the mixed side, the centered-base local route is exhausted up to the clean branch, and the source-coupled and patch-loss frontiers are now reduced to lower-core / shifted-3-point / already-quintic handoffs. Accordingly, the mixed-global four-point side is no longer a search for new local mixed geometry: by Theorem 12.408, a genuinely mixed clean branch on a nonplanar good patch cannot persist entirely inside

$$\{T_v \neq 0\} \cap \{\mathcal{R} \neq 0\},$$

except after entry into already-absorbed regimes. So the four-point side is now best viewed as captured by the centered mixed local theorem, the three-point K_v -package, and the nonmixed $\mathcal{R}$ -package, rather than as an independent mixed-global carrier problem.

- (4) **Late- v_5 degeneration is now only a handoff mechanism.** If the v_5 -patch fails and v_5 starts late, the valuation form of the septic transport constraint forces matching late vanishing of the septic transport channel v_{7,K_1} . Thus failure of the v_5 -route is not a free local cancellation but a coupled quintic-septic degeneration scheme, and in the current architecture it should be read only as a handoff into lower-core or already-quintic geometry, not as an independent mixed branch lane.
- (5) **Four-point nonmixed branch via source-level control of odd $\mathcal{W}_{\text{src}}$ -zeros.** The local nonmixed derivative geometry is now explicit: exact contacts force zeros of $\mathcal{R}$, one-sign $\mathcal{R}$ -intervals are sterile, recurrent exact contacts force affine-planar collapse, and isolated zeros are classified by parity. At source level, the same structure is encoded by the canonical Wronskian

$$\mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda,$$

which differs from $\mathcal{R}$ only by a nonvanishing analytic factor on a nonmixed arc.

For fixed source point a , repeated-source exact contacts are controlled by the quadratic-envelope scalar

$$\Gamma_a(x) = \frac{Y(x) - Y(a) - Y'(a)(x - a)}{S(x) - S(a) - S'(a)(x - a)},$$

equivalently by the source quotient

$$\Psi_a(h) = \frac{D_{u_5}(a, h)}{D_\lambda(a, h)},$$

with derivative governed by the osculatory 4×4 determinant $\mathfrak{D}(a, h)$. By Proposition 12.292, this repeated-source / first-loss mechanism is not an independent obstruction: it occurs if and only if one diagonally approaches an odd-order zero of $\mathcal{R}$, equivalently of $\mathcal{W}_{\text{src}}$.

The remaining theorem-level task on this side is therefore the global/source-level exclusion or absorption of odd zeros of $\mathcal{W}_{\text{src}}$ on opened nonmixed arcs. In the unified mixed / nonmixed picture, this is exactly the scalar $\mathcal{R}$ -lane that absorbs the interval-persistent $A = 0$ collapse from the mixed side.

Remark 12.397 (Stratified lifted local picture). The lifted local obstruction now splits into one closed regime and two genuinely live regimes.

First, exact shrinking lifted 3-point circuits concentrate into

$$K_v = 0.$$

On good v_5 -patches, this side is now blockwise globally controlled: there is no infinite-order flatness, even-order K_v -zeros are sterile, odd-order K_v -zeros seed exactly one local opposite-side branch, and that branch continues uniquely on the maximal right monotone-slope block with no interior common tangent, no source-side return, and no interior accumulation. So the three-point side contributes only possible boundary exit, not an internal recurrence mechanism.

Second, exact shrinking lifted 4-point circuits concentrate into

$$T_v = 0.$$

On the nonmixed side $K_v \neq 0$, the local derivative-side picture is now explicit. Exact contacts force zeros of

$$\mathcal{R} = \frac{T_v}{K_v^2},$$

isolated zeros are classified by parity, one-sign $\mathcal{R}$ -intervals are exactly-contact sterile, and recurrent exact contacts force affine-planar collapse. So the nonmixed 4-point side is no longer a vague local branching problem; what remains there is global/source-level control of which odd zeros of $\mathcal{R}$ actually occur on genuine opened arcs.

Third, on the mixed side $K_v = T_v = 0$, the low-order obstruction list is now finite and explicit:

$$(m, n) = (3, 4), (3, 6), (4, 6),$$

equivalently

$$(r, s) = (1, 2), (1, 4), (2, 5),$$

equivalently

$$(r, \text{ord}_0 \mathcal{Z}) = (1, 1), (1, 3), (2, 2).$$

At the pair level, however, the centered quintic fixed-sector package

$$A_5^{\dagger}(s) = u_5(s)I + v_5(s)S$$

is now known not to vanish identically on the centered diagonal branch. Near the centered base, the right-pincer package further reduces the mixed local picture: every late- v_5 branch is either locally sterile on the automatic u_5 -patch or already absorbed into the planar quintic-circuit regime, so the only genuinely new mixed local branch left there is the clean branch

$$\text{ord}_s v_5 = 1.$$

Thus the remaining lifted issues are now stratified much more sharply. On the nonmixed side, the live question is the global/source-level control of odd zeros of $\mathcal{R}$ on genuine opened arcs. On the mixed side, the centered local issue is the finite scalar descent

$$c(-s) = c(s) \pmod{s^6},$$

while the clean-patch global side has already been compressed to the scalar capture statement of Theorem 12.408. So the mixed four-point branch should no longer be viewed as an independent open geometric carrier problem away from the scalar exits.

Remark 12.398 (Current geometric bottleneck). The remaining geometric bottleneck is now split into two qualitatively different layers.

On the one-pair geometric layer, the obstruction is no longer bulk $W = 0$ geometry, and it is no longer local adjacent pair-pair birth across a generic turning boundary. On the $W = 0$, $M \neq 0$

side, Proposition 12.59 excludes open-interval flatness and Theorem 12.107 shows that every finite-order isolated $W = 0$ boundary is locally sterile for pair-pair birth. On the turning-boundary side, Theorem 12.96 removes actual adjacent pair-pair lines unconditionally.

At the lifted order-7 level, the canonical derivative-geometric determinants

$$K_v \quad \text{and} \quad T_v$$

govern shrinking exact lifted 3- and 4-point circuits on $v_5 \neq 0$ patches. On the 3-point side, this layer is stabilized: shrinking exact circuits concentrate into $\{K_v = 0\}$, finite-order isolated zeros are classified by parity, and the odd-order branch is blockwise globally rigid on good v_5 -patches.

On the 4-point side, exact shrinking lifted 4-point circuits concentrate into

$$\{T_v = 0\},$$

again with uniform small-diameter exclusion away from that zero set.

On the nonmixed side $K_v \neq 0$, generic odd-order T_v -zeros can still carry preexisting close branches, so the nonmixed branch remains genuinely live and now asks for global/source-level control of which odd zeros of

$$\mathcal{R} = \frac{T_v}{K_v^2}$$

actually arise on opened arcs.

By contrast, on the mixed side $K_v = T_v = 0$, the local geometry near the centered base is no longer the main bottleneck. There the right-pincer package shows that every late- v_5 branch is either locally sterile on the automatic u_5 -patch or already absorbed into the defect-free planar quintic-circuit regime, so the only genuinely new mixed local branch left is the clean branch

$$V_1 \neq 0.$$

Accordingly, the mixed bottleneck is no longer a diffuse local codimension-two analysis. At the local level it is the finite scalar descent

$$c(-s) = c(s) \pmod{s^6}$$

on that clean branch; at the global level, after the source-coupled and patch-loss absorptions, the clean-patch mixed problem has already compressed to the scalar alternatives

$$T_v = 0 \quad \text{or} \quad \mathcal{R} = 0,$$

as formalized in Theorem 12.408.

So the missing lifted ingredient is no longer another generic local geometric exclusion theorem. On the mixed side, the centered-base local package, the source-coupled exclusion, the patch-loss reductions, and the nonmixed $\mathcal{R}$ -handoff now combine to give the clean theorem-level conclusion formalized below in Theorem 12.408: every maximal genuinely mixed clean branch on a nonplanar good patch is forced either into the centered-base scalar exit

$$T_v = 0$$

or into the nonmixed scalar exit

$$\mathcal{R} = 0,$$

up to already-absorbed planar, lower-core, repeated-node, repeated-source, and quintic-collapse regimes. Thus the still-live frontiers are no longer mixed four-point geometry, but rather the nonmixed/source-level control of odd zeros of $\mathcal{R}$ and the separate two-point program.

Remark 12.399 (Current state of the argument). The local one-pair analysis now consists of the canonical triple

$$(c(h), v_5(h), \Delta_7(h)),$$

together with the derived quantities

$$\begin{aligned} M(h) &= c(h)v_5'(h) - c'(h)v_5(h), & N(h) &= c(h)\Delta_7'(h) - c'(h)\Delta_7(h), \\ L(h) &= v_5(h)\Delta_7'(h) - v_5'(h)\Delta_7(h), & W(h) &= \det(F(h), F'(h), F''(h)). \end{aligned}$$

On the region where $v_5 M \neq 0$, the projective graph variables

$$\mu = \frac{c}{v_5}, \quad \lambda = \frac{\Delta_7}{v_5}$$

identify the normalized curve with a graph $\lambda = \lambda(\mu)$, and

$$\lambda_{\mu\mu} = \frac{v_5^3}{M^3} W, \quad \beta := \lambda_\mu = -\frac{L}{M}, \quad T(h) = \left(-\frac{L(h)}{M(h)}, \frac{N(h)}{M(h)} \right).$$

Thus the global line-incidence problem is equivalently a problem about the recurrence of tangent slopes β , or about vertical-line re-entry on the dual tangent-line curve T .

Locally, both generic boundary types are now under control. At finite-order $W = 0$, $M \neq 0$ boundaries there is no local pair-pair birth, while at generic turning boundaries actual adjacent pair-pair lines are excluded unconditionally. The stronger symmetric directional-kernel theorem remains available only conditionally on zero-capture and now serves as a sharper but nonessential alternative route.

Accordingly, the equivalent slope/tangent/dual direct-slice formulations remain the correct geometric dictionary, but they are no longer themselves the live theorem target. The explicit direct provenance formula gives

$$\phi = A q^{(7)} \quad (A = \kappa \eta_2),$$

and on any generic one-pair block this direct slice is fully universal (Proposition 13.8). Moreover, even after imposing any finite family of linear provenance or matching conditions on q , direct-slice recurrence geometry remains formally realizable (Corollary 13.9).

Thus the remaining gap is not an abstract global no-reentry theorem for the scalar graph ϕ by itself. It is the search for a genuinely non-universal provenance law for q , or an external mixed/canonical input, that cuts down the allowed direct-slice graphs on an entire block.

On the lifted 4-point side, the mixed low-order obstruction list is now both finite and canonical:

$$(r, \text{ord}_0 \mathcal{Z}) \in \{(1, 1), (1, 3), (2, 2)\},$$

equivalently

$$(r, s) = (1, 2), (1, 4), (2, 5),$$

or

$$(m, n) = (3, 4), (3, 6), (4, 6).$$

The centered Taylor package already provides the explicit coefficients

$$U_j, V_j, \widehat{U}_j, \widehat{V}_j, D_2, D_4, D_6,$$

and the tagged quintic witness now shows that

$$A_5^{\dagger}(s) \neq 0$$

on the centered diagonal branch.

On the mixed side, however, the local theorem picture is now sharper than a mere punctured-arc patch dichotomy, and the global side is correspondingly compressed. Near the centered base, the late- v_5 sector is already removed by the u_5 -patch right-pincer package as either locally sterile or planar-lift and hence already absorbed into the defect-free quintic-circuit regime. The only genuinely new mixed local branch there is the clean branch $V_1 \neq 0$, and that branch is excluded by the canonical scalar descent together with the centered good- v_5 -patch parity package. Moreover, after the source-coupled and patch-loss reductions, the sufficiently small mixed-global problem is no longer a search for a new mixed carrier lane. At the theorem level, the clean small-branch mixed problem has already compressed to the scalar alternatives

$$T_v = 0 \quad \text{or} \quad \mathcal{R} = 0.$$

Here the $K_v = 0$ side no longer survives as an independent mixed carrier on the clean small-branch four-point side, but on the additional conditional exact fixed-shear overlap slice it still

leaves a narrower residual two-point burden. That residual slice is not a new mixed local carrier: modulo constant shear, its pointwise quotient germ is already governed by

$$\mathcal{D} = \frac{\mathcal{W}_{\text{src}}}{J_\lambda^2},$$

or equivalently on the involution quotient by

$$\widehat{\mathcal{D}}(u) = \widehat{Q}'(u).$$

Moreover, by Corollary 12.255, the ordinary local repeated-node tangent-contact mechanism on a good v_5 -patch is necessarily opposite-side, so this quotient-diagonal residual lane cannot locally merge with the standard lifted 3-point branch geometry. Thus the genuinely mixed small-branch problem is best read as mixed-to-scalar capture together with a separate quotient-diagonal residual transport problem, not as another free mixed carrier geometry.

In parallel, the finite-core endgame now splits into three layers.

First, on the two-pair side, the strengthened invariant

$$\widehat{\Psi}(h) = \left(\frac{u_5(h)}{c(h)}, \frac{v_5(h)}{c(h)}, \frac{\Delta_7(h)}{c(h)^2} \right)$$

still gives the correct algebraic bridge, and on good $M \neq 0$ patches the quantitative two-pair coincidence theorem forces bad two-point cores into a defect-scale close regime. The minimal finite-order source criterion Lemma 12.124 still isolates the exact swap / one-amplitude / diagonal-merger package that would imply the desired quadratic interaction factorization. However, the naive source-summed corrected block model does not prove this criterion: along a weighted one-atom path its extracted finite-order coefficients are even analytic in the source weight, so the required linear merger law cannot come from that model alone.

At present the two-point side is better organized by the three-front compression of Remark 12.211 than by the older language of “three conditional lanes.”

The finite-order source-level merger criterion of Lemma 12.124 remains one possible local route on good patches, but the genuinely residual mixed burden is now the quotient-diagonal exact fixed-shear corner

$$(S = 0, K_v = 0).$$

There the projective λ -shear compactness and secant-shadow analysis identify the correct residual geometry, while the collision-cancellation chart, the no-finite-local-rescue picture, and the quotient-transport package show that the remaining issue is not a missing local branch taxonomy.

So the live downstream two-point targets are now the following: a good-patch edge-law theorem for the actual corrected cubic/quintic defects in the collision-cancellation chart; a quotient-visible transport/state-local closure on the exact fixed-shear corner; or a provenance-sensitive package theorem strong enough to force coincidence/functoriality at the collision/cancellation locus. In particular, the current residual burden should be read as local blow-up regularity plus quotient transport plus package-level coincidence, not as a fourth independent mixed lane.

Second, on the lifted three-point side, shrinking exact circuits now concentrate into $K_v = 0$, and this concentration is explicitly derivative-geometric: by the osculating-determinant expansion, repeated-node limits become tangent-line contacts for the planar lifted curve

$$\Lambda_v = (x, S).$$

On good v_5 -patches this side is now blockwise globally controlled: K_v has no infinite-order zeros; even-order zeros are sterile; odd-order zeros carry exactly one local branch; and that branch continues uniquely on the maximal right monotone-slope block with no interior common-tangent event or source-side return. So the three-point side is reduced to boundary exit of the relevant slope-overlap block, not an internal recurrence mechanism. Inside the finite-core reduction, this is exactly the stabilization of the lifted $k = 3$ collinearity layer.

Third, on the lifted four-point side, shrinking exact circuits now concentrate into $T_v = 0$, repeated-node limits become tangent-plane contacts, and the local geometry splits into a nonmixed and a mixed regime.

On the nonmixed side, the local derivative-side theory is now explicit: exact contacts force zeros of $\mathcal{R} = T_v/K_v^2$, isolated zeros are classified by parity, one-sign $\mathcal{R}$ -intervals are exactly-contact sterile, and recurrent exact contacts force affine-planar collapse. At source level, the same structure is encoded by the gauge-invariant Wronskian

$$\mathcal{W}_{\text{src}} = J_\lambda J'_{u_5} - J_{u_5} J'_\lambda,$$

which differs from $\mathcal{R}$ only by a nonvanishing analytic factor on a nonmixed arc. For fixed source point a , repeated-source exact contacts factor through the quadratic-envelope scalar

$$\Gamma_a(x) = \frac{Y(x) - Y(a) - Y'(a)(x - a)}{S(x) - S(a) - S'(a)(x - a)},$$

equivalently through the source quotient

$$\Psi_a(h) = \frac{D_{u_5}(a, h)}{D_\lambda(a, h)},$$

with derivative governed by the osculatory source determinant $\mathfrak{D}(a, h)$. So the remaining nonmixed burden is global/source-level control of which odd zeros of $\mathcal{W}_{\text{src}}$, equivalently of $\mathcal{R}$, occur on a genuine opened nonmixed arc.

On the mixed side, the dangerous low-order obstruction list is the finite canonical set

$$\kappa_{1,1}, \quad \kappa_{1,3}, \quad \kappa_{2,2},$$

equivalently the order patterns

$$(r, \text{ord}_0 \mathcal{Z}) = (1, 1), (1, 3), (2, 2).$$

The remaining theorem-level task there is no longer local patch classification, and on the clean-patch side it is no longer an open global capture problem either: the sufficiently small mixed-global closure is now formalized by Theorem 12.408. Thus the mixed side should now be read as split into two parts: the centered clean-branch scalar descent

$$c(-s) = c(s) \pmod{s^6},$$

which is the remaining local centered-base task, and the already-organized mixed-to-scalar handoff showing that outside absorbed regimes every maximal genuinely mixed clean branch is forced into

$$T_v = 0 \quad \text{or} \quad \mathcal{R} = 0.$$

So any genuinely new input beyond the present lifted-variable package would now have to be RH-specific source-/matrix-level structure on the scalar side, not a new mixed local geometry theorem.

Moreover, Theorem 12.240 shows that no new sufficiently small local mixed 4-point obstruction remains near the centered diagonal base, so any external mixed input capable of constraining the direct-slice/provenance lane must be genuinely global/nonlocal rather than a new local mixed septic-jet pattern.

Remark 12.400 (Post-draft findings: established, provisional, and live). This remark summarizes ledger findings that are now partly formalized in the surrounding propositions, corollaries, and theorems, and partly retained as live WIP for the remaining package/frontier obligations.

First, on the two-point branch, the naive source-summed corrected block model still cannot prove the exact finite-order merger criterion: along a weighted one-atom path the extracted finite-order coefficients are even analytic in the source weight, so the needed linear merger law does not come from that model alone.

At the same time, the residual mixed picture is now much cleaner. The overlap bad set is fully split:

$$(K_v = 0, S \neq 0)$$

is already absorbed by the mixed low-order package,

$$(S = 0, K_v \neq 0)$$

is already transferred into the established nonmixed 4-point geometry, and only the exact fixed-shear corner

$$\{S = 0\} \cap \{K_v = 0\}$$

remains genuinely live.

On that residual corner the branch is quotient-diagonal / involutive,

$$\Phi(t) = \Phi(-t), \quad Q(t) = \widehat{Q}(t^2), \quad \widehat{D}(u) = \widehat{Q}'(u).$$

Moreover, Proposition 12.354 shows that every finite parity-normalized transport jet descends to the quotient variable $u = t^2$, while Corollary 12.355 shows that on the quartic–sextic rung the finite quotient-visible transport state has local rank one, and Corollary 12.356 rules out any third local reset inside that descended state.

So the residual exact fixed-shear burden is no longer a search for another local mixed carrier or hidden finite-order transport variable. It has become a problem of collision-cancellation regularity, quotient transport, and package/provenance-sensitive coincidence. In particular, any still-live obstruction must come from the first genuinely relational or provenance-sensitive two-atom input that survives after quotient-visible transport descent, or from non-finite-order dependence.

Moreover, by Corollary 12.255, every sufficiently small off-diagonal repeated-node tangent-contact branch on a good v_5 -patch is necessarily opposite-side. Thus the quotient-diagonal exact fixed-shear residual lane cannot locally merge with the ordinary lifted 3-point branch mechanism.

There is now also a sharper internal split of the exact fixed-shear residual lane itself, and one side of that split has now been promoted to theorem level. On the close exact side, the scalar-local compression is no longer a work-in-progress expectation: by Corollary 12.364, on a compact nonconstant quotient branch the close exact repeated-value carrier is a finite disjoint union of local fold half-arcs, one per odd zero of

$$\widehat{D} = \widehat{Q}',$$

with even zeros sterile; by Remark 12.365, the constant-quotient branch also reduces, after affine gauge, to repeated values of a single planar scalar and contributes only local fold half-arcs. So the close exact carrier no longer admits a connected nonlocal close carrier, and the remaining intrinsic obstruction on that sublane is equality of consecutive lobe amplitudes across these local fold half-arcs.

On the three-run overlap side, by contrast, the endgame is still conditional. The residual overlap regime is

$$B_0 + B_2 > B_1,$$

not

$$B_1 > B_0 + B_2.$$

In that actual outer-overlap regime, the quotient-visible problem may be reduced to the inverse-branch-gap transport problem on the two outer runs. Writing

$$\Delta(q) := \phi_2(q) - \phi_0(q),$$

any interior critical point of Δ gives a same-value / same-slope witness, and the live endgame fork becomes: either the transport-curvature scalar fails to descend across the outer runs, giving the first explicit nonlocal transport obstruction, or descent holds and the outer shells rigidify to exact translated replication. An unconditional activation theorem from open overlap alone has not yet been established. So the residual exact fixed-shear lane is no longer a vague hidden branch problem; it has split into a close scalar-local lane, now controlled at theorem level by Corollary 12.364, and a separate three-run overlap-transport lane that remains the live conditional target.

Second, on the lifted four-point branch, the explicit low-order obstruction list has been reduced to the six coefficients

$$z_2, z_4, z_6, \kappa_{1,1}, \kappa_{1,3}, \kappa_{2,2},$$

or equivalently to the order patterns

$$\text{ord}_0 \mathcal{R} \in \{1, 3, 5\}, \quad (r, \text{ord}_0 \mathcal{Z}) \in \{(1, 1), (1, 3), (2, 2)\}.$$

Third, the centered symmetric-placement Taylor package now gives explicit septic coefficients

$$D_2, \quad D_4, \quad D_6,$$

explicit bridge coefficients

$$X_1, X_3, X_5, \quad L_1, L_3, L_5,$$

and a punctured-neighborhood exclusion at the diagonal base.

Fourth, the centered fixed-sector quintic package

$$A_5^{\mathfrak{f}}(s) = u_5(s)I + v_5(s)S$$

is now firmly established as the correct pair-level object on the mixed side. The centered tagged witness survives specialization strongly enough to show

$$U_1 \neq 0, \quad u_5(s) \neq 0, \quad A_5^{\mathfrak{f}}(s) \neq 0,$$

so the centered quintic pair cannot totally collapse.

Fifth, the right-pincer package now sharpens the mixed local picture near the centered base: every late- v_5 branch is either locally sterile on the automatic u_5 -patch or already absorbed into the planar defect-free quintic-circuit regime. So the only genuinely new mixed local branch left there is the clean branch

$$V_1 \neq 0.$$

Sixth, on that clean branch the remaining theorem has become very small: the finite scalar descent

$$c(-s) = c(s) \pmod{s^6}, \quad \text{equivalently} \quad C_1 = C_3 = C_5 = 0.$$

Once this holds, the centered good- v_5 -patch bridge becomes formal:

$$Y = \frac{u_5}{c}, \quad S = \frac{\Delta_7}{cv_5}$$

are odd through the required order, and the entire explicit mixed low-order bad list

$$(r, \text{ord}_0 \mathcal{Z}) \in \{(1, 1), (1, 3), (2, 2)\}$$

is excluded there.

Seventh, the relation between the mixed and nonmixed 4-point sides is now structurally sharper. On a good v_5 -patch, an exact mixed relation

$$AY + Bx + CS + D = 0$$

with

$$Y = \frac{u_5}{c}, \quad x = \frac{v_5}{c}, \quad S = \frac{\Delta_7}{cv_5}$$

and $A = 0$ is already exact affine-planarity for the planar curve (x, S) ; by Remark 12.394, this is precisely the nonmixed $\mathcal{R}$ -lane. More generally, the sheared four-zero reduction shows that every exact genuinely mixed 4-point circuit on a good v_5 -patch already forces either

$$K_v = 0 \quad \text{or} \quad \mathcal{R} = 0$$

in its convex hull. So the mixed-global problem should now be viewed as a mixed-to-scalar capture problem rather than as a separate branch geometry.

Eighth, the source-coupled and patch-loss frontiers have both been demoted to handoff mechanisms. In the sufficiently small regime, the source-coupled lower-core walls

$$D_\lambda = 0, \quad \mathfrak{D} = 0$$

are not pass-through points. On the patch side, the quintic projective lifts

$$\tilde{Q}_v = [cv_5 : u_5v_5 : v_5^2 : \Delta_7], \quad \tilde{Q}_u = [cu_5 : u_5^2 : u_5v_5 : -\Delta_7]$$

show that single-chart failures are removable after recharting, while the genuine common boundary

$$u_5 = v_5 = 0$$

drops either to pure quintic-direction collapse or to lower-core 3-point / repeated-node quotient geometry. Thus

$$cMv_5 = 0$$

should be read as lower-core handoff, not as a fourth independent mixed-global lane.

Ninth, after the centered-base local exhaustion theorem and the preceding handoffs are combined, the sufficiently small nonplanar mixed branch has no independent carrier away from

$$T_v = 0 \quad \text{and} \quad \mathcal{R} = 0.$$

Equivalently, there is no sufficiently small genuinely mixed 4-point branch contained entirely in

$$\{T_v \neq 0\} \cap \{\mathcal{R} \neq 0\}$$

on a nonplanar good patch, except after entry into already-absorbed affine-planar or lower-core regimes.

So the present frontier is no longer a generic local geometry problem. On the residual exact fixed-shear overlap slice

$$K_v = 0,$$

the pointwise quotient structure has already compressed to the single transport scalar

$$\widehat{\mathcal{D}}(u) = \widehat{Q}'(u), \quad u = t^2.$$

The quartic–sextic rung is the first explicit case, and there the quotient branch is locally one-sheeted. Thus the residual $K_v = 0$ lane is best viewed as a quotient-diagonal / involutive transport problem rather than as a new mixed local carrier.

At the same time, this pointwise compression does *not* yet prove that the full transported overlap / critical-value state is quotient-visible. What is proved inside the quotient-visible exact fixed-shear reduction is narrower and sharper: on a connected nonconstant quotient branch, Proposition 12.348 shows that nonadjacent re-entry of regular values can arise only through a three-run lobe-amplitude pattern, and Corollary 12.349 identifies the only surviving inequality

$$B_1 \geq B_0 + B_2.$$

Moreover, on the first live quartic–sextic rung, Corollary 12.352 shows that the basepoint is locally one-sheeted in the quotient variable and does not itself contribute such a lobe obstruction. Accordingly, the live two-point burden on the $K_v = 0$ slice is now a no-hidden-reset / state-visibility problem for the actual corrected overlap state, together with the alternative source-level merger route and the already-separated higher-core issues.

Remark 12.401 (Two-point burden after mixed four-point closure). With the clean mixed four-point side now compressed to the scalar exits

$$T_v = 0 \quad \text{or} \quad \mathcal{R} = 0,$$

the mixed two-point side should no longer be read as an independent frontier of mixed local geometry.

The overlap bad set has already split as follows:

$$(K_v = 0, S \neq 0)$$

is absorbed by the mixed low-order obstruction package,

$$(S = 0, K_v \neq 0)$$

is transferred into the established nonmixed 4-point geometry, and only the exact fixed-shear residual corner

$$\{S = 0\} \cap \{K_v = 0\}$$

remains genuinely live.

On that residual corner, the quotient-visible transport package is now much more rigid than a single pointwise scalar. By Proposition 12.354, every finite parity-normalized transport jet

descends to the quotient variable $u = t^2$. On the quartic–sextic rung, Corollary 12.355 shows that every finite quotient-visible transport state is locally a function of Q alone, and Corollary 12.356 shows that no third local reset survives there.

So the residual exact fixed-shear corner is no longer best viewed as an unresolved mixed local curvature-branch problem. If the actual corrected defect/package is state-local in the descended transport state, then the corner closes by Proposition 12.357 or even by Corollary 12.358. Accordingly, any still-live input on the two-point side would have to come from the first genuinely relational or provenance-sensitive two-atom datum that survives after quotient-visible transport descent, or else from non-finite-order dependence.

Remark 12.402 (Residual mixed two-point lane: structural gain and present limitation). The present two-point picture now has three genuine structural gains on the exact fixed-shear residual corner.

First, after the off- $K_v = 0$ absorption results and the transfer of the simple-septic-zero slice into the nonmixed 4-point theory, the derivative-geometric mixed residue is sharply concentrated on the exceptional corner

$$\{S = 0\} \cap \{K_v = 0\}.$$

Second, by Proposition 12.354, every finite parity-normalized transport jet is quotient-visible in the variable

$$u = t^2.$$

So the residual corner is no longer a free two-sided t -state even at the finite transport-jet level.

Third, on the quartic–sextic rung, Corollary 12.355 shows that this descended finite transport state has local rank 1, and Corollary 12.356 shows that no finite-order hidden reset survives beyond the involution pair $t \leftrightarrow -t$.

What is *not* yet proved is that the actual corrected two-pair defect or package is already state-local in this descended transport state. If that state-locality held, then the residual branch would already be excluded by Proposition 12.357, or would collapse identically by Corollary 12.358.

Thus the live issue on the mixed residual side is now no longer local classification and no longer hidden finite-order transport dynamics. The remaining question is whether the first surviving obstruction is already quotient-visible, or instead comes from genuinely relational / provenance-sensitive two-atom structure or from non-finite-order dependence.

Proposition 12.403 (u_5 -chart exceptional-point handoff). *Work near a collapsing configuration with $u_5 \neq 0$, and use the projective lift*

$$\tilde{Q}_u = [cu_5 : u_5^2 : u_5v_5 : -\Delta_7].$$

Let

$$q_\nu = [0 : 1 : 0 : \nu]$$

be an exceptional limiting point on the exceptional line. Then every limiting hyperplane through q_ν induces the exact lower-core relation

$$Dc + Bv_5 + C(\lambda - \nu u_5) = 0, \quad \lambda = -\frac{\Delta_7}{u_5}.$$

Accordingly:

- one-node collapse hands off to exact shifted lower-core 3-point geometry;
- two-node collapse hands off to a repeated-node lower-core wall;
- if $u_5 = 0$ as well, one is on the common quintic-collapse locus already treated separately.

So the u_5 -chart side of patch loss contributes no independent mixed-global carrier lane.

Remark 12.404 (Shifted lower-core invariance on the u_5 -side). The shift

$$\lambda \longmapsto \lambda - \nu u_5$$

does not alter the derivative-geometric lower-core package: it changes only the zeroth-order affine offset, not the relevant repeated-node/tangent-contact derivative structure. Thus the handoff in Proposition 12.403 is genuinely into the existing lower-core package, not into new geometry.

Proposition 12.405 (Local clean continuation off the explicit walls). *Work on a nonplanar good v_5 -patch and write*

$$Y = \frac{u_5}{c}, \quad x = \frac{v_5}{c}, \quad S = \frac{\Delta_7}{c v_5}.$$

Let

$$\Phi_\gamma := Y - \gamma S, \quad F_{\alpha, \beta, \gamma}(x) := \Phi_\gamma(x) - \alpha - \beta x.$$

Suppose

$$h_1 < h_2 < h_3 < h_4$$

is an exact genuinely mixed clean 4-point configuration, so

$$F_{\alpha, \beta, \gamma}(x(h_i)) = 0 \quad (i = 1, 2, 3, 4),$$

the four roots $x(h_i)$ are distinct and simple, and the carrying hyperplane is genuinely mixed ($A \neq 0$). Assume moreover that the configuration does not lie on any of the walls

$$A = 0, \quad D_\lambda = 0, \quad \mathfrak{D} = 0, \quad cMv_5 = 0, \quad T_v = 0, \quad \mathcal{R} = 0.$$

Then this configuration is an interior point of a smooth local solution manifold of exact genuinely mixed clean 4-point configurations. In particular, it cannot be a terminal point of a maximal clean mixed branch.

Proof. For

$$\mathcal{F}_i(h_1, h_2, h_3, h_4, \alpha, \beta, \gamma) := Y(h_i) - \gamma S(h_i) - \alpha - \beta x(h_i),$$

the Jacobian with respect to (h_1, h_2, h_3, h_4) is diagonal:

$$\frac{\partial \mathcal{F}_i}{\partial h_j} = \delta_{ij} (Y'(h_i) - \gamma S'(h_i) - \beta x'(h_i)).$$

Since

$$Y'(h_i) - \gamma S'(h_i) - \beta x'(h_i) = x'(h_i) F'_{\alpha, \beta, \gamma}(x(h_i)),$$

the diagonal entries are nonzero on a good v_5 -patch and at simple roots. So the Jacobian is invertible, and the claim follows from the implicit function theorem. $\square$

Corollary 12.406 (Positive-scale endpoints lie on an absorbed wall or a scalar exit). *Let Γ be a maximal connected genuinely mixed clean 4-point branch on a nonplanar good v_5 -patch. Suppose $p \in \bar{\Gamma}$ is a terminal point at which the relevant node scale stays bounded below by a positive constant. Then p lies on one of*

$$A = 0, \quad D_\lambda = 0, \quad \mathfrak{D} = 0, \quad cMv_5 = 0, \quad T_v = 0, \quad \mathcal{R} = 0.$$

In particular, after the previously established handoffs are invoked, a positive-scale terminal point cannot furnish a new genuinely mixed continuation mechanism: it must either lie in the centered-base capture set $\{T_v = 0\}$, lie in the nonmixed affine-planar scalar set $\{\mathcal{R} = 0\}$, or already belong to one of the absorbed planar/lower-core/quintic-collapse regimes.

Proof. If p lay off the listed wall set, then the limiting exact genuinely mixed clean configuration at p would satisfy the hypotheses of Proposition 12.405. Hence the branch would extend smoothly through p , contradicting maximality. Therefore a positive-scale terminal point must lie on one of

$$A = 0, \quad D_\lambda = 0, \quad \mathfrak{D} = 0, \quad cMv_5 = 0, \quad T_v = 0, \quad \mathcal{R} = 0.$$

The final sentence is then exactly the previously established absorption package: $A = 0$ is the nonmixed affine-planar handoff (Remark 12.394), the source-coupled walls $D_\lambda = 0$ and $\mathfrak{D} = 0$ are locally non-carrying in the sufficiently small regime (Remark 12.400), and patch loss $cMv_5 = 0$ is absorbed into lower-core / shifted-3-point / quintic-collapse geometry (Proposition 12.403 and the surrounding post-draft findings package). $\square$

Proposition 12.407 (Small-scale non-scalar exits are absorbed). *Let Γ be a sufficiently small genuinely mixed clean 4-point branch on a nonplanar good v_5 -patch. Assume*

$$\Gamma \cap \{T_v = 0\} = \emptyset, \quad \Gamma \cap \{\mathcal{R} = 0\} = \emptyset.$$

Then Γ cannot continue through any of the non-scalar exit walls

$$A = 0, \quad D_\lambda = 0, \quad \mathfrak{D} = 0, \quad cMv_5 = 0$$

as an independent genuinely mixed clean branch.

More precisely:

- (1) *if $A = 0$ persists on an interval, then Γ has already entered the nonmixed affine-planar $\mathcal{R}$ -package (Remark 12.394, Proposition 12.267, Corollary 12.268);*
- (2) *if $D_\lambda = 0$ or $\mathfrak{D} = 0$ occurs in the sufficiently small regime, then this lies on the source-coupled lower-core marked set, which is locally non-carrying and not a pass-through mechanism (Remark 12.400);*
- (3) *if $cMv_5 = 0$ occurs, then the branch is absorbed into lower-core / shifted-3-point / quotient repeated-node / quintic-collapse geometry (Remark 12.400 and Proposition 12.403).*

Accordingly, no sufficiently small genuinely mixed clean branch can remain an independent mixed carrier while meeting only non-scalar exit walls.

Proof. The $A = 0$ case is exactly the nonmixed affine-planar handoff by Remark 12.394; interval-persistent or recurrent exact contacts are then absorbed by Proposition 12.267 and Corollary 12.268.

The source-coupled walls $D_\lambda = 0$ and $\mathfrak{D} = 0$ are, in the sufficiently small regime, only source-coupled lower-core marked points and are locally non-carrying; see Remark 12.400.

Finally, patch loss $cMv_5 = 0$ has already been reduced to lower-core / shifted-3-point / quotient repeated-node / quintic-collapse handoffs; see Remark 12.400 and Proposition 12.403.

So none of the listed non-scalar walls can provide an independent genuinely mixed clean continuation mechanism under the stated hypotheses. $\square$

Theorem 12.408 (Global mixed-to-scalar capture on the clean patch). *Let Γ be a maximal connected genuinely mixed clean 4-point branch on a nonplanar good v_5 -patch. Then Γ cannot persist as an independent genuinely mixed branch entirely inside*

$$\{T_v \neq 0\} \cap \{\mathcal{R} \neq 0\}.$$

Equivalently, every such maximal branch is forced either into

$$T_v = 0$$

or into

$$\mathcal{R} = 0,$$

up to the already-absorbed planar, lower-core, repeated-node, repeated-source, and quintic-collapse regimes.

Proof. Assume Γ is a maximal connected genuinely mixed clean 4-point branch on a nonplanar good v_5 -patch and that

$$\Gamma \cap \{T_v = 0\} = \emptyset, \quad \Gamma \cap \{\mathcal{R} = 0\} = \emptyset.$$

We show that Γ cannot persist as an independent genuinely mixed branch.

If a terminal point of Γ has positive scale, then Corollary 12.406 places it on one of

$$A = 0, \quad D_\lambda = 0, \quad \mathfrak{D} = 0, \quad cMv_5 = 0, \quad T_v = 0, \quad \mathcal{R} = 0.$$

The last two are excluded by hypothesis. The remaining three are already absorbed: $A = 0$ is the nonmixed affine-planar $\mathcal{R}$ -lane (Remark 12.394), the source-coupled walls $D_\lambda = 0$ and $\mathfrak{D} = 0$ are locally dead as pass-through mechanisms in the sufficiently small regime, and patch loss $cMv_5 = 0$ is absorbed into the lower-core / shifted-3-point / quintic-collapse package recorded in Remark 12.400 and Proposition 12.403. So no positive-scale terminal point yields an independent mixed continuation lane.

If instead the relevant scale along Γ shrinks, then Proposition 12.407 applies. Indeed, under the standing assumptions

$$\Gamma \cap \{T_v = 0\} = \emptyset, \quad \Gamma \cap \{\mathcal{R} = 0\} = \emptyset,$$

no sufficiently small genuinely mixed clean branch can continue through any of the non-scalar exit walls

$$A = 0, \quad D_\lambda = 0, \quad \mathfrak{D} = 0, \quad cMv_5 = 0$$

as an independent mixed carrier.

Thus in all cases Γ is forced either into $T_v = 0$, into $\mathcal{R} = 0$, or into one of the already-absorbed regimes. This contradicts the assumption that Γ persists as an independent genuinely mixed clean branch inside

$$\{T_v \neq 0\} \cap \{\mathcal{R} \neq 0\}.$$

□

Remark 12.409. The theorem should be read as the mixed-global four-point closure statement on a nonplanar good patch: after the source-coupled and patch-loss absorptions, no independent genuinely mixed clean carrier remains away from the scalar exits $T_v = 0$ and $\mathcal{R} = 0$.

Remark 12.410 (Unified mixed/nonmixed picture after the clean-patch closure). Theorem 12.408 should be read as the point where the mixed and nonmixed 4-point geometries fully meet.

On the mixed side, the centered-base local theorem (Theorem 12.240) closes the only genuinely new local mixed branch. On the nonmixed side, the affine-planar lane is already governed by the scalar invariant

$$\mathcal{R} = \frac{T_v}{K_v^2},$$

and interval-persistent or recurrent exact contacts force affine-planar collapse (Proposition 12.267 and Corollary 12.268).

Accordingly, the sufficiently small clean mixed-global problem is no longer a search for a new mixed carrier lane. It is a mixed-to-scalar capture theorem: every independent genuinely mixed clean branch is forced either into the centered-base scalar exit

$$T_v = 0$$

or into the already-handled nonmixed scalar exit

$$\mathcal{R} = 0,$$

with all source-coupled and patch-loss phenomena absorbed into lower-core, repeated-node, or already-quintic handoffs.

Remark 12.411 (What remains genuinely open after the clean mixed closure). After Theorem 12.408, the clean mixed four-point case is no longer an independent global frontier. What remains genuinely open is the global nonadjacent propagation problem on the nonmixed/source-level side, together with the separate two-point program.

More concretely, on a generic component the remaining one-pair theorem may be read as the absence of same-parity / distance-two 2-secant recurrence in the affine chart, equivalently as no same-parity level re-entry for

$$\beta(h) = -\frac{L(h)}{M(h)},$$

equivalently as no same-parity vertical-line re-entry for the dual curve

$$T(h) = \left(-\frac{L(h)}{M(h)}, \frac{N(h)}{M(h)} \right).$$

The current local theory already resolves the finite-order $W = 0$, $M \neq 0$ boundary behavior and the generic turning-boundary blocks. After the same-parity compression, the genuinely residual affine lane is no longer arbitrary nonadjacent recurrence but isolated same-parity sign descent, equivalently discrete peaks of the intercept sequence on triple-active windows. So the

sharpened affine frontier should reduce to middle-limited triples with a single endpoint sign test, rather than an unrestricted remote recurrence picture.

On the mixed side, the residual overlap contribution is concentrated on $K_v = 0$ and should be read there as a quotient-diagonal fixed-shear problem: a close scalar-local lane and a separate three-run overlap-transport lane, not a new local mixed carrier or hidden finite-order branch type.

Thus the remaining burden is global/nonlocal, but the one-pair derivative geometry now compresses it substantially: interior common-tangent structure is pseudoline-ordered, local birth at generic turning and finite-order $W = 0$, $M \neq 0$ boundaries is already sterile, and the affine residual has been reduced to isolated same-parity endpoint/branch configurations, equivalently to zeros of the monotone endpoint-gap functions $G_{\xi,J}$ in the middle-limited triple frontier.

On the mixed residual $K_v = 0$ lane, what remains is not another local branch-classification problem, but the separate scalar-local / transport / package questions discussed in the fixed-shear section.

13. ENDGAME ARCHITECTURE AND REMAINING PROPAGATION PROBLEM OUTSIDE THE CLEAN MIXED FOUR-POINT CASE

The remaining endgame is now a global propagation problem for the normalized line-incidence geometry rather than a local boundary-creation problem.

Remark 13.1 (Equivalent formulations of the remaining propagation problem outside the clean mixed four-point case). After Theorem 12.408, the remaining missing one-pair theorem on a generic component may be formulated in any of the following equivalent ways:

- (1) no finite-slope affine line in the chart

$$x = \frac{v_5}{c}, \quad y = \frac{\Delta_7}{c}$$

is 2-secant to two nonadjacent ECT intervals;

- (2) the slope function

$$\beta(h) := -\frac{L(h)}{M(h)}, \quad L = v_5 \Delta'_7 - v'_5 \Delta_7,$$

does not revisit a level on nonadjacent branches;

- (3) the dual tangent-line curve

$$T(h) = \left(-\frac{L(h)}{M(h)}, \frac{N(h)}{M(h)} \right)$$

has no nonadjacent same-parity re-entry against a vertical line in the (b, m) -plane;

- (4) on the highest-new slice, the scalar graph

$$\phi = \lambda^{[1]}$$

has no nonadjacent re-entry of its slope ϕ_μ .

After the same-parity compression of Remark 13.6, the equivalent global formulations in this section should be read with that sharpening built in: the genuinely residual affine lane is distance-two / same-parity only, and the remaining global propagation problem is already reduced to isolated middle-limited triples rather than arbitrary nonadjacent recurrence.

The results proved above resolve the local behavior at finite-order $W = 0$, $M \neq 0$ boundaries and at generic turning boundaries. More precisely, the one-pair support geometry now shows that fixed-slope overlap windows are one-shot on each ordered pair, interior common-tangent crossings are pseudoline-ordered, and endpoint-support incidence is controlled by the strictly monotone endpoint-gap functions

$$G_{\xi,J}(h) = y_\xi - y(h) - m(h)(\xi - x(h)).$$

Thus the remaining theorem target is no longer an arbitrary global recurrence statement. It is the exclusion of remote zeros of these endpoint-gap functions, equivalently an RH-specific sign theorem for endpoint-support incidence on ordered nonadjacent branches.

Remark 13.2 (What the present global reduction does and does not buy). The results above now force the remaining propagation problem into a much smaller combinatorial and analytic class.

On the positive side, fixed-slope incidence is interval-graph geometry, support curves are strictly concave pseudolines with single-crossing and betweenness, each ordered pair has at most one directional or pair-pair overlap window in slope, and each endpoint support point meets a given ordered target branch at most once. Thus any surviving bad core must be slope-changing and pair-nonrepeating, and any remaining one-pair obstruction must arise through some element of a finite table of remote endpoint/branch incidences.

However, it is important to record a limitation. By Remark 13.4, intrinsic one-pair affine geometry is still universal for analytic plane curves. So pure planarity, parity, interval-graph combinatorics, pseudoline support geometry, or any similar abstract strip/support formalism does *not* by itself close the remaining branch. Any final contradiction must therefore use RH-specific source/provenance/canonical structure, not only generic secant-strip geometry.

Remark 13.3 (Warning: what the current local theory does *not* imply). Several tempting globalization statements suggested by the local theory are too strong in generic models and should not be assumed without further RH-specific input.

- (1) Adjacent exclusion is not enough: exclusion of adjacent pair-pair lines does not by itself force exclusion for nonadjacent pairs.
- (2) Local convexity of the dual tangent-line curve, even together with the shared asymptote structure of Proposition 12.67, is not by itself enough to forbid nonadjacent same-parity re-entry in the dual (β, m) -plane.
- (3) The direct highest-new septic coefficient is explicit enough to prove noncancellation, but not generically one-signed. Thus no trivial sign theorem for the scalar graph $\phi = \lambda^{[1]}$ should be expected without further RH-specific structure.
- (4) By Remark 13.4, the intrinsic one-pair affine geometry is universal for analytic plane curves. So no theorem using only planarity, parity, pseudoline support order, interval-graph overlap, or similar abstract strip/support data can by itself exclude nonadjacent recurrence.

Accordingly, the remaining gap is now a genuinely global special-structure theorem: not another local birth lemma, but an RH-specific sign/ordering law for the remote endpoint-gap incidences.

Remark 13.4 (Intrinsic affine-plane universality and the ceiling of pure planar arguments). It is useful to record a sharp ceiling on what can be proved from the intrinsic one-pair affine geometry alone.

Indeed, given any real-analytic plane curve

$$h \mapsto (x(h), y(h))$$

on an interval, setting

$$c \equiv 1, \quad v_5 = x, \quad \Delta_7 = y$$

gives

$$M = x', \quad N = y', \quad W = x'y'' - y'x''.$$

So the normalized affine chart, the ECT/curvature data, the tangent-separator kernel, and the dual/support geometry reduce exactly to the usual tangent geometry of the analytic plane curve (x, y) .

Accordingly, no theorem using only planarity, parity, support-curve single-crossing, strip-overlap combinatorics, or any other intrinsic one-pair affine-topological datum can by itself exclude nonadjacent recurrence. Any final contradiction must therefore use RH-specific source/provenance/canonical structure, not only abstract plane-curve or secant-strip geometry.

Remark 13.5 (Remote endpoint-gap reduction of the remaining nonadjacent problem). Combine Lemma 12.109, Theorem 12.96, and Theorem 12.107. Then every local boundary-creation

mechanism on the generic one-pair region is already sterile: adjacent generic turning blocks contribute no pair-pair geometry, and finite-order isolated boundaries of type

$$W = 0, \quad M \neq 0$$

do not create a new opposite-side pair-pair branch.

Therefore any surviving nonadjacent pair-pair obstruction must appear as a *remote* interior zero of one of the monotone endpoint-gap functions

$$G_{\xi,J}(h) = y_{\xi} - y(h) - m(h)(\xi - x(h)),$$

where (ξ, y_{ξ}) ranges over endpoint support points of ECT branches and J ranges over ordered nonadjacent target branches.

Since each $G_{\xi,J}$ is strictly monotone on J , each endpoint/branch pair can contribute at most one such tangency. Thus the remaining propagation problem reduces to a finite remote-incidence table.

Accordingly, the natural next global target is an RH-specific sign theorem for the endpoint-gap functions $G_{\xi,J}$, not another local boundary-birth analysis.

Remark 13.6 (Work in progress: same-parity compression of the residual affine lane). A useful correction is that the adjacent branch should not be counted in the nonadjacent envelope: the genuinely residual affine lane is a distance-two / same-parity propagation problem.

For a common slope slice m , write

$$\Delta_j(m) := \tau_{I_{j+1}}(m) - \tau_{I_j}(m), \quad \varepsilon_j(m) := \operatorname{sgn} \Delta_j(m),$$

and for a triple

$$I_{j-1} < I_j < I_{j+1}$$

active on a common slope interval define

$$A_j := \Omega_{j-1} \cap \Omega_j \cap \Omega_{j+1} = (L_j, R_j).$$

Since each Δ_j is strictly decreasing, if

$$c_j := m_{j,j+1}$$

denotes the unique adjacent common-tangent slope, then on A_j

$$\Delta_j(m) > 0 \iff m < c_j, \quad \Delta_j(m) < 0 \iff m > c_j.$$

Thus the dangerous same-parity sign pattern is exactly

$$(\varepsilon_{j-1}(m), \varepsilon_j(m)) = (+, -),$$

equivalently

$$\tau_{I_{j-1}}(m) < \tau_{I_j}(m) > \tau_{I_{j+1}}(m),$$

equivalently

$$c_j < m < c_{j-1}.$$

So the residual affine obstruction is not arbitrary nonadjacent recurrence but an isolated sign descent / discrete peak of the intercept sequence $j \mapsto \tau_{I_j}(m)$. The actual triple can be dangerous only if the active window crosses that descent:

$$A_j \cap (c_j, c_{j-1}) \neq \emptyset.$$

Writing

$$e_{j-1}(m) := \sup \mathcal{B}_{I_{j-1}}(m), \quad B_j(m) := e_{j-1}(m) - \tau_{I_{j+1}}(m),$$

the residual same-parity obstruction interval is therefore

$$Q_j := \{m \in A_j \cap (c_j, c_{j-1}) : B_j(m) \geq 0\}.$$

In particular, once the active-window endpoint bookkeeping is added, the sharpened remaining affine risk should reduce to isolated middle-limited triples of the form

$$c_j < r_j < c_{j-1}, \quad B_j(r_j) > 0,$$

where r_j is the upper endpoint of the triple-active window. This is the sharpest current affine formulation, but it remains a theorem target rather than a proved closure statement.

Remark 13.7 (Further sharpening of the same-parity affine frontier). After the distance-two / same-parity correction, the residual affine lane is best viewed through the adjacent sign sequence

$$\varepsilon_j(m) := \operatorname{sgn}(\tau_{j+1}(m) - \tau_j(m)).$$

A dangerous same-parity affine defect at the triple

$$I_{j-1} < I_j < I_{j+1}$$

is exactly an isolated sign descent

$$(\varepsilon_{j-1}(m), \varepsilon_j(m)) = (+, -),$$

equivalently a discrete local peak

$$\tau_{j-1}(m) < \tau_j(m) > \tau_{j+1}(m).$$

In particular, dangerous affine defect sites are isolated: no consecutive defects can occur.

Moreover, after imposing the actual triple-active window constraint, the sharpened residual affine frontier is middle-limited:

$$c_j < r_j < c_{j+1}, \quad B_j(r_j) > 0.$$

So the unresolved affine risk is no longer arbitrary remote recurrence. It is an isolated same-parity sign descent on an actually active triple, limited on the right by the middle branch, together with one endpoint sign test.

Proposition 13.8 (Full-block universality of the direct provenance slice). *Let J be a connected generic one-pair block on which*

$$A(h) := \kappa(h)\eta_2(h)$$

is analytic and nowhere zero. On J , the direct provenance slice has the form

$$\phi(h) = A(h) q^{(7)}(h).$$

Then for every analytic function

$$\psi : J \rightarrow \mathbf{C}$$

there exists an analytic function

$$q : J \rightarrow \mathbf{C}$$

such that

$$\phi(h) = \psi(h) \quad (h \in J).$$

More explicitly, for any basepoint $h_0 \in J$ and any constants

$$a_0, \dots, a_6 \in \mathbf{C},$$

the function

$$q_\psi(h) := \sum_{j=0}^6 \frac{a_j}{j!} (h - h_0)^j + \int_{h_0}^h \frac{(h-s)^6}{6!} \frac{\psi(s)}{A(s)} ds$$

is analytic on J and satisfies

$$q_\psi^{(7)}(h) = \frac{\psi(h)}{A(h)}, \quad \phi(h) = A(h) q_\psi^{(7)}(h) = \psi(h).$$

In particular, on any generic one-pair block, the direct highest-new scalar graph ϕ is an arbitrary analytic scalar graph.

Proof. Since A is analytic and nowhere zero on J , the function

$$\frac{\psi}{A}$$

is analytic on J . The displayed formula for q_ψ is the standard variation-of-constants solution of

$$q^{(7)} = \frac{\psi}{A},$$

with arbitrary initial data at h_0 . Differentiating seven times gives

$$q_\psi^{(7)}(h) = \frac{\psi(h)}{A(h)},$$

hence

$$\phi(h) = A(h) q_\psi^{(7)}(h) = \psi(h).$$

□

Corollary 13.9 (Finite linear provenance constraints are too weak). *Let J be as in Proposition 13.8. Fix finitely many linear functionals*

$$\Lambda_r(q) = c_r, \quad r = 1, \dots, N,$$

on analytic functions q on J , where each Λ_r is built from finitely many point-evaluation jets and/or finitely many linear integral functionals with analytic kernels.

Fix also two distinct points

$$\mu_1 \neq \mu_2 \in J$$

and an affine line

$$\ell(\mu) = s\mu + b.$$

Then there exists a polynomial q on J such that:

(1) all the prescribed linear provenance conditions

$$\Lambda_r(q) = c_r \quad (r = 1, \dots, N)$$

hold;

(2) the direct-slice graph

$$\phi(\mu) = A(\mu) q^{(7)}(\mu)$$

is tangent to the line $y = \ell(\mu)$ at both μ_1 and μ_2 , i.e.

$$\phi(\mu_j) = \ell(\mu_j), \quad \phi_\mu(\mu_j) = s \quad (j = 1, 2).$$

Accordingly, no theorem using only the explicit direct formula

$$\phi = A q^{(7)}$$

together with any finite family of linear provenance or matching conditions on q can exclude bitangent, four-secant, or same-slope recurrence geometry on the direct slice.

Proof. The tangency conditions at μ_j are linear jet conditions on q :

$$A(\mu_j) q^{(7)}(\mu_j) = \ell(\mu_j),$$

and, since

$$\phi_\mu = (A q^{(7)})' = A' q^{(7)} + A q^{(8)},$$

also

$$A'(\mu_j) q^{(7)}(\mu_j) + A(\mu_j) q^{(8)}(\mu_j) = s.$$

Because $A(\mu_j) \neq 0$, these are legitimate linear conditions on the jets of q at the two points μ_1, μ_2 .

Together with the prescribed finitely many provenance constraints

$$\Lambda_r(q) = c_r,$$

this is a finite linear system on the coefficients of a polynomial q . For a polynomial of sufficiently large degree, the coefficient space has dimension larger than the number of imposed linear conditions. Standard Hermite interpolation / finite-dimensional linear algebra therefore gives a polynomial q satisfying all of them simultaneously. □

Remark 13.10 (What the direct-slice no-go results actually leave open). Proposition 13.8 shows that on a generic one-pair block the direct highest-new scalar graph

$$\phi$$

is fully universal. Corollary 13.9 shows more: even after imposing any finite family of linear provenance / matching conditions on q , the direct slice remains flexible enough to realize bitangent, four-secant, and same-slope recurrence geometry.

So the next live theorem cannot be another sign/convexity statement about the graph of ϕ , nor any theorem using only finitely many endpoint jets or finitely many linear integral conditions on q . Any successful continuation of the direct-slice lane must instead use a genuinely *continuum-strength provenance law* for q on a whole block, such as a lower-order ODE, a sign-definite integral identity, a nonlinear transport law, or some other infinite-codimension constraint.

Lemma 13.11 (Crude polynomial bound for the local value scale). *Let*

$$S(m) = \sum_{\rho} f_{\beta_{\rho}, \gamma_{\rho}}(m),$$

where

$$f_{\beta, \gamma}(m) = \frac{1 - \beta}{(1 - \beta)^2 + (2m - \gamma)^2} + \frac{\beta}{\beta^2 + (2m - \gamma)^2}.$$

Then

$$S(m) \ll Q^2, \quad Q \asymp \log |m|.$$

Consequently

$$L(m) \ll Q$$

and

$$B_{\zeta}(m)L(m)S(m) \ll Q^4.$$

Proof. By the zero-free region, for zeros at relevant height one has

$$\min(\beta, 1 - \beta) \gg \frac{1}{Q}.$$

Hence

$$f_{\beta, \gamma}(m) \ll \frac{Q}{1 + (2m - \gamma)^2}.$$

Partition the zeros into unit shells

$$\mathcal{S}_k := \{\rho : k \leq |2m - \gamma_{\rho}| < k + 1\}, \quad k = 0, 1, 2, \dots$$

Standard zero counting gives

$$\#\mathcal{S}_k \ll Q$$

for each unit shell near height m . Therefore

$$S(m) \ll \sum_{k \geq 0} \#\mathcal{S}_k \frac{Q}{1 + k^2} \ll Q \sum_{k \geq 0} \frac{Q}{1 + k^2} \ll Q^2.$$

Since

$$L(m) = \log\left(3 + |m| + \frac{1}{S(m)}\right),$$

we trivially have $L(m) \ll Q$, and the last estimate follows from $B_{\zeta}(m) \asymp Q$. $\square$

The following remark records the present status of the endgame before the final contradiction theorem.

Remark 13.12 (Work in progress: current endgame status). The contradiction argument in the next theorem is presently written in the *pair-like branch* of the local model analysis. More precisely, the current endgame is fully aligned with the case in which the actual same-scale local core is perturbatively pair-like, so that the toy-model first-odd-jet obstruction continues to govern the corrected transverse scalar.

The fixed-core analysis developed earlier in the draft shows, however, that a more general finite-core branch is also possible: the first odd coefficient may cancel while a genuine corrected anomaly survives at a higher odd order. Thus the fully general finite-core endgame would need to be formulated in terms of the *first nonzero odd jet* of the corrected transverse scalar, rather than specifically the first one.

Accordingly, the theorem below should presently be read together with the preceding work-in-progress remarks on the pair-like versus finite-core fork. In particular, the pair-based contradiction is the intended endgame under the pair-like branch, while the genuinely finite-core branch remains to be completed in full detail.

At the local algebraic level, the canonical quotient-level order-7 fixed-sector package is now closed via Theorem 12.49: the remaining order-7 ambiguity is only the choice of a raw representative inside $A_{7,K_1}^f + \mathbf{C}A_5^f$, equivalently a local gauge-fixing of (u_7, v_7) , and does not affect the quotient class $[A_7^f]$ or the canonical second residual determinant Δ_7 .

Remark 13.13 (Work in progress: asymptotic contradiction versus full RH). As presently written, the final contradiction is only asymptotic in the height parameter Q : the comparison closes after reaching a regime in which the zeta-side upper bound is dominated by the toy-side lower bound, schematically because

$$Q^4 e^{-\delta Q} \rightarrow 0 \quad (Q \rightarrow \infty).$$

Accordingly, the current argument excludes sufficiently large pair-like off-line anomalies, but does not yet by itself rule out a hypothetical low-lying off-line zero. In particular, the present proof does not furnish a uniform-in-height contradiction, nor does it presently prove that a single off-line zero at one height would force corresponding bad local anomalies at arbitrarily large heights.

A plausible repair is to split the endgame into an effective high-height theorem plus a separate finite low-height bridge.

Effective high-height step. One should first replace the soft asymptotic comparison by an explicit inequality. Concretely, the final contradiction compares a toy-side lower bound of the form

$$|\Xi_{\text{toy}}^{(N)}(u, d)| \geq c_{\text{toy}} \mathcal{L}(m)$$

against a zeta-side upper bound of the form

$$|\Xi_{\zeta}^{(N)}(m)| \leq C_{\zeta}(\rho_0, \alpha) \frac{\mathcal{U}(m)}{Q} e^{-\delta_*(\alpha, \rho_0)Q},$$

where in the present draft one has schematically

$$\mathcal{L}(m) \asymp \frac{x}{B_{\zeta}(m)} S(m)$$

(or its variable- x replacement from Remark 11.39), and

$$\mathcal{U}(m) \asymp L(m) S(m)^2.$$

To extract an actual threshold, every use of $O(\cdot)$, $\ll$, and $\asymp$ in the final comparison must be replaced by explicit constants. After substituting explicit bounds such as

$$B_{\zeta}(m) \leq C_B Q, \quad L(m) \leq C_L Q, \quad S(m) \leq C_S Q^2,$$

the contradiction should reduce to an explicit scalar inequality of the form

$$C_*(\rho_0, \alpha) Q^3 e^{-\delta_*(\alpha, \rho_0)Q} < 1 \quad (Q \geq Q_*),$$

for some computable constant $C_* > 0$. Any explicit Q_* satisfying this inequality would then yield an *effective* theorem excluding pair-like off-line anomalies for all $Q \geq Q_*$.

Height extraction. Once Q_* is known explicitly, one must convert it into an actual height threshold. Since the draft works with a midpoint parameter m satisfying

$$Q \asymp \log |m|,$$

this requires making the logarithmic comparability effective, for example in the form

$$c_Q \log(3 + |m|) \leq Q \leq C_Q \log(3 + |m|).$$

From such inequalities one obtains an explicit midpoint threshold m_* , and hence an explicit zero-height threshold T_* , since the local points under consideration satisfy

$$t_{\pm} = m \pm \frac{\sigma}{2}$$

with $|\sigma|$ microscopic relative to the height scale. Thus one would arrive at a statement of the form:

$$\text{no pair-like off-line zero can occur for } |\Im \rho| > T_*.$$

Finite low-height bridge. The effective high-height theorem alone still does not imply the full Riemann Hypothesis unless one also excludes all potential off-line zeros below the threshold T_* . A natural way to do this is to combine the structural argument above height T_* with a rigorous finite verification of RH up to height T_* . Accordingly, if one can arrange that the explicit threshold T_* lies below a rigorously verified range, then the final step would be:

$$\text{high-height exclusion above } T_* \quad + \quad \text{rigorous verification below } T_* \quad \implies \quad \text{RH}.$$

This would still be an unconditional proof, albeit a computer-assisted one rather than a purely internal asymptotic contradiction.

What still needs to be made explicit. To implement this repair rigorously, the draft must still supply all of the following ingredients in effective form:

- (1) an explicit toy-side lower constant on the admissible compact d -range and microscopic u -range;
- (2) an explicit zeta-side upper constant in the N -point cancellation estimate, including the exponential decay rate δ ;
- (3) explicit growth bounds for the auxiliary quantities appearing in the final comparison, including $B_{\zeta}(m)$, $L(m)$, and $S(m)$;
- (4) an explicit conversion between the logarithmic parameter Q and the actual height variable;
- (5) a final theorem stated as an effective exclusion above a concrete threshold T_* , rather than merely “for Q sufficiently large”;
- (6) a separate low-height bridge, either by citing a rigorous finite verification result covering the range up to T_* , or by proving an alternative uniform-in-height contradiction.

There is, in principle, a second possible repair: prove a propagation theorem showing that any off-line zero at one height would force corresponding bad local anomalies at arbitrarily large heights, so that the asymptotic contradiction would automatically apply. However, no such propagation principle is currently established in the draft, and at present this appears to be a substantially stronger theorem than the effective-threshold route described above.

Accordingly, the current endgame should be read as an asymptotic contradiction theorem in the pair-like branch. A plausible route to a complete proof of RH is available, but it requires an effective extraction of the height threshold together with a finite low-height bridge, neither of which has yet been written out in the draft.

Theorem 13.14 (Asymptotic calibrated N -point contradiction in the pair-like branch). *Assume the pair-like branch hypothesis and assume that the pair-like package-to- Ξ bridge identifies the calibrated local two-point package with the toy normal form entering Proposition 11.37. Then, for Q sufficiently large, no fixed off-line toy anomaly can occur in the pair-like branch.*

This theorem is not a finite-core closure theorem. The genuinely non-pair-like finite-core branch remains open pending the lower-transfer theorem described in Remark 12.165.

Proof. By Proposition 8.6 and the canonical calibration theorem, in the fixed small- x_0 regime one has

$$u^2 \asymp \frac{x_0}{B_\zeta(m)} S(m).$$

On the toy side, Proposition 11.37 gives

$$\Xi_{\text{toy}}^{(N)}(u, d) = u^2 \Phi_K(M(d)) + O(u^4),$$

hence on compact d -ranges,

$$|\Xi_{\text{toy}}^{(N)}(u, d)| \asymp u^2 \asymp \frac{x_0}{B_\zeta(m)} S(m).$$

On the zeta side, Corollary 11.22 gives

$$|\Xi_\zeta^{(N)}(m)| \ll_{\rho_0, \alpha} \frac{L(m)S(m)^2}{Q} e^{-\delta_*(\alpha, \rho_0)Q}.$$

By Lemma 13.11, one has $S(m) \ll Q^2$, $L(m) \ll Q$, and $B_\zeta(m)L(m)S(m) \ll Q^4$. Therefore

$$\frac{(L(m)S(m)^2/Q)e^{-\delta_*(\alpha, \rho_0)Q}}{(x_0/B_\zeta(m))S(m)} = \frac{B_\zeta(m)}{x_0} \frac{L(m)S(m)}{Q} e^{-\delta_*(\alpha, \rho_0)Q} \ll Q^3 e^{-\delta_*(\alpha, \rho_0)Q} \rightarrow 0.$$

Hence for Q sufficiently large,

$$|\Xi_\zeta^{(N)}(m)| < |\Xi_{\text{toy}}^{(N)}(u, d)|.$$

This contradicts the assumed calibrated realization of a fixed off-line toy anomaly in the pair-like branch, once Q is sufficiently large. Thus, under the pair-like branch hypothesis and the package-to-jet bridge hypothesis, the pair-like branch is excluded at sufficiently large height.

This theorem is an asymptotic branch exclusion. To obtain a complete RH proof from it one still needs: an effective extraction of the threshold Q_* , a conversion from Q_* to a height threshold T_* , a finite low-height verification or alternative low-height argument, and the non-pair-like finite-core branch closure. $\square$

Remark 13.15 (Local package results are not yet global transform closure). The local Gate B and connection-package results are package-construction results. They do not by themselves prove hidden-state equality, exceptional-divisor fiber selection, finite-core exclusion, or a full package-to-transform contradiction. The remaining global obligations are:

state-locality modulo $\ker \Phi_K$, quotient-valued diagonal merger, finite-core lower transfer.

Remark 13.16 (Current status summary after the consolidated round). Reading Lemma C.20, Theorem C.21, Definition 12.200, Definition 12.183, Theorem 12.167, and Definition 12.175 together, the present status is:

The M_A, M_C connection identities close under actual source-linear Y -middle edge parity.

The local obstruction has sharpened to the odd-background $\eta\eta$ top row inside $E_{A,Y}^{\text{BC}}$, led by $\mathcal{E}_{\eta\eta;7,0,0}^2$; the pre

The transform-safe package is $\widehat{\Psi}_{\text{red}}^+ = (\widehat{\Psi}_{\text{red}}, T)$, $T = v_7/c$.

Finite determinacy is qualitative; calibrated separation and interaction rigidity remain open.

Remark 13.17 (Current local obstruction: odd-background $\eta\eta$ top row). After correcting the Bottleneck-C bookkeeping, the first local actual-source obstruction is no longer merely an unevaluated broad finite table. The current leading row is the frozen odd-background $\eta\eta$ top row

$$\mathcal{E}_{\eta\eta;7,0,0}^2.$$

It is governed by

$$\Lambda_{2,0}^- \left[\left(\frac{\sin \Delta_0}{s^2} + \frac{q_- \cos \Delta_0}{s} \right) \eta_i \eta_j \right]_7 W_{+,0},$$

where

$$\Lambda_{2,0}^- = -\frac{1}{24\pi q^3} E_{22}.$$

Writing

$$\nu_{21,7} := [z^7] \left[\left(\frac{\sin \Delta_0}{s^2} + \frac{q_- \cos \Delta_0}{s} \right) \eta_i \eta_j \right],$$

the top-row S -coefficient is

$$f_{\eta\eta}^{\text{top}} = -\frac{u}{24\pi q^3} \nu_{21,7}.$$

Thus the local branch is now binary: either prove a zeta-specific identity

$$\nu_{21,7} = 0,$$

or supply a genuine physical aggregate S -coefficient cancellation of the row. Endpoint-diagonal source realization, even if proved, removes only endpoint-separation subchannels and does not by itself close this scalar Δ -Hessian $\eta\eta$ survivor. The only remaining local rescue is a genuine physical aggregate S -coefficient cancellation, not a formal package counterterm.

Remark 13.18 (Current outcome of the Bottleneck-C repair audits). The subsequent audits below sharpen the status of Remark 13.17. Replacing the raw far-tail source by the full non-core zero source is coherent, but the rederived top row has the same obstruction form. Replacing the scalar local readout, quotienting by the $A_5^{\mathbf{f}}$ -line, changing the finite- s carrier, or exact endpoint symmetrization likewise does not remove the S -visible $\eta\eta$ row while preserving transform faithfulness.

Consequently, within the present transform-faithful finite- s architecture, Bottleneck C still requires a genuine vanishing of the physical aggregate S -coefficient of the $\eta\eta$ top row. No such vanishing theorem is presently proved.

14. DOWNSTREAM LOCAL CLOSURE PACKAGE: GATE 2, WALL ESCAPE, AND CARRIER ROWS

This section records the current local downstream replacement for the older $D1/D2/D3$ architecture. The active Gate-2 route is no longer a Remez-Cartan or pointwise target-smallness route. It is

$$\boxed{\text{Gate 1} + |u'|_{\infty} \text{ source thickening} + L^2\text{-input real rigidity} \implies \text{Gate 2.}}$$

The active wall route is

$$\boxed{\text{Gate 2} + \text{gauge-locked wall escape} + \text{D4 carrier-quotient/filtered row conversion} + \text{row mobility} \implies \text{row energy}}.$$

The source-robust wall-row search and the scalar I -coordinate order-nine incidence wall remain fallback formulations only if the row-output architecture is forced to be exact-point or scalar-collapsed.

15. UNIFIED FINITE-SURVIVAL ARCHITECTURE

Theorem 15.1 (Unified finite-survival theorem). *The proof is organized around four finite-survival statements:*

- (1) **Source survival:** $\text{Agg}L^2(q''_{\text{can}})$ on a polynomial-weight good packet set.
- (2) **Graph survival:** Graph/O2O3_{23} after R^C -normalization, the bounded-degree Fejér-Riesz packet model, and off-block packet geometry.
- (3) **Quotient survival:** the primitive $Y12$ carrier survives in the same quotient that removes actual-source nuisances.
- (4) **Atomic contradiction:** the singular-discard observable is Type-I local-square and is bounded above on the same quotient norm.

If all four hold, the same observable is simultaneously bounded below and above by incompatible polynomial scales, giving the downstream contradiction.

$$\begin{aligned}
& \boxed{\text{Agg}L^2(q''_{\text{can}})} \\
& \Downarrow \quad (\text{packet-core micro-covering} + \text{EF completion}) \\
& \boxed{\text{Graph/O2O3}_{23}} \\
(24) \quad & \Downarrow \quad (R^C\text{-normalized Riesz basis} + \text{Fejér-Riesz charge anti-lock}) \\
& \boxed{\text{Y12-quotient survival}} \\
& \Downarrow \quad (\text{retained-space sandwich} + \text{labelled transition}) \\
& \boxed{\text{AtomicY12Contradiction}}
\end{aligned}$$

Proposition 15.2 (Proof sketch / structure). *The proof proceeds in four finite-survival layers:*

- (1) *establish aggregate source observability for $q''_{\text{can}} = B_2 + P_2 + D_2 + E_2$ on a good packet set;*
- (2) *verify the finite/O2O3 packet model and invoke Theorem E.106 to obtain finite graph nondegeneracy;*
- (3) *pass to the actual-source quotient and verify that the primitive Y12 carrier survives after removing nuisance directions;*
- (4) *compare the same quotient observable against the Type-I singular-discard upper and conclude contradiction when the lower and upper scales are separated.*

15.1. Shared finite-survival language.

Definition 15.3 (Finite-survival gaps and residuals). The downstream proof uses the following shared finite objects.

Source survival. Let

$$\mathcal{S}_X := \text{span}\{B_2, P_{2,\leq X}, D_{2,\leq X}, P_{\text{res},X}\}, \quad F_X := B_2 + P_{2,\leq X} + D_{2,\leq X} + P_{\text{res},X},$$

where the residue term is omitted if it vanishes and is absorbed into B_2 if that convention has already been adopted. Define

$$M := G_{\text{ref}}^{-1/2} G_{\text{branch}} G_{\text{ref}}^{-1/2}, \quad E := G_{\text{ref}}^{-1/2} (G_{\text{branch}}(q''_{\text{can}}) - G_{\text{branch}}(F_X)) G_{\text{ref}}^{-1/2},$$

and the source gap

$$\delta_{\text{src}} := \lambda_{\min}(M) - \|E\|_{\text{op}}.$$

Source survival means $\delta_{\text{src}} \geq Q^{-A}$. A Gershgorin lower bound for M is a sufficient way to prove this in atom-normalized coordinates, but the invariant source-frame quantity is the reference-whitened minimum eigenvalue.

Graph survival. Define

$$K := G_{00}^{-1/2} G_{0S} G_{SS}^{-1/2}, \quad \delta_{\text{graph}} := 1 - \|K\|_{\text{op}}.$$

Graph survival means $\delta_{\text{graph}} \geq Q^{-A}$, equivalently a Schur complement lower bound or a polynomial principal-angle gap.

Quotient survival. Let

$$U_p := \text{col}(T_{\text{par}}^{(\alpha)}, T_7, T_{<7}, II_{\text{arith},9}, T_{E_2}), \quad \mathcal{K}_p^Y := \text{im} \begin{bmatrix} R_{B_2,p} & R_{P,p} & R_{D,p} & A_{\alpha,p} & A_{7,p} & A_{<7,p} & A_{II,p} & A_{E_2,p} \end{bmatrix},$$

$$q_p^Y := I - P_{\mathcal{K}_p^Y}^{\text{bd}}, \quad N_p := \Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing}} \mathcal{N}_p^{\text{D}^4}, \quad Y_p := \Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing}} [\mathbf{C}_{Y12}]^{\text{tr}}.$$

The quotient residuals are

$$r_{N,p} := q_p^Y N_p, \quad r_{Y,p} := q_p^Y Y_p.$$

Quotient survival means $r_{N,p} \approx 0$ and $r_{Y,p} \neq 0$ with polynomial margin.

Graph/coercivity survival. Let

$$\mathfrak{T}_p := (I - P_{\text{sing},p})T_{\text{tr} \rightarrow \text{sing},p}J_{\text{tr},p}.$$

If $\mathcal{R}_X = \mathcal{R}_{X,\text{low},p} \oplus \mathcal{R}_{X,\text{tail},p}$ is the admissible residual split and D_p is the low-mode defect matrix for $\mathfrak{T}_p$, then

$$\delta_{\text{coerc}} := \sigma_{\min}(D_p).$$

Graph/coercivity survival means $\delta_{\text{coerc}} \geq Q^{-C}$ and the tail is absorbed by the EF/singular-discard bound.

Extraction covariance survival. On each frozen cap C_j , let

$$B_j := \sum_{p \in C_j} w_p \Gamma_p \Gamma_p^*, \quad w_p = \rho_p \frac{E_{1,p} H_p}{\mathcal{N}_p^{\text{wt}}}, \quad \Gamma_p = \begin{pmatrix} \Gamma_G(p) \\ \Gamma_{G,2}(p) \end{pmatrix}.$$

Define

$$\delta_{\text{ext}} := \frac{\det(B_j)}{(\text{tr } B_j)^2}.$$

Extraction survival means $\delta_{\text{ext}} \geq Q^{-A}$. If $\delta_{\text{ext}} \ll Q^{-A}$, the wall is effectively static and the cap is handed to the Y12 quotient module.

Atomic contradiction. The forbidden observable is

$$\Lambda_{Y,p} := q_p^Y \Pi_{\Sigma(Y12),p} T_{\text{tr} \rightarrow \text{sing},p} P_{\text{tr},p}.$$

The final contradiction compares the same quotient norm upper bound for $\Lambda_{Y,p} J_{\text{tr},p}(q_{\text{can}}'')$ with the lower bound carried by $r_{Y,p}$.

Remark 15.4 (Active interpretation of the shared residual notation). Definition 15.3 records the historical finite-survival notation used by several earlier module briefs. In the active proof branch, the residual pair $(r_{N,p}, r_{Y,p}) = (q_p^Y N_p, q_p^Y Y_p)$ is only a diagnostic for the primitive quotient route. It is not the cap-level Jacobian pair: the nuisance column N_p is quotient-killed, so any Jacobian using $q_p^Y N_p$ fails by the primitive-carrier-killed mechanism. The active same-norm final bridge uses the certified retained carrier plane and then the canonical source-graded $K9$ observable

$$\Lambda_p^{\text{gr,full}} = (I - P_p^{\text{full}}) \mathcal{L}_{K9,p}^{\text{gr}} J_{\text{tr},p},$$

or its maximal affine projection $\widehat{\Lambda}_{\text{max},p}$. Thus later references to $r_{N,p}, r_{Y,p}, q_p^Y$, or the scalar $\Lambda_{Y,p}$ should be read either as legacy diagnostics or as shorthand only after replacement by the certified live/dead, source-graded branch.

Remark 15.5 (Shared failure labels). The proof uses a fixed failure grammar:

branch holes, off-diagonal packet correlation too large, residual completion failure, Fejér–Riesz packet m

Each agent should reduce its module to one of the gaps $\delta_{\text{src}}, \delta_{\text{graph}}, \delta_{\text{coerc}}, \delta_{\text{ext}}$, or to the residuals $r_{N,p}, r_{Y,p}$.

15.2. L^2 -input real rigidity.

Lemma 15.6 (Ordinary Hardy-branch source-complexity transfer). *Let I^* be an ordinary Hardy source strip with a fixed analytic branch of $B_H = \chi^{-1/2}$. Assume the strip is source-admissible, so that Stirling bounds for the gamma quotient and the standard polynomial vertical-strip bounds for ζ and its derivatives hold through the derivative order used below. Let*

$$u_H(t) = \partial_t^3 Z_H(t), \quad \mathcal{D}_p = \lambda_p \partial_t, \quad u_{H,p}^{\text{norm}} = \lambda_p^3 u_H.$$

On every scale-admissible packet contained in I^* ,

$$\|u_{H,p}^{\text{norm}}\|_{L^\infty(I^*)} + \|\mathcal{D}_p u_{H,p}^{\text{norm}}\|_{L^\infty(I^*)} \leq Q^C.$$

Equivalently, in physical coordinates on a packet interval $I \subset I^*$,

$$\|u_H\|_{L^\infty(I)} + |I| \|u_H'\|_{L^\infty(I)} \leq Q^C.$$

Proof. The source is

$$u_H(t) = -i(\Phi_H^\#)^{(3)}(1/2 + it), \quad \Phi_H^\#(s) = B_H(s)\zeta(s).$$

A fixed number of further packet derivatives produces a finite linear combination of terms

$$B_H^{(a)}(s)\zeta^{(b)}(s), \quad a + b \leq C_0,$$

with C_0 fixed by the proof. On ordinary Hardy source strips, Stirling's formula gives polynomial bounds for the derivatives of B_H , and standard vertical-strip estimates give polynomial bounds for the corresponding $\zeta^{(b)}$. Hence

$$\|u_H\|_{L^\infty(I^*)} + \|\partial_t u_H\|_{L^\infty(I^*)} \leq Q^C.$$

Multiplication by λ_p^3 and by one additional normalized derivative $\mathcal{D}_p = \lambda_p \partial_t$ costs only a polynomial factor on scale-admissible packets. Since $|I| \asymp \lambda_p$ on such packets, the physical-coordinate form follows. $\square$

Remark 15.7 (Hardy denominator and residue guardrails for Gate-2). For Gate-2 in the active branch, the inclusive canonical convention is

$$q_{\text{can}}'' = B_2^H + P_2^H + D_2^H + E_2^H = u_H$$

in physical coordinates, and

$$q_{\text{can},p}'' = \lambda_p^3 u_H$$

in normalized packet coordinates. The term E_2^H is part of the exact Hardy-differential source decomposition unless explicitly declared to be a residual; if it is used perturbatively, then its physical and normalized packet derivatives must satisfy the stated residual ledger.

The Hardy source may contain gamma/theta amplitude derivatives, log-power Dirichlet coefficients, and residue terms from the differentiated AFE. These are part of $B_2^H, P_2^H, D_2^H, E_2^H$. Denominators such as

$$1/q, \quad 1/u, \quad 1/\Gamma_G, \quad 1/\Gamma_{G,2}, \quad 1/|\Gamma|$$

do not belong to the canonical source curvature. If such denominators appear, they belong to row-projection, carrier, or quotient algebra and must be guarded there.

Lemma 15.8 (Real-canonical Gate-2 compatibility). *Assume the actual-zeta local variables are in the real completed gauge of Definition 2.1. If this gauge is implemented through a local chart, assume throughout the packet that every chart denominator is bounded below by Q^{-C} . Thus*

$$q = q_\zeta, \quad u = q_{\text{can}}'' = \mathcal{D}^2 q_\zeta,$$

and the canonical anisotropies are

$$\Gamma_G = q_{\text{can}}'' - 6q_\zeta + 2q_\zeta^3,$$

$$\Gamma_{G,2} = (1 - q_\zeta^2)q_{\text{can}}'' - \frac{1}{6}\mathcal{D}^4 q_\zeta.$$

Then, on ordinary smooth zero-separated packets,

$$q, \quad u, \quad \Gamma_G, \quad \Gamma_{G,2}$$

are real-valued and, with $g = \Gamma_G$, $h = \Gamma_{G,2}$,

$$u = q'',$$

$$g = q'' - 6q + 2q^3,$$

and

$$h = 2q(q')^2 - \frac{1}{6}g''.$$

Proof. The real-valuedness follows from the real completed critical-line gauge. The first two identities are the definitions. For the final identity, apply Lemma 2.5 after suppressing the ζ -subscript:

$$\Gamma_{G,2} = 2q(\mathcal{D}q)^2 - \frac{1}{6}\mathcal{D}^2\Gamma_G.$$

In ordinary local prime notation this is

$$h = 2q(q')^2 - \frac{1}{6}g''.$$

□

Corollary 15.9 (Gauge-locking for the q_0 -wall). *In the real-canonical convention above, the q_0 -wall used in wall escape is the same scalar wall that appears in the Gate-2 anisotropy and in the D_4 source factor. Namely, for $q_0 = q_\zeta(m)$,*

$$u = \mathcal{D}^2q_0,$$

$$\Gamma_G = u - 6q_0 + 2q_0^3,$$

and the D_4 source factor has leading form

$$H_{\text{lead}} = -\frac{q_0u^3}{288}.$$

Thus q_0 is not an arbitrary affine primitive of u . Any affine drift $q_0 \mapsto q_0 + am + b$ changes both

$$\Gamma_G = u - 6q_0 + 2q_0^3$$

and

$$H_{\text{lead}} = -q_0u^3/288.$$

Consequently, the wall-escape argument is applied only in the canonical gauge fixed by q_ζ .

Remark 15.10 (Residual version of the compatibility package). If an approximate convention is used, define

$$r_C := u - q'',$$

$$r_G := \Gamma_G - (q'' - 6q + 2q^3),$$

and

$$r_{G,2} := \Gamma_{G,2} - \left(2q(q')^2 - \frac{1}{6}\Gamma_G''\right).$$

The main canonical convention of this draft has

$$r_C = r_G = r_{G,2} = 0.$$

If a residual convention is used instead, then the L^2 -input rigidity argument applies to q'' , while the Gate-2 source is u . Therefore one must also control the curvature discrepancy $r_C = u - q''$. A sufficient residual package on the source interval J is

$$\|r_C\|_{L^2(J)} + \|r_G\|_{L^2(J)} + \|r_{G,2}\|_{L^2(J)} \leq \varepsilon|J|^{1/2}.$$

Under this package, the residual terms enter the kinetic, potential, and small- q estimates exactly like g and h , and the conclusion of Lemma 15.12 remains valid after increasing the implicit polynomial loss.

Lemma 15.11 (Optional primitive recovery from L^2 -small g ; not used below). *Let $J_1 \Subset J$ be intervals with polynomial separation from the boundary. Assume*

$$u = q'', \quad g = u - F(q), \quad F(q) = 6q - 2q^3,$$

and

$$\|u\|_{L^\infty(J)} \leq Q^C, \quad \|g\|_{L^2(J)} \leq \varepsilon|J|^{1/2}.$$

Then on J_1 ,

$$\|q\|_{L^\infty(J_1)} \leq Q^C.$$

Moreover, on any further polynomially shrunk interval $J_2 \Subset J_1$,

$$\|q'\|_{L^\infty(J_2)} \leq Q^C.$$

Proof. Since $F(q) = u - g$,

$$\|F(q)\|_{L^2(J)} \leq \|u\|_{L^2(J)} + \|g\|_{L^2(J)} \leq Q^C |J|^{1/2}.$$

The cubic polynomial $F(q) = 6q - 2q^3$ satisfies

$$|q|^6 \leq C(1 + |F(q)|^2).$$

Hence

$$\|q\|_{L^6(J)} \leq Q^C |J|^{1/6}.$$

Since $q'' = u \in L^\infty(J)$, the one-dimensional interpolation estimate on the polynomially shrunken interval $J_1 \Subset J$ gives

$$\|q\|_{L^\infty(J_1)} \leq C \left(\|q\|_{L^6(J)} + \|q\|_{L^6(J)}^{\frac{3}{5}} \|q''\|_{L^\infty(J)}^{\frac{2}{5}} + 1 \right),$$

after rescaling J to a bounded interval and absorbing the polynomial shrinkage factor into Q^C . Hence

$$\|q\|_{L^\infty(J_1)} \leq Q^C.$$

Finally, the Landau estimate gives

$$\|q'\|_{L^\infty(J_2)} \leq C (\ell^{-1} \|q\|_{L^\infty(J_1)} + \ell \|q''\|_{L^\infty(J_1)}) \leq Q^C,$$

where $\ell = \text{dist}(J_2, \partial J_1)$ and the shrinkage is polynomial. $\square$

Lemma 15.12 (L^2 -input real rigidity; safe primitive-bounded form). *Let*

$$J = (-2, 2), \quad J' = (-1, 1).$$

Assume $q, u, g, h \in \mathbb{R}$ are sufficiently smooth on J , in the real-canonical convention of Lemma 15.8, and

$$\begin{aligned} u &= q'', \\ g &= q'' - 6q + 2q^3, \\ h &= 2q(q')^2 - \frac{1}{6}g''. \end{aligned}$$

Assume the safe local bounds

$$\|q\|_{L^\infty(J)} + \|q'\|_{L^\infty(J)} + \|q''\|_{L^\infty(J)} \leq Q^C,$$

In the active Gate-2 proof this safe bound is assumed directly on the normalized source interval. Lemma 15.11 records a possible strengthening, but it is not used to justify this lemma in the main argument. Assume also the L^2 -smallness condition

$$\|g\|_{L^2(J)} + \|h\|_{L^2(J)} \leq \varepsilon |J|^{1/2}.$$

Then, after increasing C if necessary,

$$\int_{J'} |u|^2 \leq Q^C \varepsilon^{2/5} |J'|.$$

Proof. If $\varepsilon \geq 1$, the conclusion follows immediately from $\|u\|_\infty = \|q''\|_\infty \leq Q^C$. Assume $0 < \varepsilon < 1$, and set

$$F(q) := 6q - 2q^3.$$

Since $g = q'' - 6q + 2q^3$, we have

$$u = q'' = F(q) + g.$$

Choose

$$\phi = \psi^2 \in C_c^2(J), \quad 0 \leq \phi \leq 1, \quad \phi \equiv 1 \quad \text{on } J',$$

with

$$|\phi'|^2 \leq Q^C \phi, \quad |\phi''| \leq Q^C.$$

Let $0 < \delta < 1$, to be chosen later. Let $\chi_\delta : \mathbb{R} \rightarrow [0, 1]$ be smooth with

$$\chi_\delta(x) = 0 \quad (|x| \leq \delta), \quad \chi_\delta(x) = 1 \quad (|x| \geq 2\delta),$$

and

$$|\chi'_\delta| \leq C\delta^{-1}, \quad |\chi''_\delta| \leq C\delta^{-2}.$$

Define

$$a_\delta(q) := q\chi_\delta(q).$$

Then

$$a'_\delta(q) = \chi_\delta(q) + q\chi'_\delta(q) = O(1),$$

and

$$a''_\delta(q) = 2\chi'_\delta(q) + q\chi''_\delta(q) = O(\delta^{-1}),$$

because on the transition region $\delta \leq |q| \leq 2\delta$ one has $|q| \asymp \delta$.

We first prove the weighted kinetic estimate. From

$$h = 2q(q')^2 - \frac{1}{6}g'',$$

multiply by $\phi a_\delta(q)$ and integrate over J :

$$\int_J 2\phi a_\delta(q)q(q')^2 = \int_J \phi a_\delta(q)h + \frac{1}{6} \int_J \phi a_\delta(q)g''.$$

After integrating by parts twice,

$$\int_J \phi a_\delta(q)g'' = \int_J g(\phi a_\delta(q))''.$$

Thus

$$\int_J 2\phi \chi_\delta(q)q^2(q')^2 = \int_J \phi a_\delta(q)h + \frac{1}{6} \int_J g(\phi a_\delta(q))''.$$

The first term satisfies

$$\left| \int_J \phi a_\delta(q)h \right| \leq \|\phi a_\delta(q)\|_{L^\infty(J)} \|h\|_{L^2(J)} |J|^{1/2} \leq Q^C \varepsilon |J|.$$

Next,

$$(\phi a_\delta(q))'' = \phi'' a_\delta(q) + 2\phi' a'_\delta(q)q' + \phi a''_\delta(q)(q')^2 + \phi a'_\delta(q)q''.$$

Using the bounds on $a_\delta, a'_\delta, a''_\delta$, and the safe bounds on q, q', q'' , we get

$$\|(\phi a_\delta(q))''\|_{L^2(J)} \leq Q^C \delta^{-1} |J|^{1/2}.$$

Therefore

$$\left| \int_J g(\phi a_\delta(q))'' \right| \leq Q^C \varepsilon \delta^{-1} |J|.$$

Consequently,

$$\int_J \phi \chi_\delta(q)q^2(q')^2 \leq Q^C \varepsilon \delta^{-1} |J|.$$

On $|q| \geq 2\delta$, $\chi_\delta(q) = 1$ and $q^2 \geq 4\delta^2$, so

$$(1) \quad \int_{\{|q| \geq 2\delta\}} \phi(q')^2 \leq Q^C \varepsilon \delta^{-3} |J|.$$

Let $\theta_\delta : \mathbb{R} \rightarrow [0, 1]$ be smooth with

$$\theta_\delta(q) = 0 \quad (|q| \leq 2\delta), \quad \theta_\delta(q) = 1 \quad (|q| \geq 3\delta),$$

and

$$|\theta'_\delta| \leq C\delta^{-1}.$$

Define

$$X := \int_J \phi \theta_\delta(q) F(q)^2.$$

Since $F(q) = q'' - g$,

$$X = \int_J \phi \theta_\delta F q'' - \int_J \phi \theta_\delta F g.$$

The g -term obeys

$$\left| \int_J \phi \theta_\delta F g \right| \leq X^{1/2} \|g\|_{L^2(J)} \leq X^{1/2} \varepsilon |J|^{1/2},$$

and hence, after absorption,

$$(2) \quad \left| \int_J \phi \theta_\delta F g \right| \leq \frac{1}{4} X + Q^C \varepsilon^2 |J|.$$

For the main term,

$$\int_J \phi \theta_\delta F q'' = - \int_J (\phi \theta_\delta F)' q',$$

where

$$(\phi \theta_\delta F)' = \phi' \theta_\delta F + \phi \theta'_\delta(q) q' F + \phi \theta_\delta F'(q) q'.$$

For the boundary term, using $|\phi'|^2 \leq Q^C \phi$ and (1),

$$(3) \quad \begin{aligned} \left| \int_J \phi' \theta_\delta F q' \right| &\leq Q^C \left(\int_J \phi \theta_\delta F^2 \right)^{1/2} \left(\int_{\{|q| \geq 2\delta\}} \phi(q')^2 \right)^{1/2} \\ &\leq Q^C X^{1/2} (\varepsilon \delta^{-3} |J|)^{1/2} \\ &\leq \frac{1}{4} X + Q^C \varepsilon \delta^{-3} |J|. \end{aligned}$$

For the transition term, on $\text{supp } \theta'_\delta$ one has $2\delta \leq |q| \leq 3\delta$, so

$$|F(q)| = |6q - 2q^3| \leq C\delta, \quad |\theta'_\delta| \leq C\delta^{-1}.$$

Thus $|\theta'_\delta F| \leq C$, and by (1),

$$(4) \quad \left| \int_J \phi \theta'_\delta(q) F(q) (q')^2 \right| \leq Q^C \varepsilon \delta^{-3} |J|.$$

Finally, since $F'(q) = 6 - 6q^2$ and $\|q\|_{L^\infty(J)} \leq Q^C$,

$$(5) \quad \left| \int_J \phi \theta_\delta F'(q) (q')^2 \right| \leq Q^C \int_{\{|q| \geq 2\delta\}} \phi(q')^2 \leq Q^C \varepsilon \delta^{-3} |J|.$$

Combining (2)–(5) and absorbing,

$$(6) \quad X \leq Q^C (\varepsilon \delta^{-3} + \varepsilon^2) |J|.$$

Since $\phi = 1$ on J' and $\theta_\delta = 1$ on $|q| \geq 3\delta$,

$$(7) \quad \int_{J' \cap \{|q| \geq 3\delta\}} F(q)^2 \leq Q^C (\varepsilon \delta^{-3} + \varepsilon^2) |J'|.$$

On $|q| \leq 3\delta$, $|F(q)| \leq C\delta$. Since $u = F(q) + g$,

$$(8) \quad \int_{J' \cap \{|q| \leq 3\delta\}} |u|^2 \leq Q^C (\delta^2 + \varepsilon^2) |J'|.$$

Combining (7) and (8),

$$\int_{J'} |u|^2 \leq Q^C (\delta^2 + \varepsilon \delta^{-3} + \varepsilon^2) |J'|.$$

Choose

$$\delta = \varepsilon^{1/5}.$$

Then

$$\delta^2 = \varepsilon^{2/5}, \quad \varepsilon \delta^{-3} = \varepsilon^{2/5}.$$

Since $\varepsilon^2 \leq \varepsilon^{2/5}$ for $0 < \varepsilon < 1$,

$$\int_{J'} |u|^2 \leq Q^C \varepsilon^{2/5} |J'|.$$

□

Remark 15.13 (Residual-stable form). The same estimate holds if the identities hold with L^2 -controlled residuals:

$$\begin{aligned} r_C &:= u - q'', \\ g &= q'' - 6q + 2q^3 + r_g, \end{aligned}$$

and

$$h = 2q(q')^2 - \frac{1}{6}g'' + r_h,$$

where

$$\|r_C\|_{L^2(J)} + \|r_g\|_{L^2(J)} + \|r_h\|_{L^2(J)} \leq \varepsilon |J|^{1/2}.$$

Indeed, r_h enters the weighted kinetic estimate in the same way as h , while r_g enters the potential estimate in the same way as g . The additional residual r_C is needed only at the final comparison between the ideal curvature q'' and the source u . After increasing C , the same conclusion follows:

$$\int_{J'} |u|^2 \leq Q^C \varepsilon^{2/5} |J'|.$$

In the exact real-canonical convention used in the main argument,

$$r_C = r_g = r_h = 0.$$

Remark 15.14 (Primitive recovery not used in the rigidity proof). Lemma 15.12 is used only in the safe primitive-bounded form

$$\|q\|_\infty + \|q'\|_\infty + \|q''\|_\infty \leq Q^C.$$

Lemma 15.11 records a possible way to recover these bounds on shrunken intervals from stronger source and L^2 -smallness hypotheses, but this recovery is not part of the active Gate–2 proof. The proof of Theorem 15.22 uses the displayed safe primitive bounds as hypotheses on the normalized source interval.

Remark 15.15 (Application on polynomial subintervals). Lemma 15.12 is stated on the fixed model intervals

$$J = (-2, 2), \quad J' = (-1, 1),$$

for notational convenience. In applications to a physical subinterval $J_{\text{phys}} \subset I$, the Gate–2 identities are still interpreted in the original packet-normalized derivative convention

$$\mathcal{D} = \lambda \partial_m,$$

not in a newly rescaled derivative unless the variables q, u, g, h are also renormalized and the identities are restated.

Thus, when applying the lemma on J_{phys} , we keep

$$u = q'', \quad g = q'' - 6q + 2q^3, \quad h = 2q(q')^2 - \frac{1}{6}g''$$

with primes denoting the original $\mathcal{D}$ -derivative. The cutoff functions used in the proof then have derivatives of size polynomially bounded in $|J_{\text{phys}}|^{-1}$. Since the source-thickening interval has polynomial relative length,

$$|J_{\text{phys}}| \geq Q^{-C} |I|,$$

these cutoff losses are absorbed into the Q^C factor in Lemma 15.12. No change of the canonical Gate–2 derivative convention is made.

Lemma 15.16 (Source thickening from one large source point). *Let I be a packet interval and let u be a C^1 , real- or complex-valued function on I . Assume*

$$\int_I |u|^2 dm \geq Q^{-B} |I|$$

and

$$\|u'\|_{L^\infty(I)} \leq Q^{C_u} |I|^{-1}.$$

Then there exists an interval $J \subset I$, one-sided if necessary, such that

$$|u| \geq \tau := \frac{1}{2}Q^{-B/2} \quad \text{on } J,$$

and

$$|J| \geq Q^{-B/2-C_u-O(1)}|I|.$$

Proof. Gate 1 gives $m_* \in I$ with $|u(m_*)| \geq Q^{-B/2}$. Since

$$||u(m)| - |u(m_*)|| \leq |u(m) - u(m_*)|,$$

the Lipschitz bound gives $|u| \geq \frac{1}{2}Q^{-B/2}$ whenever

$$|m - m_*| \leq cQ^{-B/2-C_u}|I|$$

with $c > 0$ sufficiently small. Intersecting with I gives the claimed one-sided or two-sided interval. $\square$

Lemma 15.17 (Source-large interval with L^2 -small anisotropy). *Assume Gate 1:*

$$\int_I |u|^2 dm \geq Q^{-B}|I|,$$

source Lipschitz control

$$\|u'\|_{L^\infty(I)} \leq Q^{C_u}|I|^{-1},$$

and Gate-2 failure at exponent A :

$$\int_I |u|^2(g^2 + h^2) dm \leq Q^{-A}|I|.$$

Then there exists an interval $J \subset I$ such that

$$|u| \geq \tau := \frac{1}{2}Q^{-B/2} \quad \text{on } J,$$

$$|J| \geq Q^{-B/2-C_u-O(1)}|I|,$$

and

$$\frac{1}{|J|} \int_J (g^2 + h^2) dm \leq Q^{-A+\frac{3}{2}B+C_u+O(1)}.$$

Proof. By Lemma 15.16, there is an interval $J \subset I$ with the stated lower bound for $|u|$ and for $|J|$. Since $|u|^2 \geq \tau^2$ on J ,

$$\tau^2 \int_J (g^2 + h^2) dm \leq \int_J |u|^2(g^2 + h^2) dm \leq \int_I |u|^2(g^2 + h^2) dm \leq Q^{-A}|I|.$$

Using $\tau^{-2} \asymp Q^B$ and

$$|J|^{-1} \leq Q^{B/2+C_u+O(1)}|I|^{-1}$$

gives the claim. $\square$

15.3. Gate 2 from source thickening and L^2 -input rigidity.

Definition 15.18 (Ordinary, singular, and transition packet split). Every Hardy packet family used by the source branch is partitioned as

$$\mathcal{P} = \mathcal{P}_{\text{ord}} \sqcup \mathcal{P}_{\text{sing}} \sqcup \mathcal{P}_{\text{trans}}.$$

The ordinary subfamily $\mathcal{P}_{\text{ord}}$ consists of packets on which the real-canonical Hardy branch is smooth, all chart denominators are bounded below by Q^{-C} , the normalized derivative convention $\mathcal{D}_p = \lambda_p \partial_t$ is fixed, and the identities

$$u_p = \mathcal{D}_p^2 q_\zeta, \quad \Gamma_G = u_p - 6q_\zeta + 2q_\zeta^3,$$

$$\Gamma_{G,2} = 2q_\zeta(\mathcal{D}_p q_\zeta)^2 - \frac{1}{6}\mathcal{D}_p^2 \Gamma_G$$

hold up to the declared residual ledger.

The singular subfamily contains packets meeting an unregularized critical-line zero, a branch denominator wall, or a chart singularity. The transition subfamily contains collar packets used to pass between ordinary charts or between scale-admissible packet regimes.

Hypothesis 15.19 (Ordinary packet no-loss routing). The active Hardy source packet family satisfies one of the following no-loss alternatives:

(a) ordinary packets carry polynomially positive source mass:

$$G_{\text{ord}} \geq Q^{-C} G_{\text{ref}};$$

- (b) the singular and transition packets are routed to certified alternative branches whose source-graded $K9/Y12$ lower bounds are polynomially comparable to the ordinary branch;
(c) the singular and transition packet contribution is $O(Q^{-B} G_{\text{ref}})$ for B larger than the final loss exponent.

Remark 15.20 (Ordinary packet no-loss is source-mass no-loss). Hypothesis 15.19 is a statement about source mass and source-graded carrier mass, not merely about the number of packets. The alternative

$$G_{\text{ord}} \geq Q^{-C} G_{\text{ref}}$$

means that the ordinary packet subfamily carries polynomially positive Hardy-source mass after the branch/reference weights and carrier projections used by the proof. It is not enough to count many ordinary packets if the source mass, Gate-2 anisotropy, or effective $K9/Y12$ carrier mass is concentrated on singular or transition packets.

Lemma 15.21 (Ordinary/singular routing no-loss consequence). *Assume Hypothesis 15.19. Then the source lower bound needed by the ordinary Gate-2 theorem may be applied to $\mathcal{P}_{\text{ord}}$ with at most polynomial loss, or else the same source-graded lower bound is supplied by a certified singular/transition branch. In particular, no source mass carried only by $\mathcal{P}_{\text{sing}} \cup \mathcal{P}_{\text{trans}}$ is silently discarded.*

Proof. The three alternatives in Hypothesis 15.19 are exhaustive by definition. In the first case the ordinary family carries the required mass. In the second case the singular and transition families are routed to an alternative certified branch with polynomially comparable lower bound. In the third case their total contribution is below the residual budget and can be absorbed into the final error ledger. Thus the active source mass is either ordinary, routed, or negligible at the declared exponent. $\square$

Theorem 15.22 (Gate 2 from Gate 1 and L^2 -input rigidity). *Assume the normalized real-canonical convention of Definitions 2.1 and 2.3. This is an ordinary smooth-packet statement. Packets containing an unregularized critical-line zero, or packets where the real completed phase is not smooth in the chosen convention, are routed to the singular/transition branch and are not covered by this theorem. Let*

$$u = q''_{\text{can}} = q'', \quad g = \Gamma_G = q'' - 6q + 2q^3, \quad h = \Gamma_{G,2} = 2q(q')^2 - \frac{1}{6}g''.$$

Assume Gate 1:

$$\int_I |u|^2 dm \geq Q^{-B} |I|.$$

Assume source Lipschitz control

$$\|u'\|_{L^\infty(I)} \leq Q^{C_u} |I|^{-1},$$

Assume moreover that every source interval produced by Lemma 15.17 can be placed in the normalized setting of Lemma 15.12, with the safe primitive bounds

$$\|q\|_\infty + \|q'\|_\infty + \|q''\|_\infty \leq Q^C$$

available on that normalized interval, and with rigidity loss Q^{C_R} . Then Gate 2 holds with any exponent

$$A > \frac{13}{2}B + C_u + 5C_R + O(1).$$

Equivalently, for such A ,

$$\boxed{\int_I |u|^2 (g^2 + h^2) dm \geq Q^{-A} |I|}.$$

Proof. Assume, toward contradiction, that

$$\int_I |u|^2 (g^2 + h^2) dm \leq Q^{-A} |I|.$$

By Lemma 15.17, there is an interval $J \subset I$ such that

$$|u| \geq \tau := \frac{1}{2} Q^{-B/2} \quad \text{on } J,$$

and

$$\frac{1}{|J|} \int_J (g^2 + h^2) dm \leq Q^{-A + \frac{3}{2}B + C_u + O(1)}.$$

Put

$$\varepsilon^2 = Q^{-A + \frac{3}{2}B + C_u + O(1)}.$$

Then

$$\|g\|_{L^2(J)} + \|h\|_{L^2(J)} \leq Q^{O(1)} \varepsilon |J|^{1/2}.$$

Absorbing the harmless factor into C_R , Lemma 15.12 gives on $J' \Subset J$

$$\int_{J'} |u|^2 dm \leq Q^{C_R} \varepsilon^{2/5} |J'|.$$

But $J' \subset J$ and $|u| \geq \tau$ on J , so

$$\int_{J'} |u|^2 dm \geq Q^{-B - O(1)} |J'|.$$

Using

$$\varepsilon^{2/5} = Q^{-\frac{1}{5}A + \frac{3}{10}B + \frac{1}{5}C_u + O(1)},$$

the two estimates are incompatible if

$$C_R - \frac{1}{5}A + \frac{3}{10}B + \frac{1}{5}C_u + O(1) < -B.$$

It suffices that

$$A > \frac{13}{2}B + C_u + 5C_R + O(1).$$

Thus Gate 2 failure is impossible for such A . $\square$

Definition 15.23 (Gate-2 residual ledger). On a normalized ordinary packet, write the exact Gate-2 identities as

$$u = \mathcal{D}_p^2 q, \quad g = u - 6q + 2q^3, \quad h = 2q(\mathcal{D}_p q)^2 - \frac{1}{6} \mathcal{D}_p^2 g.$$

If corrected coordinates or transition charts introduce errors, define

$$\begin{aligned} r_u &:= u - \mathcal{D}_p^2 q, \\ r_g &:= g - (u - 6q + 2q^3), \\ r_h &:= h - \left(2q(\mathcal{D}_p q)^2 - \frac{1}{6} \mathcal{D}_p^2 g \right). \end{aligned}$$

The Gate-2 residual ledger is the assertion that on every source-thickening interval J ,

$$\|r_u\|_{L^2(J)} + \|r_g\|_{L^2(J)} + \|r_h\|_{L^2(J)} \leq Q^{-A_{\text{res}}} |J|^{1/2},$$

with A_{res} larger than the Gate-2 rigidity threshold.

Lemma 15.24 (Residual-stable Gate–2 rigidity). *In the setting of Theorem 15.22, suppose that the exact Gate–2 identities hold up to the residuals of Definition 15.23. If*

$$\|g\|_{L^2(J)} + \|h\|_{L^2(J)} \leq \varepsilon |J|^{1/2}$$

and the residual ledger satisfies

$$\varepsilon_{\text{res}} := Q^{-A_{\text{res}}} \leq \varepsilon,$$

then on the inner interval $J' \Subset J$,

$$\int_{J'} |u|^2 \leq Q^{C_R} \left(\varepsilon^{2/5} + \varepsilon_{\text{res}}^{2/5} + \varepsilon_{\text{res}}^2 \right) |J'|.$$

Consequently the Gate–2 contradiction remains valid if

$$A_{\text{res}} > \frac{5}{2}(B + C_R) + O(1)$$

and the final downstream residual budget is chosen below the same scale.

Proof. Repeat the proof of the L^2 -input rigidity lemma with g, h replaced by their exact-model parts plus residuals. Every integration by parts term is bounded by the original ε -term plus a residual term of size $Q^{C_R} \varepsilon_{\text{res}}^{2/5}$ after the same optimization $\delta = \varepsilon^{1/5}$, and by the direct residual-square term $Q^{C_R} \varepsilon_{\text{res}}^2$. The displayed threshold makes these contributions smaller than the source-thickening lower bound. $\square$

Remark 15.25 (Derivative-convention no-go). The identity

$$h = 2q(q')^2 - \frac{1}{6}g''$$

is valid only in the same normalized derivative convention used to define $u = q''$. In packet coordinates this convention is $\mathcal{D}_p = \lambda_p \partial_t$, and all derivatives in u_p, g_p, h_p must use $\mathcal{D}_p$. Mixing physical ∂_t derivatives with normalized $\mathcal{D}_p$ -derivatives changes the powers of r_p and invalidates the Gate–2 wall-rigidity proof.

Theorem 15.26 (Carrier-visible Gate–2 output). *Let $\mathcal{P}_{\text{ord}}$ be a positive-density family of ordinary smooth Hardy-normalized packets. On each packet use the normalized derivative*

$$\mathcal{D}_p = \lambda_p \partial_t$$

and define

$$\begin{aligned} u_p &= \mathcal{D}_p^2 q_\zeta, \\ g_p &= \Gamma_G = u_p - 6q_\zeta + 2q_\zeta^3, \\ h_p &= \Gamma_{G,2} = 2q_\zeta (\mathcal{D}_p q_\zeta)^2 - \frac{1}{6} \mathcal{D}_p^2 g_p. \end{aligned}$$

Assume:

(a) *Gate–1 source survival on $\mathcal{P}_{\text{ord}}$:*

$$\sum_{p \in \mathcal{P}_{\text{ord}}} \mu_p \int_{K_p^+} |u_p|^2 d\nu_p \geq Q^{-A_1} G_{\text{ref}};$$

(b) *the ordinary Hardy primitive bounds for $q_\zeta, \mathcal{D}_p q_\zeta, \mathcal{D}_p^2 q_\zeta, \mathcal{D}_p^3 q_\zeta$;*

(c) *the residual ledger for u_p, g_p, h_p lies below the Gate–2 failure scale;*

(d) *carrier capture and the finite K9/Y12 Gram/ $\mathfrak{A}_{2Y}$ audits of Theorem E.135.*

Then the analytic Gate–2 mass

$$\sum_{p \in \mathcal{P}_{\text{ord}}} \mu_p \int_{K_p^+} |u_p|^2 (|g_p|^2 + |h_p|^2) d\nu_p \geq Q^{-A_2} G_{\text{ref}}$$

upgrades to the source-graded K9/Y12-visible lower bound

$$\sum_p \mu_p \left\| \Pi_{\text{live}} \Lambda_p^{\text{gr,full}}(u_H) \right\|_{\text{gr}}^2 \geq Q^{-A_3} G_{\text{ref}}.$$

Proof. The analytic Gate-2 lower bound follows from Theorem 15.22 on the ordinary packet subfamily, using the common normalized derivative convention and the stated primitive and residual hypotheses. The carrier-visible upgrade is exactly Theorem E.135, applied to the Hardy-source carrier generated by the differential AFE and the retained $C_7/Y12$ affine directions. The live/dead routing part of $\mathfrak{R}_{2Y}(p)$ ensures that the $Y12$ -visible component is labelled and retained rather than killed. $\square$

Remark 15.27 (Replacement of the old $D1/D2$ downstream package). Theorem 15.22 replaces the older “density/Remez propagation” and “pointwise derivative propagation” downstream inputs. The active local route is

$$\boxed{\text{Gate 1} + |u'|_\infty \text{ source thickening} + L^2\text{-input real rigidity} \Rightarrow \text{Gate 2.}}$$

No Remez–Cartan theorem, pointwise target-smallness interval, or pointwise Γ_G'' -propagation is used in this route.

15.4. Gauge-locked wall escape and row energy.

Definition 15.28 (Gauge-locked wall scalar). The wall scalar q_0 is gauge-locked to the Gate-2 and D4 package if the same scalar q_0 satisfies, in the same normalized derivative convention,

$$\begin{aligned} q_0'' &= u = q_{\text{can}}'', \\ \Gamma_G &= u - 6q_0 + 2q_0^3, \end{aligned}$$

and the D4 source factor has leading form

$$H \sim -\frac{q_0 u^3}{288}.$$

If these identities hold only up to residuals, the residuals must satisfy

$$|q_0'' - u| \leq c\tau,$$

and

$$|\Gamma_G - (u - 6q_0 + 2q_0^3)| \leq c\tau$$

on wall-escape neighborhoods.

Theorem 15.29 (Gauge-locked wall escape in normalized coordinates). *Fix the normalized derivative convention*

$$\mathcal{D} = \lambda \partial_m.$$

Let

$$q = q_0, \quad u = q_{\text{can}}'', \quad \Gamma = (\Gamma_G, \Gamma_{G,2}), \quad G := |u|^2 |\Gamma|^2.$$

Assume Gate 2 mass on I :

$$\int_I G(m) dm \geq Q^{-A} |I|.$$

Assume polynomial upper bounds

$$G \leq Q^{C_G}, \quad |\Gamma| \leq Q^{C_\Gamma}, \quad |\mathcal{D}u| \leq Q^{C_{u'}}.$$

Define the gauge residuals

$$\begin{aligned} r_C &:= u - \mathcal{D}^2 q_0, \\ r_G &:= \Gamma_G - (\mathcal{D}^2 q_0 - 6q_0 + 2q_0^3), \end{aligned}$$

and

$$r_H^* := H + \frac{q_0 u^3}{288}.$$

Assume on wall-escape neighborhoods that

$$|r_C| \leq c\tau, \quad |r_G - r_C| \leq c\tau,$$

and that r_H^* is controlled by the $D4$ residual estimate.

Choose thresholds

$$\tau = Q^{-C_\tau}, \quad \varepsilon = Q^{-C_\varepsilon},$$

with

$$2C_\tau \geq A + 2C_\Gamma + O(1),$$

and

$$C_\varepsilon \geq \max\{3C_\tau + 2C_{u'}, C_\tau + 2A + 2C_G\} + O(1).$$

Then either the non-wall region

$$\{|q_0| \geq \varepsilon\}$$

already supplies a non-wall D_4 sample, or there exists $m_1 \in I$ such that

$$|q_0(m_1)| = \varepsilon, \quad |u(m_1)| \geq c\tau, \quad |\Gamma_G(m_1)| \geq c\tau.$$

Consequently

$$G(m_1) \geq c\tau^4.$$

If D_4 row conversion holds at non-wall samples and row mobility holds, then

$$\boxed{\int_I |\mathbf{A}_N(m)|^2 dm \geq Q^{-C} |I|}.$$

Proof. Split

$$I_{\text{nd}} = \{|q_0| \geq \varepsilon\}, \quad I_{\text{wall}} = \{|q_0| < \varepsilon\}.$$

If I_{nd} carries a fixed polynomial portion of Gate 2 mass, then $G \leq Q^{C_G}$ gives a point $m_* \in I_{\text{nd}}$ with polynomially live $G(m_*)$, and D_4 applies directly.

Assume instead that wall mass dominates. Remove the low-source set

$$|u| < \tau.$$

Choosing

$$2C_\tau \geq A + 2C_\Gamma + O(1)$$

ensures this low-source set cannot carry all wall mass. Hence there is a wall-live point m_0 with

$$|q_0(m_0)| < \varepsilon, \quad |u(m_0)| \geq \tau,$$

away from the packet boundary after the standard buffered-packet reduction.

For complex q_0 , set

$$e = \frac{\overline{u(m_0)}}{|u(m_0)|}, \quad Q(m) := \Re(eq_0(m)).$$

Since

$$\mathcal{D}^2 Q(m) = \Re(e(\mathcal{D}^2 q_0)(m)) = \Re(e(u - r_C)(m)),$$

the assumptions $|r_C| \leq c\tau$ and

$$|\mathcal{D}u| r_{\text{esc}} \leq c\tau$$

imply

$$\mathcal{D}^2 Q \geq c\tau$$

on a normalized interval of radius r_{esc} . Taking

$$r_{\text{esc}} \asymp \sqrt{\varepsilon/\tau}$$

gives first exit from the strip $|q_0| < \varepsilon$ within normalized distance $O(r_{\text{esc}})$, provided

$$C_\varepsilon \geq 3C_\tau + 2C_{u'} + O(1).$$

Boundary and buffer loss are controlled by choosing

$$C_\varepsilon \geq C_\tau + 2A + 2C_G + O(1).$$

Thus there exists m_1 with

$$|q_0(m_1)| = \varepsilon, \quad |u(m_1)| \geq c\tau.$$

Finally,

$$\Gamma_G = u - 6q_0 + 2q_0^3 + (r_G - r_C).$$

At m_1 , if $\varepsilon \ll \tau$ and

$$|r_G - r_C| \leq c\tau,$$

then

$$|\Gamma_G(m_1)| \geq c\tau.$$

Thus $G(m_1) \geq c\tau^4$. D4 converts this non-wall sample to a row lower bound, and row mobility thickens the pointwise lower bound to interval row energy. $\square$

Remark 15.30 (Absolute residual corridor in wall escape). In Theorem 15.29, D4 residual control is interpreted valuation-wise. Same-valuation anisotropy-linear terms are absorbed into the effective D4 row matrix, and valuation-relative residuals are estimated relative to the leading row scale. Only absolute residual floors require a wall-threshold corridor.

Concretely, if the D4 leading valuation is

$$q_0 u^{b_*} L_* \Gamma$$

and the wall-escape sample satisfies

$$|q_0| = \varepsilon, \quad |u| \gg \tau, \quad |\Gamma| \gg \tau,$$

then an absolute residual bound

$$|\mathbf{R}_{\text{abs}}| \leq Q^{-C_R^{\text{abs}}}$$

must satisfy

$$C_R^{\text{abs}} \geq C_\varepsilon + (b_* + 1)C_\tau + C_L + O(1),$$

where Q^{-C_L} is the D4 row-rank lower bound. If D4 uses a filtered leading block through u -valuation B , replace b_* by B . This condition does not apply to residuals carrying useful q_0, u, Γ valuation.

Remark 15.31 (Packet buffer for wall escape). The first-exit wall-escape step moves by a distance

$$\rho|I|, \quad \rho \asymp \sqrt{\varepsilon/\tau}.$$

The packet used for row conversion and local-complexity estimates should therefore be chosen with a buffer of at least this scale. If the boundary layer has relative size η , and the Gate-2 density is bounded above by

$$G = |u|^2 |\Gamma|^2 \leq Q^{C_G},$$

then discarding the edge changes the Gate-2 mass by at most $Q^{C_G} \eta |I|$. Hence a boundary layer with

$$\eta \leq Q^{-A-C_G-O(1)}$$

is negligible at Gate-2 mass scale $Q^{-A}|I|$. This converts packet-edge effects into exponent bookkeeping rather than a new structural obstruction.

Remark 15.32 (D3 is replaced by wall escape plus D4). The old source-robust q_0 -wall row is not the active route in the row-energy architecture. Wall escape is the analytic version of saturating by q_0 : instead of dividing by q_0 , we move to a nearby point where

$$|q_0| = \varepsilon$$

and use the original rows. The source-robust wall-row search remains a fallback only if the final contradiction requires row conversion at a fixed preassigned point.

Lemma 15.33 (Row mobility: pointwise row output thickens to row energy). *Let $\mathbf{A}_N : I \rightarrow \mathbb{C}^M$ be a finite row vector. Suppose*

$$|\mathbf{A}_N(m_*)| \geq Q^{-C_A}$$

for some $m_ \in I$, and*

$$|\mathbf{A}'_N(m)| \leq Q^{C_D} |I|^{-1} \quad (m \in I).$$

Then

$$\int_I |\mathbf{A}_N(m)|^2 dm \geq Q^{-C'} |I|$$

for some polynomial exponent C' .

Proof. For

$$|m - m_*| \leq \frac{1}{2} Q^{-C_A - C_D} |I|,$$

the derivative bound gives

$$|\mathbf{A}_N(m) - \mathbf{A}_N(m_*)| \leq \frac{1}{2} Q^{-C_A}.$$

Thus

$$|\mathbf{A}_N(m)| \geq \frac{1}{2} Q^{-C_A}$$

on a subinterval of length at least $Q^{-C_A - C_D - O(1)} |I|$, after taking a one-sided subinterval if m_* is near the boundary. Integrating the square over that subinterval gives the claim. $\square$

Remark 15.34 (Quotient-frame version of row mobility). If the row output is measured in a moving quotient row space $\mathcal{Q}_N(m) = \mathcal{H}_N / \mathcal{C}_N(m)$, the same argument applies to

$$P_{\mathcal{C}_N(m)^\perp} \mathbf{A}_N(m)$$

provided the nuisance frame is polynomially stable:

$$\lambda_{\min}(C(m)^* C(m)) \geq Q^{-C}, \quad \|C'(m)\| \leq Q^C |I|^{-1}.$$

If raw or carrier-row energy is accepted downstream, no moving-frame derivative estimate is needed.

Remark 15.35 (Frozen quotient coordinates avoid moving-frame derivatives). If the quotient frame $\mathcal{Q}_N(m)$ varies with m , one can avoid moving-frame derivative estimates by freezing the quotient coordinates at the D4 sample point m_* . Suppose

$$|[\mathbf{A}_N(m_*)]|_{\mathcal{Q}_N(m_*)} \geq Q^{-C_A}$$

and the quotient Gram matrix at m_* is polynomially nondegenerate. Let ℓ_7^*, ℓ_9^* be dual quotient coordinates at m_* , extended as fixed linear functionals on the ambient finite row space. Then

$$|\ell_7^*(\mathbf{A}_N(m_*))|^2 + |\ell_9^*(\mathbf{A}_N(m_*))|^2 \geq Q^{-C}.$$

If the ambient row vector satisfies

$$|\mathbf{A}'_N(m)| \leq Q^{C_D} |I|^{-1},$$

then the two fixed scalar functions

$$\ell_7^*(\mathbf{A}_N(m)), \quad \ell_9^*(\mathbf{A}_N(m))$$

obey the same row-mobility argument, yielding

$$\int_I (|\ell_7^*(\mathbf{A}_N(m))|^2 + |\ell_9^*(\mathbf{A}_N(m))|^2) dm \geq Q^{-C'} |I|.$$

Thus carrier-quotient row energy can be proved using fixed dual coordinates at the sample point, without differentiating the moving quotient frame.

Theorem 15.36 (Local downstream closure from Gate-2 and D4). *Assume Gate 2 on I :*

$$\int_I |u|^2 |\Gamma|^2 dm \geq Q^{-A} |I|.$$

Assume the ordinary real-canonical local complexity bounds and the gauge-locking hypotheses needed for Theorem 15.29. Assume D4 row conversion at non-wall samples in a chosen row-output space

$$\mathcal{Q}_{\text{row}},$$

where $\mathcal{Q}_{\text{row}}$ is either the carrier quotient $\mathcal{Q}_{\text{car}}$, a raw/ambient row space, or another filtered row space with polynomially equivalent norm. Finally assume row mobility as in Lemma 15.33, or the quotient-frame version of Remark 15.34 if the quotient varies with m .

Then the sampled packet carries polynomial row energy in the chosen row-output space:

$$\boxed{\int_I \|\mathbf{A}_N(m)\|_{\mathcal{Q}_{\text{row}}}^2 dm \geq Q^{-C} |I|}.$$

In the active carrier-quotient formulation this is

$$\int_I ||[A_N^{\text{car}}(m)]||_{Q_{\text{car}}}^2 dm \geq Q^{-C}|I|.$$

If the row-output space is represented by a concrete projected coordinate system, this theorem gives projected coordinate energy only after a polynomial frame lower bound for that coordinate system. If the downstream architecture is exact-point locked or forced to use only the collapsed scalar I -coordinate, this theorem is not sufficient and the q_0 -adic row-pencil, source-robust wall-row fallback, scalar incidence avoidance, or a higher/ambient row replacement must be used.

16. CURRENT PROOF STATE AFTER THE LOCAL PACKAGE AUDITS

Remark 16.1 (Audit convention for the current proof-state section). The statements in this section distinguish three levels:

- (1) local algebraic lemmas, such as Gate A/B cancellations under fixed source/core and grading conventions;
- (2) conditional local geometric infrastructure, such as the reduced-base rank-two connection and affine forcing equation;
- (3) missing global proof gates, such as affine transport, boundary, holonomy/index, or finite-core non-concentration.

A statement at level (1) or (2) should not be read as a proof of RH. The current missing theorem is global.

Remark 16.2 (Relation to the final named-input ledger). This section preserves the local Bottleneck-C and finite-core audit history. It should not be read as the final dependency graph of the present manuscript. The final dependency graph is summarized in Section E and Remark F.1. In particular, the ordinary upstream branch is now the canonical Hardy-normalized source branch, whose proof-level input is

$$\text{Agg}L^2(q_{\text{can}}'').$$

The older O1/O2/O3 finite-row package is historical diagnostic material unless a separate transfer theorem carries it into this canonical L^2 -source norm. The local downstream branch is now routed through the certified source-graded $K9$ carrier, not through a scalar-collapsed primitive channel.

Remark 16.3 (Updated downstream status after the Gate-2 and carrier-quotient audits). The downstream local package has changed relative to earlier versions of this section.

First, the old $D1/D2$ propagation route is no longer the active Gate-2 route. In the ordinary real-canonical branch, once Gate 1 is available, Gate 2 is supplied by Theorem 15.22:

$$\text{Gate 1} + |u'|_{\infty} \text{ source thickening} + L^2\text{-input real rigidity} \implies \text{Gate 2}.$$

Second, the old source-robust q_0 -wall row is no longer the active D3 route under row-energy architecture. It is replaced by the gauge-locked wall escape theorem, Theorem 15.29, plus D4.

Third, the scalar order-nine I -coordinate wall is not the active D4 obstruction if row energy is measured in the carrier quotient. The order-seven and order-nine clean carriers survive independently in Q_{car} by Theorem 16.100.

Thus the current downstream implication is

$$\boxed{\text{Gate 1} + \text{ordinary real-canonical local complexity} + \text{carrier-quotient D4 residual control} \implies \text{row energy}.}$$

This remains a downstream theorem only. It does not discharge the upstream ordinary/singular actual-zeta inputs needed to obtain Gate 1 globally.

The remaining subsections of this section record the broader proof state after the Gate B, Bottleneck C, finite-core, and interaction audits. They are a status ledger for older and parallel proof obligations and should be read separately from the ordinary real-canonical downstream

closure package above. In particular, references below to local Bottleneck–C, finite-core sampled vector non-concentration, or dd -interaction rigidity are not meant to undo the updated downstream organization:

$$\text{Gate 1} + |u'|_\infty \text{ source thickening} + L^2\text{-input real rigidity} \Rightarrow \text{Gate 2},$$

followed by

$$\text{Gate 2} + \text{gauge-locked wall escape} + \text{carrier-quotient D4} + \text{row mobility} \Rightarrow \text{row energy}.$$

The closed branches are:

$$\boxed{M_A, M_C \text{ connection closure is complete under source-grade reflection,}}$$

$$\boxed{\text{mixed four-point dependence factors through } (c, u_5, v_5, \Delta_7),}$$

and

$$\boxed{\text{finite odd-germ determinacy holds on compact nonsingular actual-source strata.}}$$

The live branches are:

$$\boxed{\text{local Bottleneck C physical edge obstruction,}}$$

$$\boxed{\text{finite-core sampled vector non-concentration,}}$$

and

$$\boxed{dd\text{-interaction feature-balance} / \text{spectral rigidity.}}$$

The local Bottleneck–C branch has now exhausted the scalar, quotient, split, and line-bundle repair routes. The remaining viable language is the full rank-two fixed-sector bundle

$$\mathcal{F}_b = \mathfrak{f}_h(b),$$

with $E_{\eta\eta}$ treated as vector-valued affine forcing. Gate A and Gate B remain as local algebraic lemmas, but the proof now requires a global rank-two affine transport, vector-valued non-concentration, projective-plus-scale index, or comparable non-scalar theorem.

16.1. Closed branch: M_A, M_C connection closure. Let

$$M_A = \dot{r}_3 r_2 \dot{r}_4, \quad M_C = \dot{r}_3 \dot{r}_2 r_4.$$

Under the homogeneous scalar-grade reflection convention,

$$\mathbf{P}_g \circ \mathbf{S} = (-1)^g \mathbf{S} \circ \mathbf{P}_g,$$

where $\mathbf{S}$ is endpoint swap, the corrected mixed block satisfies

$$N^{\text{corr}}(h_-, h_+)^T = N^{\text{corr}}(h_+, h_-).$$

Consequently, for the source-linear middle block Y_g ,

$$Y_g(h_-, h_+)^T = (-1)^g Y_g(h_+, h_-).$$

Writing

$$Y_g(h - \delta/2, h + \delta/2) = \sum_{e \geq 0} \delta^e Y_g^{(e)}(h),$$

one obtains the edge-refined parity

$$\left(Y_{g,\beta}^{(e)} \right)^T = (-1)^{g+e} Y_{g,\beta}^{(e)}.$$

This parity closes the labelled $J = (3, 2, 4)$ pre-trivialization table:

$$Z_j[M_A] = Z_j[M_C] = 0, \quad 0 \leq j \leq 7.$$

Therefore the connection identities

$$R_7^{(A)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_A], A_i) = 0,$$

and

$$R_7^{(C)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_C], A_i) = 0$$

hold under the stated source-grade reflection convention.

Thus the M_A, M_C connection branch is no longer a live obstruction.

16.2. Live local obstruction: the frozen promoted mixed-block coefficient. The first live local actual-source/Bottleneck C carrier is the promoted quadratic mixed-block channel

$$Y_{12}^{(2)} := D^2 N_0[\xi_1, \xi_2].$$

It is coupled to one same-point linear whitening slot. The promoted channel has amplitude factor

$$a_1 a_2 (a_1 + a_2).$$

On the collision chart

$$a_2 = -a_1(1 + \kappa\delta),$$

one has

$$a_1 a_2 (a_1 + a_2) = a_1^3 \kappa \delta + O(\delta^2).$$

Thus the amplitude already supplies the first δ -edge. Therefore the first Bottleneck C coefficient is the frozen internal coefficient

$$E_{A,Y}^{\text{BC}} := \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \text{Sym}[z^7] D(G_-^{-1/2})[A_-] Y_{12}^{(2)} R_0.$$

Equivalently,

$$E_{A,Y}^{\text{BC}} = \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \text{Sym} \sum_{\chi \in \{dd, d\eta + \eta d, \eta\eta\}} \sum_{g=4}^7 \sum_{r+b=7-g} H_{r,0}^A Y_{g,0}^\chi W_{+,b}.$$

Write

$$E_{A,Y}^{\text{BC}} = e_{A,Y} I + f_{A,Y} S.$$

The quotient Bottleneck C obstruction is

$$Q_{A,Y} := e_{A,Y} v_5 - u_5 f_{A,Y}.$$

The transform-safe obstruction is

$$\Phi_K(E_{A,Y}^{\text{BC}}) = 2f_{A,Y}.$$

For comparison, the literal source-position edge

$$E_{A,Y}^{\text{pos}} := \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \text{Sym}[z^7 \delta^1] D(G_-^{-1/2})[A_-] Y_{12}^{(2)} R_0$$

belongs to the next source-motion layer. It is not the first Bottleneck C amplitude-derivative coefficient.

Remark 16.4 (Executable table for $E_{A,Y}^{\text{BC}}$ (current section)). The frozen Bottleneck C coefficient is computed from the rows

$$\mathcal{E}_{\chi;g,r,b}^a := \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \text{Sym} \left(\Lambda_{a,r}^- Y_{g,0}^\chi W_{+,b} \right),$$

where

$$\begin{aligned} a &\in \mathcal{A}_{\text{BC}}, & \chi &\in \{dd, d\eta + \eta d, \eta\eta\}, \\ 4 &\leq g \leq 7, & r + b &= 7 - g. \end{aligned}$$

Thus

$$E_{A,Y}^{\text{BC}} = \sum_{\chi \in \{dd, d\eta + \eta d, \eta\eta\}} \sum_{a \in \mathcal{A}_{\text{BC}}} \sum_{\substack{4 \leq g \leq 7 \\ r+b=7-g}} \mathcal{E}_{\chi;g,r,b}^a.$$

The skew combination

$$Y_{d\eta - \eta d}^{(2)}$$

is excluded because it is killed by fixed-sector projection.

Remark 16.5 (Status of the frozen promoted mixed-block coefficient (current section)). The finite Fréchet–Sylvester recurrences reduce $E_{A,Y}^{\text{BC}}$ to the finite symbolic table of Remark 16.4. However, the current draft does not supply the scalar midpoint source-jet data needed to evaluate

$$E_{A,Y}^{\text{BC}} = e_{A,Y}I + f_{A,Y}S.$$

The required scalar data include, safely,

$$q_0^{(k)}(m), \quad 0 \leq k \leq 9,$$

and

$$(r_i^{[a]})^{(k)}(m), \quad i = 1, 2, \quad a \in \mathcal{A}_{\text{BC}}, \quad 2 \leq k \leq 9.$$

Equivalently, the paper must supply the midpoint Taylor coefficients of

$$d_{i,\pm}(z), \quad \eta_i(z),$$

through the orders contributing to

$$Y_{g,0}^{dd}, \quad Y_{g,0}^{d\eta+\eta d}, \quad Y_{g,0}^{\eta\eta}, \quad 4 \leq g \leq 7.$$

At present no available symmetry, source-convention rebuild, quotient interpretation, transform-faithful invariant, or endpoint-symmetrized finite- s carrier forces

$$E_{A,Y}^{\text{BC}} = 0, \quad e_{A,Y}v_5 - u_5f_{A,Y} = 0, \quad f_{A,Y} = 0.$$

More specifically, the odd- $\eta\eta$ top row

$$\mathcal{E}_{\eta\eta;7,0,0}^2$$

has the S -visible coefficient recorded in Subsection 16.3. Therefore the first unresolved local actual-source/Bottleneck-C obstruction is now the top-row cancellation problem

$$\nu_{21,7} = 0$$

or an equivalent physical cancellation of

$$\mathcal{E}_{\eta\eta;7,0,0}^2.$$

Lemma 16.6 (Scalar tests for the frozen promoted mixed-block coefficient). *Let*

$$E_{A,Y}^{\text{BC}} = eI + fS, \quad A_5^{\mathfrak{f}} = u_5I + v_5S.$$

Then quotient Bottleneck C for the promoted mixed-block carrier is equivalent to

$$ev_5 - u_5f = 0.$$

The transform-safe T-horizontal condition is equivalent to

$$f = 0.$$

Proof. The quotient line is

$$\mathbf{C}A_5^{\mathfrak{f}}.$$

Thus

$$E_{A,Y}^{\text{BC}} \in \mathbf{C}A_5^{\mathfrak{f}}$$

is equivalent, on a good patch, to the vanishing of the determinant

$$ev_5 - u_5f.$$

On the other hand,

$$\Phi_K(I) = 0, \quad \Phi_K(S) = 2.$$

Hence

$$\Phi_K(E_{A,Y}^{\text{BC}}) = 2f,$$

so the transform-safe condition is exactly $f = 0$. □

Corollary 16.7 (First live T -slope carrier (current section)). *The amplitude factor of the promoted carrier is*

$$a_1 a_2 (a_1 + a_2),$$

and

$$\partial_{a_2} (a_1 a_2 (a_1 + a_2)) \big|_{a_2 = -a_1} = -a_1^2.$$

Therefore

$$\partial_\kappa T \big|_{E_{A,Y}^{\text{BC}}} = -a_1^2 \frac{\Phi_K(E_{A,Y}^{\text{BC}})}{2c} = -a_1^2 \frac{f}{c}.$$

If

$$E_{A,Y}^{\text{BC}} = \lambda A_5^f,$$

then

$$\partial_\kappa T \big|_{E_{A,Y}^{\text{BC}}} = -a_1^2 \lambda \frac{v_5}{c}.$$

Thus, on the generic patch $v_5 \neq 0$, a nonzero line-gauge $E_{A,Y}^{\text{BC}}$ closes the quotient condition but still leaks through T .

16.3. Concrete odd- $\eta\eta$ top-row obstruction. The local package audits isolate a concrete top-row survivor inside the frozen Bottleneck C coefficient. The relevant row is

$$\mathcal{E}_{\eta\eta;7,0,0}^2 = \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \text{Sym} \left(\Lambda_{2,0}^- \left[\left(\frac{\sin \Delta_0}{s^2} + \frac{q_- \cos \Delta_0}{s} \right) J \eta_i \eta_j \right]_7 W_{+,0} \right),$$

where

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

This is the actual finite Fréchet source index $a = 2$, not the older shorthand $a \in \{1, 5\}$.

The row comes from the $(2, 1)$ -off-diagonal entry

$$N_{21} = -\frac{\sin \Delta + q_- s \cos \Delta}{s^2}.$$

Twice differentiating in the scalar phase-gap variable Δ gives

$$\partial_\Delta^2 N_{21} \big|_0 = \frac{\sin \Delta_0}{s^2} + \frac{q_- \cos \Delta_0}{s}.$$

Thus the odd-background $\eta\eta$ block is not an endpoint-separation artifact. It is a scalar Δ -Hessian term.

For the active finite Fréchet row one has

$$\Lambda_{2,0}^- = -\frac{1}{24\pi q^3} E_{22}.$$

If

$$W_{+,0} = \begin{pmatrix} u & w \\ w & v \end{pmatrix},$$

then

$$\pi_{\mathfrak{f}} \text{Sym} \left(\Lambda_{2,0}^- J W_{+,0} \right) = -\frac{1}{24\pi q^3} (wI + uS).$$

Consequently, if

$$\nu_{21,7} := [z^7] \left[\left(\frac{\sin \Delta_0}{s^2} + \frac{q_- \cos \Delta_0}{s} \right) \eta_i \eta_j \right],$$

then this row contributes

$$\mathcal{E}_{\eta\eta;7,0,0}^2 = -\frac{\nu_{21,7}}{24\pi q^3} (wI + uS),$$

up to the harmless good-patch transport $\tau_{h \rightarrow m}$. In particular,

$$f_{\eta\eta}^\top = -\frac{u}{24\pi q^3} \nu_{21,7}.$$

Under the favorable phase-gap integral model,

$$\nu_{21,7} = \frac{1}{Q^{14}} \left[\frac{q_0}{480} (\rho_{i,2}\rho_{j,4} + \rho_{i,4}\rho_{j,2}) + \frac{1}{144} \left(\frac{q_0''}{6} - \frac{2q_0^3}{3} \right) \rho_{i,2}\rho_{j,2} \right].$$

For the homogeneous source $r(u) = u^2$, with $q_0'' = 0$, this gives

$$\nu_{21,7} = -\frac{q_0^3}{216Q^{14}},$$

which is nonzero on a good phase patch.

Therefore the row is generically T -visible:

$$\partial_\kappa T = -a_1^2 \frac{f_{\eta\eta}^\top}{c} \neq 0.$$

It is also quotient-visible on a generic $u_5 \neq 0$ patch; for example, if $w = 0$, then

$$Q_{\eta\eta}^\top = -u_5 f_{\eta\eta}^\top.$$

Remark 16.8 (Endpoint cleanup does not remove this row). Endpoint-diagonal realization, if proved, removes only endpoint-separation subchannels such as

$$d\eta^\Delta + \eta^\Delta d, \quad \eta^\Sigma \eta^\Delta.$$

It does not remove the scalar $\eta\eta$ term above, since that term comes from $\partial_\Delta^2 N_0[\eta_i, \eta_j]$. Thus the older route

$$[\delta^0]\eta^\Delta = 0 \implies E_{A,Y}^{\text{BC}} = 0$$

is no longer sufficient.

Remark 16.9 (No existing package mechanism cancels the row). The present package mechanisms do not cancel $\mathcal{E}_{\eta\eta;7,0,0}^2$: source-bidegree projection does not remove the promoted degree-three amplitude pattern $a_1 a_2 (a_1 + a_2)$; amplitude normalization only rescales the row; midpoint trivialization is invertible on a good patch; connection correction is post-edge and cannot cancel a raw frozen coefficient; and quotienting is insufficient because the augmented transform-safe package requires the S -component itself to vanish.

Thus the local branch is in the following binary state. Either one proves a new source-jet identity

$$\nu_{21,7} = 0,$$

or one proves a full top-row S -cancellation

$$\sum_{\text{top } \eta\eta} f_{\eta\eta}^\top = 0.$$

The only remaining local rescue is a genuine physical aggregate S -coefficient cancellation, not a formal package counterterm. Without such cancellation, the current corrected package does not close Bottleneck C or the T -horizontal condition at this row.

16.4. No-go for a transform-faithful free $\Theta_\pm$ -normal repair. The off-diagonal numerator variables

$$\Theta_+ = \sin \Delta + q_+ s \cos \Delta, \quad \Theta_- = \sin \Delta + q_- s \cos \Delta$$

make the off-diagonal entries formally affine:

$$N_{12}^\Theta = \frac{\Theta_+}{s^2}, \quad N_{21}^\Theta = -\frac{\Theta_-}{s^2}.$$

As a formal normal package this cancels the original off-diagonal $\eta\eta$ Hessian row. It does not, however, close the physical Bottleneck C obstruction.

The distinction is important. As a formal normal coordinate, $\Theta_\pm$ makes the off-diagonal entries affine. As a physical coordinate induced from (q_-, q_+, Δ) , however, $\Theta_\pm$ is constrained and its source-curvature must either remain in the raw package or be carried by the hidden physical readout. The following no-go concerns the latter, transform-faithful interpretation.

Definition 16.10 (Θ -local physical edge residual). Let

$$R_{\text{off},7}^{\Theta}$$

be the septic fixed-sector residual produced by the actual constrained source curvature of the off-diagonal variables $\Theta_{\pm}$. Let

$$R_{\text{diag},7}^{\Theta}$$

be the shifted diagonal $\eta\eta$ septic residual

$$\pi_{\mathfrak{f}} \text{Sym} \left(\Lambda_{\bullet,1}^{-} Y_6^{\eta\eta,\text{diag}} W_{+,0} + \Lambda_{\bullet,0}^{-} Y_6^{\eta\eta,\text{diag}} W_{+,1} \right).$$

Define

$$R_{\text{loc},7}^{\Theta} := R_{\text{off},7}^{\Theta} + R_{\text{diag},7}^{\Theta}.$$

Write

$$R_{\text{loc},7}^{\Theta} = \alpha_{\text{loc}}^{\Theta} I + \beta_{\text{loc}}^{\Theta} S.$$

The scalar $\beta_{\text{loc}}^{\Theta}$ is the Φ_K -visible physical edge residual left by the $\Theta_{\pm}$ -normal package.

Remark 16.11 (Constrained $\Theta_{\pm}$ variables reintroduce curvature). The variables $\Theta_{\pm}$ are not free affine source variables in the physical finite- s block. They are constrained by

$$\Theta_{\sigma} = \sin \Delta + q_{\sigma} s \cos \Delta.$$

For two source variations,

$$D_{\text{source}}^2 \Theta_{\sigma} = A_{\sigma} \eta_{ij} + sC \chi_{\sigma,ij} - \Theta_{\sigma,0} \eta_i \eta_j - sS(\chi_{\sigma,i} \eta_j + \chi_{\sigma,j} \eta_i).$$

In the pure $\eta\eta$ channel, this gives

$$D_{\text{source}}^2 N_{21}^{\Theta,\text{actual}} = \left(\frac{\sin \Delta_0}{s^2} + \frac{q_- \cos \Delta_0}{s} \right) \eta_i \eta_j,$$

which is the old off-diagonal source-curvature obstruction. Thus the $\Theta_{\pm}$ -normal form is not a physical coordinate relabeling that removes the obstruction; it is a formal package subtraction unless this curvature is carried back into the physical readout.

Proposition 16.12 (No transform-faithful septic normal-form repair). *Let K_7 be a local septic normal-form correction satisfying:*

- (1) K_7 is invisible below ordinary order 7;
- (2) K_7 is source-quadratic, hence does not alter source-linear Gate A, Gate B, or the M_A, M_C connection tables;
- (3) K_7 cancels the local septic residual

$$R_{\text{loc},7}^{\Theta} = \alpha_{\text{loc}}^{\Theta} I + \beta_{\text{loc}}^{\Theta} S;$$

- (4) the physical Φ_K -readout is unchanged, so no correction is added back into the physical hidden coordinate.

Then necessarily

$$\beta_{\text{loc}}^{\Theta} = 0.$$

Consequently, on any good patch where $\beta_{\text{loc}}^{\Theta} \neq 0$, no such transform-faithful local normal-form correction exists.

Proof. To cancel $R_{\text{loc},7}^{\Theta}$, the correction K_7 must contain the fixed-sector component

$$-\beta_{\text{loc}}^{\Theta} S.$$

But preserving the physical Φ_K -readout without adding a correction to the hidden coordinate requires

$$[z^7] \Phi_K(K_7) = 0.$$

Since $\Phi_K(S) = 2$, this forces

$$\beta_{\text{loc}}^{\Theta} = 0.$$

□

Proposition 16.13 (Physical edge obstruction survives in a good-patch model). *There exists a homogeneous good-patch local coefficient model for which*

$$\partial_\kappa \left(\frac{\beta_{\text{loc}}^\Theta}{c} \right) \Big|_{\kappa=0} \neq 0.$$

Proof. After extracting the frozen Bottleneck-C matrix coefficient, the remaining amplitude derivative is the derivative of

$$a_1 a_2 (a_1 + a_2)$$

in the transverse amplitude direction at the balanced point $a_2 = -a_1$. Equivalently, using the local edge coordinate

$$\kappa_{\text{amp}} := a_2 + a_1,$$

one has

$$a_1 a_2 (a_1 + a_2) = a_1 (-a_1 + \kappa_{\text{amp}}) \kappa_{\text{amp}} = -a_1^2 \kappa_{\text{amp}} + O(\kappa_{\text{amp}}^2),$$

so

$$\partial_{\kappa_{\text{amp}}} (a_1 a_2 (a_1 + a_2)) \Big|_{\kappa_{\text{amp}}=0} = -a_1^2.$$

This is the same scalar edge factor obtained from the collision chart

$$a_2 = -a_1 (1 + \kappa \delta)$$

after the first δ -edge has been factored into the frozen Bottleneck-C coefficient.

Since $\beta_{\text{loc}}^\Theta(0) = 0$, differentiating $1/c$ contributes nothing at $\kappa = 0$. Thus

$$\partial_\kappa \left(\frac{\beta_{\text{loc}}^\Theta}{c} \right) \Big|_{\kappa=0} = -\frac{a_1^2}{c_0} B_{\text{loc}}^\Theta,$$

where B_{loc}^Θ is the frozen physical edge coefficient.

Take a homogeneous source model

$$r_i = r_j = u^2, \quad q_0'' = 0,$$

on a good patch with

$$u q_0 q Q c_0 \neq 0.$$

The constrained off-diagonal $\Theta_\pm$ -source curvature contributes a nonzero fixed-sector S -coefficient. In the top-row normalization this has leading form

$$B_{\text{off}}^{\Theta, \text{hom}} = \frac{u q_0^3}{5184 \pi q^3 Q^{14}}.$$

The shifted diagonal term has the form

$$B_{\text{diag}}^{\Theta, \text{hom}} = -\frac{\lambda}{2\pi Q^{12}} \left(\frac{q_0 \alpha}{72} + \frac{q_0^3 \beta}{864} \right), \quad \lambda = \frac{\alpha \beta}{4\pi Q(p_1 + p_2)}.$$

Choosing

$$\alpha = \varepsilon \alpha_0, \quad \beta = \varepsilon \beta_0,$$

with $\varepsilon \neq 0$ sufficiently small gives

$$B_{\text{diag}}^{\Theta, \text{hom}} = O(\varepsilon^3),$$

while $B_{\text{off}}^{\Theta, \text{hom}}$ remains nonzero. Therefore

$$B_{\text{loc}}^{\Theta, \text{hom}} = B_{\text{off}}^{\Theta, \text{hom}} + O(\varepsilon^3) \neq 0$$

for sufficiently small ε , and the claimed derivative is nonzero. $\square$

Remark 16.14 (Status of the $\Theta_\pm$ -normal repair). A formal septic package may subtract $R_{\text{loc},7}^\Theta$ and thereby make the formal package horizontal. But the physical hidden coordinate is then

$$T^{\text{phys}} = T^\star + \frac{\beta_{\text{loc}}^\Theta}{c}.$$

Thus the transform-safe physical Bottleneck C condition is

$$\partial_\kappa \left(T^\star + \frac{\beta_{\text{loc}}^\Theta}{c} \right) = 0.$$

Proposition 16.13 shows that the second term is not currently zero. Therefore local Bottleneck C cannot be closed by $\Theta_\pm$ -normal coordinates or septic normal-form counterterms alone. The remaining local route would require a new physical edge identity

$$\partial_\kappa(\beta_{\text{loc}}^\Theta/c) = 0$$

not present in the current package calculus.

Remark 16.15 ($\Theta_\pm$ no-go versus the frozen Bottleneck-C coefficient). The no-go result for the $\Theta_\pm$ -normal repair does not replace the frozen coefficient

$$E_{A,Y}^{\text{BC}}.$$

Rather, it says that normal-form coordinates or septic counterterms cannot make the physical transform-safe Bottleneck-C obstruction disappear. The canonical local coefficient to compute remains

$$E_{A,Y}^{\text{BC}} = e_{A,Y}I + f_{A,Y}S,$$

with quotient test

$$e_{A,Y}v_5 - u_5f_{A,Y} = 0$$

and transform-safe test

$$f_{A,Y} = 0.$$

The $\Theta_\pm$ audit shows that if one removes a visible septic residual from a formal package, the same residual must be carried by the physical hidden coordinate. Thus it is an obstruction-isolation result, not a replacement for the $E_{A,Y}^{\text{BC}}$ finite table.

The concrete top-row obstruction

$$\mathcal{E}_{\eta\eta;7,0,0}^2$$

of Subsection 16.3 is the current row-level witness of this phenomenon: removing it from a formal septic package without changing the physical readout merely transfers the same obstruction to the hidden T -coordinate.

16.5. Actual exterior-pair source jets and failure of pair-level locking. The most direct possible rescue of the Bottleneck-C row would be a hidden actual-source relation forcing the midpoint post-affine source jets

$$\rho_2, \quad \rho_4$$

to satisfy the ratio required by the $\eta\eta$ top-row coefficient. This does not occur at the one-pair level.

For an exterior functional-equation zero-pair write

$$f_{\beta,\gamma}(t) = \frac{1-\beta}{(1-\beta)^2 + (2t-\gamma)^2} + \frac{\beta}{\beta^2 + (2t-\gamma)^2}.$$

Put

$$a := 1 - \beta, \quad b := \beta, \quad x := 2m - \gamma, \quad u := t - m.$$

Then

$$f_{\beta,\gamma}(m+u) = \frac{a}{a^2 + (x+2u)^2} + \frac{b}{b^2 + (x+2u)^2}.$$

Affine subtraction removes only the constant and linear terms. Thus the post-affine coefficients ρ_2, ρ_4 are the u^2, u^4 coefficients of the above expansion.

For one summand,

$$[u^2] \frac{c}{c^2 + (x+2u)^2} = \frac{4c(3x^2 - c^2)}{(c^2 + x^2)^3},$$

and

$$[u^4] \frac{c}{c^2 + (x+2u)^2} = \frac{16c(c^4 - 10c^2x^2 + 5x^4)}{(c^2 + x^2)^5}.$$

Therefore

$$\rho_2(\beta, x) = 4a \frac{3x^2 - a^2}{(a^2 + x^2)^3} + 4b \frac{3x^2 - b^2}{(b^2 + x^2)^3}$$

and

$$\rho_4(\beta, x) = 16a \frac{a^4 - 10a^2x^2 + 5x^4}{(a^2 + x^2)^5} + 16b \frac{b^4 - 10b^2x^2 + 5x^4}{(b^2 + x^2)^5}.$$

Proposition 16.16 (No one-pair jet-locking law). *For $\rho_2(\beta, x) \neq 0$, the ratio*

$$\frac{\rho_4(\beta, x)}{\rho_2(\beta, x)} = 4 \frac{a \frac{a^4 - 10a^2x^2 + 5x^4}{(a^2 + x^2)^5} + b \frac{b^4 - 10b^2x^2 + 5x^4}{(b^2 + x^2)^5}}{a \frac{3x^2 - a^2}{(a^2 + x^2)^3} + b \frac{3x^2 - b^2}{(b^2 + x^2)^3}}$$

is a nonconstant rational function of the exterior-pair parameters. In particular, the actual one-pair source class does not force

$$\frac{\rho_4}{\rho_2} = \frac{10q_0^2}{9} - \frac{5q_0''}{18q_0}.$$

Proof. The displayed formulas are obtained by Taylor expanding the two summands at $u = 0$. The ratio ρ_4/ρ_2 depends on the one-pair parameters (β, x) , while the Bottleneck-C locking ratio depends on the local baseline phase jets q_0, q_0'' . There is no source-internal identity equating these quantities. $\square$

Two useful checks are:

$$\beta = \frac{1}{2}, \quad x = 0 \quad \implies \quad \rho_2 = -64, \quad \rho_4 = 1024, \quad \frac{\rho_4}{\rho_2} = -16,$$

and, as $|x| \rightarrow \infty$,

$$\rho_2 = \frac{12}{x^4} + O(x^{-6}), \quad \rho_4 = \frac{80}{x^6} + O(x^{-8}),$$

so

$$\frac{\rho_4}{\rho_2} = \frac{20}{3x^2} + O(x^{-4}).$$

Thus far exterior source atoms approximate the formal bad germ $r(u) = u^2$ to finite jet order. Any cancellation of the aggregate Bottleneck-C coefficient must therefore be a genuine global zero-tail identity, not a pair-level source rigidity law.

Remark 16.17 (Full symmetry-packet closure does not cancel the source moments). The paired atom $f_{\beta, \gamma}$ already includes the functional-equation partner at the same ordinate:

$$\beta + i\gamma, \quad 1 - \beta + i\gamma.$$

The full symmetry packet also contains the conjugate ordinate pair

$$\beta - i\gamma, \quad 1 - \beta - i\gamma,$$

whose paired contribution is

$$f_{\beta, -\gamma}(t) = \frac{1 - \beta}{(1 - \beta)^2 + (2t + \gamma)^2} + \frac{\beta}{\beta^2 + (2t + \gamma)^2}.$$

If

$$x_+(m) = 2m - \gamma, \quad x_-(m) = 2m + \gamma,$$

then the corresponding post-affine coefficients are

$$\rho_2(\beta, x_+), \quad \rho_4(\beta, x_+), \quad \rho_2(\beta, x_-), \quad \rho_4(\beta, x_-).$$

The formulas above are even in x . Thus at a symmetric midpoint $m = 0$, the γ and $-\gamma$ packets add:

$$\rho_2(\beta, -\gamma) = \rho_2(\beta, \gamma), \quad \rho_4(\beta, -\gamma) = \rho_4(\beta, \gamma).$$

At the midpoint centered on the $+\gamma$ packet, $x_+ = 0$ while $x_- = 2\gamma$; again the conjugate packet supplies an additional contribution rather than a sign-reversed copy. Therefore functional-equation and conjugation closure do not structurally force

$$A_2 = 0, \quad A_4 = 0, \quad \nu_{\eta\eta} = 0.$$

Any cancellation after summing the full packet is a proper analytic incidence condition, not a symmetry identity.

16.6. Source-convention rebuild and persistence of the $\eta\eta$ row. The preceding obstruction used the tagged omitted-zero tail as the source object. One possible repair is to rebuild the source convention by moving all non-core zero atoms into the source. The least artificial split is

$$q = q_{\text{base}}^{\text{new}} + G^{\text{new}},$$

where

$$G^{\text{new}} := q_{\text{aux},R} + T_{\text{far},R}, \quad q_{\text{base}}^{\text{new}} := q_{\text{core}} + B_{\text{bg}} + E_{\text{est},R}.$$

Thus G^{new} remains a zero-source object; it contains the auxiliary non-core zero contribution and the far zero tail, but not the smooth background or estimation terms.

The affine-subtracted source is

$$r_G(t) := G^{\text{new}}(t) - G^{\text{new}}(m) - G^{\text{new}'}(m)(t - m).$$

Equivalently, the effective baseline in the finite- s expansion is

$$q_0^{\text{new}}(t) := q_{\text{base}}^{\text{new}}(t) + G^{\text{new}}(m) + G^{\text{new}'}(m)(t - m),$$

so that

$$q(t) = q_0^{\text{new}}(t) + r_G(t),$$

with

$$q_0^{\text{new}}(m) = q(m), \quad (q_0^{\text{new}})'(m) = q'(m), \quad (q_0^{\text{new}})''(m) = (q_{\text{base}}^{\text{new}})''(m).$$

This affine absorption is necessary: otherwise the source would contain value and linear jets and would reopen lower source-grade gates.

Remark 16.18 (Complement-source curvature). For the rebuilt source split, define

$$q_{\text{source},R}''(m) := (G^{\text{new}})''(m).$$

Since

$$r_G(t) = G^{\text{new}}(t) - G^{\text{new}}(m) - G^{\text{new}'}(m)(t - m),$$

one has

$$r_G''(m) = q_{\text{source},R}''(m).$$

This is the curvature of the affine-subtracted non-core zero complement. It is not the full canonical actual-zeta curvature

$$q_{\text{can}}''(m) = q_{\zeta}''(m),$$

which also includes the core, background, and all remaining completed-phase contributions in the fixed actual-zeta normalization.

Remark 16.19 (Admissibility conditions for the rebuilt split). The split

$$G^{\text{new}} = q_{\text{aux},R} + T_{\text{far},R}$$

is conditionally compatible with the already-closed lower gates under the following hypotheses:

- (1) G^{new} is kept as a tagged zero-source sum through the finite- s calculation, not collapsed prematurely into an unlabelled aggregate;
- (2) $r_G(m) = r_G'(m) = 0$ is imposed after the new split;
- (3) $q_{\text{base}}^{\text{new}}$ is used consistently in $q_{\pm}, \Delta_0, d_{\pm}, e_{\pm}, g_{\pm}, \eta$;
- (4) the $q_{\text{aux},R}$ -atoms satisfy the same endpoint reflection and source-grade conventions as the $T_{\text{far},R}$ -atoms;
- (5) the $q_{\text{aux},R}$ -atoms have compatible scalar-grade normalization, or else the affected coefficient tables must be regraded;
- (6) no B_{bg} or $E_{\text{est},R}$ term is treated as a zero-source variable.

Under these hypotheses the rebuild does not introduce value or linear source jets before the Bottleneck-C $\eta\eta$ row. By contrast, the more radical choices

$$G = q - q_{\text{core}} \quad \text{or} \quad G = B_{\text{bg}} + T_{\text{far},R} + E_{\text{est},R}$$

mix zero atoms with background or error terms. Those choices introduce background–source and error–source products and would require a fresh Gate A/Gate B analysis.

For a tagged exterior-zero source atom α in G^{new} , write

$$r_\alpha(t) = g_\alpha(t) - g_\alpha(m) - g'_\alpha(m)(t - m).$$

Define

$$d_{\alpha,\pm}^{\text{new}} = r_\alpha(t_\pm), \quad e_{\alpha,\pm}^{\text{new}} = r'_\alpha(t_\pm), \quad g_{\alpha,\pm}^{\text{new}} = r''_\alpha(t_\pm),$$

and let η_α^{new} be the corresponding phase-gap source variable. For two tagged source atoms i, j , the finite- s substitutions become

$$\begin{aligned} q_\pm &= q_{0,\pm}^{\text{new}} + \tau_i a_i d_{i,\pm}^{\text{new}} + \tau_j a_j d_{j,\pm}^{\text{new}}, \\ q'_\pm &= (q_0^{\text{new}})'_\pm + \tau_i a_i e_{i,\pm}^{\text{new}} + \tau_j a_j e_{j,\pm}^{\text{new}}, \\ q''_\pm &= (q_0^{\text{new}})''_\pm + \tau_i a_i g_{i,\pm}^{\text{new}} + \tau_j a_j g_{j,\pm}^{\text{new}}, \end{aligned}$$

and

$$\Delta = \Delta_0^{\text{new}} + \tau_i a_i \eta_i^{\text{new}} + \tau_j a_j \eta_j^{\text{new}}.$$

The lower-left off-diagonal channel is still

$$N_{21} = -\frac{\sin \Delta + q_- s \cos \Delta}{s^2}.$$

Therefore its pure $\eta_i \eta_j$ second phase-gap variation is

$$\partial_\Delta^2 N_{21}|_{\Delta=\Delta_0^{\text{new}}} = \frac{\sin \Delta_0^{\text{new}}}{s^2} + \frac{q_{0,-}^{\text{new}} \cos \Delta_0^{\text{new}}}{s}.$$

Thus the rebuilt top-row scalar is

$$\nu_{21,7}^{\text{new},ij} := [z^7] \left[\left(\frac{\sin \Delta_0^{\text{new}}}{s^2} + \frac{q_{0,-}^{\text{new}} \cos \Delta_0^{\text{new}}}{s} \right) \eta_i^{\text{new}} \eta_j^{\text{new}} \right].$$

Let $\rho_{\alpha,2}^{\text{new}}$ and $\rho_{\alpha,4}^{\text{new}}$ be the midpoint source coefficients of r_α in the same normalization as before. Then the rederived pair coefficient has the same form as the old obstruction:

$$\boxed{\nu_{21,7}^{\text{new},ij} = \frac{1}{Q^{14}} \left[\frac{q_0^{\text{new}}}{480} (\rho_{i,2}^{\text{new}} \rho_{j,4}^{\text{new}} + \rho_{i,4}^{\text{new}} \rho_{j,2}^{\text{new}}) + \frac{1}{144} \left(\frac{(q_0^{\text{new}})''}{6} - \frac{2(q_0^{\text{new}})^3}{3} \right) \rho_{i,2}^{\text{new}} \rho_{j,2}^{\text{new}} \right]}.$$

Hence the source-convention rebuild does not algebraically kill the top row.

Aggregating over the full non-core zero source, define

$$A_2^{\text{new}} := \sum_{\alpha \in \mathcal{I}_{\text{ext}}^{\text{new}}} w_\alpha \rho_{\alpha,2}^{\text{new}} = \frac{(G^{\text{new}})''(m)}{2},$$

and

$$A_4^{\text{new}} := \sum_{\alpha \in \mathcal{I}_{\text{ext}}^{\text{new}}} w_\alpha \rho_{\alpha,4}^{\text{new}} = \frac{(G^{\text{new}})^{(4)}(m)}{24}.$$

Then

$$\boxed{\nu_{21,7}^{\text{new,agg}} = \frac{1}{Q^{14}} \left[\frac{q_0^{\text{new}}}{240} A_2^{\text{new}} A_4^{\text{new}} + \frac{1}{144} \left(\frac{(q_0^{\text{new}})''}{6} - \frac{2(q_0^{\text{new}})^3}{3} \right) (A_2^{\text{new}})^2 \right]}.$$

With

$$W_{+,0}^{\text{new}} = \begin{pmatrix} u^{\text{new}} & w^{\text{new}} \\ w^{\text{new}} & v^{\text{new}} \end{pmatrix},$$

the corresponding fixed-sector row is

$$\mathcal{E}_{\eta\eta;7,0,0}^{2,\text{new}} = -\frac{\nu_{21,7}^{\text{new}}}{24\pi(q_{\text{wh}}^{\text{new}})^3} (w^{\text{new}} I + u^{\text{new}} S),$$

so

$$f_{\eta\eta}^{\text{new},\top} = -\frac{u^{\text{new}}}{24\pi(q_{\text{wh}}^{\text{new}})^3}\nu_{21,7}^{\text{new}}.$$

Consequently, on a good patch with $u^{\text{new}}q_{\text{wh}}^{\text{new}} \neq 0$, physical Bottleneck C closure still requires

$$\nu_{21,7}^{\text{new,agg}} = 0.$$

Remark 16.20 (Outcome of the source-convention rebuild). The split

$$G^{\text{new}} = q_{\text{aux},R} + T_{\text{far},R}$$

is cleaner than the raw far-tail source convention because all non-core zero atoms are placed on the source side. Nevertheless, the rederived $\eta\eta$ top row has the same obstruction form, with

$$q_0, q_0'', A_2, A_4$$

replaced by

$$q_0^{\text{new}}, (q_0^{\text{new}})'', A_2^{\text{new}}, A_4^{\text{new}}.$$

Thus the rebuild merely shifts the needed cancellation theorem to the full non-core zero complement:

$$\frac{q_0^{\text{new}}}{240}A_2^{\text{new}}A_4^{\text{new}} + \frac{1}{144}\left(\frac{(q_0^{\text{new}})''}{6} - \frac{2(q_0^{\text{new}})^3}{3}\right)(A_2^{\text{new}})^2 = 0.$$

If $A_2^{\text{new}} \neq 0$, this is equivalently

$$\frac{A_4^{\text{new}}}{A_2^{\text{new}}} = \frac{10(q_0^{\text{new}})^2}{9} - \frac{5(q_0^{\text{new}})''}{18q_0^{\text{new}}}.$$

No such global zero-complement identity is presently proved.

Remark 16.21 (Source-tagging and renormalized-complement ODEs). The quantities A_2^{new} and A_4^{new} in the rebuilt Bottleneck-C coefficient are the second and fourth midpoint jets of the tagged zero-source object

$$G^{\text{new}} = q_{\text{aux},R} + T_{\text{far},R},$$

after affine subtraction. They are not the corresponding jets of

$$q - q_{\text{core}}, \quad B_{\text{bg}} + T_{\text{far},R} + E_{\text{est},R}, \quad q,$$

or any other renormalized complement unless the entire finite- s row is rebuilt.

Thus an ODE for a different complement does not affect the already-derived row. In the present source convention, the only relevant derivative-ratio identity is

$$\frac{(G^{\text{new}})^{(4)}(m)}{(G^{\text{new}})''(m)} = \frac{40(q_0^{\text{new}})^2}{3} - \frac{10(q_0^{\text{new}})''}{3q_0^{\text{new}}}.$$

No ODE, meromorphic identity, moving-cutoff identity, or standard ξ'/ξ phase-calculus identity currently supplies this condition.

Moreover, near an exterior zero-source pole p in G^{new} ,

$$G^{\text{new}}(t) \sim \frac{C}{t-p}, \quad (G^{\text{new}})''(t) \sim \frac{2C}{(t-p)^3}, \quad (G^{\text{new}})^{(4)}(t) \sim \frac{24C}{(t-p)^5},$$

so

$$\frac{(G^{\text{new}})^{(4)}}{(G^{\text{new}})''} \sim \frac{12}{(t-p)^2}.$$

But $q_{\text{base}}^{\text{new}} = q_{\text{core}} + B_{\text{bg}} + E_{\text{est},R}$ does not contain that exterior pole, so the right-hand side

$$\frac{40(q_{\text{base}}^{\text{new}})^2}{3} - \frac{10(q_{\text{base}}^{\text{new}})''}{3q_{\text{base}}^{\text{new}}}$$

has no matching pole there. Hence the required equation is not a meromorphic identity for the source-compatible complement.

16.7. Invariant and quotient no-go for the S -visible defect. The source rebuild still leaves a physical S -visible defect. One might therefore ask whether replacing the local invariant or quotienting the fixed sector can make this defect irrelevant. This subsection records the negative answer.

Let

$$E = eI + fS$$

be a Bottleneck-C fixed-sector residual. The current physical readout has

$$\Phi_K(I) = 0, \quad \Phi_K(S) = 2,$$

so

$$\Phi_K(E) = 2f.$$

The hidden-state slope is

$$\partial_\kappa T = -a_1^2 \frac{f}{c}.$$

Proposition 16.22 (Local invariant no-go). *Let $\mathcal{J}$ be a smooth local invariant or readout defined near the exceptional divisor. Suppose $\mathcal{J}$ is transform-faithful in the sense that its first variation determines the physical T -direction. Then*

$$D\mathcal{J}[S] \neq 0.$$

Consequently, whenever the Bottleneck-C defect has $f \neq 0$, the first exceptional-divisor slope of $\mathcal{J}$ contains a nonzero contribution from the same S -component. Thus replacing Φ_K by $\mathcal{J}$ does not remove the $\eta\eta$ obstruction.

Conversely, if

$$D\mathcal{J}[S] = 0,$$

then $\mathcal{J}$ may hide the $\eta\eta$ row at first order, but it cannot reconstruct or control the physical hidden-state slope $\partial_\kappa T$. Such a readout is not a valid Bottleneck-C replacement unless an independent theorem proves that T is pure gauge.

Proof. At the exceptional divisor,

$$\partial_\kappa \mathcal{J} = D\mathcal{J}[E] = D\mathcal{J}[eI + fS].$$

If $D\mathcal{J}[S] \neq 0$, then the fS -component contributes to $\partial_\kappa \mathcal{J}$. After any normalization that removes scalar I -data, the S -term remains visible. If $D\mathcal{J}[S] = 0$, then $\mathcal{J}$ cannot distinguish eI from $eI + \varepsilon S$ to first order. But this is precisely the distinction measured by

$$\partial_\kappa T = -a_1^2 f/c.$$

Thus $D\mathcal{J}[S] = 0$ loses the physical T -direction, while $D\mathcal{J}[S] \neq 0$ inherits the obstruction. $\square$

Remark 16.23 (Examples of non-faithful and faithful invariants). Trace, determinant to first order at the scalar merger, unordered conjugacy class, and symplectic area are blind to f at first order. They hide the bad row only by discarding the oriented S -direction. Oriented eigenvalue ratios, oriented projective cross-ratios, and oriented curvature or Schwarzian-type readouts are sensitive to f , and therefore inherit the same obstruction.

Next consider the quotient line generated by

$$A_5^f = u_5 I + v_5 S.$$

The quotient determinant coordinate is

$$\Delta_7 = u_7 v_5 - u_5 v_7.$$

A residual $E = eI + fS$ changes this coordinate by

$$\omega_h(E, A_5^f) = ev_5 - u_5 f.$$

Thus the quotient condition is

$$ev_5 - u_5 f = 0.$$

Proposition 16.24 (Quotient invisibility is not transform invisibility). *Assume $v_5 \neq 0$. The quotient condition*

$$ev_5 - u_5 f = 0$$

does not imply the transform-safe condition $f = 0$. In particular,

$$E = \chi A_5^f$$

is invisible to the quotient determinant, but

$$\Phi_K(E) = 2\chi v_5$$

and

$$T \mapsto T + \chi \frac{v_5}{c}.$$

Therefore the quotient

$$\mathfrak{f}_h / \mathbf{C} A_5^f$$

is a valid moduli object for mixed four-point scalar exits, but not for the full Bottleneck-C physical transform state.

Proof. If $E = \chi A_5^f$, then

$$E = \chi u_5 I + \chi v_5 S,$$

so

$$\omega_h(E, A_5^f) = (\chi u_5) v_5 - u_5 (\chi v_5) = 0.$$

However

$$\Phi_K(E) = 2\chi v_5.$$

On a $v_5 \neq 0$ patch this is nonzero for $\chi \neq 0$. The hidden coordinate changes by

$$T = \frac{v_7}{c} \mapsto T + \chi \frac{v_5}{c}.$$

Thus the quotient identifies states with different Φ_K -visible transform data. $\square$

Remark 16.25 (Layer separation). The quotient determinant layer and the physical transform layer see different functionals:

$$\omega_h(E, A_5^f) = ev_5 - u_5 f, \quad \Phi_K(E) = 2f.$$

The kernel of the former is not contained in the kernel of the latter. Hence quotient-horizontal Bottleneck C can close the mixed four-point scalar-exit branch while still leaking in the physical T -coordinate.

16.8. Finite- s carrier and endpoint-symmetrization no-go. The remaining possible repair is to alter the finite- s carrier before projection. This also fails inside the present transform-faithful framework.

Write

$$N_{12} = \frac{n_+}{s^2}, \quad n_+ = \sin \Delta + q_+ s \cos \Delta,$$

and

$$N_{21} = -\frac{n_-}{s^2}, \quad n_- = \sin \Delta + q_- s \cos \Delta.$$

For a pure $\eta_i \eta_j$ source variation, the relevant variation is through Δ . Holding $q_{\pm}$ fixed in this Hessian row,

$$\partial_{\Delta} n_{\pm} = \cos \Delta - q_{\pm} s \sin \Delta, \quad \partial_{\Delta}^2 n_{\pm} = -\sin \Delta - q_{\pm} s \cos \Delta = -n_{\pm}.$$

Therefore

$$\partial_{\Delta}^2 N_{12} = -\frac{\sin \Delta}{s^2} - \frac{q_+ \cos \Delta}{s},$$

whereas

$$\partial_{\Delta}^2 N_{21} = \frac{\sin \Delta}{s^2} + \frac{q_- \cos \Delta}{s}.$$

These are exact finite- s identities, not artifacts of Taylor truncation.

For a general linear endpoint carrier

$$C_{a,b} := aN_{12} + bN_{21},$$

one obtains

$$\partial_\Delta^2 C_{a,b} = \frac{(b-a)\sin\Delta}{s^2} + \frac{(bq_- - aq_+)\cos\Delta}{s}.$$

Thus identical cancellation requires

$$b - a = 0, \quad bq_- - aq_+ = 0.$$

For a nonzero carrier this forces $q_- = q_+$, which is not a generic finite- s identity.

The common endpoint combinations illustrate the obstruction:

$$N_{12} + N_{21} = \frac{(q_+ - q_-)\cos\Delta}{s},$$

so

$$\partial_\Delta^2 (N_{12} + N_{21}) = -\frac{(q_+ - q_-)\cos\Delta}{s}.$$

This cancels the $\sin\Delta/s^2$ part but leaves the endpoint-gradient term. Similarly,

$$q_+N_{21} + q_-N_{12} = \frac{(q_- - q_+)\sin\Delta}{s^2},$$

so

$$\partial_\Delta^2 (q_+N_{21} + q_-N_{12}) = \frac{(q_+ - q_-)\sin\Delta}{s^2}.$$

This cancels the $q_\pm \cos\Delta/s$ part but leaves the phase-gap term. Endpoint balancing can remove one component of the Hessian, but not both.

At the matrix level, define

$$B_\pm := \frac{\sin\Delta}{s^2} + \frac{q_\pm \cos\Delta}{s}.$$

The off-diagonal second-Hessian matrix is

$$H_{\eta\eta} = \begin{pmatrix} 0 & -B_+ \\ B_- & 0 \end{pmatrix}.$$

Writing

$$J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

one has

$$H_{\eta\eta} = AJ + DX,$$

where

$$A = \frac{B_+ + B_-}{2} = \frac{\sin\Delta}{s^2} + \frac{(q_+ + q_-)\cos\Delta}{2s},$$

and

$$D = \frac{B_- - B_+}{2} = \frac{(q_- - q_+)\cos\Delta}{2s}.$$

Neither A nor D is generically zero. The skew part AJ is the endpoint-average obstruction, while the symmetric part DX removes the $\sin\Delta/s^2$ term but retains endpoint-gradient data.

The active left whitening coefficient is

$$\Lambda_{2,0}^- = -\frac{1}{24\pi q^3} E_{22}.$$

Since

$$E_{22}J = E_{21}, \quad E_{22}X = E_{21},$$

both endpoint-skew and endpoint-symmetric Hessian pieces feed the same leading row after left multiplication. Thus

$$\Lambda_{2,0}^-(AJ)W_{+,0} = -\frac{A}{24\pi q^3} E_{21}W_{+,0},$$

and

$$\Lambda_{2,0}^-(DX)W_{+,0} = -\frac{D}{24\pi q^3}E_{21}W_{+,0}.$$

After fixed-sector symmetrization,

$$\pi_{\mathfrak{f}} \text{Sym}(E_{21}W_{+,0}) = wI + uS,$$

up to the same normalization convention used in Subsection 16.3. Hence any nonzero endpoint-symmetrized coefficient remains S -visible on a patch with $u \neq 0$.

Proposition 16.26 (Endpoint-symmetrized finite- s carrier no-go). *Let a finite- s endpoint carrier be formed from N_{12}, N_{21} by smooth endpoint-symmetric linear combinations and endpoint or Gram normalizations that do not explicitly depend on the source variation direction. If the carrier remains transform-faithful, then its pure $\eta_i\eta_j$ second Δ -Hessian has a nonzero lower-row component. After the active left-whitening slot $\Lambda_{2,0}^-$, this produces a fixed-sector term*

$$-\frac{\mu_{\eta\eta}}{24\pi q^3}(wI + uS)$$

for some endpoint coefficient $\mu_{\eta\eta}$. In particular, unless $\mu_{\eta\eta}$ is forced to vanish by a genuine zeta identity, the physical S -coefficient

$$f_{\eta\eta} = -\frac{u}{24\pi q^3}\mu_{\eta\eta}$$

survives.

Remark 16.27 (Nonlinear carriers do not repair the obstruction). Ratio, logarithmic, determinant-normalized, and phase-coordinate carriers do not supply a transform-faithful repair. A ratio or logarithmic carrier normalizes the Hessian but leaves a nonzero endpoint-curvature term. A phase coordinate such as Δ is affine in the source and therefore has zero second Δ -Hessian, but reconstructing the physical finite- s kernel requires applying

$$\Delta \mapsto \sin \Delta + q_{\pm}s \cos \Delta,$$

whose second derivative is exactly the missing Hessian. Thus the bad row is either visible in the carrier, reintroduced by reconstruction, or hidden only by losing the physical S -visible transform readout.

16.9. Local Bottleneck-C ledger after repair attempts. Combining Subsections 16.6, 16.7, and 16.8, the present local ledger is as follows.

Changing the source convention from the raw far tail to the full non-core zero source

$$G^{\text{new}} = q_{\text{aux},R} + T_{\text{far},R}$$

is coherent under the stated tagging and grading hypotheses, but it reproduces the same $\eta\eta$ top-row obstruction form. Replacing the scalar local invariant hides the defect only if it discards the physical T -direction. Quotienting by $\mathbf{CA}_5^I$ is legitimate for mixed four-point scalar exits but not for the full transform state. Finally, endpoint symmetrization and alternative finite- s carriers either preserve the second- Δ Hessian, reintroduce it during reconstruction, or lose transform faithfulness.

Therefore, within the present transform-faithful finite- s off-diagonal kernel architecture, Bottleneck C closes only if the physical aggregate S -visible coefficient of the $\eta\eta$ top row vanishes. In the rebuilt full non-core zero-source convention this means

$$\nu_{21,7}^{\text{new,agg}} = 0,$$

or equivalently

$$\frac{q_0^{\text{new}}}{240}A_2^{\text{new}}A_4^{\text{new}} + \frac{1}{144}\left(\frac{(q_0^{\text{new}})''}{6} - \frac{2(q_0^{\text{new}})^3}{3}\right)(A_2^{\text{new}})^2 = 0.$$

No currently proved identity supplies this vanishing. The subsequent subsections record the present interpretation of this obstruction: it is not a natural boundary term or residue theorem, and it is best viewed as the vertical defect of the T -horizontal Bottleneck-C connection. Thus

the present local route remains open-negative: either one proves a new pointwise global zero-source cancellation theorem, or else one pivots to a different architecture, such as global curvature/holonomy control or a quotient-only no- T program.

16.10. Last-chance pointwise flatness in full phase variables. The rebuilt source convention removes the unstable-cutoff objection by using the full non-core zero complement

$$G^{\text{new}} = q_{\text{aux},R} + T_{\text{far},R}.$$

A final possible escape is that the exact decomposition of the full phase derivative into rebuilt baseline and non-core complement might itself force the Bottleneck-C flatness identity. This subsection records that it does not.

Write

$$Q(t) := q(t)$$

for the full phase derivative, and decompose

$$Q(t) = b(t) + G(t),$$

where

$$b(t) := q_{\text{base}}^{\text{new}}(t) = q_{\text{core}}(t) + B_{\text{bg}}(t) + E_{\text{est},R}(t),$$

and

$$G(t) := G^{\text{new}}(t) = q_{\text{aux},R}(t) + T_{\text{far},R}(t).$$

At the midpoint m , put

$$b_0 := b(m), \quad b_2 := b''(m),$$

and

$$A_2 := \frac{G''(m)}{2}, \quad A_4 := \frac{G^{(4)}(m)}{24}.$$

Then

$$A_2 = \frac{Q''(m) - b''(m)}{2}, \quad A_4 = \frac{Q^{(4)}(m) - b^{(4)}(m)}{24}.$$

The rebuilt Bottleneck-C flatness condition is

$$\frac{b_0}{240} A_2 A_4 + \frac{1}{144} \left(\frac{b_2}{6} - \frac{2b_0^3}{3} \right) A_2^2 = 0.$$

Substitution gives

$$\frac{b_0}{11520} (Q'' - b'')(Q^{(4)} - b^{(4)}) + \frac{1}{576} \left(\frac{b''}{6} - \frac{2b_0^3}{3} \right) (Q'' - b'')^2 = 0,$$

with all derivatives evaluated at m . Equivalently, if

$$D_2 := Q'' - b'' = G'', \quad D_4 := Q^{(4)} - b^{(4)} = G^{(4)},$$

then the condition is

$$D_2 \left[b_0 D_4 + 20 \left(\frac{b''}{6} - \frac{2b_0^3}{3} \right) D_2 \right] = 0.$$

Proposition 16.28 (The rebuilt flatness equation is an extra fourth-order constraint). *The identity*

$$Q = b + G$$

does not imply

$$\nu_{21,7}^{\text{new,agg}} = 0.$$

On a good patch with $G''(m) \neq 0$, pointwise flatness is equivalent to the additional derivative-ratio constraint

$$\frac{G^{(4)}(m)}{G''(m)} = \frac{40b(m)^2}{3} - \frac{10b''(m)}{3b(m)}.$$

Equivalently,

$$b^{(4)}(m) = Q^{(4)}(m) - \left(\frac{40b(m)^2}{3} - \frac{10b''(m)}{3b(m)} \right) (Q''(m) - b''(m)).$$

Thus the desired flatness is not a consequence of the tautological full-phase decomposition. It is an extra nonlinear fourth-order pointwise condition on the selected baseline/source split.

Proof. The displayed factorization shows that, if $D_2 = G''(m) \neq 0$, the vanishing condition is precisely

$$b_0 D_4 + 20 \left(\frac{b''}{6} - \frac{2b_0^3}{3} \right) D_2 = 0.$$

Dividing by $b_0 D_2$ gives

$$\frac{G^{(4)}(m)}{G''(m)} = \frac{40b_0^2}{3} - \frac{10b''(m)}{3b_0}.$$

The equivalent formula for $b^{(4)}$ follows from

$$G^{(4)} = Q^{(4)} - b^{(4)}.$$

The decomposition $Q = b + G$ supplies only

$$G'' = Q'' - b'', \quad G^{(4)} = Q^{(4)} - b^{(4)}.$$

It supplies no relation fixing $G^{(4)}/G''$ in terms of b, b'' . □

Remark 16.29 (Why defining the baseline by flatness is circular). One could formally choose b by requiring the fourth-order equation in Proposition 16.28. Such a choice would kill the Bottleneck-C row by definition. It is not a proof unless an independent zeta-side principle selects exactly this baseline, for example through the functional equation, a canonical scattering law, a zero-counting conservation principle, or a functorial fixed-core construction. No such principle is currently present.

Remark 16.30 (Asymptotic scale is not an identity). If b is dominated by the smooth phase/background, one expects roughly

$$b(t) \sim \frac{1}{2} \log(t/2\pi).$$

Then the required ratio has scale

$$\frac{G^{(4)}}{G''} \sim \frac{40}{3} b^2 \asymp \log^2 t.$$

A non-core zero complement with pole distances of order $1/\log t$ can have the same dimensional scale. This is not enough: the identity requires the exact constant and sign

$$\frac{40b^2}{3} - \frac{10b''}{3b}$$

pointwise. Local one-pair and finite-tail tests show that source geometry can produce different fourth-to-second derivative ratios. Thus a proof of flatness would have to be a genuinely new global zero-complement rigidity theorem.

16.11. Global obstruction-class interpretation of the Bottleneck-C defect. The preceding local ledger shows that the present transform-faithful finite- s Bottleneck-C architecture closes only if the rebuilt aggregate coefficient

$$\nu_{21,7}^{\text{new,agg}}$$

vanishes pointwise. The next question is whether this coefficient is a hidden boundary term, residue identity, or global conservation-law density. The current evidence is negative: it is better understood as a bulk vertical connection-defect density for the T -horizontal Bottleneck-C connection.

For this subsection write, for readability,

$$G := G^{\text{new}} = q_{\text{aux},R} + T_{\text{far},R}, \quad A_2 := \frac{G''(m)}{2} = \frac{1}{2} q''_{\text{source},R}(m), \quad A_4 := \frac{G^{(4)}(m)}{24},$$

and use the baseline-local notation

$$q_b := q_0^{\text{new}}, \quad q_b^{(2)} := (q_0^{\text{new}})''.$$

The notation $q_b^{(2)}$ is deliberately not written as q'' , because q'' is reserved in the rank-two arithmetic sections for the full canonical curvature q''_{can} . The rebuilt obstruction is

$$\nu_{21,7}^{\text{new,agg}} = \frac{1}{Q^{14}} \left[\frac{q_b}{240} A_2 A_4 + \frac{1}{144} \left(\frac{q_b^{(2)}}{6} - \frac{2q_b^3}{3} \right) A_2^2 \right].$$

Proposition 16.31 (No natural exactness for the rebuilt Bottleneck-C defect). *Assume the source set is fixed as m varies, so that*

$$A_4 = \frac{1}{12} A_2''.$$

Then

$$2880Q^{14}\nu_{21,7}^{\text{new,agg}} = q_b A_2 A_2'' + 20 \left(\frac{q_b^{(2)}}{6} - \frac{2q_b^3}{3} \right) A_2^2.$$

This density is not an exact first-order local derivative in the natural midpoint variables (A_2, A_2', q_b, q_b') . More precisely,

$$2880Q^{14}\nu_{21,7}^{\text{new,agg}} = \left(q_b A_2 A_2' - \frac{1}{2} q_b' A_2^2 \right)' - q_b (A_2')^2 + \left(\frac{23}{6} q_b^{(2)} - \frac{40}{3} q_b^3 \right) A_2^2.$$

The final two terms are genuine bulk terms. Hence

$$\nu_{21,7}^{\text{new,agg}}$$

is not a boundary term or conservation-law density in the ordinary midpoint calculus.

Proof. Since $A_2 = G''/2$, fixed source tagging gives

$$A_2'' = \frac{G^{(4)}}{2}, \quad A_4 = \frac{G^{(4)}}{24} = \frac{1}{12} A_2''.$$

Substitution into the formula for $\nu_{21,7}^{\text{new,agg}}$ gives the first displayed identity. For the integration-by-parts decomposition, compute

$$(q_b A_2 A_2')' = q_b A_2 A_2'' + q_b (A_2')^2 + q_b' A_2 A_2',$$

and

$$(q_b' A_2^2)' = q_b^{(2)} A_2^2 + 2q_b' A_2 A_2'.$$

Therefore

$$q_b A_2 A_2'' = \left(q_b A_2 A_2' - \frac{1}{2} q_b' A_2^2 \right)' - q_b (A_2')^2 + \frac{1}{2} q_b^{(2)} A_2^2.$$

Adding

$$20 \left(\frac{q_b^{(2)}}{6} - \frac{2q_b^3}{3} \right) A_2^2$$

gives the claimed residual term

$$-q_b (A_2')^2 + \left(\frac{23}{6} q_b^{(2)} - \frac{40}{3} q_b^3 \right) A_2^2.$$

□

Remark 16.32 (Moving source windows). If the source set, cutoff, or core window depends on m , write it as $G_m(t)$. Then

$$A_2(m) = \frac{1}{2} \partial_t^2 G_m(t)|_{t=m},$$

and differentiation gives

$$A_2''(m) = \frac{1}{2} (\partial_m^2 \partial_t^2 + 2\partial_m \partial_t^3 + \partial_t^4) G_m(t)|_{t=m}.$$

Consequently

$$A_4 = \frac{1}{12} A_2'' - \frac{1}{24} (\partial_m^2 \partial_t^2 + 2\partial_m \partial_t^3) G_m(t)|_{t=m}.$$

Thus moving cutoffs or moving core windows introduce extra correction terms. They do not by themselves create a conservation law; they add another identity requirement.

Remark 16.33 (The induced second-order equation). If $A_2 \neq 0$, pointwise vanishing of $\nu_{21,7}^{\text{new,agg}}$ is equivalent, in the fixed-source-set calculus, to

$$q_b A_2'' + 20 \left(\frac{q_b^{(2)}}{6} - \frac{2q_b^3}{3} \right) A_2 = 0,$$

or

$$A_2'' + \left(\frac{10}{3} \frac{q_b^{(2)}}{q_b} - \frac{40}{3} q_b^2 \right) A_2 = 0.$$

This should be read as the unproved pointwise equation demanded by the current Bottleneck-C gate, not as an already available zeta-side ODE.

16.12. Meromorphic residue no-go for the rebuilt obstruction. There is also no current residue-theoretic mechanism forcing

$$\nu_{21,7}^{\text{new,agg}}(m) = 0.$$

The rebuilt source is a zero-pair pole field. Locally write

$$G(m) = \sum_{p \in P_{\text{ext}}} \frac{C_p}{m-p},$$

where P_{ext} is the set of non-core zero-source poles and C_p includes the residue and source weight. Then

$$A_2(m) = \sum_p \frac{C_p}{(m-p)^3}, \quad A_4(m) = \sum_p \frac{C_p}{(m-p)^5}.$$

If

$$K(m) := \frac{q''(m)}{6} - \frac{2q(m)^3}{3},$$

then, up to the global factor Q^{-14} ,

$$\nu(m) := \frac{q(m)}{240} A_2 A_4 + \frac{K(m)}{144} A_2^2$$

is the quadratic pole-interaction field

$$\nu(m) = \frac{q(m)}{240} \sum_{p,r} \frac{C_p C_r}{(m-p)^3 (m-r)^5} + \frac{K(m)}{144} \sum_{p,r} \frac{C_p C_r}{(m-p)^3 (m-r)^3}.$$

Proposition 16.34 (The rebuilt obstruction is not residue-killed). *The meromorphic differential $\nu(m) dm$ is not, in general, an exact meromorphic derivative. Its local residues at zero-source poles are generically nonzero. Hence the global residue theorem can at most give cancellation of the sum of residues; it does not imply the pointwise condition*

$$\nu(m) = 0$$

required by Bottleneck C.

Proof. Fix a pole p and write $\tau = m - p$. Decompose

$$A_2 = C_p \tau^{-3} + a(\tau), \quad A_4 = C_p \tau^{-5} + b(\tau),$$

where a, b are holomorphic at $\tau = 0$. Then

$$A_2 A_4 = C_p^2 \tau^{-8} + C_p b \tau^{-3} + C_p a \tau^{-5} + ab,$$

and

$$A_2^2 = C_p^2 \tau^{-6} + 2C_p a \tau^{-3} + a^2.$$

Thus $\nu(m) dm$ has high-order poles, and its residue depends on Taylor coefficients of q, K, a, b at p . There is no structural reason for this residue to vanish.

Already in a two-pole constant- (q, K) model, with poles p_1, p_2 , residues C_1, C_2 , and $d = p_1 - p_2$, the analytic tail at p_1 gives

$$\text{Res}_{p_1}(\nu dm) = C_1 C_2 \left(\frac{q}{8d^7} + \frac{K}{12d^5} \right),$$

which is generically nonzero. Therefore $\nu(m) dm$ is not an exact meromorphic derivative. $\square$

Remark 16.35 (What residue calculus can and cannot give). Functional-equation pairing and conjugation may pair or cancel residues in a global sum. Such cancellation is compatible with the global residue theorem, but it is much weaker than the pointwise identity

$$\nu_{21,7}^{\text{new,agg}}(m) = 0.$$

The Bottleneck-C gate is pointwise because

$$f_{\eta\eta}^{\text{new,agg}} = -\frac{u^{\text{new}}}{24\pi(q_{\text{wh}}^{\text{new}})^3} \nu_{21,7}^{\text{new,agg}}, \quad \partial_\kappa T = -a_1^2 f/c.$$

Thus an integrated or cyclic residue cancellation cannot close the present T -horizontal gate.

16.13. Meromorphic ODE principal-part no-go. A related failed repair is to seek a meromorphic differential equation for the source complement or for the full phase derivative. The local flatness condition would be implied by an equation of the form

$$\frac{G^{(4)}}{G''} = \frac{40q^2}{3} - \frac{10q''}{3q},$$

or by the same equation with $G = q_\zeta$, the full actual-zeta phase derivative. This is incompatible with the principal parts of the actual zero-pair pole fields.

Let

$$H(t) = \frac{C}{t-p} + O(1)$$

have a simple pole. Then

$$H''(t) = \frac{2C}{(t-p)^3} + O(1), \quad H^{(4)}(t) = \frac{24C}{(t-p)^5} + O(1),$$

so

$$\frac{H^{(4)}}{H''} = \frac{12}{(t-p)^2} + O\left(\frac{1}{t-p}\right).$$

On the other hand, if the full phase derivative has

$$q_\zeta(t) = \frac{R}{t-p} + O(1),$$

then

$$q_\zeta^2(t) = \frac{R^2}{(t-p)^2} + O\left(\frac{1}{t-p}\right), \quad \frac{q_{\text{can}}''}{q_\zeta} = \frac{2}{(t-p)^2} + O\left(\frac{1}{t-p}\right).$$

Hence

$$\frac{40q^2}{3} - \frac{10q''}{3q} = \left(\frac{40R^2}{3} - \frac{20}{3} \right) \frac{1}{(t-p)^2} + O\left(\frac{1}{t-p}\right).$$

Matching the left-hand double-pole coefficient 12 would require

$$R^2 = \frac{7}{5}.$$

Proposition 16.36 (Zero-pair residues rule out the meromorphic ODE repair). *The flatness-forcing ODE*

$$\frac{H^{(4)}}{H''} = \frac{40q^2}{3} - \frac{10q''}{3q}$$

cannot hold as a meromorphic identity for the raw non-core zero source, for the full phase derivative, or for any complement retaining the zero-pair poles.

Proof. For a zero-pair summand

$$f_a(t) = \frac{a}{a^2 + (2t - \gamma)^2},$$

the poles are

$$p_a^\pm = \frac{\gamma \pm ia}{2},$$

with residues

$$\text{Res}_{p_a^+} f_a = -\frac{i}{4}, \quad \text{Res}_{p_a^-} f_a = \frac{i}{4}.$$

Thus the phase-pole residues are purely imaginary. For a simple pole,

$$R = \pm \frac{i}{4},$$

so

$$R^2 = -\frac{1}{16},$$

not $7/5$. If several same-sign pole contributions coincide, then

$$R = \pm \frac{ni}{4},$$

and still $R^2 < 0$. Therefore the right-hand double-pole coefficient cannot match the coefficient 12 of $H^{(4)}/H''$. $\square$

Remark 16.37 (Pole-cancelled complements are the wrong object). A pole-cancelled complement can evade the principal-part contradiction at the cancelled pole. But Bottleneck-C uses the source moments

$$A_2 = \frac{G''(m)}{2}, \quad A_4 = \frac{G^{(4)}(m)}{24}.$$

An ODE for a pole-cancelled complement is therefore an ODE for the wrong object, unless the entire order-seven row is rederived with different moments.

16.14. Connection-defect interpretation and possible architecture pivot. The preceding no-go results suggest a different interpretation. The coefficient

$$\nu_{21,7}^{\text{new,agg}}$$

is not merely a bad scalar coefficient; it is the vertical defect of the current T -horizontal Bottleneck-C connection.

Recall that

$$A_7^f = u_7 I + v_7 S, \quad T = \frac{v_7}{c}.$$

For a residual

$$E = eI + fS,$$

the physical readout satisfies

$$\Phi_K(E) = 2f,$$

and the Bottleneck-C exceptional slope is

$$\partial_\kappa T = -a_1^2 \frac{f}{c}.$$

For the rebuilt non-core zero-source convention, the active $\eta\eta$ row gives

$$f_{\eta\eta}^{\text{new,agg}} = -\frac{u^{\text{new}}}{24\pi(q_{\text{wh}}^{\text{new}})^3} \nu_{21,7}^{\text{new,agg}}.$$

Hence, on a good patch,

$$\partial_\kappa T|_{\eta\eta} = a_1^2 \frac{u^{\text{new}}}{24\pi c (q_{\text{wh}}^{\text{new}})^3} \nu_{21,7}^{\text{new,agg}}.$$

Definition 16.38 (Bottleneck-C vertical defect density). Define

$$\mathcal{K}_{\eta\eta;\kappa} := \partial_\kappa T|_{\eta\eta}.$$

Equivalently,

$$\mathcal{K}_{\eta\eta;\kappa} = C_{\text{geom}} \nu_{21,7}^{\text{new,agg}}, \quad C_{\text{geom}} = a_1^2 \frac{u^{\text{new}}}{24\pi c (q_{\text{wh}}^{\text{new}})^3}.$$

On a good patch,

$$C_{\text{geom}} \neq 0.$$

Remark 16.39 (Curvature versus second fundamental form). The active row is symmetric in the source directions η_i, η_j . Thus $\mathcal{K}_{\eta\eta;\kappa}$ is not ordinary antisymmetric gauge curvature by itself. It is better regarded as the vertical second fundamental form, or quadratic curvature-density, of the diagonal-merger horizontal distribution in the T -coordinate.

Proposition 16.40 (Local flatness equivalence). *Within the current transform-faithful Bottleneck-C architecture, local T -horizontal closure of the rebuilt $\eta\eta$ row is equivalent to*

$$\nu_{21,7}^{\text{new,agg}} = 0.$$

Proof. By Definition 16.38,

$$\partial_\kappa T|_{\eta\eta} = C_{\text{geom}} \nu_{21,7}^{\text{new,agg}}.$$

On a good patch $C_{\text{geom}} \neq 0$. Therefore

$$\partial_\kappa T|_{\eta\eta} = 0$$

if and only if

$$\nu_{21,7}^{\text{new,agg}} = 0.$$

□

Remark 16.41 (Possible curvature/holonomy pivot). The current Bottleneck-C architecture demands pointwise flatness:

$$\mathcal{K}_{\eta\eta;\kappa} = 0.$$

A different architecture could attempt to treat

$$C_{\text{geom}} \nu_{21,7}^{\text{new,agg}} dm$$

as a global curvature or holonomy density, replacing pointwise T -flatness by a global statement such as

$$\int \mathcal{K}_{\eta\eta;\kappa} dm = 0, \quad \text{Hol}_T(\mathcal{K}) = 0, \quad \text{or} \quad \text{Index}(\mathcal{K}) = 0.$$

This would be a genuine architecture pivot, not a repair of the present local Bottleneck-C gate. To make such a pivot meaningful, one would first need to prove transformation laws showing that

$$C_{\text{geom}} \nu_{21,7}^{\text{new,agg}} dm$$

is a globally defined connection or curvature density under changes of patch, core window, source convention, and normalization.

16.15. No- T quotient-determinant architecture as a possible pivot. A second possible pivot is to abandon the T -horizontal Bottleneck-C gate entirely and formulate the local-to-global argument using only quotient determinant data. This is not a repair of Bottleneck C; it is a different proof architecture.

The fixed-sector data are

$$A_5^{\mathfrak{f}} = u_5 I + v_5 S, \quad A_7^{\mathfrak{f}} = u_7 I + v_7 S.$$

The quotient determinant coordinate is

$$\Delta_7 = u_7 v_5 - u_5 v_7.$$

A shift

$$A_7^{\mathfrak{f}} \mapsto A_7^{\mathfrak{f}} + \chi A_5^{\mathfrak{f}}$$

preserves Δ_7 , but changes

$$T = \frac{v_7}{c} \quad \mapsto \quad T + \chi \frac{v_5}{c}.$$

It also changes

$$\Phi_K(A_7^f)$$

by $2\chi v_5$. Thus T is not quotient-invariant.

A quotient-only architecture would use as local moduli

$$\mathcal{Q}(h) = (c(h), u_5(h), v_5(h), \Delta_7(h)),$$

or equivalently the quotient class

$$[A_7^f] \in \mathfrak{f}_h / \mathbf{C}A_5^f.$$

Its local gate would be

$$ev_5 - u_5 f = 0,$$

not

$$f = 0.$$

Remark 16.42 (What a no- T architecture would require). A no- T architecture can make the T -leak irrelevant only if every downstream RH-relevant detection step is reformulated through quotient data:

$$(c, u_5, v_5, \Delta_7),$$

mixed four-point scalar exits, and finite-core determinant geometry. It cannot use

$$T = \frac{v_7}{c}, \quad \Phi_K(A_7^f),$$

or hidden-state reconstruction as independent data.

Proposition 16.43 (Conditional no-go for quotient-only architecture). *If a downstream local-to-global step requires reconstruction of T or $\Phi_K(A_7^f)$, then no quotient-only architecture can replace the current Bottleneck-C gate.*

Proof. The quotient identifies representatives

$$A_7^f \quad \text{and} \quad A_7^f + \chi A_5^f.$$

These representatives have the same Δ_7 , but their T -coordinates differ by

$$\chi \frac{v_5}{c}.$$

On a $v_5 \neq 0$ patch this is nonzero for $\chi \neq 0$. Therefore no function of the quotient data can recover T or $\Phi_K(A_7^f)$. Any proof step depending on those quantities cannot be performed in the quotient-only architecture. $\square$

Remark 16.44 (Relation to the Bottleneck-C obstruction). The no- T quotient program would make the obstruction

$$\nu_{21,7}^{\text{new,agg}}$$

irrelevant only by removing T -horizontal closure from the proof. It would not prove

$$\nu_{21,7}^{\text{new,agg}} = 0.$$

The burden would instead move to quotient finite-core non-concentration and quotient interaction rigidity using only (c, u_5, v_5, Δ_7) and the mixed four-point scalar exits.

16.16. Post-quotient audit: split and line-bundle repairs. The no- T quotient architecture keeps the determinant data

$$D_{2r+1} = \omega_h(A_{2r+1}^{\mathfrak{f}}, A_5^{\mathfrak{f}}) = u_{2r+1}v_5 - u_5v_{2r+1}.$$

This removes the representative-level T -coordinate, but it also discards the full $A_5^{\mathfrak{f}}$ -line. Thus the pure line anomaly

$$A_{2r+1}^{\mathfrak{f}} = \lambda_{2r+1} A_5^{\mathfrak{f}}$$

has

$$D_{2r+1} = 0,$$

while still giving

$$\Phi_K(A_{2r+1}^{\mathfrak{f}}) = 2\lambda_{2r+1}v_5, \quad T_{2r+1} = \lambda_{2r+1} \frac{v_5}{c}.$$

Consequently the quotient scalar

$$H_m^{\text{quot}}(s) = \omega_h\left(\pi_{\mathfrak{f}}\widehat{\Omega}_{\zeta}^{\text{corr}}(s; m), A_5^{\mathfrak{f}}\right)$$

is analytically coherent but incomplete for the original transform-state detection problem.

A natural repair is the split readout

$$\mathcal{H}_m^{\text{split}}(s) = \left(H_m^{\text{quot}}(s), H_m^{\parallel}(s)\right),$$

where

$$H_m^{\parallel}(s) = \frac{\left\langle \pi_{\mathfrak{f}}\widehat{\Omega}_{\zeta}^{\text{corr}}(s; m), A_5^{\mathfrak{f}} \right\rangle_{\text{wh}}}{\left\langle A_5^{\mathfrak{f}}, A_5^{\mathfrak{f}} \right\rangle_{\text{wh}}}.$$

For each coefficient

$$A_{2r+1}^{\mathfrak{f}} = u_{2r+1}I + v_{2r+1}S,$$

this gives the pair

$$(D_{2r+1}, \lambda_{2r+1}),$$

with

$$\lambda_{2r+1} = \frac{\langle A_{2r+1}^{\mathfrak{f}}, A_5^{\mathfrak{f}} \rangle_{\text{wh}}}{\langle A_5^{\mathfrak{f}}, A_5^{\mathfrak{f}} \rangle_{\text{wh}}}.$$

On a good patch, if the whitening metric is fixed and nondegenerate, the pair

$$(D_{2r+1}, \lambda_{2r+1})$$

recovers the full fixed-sector vector $A_{2r+1}^{\mathfrak{f}}$. Thus the split readout repairs the fatal incompleteness of the pure quotient readout.

However, this completeness reintroduces the Bottleneck-C obstruction through the longitudinal coordinate. For

$$E_{\eta\eta} = -\frac{\nu_{\eta\eta}}{24\pi q^3}(wI + uS),$$

one has

$$D(E_{\eta\eta}) = -\frac{\nu_{\eta\eta}}{24\pi q^3}(wv_5 - uu_5),$$

and

$$\lambda(E_{\eta\eta}) = -\frac{\nu_{\eta\eta}}{24\pi q^3} \frac{\langle wI + uS, A_5^{\mathfrak{f}} \rangle_{\text{wh}}}{\langle A_5^{\mathfrak{f}}, A_5^{\mathfrak{f}} \rangle_{\text{wh}}}.$$

Thus if the split readout is used as a pointwise vanishing gate, then on a good patch it requires

$$E_{\eta\eta} = 0,$$

and hence again requires

$$\nu_{\eta\eta} = 0$$

unless $wI + uS$ degenerates. The split readout is therefore a complete coordinate system for the fixed-sector vector, not a new closure mechanism.

Remark 16.45 (Line-bundle transport attempt). One can try to avoid pointwise vanishing by treating the longitudinal coordinate as a section of

$$L = \mathbf{C}A_5^{\mathfrak{f}}.$$

With a fixed whitening metric, the natural local connection is

$$\alpha_L = \frac{\langle dA_5^{\mathfrak{f}}, A_5^{\mathfrak{f}} \rangle_{\text{wh}}}{\langle A_5^{\mathfrak{f}}, A_5^{\mathfrak{f}} \rangle_{\text{wh}}}.$$

If

$$s = \lambda A_5^{\mathfrak{f}},$$

then

$$\nabla^L s = (d\lambda + \lambda \alpha_L) A_5^{\mathfrak{f}}.$$

Locally this is coherent: under

$$A_5^{\mathfrak{f}} \mapsto \mu A_5^{\mathfrak{f}}$$

one has

$$\alpha_L \mapsto \alpha_L + d \log \mu,$$

and the covariant derivative transforms correctly.

The obstruction is global. The construction requires the $A_5^{\mathfrak{f}}$ line and the whitening metric to transform canonically under changes of patch, source/core window, and whitening convention. This is not currently proved. Moreover, no canonical path, boundary condition, or holonomy theorem is available for (L, ∇^L) , and the longitudinal forcing is not exact in the natural variables. Thus line-bundle transport is a useful diagnostic language but not presently a proof gate.

Remark 16.46 (Current status of quotient, split, and line-bundle repairs). The quotient scalar is invariant but incomplete. The split readout is complete but restores the original fixed-sector obstruction. The line-bundle connection makes the longitudinal defect locally covariant, but lacks global covariance, canonical paths, boundary data, exactness, or holonomy control. Therefore none of these repairs currently replaces pointwise Bottleneck-C closure.

16.17. Reduced core, packet visibility, and the visibility-positive base. The rank-two architecture is only meaningful after fixing the source/core convention. Arbitrary transfers of zero atoms between source and core are not currently functorial. Moving a single zero contribution q_ρ from source to core changes the local corrected object, not merely its coordinates. At coefficient level one should expect

$$A_{2r+1,j}^{\mathfrak{f}} = G_{ij}^{-1} A_{2r+1,i}^{\mathfrak{f}} + B_{\rho,2r+1},$$

not a pure GL_2 -transition. Therefore source/core moves are excluded from the atlas rather than transported.

We use the canonical reduced-core rule

$$\mathcal{C}_k(m) = \text{the unique symmetry-closed, same-scale } k\text{-packet nearest } m.$$

Let

$$\mathcal{E}_{\text{geom}}^{\text{core}} = \mathcal{E}_{\text{geom}} \cup \mathcal{E}_{\text{core boundary}},$$

where $\mathcal{E}_{\text{core boundary}}$ includes nearest-packet ties, same-scale boundary, symmetry-closure ambiguity, endpoint-gap collapse, zero collision or multiplicity, Gram/whitening singularity, and patch-boundary failure. Define

$$\mathcal{B}_{\text{red}} = \mathcal{B} \setminus \mathcal{E}_{\text{geom}}^{\text{core}}.$$

On $\mathcal{B}_{\text{red}}$, the selected core is locally constant as a labelled symmetry-closed packet.

Hypothesis 16.47 (Reduced good-patch reach). Every nondegenerate RH-relevant off-line symmetry packet

$$\mathcal{P}(\rho) = \{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}, \quad \rho = \beta + i\gamma, \quad \beta \neq \frac{1}{2},$$

admits a nearby reduced good patch

$$b = (m, s, \mathcal{C}_k(m)) \in \mathcal{B}_{\text{red}},$$

or else the failure lies in a controlled component of $\mathcal{E}_{\text{geom}}^{\text{core}}$.

Remark 16.48 (Status of the reach hypothesis). The strict-gap, same-scale, endpoint, symmetry-closure, and Gram/whitening conditions are open once they hold at one point. Thus Hypothesis 16.47 is a non-swallowing assertion: the enlarged exceptional set

$$\mathcal{E}_{\text{geom}}^{\text{core}}$$

must not contain all RH-relevant off-line packet configurations. This is a global reach condition for the rank-two route, not a consequence of Gate A or Gate B.

The RH-relevant local object is not a single zero but the symmetry packet

$$\mathcal{P}(\rho) = \{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}, \quad \rho = \beta + i\gamma, \quad \beta \neq \frac{1}{2}.$$

Packet-level ordering is essential: functional-equation and conjugation symmetries force internal equalities inside $\mathcal{P}(\rho)$, and these should not be counted as core-selection failures. The dangerous tie walls are ties between distinct packets.

For packet visibility, set

$$\mathcal{B}_{\text{red}}^+ = \mathcal{B}_{\text{red}} \setminus \mathcal{E}_{\text{invis}}^{\text{core},7}.$$

For the centered one-whitening-slot route, the incidence walls include

$$\{y = 0\} \cup \{q'' = 0\} \cup \{g = h\},$$

and, when quotient isolation is used, also $\{v_5 = 0\}$. For the non-centered $q^{(7)}$ -transport route, include

$$\{q^{(7)} = 0\} \cup \{\Xi_{K_1} = 0\}.$$

On $\mathcal{B}_{\text{red}}^+$, the off-line packet fixed-sector visibility vector

$$C_{\text{fd}}^{\bar{f}} = (A_7^{\bar{f}}, A_9^{\bar{f}}, \dots)$$

is nonzero modulo these explicit incidence exclusions.

16.18. Rank-two fixed-sector bundle as the remaining viable language. The preceding failures suggest that the defect should not be scalarized, nor should one single out the $A_5^{\bar{f}}$ -line as the primary geometric object. The best remaining local language is the full rank-two fixed-sector bundle

$$\mathcal{F} \rightarrow \mathcal{B}, \quad \mathcal{F}_b = \mathfrak{f}_h(b) = \mathbf{C}I_b \oplus \mathbf{C}S_b.$$

This formulation keeps the whole fixed-sector vector

$$A_{2r+1}^{\bar{f}} \in \mathcal{F}_b$$

and therefore avoids the incompleteness of the quotient scalar and the noncanonical line-preservation problem of $L = \mathbf{C}A_5^{\bar{f}}$.

The local candidate connection is the projected ambient derivative

$$\nabla_X^{\bar{f}} A = \Pi_{\bar{f}}(d_X^{\mathcal{J}} A),$$

where $\mathcal{J}$ is the ambient corrected finite- s jet/kernel-state bundle, $d^{\mathcal{J}}$ is the ambient derivative, and

$$\Pi_{\bar{f}} : \mathcal{J}_b \rightarrow \mathfrak{f}_h(b)$$

is the fixed-sector projection. In a local frame

$$E = (I, S), \quad A = Ea,$$

the connection has the form

$$\nabla^{\bar{f}} A = E(da + \Gamma^{\bar{f}} a).$$

If

$$E = E^{\text{raw}} W,$$

then formally

$$\Gamma^{\bar{f}} = W^{-1} \Gamma^{\text{raw}} W + W^{-1} dW.$$

In particular, if the raw frame is flat in the chart, then

$$\Gamma^\natural = W^{-1}dW.$$

On an overlap with

$$E_j = E_i G_{ij}, \quad G_{ij} \in GL_2(\mathbf{C}),$$

the connection matrices must satisfy

$$\Gamma_j = G_{ij}^{-1} \Gamma_i G_{ij} + G_{ij}^{-1} dG_{ij}.$$

Thus the rank-two connection is canonical if the fixed-sector frames glue by canonical GL_2 -transition maps and the fixed-sector projection is functorial under the ambient transition maps.

Proposition 16.49 (Conditional rank-two connection theorem). *Assume that on a reduced good-patch base $\mathcal{B}_{\text{red}}^+$:*

- (1) *the core/source convention is fixed and locally stable;*
- (2) *no zero moves into or out of q_{core} ;*
- (3) *the fixed-sector frames satisfy transition laws*

$$E_j = E_i G_{ij}, \quad G_{ij} \in GL_2(\mathbf{C}), \quad G_{ij} G_{jk} = G_{ik};$$

- (4) *the ambient corrected jet bundle $\mathcal{J}$ and projection $\Pi_{\mathfrak{f}}$ are functorial under these transitions;*
- (5) *the whitening branch and Gram normalization are fixed or functorial.*

Then

$$\nabla_X^{\mathfrak{f}} A = \Pi_{\mathfrak{f}}(d_X^{\mathcal{J}} A)$$

defines a canonical connection on

$$\mathcal{F}_{\text{red}}^+ \rightarrow \mathcal{B}_{\text{red}}^+.$$

The reduced-base hypotheses are essential. On the unreduced base, changing the source/core convention can produce affine corrections

$$A_j^{\mathfrak{f}} = G_{ij}^{-1} A_i^{\mathfrak{f}} + B_{ij}.$$

Such transformations do not define an ordinary vector-bundle connection. Thus the present rank-two connection is currently a fixed-convention, reduced-base object unless an affine torsor transition theorem is proved.

Remark 16.50 (Why rank two improves on the line bundle). The $A_5^{\mathfrak{f}}$ -line may rotate under legitimate changes of trivialization, source/core convention, or whitening. This blocked the line-bundle route. The full plane $\mathfrak{f}_h = \mathbf{C}I \oplus \mathbf{C}S$ can still transform by a GL_2 -matrix. Hence the rank-two bundle is the right level at which to formulate covariance.

Definition 16.51 (Row-normal form for the 2Y live fiber). Let ℓ_7 be the covariant order-seven live row and let ω_Y be a primitive Y12-row with $\omega_Y(e_Y) = 1$. Set

$$\kappa := \omega_Y(e_7), \quad \ell_9^Y := \omega_Y - \kappa \ell_7, \quad \beta_9^{\text{tri}} := \beta_9 - \kappa c_7.$$

The 2Y-fiber is in row-normal form if

$$(\ell_7, \ell_9^Y)(e_7) = (1, 0), \quad (\ell_7, \ell_9^Y)(e_Y) = (0, 1).$$

All branch classifications below are made in this triangularized row basis.

Remark 16.52 (Triangularized rank-one fallback ratio). If the live obstruction collapses to a graph line

$$B_Y = \mathbf{C}(1, \rho_Y)$$

in the triangularized basis, then the graph-killing row gives the scalar rank-one wall

$$H((\beta_9^{\text{tri}} - \rho_Y c_7) \Gamma_G + \alpha_9 \Gamma_{G,2}).$$

Hence the corrected rank-one ratio is

$$\rho_{\Gamma}^{\text{tri}} := \frac{\beta_9^{\text{tri}} - \rho_Y c_7}{\alpha_9}.$$

The raw- β_9 expression is historical only and is not used in live row-normal formulas.

Theorem 16.53 (Two-row exact sequence for the $2Y$ live fiber). *Let the septic row be the covariant quotient row*

$$\ell_7 = \frac{\Delta_7^\nabla}{\Delta_7^\nabla(C_7)}.$$

Let the primitive $Y12$ -whitening row be a covector

$$\omega_Y : W \rightarrow \mathbf{C}$$

such that

$$\omega_Y(a_Y) = 1, \quad W_{\text{dead}}^{\text{fixed}} \subseteq \ker \omega_Y.$$

Set

$$\kappa := \omega_Y(e_7), \quad \ell_9^Y := \omega_Y - \kappa \ell_7, \quad \beta_9^{\text{tri}} := \beta_9 - \kappa c_7.$$

Assume the septic exact-sequence conditions

$$(R7) \quad \mathcal{A}_7^{\text{loc}} \subseteq \ker \Delta_7^\nabla, \quad \Delta_7^\nabla(C_7) \neq 0,$$

and the primitive $Y12$ -whitening conditions

$$(RY) \quad P_W \mathcal{A}_Y^{\text{loc}} \subseteq W_{\text{dead}}^{\text{fixed}}, \quad a_Y \notin W_{\text{dead}}^{\text{fixed}} \quad \text{with polynomial separation.}$$

Assume also the grade-separation condition

$$(G) \quad \ell_7(e_Y) = 0$$

and the polynomial row bound

$$|\omega_Y(e_7)| \leq Q^A.$$

Then (ℓ_7, ℓ_9^Y) is a polynomially bounded live row pair satisfying

$$(\ell_7, \ell_9^Y)(e_7) = (1, 0), \quad (\ell_7, \ell_9^Y)(e_Y) = (0, 1).$$

Moreover, if

$$\ell_7(A_{2Y}) = 0, \quad \omega_Y(A_{2Y}) = 0,$$

then

$$\ell_7(A_{2Y}) = 0, \quad \ell_9^Y(A_{2Y}) = 0.$$

Thus the $2Y$ -local image is zero in the row-normalized live fiber:

$$B_Y = 0.$$

Proof sketch. The septic exact sequence gives a bounded row ℓ_7 detecting C_7 and killing all admissible s_7 -dead columns. The whitening exact sequence gives a bounded primitive row ω_Y detecting a_Y and killing the fixed- α actual $Y12$ -slot image. Triangularization removes the allowed leakage $\omega_Y(e_7) = \kappa$ by setting

$$\ell_9^Y = \omega_Y - \kappa \ell_7.$$

The condition $\ell_7(e_Y) = 0$ is the non-negotiable grade-separation condition: the primitive order-nine $Y12$ -carrier has no covariant order-seven septic shadow. The row-normalized conclusions follow immediately. $\square$

Theorem 16.54 (Rank-two closure spine). *Assume*

$$\text{TwoRowExactSequence}_{2Y}, \quad \text{RowNormalForm}_{2Y}, \quad \text{NuisancePreservesRows}_{2Y}.$$

Then

$$B_Y = 0.$$

Consequently the row-normal rank selector of Definition 16.55 satisfies

$$r_Y = \text{rank}(M_{D4}^{\text{tri}}).$$

If moreover $c_7 \alpha_9 \neq 0$, then $r_Y = 2$, and the existing rank-two $D4$ branch activates. The $D4$ matrix in row-normal coordinates is

$$M_{D4}^{\text{tri}} = \begin{pmatrix} c_7 & 0 \\ \beta_9^{\text{tri}} & \alpha_9 \end{pmatrix},$$

and $D4$ supplies the plain wall

$$F_{2Y}^{\text{tri}} = H \begin{pmatrix} c_7 \Gamma_G \\ \beta_9^{\text{tri}} \Gamma_G + \alpha_9 \Gamma_{G,2} \end{pmatrix}.$$

On coefficient-good cores this is polynomially equivalent to

$$H(\Gamma_G, \Gamma_{G,2}).$$

No derivative wall components $D\Gamma_G$ or $D^2\Gamma_G$ are supplied by this rank-two $D4$ statement.

Definition 16.55 (Row-normal rank selector for the $2Y$ branch). Let $B_Y \subset \mathbf{C}^2$ be the actual/local live-fiber image in the row-normal coordinates

$$(s_7, s_9^Y) := (\ell_7, \ell_9^Y), \quad \ell_9^Y = \omega_Y - \omega_Y(e_7)\ell_7.$$

Let

$$M_{D4}^{\text{tri}} := \begin{pmatrix} c_7 & 0 \\ \beta_9^{\text{tri}} & \alpha_9 \end{pmatrix}, \quad \beta_9^{\text{tri}} := \beta_9 - \omega_Y(e_7)c_7.$$

The $2Y$ -rank selector is

$$r_Y := \text{rank}(\Pi_{\mathbf{C}^2/B_Y} M_{D4}^{\text{tri}}).$$

Thus the relevant rank is not the raw rank of formal $D4$ carriers. It is the rank of the row-normal $D4$ map after quotienting by actual/local source freedom.

Theorem 16.56 (Rank-two bridge to the existing $D4$ branch). Assume the live fiber is in row-normal form:

$$(\ell_7, \ell_9^Y)(e_7) = (1, 0), \quad (\ell_7, \ell_9^Y)(e_Y) = (0, 1),$$

with polynomially bounded rows. If the local row-exactness package gives

$$B_Y = 0,$$

then

$$r_Y = \text{rank}(M_{D4}^{\text{tri}}) = 2 \iff c_7 \alpha_9 \neq 0.$$

In this case the new $2Y$ -rank-two branch is exactly the existing rank-two $D4$ branch, interpreted as two surviving $D4$ coordinates after quotienting by actual/local source freedom. The $D4$ output is

$$F_{2Y}^{\text{tri}} = H \begin{pmatrix} c_7 \Gamma_G \\ \beta_9^{\text{tri}} \Gamma_G + \alpha_9 \Gamma_{G,2} \end{pmatrix},$$

and on coefficient-good cores,

$$|F_{2Y}^{\text{tri}}|^2 \asymp_{Q^A} |H|^2 (|\Gamma_G|^2 + |\Gamma_{G,2}|^2).$$

Equivalently, the honest rank-two wall is the plain wall

$$H(\Gamma_G, \Gamma_{G,2}).$$

No derivative wall components $D\Gamma_G$ or $D^2\Gamma_G$ are supplied by this rank selector.

Remark 16.57 (Branch classification by B_Y). The row-normal selector gives the following branch table:

B_Y	r_Y	paper branch
0	2	rank-two $D4$ vector wall
$\mathbf{C}(1, \rho_Y)$	1	rank-one moving-wall fallback
$\mathbf{C}(0, 1)$	0 for the $Y12$ column	vertical $Y12$ death
$\mathbf{C}^2$	0	live fiber absorbed

In the graph case $B_Y = \mathbf{C}(1, \rho_Y)$, the graph-killing row gives the rank-one scalar wall

$$H((\beta_9^{\text{tri}} - \rho_Y c_7) \Gamma_G + \alpha_9 \Gamma_{G,2}).$$

Thus the corrected triangularized fallback ratio is

$$\rho_\Gamma^{\text{tri}} = \frac{\beta_9^{\text{tri}} - \rho_Y c_7}{\alpha_9}.$$

The raw- β_9 expression is historical only and is not used in live row-normal formulas.

Remark 16.58 (Do not conflate with the old rank-two norm route). The $2Y$ -rank-two bridge is not the old abstract rank-two norm route. It is the concrete statement

$$\text{rank}(\Pi_{\mathbf{C}^2/B_Y} M_{D4}^{\text{tri}}) = 2.$$

It may supersede or bypass older rank-two norm language, but it should not be identified with any no-go attached to a different rank-two norm-control route unless the objects are explicitly the same.

Remark 16.59 (Three normal forms for the rank-two branch). The rank-two branch is organized by three internal normal forms. If the final contradiction is a global forbidden identity rather than the scalar plain-wall route, a fourth forbidden-channel normal form is also required.

First, the local live fiber is always put in row normal form:

$$\kappa = \omega_Y(e_7), \quad \ell_9^Y = \omega_Y - \kappa \ell_7, \quad \beta_9^{\text{tri}} = \beta_9 - \kappa c_7.$$

All branch classifications are made in the triangularized basis

$$(\ell_7, \ell_9^Y),$$

not in the raw row pair (ℓ_7, ω_Y) .

Second, the analytic wall is put in the source-curvature variable

$$p = q'',$$

so that, up to fixed nonzero normalization constants,

$$\Gamma_G = -6p + 36q - 12q^3, \quad \Gamma_{G,2} = p'' - 6p + 6q^2p.$$

Thus the finite wall spaces are built from a finite source space $\mathcal{P}_I$ for p , together with the affine reconstruction

$$\mathcal{V}_I = \mathcal{A}_1 + \mathcal{I}^2 \mathcal{P}_I.$$

Third, weighted transfer is made through predetermined coefficient/smooth-good cores

$$E_I^{\text{core}} \subset I, \quad |E_I^{\text{core}}| \geq R^{-C} |I|,$$

chosen before wall extraction and independent of

$$\Gamma_G, \quad \Gamma_{G,2}, \quad W_2, \quad W_2 := |\Gamma_G|^2 + |\Gamma_{G,2}|^2.$$

This prevents post-hoc wall-dependent selection.

If the route uses a global forbidden channel, the fourth normal form is forbidden-channel normal form. One fixes

$$M_{\text{forb}} := \Lambda E_{\text{car}} L_{\text{clean}}$$

on the $D4$ live plane and extracts against

$$E_{\text{wall}}^{\text{forb}} = \frac{|H|^2 |M_{\text{forb}} \Gamma|^2}{\mathcal{N}_{D4}}, \quad \Gamma = (\Gamma_G, \Gamma_{G,2})^T.$$

The current concrete Type-I candidate is the transition-singular discard operator

$$\Lambda_{\text{tr,sing},p} = (I - P_{\text{sing},p}) T_{\text{tr} \rightarrow \text{sing},p} P_{\text{tr},p}.$$

Remark 16.60 (Admissibility before quotient, projection, and selection). The rank-two branch uses an admissibility-first convention at every layer.

A direction may enter the exact local denominator only after it has been proved row-dead and live-pair preserving. A finite coherent direction is routed to $\mathcal{C}_{2Y}^{\text{arith}}$, and a small direction is routed to R_X^{2Y} ; neither is silently quotiented as exact nuisance.

Similarly, finite-atlas Riesz estimates are applied only after wall-variable completion has produced finite representatives

$$G_X \in \mathcal{W}_G, \quad G_{2,X} \in \mathcal{W}_{G2}$$

with

$$\|\Gamma_G - G_X\|_{H_{\text{sc}}^2(I)} + \|\Gamma_{G,2} - G_{2,X}\|_{L^2(I)} \leq Q^{-B}.$$

Finally, core sets E_I^{core} for weighted transfer must be chosen from coefficient, smooth, and chart data before observing the wall W_2 .

Remark 16.61 (Current architecture after the L^2 Gate–1 repair). The active architecture is now

$$\boxed{\text{Agg}L^2(q''_{\text{can}}) + \text{Graph/O2O3}_{23} + \text{NormalD4}_{K9}^{\text{gr}}}.$$

Here Gate–1 is local scalar L^2 -energy of the full canonical source

$$q''_{\text{can}} = B_2 + P_2 + D_2 + E_2.$$

Old branch observables such as $T_{\text{can}}X_{\text{can}}$, transition jets, raw singular sine moments, and projected singular low modes are proof tools or diagnostics. They are not the current Gate–1 definition.

The local rank-two/D4 cleanup remains valid: after $B_Y = 0$, D4 supplies only the plain wall

$$H(\Gamma_G, \Gamma_{G,2}),$$

not derivative wall components.

The finite/O2O3 module uses the graph/O2O3 package supplied by Theorem E.106. The downstream forbidden channel is the canonical source-graded $K9$ channel

$$\Lambda_p^{\text{gr,full}} = (I - P_p^{\text{full}})\mathcal{L}_{K9,p}^{\text{gr}}J_{\text{tr},p},$$

and its maximal affine projection. Raw tests of

$$(I - P_{\text{sing}}^{\text{raw}})T_{\text{tr} \rightarrow \text{sing}} \text{span}\{[C_7]^{\text{tr}}, [C_{Y12}]^{\text{tr}}\}$$

are archived diagnostics for noncanonical singular projections; they are not active finite audits in the canonical source-graded branch.

Remark 16.62 (D4 visibility and the canonical forbidden channel). Carrier-quotient D4 supplies local visibility of the row-normalized plain wall:

$$F_{2Y}^{\text{tri}} = H \left(\begin{matrix} c_7 \Gamma_G \\ \beta_9^{\text{tri}} \Gamma_G + \alpha_9 \Gamma_{G,2} \end{matrix} \right).$$

This local visibility was not, by itself, a contradiction; it required a forbidden channel with a same-norm upper. In the active proof branch that channel is now fixed:

$$\Lambda_p^{\text{gr,full}} = (I - P_p^{\text{full}})\mathcal{L}_{K9,p}^{\text{gr}}J_{\text{tr},p}, \quad \widehat{\Lambda}_{\text{max},p} = \Pi_{\text{max},p}^{\text{gr}}\Lambda_p^{\text{gr,full}}.$$

The local D4 lower is seen by this channel through the maximal affine row realization, while Theorem 16.189 gives the same-norm upper. Therefore the warning in this remark applies only to carrier-quotient D4 used without the canonical source-graded $K9$ channel.

Definition 16.63 (Covariant septic row). Let

$$A_5^{\mathfrak{f}}(m) = u_5 I + v_5 S, \quad L_5(m) := \mathbf{C}A_5^{\mathfrak{f}}(m) \subset \mathfrak{f}_m.$$

Define

$$\Delta_7^{\nabla}(u) := \omega_m \left(Q_7^{\text{pkg}, \nabla} u, A_5^{\mathfrak{f}}(m) \right),$$

as a covector on the moving quotient

$$\mathfrak{f}_m / L_5(m).$$

Thus

$$\Delta_7^{\nabla}(u) = 0 \quad \Longleftrightarrow \quad Q_7^{\text{pkg}, \nabla} u \in \mathbf{C}A_5^{\mathfrak{f}}(m).$$

The normalized septic row is

$$\ell_7(u) := \frac{\Delta_7^{\nabla}(u)}{\Delta_7^{\nabla}(C_7)}.$$

Raw determinant drift in $u_7 v_5 - u_5 v_7$ is not a live s_7 -motion unless it survives this covariant quotient.

Definition 16.64 (Typed septic denominator). Define

$$\begin{aligned} \mathcal{D}_7^{\text{typed}} := & \mathbf{C}A_5^f + \mathcal{D}_{7,\text{EF}} + \mathcal{D}_{7,\text{chart}} + \mathcal{D}_{7,\text{norm}} + \mathcal{D}_{7,\gamma/\text{bg}} \\ & + \mathcal{D}_{7,\text{tail/res}} + \mathcal{D}_{7,\text{wrong}} + \mathcal{N}_{7,\text{exact}}. \end{aligned}$$

Here $\mathcal{N}_{7,\text{exact}}$ consists only of exact nuisance directions that have already passed the row-deadness and live-pair preservation tests.

Theorem 16.65 (Septic exact sequence audit). *Let*

$$\mathcal{A}_7^{\text{loc}} := \mathcal{I}_{\text{can}}^{(2)} + T_{\text{lower}}^{\text{arith}} + T_{E_2/R_{\text{lab}},\text{loc}}^{s_7} + \mathcal{N}_{7,\text{exact}}.$$

Assume

$$(SE1) \quad \Delta_7^\nabla(\mathcal{A}_7^{\text{loc}}) = 0,$$

and

$$(SE2) \quad |\Delta_7^\nabla(C_7)| \geq Q^{-A}.$$

Equivalently,

$$\text{rank}[\mathcal{D}_7^{\text{typed}}, \mathcal{A}_7^{\text{loc}}] = \text{rank } \mathcal{D}_7^{\text{typed}},$$

while

$$\text{dist}(C_7, \mathcal{D}_7^{\text{typed}}) \geq Q^{-A} \|C_7\|.$$

Then

$$\ell_7(\mathcal{A}_7^{\text{loc}}) = 0, \quad \ell_7(C_7) = 1.$$

Remark 16.66 (Actual lower kernel audit). The actual-lower part of (SE1) is the finite kernel condition

$$\Delta_7^\nabla Q_7^{\text{pkg},\nabla} D\mathcal{S}_{\text{lower}} = 0.$$

Equivalently, the actual lower parameter space has no corrected septic package-coordinate summand:

$$\Theta_{\text{pkg},7} = 0.$$

Formal C_7 -carriers and formal $C_7 \rightarrow Y12$ connections do not enter B_Y unless actualized by a genuine actual-source, exact nuisance, or local fitting tangent.

Definition 16.67 (Actual lower parameter package map). Let $\Theta_{\text{lower}}^{\text{arith}}$ denote the actual lower-source parameter space. It contains only actual EF/gamma/chart/tail lower deformations, actual normalization or representative lower freedoms, and genuine exact nuisance lower directions. It excludes formal D4 carrier variables, formal C_7 labels, formal $C_7 \rightarrow Y12$ connections without actual tangent, proof thickening parameters, and free corrected-package coordinates not present in the actual source.

Define the lower package map

$$\mathfrak{d}_{7,\text{lower}} := \Delta_7^\nabla Q_7^{\text{pkg},\nabla} D\mathcal{S}_{\text{lower}} : \Theta_{\text{lower}}^{\text{arith}} \rightarrow \mathbf{C}.$$

Lemma 16.68 (Actual lower no-package-coordinate audit). *The actual lower contribution to the s_7 -row is zero if and only if*

$$\mathfrak{d}_{7,\text{lower}} = 0.$$

Equivalently, the actual lower parameter space has no corrected septic package-coordinate summand:

$$\Theta_{\text{pkg},7} = 0.$$

A formal C_7 -carrier or formal $C_7 \rightarrow Y12$ connection contributes to B_Y only after actualization by a genuine actual-source, exact nuisance, or atom-local tangent with nonzero Δ_7^∇ .

Remark 16.69 (Failure routing for actual lower package leakage). If $u_{\text{lower}} \in T_{\text{lower}}^{\text{arith}}$ satisfies

$$\Delta_7^\nabla(u_{\text{lower}}) \neq 0,$$

then compute its triangularized live-fiber vector

$$(\ell_7(u_{\text{lower}}), \ell_9^Y(u_{\text{lower}})).$$

If all such nonzero vectors span a graph line $\mathbf{C}(1, \rho_Y)$, the rank-one moving-wall fallback remains available. If a vertical vector $\mathbf{C}(0, 1)$, a second slope, or a full rank-two image appears, the current $Y12$ route dies locally or requires a different carrier.

Definition 16.70 (Whitening quotient for the primitive $Y12$ row). Let W be the finite same-point whitening-input space seen by the order-nine singular extractor. Define

$$W_{\text{dead}}^{\text{fixed}} := W_{\text{global}} + W_{\text{chart}} + W_{\text{norm}} + W_{\text{bg}} + W_{\text{trans}} + W_{\text{wrong}}.$$

Here W_{trans} denotes transported same-point whitening data and W_{wrong} denotes wrong-origin terms, such as $DW[DN_0]$, $D^2W[N_0]$, or whitening applied to residual/completion data rather than to the primitive D^2N_0 -carrier. Let

$$a_Y \in W$$

be the live same-point whitening input in the primitive carrier

$$DW_{\text{same}}[a_Y] D^2N_0[\xi_1, \xi_2]$$

with mixed branch 12, same-point incidence, and one whitening slot.

Theorem 16.71 (Whitening exact sequence for fixed- α $Y12$). *Assume*

$$(W1) \quad P_W V_{\text{arith}}^{\text{fixed}} \subseteq W_{\text{dead}}^{\text{fixed}}$$

and

$$(W2) \quad \text{dist}(a_Y, W_{\text{dead}}^{\text{fixed}}) \geq Q^{-A}|a_Y|.$$

Then there is a polynomially bounded covector

$$\omega_Y : W \rightarrow \mathbf{C}$$

such that

$$\omega_Y(a_Y) = 1, \quad \omega_Y(W_{\text{dead}}^{\text{fixed}}) = 0, \quad \omega_Y(P_W V_{\text{arith}}^{\text{fixed}}) = 0.$$

Consequently fixed- α actual first and second canonical variations do not produce a primitive $Y12$ -row column.

If $W1$ fails, the fallback audit is the ordered slot cascade:

same-point whitening slot $\longrightarrow$ source slot 1 $\longrightarrow$ source slot 2 $\longrightarrow$ coupled bilinear image.

Only failure of the final coupled bilinear image creates an actual $Y12$ -live column.

Corollary 16.72 (Fixed- α whitening kills primitive $Y12$ rows). *Under assumptions (W1) and (W2), the actual first and second canonical variations do not produce a primitive $Y12$ row column. The whitening quotient is therefore a derived consequence of Theorem 16.71, not a separate hypothesis.*

Remark 16.73 (Whitening naturality also controls labelled transition). The same whitening quotient controls the $Y12$ -part of the labelled/raw transition defect. Let

$$\mathfrak{W}_Y := DW_{\text{sing, same}} - S_W DW_{\text{tr, same}} S^{-1}.$$

The $Y12$ -safe labelled-transition condition is

$$D_{\text{loc}} P_{(W_{\text{dead}}^{\text{fixed}})^\perp} (\mathfrak{W}_Y[D^2N_0]) = 0.$$

Equivalently, same-point whitening may be transported or vary coherently, but it must not be refit atom-by-atom on the primitive carrier. Fixed coherent defects are routed to $\mathcal{C}_{2Y}^{\text{arith}}$; residual defects are routed to R_X^{2Y} ; atom-local live defects enter B_Y .

Definition 16.74 (Locality filtration for completion and labelled defects). For any $2Y$ -fiber-valued expression X , write

$$X = X_{\text{fixed}} + X_{\text{coh}} + X_{\text{res}} + X_{\text{loc}}.$$

Only X_{loc} contributes to B_Y . The fixed part is absorbed into basepoint/recentering data, the coherent part is routed to $\mathcal{C}_{2Y}^{\text{arith}}$, and the residual part is routed to R_X^{2Y} .

Theorem 16.75 (No atom-fitting for E_2 and labelled transition). *Assume the following three guardrails.*

(1) *No local live completion counterterms:*

$$\text{Loc } \Pi_{\text{fib}, Y} E_{\text{model}} = 0.$$

In particular, no local terms of the forms

$$\eta_7(p)e_7(p) + \eta_Y(p)e_Y(p), \quad \eta(p)(e_7(p) + \rho_Y e_Y(p))$$

are admitted.

(2) *Normal form data are not atom-fitted:*

$$D_{\text{loc}} \Pi_{\text{fib}, Y} (D^2 N_{0, \text{sing}} \circ (L, L) - S D^2 N_{0, \text{tr}}) = 0.$$

(3) *Same-point whitening is not atom-fitted:*

$$D_{\text{loc}} \Pi_{\text{fib}, Y} [(DW_{\text{sing}, \text{same}} - S_W DW_{\text{tr}, \text{same}} S^{-1}) [D^2 N_0]] = 0.$$

Then

$$D_{\text{loc}} \Pi_{\text{fib}, Y} (E_2 + R_{\text{lab}}) = 0.$$

Thus E_2/R_{lab} affects B_Y only through atom-local live-fiber fitting. Fixed, coherent, and residual effects do not enter B_Y .

Corollary 16.76 (Exact nuisance and labelled transition are routed out of B_Y). *Under the three guardrails of Theorem 16.75, one has*

$$D_{\text{loc}} \Pi_{\text{fib}, Y} (E_2 + R_{\text{lab}}) = 0.$$

Thus exact nuisance, coherent nuisance, and residual nuisance are not live inputs to B_Y ; they are eliminated or rerouted before the local row test.

Remark 16.77 (Amplitude projection is downstream of $B_Y = 0$). The proof of $B_Y = 0$ must precede any amplitude projection. A finite coherent space $\mathcal{C}_{2Y}^{\text{arith}}$ may be introduced only after the local live-fiber quotient has been fixed. In particular, a projected wall

$$(I - P_{\mathcal{C}_{2Y}^{\text{arith}}}) F_{2Y}$$

must not be used to define D_{2Y} , the live rows, or the rank-two quotient, unless the projection is itself proved to have bounded local branch complexity.

Definition 16.78 (Exact/coherent/residual nuisance split). Split the declared singular nuisance space as

$$\mathcal{N}_{\text{sing}} = \mathcal{N}_{\text{exact}} + \mathcal{N}_{\text{coh}} + \mathcal{N}_{\text{res}}.$$

Only $\mathcal{N}_{\text{exact}}$ enters the exact local denominator. The coherent part is routed to $\mathcal{C}_{2Y}^{\text{arith}}$, and the residual part is routed to R_X^{2Y} . Thus

$$D_{2Y} := D_{2Y}^0 + \mathcal{N}_{\text{exact}},$$

not $D_{2Y}^0 + \mathcal{N}_{\text{sing}}$.

Definition 16.79 (Admissible exact nuisance). An exact nuisance generator $n \in \mathcal{N}_{\text{exact}}$ is admissible for the rank-two quotient only if

$$\Delta_7^\nabla(n) = 0, \quad \omega_Y(P_W n) = 0.$$

Any declared nuisance generator failing either row test is not exact nuisance. It must instead be routed to one of:

$$\mathcal{C}_{2Y}^{\text{arith}}, \quad R_X^{2Y}, \quad \text{quotient repair}, \quad \text{local rank failure}.$$

Remark 16.80 (Two-grade nuisance homogeneity). The exact nuisance denominator must not identify the two live grades. In particular, one must rule out

$$e_Y - \lambda e_7 \in D_{2Y}^0 + \mathcal{N}_{\text{exact}}$$

with polynomially small angle. Equivalently, exact nuisance should be grade-homogeneous,

$$\mathcal{N}_{\text{exact}} = \mathcal{N}_{7,\text{exact}} \oplus \mathcal{N}_{9,\text{exact}},$$

or at least block-triangular with no live $7 \leftrightarrow 9$ relation.

Lemma 16.81 (Nuisance preservation of the live rows). *Assume*

$$\Delta_7^\nabla(\mathcal{N}_{\text{exact}}) = 0, \quad \omega_Y(P_W \mathcal{N}_{\text{exact}}) = 0.$$

Assume also the live-pair Schur condition

$$\lambda_{\min} \left([e_7, e_Y]^* (I - P_{D_{2Y}^0 + \mathcal{N}_{\text{exact}}}) [e_7, e_Y] \right) \geq Q^{-A}.$$

Then

$$(\ell_7, \ell_9^Y)(\mathcal{N}_{\text{exact}}) = 0,$$

and the live pair remains two-dimensional and polynomially conditioned modulo $D_{2Y}^0 + \mathcal{N}_{\text{exact}}$.

Proof sketch. The first row-deadness condition gives $\ell_7(\mathcal{N}_{\text{exact}}) = 0$. The second gives $\omega_Y(P_W \mathcal{N}_{\text{exact}}) = 0$. Since

$$\ell_9^Y = \omega_Y - \kappa \ell_7,$$

we get $\ell_9^Y(\mathcal{N}_{\text{exact}}) = 0$. The Schur condition rules out

$$e_7 \in D_{2Y}^0 + \mathcal{N}_{\text{exact}}, \quad e_Y \in D_{2Y}^0 + \mathcal{N}_{\text{exact}},$$

and

$$e_Y - \lambda e_7 \in D_{2Y}^0 + \mathcal{N}_{\text{exact}}.$$

Thus exact nuisance is both row-dead and live-pair preserving. $\square$

Corollary 16.82 (Nuisance-safe contribution to $B_Y = 0$). *Assume $\mathcal{N}_{\text{sing}}$ has been split as*

$$\mathcal{N}_{\text{sing}} = \mathcal{N}_{\text{exact}} + \mathcal{N}_{\text{coh}} + \mathcal{N}_{\text{res}},$$

and that only $\mathcal{N}_{\text{exact}}$ enters the exact denominator. Suppose $\mathcal{N}_{\text{exact}}$ is row-dead:

$$\Delta_7^\nabla(\mathcal{N}_{\text{exact}}) = 0, \quad \omega_Y(P_W \mathcal{N}_{\text{exact}}) = 0,$$

and preserves the live pair:

$$e_7 \notin D_{2Y}^0 + \mathcal{N}_{\text{exact}}, \quad e_Y \notin D_{2Y}^0 + \mathcal{N}_{\text{exact}},$$

and

$$e_Y - \lambda e_7 \notin D_{2Y}^0 + \mathcal{N}_{\text{exact}} \quad \text{for every } \lambda \in \mathbf{C},$$

with polynomial angle. Then nuisance contributes no local column to the row-normal live fiber:

$$(\ell_7, \ell_9^Y)(\mathcal{N}_{\text{exact}}) = 0.$$

The coherent part $\mathcal{N}_{\text{coh}}$ is routed to $\mathcal{C}_{2Y}^{\text{arith}}$, and the residual part $\mathcal{N}_{\text{res}}$ is routed to R_X^{2Y} . Neither coherent nor residual nuisance may be placed in the exact denominator merely by name.

Theorem 16.83 (Row-normal live-fiber avoidance). *Let the live rows be in row-normal form:*

$$\ell_7 = \frac{\Delta_7^\nabla}{\Delta_7^\nabla(C_7)}, \quad \ell_9^Y = \omega_Y - \kappa \ell_7, \quad \kappa = \omega_Y(e_7).$$

Assume

$$\ell_7(e_7) = 1, \quad \ell_7(e_Y) = 0, \quad \omega_Y(e_Y) = 1, \quad |\omega_Y(e_7)| \leq Q^A,$$

and polynomial row bounds. Assume further that the five row blockers vanish:

$$\begin{aligned} \Delta_7^\nabla(D_{\text{actual}} T \mathcal{M}_Y^{\text{arith}}) &= 0, \\ \omega_Y(P_W D_{\text{actual}} T \mathcal{M}_Y^{\text{arith}}) &= 0, \\ D_{\text{loc}}(\ell_7, \ell_9^Y)(E_2 + R_{\text{lab}}) &= 0, \end{aligned}$$

$$\Delta_7^\nabla(\mathcal{N}_{\text{exact}}) = 0,$$

and

$$\omega_Y(P_W \mathcal{N}_{\text{exact}}) = 0.$$

Then

$$B_Y = 0.$$

Consequently, if the triangularized D_4 coefficients are polynomially conditioned, D_4 gives the plain rank-two wall

$$F_{2Y}^{\text{tri}} = H \left(\begin{matrix} c_7 \Gamma_G \\ \beta_9^{\text{tri}} \Gamma_G + \alpha_9 \Gamma_{G,2} \end{matrix} \right) \asymp_{Q^A} H(\Gamma_G, \Gamma_{G,2}).$$

No derivative wall components are produced by this theorem.

16.19. Order-seven packet visibility on the reduced base. For this and the following subsections, write the centered diagonal Gram data as

$$g = \frac{q_0}{\pi}, \quad h = \frac{1}{\pi} \left(\frac{1}{3} q_0^3 + \frac{1}{6} q'' \right),$$

so that the centered same-point Gram block at $s = 0$ is, up to the fixed ambient normalization,

$$G_0 = \text{diag}(2g, 2h).$$

Definition 16.84 (First and second Gram-anisotropy functionals). On a centered reduced patch, unless explicitly stated otherwise, q_0 , q'' , and $q^{(4)}$ mean the full actual-zeta centered jets

$$q_0 = q_\zeta(m), \quad q'' = q''_{\text{can}}(m) = q''_\zeta(m), \quad q^{(4)} = (q_\zeta)^{(4)}(m).$$

Define

$$\Gamma_G^{\text{can}} := q''_{\text{can}} - 6q_0 + 2q_0^3.$$

For readability we usually suppress the superscript and write $\Gamma_G = \Gamma_G^{\text{can}}$. Then

$$g - h = -\frac{\Gamma_G}{6\pi}.$$

Thus the Gram-isotropy wall $g = h$ is equivalently

$$\Gamma_G = 0.$$

We also define the second Gram-anisotropy functional

$$\Gamma_{G,2}^{\text{can}} := (1 - q_0^2)q''_{\text{can}} - \frac{1}{6}(q_\zeta)^{(4)}.$$

Again, when the canonical convention is fixed, we write $\Gamma_{G,2} = \Gamma_{G,2}^{\text{can}}$. This is the anisotropy of the second centered Gram variation. In particular, on $g = h$, the first Gram anisotropy has vanished, but the second one need not vanish.

The first admissible visibility coefficient is not A_7^{f} , which defines the reference line, but A_7^{f} . The direct raw off-diagonal order-seven term is not itself the witness. At a centered even packet,

$$N_7^{\text{off}} = C_7 J, \quad J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

Since $X_{-,0} = X_{+,0} = X_0$,

$$X_0 N_7^{\text{off}} X_0 = C_7 (\det X_0) J,$$

so

$$\pi_{\text{f}}(X_0 N_7^{\text{off}} X_0) = 0.$$

The live centered channel is instead the one-whitening-slot term

$$T_7^{(1)} = X_{-,1} Y X_{+,0} + X_{-,0} Y X_{+,1}, \quad Y = yJ.$$

At centered parity,

$$X_{+,0} = X_{-,0} = X_0, \quad X_{+,1} = -X_{-,1}.$$

Write

$$X_0 = pI + rD, \quad X_{-,1} = BS.$$

Then

$$T_7^{(1)} = yB(SJX_0 - X_0JS) = -2yB(pD + rI),$$

and hence

$$\boxed{\pi_{\mathfrak{f}}(T_7^{(1)}) = -2yBr I.}$$

For the centered off-line packet,

$$q'' = 2A_2, \quad A_2 = -4(a^{-3} + b^{-3}) \neq 0,$$

and

$$y_{\text{lead}} = -\frac{q_0 A_2^2}{72}.$$

Moreover

$$B = \frac{q''}{4\pi\sqrt{gh}(\sqrt{g} + \sqrt{h})}, \quad r = \frac{1}{2}(g^{-1/2} - h^{-1/2}).$$

Thus the one-whitening-slot channel is nonzero outside the explicit incidence walls

$$y = 0, \quad q'' = 0, \quad g = h.$$

Lemma 16.85 (Centered order-seven fixed-sector row and first Gram anisotropy). *At centered parity,*

$$X_{+,0} = X_{-,0} = X_0, \quad X_{+,1} = -X_{-,1}.$$

Write

$$X_0 = pI + rD, \quad X_{-,1} = BS, \quad Y = yJ,$$

where

$$p = \frac{1}{2}(g^{-1/2} + h^{-1/2}), \quad r = \frac{1}{2}(g^{-1/2} - h^{-1/2}).$$

Then the centered one-whitening-slot row

$$T_7^{(1)} := X_{-,1}YX_{+,0} + X_{-,0}YX_{+,1}$$

satisfies

$$T_7^{(1)} = -2yB(pD + rI).$$

Consequently

$$\pi_{\mathfrak{f}}T_7^{(1)} = -2yBr I.$$

Equivalently,

$$\pi_{\mathfrak{f}}T_7^{(1)} = -y \frac{q''(h - g)}{4\pi gh(\sqrt{g} + \sqrt{h})^2} I.$$

Thus, away from the source walls $y = 0$ and $q'' = 0$, the centered order-seven fixed-sector signal is nonzero exactly off the first Gram-isotropy wall $g = h$, or equivalently off $\Gamma_G = 0$.

Remark 16.86 (Why order seven alone is insufficient on $g = h$). On $g = h$ one has $r = 0$. The order-seven row does not disappear in the ambient 2×2 algebra; rather, it becomes

$$T_7^{(1)}|_{g=h} = -2yBpD.$$

This component is anti-fixed and is killed by $\pi_{\mathfrak{f}}$. Hence the centered order-seven fixed-sector argument requires either quantitative separation from $g = h$, an ambient D -sector enlargement, or a higher-order fixed-sector replacement. The next replacement is the order-nine row below.

Lemma 16.87 (Second whitening anisotropy on the Gram-isotropy wall). *Assume $g = h$, so that*

$$G_0 = gI.$$

Write

$$G_{-}(s) = G_0 + sG_1 + s^2G_2 + O(s^3), \quad X_{-}(s) = G_{-}(s)^{-1/2} = X_0 + sX_1 + s^2X_2 + O(s^3),$$

with the convention $G_{-}(s) = G(m - s/2)$, so

$$G_1 = -\frac{1}{2}G'(m), \quad G_2 = \frac{1}{8}G''(m).$$

At a scalar base,

$$X_2 = -\frac{1}{2}g^{-3/2}G_2 + \frac{3}{8}g^{-5/2}G_1^2.$$

At centered parity, $G_1 \in \mathbf{CS}$, hence

$$G_1^2 \in \mathbf{CI}.$$

Therefore

$$(X_2)_D = -\frac{1}{2}g^{-3/2}(G_2)_D.$$

Using the centered Gram model

$$G(t) = \frac{1}{\pi} \begin{pmatrix} q(t) & \frac{1}{2}q'(t) \\ \frac{1}{2}q'(t) & \frac{1}{3}q(t)^3 + \frac{1}{6}q''(t) \end{pmatrix},$$

and $q'(m) = q'''(m) = 0$, one has

$$G''(m) = \frac{1}{\pi} \begin{pmatrix} q'' & 0 \\ 0 & q_0^2 q'' + \frac{1}{6}q^{(4)} \end{pmatrix}.$$

Thus

$$(G_2)_D = \frac{1}{16\pi} \left((1 - q_0^2)q'' - \frac{1}{6}q^{(4)} \right) = \frac{\Gamma_{G,2}}{16\pi}.$$

Consequently

$$(X_2)_D = -\frac{1}{32\pi}g^{-3/2}\Gamma_{G,2}.$$

Proposition 16.88 (Order-nine fixed-sector survivor on $g = h$). *Assume $g = h$. At centered parity,*

$$X_{+,1} = -X_{-,1}, \quad X_{+,2} = X_{-,2}.$$

Let

$$X_1 := X_{-,1} = BS, \quad Y = yJ, \quad X_2 := X_{-,2} = aI + dD,$$

with $d = (X_2)_D$. Consider the order-nine row

$$T_9^{(1,2)} := X_{-,1}YX_{+,2} + X_{-,2}YX_{+,1}.$$

Then

$$T_9^{(1,2)} = X_1YX_2 - X_2YX_1.$$

Using $SJ = -D$ and $JS = D$, we obtain

$$T_9^{(1,2)} = -2yB(aD + dI).$$

Therefore

$$\pi_{\mathfrak{f}}T_9^{(1,2)} = -2yBdI.$$

At $g = h$,

$$B = \frac{q''}{8\pi g^{3/2}}$$

and, by Lemma 16.87,

$$d = -\frac{1}{32\pi}g^{-3/2}\Gamma_{G,2}.$$

Hence

$$\pi_{\mathfrak{f}}T_9^{(1,2)} = \frac{yq''\Gamma_{G,2}}{128\pi^2g^3}I.$$

In particular, on the source-good locus $yq'' \neq 0$, the order-nine fixed-sector row survives on $g = h$ whenever

$$\Gamma_{G,2} \neq 0,$$

modulo the correction incidence defined below.

Lemma 16.89 (The second anisotropy wall is not forced by $g = h$). *The first Gram-isotropy relation*

$$g = h$$

is equivalent to

$$q'' = 6q_0 - 2q_0^3.$$

It does not imply

$$\Gamma_{G,2} = 0.$$

Indeed, on $g = h$,

$$\Gamma_{G,2} = (1 - q_0^2)(6q_0 - 2q_0^3) - \frac{1}{6}q^{(4)}.$$

Thus $\Gamma_{G,2} = 0$ imposes the additional fourth-jet relation

$$q^{(4)} = 6(1 - q_0^2)(6q_0 - 2q_0^3).$$

This is an independent higher-jet incidence condition.

For a centered paired packet

$$f_{\beta,\gamma}(t) = \frac{a}{a^2 + x^2} + \frac{b}{b^2 + x^2}, \quad x = 2t - \gamma,$$

evaluated at $x = 0$, one has

$$q_0 = a^{-1} + b^{-1}, \quad q'' = -8(a^{-3} + b^{-3}), \quad q^{(4)} = 384(a^{-5} + b^{-5}).$$

Hence

$$\Gamma_{G,2} = (1 - q_0^2)q'' - \frac{1}{6}q^{(4)}.$$

For example, for $a = \frac{1}{4}$, $b = \frac{3}{4}$,

$$q_0 = \frac{16}{3}, \quad q'' = -\frac{14336}{27}, \quad q^{(4)} = \frac{31981568}{81},$$

and therefore

$$\Gamma_{G,2} = -\frac{12449792}{243} \neq 0.$$

Thus the order-nine wall $\Gamma_{G,2} = 0$ is a proper additional analytic wall, not a consequence of centered parity or of $g = h$.

Proposition 16.90 (Raw order-nine I -row audit). *Under the centered raw two-point convention, the raw odd carriers are skew off-diagonal:*

$$N_{2j+1} = F_{2j+1}J.$$

In particular,

$$N_9 = F_9J, \quad N_7 = F_7J, \quad N_5 = F_5J, \quad N_3 = F_3J, \quad N_1 = F_1J.$$

Consequently their I - and D -components vanish:

$$N_{2j+1}^I = N_{2j+1}^D = 0.$$

The raw even carriers are fixed under endpoint swap. In the centered raw two-point convention, their symmetric off-diagonal S -parts do not contribute to the order-nine I -coordinate:

$$N_4^S = N_2^S = N_0^S = 0$$

for the raw lower-carrier row relevant here.

Therefore the raw order-nine I -coordinate receives no contribution from the raw carriers other than the order-nine survivor

$$T_9^{(1,2)} = X_{-,1}YX_{+,2} + X_{-,2}YX_{+,1}.$$

Equivalently,

$$R_{\leq 4}^{I,\text{raw}} = 0,$$

and

$$R_9^I[N_9] = R_7^I[N_7] = R_5^I[N_5] = 0.$$

Remark 16.91 (Convention caveat for corrected rows). Proposition 16.90 concerns the raw pre-whitening two-point carriers N_b . If a later notation folds normalization, subtraction, background/core, line, or connection correction terms into a symbol also called N_b , those terms should be separated from the raw carrier and placed into the correction remainder below. The raw matrix-parity audit does not by itself eliminate such correction rows.

Definition 16.92 (Order-nine correction incidence). Let $R_{\text{corr}}^{I,(9)}$ denote the residual order-nine I -coordinate coming from correction rows not included in the raw two-point carrier table. After covariantization and the fixed-convention local gates, this residual may only include line, background/core, or normalization/subtraction correction data:

$$R_{\text{corr}}^{I,(9)} = R_{\text{line}}^{I,(9)} + R_{\text{bg/core}}^{I,(9)} + R_{\text{norm/sub}}^{I,(9)}.$$

Connection/frame/projection/whitening-gauge terms are absorbed by $\nabla^{\mathbf{f}}$; Gate A/B rows are closed under the fixed-convention hypotheses; shadow rows are projected away; and pure same-point odd rows vanish by centered parity.

For the collapsed scalar I -coordinate fallback, define

$$F_9^I := R_{\text{corr}}^{I,(9)} + \frac{yq''\Gamma_{G,2}}{128\pi^2g^3}.$$

Thus the order-nine scalar I -coordinate is

$$(A_9^{\mathbf{f}})_I = F_9^I.$$

Define the order-nine correction incidence wall by

$$\mathcal{E}_{\text{corr}}^{(9)} := \{F_9^I = 0, \quad yq''\Gamma_{G,2} \neq 0\}.$$

Equivalently,

$$\mathcal{E}_{\text{corr}}^{(9)} = \left\{ R_{\text{corr}}^{I,(9)} = -\frac{yq''\Gamma_{G,2}}{128\pi^2g^3}, \quad yq''\Gamma_{G,2} \neq 0 \right\}.$$

On the complement of this wall, the collapsed order-nine I -coordinate is nonzero whenever $yq''\Gamma_{G,2} \neq 0$.

Theorem 16.93 (Layered fixed-sector visibility through order nine). *Work on a centered reduced good patch in the fixed-core convention. Assume*

$$g > 0, \quad h > 0, \quad yq'' \neq 0.$$

Let

$$\Gamma_G = q'' - 6q_0 + 2q_0^3, \quad \Gamma_{G,2} = (1 - q_0^2)q'' - \frac{1}{6}q^{(4)}.$$

Assume also that the point is not in the order-seven cross-channel incidence wall and not in the order-nine correction incidence wall:

$$\theta \notin \mathcal{E}_{\text{cc}}^{(7)} \cup \mathcal{E}_{\text{corr}}^{(9)}.$$

If

$$\Gamma_G \neq 0 \quad \text{or} \quad \Gamma_{G,2} \neq 0,$$

then

$$(A_7^{\mathbf{f}}, A_9^{\mathbf{f}}) \neq 0.$$

More precisely:

$$\Gamma_G \neq 0 \implies A_7^{\mathbf{f}} \neq 0,$$

where the nonzero fixed-sector row is the centered order-seven row

$$\pi_{\mathbf{f}} T_7^{(1)} = -2yBr I.$$

On the first Gram-isotropy wall

$$\Gamma_G = 0,$$

if

$$\Gamma_{G,2} \neq 0$$

and

$$\theta \notin \mathcal{E}_{\text{corr}}^{(9)},$$

then

$$A_9^{\mathfrak{f}} \neq 0,$$

with order-nine I -coordinate

$$(A_9^{\mathfrak{f}})_I = \frac{yq''\Gamma_{G,2}}{128\pi^2g^3} + R_{\text{corr}}^{I,(9)}.$$

Proof. If $\Gamma_G \neq 0$, then $g - h \neq 0$. By Lemma 16.85,

$$\pi_{\mathfrak{f}}T_7^{(1)} = -y \frac{q''(h-g)}{4\pi gh(\sqrt{g} + \sqrt{h})^2} I,$$

which is nonzero under $yq'' \neq 0$, $g, h > 0$, and $g - h \neq 0$, outside the named order-seven cross-channel incidence.

It remains to consider $\Gamma_G = 0$, equivalently $g = h$. By Proposition 16.88,

$$\pi_{\mathfrak{f}}T_9^{(1,2)} = \frac{yq''\Gamma_{G,2}}{128\pi^2g^3} I.$$

By Proposition 16.90, the raw order-nine I -row has no additional contribution from N_9, N_7, N_5 , or from the raw lower carriers. Thus the only possible remaining I -coordinate correction is $R_{\text{corr}}^{I,(9)}$. Outside $\mathcal{E}_{\text{corr}}^{(9)}$, this correction does not cancel the displayed candidate row. Hence $A_9^{\mathfrak{f}} \neq 0$. $\square$

Remark 16.94 (What the layered theorem does and does not prove). Theorem 16.93 is a local visibility theorem. It improves the centered order-seven story by replacing the single excluded wall

$$\Gamma_G = 0$$

with the layered incidence

$$\Gamma_G = 0, \quad \Gamma_{G,2} = 0,$$

together with the named correction incidence. In particular, the order-seven fixed-sector signal detects off first Gram isotropy, while the order-nine fixed-sector signal detects on the first Gram-isotropy wall unless the second Gram anisotropy also vanishes.

This is still not an RH-scale quantitative lower bound. For a global proof one would need a non-concentration estimate of the form

$$|yq''| \max\{|\Gamma_G|, |\Gamma_{G,2}|\} \geq Q^{-A},$$

or an equivalent lower bound for

$$\|(A_7^{\mathfrak{f}}, A_9^{\mathfrak{f}})\|$$

outside the layered incidence and correction walls. Qualitative nonvanishing alone does not rule out super-polynomial approach to the layered incidence set.

Remark 16.95 (No-go ledger for the centered order-seven/nine analysis). The following points should be kept separate.

- (1) The centered order-seven fixed-sector row is structurally proportional to the first Gram anisotropy $g - h$. Hence it cannot detect the wall $g = h$ inside the fixed sector.
- (2) The ambient order-seven D -component survives on $g = h$, but using it would require an ambient, non-fixed-sector enlargement. The present theorem instead stays within the rank-two fixed sector and uses order nine.
- (3) The order-nine survivor is not a raw-carrier artifact. It arises from the channel

$$Y_J X_{1,S} X_{2,D},$$

namely

$$X_{-,1} Y X_{+,2} + X_{-,2} Y X_{+,1}.$$

- (4) The raw carriers N_9, N_7, N_5, N_3, N_1 are skew off-diagonal at centered parity:

$$N_{2j+1} = F_{2j+1} J.$$

They do not contribute to the order-nine I -coordinate. The raw even lower carriers are likewise harmless for the order-nine I -row under the centered raw convention.

- (5) Any remaining order-nine cancellation is therefore a correction-row or cross-channel analytic incidence issue, not a sector/parity identity of the raw row table.

Definition 16.96 (Carrier-graded row module through order nine). Let $\mathcal{M}_{\text{car}}$ be the pre-collapse carrier row module through order nine, freely generated over the local coefficient ring by

$$\mathbf{C}_7, \quad \mathbf{C}_{Y12}, \quad \mathbf{C}_{N7,I}, \quad \mathbf{C}_{N7,D}, \quad \mathbf{C}_{N5,I}, \quad \mathbf{C}_{N5,D},$$

together with line, background/core, and normalization/subtraction carriers

$$\mathbf{C}_{\text{line}}, \quad \mathbf{C}_{\text{bg/core}}, \quad \mathbf{C}_{\text{norm/sub}}.$$

Here $\mathbf{C}_7$ denotes the order-seven live carrier $T_7^{(1)}$, and $\mathbf{C}_{Y12}$ denotes the clean centered order-nine carrier

$$T_9^{(1,2)} = X_{-,1} Y X_{+,2} + X_{-,2} Y X_{+,1}.$$

This is the same carrier denoted $T_9^{(1,2)}$ in Proposition 16.90 and Theorem 16.93.

Define the nuisance submodule

$$\mathcal{C}_{\text{car}} := \text{span}\{\mathbf{C}_{N7,I}, \mathbf{C}_{N7,D}, \mathbf{C}_{N5,I}, \mathbf{C}_{N5,D}, \mathbf{C}_{\text{line}}, \mathbf{C}_{\text{bg/core}}, \mathbf{C}_{\text{norm/sub}}\}.$$

The carrier quotient is

$$\mathcal{Q}_{\text{car}} := \mathcal{M}_{\text{car}} / \mathcal{C}_{\text{car}}.$$

Lemma 16.97 (Survival of the order-nine clean carrier in the carrier quotient). *In $\mathcal{Q}_{\text{car}}$,*

$$[\mathbf{C}_{Y12}] \neq 0,$$

and

$$[\mathbf{C}_7] \wedge [\mathbf{C}_{Y12}] \neq 0 \quad \text{in } \wedge^2 \mathcal{Q}_{\text{car}}.$$

Proof. By definition, $\mathcal{M}_{\text{car}}$ is free on the displayed carrier basis. The nuisance submodule $\mathcal{C}_{\text{car}}$ is generated by the N_7 , N_5 , line, background/core, and normalization/subtraction carriers. It does not contain $\mathbf{C}_{Y12}$ or $\mathbf{C}_7$. Therefore their quotient classes are nonzero and linearly independent, unless an additional row relation is imposed identifying $\mathbf{C}_{Y12}$ or $\mathbf{C}_7$ with the nuisance carriers. No such relation is part of the carrier-graded associated row module. $\square$

Remark 16.98 (Carrier freeness is the row-output choice). Lemma 16.97 is a statement in the pre-collapse carrier module. It uses the row-output choice that carrier labels are retained until after quotienting nuisance directions. If one instead defines the row output only after applying the scalar collapse map

$$S_I : \mathcal{M}_{\text{car}} \rightarrow \mathcal{O} \cdot I,$$

then the freeness assertion is no longer available and the scalar cross-channel incidence problem for F_9^I returns. Thus carrier freeness is not a hidden scalar non-cancellation theorem; it is the explicit carrier-quotient row-energy architecture.

Lemma 16.99 (Leading source factor). *In the centered reduced-patch normalization,*

$$H := yq'',$$

and the leading source term satisfies

$$H_{\text{lead}} = -\frac{q_0(q'')^3}{288}.$$

Equivalently, with $u = q''_{\text{can}} = q''$,

$$H_{\text{lead}} = -\frac{q_0 u^3}{288}.$$

Proof. The centered source calculation gives

$$y_{\text{lead}} = -\frac{q_0 A_2^2}{72}, \quad q'' = 2A_2.$$

Hence

$$H_{\text{lead}} = y_{\text{lead}} q'' = -\frac{q_0 A_2^2}{72} \cdot 2A_2 = -\frac{q_0 A_2^3}{36}.$$

Since $A_2 = q''/2$,

$$H_{\text{lead}} = -\frac{q_0 (q'')^3}{288}.$$

□

Theorem 16.100 (D4 carrier-quotient row conversion through order nine). *This theorem is a quotient row-energy theorem. It does not assert coercivity of the collapsed scalar I -coordinate. If the final downstream norm is forced to be the scalar I -coordinate norm, then one must instead control the scalar incidence wall in Definition 16.92, or pass to a higher or ambient row architecture.*

Assume the carrier-quotient row-energy formulation is admissible downstream. Let

$$W_{\text{car}} := \text{span}\{[C_7], [C_{Y12}]\} \subset Q_{\text{car}}.$$

Assume the quotient Gram matrix of the ordered pair

$$[C_7], \quad [C_{Y12}]$$

has determinant bounded below by Q^{-C} .

Assume that, in Q_{car} , the order-seven and clean order-nine live carrier classes have the leading-symbol form, modulo nuisance directions and $D4$ residual terms,

$$[A_7] = H\Gamma_G[C_7] + \mathcal{E}_7,$$

$$[A_9] = H\Gamma_{G,2}[C_{Y12}] + \mathcal{E}_9,$$

and that the quotient projection of the residual into W_{car} is half-leading:

$$|P_{W_{\text{car}}}(\mathcal{E}_7 + \mathcal{E}_9)|_{Q_{\text{car}}} \leq \frac{1}{2}|H|(\Gamma_G^2 + \Gamma_{G,2}^2)^{1/2}.$$

Then

$$|[A_7] + [A_9]|_{Q_{\text{car}}} \geq Q^{-C}|H|(\Gamma_G^2 + \Gamma_{G,2}^2)^{1/2}.$$

Consequently, combined with Gate 2, gauge-locked wall escape, and Lemma 15.33, one obtains carrier-quotient row energy

$$\int_I |[A_N^{\text{car}}(m)]|_{Q_{\text{car}}}^2 dm \geq Q^{-C}|I|.$$

Proof. The Gram determinant lower bound gives a polynomial lower frame bound for the two quotient carrier directions. Therefore

$$|H\Gamma_G[C_7] + H\Gamma_{G,2}[C_{Y12}]|_{Q_{\text{car}}} \geq Q^{-C}|H|(\Gamma_G^2 + \Gamma_{G,2}^2)^{1/2}.$$

The residual bound is half-leading in the only quotient directions that can cancel the leading response. The triangle inequality gives the pointwise row lower bound. Row mobility upgrades the pointwise lower bound to row energy.

This proves carrier-quotient row conversion only. It does not assert coercivity of any collapsed scalar coordinate such as the I -coordinate order-nine readout. □

Remark 16.101 (Finite D4 audit ledger). The carrier-quotient theorem separates the structural rank statement from the finite row-algebra audit. The finite D4 audit consists of the following checks.

(i) Verify the principal source factor

$$\text{gr}(H) = -\frac{1}{288}q_0 u^3.$$

(ii) Verify the order-seven carrier detecting Γ_G :

$$[\mathcal{A}_7] \sim H\Gamma_G[\mathcal{C}_7].$$

(iii) Verify the clean order-nine carrier detecting $\Gamma_{G,2}$:

$$[\mathcal{A}_9] \sim H\Gamma_{G,2}[\mathcal{C}_{Y12}].$$

(iv) Identify the nuisance carrier subspace being quotiented.

(v) Prove quotient coercivity:

$$[\mathcal{C}_7] \wedge [\mathcal{C}_{Y12}] \neq 0$$

with polynomial quantitative lower bound in the chosen quotient norm.

(vi) Bound only the residual components that can cancel the leading response:

$$|P_{W_{\text{car}}}(\mathcal{E}_7 + \mathcal{E}_9)|_{\mathcal{Q}_{\text{car}}} \leq \frac{1}{2}|H|(\Gamma_G^2 + \Gamma_{G,2}^2)^{1/2}.$$

In scalar I -coordinate output, additional cross-channel incidence walls may appear after independent carrier directions are collapsed into the same scalar coordinate. These are scalar-output artifacts unless the downstream theorem explicitly requires scalar I -coordinate energy.

16.20. Forbidden singular-discard D4 candidate. The best current finite-D4 forbidden operator is the local singular-discard operator, not the full three-chart atlas loop.

Definition 16.102 (Certified singular retained convention and discard operator). Let $H_{\text{sing},p}$ be the singular Hilbert/module space in which

$$T_{\text{tr} \rightarrow \text{sing},p} J_{\text{tr},p}(q''_{\text{can}})$$

is represented. A pre-certification retained space, when it is mentioned for comparison or no-go purposes, is denoted $R_{\text{sing},p}^{\text{raw}} \subset H_{\text{sing},p}$, with orthogonal projection $P_{\text{sing},p}^{\text{raw}}$.

The proof-level singular-discard branch uses the certified retained space $R_{\text{sing},p}^{\text{cert}}$ of Definition 16.126; this space is obtained from the same dead/live ledger as the certified $C_7/Y12$ quotient. Unless a local paragraph explicitly says “raw”, the unsuperscripted notation in the forbidden-channel branch means

$$R_{\text{sing},p} := R_{\text{sing},p}^{\text{cert}}, \quad P_{\text{sing},p} := P_{\text{sing},p}^{\text{cert}}.$$

Thus the active discarded operator is

$$\Lambda_{\text{tr},\text{sing},p} := (I - P_{\text{sing},p}^{\text{cert}}) T_{\text{tr} \rightarrow \text{sing},p} P_{\text{tr},p}.$$

The Taylor-jet-complete or carrier-complete retained space is never an admissible proof-level choice for this branch: it transports the formal $C_7/Y12$ carrier plane functorially and makes the discarded D4 block rank zero. Such spaces remain useful only as no-go diagnostics and must be written with the superscript raw.

Definition 16.103 (Combined nuisance block and projected transition operator). For each packet p , define the combined $Y12$ nuisance block

$$U_p := \text{span}(T_{\text{par}}^{(\alpha)}, T_7, T_{<7}, II_{\text{arith},9}, T_{E_2}),$$

with the understanding that T_{E_2} is included only when E_2 is admitted in the current quotient layer.

Define the projected transition operator

$$\mathfrak{T}_p := (I - P_{\text{sing},p}) T_{\text{tr} \rightarrow \text{sing},p} J_{\text{tr},p}.$$

Lemma 16.104 (Finite-codimension $Y12$ graph/coercivity lemma). *Assume that the $Y12$ quotient annihilates the combined nuisance block,*

$$U_p \subseteq \ker \Lambda_{Y,p},$$

and that the admissible residual class $\mathcal{R}_X$ splits as a finite-codimension graph space

$$\mathcal{R}_X = \mathcal{R}_{X,\text{low},p} \oplus \mathcal{R}_{X,\text{tail},p}, \quad \dim \mathcal{R}_{X,\text{low},p} < \infty.$$

Let $\{e_{1,p}, \dots, e_{r,p}\}$ be a basis of $\mathcal{R}_{X,\text{low},p}$, and define the defect matrix

$$D_p := [\mathfrak{T}_p e_{1,p} \quad \cdots \quad \mathfrak{T}_p e_{r,p}].$$

If

$$\sigma_{\min}(D_p) \geq Q^{-C} \quad \text{and} \quad \sum_p \|\mathfrak{T}_p(R_X)\|^2 \leq Q^{-B} \mu(\mathcal{C}),$$

then $\mathfrak{T}_p$ is coercive on $\mathcal{R}_X$ up to the tail norm, and the Y12 discarded observable survives on a polynomial-weight good set.

Definition 16.105 (Transition–singular low-block determinant). Let

$$V_p := \mathcal{R}_{X,\text{low},p}, \quad A_p := T_{\text{tr} \rightarrow \text{sing},p} J_{\text{tr},p}|_{V_p}, \quad \Pi_p^\perp := I - P_{\text{sing},p}.$$

Let $\{e_{1,p}, \dots, e_{r,p}\}$ be a basis of V_p , and set

$$C_p := [A_p e_{1,p} \quad \cdots \quad A_p e_{r,p}] \subset H_{\text{sing},p}, \quad D_p := \Pi_p^\perp C_p.$$

Let $B_p = (\langle e_{i,p}, e_{j,p} \rangle_{V_p})_{i,j}$ be the low-block basis Gram matrix. The coordinate-free coercivity constant is

$$\delta_{\text{coerc},p} := \inf_{0 \neq v \in V_p} \frac{\|\Pi_p^\perp A_p v\|_{H_{\text{sing},p}}}{\|v\|_{V_p}}.$$

Equivalently,

$$\delta_{\text{coerc},p}^2 = \lambda_{\min}(B_p^{-1/2} D_p^* D_p B_p^{-1/2}).$$

In particular, if the low basis is retained-metric orthonormal, then $\delta_{\text{coerc},p}^2 = \lambda_{\min}(D_p^* D_p)$.

Lemma 16.106 (Schur formula for the discarded Gram matrix). Let $R_p = [r_{1,p} \quad \cdots \quad r_{s,p}]$ be any full-rank basis matrix for the retained singular space $R_{\text{sing},p} \subset H_{\text{sing},p}$. Let M_p be the Gram matrix of the ambient singular inner product in the chosen finite coordinates. Then

$$D_p^* D_p = C_p^* M_p C_p - C_p^* M_p R_p (R_p^* M_p R_p)^{-1} R_p^* M_p C_p.$$

Equivalently, $D_p^* D_p$ is the Schur complement of $R_p^* M_p R_p$ in the bordered Gram matrix

$$\mathcal{G}_p := \begin{pmatrix} R_p^* M_p R_p & R_p^* M_p C_p \\ C_p^* M_p R_p & C_p^* M_p C_p \end{pmatrix}.$$

Consequently, when the low block has dimension r ,

$$\det(D_p^* D_p) = \frac{\det \mathcal{G}_p}{\det(R_p^* M_p R_p)}.$$

Thus $\sigma_{\min}(D_p) \geq Q^{-C}$ is equivalent, up to the already required polynomial upper conditioning of B_p , M_p , C_p , and R_p , to the finite Schur determinant lower bound $\det \mathcal{G}_p \geq Q^{-C} \det(R_p^* M_p R_p)$ together with the polynomial upper bound for $D_p^* D_p$.

Proof. The orthogonal projection onto $R_{\text{sing},p}$ is $P_{\text{sing},p} = R_p (R_p^* M_p R_p)^{-1} R_p^* M_p$. Therefore $D_p^* D_p = C_p^* M_p (I - P_{\text{sing},p}) C_p$, and the displayed formula follows. The bordered determinant identity is the standard Schur-complement formula. In fixed finite dimension and under polynomial upper conditioning, a polynomial determinant lower bound is equivalent to a polynomial lower bound for the smallest eigenvalue. $\square$

Definition 16.107 (Two-carrier transition–singular determinant). On the live carrier plane set

$$c_{7,p} := T_{\text{tr} \rightarrow \text{sing},p} [\mathcal{C}_7]^{\text{tr}}, \quad c_{Y,p} := T_{\text{tr} \rightarrow \text{sing},p} [\mathcal{C}_{Y12}]^{\text{tr}},$$

and

$$c_{7,p}^\perp := \Pi_p^\perp c_{7,p}, \quad c_{Y,p}^\perp := \Pi_p^\perp c_{Y,p}.$$

Define

$$W_p := [c_{7,p}^\perp \quad c_{Y,p}^\perp], \quad S_p := W_p^* W_p.$$

The two-carrier transition–singular determinant is

$$\Delta_{\text{tr},\text{sing},p} := \det S_p = \|c_{7,p}^\perp\|^2 \|c_{Y,p}^\perp\|^2 - |\langle c_{7,p}^\perp, c_{Y,p}^\perp \rangle|^2.$$

Equivalently, $\Delta_{\text{tr},\text{sing},p} = \|c_{7,p}^\perp\|^2 \|c_{Y,p}^\perp\|^2 \sin^2 \angle(c_{7,p}^\perp, c_{Y,p}^\perp)$.

Lemma 16.108 (Two-carrier determinant criterion). *Assume $\|c_{7,p}^\perp\| + \|c_{Y,p}^\perp\| \leq Q^C$. Then $\sigma_{\min}(W_p) \geq Q^{-C'}$ for some polynomial exponent C' if and only if $\Delta_{\text{tr,sing},p} \geq Q^{-C''}$ for some polynomial exponent C'' . In particular, a rank-one Y12-signature test gives only $\text{rank } W_p \geq 1$; it does not imply $\sigma_{\min}(W_p) > 0$ on the two-dimensional carrier plane.*

Proof. The eigenvalues of S_p are nonnegative and satisfy $\lambda_{\min}(S_p) = \det S_p / \lambda_{\max}(S_p)$. The polynomial upper bound gives $\lambda_{\max}(S_p) \leq Q^C$, hence $\det S_p \geq Q^{-C''}$ implies $\lambda_{\min}(S_p) \geq Q^{-C'}$. The converse is immediate from $\det S_p = \lambda_{\min}(S_p) \lambda_{\max}(S_p)$ and polynomial upper conditioning. If only one normal functional detects the carrier plane, its kernel is a nontrivial line in the two-dimensional carrier plane, so the smallest singular value on that plane is zero. $\square$

Lemma 16.109 (Normal-row certificate for transition-singular coercivity). *Suppose there exists a finite normal row map $L_p : H_{\text{sing},p} \rightarrow \mathbb{C}^r$ such that*

$$L_p R_{\text{sing},p} = 0, \quad \|L_p\|_{\text{op}} \leq Q^C, \quad \sigma_{\min}(L_p A_p [e_{1,p} \cdots e_{r,p}] B_p^{-1/2}) \geq Q^{-C}.$$

Then $\delta_{\text{coerc},p} \geq Q^{-C'}$. For the two-carrier live plane it is enough to find two polynomially bounded normal rows $\ell_{7,p}^\perp, \ell_{Y,p}^\perp$ annihilating $R_{\text{sing},p}$ and satisfying the triangular lower bound

$$\begin{pmatrix} \ell_{7,p}^\perp c_{7,p} & \ell_{7,p}^\perp c_{Y,p} \\ \ell_{Y,p}^\perp c_{7,p} & \ell_{Y,p}^\perp c_{Y,p} \end{pmatrix} = \begin{pmatrix} \eta_{7,p} & * \\ 0 & \eta_{Y,p} \end{pmatrix}, \quad |\eta_{7,p} \eta_{Y,p}| \geq Q^{-C}.$$

Proof. Since $L_p R_{\text{sing},p} = 0$, one has $L_p = L_p \Pi_p^\perp$. Hence, for every coefficient vector x , $\|L_p A_p [e_{1,p} \cdots e_{r,p}] x\| = \|L_p \Pi_p^\perp A_p [e_{1,p} \cdots e_{r,p}] x\| \leq \|L_p\|_{\text{op}} \|D_p x\|$. After normalizing the domain by $B_p^{-1/2}$, the assumed lower bound for the left-hand side gives the claimed lower bound for D_p . In the two-carrier case, the displayed triangular matrix has polynomially nonzero determinant and polynomially bounded entries, hence has polynomial smallest singular value. $\square$

16.21. Singular row lift for the low-block coercivity gap. The boundary certified quotient and the singular-discard projection are not the same operation. The boundary quotient proves that the live $C_7/Y12$ pair survives after quotienting dead boundary columns. The low-block coercivity gap requires a stronger statement: the same live pair must survive after orthogonal projection away from the retained singular coherence space. This is a finite row-lift problem.

Definition 16.110 (Singular two-row lift certificate). Fix a good packet p . Let

$$E_{\text{sing},p}^{2Y} := [c_{7,p} \ c_{Y,p}] := T_{\text{tr} \rightarrow \text{sing},p} [[C_7]^{\text{tr}} \ [C_{Y12}]^{\text{tr}}] : \mathbb{C}^2 \rightarrow H_{\text{sing},p}.$$

A singular two-row lift certificate consists of a row map

$$L_{\text{sing},p} = \begin{pmatrix} \ell_{7,p}^{\text{sing}} \\ \ell_{Y,p}^{\text{sing}} \end{pmatrix} : H_{\text{sing},p} \rightarrow \mathbb{C}^2$$

with polynomial norm and polynomially conditioned carrier matrix:

$$\|L_{\text{sing},p}\|_{\text{op}} \leq Q^A, \quad L_{\text{sing},p} R_{\text{sing},p} = 0, \quad \sigma_{\min}(L_{\text{sing},p} E_{\text{sing},p}^{2Y}) \geq Q^{-A}.$$

In row-normal coordinates this may be verified in triangular form:

$$L_{\text{sing},p} E_{\text{sing},p}^{2Y} = \begin{pmatrix} \eta_{7,p} & * \\ 0 & \eta_{Y,p} \end{pmatrix}, \quad |\eta_{7,p} \eta_{Y,p}| \geq Q^{-A},$$

with polynomial upper bounds for the remaining entries.

Lemma 16.111 (Singular row lift implies projected carrier coercivity). *Assume the singular two-row lift certificate of Definition 16.110. Let $P_{\text{sing},p}$ be the orthogonal projection onto $R_{\text{sing},p}$, and set $W_p := (I - P_{\text{sing},p}) E_{\text{sing},p}^{2Y}$. Then $\sigma_{\min}(W_p) \geq Q^{-A'}$. Equivalently,*

$$\det(W_p^* W_p) = \|c_{7,p}^\perp\|^2 \|c_{Y,p}^\perp\|^2 - |\langle c_{7,p}^\perp, c_{Y,p}^\perp \rangle|^2 \geq Q^{-A'}$$

under the standing polynomial upper bounds.

Proof. Since $L_{\text{sing},p}$ annihilates $R_{\text{sing},p}$, it annihilates the orthogonal projection onto that space, hence $L_{\text{sing},p} = L_{\text{sing},p}(I - P_{\text{sing},p})$. For every $a \in \mathbb{C}^2$, $\|L_{\text{sing},p}E_{\text{sing},p}^{2Y}a\| = \|L_{\text{sing},p}(I - P_{\text{sing},p})E_{\text{sing},p}^{2Y}a\| \leq \|L_{\text{sing},p}\|_{\text{op}} \|(I - P_{\text{sing},p})E_{\text{sing},p}^{2Y}a\|$. Taking the infimum over $\|a\| = 1$ gives $\sigma_{\min}(W_p) \geq \sigma_{\min}(L_{\text{sing},p}E_{\text{sing},p}^{2Y})/\|L_{\text{sing},p}\|_{\text{op}} \geq Q^{-A'}$. The determinant form follows from the two-dimensional singular-value identity $\det(W_p^*W_p) = \sigma_1(W_p)^2\sigma_2(W_p)^2$ and the standing polynomial upper bound for W_p . $\square$

Definition 16.112 (Signature extractor for the singular finite jet). Let $\Theta_{2Y,p} : H_{\text{sing},p} \rightarrow \mathcal{H}_p^Y$ be the finite signature extractor which records the covariant septic row and the primitive fixed-whitened $Y12$ row in the singular chart. It is admissible if

$$\|\Theta_{2Y,p}\|_{\text{op}} \leq Q^A,$$

$$\Theta_{2Y,p}T_{\text{tr} \rightarrow \text{sing},p} \begin{bmatrix} [C_7]^{\text{tr}} & [C_{Y12}]^{\text{tr}} \end{bmatrix} = E_p + O(Q^{-B}),$$

and $\Theta_{2Y,p}R_{\text{sing},p} \subseteq \ker L_p^{\text{row}} + O(Q^{-B})$, where $E_p = [C_{7,p}^{\text{tar}} \ C_{Y,p}^{\text{tar}}]$ and $L_p^{\text{row}} = \begin{pmatrix} \ell_{7,p}^Y \\ \ell_{9,p}^Y \end{pmatrix}$ is the concrete boundary row certificate.

16.22. Retained-generator verification for the singular row lift. The singular row-lift condition can be verified without expanding the full Schur determinant, provided the retained singular generators are row-dead for exactly the same two rows that detect the live $C_7/Y12$ carrier pair. The following formulation is finite-dimensional and is invariant under a change of basis in the retained singular space.

Definition 16.113 (Retained-generator row-deadness certificate). Let

$$E_{\text{sing},p}^{2Y} := T_{\text{tr} \rightarrow \text{sing},p} \begin{bmatrix} [C_7]^{\text{tr}} & [C_{Y12}]^{\text{tr}} \end{bmatrix} : \mathbb{C}^2 \rightarrow H_{\text{sing},p}.$$

Let $L_p^{\text{row}} = \begin{pmatrix} \ell_{7,p}^Y \\ \ell_{9,p}^Y \end{pmatrix} : \mathcal{H}_p^Y \rightarrow \mathbb{C}^2$ be the concrete live row map of the $C_7/Y12$ certificate, and let $\Theta_{2Y,p} : H_{\text{sing},p} \rightarrow \mathcal{H}_p^Y$ be the singular finite-jet signature extractor. Put $\widehat{L}_{\text{sing},p} := L_p^{\text{row}}\Theta_{2Y,p}$. Let $R_p = [r_{1,p} \ \cdots \ r_{s,p}]$ be a full-rank generator matrix for $R_{\text{sing},p}$, and let M_p be the singular ambient metric. The retained singular space satisfies the *retained-generator row-deadness certificate* if, for some fixed polynomial exponent A and arbitrarily large residual exponent B , the following finite estimates hold:

$$\|L_p^{\text{row}}\|_{\text{op}} + \|\Theta_{2Y,p}\|_{\text{op}} + \|E_{\text{sing},p}^{2Y}\|_{\text{op}} \leq Q^A, \quad \kappa(R_p^*M_pR_p) \leq Q^A,$$

$$(RD) \quad \|\widehat{L}_{\text{sing},p}R_p\|_{\text{op}} \leq Q^{-B},$$

while on the live carrier pair

$$(LC) \quad \widehat{L}_{\text{sing},p}E_{\text{sing},p}^{2Y} = I_2 + \mathcal{E}_{\text{car},p}, \quad \|\mathcal{E}_{\text{car},p}\|_{\text{op}} \leq Q^{-B}.$$

The exact version has $\widehat{L}_{\text{sing},p}R_p = 0$ and $\mathcal{E}_{\text{car},p} = 0$.

Lemma 16.114 (Finite correction to exact singular-normal rows). *Assume the retained-generator row-deadness certificate. Let $P_{\text{sing},p}$ be the M_p -orthogonal projection onto $R_{\text{sing},p}$, and define $L_{\text{sing},p} := \widehat{L}_{\text{sing},p}(I - P_{\text{sing},p})$. If B is larger than the polynomial losses in the conditioning of $R_p, M_p, \Theta_{2Y,p}$, and $E_{\text{sing},p}^{2Y}$, then $L_{\text{sing},p}R_{\text{sing},p} = 0$, $\|L_{\text{sing},p}\|_{\text{op}} \leq Q^{A'}$, and $L_{\text{sing},p}E_{\text{sing},p}^{2Y} = I_2 + O(Q^{-B+A'})$. In particular, $\sigma_{\min}(L_{\text{sing},p}E_{\text{sing},p}^{2Y}) \geq Q^{-A'}$.*

Proof. The first identity is immediate from the definition of $L_{\text{sing},p}$. The norm bound follows from $\|I - P_{\text{sing},p}\| \leq 1$ in the singular metric and the polynomial bounds on L_p^{row} and $\Theta_{2Y,p}$. Since $P_{\text{sing},p} = R_p(R_p^*M_pR_p)^{-1}R_p^*M_p$, the polynomial conditioning of $R_p^*M_pR_p$ and the row-deadness estimate (RD) give $\|\widehat{L}_{\text{sing},p}P_{\text{sing},p}\|_{\text{op}} \leq Q^{-B+A_1}$. Therefore

$$L_{\text{sing},p}E_{\text{sing},p}^{2Y} = \widehat{L}_{\text{sing},p}E_{\text{sing},p}^{2Y} - \widehat{L}_{\text{sing},p}P_{\text{sing},p}E_{\text{sing},p}^{2Y} = I_2 + \mathcal{E}_{\text{car},p} + O(Q^{-B+A_1}).$$

Taking B above the accumulated polynomial losses gives a perturbation of I_2 with operator norm at most $1/2$, hence a polynomial lower bound for the smallest singular value. $\square$

Theorem 16.115 (Singular finite-jet row lift from retained-generator exactness). *Assume the concrete $C_7/Y12$ row certificate $L_p^{\text{row}}[C_{7,p}^{\text{tar}} C_{Y,p}^{\text{tar}}] = I_2$, $\|L_p^{\text{row}}\|_{\text{op}} \leq Q^A$, and assume the retained-generator row-deadness certificate of Definition 16.113. Then the map $L_{\text{sing},p} = L_p^{\text{row}} \Theta_{2Y,p}(I - P_{\text{sing},p})$ is a singular two-row lift certificate: it annihilates $R_{\text{sing},p}$, has polynomial operator norm, and detects the transitioned live carrier pair with polynomial singular-value margin $\sigma_{\min}(L_{\text{sing},p} E_{\text{sing},p}^{2Y}) \geq Q^{-A'}$. Consequently $\sigma_{\min}((I - P_{\text{sing},p}) E_{\text{sing},p}^{2Y}) \geq Q^{-A''}$.*

Proof. The first three assertions are Lemma 16.114. Since $L_{\text{sing},p} = L_{\text{sing},p}(I - P_{\text{sing},p})$, for every $a \in \mathbb{C}^2$, $\|L_{\text{sing},p} E_{\text{sing},p}^{2Y} a\| \leq \|L_{\text{sing},p}\|_{\text{op}} \|(I - P_{\text{sing},p}) E_{\text{sing},p}^{2Y} a\|$. Taking the infimum over $\|a\| = 1$ gives the projected-carrier singular-value lower bound. $\square$

Corollary 16.116 (Low-block coercivity from retained-generator exactness). *Let $V_p = \mathcal{R}_{X,\text{low},p}$, $A_p = T_{\text{tr} \rightarrow \text{sing},p} J_{\text{tr},p}|_{V_p}$, and let $E_p^{\text{low}} : \mathbb{C}^r \rightarrow V_p$ be the low-mode basis map with Gram matrix B_p . Assume that the transitioned low block is polynomially identified with the live $C_7/Y12$ carrier plane:*

$$A_p E_p^{\text{low}} = E_{\text{sing},p}^{2Y} M_{\text{low},p} + \mathcal{E}_{\text{low},p}, \quad \sigma_{\min}(M_{\text{low},p} B_p^{-1/2}) \geq Q^{-A}, \quad \|\mathcal{E}_{\text{low},p} B_p^{-1/2}\| \leq Q^{-B}.$$

Under the hypotheses of Theorem 16.115, $\lambda_{\min}(B_p^{-1/2} D_p^ D_p B_p^{-1/2}) \geq Q^{-A'}$ with $D_p = (I - P_{\text{sing},p}) A_p E_p^{\text{low}}$, hence $\delta_{\text{coerc},p} \geq Q^{-A'}$.*

Proof. By Theorem 16.115, $\sigma_{\min}((I - P_{\text{sing},p}) E_{\text{sing},p}^{2Y}) \geq Q^{-A_1}$. Multiplying by the conditioned matrix $M_{\text{low},p} B_p^{-1/2}$ gives a polynomial lower bound for the main term in $D_p B_p^{-1/2}$. The perturbation $(I - P_{\text{sing},p}) \mathcal{E}_{\text{low},p} B_p^{-1/2}$ has norm $O(Q^{-B})$; the standard singular-value perturbation inequality gives the claim after increasing B . $\square$

Lemma 16.117 (Retained generators are row-dead under the existing row-exactness tests). *Assume the row-exactness certificate $\mathcal{E}_{2Y}(p)$, the no-atom-fitting guardrails for E_2/R_{lab} , and the admissible exact-nuisance split. Assume further that every generator of $R_{\text{sing},p}$ is one of the following classes after applying the singular finite-jet extractor $\Theta_{2Y,p}$:*

$$\mathcal{K}_{p,0}^{Y,\text{dead}}, \quad \mathcal{D}_{7,\text{chart}}, \quad \mathcal{D}_{7,\text{norm}}, \quad \mathcal{D}_{7,\gamma/\text{bg}}, \quad \mathcal{D}_{7,<7}, \quad W_{\text{dead}}^{\text{fixed}}, \quad \mathcal{N}_{\text{exact}},$$

or a fixed/coherent generator whose live row signature is routed into the effective carrier matrix and is not included in the retained singular space. Then $L_p^{\text{row}} \Theta_{2Y,p} R_{\text{sing},p} = 0$ up to the residual ledger. Hence the row-deadness estimate (RD) holds.

Proof. The septic row kills the typed septic denominator and the exact nuisance space by septic exactness. The primitive Y12 row kills the fixed-whitening dead space and the transported wrong-origin data by the whitening exact sequence. The no-atom-fitting guardrails exclude local live contributions from completion and labelled-transition terms. Therefore every listed exact generator has zero row signature under $L_p^{\text{row}} \Theta_{2Y,p}$. Residual generators are not retained exactly; they enter the residual ledger and contribute only the allowed $O(Q^{-B})$ error. Fixed/coherent generators with nonzero live signature are not row-dead and therefore cannot be included in $R_{\text{sing},p}$; by hypothesis they are routed into the effective carrier matrix instead. $\square$

Proposition 16.118 (Live retained generators obstruct the singular row lift). *If, for some nonzero $a \in \mathbb{C}^2$, $E_{\text{sing},p}^{2Y} a \in R_{\text{sing},p}$, then $(I - P_{\text{sing},p}) E_{\text{sing},p}^{2Y} a = 0$, and the rank-two projected carrier estimate fails. In particular, no pair of normal rows annihilating $R_{\text{sing},p}$ can have polynomially invertible restriction to $E_{\text{sing},p}^{2Y} \mathbb{C}^2$.*

Proof. The first claim follows from the definition of the orthogonal projection onto $R_{\text{sing},p}$. If normal rows annihilate $R_{\text{sing},p}$, they annihilate $E_{\text{sing},p}^{2Y} a$, so their restriction to the live carrier plane has a nontrivial kernel. $\square$

Remark 16.119 (Retained-generator row-deadness checklist). This checklist is retained for auditability but is no longer an open item in the certified branch. The row-lift theorem reduces the determinant computation to showing that retained singular generators are row-dead for

$L_p^{\text{row}} \Theta_{2Y,p}$ and that the transitioned $C_7/Y12$ carriers have identity row signature. The certified retained space of Definition 16.126 enforces this by construction: every retained generator is replaced by its row-dead part, while live coherent components are routed into the effective carrier and atom-local live components are excluded by the no-atom-fitting guardrails. The same caveat remains as a no-go diagnostic: a raw retained generator with a nonzero $C_7/Y12$ live row signature cannot be retained in the proof-level R_{sing} .

Proposition 16.120 (Boundary row certificate lifts to singular normal rows). *Assume the concrete boundary row certificate $L_p^{\text{row}} E_p = I_2$, $\|L_p^{\text{row}}\|_{\text{op}} \leq Q^A$, and assume an admissible singular finite-jet signature extractor in the sense of Definition 16.112. Define $L_{\text{sing},p} := L_p^{\text{row}} \Theta_{2Y,p}$. Then, after increasing B relative to the polynomial losses, $L_{\text{sing},p}$ is a singular two-row lift certificate. In particular, $L_{\text{sing},p} R_{\text{sing},p} = O(Q^{-B})$ and $\sigma_{\min}(L_{\text{sing},p} E_{\text{sing},p}^{2Y}) \geq Q^{-A}$. If the extractor identities are exact, the annihilation is exact.*

Proof. The norm bound follows from $\|L_{\text{sing},p}\|_{\text{op}} \leq \|L_p^{\text{row}}\|_{\text{op}} \|\Theta_{2Y,p}\|_{\text{op}} \leq Q^A$. The retained-space condition gives $L_{\text{sing},p} R_{\text{sing},p} = L_p^{\text{row}} \Theta_{2Y,p} R_{\text{sing},p} \subseteq L_p^{\text{row}} (\ker L_p^{\text{row}}) + O(Q^{-B}) = O(Q^{-B})$. On the live carrier pair, $L_{\text{sing},p} E_{\text{sing},p}^{2Y} = L_p^{\text{row}} \Theta_{2Y,p} E_{\text{sing},p}^{2Y} = L_p^{\text{row}} (E_p + O(Q^{-B})) = I_2 + O(Q^{A-B})$. For B larger than the polynomial losses, this matrix has smallest singular value at least Q^{-A} ; the exact case is immediate. $\square$

Theorem 16.121 (Low-block coercivity from the singular row lift). *Let $V_p = \mathcal{R}_{X,\text{low},p}$, $A_p = T_{\text{tr} \rightarrow \text{sing},p} J_{\text{tr},p}|_{V_p}$, $\Pi_p^\perp = I - P_{\text{sing},p}$, and let $E_p^{\text{low}} : \mathbb{C}^r \rightarrow V_p$ be the chosen low-mode basis map and $D_p := \Pi_p^\perp A_p E_p^{\text{low}}$. Assume that the low block is identified with the live transition carrier plane with polynomial conditioning: there is a matrix $M_{\text{low},p}$ such that*

$$A_p E_p^{\text{low}} = E_{\text{sing},p}^{2Y} M_{\text{low},p} + \mathcal{E}_{\text{low},p},$$

$$\sigma_{\min}(M_{\text{low},p} B_p^{-1/2}) \geq Q^{-A}, \quad \|M_{\text{low},p}\|_{\text{op}} \leq Q^A, \quad \|\Pi_p^\perp \mathcal{E}_{\text{low},p} B_p^{-1/2}\|_{\text{op}} \leq Q^{-B}.$$

If the singular two-row lift certificate holds, then $\lambda_{\min}(B_p^{-1/2} D_p^ D_p B_p^{-1/2}) \geq Q^{-A'}$, hence $\delta_{\text{coerc},p} = \sigma_{\min}(D_p B_p^{-1/2}) \geq Q^{-A'}$.*

Proof. By Lemma 16.111, $\sigma_{\min}(\Pi_p^\perp E_{\text{sing},p}^{2Y}) \geq Q^{-A_1}$. Hence $\sigma_{\min}(\Pi_p^\perp E_{\text{sing},p}^{2Y} M_{\text{low},p} B_p^{-1/2}) \geq Q^{-A_1} \sigma_{\min}(M_{\text{low},p} B_p^{-1/2}) \geq Q^{-A_2}$. The perturbation term $\Pi_p^\perp \mathcal{E}_{\text{low},p} B_p^{-1/2}$ has operator norm at most Q^{-B} ; taking $B > A_2 + 10$ and applying the elementary singular-value perturbation inequality gives $\sigma_{\min}(D_p B_p^{-1/2}) \geq Q^{-A'}$. Squaring gives the displayed Gram lower bound. $\square$

Proposition 16.122 (Retained live directions are the obstruction). *Suppose the retained singular space contains a nonzero live carrier combination: $E_{\text{sing},p}^{2Y} a \in R_{\text{sing},p}$ for some $0 \neq a \in \mathbb{C}^2$. Then $(I - P_{\text{sing},p}) E_{\text{sing},p}^{2Y} a = 0$, so the two-carrier projected matrix has a kernel. In particular, full rank-two low-block coercivity fails on the live carrier plane. Thus any coherent live column which is retained inside $R_{\text{sing},p}$, rather than subtracted or routed out before forming the singular retained space, destroys the rank-two singular-discard proof.*

Proof. If $E_{\text{sing},p}^{2Y} a \in R_{\text{sing},p}$, then its orthogonal-complement projection is zero, so the projected carrier matrix has a nontrivial kernel. The final assertion is this statement applied to a retained coherent live column in the $C_7/Y12$ plane. $\square$

Remark 16.123 (Certified discharge of the finite row-lift item). The pre-certification finite item was the packetwise test

$$\Theta_{2Y,p} R_{\text{sing},p} \subseteq \ker L_p^{\text{row}} + O(Q^{-B}), \quad \Theta_{2Y,p} E_{\text{sing},p}^{2Y} = E_p + O(Q^{-B}),$$

together with polynomial conditioning of the low-mode-to-carrier map. In the active branch these tests are discharged by the certified singular retained space and the labelled-transition identities:

$$L_p^{\text{row}} \Theta_{2Y,p} R_{\text{sing},p}^{\text{cert}} = 0, \quad L_p^{\text{row}} \Theta_{2Y,p} E_{\text{sing},p}^{2Y} = I_2 + O(Q^{A-B}).$$

Thus the raw finite item is archived; the proof-level conclusion is Theorem 16.130. If one returns to a raw retained space, the four old failure alternatives remain valid: retained live signature, collapsed transition carrier plane, bad low-mode/carrier identification, or ill-conditioned singular metric/extractor.

16.23. Certified singular retention and low-block coercivity. The transition–singular determinant is fixed once the retained singular space is tied to the same certified dead/live split used in the rank-two $C_7/Y12$ quotient. The retained singular space must contain the actual-source dead and coherent-denominator directions needed for the upper bound, but it must not retain the two live $C_7/Y12$ carrier directions. The following construction performs this split inside the singular finite jet.

Definition 16.124 (Singular row-signature matrix). Fix a good packet p . Let

$$E_{\text{sing},p}^{2Y} := T_{\text{tr} \rightarrow \text{sing},p} \begin{bmatrix} [C_7]^{\text{tr}} & [C_{Y12}]^{\text{tr}} \end{bmatrix} : \mathbb{C}^2 \rightarrow H_{\text{sing},p}.$$

Let $\Theta_{2Y,p} : H_{\text{sing},p} \rightarrow \mathcal{H}_p^Y$ be the singular finite-jet signature extractor and let $L_p^{\text{row}} := \begin{pmatrix} \ell_{7,p}^Y \\ \ell_{9,p}^Y \end{pmatrix} : \mathcal{H}_p^Y \rightarrow \mathbb{C}^2$ be the certified live row pair obtained from the septic and fixed-whitening exact sequences. Define $C_{\text{sing},p}^{2Y} := L_p^{\text{row}} \Theta_{2Y,p} E_{\text{sing},p}^{2Y}$.

Lemma 16.125 (Transition labels reproduce the certified live rows). *Assume the certified $C_7/Y12$ row certificate, the labelled-transition no-atom-fitting guardrails, and fixed-whitening naturality in the transition–singular overlap chart. Then*

$$\Theta_{2Y,p} E_{\text{sing},p}^{2Y} = E_p + \mathcal{R}_{E,p}, \quad E_p := [C_{7,p}^{\text{tar}} \ C_{Y,p}^{\text{tar}}], \quad \|\mathcal{R}_{E,p}\|_{\text{op}} \leq Q^{-B}.$$

Consequently, after taking B larger than the fixed polynomial losses, $C_{\text{sing},p}^{2Y} = I_2 + O(Q^{A-B})$, $\sigma_{\min}(C_{\text{sing},p}^{2Y}) \geq Q^{-A}$, and $\|(C_{\text{sing},p}^{2Y})^{-1}\|_{\text{op}} \leq Q^A$. In the exact labelled finite-jet normalization the error term is zero.

Proof. The extractor $\Theta_{2Y,p}$ records exactly the covariant septic signature and the primitive fixed-whitened $Y12$ signature in the singular chart. The labelled-transition no-atom-fitting guardrails exclude atom-by-atom refitting of these two coordinates, and fixed-whitening naturality identifies the same-point whitening slot in the transition and singular charts. Therefore the transition image of $[C_7]^{\text{tr}}$ has the certified target signature $C_{7,p}^{\text{tar}}$, and the transition image of $[C_{Y12}]^{\text{tr}}$ has the certified target signature $C_{Y,p}^{\text{tar}}$, up to the residual ledger. The certified row certificate gives $L_p^{\text{row}} E_p = I_2$ and $\|L_p^{\text{row}}\|_{\text{op}} \leq Q^A$, so $C_{\text{sing},p}^{2Y} = L_p^{\text{row}} (E_p + \mathcal{R}_{E,p}) = I_2 + O(Q^{A-B})$. For B above the accumulated polynomial losses, this is an invertible perturbation of the identity with polynomially bounded inverse. $\square$

Definition 16.126 (Certified singular retained space). Let $\mathcal{G}_{\text{sing},p}^{\text{act}} \subset H_{\text{sing},p}$ be the finite list of singular generators retained for the EF-coherent upper: the transitioned localized finite canonical model, together with the mandatory chart, normalization, gamma/background, lower-order, exact-nuisance, and coherent generators admitted by the certified routing ledger. For $h \in H_{\text{sing},p}$, define its certified dead part by

$$h^{\text{dead}} := h - E_{\text{sing},p}^{2Y} (C_{\text{sing},p}^{2Y})^{-1} L_p^{\text{row}} \Theta_{2Y,p} h.$$

The certified retained singular space is

$$R_{\text{sing},p}^{\text{cert}} := \text{span}\{h^{\text{dead}} : h \in \mathcal{G}_{\text{sing},p}^{\text{act}}\} + R_{\text{sing},p}^{\text{tail/dead}},$$

where $R_{\text{sing},p}^{\text{tail/dead}}$ denotes the already row-dead tail/completion part of the singular retained model. Let $P_{\text{sing},p}^{\text{cert}}$ be the orthogonal projection onto $R_{\text{sing},p}^{\text{cert}}$ in the singular Hilbert norm.

Lemma 16.127 (Certified retention is row-dead and transverse to the live plane). *Under the hypotheses of Lemma 16.125, $L_p^{\text{row}} \Theta_{2Y,p} R_{\text{sing},p}^{\text{cert}} = 0$. Moreover, $R_{\text{sing},p}^{\text{cert}} \cap E_{\text{sing},p}^{2Y} \mathbb{C}^2 = \{0\}$, and the angle between $R_{\text{sing},p}^{\text{cert}}$ and the live carrier plane is polynomially bounded below: $\text{dist}(E_{\text{sing},p}^{2Y} a, R_{\text{sing},p}^{\text{cert}}) \geq Q^{-A} \|a\|$ for $a \in \mathbb{C}^2$.*

Proof. For every retained generator h ,

$$L_p^{\text{row}} \Theta_{2Y,p} h^{\text{dead}} = L_p^{\text{row}} \Theta_{2Y,p} h - C_{\text{sing},p}^{2Y} (C_{\text{sing},p}^{2Y})^{-1} L_p^{\text{row}} \Theta_{2Y,p} h = 0.$$

The tail/dead retained part is row-dead by the origin, grade, exact-nuisance, and no-atom-fitting row-deadness lemmas. Hence the whole certified retained space is killed by $L_p^{\text{row}} \Theta_{2Y,p}$. If $E_{\text{sing},p}^{2Y} a \in R_{\text{sing},p}^{\text{cert}}$, applying $L_p^{\text{row}} \Theta_{2Y,p}$ gives $C_{\text{sing},p}^{2Y} a = 0$. Since $C_{\text{sing},p}^{2Y}$ is polynomially invertible, $a = 0$. More generally, for any $r \in R_{\text{sing},p}^{\text{cert}}$, $\|L_p^{\text{row}} \Theta_{2Y,p} (E_{\text{sing},p}^{2Y} a - r)\| = \|C_{\text{sing},p}^{2Y} a\| \geq Q^{-A} \|a\|$; using $\|L_p^{\text{row}} \Theta_{2Y,p}\|_{\text{op}} \leq Q^A$ and taking the infimum over r gives the displayed polynomial angle bound. $\square$

Theorem 16.128 (Singular finite-jet row lift). *Assume the certified $C_7/Y12$ row certificate, the certified routing ledger for actual singular generators, the labelled-transition no-atom-fitting guardrails, and fixed-whitening naturality. Let $L_{\text{sing},p} := L_p^{\text{row}} \Theta_{2Y,p}$. Then $L_{\text{sing},p}$ is a singular two-row lift certificate for the certified retained singular space $R_{\text{sing},p}^{\text{cert}}$:*

$$\|L_{\text{sing},p}\|_{\text{op}} \leq Q^A, \quad L_{\text{sing},p} R_{\text{sing},p}^{\text{cert}} = 0, \quad \sigma_{\min}(L_{\text{sing},p} E_{\text{sing},p}^{2Y}) \geq Q^{-A}.$$

Consequently $\sigma_{\min}((I - P_{\text{sing},p}^{\text{cert}}) E_{\text{sing},p}^{2Y}) \geq Q^{-A}$.

Proof. The operator norm bound follows from the polynomial bounds for L_p^{row} and $\Theta_{2Y,p}$. The annihilation statement is Lemma 16.127. Finally, $L_{\text{sing},p} E_{\text{sing},p}^{2Y} = C_{\text{sing},p}^{2Y}$, which has polynomial smallest singular value by Lemma 16.125. Since $L_{\text{sing},p}$ annihilates $R_{\text{sing},p}^{\text{cert}}$, $L_{\text{sing},p} = L_{\text{sing},p} (I - P_{\text{sing},p}^{\text{cert}})$, so for every $a \in \mathbb{C}^2$, $\|C_{\text{sing},p}^{2Y} a\| \leq \|L_{\text{sing},p}\|_{\text{op}} \|(I - P_{\text{sing},p}^{\text{cert}}) E_{\text{sing},p}^{2Y} a\|$. Taking the infimum over $\|a\| = 1$ gives the projected-carrier lower bound. $\square$

Definition 16.129 (Certified two-dimensional low block). For the rank-two $2Y$ branch, define the certified finite low block by $V_{\text{low},p}^{\text{cert}} := \text{span}\{[C_7]^{\text{tr}}, [C_{Y12}]^{\text{tr}}\}$. Let $E_{\text{low},p}^{\text{cert}} : \mathbb{C}^2 \rightarrow V_{\text{low},p}^{\text{cert}}$, $E_{\text{low},p}^{\text{cert}} a = a_7 [C_7]^{\text{tr}} + a_Y [C_{Y12}]^{\text{tr}}$. All other finite actual-source singular directions are treated by the certified routing ledger: row-dead directions are retained in $R_{\text{sing},p}^{\text{cert}}$, coherent live directions are absorbed into the effective carrier coordinates, and residual directions enter the singular-discard tail/EF upper.

Theorem 16.130 (Low-block coercivity in the certified singular quotient). *Assume the hypotheses of Theorem 16.128, the polynomial conditioning of the certified carrier basis $E_{\text{low},p}^{\text{cert}}$, and the singular-discard tail/EF upper for all directions outside $V_{\text{low},p}^{\text{cert}}$. Set*

$$D_p^{\text{cert}} := (I - P_{\text{sing},p}^{\text{cert}}) T_{\text{tr} \rightarrow \text{sing},p} J_{\text{tr},p} E_{\text{low},p}^{\text{cert}}.$$

Then $\sigma_{\min}(D_p^{\text{cert}}) \geq Q^{-A}$. Equivalently, $\lambda_{\min}((B_p^{\text{cert}})^{-1/2} (D_p^{\text{cert}})^* D_p^{\text{cert}} (B_p^{\text{cert}})^{-1/2}) \geq Q^{-A}$, where $B_p^{\text{cert}} = (E_{\text{low},p}^{\text{cert}})^* E_{\text{low},p}^{\text{cert}}$. Hence the $Y12$ graph/coercivity low-block gap $\delta_{\text{coerc},p} \geq Q^{-A}$ is proved for the certified singular-discard quotient.

Proof. By construction, $T_{\text{tr} \rightarrow \text{sing},p} J_{\text{tr},p} E_{\text{low},p}^{\text{cert}} = E_{\text{sing},p}^{2Y} M_{\text{low},p}$, where $M_{\text{low},p}$ is the finite basis-change matrix from the chosen low-block coordinates to the labelled carrier coordinates. The certified basis-conditioning gives $\sigma_{\min}(M_{\text{low},p} (B_p^{\text{cert}})^{-1/2}) \geq Q^{-A}$. Theorem 16.128 gives $\sigma_{\min}((I - P_{\text{sing},p}^{\text{cert}}) E_{\text{sing},p}^{2Y}) \geq Q^{-A}$. Therefore $\sigma_{\min}(D_p^{\text{cert}} (B_p^{\text{cert}})^{-1/2}) \geq Q^{-A}$, and squaring gives the Gram lower bound. Directions outside the certified carrier low block are not part of the finite defect block: row-dead finite directions are retained in $R_{\text{sing},p}^{\text{cert}}$, coherent live directions have been absorbed into the effective carrier coordinates, and residual directions are controlled by the singular-discard tail/EF upper. $\square$

Corollary 16.131 (Transition-singular determinant for the certified quotient). *With $P_{\text{sing},p} = P_{\text{sing},p}^{\text{cert}}$ and $V_p = V_{\text{low},p}^{\text{cert}}$, the Schur determinant in Lemma 16.106 has a polynomial lower bound: $\det \mathcal{G}_p \geq Q^{-A} \det(R_p^* M_p R_p)$, where R_p is any full-rank basis for $R_{\text{sing},p}^{\text{cert}}$. Equivalently,*

$$\det \begin{pmatrix} \|c_{7,p}^\perp\|^2 & \langle c_{7,p}^\perp, c_{Y,p}^\perp \rangle \\ \langle c_{Y,p}^\perp, c_{7,p}^\perp \rangle & \|c_{Y,p}^\perp\|^2 \end{pmatrix} \geq Q^{-A},$$

where $c_{7,p}^\perp, c_{Y,p}^\perp$ are computed using $P_{\text{sing},p}^{\text{cert}}$.

Proof. Theorem 16.130 gives a polynomial lower bound for the smallest singular value of the projected carrier matrix, while the finite atlas gives polynomial upper bounds for its operator norm; hence the two-dimensional Gram determinant is polynomially bounded below. The bordered Schur determinant identity of Lemma 16.106 gives the displayed $\det \mathcal{G}_p$ lower bound. $\square$

Remark 16.132 (Replacement of the raw finite-audit item). The theorem above closes the low-block coercivity item for the certified singular retained space. It does not assert the same conclusion for an older raw retained space that contains a nonzero vector in $T_{\text{tr} \rightarrow \text{sing},p} \text{span}\{[C_7]^{\text{tr}}, [C_{Y12}]^{\text{tr}}\}$. Such a raw space makes the projected carrier matrix singular. Therefore the proof-level formulation must use $R_{\text{sing},p}^{\text{cert}}$ and $P_{\text{sing},p}^{\text{cert}}$ in the definition of the transition–singular forbidden channel. With this certified replacement, the legacy transition–singular low-block determinant item, recorded under label 16.134, is discharged by Corollary 16.131.

Remark 16.133 (Certified low-block coercivity status after certification). The low-block $Y12$ graph/coercivity item is closed for the certified singular retained space by Theorem 16.130 and Corollary 16.131. Older remarks that describe δ_{coerc} as a raw finite audit should be read as warnings about non-certified retained spaces only. In the active proof route all transition–singular low-block objects must use $R_{\text{sing},p}^{\text{cert}}$, $P_{\text{sing},p}^{\text{cert}}$, and D_p^{cert} . With these certified objects, the low-block gap $\delta_{\text{coerc},p} \geq Q^{-A}$ is supplied by the certified singular quotient. It is not an additional open cap-side item.

Corollary 16.134 (Transition–singular low-block determinant is certified). *With the convention $P_{\text{sing},p} = P_{\text{sing},p}^{\text{cert}}$ and $V_p = V_{\text{low},p}^{\text{cert}}$, the transition–singular low-block Schur determinant has a polynomial lower bound. Equivalently,*

$$\lambda_{\min}\left((B_p^{\text{cert}})^{-1/2}(D_p^{\text{cert}})^* D_p^{\text{cert}}(B_p^{\text{cert}})^{-1/2}\right) \geq Q^{-A},$$

where

$$D_p^{\text{cert}} = (I - P_{\text{sing},p}^{\text{cert}})T_{\text{tr} \rightarrow \text{sing},p}J_{\text{tr},p}E_{\text{low},p}^{\text{cert}}.$$

In the two-carrier normalization,

$$\det \begin{pmatrix} \|c_{7,p}^\perp\|^2 & \langle c_{7,p}^\perp, c_{Y,p}^\perp \rangle \\ \langle c_{Y,p}^\perp, c_{7,p}^\perp \rangle & \|c_{Y,p}^\perp\|^2 \end{pmatrix} \geq Q^{-A},$$

with both perpendicular vectors computed using $P_{\text{sing},p}^{\text{cert}}$. Consequently the former finite-audit hypothesis “Transition–singular low-block determinant” is discharged in the certified singular quotient by Theorem 16.130 and Corollary 16.131.

Proof. This is Corollary 16.131, rewritten in the legacy label used by downstream citations. The certified row lift gives $\sigma_{\min}((I - P_{\text{sing},p}^{\text{cert}})E_{\text{sing},p}^{2Y}) \geq Q^{-A}$, and the certified low-mode basis is polynomially identified with the labelled $C_7/Y12$ transition carrier plane. The Gram and Schur determinant forms are then equivalent by Lemma 16.106 and the polynomial conditioning of the finite atlas. $\square$

Theorem 16.135 (Low-block coercivity in the active singular-discard branch). *In the proof-level branch, where $P_{\text{sing},p} = P_{\text{sing},p}^{\text{cert}}$, the finite low-block part of the $Y12$ graph/coercivity module is closed:*

$$\sigma_{\min}(D_p^{\text{cert}}) \geq Q^{-A}, \quad \delta_{\text{coerc},p} \geq Q^{-A}.$$

The only remaining contribution in the graph/coercivity module is the residual singular-discard tail, which is controlled by the separate EFCompletion_{K9} or targeted $\text{EFCompletion}_{Y12}$ upper appropriate to the chosen forbidden channel.

Proof. The singular-value lower bound is Theorem 16.130. The passage from the low-block matrix to the coordinate-free $\delta_{\text{coerc},p}$ uses the Gram identity in Definition 16.105. The tail is not part of the finite defect matrix; it is exactly the residual term in the singular-discard/EF upper ledger. $\square$

Remark 16.136 (Raw retained-space caveat). The corollary above does not assert coercivity for an arbitrary raw retained space. If a raw choice satisfies

$$T_{\text{tr} \rightarrow \text{sing}, p} \text{span}\{[C_7]^{\text{tr}}, [C_{Y12}]^{\text{tr}}\} \subseteq R_{\text{sing}, p}^{\text{raw}},$$

then the projected carrier matrix is zero and $\delta_{\text{coerc}, p} = 0$ on the live carrier plane. This is the reason for the certified convention in Definition 16.102: exact/dead singular directions are retained, coherent live components are routed into the effective carrier, and atom-local live components are excluded by the no-atom-fitting guardrails. All proof-level uses of $\Lambda_{\text{tr}, \text{sing}}$, D_p , and $K_{D4, p}$ therefore use $P_{\text{sing}}^{\text{cert}}$, not a raw or Taylor-complete projection.

16.24. Rank-one Y12 normality versus full low-block coercivity. The current downstream repair supplies source-weighted D4 amplitude mass after Gate-1/Gate-2 and identifies a targeted primitive Y12-signature normal route. This is enough for a rank-one forbidden wall, but it is not the same statement as the low-block singular value bound

$$\delta_{\text{coerc}, p} := \sigma_{\min}\left((I - P_{\text{sing}, p})T_{\text{tr} \rightarrow \text{sing}, p}J_{\text{tr}, p}|_{\mathcal{R}_{X, \text{low}, p}}\right) \geq Q^{-C}.$$

The latter is a full principal-angle statement between the transitioned low block and the retained singular space.

Definition 16.137 (Low-block singular-discard Gram). Let

$$V_p := \mathcal{R}_{X, \text{low}, p}, \quad A_p := T_{\text{tr} \rightarrow \text{sing}, p}J_{\text{tr}, p}|_{V_p}, \quad \Pi_p^\perp := I - P_{\text{sing}, p}.$$

For a basis $e_{1, p}, \dots, e_{r, p}$ of V_p , put

$$C_p = [A_p e_{1, p} \ \cdots \ A_p e_{r, p}], \quad D_p = \Pi_p^\perp C_p, \quad G_p = D_p^* D_p.$$

If $B_p = (\langle e_{i, p}, e_{j, p} \rangle)_{ij}$ is the Gram matrix of the chosen low basis, then

$$\delta_{\text{coerc}, p}^2 = \lambda_{\min}\left(B_p^{-1/2} G_p B_p^{-1/2}\right).$$

Equivalently, $\delta_{\text{coerc}, p} = \inf_{0 \neq v \in V_p} \text{dist}(A_p v, R_{\text{sing}, p}) / \|v\|$.

Lemma 16.138 (Rank-one normal visibility is not full coercivity). *Suppose the low block contains a two-dimensional live carrier plane $W_p = \text{span}\{c_{7, p}, c_{Y, p}\} \subset V_p$, and suppose there is a single polynomially bounded normal row $\ell_{Y, p}$ such that*

$$\ell_{Y, p} R_{\text{sing}, p} = 0, \quad \ell_{Y, p} A_p c_{Y, p} = b_p, \quad |b_p| \geq Q^{-C}.$$

Then the Y12-normal channel is nonzero, but this does not imply $\sigma_{\min}(D_p|_{W_p}) \geq Q^{-C'}$: if $\ell_{Y, p} A_p c_{7, p} = a_p$, the row sees only $a_p x_7 + b_p x_Y$, and the nonzero vector $x = (b_p, -a_p)$ is killed. Therefore a second independent normal row, or a one-dimensional reduction of the low block, is necessary to prove a positive smallest singular value on W_p .

Proof. The assertion is the rank computation for the 1×2 matrix $[a_p \ b_p]$: its rank is at most one, hence it has a nontrivial kernel on a two-dimensional plane. This kernel is compatible with $b_p \neq 0$, so primitive Y12-signature survival gives rank-one normal visibility only. $\square$

Proposition 16.139 (Two-normal-row certificate for δ_{coerc}). *Let $V_p = \mathcal{R}_{X, \text{low}, p}$, let $A_p = T_{\text{tr} \rightarrow \text{sing}, p}J_{\text{tr}, p}|_{V_p}$, and let $\Pi_p^\perp = I - P_{\text{sing}, p}$. Suppose there is a matrix of normal rows $L_p : H_{\text{sing}, p} \rightarrow \mathbb{C}^r$ with*

$$L_p P_{\text{sing}, p} = 0, \quad \|L_p\|_{\text{op}} \leq Q^{C_L}, \quad \sigma_{\min}\left(L_p A_p|_{V_p}\right) \geq Q^{-C_M}$$

in the same low-basis metric. Then

$$\sigma_{\min}\left((I - P_{\text{sing}, p})A_p|_{V_p}\right) \geq Q^{-(C_L + C_M)}.$$

Proof. Because $L_p P_{\text{sing}, p} = 0$, $L_p = L_p(I - P_{\text{sing}, p}) = L_p \Pi_p^\perp$. Thus $\|L_p A_p v\| = \|L_p \Pi_p^\perp A_p v\| \leq \|L_p\|_{\text{op}} \|\Pi_p^\perp A_p v\|$ for every $v \in V_p$. The lower bound on $L_p A_p$ gives $Q^{-C_M} \|v\| \leq \|L_p A_p v\| \leq Q^{C_L} \|\Pi_p^\perp A_p v\|$. $\square$

Corollary 16.140 (Triangular $C_7/Y12$ normal-row closure). *Assume the low block is the live carrier plane with a polynomially conditioned basis $V_p = \text{span}\{c_{7,p}, c_{Y,p}\}$. Suppose there are polynomially bounded rows $\ell_{7,p}^\perp, \ell_{Y,p}^\perp : H_{\text{sing},p} \rightarrow \mathbb{C}$ with $\ell_{7,p}^\perp R_{\text{sing},p} = 0$, $\ell_{Y,p}^\perp R_{\text{sing},p} = 0$, and (after polynomially bounded row operations)*

$$\begin{pmatrix} \ell_{7,p}^\perp A_p c_{7,p} & \ell_{7,p}^\perp A_p c_{Y,p} \\ \ell_{Y,p}^\perp A_p c_{7,p} & \ell_{Y,p}^\perp A_p c_{Y,p} \end{pmatrix} = \begin{pmatrix} \eta_{7,p} & * \\ 0 & \eta_{Y,p} \end{pmatrix}, \quad |\eta_{7,p} \eta_{Y,p}| \geq Q^{-C}.$$

Then $\sigma_{\min}\left((I - P_{\text{sing},p})T_{\text{tr} \rightarrow \text{sing},p}J_{\text{tr},p}|_{V_p}\right) \geq Q^{-C'}$.

Proof. Take $L_p = \begin{pmatrix} \ell_{7,p}^\perp \\ \ell_{Y,p}^\perp \end{pmatrix}$. The displayed triangular matrix has polynomial determinant and polynomial upper norm, hence polynomial smallest singular value. Apply Proposition 16.139. $\square$

Remark 16.141 (What the present $Y12$ -signature route proves). The primitive $Y12$ -signature route supplies a candidate for $\ell_{Y,p}^\perp$: a conditioned normal row detecting $T_{\text{tr} \rightarrow \text{sing},p}[\mathbf{C}_{Y12}]^{\text{tr}}$ after quotienting actual-source tangents, lower connections, and actual second-curvature directions. This is a rank-one statement. It gives a forbidden wall $a_p \Gamma_G + b_p \Gamma_{G,2}$ with $|b_p| \geq Q^{-C}$, which must then be paired with moving-wall anti-concentration. It does not by itself supply a positive smallest singular value for a two-dimensional $C_7/Y12$ low block.

Remark 16.142 (Missing compatibility for the septic row). The covariant septic row ℓ_7 detects C_7 in the typed septic carrier quotient. To use it in Corollary 16.140 one still needs the singular-normal compatibility theorem

$$\ell_{7,p}^\perp R_{\text{sing},p} = 0, \quad \ell_{7,p}^\perp T_{\text{tr} \rightarrow \text{sing},p}[\mathbf{C}_7]^{\text{tr}} \neq 0,$$

with polynomial conditioning and with $\ell_{7,p}^\perp$ defined on the same singular discarded quotient as $I - P_{\text{sing},p}$. This is not a formal consequence of the carrier quotient or of primitive $Y12$ -signature survival; it is a separate transition-singular row-lift theorem.

Theorem 16.143 (Archived raw determinant test for non-certified δ_{coerc}). *For an arbitrary raw retained singular space, the low-block gap would be closed if one proved the finite determinant estimate*

$$\lambda_{\min}\left(B_p^{-1/2} D_p^* D_p B_p^{-1/2}\right) \geq Q^{-C}, \quad D_p = (I - P_{\text{sing},p})T_{\text{tr} \rightarrow \text{sing},p}J_{\text{tr},p}[e_{1,p} \cdots e_{r,p}].$$

In the two-carrier reduction this is equivalently

$$\det \begin{pmatrix} \|v_{7,p}^\perp\|^2 & \langle v_{7,p}^\perp, v_{Y,p}^\perp \rangle \\ \langle v_{Y,p}^\perp, v_{7,p}^\perp \rangle & \|v_{Y,p}^\perp\|^2 \end{pmatrix} \geq Q^{-C},$$

with $v_{7,p}^\perp := (I - P_{\text{sing},p})T_{\text{tr} \rightarrow \text{sing},p}[\mathbf{C}_7]^{\text{tr}}$, $v_{Y,p}^\perp := (I - P_{\text{sing},p})T_{\text{tr} \rightarrow \text{sing},p}[\mathbf{C}_{Y12}]^{\text{tr}}$. A rank-one $Y12$ -signature normal theorem proves only that $v_{Y,p}^\perp \neq 0$ in a quotient direction. For the active certified branch this raw determinant test is superseded by Theorem 16.130. The displayed criterion remains only as an archive explaining why arbitrary raw singular projections are not proof-level choices.

Remark 16.144 (Status after certified singular retention). *Status: PROVED for the certified singular-discard branch.* The rank-one warning remains mathematically correct for an arbitrary normal row, but it is no longer an active gap in the proof-level branch: the certified singular retained space supplies two normal rows through $L_p^{\text{row}} \Theta_{2Y,p}$, and Theorem 16.128 gives a polynomially invertible 2×2 carrier matrix. Hence the full rank-two low-block coercivity gap is supplied by Theorem 16.130. The caveat is only that raw or Taylor-complete R_{sing} choices are inadmissible for proof-level $\Lambda_{\text{tr},\text{sing}}$; see Remark 16.136.

Remark 16.145 (Archived low-block Gershgorin sufficient test). The coordinate Gershgorin margin is no longer an active hypothesis in the proof-level branch. It remains a useful sufficient test for an arbitrary raw finite basis:

$$D_p = [\mathfrak{T}_p e_{1,p} \cdots \mathfrak{T}_p e_{r,p}], \quad G_p := D_p^* D_p, \quad \mathfrak{T}_p := (I - P_{\text{sing},p})T_{\text{tr} \rightarrow \text{sing},p}J_{\text{tr},p}.$$

If a chosen raw basis satisfies $(G_p)_{ii} \geq \sum_{j \neq i} |(G_p)_{ij}| + c_0 Q^{-C}$, then Gershgorin gives $\lambda_{\min}(G_p) \geq c_0 Q^{-C}$ and $\sigma_{\min}(D_p) \geq \sqrt{c_0} Q^{-C/2}$. However, in the active proof route the finite low-block gap is supplied instead by the certified singular retained space: $P_{\text{sing},p} = P_{\text{sing},p}^{\text{cert}}$, $D_p = D_p^{\text{cert}}$, and by Theorem 16.130. Thus the Gershgorin criterion is an archived coordinate sufficient test, not a remaining open assumption.

Remark 16.146 (Certified Y12 graph/coercivity status). The Y12 graph/coercivity low-block gap is closed on the certified singular-discard branch. Any statement referring to an unresolved raw Gershgorin margin concerns only non-certified or Taylor-complete retained spaces and is not part of the active proof branch. In the active canonical route, the residual upper is not a scalar certified-quotient assumption: it is the source-graded full-vector $K9$ upper of Definition 16.159 and Theorem 16.189. Thus no low-block determinant or scalar residual-upper audit remains in δ_{coerc} .

Remark 16.147 (Defect matrix and $D4$ transversality). The finite defect matrix D_p is the low-mode representation of the same discarded operator encoded by

$$K_{D4,p} = (I - P_{\text{sing},p})T_{\text{tr} \rightarrow \text{sing},p}E_{\text{car},p}^{\text{tr}}L_{\text{clean},p}^{\text{tr}}.$$

Thus the coercivity of $\mathfrak{T}_p$ on the low block and the rank condition for $K_{D4,p}$ are the same finite test in two coordinate systems.

Definition 16.148 (Coordinate-free low-block coercivity gap). Let

$$V_p := \mathcal{R}_{X,\text{low},p}, \quad A_p := T_{\text{tr} \rightarrow \text{sing},p}J_{\text{tr},p}|_{V_p}, \quad \Pi_p^\perp := I - P_{\text{sing},p}.$$

The intrinsic low-block coercivity gap is

$$\delta_{\text{coerc},p} := \inf_{0 \neq v \in V_p} \frac{\|\Pi_p^\perp A_p v\|_{H_{\text{sing},p}}}{\|v\|_{X,p}}.$$

Equivalently,

$$\delta_{\text{coerc},p} = \inf_{0 \neq v \in V_p} \frac{\text{dist}(A_p v, R_{\text{sing},p})}{\|v\|_{X,p}}.$$

Thus the statement $\delta_{\text{coerc},p} \geq Q^{-C}$ is precisely the quantitative transversality of the transported low residual space $A_p V_p$ to the retained singular space $R_{\text{sing},p}$.

Lemma 16.149 (Generalized Gram form of δ_{coerc}). Let

$$E_p^{\text{low}} : \mathbb{C}^r \rightarrow V_p, \quad E_p^{\text{low}} a := \sum_{k=1}^r a_k e_{k,p},$$

where $\{e_{1,p}, \dots, e_{r,p}\}$ is any basis of V_p . Set

$$B_p := (E_p^{\text{low}})^* E_p^{\text{low}}, \quad D_p := \Pi_p^\perp A_p E_p^{\text{low}}, \quad G_p := D_p^* D_p.$$

Then

$$\delta_{\text{coerc},p}^2 = \lambda_{\min}(B_p^{-1/2} G_p B_p^{-1/2}).$$

In particular, if the chosen basis is orthonormal in the domain metric, then $\delta_{\text{coerc},p}^2 = \lambda_{\min}(G_p)$. More generally, if B_p is polynomially conditioned, then a polynomial lower bound for $\lambda_{\min}(G_p)$ is equivalent, up to polynomial loss, to $\delta_{\text{coerc},p} \geq Q^{-C}$.

Proof. For $a \in \mathbb{C}^r$, write $v = E_p^{\text{low}} a$. Then $\|v\|_{X,p}^2 = a^* B_p a$ and $\|\Pi_p^\perp A_p v\|_{H_{\text{sing},p}}^2 = a^* G_p a$. Taking the infimum of the Rayleigh quotient over $a \neq 0$ gives $\delta_{\text{coerc},p}^2 = \inf_{a \neq 0} (a^* G_p a) / (a^* B_p a) = \lambda_{\min}(B_p^{-1/2} G_p B_p^{-1/2})$. The orthonormal and polynomially conditioned conclusions are immediate. $\square$

Proposition 16.150 (Adjacent finite gates do not imply low-block coercivity). The already separate estimates for source survival, Graph/O2O3 survival, and certified Y12-quotient survival do not logically imply $\delta_{\text{coerc},p} \geq Q^{-C}$. A further transition-singular transversality input is necessary.

Proof. It is enough to give a finite model satisfying the adjacent algebraic conclusions while forcing $D_p = 0$. Fix $r \geq 1$, let

$$V_p = \mathbb{C}^r, \quad H_{\text{sing},p} = \mathbb{C}^r \oplus \mathbb{C}^r, \quad A_p v = (v, 0),$$

and take $R_{\text{sing},p} = \mathbb{C}^r \oplus \{0\}$, $P_{\text{sing},p} = P_{R_{\text{sing},p}}$. Then $(I - P_{\text{sing},p})A_p v = 0$ for $v \in V_p$, so $D_p = 0$ and $\delta_{\text{coerc},p} = 0$. The adjacent finite gates may simultaneously be made perfect: choose a source frame with $M = I$ and no residual error, so the source gap is one; choose Graph/O2O3 Gram blocks $G_{00} = I, G_{SS} = I, G_{0S} = 0$, so $1 - \|G_{00}^{-1/2} G_{0S} G_{SS}^{-1/2}\|_{\text{op}} = 1$; choose a certified quotient model with $q_{p,\text{cert}}^Y = I$ and live carrier matrix $E_p = I_2$, so $\sigma_{\min}(q_{p,\text{cert}}^Y E_p) = 1$. These choices satisfy the stated adjacent conclusions but leave $\delta_{\text{coerc},p} = 0$. Therefore the coercivity bound is not a formal consequence of those gates; it must come from a theorem controlling the relative position of $A_p V_p$ and $R_{\text{sing},p}$. $\square$

Lemma 16.151 (Two-carrier determinant criterion for singular-discard leakage). *Assume the low block is identified, with polynomially conditioned coordinates, with the live transition carrier plane*

$$W_{\text{tr},p} = \text{span}\{[C_7]^{\text{tr}}, [C_{Y12}]^{\text{tr}}\}.$$

Set

$$v_{7,p}^\perp := (I - P_{\text{sing},p})T_{\text{tr} \rightarrow \text{sing},p}[C_7]^{\text{tr}}, \quad v_{Y,p}^\perp := (I - P_{\text{sing},p})T_{\text{tr} \rightarrow \text{sing},p}[C_{Y12}]^{\text{tr}},$$

and $W_p := [v_{7,p}^\perp \ v_{Y,p}^\perp]$, $S_p := W_p^* W_p$. If $\text{tr } S_p \leq Q^{C_0}$ and $\det S_p \geq Q^{-C_1}$, then $\sigma_{\min}(W_p) \geq Q^{-(C_0+C_1)/2}$. Equivalently, if

$$\|v_{7,p}^\perp\| \geq Q^{-A_1}, \quad \|v_{Y,p}^\perp\| \geq Q^{-A_1}, \quad \sin \angle(v_{7,p}^\perp, v_{Y,p}^\perp) \geq Q^{-A_2},$$

and $\text{tr } S_p \leq Q^{C_0}$, then $\sigma_{\min}(W_p) \geq Q^{-C'}$ for a polynomial exponent C' .

Proof. The two eigenvalues of S_p are nonnegative and satisfy $\lambda_{\min}(S_p) = \det S_p / \lambda_{\max}(S_p) \geq \det S_p / \text{tr } S_p$. Thus $\lambda_{\min}(S_p) \geq Q^{-(C_0+C_1)}$, and $\sigma_{\min}(W_p) = \lambda_{\min}(S_p)^{1/2}$. The equivalent angular criterion follows from

$$\det S_p = \|v_{7,p}^\perp\|^2 \|v_{Y,p}^\perp\|^2 - |\langle v_{7,p}^\perp, v_{Y,p}^\perp \rangle|^2 = \|v_{7,p}^\perp\|^2 \|v_{Y,p}^\perp\|^2 \sin^2 \angle(v_{7,p}^\perp, v_{Y,p}^\perp).$$

$\square$

Theorem 16.152 (Conditional closure of δ_{coerc} from transition-singular leakage). *Let the low-mode coefficient map into transition live-carrier coordinates be polynomially conditioned. More precisely, suppose there is an isomorphism $R_p : \mathbb{C}^r \rightarrow \mathbb{C}^r$ such that the defect matrix factors as $D_p = W_p R_p$ with $\sigma_{\min}(R_p) \geq Q^{-C_R}$ and $\|R_p\|_{\text{op}} \leq Q^{C_R}$. If the transition-singular leakage matrix satisfies $\sigma_{\min}(W_p) \geq Q^{-C_W}$, then $\sigma_{\min}(D_p) \geq Q^{-(C_R+C_W)}$. Consequently $\delta_{\text{coerc},p} \geq Q^{-C}$, after the polynomial basis-conditioning loss of Lemma 16.149.*

Proof. For every coefficient vector a , $\|D_p a\| = \|W_p R_p a\| \geq \sigma_{\min}(W_p) \|R_p a\| \geq Q^{-C_W} Q^{-C_R} \|a\|$. Taking the infimum over $\|a\| = 1$ gives the claimed singular-value lower bound. The intrinsic δ_{coerc} statement follows from the generalized Gram representation. $\square$

Remark 16.153 (Archived raw low-block determinant test). Before certification, the low-block determinant had to be proved directly as

$$\lambda_{\min}(B_p^{-1/2} D_p^* D_p B_p^{-1/2}) \geq Q^{-C}.$$

This remains the correct test for an arbitrary raw singular projection, and it explains why raw Taylor-complete choices fail. In the proof-level branch the test is no longer missing: the certified singular row lift supplies two normal rows, and Corollary 16.134 gives the same Schur determinant lower bound with $P_{\text{sing},p} = P_{\text{sing},p}^{\text{cert}}$. The rank-one warning remains only as an archive: one $Y12$ -normal row detects one wall, whereas the certified $C_7/Y12$ two-row lift gives the full rank-two low-block singular-value bound.

Remark 16.154 (Graph-space no-go). If $U_p \not\subseteq \ker \Lambda_{Y,p}$, or if $\mathcal{R}_X$ fails to admit a finite-codimension graph decomposition with tail norm $\|\cdot\|_{\text{EF}}$, then the $Y12$ discarded observable is not coercive and the downstream atomic contradiction cannot be derived in the current norm.

Remark 16.155 (Legacy scalar singular-discard upper). This legacy label is retained only for citations to the old scalar transition–singular formulation. It is not the active proof-level upper in the canonical source-graded branch. The active upper is Theorem 16.189, namely the estimate in the fixed labelled Hilbert space $H_{K9,p}^{\text{gr}}$ for

$$\Lambda_p^{\text{gr,full}} = (I - P_p^{\text{full}}) \mathcal{L}_{K9,p}^{\text{gr}} J_{\text{tr},p}.$$

The old scalar display

$$(I - P_{\text{sing},p}) T_{\text{tr} \rightarrow \text{sing},p} J_{\text{tr},p}(q''_{\text{can}})$$

may be used only as a collapsed diagnostic after the source-graded same-norm contradiction has already been formulated. In particular, the proof no longer assumes a scalar singular-discard upper for q''_{can} as a separate hypothesis. The upper is obtained by retaining the full finite-main labelled model $H_{F,p}$, using $q''_{\text{can}} = F_X + R_X$, and applying full-vector source-graded $K9$ completion to R_X .

Definition 16.156 (Active meaning of EFCompletion_{K9}). The active $K9$ -completion estimate is the full-vector, source-graded tail upper before scalar collapse. In the canonical branch it means

$$\sum_p \left\| (I - P_{R_p}) \mathcal{L}_{K9,p}^{\text{gr}} J_{\text{tr},p}(R_X) \right\|_{\text{gr}}^2 \leq Q^{-B} \mu(\mathcal{C}),$$

for arbitrarily large prescribed B , after choosing the smooth AFE cutoff depth sufficiently deep. Here R_p is the certified retained/dead source-graded space before full finite-main retention, and $\mathcal{L}_{K9,p}^{\text{gr}}$ is the fixed labelled lift of Definition 16.161. Its cap-local form is

$$\sum_{p \in C_j} \mu_p \left\| (I - P_{R_p}) \mathcal{L}_{K9,p}^{\text{gr}} J_{\text{tr},p}(R_X) \right\|_{\text{gr}}^2 \leq Q^{-B} \sum_{p \in C_j} \mu_p \|v_p\|^2.$$

The older scalar notation

$$\Lambda_{\text{tr},\text{sing},p} = (I - P_{\text{sing},p}) T_{\text{tr} \rightarrow \text{sing},p} P_{\text{tr},p}$$

is an archived shorthand for a collapsed diagnostic. It is not sufficient for the affine-forcing branch, because scalar collapse can hide cancellation between finite-main and affine live labels. The proof-level upper is always measured in the same source-graded norm used for the lower bound.

Definition 16.157 ($\text{EFCompletion}_{Y12}$). The targeted $Y12$ -signature completion estimate is

$$\sum_p \left| \Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing},p} J_{\text{tr},p}(R_X) \right|^2 \leq Q^{-B} \mu(\mathcal{C}).$$

This is weaker than EFCompletion_{K9} , which controls the full order-nine residual after applying $\mathcal{K}_9$. It is sufficient for the $Y12$ -signature D4 contradiction if Hypothesis E.116 holds.

Remark 16.158 (Three completion ledgers). The proof now uses three different completion strengths:

$$\boxed{\text{EFCompletion}_{L^2}}$$

for aggregate Gate–1 source energy,

$$\boxed{\text{EFCompletion}_{K9}}$$

for a full order-nine residual upper, and

$$\boxed{\text{EFCompletion}_{Y12}}$$

for the targeted primitive $Y12$ -signature channel. These must not be conflated. In particular,

$$\text{EFCompletion}_{L^2} \not\Rightarrow \text{EFCompletion}_{K9}$$

unless the operator

$$\mathcal{K}_9 T_{\text{tr} \rightarrow \text{sing}} J_{\text{tr}}$$

is polynomially bounded from the L^2 -completion norm. The targeted Y12-estimate may be substantially easier than full K_9 -completion.

Definition 16.159 (Full-vector source-graded certified K_9 completion). On the proof-level canonical source-graded branch, define

$$\Lambda_{K_9,p}^{\text{gr},\text{vec}} := (I - P_{R_p}) \mathcal{L}_{K_9,p}^{\text{gr}} J_{\text{tr},p},$$

where $R_p \subset H_{K_9,p}^{\text{gr}}$ is the certified retained/dead source-graded space before adjoining the finite-AFE main model. The assertion $\text{EFCompletion}_{K_9}^{\text{gr},\text{vec}}$ is the estimate

$$\sum_p \|\Lambda_{K_9,p}^{\text{gr},\text{vec}}(R_X)\|_{\text{gr}}^2 \leq Q^{-B} \mu(\mathcal{C})$$

for arbitrarily large prescribed B , after choosing the smooth AFE cutoff sufficiently deep. Its cap-local weighted form is

$$\sum_{p \in C_j} \mu_p \|\Lambda_{K_9,p}^{\text{gr},\text{vec}}(R_X)\|_{\text{gr}}^2 \leq Q^{-B} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2.$$

This is the full labelled/source-graded row norm before scalar I -coordinate collapse and before projection to a primitive Y12 row. The older collapsed notation

$$(I - P_{\text{sing},p}^{\text{cert}}) T_{\text{tr} \rightarrow \text{sing},p} P_{\text{tr},p}$$

is the image of this source-graded row under the fixed collapse map only after the same-norm comparison has already been made. It is not the active norm for the canonical affine-forcing branch.

Remark 16.160 (Legacy collapsed K_9 notation). If an older citation writes $\Lambda_{K_9,p}^{\text{vec}}$ without the superscript gr , it should be read in the canonical branch as $\Lambda_{K_9,p}^{\text{gr},\text{vec}}$. Scalar/collapsed K_9 or primitive-only Y12 completion is insufficient for the affine-forcing repair because it can hide cancellation between orthogonal source-grade components. The upper used in Theorem 16.172 is the source-graded full-vector estimate above.

Definition 16.161 (Canonical source-graded K_9 lift). For each selected ordinary real-canonical packet p , let $\mathfrak{L}_{K_9,p}$ be the finite set of labelled order-nine source rows appearing in the certified transition-singular row table before scalar I -coordinate collapse. Define the source-graded Hilbert space

$$H_{K_9,p}^{\text{gr}} := \bigoplus_{\lambda \in \mathfrak{L}_{K_9,p}} H_{p,\lambda}, \quad \|(x_\lambda)_\lambda\|_{\text{gr}}^2 := \sum_\lambda \|x_\lambda\|^2.$$

The canonical source-graded lift is the fixed linear map

$$\mathcal{L}_{K_9,p}^{\text{gr}} J_{\text{tr},p}(f) := \bigoplus_{\lambda \in \mathfrak{L}_{K_9,p}} \Lambda_{\lambda,p} J_{\text{tr},p}(f),$$

where $\Lambda_{\lambda,p}$ is the labelled row functional determined by the transition-singular row table. The scalar/collapsed transition-singular map is recovered by a fixed linear collapse map

$$\pi_{\text{coll},p} : H_{K_9,p}^{\text{gr}} \rightarrow H_{\text{sing},p}, \quad \pi_{\text{coll},p} \mathcal{L}_{K_9,p}^{\text{gr}} = T_{\text{tr} \rightarrow \text{sing},p} J_{\text{tr},p}.$$

The lift is part of the row model, not a decomposition-dependent post-processing device: for every decomposition $f = f_1 + f_2$,

$$\mathcal{L}_{K_9,p}^{\text{gr}} J_{\text{tr},p}(f) = \mathcal{L}_{K_9,p}^{\text{gr}} J_{\text{tr},p}(f_1) + \mathcal{L}_{K_9,p}^{\text{gr}} J_{\text{tr},p}(f_2).$$

The labelled space is decomposed orthogonally as

$$H_{K_9,p}^{\text{gr}} = H_{p,\text{main}} \oplus H_{p,\text{aff}} \oplus H_{p,\text{dead}} \oplus H_{p,\text{res}},$$

where $H_{p,\text{main}}$ is spanned by the finite-AFE main labels, $H_{p,\text{aff}}$ is spanned by the live affine C_7 /Y12 labels, including the clean D4 carrier and every retained live source-Hessian affine forcing label, $H_{p,\text{dead}}$ is the certified dead/retained label space, and $H_{p,\text{res}}$ is the residual ledger.

Remark 16.162 (Why the source-graded lift is load-bearing). The source-graded norm is an observable norm only because Definition 16.161 gives a fixed linear lift of the transition–singular row before scalar collapse. If the grading were assigned after choosing the proof decomposition $q''_{\text{can}} = F_X + R_X$, then the same-norm upper would not be a statement about the actual canonical source. Thus scalar collapse may be used only after the same-norm contradiction has been obtained in $H_{K9,p}^{\text{gr}}$.

Definition 16.163 (Maximal source-graded affine target). Let $E_p = [C_{7,p}^{\text{tar}} \ C_{Y,p}^{\text{tar}}]$ and let $\mathfrak{A}_{K9,p} \subset \mathfrak{L}_{K9,p}$ be the finite set of all live affine $C_7/Y12$ -labelled source rows in the certified $K9$ table, excluding the clean D4 linear carrier label. This set includes the odd- $\eta\eta$ / Bottleneck-C forcing label and any same-grade live source-Hessian companion that survives the fixed-sector projection. For each $\lambda \in \mathfrak{A}_{K9,p}$, let $m_{\lambda,p} \in \mathbb{C}^2$ be its live coordinate, i.e. $E_p m_{\lambda,p}$ is the $C_7/Y12$ -live part of the row with label λ . Define

$$\hat{\mathcal{H}}_p^{Y,\max} := \mathcal{H}_{p,\text{lin}}^Y \oplus \bigoplus_{\lambda \in \mathfrak{A}_{K9,p}} \mathcal{H}_{p,\lambda}^Y,$$

with the orthogonal source-graded norm. The maximal affine observable is

$$\hat{\Phi}_{p,\max}^Y(v) := \left(q_{p,\text{aff}}^Y E_p H_p L_{\text{clean},p} v, (q_{p,\text{aff}}^Y E_p m_{\lambda,p})_{\lambda \in \mathfrak{A}_{K9,p}} \right).$$

The earlier two-component affine target is the special case $\mathfrak{A}_{K9,p} = \{\eta\eta\}$. The maximal target is used only to avoid a hidden same-grade companion; adding orthogonal affine summands can only increase the lower-side norm.

Proposition 16.164 (No-go for quotient-killing a live affine arithmetic row). *Let*

$$A_{II,p}^{\text{raw}} = A_{II,p}^{\text{dead}} + E_p m_{\eta,p}, \quad A_{II,p}^{\text{dead}} \in K_p,$$

with $m_{\eta,p} \neq 0$. If the full unsplit column $A_{II,p}^{\text{raw}}$ is placed in the killed quotient space, then

$$\sigma_{\min}((I - P_{K_p + \text{span}\{A_{II,p}^{\text{raw}}\}}^{\text{bd}})E_p) = 0.$$

Thus a nonzero odd- $\eta\eta$ / Bottleneck-C live component must be retained as affine forcing; it cannot be quotient-killed as nuisance.

Proof. Since both $A_{II,p}^{\text{raw}}$ and $A_{II,p}^{\text{dead}}$ lie in the enlarged killed space, their difference $E_p m_{\eta,p}$ also lies there. Hence the quotient annihilates $E_p m_{\eta,p}$, giving a nonzero kernel vector for the map on the live carrier plane. $\square$

Definition 16.165 (Finite-main Schur complement). Let $R_p \subset H_{K9,p}^{\text{gr}}$ be the certified retained/dead space before retaining the finite-AFE main model, and set $q_p := I - P_{R_p}$. Let

$$E_p^{\text{aff}} : \mathbb{C}^d \rightarrow H_{K9,p}^{\text{gr}}$$

be the maximal affine live carrier map, and let

$$H_{F,p} : \mathbb{C}^m \rightarrow H_{K9,p}^{\text{gr}}$$

be the finite-main singular map whose image spans the source-graded rows of $B_2, P_{2,\leq X}, D_{2,\leq X}$. The finite-main Schur complement is

$$S_{F,p} := (E_p^{\text{aff}})^*(I - P_{R_p + \text{im } H_{F,p}})E_p^{\text{aff}}.$$

Equivalently,

$$S_{F,p} = G_{EE,p} - G_{EH,p} G_{HH,p}^{\dagger} G_{EH,p}^*,$$

where

$$G_{EE,p} := (E_p^{\text{aff}})^* q_p E_p^{\text{aff}}, \quad G_{EH,p} := (E_p^{\text{aff}})^* q_p H_{F,p}, \quad G_{HH,p} := H_{F,p}^* q_p H_{F,p}.$$

Here $G_{HH,p}^{\dagger}$ denotes the Moore–Penrose inverse on $\text{ran}(q_p H_{F,p})$.

Lemma 16.166 (Finite-main Schur complement formula). *Definition 16.165 gives the exact Schur formula*

$$(E_p^{\text{aff}})^*(I - P_{R_p + \text{im } H_{F,p}})E_p^{\text{aff}} = G_{EE,p} - G_{EH,p}G_{HH,p}^\dagger G_{EH,p}^*.$$

Proof. On $R_p^\perp$, the projection onto $R_p + \text{im } H_{F,p}$ is the projection onto $\text{im}(q_p H_{F,p})$, namely

$$P_{q_p H_{F,p}} = q_p H_{F,p} (H_{F,p}^* q_p H_{F,p})^\dagger H_{F,p}^* q_p.$$

Thus

$$I - P_{R_p + \text{im } H_{F,p}} = q_p - P_{q_p H_{F,p}}$$

on the range relevant to E_p^{aff} . Multiplying on the left and right by $(E_p^{\text{aff}})^*$ and E_p^{aff} gives the stated formula. $\square$

Theorem 16.167 (Canonical grade separation closes the finite-main Schur gap). *Assume the canonical source-graded lift of Definition 16.161. Suppose that the retained/dead space R_p is source-grade saturated and that, modulo R_p ,*

$$q_p H_{F,p} \subset H_{p,\text{main}}, \quad q_p E_p^{\text{aff}} \subset H_{p,\text{aff}}, \quad H_{p,\text{main}} \perp H_{p,\text{aff}}.$$

If certified affine coercivity gives

$$(E_p^{\text{aff}})^* q_p E_p^{\text{aff}} \geq Q^{-C} I,$$

then the finite-main Schur complement satisfies

$$S_{F,p} = (E_p^{\text{aff}})^*(I - P_{R_p + \text{im } H_{F,p}})E_p^{\text{aff}} \geq Q^{-C} I.$$

Proof. The source-grade orthogonality assumptions imply

$$G_{EH,p} = (E_p^{\text{aff}})^* q_p H_{F,p} = 0.$$

Lemma 16.166 therefore gives

$$S_{F,p} = G_{EE,p} = (E_p^{\text{aff}})^* q_p E_p^{\text{aff}},$$

which is bounded below by the certified affine coercivity assumption. $\square$

Corollary 16.168 (Finite-main Schur obstruction discharged by the canonical lift). *Assume Definition 16.161. In particular, $\mathcal{L}_{K9,p}^{\text{gr}}$ is a fixed linear labelled lift before scalar collapse, the decomposition*

$$H_{K9,p}^{\text{gr}} = H_{p,\text{main}} \oplus H_{p,\text{aff}} \oplus H_{p,\text{dead}} \oplus H_{p,\text{res}}$$

is orthogonal, the finite-main labels of $B_2, P_{2,\leq X}, D_{2,\leq X}$ lie in $H_{p,\text{main}}$, and the maximal affine carrier image lies in $H_{p,\text{aff}}$. Let $R_p \subset H_{K9,p}^{\text{gr}}$ be the source-grade saturated certified retained/dead space, let $q_p := I - P_{R_p}$, let $E_p^{\text{aff}} : \mathbb{C}^d \rightarrow H_{K9,p}^{\text{gr}}$ be the maximal affine live carrier map, and let $H_{F,p} : \mathbb{C}^m \rightarrow H_{K9,p}^{\text{gr}}$ be the finite-main singular map. If certified affine coercivity gives

$$(E_p^{\text{aff}})^* q_p E_p^{\text{aff}} \geq Q^{-C} I_d,$$

then

$$\boxed{(E_p^{\text{aff}})^*(I - P_{R_p + \text{im } H_{F,p}})E_p^{\text{aff}} \geq Q^{-C} I_d.}$$

Consequently the finite-main Schur complement

$$G_{EE,p} - G_{EH,p}G_{HH,p}^\dagger G_{EH,p}^*$$

is polynomially positive. The finite-main obstruction is therefore discharged in the canonical source-graded branch.

Proof. By source-grade saturation, projecting by q_p preserves the orthogonal separation of the main and affine grades modulo the retained/dead space:

$$q_p H_{F,p} \subset H_{p,\text{main}}, \quad q_p E_p^{\text{aff}} \subset H_{p,\text{aff}}, \quad H_{p,\text{main}} \perp H_{p,\text{aff}}.$$

Hence

$$G_{EH,p} := (E_p^{\text{aff}})^* q_p H_{F,p} = 0.$$

The Schur formula of Lemma 16.166 gives

$$(E_p^{\text{aff}})^*(I - P_{R_p + \text{im } H_{F,p}})E_p^{\text{aff}} = G_{EE,p} = (E_p^{\text{aff}})^*q_pE_p^{\text{aff}},$$

which is bounded below by certified affine coercivity. $\square$

Remark 16.169 (No remaining finite-main Schur audit in the canonical branch). The warning in Proposition 16.170 remains important for noncanonical or scalar-collapsed formulations. In the active proof branch, however, Definition 16.161 commits to the labelled source-graded lift before scalar collapse. Thus the finite-main rows and affine live rows are orthogonal source labels, not two post hoc decompositions of the same scalar row. The possible obstruction

$$G_{EE,p} - G_{EH,p}G_{HH,p}^\dagger G_{EH,p}^* \not\geq Q^{-C}I$$

reduces to $G_{EH,p} = 0$, and is discharged by the canonical grade separation.

Proposition 16.170 (No-go without finite-main Schur gap). *If the finite-main Schur complement has a nontrivial kernel, then retaining the finite-AFE main model destroys the affine live lower bound in that direction. In particular, the tail estimate for R_X cannot by itself imply the same-norm Type-I upper/lower contradiction.*

Proof. If $S_{F,p}v = 0$ for some nonzero v , then

$$(I - P_{R_p + \text{im } H_{F,p}})E_p^{\text{aff}}v = 0.$$

Thus the affine live carrier map has a nonzero kernel after full-main retention. The lower bound in that affine direction is lost. Conversely, if the finite main model is not retained, the identity $q''_{\text{can}} = F_X + R_X$ leaves an uncontrolled F_X -contribution in the same source-graded discarded norm; the R_X -tail EFCompletion estimate cannot control that contribution. $\square$

Lemma 16.171 (Maximal affine row realization in the canonical labelled lift). *Let $\Pi_{\text{max},p}^{\text{gr}}$ denote the orthogonal projection in $H_{K9,p}^{\text{gr}}$ onto the maximal affine labelled target of Definition 16.163. Let*

$$P_p^{\text{full}} := P_{R_p + \text{im } H_{F,p}}.$$

For the canonical source-graded lift,

$$\Pi_{\text{max},p}^{\text{gr}}(I - P_p^{\text{full}})\mathcal{L}_{K9,p}^{\text{gr}}J_{\text{tr},p}(q''_{\text{can}}) = \Pi_{\text{max},p}^{\text{gr}}(I - P_p^{\text{full}})E_p^{\text{aff}}z_p + \mathcal{R}_{p,\text{max}}^{\text{id}},$$

where z_p is the vector of maximal affine live coefficients: its clean component is $H_pL_{\text{clean},p}v_p$, and its remaining components are the retained affine forcing coefficients $m_{\lambda,p}$, including the odd- $\eta\eta$ / Bottleneck- C coefficient. The error $\mathcal{R}_{p,\text{max}}^{\text{id}}$ is the projection of the residual-labelled part of the row table. In particular, if the residual ledger gives

$$\sum_{p \in C_j} \mu_p \|\mathcal{R}_{p,\text{max}}^{\text{id}}\|_{\text{gr}}^2 \leq Q^{-B_{\text{id}}} \sum_{p \in C_j} \mu_p \|v_p\|^2,$$

then the same-norm row-realization estimate holds in the maximal affine target.

Proof. The labelled lift is linear and decomposition-independent. The labelled $K9$ row table is a finite orthogonal direct sum of main labels, affine live labels, dead/retained labels, and residual labels. By definition, $H_{F,p}$ spans the finite-main labels, so these labels are annihilated by $I - P_p^{\text{full}}$. The dead labels are contained in R_p and are also annihilated. The projection $\Pi_{\text{max},p}^{\text{gr}}$ keeps exactly the affine live labels; since the target is maximal, no further same-grade $K9$ affine companion remains outside it. What is left is precisely $(I - P_p^{\text{full}})E_p^{\text{aff}}z_p$, plus the residual-labelled part. The displayed cap estimate is exactly the residual ledger measured in this same source-graded Hilbert norm. $\square$

Theorem 16.172 (Canonical source-graded affine branch closure). *Assume the following already-proved inputs on the selected ordinary real-canonical routed cap family:*

- (1) *the canonical source-graded $K9$ lift of Definition 16.161;*
- (2) *certified affine coercivity for E_p^{aff} ;*
- (3) *affine-live quotient survival and the no-kill repair for the live odd- $\eta\eta$ arithmetic row;*

- (4) clean row-normal carrier invertibility and source-weighted D4 amplitude mass;
- (5) full-vector source-graded K9 EFCompletion for the AFE tail R_X ;
- (6) the residual estimate in Lemma 16.171 with exponent larger than the lower-bound losses.

Then the source-graded affine branch gives a same-norm Type-I contradiction. Equivalently, no nonzero routed cap core can survive in this branch; the static-line case is the specialization $v_p = u_j$.

Proof. By Corollary 16.168,

$$\|(I - P_p^{\text{full}})E_p^{\text{aff}}w\|^2 \geq Q^{-C}\|w\|^2$$

for every maximal affine coefficient vector w . Apply this to the affine coefficient vector z_p . Its clean component is $H_p L_{\text{clean},p} v_p$, hence clean carrier invertibility and D4 amplitude mass imply

$$\sum_{p \in C_j} \mu_p \|z_p\|^2 \geq Q^{-A} \sum_{p \in C_j} \mu_p \|v_p\|^2.$$

Lemma 16.171, together with $\|x + y\|^2 \geq \frac{1}{2}\|x\|^2 - \|y\|^2$, therefore gives the lower bound

$$\sum_{p \in C_j} \mu_p \|(I - P_p^{\text{full}})\mathcal{L}_{K9,p}^{\text{gr}} J_{\text{tr},p}(q''_{\text{can}})\|_{\text{gr}}^2 \geq Q^{-A'} \sum_{p \in C_j} \mu_p \|v_p\|^2.$$

For the upper, the finite main model is retained by P_p^{full} , so linearity of the canonical lift and the AFE identity $q''_{\text{can}} = F_X + R_X$ give

$$(I - P_p^{\text{full}})\mathcal{L}_{K9,p}^{\text{gr}} J_{\text{tr},p}(q''_{\text{can}}) = (I - P_p^{\text{full}})\mathcal{L}_{K9,p}^{\text{gr}} J_{\text{tr},p}(R_X).$$

Full-vector source-graded K9 EFCompletion controls the right-hand side; enlarging the retained space only decreases the discarded norm. Thus

$$\sum_{p \in C_j} \mu_p \|(I - P_p^{\text{full}})\mathcal{L}_{K9,p}^{\text{gr}} J_{\text{tr},p}(q''_{\text{can}})\|_{\text{gr}}^2 \leq Q^{-B'} \sum_{p \in C_j} \mu_p \|v_p\|^2.$$

Choosing the completion and residual exponents with $B' > A'$ gives the contradiction. $\square$

Definition 16.173 (Maximal source-graded affine discarded target). This label is retained for compatibility with earlier citations. Define the legacy two-summand target

$$\widehat{\mathcal{H}}_p^{Y,\text{aff}} := \mathcal{H}_{p,\text{lin}}^Y \oplus \mathcal{H}_{p,\eta\eta}^Y,$$

which is the special case $\mathfrak{A}_{K9,p} = \{\eta\eta\}$ of Definition 16.163; it embeds as an orthogonal sub-target of $\widehat{\mathcal{H}}_p^{Y,\text{max}}$. In the active branch the same-norm comparison is carried out in the maximal target $\widehat{\mathcal{H}}_p^{Y,\text{max}}$, not in this two-summand sub-target.

Set

$$P_p^{\text{full}} := P_{R_p + \text{im } H_{F,p}}$$

in the canonical source-graded Hilbert space $H_{K9,p}^{\text{gr}}$, and let

$$\widehat{\Lambda}_{\text{aff},p} := \Pi_{\text{max},p}^{\text{gr}}(I - P_p^{\text{full}})\mathcal{L}_{K9,p}^{\text{gr}} J_{\text{tr},p}$$

denote the canonical-branch affine discarded observable. When the symbol $\widehat{\Lambda}_{\text{aff},p}$ appears in the active proof it refers to this canonical-branch operator; the legacy two-summand observable $\widehat{\Phi}_{p,\text{aff}}^Y$ of Definition 16.359 is the corresponding sub-target projection.

Remark 16.174 (Scalar completion is not enough). The source-graded affine target is deliberately not a scalar collapse. The same-norm branch is carried out in $H_{K9,p}^{\text{gr}}$, with finite-main labels in $H_{p,\text{main}}$ and affine live labels in $H_{p,\text{aff}}$. A scalar functional may see $x + y$, whereas the source-graded norm sees $\|x\|^2 + \|y\|^2$. Consequently a primitive-only Y12 completion or scalar I -coordinate completion cannot replace the source-graded full-vector K9 completion used below.

Remark 16.175 (Projection notation in the active affine branch). In the active branch the old symbol $\Pi_{\text{aff},p}^{\text{gr}}$ is only a compatibility name for $\Pi_{\text{max},p}^{\text{gr}}$. All same-norm upper/lower estimates are made with the maximal source-graded affine projection

$$\Pi_{\text{max},p}^{\text{gr}} : H_{K9,p}^{\text{gr}} \rightarrow \widehat{\mathcal{H}}_p^{Y,\text{max}}.$$

This prevents accidental reversion to the earlier two-summand affine target or to a scalar-collapsed primitive $Y12$ projection.

Lemma 16.176 (Finite source-graded direct-sum amplification of certified coercivity). *Let $A_p : \mathbb{C}^2 \rightarrow H_p$ and $q_p := I - P_{R_p}$. Suppose*

$$\sigma_{\min}(q_p A_p) \geq Q^{-C}.$$

Let Λ_p be a finite source-label set with $|\Lambda_p| \leq Q^C$, and let $M_{\lambda,p}$ be polynomially conditioned live-coordinate maps:

$$\sigma_{\min}(M_{\lambda,p}) \geq Q^{-C}, \quad \|M_{\lambda,p}\|_{\text{op}} \leq Q^C.$$

For the orthogonal direct sum

$$A_p^\Lambda(a_\lambda)_\lambda := (q_p A_p M_{\lambda,p} a_\lambda)_{\lambda \in \Lambda_p}$$

one has

$$\sigma_{\min}(A_p^\Lambda) \geq Q^{-C'}.$$

Proof. By orthogonality of the source labels,

$$\|A_p^\Lambda(a_\lambda)_\lambda\|_{\text{gr}}^2 = \sum_{\lambda \in \Lambda_p} \|q_p A_p M_{\lambda,p} a_\lambda\|^2 \geq Q^{-C'} \sum_{\lambda \in \Lambda_p} \|a_\lambda\|^2.$$

Taking the infimum over unit source-graded coefficient vectors gives the claim. The polynomial loss absorbs the finite cardinality and conditioning of the maps $M_{\lambda,p}$. $\square$

Definition 16.177 (Certified full-main retained space). In the active canonical branch the full-main retained space is defined in the source-graded $K9$ Hilbert space by

$$R_p^{\text{full}} := R_p + \text{im } H_{F,p}, \quad P_p^{\text{full}} := P_{R_p^{\text{full}}}.$$

Here $H_{F,p}$ is the finite-main labelled map whose image is spanned by the $K9$ -labels of $B_2, P_{2,\leq X}, D_{2,\leq X}$. The older construction that first replaced finite-main generators by their $C_7/Y12$ -row-dead parts is archived as a fallback for noncanonical or scalar-collapsed quotients. It is not the active branch once Definition 16.161 is in force.

Lemma 16.178 (Full-main retention preserves the affine Schur gap in the canonical branch). *Assume the canonical source-graded lift of Definition 16.161. Let $q_p := I - P_{R_p}$, let $E_p^{\text{aff}} : \mathbb{C}^d \rightarrow H_{K9,p}^{\text{gr}}$ be the maximal affine live carrier map, and let $H_{F,p}$ be the finite-main labelled map. If*

$$(E_p^{\text{aff}})^* q_p E_p^{\text{aff}} \geq Q^{-C} I_d,$$

then

$$(E_p^{\text{aff}})^* (I - P_p^{\text{full}}) E_p^{\text{aff}} \geq Q^{-C} I_d.$$

Proof. By source-grade saturation,

$$q_p H_{F,p} \subset H_{p,\text{main}}, \quad q_p E_p^{\text{aff}} \subset H_{p,\text{aff}}, \quad H_{p,\text{main}} \perp H_{p,\text{aff}}.$$

Thus the finite-main cross Gram block is zero:

$$G_{EH,p} := (E_p^{\text{aff}})^* q_p H_{F,p} = 0.$$

The Schur formula of Lemma 16.166 gives

$$(E_p^{\text{aff}})^* (I - P_p^{\text{full}}) E_p^{\text{aff}} = G_{EE,p} - G_{EH,p} G_{HH,p}^\dagger G_{EH,p}^* = G_{EE,p},$$

and $G_{EE,p} \geq Q^{-C} I_d$ by hypothesis. $\square$

Definition 16.179 (Affine full-main carrier map). The active affine carrier map is the maximal source-graded live carrier map

$$E_p^{\text{aff}} : \mathbb{C}^d \longrightarrow H_{K9,p}^{\text{gr}}$$

whose image lies in $H_{p,\text{aff}}$. Its first labelled component is the clean D4 $C_7/Y12$ carrier coefficient $H_p L_{\text{clean},p} v_p$; the remaining labelled components are the live affine forcing coordinates $m_{\lambda,p}$, including the odd- $\eta\eta$ / Bottleneck-C row. This maximal map supersedes the older two-copy map $(a, b) \mapsto (E_{\text{sing},p}^{2Y} a, E_{\text{sing},p}^{2Y} b)$, which is only its two-label special case.

Theorem 16.180 (Affine full-main Schur gap from certified δ_{coerc}). *In the canonical source-graded branch,*

$$\sigma_{\min}((I - P_p^{\text{full}})E_p^{\text{aff}}) \geq Q^{-A}.$$

Equivalently,

$$(E_p^{\text{aff}})^*(I - P_p^{\text{full}})E_p^{\text{aff}} \geq Q^{-A}I_d.$$

Proof. Certified δ_{coerc} supplies the two-carrier normal-row gap on $E_{\text{sing},p}^{2Y}$. Lemma 16.176 amplifies that gap over the finite source-label set that forms E_p^{aff} , giving $(E_p^{\text{aff}})^* q_p E_p^{\text{aff}} \geq Q^{-A}I_d$. Lemma 16.178 then passes the lower bound through full finite-main retention, because $G_{EH,p} = 0$ in the canonical labelled lift. $\square$

Proposition 16.181 (No-go for noncanonical full-main retention). *The old warning remains valid only outside the canonical source-graded branch: if a scalar-collapsed or non-source-graded retained space contains a nonzero live carrier direction $E_p^{\text{aff}} a$, then the projected affine carrier map has a kernel and the Schur gap fails.*

Proof. If $E_p^{\text{aff}} a \in R$ for some $a \neq 0$, then $(I - P_R)E_p^{\text{aff}} a = 0$. Hence $\sigma_{\min}((I - P_R)E_p^{\text{aff}}) = 0$. This is precisely why the active branch keeps finite-main and affine live directions in orthogonal source-grade summands before scalar collapse. $\square$

Remark 16.182 (Superseded row-dead finite-main formulation). The earlier row-dead finite-main formulation is no longer the active same-norm branch. In the canonical source-graded $K9$ lift, $F_X = B_2 + P_{2,\leq X} + D_{2,\leq X}$ is retained as a finite-main labelled subspace, while the affine live $C_7/Y12$ directions lie in orthogonal affine source-grade labels. Therefore the finite-main Schur complement is

$$S_{F,p} = G_{EE,p} - G_{EH,p} G_{HH,p}^{\dagger} G_{EH,p}^*, \quad G_{EH,p} = 0,$$

and hence

$$S_{F,p} = G_{EE,p} \geq Q^{-C}I$$

by certified affine coercivity. The proof does not require the finite main rows to be row-dead in the affine row extractor. It requires the canonical labelled lift before scalar collapse and the orthogonality of finite-main and affine labels. Thus any statement saying that the finite-main obstruction remains open, or that full finite-main retention is allowed only after replacing each generator by a row-dead part, should be deleted or replaced by Corollary 16.168 and Theorem 16.172.

Remark 16.183 (Legacy row-dead finite-main route superseded). This legacy label is retained for citations to the older finite-main row-deadification route. Its content is now subsumed by Remark 16.182 and Corollary 16.168. In the active branch, the full finite-main labelled model is retained in $H_{p,\text{main}}$, while affine live labels remain in the orthogonal source-grade $H_{p,\text{aff}}$.

Remark 16.184 (Scope of certified coercivity after the canonical lift). Certified low-block coercivity supplies the affine carrier lower bound

$$(E_p^{\text{aff}})^*(I - P_{R_p})E_p^{\text{aff}} \geq Q^{-C}I$$

in the source-graded singular Hilbert space. The canonical labelled lift and finite-main/affine grade separation imply

$$(E_p^{\text{aff}})^*(I - P_{R_p + \text{im } H_{F,p}})E_p^{\text{aff}} = (E_p^{\text{aff}})^*(I - P_{R_p})E_p^{\text{aff}},$$

because the mixed Gram block

$$(E_p^{\text{aff}})^*(I - P_{R_p})H_{F,p}$$

vanishes by orthogonality of $H_{p,\text{aff}}$ and $H_{p,\text{main}}$. Hence certified coercivity closes the Schur-gap side of the final affine branch after full finite-main retention. The row realization is also theorem-level in the canonical branch, because the maximal affine target contains every live affine $K9$ source-grade label; see Lemma 16.171.

Lemma 16.185 (Source-graded $K9$ completion implies affine-graded completion). *Assume $\text{EFCompletion}_{K9}^{\text{gr},\text{vec}}$. Then*

$$\sum_p \|\widehat{\Lambda}_{\text{aff},p}(R_X)\|_{\text{gr}}^2 \leq Q^{-B} \mu(\mathcal{C})$$

with the same exponent up to harmless polynomial loss. The same implication holds in the cap-local weighted form.

Proof. By Definition 16.173,

$$\widehat{\Lambda}_{\text{aff},p} = \Pi_{\text{max},p}^{\text{gr}}(I - P_p^{\text{full}})\mathcal{L}_{K9,p}^{\text{gr}}J_{\text{tr},p}.$$

Because P_p^{full} contains R_p , enlarging the retained space can only decrease the discarded norm:

$$\|(I - P_p^{\text{full}})x\|_{\text{gr}} \leq \|(I - P_{R_p})x\|_{\text{gr}}.$$

Since $\Pi_{\text{max},p}^{\text{gr}}$ is an orthogonal projection,

$$\|\widehat{\Lambda}_{\text{aff},p}(R_X)\|_{\text{gr}} \leq \|\Lambda_{K9,p}^{\text{gr},\text{vec}}(R_X)\|_{\text{gr}}.$$

Squaring and summing, with or without cap weights, gives the claim. $\square$

Lemma 16.186 (Polynomial finite-jet bound for source-graded certified $K9$ rows). *On an ordinary real-canonical, Gram-good, boundary-collar-excluded packet, the source-graded certified transition-singular row map satisfies*

$$\|\Lambda_{K9,p}^{\text{gr},\text{vec}}(f)\|_{\text{gr}}^2 \leq Q^{C_J} \sum_{0 \leq r \leq J} \int_{K_p^+} |\partial_t^r f(t)|^2 W_{\text{br}}(t) dt$$

for some fixed finite J and an absolute exponent C_J . Here W_{br} is any branch/reference weight in the packet class which is polynomially comparable to Lebesgue measure on K_p^+ .

Proof. The canonical lift $\mathcal{L}_{K9,p}^{\text{gr}}$ is a finite direct sum of labelled transition-singular rows. Each labelled row uses only finitely many transition jets, and the certified projections are finite-dimensional matrices with entries rational/analytic in the packet chart data. On the ordinary real-canonical, Gram-good, boundary-collar-excluded set all normalizing denominators are bounded below by a power of Q , while all numerators have polynomial size. The direct-sum norm has only polynomially many source labels, so the row-map norm is $O(Q^C)$ times the norm of finitely many point jets of f . The one-dimensional local Sobolev inequality on K_p^+ , together with polynomial scale bounds and polynomial weight comparability on good packets, gives the displayed estimate after choosing J larger than the maximum jet order used by the certified $K9$ table. $\square$

Theorem 16.187 (Source-graded full-vector $K9$ EF completion from the smooth AFE tail). *Assume the smooth finite AFE tail estimate gives, for every fixed J and every $A_0 > 0$,*

$$\sum_{0 \leq r \leq J} \|\partial_t^r R_X\|_{L^2(W_{\text{br}})}^2 \leq X^{-A_0}.$$

Assume the certified packet family has bounded overlap. Then $\text{EFCompletion}_{K9}^{\text{gr},\text{vec}}$ holds. More precisely, for every prescribed B one can choose the AFE cutoff depth so that

$$\sum_p \|\Lambda_{K9,p}^{\text{gr},\text{vec}}(R_X)\|_{\text{gr}}^2 \leq Q^{-B} \mu(\mathcal{C}).$$

The same conclusion holds in the cap-local weighted form whenever the cap quadrature/core-transfer estimate compares the discrete cap measure $\sum_{p \in C_j} \mu_p \delta_{m_p}$ with $W_{\text{br}}(t) dt$ for the finite collection of functions $|\partial_t^r R_X|^2$, $0 \leq r \leq J$, with polynomial loss.

Proof. Apply Lemma 16.186 with $f = R_X$, and sum over p . Bounded overlap of the enlarged packet cores gives

$$\sum_p \|\Lambda_{K9,p}^{\text{gr,vec}}(R_X)\|_{\text{gr}}^2 \leq Q^{C'_J} \sum_{0 \leq r \leq J} \|\partial_t^r R_X\|_{L^2(W_{\text{br}})}^2 \leq Q^{C'_J} X^{-A_0}.$$

Choose A_0 and the cutoff depth so that $Q^{C'_J} X^{-A_0} \leq Q^{-B} \mu(\mathcal{C})$. The cap-local weighted version is the same argument after replacing bounded overlap by the stated cap quadrature/core-transfer estimate; the polynomial transfer loss is absorbed into the exponent. $\square$

Definition 16.188 (Proof-level source-graded discard). In the canonical source-graded $K9$ branch set

$$P_p^{\text{full}} := P_{R_p + \text{im } H_{F,p}}, \quad \Lambda_p^{\text{gr,full}} := (I - P_p^{\text{full}}) \mathcal{L}_{K9,p}^{\text{gr}} J_{\text{tr},p}.$$

The maximal affine projection of this observable is

$$\widehat{\Lambda}_{\text{max},p} := \Pi_{\text{max},p}^{\text{gr}} \Lambda_p^{\text{gr,full}}.$$

The comparison norm is the Hilbert norm of $H_{K9,p}^{\text{gr}}$, or the orthogonal projected norm on the maximal affine summand. No scalar I -coordinate or primitive-only $Y12$ collapse is made before the contradiction.

Theorem 16.189 (Source-graded same-norm Type-I upper). Assume the canonical source-graded $K9$ lift, full finite-main retention through P_p^{full} , and the cap-local form of $\text{EFCompletion}_{K9}^{\text{gr,vec}}$ for the AFE tail R_X . Then, for every prescribed upper exponent B_{up} , after choosing the AFE cutoff depth sufficiently large,

$$\sum_{p \in C_j} \mu_p \|\Lambda_p^{\text{gr,full}}(q''_{\text{can}})\|_{\text{gr}}^2 \leq Q^{-B_{\text{up}}} \sum_{p \in C_j} \mu_p \|v_p\|^2.$$

The same estimate holds with $\Lambda_p^{\text{gr,full}}$ replaced by $\widehat{\Lambda}_{\text{max},p}$. Here $v_p = \Gamma_p$ in the nonstatic branch, and v_p is the normalized static direction in the typed static-line branch.

Proof. By definition of $H_{F,p}$, the source-graded finite-main row satisfies

$$\mathcal{L}_{K9,p}^{\text{gr}} J_{\text{tr},p}(F_X) \in \text{im } H_{F,p}.$$

Therefore

$$\Lambda_p^{\text{gr,full}}(F_X) = 0.$$

Using the EF split $q''_{\text{can}} = F_X + R_X$ and linearity of the canonical lift,

$$\Lambda_p^{\text{gr,full}}(q''_{\text{can}}) = \Lambda_p^{\text{gr,full}}(R_X).$$

Since P_p^{full} retains at least the certified retained space used in $\text{EFCompletion}_{K9}^{\text{gr,vec}}$, enlarging the retained space can only decrease the discarded norm. Hence the cap-local full-vector $K9$ completion estimate for R_X gives the displayed upper. The maximal affine estimate follows because $\Pi_{\text{max},p}^{\text{gr}}$ is an orthogonal projection. $\square$

Theorem 16.190 (Maximal affine row realization in the canonical branch). Let

$$z_p := (H_p L_{\text{clean},p} v_p, (m_{\lambda,p})_{\lambda \in \mathfrak{A}_{K9,p}})$$

be the maximal affine coefficient vector. In the canonical labelled lift, on the selected cap core,

$$\widehat{\Lambda}_{\text{max},p}(q''_{\text{can}}) = \Pi_{\text{max},p}^{\text{gr}} (I - P_p^{\text{full}}) E_p^{\text{aff}} z_p + \mathcal{R}_{p,\text{max}}^{\text{id}},$$

and the residual ledger gives

$$\sum_{p \in C_j} \mu_p \|\mathcal{R}_{p,\text{max}}^{\text{id}}\|_{\text{gr}}^2 \leq Q^{-B_{\text{id}}} \sum_{p \in C_j} \mu_p \|v_p\|^2$$

for arbitrarily large prescribed B_{id} , after the standard residual-depth choices.

Proof. The labelled $K9$ row table is a finite orthogonal direct sum of four classes: finite-main labels, affine live labels, dead/retained labels, and residual labels. The finite-main labels lie in $\text{im } H_{F,p}$ and are killed by $I - P_p^{\text{full}}$. The dead/retained labels lie in R_p and are also killed. The projection $\Pi_{\text{max},p}^{\text{gr}}$ keeps exactly the maximal affine live label set $\mathfrak{A}_{K9,p}$ together with the clean D4 affine label. Since this set is maximal, no same-grade affine companion remains outside the target. The surviving affine coefficients are precisely the clean coefficient $H_p L_{\text{clean},p} v_p$ and the retained forcing coefficients $(m_{\lambda,p})_{\lambda \in \mathfrak{A}_{K9,p}}$. Everything else is in the residual-labelled ledger, which is measured in the same source-graded Hilbert norm and gives the displayed bound. $\square$

Corollary 16.191 (Localized source-graded singular-overlap completion). *In the canonical source-graded $K9$ branch, localized singular-overlap completion is a theorem:*

$$\sum_{p \in C_j} \mu_p \|\hat{\Lambda}_{\text{max},p}(q''_{\text{can}})\|_{\text{gr}}^2 \leq Q^{-B_{\text{up}}} \sum_{p \in C_j} \mu_p \|v_p\|^2.$$

It is the orthogonal maximal-affine projection of Theorem 16.189.

Proof. Apply Theorem 16.189 and project by $\Pi_{\text{max},p}^{\text{gr}}$. $\square$

Remark 16.192 (Tail-only no-go retained only for noncanonical formulations). The finite-dimensional warning that R_X -completion alone does not control $q''_{\text{can}} = F_X + R_X$ remains valid if the finite-main model is not retained in the same source-graded norm. It no longer represents an active gap in the canonical branch, because P_p^{full} retains the full finite-main labelled model and Corollary 16.168 shows that this retention does not destroy the affine live Schur gap.

Remark 16.193 (Consolidated status of the affine-forcing repair: discharged). The nonzero primitive arithmetic coefficient is no longer a quotient-rank obstruction: the live odd- $\eta\eta$ row is retained as an affine source-grade component rather than killed. The six-row effective-carrier table remains separate and is not used to absorb this affine forcing. The apparent finite-main obstruction is the Schur complement

$$(E_p^{\text{aff}})^*(I - P_{R_p + \text{im } H_{F,p}})E_p^{\text{aff}} = G_{EE,p} - G_{EH,p}G_{HH,p}^{\dagger}G_{EH,p}^*.$$

In the active proof branch, the source-graded $K9$ lift is a genuine fixed linear labelled lift before scalar collapse. Therefore finite-main labels and affine live labels are orthogonal, $G_{EH,p} = 0$, and the finite-main Schur complement equals $G_{EE,p}$. Certified affine coercivity gives the polynomial lower bound. Hence the affine-forcing same-norm branch is closed by Theorem 16.172.

Remark 16.194 (Ledger update: finite-main obstruction closed in the canonical branch). In the canonical source-graded $K9$ branch, the following items are now closed:

live arithmetic row torsion repair, finite-main Schur complement, same-norm affine upper/lower comp

The closure depends essentially on keeping the proof in the labelled source-graded norm until the contradiction is obtained. A scalar I -coordinate or primitive-only $Y12$ completion is not a substitute for this same-norm comparison.

Remark 16.195 ($\text{EFCompletion}_{L^2}$ does not automatically imply EFCompletion_{K9}). The implication

$$\text{EFCompletion}_{L^2} \implies \text{EFCompletion}_{K9}$$

requires a polynomial operator bound for

$$\Lambda_{\text{tr},\text{sing}} J_{\text{tr}}$$

from the L^2 -completion norm to the discarded singular norm. If such a bound is unavailable, the singular-discard upper remains open even when aggregate L^2 Gate-1 is well-defined. This is a separate completion ledger and should not be merged with the source-energy Gate-1 completion.

Definition 16.196 (Certified source-graded D4 transversality matrix). In the active source-graded branch the D4 transversality matrix is

$$K_{\text{D4},p}^{\text{gr}} := (I - P_p^{\text{full}})\mathcal{L}_{K9,p}^{\text{gr}}E_{\text{car},p}^{\text{tr}}L_{\text{eff},p}^{\text{tr}}.$$

Equivalently, after projecting to the maximal affine labels,

$$\Pi_{\max,p}^{\text{gr}} K_{\text{D4},p}^{\text{gr}} = \Pi_{\max,p}^{\text{gr}} (I - P_p^{\text{full}}) E_p^{\text{aff}} (H_p L_{\text{clean},p} \quad (m_{\lambda,p})_{\lambda}),$$

with the evident interpretation of the affine coefficient vector. The decisive rank/singular-value lower bound is the source-graded Schur gap of Corollary 16.168, combined with the effective-carrier determinant. The older raw matrix $(I - P_{\text{sing},p}) T_{\text{tr} \rightarrow \text{sing},p} E_{\text{car},p}^{\text{tr}} L_{\text{clean},p}^{\text{tr}}$ is only an archived diagnostic for noncertified singular projections.

Remark 16.197 (Legacy scalar-discard and primitive-Y12 branches). The definitions and hypotheses below concerning scalar K_9 , primitive Y12, moving Y12-walls, and amplitude Schur tests are retained as archived diagnostics and possible fallback channels. They are not the active same-norm final bridge after Definition 16.161. The active proof-level observable is $\Lambda_p^{\text{gr},\text{full}}$ and its maximal affine projection $\hat{\Lambda}_{\max,p}$. Any scalar or primitive-only statement must be reproved as an orthogonal projection of the canonical labelled K_9 observable before it can be used in the final contradiction.

Definition 16.198 (Singular order-nine connection residual). Let $s = (s_0, \dots, s_9)$ denote the singular coefficient vector in the transition-to-singular overlap chart. Suppose the actual EF-coherent singular source satisfies an order-nine connection relation of the form

$$s_9 = \mathcal{R}_{7 \rightarrow 9} s_7 + \mathcal{R}_{<7 \rightarrow 9}(s_{<7}) + \text{Tail}_9.$$

Define the singular order-nine residual

$$\mathcal{K}_9(s) := s_9 - \mathcal{R}_{7 \rightarrow 9} s_7 - \mathcal{R}_{<7 \rightarrow 9}(s_{<7}).$$

The associated forbidden operator is

$$\Lambda_{K_9,p} := \mathcal{K}_9 \circ T_{\text{tr} \rightarrow \text{sing},p} P_{\text{tr},p}.$$

It is a valid singular-discard channel only if actual EF-coherent rows satisfy

$$\sum_p |\mathcal{K}_9(T_{\text{tr} \rightarrow \text{sing},p} J_{\text{tr},p}(q_{\text{can}}''))|^2 \leq Q^{-B} \mu(\mathcal{C}),$$

and if the D4 live plane has nonzero normal image:

$$\text{rank}(\mathcal{K}_9 T_{\text{tr} \rightarrow \text{sing},p} E_{\text{car},p}^{\text{tr}} L_{\text{clean},p}^{\text{tr}}) \geq 1$$

on a polynomial-weight set of packets.

Definition 16.199 (Lower-order tangent observability for K_9). Let the actual-source tangent matrix in the singular order-nine chart be written in block form

$$T = \begin{pmatrix} T_L \\ T_9 \end{pmatrix},$$

where T_L records the lower-order coordinates and T_9 records the order-nine coordinate. The residual

$$\mathcal{K}_9(s) = s_9 - \mathcal{R}_{7 \rightarrow 9} s_7 - \mathcal{R}_{<7 \rightarrow 9}(s_{<7})$$

is polynomially conditioned if the lower-order block observes every actual-source tangent:

$$\boxed{|T_L a| \geq Q^{-A} |T_9 a| \quad \text{for every actual-source tangent } a.}$$

Equivalently, the connection operator

$$\mathcal{R} := T_9 T_L^\dagger$$

exists on the relevant tangent image with

$$\|\mathcal{R}\| \leq Q^A.$$

Remark 16.200 (Why K_9 conditioning is separate from $Y12$ -nonvanishing). The condition

$$K_9(T_{\text{tr} \rightarrow \text{sing}}[C_{Y12}]^{\text{tr}}) \neq 0$$

is a signature nonvanishing test. It does not by itself ensure that the normal functional K_9 is quantitatively usable. If an actual-source tangent is almost purely order nine, then K_9 may have super-polynomial norm after quotienting lower-order directions. Thus the K_9 -route requires both:

$$\boxed{K_9(T_{\text{tr} \rightarrow \text{sing}}[C_{Y12}]^{\text{tr}}) \neq 0} \quad \text{and} \quad \boxed{\text{LowerOrderTangentObservability}_{K_9}}.$$

Remark 16.201 (The $Y12$ channel is the preferred K_9 test). For the rank-two D4 output

$$H \left(\begin{matrix} c_7 \Gamma_G \\ \beta_9^{\text{tri}} \Gamma_G + \alpha_9 \Gamma_{G,2} \end{matrix} \right),$$

the cleanest singular-discard test is

$$\mathcal{K}_9(T_{\text{tr} \rightarrow \text{sing}}[C_{Y12}]^{\text{tr}}) \neq 0.$$

If this holds, then the pure $\Gamma_{G,2}$ channel survives in the discarded singular normal direction, because $\alpha_9 \neq 0$ on D4 coefficient-good charts. If it fails, one should next test the Γ_G -channel through

$$\mathcal{K}_9(T_{\text{tr} \rightarrow \text{sing}}[C_7]^{\text{tr}}).$$

If both tests vanish, the order-nine singular residual $\mathcal{K}_9$ is rank-zero for finite D4 and cannot serve as the downstream forbidden channel.

Remark 16.202 (E_2^{Y12} honesty audit). Before claiming that the $Y12$ -normal direction is forbidden, one must audit the primitive $Y12$ -signature of the actual completion/curvature term:

$$\Pi_{Y12} E_2.$$

There are three cases.

$$\boxed{\text{Case A: } \Pi_{Y12} E_2 = 0 \text{ or } Q^{-B}\text{-small.}}$$

Then $Y12$ is a strong forbidden-channel candidate, and the actual-row upper requires only

$$\Pi_{Y12} T_{\text{tr} \rightarrow \text{sing}} J_{\text{tr}}(R_X)$$

to be small.

$$\boxed{\text{Case B: } \Pi_{Y12} E_2 \neq 0 \text{ but fixed/coherent.}}$$

Then this piece must be included in the retained actual-source model or in R_{sing} . The audit becomes whether its admissible variations span the same primitive $Y12$ direction as the D4 carrier.

$$\boxed{\text{Case C: } \Pi_{Y12} E_2 \text{ supplies a free primitive } Y12 \text{ amplitude.}}$$

Then the $Y12$ -normal D4 channel is tangent/coherent and the rank-one $Y12$ -forbidden route is dead.

Thus the finite audit is not merely

$$K_9(T_{\text{tr} \rightarrow \text{sing}}[C_{Y12}]) \neq 0,$$

but also whether actual E_2 -coherence already contains that same primitive direction.

Remark 16.203 (Coherent E_2^Y and the final $Y12$ -normal wall). The $Y12$ -normal D4 channel is safe only after the E_2^Y component is classified. There are three cases.

First, E_2^Y may be absent, fixed, or residual-controlled. Then it does not modify the local forbidden wall.

Second, E_2^Y may be coherent and graph-like. In this case it modifies the fiber graph slope

$$\rho_Y \longmapsto \rho_Y^{\text{final}},$$

and hence modifies the rank-one wall ratio to

$$\rho_{\Gamma}^{\text{final}} = \frac{\beta_9^{\text{tri}} - \rho_Y^{\text{final}} c_7}{\alpha_9}.$$

This is safe only if ρ_Y^{final} remains low-order/coherent and does not depend on the actual $\Gamma_{G,2}$ -wall value.

Third, E_2^Y may generate an independent finite-dimensional coherent amplitude space

$$\mathcal{C}_Y^{E_2}.$$

Then the actual forbidden wall is the Schur projection

$$F_Y^{\text{eff}} = (I - P_{\mathcal{C}_Y^{E_2}}^{\text{bd}}) [H((\beta_9^{\text{tri}} - \rho_Y c_7)\Gamma_G + \alpha_9 \Gamma_{G,2})].$$

The branch-charge bandwidth d_0 used in finite/O2/O3 must be computed from F_Y^{eff} , not from the unprojected wall. If E_2^Y is locally free at extraction scale, then $Y12$ is actual-source coherent and the normal-D4 channel is killed.

Hypothesis 16.204 (AmplitudeSchur $_{Y12}$). Let $\mathcal{C}_Y^{\text{arith}}$ be the finite coherent $Y12$ -amplitude space remaining after exact nuisance and residual routing. Under the bounded-degree Stack weight $d\nu_{\text{bd}}$, the effective $Y12$ -wall satisfies

$$\text{dist}_{L^2(d\nu_{\text{bd}})} \left(H((\beta_9^{\text{tri}} - \rho_Y c_7)\Gamma_G + \alpha_9 \Gamma_{G,2}), \mathcal{C}_Y^{\text{arith}} \right)^2 \geq Q^{-A} \|H((\beta_9^{\text{tri}} - \rho_Y c_7)\Gamma_G + \alpha_9 \Gamma_{G,2})\|_{L^2(d\nu_{\text{bd}})}^2.$$

If this fails, the coherent amplitude model absorbs the $Y12$ -normal wall.

Definition 16.205 (Order-nine $Y12$ normal quotient). Let Y_9 denote the order-nine singular carrier space containing the transitioned primitive $Y12$ -carrier. Define the forbidden order-nine quotient

$$\mathcal{Q}_9^{\text{forb}} := Y_9 / (T_{\text{par}} + T_7 + T_{<7}),$$

where T_{par} denotes actual parameter tangents, T_7 denotes actual or lower order-seven connection tangents, and $T_{<7}$ denotes lower typed dead directions. The primitive $Y12$ -normal D4 channel survives exactly when

$$[\mathcal{C}_{Y12}] \neq 0 \quad \text{in} \quad \mathcal{Q}_9^{\text{forb}}.$$

Remark 16.206 (Relation between singular-discard and $K_9/Y12$). The $K_9/Y12$ channel is a specialization of the singular-discard candidate. The finite test

$$(I - P_{\text{sing}})T_{\text{tr} \rightarrow \text{sing}}[\mathcal{C}_{Y12}]^{\text{tr}} \neq 0$$

is equivalent to the existence of some normal K_9 detecting the transitioned $Y12$ -class. The quotient $\mathcal{Q}_9^{\text{forb}}$ records precisely which actual-source and lower connection directions must be divided out before $Y12$ is considered a genuine forbidden normal carrier.

If $[\mathcal{C}_{Y12}] = 0$ in $\mathcal{Q}_9^{\text{forb}}$, then the rank-one $Y12$ -normal route is dead, even if D4 remains locally visible in a larger carrier quotient. If $[\mathcal{C}_{Y12}] \neq 0$, then finite D4 has a concrete atomic forbidden channel.

Hypothesis 16.207 (Moving $Y12$ -wall anti-concentration). Let

$$\ell_p(\Gamma) := a_p \Gamma_G + b_p \Gamma_{G,2}$$

be the moving linear form obtained from the $Y12$ -signature D4 channel, with

$$|b_p| \geq Q^{-A}$$

on the extracted set. Under the post-Gate/O2/O3 measure ν , assume

$$\boxed{\int |\ell_p(\Gamma)|^2 d\nu \geq Q^{-A} \int (|\Gamma_G|^2 + |\Gamma_{G,2}|^2) d\nu.}$$

Equivalently, the post-Gate/O2/O3 measure is not concentrated on the moving sublevel set

$$\{p : |a_p \Gamma_G + b_p \Gamma_{G,2}| \leq Q^{-A} |\Gamma|\}.$$

In the normalized case $a_p = 0$, this reduces to anti-concentration of $\Gamma_{G,2}$.

Remark 16.208 (Rank-one Y12 normal channel is archived in the canonical branch). Hypothesis 16.204, Hypothesis E.116, and Hypothesis 16.207 belong to the older rank-one primitive Y12-normal route. They are useful diagnostics and may still support an alternate one-row forbidden wall, but they are not active assumptions in the canonical source-graded K9 branch. The proof-level same-norm contradiction instead uses the maximal affine target, Corollary 16.168, Theorem 16.189, and Theorem F.5. Thus scalar or primitive-only Y12 hypotheses should not be listed as remaining obligations for the canonical branch.

Remark 16.209 (Signed cap weights are not the active covariance object). The formula

$$w_p = \rho_p \frac{E_{1,p} H_p}{\mathcal{N}_p^{\text{wt}}}$$

may occur naturally as a wall or row coefficient, but it is not the active covariance density unless the packet branch is sign-coherent. In the active branch all covariance statements use the positive packet covariance matrix

$$B_j^{\text{cov}} = \sum_{p \in C_j} \mu_p^{\text{pkt}} \Gamma_p \Gamma_p^*.$$

Consequently, references to $B_j = \sum w_p \Gamma_p \Gamma_p^*$ in covariance arguments should be read as the sign-coherent special case, or replaced by B_j^{cov} . The signed matrix has no variance meaning in general.

Remark 16.210 (Legacy sign-coherent covariance convention). The formula

$$B_j = \sum_{p \in C_j} w_p \Gamma_p \Gamma_p^*, \quad w_p = \rho_p \frac{E_{1,p} H_p}{\mathcal{N}_p^{\text{wt}}},$$

defines a positive covariance matrix only on a sign-coherent branch where $w_p \geq 0$ and $\sum_{p \in C_j} w_p |\Gamma_G(p)|^2 > 0$. This sign-coherent case is a legacy special case of the active positive-density formulation below. In the canonical proof branch the covariance density is the positive packet quadrature density $\omega_p = \mu_p^{\text{pkt}}$, not the signed wall coefficient w_p , unless the branch has already been restricted so that w_p is nonnegative up to a common phase.

Lemma 16.211 (Exact cap covariance identity). *Let C_j be a frozen cap and suppose $w_p \geq 0$. Write*

$$\Gamma_p = \begin{pmatrix} g_p \\ h_p \end{pmatrix}, \quad s_p := h_p / g_p$$

on a chart where $g_p \neq 0$, and put

$$\mu_p := w_p |g_p|^2, \quad S_j := \sum_{p \in C_j} \mu_p.$$

Assume $S_j > 0$, and define

$$\bar{s}_j := S_j^{-1} \sum_{p \in C_j} \mu_p s_p.$$

Then

$$\det \left(\sum_{p \in C_j} w_p \Gamma_p \Gamma_p^* \right) = S_j \sum_{p \in C_j} \mu_p |s_p - \bar{s}_j|^2.$$

If in addition

$$|g_p| \geq \frac{1}{2} \|\Gamma_p\| \quad (p \in C_j),$$

then

$$\frac{1}{16} \frac{\sum_p \mu_p |s_p - \bar{s}_j|^2}{\sum_p \mu_p} \leq \frac{\det(B_j)}{(\text{tr } B_j)^2} \leq \frac{\sum_p \mu_p |s_p - \bar{s}_j|^2}{\sum_p \mu_p}.$$

Proof. Since $h_p = g_p s_p$,

$$\Gamma_p \Gamma_p^* = |g_p|^2 \begin{pmatrix} 1 & \bar{s}_p \\ s_p & |s_p|^2 \end{pmatrix}.$$

Thus

$$B_j = \begin{pmatrix} S_j & \bar{M}_j \\ M_j & T_j \end{pmatrix}, \quad M_j = \sum_p \mu_p s_p, \quad T_j = \sum_p \mu_p |s_p|^2.$$

Therefore

$$\det(B_j) = S_j T_j - |M_j|^2 = S_j \sum_p \mu_p |s_p - \bar{s}_j|^2.$$

The chart condition gives

$$S_j \leq \text{tr } B_j = \sum_p \mu_p (1 + |s_p|^2) \leq 4S_j.$$

Dividing by $(\text{tr } B_j)^2$ gives the claimed comparison. $\square$

Lemma 16.212 (Exact extraction covariance dichotomy). *Let C_j be a good frozen cap. Assume $w_p \geq 0$, write*

$$\Gamma_p = \begin{pmatrix} g_p \\ h_p \end{pmatrix} = g_p \begin{pmatrix} 1 \\ s_p \end{pmatrix}, \quad \mu_p := w_p |g_p|^2, \quad S_j := \sum_{p \in C_j} \mu_p > 0,$$

and set

$$\bar{s}_j := S_j^{-1} \sum_{p \in C_j} \mu_p s_p.$$

By Lemma 16.211,

$$\det(B_j) = S_j \sum_{p \in C_j} \mu_p |s_p - \bar{s}_j|^2.$$

If $|g_p| \geq \frac{1}{2} \|\Gamma_p\|$ on C_j , then

$$\frac{1}{16} \frac{\sum_p \mu_p |s_p - \bar{s}_j|^2}{S_j} \leq \delta_{\text{ext}} \leq \frac{\sum_p \mu_p |s_p - \bar{s}_j|^2}{S_j}.$$

Consequently, for every threshold Q^{-A} , either

$$\delta_{\text{ext}} \geq Q^{-A'}$$

or the cap is quantitatively static:

$$\sum_{p \in C_j} \mu_p |s_p - \bar{s}_j|^2 \leq Q^{-A} S_j,$$

in which case the static slope is $s_j^* := \bar{s}_j$.

Remark 16.213 (Legacy warning: scalar static branch). This warning is retained only for the older scalar or primitive-only Y12 route. In that route, the covariance identity by itself gives only a nonstatic/static dichotomy, and one-wall anti-concentration does not rule out a rank-one cap

$$\Gamma_p = A_p(1, s_0)^T.$$

In the active canonical branch this issue is superseded by the positive packet-density formulation and the source-graded affine comparison:

$$B_j^{\text{cov}} = \sum_{p \in C_j} \omega_p \Gamma_p \Gamma_p^*, \quad \omega_p = \mu_p^{\text{pkt}},$$

together with Theorem 16.219 and Theorem 16.172. Thus the static branch is not an active obstruction in the canonical source-graded K9 branch.

Corollary 16.214 (Static-line Y12 quotient survival in the certified quotient). *Let C_j be a quantitatively static good cap and let $u_j = (1, s_j^*)^T / (1 + |s_j^*|^2)^{1/2}$ be its static direction. In the certified branch set*

$$\Phi_{p,\text{cert}}^Y := \Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing},p} E_{\text{car},p}^{\text{tr}} L_{\text{eff},p}^{\text{tr}}.$$

Assume the coefficient-good expansion of the D4/Y12 carrier and the certified quotient/effective-carrier bounds from Theorem 16.280. Then

$$\sum_{p \in C_j} \mu_p \|q_{p,\text{cert}}^Y \Phi_{p,\text{cert}}^Y u_j\|^2 \geq Q^{-A} \sum_{p \in C_j} \mu_p \|\Phi_{p,\text{cert}}^Y u_j\|^2.$$

Thus the former static-line survival hypothesis is discharged in the certified quotient. In the active proof branch this relative quotient-survival statement is superseded by the source-graded affine comparison of Theorem 16.172, which supplies the same-norm Type-I contradiction after the static direction is routed into the maximal affine target.

Proof. For each packet, Theorem 16.288 gives $\sigma_{\min}(q_{p,\text{cert}}^Y E_p) \geq Q^{-A}$, and the effective carrier certificate gives $\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-A}$. The carrier expansion writes $\Phi_{p,\text{cert}}^Y = H_p E_p L_{\text{eff},p} + \varepsilon_p$, with the same quotient-error ledger used in Theorem 16.252. Applying the certified lower bound to the fixed unit vector u_j , summing over C_j , and absorbing the quotient error for sufficiently large residual exponent gives the displayed inequality. The statement is typed because both sides live in the Y12 boundary quotient, not in the abstract line $\mathbb{C}u_j$. $\square$

16.24.1. *Positive-density extraction covariance and static-line handoff.* The scalar covariance argument is an order argument. It is therefore made with a nonnegative cap density. On a frozen cap set

$$\omega_p \geq 0, \quad B_j^{\text{cov}} := \sum_{p \in C_j} \omega_p \Gamma_p \Gamma_p^*, \quad \delta_{\text{ext}}^{\text{cov}} := \frac{\det B_j^{\text{cov}}}{(\text{tr } B_j^{\text{cov}})^2}.$$

If the original extraction coefficient $w_p = \rho_p E_{1,p} H_p / \mathcal{N}_p^{\text{wt}}$ is already nonnegative on the selected branch, take $\omega_p = w_p$. If it is signed or complex, the covariance object is not the signed matrix $\sum w_p \Gamma_p \Gamma_p^*$; it is the positive-density replacement above, with ω_p supplied either by a sign-coherent branch or by a positive packet density polynomially comparable to the extracted mass.

Theorem 16.215 (Positive extraction density identification). *On every selected ordinary real-canonical good frozen cap C_j , the covariance density may be taken to be the positive cap/quadrature density already used in the source-graded K9 upper and in the D4 amplitude mass estimates. More precisely, let $\mu_p^{\text{pkt}} > 0$ denote that packet density and set*

$$\omega_p := \mu_p^{\text{pkt}}, \quad B_j^{\text{cov}} := \sum_{p \in C_j} \omega_p \Gamma_p \Gamma_p^*, \quad \mu_p^{\text{cov}} := \omega_p |g_p|^2.$$

On a good chart with

$$|g_p| \geq \frac{1}{2} \|\Gamma_p\|,$$

one has

$$\frac{1}{4} \mu_p^{\text{pkt}} \|\Gamma_p\|^2 \leq \mu_p^{\text{cov}} \leq \mu_p^{\text{pkt}} \|\Gamma_p\|^2.$$

Consequently

$$S_j^{\text{cov}} := \sum_{p \in C_j} \mu_p^{\text{cov}} > 0$$

for every selected cap carrying nonzero live mass. Moreover, the downstream estimates needed in the extraction/static-line branch transfer from the packet density to μ_p^{cov} by the displayed chart comparison, after only polynomial changes of exponents. In particular, source-weighted D4 amplitude survival in the packet density,

$$\sum_{p \in C_j} \mu_p^{\text{pkt}} |H_p|^2 \|\Gamma_p\|^2 \geq Q^{-A} \sum_{p \in C_j} \mu_p^{\text{pkt}} \|\Gamma_p\|^2,$$

implies the covariance-weighted form

$$\sum_{p \in C_j} \mu_p^{\text{cov}} |H_p|^2 \geq Q^{-A'} \sum_{p \in C_j} \mu_p^{\text{cov}}.$$

The signed extraction coefficient

$$w_p = \rho_p \frac{E_{1,p} H_p}{\mathcal{N}_p^{\text{wt}}}$$

is not the covariance density unless the selected branch is already sign-coherent. If w_p is signed or complex, it remains only a wall or row coefficient; it is not used to define a positive covariance matrix.

Proof. The packet density μ_p^{pkt} is positive by construction of the cap quadrature measure. On the selected good chart, $|g_p| \geq \frac{1}{2} \|\Gamma_p\|$, while trivially $|g_p| \leq \|\Gamma_p\|$. This gives the displayed pointwise comparison between μ_p^{cov} and $\mu_p^{\text{pkt}} \|\Gamma_p\|^2$. Since selected caps have nonzero live mass, $S_j^{\text{cov}} > 0$.

For the extraction/static-line branch we only need the carrier-mass estimates obtained from the same positive packet density. The displayed chart comparison transfers those estimates to μ_p^{cov} . Applying the pointwise comparison to the D4 amplitude estimate gives

$$\sum_p \mu_p^{\text{cov}} |H_p|^2 \geq \frac{1}{4} \sum_p \mu_p^{\text{pkt}} \|\Gamma_p\|^2 |H_p|^2 \geq Q^{-A-O(1)} \sum_p \mu_p^{\text{pkt}} \|\Gamma_p\|^2 \geq Q^{-A'} \sum_p \mu_p^{\text{cov}}.$$

This proves admissibility of the positive covariance density. The final sentence follows because a signed or complex weighted sum of rank-one matrices need not be positive semidefinite and therefore has no covariance or variance interpretation. $\square$

Lemma 16.216 (Exact positive-density cap covariance). *Let C_j be a frozen cap satisfying Theorem 16.215. On a chart with $g_p \neq 0$, write $\Gamma_p = g_p(1, s_p)^T$, $\mu_p^{\text{cov}} := \omega_p |g_p|^2$, $S_j^{\text{cov}} := \sum_{p \in C_j} \mu_p^{\text{cov}}$, and $\bar{s}_j^{\text{cov}} := (S_j^{\text{cov}})^{-1} \sum_{p \in C_j} \mu_p^{\text{cov}} s_p$. Then*

$$\det B_j^{\text{cov}} = S_j^{\text{cov}} \sum_{p \in C_j} \mu_p^{\text{cov}} |s_p - \bar{s}_j^{\text{cov}}|^2.$$

If $|g_p| \geq \frac{1}{2} \|\Gamma_p\|$ on C_j , then

$$\frac{1}{16} \frac{\sum_p \mu_p^{\text{cov}} |s_p - \bar{s}_j^{\text{cov}}|^2}{S_j^{\text{cov}}} \leq \delta_{\text{ext}}^{\text{cov}} \leq \frac{\sum_p \mu_p^{\text{cov}} |s_p - \bar{s}_j^{\text{cov}}|^2}{S_j^{\text{cov}}}.$$

Proof. The proof is the 2×2 covariance computation: $B_j^{\text{cov}} = \begin{pmatrix} S_j^{\text{cov}} & \overline{M_j^{\text{cov}}} \\ M_j^{\text{cov}} & T_j^{\text{cov}} \end{pmatrix}$ with $M_j^{\text{cov}} = \sum_p \mu_p^{\text{cov}} s_p$ and $T_j^{\text{cov}} = \sum_p \mu_p^{\text{cov}} |s_p|^2$. Thus $\det B_j^{\text{cov}} = S_j^{\text{cov}} T_j^{\text{cov}} - |M_j^{\text{cov}}|^2 = S_j^{\text{cov}} \sum_p \mu_p^{\text{cov}} |s_p - \bar{s}_j^{\text{cov}}|^2$. The chart condition gives $S_j^{\text{cov}} \leq \text{tr } B_j^{\text{cov}} = \sum_p \mu_p^{\text{cov}} (1 + |s_p|^2) \leq 4S_j^{\text{cov}}$, which gives the displayed comparison after division by $(\text{tr } B_j^{\text{cov}})^2$. $\square$

Corollary 16.217 (Positive-density nonstatic/static dichotomy). *Under the hypotheses of Lemma 16.216, for every threshold Q^{-A} either $\delta_{\text{ext}}^{\text{cov}} \geq Q^{-A'}$ or the cap is quantitatively static in the positive covariance density:*

$$\sum_{p \in C_j} \mu_p^{\text{cov}} |s_p - \bar{s}_j^{\text{cov}}|^2 \leq Q^{-A} S_j^{\text{cov}}.$$

In the static alternative set $s_j^* := \bar{s}_j^{\text{cov}}$, $u_j := (1, s_j^*)^T / (1 + |s_j^*|^2)^{1/2}$. On the same chart, after increasing the exponent if necessary,

$$\sum_{p \in C_j} \mu_p^{\text{cov}} \left\| \frac{(1, s_p)^T}{(1 + |s_p|^2)^{1/2}} - u_j \right\|^2 \leq Q^{-A'} S_j^{\text{cov}}.$$

Proof. The first assertion is Lemma 16.216. The final estimate follows because, on the chart $|g_p| \geq \frac{1}{2} \|\Gamma_p\|$, the map $s \mapsto (1, s)^T / (1 + |s|^2)^{1/2}$ is uniformly Lipschitz. $\square$

Proposition 16.218 (Static branch in the certified quotient). *Let C_j be in the static alternative of Corollary 16.217. Assume that all packets in C_j satisfy the concrete row/routing certificate $\mathfrak{B}_{2Y,p}$, so that Theorem 16.250 applies. Assume also the effective-carrier lower bound, the coefficient-good expansion $\Phi_{p,\text{cert}}^Y = H_p E_p L_{\text{eff},p} + \varepsilon_p$, the quotient-error estimate, and the source-weighted D4 amplitude survival with the positive covariance weights μ_p^{cov} . Then*

$$\sum_{p \in C_j} \mu_p^{\text{cov}} \|q_{p,\text{cert}}^Y \Phi_{p,\text{cert}}^Y u_j\|^2 \geq Q^{-A} \sum_{p \in C_j} \mu_p^{\text{cov}} \|\Phi_{p,\text{cert}}^Y u_j\|^2.$$

Proof. The rank-two row/routing certificate gives $\sigma_{\min}(q_{p,\text{cert}}^Y E_p) \geq Q^{-C}$. The effective carrier lower bound gives $\|L_{\text{eff},p} v\| \geq Q^{-C} \|v\|$ for every $v \in \mathbb{C}^2$. Applying these estimates to the fixed unit vector u_j , using the carrier expansion, summing with the weights μ_p^{cov} , applying source-weighted D4 amplitude survival, and absorbing the quotient error yields the stated inequality. The perturbative static estimate from Corollary 16.217 transfers the same lower bound from the actual directions to the frozen direction u_j . $\square$

Theorem 16.219 (Extraction covariance closure in positive packet density). *On every selected good frozen cap satisfying the good chart condition, with the positive packet covariance density of Theorem 16.215, at least one of the following alternatives applies. The alternatives may overlap at the threshold after exponent renaming; this harmless overlap is resolved by choosing either available branch.*

(1) *The nonstatic branch holds:*

$$\delta_{\text{ext}}^{\text{cov}} := \frac{\det B_j^{\text{cov}}}{(\text{tr } B_j^{\text{cov}})^2} \geq Q^{-A}.$$

(2) *The cap is quantitatively static: for*

$$s_j^* := \bar{s}_j^{\text{cov}}, \quad u_j := \frac{(1, s_j^*)^T}{(1 + |s_j^*|^2)^{1/2}},$$

one has

$$\sum_{p \in C_j} \mu_p^{\text{cov}} \left\| \frac{(1, s_p)^T}{(1 + |s_p|^2)^{1/2}} - u_j \right\|^2 \leq Q^{-A'} S_j^{\text{cov}},$$

and the static direction is routed into the canonical source-graded affine branch. Under the certified rank-two row/routing certificate, source-weighted D4 amplitude mass, full-vector source-graded K9 EFCompletion, and the canonical labelled-lift closure of Theorem 16.172, this static branch gives the same-norm Type-I contradiction.

Thus the extraction-covariance module is closed in the positive packet covariance formulation. It does not assert that the signed matrix $\sum w_p \Gamma_p \Gamma_p^$ is a covariance matrix, and it does not assert the false all-caps scalar statement $\delta_{\text{ext}} \geq Q^{-A}$ on static rank-one caps.*

Proof. The nonstatic/static dichotomy is Corollary 16.217, using the positive density supplied by Theorem 16.215. In the nonstatic alternative, Lemma 16.216 gives the lower bound for $\delta_{\text{ext}}^{\text{cov}}$, after renaming exponents.

In the static alternative, the Lipschitz chart estimate in Corollary 16.217 gives the fixed direction u_j . The D4 amplitude estimate in Theorem 16.215, together with the certified rank-two row/routing and effective carrier estimates, gives the source-graded lower bound for the affine coefficient vector whose clean component is $H_p L_{\text{clean},p} u_j$. The live affine forcing components are kept in orthogonal source-grade summands, so they cannot cancel the clean component. The same-norm upper and the finite-main Schur gap are exactly the content of Theorem 16.172, which uses the canonical source-graded K9 lift before scalar collapse. Choosing the upper and residual exponents larger than the lower losses contradicts nonzero cap mass. $\square$

Remark 16.220 (Signed extraction coefficients). If the original coefficients w_p are signed or complex, then $\sum_p w_p \Gamma_p \Gamma_p^*$ is not a positive covariance matrix, and $\det(B_j)/(\text{tr } B_j)^2$ has no variance interpretation. The literal signed matrix can be used only after a sign-coherent restriction

that makes it positive up to a common phase. Otherwise all cap covariance, static-slope, and cap-core estimates must be written with B_j^{cov} , $\delta_{\text{ext}}^{\text{cov}}$, and μ_p^{cov} .

Remark 16.221 (Status of the extraction covariance module). *Status: closed in the positive packet-density formulation.* The only covariance object used in the active proof is $B_j^{\text{cov}} = \sum \omega_p \Gamma_p \Gamma_p^*$, with $\omega_p = \mu_p^{\text{pkt}}$. Older wording that refers to a “positive extraction density selection” as an open finite audit should be read as superseded by Theorem 16.215. Older wording that refers to the signed coefficient w_p as a covariance weight is valid only on a sign-coherent branch; otherwise it must be replaced by ω_p throughout the covariance identity, static slope, and cap core estimates.

This status update is local to extraction covariance. Any remaining raw row/routing verification for $\mathfrak{B}_{2Y,p}$, or any upstream Gate-1/Gate-2 source input, remains recorded in its own ledger and is not part of the covariance identity.

Remark 16.222 (Legacy scalar static-line transfer lemmas). The following scalar Y12-transfer and static-wall escape lemmas are retained as fallback diagnostics for noncanonical or primitive-only routes. They are not active assumptions in the canonical source-graded $K9$ branch. The active static-line closure is supplied by Theorem 16.219, Theorem 16.172, and Corollary F.6.

Lemma 16.223 (Static-line transfer from actual Y12-survival). *Let C_j be a quantitatively static good cap. Put*

$$e_p := \frac{(1, s_p)}{\sqrt{1 + |s_p|^2}}, \quad u_j := \frac{(1, s_j^*)}{\sqrt{1 + |s_j^*|^2}},$$

and let

$$A_p := \Phi_p^Y = \Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing}, p} E_{\text{car}, p}^{\text{tr}} L_{\text{clean}, p}^{\text{tr}}, \quad Q_p := I - P_{\mathcal{K}_p^Y}^{\text{bd}}.$$

Assume the polynomial operator bounds

$$\|A_p\|_{\text{op}} + \|Q_p A_p\|_{\text{op}} \leq Q^{C_0},$$

the actual-vector Y12-quotient survival

$$(QS) \quad \sum_{p \in C_j} \mu_p \|Q_p A_p e_p\|^2 \geq Q^{-A_0} \sum_{p \in C_j} \mu_p \|A_p e_p\|^2,$$

and carrier noncollapse on the frozen static direction

$$(NC) \quad \sum_{p \in C_j} \mu_p \|A_p u_j\|^2 \geq Q^{-C_1} \sum_{p \in C_j} \mu_p.$$

If

$$(ST) \quad \sum_{p \in C_j} \mu_p \|e_p - u_j\|^2 \leq Q^{-B} \sum_{p \in C_j} \mu_p$$

with B sufficiently large depending on A_0, C_0, C_1 , then

$$\sum_{p \in C_j} \mu_p \|Q_p A_p u_j\|^2 \geq Q^{-A_2} \sum_{p \in C_j} \mu_p \|A_p u_j\|^2.$$

Proof. By the operator bound,

$$\sum_p \mu_p \|Q_p A_p (e_p - u_j)\|^2 \leq Q^{2C_0} \sum_p \mu_p \|e_p - u_j\|^2 \leq Q^{2C_0 - B} S_j,$$

and similarly

$$\sum_p \mu_p \|A_p (e_p - u_j)\|^2 \leq Q^{2C_0 - B} S_j.$$

Using $\|x + y\|^2 \geq \frac{1}{2}\|x\|^2 - \|y\|^2$,

$$\sum_p \mu_p \|Q_p A_p u_j\|^2 \geq \frac{1}{2} \sum_p \mu_p \|Q_p A_p e_p\|^2 - Q^{2C_0 - B} S_j.$$

By (QS),

$$\sum_p \mu_p \|Q_p A_p u_j\|^2 \geq \frac{1}{2} Q^{-A_0} \sum_p \mu_p \|A_p e_p\|^2 - Q^{2C_0-B} S_j.$$

A second perturbation gives

$$\sum_p \mu_p \|A_p e_p\|^2 \geq \frac{1}{2} \sum_p \mu_p \|A_p u_j\|^2 - Q^{2C_0-B} S_j,$$

hence

$$\sum_p \mu_p \|Q_p A_p u_j\|^2 \geq \frac{1}{4} Q^{-A_0} \sum_p \mu_p \|A_p u_j\|^2 - O(Q^{2C_0-B} S_j).$$

By (NC),

$$Q^{2C_0-B} S_j \leq Q^{2C_0+C_1-B} \sum_p \mu_p \|A_p u_j\|^2.$$

Choosing B so that $O(Q^{2C_0+C_1-B}) \leq \frac{1}{8} Q^{-A_0}$ gives

$$\sum_p \mu_p \|Q_p A_p u_j\|^2 \geq \frac{1}{8} Q^{-A_0} \sum_p \mu_p \|A_p u_j\|^2,$$

which is the claim after exponent renaming. $\square$

Lemma 16.224 (Static-line survival from primitive quotient and wall escape). *Let C_j be a quantitatively static good cap with static direction*

$$u_j = \frac{(1, s_j^*)}{\sqrt{1 + |s_j^*|^2}},$$

let

$$Q_p := I - P_{\mathcal{K}_p^Y}^{\text{bd}}, \quad \Phi_p^Y := \Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing}, p} E_{\text{car}, p}^{\text{tr}} L_{\text{clean}, p}^{\text{tr}},$$

and assume the row-normal carrier expansion

$$\Phi_p^Y(1, s) = H_p(c_{7,p} N_p + (\beta_{9,p}^{\text{tri}} + \alpha_{9,p} s) Y_p) + \varepsilon_p(s),$$

with $\|H_p\|$, $\|N_p\|$, $\|Y_p\|$ polynomially bounded above and below in Q , with quotient annihilation of the N -channel

$$\|Q_p N_p\| \leq Q^{-B} \|N_p\|,$$

with carrier-error control

$$\|\varepsilon_p(s_j^*)\| + \|Q_p \varepsilon_p(s_j^*)\| \leq Q^{-B},$$

and primitive $Y12$ quotient survival

$$\|Q_p Y_p\| \geq Q^{-C} \|Y_p\|.$$

Assume in addition the static wall-escape estimate

$$\sum_{p \in C_j} \mu_p |H_p|^2 |\beta_{9,p}^{\text{tri}} + \alpha_{9,p} s_j^*|^2 \|Q_p Y_p\|^2 \geq Q^{-A} \sum_{p \in C_j} \mu_p \|\Phi_p^Y u_j\|^2.$$

Then for B sufficiently large depending polynomially on A , C , and the operator bounds,

$$\sum_{p \in C_j} \mu_p \|Q_p \Phi_p^Y u_j\|^2 \geq Q^{-A'} \sum_{p \in C_j} \mu_p \|\Phi_p^Y u_j\|^2.$$

Proof. Apply Q_p to the row-normal expansion at $s = s_j^*$:

$$Q_p \Phi_p^Y(1, s_j^*) = H_p(\beta_{9,p}^{\text{tri}} + \alpha_{9,p} s_j^*) Q_p Y_p + H_p c_{7,p} Q_p N_p + Q_p \varepsilon_p(s_j^*).$$

By the quotient annihilation $\|Q_p N_p\| \leq Q^{-B} \|N_p\|$ and the carrier error bound, the last two terms are $O(Q^{-B})$ in the weighted cap norm. Using $\|x + y\|^2 \geq \frac{1}{2} \|x\|^2 - \|y\|^2$ and choosing B larger than the polynomial operator losses gives

$$\sum_{p \in C_j} \mu_p \|Q_p \Phi_p^Y u_j\|^2 \geq \frac{1}{2} \sum_{p \in C_j} \mu_p |H_p|^2 |\beta_{9,p}^{\text{tri}} + \alpha_{9,p} s_j^*|^2 \|Q_p Y_p\|^2 - O(Q^{-B'}).$$

The wall-escape hypothesis lower-bounds the first term by $Q^{-A} \sum_p \mu_p \|\Phi_p^Y u_j\|^2$, and the carrier noncollapse implicit in the same hypothesis dominates the $O(Q^{-B'})$ error for B large enough, which gives the conclusion after exponent renaming. $\square$

Remark 16.225 (Static wall as the residual finite obstruction). This is a residual finite input only for the legacy scalar Y12-wall route. It is superseded in the canonical source-graded branch, where the affine forcing components are retained in orthogonal source-grade summands and the same-norm Type-I contradiction is supplied by Theorem F.5. Lemma 16.224 sharpens Hypothesis 16.214 by isolating the remaining finite input as a one-dimensional wall-escape estimate on the static line:

$$\sum_{p \in C_j} \mu_p |\beta_{9,p}^{\text{tri}} + \alpha_{9,p} s_j^*|^2 \geq Q^{-A} \sum_{p \in C_j} \mu_p,$$

under uniform polynomial lower bounds on $|H_p|$, $|\alpha_{9,p}|$, and $\|Q_p Y_p\|/\|Y_p\|$. Equivalently, the static slope s_j^* must not align with the killed row-normal direction $s = -\beta_{9,p}^{\text{tri}}/\alpha_{9,p}$ on a polynomially large μ -mass of C_j . A two-dimensional countermodel $\Phi(1, s) = c_7 N + (\beta + \alpha s) Y$ with $qN = 0$, $qY = Y$, and $s_j^* = -\beta/\alpha$ satisfies primitive Y12 quotient survival $qY \neq 0$ but kills static-line survival $q\Phi(1, s_j^*) = 0$; thus static wall escape is not a formal consequence of Y12 quotient survival alone. This is the residual finite input for the static branch: quotient-descent linear algebra plus quotient survival of the primitive Y -carrier reduce static-line survival to the wall-escape estimate above, and no further.

Remark 16.226 (Legacy signed covariance bottleneck). The signed covariance bottleneck

$$B_j = \sum_{p \in C_j} w_p \Gamma_p \Gamma_p^*$$

is retained only as a diagnostic for sign-coherent branches. In the active proof branch the covariance object is

$$B_j^{\text{cov}} = \sum_{p \in C_j} \omega_p \Gamma_p \Gamma_p^*, \quad \omega_p = \mu_p^{\text{pkt}} > 0.$$

The active extraction statement is therefore Theorem 16.219, not the legacy signed determinant condition.

Remark 16.227 (Extraction bridge status). The extraction covariance bridge is closed in the positive packet-density formulation by Theorem 16.219. Any remaining references to a cap-wise signed determinant test should be read as legacy diagnostics unless the branch is explicitly sign-coherent.

Definition 16.228 (Affine gauge slice and regular locus). Let $T_{\alpha,\beta}(x) = \alpha x + \beta$ with $\alpha > 0$. Let $\mathcal{N}_1, \mathcal{N}_2$ be the two normalization functionals already fixed in the draft, and define

$$\mathcal{F}_f(\alpha, \beta) := (\mathcal{N}_1(T_{\alpha,\beta} f), \mathcal{N}_2(T_{\alpha,\beta} f) - 1).$$

We call f *regular* if there exists a solution (α_0, β_0) to $\mathcal{F}_f(\alpha, \beta) = 0$ such that

$$\det D_{(\alpha,\beta)} \mathcal{F}_f(\alpha_0, \beta_0) \neq 0.$$

Let $\mathcal{U}$ denote the set of regular classes.

Lemma 16.229 (Local affine quotient normal form). *On the regular locus $\mathcal{U}$, every equivalence class has a unique representative of the form $T_{\alpha,\beta} f$ satisfying the normalization conditions*

$$\mathcal{N}_1(T_{\alpha,\beta} f) = 0, \quad \mathcal{N}_2(T_{\alpha,\beta} f) = 1,$$

with $\alpha > 0$. Moreover, the map

$$[f] \longmapsto (\alpha(f), \beta(f))$$

is C^1 (indeed smooth if the data are smooth) on $\mathcal{U}$.

Proof. Fix a regular class $[f_0] \in \mathcal{U}$, and let (α_0, β_0) be a solution of $\mathcal{F}_{f_0}(\alpha, \beta) = 0$ with

$$\det D_{(\alpha, \beta)} \mathcal{F}_{f_0}(\alpha_0, \beta_0) \neq 0.$$

Since the affine group is two-dimensional, the equation $\mathcal{F}_f(\alpha, \beta) = 0$ is a square system in the two variables (α, β) . By the implicit function theorem, there is a neighborhood $\mathcal{V}$ of f_0 in the regular locus and unique C^1 functions

$$\alpha(\cdot), \beta(\cdot) : \mathcal{V} \rightarrow \mathbb{R}$$

such that

$$\mathcal{F}_f(\alpha(f), \beta(f)) = 0 \quad \text{for all } f \in \mathcal{V}.$$

Uniqueness follows because any two solutions in $\mathcal{V}$ would define two points on the same transverse local slice, hence coincide by the implicit-function theorem uniqueness statement. Since the reference solution has $\alpha_0 > 0$, shrinking $\mathcal{V}$ if necessary keeps $\alpha(f) > 0$ throughout the neighborhood. $\square$

Corollary 16.230 (Local affine quotient angle). *On the regular locus $\mathcal{U}$, the quotient angle*

$$\theta([f]) := \arg(\alpha(f) + i\beta(f))$$

is well-defined and C^1 (indeed smooth if the data are smooth).

Proof. By Lemma 16.229, the normalized pair $(\alpha(f), \beta(f))$ depends smoothly on the quotient class. Since $\alpha(f) > 0$ on $\mathcal{U}$, the complex number $\alpha(f) + i\beta(f)$ is never zero on the regular locus, so $\arg$ is smooth there. Therefore $\theta([f])$ is well-defined and smooth. $\square$

Remark 16.231 (Two Jacobian notions). The determinant condition in Definition 16.228 is the slice-regularity condition for the abstract quotient theorem. The separate lower bound in Corollary 16.315 is a cap-level transversality certificate and is not implied by the abstract slice theorem alone.

Remark 16.232 (Compatibility with the D_4 -symmetry). If the D_4 -action preserves the normalization slice $\mathcal{N}_1 = 0$, $\mathcal{N}_2 = 1$, then the induced action on θ is the finite symmetry identification already built into the quotient.

Remark 16.233 (Nuisance N_p^{nuis} versus retained $C_{7,p}^{\text{tar}}$: kill-vs-preserve obstruction). The corollary 16.315 sets

$$N_p := \Pi_{\Sigma(Y_{12})} T_{\text{tr} \rightarrow \text{sing}} \mathcal{N}_p^{\text{D4}}$$

and uses $r_{N,p} = q_p^Y N_p$ as the first row of a putative rank-two Jacobian. But quotient survival as stated requires $U_p \subseteq \ker \Lambda_{Y,p}$, so every column of $\mathcal{N}_p^{\text{D4}}$ maps into $\mathcal{K}_p^Y$ under $\Pi_{\Sigma(Y_{12})} T_{\text{tr} \rightarrow \text{sing}}$, and consequently $q_p^Y N_p = 0$. With N_p in the kernel of q_p^Y the Jacobian determinant $H_{j,p}^2 c_7 \alpha_9$ is identically zero on the nuisance image and the rank-two normal form cannot be derived. The cap bridge therefore needs the role split: the quotient kills

$$N_p^{\text{nuis}} := \Pi_{\Sigma(Y_{12})} T_{\text{tr} \rightarrow \text{sing}} \mathcal{N}_p^{\text{D4}},$$

while it must *retain* the two carrier columns

$$C_{7,p}^{\text{tar}} := \Pi_{\Sigma(Y_{12})} T_{\text{tr} \rightarrow \text{sing}, p} [C_7]^{\text{tr}}, \quad C_{Y,p}^{\text{tar}} := \Pi_{\Sigma(Y_{12})} T_{\text{tr} \rightarrow \text{sing}, p} [C_{Y_{12}}]^{\text{tr}}.$$

The cap-level rank-two Jacobian then runs on $(r_{C_{7,p}}, r_{C_{Y,p}})$ with $r_{C_{7,p}} := q_p^Y C_{7,p}^{\text{tar}}$ and $r_{C_{Y,p}} := q_p^Y C_{Y,p}^{\text{tar}}$, not on $(r_{N,p}, r_{Y,p})$. This is the typed precondition for Corollary 16.239 below; without it, quotient descent and the cap bridge demand q_p^Y to both kill and preserve the same channel.

Lemma 16.234 (Static-line survival from rank-two quotient survival). *Assume Corollary 16.239 and the row-normal carrier expansion*

$$\Phi_p^Y(1, s) = H_p \begin{bmatrix} C_{7,p}^{\text{tar}} & C_{Y,p}^{\text{tar}} \end{bmatrix} \begin{pmatrix} c_7 & 0 \\ \beta_9^{\text{tri}} & \alpha_9 \end{pmatrix} \begin{pmatrix} 1 \\ s \end{pmatrix} + \varepsilon_p(s),$$

with

$$|H_p| \geq Q^{-C}, \quad \sigma_{\min} \begin{pmatrix} c_7 & 0 \\ \beta_9^{\text{tri}} & \alpha_9 \end{pmatrix} \geq Q^{-C}, \quad \|\varepsilon_p(s)\| + \|(I - P_{\mathcal{K}_p^Y}^{\text{bd}}) \varepsilon_p(s)\| \leq Q^{-B}.$$

Then for B sufficiently large depending polynomially on C and the operator bounds, and for every unit vector $u \in \mathbb{C}^2$,

$$\|(I - P_{\mathcal{K}_p^Y}^{\text{bd}})\Phi_p^Y u\| \geq Q^{-A}\|\Phi_p^Y u\|.$$

Consequently the static-line handoff

$$\sum_{p \in C_j} \mu_p \|(I - P_{\mathcal{K}_p^Y}^{\text{bd}})\Phi_p^Y u_j\|^2 \geq Q^{-A} \sum_{p \in C_j} \mu_p \|\Phi_p^Y u_j\|^2$$

holds for every static direction $u_j = (1, s_j^*)/\sqrt{1 + |s_j^*|^2}$ without a separate static wall-escape estimate on $|\beta_9^{\text{tri}} + \alpha_9 s_j^*|$.

Proof. Write $R_p := (I - P_{\mathcal{K}_p^Y}^{\text{bd}})[C_{7,p}^{\text{tar}} \ C_{Y,p}^{\text{tar}}]$. Corollary 16.239 gives $\sigma_{\min}(R_p) \geq Q^{-C}$, so for every $v \in \mathbb{C}^2$, $\|R_p v\| \geq Q^{-C}\|v\|$ and $\|[C_{7,p}^{\text{tar}} \ C_{Y,p}^{\text{tar}}]v\| \leq Q^{C_0}\|v\|$ by the operator bounds. Apply $(I - P_{\mathcal{K}_p^Y}^{\text{bd}})$ to the row-normal expansion at a unit $u \in \mathbb{C}^2$:

$$(I - P_{\mathcal{K}_p^Y}^{\text{bd}})\Phi_p^Y u = H_p R_p L_{\text{clean}} u + (I - P_{\mathcal{K}_p^Y}^{\text{bd}})\varepsilon_p(s_u),$$

with $L_{\text{clean}} = \begin{pmatrix} c_7 & 0 \\ \beta_9^{\text{tri}} & \alpha_9 \end{pmatrix}$ and s_u the affine coordinate of u . By the singular-value bounds, $\|H_p R_p L_{\text{clean}} u\| \geq Q^{-3C}\|u\|$; the error contributes $O(Q^{-B})$, absorbed for B large. Thus $\|(I - P_{\mathcal{K}_p^Y}^{\text{bd}})\Phi_p^Y u\| \geq Q^{-A}\|u\|$. The operator bound also gives $\|\Phi_p^Y u\| \leq Q^{C_0}\|u\|$, hence $\|(I - P_{\mathcal{K}_p^Y}^{\text{bd}})\Phi_p^Y u\| \geq Q^{-A'}\|\Phi_p^Y u\|$. Squaring and weighted-summing over $p \in C_j$ gives the static-line handoff for any u_j (no static wall-escape required, since the lower bound is uniform in u). $\square$

Remark 16.235 (Comparison with Lemma 16.224). Lemma 16.224 reduces the static branch to a one-dimensional wall-escape estimate $\sum_p \mu_p |\beta_9^{\text{tri}} + \alpha_9 s_j^*|^2 \geq Q^{-A} \sum_p \mu_p$ under primitive Y -only quotient survival $\|q_p^Y Y_p\| \geq Q^{-C}\|Y_p\|$ and quotient annihilation of N_p . Lemma 16.234 closes the static branch unconditionally on the static slope, but at the cost of a strictly stronger quotient hypothesis: both $C_{7,p}^{\text{tar}}$ and $C_{Y,p}^{\text{tar}}$ must survive the quotient with polynomial principal angle. The two routes are not interchangeable: the rank-two route requires the quotient to retain the C_7 channel, which contradicts treating it as nuisance. Per Remark 16.233, the live quotient must distinguish nuisance N_p^{nuis} from retained $C_{7,p}^{\text{tar}}$ before the rank-two route is available.

Lemma 16.236 (Killed nuisance cannot be a live Jacobian row). *Assume*

$$N_p^{\text{nuis}} := \Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing}} \mathcal{N}_p^{\text{D4}} \subseteq \mathcal{K}_p^Y.$$

Then $q_p^Y N_p^{\text{nuis}} = 0$. Consequently no cap-level Jacobian lower bound can be derived from the pair $(q_p^Y N_p^{\text{nuis}}, q_p^Y Y_p)$: a putative normal form

$$\begin{pmatrix} q_p^Y N_p^{\text{nuis}} \\ q_p^Y Y_p \end{pmatrix} = H_{j,p} \begin{pmatrix} c_7 & 0 \\ \beta_9^{\text{tri}} & \alpha_9 \end{pmatrix} \begin{pmatrix} 1 \\ s \end{pmatrix} + \varepsilon_{j,p}(s), \quad \|\partial_s \varepsilon_{j,p}\| \leq Q^{-B},$$

forces $|H_{j,p} c_7| \leq Q^{-B}$, incompatible with $|H_{j,p}| \geq Q^{-C}$ and $|c_7| \geq Q^{-C}$ for B sufficiently large. Equivalently, the Wronskian $W_{j,p} = r'_{Y,p} r_{N,p} - r'_{N,p} r_{Y,p}$ of any pair containing $r_{N,p} \equiv 0$ is identically zero. The cap-level rank-two Jacobian must therefore be carried on retained live carriers $C_{7,p}^{\text{tar}}, C_{Y,p}^{\text{tar}}$, not on N_p^{nuis} ; see Corollary 16.239 and Corollary 16.295 below.

Proof. $N_p^{\text{nuis}} \subseteq \mathcal{K}_p^Y$ gives $P_{\mathcal{K}_p^Y}^{\text{bd}} N_p^{\text{nuis}} = N_p^{\text{nuis}}$, hence $q_p^Y N_p^{\text{nuis}} = (I - P_{\mathcal{K}_p^Y}^{\text{bd}}) N_p^{\text{nuis}} = 0$. The first row of the assumed normal form then reads $0 = H_{j,p} c_7 + \varepsilon_{1,j,p}(s)$, so $|H_{j,p} c_7| = |\varepsilon_{1,j,p}| \leq Q^{-B}$, contradicting the polynomial lower bounds on $|H_{j,p}|$ and $|c_7|$ for $B > 2C$. $\square$

Lemma 16.237 (Live-coefficient obstruction to rank-two quotient survival). *Let*

$$E_p := [C_{7,p}^{\text{tar}} \ C_{Y,p}^{\text{tar}}]$$

and let $K_{0,p} \subseteq \mathcal{H}_p^Y$ be a dead nuisance space such that

$$\sigma_{\min}((I - P_{K_{0,p}})E_p) \geq Q^{-C}.$$

Suppose a declared nuisance column has the live decomposition

$$A_p = D_p + E_p m_p, \quad D_p \in K_{0,p}, \quad m_p = \begin{pmatrix} m_{7,p} \\ m_{Y,p} \end{pmatrix} \in \mathbb{C}^2.$$

Let $K_p := K_{0,p} + \text{span}\{A_p\}$. If $m_p \neq 0$, then

$$\sigma_{\min}((I - P_{K_p})E_p) = 0.$$

Consequently, rank-two C_7/Y_{12} quotient survival is compatible with placing A_p in the nuisance quotient only if its live coefficient m_p vanishes, up to the desired polynomial error threshold. More generally, if a finite family $A_p^{(a)} = D_p^{(a)} + E_p m_p^{(a)}$ is placed in the nuisance quotient, the live-coefficient matrix $M_{\text{live},p} := [m_p^{(1)} \ m_p^{(2)} \ \cdots]$ must satisfy $\text{rank } M_{\text{live},p} = 0$ up to the polynomial error; $\text{rank} = 1$ leaves only one live direction surviving, and $\text{rank} = 2$ collapses both retained carriers.

Proof. Since $D_p \in K_{0,p} \subseteq K_p$ and $A_p \in K_p$, $E_p m_p = A_p - D_p \in K_p$, hence $(I - P_{K_p})E_p m_p = 0$. If $m_p \neq 0$ the map $(I - P_{K_p})E_p : \mathbb{C}^2 \rightarrow K_p^\perp$ has a nontrivial kernel and minimum singular value 0. The general statement follows by induction on the family, since each successive nuisance column adds at most one column to the live-coefficient matrix and at most one direction to the kernel. $\square$

Corollary 16.238 (Primitive arithmetic coefficient must be dead or rerouted). *Suppose the arithmetic nuisance column has the live decomposition*

$$A_{II,p} = A_{II,p}^{\text{dead}} + c_{\text{arith},7,p} C_{7,p}^{\text{tar}} + c_{\text{arith},Y,p} C_{Y,p}^{\text{tar}},$$

and suppose the full $A_{II,p}$ is placed in $\mathcal{K}_p^Y$. Then rank-two C_7/Y_{12} quotient survival forces

$$c_{\text{arith},7,p} = c_{\text{arith},Y,p} = 0$$

up to the required polynomial error. In particular, the named arithmetic coefficient

$$c_{\text{arith},Y} := [II_{\text{arith},9}^{\text{exact}}]_{D^2 N_0, \text{ same, } w=1, b=12}$$

must vanish if $II_{\text{arith},9}^{\text{exact}}$ is treated as exact nuisance. If $c_{\text{arith},Y} \neq 0$, the only consistent options are: (a) remove the live $C_{Y_{12}}$ -part of $A_{II,p}$ before forming $\mathcal{K}_p^Y$; (b) route that live part to a coherent or residual carrier channel rather than to exact nuisance; or (c) abandon rank-two C_7/Y_{12} survival in this quotient. A residual L^2 -bound on the column does not fix the obstruction: once the column is quotient-killed, the corresponding live direction is gone. The same analysis applied to $A_{7,p} = A_{7,p}^{\text{dead}} + \lambda_{77,p} C_{7,p}^{\text{tar}} + \lambda_{7Y,p} C_{Y,p}^{\text{tar}}$ forces $\lambda_{77,p} = \lambda_{7Y,p} = 0$ and is particularly load-bearing because killing C_7 collapses the rank-two bridge to a primitive Y_{12} -only route, which does not close the static-line branch.

Corollary 16.239 (Conditional torsion-free rank-two C_7/Y_{12} quotient survival). *Assume the concrete row/routing coefficient certificate $\mathfrak{R}_{2Y}(p)$, equivalently the verified vanishing/lower-bound conditions encoded by $\mathfrak{B}_{2Y,p}$ (Definition 16.282). For the certified dead-only nuisance quotient $q_{p,\text{cert}}^Y = I - P_{\mathcal{K}_{p,\text{dead}}^{Y,\text{cert}}}^{\text{bd}}$ and the live carrier matrix $E_p = [C_{7,p}^{\text{tar}} \ C_{Y,p}^{\text{tar}}]$, one has*

$$\sigma_{\min}(q_{p,\text{cert}}^Y E_p) \geq Q^{-C}.$$

Equivalently, the certified quotient imposes no hidden torsion relation between C_7 and $C_{Y_{12}}$: no nonzero polynomially bounded pair (a, b) satisfies

$$a q_{p,\text{cert}}^Y C_{7,p}^{\text{tar}} + b q_{p,\text{cert}}^Y C_{Y,p}^{\text{tar}} = 0$$

up to a polynomially small angle. This is the proof-level replacement for the old rank-two quotient-survival hypothesis once the raw coefficient certificate $\mathfrak{R}_{2Y}(p)$ has been verified.

Proof. This is Theorem 16.288; in its formula-level packet form it is also included in Theorem 16.280. The dead/live split removes every live component from the quotient kernel before projection, so L_p^{live} annihilates the certified dead space while $L_p^{\text{live}} E_p = I_2$. The bounded dual-row argument gives the stated singular-value lower bound. $\square$

Definition 16.240 (Dead-only quotient space and live carrier matrix). Let $\mathcal{H}_p^Y$ be the finite-dimensional Hilbert space in which the $Y12$ -boundary outputs live. Write

$$K_p := \mathcal{K}_{p,\text{dead}}^Y \subset \mathcal{H}_p^Y, \quad q_p := I - P_{K_p}^{\text{bd}},$$

for the dead-only nuisance space and its boundary projector (Corollary 16.239), and

$$E_p := [C_{7,p}^{\text{tar}} \quad C_{Y,p}^{\text{tar}}] : \mathbb{C}^2 \rightarrow \mathcal{H}_p^Y$$

for the retained live carrier matrix. The rank-two live-pair quotient gap

$$\delta_{2Y,p} := \sigma_{\min}(q_p E_p)$$

is the standing finite-dimensional invariant of the cap/ $Y12$ bridge.

Lemma 16.241 (Dual-row certificate for rank-two $C_7/Y12$ quotient survival). *Let*

$$E_p := [C_{7,p}^{\text{tar}} \quad C_{Y,p}^{\text{tar}}], \quad q_p := I - P_{\mathcal{K}_p^Y}^{\text{bd}},$$

where $\mathcal{K}_p^Y$ is the dead-only nuisance space of Corollary 16.239. Suppose there exist linear functionals $\ell_{7,p}, \ell_{Y,p} : \mathcal{H}_p^Y \rightarrow \mathbb{C}$ on the ambient $Y12$ -boundary quotient space such that

$$\ell_{7,p}(\mathcal{K}_p^Y) = \ell_{Y,p}(\mathcal{K}_p^Y) = 0, \quad \begin{pmatrix} \ell_{7,p}(C_{7,p}^{\text{tar}}) & \ell_{7,p}(C_{Y,p}^{\text{tar}}) \\ \ell_{Y,p}(C_{7,p}^{\text{tar}}) & \ell_{Y,p}(C_{Y,p}^{\text{tar}}) \end{pmatrix} = I_2,$$

and $\|\ell_{7,p}\| + \|\ell_{Y,p}\| \leq Q^C$. Then

$$\sigma_{\min}(q_p E_p) \geq Q^{-C'}$$

for some C' depending polynomially on C . Conversely, if $\sigma_{\min}(q_p E_p) \geq Q^{-C}$ then the rows $\ell_{i,p} := ((q_p E_p)^*(q_p E_p))^{-1} (q_p E_p)^* e_i$ are bounded by Q^C and vanish on $\mathcal{K}_p^Y$, giving a dual certificate.

Proof. For the forward direction, fix $v = (a, b)^T \in \mathbb{C}^2$. Each row vanishes on $\mathcal{K}_p^Y$, hence $\ell_{i,p}(q_p E_p v) = \ell_{i,p}(E_p v)$, and the normalization gives $(\ell_{7,p}(E_p v), \ell_{Y,p}(E_p v))^T = v$. Therefore

$$|v| \leq \sqrt{\|\ell_{7,p}\|^2 + \|\ell_{Y,p}\|^2} \|q_p E_p v\| \leq \sqrt{2} Q^C \|q_p E_p v\|,$$

so $\|q_p E_p v\| \geq Q^{-C-1} |v|$, which is the claim with $C' := C+1$. For the converse, the Moore–Penrose pseudoinverse rows have operator norm $\sigma_{\min}(q_p E_p)^{-1} \leq Q^C$, and they act after q_p , hence vanish on $\mathcal{K}_p^Y$; the diagonal normalization is the standard Moore–Penrose identity $L_p^\dagger(q_p E_p) = I_2$. $\square$

Lemma 16.242 (Canonical dead-column split relative to detecting rows). *Assume the equivalent conditions of Lemma 16.241 for a base nuisance space $K_{0,p} \subset \mathcal{H}_p^Y$, and let $L_{0,p} : \mathcal{H}_p^Y \rightarrow \mathbb{C}^2$ be a bounded row certificate with*

$$L_{0,p} K_{0,p} = 0, \quad L_{0,p} E_p = I_2, \quad \|L_{0,p}\| \leq Q^C.$$

For every declared nuisance column $A_{a,p} \in \mathcal{H}_p^Y$, define its canonical live coefficient and dead remainder by

$$m_{a,p} := L_{0,p} A_{a,p} \in \mathbb{C}^2, \quad A_{a,p}^{\text{dead}} := A_{a,p} - E_p m_{a,p}.$$

Then $L_{0,p} A_{a,p}^{\text{dead}} = 0$. Consequently, if the nuisance space is enlarged only by the dead remainders,

$$K_{p,\text{dead}} := K_{0,p} + \text{span}\{A_{a,p}^{\text{dead}} : a \in \mathcal{A}\},$$

then the same rows $L_{0,p}$ still kill $K_{p,\text{dead}}$ and still normalize E_p , so that

$$\sigma_{\min}((I - P_{K_{p,\text{dead}}}^{\text{bd}}) E_p) \geq Q^{-C'}$$

for some C' depending polynomially on C .

Proof. By definition, $L_{0,p} A_{a,p}^{\text{dead}} = L_{0,p} A_{a,p} - L_{0,p} E_p m_{a,p} = m_{a,p} - m_{a,p} = 0$. Thus $L_{0,p}$ kills both $K_{0,p}$ and every added dead remainder, so it kills $K_{p,\text{dead}}$. Since $L_{0,p} E_p = I_2$ and $\|L_{0,p}\| \leq Q^C$, the dual-row implication of Lemma 16.241 gives the stated singular-value lower bound, with only polynomial loss. $\square$

Remark 16.243 (What the dead-column split proves). Lemma 16.242 is not a new analytic estimate. It is the algebraic bookkeeping needed to assemble a nuisance space that is row-dead and live-pair preserving. The actual analytic content is the existence of bounded rows $L_{0,p}$, equivalently the rank-two Schur gap of Lemma 16.241. Once those rows exist, every declared nuisance column has the canonical live coefficient $m_{a,p} = L_{0,p}A_{a,p}$; placing the unsplit column into the kernel is safe only when $m_{a,p} = 0$ or when the live part has been rerouted out of the dead quotient, as recorded in Corollary 16.238.

Remark 16.244 (Necessity of dead-column split for rank-two survival). Suppose the arithmetic nuisance column has the live decomposition

$$A_{II,p} = A_{II,p}^{\text{dead}} + c_{\text{arith},Y} C_{Y,p}^{\text{tar}} + c_{\text{arith},7} C_{7,p}^{\text{tar}}.$$

If the unsplit $A_{II,p}$ is placed in $\mathcal{K}_p^Y$, then any row ℓ annihilating $\mathcal{K}_p^Y$ satisfies $0 = \ell(A_{II,p}) = \ell(A_{II,p}^{\text{dead}}) + c_{\text{arith},Y} \ell(C_{Y,p}^{\text{tar}}) + c_{\text{arith},7} \ell(C_{7,p}^{\text{tar}})$. Since $\ell(A_{II,p}^{\text{dead}}) = 0$ on a genuine dead column, this forces a linear relation among $\ell(C_{7,p}^{\text{tar}}), \ell(C_{Y,p}^{\text{tar}})$; for the normalized detecting rows of Lemma 16.241 this collapses to $c_{\text{arith},Y} = c_{\text{arith},7} = 0$. The same argument applied to the $A_{7,p}$ column with live coefficient $\lambda_{7,p} \neq 0$ forces $q_p C_{7,p}^{\text{tar}} = 0$, violating the rank-two survival. Thus the dead-only definition of $\mathcal{K}_p^Y$ in Corollary 16.239 is not optional: any unsplit live column placed in the quotient kernel breaks the dual-row certificate.

Theorem 16.245 (Rank-two quotient survival: primal–dual–Schur equivalence). *Let $K_p \subset \mathcal{H}_p^Y$ be a finite-dimensional nuisance space of the Y12-boundary ambient space, let*

$$q_p := I - P_{K_p}^{\text{bd}}, \quad E_p := [C_{7,p}^{\text{tar}} \quad C_{Y,p}^{\text{tar}}] : \mathbb{C}^2 \rightarrow \mathcal{H}_p^Y.$$

The following three statements are equivalent up to polynomial loss in Q :

- (1) *the singular-value form $\sigma_{\min}(q_p E_p) \geq Q^{-C}$;*
- (2) *the Schur–Gram form $E_p^* q_p E_p \geq Q^{-C} I_2$ as Hermitian forms on $\mathbb{C}^2$;*
- (3) *the bounded dual-row form: there exist linear functionals $\ell_{7,p}, \ell_{Y,p} : \mathcal{H}_p^Y \rightarrow \mathbb{C}$ with*

$$\ell_{7,p}(K_p) = \ell_{Y,p}(K_p) = 0, \quad \begin{pmatrix} \ell_{7,p}(C_{7,p}^{\text{tar}}) & \ell_{7,p}(C_{Y,p}^{\text{tar}}) \\ \ell_{Y,p}(C_{7,p}^{\text{tar}}) & \ell_{Y,p}(C_{Y,p}^{\text{tar}}) \end{pmatrix} = I_2, \quad \|\ell_{7,p}\| + \|\ell_{Y,p}\| \leq Q^C.$$

Moreover, each of these is equivalent to the distance form $\text{dist}(E_p v, K_p) \geq Q^{-C} \|v\|$ for every $v \in \mathbb{C}^2$.

Proof. The identity $\|q_p E_p v\|^2 = \langle E_p^* q_p E_p v, v \rangle$ gives $\sigma_{\min}(q_p E_p)^2 = \lambda_{\min}(E_p^* q_p E_p)$, so (1) and (2) are equivalent. Since q_p is the orthogonal projection onto $K_p^\perp$, $\|q_p E_p v\| = \text{dist}(E_p v, K_p)$, giving the distance form. For (1) $\Rightarrow$ (3), set $L_p := ((q_p E_p)^*(q_p E_p))^{-1} (q_p E_p)^* q_p$; then $L_p E_p = I_2$, $L_p K_p = 0$ (from $q_p K_p = 0$), and $\|L_p\|_{\text{op}} \leq \sigma_{\min}(q_p E_p)^{-1} \leq Q^C$; the rows of L_p supply $(\ell_{7,p}, \ell_{Y,p})$. For (3) $\Rightarrow$ (1), the dual rows kill K_p , so $\ell_{i,p}(E_p v) = \ell_{i,p}(q_p E_p v)$, and the normalization gives $(\ell_{7,p}(E_p v), \ell_{Y,p}(E_p v))^T = v$; hence $\|v\| \leq \sqrt{2} Q^C \|q_p E_p v\|$, i.e. $\sigma_{\min}(q_p E_p) \geq Q^{-C-1}$. This bidirectional argument is the same as Lemma 16.241, promoted to include the Schur and distance forms. $\square$

Remark 16.246 (Two irreducible cap-bridge inputs). After Theorem 16.245, the four heuristic targets

dead-column split + bounded dual rows + analytic live-pair Schur gap + D4 amplitude survival collapse to two genuinely irreducible finite obligations:

- (1) *the rank-two quotient Schur gap $\sigma_{\min}(q_p E_p) \geq Q^{-C}$, in any of its equivalent primal/dual/Schur/distance forms; and*
- (2) *the D4 amplitude mass on the cap core*

$$\sum_{p \in C_j} \mu_p |H_p|^2 \|\Gamma_p\|^2 \geq Q^{-A} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2 \quad (\text{Hypothesis 16.297}).$$

The dead-column split (Lemma 16.237, Corollary 16.238) is the coefficient-level condition needed to build the nuisance space without killing the live pair: it is not a separate input but a constraint on how K_p is assembled. The bounded dual rows are the dual form of the Schur gap, not an independent target. The D4 amplitude mass is genuinely independent: a configuration with strongly varying $s_p = \Gamma_{G,2}(p)/\Gamma_G(p)$ and $H_p \equiv 0$ satisfies $\delta_{\text{ext}} \geq Q^{-A}$ and $\sigma_{\min}(q_p E_p) \geq Q^{-C}$ simultaneously while violating the amplitude mass.

16.24.2. *The actual rank-two quotient certificate.* The preceding primal–dual–Schur theorem proves that the rank-two quotient condition is equivalent to the existence of bounded live detecting rows. It must not be read as a proof that such rows exist for the actual quotient. The missing object is the following finite coefficient certificate.

Definition 16.247 (Concrete row certificate for the live $C_7/Y12$ plane). Let

$$E_p := [C_{7,p}^{\text{tar}} \quad C_{Y,p}^{\text{tar}}] : \mathbb{C}^2 \rightarrow \mathcal{H}_p^Y.$$

A concrete row certificate for the live $C_7/Y12$ -plane is a pair of bounded rows

$$L_p^{\text{row}} = \begin{pmatrix} \ell_{7,p} \\ \ell_{Y,p} \end{pmatrix} : \mathcal{H}_p^Y \rightarrow \mathbb{C}^2$$

with

$$L_p^{\text{row}} E_p = I_2, \quad \|L_p^{\text{row}}\|_{\text{op}} \leq Q^C.$$

In the row-normal construction these rows are

$$\ell_{7,p} = \frac{\Delta_7^\nabla}{\Delta_7^\nabla(C_7)}, \quad \ell_{Y,p} := \ell_{9,p}^Y = \omega_Y - \omega_Y(e_7)\ell_{7,p}.$$

Thus the certificate is concrete only after the following coefficient facts have been verified:

$$\Delta_7^\nabla(C_7) \neq 0, \quad \ell_{7,p}(C_{Y,p}^{\text{tar}}) = 0, \quad \omega_Y(C_{Y,p}^{\text{tar}}) = 1, \quad \omega_Y(W_{\text{dead}}^{\text{fixed}}) = 0,$$

with polynomial margins, and with polynomial row norm. These are precisely the septic exactness, grade-separation, and fixed-whitening separation tests.

Definition 16.248 (Column routing certificate for the $Y12$ quotient). Fix a concrete row certificate L_p^{row} . Let the raw nuisance list be

$$\mathcal{A}_p^{\text{raw}} := \{R_{B_2,p}, R_{P,p}, R_{D,p}, A_{\alpha,p}, A_{7,p}, A_{<7,p}, A_{II,p}, A_{E_2,p}\}.$$

For each raw column $A_{a,p}^{\text{raw}} \in \mathcal{A}_p^{\text{raw}}$, define its live coefficient and dead part by

$$m_{a,p} := L_p^{\text{row}} A_{a,p}^{\text{raw}} \in \mathbb{C}^2, \quad A_{a,p}^{\text{dead}} := A_{a,p}^{\text{raw}} - E_p m_{a,p}.$$

The column is classified as follows.

- (1) It is *row-dead* if $m_{a,p} = 0$.
- (2) It is *coherent live* if $m_{a,p} \neq 0$ but the coefficient is part of the fixed or coherently transported row data. In this case the live part $E_p m_{a,p}$ is not put in the dead quotient; it is routed into the effective live carrier matrix.
- (3) It is *residual live* if its weighted cap norm is $O(Q^{-B})$ after the same projection. It is routed to the residual error.
- (4) It is *atom-local live* if $m_{a,p} \neq 0$ is a free local packet parameter. This is a structural failure of the rank-two $Y12$ quotient route: the column cannot be killed while preserving the live $C_7/Y12$ -plane.

The column routing certificate holds on packet p if no raw nuisance column is atom-local live and if all coherent live columns are included in the finite matrix

$$M_{\text{live},p} := \sum_{a \in \mathcal{A}_{\text{coh}}} m_{a,p} \eta_a^*,$$

where the fixed vectors $\eta_a \in \mathbb{C}^2$ encode how the coherent column enters the D4 two-row carrier. The dead quotient is then

$$\mathcal{K}_{p,\text{dead}}^{Y,\text{cert}} := \mathcal{K}_{p,0}^{Y,\text{dead}} + \text{span}\{A_{a,p}^{\text{dead}} : A_{a,p}^{\text{raw}} \in \mathcal{A}_p^{\text{raw}}\},$$

where $\mathcal{K}_{p,0}^{Y,\text{dead}} \subseteq \ker L_p^{\text{row}}$ is the base source-linear/dead quotient space.

Lemma 16.249 (Column-by-column consequences of the routing certificate). *Assume the concrete row certificate of Definition 16.247. Then the raw nuisance list has the following exact tests.*

raw column	live coefficient	required disposition
$R_{B_2,p}, R_{P,p}, R_{D,p}$	$L_p^{\text{row}} A_{a,p}^{\text{raw}}$	0, by base source-linearity/deadness
$A_{\alpha,p}$	$L_p^{\text{row}} A_{\alpha,p}^{\text{raw}}$	0, or coherent affine recentering
$A_{<7,p}$	$L_p^{\text{row}} A_{<7,p}^{\text{raw}}$	0, by grade/order separation
$A_{7,p}$	$(\lambda_{77,p}, \lambda_{7Y,p})^T$	0 or coherent live routing into $L_{\text{eff},p}$
$A_{II,p}$	$(c_{\text{arith},7,p}, c_{\text{arith},Y,p})^T$	0 or coherent live routing into $L_{\text{eff},p}$
$A_{E_2,p}$	$L_p^{\text{row}} A_{E_2,p}^{\text{raw}}$	0, $O(Q^{-B})$, or coherent live routing

If any entry in the last three rows is atom-local live, the rank-two $C_7/Y12$ quotient route fails on that packet.

Proof. The first three rows are exactly the assertion that source-linear, affine, and lower-order chart/transport columns are in the base dead space or have no same-point one-whitening $Y12$ -signature. The A_7 , A_{II} , and A_{E_2} rows are not automatically dead: applying the concrete row certificate extracts precisely their live $C_7/Y12$ -coordinates. If such a coordinate is coherent, quotienting the unsplit column would kill a retained carrier direction; hence it must be removed from the dead quotient and added to the effective live carrier. If it is atom-local, then the actual tangent space contains a local free live $C_7/Y12$ direction, so no quotient that kills that tangent can preserve the two-dimensional live plane. $\square$

Theorem 16.250 (Rank-two quotient survival from the concrete row and column certificates). *Assume the concrete row certificate L_p^{row} of Definition 16.247 with $\|L_p^{\text{row}}\|_{\text{op}} \leq Q^C$, and assume the column routing certificate of Definition 16.248. Let*

$$q_{p,\text{cert}}^Y := I - P_{\mathcal{K}_{p,\text{dead}}^{Y,\text{cert}}}^{\text{bd}}.$$

Then $\sigma_{\min}(q_{p,\text{cert}}^Y E_p) \geq Q^{-C'}$. Moreover the quotient descent statement holds in the dead-only form

$$U_p^{\text{dead}} \subseteq \ker(q_{p,\text{cert}}^Y \Pi_{\Sigma(Y12),p} T_{\text{tr} \rightarrow \text{sing},p} P_{\text{tr},p}),$$

and no live $C_7/Y12$ torsion relation is introduced by quotienting.

Proof. By construction, $L_p^{\text{row}} A_{a,p}^{\text{dead}} = L_p^{\text{row}} A_{a,p}^{\text{raw}} - L_p^{\text{row}} E_p m_{a,p} = m_{a,p} - m_{a,p} = 0$. The base dead space is also contained in $\ker L_p^{\text{row}}$. Hence $L_p^{\text{row}} \mathcal{K}_{p,\text{dead}}^{Y,\text{cert}} = 0$. For every $v \in \mathbb{C}^2$,

$$v = L_p^{\text{row}} E_p v = L_p^{\text{row}} q_{p,\text{cert}}^Y E_p v,$$

because $E_p v - q_{p,\text{cert}}^Y E_p v \in \mathcal{K}_{p,\text{dead}}^{Y,\text{cert}}$. Therefore $\|v\| \leq \|L_p^{\text{row}}\|_{\text{op}} \|q_{p,\text{cert}}^Y E_p v\| \leq Q^C \|q_{p,\text{cert}}^Y E_p v\|$. Taking the infimum over $\|v\| = 1$ gives the singular-value lower bound. The quotient-descent statement follows because only row-dead columns are put into U_p^{dead} , and these columns span the certified dead quotient. $\square$

Corollary 16.251 (Treatment of the primitive arithmetic coefficient). *Let*

$$c_{\text{arith},Y} := [II_{\text{arith},9}^{\text{exact}}]_{D^2 N_0, \text{ same, } w=1, b=12}.$$

Under the concrete row certificate, $c_{\text{arith},Y}$ is classified as follows.

- (1) If $c_{\text{arith},Y} = 0$, the primitive arithmetic column is row-dead in the $Y12$ slot and may enter the exact dead quotient after the remaining C_7 -coordinate is also dead.
- (2) If $c_{\text{arith},Y} \neq 0$ but is coherent/fixed, the term is not dead. It must be routed into $M_{\text{live},p}$ and hence into $L_{\text{eff},p} = L_{\text{clean},p} + M_{\text{live},p}$.
- (3) If $c_{\text{arith},Y} \neq 0$ and is atom-local/free, then $C_{Y,p}^{\text{tar}}$ lies in an admitted local tangent direction and the rank-two quotient survival route is invalid on that packet.

Thus the draft no longer needs a global assertion $c_{\text{arith},Y} = 0$ if the coefficient is coherent and the effective carrier remains polynomially invertible. What is forbidden is quotient-killing an unsplit arithmetic column containing a live primitive $Y12$ -component.

Theorem 16.252 (Cap/ $Y12$ bridge after concrete row certification). *Assume the hypotheses of Theorem 16.250. Let*

$$L_{\text{eff},p} := L_{\text{clean},p} + M_{\text{live},p}$$

be the effective carrier matrix after coherent live routing, and assume $\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-C}$. Assume also the already-proved actual-source $D4$ amplitude survival on the selected cap core and the error bound

$$\Phi_p^Y \Gamma_p = H_p E_p L_{\text{eff},p} \Gamma_p + \varepsilon_p, \quad \sum_{p \in C_j} \mu_p \|q_{p,\text{cert}}^Y \varepsilon_p\|^2 \leq Q^{-B} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2.$$

Then

$$\sum_{p \in C_j} \mu_p \|q_{p,\text{cert}}^Y \Phi_p^Y \Gamma_p\|^2 \geq Q^{-A} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2.$$

In particular, the static-line handoff and the nonstatic cap bridge are both closed in the certified quotient.

Proof. Theorem 16.250 gives $\sigma_{\min}(q_{p,\text{cert}}^Y E_p) \geq Q^{-C_1}$, so $\|q_{p,\text{cert}}^Y E_p L_{\text{eff},p} \Gamma_p\| \geq Q^{-C_1-C} \|\Gamma_p\|$. Using $\|x+y\|^2 \geq \frac{1}{2}\|x\|^2 - \|y\|^2$, $\|q_{p,\text{cert}}^Y \Phi_p^Y \Gamma_p\|^2 \geq \frac{1}{2}Q^{-2C_1-2C} |H_p|^2 \|\Gamma_p\|^2 - \|q_{p,\text{cert}}^Y \varepsilon_p\|^2$. Summing over C_j , applying $D4$ amplitude survival, and taking B larger than the polynomial losses gives the claim. $\square$

Proposition 16.253 (Atom-local live column obstruction). *If any raw nuisance column has an atom-local live component $L_p^{\text{row}} A_{a,p}^{\text{raw}} = m_{a,p} \neq 0$ and the full column is required to be killed by the quotient, then no rank-two quotient preserving $E_p = [C_{7,p}^{\text{tar}}, C_{Y,p}^{\text{tar}}]$ can exist on that packet.*

Proof. If the full column is killed and its non-dead part is $E_p m_{a,p}$, then $E_p m_{a,p} \in \mathcal{K}_p^Y$, hence $q_p^Y E_p m_{a,p} = 0$. Since $m_{a,p} \neq 0$, the map $q_p^Y E_p$ has a nontrivial kernel and so its smallest singular value is zero. $\square$

Remark 16.254 (Status after the coordinator review). *Status: SUPERSEDED BY CERTIFICATION.* This paragraph records the pre-verification target that led to the concrete certificate $\mathfrak{C}_{2Y}(p)$. The abstract rank-two $C_7/Y12$ quotient-survival hypothesis has been retired in favour of the certified live-row quotient and the concrete row/column/effective-carrier certificate. The required packet facts are now collected in Theorem 16.280; when that theorem's coefficient hypotheses hold, it supplies

$$\sigma_{\min}(q_{p,\text{cert}}^Y E_p) + \sigma_{\min}(L_{\text{eff},p}) \geq Q^{-A}$$

up to polynomial constants and closes the rank-two quotient/effective-carrier part of the cap/ $Y12$ bridge. If an atom-local live column remains, the route is false on that packet; otherwise live coherent columns are routed into $L_{\text{eff},p}$, not quotient-killed.

Definition 16.255 (Explicit coefficient certificate for the concrete rank-two quotient). For a selected packet p , let $E_p := [C_{7,p}^{\text{tar}}, C_{Y,p}^{\text{tar}}]$ and

$$L_p^{\text{row}} = \begin{pmatrix} \ell_{7,p}^Y \\ \ell_{9,p}^Y \end{pmatrix}, \quad \ell_{7,p} := \frac{\Delta_7^Y}{\Delta_7^Y(C_7)}, \quad \ell_{9,p}^Y := \omega_Y - \omega_Y(e_7)\ell_{7,p}.$$

The explicit coefficient certificate $\mathfrak{R}_{2Y}(p)$ is the following list.

(1) *Row construction and conditioning:*

$$|\Delta_7^Y(C_7)| \geq Q^{-A}, \quad \|\Delta_7^Y\| \leq Q^A,$$

$$\ell_{7,p}(C_{Y,p}^{\text{tar}}) = 0, \quad \omega_Y(C_{Y,p}^{\text{tar}}) = 1,$$

$$\omega_Y(W_{\text{dead}}^{\text{fixed}}) = 0, \quad \|\omega_Y\| \leq Q^A, \quad |\omega_Y(e_7)| \leq Q^A.$$

Equivalently, the septic exactness, grade-separation, and fixed-whitening separation tests hold with polynomial margins.

(2) *Base source-linear/dead rows:*

$$L_p^{\text{row}} R_{B_2,p} = 0, \quad L_p^{\text{row}} R_{P,p} = 0, \quad L_p^{\text{row}} R_{D,p} = 0.$$

(3) *Affine and lower-order rows:*

$$L_p^{\text{row}} A_{\alpha,p}^{\text{raw}} \in \mathcal{C}_{\text{coh},p}^{2Y}, \quad L_p^{\text{row}} A_{<7,p}^{\text{raw}} = 0,$$

where coherent affine recentering is routed to the effective carrier and is not placed in the dead quotient.

(4) *Order-seven, arithmetic, and completion rows:*

$$L_p^{\text{row}} A_{7,p}^{\text{raw}} = (\lambda_{77,p}, \lambda_{7Y,p})^T,$$

$$L_p^{\text{row}} A_{II,p}^{\text{raw}} = (c_{\text{arith},7,p}, c_{\text{arith},Y,p})^T,$$

$$L_p^{\text{row}} A_{E_2,p}^{\text{raw}} = m_{E_2,p}.$$

Each of these live coefficient vectors is either zero, residual of weighted cap norm $O(Q^{-B})$, or fixed/coherent and routed into $M_{\text{live},p}$. No atom-local live coefficient is quotient-killed.

(5) *Effective carrier noncancellation:* after all coherent live vectors have been routed into $L_{\text{eff},p} := L_{\text{clean},p} + M_{\text{live},p}$,

$$\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-A}.$$

Equivalently, after triangular row operations,

$$L_{\text{eff},p} \sim \begin{pmatrix} c_{7,p}^{\text{eff}} & 0 \\ \beta_{9,p}^{\text{eff}} & \alpha_{9,p}^{\text{eff}} \end{pmatrix}, \quad |c_{7,p}^{\text{eff}} \alpha_{9,p}^{\text{eff}}| \geq Q^{-A}.$$

Lemma 16.256 (Verification of the concrete row certificate from $\mathfrak{R}_{2Y}$). *If the row-construction part of $\mathfrak{R}_{2Y}(p)$ holds, then*

$$L_p^{\text{row}} E_p = I_2, \quad \|L_p^{\text{row}}\|_{\text{op}} \leq Q^A.$$

Thus L_p^{row} is a concrete row certificate in the sense of Definition 16.247.

Proof. By definition, $\ell_{7,p}(C_{7,p}^{\text{tar}}) = 1$. The grade-separation condition gives $\ell_{7,p}(C_{Y,p}^{\text{tar}}) = 0$. For the second row, $\ell_{9,p}^Y(C_{7,p}^{\text{tar}}) = \omega_Y(e_7) - \omega_Y(e_7) \cdot 1 = 0$ and $\ell_{9,p}^Y(C_{Y,p}^{\text{tar}}) = \omega_Y(C_{Y,p}^{\text{tar}}) - \omega_Y(e_7) \cdot 0 = 1$. Hence $L_p^{\text{row}} E_p = I_2$. The norm bound follows from $\|\ell_{7,p}\| \leq \|\Delta_7^{\nabla}\|/|\Delta_7^{\nabla}(C_7)| \leq Q^A$ and $\|\ell_{9,p}^Y\| \leq \|\omega_Y\| + |\omega_Y(e_7)| \|\ell_{7,p}\| \leq Q^A$ after enlarging the exponent. $\square$

Proposition 16.257 (Verification of the column routing certificate from $\mathfrak{R}_{2Y}$). *Assume $\mathfrak{R}_{2Y}(p)$. Define, for every raw nuisance column, $m_{a,p} := L_p^{\text{row}} A_{a,p}^{\text{raw}}$ and $A_{a,p}^{\text{dead}} := A_{a,p}^{\text{raw}} - E_p m_{a,p}$. Then $L_p^{\text{row}} A_{a,p}^{\text{dead}} = 0$ for every raw nuisance column, no atom-local live vector is placed into the dead quotient, and every coherent live vector is included in $M_{\text{live},p}$. Hence the column routing certificate of Definition 16.248 holds.*

Proof. The identity $L_p^{\text{row}} A_{a,p}^{\text{dead}} = L_p^{\text{row}} A_{a,p}^{\text{raw}} - L_p^{\text{row}} E_p m_{a,p} = m_{a,p} - m_{a,p} = 0$ is formal. The substantive content is precisely the classification asserted in $\mathfrak{R}_{2Y}(p)$: source-linear, affine-dead, and lower-order columns are row-dead or coherent, while the order-seven, arithmetic, and completion columns have no atom-local live part and any coherent live part is routed to $M_{\text{live},p}$, not to the dead quotient. $\square$

Theorem 16.258 (Concrete certificate closes the rank-two cap/Y12 bridge). *If $\mathfrak{R}_{2Y}(p)$ holds on every selected packet in the frozen cap, then*

$$\sigma_{\min}(q_{p,\text{cert}}^Y E_p) \geq Q^{-A},$$

the dead-only quotient descent holds, and the effective carrier is polynomially invertible. Consequently, together with the already established actual-source D4 amplitude survival and the small quotient-error estimate, Theorem 16.252 applies.

Proof. Lemma 16.256 gives the bounded detecting rows. Proposition 16.257 gives the certified dead-only nuisance quotient. Hence Theorem 16.250 gives $\sigma_{\min}(q_{p,\text{cert}}^Y E_p) \geq Q^{-A}$ and dead-only quotient descent. The effective carrier lower bound is the last clause of $\mathfrak{R}_{2Y}(p)$. The cap/Y12 lower bound is then Theorem 16.252. $\square$

Remark 16.259 (Verification status of $\mathfrak{R}_{2Y}$ in the present draft). *Status: reduced to concrete coefficient verification.* The finite-dimensional quotient algebra and the effective-carrier algebra are now theorem-level once the packet certificate $\mathfrak{R}_{2Y}(p)$, equivalently $\mathfrak{C}_{2Y}(p)$, has been verified or imported on the selected packets. The checklist below is therefore not a downstream same-norm obstruction; it is the raw row-table entry condition for the certified branch.

The entries are:

- (i) septic nonvanishing and grade separation;
- (ii) fixed-whitening separation;
- (iii) row-deadness of $R_{B_2}, R_P, R_D, A_\alpha, A_{<7}$;
- (iv) classification of A_7, A_{II}, A_{E_2} as zero/residual/coherent, not atom-local;
- (v) effective carrier noncancellation $\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-A}$.

Items (i)–(iv) are actual coefficient checks. Item (v) is supplied in the active carrier-quotient route by the six-row carrier correction table and the effective-carrier theorem, rather than by an independent anti-cancellation hypothesis.

The subsequent verification subsection proves the implications

$$\mathfrak{R}_{2Y}(p) \implies \mathfrak{C}_{2Y}(p) \implies \sigma_{\min}(q_{p,\text{cert}}^Y E_p) \geq Q^{-A}, \quad \sigma_{\min}(L_{\text{eff},p}) \geq Q^{-A}.$$

It should not be read as a proof of the raw coefficient facts for arbitrary packets. The primitive arithmetic coefficient is handled by the same typed classification: if it is nonzero and coherent, it is routed into the live affine/source-graded ledger; if it is atom-local and quotient-killed, the certified quotient route is inadmissible on that packet.

16.25. Verification of the concrete $C_7/Y12$ row certificate. The preceding section reduces rank-two quotient survival to the concrete row certificate and the column routing certificate. This subsection records which parts of that certificate are now finite-dimensional consequences of already stated exact-sequence hypotheses, and which parts remain genuine coefficient verifications. Throughout, set

$$E_p := [C_{7,p}^{\text{tar}} \quad C_{Y,p}^{\text{tar}}], \quad L_p^{\text{row}} := \begin{pmatrix} \ell_{7,p} \\ \ell_{9,p}^Y \end{pmatrix}.$$

Definition 16.260 (Row-exactness certificate). We say that packet p satisfies the *row-exactness certificate* $\mathcal{E}_{2Y}(p)$ if the following hold.

- (1) The septic exact sequence hypotheses (SE1)–(SE2) from Theorem 16.65 hold, and

$$\|\Delta_7^\nabla\| \leq Q^A, \quad |\Delta_7^\nabla(C_{7,p}^{\text{tar}})| \geq Q^{-A}.$$

- (2) The grade separation condition $\Delta_7^\nabla(C_{Y,p}^{\text{tar}}) = 0$ holds.

- (3) The whitening exact sequence hypotheses (W1)–(W2) from Theorem 16.71 hold after applying the same trace-to-singular and Y12-projection maps, and the corresponding covector satisfies

$$\omega_Y(C_{Y,p}^{\text{tar}}) = 1, \quad \|\omega_Y\| \leq Q^A, \quad |\omega_Y(C_{7,p}^{\text{tar}})| \leq Q^A.$$

All constants are allowed to change from line to line, but only by polynomial losses.

Theorem 16.261 (Concrete live rows from septic and whitening exactness). *Assume $\mathcal{E}_{2Y}(p)$. Define*

$$\ell_{7,p} := \frac{\Delta_7^\nabla}{\Delta_7^\nabla(C_{7,p}^{\text{tar}})}, \quad \ell_{9,p}^Y := \omega_Y - \omega_Y(C_{7,p}^{\text{tar}})\ell_{7,p}.$$

Then $L_p^{\text{row}} E_p = I_2$ and $\|L_p^{\text{row}}\|_{\text{op}} \leq Q^{A'}$. Moreover, every vector lying in the septic exact denominator and in the fixed-whitening dead space is killed by both rows.

Proof. The lower bound on $\Delta_7^\nabla(C_{7,p}^{\text{tar}})$ and the norm bound on Δ_7^∇ give $\|\ell_{7,p}\| \leq \|\Delta_7^\nabla\| |\Delta_7^\nabla(C_{7,p}^{\text{tar}})|^{-1} \leq Q^{A'}$. By definition, $\ell_{7,p}(C_{7,p}^{\text{tar}}) = 1$ and $\ell_{7,p}(C_{Y,p}^{\text{tar}}) = 0$. The whitening normalization gives $\omega_Y(C_{Y,p}^{\text{tar}}) = 1$, and the grade separation gives $\ell_{9,p}^Y(C_{Y,p}^{\text{tar}}) = \omega_Y(C_{Y,p}^{\text{tar}}) - \omega_Y(C_{7,p}^{\text{tar}}) \cdot 0 = 1$, $\ell_{9,p}^Y(C_{7,p}^{\text{tar}}) = \omega_Y(C_{7,p}^{\text{tar}}) - \omega_Y(C_{7,p}^{\text{tar}}) \cdot 1 = 0$. Thus $L_p^{\text{row}} E_p = I_2$. The norm bound for $\ell_{9,p}^Y$ follows from $\|\ell_{9,p}^Y\| \leq \|\omega_Y\| + |\omega_Y(C_{7,p}^{\text{tar}})| \|\ell_{7,p}\| \leq Q^{A'}$. The row-deadness assertion follows from (SE1) and from $\omega_Y(W_{\text{dead}}^{\text{fixed}}) = 0$. $\square$

Remark 16.262 (What is no longer a black box). The existence of bounded detecting rows is not a free choice. It is proved by Theorem 16.261 once the septic exactness/separation and fixed-whitening exactness/separation assertions are verified. Thus the row part of $\mathfrak{C}_{2Y}(p)$ has been reduced to the explicit coefficient tests $\mathcal{E}_{2Y}(p)$, rather than to an abstract Schur-gap hypothesis.

Definition 16.263 (Row signature of a raw nuisance column). Assume $\mathcal{E}_{2Y}(p)$, and let

$$\mathfrak{r}_{2Y,p}(A) := L_p^{\text{row}} A = \begin{pmatrix} \ell_{7,p}(A) \\ \ell_{9,p}^Y(A) \end{pmatrix} \in \mathbb{C}^2.$$

For a raw nuisance column A , the vector $\mathfrak{r}_{2Y,p}(A)$ is its *live row signature*. The column is row-dead exactly when $\mathfrak{r}_{2Y,p}(A) = 0$. If $\mathfrak{r}_{2Y,p}(A) \neq 0$, the column is not exact nuisance and must be routed as coherent live, residual live, or atom-local live, according to the locality filtration of Definition 16.74.

Lemma 16.264 (Origin and grade row-deadness). Assume $\mathcal{E}_{2Y}(p)$. Let A be any raw column whose projected trace-to-singular $Y12$ -component belongs to the sum of the base source-linear space, the fixed affine/chart normalization space, the wrong-origin whitening space, the lower-order < 7 space, and the transported fixed-whitening dead space, namely

$$A \in \mathcal{K}_{p,0}^{Y,\text{dead}} + \mathcal{D}_{7,\text{chart}} + \mathcal{D}_{7,\text{norm}} + \mathcal{D}_{7,\gamma/\text{bg}} + \mathcal{D}_{7,<7} + W_{\text{dead}}^{\text{fixed}}.$$

Then $\mathfrak{r}_{2Y,p}(A) = 0$. Consequently the columns $R_{B_2,p}, R_{P,p}, R_{D,p}$ are row-dead whenever their images are included in the base source-linear/dead quotient space, and $A_{<7,p}$ is row-dead whenever its image is genuinely of order < 7 with no corrected septic package-coordinate summand.

Proof. On the septic row, all listed spaces are contained in the typed denominator $\mathcal{D}_7^{\text{typed}}$ or in the exact kernel of Δ_7^∇ by Theorem 16.65; hence $\ell_{7,p}(A) = 0$. On the primitive $Y12$ row, the same spaces project into $W_{\text{dead}}^{\text{fixed}}$ or into wrong-origin / transported whitening data. Theorem 16.71 gives $\omega_Y(P_W A) = 0$. Since $\ell_{9,p}^Y = \omega_Y - \omega_Y(C_{7,p}^{\text{tar}})\ell_{7,p}$, both terms vanish and $\ell_{9,p}^Y(A) = 0$. $\square$

Lemma 16.265 (Affine and coherent-recentering columns are not quotient-killed). Assume $\mathcal{E}_{2Y}(p)$. Let $A_{\alpha,p}^{\text{raw}}$ be an affine/recentering column. Then exactly one of the following holds.

- (1) $\mathfrak{r}_{2Y,p}(A_{\alpha,p}^{\text{raw}}) = 0$, in which case the column is row-dead and may enter the exact quotient.
- (2) $\mathfrak{r}_{2Y,p}(A_{\alpha,p}^{\text{raw}}) \neq 0$, but the signature is determined by fixed affine chart data. Then the column is coherent live and must be routed into $M_{\text{live},p}$; it may not be included unsplit in the exact quotient.

In particular, an affine/recentering column is atom-local live only if the affine chart itself has been refit packet-by-packet after the live rows are chosen. Such refitting is excluded by the fixed-normalization convention in the quotient slice.

Proof. The affine action and its normalization functionals are fixed before the local $Y12$ live-row quotient is formed. Hence the row signature of an affine column is a function of the fixed affine chart data and not of an atom-local free parameter. If the signature vanishes the column is dead. If it does not vanish, quotienting the unsplit column would force a live relation in the plane spanned by $C_{7,p}^{\text{tar}}, C_{Y,p}^{\text{tar}}$. The live-coefficient obstruction of Lemma 16.237 therefore forces the coherent-routing alternative. $\square$

Lemma 16.266 (Order-seven raw column classification). *Assume $\mathcal{E}_{2Y}(p)$. Let $A_{7,p}^{\text{raw}}$ be the raw order-seven nuisance column. Decompose it by the locality filtration*

$$A_{7,p}^{\text{raw}} = A_{7,p}^{\text{fixed}} + A_{7,p}^{\text{coh}} + A_{7,p}^{\text{res}} + A_{7,p}^{\text{loc}}.$$

If $\Delta_7^\nabla(A_{7,p}^{\text{loc}}) = 0$ and $\omega_Y(P_W A_{7,p}^{\text{loc}}) = 0$, then $A_{7,p}^{\text{raw}}$ has no atom-local live component. Its fixed and coherent live signatures, if nonzero, are routed into $M_{\text{live},p}$, and its residual component is routed to the error term. If either displayed row evaluation is nonzero, the rank-two $C_7/Y12$ quotient route fails on that packet unless that local component is removed from the nuisance tangent.

Proof. The two displayed equalities say precisely that the local part is killed by the septic row and by the primitive Y12 row, hence is not an atom-local live tangent. Fixed/coherent terms with nonzero row signature are not exact nuisance; they are retained as coherent carrier corrections by Definition 16.248. Residual terms are routed to the error ledger. If the local part has a nonzero row signature, then quotient-killing it would annihilate a nonzero vector in the live carrier plane, contradicting rank-two survival by Lemma 16.237. $\square$

Theorem 16.267 (Primitive arithmetic coefficient classification). *Assume $\mathcal{E}_{2Y}(p)$. Let $A_{II,p}^{\text{raw}} = A_{II,p}^{\text{fixed}} + A_{II,p}^{\text{coh}} + A_{II,p}^{\text{res}} + A_{II,p}^{\text{loc}}$ be its locality filtration, and set*

$$\mathfrak{r}_{2Y,p}(A_{II,p}^{\text{raw}}) = \begin{pmatrix} c_{\text{arith},7,p} \\ c_{\text{arith},Y,p} \end{pmatrix}.$$

If

$$(AI\text{-loc}) \quad A_{II,p}^{\text{loc}} = 0 \quad \text{in the primitive same-point one-whitening slot,}$$

and the fixed arithmetic variations satisfy

$$(AI\text{-fix}) \quad P_W V_{\text{arith}}^{\text{fixed}} \subseteq W_{\text{dead}}^{\text{fixed}},$$

then the primitive atom-local obstruction vanishes, and $c_{\text{arith},Y,p} = 0$ for the exact fixed part; any nonzero Y12-row signature of $A_{II,p}^{\text{raw}}$ is coherent or residual and must be routed to $M_{\text{live},p}$ or to the residual error, not to the exact dead quotient. If (AI-loc) fails, the nonzero primitive coefficient $c_{\text{arith},Y,p}$ is atom-local and the rank-two quotient route is false on that packet.

Proof. The fixed part is killed by ω_Y by Theorem 16.71 and (AI-fix). Its septic component is killed by the septic exact denominator when admitted as exact nuisance. Hence the exact fixed part has zero row signature. The local primitive same-point one-whitening slot is exactly the slot detected by a_Y and normalized by $\omega_Y(a_Y) = 1$. If (AI-loc) fails, applying ω_Y detects a nonzero local Y12-coordinate; quotient-killing that column would kill $C_{Y,p}^{\text{tar}}$ or a nonzero linear combination of the two live carriers. If (AI-loc) holds, the remaining nonzero signatures are fixed/coherent or residual by the locality filtration, and are routed accordingly. $\square$

Corollary 16.268 (Safe treatment of $c_{\text{arith},Y}$). *The coefficient*

$$c_{\text{arith},Y} := [II_{\text{arith},9}^{\text{exact}}]_{D^2 N_0, \text{ same, } w=1, b=12}$$

may be placed in the exact dead quotient only in the zero case. If it is nonzero and coherent, it is absorbed into $L_{\text{eff},p}$. If it is nonzero and atom-local, the local rank-two quotient certificate fails. Thus “ $c_{\text{arith},Y} = 0$ ” is no longer a global necessity; the necessary condition is that no nonzero atom-local primitive Y12-coefficient be quotient-killed.

Theorem 16.269 (No atom-local live contribution from E_2/R_{lab}). *Assume the three guardrails of Theorem 16.75. Then $\mathfrak{r}_{2Y,p}(A_{E_2,p}^{\text{raw}})$ is zero, residual, or coherent. In particular $A_{E_2,p}^{\text{raw}}$ has no atom-local live $C_7/Y12$ component and is safe for the certified quotient after the canonical dead/coherent/residual split.*

Proof. The guardrails give $D_{\text{loc}} \Pi_{\text{fib},Y}(E_2 + R_{\text{lab}}) = 0$ by Theorem 16.75. Hence the local component of $A_{E_2,p}^{\text{raw}}$ has zero image under both live rows. The fixed part is part of the base row data, the coherent part is routed to C_{2Y}^{arith} and then to $M_{\text{live},p}$, and the residual part is routed to R_X^{2Y} . No atom-local live coefficient remains to be killed by the quotient. $\square$

Definition 16.270 (Coordinate-coherent live correction). A coherent live correction $M_{\text{live},p}$ is called *coordinate-coherent* if there exist polynomially conditioned matrices $U_p, V_p \in GL_2(\mathbb{C})$ and a residual matrix R_p with $\|R_p\| \leq Q^{-B}$ such that

$$L_{\text{clean},p} + M_{\text{live},p} = U_p L_{\text{clean},p} V_p + R_p, \quad \sigma_{\min}(U_p), \sigma_{\min}(V_p) \geq Q^{-A}, \quad \|U_p\| + \|V_p\| \leq Q^A.$$

This is the exact condition that the coherent live terms change the live-row basis and/or the domain slope chart rather than introduce an arbitrary cancelling additive perturbation.

Lemma 16.271 (Effective carrier noncancellation from coordinate coherence). Assume $\sigma_{\min}(L_{\text{clean},p}) \geq Q^{-A}$ and that $M_{\text{live},p}$ is coordinate-coherent in the sense of Definition 16.270. If B is larger than the polynomial losses, then $\sigma_{\min}(L_{\text{eff},p}) = \sigma_{\min}(L_{\text{clean},p} + M_{\text{live},p}) \geq Q^{-A'}$. Equivalently, after triangularization, $|c_{7,p}^{\text{eff}} \alpha_{9,p}^{\text{eff}}| \geq Q^{-A'}$.

Proof. For unit v , $\|U_p L_{\text{clean},p} V_p v\| \geq \sigma_{\min}(U_p) \sigma_{\min}(L_{\text{clean},p}) \sigma_{\min}(V_p) \geq Q^{-A'}$. Therefore $\|L_{\text{eff},p} v\| \geq Q^{-A'} - \|R_p\|$; taking B large gives the asserted bound. The triangular determinant statement follows from the equivalence between a polynomial singular-value lower bound for a 2×2 matrix and a polynomial lower bound for the product of its triangular pivots after polynomially conditioned row operations. $\square$

Proposition 16.272 (Unrestricted coherent live routing does not imply effective carrier coercivity). The concrete row certificate, the dead-column split, and dead-only quotient survival do not by themselves imply $\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-C}$. More precisely, there is a two-dimensional finite model in which $L_p^{\text{row}} E_p = I_2$, $\sigma_{\min}(q_{p,\text{cert}}^Y E_p) = 1$, all live nuisance pieces are coherently routed rather than quotient-killed, but $L_{\text{eff},p} = 0$.

Proof. Take $\mathcal{H}_p^Y = \mathbb{C}^2$, $E_p = I_2$, $L_p^{\text{row}} = I_2$. The concrete row certificate is exact and polynomially conditioned. Let

$$L_{\text{clean},p} = \begin{pmatrix} c & 0 \\ \beta & \alpha \end{pmatrix}, \quad c\alpha \neq 0.$$

Choose two coherent live raw columns $A_{1,p}^{\text{raw}} = E_p \begin{pmatrix} -c \\ -\beta \end{pmatrix}$ and $A_{2,p}^{\text{raw}} = E_p \begin{pmatrix} 0 \\ -\alpha \end{pmatrix}$. Their live coefficients are $m_{1,p} = L_p^{\text{row}} A_{1,p}^{\text{raw}} = \begin{pmatrix} -c \\ -\beta \end{pmatrix}$ and $m_{2,p} = L_p^{\text{row}} A_{2,p}^{\text{raw}} = \begin{pmatrix} 0 \\ -\alpha \end{pmatrix}$, and their dead parts are zero $A_{i,p}^{\text{dead}} = A_{i,p}^{\text{raw}} - E_p m_{i,p} = 0$. Thus the certified dead quotient introduces no live torsion relation; indeed $q_{p,\text{cert}}^Y E_p = I_2$. If the fixed carrier-entry rows are $\eta_1^* = (1, 0)$ and $\eta_2^* = (0, 1)$, then

$$M_{\text{live},p} = m_{1,p} \eta_1^* + m_{2,p} \eta_2^* = \begin{pmatrix} -c & 0 \\ -\beta & -\alpha \end{pmatrix} = -L_{\text{clean},p},$$

hence $L_{\text{eff},p} = L_{\text{clean},p} + M_{\text{live},p} = 0$. This proves that effective carrier coercivity is a separate coefficient anti-cancellation assertion, not a formal consequence of the live-row quotient certificate. $\square$

Definition 16.273 (Effective $2Y$ carrier determinant certificate). Write

$$L_{\text{clean},p} = \begin{pmatrix} c_{7,p} & 0 \\ \beta_{9,p}^{\text{tri}} & \alpha_{9,p} \end{pmatrix}, \quad M_{\text{live},p} = \begin{pmatrix} m_{11,p} & m_{12,p} \\ m_{21,p} & m_{22,p} \end{pmatrix}.$$

The packet p satisfies the *effective $2Y$ carrier determinant certificate*, denoted $\mathfrak{C}_{2Y}^{\text{det}}(p)$, if $\|L_{\text{eff},p}\|_{\text{op}} = \|L_{\text{clean},p} + M_{\text{live},p}\|_{\text{op}} \leq Q^C$ and $|\det(L_{\text{clean},p} + M_{\text{live},p})| \geq Q^{-C}$. Equivalently, in entries,

$$\begin{aligned} \det L_{\text{eff},p} &= (c_{7,p} + m_{11,p})(\alpha_{9,p} + m_{22,p}) - m_{12,p}(\beta_{9,p}^{\text{tri}} + m_{21,p}) \\ &= c_{7,p} \alpha_{9,p} + c_{7,p} m_{22,p} + \alpha_{9,p} m_{11,p} + m_{11,p} m_{22,p} - \beta_{9,p}^{\text{tri}} m_{12,p} - m_{12,p} m_{21,p}, \end{aligned}$$

and this determinant is polynomially bounded away from zero.

Lemma 16.274 (Determinant certificate implies effective carrier coercivity). If $\mathfrak{C}_{2Y}^{\text{det}}(p)$ holds, then $\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-C'}$.

Proof. For every 2×2 matrix L , $\sigma_{\min}(L)\sigma_{\max}(L) = |\det L|$. Applying this to $L = L_{\text{eff},p}$ gives $\sigma_{\min}(L_{\text{eff},p}) = |\det L_{\text{eff},p}|/\sigma_{\max}(L_{\text{eff},p}) \geq Q^{-C}/Q^C = Q^{-C'}$. $\square$

Definition 16.275 (Triangular effective carrier certificate). A stronger, triangular form of $\mathfrak{C}_{2Y}^{\det}(p)$ is the following set of coefficient conditions: $m_{12,p} = 0$, $|c_{7,p} + m_{11,p}| \geq Q^{-C}$, $|\alpha_{9,p} + m_{22,p}| \geq Q^{-C}$, and $|\beta_{9,p}^{\text{tri}} + m_{21,p}| \leq Q^C$. Equivalently,

$$L_{\text{eff},p} = \begin{pmatrix} c_{7,p}^{\text{eff}} & 0 \\ \beta_{9,p}^{\text{eff}} & \alpha_{9,p}^{\text{eff}} \end{pmatrix}, \quad |c_{7,p}^{\text{eff}}| \geq Q^{-C}, \quad |\alpha_{9,p}^{\text{eff}}| \geq Q^{-C}, \quad |\beta_{9,p}^{\text{eff}}| \leq Q^C.$$

Lemma 16.276 (Triangular certificate closes the effective carrier). *The triangular effective carrier certificate implies $\mathfrak{C}_{2Y}^{\det}(p)$, and hence $\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-C'}$.*

Proof. Under Definition 16.275, $\det L_{\text{eff},p} = c_{7,p}^{\text{eff}}\alpha_{9,p}^{\text{eff}}$, so $|\det L_{\text{eff},p}| \geq Q^{-2C}$. The displayed upper bounds on the three entries give $\|L_{\text{eff},p}\|_{\text{op}} \leq Q^{C_1}$. Lemma 16.274 then gives the singular-value lower bound. $\square$

Corollary 16.277 (Vanishing carrier correction closes the bottom item). *If all coherent live correction rows have zero projection to the retained carrier quotient, so that $M_{\text{live},p} = 0$, then $L_{\text{eff},p} = L_{\text{clean},p}$ and the bottom item of $\mathfrak{C}_{2Y}(p)$ is proved on every Gram-good packet.*

Proof. For the clean row-normal matrix,

$$L_{\text{clean},p} = \begin{pmatrix} c_{7,p} & 0 \\ \beta_{9,p}^{\text{tri}} & \alpha_{9,p} \end{pmatrix},$$

with

$$c_{7,p} = -\frac{1}{24\pi^2 g_{\text{Gr},p} h_{\text{Gr},p} (\sqrt{g_{\text{Gr},p}} + \sqrt{h_{\text{Gr},p}})^2},$$

$$\alpha_{9,p} = \frac{1}{64\pi^2 \sqrt{g_{\text{Gr},p} h_{\text{Gr},p}} h_{\text{Gr},p}^{3/2} (\sqrt{g_{\text{Gr},p}} + \sqrt{h_{\text{Gr},p}})}.$$

Hence

$$\det L_{\text{clean},p} = c_{7,p}\alpha_{9,p} = -\frac{1}{1536\pi^4 g_{\text{Gr},p}^{3/2} h_{\text{Gr},p}^3 (\sqrt{g_{\text{Gr},p}} + \sqrt{h_{\text{Gr},p}})^3}.$$

On a Gram-good packet, $Q^{-C} \leq g_{\text{Gr},p}, h_{\text{Gr},p} \leq Q^C$, so $|\det L_{\text{clean},p}| \geq Q^{-C}$ and $\|L_{\text{clean},p}\|_{\text{op}} \leq Q^C$. The determinant/singular-value identity gives $\sigma_{\min}(L_{\text{clean},p}) \geq Q^{-C'}$. Since $L_{\text{eff},p} = L_{\text{clean},p}$, the claim follows. $\square$

Remark 16.278 (Bottom line of $\mathfrak{R}_{2Y}$ after the carrier table). The final clause $\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-A}$ in Definition 16.255 is now supplied, in the active carrier-quotient route, by the six-row carrier correction table: $L_{\text{eff},p} = L_{\text{clean},p}^{\text{tri}}$, $\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-A}$. Equivalently, item (v) of the older effective-carrier audit is discharged by Lemma 16.326 and Theorem 16.329. The more general determinant certificate $\mathfrak{C}_{2Y}^{\det}(p)$ and the triangular certificate remain recorded as fallback tools for a different routing convention in which coherent live correction matrices are not forced to vanish in $\mathcal{Q}_{\text{car}}$. They are not additional assumptions in the active route.

Remark 16.279 (Status of the effective carrier item). *Status: PROVED under the packet coefficient certificate.* The generic warning in Proposition 16.272 remains useful: dead-column splitting alone does not force $\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-C}$. The active draft no longer uses that implication. Instead, effective-carrier coercivity is supplied by Theorem 16.280, whose eighth item is proved either from coordinate-coherent live corrections via Lemma 16.271, from the determinant certificate $\mathfrak{C}_{2Y}^{\det}(p)$ via Lemma 16.274, from the triangular/vanishing correction certificate via Lemma 16.276 and Corollary 16.277, or from the six-row carrier correction table via Theorem 16.329. Thus the effective-carrier item is not an open structural module; it is a finite coefficient certificate already encoded in $\mathfrak{C}_{2Y}(p)$.

Theorem 16.280 (Verification theorem for $\mathfrak{C}_{2Y}(p)$). *Assume the following packet-level coefficient facts.*

- (1) The row-exactness certificate $\mathcal{E}_{2Y}(p)$ holds.
- (2) The images of $R_{B_2,p}, R_{P,p}, R_{D,p}$ lie in the base source-linear/dead quotient space.
- (3) The affine/recentering column is fixed or coherent in the sense of Lemma 16.265.
- (4) The lower-order column $A_{<7,p}$ has no corrected septic package-coordinate summand.
- (5) The order-seven local component satisfies the two row-vanishing tests of Lemma 16.266.
- (6) The arithmetic column satisfies (AI-loc) and (AI-fix) of Theorem 16.267.
- (7) The E_2/R_{lab} no-atom-fitting guardrails of Theorem 16.75 hold.
- (8) The associated-graded six-row carrier correction table holds, namely Lemma 16.326; equivalently the line, background/core, and normalization/subtraction correction rows have zero retained $C_7/Y12$ -projection in $\mathcal{Q}_{\text{car}}$. The clean triangular D_4 carrier is Gram-good.

Then the full concrete certificate $\mathfrak{C}_{2Y}(p)$ holds:

$$\begin{aligned} L_p^{\text{row}} E_p &= I_2, \quad \|L_p^{\text{row}}\|_{\text{op}} \leq Q^A, \quad \sigma_{\min}(q_{p,\text{cert}}^Y E_p) \geq Q^{-A}, \\ U_p^{\text{dead}} &\subseteq \ker(q_{p,\text{cert}}^Y \Pi_{\Sigma(Y12),p} T_{\text{tr} \rightarrow \text{sing},p} P_{\text{tr},p}), \\ \sigma_{\min}(L_{\text{eff},p}) &\geq Q^{-A}. \end{aligned}$$

Thus the rank-two $Y12$ quotient survival and the effective row-normal D_4 carrier survival are proved on packet p .

Proof. The first assertion follows from Theorem 16.261. Items (2)–(4) and Lemmas 16.264 and 16.265 show that the base, affine, and lower-order columns are row-dead or coherent-reroutable. Item (5) and Lemma 16.266 eliminate atom-local live contributions from A_7 . Item (6) and Theorem 16.267 eliminate the primitive atom-local arithmetic obstruction, including the dangerous $c_{\text{arith},Y}$ slot. Item (7) and Theorem 16.269 eliminate atom-local live contributions from E_2/R_{lab} . Consequently every raw nuisance column is either row-dead, coherent-reroutable, or residual; no atom-local live column is quotient-killed. The certified dead quotient is therefore formed only from row-dead remainders, and Theorem 16.250 gives $\sigma_{\min}(q_{p,\text{cert}}^Y E_p) \geq Q^{-A}$ and the dead-only descent statement. Finally, item (8), Lemma 16.326, and Theorem 16.329 give $L_{\text{eff},p} = L_{\text{clean},p}^{\text{tri}}$ and $\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-A}$. Thus the effective carrier lower bound is supplied by the carrier-quotient six-row computation rather than by an independent coherent-live anti-cancellation hypothesis. $\square$

Remark 16.281 (Updated smallest remaining theorem). After Theorem 16.280 and the six-row carrier correction table, the remaining work is no longer the abstract construction of live rows, the Schur-gap linear algebra, or the effective-carrier determinant. The smallest coefficient theorem still to be checked from the actual formulas is the row/routing certificate:

- (a) $\Delta_7^{\nabla}(C_{7,p}^{\text{tar}})$ has a polynomial lower bound and the C_Y grade is separated;
- (b) $P_W V_{\text{arith}}^{\text{fixed}} \subseteq W_{\text{dead}}^{\text{fixed}}$ and no primitive atom-local $II_{\text{arith},9}$ slot appears;
- (c) E_2/R_{lab} satisfies the no-atom-fitting guardrails;
- (d) all nonzero live signatures of A_7, A_{II}, A_{E_2} are residual or coherent/rerouted, never atom-local quotient-

The coherent-live determinant item has been removed from this list in the active route: the six correction rows vanish in $\mathcal{Q}_{\text{car}}$, so $L_{\text{eff},p} = L_{\text{clean},p}^{\text{tri}}$. If any of (a)–(d) fails with an atom-local live vector, the rank-two $Y12$ quotient route is false on that packet, not merely incomplete.

Definition 16.282 (Actual finite row obstruction vector for the certified $2Y$ quotient). Assume the notation of Subsection 16.25. On a selected ordinary real-canonical packet define the actual

row obstruction vector

$$\mathfrak{B}_{2Y,p} := \begin{pmatrix} \Delta_7^\nabla(C_{Y,p}^{\text{tar}}) \\ Q^A \text{dist}(C_{7,p}^{\text{tar}}, \ker \Delta_7^\nabla)^{-1} \text{ failure flag} \\ P_{(W_{\text{dead}}^{\text{fixed}})^\perp} a_Y \\ P_{(W_{\text{dead}}^{\text{fixed}})^\perp} P_W V_{\text{arith}}^{\text{fixed}} \\ \Delta_7^\nabla(A_{7,p}^{\text{loc}}) \\ \omega_Y(P_W A_{7,p}^{\text{loc}}) \\ D_{\text{loc}} P_{(W_{\text{dead}}^{\text{fixed}})^\perp} [II_{\text{arith},9}^{\text{exact}}]_{D^2 N_0, \text{same}, w=1, b=12} \\ D_{\text{loc}} \Pi_{\text{fib},Y}(E_2 + R_{\text{lab}}) \end{pmatrix}.$$

The second entry is shorthand for the required polynomial lower bound $|\Delta_7^\nabla(C_{7,p}^{\text{tar}})| \geq Q^{-A} \|\Delta_7^\nabla\| \|C_{7,p}^{\text{tar}}\|$.

The third entry is required to be polynomially nonzero: $\|P_{(W_{\text{dead}}^{\text{fixed}})^\perp} a_Y\| \geq Q^{-A} \|a_Y\|$, whereas the fourth, fifth, sixth, seventh, and eighth entries are required to vanish, or to be already routed as residual/coherent data before forming the certified dead quotient. The seventh entry is the primitive arithmetic obstruction; in scalar notation it is precisely the atom-local part of

$$c_{\text{arith},Y} := [II_{\text{arith},9}^{\text{exact}}]_{D^2 N_0, \text{same}, w=1, b=12}.$$

Theorem 16.283 (Verification audit for $\mathfrak{R}_{2Y}(p)$ from raw row algebra). *The finite-dimensional quotient algebra in Theorem 16.250, Theorem 16.261, and Theorem 16.280 proves the certified rank-two quotient survival*

$$\sigma_{\min}(q_{p,\text{cert}}^Y [C_{7,p}^{\text{tar}} \ C_{Y,p}^{\text{tar}}]) \geq Q^{-A}$$

provided the actual obstruction vector of Definition 16.282 satisfies the following coefficient conditions:

- (1) $\Delta_7^\nabla(C_{Y,p}^{\text{tar}}) = 0$ and $|\Delta_7^\nabla(C_{7,p}^{\text{tar}})| \geq Q^{-A} \|\Delta_7^\nabla\| \|C_{7,p}^{\text{tar}}\|$.
- (2) $\text{dist}(a_Y, W_{\text{dead}}^{\text{fixed}}) \geq Q^{-A} |a_Y|$ and $P_W V_{\text{arith}}^{\text{fixed}} \subseteq W_{\text{dead}}^{\text{fixed}}$.
- (3) The actual local order-seven component is row-dead:

$$\Delta_7^\nabla(A_{7,p}^{\text{loc}}) = 0, \quad \omega_Y(P_W A_{7,p}^{\text{loc}}) = 0.$$

- (4) The primitive arithmetic same-point one-whitening slot is not atom-local:

$$D_{\text{loc}} P_{(W_{\text{dead}}^{\text{fixed}})^\perp} [II_{\text{arith},9}^{\text{exact}}]_{D^2 N_0, \text{same}, w=1, b=12} = 0.$$

- (5) The completion/labelled-transition contribution is not atom-local: $D_{\text{loc}} \Pi_{\text{fib},Y}(E_2 + R_{\text{lab}}) = 0$.

If all five items hold, then the concrete coefficient certificate $\mathfrak{R}_{2Y}(p)$, and hence $\mathfrak{C}_{2Y}(p)$, holds on the packet. If item (4) fails, then the primitive coefficient $c_{\text{arith},Y}$ is atom-local live and the certified rank-two $C_7/Y12$ quotient route is false on that packet unless that live component is removed from the exact nuisance quotient and routed as coherent/residual data.

Proof. Items (1)–(2) are exactly the row-exactness certificate $\mathcal{E}_{2Y}(p)$: item (1) gives the normalized septic row $\ell_{7,p}$, and item (2) gives the primitive whitening row ω_Y . The triangular definition $\ell_{9,p}^Y = \omega_Y - \omega_Y(C_{7,p}^{\text{tar}})\ell_{7,p}$ then gives $L_p^{\text{row}} E_p = I_2$ and $\|L_p^{\text{row}}\|_{\text{op}} \leq Q^A$ by Theorem 16.261.

The base source-linear columns $R_{B_{2,p}}, R_{P,p}, R_{D,p}$, the affine fixed/recentering column, and the lower-order column $A_{<7,p}$ are killed or coherently routed by Lemmas 16.264 and 16.265, once they are placed in the origin/grade spaces listed there. Item (3) is precisely the atom-local part of the order-seven test in Lemma 16.266. Item (4), together with item (2), is precisely the primitive arithmetic test in Theorem 16.267. Item (5) is the hypothesis needed in Theorem 16.269.

Therefore every raw nuisance column is row-dead, residual, or coherent/reroutable; no atom-local live column is quotient-killed. The certified dead quotient is formed only from row-dead remainders, and the singular-value lower bound follows from Theorem 16.250. The effective carrier item is already supplied in the active route by the six-row carrier correction table and Theorem 16.329.

If item (4) fails, the vector $D_{\text{loc}} P_{(W_{\text{dead}}^{\text{fixed}})^\perp} [II_{\text{arith},9}^{\text{exact}}]_{D^2 N_0, \text{same}, w=1, b=12}$ is a nonzero local primitive $Y12$ -coordinate. Placing the unsplit arithmetic column in the exact dead quotient then

kills a nonzero vector in the live plane, contradicting Lemma 16.237. Hence the certified route is invalid on that packet unless the component is rerouted out of the exact quotient. $\square$

Remark 16.284 (Current status of the row/routing certificate). The certified quotient/effective-carrier implication is proved by Theorem 16.280: once the concrete packet certificate $\mathfrak{C}_{2Y}(p)$ is verified, it supplies

$$\sigma_{\min}(q_{p,\text{cert}}^Y E_p) \geq Q^{-A}, \quad \sigma_{\min}(L_{\text{eff},p}) \geq Q^{-A}.$$

This remark is the sole remaining ledger entry for the *raw* verification of that packet certificate. The canonical source-graded $K9$ repair does not prove the septic exact sequence, whitening exact sequence, or no-atom-fitting rows from first principles; those raw coefficient facts are precisely the entries of $\mathfrak{B}_{2Y,p}$. Therefore any downstream statement saying “the certified quotient is proved” means “proved after $\mathfrak{C}_{2Y}(p)$ has been verified on the selected packets.” It should not be read as an independent proof of the raw coefficient table.

If the raw formulas establish the entries of $\mathfrak{B}_{2Y,p}$ on every selected packet, this ledger entry may be upgraded to “closed”. Until then, this is an upstream certificate audit, not a same-norm final-bridge obstruction.

Remark 16.285 (Consistency of the certified branch with the row/routing audit). The canonical source-graded closure does not remove the need for the concrete row/routing certificate $\mathfrak{R}_{2Y}(p)$. It removes the downstream finite-main Schur and same-norm affine-forcing obstructions once the certified branch is entered. Thus statements saying “the certified quotient is proved” mean: proved on packets for which $\mathfrak{R}_{2Y}(p)$ has been verified or explicitly imported. Statements saying the raw row/routing certificate remains a finite audit refer only to the coefficient-table verification that enters the branch; they do not reopen the positive-density extraction covariance, the finite-main Schur complement, or the source-graded same-norm comparison.

Remark 16.286 (Superseding the older primitive-only quotient ledger). Older ledgers that say the only unresolved quotient point is $c_{\text{arith},Y} = 0$ should be read in the certified sense of Theorem 16.283. The active proof does not require a global scalar identity $c_{\text{arith},Y} = 0$. It requires the stronger typed statement that the atom-local primitive component

$$D_{\text{loc}} P_{(W_{\text{dead}}^{\text{fixed}})^{\perp}} [II_{\text{arith},9}^{\text{exact}}]_{D^2 N_0, \text{same}, w=1, b=12}$$

vanishes, or else is not placed in the exact dead quotient. Coherent or residual primitive terms are allowed only after being routed out of the exact quotient. A nonzero atom-local primitive term is a genuine local failure of rank-two $C_7/Y12$ quotient survival.

Definition 16.287 (Certified live-row quotient). Let $\mathcal{H}_p^Y$ be the finite-dimensional $Y12$ -boundary output space and let

$$E_p := [C_{7,p}^{\text{tar}} \quad C_{Y,p}^{\text{tar}}] : \mathbb{C}^2 \rightarrow \mathcal{H}_p^Y.$$

A *certified live-row quotient* consists of two rows collected as a map

$$L_p^{\text{live}} : \mathcal{H}_p^Y \rightarrow \mathbb{C}^2$$

such that

$$L_p^{\text{live}} E_p = I_2, \quad \|L_p^{\text{live}}\|_{\text{op}} \leq Q^C.$$

Given a raw nuisance list $\{A_{a,p}^{\text{raw}}\}_{a \in \mathcal{A}}$, define its live coefficient and dead part by

$$m_{a,p} := L_p^{\text{live}} A_{a,p}^{\text{raw}} \in \mathbb{C}^2, \quad A_{a,p}^{\text{dead}} := A_{a,p}^{\text{raw}} - E_p m_{a,p}.$$

The certified dead nuisance space is

$$\mathcal{K}_{p,\text{dead}}^{Y,\text{cert}} := \text{span}\{A_{a,p}^{\text{dead}} : a \in \mathcal{A}\} + \mathcal{K}_{p,0}^{Y,\text{dead}},$$

where $\mathcal{K}_{p,0}^{Y,\text{dead}} \subset \ker L_p^{\text{live}}$ is the base source-linear/dead quotient space. The quotient projector is

$$q_{p,\text{cert}}^Y := I - P_{\mathcal{K}_{p,\text{dead}}^{Y,\text{cert}}}^{\text{bd}}.$$

The discarded live matrix is

$$M_{\text{live},p} := [m_{a,p} : a \in \mathcal{A}].$$

All live coefficients are retained and routed into the effective carrier; no unsplit column with nonzero $m_{a,p}$ is placed in the dead quotient.

Theorem 16.288 (Certified quotient gives rank-two survival). *For a certified live-row quotient in the sense of Definition 16.287,*

$$\sigma_{\min}(q_{p,\text{cert}}^Y E_p) \geq Q^{-C'}.$$

Moreover every certified dead nuisance column is killed by the live rows:

$$L_p^{\text{live}} A_{a,p}^{\text{dead}} = 0,$$

and therefore no live $C_7/Y12$ torsion relation is introduced by forming $\mathcal{K}_{p,\text{dead}}^{Y,\text{cert}}$.

Proof. By construction,

$$L_p^{\text{live}} A_{a,p}^{\text{dead}} = L_p^{\text{live}} A_{a,p}^{\text{raw}} - L_p^{\text{live}} E_p m_{a,p} = m_{a,p} - m_{a,p} = 0,$$

and $L_p^{\text{live}} \mathcal{K}_{p,0}^{Y,\text{dead}} = 0$. Hence $L_p^{\text{live}} \mathcal{K}_{p,\text{dead}}^{Y,\text{cert}} = 0$. For every $v \in \mathbb{C}^2$,

$$v = L_p^{\text{live}} E_p v = L_p^{\text{live}} q_{p,\text{cert}}^Y E_p v,$$

because the difference $E_p v - q_{p,\text{cert}}^Y E_p v$ lies in $\mathcal{K}_{p,\text{dead}}^{Y,\text{cert}}$, which is annihilated by L_p^{live} . Thus

$$\|v\| \leq \|L_p^{\text{live}}\|_{\text{op}} \|q_{p,\text{cert}}^Y E_p v\| \leq Q^C \|q_{p,\text{cert}}^Y E_p v\|.$$

Taking the infimum over unit v gives the asserted singular-value lower bound. $\square$

Remark 16.289 (Edit to the quotient-descent statement). Replace the old descent statement

$$U_p \subseteq \ker(q_p^Y \Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing}} P_{\text{tr}})$$

by the dead-only statement

$$U_p^{\text{dead}} \subseteq \ker(q_{p,\text{cert}}^Y \Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing}} P_{\text{tr}}),$$

where each raw column of U_p has first been split as

$$A_{a,p}^{\text{raw}} = A_{a,p}^{\text{dead}} + E_p m_{a,p}.$$

The live summand $E_p m_{a,p}$ is not nuisance; it is retained and absorbed into the effective two-row carrier matrix. In particular a nonzero $c_{\text{arith},Y}$ is no longer allowed to be quotient-killed. It must either vanish by a coefficient theorem or be routed into the live matrix.

Definition 16.290 (Effective live carrier matrix after dead split). Let $L_{\text{clean},p}$ be the row-normal D4 matrix before raw nuisance splitting. The live coefficients removed from the raw nuisance list contribute a finite matrix $M_{\text{live},p}$ to the same retained $C_7/Y12$ carrier plane. Define

$$L_{\text{eff},p} := L_{\text{clean},p} + M_{\text{live},p}.$$

The row-normal D4 carrier condition is imposed on $L_{\text{eff},p}$:

$$\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-C}.$$

After elementary row operations in $\mathbb{C}^2$, this is equivalent to the triangular form

$$L_{\text{eff},p} \sim \begin{pmatrix} c_{7,p}^{\text{eff}} & 0 \\ \beta_{9,p}^{\text{eff}} & \alpha_{9,p}^{\text{eff}} \end{pmatrix}, \quad |c_{7,p}^{\text{eff}} \alpha_{9,p}^{\text{eff}}| \geq Q^{-C'}.$$

Proposition 16.291 (Cap lower bound from the certified quotient and effective carrier). *Assume the certified quotient of Definition 16.287, the effective carrier coercivity $\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-C}$, and the expansion*

$$\Phi_p^Y \Gamma_p = H_p E_p L_{\text{eff},p} \Gamma_p + \varepsilon_p, \quad \sum_{p \in C_j} \mu_p \|q_{p,\text{cert}}^Y \varepsilon_p\|^2 \leq Q^{-B} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2.$$

If the amplitude mass estimate

$$\sum_{p \in C_j} \mu_p |H_p|^2 \|\Gamma_p\|^2 \geq Q^{-A} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2$$

holds, then

$$\sum_{p \in C_j} \mu_p \|q_{p,\text{cert}}^Y \Phi_p^Y \Gamma_p\|^2 \geq Q^{-A'} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2.$$

Proof. By Theorem 16.288, $\sigma_{\min}(q_{p,\text{cert}}^Y E_p) \geq Q^{-C_1}$. Hence

$$\|q_{p,\text{cert}}^Y E_p L_{\text{eff},p} \Gamma_p\| \geq Q^{-C_1} Q^{-C} \|\Gamma_p\|.$$

The inequality $\|x + y\|^2 \geq \frac{1}{2} \|x\|^2 - \|y\|^2$ gives

$$\|q_{p,\text{cert}}^Y \Phi_p^Y \Gamma_p\|^2 \geq \frac{1}{2} Q^{-2C_1-2C} |H_p|^2 \|\Gamma_p\|^2 - \|q_{p,\text{cert}}^Y \varepsilon_p\|^2.$$

Summing with weights μ_p , applying amplitude mass, and choosing B larger than the accumulated polynomial losses proves the claim. $\square$

Theorem 16.292 (Amplitude mass from source-amplitude nonflatness and core transfer). *Let I be a routed block and suppose the scalar D_4 amplitude H belongs to a finite atlas $\mathcal{A}_H(I)$ with polynomial Bernstein/Riesz bounds. Assume the source-amplitude nonflatness estimate*

$$\int_I |H|^2 d\nu \geq Q^{-C_H} \nu(I), \quad \|H\|_{L^\infty(I)} \leq Q^{C_H},$$

and the predetermined-core transfer estimate: for every set $E \subset I$ selected before wall extraction with $\nu(E) \geq Q^{-C_E} \nu(I)$,

$$\sum_{p \in C_j \cap E} \mu_p \|\Gamma_p\|^2 \geq Q^{-C_R} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2.$$

Then there exists a predetermined coefficient-good core

$$E_I^{\text{core}} := \{t \in I : |H(t)| \geq Q^{-C_*}\}$$

with $\nu(E_I^{\text{core}}) \geq Q^{-C_E} \nu(I)$, and on every frozen cap $C_j \subset I$,

$$\sum_{p \in C_j} \mu_p |H_p|^2 \|\Gamma_p\|^2 \geq Q^{-A_*} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2.$$

Proof. Choose C_* much larger than C_H . If $E_I^{\text{core}} = \{|H| \geq Q^{-C_*}\}$ had measure smaller than $Q^{-C_E} \nu(I)$, then

$$\int_I |H|^2 d\nu \leq Q^{-2C_*} \nu(I) + Q^{2C_H} Q^{-C_E} \nu(I).$$

Taking C_*, C_E sufficiently large relative to C_H contradicts $\int_I |H|^2 d\nu \geq Q^{-C_H} \nu(I)$. Thus the core has polynomial measure and is selected using only H . On the core, $|H|^2 \geq Q^{-2C_*}$. The core-transfer estimate gives

$$\sum_{p \in C_j} \mu_p |H_p|^2 \|\Gamma_p\|^2 \geq Q^{-2C_*} \sum_{p \in C_j \cap E_I^{\text{core}}} \mu_p \|\Gamma_p\|^2 \geq Q^{-2C_*-C_R} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2.$$

$\square$

Proposition 16.293 (Why source-amplitude nonflatness is necessary). *The amplitude mass estimate in Theorem 16.292 is not a consequence of the rank-two quotient Schur gap, of the dead-column split, or of extraction covariance. In the local row algebra one may have*

$$q_0 = 2, \quad q_{\text{can}}'' = 0, \quad (q_\zeta)^{(4)} = 0,$$

so that

$$\Gamma_G = q_{\text{can}}'' - 6q_0 + 2q_0^3 = 4, \quad \Gamma_{G,2} = (1 - q_0^2)q_{\text{can}}'' - \frac{1}{6}(q_\zeta)^{(4)} = 0,$$

while

$$H = yq_{\text{can}}'' \sim -\frac{q_0(q_{\text{can}}'')^3}{288} = 0.$$

Thus the live anisotropy vector is nonzero but the D_4 amplitude vanishes. A polynomially small perturbation with $|q_{\text{can}}''| \leq Q^{-C}$ keeps $\max\{|\Gamma_G|, |\Gamma_{G,2}|\} \asymp 1$ while making $|H| \max\{|\Gamma_G|, |\Gamma_{G,2}|\} \ll$

Q^{-3C} . Therefore an unconditional proof must supply source-amplitude nonflatness, a source-robust row not divisible by H , or a separate branch that handles this source-flat regime.

Corollary 16.294 (Fully unconditional cap/Y12 bridge after source-amplitude supply). *Assume the quotient is replaced by the certified dead-only quotient of Definition 16.287; assume the effective D4 carrier matrix satisfies $\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-C}$; and assume source-amplitude nonflatness plus predetermined-core transfer as in Theorem 16.292. Then the four-item package*

dead-column split + bounded dual rows + analytic live-pair Schur gap + D4 amplitude survival is fully proved, and the cap/Y12 lower bound

$$\sum_{p \in C_j} \mu_p \|q_{p,\text{cert}}^Y \Phi_p^Y \Gamma_p\|^2 \geq Q^{-A} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2$$

holds on every selected good cap.

Proof. The certified quotient proves the dead-column split and the rank-two Schur gap by Theorem 16.288; bounded dual rows are part of the certificate and are equivalent to the Schur gap by Theorem 16.245. The source nonflatness/core-transfer theorem gives D4 amplitude survival. Proposition 16.291 then gives the cap/Y12 lower bound. $\square$

Corollary 16.295 (Cap transversality in certified live $C_7/Y12$ coordinates). Status: certified-input conditional. *This is the corrected cap bridge in certified live coordinates, replacing Corollary 16.315 per Lemma 16.236. The rank-two quotient and effective-carrier inputs are supplied by the certified quotient and $\mathfrak{C}_{2Y}(p)$; the remaining branch choice is the extraction covariance alternative $\delta_{\text{ext}} \geq Q^{-A}$ versus the certified static-line handoff of Corollary 16.214.*

Work on the proof-level certified branch. Set

$$Q_p^Y := q_{p,\text{cert}}^Y = I - P_{\mathcal{K}_{p,\text{dead}}^{Y,\text{cert}}}^{\text{bd}}, \quad E_p := [C_{7,p}^{\text{tar}} \quad C_{Y,p}^{\text{tar}}], \quad R_p^{\text{cert}} := Q_p^Y E_p.$$

Assume the certified row/routing certificate on the selected packet, so that

$$\sigma_{\min}(R_p^{\text{cert}}) \geq Q^{-C}, \quad \sigma_{\min}(L_{\text{eff},p}) \geq Q^{-C},$$

and assume the certified row-normal expansion

$$\Phi_p^Y(1, s) = H_p E_p L_{\text{eff},p} \begin{pmatrix} 1 \\ s \end{pmatrix} + \varepsilon_p(s), \quad \|\partial_s Q_p^Y \varepsilon_p(s)\| \leq Q^{-B}.$$

Define live scalar coordinates by

$$\rho_p(s) := \begin{pmatrix} \rho_{7,p}(s) \\ \rho_{Y,p}(s) \end{pmatrix} := (R_p^{\text{cert}})^\dagger Q_p^Y \Phi_p^Y(1, s).$$

Then, for B sufficiently large,

$$\rho_p(s) = H_p L_{\text{eff},p} \begin{pmatrix} 1 \\ s \end{pmatrix} + O(Q^{-B+C}).$$

After polynomially conditioned row operations on the two live coordinates,

$$L_{\text{eff},p} \sim \begin{pmatrix} c_{7,p}^{\text{eff}} & 0 \\ \beta_{9,p}^{\text{eff}} & \alpha_{9,p}^{\text{eff}} \end{pmatrix}, \quad |c_{7,p}^{\text{eff}} \alpha_{9,p}^{\text{eff}}| \geq Q^{-C}.$$

Consequently, on the coefficient-good core where the amplitude factor is pointwise polynomially nonzero,

$$\left| \det \frac{\partial(\rho_{7,p}, \rho_{Y,p})}{\partial(1, s)} \right| = |H_p|^2 |c_{7,p}^{\text{eff}} \alpha_{9,p}^{\text{eff}}| + O(Q^{-B+C}) \geq Q^{-A}.$$

Equivalently, in the certified quotient Hilbert norm, the vector-valued area Jacobian satisfies

$$\sqrt{\det((D_{(1,s)} Q_p^Y \Phi_p^Y)^* (D_{(1,s)} Q_p^Y \Phi_p^Y))} \geq Q^{-A}.$$

The projective Wronskian in the certified live coordinates obeys

$$W_p^{\text{cert}}(s) := \rho'_{Y,p}(s) \rho_{7,p}(s) - \rho'_{7,p}(s) \rho_{Y,p}(s) = H_p^2 c_{7,p}^{\text{eff}} \alpha_{9,p}^{\text{eff}} + O(Q^{-B+C}),$$

and the corresponding certified quotient Gram angle satisfies $\sin^2 \theta_p^{\text{cert}} \geq Q^{-A}$ under the standard polynomial upper-conditioning bounds for the finite carrier atlas.

For a quantitatively static cap and a static unit direction u_j , the same certified rank-two bound gives the typed static-line handoff

$$\sum_{p \in C_j} \mu_p \|Q_p^Y \Phi_p^Y u_j\|^2 \geq Q^{-A} \sum_{p \in C_j} \mu_p |H_p|^2 \|L_{\text{eff},p} u_j\|^2 - Q^{-B} \sum_{p \in C_j} \mu_p \|u_j\|^2.$$

Together with source-weighted $D4$ amplitude survival, this gives the same absolute lower bound needed by the certified atomic comparison. No separate static wall-escape estimate on $|\beta_{9,p}^{\text{eff}} + \alpha_{9,p}^{\text{eff}} s_j^*|$ is required on the certified rank-two branch.

Proof. The certified quotient gives $\|(R_p^{\text{cert}})^\dagger\| \leq Q^C$. Applying this left inverse to $Q_p^Y \Phi_p^Y(1, s)$ gives the displayed scalar coordinate expansion. The Jacobian determinant in these coordinates is $\det(H_p L_{\text{eff},p} + O(Q^{-B+C})) = H_p^2 \det L_{\text{eff},p} + O(Q^{-B+C})$. The effective-carrier certificate gives $|\det L_{\text{eff},p}| \geq Q^{-C}$, after the displayed triangularization, and the amplitude condition gives the pointwise lower bound on the core. The vector-valued area lower bound follows because R_p^{cert} has smallest singular value at least Q^{-C} . The Wronskian identity is the same determinant written in projective live coordinates; common derivatives of the scalar amplitude cancel in $\rho'_Y \rho_7 - \rho'_7 \rho_Y$, and the residual is absorbed by the Q^{-B} ledger. The Gram-angle bound is $\det G = \|v_1\|^2 \|v_2\|^2 \sin^2 \theta$ with polynomial upper conditioning. For the static direction, use $\|Q_p^Y E_p L_{\text{eff},p} u_j\| \geq \sigma_{\min}(Q_p^Y E_p) \sigma_{\min}(L_{\text{eff},p}) \|u_j\| \geq Q^{-C} \|u_j\|$, square, sum over the cap, and absorb the certified quotient error. The final absolute mass is supplied by the same source-weighted amplitude survival used in the nonstatic branch. $\square$

Hypothesis 16.296 (Cap amplitude noncollapse). *Status: cap-side condition; its carrier-mass strengthening is discharged by Corollary 16.307 on ordinary real-canonical routed blocks.* On every good frozen cap C_j , the scalar amplitude $H_{j,p}$ of the row-normal carrier expansion in Corollary 16.295 satisfies either the pointwise lower bound

$$|H_{j,p}| \geq Q^{-C} \quad \text{on the extracted cap core,}$$

or the weighted average

$$\sum_{p \in C_j} \mu_p |H_{j,p}|^2 \geq Q^{-A} \sum_{p \in C_j} \mu_p.$$

This is independent of the slope-variance identity $\delta_{\text{ext}} \geq Q^{-A}$: the latter constrains the spread of $s_p = \Gamma_{G,2}(p)/\Gamma_G(p)$ across the cap but does not lower-bound the scalar amplitude that multiplies $L_{\text{clean}}(1, s)$ inside Φ_p^Y . Without amplitude noncollapse, the cap Jacobian determinant $H_{j,p}^2 c_7 \alpha_9$ can be polynomially small even when the cap is nonstatic. Amplitude noncollapse is therefore an additional finite obligation of the cap bridge, on top of the rank-two quotient and row-normal carrier hypotheses.

Hypothesis 16.297 ($D4$ amplitude survival on the cap core (carrier-mass form)). *Status: exact cap-side condition, discharged on ordinary real-canonical blocks by Corollary 16.304 once Gate-1, Gate-2, source-factor residual control, and packet quadrature have been proved.* On every selected good cap C_j ,

$$\sum_{p \in C_j} \mu_p |H_p|^2 \|\Gamma_p\|^2 \geq Q^{-A} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2.$$

Equivalently, the $D4$ scalar amplitude H_p is not lost on the cap mass that carries the extraction covariance. This is the carrier-mass weighted strengthening of Hypothesis 16.296: it controls the actual contribution of the $D4$ carrier residual to the absolute atomic lower bound, not just the unweighted average of $|H_p|^2$. The slope-variance lower bound $\delta_{\text{ext}} \geq Q^{-A}$ does not imply this estimate, since δ_{ext} is independent of H_p and a configuration with strongly varying s_p and $H_p \equiv 0$ is consistent with nonstatic extraction covariance.

Definition 16.298 (D4 coefficient-good cap core). Let C_j be a frozen cap with nonnegative weights μ_p and extraction vectors $\Gamma_p \in \mathbb{C}^2$. A subset $C_j^{\text{core}} \subset C_j$ is called a *D4 coefficient-good cap core* if there are constants C_H, C_M such that the pointwise amplitude bound

$$|H_p| \geq Q^{-C_H} \quad (p \in C_j^{\text{core}})$$

holds and the core carries polynomially much of the weighted carrier mass:

$$\sum_{p \in C_j^{\text{core}}} \mu_p \|\Gamma_p\|^2 \geq Q^{-C_M} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2.$$

Lemma 16.299 (Buffered weighted point extraction). *Let I be a normalized packet interval, let*

$$I_\eta := \{m \in I : \text{dist}_{\text{norm}}(m, \partial I) \geq \eta\},$$

and let $d\nu = w_\nu(m) dm$ with $Q^{-C_\nu} \leq w_\nu \leq Q^{C_\nu}$. If

$$\int_{I_\eta} |f(m)|^2 d\nu(m) \geq Q^{-A} \nu(I),$$

then there exists $m_0 \in I_\eta$ such that

$$|f(m_0)| \geq Q^{-A/2}.$$

After increasing the exponent, the same conclusion holds if the right side is $Q^{-A} \nu(I_\eta)$ or if I_η has only polynomially comparable ν -mass to I .

Proof. If $|f| < Q^{-A/2}$ everywhere on I_η , then

$$\int_{I_\eta} |f|^2 d\nu < Q^{-A} \nu(I_\eta) \leq Q^{-A} \nu(I),$$

contradicting the hypothesis. The variants follow from $\nu(I_\eta) \asymp_{Q^C} \nu(I)$. $\square$

Lemma 16.300 (Buffered simultaneous point extraction). *Let I_η , $d\nu$, u , and Γ be as above. Assume*

$$\int_{I_\eta} |u|^2 \|\Gamma\|^2 d\nu \geq Q^{-A} \nu(I),$$

and the pointwise polynomial bounds

$$|u| \leq Q^{C_u}, \quad \|\Gamma\| \leq Q^{C_\Gamma}$$

on I_η . Then there is $m_0 \in I_\eta$ such that

$$|u(m_0)| \geq Q^{-B_u}, \quad \|\Gamma(m_0)\| \geq Q^{-B_\Gamma},$$

with B_u, B_Γ depending only on A, C_u, C_Γ and the density regularity exponent.

Proof. Choose $\tau := Q^{-(A+2C_\Gamma+3)/2}$ and $\gamma := Q^{-(A+2C_u+3)/2}$. If every point of I_η satisfied either $|u| < \tau$ or $\|\Gamma\| < \gamma$, then pointwise

$$|u|^2 \|\Gamma\|^2 \leq \tau^2 Q^{2C_\Gamma} + Q^{2C_u} \gamma^2 \leq 2Q^{-A-3} < Q^{-A},$$

for Q sufficiently large after harmless exponent adjustment. Integrating would contradict the lower bound. Hence some $m_0 \in I_\eta$ satisfies both lower bounds. Rename the exponents. $\square$

Lemma 16.301 (Quadratic escape from the source wall). *Let I be a normalized packet interval and let $\mathcal{D}$ denote the normalized derivative. Let q, u be complex-valued C^1 functions on I satisfying*

$$\mathcal{D}^2 q = u$$

in the weak, hence classical, sense on the packet. Assume

$$|u(m_0)| \geq \tau, \quad |\mathcal{D}u| \leq M$$

on the buffered subinterval containing m_0 . Let $0 < \varepsilon < 1$, and suppose

$$|q(m_0)| < \varepsilon.$$

If

$$r_\varepsilon := 4\sqrt{\varepsilon/\tau} \leq \min\{\text{dist}(m_0, \partial I), \tau/(8M)\},$$

then there is a point $m_1 \in I$, with $|m_1 - m_0| \leq r_\varepsilon$ in normalized distance, such that

$$|q(m_1)| \geq \varepsilon, \quad |u(m_1)| \geq \tau/2.$$

If, in addition, a vector field Γ satisfies

$$|\Gamma(m_0)| \geq \gamma, \quad |\mathcal{D}\Gamma| \leq M_\Gamma, \quad r_\varepsilon \leq \gamma/(2M_\Gamma),$$

then

$$|\Gamma(m_1)| \geq \gamma/2.$$

Proof. Set $e := \overline{u(m_0)}/|u(m_0)|$ and $Q_e(m) := \Re(eq(m))$. Then $\mathcal{D}^2 Q_e(m_0) = |u(m_0)| \geq \tau$. For $|m - m_0| \leq \tau/(8M)$, the bound on $\mathcal{D}u$ gives $\mathcal{D}^2 Q_e(m) \geq \tau/2$. Choose one of the two signs $\sigma \in \{\pm 1\}$ so that $\sigma \mathcal{D}Q_e(m_0) \geq 0$, and put $m_1 = m_0 + \sigma r_\varepsilon$. Taylor's formula in the normalized coordinate gives

$$Q_e(m_1) \geq Q_e(m_0) + \frac{\tau}{4} r_\varepsilon^2 \geq -\varepsilon + 4\varepsilon \geq \varepsilon,$$

so $|q(m_1)| \geq |Q_e(m_1)| \geq \varepsilon$. The Lipschitz bound on u and the size condition on r_ε give $|u(m_1)| \geq |u(m_0)| - Mr_\varepsilon \geq \tau/2$. The estimate for Γ is identical. $\square$

Theorem 16.302 (Actual-source D4 amplitude nonflatness). *Let I be a routed ordinary real-canonical block, with normalized measure $d\nu = w_\nu(m) dm$ satisfying the packet regularity bounds*

$$Q^{-C_\nu} \leq w_\nu(m) \leq Q^{C_\nu} \quad (m \in I).$$

Put

$$q = q_\zeta, \quad u = q''_{\text{can}} = \mathcal{D}^2 q, \quad H = yq''_{\text{can}}.$$

Assume the Gate-1 source-curvature lower bound

$$\int_I |u(m)|^2 d\nu(m) \geq Q^{-A_1} \nu(I).$$

Assume the source local-complexity bounds

$$|u| + |\mathcal{D}u| \leq Q^{C_u}, \quad |\mathcal{D}H| \leq Q^{C_H}$$

on the buffered block. Assume also the gauge-locked D4 source-factor identity with residual

$$H(m) = -\frac{q(m)u(m)^3}{288} + R_H(m),$$

and suppose that, whenever $|q(m)| \geq Q^{-B_q}$ and $|u(m)| \geq Q^{-B_u}$, one has the relative residual estimate

$$|R_H(m)| \leq \frac{1}{576} |q(m)| |u(m)|^3.$$

Then there is an exponent A_H , depending only on the displayed polynomial exponents, such that

$$\int_I |H(m)|^2 d\nu(m) \geq Q^{-A_H} \nu(I).$$

More precisely, there exists a subinterval $J_H \subset I$, of normalized length at least Q^{-A_H} , on which $|H(m)| \geq Q^{-A_H}$.

Proof. By the boundary-collar/buffer exclusion already used in the packet construction, we may replace I by a buffered subinterval $I_\eta \subset I$, $\eta \geq Q^{-C_\eta}$, with

$$\int_{I_\eta} |u|^2 d\nu \geq Q^{-A'_1} \nu(I).$$

Lemma 16.299 gives $m_0 \in I_\eta$ with $|u(m_0)| \geq \tau := Q^{-B_u}$. Choose $\varepsilon = Q^{-B_q}$ so large in exponent that

$$r_\varepsilon := 4\sqrt{\varepsilon/\tau} \leq \min\{\eta/2, \tau/(8Q^{C_u})\}.$$

If $|q(m_0)| \geq \varepsilon$, put $m_1 = m_0$; otherwise apply Lemma 16.301 with $M = Q^{C_u}$. In both cases $m_1 \in I$ and $|q(m_1)| \geq \varepsilon$, $|u(m_1)| \geq \tau/2$. The gauge-locked D4 source-factor identity and the residual hypothesis yield

$$|H(m_1)| \geq \frac{1}{576} |q(m_1)| |u(m_1)|^3 \geq Q^{-B_H}.$$

Since $|\mathcal{D}H| \leq Q^{C_H}$, the lower bound persists on a normalized interval $J_H \subset I$ of length at least $Q^{-B_H - C_H - O(1)}$:

$$|H(m)| \geq \frac{1}{2} Q^{-B_H} \quad (m \in J_H).$$

The density bounds give

$$\int_I |H|^2 d\nu \geq \int_{J_H} |H|^2 d\nu \geq Q^{-A_H} \nu(I)$$

for a polynomial exponent A_H . □

Theorem 16.303 (Source-weighted D4 amplitude nonflatness). *In addition to the hypotheses of Theorem 16.302, let $\Gamma = (\Gamma_G, \Gamma_{G,2})$, $W_2 := \|\Gamma\|^2$, and assume the Gate-2 source-anisotropy mass bound*

$$\int_I |u|^2 W_2 d\nu \geq Q^{-A_2} \nu(I),$$

together with the local-complexity bounds $W_2 \leq Q^{C_\Gamma}$, $|\mathcal{D}\Gamma| \leq Q^{C_\Gamma}$, $|\mathcal{D}(H\Gamma)| \leq Q^{C_{H\Gamma}}$. Then there is an exponent $A_{H\Gamma}$ such that

$$\int_I |H|^2 W_2 d\nu \geq Q^{-A_{H\Gamma}} \nu(I).$$

Consequently, since $W_2 \leq Q^{C_\Gamma}$,

$$\int_I |H|^2 W_2 d\nu \geq Q^{-A_{H\Gamma} - C_\Gamma} \int_I W_2 d\nu.$$

Proof. After applying the same buffered-packet reduction, assume

$$\int_{I_\eta} |u|^2 \|\Gamma\|^2 d\nu \geq Q^{-A'_2} \nu(I), \quad \eta \geq Q^{-C_\eta}.$$

Lemma 16.300 gives $m_0 \in I_\eta$ with $|u(m_0)| \geq \tau := Q^{-B_u}$ and $\|\Gamma(m_0)\| \geq \gamma := Q^{-B_\Gamma}$. Choose $\varepsilon = Q^{-B_q}$ so that

$$r_\varepsilon := 4\sqrt{\varepsilon/\tau} \leq \min\{\eta/2, \tau/(8Q^{C_u}), \gamma/(2Q^{C_\Gamma})\}.$$

If $|q(m_0)| \geq \varepsilon$, put $m_1 = m_0$; otherwise apply Lemma 16.301; its vector-field clause gives

$$|q(m_1)| \geq \varepsilon, \quad |u(m_1)| \geq \tau/2, \quad \|\Gamma(m_1)\| \geq \gamma/2.$$

The D4 residual estimate gives $|H(m_1)| \|\Gamma(m_1)\| \geq Q^{-B_{H\Gamma}}$. The derivative bound $|\mathcal{D}(H\Gamma)| \leq Q^{C_{H\Gamma}}$ thickens this to a subinterval $J_{H\Gamma} \subset I$ of polynomial normalized length on which $|H\Gamma| \geq Q^{-B_{H\Gamma} - O(1)}$. Hence

$$\int_I |H|^2 \|\Gamma\|^2 d\nu = \int_I |H\Gamma|^2 d\nu \geq Q^{-A_{H\Gamma}} \nu(I).$$

Finally, $\|\Gamma\|^2 \leq Q^{C_\Gamma}$ gives $\int_I |H|^2 \|\Gamma\|^2 d\nu \geq Q^{-A_{H\Gamma} - C_\Gamma} \int_I \|\Gamma\|^2 d\nu$. □

Corollary 16.304 (Discrete cap D4 amplitude survival). *Let C_j be a frozen cap inside a routed block I . Suppose the packet quadrature and cap-transfer estimates compare the discrete cap measure with $d\nu$ for the two functions W_2 and $|H|^2 W_2$, with only polynomial loss. Then*

$$\sum_{p \in C_j} \mu_p |H_p|^2 \|\Gamma_p\|^2 \geq Q^{-A} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2.$$

Thus Hypothesis 16.297 is discharged on the ordinary real-canonical blocks for which Gate-1, Gate-2, gauge-locking, D4 residual control, and packet quadrature/core-transfer have already been proved.

Proof. Apply Theorem 16.303 on the routed block and transfer the continuous inequalities to the packet cap using the stated quadrature/core-transfer estimates. The polynomial losses are absorbed into the exponent A . $\square$

Remark 16.305 (What this theorem does and does not prove). Theorem 16.302 is not a substitute for Gate-1. Its first hypothesis is precisely source-curvature observability for the actual canonical object q''_{can} . Without that hypothesis the statement is false in the local model: one may have $q''_{\text{can}} \approx 0$ and therefore $H = yq''_{\text{can}} \approx 0$ while the anisotropy vector remains live (Proposition 16.293). The theorem proves the downstream implication

$$\text{Gate-1} + \text{Gate-2} + \text{gauge-locked D4 residual control} \implies \text{D4 amplitude survival},$$

so the D4 amplitude item is no longer an independent cap-side finite-audit hypothesis once the upstream actual-source Gate-1 theorem and the already-stated Gate-2 rigidity theorem are available.

Remark 16.306 (D4 amplitude survival status after actual-source nonflatness). Hypotheses 16.296 and 16.297 should no longer be listed as independent cap-side finite-audit assumptions on ordinary real-canonical routed blocks. They are discharged by the chain

Gate-1+Gate-2+gauge-locked D4 source-factor identity+D4 residual control+packet quadrature/core transfer, namely Theorems 16.302 and 16.303 followed by Corollary 16.304. They remain useful labels for the exact estimate consumed by the cap bridge, but the ledger entry should read “proved downstream of Gate-1/Gate-2 and quadrature”, not “open cap finite audit.” This update does not prove Gate-1, Gate-2, or the quadrature theorem themselves; it removes only the previously independent scalar-amplitude obstruction.

Corollary 16.307 (Cap amplitude noncollapse from actual-source D4 nonflatness). *On every ordinary real-canonical cap for which the hypotheses of Corollary 16.304 hold, Hypothesis 16.297 holds. In particular the weighted alternative in Hypothesis 16.296 holds after replacing the unweighted mass by the carrier mass $\mu_p \|\Gamma_p\|^2$.*

Proof. This is exactly the conclusion of Corollary 16.304. The carrier-mass form is stronger for the cap-side absolute lower bound than an unweighted average of $|H_p|^2$, because the quotient residual is paired with $L_{\text{clean}}\Gamma_p$. $\square$

Theorem 16.308 (D4 amplitude survival from a coefficient-good cap core). *If $C_j^{\text{core}} \subset C_j$ is a D4 coefficient-good cap core in the sense of Definition 16.298, then the carrier-mass form of D4 amplitude survival holds on C_j :*

$$\sum_{p \in C_j} \mu_p |H_p|^2 \|\Gamma_p\|^2 \geq Q^{-A_H} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2, \quad A_H = 2C_H + C_M.$$

In particular, Hypothesis 16.297 is discharged once a coefficient-good cap core has been exhibited.

Proof. By nonnegativity of the weights and the pointwise core lower bound on $|H_p|$,

$$\sum_{p \in C_j} \mu_p |H_p|^2 \|\Gamma_p\|^2 \geq \sum_{p \in C_j^{\text{core}}} \mu_p |H_p|^2 \|\Gamma_p\|^2 \geq Q^{-2C_H} \sum_{p \in C_j^{\text{core}}} \mu_p \|\Gamma_p\|^2.$$

The core mass condition gives $\sum_{C_j^{\text{core}}} \mu_p \|\Gamma_p\|^2 \geq Q^{-C_M} \sum_{C_j} \mu_p \|\Gamma_p\|^2$; combining the two estimates proves the result. $\square$

Corollary 16.309 (D4 amplitude survival from admissible cores and vector Remez). *Suppose that on each routed block I there is a predetermined coefficient-good core E_I^{core} , chosen independently of Γ_p , of the hard atoms, and of the live residuals, such that $|H_p| \geq Q^{-C_H}$ for $p \in E_I^{\text{core}}$; and suppose the vector-Remez / core-transfer estimate gives, after restriction to every frozen cap C_j ,*

$$\sum_{p \in C_j \cap E_I^{\text{core}}} \mu_p \|\Gamma_p\|^2 \geq Q^{-C_M} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2.$$

Then the D4 amplitude survival estimate of Theorem 16.308 holds on C_j with $C_j^{\text{core}} := C_j \cap E_I^{\text{core}}$.

Proof. Apply Theorem 16.308 with $C_j^{\text{core}} := C_j \cap E_I^{\text{core}}$: the pointwise amplitude bound is the hypothesis on E_I^{core} , and the core mass condition is exactly the Remez/core-transfer estimate. $\square$

Remark 16.310 (Relative versus absolute static-line survival). Rank-two quotient survival (Corollary 16.239) together with $\sigma_{\min}(L_{\text{clean}}) \geq Q^{-C}$ closes the static-line handoff as a *relative* lower bound,

$$|q_p^Y \Phi_p^Y u_j| \geq Q^{-A} \|\Phi_p^Y u_j\|,$$

in which the scalar H_p cancels. The final atomic contradiction needs an *absolute* lower bound in the same quotient norm as the upper bound, namely

$$\sum_{p \in C_j} \mu_p |H_p|^2 |L_{\text{clean}} \Gamma_p|^2 \geq Q^{-A} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2,$$

which is supplied by Hypothesis 16.297. Relative survival without amplitude survival is not enough: a cap with strongly varying s_p and $H_p \equiv 0$ has $\delta_{\text{ext}} \geq Q^{-A}$ and $q_p^Y \Phi_p^Y u_j = 0$ simultaneously.

The absolute lower bound is now supplied on ordinary real-canonical routed blocks by Corollary 16.304, not by quotient geometry. The distinction remains important: rank-two quotient survival gives a relative static-line handoff, while the same-norm atomic lower bound requires $D4$ amplitude mass.

Proposition 16.311 (Cap $D4$ lower bound from quotient Schur gap and amplitude mass). *Assume the rank-two quotient Schur gap (Theorem 16.245, in any of its equivalent forms),*

$$\sigma_{\min}(q_p^Y E_p) \geq Q^{-C},$$

the row-normal carrier coercivity $\sigma_{\min}(L_{\text{clean}}) \geq Q^{-C}$, and the $D4$ amplitude mass on the cap core,

$$\sum_{p \in C_j} \mu_p |H_p|^2 \|\Gamma_p\|^2 \geq Q^{-A} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2.$$

Assume the carrier expansion

$$\Phi_p^Y \Gamma_p = H_p E_p L_{\text{clean}} \Gamma_p + \varepsilon_p, \quad \sum_{p \in C_j} \mu_p \|q_p^Y \varepsilon_p\|^2 \leq Q^{-B} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2.$$

Then for B sufficiently large depending polynomially on A and C ,

$$\sum_{p \in C_j} \mu_p \|q_p^Y \Phi_p^Y \Gamma_p\|^2 \geq Q^{-A'} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2.$$

This is the absolute cap-side $D4$ lower bound: the upper bound from the cap covariance and the lower bound from the quotient residual now live in the same μ -weighted norm, supplying the comparison required for the atomic contradiction.

Proof. By the row-normal coercivity and the Schur gap,

$$\|q_p^Y E_p L_{\text{clean}} \Gamma_p\| \geq \sigma_{\min}(q_p^Y E_p) \sigma_{\min}(L_{\text{clean}}) \|\Gamma_p\| \geq Q^{-2C} \|\Gamma_p\|.$$

Hence

$$\|q_p^Y \Phi_p^Y \Gamma_p\|^2 \geq \frac{1}{2} |H_p|^2 Q^{-4C} \|\Gamma_p\|^2 - \|q_p^Y \varepsilon_p\|^2.$$

Weighted-summing over $p \in C_j$,

$$\sum_{p \in C_j} \mu_p \|q_p^Y \Phi_p^Y \Gamma_p\|^2 \geq \frac{1}{2} Q^{-4C} \sum_{p \in C_j} \mu_p |H_p|^2 \|\Gamma_p\|^2 - \sum_{p \in C_j} \mu_p \|q_p^Y \varepsilon_p\|^2.$$

The amplitude mass hypothesis bounds the first sum below by $Q^{-A-4C} \sum_p \mu_p \|\Gamma_p\|^2$; the error sum is $\leq Q^{-B} \sum_p \mu_p \|\Gamma_p\|^2$, absorbed for $B > A + 4C + 1$. Renaming the exponent gives the stated lower bound. $\square$

Corollary 16.312 (Four-item cap/Y12 bridge package). *Assume the bounded dual-row certificate of Lemma 16.241, equivalently the rank-two quotient Schur gap of Theorem 16.245. Form the nuisance quotient using the canonical dead-column split of Lemma 16.242, and assume the coefficient-good cap core of Theorem 16.308. If the row-normal D4 expansion has small quotient error and $\sigma_{\min}(L_{\text{clean}}) \geq Q^{-C}$, then the cap-side absolute D4 lower bound*

$$\sum_{p \in C_j} \mu_p \|q_p^Y \Phi_p^Y \Gamma_p\|^2 \geq Q^{-A} \sum_{p \in C_j} \mu_p \|\Gamma_p\|^2$$

holds. Equivalently, the four heuristic items

dead-column split, bounded dual rows, analytic live-pair Schur gap, D4 amplitude survival form a closed referee-safe cap/Y12 package once the dual rows and the coefficient-good cap core have been established by the preceding modules.

Proof. The dual rows give the Schur gap by Theorem 16.245. The canonical split of Lemma 16.242 keeps the quotient dead-only and preserves those rows, so the Schur gap survives the formation of $\mathcal{K}_p^Y$. The cap core gives D4 amplitude survival on C_j by Theorem 16.308. The absolute D4 lower bound is then exactly Proposition 16.311. $\square$

Corollary 16.313 (Cap transversality in live carrier coordinates: projective Wronskian form). Status: proved on the D4-good core under the live-coordinate normal form, rank-two quotient Schur gap, row-normal carrier lower bound, and Corollary 16.314. Sharper projective form of Corollary 16.295. *Under the hypotheses of Corollary 16.295 together with Hypothesis 16.296,*

$$\rho_{7,p}(s) = H_{j,p} c_7 + O(Q^{-B'}), \quad \rho_{Y,p}(s) = H_{j,p}(\beta_9^{\text{tri}} + \alpha_9 s) + O(Q^{-B'}),$$

and the projective Wronskian

$$W_p^{\text{proj}}(s) := \rho'_{Y,p}(s) \rho_{7,p}(s) - \rho'_{7,p}(s) \rho_{Y,p}(s)$$

satisfies the absolute lower bound

$$|W_p^{\text{proj}}(s_j)| \geq Q^{-A}$$

on the good cap core. In particular the lower bound does not contain a $|\rho_{7,p}(s_j)|^2$ factor.

Proof. The displayed formulas for $\rho_{7,p}, \rho_{Y,p}$ are the entrywise unpacking of the row-normal scalar normal form proved in Corollary 16.295. At the leading order,

$$W_p^{\text{proj}}(s) = (H'_{j,p}(\beta_9^{\text{tri}} + \alpha_9 s) + H_{j,p} \alpha_9) H_{j,p} c_7 - H'_{j,p} c_7 H_{j,p}(\beta_9^{\text{tri}} + \alpha_9 s) = H_{j,p}^2 c_7 \alpha_9,$$

the $H'_{j,p}$ -terms cancelling because both rows share the same scalar amplitude $H_{j,p}$. By Hypothesis 16.296 and the row-normal carrier bound $|c_7 \alpha_9| \geq Q^{-C}$, $|H_{j,p}^2 c_7 \alpha_9| \geq Q^{-3C}$; the $O(Q^{-B'})$ error from the quotient residual $R_p^\dagger q_p \varepsilon_p$ is absorbed for B' sufficiently large. $\square$

Corollary 16.314 (Projective Wronskian lower bound on the D4-good core). *Assume the live-coordinate normal form of Corollary 16.295, the rank-two quotient Schur gap $\sigma_{\min}(q_p^Y E_p) \geq Q^{-C}$, the row-normal lower bound $|c_7 \alpha_9| \geq Q^{-C}$, and the pointwise D4-good core bound $|H_{j,p}| \geq Q^{-C}$ on the selected cap core. Then the projective live Wronskian*

$$W_p^{\text{proj}}(s) := \rho'_{Y,p}(s) \rho_{7,p}(s) - \rho'_{7,p}(s) \rho_{Y,p}(s)$$

satisfies

$$|W_p^{\text{proj}}(s)| \geq Q^{-A}$$

on the core, after absorbing the quotient error.

Proof. In live coordinates, $\rho_{7,p}(s) = H_{j,p} c_7 + O(Q^{-B'})$ and $\rho_{Y,p}(s) = H_{j,p}(\beta_9^{\text{tri}} + \alpha_9 s) + O(Q^{-B'})$. Differentiating in the projective parameter and forming the Wronskian gives

$$\begin{aligned} W_p^{\text{proj}}(s) &= (H'_{j,p}(\beta_9^{\text{tri}} + \alpha_9 s) + H_{j,p} \alpha_9) H_{j,p} c_7 - H'_{j,p} c_7 H_{j,p}(\beta_9^{\text{tri}} + \alpha_9 s) + O(Q^{-B''}) \\ &= H_{j,p}^2 c_7 \alpha_9 + O(Q^{-B''}). \end{aligned}$$

The $H'_{j,p}$ -terms cancel because the two live rows share the same scalar amplitude. The pointwise core bound and $|c_7 \alpha_9| \geq Q^{-C}$ give the claim for B'' sufficiently large. $\square$

Corollary 16.315 (Cap-level Jacobian transversality (superseded conditional bridge)). Status: SUPERSEDED. *The original cap bridge ran the rank-two Jacobian on the residual pair $(r_{N,p}, r_{Y,p})$ with*

$$N_p := \Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing}} \mathcal{N}_p^{\text{D4}}, \quad Y_p := \Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing}} [C_{Y12}]^{\text{tr}}, \quad q_p^Y := I - P_{\mathcal{K}_p^Y}^{\text{bd}}.$$

Per Lemma 16.236, quotient descent $\Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing}} \mathcal{N}_p^{\text{D4}} \subseteq \mathcal{K}_p^Y$ forces $r_{N,p} = q_p^Y N_p = 0$, so the determinant $H_{j,p}^2 c_7 \alpha_9$ is identically zero on the nuisance image and the rank-two normal form cannot be derived. The corrected cap-level Jacobian transversality bridge runs in live coordinates $(\rho_{7,p}, \rho_{Y,p})$ built from the retained carrier columns $C_{7,p}^{\text{tar}}, C_{Y,p}^{\text{tar}}$ under the torsion-free rank-two hypothesis (Corollary 16.239); see Corollary 16.295. The label is retained here for citation stability with prior text and downstream references; new uses should cite Corollary 16.295.

Remark 16.316 (Downstream inputs after certified quotient, coercivity, D4 carrier, and canonical closure). This legacy status label is retained for citations, but its pre-canonical list has been superseded in the active branch. In the canonical source-graded K9 proof route, the singular-discard upper is the same-norm estimate of Theorem 16.189, the finite-main Schur component is discharged by Corollary 16.168, and the maximal affine row realization is supplied by Theorem 16.190.

The active cap-level fork is therefore the positive-density fork of Theorem 16.219: either

$$\delta_{\text{ext}}^{\text{cov}} = \frac{\det B_j^{\text{cov}}}{(\text{tr } B_j^{\text{cov}})^2} \geq Q^{-A},$$

or the cap is quantitatively static and is routed to the canonical source-graded affine branch. The signed matrix $\sum_p w_p \Gamma_p \Gamma_p^*$ is not used as a covariance object unless a sign-coherent branch has already made it positive.

Thus the earlier phrases “singular-discard/EF upper”, “ δ_{ext} or typed static-line handoff”, and “final Type-I local-square atomic comparison” are no longer active downstream gaps in the canonical branch. Remaining obligations recorded here are upstream or entry obligations: the raw row/routing coefficient certificate $\mathfrak{R}_{2Y}(p)$, the packet-model hypotheses for the graph step, and the source Gate-1/Gate-2 and quadrature inputs needed to enter the certified branch.

Proposition 16.317 (Source-closed downstream handoff in the canonical branch). *Assume Gate-1 source survival, Gate-2 source-anisotropy mass, the source-factor residual control for $H = -qu^3/288 + R_H$, and packet quadrature. Assume also the graph step of Theorem E.106 under its finite/O2O3 packet-model hypotheses, and assume the selected packets satisfy or import the certified raw row/routing certificate $\mathfrak{R}_{2Y}(p)$. Then the downstream handoff is the canonical source-graded chain*

$Y12\text{-rank-two quotient survival} \implies Y12\text{-graph/coercivity} \implies \text{row-normal effective D4 carrier} \implies \text{positive-den}$

On each frozen good cap C_j the active fork is

$$\delta_{\text{ext}}^{\text{cov}}(C_j) = \frac{\det B_j^{\text{cov}}}{(\text{tr } B_j^{\text{cov}})^2} \geq Q^{-A},$$

or the cap is quantitatively static in the positive covariance density and the static direction is routed through Theorem 16.172 and Corollary F.6. The older signed expression $\det B_j / (\text{tr } B_j)^2$ is only a sign-coherent special case of $\delta_{\text{ext}}^{\text{cov}}$.

Proof. The source hypotheses imply the carrier-mass D4 amplitude estimate by Corollary 16.304. The graph theorem supplies the upstream principal-angle source/graph gap under its structural hypotheses. On packets satisfying $\mathfrak{R}_{2Y}(p)$, the certified quotient, coercivity, and effective-carrier estimates give the rank-two D4 carrier in the certified branch. The cap covariance step is Theorem 16.219. In the nonstatic alternative it gives the covariance lower bound directly; in the static alternative it supplies the fixed direction u_j , and the canonical source-graded branch gives the same-norm upper/lower contradiction by Theorem 16.172. This is the same final comparison as Theorem F.5. $\square$

Theorem 16.318 (Normal D4 through the Y12-signature K_9 channel). *Assume the following.*

(1) $\text{EFCompletion}_{Y12}$ holds:

$$\sum_p |\Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing}, p} J_{\text{tr}, p}(R_X)|^2 \leq Q^{-B} \mu(\mathcal{C}).$$

(2) Lower-order tangent observability holds, so the covariant residual K_9 is polynomially conditioned.

(3) The Y12-signature separation Hypothesis E.116 holds.

(4) The singular actual-source manifold has polynomial tubular control as in Definition 16.321.

(5) On a polynomial-weight extracted set, the D4 coefficient package is good:

$$|H|, |\alpha_9|, |\alpha_9^{-1}|, |c_7|, |c_7^{-1}|, |\beta_9^{\text{tri}}| \leq Q^A,$$

with $|H| \geq Q^{-A}$.

(6) The moving Y12-wall anti-concentration estimate Hypothesis 16.207 holds.

Then the D4 displacement has a forbidden Y12-signature component. More precisely,

$$\Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing}} E_{\text{car}}^{\text{tr}} L_{\text{clean}}^{\text{tr}} \Gamma = H (a \Gamma_G + b \Gamma_{G,2})$$

with

$$\boxed{b \neq 0}$$

on the extracted set, up to Q^{-A} -controlled normalization. Hence the forbidden wall

$$E_{\text{wall}}^{Y12} := \frac{|H|^2 |a \Gamma_G + b \Gamma_{G,2}|^2}{\mathcal{N}_{D4}}$$

has polynomial mass. Summing over packets contradicts the $\text{EFCompletion}_{Y12}$ upper, after the tubular quadratic error is absorbed.

Proof sketch. The Y12-signature separation removes actual-source tangents, lower connection residuals, and actual second-curvature effects from the primitive Y12-signature quotient. Therefore

$$\Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing}} [\mathbf{C}_{Y12}]$$

is nonzero. Applying the triangular D4 matrix gives

$$H (\beta_9^{\text{tri}} \Gamma_G + \alpha_9 \Gamma_{G,2}) \Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing}} [\mathbf{C}_{Y12}],$$

so the $\Gamma_{G,2}$ -coefficient is nonzero because $\alpha_9 \neq 0$. Tubular control turns the linear normal component into an actual distance from the singular actual-source manifold. The Y12-completion upper bounds the same normal component for actual rows. The moving-wall anti-concentration gives positive mass for the D4 normal component, producing the contradiction. $\square$

Proposition 16.319 (Archived scalar certified singular-discard D4 visibility). *For the legacy scalar-certified branch define*

$$K_{D4,p}^{\text{cert}} := (I - P_{\text{sing},p}^{\text{cert}}) T_{\text{tr} \rightarrow \text{sing},p} E_{\text{car},p}^{\text{tr}} L_{\text{eff},p}^{\text{tr}}.$$

Assume the certified low-block identification with the $C_7/Y12$ transition carrier plane and the effective-carrier bound $\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-A}$. Then on the routed good packet set

$$\sigma_{\min}(K_{D4,p}^{\text{cert}}) \geq Q^{-A'}, \quad \text{rank } K_{D4,p}^{\text{cert}} = 2.$$

Consequently the finite D4 wall energy may be formed in that scalar-certified projection. In the active canonical source-graded branch this archived matrix is replaced by $K_{D4,p}^{\text{gr}}$ from Definition 16.196; the final upper/lower comparison is made in $H_{K9,p}^{\text{gr}}$, not in the scalar certified quotient. The archived scalar wall energy is

$$E_{\text{wall}}^{\text{forb}}(p) := \frac{|H(p)|^2 \|K_{D4,p}^{\text{cert}} \Gamma(p)\|^2}{\mathcal{N}_{D4}(p)}, \quad \Gamma = (\Gamma_G, \Gamma_{G,2})^T.$$

Proof. The carrier part of $E_{\text{car},p}^{\text{tr}}$ is the labelled transition $C_7/Y12$ plane up to polynomially conditioned basis change. Theorem 16.128 gives a polynomial lower bound for $(I - P_{\text{sing},p}^{\text{cert}})T_{\text{tr} \rightarrow \text{sing},p}$ on that plane. Multiplication by the polynomially invertible $L_{\text{eff},p}^{\text{tr}}$ preserves a polynomial smallest singular value. Hence the rank is two and the lower wall is in the same certified singular-discard norm as the upper ledger. $\square$

Remark 16.320 (Failure routes for raw singular-discard choices). The following are no-go routes for raw or noncertified singular-discard choices. They are not active gaps in the certified branch.

- (1) If $R_{\text{sing}}^{\text{raw}}$ is Taylor-jet-complete or carrier-complete, then

$$T_{\text{tr} \rightarrow \text{sing}} \text{span}\{[C_7], [C_{Y12}]\} \subset R_{\text{sing}}^{\text{raw}},$$

so the discarded carrier matrix has rank zero.

- (2) If the EF-coherent upper fails in the certified norm,

$$(I - P_{\text{sing}}^{\text{cert}})T_{\text{tr} \rightarrow \text{sing}}J_{\text{tr}}(q_{\text{can}}'')$$

is not small for actual rows. This is a same-norm upper failure, not a δ_{coerc} failure.

- (3) If transition stability fails in the singular overlap norms, the bounded row and Schur estimates lose polynomial conditioning.
- (4) If the D4 live plane is not coefficient-good, the wall energy loses polynomial normalization even though the certified projection is transverse.

With the certified scalar projection and coefficient-good packets, the archived rank-two finite D4 visibility item is supplied by Proposition 16.319. In the active canonical branch, this role is taken by the source-graded matrix $K_{D4,p}^{\text{gr}}$, while the same-norm upper and atomic comparison are closed by Theorem 16.172 and the canonical source-graded atomic contradiction. Thus the listed failure routes are diagnostics for noncanonical singular-discard choices, not active obstructions in the source-graded branch.

Definition 16.321 (Tubular normality for the singular actual-source manifold). Let $\mathcal{M}_{\text{sing}}^{\text{arith}}$ be the finite-dimensional actual-source singular coherence manifold in the retained singular chart. We say it has polynomial tubular control on a packet family if its normal projection is defined on a Q^{-C} -tube and the second fundamental form satisfies

$$\|\Pi_{\mathcal{M}_{\text{sing}}^{\text{arith}}}\| \leq Q^C.$$

Equivalently, for every tangent point $u \in \mathcal{M}_{\text{sing}}^{\text{arith}}$ and small displacement D ,

$$\text{dist}(u + D, \mathcal{M}_{\text{sing}}^{\text{arith}}) \geq \|\Pi_u^\perp D\| - Q^C \|D\|^2 - \text{Err}_{\text{chart}}.$$

Remark 16.322 (Why tubular control is needed). A nominally normal D4 displacement can be absorbed by a nearby nonlinear actual-source parameter shift if the singular actual-source manifold has large curvature at the relevant scale. Therefore the singular-discard contradiction requires both a linear transversality audit

$$\text{rank}(\Lambda_{\text{tr},\text{sing}} E_{\text{car}}^{\text{tr}} L_{\text{clean}}^{\text{tr}}) \geq 1$$

and polynomial tubular control for $\mathcal{M}_{\text{sing}}^{\text{arith}}$. The latter is finite-geometric, not an arithmetic Gate-1 statement.

Remark 16.323 (Partial explicit-formula residuals are upstream completion tools). The full canonical explicit-formula identity

$$q_{\text{can}}'' = B_2 + P_2 + D_2 + E_2$$

is definitional for the canonical source. Its full residual

$$q_{\text{can}}'' - (B_2 + P_2 + D_2 + E_2)$$

is therefore expected to be D4-invisible if all terms are varied coherently. It should not be used as the downstream forbidden channel Λ .

Partial residuals such as

$$R_X := q''_{\text{can}} - (B_2 + P_{2,\leq X} + D_{2,\leq X})$$

are useful upstream for source completion and for proving aggregate L^2 -observability of q''_{can} . A typical use is:

$$q''_{\text{can}} = F_X + R_X, \quad F_X := B_2 + P_{2,\leq X} + D_{2,\leq X} + P_{\text{res},X},$$

with

$$\|R_X\|_{L^2(W)} \leq \frac{1}{2} \|F_X\|_{L^2(W)}$$

on the weighted branch window W . Then the full canonical source inherits the finite approximant's L^2 -mass.

A partial residual becomes a downstream forbidden channel only if both of the following are proved:

$$\|\Lambda_{X,\psi} V_{\text{glob}}\|^2 \leq Q^{-B} \|\psi\|^2$$

for localized cutoffs ψ , and

$$\text{rank}(\Lambda_{X,\psi} E_{\text{car}} L_{\text{clean}}) \geq 1$$

on the D4 live plane. In the absence of this upper/rank pair, partial EF residuals are completion-coherence inputs, not the proof-closing Λ .

Remark 16.324 (Final D4 residual status in the carrier quotient). The D4 residual statement is a carrier-quotient row-energy statement for the canonical scalar source

$$u = q''_{\text{can}}.$$

It is not a scalar I -coordinate coercivity statement.

In the carrier quotient, the D4 row has the effective form

$$P_{W_{\text{car}}}([\mathcal{A}_7] + [\mathcal{A}_9]) = H E_{\text{car}} L_{\text{eff}} \begin{pmatrix} \Gamma_G \\ \Gamma_{G,2} \end{pmatrix} + \varepsilon_{\text{rem}}.$$

The clean row-normal matrix is

$$L_{\text{clean}}^{\text{tri}} = \begin{pmatrix} c_7 & 0 \\ \beta_9^{\text{tri}} & \alpha_9 \end{pmatrix}, \quad \beta_9^{\text{tri}} := \beta_9 - \omega_Y(e_7)c_7,$$

with c_7, α_9 as in Lemma 16.342. The same-scale $\beta_9^{\text{tri}} \Gamma_G$ term is part of L_{eff} , not a residual. All D4 coefficient-good and rank-one fallback statements in this branch use the triangularized coefficient β_9^{tri} , not the raw β_9 .

The remaining correction-row issue is purely carrier-theoretic. If, before scalar collapse, the order-seven and order-nine line, background/core, and normalization/subtraction correction rows satisfy

$$R_\nu^{(b)} \in \mathcal{C}_{\text{car}}, \quad \nu \in \{\text{line, bg/core, norm/sub}\}, \quad b \in \{7, 9\},$$

then

$$P_{W_{\text{car}}} R_\nu^{(b)} = 0$$

for all six correction rows. Consequently

$$L_{\text{eff}} = L_{\text{clean}}^{\text{tri}}, \quad \varepsilon_{\text{rem}} = 0.$$

Thus the half-leading residual estimate closes automatically in the active carrier quotient.

Definition 16.325 (Associated-graded correction-row convention). Work in the carrier-graded row module $\mathcal{M}_{\text{car}}$ of Definition 16.96. The clean live rows through order nine are the two components already extracted in Lemma 16.342, namely the $\mathbf{C}_7$ -row and the clean centered $\mathbf{C}_{Y_{12}}$ -row. After these two live rows are removed, the remaining correction rows through order nine are, by definition, the associated-graded remainder in the three non-live correction origins $\nu \in \{\text{line, bg/core, norm/sub}\}$. Thus, for $b \in \{7, 9\}$, the correction rows have the associated-graded form

$$R_{\text{line}}^{(b)} = a_{\text{line},b} \mathbf{C}_{\text{line}}, \quad R_{\text{bg/core}}^{(b)} = a_{\text{bg/core},b} \mathbf{C}_{\text{bg/core}}, \quad R_{\text{norm/sub}}^{(b)} = a_{\text{norm/sub},b} \mathbf{C}_{\text{norm/sub}},$$

with coefficients in the local coefficient ring. Any term with a C_7 - or C_{Y12} -component is not part of $R_\nu^{(b)}$; it is, by this convention, part of the live carrier matrix. Equivalently, the symbols $R_\nu^{(b)}$ denote the non-live residual after projection away from the retained carrier plane $W_{\text{car}} := \text{span}\{[C_7], [C_{Y12}]\}$.

Lemma 16.326 (Six-row carrier correction table). *In the associated-graded correction-row convention, the six correction rows through order nine have the following carrier table:*

row	associated carrier	class in $\mathcal{Q}_{\text{car}}$	$P_{W_{\text{car}}} R_\nu^{(b)}$	$L_\nu^{(b)}$
$R_{\text{line}}^{(7)}$	C_{line}	0	0	0
$R_{\text{line}}^{(9)}$	C_{line}	0	0	0
$R_{\text{bg/core}}^{(7)}$	$C_{\text{bg/core}}$	0	0	0
$R_{\text{bg/core}}^{(9)}$	$C_{\text{bg/core}}$	0	0	0
$R_{\text{norm/sub}}^{(7)}$	$C_{\text{norm/sub}}$	0	0	0
$R_{\text{norm/sub}}^{(9)}$	$C_{\text{norm/sub}}$	0	0	0

Consequently all six correction rows lie in $\mathcal{C}_{\text{car}}$, have zero retained carrier projection, and contribute no coherent live correction matrix: $M_{\text{live,p}}^{\text{corr}} = 0$.

Proof. By Definition 16.96, the nuisance submodule is $\mathcal{C}_{\text{car}} = \text{span}\{C_{N7,I}, C_{N7,D}, C_{N5,I}, C_{N5,D}, C_{\text{line}}, C_{\text{bg/core}}, C_{\text{norm/sub}}\}$. Definition 16.325 writes the six correction rows as coefficient multiples of C_{line} , $C_{\text{bg/core}}$, or $C_{\text{norm/sub}}$, so each belongs to $\mathcal{C}_{\text{car}}$ and its class in $\mathcal{Q}_{\text{car}} = \mathcal{M}_{\text{car}}/\mathcal{C}_{\text{car}}$ is zero. Since $W_{\text{car}} \subset \mathcal{Q}_{\text{car}}$, the projection to W_{car} is zero as well. The same-scale live matrix contribution $L_\nu^{(b)}$ is precisely the (C_7, C_{Y12}) -coordinate of the row, which is zero by the associated-graded convention. $\square$

Remark 16.327 (No double counting of the triangularization row). The row-normal subtraction $\ell_9^Y = \omega_Y - \omega_Y(e_7)\ell_7$ and the corresponding coefficient $\beta_9^{\text{tri}} := \beta_9 - \omega_Y(e_7)c_7$ are part of the clean live row-normal matrix

$$L_{\text{clean}}^{\text{tri}} = \begin{pmatrix} c_7 & 0 \\ \beta_9^{\text{tri}} & \alpha_9 \end{pmatrix}.$$

They are not counted as one of the six correction rows $R_\nu^{(b)}$. Thus the table in Lemma 16.326 does not remove or hide the same-scale $\beta_9^{\text{tri}}\Gamma_G$ contribution; it records only that line, background/core, and normalization/subtraction remainders have no retained $C_7/Y12$ projection in the carrier quotient.

Remark 16.328 (Scope of the six-row carrier correction table). Lemma 16.326 and Theorem 16.329 are statements in the associated-graded carrier quotient $\mathcal{Q}_{\text{car}} = \mathcal{M}_{\text{car}}/\mathcal{C}_{\text{car}}$. They are not scalar I -coordinate non-cancellation theorems. The proof uses the fact that the retained carrier classes $[C_7]$ and $[C_{Y12}]$ remain distinct from the line, background/core, and normalization/subtraction correction classes before scalar collapse. Thus

$$P_{W_{\text{car}}} R_\nu^{(b)} = 0, \quad \nu \in \{\text{line, bg/core, norm/sub}\}, \quad b \in \{7, 9\},$$

is a carrier-quotient statement. If one applies a scalar collapse map $S_I : \mathcal{M}_{\text{car}} \rightarrow \mathcal{O} \cdot I$, distinct carrier labels may acquire scalar incidences; those incidences belong to the quarantined scalar- I fallback audit and are not used in the active D4 route.

Theorem 16.329 (Effective carrier closes after the six-row table). *On every Gram-good packet in the carrier-quotient route, the six-row carrier correction table gives*

$$L_{\text{eff,p}} = L_{\text{clean,p}}^{\text{tri}} = \begin{pmatrix} c_{7,p} & 0 \\ \beta_{9,p}^{\text{tri}} & \alpha_{9,p} \end{pmatrix},$$

and hence $\sigma_{\min}(L_{\text{eff,p}}) \geq Q^{-C}$. Consequently the bottom item of $\mathfrak{C}_{2Y}(p)$ is proved in the carrier-quotient formulation.

Proof. Lemma 16.326 gives $M_{\text{live},p}^{\text{corr}} = 0$, so $L_{\text{eff},p} = L_{\text{clean},p}^{\text{tri}}$. By Lemma 16.342,

$$\det L_{\text{clean},p}^{\text{tri}} = c_{7,p} \alpha_{9,p} = -\frac{1}{1536\pi^4 g_{\text{Gr},p}^{3/2} h_{\text{Gr},p}^3 (\sqrt{g_{\text{Gr},p}} + \sqrt{h_{\text{Gr},p}})^3}.$$

On a Gram-good packet, $Q^{-C_0} \leq g_{\text{Gr},p}, h_{\text{Gr},p} \leq Q^{C_0}$, so $|\det L_{\text{clean},p}^{\text{tri}}| \geq Q^{-C_1}$ and $\|L_{\text{clean},p}^{\text{tri}}\|_{\text{op}} \leq Q^{C_1}$. For a 2×2 matrix L , $\sigma_{\min}(L) = |\det L|/\sigma_{\max}(L)$, giving $\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-C}$. $\square$

Corollary 16.330 (D4 residual audit closes in the active carrier quotient). *Assume the retained carrier frame lower bound $\sigma_{\min}(E_{\text{car}}) \geq Q^{-C}$ and the Gram-good packet bounds. Then $\sigma_{\min}(E_{\text{car}}L_{\text{eff},p}) \geq Q^{-C}$ and $\varepsilon_{\text{rem}} = 0$ in the active carrier quotient. In particular, the effective-carrier finite-audit item is closed; no determinant anti-cancellation condition remains for the six correction rows.*

Proof. Theorem 16.329 gives $\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-C}$, and the carrier-frame assumption then gives $\sigma_{\min}(E_{\text{car}}L_{\text{eff},p}) \geq \sigma_{\min}(E_{\text{car}})\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-C}$. By Lemma 16.326, the six correction rows have zero carrier-quotient class, so the correction residual in the active carrier quotient is zero. $\square$

Remark 16.331 (Status update for the effective-carrier ledger). This supersedes Remark 16.279, which treated $\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-C}$ as a finite-audit item. In the active carrier-quotient formulation, the six correction rows are not arbitrary coherent live matrices. They are associated-graded line, background/core, and normalization/subtraction rows, hence lie in $\mathcal{C}_{\text{car}}$ and have zero retained $C_7/Y12$ projection. Therefore $M_{\text{live},p}^{\text{corr}} = 0$, $L_{\text{eff},p} = L_{\text{clean},p}^{\text{tri}}$, and the clean triangular determinant closes the effective-carrier item. The only way this conclusion could fail would be to work after scalar I -coordinate collapse, where the carrier labels are no longer independent. That scalar formulation is not the active D4 route.

Lemma 16.332 (Nuisance correction rows vanish in the carrier quotient). *Assume that the finite correction rows through order nine satisfy*

$$R_{\text{line}}^{(b)} \in \mathcal{C}_{\text{car}}, \quad R_{\text{bg/core}}^{(b)} \in \mathcal{C}_{\text{car}}, \quad R_{\text{norm/sub}}^{(b)} \in \mathcal{C}_{\text{car}},$$

for $b \in \{7, 9\}$. Then, in

$$\mathcal{Q}_{\text{car}} = \mathcal{M}_{\text{car}}/\mathcal{C}_{\text{car}},$$

one has

$$[R_{\text{line}}^{(b)}] = [R_{\text{bg/core}}^{(b)}] = [R_{\text{norm/sub}}^{(b)}] = 0.$$

Therefore

$$P_{W_{\text{car}}} R_{\nu}^{(b)} = 0, \quad \nu \in \{\text{line, bg/core, norm/sub}\}, \quad b \in \{7, 9\}.$$

Equivalently, in the residual-table notation,

$$L_{\nu}^{(b)} = 0, \quad R_{\nu, \text{val}}^{(b)} = 0, \quad R_{\nu, \text{abs}}^{(b)} = 0.$$

Proof. The quotient map

$$\mathcal{M}_{\text{car}} \longrightarrow \mathcal{Q}_{\text{car}} = \mathcal{M}_{\text{car}}/\mathcal{C}_{\text{car}}$$

kills every element of $\mathcal{C}_{\text{car}}$. The stated carrier-routing hypothesis places all six correction rows in $\mathcal{C}_{\text{car}}$, so their classes vanish in $\mathcal{Q}_{\text{car}}$. Since W_{car} is the live carrier plane in the quotient, their projection to W_{car} is also zero. The residual-table identities follow by taking all same-scale matrices and residual components equal to zero. $\square$

Corollary 16.333 (D4 residual audit closes in the formal carrier quotient). *Assume Lemma 16.332, the Gram-good bounds*

$$Q^{-C_0} \leq g_{\text{Gr}}, h_{\text{Gr}} \leq Q^{C_0},$$

and the carrier-frame lower bound

$$\sigma_{\min}(E_{\text{car}}) \geq Q^{-C_0}.$$

Then

$$L_{\text{eff}} = L_{\text{clean}}^{\text{tri}} = \begin{pmatrix} c_7 & 0 \\ \beta_9^{\text{tri}} & \alpha_9 \end{pmatrix}, \quad \beta_9^{\text{tri}} := \beta_9 - \omega_Y(e_7)c_7,$$

and

$$\varepsilon_{\text{rem}} = 0.$$

Moreover

$$\det L_{\text{eff}} = -\frac{1}{1536\pi^4 g_{\text{Gr}}^{3/2} h_{\text{Gr}}^3 (\sqrt{g_{\text{Gr}}} + \sqrt{h_{\text{Gr}}})^3},$$

so

$$\sigma_{\min}(E_{\text{car}} L_{\text{eff}}) \geq Q^{-C}$$

on Gram-good patches. Finally,

$$\|\varepsilon_{\text{rem}}\|_{\mathcal{Q}_{\text{car}}} = 0 \leq \frac{1}{2}|H| \sigma_{\min}(E_{\text{car}} L_{\text{eff}}) (\Gamma_G^2 + \Gamma_{G,2}^2)^{1/2}.$$

Thus D4_{res} is Green in the formal carrier quotient for the canonical scalar source $u = q''_{\text{can}}$.

Proof. By Lemma 16.332, all six correction-row projections vanish in the carrier quotient. Hence

$$L_{\text{eff}} = L_{\text{clean}}^{\text{tri}} + \sum_{\nu, b} L_{\nu}^{(b)} = L_{\text{clean}}^{\text{tri}}, \quad \varepsilon_{\text{rem}} = 0.$$

The determinant formula is the clean triangular determinant from Lemma 16.342. On the Gram-good patch this determinant is polynomially bounded below, while the entries of $L_{\text{clean}}^{\text{tri}}$ are polynomially bounded. Therefore

$$\sigma_{\min}(L_{\text{clean}}^{\text{tri}}) \geq Q^{-C}.$$

Combining this with

$$\sigma_{\min}(E_{\text{car}}) \geq Q^{-C_0}$$

gives

$$\sigma_{\min}(E_{\text{car}} L_{\text{eff}}) \geq Q^{-C}.$$

The residual estimate is immediate from $\varepsilon_{\text{rem}} = 0$. $\square$

Remark 16.334 (Scalar I -coordinate D4 is quarantined). The scalar I -coordinate D4 formulation is not an active proof route. It collapses distinct carrier classes and can reintroduce cancellations that are absent in the associated-graded carrier quotient. The active D4 statement is the carrier-quotient row-energy statement for the canonical scalar source $u = q''_{\text{can}}$. In particular, Corollary 16.333 and Theorem 16.336 do not assert a lower bound after applying a scalar collapse map S_I . Any future scalar I -coordinate proof would require its own incidence-avoidance theorem; it is not a fallback consequence of the present carrier-quotient D4 package.

Remark 16.335 (Compatibility with high-dimensional Gate-1 redesigns). The D4 carrier-quotient mechanism is tied to the canonical scalar curvature source

$$u = q''_{\text{can}}.$$

It is not a generic converter for arbitrary high-dimensional or family-valued Gate-1 observables.

Thus a redesigned upstream Gate-1 may feed the present downstream D4 route only if it certifies the same canonical source q''_{can} , for example by implying the required canonical jet or row lower bound for $u = q''_{\text{can}}$. Arbitrary projected observables

$$G_{\nu} = L_{\nu} X$$

are not themselves D4 sources unless each G_{ν} comes with its own canonical phase, Gram matrix, whitening tower, carrier quotient, and order-seven and order-nine D4 identities. No such generic transfer is asserted here.

Theorem 16.336 (Effective-matrix carrier-quotient D4 conversion). *Let*

$$E_{\text{car}} := ([\mathbf{C}_7] \ [\mathbf{C}_{Y12}])$$

be the two-column carrier response map into $\mathcal{Q}_{\text{car}}$. Assume the carrier frame is polynomially conditioned:

$$\sigma_{\min}(E_{\text{car}}) \geq Q^{-C}.$$

Assume the carrier-row class satisfies

$$P_{W_{\text{car}}}([\mathcal{A}_7] + [\mathcal{A}_9]) = H E_{\text{car}} L_{\text{eff}} \begin{pmatrix} \Gamma_G \\ \Gamma_{G,2} \end{pmatrix} + \varepsilon_{\text{rem}},$$

where

$$\sigma_{\min}(L_{\text{eff}}) \geq Q^{-C}$$

and

$$\|\varepsilon_{\text{rem}}\|_{\mathcal{Q}_{\text{car}}} \leq \frac{1}{2} |H| \sigma_{\min}(E_{\text{car}} L_{\text{eff}}) (\Gamma_G^2 + \Gamma_{G,2}^2)^{1/2}.$$

Then

$$|P_{W_{\text{car}}}([\mathcal{A}_7] + [\mathcal{A}_9])|_{\mathcal{Q}_{\text{car}}} \geq Q^{-C} |H| (\Gamma_G^2 + \Gamma_{G,2}^2)^{1/2}.$$

Consequently, combined with Gate-2, wall escape, and row mobility, the local carrier-quotient D4 output gives interval carrier-quotient row energy.

In the formal carrier quotient of Corollary 16.333, one has

$$L_{\text{eff}} = L_{\text{clean}}^{\text{tri}}, \quad \varepsilon_{\text{rem}} = 0,$$

so the hypotheses above reduce to the clean triangular determinant and carrier frame lower bounds.

Proof. The polynomial lower bound for E_{car} and L_{eff} gives

$$\left\| H E_{\text{car}} L_{\text{eff}} \begin{pmatrix} \Gamma_G \\ \Gamma_{G,2} \end{pmatrix} \right\|_{\mathcal{Q}_{\text{car}}} \geq Q^{-C} |H| (\Gamma_G^2 + \Gamma_{G,2}^2)^{1/2}.$$

The residual estimate is half-leading, so the triangle inequality gives the claimed lower bound. Row mobility then converts the pointwise or packet-level carrier lower bound into interval carrier-quotient energy. $\square$

Remark 16.337 (Why the unit-coordinate D4 form is no longer used). The carrier coordinates should not be normalized as if the D4 map were

$$H I_2 \begin{pmatrix} \Gamma_G \\ \Gamma_{G,2} \end{pmatrix}.$$

The clean finite computation gives the triangular row-normal matrix

$$L_{\text{clean}}^{\text{tri}} = \begin{pmatrix} c_7 & 0 \\ \beta_9^{\text{tri}} & \alpha_9 \end{pmatrix}, \quad \beta_9^{\text{tri}} := \beta_9 - \omega_Y(e_7)c_7.$$

The same-scale Γ_G -term in the order-nine row is part of the effective leading matrix. It is not a residual.

Remark 16.338 (Scalar I -coordinate incidence is a collapse artifact). The scalar obstruction

$$(A_9^{\text{f}})_I = \frac{yq''\Gamma_{G,2}}{128\pi^2g^3} + R_9^I$$

arises after the collapse map

$$S_I : \mathcal{M}_{\text{car}} \rightarrow \mathcal{O} \cdot I$$

has sent several independent carrier directions to the same scalar I -coordinate. The active D4 theorem in this draft is the effective-matrix carrier-quotient conversion theorem, Theorem 16.336, together with the formal residual closure of Corollary 16.333 and the six-row correction table

of Lemma 16.326. The scalar I -coordinate formulation below is retained only as a quarantined fallback audit. In the carrier quotient, the lower-carrier, line, background/core, and normalization/subtraction contributions are nuisance directions. Thus the scalar incidence wall

$$F_9^I = 0$$

is relevant only if the downstream row theorem insists on collapsed scalar I -coordinate energy. It is not an obstruction to carrier/quotient row energy.

If one nevertheless imposes scalar I -coordinate row energy as the downstream output, the remaining finite obstruction is the incidence wall

$$\mathcal{E}_{cc}^{(9)} = \{F_9^I = 0, \quad yq''\Gamma_{G,2} \neq 0\},$$

where

$$F_9^I := R_9^I + \frac{yq''\Gamma_{G,2}}{128\pi^2g^3}.$$

Closing that scalar formulation would require either proving $F_9^I \neq 0$ together with a quantitative avoidance theorem for $\mathcal{E}_{cc}^{(9)}$, proving that R_9^I is zero, H -divisible, or nuisance-quotient, or replacing the scalar readout by a higher-order or ambient row. Under the carrier-quotient formulation this scalar incidence is not a D4 obstruction. We refer to this fallback scalar obstruction as the order-nine scalar cross-channel incidence problem. It is not part of the active carrier-quotient D4 route.

Remark 16.339 (Admissibility of carrier-quotient row energy). The downstream row obstruction is formulated at the row-module or quotient-row level, not only after scalar collapse to the I -coordinate. Therefore a lower bound of the form

$$\int_I ||[\mathcal{A}_N^{\text{car}}(m)]||_{\mathcal{Q}_{\text{car}}}^2 dm \geq Q^{-C}|I|$$

is an admissible row-energy output. The scalar coordinate

$$S_I(\mathcal{A}_N)$$

is only one coordinate readout and is not faithful on the carrier quotient. If one were to require scalar I -coordinate energy only, then the scalar cross-channel wall would remain a genuine obstruction. Under the carrier quotient formulation it is quotiented away.

Definition 16.340 (D4 residual valuation ledger). After extracting the carrier-quotient leading rows, write the remaining D4 row contribution schematically as

$$\mathbf{R} = \mathbf{R}_{\text{same}} + \mathbf{R}_{\text{val}} + \mathbf{R}_{\text{abs}}.$$

Terms with the same anisotropy-linear valuation as the leading row are not automatically errors. They must first be absorbed into the effective leading matrix L_{eff} . Only the component remaining after this absorption is eligible to be estimated as a residual. A residual term T has valuation

$$T = q_0^a u^b \Gamma^c E_T, \quad \Gamma = (\Gamma_G, \Gamma_{G,2}),$$

where E_T is polynomially bounded or separately controlled.

Terms with the same anisotropy-linear valuation as the chosen leading D4 row are not errors. In the expected carrier-quotient case this valuation is

$$(a, b, c) = (1, 3, 1),$$

corresponding to $q_0 u^3 \Gamma$. If the exact row algebra contains a lower or delayed u -valuation anisotropy-linear layer, the leading valuation is chosen according to the filtered row-observability rule in Remark 16.341. They are absorbed into the effective leading matrix or the carrier-quotient coordinate map.

Terms with higher q_0 -valuation, higher anisotropy valuation, or higher source valuation belong to $\mathbf{R}_{\text{val}}$ and are controlled by the chosen non-wall and wall-escape thresholds.

Terms without sufficient factors of q_0, u, Γ belong to $\mathbf{R}_{\text{abs}}$. These are the only residuals requiring a hard absolute smallness condition in the wall-escape corridor.

Equivalently, for a schematic term

$$T = q_0^a u^b \Gamma^\ell D, \quad |D| \leq Q^C,$$

compare to the leading scale

$$q_0 u^3 \Gamma.$$

The relative size is bounded by

$$\frac{|T|}{|q_0 u^3 \Gamma|} \leq Q^C |q_0|^{a-1} |u|^{b-3} |\Gamma|^{\ell-1}.$$

Terms with $(a, b, \ell) = (1, 3, 1)$ belong in the effective leading matrix L , not in the residual. Terms with extra useful q_0, u, Γ -valuation are relative residuals. Only terms without enough useful valuation are absolute floors and must satisfy the wall-escape corridor estimates.

Remark 16.341 (Filtered D4 row observability). The notation $q_0 u^3 L \Gamma$ records the expected leading row scale. It should not be hard-coded if the exact row algebra contains lower u -valuation anisotropy-linear terms. In general one should write

$$\mathbf{A}_N^{\text{lin}} = q_0 \sum_{b \geq b_0} u^b L_b \Gamma.$$

D4 closes once the stacked filtered response

$$\begin{pmatrix} L_{b_0} \\ u L_{b_0+1} \\ \vdots \\ u^{B-b_0} L_B \end{pmatrix}$$

has second singular value bounded below by Q^{-C} on the sampled branch. The expected carrier-quotient case is $b_0 = B = 3$ and

$$L_3 = -\frac{1}{288} I_2.$$

If a lower u -valuation anisotropy-linear layer appears, it should be promoted into the leading filtered matrix rather than treated as an error.

Lemma 16.342 (Clean triangular D4 carrier matrix). *In the D4 finite carrier calculation, write the centered Gram diagonal scalars as*

$$g_{\text{Gr}} := \frac{q_0}{\pi}, \quad h_{\text{Gr}} := \frac{1}{\pi} \left(\frac{1}{3} q_0^3 + \frac{1}{6} u \right), \quad u = q_{\text{can}}''.$$

Let $\kappa = \omega_Y(e_7)$, and define the row-normal coefficient

$$\beta_9^{\text{tri}}(g, h, u) := \beta_9(g, h, u) - \kappa c_7(g, h).$$

Then

$$g_{\text{Gr}} - h_{\text{Gr}} = -\frac{\Gamma_G}{6\pi}.$$

In this lemma set $g = g_{\text{Gr}}$, $h = h_{\text{Gr}}$. The clean order-seven carrier coordinate is

$$\pi_{\mathfrak{f}} T_7^{(1)} = H c_7(g, h) \Gamma_G I, \quad H = yu,$$

where

$$c_7(g, h) = -\frac{1}{24\pi^2 g h (\sqrt{g} + \sqrt{h})^2}.$$

For the clean order-nine channel, write

$$X_{-,1} = BS, \quad B = \frac{u}{4\pi \sqrt{gh} (\sqrt{g} + \sqrt{h})}.$$

Let

$$G_2 = \text{diag}(K_1, K_2), \quad K_1 = \frac{u}{8\pi}, \quad K_2 = \frac{q_0^2 u + \frac{1}{6} q^{(4)}}{8\pi}.$$

If $X_{-,2} = aI + dD$, then

$$d = \frac{1}{4} \left[\frac{B^2 h + 2B^2 \sqrt{gh} - K_1/g}{\sqrt{g}} - \frac{B^2 g + 2B^2 \sqrt{gh} - K_2/h}{\sqrt{h}} \right].$$

Equivalently,

$$d = -\frac{\Gamma_{G,2}}{32\pi h^{3/2}} + \Gamma_G \rho_9(g, h, u),$$

where

$$\rho_9(g, h, u) = \frac{B^2(g + 3\sqrt{gh} + h)}{24\pi(\sqrt{h} + \sqrt{g})\sqrt{gh}} - \frac{K_1(g + \sqrt{gh} + h)}{24\pi(\sqrt{h} + \sqrt{g})g^{3/2}h^{3/2}}.$$

Since

$$\pi_{\mathfrak{f}} T_9^{(1,2)} = -2yBdI,$$

in the row-normal triangular coordinates the clean order-nine carrier coordinate has the form

$$\pi_{\mathfrak{f}} T_9^{(1,2)} = H(\beta_9^{\text{tri}}(g, h, u)\Gamma_G + \alpha_9(g, h)\Gamma_{G,2})I,$$

where

$$\alpha_9(g, h) = \frac{1}{64\pi^2 \sqrt{gh} h^{3/2} (\sqrt{g} + \sqrt{h})}$$

and

$$\beta_9^{\text{tri}}(g, h, u) = -\frac{\rho_9(g, h, u)}{2\pi \sqrt{gh}(\sqrt{g} + \sqrt{h})} - \kappa c_7(g, h).$$

In particular, on $g = h$,

$$\alpha_9(g, g) = \frac{1}{128\pi^2 g^3},$$

recovering the clean wall formula

$$\pi_{\mathfrak{f}} T_9^{(1,2)} = \frac{H\Gamma_{G,2}}{128\pi^2 g^3} I.$$

Thus, in the clean carrier coordinates,

$$\begin{pmatrix} B_7 \\ B_9 \end{pmatrix} = H \begin{pmatrix} c_7 & 0 \\ \beta_9^{\text{tri}} & \alpha_9 \end{pmatrix} \begin{pmatrix} \Gamma_G \\ \Gamma_{G,2} \end{pmatrix}.$$

The same-scale $\beta_9^{\text{tri}}\Gamma_G$ term belongs to the effective leading matrix, not a residual. Moreover

$$\det \begin{pmatrix} c_7 & 0 \\ \beta_9^{\text{tri}} & \alpha_9 \end{pmatrix} = c_7 \alpha_9 = -\frac{1}{1536\pi^4 g^{3/2} h^3 (\sqrt{g} + \sqrt{h})^3}.$$

Consequently this clean D_4 carrier matrix is polynomially invertible on every Gram-good patch on which

$$Q^{-C} \leq g_{\text{Gr}}, h_{\text{Gr}} \leq Q^C.$$

Remark 16.343 (Associated-graded D_4 determinant certificate). *Historical pre-triangular bookkeeping.* This remark is retained only as a reference to the pre-normalization carrier. All live row-normal formulas use the triangularized coefficient β_9^{tri} from the row-normal D_4 branch.

In the clean carrier normalization of Lemma 16.342,

$$L_{\text{clean}} = \begin{pmatrix} c_7 & 0 \\ \beta_9^{\text{tri}} & \alpha_9 \end{pmatrix},$$

and the corresponding clean carrier image is

$$F_{2Y}^{\text{tri}} = H \begin{pmatrix} c_7 \Gamma_G \\ \beta_9^{\text{tri}} \Gamma_G + \alpha_9 \Gamma_{G,2} \end{pmatrix}.$$

The same-scale $\beta_9^{\text{tri}}\Gamma_G$ contribution belongs to L_{eff} , not to the residual. Consequently,

$$\det L_{\text{clean}} = c_7 \alpha_9 = -\frac{1}{1536\pi^4 g_{\text{Gr}}^{3/2} h_{\text{Gr}}^3 (\sqrt{g_{\text{Gr}}} + \sqrt{h_{\text{Gr}}})^3},$$

which is polynomially bounded below on Gram-good patches. Hence the clean D4 carrier matrix is triangularly invertible before correction-row residuals are added.

Remark 16.344 (Nonzero D4 root traps are harmless under energetic sampling). If a triangularized row coordinate has the form

$$B_7 = X + E_7, \quad B_9 = Y R_Y(Y) + X(\cdots) + E_9,$$

where

$$X = \Gamma_G, \quad Y = \Gamma_{G,2},$$

then a possible finite obstruction is a nonzero root

$$R_Y(\zeta) = 0, \quad \zeta \neq 0.$$

The corresponding bad branch is

$$X \approx 0, \quad Y \approx \zeta.$$

This branch cannot persist on a polynomial-size real interval under the Gate-2 ODE identities. Indeed,

$$X = q'' - 6q + 2q^3, \quad Y = 2q(q')^2 - \frac{1}{6}X''.$$

Testing the second identity against a cutoff and integrating X'' by parts twice converts the apparent derivative obstruction into an L^2 -small X -term, just as in the L^2 -input rigidity proof. Thus an interval on which $X \approx 0$ and $Y \approx \zeta \neq 0$ would force Y small on a shrunken interval, contradicting $|\zeta| \geq Q^{-C}$.

Consequently, under existential or energetic row sampling, nonzero row roots are not fatal: isolated occurrences can be avoided, and interval traps are ruled out by the same local dynamics behind Gate-2 rigidity. If the proof were exact-point locked, such roots would again become relevant.

Remark 16.345 (Raw skew coefficients F_5 and F_7). For the centered raw off-diagonal scalar

$$F(s) = \frac{\sin \Delta(s) + q_-(s)s \cos \Delta(s)}{s^2},$$

with

$$\Delta(s) = q_0 s + \frac{q_2}{24} s^3 + \frac{q_4}{1920} s^5 + \frac{q_6}{322560} s^7 + \cdots,$$

and

$$q_-(s) = q_0 + \frac{q_2}{8} s^2 + \frac{q_4}{384} s^4 + \frac{q_6}{46080} s^6 + \cdots,$$

the raw odd carriers are skew:

$$N_{2j+1} = F_{2j+1} J.$$

In the normalization used here,

$$F_5 = -\frac{q_0^7}{630} + \frac{q_0^4 q_2}{72} - \frac{q_0^2 q_4}{480} - \frac{q_0 q_2^2}{144} + \frac{q_6}{40320},$$

and

$$F_7 = \frac{q_0^9}{36288} - \frac{q_0^6 q_2}{1728} + \frac{q_0^4 q_4}{4608} + \frac{5q_0^3 q_2^2}{3456} - \frac{q_0^2 q_6}{64512} - \frac{q_0 q_2 q_4}{4608} - \frac{5q_2^3}{41472} + \frac{q_8}{9289728}.$$

These coefficients are useful for auditing the raw row table, but they multiply J and hence do not contribute directly to the fixed I -coordinate.

There is also a non-centered visibility route. The local construction uses

$$t_{\pm} = m \pm \frac{s}{2},$$

so m is a sampling midpoint, not necessarily the zero-packet center. For the paired packet

$$f_{\beta, \gamma}(t) = \frac{a}{a^2 + x^2} + \frac{b}{b^2 + x^2}, \quad x = 2t - \gamma,$$

one has, near $x_m = 2m - \gamma = 0$,

$$q^{(7)}(m) = 5160960(a^{-9} + b^{-9})x_m + O(x_m^3).$$

Thus $x_m = 0$ is a simple zero of $q^{(7)}(m)$, and every sufficiently small nonzero perturbation of m activates the septic K_1 -transport slice, outside

$$q^{(7)} = 0, \quad \Xi_{K_1} = 0.$$

Proposition 16.346 (Packet visibility on $\mathcal{B}_{\text{red}}^+$). *On $\mathcal{B}_{\text{red}}^+$, the finite fixed-sector vector*

$$C_{\text{fd}}^{\vec{f}} = (A_7^{\mathbf{f}}, A_9^{\mathbf{f}}, \dots)$$

is nonzero for the reduced off-line packet family.

Proof. For the centered route, the direct skew N_7^{off} -term projects to zero, but the one-whitening-slot K_1 -channel has fixed-sector projection

$$-2yBr I.$$

This vector is nonzero outside

$$\{y = 0\} \cup \{q'' = 0\} \cup \{g = h\}.$$

To rule out cancellation by the K_5 -line sector using quotient isolation, we also exclude

$$v_5 = 0.$$

Indeed,

$$\omega_h(-2yBr I, A_5^{\mathbf{f}}) = -2yBr v_5,$$

so K_5 -line terms cannot cancel the K_1 -channel on the generic patch

$$yBr v_5 \neq 0.$$

The branch $v_5 = 0$ is therefore either included in $\mathcal{E}_{\text{invis}}^{\text{core},7}$ or must be treated by a separate full-vector K_5 -coefficient computation.

For the non-centered route, m is a free sampling midpoint. The expansion of $q^{(7)}(m)$ shows that $q^{(7)} = 0$ is a proper analytic wall near the packet center. Away from this wall and from $\{\Xi_{K_1} = 0\}$, the septic K_1 -transport slice gives $A_7^{\mathbf{f}} \neq 0$. $\square$

Remark 16.347 (Order-seven channel independence ledger). The full order-seven corrected-whitening coefficient is

$$A_{7,\text{core}}^{\mathbf{f}} = \pi_{\mathfrak{f}} \sum_{a+g+b=7} X_{-,a} N_g X_{+,b}.$$

The direct skew N_7^{off} -term is killed by $\pi_{\mathfrak{f}}$. The live one-whitening-slot term belongs to the K_1 -transport sector. K_3 shadow terms are projected away, pure same-point septic terms vanish in the odd two-point germ, and K_5 -terms lie in

$$\mathbf{C}A_5^{\mathbf{f}}.$$

Thus exact cancellation of the live $-2yBr I$ channel can only come from a same K_1 -transport contribution or a direct diagonal N_7 -term with the same jet monomial. No structural identity forcing such cancellation is currently present; such cancellation is included in the visibility incidence locus.

16.26. The $\eta\eta$ row as rank-two affine forcing. In the rank-two formulation, the active Bottleneck-C row is no longer a scalar defect by itself. It is a vector-valued section:

$$E_{\eta\eta} = -\frac{\nu_{\eta\eta}}{24\pi q^3}(wI + uS) \in \Gamma(\mathcal{F}_{\text{red}}^+).$$

Under a frame change

$$E_j = E_i G_{ij},$$

the coordinate vector of $wI + uS$ transforms by

$$[w, u]_j^T = G_{ij}^{-1} [w, u]_i^T.$$

The scalar coefficient

$$-\frac{\nu_{\eta\eta}}{24\pi q^3}$$

is fixed-convention scalar data. Therefore $E_{\eta\eta}$ transforms as a section of the rank-two fixed-sector bundle, provided the reduced atlas is functorial.

This row should not be identified with ordinary curvature of ∇^f . Curvature is antisymmetric in two tangent directions:

$$F_{\nabla^f}(X, Y) = [\nabla_X^f, \nabla_Y^f] - \nabla_{[X, Y]}^f.$$

The active row is symmetric in the source directions η_i, η_j . It is better classified as a symmetric source-Hessian forcing term, or a second-fundamental-form component, in the fixed-sector transport equation.

The natural local replacement for scalar Bottleneck-C flatness is therefore an affine transport equation of the form

$$\nabla_\kappa^f A_7^f = E_{\eta\eta} + E_{\text{lower}}.$$

Here E_{lower} denotes all lower, line, shadow, pure, connection, and background/core terms. The expected decomposition is

$$E_{\text{lower}} = E_{\text{GateA}} + E_{\text{GateB}} + E_{\text{line}} + E_{K_3} + E_{\text{pure}} + E_{\text{conn}} + E_{\text{bg/core}} + E_{K_1}^{\text{comp}}.$$

Under the current local ledger,

$$E_{\text{GateA}} = 0, \quad E_{\text{GateB}} = 0$$

under the accepted Gate A/B hypotheses,

$$E_{\text{conn}} = 0$$

after passing to the covariant derivative,

$$E_{K_3} = 0$$

after fixed-sector projection,

$$E_{\text{pure}} = 0$$

in the odd two-point germ, and

$$E_{\text{line}} \in \mathcal{CA}_5^f.$$

The remaining table dependency is to verify that no additional uncontrolled same-grade K_1 source-Hessian companion survives outside the already identified $E_{\eta\eta}$ row.

The current local row table is:

row class	source	fixed-sector status	present row
E_{GateA}	aggregate homogeneous scalar-grade rows	0	closed by Gate A
E_{GateB}	ordinary order ≤ 7 local rows	0	closed by Gate B
$E_{\eta\eta}$	symmetric second-source Hessian / mixed-block row	$-\frac{\nu_{\eta\eta}}{24\pi q^3}(wI + uS)$	retained as affine
E_{line}	K_5 normalization shadows	$\in \mathcal{CA}_5^f$	transported line
E_{K_3}	K_3 shadow sector	0	projected away
E_{pure}	pure same-point septic sector	0	vanishes in odd two-
E_{conn}	frame/projection/whitening gauge motion	0	absorbed by
$E_{\text{bg/core}}$	fixed background/core baseline terms	fixed analytic section	included in affine

The visibility-positive incidence exclusions used to define $\mathcal{B}_{\text{red}}^+$ do not alter Gate A/B; they only restrict the base to points where the packet-visibility channel is active.

Theorem 16.348 (Conditional local rank-two affine forcing equation). *On $\mathcal{B}_{\text{red}}^+$, assume the rank-two connection theorem and the row-table classifications above. Then*

$$\nabla_{\kappa}^{\mathfrak{f}} A_7^{\mathfrak{f}} = E_{\eta\eta} + E_{\text{lower}},$$

where

$$E_{\eta\eta} = -\frac{\nu_{\eta\eta}}{24\pi q^3}(wI + uS)$$

is a tensorial section of $\mathcal{F}_{\text{red}}^+$, and E_{lower} contains only Gate A/B-closed, connection-absorbed, line, shadow, pure, background/core, and already-classified analytic forcing terms.

Proof. By definition,

$$A_7^{\mathfrak{f}} = [s^7] \Pi_{\mathfrak{f}} \widehat{\Omega}_{\zeta}^{\text{corr}}(s; m).$$

The projected connection gives

$$\nabla_{\kappa}^{\mathfrak{f}} A_7^{\mathfrak{f}} = \Pi_{\mathfrak{f}} d_{\kappa}^{\mathcal{J}} \left([s^7] \widehat{\Omega}_{\zeta}^{\text{corr}} \right),$$

so frame, projection, and whitening-gauge variations are absorbed into $\nabla^{\mathfrak{f}}$. The $\eta_i \eta_j$ second-source Hessian row of the promoted mixed block gives $E_{\eta\eta}$. All remaining terms are classified by the displayed table. $\square$

Remark 16.349 (The equation is local, not yet global). Theorem 16.348 does not prove a global contradiction. It changes the local interpretation of the Bottleneck-C row: the row is no longer a scalar defect demanding pointwise vanishing, but an inhomogeneous forcing term in a rank-two transport equation. A global boundary, holonomy, exactness, index, or transported finite-core non-concentration theorem is still required.

Remark 16.350 (What the affine equation changes). The old scalar Bottleneck-C gate demanded local pointwise vanishing of the S -visible component. The rank-two equation instead allows the same row to be an inhomogeneous forcing term in a vector-valued transport equation. This is a genuine conceptual improvement only if a later global theorem controls the forced transport. If one still demands

$$E_{\eta\eta} = 0$$

pointwise, then the rank-two formulation collapses back to the old obstruction

$$\nu_{\eta\eta} = 0$$

on a good patch.

Proposition 16.351 (No-go for quotient-killing an unsplit live arithmetic row). *Let $K_{0,p}$ be a dead nuisance space and let $E_p = [C_{7,p}^{\text{tar}} \ C_{Y,p}^{\text{tar}}]$. Suppose the raw arithmetic column has the decomposition*

$$A_{II,p}^{\text{raw}} = A_{II,p}^{\text{dead}} + E_p m_{\eta,p}, \quad A_{II,p}^{\text{dead}} \in K_{0,p}, \quad m_{\eta,p} \neq 0.$$

If the full unsplit column $A_{II,p}^{\text{raw}}$ is placed in the killed quotient space $K_p := K_{0,p} + \text{span}\{A_{II,p}^{\text{raw}}\}$, then $\sigma_{\min}((I - P_{K_p}^{\text{bd}})E_p) = 0$.

Proof. Since $A_{II,p}^{\text{raw}} \in K_p$ and $A_{II,p}^{\text{dead}} \in K_p$, their difference lies in K_p : $E_p m_{\eta,p} = A_{II,p}^{\text{raw}} - A_{II,p}^{\text{dead}} \in K_p$. Thus $(I - P_{K_p}^{\text{bd}})E_p m_{\eta,p} = 0$. Since $m_{\eta,p} \neq 0$, the quotient map has a nontrivial kernel on the retained live plane. $\square$

Remark 16.352 (What triggers the affine repair). The vanishing route for the primitive arithmetic row is not available from the current local row algebra. The active odd- $\eta\eta$ top row has $f_{\eta\eta}^{\text{top}} = -u \nu_{21,7}/(24\pi q^3)$, and in the homogeneous source model one obtains $\nu_{21,7} = -q_0^3/(216Q^{14})$, which is generically nonzero on ordinary good patches. Therefore the correct repair is not to force $c_{\text{arith},Y} = 0$, but to split the arithmetic column and retain its live $C_7/Y12$ part as affine forcing.

Definition 16.353 (Affine-live certified quotient). Construct the quotient in the following order. First fix the retained live plane $E_p = [C_{7,p}^{\text{tar}} \ C_{Y,p}^{\text{tar}}]$ and bounded rows L_p^{row} satisfying $L_p^{\text{row}} E_p = I_2$ and $\|L_p^{\text{row}}\|_{\text{op}} \leq Q^C$. For each raw nuisance column set

$$m_{a,p} := L_p^{\text{row}} A_{a,p}^{\text{raw}}, \quad A_{a,p}^{\text{dead}} := A_{a,p}^{\text{raw}} - E_p m_{a,p}.$$

In particular, $A_{II,p}^{\text{raw}} = A_{II,p}^{\text{dead}} + E_p m_{\eta,p}$. The affine-live killed space is

$$\mathcal{K}_{p,\text{aff}}^Y := \text{span}\{A_{a,p}^{\text{dead}} : a \text{ in the nuisance list}\} + K_{p,0}^{Y,\text{dead}}, \quad q_{p,\text{aff}}^Y := I - P_{\mathcal{K}_{p,\text{aff}}^Y}^{\text{bd}}.$$

The live vectors $E_p m_{a,p}$, and in particular $E_p m_{\eta,p}$, are not quotient-killed; they are retained in the affine discarded observable.

Lemma 16.354 (Affine-live quotient preserves the retained two-plane). Assume $L_p^{\text{row}} \mathcal{K}_{p,\text{aff}}^Y = 0$, $L_p^{\text{row}} E_p = I_2$, and $\|L_p^{\text{row}}\|_{\text{op}} \leq Q^C$. Then $\sigma_{\min}(q_{p,\text{aff}}^Y E_p) \geq Q^{-C}$.

Proof. For $v \in \mathbb{C}^2$, $v = L_p^{\text{row}} E_p v = L_p^{\text{row}} q_{p,\text{aff}}^Y E_p v$, because L_p^{row} kills $\mathcal{K}_{p,\text{aff}}^Y$. Hence $\|v\| \leq \|L_p^{\text{row}}\|_{\text{op}} \|q_{p,\text{aff}}^Y E_p v\| \leq Q^C \|q_{p,\text{aff}}^Y E_p v\|$. Taking the infimum over $\|v\| = 1$ gives the singular-value bound. $\square$

Definition 16.355 (Exhaustive maximal labelled $K9$ live/dead routing). In the canonical labelled $K9$ branch, every raw row or nuisance column with possible $C_7/Y12$ visibility is processed before forming the killed quotient. Let

$$E_p = [C_{7,p}^{\text{tar}} \ C_{Y,p}^{\text{tar}}], \quad L_p^{\text{row}} E_p = I_2, \quad \|L_p^{\text{row}}\|_{\text{op}} \leq Q^C.$$

For each raw column $A_{a,p}^{\text{raw}}$, set

$$m_{a,p} := L_p^{\text{row}} A_{a,p}^{\text{raw}}, \quad A_{a,p}^{\text{dead}} := A_{a,p}^{\text{raw}} - E_p m_{a,p}.$$

Only the dead remainders $A_{a,p}^{\text{dead}}$ are allowed to enter the killed quotient. Every nonzero live part $E_p m_{a,p}$ is either:

- (1) one of the clean retained $C_7/Y12$ carrier components;
- (2) a labelled affine forcing summand included in the maximal affine target $\hat{\mathcal{H}}_p^{Y,\text{max}}$;
- (3) or a term assigned to the residual ledger with a stated Q^{-B} -small bound in the source-graded norm.

No atom-local live $C_7/Y12$ component is quotient-killed. Equivalently, the exact killed quotient is formed only after the canonical labelled live/dead split.

Theorem 16.356 (Raw $2Y$ certificate from exhaustive labelled routing). Assume Definition 16.355 on a selected ordinary packet p . Let $\mathcal{K}_{p,\text{aff}}^Y$ be the killed space generated by the row-dead remainders and by the fixed dead quotient space, and let

$$q_{p,\text{aff}}^Y := I - P_{\mathcal{K}_{p,\text{aff}}^Y}^{\text{bd}}.$$

Then

$$\sigma_{\min}(q_{p,\text{aff}}^Y E_p) \geq Q^{-C}.$$

Consequently the rank-two $C_7/Y12$ quotient survival required by the certified $2Y$ bridge is proved in the canonical labelled branch. In particular, a nonzero primitive arithmetic coefficient such as $c_{\text{arith},Y}$ is not an obstruction when its live component is retained as a labelled affine forcing summand rather than killed in the exact quotient.

Proof. By construction,

$$L_p^{\text{row}} A_{a,p}^{\text{dead}} = L_p^{\text{row}} A_{a,p}^{\text{raw}} - L_p^{\text{row}} E_p m_{a,p} = m_{a,p} - m_{a,p} = 0.$$

The fixed dead quotient space is also killed by L_p^{row} . Hence

$$L_p^{\text{row}} \mathcal{K}_{p,\text{aff}}^Y = 0.$$

For every $v \in \mathbb{C}^2$,

$$v = L_p^{\text{row}} E_p v = L_p^{\text{row}} q_{p,\text{aff}}^Y E_p v.$$

Therefore

$$\|v\| \leq \|L_p^{\text{row}}\|_{\text{op}} \|q_{p,\text{aff}}^Y E_p v\| \leq Q^C \|q_{p,\text{aff}}^Y E_p v\|.$$

Taking the infimum over $\|v\| = 1$ gives the singular-value lower bound. The final statement follows because every nonzero live coefficient has been removed from the killed quotient before the quotient is formed. $\square$

Remark 16.357 (Certified labelled routing versus raw row verification). Theorem 16.356 is a certified-branch implication. It proves the rank-two quotient lower bound once all of the following data have been supplied on the selected packet:

$$L_p^{\text{row}} E_p = I_2, \quad \|L_p^{\text{row}}\|_{\text{op}} \leq Q^C,$$

polynomial conditioning of the labelled live plane, and an exhaustive labelled split of every raw nuisance column before the killed quotient is formed.

It does not, by itself, compute the concrete rows $\ell_{7,p}, \ell_{9,p}^Y$ from the raw formulas, nor does it prove the septic exactness, fixed-whitening separation, order-seven local row-deadness, primitive-arithmetic routing, or E_2/R_{lab} no-atom-locality tests. Those raw coefficient checks are exactly the content of the raw row/routing certificate $\mathfrak{R}_{2Y}(p)$, equivalently the obstruction vector $\mathfrak{B}_{2Y,p}$, unless they are imported from a separate verified table.

Thus the active logical implication is

$$\mathfrak{R}_{2Y}(p) \implies \text{exhaustive labelled live/dead routing} \implies \sigma_{\min}(q_{p,\text{aff}}^Y E_p) \geq Q^{-C}.$$

The affine repair removes the obsolete target $c_{\text{arith},Y} = 0$: a nonzero primitive arithmetic row is harmless if it is retained as a labelled affine forcing term or residual-routed before quotienting. It is still fatal if an atom-local live $C_7/Y12$ component is quotient-killed before the labelled split.

Remark 16.358 (Reduced raw $2Y$ obstruction after maximal routing). After the affine-live repair, the active raw $2Y$ obstruction is not the scalar identity $c_{\text{arith},Y} = 0$. The active obstruction is the existence of an atom-local live $C_7/Y12$ row outside the maximal labelled target, or a live row which is killed before the live/dead split. Thus older obstruction vectors $\mathfrak{B}_{2Y,p}$ should be read as the finite row-table test for:

- (1) polynomially bounded detecting rows with $L_p^{\text{row}} E_p = I_2$;
- (2) septic exactness and fixed-whitening separation;
- (3) row-deadness or residual/coherent routing of the A_7 , $II_{\text{arith},9}$, and E_2/R_{lab} entries;
- (4) exhaustiveness of the labelled target for all atom-local live $C_7/Y12$ components.

If these raw facts are verified, then Theorem 16.356 discharges the certified quotient survival. If they are not verified, then $\mathfrak{B}_{2Y,p}$ remains the finite audit item for the $2Y$ bridge.

Definition 16.359 (Affine-graded cap observable). For $v \in \mathbb{C}^2$, define

$$\hat{\Phi}_{p,\text{aff}}^Y(v) := (q_{p,\text{aff}}^Y E_p H_p L_{\text{clean},p} v, q_{p,\text{aff}}^Y E_p m_{\eta,p}) \in \hat{\mathcal{H}}_p^{Y,\text{aff}}.$$

Equivalently, the scalar-collapsed affine expression $q_{p,\text{aff}}^Y E_p (H_p L_{\text{clean},p} v + m_{\eta,p})$ is replaced by the source-graded vector with separate clean and $\eta\eta$ components.

Proposition 16.360 (Static-line lower bound in the affine-graded norm). *Let $u_j \in \mathbb{C}^2$ be a unit static direction. Assume $\sigma_{\min}(q_{p,\text{aff}}^Y E_p) \geq Q^{-C_E}$, $\sigma_{\min}(L_{\text{clean},p}) \geq Q^{-C_L}$, and the $D4$ amplitude mass estimate $\sum_{p \in C_j} \mu_p |H_p|^2 \geq Q^{-A_H} \sum_{p \in C_j} \mu_p$ on the selected static cap core. Then*

$$\sum_{p \in C_j} \mu_p \|\hat{\Phi}_{p,\text{aff}}^Y(u_j)\|_{\text{gr}}^2 \geq Q^{-A'} \sum_{p \in C_j} \mu_p.$$

More generally, replacing u_j by arbitrary Γ_p gives the carrier-mass form with $\sum \mu_p \|\Gamma_p\|^2$ on the right, provided the source-weighted $D4$ amplitude mass estimate is used.

Proof. By definition of the graded norm, $\|\hat{\Phi}_{p,\text{aff}}^Y(u_j)\|_{\text{gr}}^2 = \|q_{p,\text{aff}}^Y E_p H_p L_{\text{clean},p} u_j\|^2 + \|q_{p,\text{aff}}^Y E_p m_{\eta,p}\|^2 \geq Q^{-2C_E} |H_p|^2 \|L_{\text{clean},p} u_j\|^2 \geq Q^{-2C_E-2C_L} |H_p|^2$. Summing against μ_p and using amplitude mass gives the claim. The carrier-mass form is identical with u_j replaced by Γ_p , using $\|L_{\text{clean},p} \Gamma_p\| \geq Q^{-C_L} \|\Gamma_p\|$. $\square$

16.27. Global affine transport dependency. Assuming the local affine equation

$$\nabla^{\dagger} A = E$$

is available, parallel transport along a path $\Gamma : a \rightarrow b$ gives

$$A(b) = U_{a \rightarrow b} A(a) + \int_a^b U_{t \rightarrow b} E(t) dt.$$

For the order-seven fixed-sector section this becomes

$$A_7^{\dagger}(b) = U_{a \rightarrow b} A_7^{\dagger}(a) + \int_a^b U_{t \rightarrow b} (E_{\eta\eta} + E_{\text{lower}})(t) dt.$$

Thus the local Bottleneck-C obstruction becomes an affine displacement problem.

At present no canonical global proof gate for this displacement is available. The natural path and boundary candidates have the following status:

- κ -paths are locally natural but have no known endpoint values for $A_7^{\dagger}$.
- midpoint paths $m \mapsto b(m)$ are natural locally, but good-patch boundaries are exceptional walls rather than known boundary conditions.
- source/core deformation paths are excluded on the reduced base, since source/core transfer is not functorial as an ordinary vector-bundle transition.
- Gram or whitening loops may measure gauge/atlas data rather than zeta geometry unless a global atlas theorem is proved.
- zero-counting contours are canonical for ξ or ζ , but no map from those contours into the rank-two fixed-sector base is currently constructed.

Nor is the forcing currently exact. In the fixed-source midpoint calculus, the active scalar density satisfies

$$2880Q^{14}\nu_{\eta\eta} = qAA'' + 20\left(\frac{q''}{6} - \frac{2q^3}{3}\right)A^2, \quad A = A_2.$$

Integration by parts yields

$$qAA'' + 20\left(\frac{q''}{6} - \frac{2q^3}{3}\right)A^2 = \left(qAA' - \frac{1}{2}q'A^2\right)' - q(A')^2 + \left(\frac{23}{6}q'' - \frac{40}{3}q^3\right)A^2.$$

The last two terms are genuine bulk terms. Therefore the forcing is not presently a boundary term in the natural variables.

For a loop Γ based at b , the corresponding affine holonomy data are

$$\text{AffHol}_{\Gamma} = \left(U_{\Gamma}, \Delta_{\Gamma}^{\text{aff}}\right),$$

where

$$\Delta_{\Gamma}^{\text{aff}} = \int_{\Gamma} U_{t \rightarrow b} (E_{\eta\eta} + E_{\text{lower}})(t) dt.$$

A holonomy proof gate would need all of:

- (1) a canonical loop $\Gamma \subset \mathcal{B}_{\text{red}}^+$;
- (2) a global theorem constraining U_{Γ} ;
- (3) a global zeta-theoretic law forcing $\Delta_{\Gamma}^{\text{aff}} = 0$, or a prescribed value;
- (4) an off-line-packet theorem forcing an incompatible value of $\Delta_{\Gamma}^{\text{aff}}$.

None of these is presently available.

Remark 16.361 (Affine holonomy is not presently a proof gate). Loops around core-boundary walls, packet-visibility incidence walls, Gram singularities, or whitening branch loci are diagnostic but not currently canonical RH objects. A loop around a core-boundary wall may change labels or source/core convention; a Gram loop may measure only gauge or atlas monodromy; and zero-counting contours live in the complex zeta plane, not yet in the rank-two fixed-sector base $\mathcal{B}_{\text{red}}^+$. Thus affine holonomy cannot presently replace the local affine forcing equation. The most plausible global successors are real path/boundary transport or a transported finite-core non-concentration theorem.

Remark 16.362 (Current global dependency). The rank-two affine equation is currently descriptive rather than contradictory. To turn it into a proof gate one needs one of:

- (1) canonical boundary values for A_7^f ;
- (2) a canonical path or loop with prescribed affine holonomy;
- (3) a path-independent affine displacement theorem;
- (4) a projective/scale index theorem derived from the rank-two transport;
- (5) a transported finite-core non-concentration theorem for the full fixed-sector vector.

None of these is presently proved.

Hypothesis 16.363 (Global rank-two affine transport gate). There exists a canonical path, loop, boundary condition, index, or transported finite-core non-concentration theorem on $\mathcal{B}_{\text{red}}^+$ that controls

$$\Delta_{\Gamma}^{\text{aff}} = \int_{\Gamma} U_{t \rightarrow b}(E_{\eta\eta} + E_{\text{lower}})(t) dt.$$

Equivalently, the rank-two affine forcing equation must be coupled to a global law that distinguishes RH-compatible configurations from off-line-packet configurations. Without such a law, the rank-two architecture is locally coherent but globally non-closing.

16.28. Derived diagnostics: projective, vector-valued, and energy data. Several objects remain useful as diagnostics, but none currently replaces the rank-two affine transport theorem.

First, projectivizing the fixed-sector vector gives

$$\ell_{2r+1}(m) = [A_{2r+1}^f(m)] \in \mathbf{P}(\mathfrak{f}_h(m)).$$

The quotient determinant is exactly projective incidence:

$$D_{2r+1} = \omega_h(A_{2r+1}^f, A_5^f),$$

and

$$D_{2r+1} = 0 \iff \ell_{2r+1} = \ell_5.$$

Thus projective/Maslov/spectral-flow data recover the quotient determinant geometry. They remain incomplete by themselves because pure line anomalies

$$A_{2r+1}^f = \lambda A_5^f$$

are projectively invisible but still visible to T and Φ_K . A projective index would therefore need to be paired with scale, energy, or the full rank-two connection.

Second, the complete local fixed-sector object is the vector-valued odd germ

$$\mathcal{A}_{\text{odd}}^f(s; m) = \sum_{r \geq 0} A_{2r+1}^f(m) s^{2r+1} \in \mathfrak{f}_h(m).$$

It retains all fixed-sector information, preserves componentwise holomorphy and Cauchy recovery, and avoids the incompleteness of quotient scalarization. However, it is essentially the full fixed-sector version of the split readout. It does not by itself prove the required finite-core lower bound

$$\theta(m) \notin \mathcal{Z}_k^f \cup \mathcal{E}_{\text{geom}}^f \implies \left\| P_{\text{fd}}^f(\theta(m)) + J_{\text{fd}}^f(\theta(m)) \right\|_{\infty} \geq \Lambda_f(m, Q).$$

The aligned branch

$$J_{\text{fd}}^f \approx -P_{\text{fd}}^f$$

therefore remains a genuine downstream dependency.

Third, one may define fixed-sector energies such as

$$\mathcal{E}_7 = \|A_7^f\|_{\text{wh}}^2$$

or

$$\mathcal{E}_{\text{odd}} = \sum_r \omega_r \|A_{2r+1}^f\|_{\text{wh}}^2.$$

For

$$E_{\eta\eta} = \gamma(wI + uS),$$

the first variation of $\mathcal{E}_7$ contains

$$2 \operatorname{Re} \left[\gamma \langle A_7^\dagger, wI + uS \rangle_{\text{wh}} \right],$$

which is sign-indefinite. The second variation is positive but tautological:

$$2|\gamma|^2 \|wI + uS\|_{\text{wh}}^2 \geq 0.$$

No zero-energy, monotonicity, or endpoint theorem is presently available. Thus energy detects the defect but does not yet provide a contradiction.

Remark 16.364 (Role of Gate A and Gate B after the downstream repairs). Gate A and Gate B remain useful local algebraic lemmas:

Gate A: aggregate homogeneous cancellation,

Gate B: conditional local quotient/fixed-sector algebra through order 7.

They simplify the local expansion and classify lower terms, but they are not proof-closing global gates.

After the L^2 -rigidity repair, Gate 2 source-anisotropy reintegration is no longer an independent downstream input in the ordinary real-canonical branch; it is supplied by Theorem 15.22, conditional on the real-canonical definitions and source local complexity. The remaining burden is therefore sharper: prove global branch-entry and no-loss routing from a hypothetical off-line zero into the ordinary Hardy-normalized canonical packet branch, prove Fejér–Riesz persistence for the active graph packets, verify or import the raw $2Y$ row/routing certificate, and transfer the surviving packet mass into the certified source-graded $K9$ norm.

Remark 16.365 (Coherent graph versus forbidden vertical residual). The current architecture repeatedly has the same form:

$$\boxed{\text{coherent graph}} \quad \text{versus} \quad \boxed{\text{forbidden vertical residual.}}$$

For the $Y12$ -fiber, the coherent graph has the form

$$s_9^Y = \rho_Y s_7 + \eta_{\text{coh}},$$

and the forbidden normal component is

$$s_9^Y - \rho_Y s_7.$$

For finite/O2/O3, the coherent graph is the near-pure determinant span

$$D_{-n+1}, \dots, D_{-n+d_0},$$

and the forbidden residual is the Schur component

$$D_{-n} \notin \operatorname{span}\{D_{-n+1}, \dots, D_{-n+d_0}\}.$$

For canonical source survival, the coherent graph is the Hardy-differential AFE cancellation model, and the forbidden residual is the surviving Hardy-curvature source energy

$$q_{\text{can}}'' = u_H = \partial_t^3 Z_H \not\approx 0$$

on the canonical branch-complete window $W_{\text{branch}}^{\text{can}}$.

This language should be used to keep the three live modules aligned:

$$\operatorname{Agg}L^2(q_{\text{can}}''), \quad \text{Graph/O2O3}_{23}, \quad Y12\text{NormalD4}.$$

The finite-core problem below should therefore be read in a further updated role. It is no longer merely a downstream step after a closed Bottleneck-C gate, nor is it only a generic vector-valued non-concentration problem. The source-collapse audits show that anisotropy non-concentration alone is insufficient, because the known fixed-sector rows carry the common source factor

$$H = yq_{\text{can}}''.$$

The local downstream package supplies source–anisotropy reintegration from Gate 1 by Theorem 15.22. Thus the remaining proof-relevant finite-core theorem must include Gate 1 source-level nonflatness,

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 dm \geq Q^{-B} |I_\rho|,$$

and then connect the resulting Gate 2 source-weighted anisotropy to the D4 carrier-quotient row-energy output.

Remark 16.366 (Visibility is not non-concentration). The order-seven packet-visibility result proves that the relevant fixed-sector vector is not identically zero on the reduced visibility-positive base. It does not prove the sampled finite-core lower bound. The aligned branch

$$J_{\text{fd}}^{\vec{f}} \approx -P_{\text{fd}}^{\vec{f}}$$

remains possible until a vector-valued non-concentration, cone-control, interaction-rigidity, or transport-index theorem rules it out. Thus visibility closes the local detection problem but not the global sampling problem.

Remark 16.367 (Transported finite-core data do not remove alignment). One may try to strengthen the finite-core vector by adding covariant transport data, for instance

$$C_{\text{fd}}^{\text{tr}} = (A_7^{\text{f}}, \nabla^{\text{f}} A_7^{\text{f}}, A_9^{\text{f}}, \nabla^{\text{f}} A_9^{\text{f}}, \dots).$$

Using the local affine equation, this is equivalent to adding

$$E_{\eta\eta} + E_{\text{lower}}$$

as additional coordinates. This is coherent, but it does not by itself prove sampled non-concentration. The old aligned branch simply becomes

$$J_{\text{fd}}^{\text{tr}} \approx -P_{\text{fd}}^{\text{tr}}.$$

Adding derivative or forcing coordinates enlarges the finite vector but does not impose a sign, cone, monotonicity, or endpoint constraint that forbids alignment. A transported finite-core theorem would therefore require a new coercivity, boundary, incidence, or cone-control input.

16.29. Source-weighted interval non-concentration after the Gate–2 repair. The local fixed-sector visibility result is layered through order nine:

$$\Gamma_G \neq 0 \implies A_7^{\text{f}} \neq 0,$$

while

$$\Gamma_G = 0, \quad \Gamma_{G,2} \neq 0 \implies A_9^{\text{f}} \neq 0$$

outside the named correction and cross-channel incidence walls. This local visibility theorem is not the remaining proof gate.

After Theorem 15.22, the old two-stage downstream problem should be read as follows. First one needs Gate 1 source-curvature observability,

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 dm \geq Q^{-B} |I_\rho|.$$

In the ordinary real-canonical branch, Gate 2 source–anisotropy reintegration is then supplied locally:

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 (\Gamma_G(m)^2 + \Gamma_{G,2}(m)^2) dm \geq Q^{-A} |I_\rho|.$$

The remaining downstream conversion is D4: convert this source-weighted anisotropy into carrier-quotient row energy. In scalar language the target scale is

$$\boxed{\sup_{m \in I_\rho} |H(m)| \max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \geq Q^{-A}, \quad H = yq''_{\text{can}}.}$$

In the carrier quotient, this is replaced by the quotient row-energy object in Remark 16.369.

In the transported version, one may instead work with

$$C_{\text{fd}}^{\text{tr}} = \left(A_7^{\text{f}}, \nabla^{\text{f}} A_7^{\text{f}}, A_9^{\text{f}}, \nabla^{\text{f}} A_9^{\text{f}}, \dots \right),$$

or minimally, using the affine equation,

$$C_{\text{fd}}^{\text{tr}} = \left(A_7^{\text{f}}, E_{\eta\eta} + E_{\text{lower}}, A_9^{\text{f}}, \dots \right).$$

The remaining global obstruction is that pair and interaction pieces may still align:

$$J_{\text{fd}}^{\text{tr}} \approx -P_{\text{fd}}^{\text{tr}}.$$

Thus the next proof-closing theorem must be a quantitative layered non-concentration, incidence-distance, cone, or arithmetic separation result.

Definition 16.368 (Source-weighted layered residual). On a centered reduced good patch, in the canonical rank-two convention define

$$H := yq_{\text{can}}''.$$

The order-seven/order-nine layered fixed-sector rows have the schematic form

$$A_7^{\text{f}} \sim H\Gamma_G, \quad A_9^{\text{f}} \sim H\Gamma_{G,2},$$

where

$$\Gamma_G = q_{\text{can}}'' - 6q_0 + 2q_0^3$$

and

$$\Gamma_{G,2} = (1 - q_0^2)q_{\text{can}}'' - \frac{1}{6}(q_\zeta)^{(4)}.$$

Thus the relevant norm-scale residual is not merely

$$(\Gamma_G, \Gamma_{G,2}),$$

but the source-weighted vector

$$\mathcal{R}_{\text{sw}} := (H\Gamma_G, H\Gamma_{G,2}).$$

Equivalently, the scalar lower-bound target is

$$\sup_{m \in I_\rho} |H(m)| \max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \geq Q^{-A}.$$

At leading centered source order,

$$y_{\text{lead}} = -\frac{q_0 A_2^2}{72}, \quad q_{\text{can}}'' = 2A_2,$$

and hence

$$H = yq_{\text{can}}'' \sim -\frac{q_0 (q_{\text{can}}'')^3}{288} = -\frac{q_0 A_2^3}{36}.$$

This cubic source factor is the main vulnerability of the rank-two norm route.

Remark 16.369 (Carrier quotient version of the source-weighted residual). Definition 16.368 records the scalar source-weighted scale

$$(H\Gamma_G, H\Gamma_{G,2}).$$

In the carrier-quotient row formulation, the corresponding object is

$$\mathcal{R}_{\text{sw}}^{\text{car}} := (H\Gamma_G[\mathbf{C}_7], H\Gamma_{G,2}[\mathbf{C}_{Y12}]) \in \mathcal{Q}_{\text{car}}^{\oplus 2}.$$

This is the proof-relevant row-energy object whenever the downstream contradiction is stated before scalar I -coordinate collapse.

Lemma 16.370 (Source-curvature completion for the rank-two plain wall). *Let $I \Subset I^*$ be a chart-good interval with R^{-C} -scale buffer, and set*

$$p := q''.$$

Assume there is a finite source space $\mathcal{P}_I$ and $p_X \in \mathcal{P}_I$ such that

$$\|p - p_X\|_{H_{\text{sc}}^2(I^*)} \leq Q^{-B_0}.$$

Let

$$\mathcal{V}_I := \mathcal{A}_1 + \mathcal{I}^2 \mathcal{P}_I, \quad \mathcal{A}_1 := \text{span}\{1, t\},$$

and choose $q_X \in \mathcal{V}_I$ with $q_X'' = p_X$, with affine modes included or normalized. Define

$$\mathcal{W}_G := \mathcal{P}_I + \mathcal{V}_I + \text{Sym}^3 \mathcal{V}_I,$$

and

$$\mathcal{W}_{G2} := D^2 \mathcal{P}_I + \mathcal{P}_I + \text{Sym}^2 \mathcal{V}_I \cdot \mathcal{P}_I.$$

If

$$\|q_X\|_{C_{\text{sc}}^4(I^*)} \leq Q^{C_0},$$

then

$$G_X := \Gamma_G(q_X) \in \mathcal{W}_G, \quad G_{2,X} := \Gamma_{G,2}(q_X) \in \mathcal{W}_{G2},$$

and, after increasing B_0 ,

$$\|\Gamma_G(q) - G_X\|_{H_{\text{sc}}^2(I)} + \|\Gamma_{G,2}(q) - G_{2,X}\|_{L^2(I)} \leq Q^{-B}.$$

Proof sketch. Use

$$\Gamma_G(q) = -6q'' + 36q - 12q^3 = -6p + 36q - 12q^3,$$

and

$$\Gamma_{G,2}(q) = q^{(4)} - 6(1 - q^2)q'' = p'' - 6p + 6q^2p.$$

Since $q_X'' = p_X$, the claimed finite-space inclusions are immediate.

Set $r = q - q_X$. Then $r'' = p - p_X$, and affine reconstruction gives

$$\|r\|_{H_{\text{sc}}^4(I)} \leq Q^C \|p - p_X\|_{H_{\text{sc}}^2(I^*)}.$$

The residuals are

$$R_G = -6r'' + (36 - 36q_X^2)r - 36q_X r^2 - 12r^3,$$

and

$$R_{G2} = r^{(4)} - 6r'' + 6q_X^2 r'' + 12q_X r q_X'' + 12q_X r r'' + 6r^2 q_X'' + 6r^2 r''.$$

One-dimensional product estimates yield the stated $H_{\text{sc}}^2 \times L^2$ residual bound. $\square$

Remark 16.371 (Wall-variable completion contract). The completion needed for the plain rank-two wall is a wall-variable completion, not necessarily a primitive $H^4(q)$ -completion. It is enough to produce finite wall representatives

$$G_X \in \mathcal{W}_G, \quad G_{2,X} \in \mathcal{W}_{G2},$$

such that

$$\|\Gamma_G - G_X\|_{H_{\text{sc}}^2(I)} + \|\Gamma_{G,2} - G_{2,X}\|_{L^2(I)} \leq Q^{-B}.$$

The finite-atlas Riesz/Markov estimates are applied to $G_X, G_{2,X}$, and the residual is inserted only after the finite-atlas estimates are made.

A sufficient source-level route is to work with

$$p = q'',$$

to approximate p in $H_{\text{sc}}^2(I^*)$ by $p_X \in \mathcal{P}_I$, and to reconstruct

$$q_X \in \mathcal{V}_I := \mathcal{A}_1 + \mathcal{I}^2 \mathcal{P}_I$$

with affine modes included or normalized. Then, up to fixed harmless normalization constants,

$$\Gamma_G = -6p + 36q - 12q^3, \quad \Gamma_{G,2} = p'' - 6p + 6q^2p,$$

and the finite wall spaces are

$$\mathcal{W}_G = \mathcal{P}_I + \mathcal{V}_I + \text{Sym}^3 \mathcal{V}_I,$$

$$\mathcal{W}_{G2} = D^2\mathcal{P}_I + \mathcal{P}_I + \text{Sym}^2 \mathcal{V}_I \cdot \mathcal{P}_I.$$

This is the source-curvature price of the rank-two plain-wall route.

Remark 16.372 (Two-derivative tax for B_2 and background curvature). If the canonical gamma/background curvature term B_2 is included in

$$p = q'',$$

then the second rank-two wall component

$$\Gamma_{G,2} = p'' - 6p + 6q^2p$$

sees two scaled derivatives of B_2 . Hence the wall-variable completion ledger must include B_2 , DB_2 , and D^2B_2 , either exactly or to Q^{-B} -accuracy in the scaled norms used on I^* . A zeroth-order approximation to B_2 is not sufficient for the rank-two wall completion.

Theorem 16.373 (FiniteAtlasRiesz in $p = q''$ form). *Assume on each chart-good $I \in I^*$:*

(1) $\dim \mathcal{P}_I \leq R^C$ and $\mathcal{P}_I$ has a Q^C -Riesz basis in $L^2(I^*)$.

(2)

$$\|D^2p_X\|_{L^2(I)} \leq Q^C \|p_X\|_{L^2(I^*)}, \quad p_X \in \mathcal{P}_I.$$

(3) Affine reconstruction into $\mathcal{V}_I = \mathcal{A}_1 + \mathcal{I}^2\mathcal{P}_I$ is polynomially controlled.

(4) $\text{Sym}^3 \mathcal{V}_I$ and $\text{Sym}^2 \mathcal{V}_I \cdot \mathcal{P}_I$ have Q^C -conditioned quotient bases and bounded leverage.

(5) All relevant chart denominators are Q^{-C} -bounded below on I^* .

(6) For every predetermined $E \subset I$ with $|E| \geq R^{-C}|I|$, Remez holds for

$$\mathcal{W}_G = \mathcal{P}_I + \mathcal{V}_I + \text{Sym}^3 \mathcal{V}_I$$

and

$$\mathcal{W}_{G2} = D^2\mathcal{P}_I + \mathcal{P}_I + \text{Sym}^2 \mathcal{V}_I \cdot \mathcal{P}_I.$$

Then

$$\|w\|_{H_{\text{sc}}^2(I)} \leq Q^C \|w\|_{L^2(I^*)}, \quad w \in \mathcal{W}_G.$$

Moreover, after Lemma 16.370,

$$|E| \geq R^{-C}|I| \implies \int_E W_2 \geq Q^{-C} \int_I W_2 - Q^{-B},$$

where

$$W_2 := |\Gamma_G|^2 + |\Gamma_{G,2}|^2.$$

Definition 16.374 (Product-Riesz audit for the $p = q''$ wall atlas). Let $\{p_i\}$ be a basis of $\mathcal{P}_I$ and let $\{v_i\}$ be the corresponding reconstructed basis of

$$\mathcal{V}_I = \mathcal{A}_1 + \mathcal{I}^2\mathcal{P}_I.$$

The product-Riesz audit consists of the raw product spanning lists

$$v_i v_j v_k \quad \text{for} \quad \text{Sym}^3 \mathcal{V}_I,$$

and

$$v_i v_j p_k \quad \text{for} \quad \text{Sym}^2 \mathcal{V}_I \cdot \mathcal{P}_I.$$

For each list, form the $L^2(I^*)$ -Gram matrix, remove singular directions below Q^{-C} , and retain a quotient basis. The audit passes if the retained quotient basis is Q^C -Riesz and has Q^C -bounded leverage on I .

Remark 16.375 (FiniteAtlasRiesz failure modes). The theorem FiniteAtlasRiesz_{2Y}^p fails only through explicit finite-atlas mechanisms:

$$\boxed{D^2\mathcal{P}_I \text{ not controlled,}}$$

$$\boxed{\text{affine reconstruction unstable,}}$$

$$\boxed{\text{product-Riesz or product-leverage spike,}}$$

$$\boxed{\text{chart denominator singularity on } I^*,}$$

lack of R^{-C} -Remez on predetermined cores.

These are finite-model failures, not rank-one wall finite-type failures.

Proposition 16.376 (Source-divisibility of the established fixed-sector tower). *The established fixed-sector escape tower generated by the channels*

$$X_1 Y X_{2k} - X_{2k} Y X_1, \quad Y = yJ, \quad X_1 = BS,$$

is divisible by the common source factor $H = yq''_{\text{can}}$.

More precisely, write

$$X_{2k} = a_k I + d_k D + \cdots.$$

Then

$$X_1 Y X_{2k} - X_{2k} Y X_1 = yB(SJX_{2k} - X_{2k}JS).$$

Using

$$SJ = -D, \quad JS = D,$$

one obtains

$$X_1 Y X_{2k} - X_{2k} Y X_1 = -yB(DX_{2k} + X_{2k}D).$$

The I -component of $DX_{2k} + X_{2k}D$ is $2d_k I$. Hence

$$\pi_{\dagger}(X_1 Y X_{2k} - X_{2k} Y X_1) = -2yB d_k I.$$

Since

$$B \propto q''_{\text{can}},$$

every row in this tower contains

$$yB \sim yq''_{\text{can}} = H.$$

Thus increasing the anisotropy order replaces

$$\Gamma_G, \Gamma_{G,2}, \dots$$

by higher anisotropy factors, but it does not remove the common source factor H .

Remark 16.377 (Why the anisotropy tower does not fix source collapse). The order-seven row detects first Gram anisotropy:

$$A_7^{\dagger} \sim H\Gamma_G.$$

The order-nine row detects the second anisotropy on the first wall:

$$A_9^{\dagger} \sim H\Gamma_{G,2}.$$

A higher row in the same tower would have the schematic form

$$A_{2k+7}^{\dagger} \sim H\Gamma_{G,k+1}.$$

This improves the local layered visibility against the walls

$$\Gamma_G = 0, \quad \Gamma_{G,2} = 0, \quad \dots,$$

but it leaves the source-collapse wall

$$H = yq''_{\text{can}} = 0$$

untouched. Therefore the rank-two norm route requires either a source-robust row not divisible by H , or an external quantitative theorem preventing source collapse.

Proposition 16.378 (Source-collapse countermodel for the rank-two norm route). *The source-weighted lower bound*

$$\sup_{m \in I_\rho} |H(m)| \max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \geq Q^{-A}$$

is not a consequence of the current local rank-two structure.

Indeed, consider the local jet target

$$q_0 = 2, \quad q''_{\text{can}} = 0, \quad (q_\zeta)^{(4)} = 0.$$

Then

$$g = \frac{q_0}{\pi} = \frac{2}{\pi} > 0,$$

and

$$h = \frac{1}{\pi} \left(\frac{1}{3} q_0^3 + \frac{1}{6} q_{\text{can}}'' \right) = \frac{8}{3\pi} > 0.$$

Moreover

$$\Gamma_G = q_{\text{can}}'' - 6q_0 + 2q_0^3 = 0 - 12 + 16 = 4,$$

while

$$\Gamma_{G,2} = (1 - q_0^2) q_{\text{can}}'' - \frac{1}{6} (q_\zeta)^{(4)} = 0.$$

Thus the layered anisotropy vector is live:

$$\max\{|\Gamma_G|, |\Gamma_{G,2}|\} = 4.$$

However,

$$H = y q_{\text{can}}'' \sim -\frac{q_0 (q_{\text{can}}'')^3}{288} = 0.$$

Perturbing the target to

$$\sup_{m \in I_\rho} |q_{\text{can}}''(m)| \leq Q^{-C}$$

while keeping

$$q_0(m) \asymp 1, \quad g(m), h(m) > 0, \quad \max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \asymp 1$$

gives

$$\sup_{m \in I_\rho} |H(m)| \max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \ll Q^{-3C}.$$

Choosing $C > A/3$ defeats the desired Q^{-A} lower bound.

Remark 16.379 (Interpretation of the source-collapse countermodel). Proposition 16.378 does not assert that actual zeta realizes this jet configuration. It says that the present local rank-two row algebra, Gram positivity, and layered anisotropy structure do not prevent it. A proof must therefore supply a new input such as

$$\sup_{m \in I_\rho} |q''(m)| \geq Q^{-B},$$

or a stronger source-jet / interval nonflatness theorem for actual zeta data.

Definition 16.380 (D4 coefficient-good core). For each routed block I , choose before wall extraction a core

$$E_I^{\text{core}} \subset I$$

depending only on

$$H, c_7, \alpha_9, \beta_9^{\text{tri}}, N_{\text{sm}}, \quad \text{and chart data,}$$

not on

$$\Gamma_G, \Gamma_{G,2}, W_2, \text{ hard atoms, or live residuals.}$$

The core is admissible if

$$|E_I^{\text{core}}| \geq R^{-C} |I|,$$

and on E_I^{core} ,

$$|H| \geq Q^{-A},$$

$$|c_7|, |c_7^{-1}|, |\alpha_9|, |\alpha_9^{-1}|, |\beta_9^{\text{tri}}| \leq Q^A,$$

and

$$N_{\text{sm}} \geq Q^{-A}.$$

Theorem 16.381 (D4 coefficient-core routing). *For every selected block I , suppose there is a predetermined core*

$$E_I^{\text{core}} \subset I, \quad |E_I^{\text{core}}| \geq R^{-C}|I|,$$

chosen before wall extraction. The core may depend on

$$H, \quad c_7, \quad \alpha_9, \quad \beta_9^{\text{tri}}, \quad N_{\text{sm}}, \quad \text{and chart data,}$$

but not on

$$\Gamma_G, \quad \Gamma_{G,2}, \quad W_2 := |\Gamma_G|^2 + |\Gamma_{G,2}|^2,$$

hard extracted atoms, or live residuals. Assume on E_I^{core} that

$$|H| \geq Q^{-A},$$

$$|c_7|, \quad |c_7^{-1}|, \quad |\alpha_9|, \quad |\alpha_9^{-1}|, \quad |\beta_9^{\text{tri}}| \leq Q^A,$$

and

$$N_{\text{sm}} \geq Q^{-A}.$$

Then on the core

$$\left\| H \left(\beta_9^{\text{tri}} \Gamma_G + \alpha_9 \Gamma_{G,2} \right) \right\|^2 \asymp_{Q^A} |H|^2 (|\Gamma_G|^2 + |\Gamma_{G,2}|^2).$$

After multiplying by N_{sm} , the same core supplies the weighted wall

$$N_{\text{sm}} |H|^2 (|\Gamma_G|^2 + |\Gamma_{G,2}|^2).$$

Thus coefficient-good routing is a core-selection statement, not a post-hoc selection of a wall-large set.

Lemma 16.382 (Core weighted plain-wall transfer). *Assume admissible cores and vector Remez:*

$$|E| \geq R^{-C}|I| \implies \int_E W_2 d\nu \geq Q^{-C} \int_I W_2 d\nu.$$

Then

$$\int_I N_{\text{sm}} |H|^2 W_2 d\nu \geq Q^{-A'} \int_I W_2 d\nu.$$

Remark 16.383 (Non-circularity of coefficient-good cores). The core

$$E_I^{\text{core}}$$

must be selected before wall extraction. It may depend on

$$H, \quad c_7, \quad \alpha_9, \quad \beta_9^{\text{tri}}, \quad N_{\text{sm}}, \quad \text{and chart data,}$$

but not on

$$\Gamma_G, \quad \Gamma_{G,2}, \quad W_2 = |\Gamma_G|^2 + |\Gamma_{G,2}|^2,$$

nor on hard-extracted atoms or live residuals. This is the transfer analogue of the local no-atom-fitting rule. If E_I^{core} is chosen after seeing W_2 , the weighted transfer becomes circular.

Remark 16.384 (Symbolic rank-two O2/O3 degree). The scalar O2/O3 input is degree-parametric. After scalar component selection, the weight is

$$W_\star = N_{\text{sm}} |H|^2 |G_\star|^2, \quad G_\star \in \{\Gamma_G, \Gamma_{G,2}\}.$$

The required branch-charge bandwidth is

$$d_{\text{max}}^{(2)} = d_N + 2d_H + 2 \max(d_{\Gamma_G}, d_{\Gamma_{G,2}}).$$

One should not hard-code $d_{\text{max}} = 14$ unless this degree audit has been performed in the active convention. If H is a smooth coefficient or normalization denominator absorbed into the measure, then $d_H = 0$. If H is branch-bearing, its fixed branch degree must be counted. If H is locally adjustable, the present D4 normalization is not valid without renormalization or an alternate chart.

Lemma 16.385 (Scalar component selection). *If*

$$\int N_{\text{sm}} |H|^2 (|\Gamma_G|^2 + |\Gamma_{G,2}|^2) d\nu \geq Q^{-A},$$

then there exists a fixed component

$$G_\star \in \{\Gamma_G, \Gamma_{G,2}\}$$

chosen at the smooth aggregate stage such that

$$\int N_{\text{sm}} |H|^2 |G_\star|^2 d\nu \geq Q^{-A'}.$$

Definition 16.386 (Rank-two wall degree). For

$$G_\star \in \{\Gamma_G, \Gamma_{G,2}\},$$

define

$$d_0(G_\star) := d_N + 2d_H + 2d_{G_\star}.$$

Set

$$d_{\max}^{(2)} := d_N + 2d_H + 2 \max(d_{\Gamma_G}, d_{\Gamma_{G,2}}).$$

Conservatively,

$$d_{\Gamma_G}, d_{\Gamma_{G,2}} \leq 3d_q,$$

so

$$d_{\max}^{(2)} \leq d_N + 2d_H + 6d_q.$$

If H is smooth-renormalizable and carries no branch charge, set $d_H = 0$.

Theorem 16.387 (Scalar O2/O3 interface for rank two). *Assume weighted rank-two plain wall mass has been transferred, and assume*

$$\mathbb{C}_{d_{\max}^{(2)}}[y] \subset V_{\text{tr}}.$$

Assume the scalar O2/O3 near-pure separation holds uniformly for every fixed $d \leq d_{\max}^{(2)}$:

$$\text{dist}_{L^2(\mu_{\text{sub}})} \left(1, \text{span}\{R_{\text{fold}}, R_{\text{fold}}^2, \dots, R_{\text{fold}}^d\} \right)^2 \geq Q^{-A(d_{\max}^{(2)})}.$$

Then the selected scalar wall

$$W_\star = N_{\text{sm}} |H|^2 |G_\star|^2$$

is a valid bounded-charge O2/O3 input.

Remark 16.388 (Rank-two no-go and guardrail register). The following rules are active guardrails for the rank-two branch.

- (1) D4 supplies only the plain wall

$$H(\Gamma_G, \Gamma_{G,2}).$$

The derivative family

$$\{\Gamma_G, D\Gamma_G, D^2\Gamma_G, \Gamma_{G,2}\}$$

is analytic auxiliary data only. It is not a D4/O2/O3 wall unless a separate DerivativeD4Channels_{2Y} or block-difference derivative-channel theorem is proved.

- (2) All branch classification is made in row-normal coordinates:

$$\ell_9^Y = \omega_Y - \omega_Y(e_7)\ell_7, \quad \beta_9^{\text{tri}} = \beta_9 - \omega_Y(e_7)c_7.$$

Any rank-one fallback must use

$$\rho_\Gamma^{\text{tri}} = \frac{\beta_9^{\text{tri}} - \rho_Y c_7}{\alpha_9}.$$

- (3) The rank-two selector is the quotient rank

$$r_Y = \text{rank} \left(\Pi_{\mathcal{C}^2/B_Y} M_{D4}^{\text{tri}} \right),$$

not the raw rank of formal D4 carriers. Therefore formal $C_7 \rightarrow Y12$ connections, fixed basepoint offsets, and coherent amplitude remnants do not reduce r_Y unless they produce actual/local columns in B_Y .

- (4) Formal C_7 carriers and formal $C_7 \rightarrow Y12$ connections do not contribute to B_Y unless actualized by a genuine actual-source, exact nuisance, or local fitting direction.
- (5) Exact nuisance must be row-dead and live-pair preserving before it is placed in the denominator. Coherent nuisance and residual nuisance are routed to $\mathcal{C}_{2Y}^{\text{arith}}$ and R_X^{2Y} , respectively.
- (6) E_2 and R_{lab} affect B_Y only through atom-local live-fiber fitting. Fixed, coherent, and residual terms do not define local columns of B_Y .
- (7) Amplitude projection occurs only after $B_Y = 0$ is proved. A projected wall is not fed back into the local quotient, the row definitions, the live pair, or the D4 chart unless its branch complexity has been separately controlled.
- (8) The weighted-transfer core E_I^{core} must be selected from coefficient, smooth, and chart data before observing

$$W_2 = |\Gamma_G|^2 + |\Gamma_{G,2}|^2.$$

Post-hoc wall-dependent core selection is not allowed.

- (9) The scalar O2/O3 theorem is degree-parametric. Do not hard-code a specific value such as 14 until the rank-two wall degree audit has fixed

$$d_{\max}^{(2)} = d_N + 2d_H + 2 \max(d_{\Gamma_G}, d_{\Gamma_{G,2}}).$$

- (10) An α -twisted determinant average is not proof-critical unless AlphaAdmissibility⁺ is proved. In particular, one needs regular zero persistence, badness persistence, twist nondegeneracy, Stack uniformity, and α -D4 compatibility:

$$[\mathcal{C}_{Y12}] \neq 0 \quad \text{in} \quad \mathcal{Q}_9^{\text{forb}, \alpha}.$$

- (11) A pure branch determinant is not a scalar O2/O3 theorem unless it survives inside the full determinant of the canonical matrix:

$$\int |D_{\text{full}}|^2 d\nu \geq Q^{-A} \int |D_{\text{dom}}|^2 d\nu.$$

The Stack measure is

$$d\nu = F_1 F_{\text{wall}}^{\text{forb}} N d\rho,$$

so neutral-measure branch orthogonality is not enough.

- (12) A forbidden Λ must be non-functorial on at least one D4 live carrier. If every natural carrier lift is functorial through order 9, then

$$\Lambda[\mathcal{C}_7] = \Lambda[\mathcal{C}_{Y12}] = 0,$$

and finite D4 has no contradiction channel.

- (13) Carrier-quotient D4 is a visibility theorem, not a contradiction theorem, unless a forbidden channel Λ is supplied. In the active canonical source-graded branch this channel is $\Lambda_p^{\text{gr}, \text{full}}$ and its maximal affine projection, so the old rank audit

$$\text{rank}(\Lambda E_{\text{car}} L_{\text{clean}}) \geq 1$$

is retained only as a noncanonical diagnostic.

Remark 16.389 (Interval-local source collapse). The pointwise source-collapse target becomes an interval-local obstruction because the proof requires a supremum over an admissible packet interval

$$I_\rho = [m_0 - \Delta, m_0 + \Delta].$$

The scale audit gives the natural estimates

$$\Omega_{\text{eff}}(q'') \sim Q, \quad \Delta \sim Q^{-1}, \quad \Omega_{\text{eff}}(q'')\Delta \sim 1.$$

Thus q'' has only $O(1)$ effective oscillatory complexity on the natural microscopic packet interval.

Let

$$z(m) := q''(m).$$

To force

$$\sup_{|m-m_0| \leq c/Q} |z(m)| \leq Q^{-C},$$

it is enough to flatten finitely many derivatives:

$$z^{(r)}(m_0) \approx 0, \quad 0 \leq r \leq R.$$

A Taylor–Bernstein estimate gives, schematically,

$$\sup_{|u| \leq c/Q} |z(m_0 + u)| \lesssim Q^{M_0} \frac{C_c^{R+1}}{(R+1)!}.$$

Hence a safe choice is

$$R \asymp (C + M_0) \frac{\log Q}{\log \log Q},$$

or more conservatively $R = O(\log Q)$. This order is far below the recurrence-danger threshold $Q/\log Q$. Therefore the interval-local source-collapse countermodel is not defeated by interval length or bandwidth alone.

Remark 16.390 (Fixed-prime and AFE derivative-flat surrogates). A fixed-prime surrogate for the source-collapse mechanism is

$$L_P(m) = \sum_{p \leq P} (\log p) p^{-1/2} e^{-im \log p}.$$

Define

$$\begin{aligned} J_0(m) &= B_0 + \Re L_P(m), \\ J_2(m) &= B_2 + \Re L_P''(m), \\ J_4(m) &= B_4 + \Re L_P^{(4)}(m). \end{aligned}$$

The model residuals are

$$\begin{aligned} R_1 &= J_2 - 6J_0 + 2J_0^3, \\ R_2 &= (1 - J_0^2)J_2 - \frac{1}{6}J_4, \end{aligned}$$

and

$$H_{\text{model}} = -\frac{J_0 J_2^3}{288}.$$

The interval-local source-collapse target is

$$J_2^{(r)}(m_0) \approx 0, \quad 0 \leq r \leq R,$$

with side constraints such as

$$J_0(m_0) \approx 2, \quad J_4(m_0) \approx 0.$$

At the exact target

$$(J_0, J_2, J_4) = (2, 0, 0)$$

one has

$$R_1 = 4, \quad R_2 = 0,$$

and

$$g = \frac{2}{\pi} > 0, \quad h = \frac{8}{3\pi} > 0.$$

Thus the source is collapsed while anisotropy and Gram positivity remain live.

The phase-relaxed finite-prime problem has no rank obstruction: the rows

$$J_2^{(r)}(m_0)$$

are governed by powers of the frequencies

$$\lambda_j = \log p_j \quad \text{or} \quad \lambda_j = k \log p,$$

so the derivative-jet matrix has a Vandermonde-type structure. The remaining arithmetic issue is whether the actual one-parameter prime-log flow

$$m_0 \mapsto (m_0 \log p_j)_j \pmod{2\pi}$$

recurs to such a derivative-flat target. The expected codimension is approximately

$$K = R + 3,$$

so a heuristic recurrence cost at tolerance $\varepsilon = Q^{-C}$ is

$$T_{\text{hit}} \sim Q^{CK}.$$

For $R = O(\log Q)$, this is

$$T_{\text{hit}} = \exp(O((\log Q)^2)) \ll e^Q.$$

Thus scale considerations do not presently block the fixed-prime source-collapse surrogate.

The phase-relaxed derivative-flat target can be sharpened further. In the fixed-amplitude prime model

$$L_P(m) = \sum_{p \leq P} a_p e^{-im \log p}, \quad a_p = \frac{\log p}{\sqrt{p}},$$

one obtains

$$(J_0, J_2, J_4) = \sum_{p \leq P} x_p a_p (1, -(\log p)^2, (\log p)^4), \quad x_p = \cos(m \log p).$$

Thus the phase-relaxed attainable jet values form a fixed-amplitude prime zonotope. The source-collapse target

$$(J_0, J_2, J_4) = (2, 0, 0)$$

is compatible with good-patch positivity:

$$g = \frac{2}{\pi} > 0, \quad h = \frac{8}{3\pi} > 0,$$

and has live anisotropy

$$R_1 = 4, \quad R_2 = 0.$$

Finite-dimensional feasibility tests for the moment system

$$\sum_{p \leq P} a_p x_p = 2, \quad \sum_{p \leq P} a_p (\log p)^{2k} x_p = 0$$

show interior solutions for low and moderate finite orders. For example:

moments imposed	P	$\#\{p \leq P\}$	$\min \max_{p \leq P} x_p $
0, 2, 4	100	25	0.699
0, 2, 4, 6	300	62	0.679
0, 2, 4, 6, 8	1000	168	0.637
0, 2, 4, 6, 8, 10	1000	168	0.888
0, 2, 4, 6, 8, 10, 12	5000	669	0.718.

So the target lies in the interior of the fixed-prime phase-relaxed cosine zonotope, not on a boundary. The remaining issue is therefore not amplitude feasibility. It is the one-parameter recurrence problem

$$m_0 \log p \equiv \theta_p \pmod{2\pi}$$

for selected phases θ_p .

AFE-compatible models add prime powers, smoothing, a dual sum, and gamma background terms. These shift or couple the target, but no algebraic obstruction is presently known that forces

$$\max_{0 \leq r \leq R} |J_2^{(r)}(m_0)| \geq Q^{-B}.$$

Therefore the exact missing Gate 1 theorem is actual-zeta source-curvature observability, and it must later be paired with Gate 2 source–anisotropy reintegration. It is not another local row identity.

Remark 16.391 (Prime-log recurrence to derivative-flat tubes). Let

$$\lambda = (\log p_1, \dots, \log p_N), \quad \Theta(m) = m\lambda \pmod{2\pi}.$$

Let $\mathcal{T}_b(\varepsilon) \subset \mathbb{T}^N$ denote the ε -tube around the derivative-flat target

$$|J_2^{(r)}| \leq \varepsilon, \quad 0 \leq r \leq R,$$

with side constraints

$$|J_0 - 2| \leq \varepsilon, \quad |J_4| \leq \varepsilon.$$

If this target is regular, then

$$\mu(\mathcal{T}_b(\varepsilon)) \asymp \varepsilon^K, \quad K \sim R + 3.$$

This is a positive-measure analytic tube, not a prescribed point in $\mathbb{T}^N$.

For polynomial tolerance

$$\varepsilon = Q^{-C},$$

the random/equidistribution hitting scale is

$$T_{\text{hit}} \sim \varepsilon^{-K} = Q^{CK}.$$

For the derivative orders relevant to Taylor–Bernstein source flattening,

$$R \asymp (C + M_0) \frac{\log Q}{\log \log Q}$$

or even $R = O(\log Q)$, this gives

$$T_{\text{hit}} = \exp\left(O\left(\frac{(\log Q)^2}{\log \log Q}\right)\right) \quad \text{or} \quad T_{\text{hit}} = \exp(O((\log Q)^2)).$$

Both are far below the available height scale e^Q .

Thus recurrence cost does not presently block the interval-local source-collapse surrogate. A proof of the desired source-weighted lower bound would need a nonstandard quantitative nonrecurrence theorem for the prime-log flow against these derivative-flat tubes, or actual-zeta Gate 1 source-curvature observability together with Gate 2 source–anisotropy reintegration.

Remark 16.392 (Current Gate–1 attack ledger: finite-jet and Wronskian forms). The current Gate–1 target is

$$\int_{I_\rho} |q_{\text{can}}''(m)|^2 dm \geq Q^{-B} |I_\rho|.$$

In normalized coordinates

$$m = m_0 + \lambda x, \quad F_\lambda(x) = \lambda^3 q_{\text{can}}''(m_0 + \lambda x),$$

it is enough, subject to the finite-jet-to-energy transfer, to prove a finite-jet anti-design estimate of the form

$$\max_{0 \leq r \leq R(Q)} |\partial_x^r F_\lambda(x_0)| \geq Q^{-B_{\text{jet}}} \lambda^3.$$

The natural depth is

$$R(Q) \asymp \frac{\log Q}{\log \log Q},$$

with $R = O(\log Q)$ as a safer but more expensive depth.

A promising formulation is a normalized Wronskian/transversality package. Let

$$G_0 = Z_\alpha, \quad G_{j+1} = \Delta^j \partial_m^j q_{\text{can}}'', \quad 0 \leq j \leq R.$$

The associated normalized Wronskian is

$$\widetilde{W}_R(m, \alpha) = \det(\mathcal{D}^\ell G_j(m, \alpha))_{0 \leq j, \ell \leq R+1}, \quad \mathcal{D} = \Delta \partial_m.$$

A polynomial lower bound

$$|\widetilde{\mathcal{W}}_R(m, \alpha)| \geq Q^{-A}$$

at admissible centers would imply finite-jet anti-design and hence Gate 1.

At present this route is not Green. The Z_α -symbol appears to add genuine rank to the q''_{can} -jet system, but the Wronskian expands into a fixed-coefficient subset-product exponential sum. Cauchy–Binet nonvanishing proves only that the determinant is not identically zero; it does not give the required pointwise lower bound on the actual prime-log orbit. Thus the current missing theorem is a fixed actual-zeta determinant or nonrecurrence lower bound, not another formal rank calculation.

Remark 16.393 (Z_α -symbol transversality subproblem). The first finite-dimensional test for the Wronskian route is whether the off-line equation contributes a row independent from the q''_{can} -jet rows. In a simplified AFE block, write

$$Z_\alpha(m) \sim \sum_n z_n(\alpha) e^{-im \log n},$$

and

$$\partial_m^r q''_{\text{can}}(m) \sim \sum_n c_n (-i \log n)^r e^{-im \log n}.$$

On an active block, the relevant symbolic test is whether

$$\frac{z_n(\alpha)}{c_n}$$

is approximable by a polynomial in $\log n$ of degree at most R . The expected model is

$$\frac{z_n(\alpha)}{c_n} \approx \frac{n^{-\alpha}}{(\log n)^3} \frac{V_\alpha(n/N)}{W_2(n/N)},$$

or, with $u = \log n$,

$$f_\alpha(u) \approx e^{-\alpha u} u^{-3} A_\alpha(u).$$

This is not polynomial-like on a thick active block unless the weights are degenerate. Thus Z_α likely adds genuine symbolic rank. The missing step is quantitative: one needs a lower bound

$$\inf_{P \in \mathcal{P}_{\leq R}} \max_{n \in \mathcal{B}} \left| \frac{z_n(\alpha)}{c_n} - P(\log n) \right| \geq Q^{-C}$$

for the actual AFE coefficients and active block. Symbolic transversality alone still does not imply the required pointwise Wronskian lower bound.

Remark 16.394 (Derivative-order and error budget for the Wronskian route). The normalized Wronskian route is sensitive to derivative order. Since

$$G_{j+1} = \Delta^j \partial_m^j q''_{\text{can}},$$

and the Wronskian differentiates G_j up to order $R+1$, the argument requires derivatives of q''_{can} through order approximately $2R+1$, equivalently derivatives of q_ζ through order $2R+3$. The Z_α -row requires derivatives through order $R+1$.

Therefore the canonical error term in

$$q''_{\text{can}} = B_2 + P_2 + D_2 + E_2$$

must be controlled through the same derivative order. The safe choice

$$R \asymp \frac{\log Q}{\log \log Q}$$

keeps the normalized factorial and AFE derivative losses polynomial. Taking $R = O(\log Q)$ is dangerous unless a separate lemma removes the resulting factorial and differentiated-error losses.

Remark 16.395 (Zero-side same-scale source flattening). The source-collapse mechanism is not merely a prime-side artefact. On the zero side, one can seek same-scale positive packet configurations for which the aggregate source curvature is flat to high finite order.

For a centered Cauchy packet

$$k_c(t) = \frac{c}{c^2 + 4(t - m)^2},$$

the centered jets are

$$\begin{aligned} k_c(m) &= c^{-1}, \\ k_c''(m) &= -8c^{-3}, \end{aligned}$$

and

$$k_c^{(4)}(m) = 384c^{-5}.$$

A single centered packet cannot realize

$$q'' = q^{(4)} = 0$$

with $q_0 > 0$. However, once off-center same-scale packets and positive linear combinations are allowed, the problem becomes a finite moment-matching problem.

In particular, one can use same-scale packet families to impose finitely many conditions of the form

$$q^{(2)}(m_0) = q^{(3)}(m_0) = \dots = q^{(R+2)}(m_0) = 0$$

while preserving

$$q_0(m_0) > 0.$$

At the target

$$q_0 = 2, \quad q'' = 0, \quad q^{(4)} = 0,$$

Gram positivity and live anisotropy remain:

$$g = \frac{2}{\pi} > 0, \quad h = \frac{8}{3\pi} > 0, \quad \Gamma_G = 4.$$

Thus same-scale packet geometry alone does not provide Gate 1 source-curvature observability, and a fortiori does not provide the full two-gate package. A proof must use information not already contained in finite same-scale positive packet geometry.

Remark 16.396 (Local-kernel notation in the same-scale model). In the same-scale packet model below, the unadorned $q, q'', q^{(4)}$ denote the local positive source-kernel aggregate being constructed, not the full actual-zeta canonical curvature q''_{can} . The model is used only as a local/Hadamard-side no-go: it shows that finite same-scale positive packet geometry alone does not force nonflatness of the canonical actual-zeta curvature.

Theorem 16.397 (Continuous same-scale source-flattening model). *Fix $d > 0$. For*

$$c = d \sin \theta, \quad x = d \cos \theta, \quad 0 < \theta < \pi,$$

consider the same-distance packet kernel

$$K_{\theta,d}(u) = \frac{c}{c^2 + (x + 2u)^2}.$$

Let

$$d\mu(\theta) = \frac{2}{\pi} \sin \theta \, d\theta.$$

Then, for $|2u/d| < 1$,

$$\boxed{\int_0^\pi K_{\theta,d}(u) \, d\mu(\theta) = \frac{1}{d}.}$$

Consequently,

$$q_\mu(u) := \int_0^\pi K_{\theta,d}(u) \, d\mu(\theta)$$

satisfies

$$q_\mu(0) = \frac{1}{d} > 0, \quad q_\mu^{(r)}(0) = 0 \quad (r \geq 1).$$

In particular,

$$q_\mu^{(2)}(0) = q_\mu^{(4)}(0) = q_\mu^{(6)}(0) = \cdots = 0.$$

Proof. Put

$$t = \frac{2u}{d}.$$

Then

$$K_{\theta,d}(u) = \frac{1}{d} \frac{\sin \theta}{1 + 2t \cos \theta + t^2}.$$

Hence

$$\int_0^\pi K_{\theta,d}(u) d\mu(\theta) = \frac{1}{d} \int_0^\pi \frac{2}{\pi} \frac{\sin^2 \theta}{1 + 2t \cos \theta + t^2} d\theta.$$

For $|t| < 1$, the elementary Poisson-kernel identity

$$\int_0^\pi \frac{2}{\pi} \frac{\sin^2 \theta}{1 + 2t \cos \theta + t^2} d\theta = 1$$

gives the claim. Therefore $q_\mu(u) = 1/d$ on a neighborhood of the origin, and all positive-order derivatives at the origin vanish. $\square$

Corollary 16.398 (Finite same-scale source flattening to arbitrary finite jet order). *For every $M \geq 1$ and every $\epsilon > 0$, there exist finitely many angles*

$$0 < \theta_1, \dots, \theta_N < \pi$$

and positive weights $w_i > 0$ such that, with

$$c_i = d \sin \theta_i, \quad x_i = d \cos \theta_i,$$

one has

$$x_i^2 + c_i^2 = d^2$$

for every i , and

$$\left| \sum_i w_i K_{\theta_i,d}^{(r)}(0) \right| \leq \epsilon, \quad 1 \leq r \leq M,$$

while

$$\sum_i w_i K_{\theta_i,d}(0) > 0.$$

Equivalently, finite positive same-distance packet clusters can make

$$q^{(r)}(0), \quad 1 \leq r \leq M,$$

arbitrarily small while retaining

$$q_0 = q(0) > 0.$$

In particular, taking $M \geq R + 2$ gives the finite source-flatness target

$$(q'')^{(r)}(0) \approx 0, \quad 0 \leq r \leq R.$$

Proof. The continuous measure μ in Theorem 16.397 satisfies

$$q_\mu(u) = \int K_{\theta,d}(u) d\mu(\theta) = \frac{1}{d}$$

near $u = 0$. Hence all derivatives $q_\mu^{(r)}(0)$, $r \geq 1$, vanish.

Apply Tchakaloff's theorem, or finite-dimensional Carathéodory reduction, to the positive measure μ and the finite list of continuous functions

$$K_{\theta,d}(0), \quad K_{\theta,d}^{(1)}(0), \quad K_{\theta,d}^{(2)}(0), \dots, K_{\theta,d}^{(M)}(0).$$

This gives a finite positive quadrature rule reproducing the required finite moment vector on the closed support. If an atom occurs at $\theta = 0$ or $\theta = \pi$, replace μ by its restriction to

$$[\varepsilon, \pi - \varepsilon]$$

and renormalize. Since

$$d\mu(\theta) = \frac{2}{\pi} \sin \theta d\theta$$

has vanishing endpoint mass, the corresponding finite moment vector converges to the original one as $\varepsilon \downarrow 0$. Applying finite quadrature on $[\varepsilon, \pi - \varepsilon]$ and choosing ε sufficiently small gives interior atoms $0 < \theta_i < \pi$ and the stated finite-order approximation. This approximation is the form needed below for interval source-collapse, where the target tolerance is polynomial in Q^{-1} . $\square$

Remark 16.399 (Uniform same-scale caveat). The construction above is same-scale in the strong fixed-distance sense $x_i^2 + c_i^2 = d^2$. If a later argument requires the additional uniform angular condition

$$\epsilon \leq \frac{c_i}{d} \leq 1 - \epsilon$$

with $\epsilon > 0$ independent of M , then one still needs a strengthened annular positive-cone lemma. The present result is already enough to show that finite source-jet flattening is not excluded by positive same-distance packet geometry. Thus any proof-closing lower bound for q'' must use additional actual-zeta arithmetic, not merely same-scale positivity.

Remark 16.400 (Source-kernel versus actual-zeta packet caveat). The same-scale flattening construction is a statement about positive local Cauchy source kernels at a fixed distance scale. It should not be read as a claim that actual zeta zeros can realize the same quadrature configuration. Actual zeta imposes functional-equation symmetry, conjugation, discreteness, zero-counting constraints, and possibly Euler-product arithmetic not present in the continuous source-kernel model.

The point of Theorem 16.397 and Corollary 16.398 is therefore negative but precise: local same-scale positivity does not by itself force source-curvature nonflatness. Any proof of

$$\sup_{m \in I_\rho} |q''_{\text{can}}(m)| \geq Q^{-B}$$

or of the stronger interval estimate

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 dm \geq Q^{-B} |I_\rho|$$

must use actual-zeta arithmetic or a stronger packet-realizability theorem, not merely same-scale positive kernel geometry.

Corollary 16.401 (Interval-local source collapse from finite quadrature). *All unadorned $q, q'', q^{(4)}$ in this corollary denote the local same-scale source-kernel aggregate from Remark 16.396, not the actual-zeta canonical curvature. In the actual-zeta notation, this is a model for the hypothetical regime $q''_{\text{can}} \approx 0$, but it is not a construction of actual zeta data.*

Fix $A > 0$. Assume the packet scale is microscopic,

$$d \asymp Q^{-1},$$

and let

$$I = [m_0 - c/Q, m_0 + c/Q]$$

with $c > 0$ fixed sufficiently small. For every $C > A/3$, a finite same-distance positive packet cluster can be chosen so that

$$\sup_{m \in I} |q''(m)| \leq Q^{-C},$$

while

$$q_0(m) \asymp 1, \quad g(m) > 0, \quad h(m) > 0,$$

and

$$\max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \asymp 1$$

on I , after a harmless normalization of the total weight.

Consequently, at leading source order,

$$H_{\text{loc}} = yq'' \sim -\frac{q_0(q'')^3}{288}$$

satisfies

$$\sup_{m \in I} |H_{\text{loc}}(m)| \max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \ll Q^{-3C},$$

and therefore can be made smaller than Q^{-A} .

Proof. By Corollary 16.398, choose the packet cluster so that, for a prescribed polynomial tolerance,

$$q^{(r)}(m_0) = O(Q^{-C'}) \quad 1 \leq r \leq M,$$

with M large enough relative to the desired interval bound. In particular,

$$(q'')^{(r)}(m_0) = O(Q^{-C'}) \quad 0 \leq r \leq R$$

for the required finite order R .

The standard Taylor–Bernstein estimate at scale $d \asymp Q^{-1}$ then gives

$$\sup_I |q''| \leq Q^{-C}$$

after taking C' and M sufficiently large. For example, the usual Taylor bookkeeping permits

$$M \asymp \frac{C \log Q}{\log \log Q}$$

or the safer choice $M = O(\log Q)$, with constants depending on the fixed interval parameter and same-scale annulus. Normalizing the total mass so that q_0 is close to 2, one obtains

$$g = \frac{q_0}{\pi} > 0, \quad h = \frac{1}{\pi} \left(\frac{1}{3} q_0^3 + \frac{1}{6} q'' \right) > 0,$$

and

$$\Gamma_G = q'' - 6q_0 + 2q_0^3 = 4 + O(Q^{-C}).$$

Hence the anisotropy factor remains bounded below while the source factor collapses cubically in q'' .

This is a finite-order local obstruction model. It shows that source-collapse is compatible with positive same-scale packet geometry to every prescribed polynomial accuracy. It is not a claim that actual zeta zeros realize such a quadrature configuration. $\square$

Proposition 16.402 (No-go for the current rank-two norm route). *The current rank-two norm route cannot close RH from the established local row theory and same-scale positive packet geometry alone.*

Here the displayed q'' and $H = yq''$ first refer to the local same-scale source-kernel model of Corollary 16.401. The proposition's conclusion for the actual rank-two route is obtained by comparison: if actual zeta permits the analogous canonical regime

$$\sup_{m \in I_\rho} |q''_{\text{can}}(m)| \leq Q^{-C},$$

then the actual source factor

$$H = yq''_{\text{can}}$$

collapses in the same way.

More precisely, the following data are compatible with the current local rank-two structure:

$$q_0 \asymp 1, \quad g, h > 0, \quad \max\{|\Gamma_G|, |\Gamma_{G,2}|\} \asymp 1,$$

and

$$\sup_{m \in I_\rho} |q''(m)| \leq Q^{-C}.$$

For such data,

$$H = yq'' \sim -\frac{q_0(q'')^3}{288}$$

satisfies

$$\sup_{m \in I_\rho} |H(m)| \ll Q^{-3C}.$$

Consequently,

$$\sup_{m \in I_\rho} |H(m)| \max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \ll Q^{-3C}.$$

Taking $C > A/3$ violates the desired Q^{-A} source-weighted lower bound.

Thus the local rank-two norm route requires the actual-zeta Gate 1 theorem excluding canonical source-flat intervals,

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 dm \geq Q^{-B}|I_\rho|,$$

and then the Gate 2 source–anisotropy reintegration theorem connecting that source mass back to the row vector. By Corollary 16.401, the source-flat interval model is compatible with finite positive same-distance packet quadrature to arbitrary finite order. Therefore the missing input cannot be a finite local packet-geometry lemma; it must use actual zeta arithmetic, a source-robust row, or a genuinely global transport/projective constraint.

This proposition is not a counterexample to RH or to actual zeta; it is a no-go for closing the present rank-two norm route from local packet geometry alone.

Remark 16.403 (What would be needed to save the norm route). A necessary new input is actual-zeta source-curvature observability, for example

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 dm \geq Q^{-B}|I_\rho|.$$

A finite-jet version would be

$$\max_{0 \leq r \leq R(Q)} |\partial_m^r q''_{\text{can}}(m_0)| \geq Q^{-B}.$$

The currently available local framework does not prove either statement. The distinguished packet contributes nontrivially to q''_{can} , but rest/core/background/dual data can cancel that contribution. Same-scale selection, strict core gap, endpoint separation, Gram positivity, finite Hadamard symmetry, and the existing fixed-sector row algebra do not prevent such cancellation.

Even this source-curvature input is not sufficient by itself. The norm route also needs source–anisotropy reintegration:

$$\int_{I_\rho} |q''_{\text{can}}|^2 (\Gamma_G^2 + \Gamma_{G,2}^2) dm \geq Q^{-A}|I_\rho|.$$

Thus the required new input is either the two-gate pair above, a single stronger theorem implying both, a genuinely source-robust packet-visible row, or a global projective/trace/transport theorem.

Remark 16.404 (No-go ledger after the source-collapse push). The current push rules out several tempting repairs of the source-collapse obstruction.

(1) *Higher anisotropy rows.* The tower

$$X_1 Y X_{2k} - X_{2k} Y X_1$$

is divisible by $H = yq''_{\text{can}}$. Higher order can detect higher Gram anisotropies, but it does not remove source collapse.

- (2) *Ambient D-sector at order seven.* At $g = h$, the ambient row

$$T_7^{(1)} = -2yBpD$$

survives the fixed-sector Gram-isotropy failure, but still contains

$$yB \sim yq''_{\text{can}}.$$

It therefore does not solve Regime B source collapse.

- (3) *Transported derivatives.* If

$$A_j = HG_j,$$

then

$$\nabla A_j = (\nabla H)G_j + H\nabla G_j.$$

Since

$$H \sim -\frac{q_0(q''_{\text{can}})^3}{288},$$

one has

$$\nabla H = O((q''_{\text{can}})^2)$$

under good-patch derivative control. Thus finitely many transported derivatives do not presently defeat an interval where q''_{can} and finitely many of its derivatives are small.

- (4) *Projectivization.* The source-normalized projective ratio

$$[A_7 : A_9] = [\Gamma_G : \Gamma_{G,2}]$$

formally cancels H , but it loses norm-scale information, is singular at

$$\Gamma_G = \Gamma_{G,2} = 0,$$

and requires correction-row control plus a new global projective index or transport theorem. No such theorem is presently known.

- (5) *Correction rows.* Line, background/core, and normalization/subtraction rows are not currently certified as source-robust off-line packet detectors. They are either gauge/baseline/normalization data or must be carried as enlarged projective coordinates. They do not provide a clean replacement for the source-divisible rows.
- (6) *Explicit formula / prime-side coercivity.* The explicit formula sees the linear ingredients

$$q_0, \quad q''_{\text{can}}, \quad (q_\zeta)^{(4)},$$

but the residual

$$(yq''_{\text{can}})^2(\Gamma_G^2 + \Gamma_{G,2}^2)$$

is nonlinear. Tensorizing the explicit formula produces oscillatory multi-prime sums with no identified positivity or coercive lower bound.

Accordingly, the current route is non-closing unless one proves the two-gate actual-zeta package

$$\int_{I_\rho} |q''_{\text{can}}|^2 dm \geq Q^{-B} |I_\rho| \quad \text{and} \quad \int_{I_\rho} |q''_{\text{can}}|^2 (\Gamma_G^2 + \Gamma_{G,2}^2) dm \geq Q^{-A} |I_\rho|,$$

or else proves a source-robust row theorem or a genuinely global arithmetic/projective/transport index theorem.

Remark 16.405 (Status ledger for candidate routes). The source-collapse push leaves the following status for the main candidate routes considered so far.

- (1) *Order eleven and the finite anisotropy tower.* A further local row computation may identify the next anisotropy factor $\Gamma_{G,3}$ and may be useful for documentation. However, within the established tower

$$X_1 Y X_{2k} - X_{2k} Y X_1, \quad Y = yJ, \quad X_1 = BS, \quad B \propto q''_{\text{can}},$$

the fixed-sector row is always divisible by

$$H = yq''_{\text{can}}.$$

Thus an order-eleven row of the same kind would at best have the schematic form

$$A_{11}^f \sim H\Gamma_{G,3}.$$

It may help with further anisotropy walls, but it does not address the dominant source-collapse regime $H \approx 0$. Order eleven is therefore diagnostic / documentary, not presently proof-closing.

- (2) *Norm-based rank-two finite-core route.* The known rows have the form

$$A_7^f \sim H\Gamma_G, \quad A_9^f \sim H\Gamma_{G,2}.$$

Since

$$H \sim -\frac{q_0(q''_{\text{can}})^3}{288},$$

an interval on which q''_{can} is small collapses the norm vector even when Γ_G remains bounded away from zero. This route is blocked unless one proves both source-curvature observability and source-anisotropy reintegration, or finds a source-robust row.

- (3) *Transported derivative tower.* Differentiation gives

$$\nabla A_j = (\nabla H)G_j + H\nabla G_j.$$

Near $q'' = 0$,

$$\nabla H = O((q'')^2)$$

under good-patch derivative control. Thus first derivatives still collapse. For every fixed finite derivative order, finite local models can plausibly flatten $H, \nabla H, \dots, \nabla^R H$. A derivative-tower rescue would require a genuinely new growing-order source-jet theorem.

- (4) *Projectivization / source-normalized ratios.* The ratio

$$[A_7 : A_9] = [\Gamma_G : \Gamma_{G,2}]$$

formally cancels H in the idealized factorized model. However, this loses magnitude, remains singular at

$$\Gamma_G = \Gamma_{G,2} = 0,$$

and is not cleanly defined for the corrected row vector unless correction rows vanish or are H -divisible. No global projective index, endpoint, winding, or holonomy theorem is presently available. The projective branch is therefore diagnostic only.

- (5) *Correction rows as source-robust detectors.* The current audits did not identify a correction row that is simultaneously non- H -divisible, packet-visible, and absent in RH-compatible/on-line models. Line rows are gauge/reference-line terms, background/core rows are baseline data, and normalization/subtraction rows are convention-dependent unless separately packet-isolated. Correction rows therefore do not currently provide the missing source-robust detector.

- (6) *Explicit formula / prime-side coercivity.* The explicit formula represents the linear ingredients

$$q_0, \quad q'', \quad q^{(4)},$$

but the source-weighted residual

$$(yq'')^2(\Gamma_G^2 + \Gamma_{G,2}^2)$$

is nonlinear. Tensorizing the explicit formula produces oscillatory multi-prime sums with no identified positivity or coercive lower bound. Thus the explicit formula is representational here, not currently coercive.

- (7) *Fixed-prime and AFE-style adversaries.* Fixed-prime and AFE-compatible surrogate models support the interval-local source-collapse mechanism. In such models one attempts to force

$$J_2^{(r)}(m_0) \approx 0, \quad 0 \leq r \leq R,$$

with $J_2 \sim q''$, while maintaining

$$J_0 \asymp 1, \quad g, h > 0, \quad \Gamma_G \asymp 1.$$

The AFE main sum, dual sum, smoothing, and gamma/background terms shift or couple the finite target, but no algebraic blocker is currently known.

- (8) *Same-scale packet geometry.* The continuous same-scale source-flattening model and its finite quadrature corollary show that positive same-distance packet geometry can make finite source jets vanish while preserving $q_0 > 0$, Gram positivity, and live Γ_G . Therefore source-collapse cannot be excluded by local same-scale packet geometry alone.
- (9) *Interval L^2 / source-curvature observability.* The sharpest remaining positive target is not the full nonlinear residual but the source-curvature estimate

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 dm \geq Q^{-B} |I_\rho|,$$

or at least

$$\sup_{m \in I_\rho} |q''_{\text{can}}(m)| \geq Q^{-B}.$$

This is the least nonlinear form of the missing arithmetic input.

- (10) *Source-anisotropy reintegration.* Even if the interval source-curvature estimate

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 dm \geq Q^{-B} |I_\rho|$$

is proved, it does not by itself imply the source-weighted rank-two lower bound. One still needs to exclude concentration of q''_{can} -mass near

$$\Gamma_G = 0, \quad \Gamma_{G,2} = 0.$$

Equivalently, the architecture needs the vector estimate

$$\int_{I_\rho} |q''_{\text{can}}|^2 (\Gamma_G^2 + \Gamma_{G,2}^2) dm \geq Q^{-A} |I_\rho|.$$

The identity

$$\Gamma_{G,2} = 2q(q')^2 - \frac{1}{6}\Gamma_G''$$

turns this into a local analytic/ODE rigidity problem coupled to the arithmetic source-observability problem. This bridge is currently a missing theorem, not a consequence of source observability alone.

- (11) *Global trace / projective index / Connes-style imbalance.* A global trace, projective index, or spectral accounting theorem is not ruled out. However, it would be a new global proof architecture, not a continuation of the present local rank-two norm route.

Thus the current proof state is: the local rank-two machinery is a precise obstruction and diagnostic framework, but it is non-closing unless one supplies both actual-zeta source-curvature observability and source-anisotropy reintegration, or else supplies a genuinely source-robust row or a new global arithmetic/projective/trace theorem.

Remark 16.406 (How the downstream repairs modify the source-collapse ledger). The source-collapse ledger remains valid as a no-go statement for purely local rank-two norm arguments and for collapsed scalar I -coordinate row energy. It should not be read as a no-go for the repaired downstream package. The repaired package first proves the source-anisotropy Gate 2 estimate from Gate 1 by L^2 -input rigidity, then uses wall escape and carrier-quotient row energy to convert the resulting interval mass into row energy. Thus the source-collapse examples remain diagnostics and negative controls, but they no longer constitute the active downstream obstruction once the hypotheses of Theorems 15.22, 15.29, and 16.100 are assumed.

Remark 16.407 (Corrected projective transport obstruction). One possible response to source divisibility is to pass to the projective ratio. In the idealized factorized model

$$A_7 = HG_1, \quad A_9 = HG_2, \quad H = yq''_{\text{can}},$$

one has

$$[A_7 : A_9] = [G_1 : G_2].$$

In the present setting G_1 and G_2 are represented by

$$G_1 \sim \Gamma_G, \quad G_2 \sim \Gamma_{G,2}.$$

Thus projectivization formally cancels the common source factor H .

However, the actual corrected rows may have the form

$$A_7 = HG_1 + R_7, \quad A_9 = HG_2 + R_9,$$

where R_7, R_9 denote line, background/core, normalization, or other correction-row contributions not known to be H -divisible. On the chart $A_7 \neq 0$, put

$$Z = \frac{A_9}{A_7}.$$

Then

$$\nabla Z = \frac{A_7 \nabla A_9 - A_9 \nabla A_7}{A_7^2},$$

and substitution gives

$$\begin{aligned} A_7 \nabla A_9 - A_9 \nabla A_7 &= H^2(G_1 \nabla G_2 - G_2 \nabla G_1) \\ &\quad + H(G_1 \nabla R_9 - G_2 \nabla R_7) \\ &\quad + H(R_7 \nabla G_2 - R_9 \nabla G_1) \\ &\quad + (\nabla H)(R_7 G_2 - R_9 G_1) \\ &\quad + (R_7 \nabla R_9 - R_9 \nabla R_7). \end{aligned}$$

Therefore the clean source-normalized identity

$$\nabla \left(\frac{A_9}{A_7} \right) = \nabla \left(\frac{G_2}{G_1} \right)$$

is valid only if the correction rows vanish or are themselves H -divisible. If non- H -divisible corrections remain, then near $H = 0$

$$[A_7 : A_9] \rightarrow [R_7 : R_9],$$

not

$$[\Gamma_G : \Gamma_{G,2}].$$

Thus the projective branch is coherent only after a correction-divisibility or correction-absorption theorem, or after enlarging the projective coordinate to

$$[A_7 : A_9 : R_7 : R_9 : \cdots].$$

No global projective endpoint, holonomy, or index theorem is presently known for that enlarged object.

16.30. Finite-core sampled vector non-concentration.

Remark 16.408 (Updated role of finite-core non-concentration). The finite-core non-concentration problem below is now secondary to the source-collapse obstruction. The known fixed-sector rows are source-divisible, so a lower bound for a transported finite-core vector cannot be proved from anisotropy non-concentration alone. Any proof-relevant finite-core theorem must either include the two-gate actual-zeta package

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 dm \geq Q^{-B} |I_\rho|$$

and

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 (\Gamma_G(m)^2 + \Gamma_{G,2}(m)^2) dm \geq Q^{-A} |I_\rho|,$$

or else use a genuinely source-robust row/invariant not divisible by

$$H = yq''_{\text{can}}.$$

Without that additional source-level input or source-robust replacement, the aligned/non-concentration language below remains a diagnostic framework rather than a proof-closing theorem.

The notation in this subsection originally arose in the scalar Φ_K -sampled formulation. In the current rank-two architecture, the corresponding fixed-sector vector is

$$C_{\text{fd}}^{\vec{f}}(\theta) = (A_7^{\text{f}}, A_9^{\text{f}}, \dots, A_{2R_{\Theta_k}+1}^{\text{f}})(\theta).$$

The old scalar vector

$$C_{\text{fd}}(\theta) = (c_{\text{core},1}, c_{\text{core},3}, \dots, c_{\text{core},2R_{\Theta_k}+1})(\theta)$$

should be regarded as its Φ_K -readout, not as the full fixed-sector object. The rank-two non-concentration theorem needed by the present architecture is therefore a lower bound for

$$P_{\text{fd}}^{\vec{f}}(\theta) + J_{\text{fd}}^{\vec{f}}(\theta),$$

or for a transported enhancement of this vector, rather than merely for the scalar Φ_K -projection.

Let Θ_k be a compact nonsingular actual-source k -core stratum. Define the finite-determinacy vector

$$C_{\text{fd}}(\theta) = (c_{\text{core},1}, c_{\text{core},3}, \dots, c_{\text{core},2R_{\Theta_k}+1})(\theta).$$

Decompose it as

$$C_{\text{fd}}(\theta) = P_{\text{fd}}(\theta) + J_{\text{fd}}(\theta),$$

where P_{fd} is the pair-like or one-body finite vector and

$$J_{\text{fd}} = (J_1, J_3, \dots, J_{2R_{\Theta_k}+1}), \quad J_{2r+1} = [s^{2r+1}] \Phi_K(\mathcal{I}_{\geq 2}),$$

is the finite interaction vector.

For $1 \leq N \leq R_{\Theta_k} + 1$, define

$$F_N(\theta, Q) = c_{\text{core},2N-1}(\theta) + \sum_{j \geq 1} c_{\text{core},2N-1+2j}(\theta) h_j(1^2, \dots, N^2) Q^{-2j}.$$

Set

$$\Lambda(m, Q) = C_* \max_{1 \leq N \leq R_{\Theta_k}+1} \left(\frac{Q^{2N+2}}{(2N-1)N!(N-1)!} \frac{x}{B_\zeta(m)} S(m) \right).$$

Hypothesis 16.409 (Bounded-depth projective detection). On the compact nonsingular actual-source stratum Θ_k , there exist constants

$$Q_0(\Theta_k), \quad c_0 > 0$$

such that for every $Q \geq Q_0(\Theta_k)$ and every $\theta \notin \mathcal{Z}_k$, there is an index

$$1 \leq N \leq R_{\Theta_k} + 1$$

with

$$|F_N(\theta, Q)| \geq c_0 \|C_{\text{fd}}(\theta)\|_\infty.$$

This is the bounded-depth projective detection and Q^{-2} -tail-control input. It is separate from sampled vector non-concentration: it detects the finite-determinacy vector once that vector has calibrated size, but it does not prove that calibrated size.

Hypothesis 16.410 (Sampled finite-core vector non-concentration). For every actual zeta finite-core parameter

$$\theta(m) \in \Theta_k,$$

one needs the following alternative:

$$\theta(m) \in \mathcal{Z}_k,$$

or

$$\theta(m) \in \mathcal{E}_{\text{geom}},$$

or

$$\|P_{\text{fd}}(\theta(m)) + J_{\text{fd}}(\theta(m))\|_\infty \geq \Lambda(m, Q).$$

Definition 16.411 (Interaction-aligned finite-core branch). Define

$$\mathcal{N}_k^{\text{align}}(m, Q)$$

to be the set of actual zeta finite-core parameters

$$\theta(m) \in \Theta_k \setminus (\mathcal{Z}_k \cup \mathcal{E}_{\text{geom}})$$

such that

$$0 < \|P_{\text{fd}}(\theta(m)) + J_{\text{fd}}(\theta(m))\|_{\infty} < \Lambda(m, Q).$$

Equivalently, Hypothesis 16.410 asserts

$$\mathcal{N}_k^{\text{align}}(m, Q) = \emptyset.$$

Remark 16.412 (Relation among finite-core bad branches). The branch

$$\mathcal{N}_k^{\text{align}}(m, Q)$$

is the current sharp form of the finite-core obstruction. Earlier formulations used either a distance-to- $\mathcal{Z}_k$ tube

$$\mathcal{N}_k(m, Q),$$

or a coefficient-small branch

$$\mathcal{N}_k^{\text{coef}}(m, Q) = \left\{ \theta(m) \notin \mathcal{Z}_k \cup \mathcal{E}_{\text{geom}} : \max_N |F_N(\theta(m), Q)| < \Lambda(m, Q) \right\}.$$

For $Q \geq Q_0(\Theta_k)$, bounded-depth projective detection and Q^{-2} -tail control identify the coefficient-small branch with radial smallness of the finite-determinacy vector, up to fixed compact-patch constants:

$$0 < \|C_{\text{fd}}(\theta(m))\|_{\infty} < \Lambda(m, Q).$$

Since

$$C_{\text{fd}} = P_{\text{fd}} + J_{\text{fd}},$$

this is exactly the interaction-aligned branch

$$\mathcal{N}_k^{\text{align}}(m, Q).$$

Accordingly, subsequent finite-core arguments should target $\mathcal{N}_k^{\text{align}}$, not repeat finite determinacy or projective detection.

Lemma 16.413 (Finite-core lower transfer from sampled vector non-concentration). *Assume Hypothesis 16.410 and Hypothesis 16.409. Then every actual zeta finite-core sample*

$$\theta(m) \notin \mathcal{Z}_k \cup \mathcal{E}_{\text{geom}}$$

has some $1 \leq N \leq R_{\Theta_k} + 1$ satisfying

$$|\Xi_{\text{core}}^{(N)}(m)| \geq c \frac{x}{B_{\zeta}(m)} S(m).$$

Proof. Choose N by projective detection. Since

$$C_{\text{fd}} = P_{\text{fd}} + J_{\text{fd}},$$

Hypothesis 16.410 gives

$$|F_N(\theta(m), Q)| \geq c_0 \Lambda(m, Q).$$

By the definition of $\Lambda(m, Q)$,

$$\Lambda(m, Q) \geq C_* \frac{Q^{2N+2}}{(2N-1)N!(N-1)!} \frac{x}{B_{\zeta}(m)} S(m).$$

Substituting this into

$$\Xi_{\text{core}}^{(N)}(m) = (-1)^{N+1} (2N-1)N!(N-1)! Q^{-(2N+2)} F_N(\theta(m), Q)$$

gives the asserted lower bound. $\square$

Remark 16.414 (Why sampled vector non-concentration is the exact finite-core dependency). Lemma 16.413 separates the finite-core problem into a qualitative part and a sampled-size part. The qualitative part is finite odd-germ determinacy: on a compact nonsingular actual-source stratum,

$$C_{\text{fd}}(\theta) = 0 \iff H_{\text{core},\theta} \equiv 0.$$

Together with bounded-depth Q^{-2} -tail control, this gives projective detection of every nonzero finite-core germ by some

$$1 \leq N \leq R_{\Theta_k} + 1.$$

What remains is the sampled finite-vector lower bound

$$\|C_{\text{fd}}(\theta(m))\|_{\infty} = \|P_{\text{fd}}(\theta(m)) + J_{\text{fd}}(\theta(m))\|_{\infty} \geq \Lambda(m, Q)$$

outside

$$\mathcal{Z}_k \cup \mathcal{E}_{\text{geom}}.$$

Equivalently, the remaining finite-core branch is the interaction-aligned small-vector branch

$$\mathcal{N}_k^{\text{align}}(m, Q).$$

Thus any further finite-core proof must either prove Hypothesis 16.410, or add an independent theorem forcing $P_{\text{fd}} + J_{\text{fd}}$ -smallness into a controlled incidence stratum.

Remark 16.415 (Status of the finite-core branch). Finite odd-germ determinacy supplies qualitative detection on compact nonsingular strata. It does not supply the sampled vector lower bound in Hypothesis 16.410. The live finite-core obstruction is calibrated interaction alignment:

$$J_{\text{fd}}(\theta(m)) \approx -P_{\text{fd}}(\theta(m)),$$

making $C_{\text{fd}} = P_{\text{fd}} + J_{\text{fd}}$ nonzero but smaller than the N -point scale $\Lambda(m, Q)$, without entering any currently listed controlled incidence stratum.

16.31. dd -interaction feature-balance obstruction. In the covariance-weighted dd -interaction model, write

$$K_{ij}^{dd} = (u_i - u_j)^2 \langle v_i, v_j \rangle, \quad v_i = v(u_i).$$

For balanced amplitudes

$$\sum_i a_i = 0,$$

the associated quadratic form is

$$Q_{dd}(a) = 2 \left\langle \sum_i a_i u_i^2 v_i, \sum_j a_j v_j \right\rangle - 2 \left| \sum_i a_i u_i v_i \right|^2.$$

The second term is one-sided. The first term has no fixed sign unless one has feature balance:

$$\sum_i a_i v_i = 0.$$

Ordinary amplitude balance does not imply feature balance. Even rank-one locking

$$v_i = \lambda_i v_0$$

only changes the missing condition to

$$\sum_i a_i \lambda_i = 0.$$

Lemma 16.416 (Feature balance is stronger than amplitude balance). *Suppose that, for a fixed finite core,*

$$\sum_i a_i v_i = 0$$

for every balanced amplitude vector

$$\sum_i a_i = 0.$$

Then

$$v_1 = v_2 = \cdots = v_k.$$

Thus ordinary amplitude balance implies feature balance only when the transverse dd -features are constant on the core.

Proof. Take $a = e_i - e_j$. Then $\sum_{\ell} a_{\ell} = 0$, so feature balance gives $v_i - v_j = 0$. Since i, j are arbitrary, all v_i are equal. $\square$

Remark 16.417 (Rank-one locking does not close feature balance). If

$$v_i = \lambda_i v_0,$$

then feature balance becomes

$$\sum_i a_i \lambda_i = 0.$$

This is not implied by

$$\sum_i a_i = 0$$

unless the weights λ_i are constant on the core. Thus rank-one transverse locking merely transfers the missing condition from the vector features v_i to the scalar weights λ_i .

Definition 16.418 (dd -interaction branch). The centered dd -interaction branch is

$$\mathfrak{Z}_{\text{int}}^{dd} = \left\{ \sum_{\nu} a_{\nu} K(d_{\nu}) + \sum_{i,j} a_i a_j (u_i - u_j)^2 \langle v_i, v_j \rangle = 0 \right\} \setminus \mathcal{E}_{\text{geom}}.$$

Definition 16.419 (Transported dd -kernel Taylor coefficients). Assume the favorable transported-feature reduction

$$K_{dd}(t; m) = t^2 \left\langle v \left(m + \frac{t}{2} \right), v \left(m - \frac{t}{2} \right) \right\rangle.$$

Write

$$v(m + h) = v_0 + h v_1 + \frac{h^2}{2} v_2 + O(h^3).$$

Then

$$K_{dd}(t; m) = \kappa_2 t^2 + \kappa_4 t^4 + O(t^6),$$

where

$$\kappa_2 = \langle v_0, v_0 \rangle,$$

and

$$\kappa_4 = \frac{1}{4} (\langle v_0, v_2 \rangle - \langle v_1, v_1 \rangle).$$

Remark 16.420 (First Taylor obstruction to dd -kernel conditional definiteness). A positive spectral representation of the form

$$\text{Re}(e^{-i\theta_0} K_{dd}(t; m)) = \alpha(m) t^2 + \int_{\mathbb{R}} (1 - \cos(t\xi)) d\mu_m(\xi), \quad \mu_m \geq 0,$$

or its negative, imposes sign restrictions on the Taylor coefficients. In particular, if the phase is chosen so that

$$\text{Re}(e^{-i\theta_0} \kappa_2) > 0,$$

then the first nonquadratic coefficient must satisfy

$$\text{Re}(e^{-i\theta_0} \kappa_4) \leq 0.$$

By Definition 16.419, this requires

$$\text{Re} \left(e^{-i\theta_0} (\langle v_0, v_2 \rangle - \langle v_1, v_1 \rangle) \right) \leq 0.$$

The current one-pair incidence package does not prove this transverse feature-curvature inequality, nor the higher moment/Hankel positivity conditions needed for a full positive spectral representation.

Proposition 16.421 (Unit transported feature curves need not be spectrally positive). *Let*

$$v(u) = (\cos \phi(u), \sin \phi(u))$$

be a unit transported transverse feature curve, and set

$$K_{dd}(t) = t^2 \left\langle v\left(\frac{t}{2}\right), v\left(-\frac{t}{2}\right) \right\rangle.$$

If

$$\phi(u) = au + bu^3,$$

then

$$K_{dd}(t) = t^2 - \frac{a^2}{2}t^4 + \left(\frac{a^4}{24} - \frac{ab}{4}\right)t^6 + O(t^8).$$

Consequently, even unit normalization of the transverse feature curve does not imply the coefficient signs required by a conditionally one-sided spectral representation. In the negative-type convention, after $\kappa_2 > 0$ and $\kappa_4 \leq 0$, one still needs

$$\kappa_6 = \frac{a^4}{24} - \frac{ab}{4} \geq 0,$$

which fails whenever

$$b > \frac{a^3}{6}.$$

Proof. Since

$$\left\langle v\left(\frac{t}{2}\right), v\left(-\frac{t}{2}\right) \right\rangle = \cos\left(\phi\left(\frac{t}{2}\right) - \phi\left(-\frac{t}{2}\right)\right),$$

and

$$\phi\left(\frac{t}{2}\right) - \phi\left(-\frac{t}{2}\right) = at + \frac{b}{4}t^3,$$

we have

$$K_{dd}(t) = t^2 \cos\left(at + \frac{b}{4}t^3\right).$$

Expanding the cosine through order t^6 gives the stated formula. $\square$

Remark 16.422 (Needed input for the dd -interaction branch). On each compact nonsingular same-scale finite-core patch away from the controlled degeneracy set, let $v(u)$ be the transverse dd -source feature after same-side whitening normalization. To close the dd -interaction branch, one needs at least one of the following:

(1) *Feature balance:*

$$\sum_i a_i v(u_i) = 0$$

for every admissible balanced finite-core amplitude vector

$$\sum_i a_i = 0.$$

By Lemma 16.416, this is much stronger than ordinary amplitude balance.

(2) *Conditional spectral rigidity:* the reduced kernel

$$K_{dd}(t; m) = t^2 \left\langle v\left(m + \frac{t}{2}\right), v\left(m - \frac{t}{2}\right) \right\rangle$$

admits, after phase rotation, a conditionally one-sided spectral representation, or its negative:

$$\operatorname{Re}(e^{-i\theta_0} K_{dd}(t; m)) = \alpha(m)t^2 + \int_{\mathbb{R}} (1 - \cos(t\xi)) d\mu_m(\xi), \quad \alpha(m) \geq 0, \quad \mu_m \geq 0,$$

or the negative of such a kernel. At minimum this route requires the Taylor coefficient sign condition of Remark 16.420. Proposition 16.421 shows that unit normalization of $v(u)$ alone is insufficient.

- (3) *Incidence containment*: all feature-balance failures and centered kernel signature-changing loci lie in the controlled incidence set $\mathcal{E}_{\text{geom}}$.

Absent one of these inputs, the branch

$$\mathfrak{Z}_{\text{int}}^{dd}$$

remains independent. In particular, pair-like positivity of the one-body kernel is not by itself a centered-kernel spectral theorem for the covariance-weighted dd -interaction.

16.32. The remaining global gate: quantitative transported non-concentration. The reduced rank-two construction should be regarded as local infrastructure. Under the reach, atlas, and row-table hypotheses, it gives a visibility-positive fixed-sector bundle

$$\mathcal{F}_{\text{red}}^+ \rightarrow \mathcal{B}_{\text{red}}^+$$

with connection

$$\nabla_X^{\mathfrak{f}} A = \Pi_{\mathfrak{f}}(d_X^{\mathcal{J}} A),$$

and a local affine forcing equation

$$\nabla_{\kappa}^{\mathfrak{f}} A_7^{\mathfrak{f}} = E_{\eta\eta} + E_{\text{lower}}.$$

This does not by itself imply RH.

The remaining global problem is quantitative. For a transported finite-core vector

$$C_{\text{fd}}^{\text{tr}} = (A_7^{\mathfrak{f}}, \nabla^{\mathfrak{f}} A_7^{\mathfrak{f}}, A_9^{\mathfrak{f}}, \nabla^{\mathfrak{f}} A_9^{\mathfrak{f}}, \dots),$$

one must prove a lower bound of the form

$$\|C_{\text{fd}}^{\text{tr}}(\theta(m))\|_{\infty} \geq Q^{-B}$$

outside controlled transported incidence.

A qualitative nonvanishing statement is insufficient. Indeed, in the pair/interaction decomposition

$$C_{\text{fd}}^{\text{tr}} = P_{\text{fd}}^{\text{tr}} + J_{\text{fd}}^{\text{tr}},$$

the aligned branch

$$J_{\text{fd}}^{\text{tr}} \approx -P_{\text{fd}}^{\text{tr}}$$

is not excluded by the local rank-two formalism. Equivalently, one needs a quantitative incidence-distance theorem

$$\text{dist}(\theta(m), \mathcal{Z}_k^{\text{tr}} \cup \mathcal{E}_{\text{geom}}^{\text{tr}}) \geq Q^{-A},$$

A more concrete form of the same obstruction is the source-weighted layered lower bound

$$\sup_{m \in I_{\rho}} |y(m) q_{\text{can}}''(m)| \max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \geq Q^{-A}.$$

The source-collapse countermodels above show that this estimate cannot be deduced from local rank-two row algebra, finite Hadamard/quartet symmetry, same-scale packet geometry, Gram positivity, or finite-prime/AFE surrogate structure alone. It now requires the two-gate actual-zeta package:

$$\int_{I_{\rho}} |q_{\text{can}}''(m)|^2 dm \geq Q^{-B} |I_{\rho}|$$

together with

$$\int_{I_{\rho}} |q_{\text{can}}''(m)|^2 (\Gamma_G(m)^2 + \Gamma_{G,2}(m)^2) dm \geq Q^{-A} |I_{\rho}|.$$

Alternatively, a zeta-specific rigidity, cone-separation, source-robust row, or global arithmetic non-concentration theorem implying the displayed source-weighted lower bound would suffice.

Remark 16.423 (Numerical fixed-prime and AFE-style evidence). Fixed-prime and AFE-style surrogate experiments support the source-collapse mechanism. In these models one defines

$$L_P(m) = \sum_{p \leq P} (\log p) p^{-1/2} e^{-im \log p},$$

$$J_0 = B_0 + \Re L_P, \quad J_2 = B_2 + \Re L_P'', \quad J_4 = B_4 + \Re L_P^{(4)},$$

and

$$H_{\text{model}} = -\frac{J_0 J_2^3}{288}.$$

The interval-local source-collapse search minimizes

$$J_2^{(r)}(m_0) \approx 0, \quad 0 \leq r \leq R,$$

with side constraints

$$J_0(m_0) \approx 2, \quad J_4(m_0) \approx 0,$$

and then verifies on

$$I = [m_0 - c/Q, m_0 + c/Q]$$

that

$$\sup_I |J_2|$$

is small while

$$\max(|R_1|, |R_2|)$$

remains bounded below.

Such experiments are not evidence about actual zeta by themselves, because they use simplified backgrounds, finite prime cutoffs, and surrogate AFE terms. They do, however, show that the source-collapse mechanism is not a mere symbolic artefact: it survives fixed-prime phases, prime powers, smoothing, dual oscillatory blocks, and interval verification in the surrogate models. Therefore the positive route must identify a genuine actual-zeta arithmetic obstruction, not another local rank-two algebraic identity.

For orientation, the latest normalized fixed-prime source-collapse run found an interval with

$$\sup_I |J_2| \approx 5.3 \times 10^{-3},$$

$$\inf_I \max(|R_1|, |R_2|) \approx 4.49,$$

and

$$\sup_I |H| \max(|R_1|, |R_2|) \approx 4.7 \times 10^{-9}.$$

Adding prime powers did not remove the mechanism; a reported prime-power run gave interval product about

$$8.52 \times 10^{-8}.$$

A smoothed main sum plus frozen-dual AFE surrogate gave a best interval product near

$$8.77 \times 10^{-9}.$$

Including moving differentiated $\chi(t)$, followed by exact $\chi(1/2 + it)$ -derivatives, still left source-collapse examples; the reported exact- χ interval product was about

$$1.93 \times 10^{-7}.$$

These values are retained only as diagnostics. They remove several easy objections to the source-collapse mechanism, but they are not a theorem about actual ζ or its exact AFE normalization.

The numerical evidence is not used as proof. Its role is to identify the precise missing theorem: actual-zeta source-curvature observability on I_ρ , as formulated in Subsection 16.33.

Remark 16.424 (Exact χ -derivatives in AFE-style source-collapse tests). For the exact critical-line factor

$$\chi(t) := \chi\left(\frac{1}{2} + it\right) = \pi^{it} \frac{\Gamma\left(\frac{1}{4} - \frac{it}{2}\right)}{\Gamma\left(\frac{1}{4} + \frac{it}{2}\right)},$$

write $g(t) = \log \chi(t)$. Then

$$g(t) = it \log \pi + \log \Gamma\left(\frac{1}{4} - \frac{it}{2}\right) - \log \Gamma\left(\frac{1}{4} + \frac{it}{2}\right),$$

and, for $n \geq 1$,

$$g^{(n)}(t) = \mathbf{1}_{n=1} i \log \pi + \left(-\frac{i}{2}\right)^n \psi^{(n-1)}\left(\frac{1}{4} - \frac{it}{2}\right) - \left(\frac{i}{2}\right)^n \psi^{(n-1)}\left(\frac{1}{4} + \frac{it}{2}\right).$$

If

$$E_0 = 1, \quad E_n = \sum_{k=0}^{n-1} \binom{n-1}{k} E_k g^{(n-k)},$$

then

$$\chi^{(n)} = \chi E_n.$$

Consequently,

$$\frac{d^k}{dt^k} \left[\chi(t) \sum_n b_n e^{it \log n} \right] = \sum_{a=0}^k \binom{k}{a} \chi^{(a)}(t) \sum_n b_n (i \log n)^{k-a} e^{it \log n}.$$

In the staged AFE-style tests, replacing the leading asymptotic phase by this exact χ -derivative calculation did not remove interval-local source collapse. It changed the finite derivative target, but did not create a Gate 1 source-curvature observability mechanism, let alone the full two-gate package.

Remark 16.425 (Gamma/background terms as shifted derivative targets). Let

$$J_2(t) = B_2(t) + P_2(t),$$

where B_2 is the gamma/Riemann–Siegel/far-tail background contribution and P_2 is the oscillatory prime/AFE contribution. Interval source collapse requires

$$J_2^{(r)}(m_0) \approx 0, \quad 0 \leq r \leq R.$$

Equivalently,

$$P_2^{(r)}(m_0) \approx -B_2^{(r)}(m_0), \quad 0 \leq r \leq R.$$

Thus exact gamma/background terms shift the finite derivative-flat target. They do not change the type of the problem unless one proves a range, sign-cone, or arithmetic nonflatness theorem of the form

$$\max_{0 \leq r \leq R} |B_2^{(r)}(m_0) + P_2^{(r)}(m_0)| \geq Q^{-B}.$$

No such theorem is currently supplied by the local rank-two package. On the microscopic interval $|t - m_0| \leq c/Q$, the background is smooth and is captured by finite Taylor data, so it behaves as a shifted target for the oscillatory AFE part.

Remark 16.426 (Next numerical test: actual-zeta source curvature). The surrogate computations should now be followed by an actual-zeta numerical scan of the canonical q'' defined in Definition 16.429. The diagnostics should be interval-local:

$$\sup_{m \in I} |q''_{\text{can}}(m)|, \quad \int_I |q''_{\text{can}}(m)|^2 dm,$$

on intervals

$$I = [m_0 - c/Q, m_0 + c/Q].$$

The purpose is not to prove anything numerically, but to decide whether source-flat intervals are common in actual zeta data or are an artefact of the fixed-prime/AFE surrogates. If source-flat intervals appear in actual data, Hypothesis 16.431 is likely false in its present form. If they do

not appear, the first theorem to attack is the interval L^2 source-curvature observability estimate for the full canonical object

$$q''_{\text{can}} = B_2 + P_2 + D_2 + E_2.$$

In the ordinary real-canonical branch, the source–anisotropy step is no longer a separate next theorem; it is supplied by Theorem 15.22.

Remark 16.427 (Current branch decision). This is a historical source-collapse fork retained for the interval/local norm route. After the L^2 -Gate-1 repair, the active proof-closing upstream branch is the aggregate observability theorem

$$\text{Agg}L^2(q''_{\text{can}}).$$

The interval statements below should be read as localized proof tools or diagnostics for that theorem. In that older interval formulation, the sharp fork was:

- (1) Prove actual-zeta source-curvature observability,

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 dm \geq Q^{-B} |I_\rho|.$$

In the ordinary real-canonical branch, source–anisotropy reintegration then follows from Theorem 15.22. The remaining downstream burden is to convert the resulting Gate-2 mass into row energy through D4 carrier-quotient row conversion and residual control;

- (2) or construct/verify an actual-zeta source-collapse model satisfying

$$0 < \sup_{m \in I_\rho} |yq''_{\text{can}}| \max\{|\Gamma_G|, |\Gamma_{G,2}|\} < Q^{-A};$$

- (3) or find a genuinely source-robust packet-visible row/invariant not divisible by $H = yq''_{\text{can}}$;
 (4) or prove a global projective/holonomy/trace/index theorem that gives a contradiction without a local norm lower bound.

Absent one of these, the present rank-two norm/projective architecture remains a strong local obstruction framework but not a proof of RH.

Remark 16.428 (Current proof frontier after the local package audits). This is the proof frontier after the local package audits and before the final source-collapse reduction. It is retained as a ledger of local dependencies; the terminal status is given in Remark 16.440.

- (1) The M_A, M_C connection branch is closed under the source-grade reflection convention.
 (2) The promoted quadratic mixed-block Bottleneck C coefficient

$$E_{A,Y}^{\text{BC}}$$

is now blocked at the concrete odd-background top row

$$\mathcal{E}_{\eta\eta;7,0,0}^2.$$

The associated S -coefficient is

$$f_{\eta\eta}^{\text{top}} = -\frac{u}{24\pi q^3} \nu_{21,7}.$$

Unless one proves

$$\nu_{21,7} = 0,$$

finds a cancelling active top-row term, or adds a principled second-order correction, this row is generically T -visible and can also be quotient visible on a generic $u_5 \neq 0$ patch.

- (3) The $\Theta_\pm$ -normal form and septic normal-form counterterms do not close the physical local obstruction. They can move the obstruction into a formal package, but the physical hidden coordinate carries it back through

$$T^{\text{phys}} = T^* + \frac{\beta_{\text{loc}}^\Theta}{c}.$$

The known good-patch models have

$$\partial_\kappa(\beta_{\text{loc}}^\Theta/c) \neq 0.$$

- (4) The finite-core lower-transfer branch is now subordinate to the source-collapse problem. Earlier formulations reduced it to sampled vector non-concentration, for example

$$\mathcal{N}_k^{\text{align}}(m, Q) = \emptyset.$$

The later audits show that such a theorem must include source-level nonflatness. In particular, anisotropy or vector non-concentration alone cannot control the known rows when

$$H = yq''_{\text{can}} \approx 0.$$

Thus the remaining source-side input is Gate 1 for the full canonical object on I_ρ , not merely finite determinacy, compactness, pairwise zero-spacing, endpoint-gap control, or Remez–Łojasiewicz volume estimates. In the ordinary real-canonical branch, the former source–anisotropy step is supplied downstream by Theorem 15.22.

- (5) The source-weighted layered route is blocked by source collapse. The known order-seven/order-nine rows have the form

$$A_7^{\mathfrak{f}} \sim H\Gamma_G, \quad A_9^{\mathfrak{f}} \sim H\Gamma_{G,2}, \quad H = yq''_{\text{can}}.$$

At leading order,

$$H \sim -\frac{q_0(q''_{\text{can}})^3}{288}.$$

Thus an admissible interval on which

$$q''_{\text{can}}(m) \approx 0$$

while

$$q_0(m) \asymp 1, \quad g(m), h(m) > 0, \quad \max\{|\Gamma_G(m)|, |\Gamma_{G,2}(m)|\} \asymp 1$$

makes the norm vector

$$(A_7^{\mathfrak{f}}, A_9^{\mathfrak{f}})$$

super-polynomially small. Fixed-prime and AFE-style derivative-flat surrogates, as well as same-scale packet moment models, do not presently supply an obstruction. The missing positive source-side input is Gate 1:

$$\int_{I_\rho} |q''_{\text{can}}|^2 dm \geq Q^{-B} |I_\rho|.$$

Once this is available in the ordinary real-canonical branch, the source-weighted anisotropy estimate

$$\int_{I_\rho} |q''_{\text{can}}|^2 (\Gamma_G^2 + \Gamma_{G,2}^2) dm \geq Q^{-A} |I_\rho|$$

is supplied by Theorem 15.22.

- (6) The dd -interaction branch remains independent unless feature balance, conditional spectral positivity, or incidence containment is proved. In the transported-kernel reduction, the first nonquadratic Taylor coefficient

$$\kappa_4 = \frac{1}{4}(\langle v_0, v_2 \rangle - \langle v_1, v_1 \rangle)$$

already gives a necessary sign test for conditional definiteness.

16.33. Arithmetic source-curvature observability: the next attack. The preceding source-collapse countermodels show that the current rank-two norm route cannot close from local row algebra, same-scale packet geometry, or finite Hadamard/quartet symmetry alone. The first missing input is an arithmetic source-curvature theorem for actual zeta data. In the notation fixed below, the minimal useful pointwise form is

$$\sup_{m \in I_\rho} |q''_{\text{can}}(m)| \geq Q^{-B}.$$

A stronger and more stable form is the interval energy estimate

$$\boxed{\int_{I_\rho} |q''_{\text{can}}(m)|^2 dm \geq Q^{-B} |I_\rho|}.$$

This estimate would block the pure source-collapse branch, but it is now only the first gate. It does not by itself recover the source-weighted rank-two lower bound, because the q''_{can} -mass could still concentrate near the anisotropy base locus

$$\Gamma_G \approx 0, \quad \Gamma_{G,2} \approx 0.$$

The second missing input is therefore a source–anisotropy reintegration theorem, formulated below.

This subsection is the positive-theorem counterpart of Remark 16.405: all currently explored local, projective, correction-row, higher-anisotropy, and finite-Hadamard routes have either been classified as diagnostic or shown vulnerable to source collapse.

Definition 16.429 (Canonical source-curvature representation problem). In this subsection and in every source-observability statement below, the canonical source-curvature object is

$$\boxed{q''_{\text{can}}(m) = q''_\zeta(m) = \Phi'''_\zeta(m)}.$$

This is the full actual-zeta phase-derivative curvature in the same Gram normalization used by the local rank-two rows. It is not the affine-subtracted complement curvature. The latter is denoted separately by

$$q''_{\text{source},R}(m) := (G^{\text{new}})''(m), \quad G^{\text{new}} = q_{\text{aux},R} + T_{\text{far},R}.$$

The affine-subtracted complement curvature is a component in a chosen source/core split; it is not the canonical arithmetic object whose nonflatness is required.

The arithmetic representation problem is to write q''_{can} on both sides of the explicit formula. On the zero side one needs a representation

$$q''_{\text{can}}(m) = B_2(m) + \sum_{\rho}^{\text{reg}} K_2(m; \rho),$$

with all background, tail, source/core, and regularization conventions fixed. The background curvature is fixed canonically by

$$\boxed{B_2(t) = \theta'''(t) = -\frac{1}{8} \Re \psi^{(2)}\left(\frac{1}{4} + \frac{it}{2}\right)}.$$

Equivalently, for large t ,

$$B_2(t) = -\frac{1}{2t^2} - \frac{1}{8t^4} - \frac{7}{64t^6} + O(t^{-8}).$$

The background term is included by default in zero-side proxy and canonical scans. A B_2 -only mode is diagnostic only; it is never a Gate 1 candidate because Gate 1 concerns the full object q''_{can} , not the gamma/background term alone. On the prime/AFE side one needs a representation of the form

$$q''_{\text{can}}(m) = B_2(m) + P_2(m) + D_2(m) + E_2(m),$$

where

$$P_2(m) = \Re \sum_{n \leq X} b_n (\log n)^2 n^{-1/2-im} W(n/T)$$

has fixed actual-zeta coefficients, D_2 denotes the dual or functional-equation contribution, and E_2 denotes a controlled differentiated error term. The coefficients b_n , smoothing, dual term, gamma/background contribution, and error scale must be specified in the same normalization used by the rank-two local rows. The point of this convention is to forbid replacing q''_{can} by an arbitrary Dirichlet polynomial or by the affine-subtracted complement $q''_{\text{source},R}$. The displayed $(\log n)^2$ -weight corresponds to the literal q''_{can} curvature convention. Some numerical determinant

stress tests also use a script-convention source row with $k_q = 3$, hence a $\Lambda(n)(\log n)^3 n^{-1/2}$ -type coefficient. That $k_q = 3$ convention is a numerical stress-test convention recorded in Remark 16.430; it is not a replacement for the literal q''_{can} representation displayed here.

Remark 16.430 (Log-derivative AFE convention for numerical stress tests). The packet-frequency determinant and local Z_α -amplitude stress tests use the logarithmic-derivative convention

$$A(s) = -\frac{\zeta'}{\zeta}(s), \quad C_k(s) = (-1)^k \partial_s^k A(s).$$

The reflected functional-equation sign is

$$\sigma_k = (-1)^{k+1}.$$

For the current Z_α -row convention one has $k_Z = 0$, so the plus and minus branch coefficients are

$$z_n^+(\alpha) = \Lambda(n)n^{-1/2-\alpha}V_Z^+(n/N_Z^+),$$

$$z_n^-(\alpha) = -\Lambda(n)n^{-1/2+\alpha}V_Z^-(n/N_Z^-).$$

For the script-convention source row one uses $k_q = 3$:

$$c_n^+ = \Lambda(n)(\log n)^3 n^{-1/2}V_q^+(n/N_q^+),$$

$$c_n^- = +\Lambda(n)(\log n)^3 n^{-1/2}V_q^-(n/N_q^-).$$

For a literal q''_{can} -row test, replace $k_q = 3$ by $k_q = 2$, giving

$$c_n^+ = \Lambda(n)(\log n)^2 n^{-1/2}V_q^+, \quad c_n^- = -\Lambda(n)(\log n)^2 n^{-1/2}V_q^-.$$

This convention switch must not be made inside a single proof-level Wronskian. A given proof run must fix one convention for q''_{can} , and then use the corresponding Z_α -symbol ratios, dual branch, E_2 budget, and local row identities throughout.

The Wronskian operator node is the signed differential-frequency node, not the folded conductor diagnostic node:

$$\text{plus operator node} = +\log n, \quad \text{minus operator node} = -\log n.$$

Folded quantities such as

$$\log \frac{m}{2\pi n}$$

may be recorded as conductor/saddle diagnostics, but they are not substituted for the eigenvalue of $\lambda\partial_m$ unless a different folded differential operator has been explicitly defined.

These conventions are numerical guardrails. Proof-level canonical use still requires the exact Mellin smoothing, the paired $V^\pm$ -weights, derivatives of m -dependent weights, the E_2 -tail budget, and the selected real/phase projection.

Hypothesis 16.431 (Gate 1: off-line-packet source-curvature observability). Let I_ρ be an admissible reduced good-patch interval attached to a hypothetical off-line packet ρ . Actual zeta data satisfy

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 dm \geq Q^{-B}|I_\rho|$$

for some fixed B , outside the already named reduced-base and correction incidence loci. Equivalently, in a weaker pointwise form,

$$\sup_{m \in I_\rho} |q''_{\text{can}}(m)| \geq Q^{-B}.$$

This is a necessary source-level nonflatness input for the current rank-two norm route. It is not sufficient by itself to recover the desired lower bound for

$$|H| \max\{|\Gamma_G|, |\Gamma_{G,2}|\}.$$

Remark 16.432 (Gate 2 is no longer an independent ordinary-branch hypothesis). In earlier versions, the source–anisotropy lower bound

$$\int_{I_\rho} |q''_{\text{can}}(m)|^2 (\Gamma_G(m)^2 + \Gamma_{G,2}(m)^2) dm \geq Q^{-A} |I_\rho|$$

was stated as a separate Gate–2 hypothesis. In the ordinary smooth real-canonical branch, this is now supplied by Theorem 15.22, assuming Gate 1, source Lipschitz control for $u = q''_{\text{can}}$, and the L^2 -input real rigidity lemma.

Thus the displayed source–anisotropy estimate is retained as the Gate–2 conclusion, not as an independent ordinary-branch input. Singular packets, transition packets, or packets outside the real-canonical smooth convention still require their own routing.

Remark 16.433 (ODE form of the reintegration problem). Let

$$u = q''_{\text{can}}, \quad \Gamma_G = u - 6q + 2q^3, \quad \Gamma_{G,2} = (1 - q^2)u - \frac{1}{6}q^{(4)}.$$

Differentiating Γ_G gives

$$\Gamma''_G = u'' - 6u + 12q(q')^2 + 6q^2u.$$

Since $u'' = q^{(4)}$, one obtains the identity

$$\boxed{\Gamma_{G,2} = 2q(q')^2 - \frac{1}{6}\Gamma''_G.}$$

Thus the historical reintegration problem has a local analytic/ODE formulation: one must prevent source mass for $u = q''_{\text{can}}$ from concentrating where both

$$\Gamma_G \approx 0 \quad \text{and} \quad 2q(q')^2 - \frac{1}{6}\Gamma''_G \approx 0.$$

In the present ordinary real-canonical branch this prevention is supplied by the L^2 -input rigidity mechanism of Lemma 15.12 and Theorem 15.22. The ODE formulation is therefore retained as explanatory structure and as a fallback diagnostic, not as an independent active proof gate.

Remark 16.434 (Status of the Gate–2 analytic package after the L^2 -rigidity repair). The previous conditional route through thick-set-to-interval propagation and pointwise control of Γ''_G is no longer the active Gate–2 route. In the ordinary real-canonical branch, Gate 2 is supplied by Theorem 15.22.

The remaining local hypotheses are now sharply reduced:

- (1) the real-canonical definition of q_ζ , q''_{can} , Γ_G , and $\Gamma_{G,2}$;
- (2) the source Lipschitz bound

$$\|u'\|_{L^\infty(I)} \leq Q^{C_u} |I|^{-1}, \quad u = q''_{\text{can}};$$

- (3) the safe primitive-bounded hypotheses in Lemma 15.12, or a later replacement proving the primitive-free strengthening;
- (4) residual-stable identities if the displayed formulas for g and h are used with controlled L^2 -errors.

Thus the old $D1/D2$ status should be read as historical/fallback for this branch, not as the current downstream proof gate.

Remark 16.435 (Off-line-packet conditional form). A more targeted version of Hypothesis 16.431 uses the hypothetical off-line packet itself. Write

$$q''_{\text{can}} = q''_P + q''_{\text{rest}},$$

where q''_P is the contribution of the distinguished off-line packet and q''_{rest} is the contribution of all remaining actual-zeta data: other core packets, auxiliary/tail terms, background, dual terms, and controlled errors in the fixed convention. This q''_{rest} should not be confused with the affine-subtracted complement curvature $q''_{\text{source},R} = (G^{\text{new}})''$ unless the source/core split has explicitly identified the two.

The interval observability claim may be sharpened to

$$\int_{I_\rho} |q''_P(m) + q''_{\text{rest}}(m)|^2 dm \geq Q^{-B} |I_\rho|.$$

Equivalently, the rest term cannot satisfy

$$q''_{\text{rest}}(m) = -q''_P(m) + O(Q^{-B})$$

throughout the packet interval. This formulation uses the off-line-packet hypothesis directly and is stronger than a generic short-interval lower bound for q''_{can} . The same-scale source-flattening construction shows that this cannot be proved from arbitrary positive same-scale packet geometry alone; it must use actual-zeta arithmetic or a restriction on how actual zero packets may be arranged around ρ .

Remark 16.436 (Why this theorem is genuinely arithmetic). Hypothesis 16.431 cannot follow from the local packet geometry developed above. The same-scale source-flattening theorem and finite quadrature corollary show that positive same-distance packet configurations can make

$$q''_{\text{can}}(m_0), (q_\zeta)^{(4)}(m_0), \dots, (q_\zeta)^{(M)}(m_0)$$

vanish to arbitrary finite order while keeping

$$q_0 > 0, \quad g > 0, \quad h > 0,$$

and while keeping

$$\Gamma_G = q''_{\text{can}} - 6q_0 + 2q_0^3$$

bounded away from zero after normalization, for example near $q_0 = 2$. Thus any proof of Hypothesis 16.431 must use actual zeta arithmetic beyond finite same-scale positive packet geometry.

Remark 16.437 (Zero-side quadrature obstruction to local proofs). The same-scale source-flattening model gives a concrete obstruction to any purely local zero-side proof of Hypothesis 16.431. The continuous measure in Theorem 16.397 lives on the same-distance curve

$$x^2 + c^2 = d^2$$

and has positive density

$$d\mu(\theta) = \frac{2}{\pi} \sin \theta d\theta.$$

It makes the averaged packet profile locally constant, and finite quadrature then approximates arbitrary finite source jets while keeping positive weights and same-distance packet scale. Therefore actual zeta must exclude, or fail to approximate, this quadrature pattern if the rank-two norm route is to close.

A plausible arithmetic formulation is: actual zeta zeros near a hypothetical off-line packet cannot realize the finite same-scale quadrature needed to make

$$q''(m_0), q^{(4)}(m_0), \dots, q^{(2M)}(m_0)$$

all small while $q_0 > 0$ and the Gram block remains positive. This is the zero-side form of the source-curvature observability problem.

Remark 16.438 (Prime-side observability formulation). After fixing the representation in Definition 16.429, the source-curvature problem becomes a short-interval fixed-coefficient non-approximation problem. Writing

$$q''_{\text{can}}(m) = B_2(m) + P_2(m) + D_2(m) + E_2(m),$$

source collapse means

$$P_2(m) \approx -B_2(m) - D_2(m) \quad \text{throughout } I_\rho,$$

up to the controlled error E_2 . The desired arithmetic theorem is therefore an observability estimate of the form

$$\int_{I_\rho} |B_2(m) + P_2(m) + D_2(m)|^2 dm \geq Q^{-B} |I_\rho|,$$

with E_2 absorbed into the final error budget.

This is not a generic Dirichlet-polynomial problem. The coefficients in P_2 are fixed by the actual zeta/AFE normalization. Standard localized large-sieve, Montgomery–Vaughan, explicit-formula energy, or uncertainty principles are useful diagnostics, but no theorem currently proves this fixed-coefficient lower bound. If interval estimates remain out of reach, the equivalent finite-jet formulation below is the sharper arithmetic target.

Remark 16.439 (Finite-jet version of the arithmetic problem). The interval energy estimate may also be attacked through finite jets. On an interval

$$I_\rho = [m_0 - c/Q, m_0 + c/Q],$$

source collapse requires simultaneous smallness of

$$\partial_m^r q_{\text{can}}''(m_0), \quad 0 \leq r \leq R,$$

where the scale audit gives

$$R \asymp \frac{\log Q}{\log \log Q}$$

or more conservatively $R = O(\log Q)$. In an AFE model this becomes a finite system

$$\partial_m^r (B_2 + D_2 + P_2)(m_0) \approx 0, \quad 0 \leq r \leq R,$$

or equivalently

$$P_2^{(r)}(m_0) \approx - (B_2 + D_2)^{(r)}(m_0).$$

The positive theorem would be a fixed-coefficient prime-log derivative-jet nonrecurrence statement:

$$\max_{0 \leq r \leq R} |\partial_m^r q_{\text{can}}''(m_0)| \geq Q^{-B}.$$

The recurrence heuristics developed above suggest that this is a difficult arithmetic theorem and may be false in fixed-prime or AFE surrogates. It is therefore not currently proved by large-sieve or generic Dirichlet-polynomial approximation estimates.

Remark 16.440 (Archived terminal-status register). This register predates the Hardy source repair, the pre-extraction Fejér–Riesz Stack, the certified live/dead quotient, and the canonical source-graded $K9$ final bridge. It is retained for citation stability, not as the current terminal status.

The active canonical chain is

$$\text{Agg}L^2(q_{\text{can}}'') \implies \text{Graph/O2O3}_{23} \implies \text{certified } C_7/Y12 \text{ quotient and } L_{\text{eff},p} \implies \text{NormalD4}_{K9}^{\text{gr}} \implies \text{AtomicY12Co}$$

Here $\text{Agg}L^2(q_{\text{can}}'')$ is supplied in the ordinary Hardy canonical branch by the Hardy source theorem and the canonical branch-complete window. The graph step is supplied by the pre-extraction Fejér–Riesz charge anti-lock theorem under its structural packet-model hypotheses. The downstream branch uses the fixed linear source-graded $K9$ lift and the full-vector source-graded tail completion; primitive-only $Y12$ completion and scalar collapsed $K9$ completion are archived fallback diagnostics.

The current remaining qualifications are not the finite-main Schur gap, affine-forcing cancellation, positive extraction density, or same-norm atomic comparison. They are recorded in the current ledgers: the global RH-entry into the ordinary Hardy canonical branch, the structural hypotheses of the Fejér–Riesz/O2O3 packet model, and the raw row/routing coefficient certificate $\mathfrak{R}_{2Y}(p)$ if that certificate has not been imported.

17. HOW THE OLDER FRAMING MAPS TO THE NEW FRAMING

Older framing	New framing	Status
Finite-row O1/O2/O3 Gate–1	$\text{Agg}L^2(q''_{\text{can}})$ source survival	historical tool only
S1/T1 transition / singular diagnostics	proof tools for scalar source observability	not the live Gate–1 definition
Scalar I -coordinate D4	carrier quotient / row-normal plain wall	dead fallback
Derivative-wall D4	separate no-go / future derivative-channel theorem	not active
Optimized Ξ_N route	rank-zero on the current finite D4 live plane	no-go
One-parameter row mobility	not a forbidden identity	no-go
Scalar/collapsed K_9 -or-bust route	stale	replaced by canonical source-graded K_9 final bridge

The current live chain is

$$\text{Agg}L^2(q''_{\text{can}}) \Rightarrow \text{Graph/O2O3}_{23} \Rightarrow \text{Y12-certified quotient/effective carrier} \Rightarrow \text{NormalD4}_{K9,Y12} \Rightarrow \text{AtomicY12}$$

where the final comparison is made in the canonical source-graded K_9 norm, not in a scalar-collapsed primitive Y12 norm.

18. NO-GO AND AUDIT LEDGER

All later wip-remarks in this section use the shared finite-survival notation of Subsection 15.1 (Definition 15.3).

Remark 18.1 (Archived finite-row nonrecurrence obstruction). The older O1a, O1b, O2–BZ, and O3 route is archived as a finite-row diagnostic. It records why several pointwise actual-orbit incidence exclusions are not supplied by continuous mean-square estimates, relaxed torus models, generic determinant nonvanishing, exterior coefficient mass, or averaged large-sieve estimates.

This is no longer the active upstream proof target. The active source target is the canonical Hardy-normalized aggregate estimate

$$\text{Agg}L^2(q''_{\text{can}}), \quad q''_{\text{can}} = u_H = \partial_t^3 Z_H.$$

The remaining global burden is not to revive the finite-row O1/O2/O3 incidence package, but to prove branch-entry/no-loss routing:

off-line zero $\implies$ ordinary Hardy-normalized canonical packet branch

with mass-good packets, Fejér–Riesz persistence, raw $2Y$ certification, and packet-to- K_9 transfer, all at polynomial cost.

Remark 18.2 (Historical route register). The following routes are historical diagnostics or no-go branches unless explicitly reactivated by a transfer theorem to the canonical Hardy/ K_9 branch:

- O1/O2/O3 ordinary finite-row Gate–1;
- S1/T1 transition and singular point-jet or sine-moment Gate–1 proxies;
- scalar I -coordinate D4 and derivative-wall D4;
- optimized Ξ_N as a finite D4 forbidden channel;
- one-parameter row mobility as a contradiction mechanism;
- functorial carrier lifts when $R_{\text{sing}}^{\text{raw}}$ is Taylor-complete, carrier-complete, or otherwise retains the live $C_7/Y12$ transition plane;
- pure-branch determinant-only proofs for full O2/O3.

The live architecture is:

$$\text{Agg}L^2(q''_{\text{can}}) + \text{Graph/O2O3}_{23} + \text{NormalD4}_{K9,Y12} \implies \text{AtomicY12Contradiction}.$$

Remark 18.3 (Unified live/dead dictionary). Whenever the active canonical text says Gate-1, it means $\text{Agg}L^2(q''_{\text{can}})$ for the Hardy canonical source and the canonical branch-complete window. Whenever the final bridge uses a discarded observable, it means

$$\Lambda_p^{\text{gr,full}} = (I - P_p^{\text{full}})\mathcal{L}_{K9,p}^{\text{gr}}J_{\text{tr},p}$$

or its orthogonal maximal-affine projection $\hat{\Lambda}_{\text{max},p}$, unless a local section explicitly declares that it is discussing an archived scalar or primitive-only diagnostic. The collapsed operator $(I - P_{\text{sing}})T_{\text{tr} \rightarrow \text{sing}}P_{\text{tr}}$ is not a substitute for the active same-norm final bridge. D4 is local visibility until it is coupled to this source-graded forbidden channel and the corresponding source-graded upper.

Remark 18.4 (Archived primitive-only Y12 quotient audit status). This remark is an archive of the primitive-only quotient route, not the active source-graded final bridge. In that older route one formed

$$q_p^Y = I - P_{\mathcal{K}_p^Y}^{\text{bd}}, \quad r_{N,p} = q_p^Y N_p, \quad r_{Y,p} = q_p^Y Y_p,$$

and attempted to use $(r_{N,p}, r_{Y,p})$ as the cap residual pair. The active certified route has superseded this for two reasons. First, N_p is an exact nuisance column in the quotient-killed space, so $q_p^Y N_p = 0$ and the pair cannot carry a rank-two Jacobian. Second, the primitive arithmetic slot

$$c_{\text{arith},Y} := [II_{\text{arith},9}^{\text{exact}}]_{D^2 N_0, \text{ same, } w=1, b=12}$$

is not required to vanish globally in the canonical branch. A live primitive component must not be placed in the exact dead quotient; it is either ruled out by the raw row/routing certificate or retained/routed as a live affine source-grade component in the canonical K9 lift.

Consequently, statements below that use the primitive quotient notation $\mathcal{K}_p^Y, q_p^Y, r_{N,p}, r_{Y,p}$ are retained for no-go diagnostics and citation stability. They should not be read as the proof-level final bridge. The active requirements are the raw packet certificate $\mathfrak{B}_{2Y,p}$ for entering the certified live/dead quotient, followed by the source-graded same-norm comparison of $\Lambda_p^{\text{gr,full}}$ or $\hat{\Lambda}_{\text{max},p}$.

Lemma 18.5 (Y12 quotient descent and primitive survival). *For each packet p , define*

$$\mathcal{K}_p^Y := \text{im} [R_{B_2,p} \ R_{P,p} \ R_{D,p} \ A_{\alpha,p} \ A_{7,p} \ A_{<7,p} \ A_{II,p} \ A_{E_2,p}], \quad q_p^Y := I - P_{\mathcal{K}_p^Y}^{\text{bd}},$$

and let

$$\Lambda_{Y,p} := q_p^Y \Pi_{\Sigma(Y12),p} T_{\text{tr} \rightarrow \text{sing},p} P_{\text{tr},p}.$$

Assume the D4 nuisance image is killed by the same quotient:

$$\Pi_{\Sigma(Y12),p} T_{\text{tr} \rightarrow \text{sing},p} \mathcal{N}_p^{\text{D4}} \subseteq \mathcal{K}_p^Y.$$

Then $\Lambda_{Y,p}$ factors through the D4 quotient V_p/N_p^{D4} . Moreover, if

$$Y_{Y,p}^{\text{tar}} := \Pi_{\Sigma(Y12),p} T_{\text{tr} \rightarrow \text{sing},p} [\mathbf{C}_{Y12}]^{\text{tr}},$$

then the Schur residual

$$r_{Y,p} := (I - P_{\mathcal{K}_p^Y}^{\text{bd}})Y_{Y,p}^{\text{tar}}$$

vanishes if and only if $Y_{Y,p}^{\text{tar}} \in \mathcal{K}_p^Y$.

Proof. If $v - v' \in N_p^{\text{D4}}$, then by hypothesis

$$\Pi_{\Sigma(Y12),p} T_{\text{tr} \rightarrow \text{sing},p} (v - v') \in \mathcal{K}_p^Y,$$

hence

$$q_p^Y \Pi_{\Sigma(Y12),p} T_{\text{tr} \rightarrow \text{sing},p} (v - v') = 0.$$

Therefore $\Lambda_{Y,p}(v) = \Lambda_{Y,p}(v')$, so $\Lambda_{Y,p}$ is constant on cosets of N_p^{D4} and descends to a unique induced map on V_p/N_p^{D4} .

For the residual statement, let A_p be any matrix whose columns span $\mathcal{K}_p^Y$. Then

$$r_{Y,p} = (I - P_{\mathcal{K}_p^Y}^{\text{bd}})Y_{Y,p}^{\text{tar}}$$

and

$$\|r_{Y,p}\|^2 = \|Y_{Y,p}^{\text{tar}}\|^2 - (Y_{Y,p}^{\text{tar}})^* A_p (A_p^* A_p)^\dagger A_p^* Y_{Y,p}^{\text{tar}}.$$

Hence $r_{Y,p} = 0$ precisely when $Y_{Y,p}^{\text{tar}} \in \mathcal{K}_p^Y$. $\square$

Corollary 18.6 (Rank form). *With the notation of Lemma 18.5,*

$$\text{rank} \begin{bmatrix} \mathcal{K}_p^Y & \Pi_{\Sigma(Y12),p} T_{\text{tr} \rightarrow \text{sing},p} \mathcal{N}_p^{\text{D4}} \end{bmatrix} = \text{rank}(\mathcal{K}_p^Y)$$

if and only if the Y12 observable descends through the D4 quotient, and

$$\text{rank} \begin{bmatrix} \mathcal{K}_p^Y & Y_{Y,p}^{\text{tar}} \end{bmatrix} = \text{rank}(\mathcal{K}_p^Y) + 1$$

if and only if the primitive Y12 carrier survives the quotient.

In particular, the current $D4 \rightarrow Y12$ route reduces to the two finite checks

$$\Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing}} \mathcal{N}_p^{\text{D4}} \subseteq \mathcal{K}_p^Y \quad \text{and} \quad Y_{Y,p}^{\text{tar}} \notin \mathcal{K}_p^Y.$$

Remark 18.7 (wip-zeta-bridge-proof-y12-descent). What remains is the quotient compatibility test:

$$\Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing}} \mathcal{N}_p^{\text{D4}} \subseteq \mathcal{K}_p^Y,$$

together with primitive carrier survival:

$$Y_{Y,p}^{\text{tar}} \notin \mathcal{K}_p^Y.$$

Remark 18.8 (No-go: Taylor-complete singular retained space). If the retained singular space R_{sing} contains all Taylor/jet data through the D4 carrier orders, then

$$T_{\text{tr} \rightarrow \text{sing}} \text{span}\{[C_7]^{\text{tr}}, [C_{Y12}]^{\text{tr}}\} \subset R_{\text{sing}}.$$

Consequently

$$(I - P_{\text{sing}}) T_{\text{tr} \rightarrow \text{sing}} [C_7]^{\text{tr}} = 0, \quad (I - P_{\text{sing}}) T_{\text{tr} \rightarrow \text{sing}} [C_{Y12}]^{\text{tr}} = 0,$$

and the singular-discard forbidden channel has rank zero. Thus R_{sing} must be an actual EF-coherent retained space, not a Taylor-complete carrier space, if this Λ -route is to remain viable.

Remark 18.9 (Active no-go boundary). The current $D4 \rightarrow Y12$ bridge is only valid if the forbidden observable descends through the same quotient that kills the $D4$ nuisance block and the primitive Y12 carrier survives that quotient. If either condition fails, the bridge is structurally incompatible, not merely unfinished.

Remark 18.10 (Archived rank-two closure register). This register records the older scalar/primitive rank-two route. It is kept for citation stability and for diagnosing noncanonical branches. The active canonical branch no longer uses the scalar wall-completion pipeline displayed here as its final bridge. Instead, the rank-two output enters the certified live/dead quotient after the raw certificate $\mathfrak{R}_{2Y}(p)$ is verified or imported, and the final contradiction is made in the canonical source-graded $K9$ norm.

In the active branch, the relevant replacements are:

$$\text{raw row table } \mathfrak{R}_{2Y}(p) \implies \text{certified live/dead quotient and } L_{\text{eff},p},$$

actual-source D4 amplitude mass+positive-density extraction fork $\implies$ source-graded affine lower,

$$\text{EFCompletion}_{K9}^{\text{gr,vec}} + \text{full finite-main retention} \implies \text{same-norm upper}.$$

Consequently the unresolved local and analytic blockers listed in older versions of this register apply only to the archived scalar or primitive-only routes. The current terminal ledgers are Remarks F.10 and F.2.

Remark 18.11 (Archived finite-audit checklist after the L^2 -Gate-1 repair). This checklist records older finite-audit items and their current routing. It is not the active terminal ledger; see Remark F.10 for the current source-graded status. The archived items are:

(1) **Finite AFE anti-lock**. Prove

$$\|B_2 + P_{2,\leq X} + D_{2,\leq X}\|_{L^2(W_{\text{branch}})}^2 \geq Q^{-A} \|B_2 + P_{2,\leq X} + D_{2,\leq X}\|_{L^2(W_{\text{ref}})}^2.$$

- (2) **Central tube routing.** For $\delta = R^{-C}$, route

$$\Omega_{\text{cen}} = \{|x - \tfrac{1}{2}| < \delta\}$$

as a rescaled branch and verify both the L^2 -anti-lock and the finite/O2O3 packet-model hypotheses there.

- (3) **Canonical quadratic multiplier.** Prove

$$m_{\text{can},+}(x) = \kappa(x - z_0)^2 u(x)$$

with R^C -controlled divided differences for u , giving

$$\text{dist}(1/m_{\text{can},+}, \mathbf{C}_R[x]) \geq Q^{-A}.$$

- (4) **Near-pure Schur block.** Verify that the active Stack weight and finite/O2O3 packets satisfy the bounded-degree Fejér–Riesz charge model used in Theorem E.106. Under those structural hypotheses, the theorem proves

$$D_{-n} \notin_{Q^{-A}} \text{span}\{D_{-n+1}, \dots, D_{-n+d_0}\}$$

with polynomial loss.

- (5) **EF residual split.** Keep the three completion ledgers distinct:

$$\text{EFCompletion}_{L^2}, \quad \text{EFCompletion}_{K_9}^{\text{gr,vec}}, \quad \text{EFCompletion}_{Y_{12}}.$$

In the canonical source-graded branch the K_9 item is the full-vector tail estimate of Definition 16.159; it is not the older scalar singular-discard estimate.

- (6) **Y12 honesty.** Classify

$$\Pi_{Y_{12}} E_2$$

as absent/small, fixed/coherent, or freely tangent.

- (7) **Y12 normal quotient.** Compute whether

$$[\mathbf{C}_{Y_{12}}] \neq 0 \quad \text{in} \quad \mathcal{Q}_9^{\text{forb}}.$$

- (8) **Certified transition–singular branch.** The finite low-block leakage of the C_7/Y_{12} carrier pair is closed on the certified branch by Theorem 16.130. In the active same-norm branch, the residual upper is the source-graded full-vector estimate

$$\sum_p \|(I - P_{R_p}) \mathcal{L}_{K_9,p}^{\text{gr}} J_{\text{tr},p}(R_X)\|_{\text{gr}}^2 \leq Q^{-B} \mu(\mathcal{C}),$$

followed by full finite-main retention through P_p^{full} . Raw computations of $(I - P_{\text{sing}}^{\text{raw}}) T_{\text{tr} \rightarrow \text{sing}} [\mathbf{C}_7]^{\text{tr}}$ and $(I - P_{\text{sing}}^{\text{raw}}) T_{\text{tr} \rightarrow \text{sing}} [\mathbf{C}_{Y_{12}}]^{\text{tr}}$ remain diagnostic only.

- (9) **K_9 -conditioning.** Prove lower-order tangent observability:

$$|T_L a| \geq Q^{-A} |T_9 a| \quad \text{for every actual-source tangent } a.$$

- (10) **Tubular singular coherence.** Prove a Q^C -bounded second fundamental form / tubular radius estimate for

$$\mathcal{M}_{\text{sing}}^{\text{arith}}.$$

Remark 18.12 (Final unresolved finite decisions). After the current cleanup, the remaining decisive finite decisions are:

- (1) The proof-level singular-discard branch uses the certified retained space $R_{\text{sing}}^{\text{cert}}$, and the active final comparison uses the canonical source-graded lift. The rank-two low-block leakage is closed by Theorem 16.130. The residual upper is supplied, in the active branch, by $\text{EFCompletion}_{K_9}^{\text{gr,vec}}$ and Theorem 16.189; it is no longer a separate scalar certified-norm decision. Raw choices of $R_{\text{sing}}^{\text{raw}}$ may still have rank 0, 1, or 2, but those cases are diagnostic archives rather than active proof branches.
- (2) Decide whether $T_{\text{tr} \rightarrow \text{sing}}$ transports formed labelled carriers or recomputes $N_0/\mathcal{W}$. Transport preserves the Y_{12} -origin filter; recomputation forces a direct Schur separation theorem.

- (3) O2/O3 is now routed through full graph-block Riesz survival via the Fejér–Riesz charge anti-lock theorem. The remaining audit is structural: check the bounded-degree charge model, Fejér–Riesz factorization, and off-block packet decay for the active Stack weight. Determinant branch survival and α -admissible determinant averaging are fallback routes; pure-branch graph non-polynomiality alone still does not certify the full canonical augmented matrix.
- (4) Decide whether Y12-normality is proved by the full K_9 -residual or by the targeted Y12-signature quotient

$$\mathcal{Q}_{9,Y12}^{\text{forb},\alpha}.$$

The targeted route is weaker and should be attempted first.

Remark 18.13 (No forbidden channel, no D4 contradiction). Carrier-quotient D4 is a local visibility statement. It becomes a contradiction only after a globally forbidden channel Λ is identified. A total Bessel or row-energy upper of the form

$$\|V_{\text{glob}}\|^2 \leq Q^C \mu(\mathcal{C})$$

does not contradict a lower bound of size $Q^{-C} \mu(\mathcal{C})$.

The acceptable forbidden-channel mechanisms are:

- (1) a positive atomic upper

$$\sum_a |\Lambda_a V_{\text{glob}}|^2 \leq Q^{-B} \mu(\mathcal{C});$$

- (2) a localizable identity family

$$|\Lambda_\psi V_{\text{glob}}|^2 \leq Q^{-B} \|\psi\|^2;$$

- (3) a coboundary/curl identity such as $\delta^2 = 0$;

- (4) an approximate identity

$$\Lambda V_{\text{glob}} = \text{Err}, \quad \|\text{Err}\|^2 \leq Q^{-B} \mu(\mathcal{C}).$$

Without one of these, finite D4 remains a detector, not a proof-closing contradiction. In the active canonical branch the proof-level forbidden channel is no longer the scalar certified discard alone, but the fixed source-graded observable

$$\Lambda_p^{\text{gr},\text{full}} = (I - P_p^{\text{full}}) \mathcal{L}_{K9,p}^{\text{gr}} J_{\text{tr},p},$$

and its maximal affine projection $\hat{\Lambda}_{\text{max},p}$. The scalar certified channel $(I - P_{\text{sing}}^{\text{cert}}) T_{\text{tr} \rightarrow \text{sing}} P_{\text{tr}}$ remains an archived collapsed diagnostic; it is not the active same-norm forbidden channel.

Remark 18.14 (Current post-closure architecture). After the canonical source-graded cleanup, the live proof architecture is best recorded as

$$\boxed{\text{Agg} L^2(q''_{\text{can}}) + \text{Graph/O2O3}_{23} + \text{NormalD4}_{K9}^{\text{gr}}}.$$

The first term is branch-complete local scalar source energy:

$$E_{1,b}^{L^2}(m) = \int_{I_b} |\lambda_b^3 q''_{\text{can}}(m + \lambda_b x)|^2 dx.$$

The second term is the finite O2/O3 nondegeneracy package. Its normalized graph component is supplied by Theorem E.106 after the active packets are expressed in the bounded-degree Fejér–Riesz charge model:

$$\mathbf{C} \cdot 1 + m_{\text{can}} \mathbf{C}_R[x] \simeq m_{\text{can}} (\mathbf{C} \cdot (1/m_{\text{can}}) + \mathbf{C}_R[x]).$$

The third term is the canonical source-graded forbidden D4 channel

$$\Lambda_p^{\text{gr},\text{full}} = (I - P_p^{\text{full}}) \mathcal{L}_{K9,p}^{\text{gr}} J_{\text{tr},p},$$

together with its maximal affine projection $\hat{\Lambda}_{\text{max},p}$. The rank-two carrier algebra, effective-carrier determinant, source-weighted D4 amplitude mass, finite-main Schur complement, same-norm

upper, and atomic comparison are closed in this canonical branch. The older scalar transition–singular discard operator is retained only as a collapsed diagnostic for noncanonical routes.

The following negative results and audit boundaries are part of the current proof state.

- (i) *Pointwise base-locus rigidity is false.* At a single point one may have

$$\Gamma_G = 0, \quad \Gamma_{G,2} = 0, \quad u \neq 0.$$

Gate 2 must use interval or L^2 structure.

- (ii) *Finite same-scale packet geometry does not prove Gate 1.* Positive finite quadrature can annihilate finitely many source-curvature jets. Gate 1 must use actual-zeta arithmetic, not only local Hadamard packet geometry.
- (iii) *Band-limited singular probes do not avoid long-prime control.* In the deep singular branch, detecting the R sine moments

$$\sin(k\theta), \quad 3 \leq k \leq R+3,$$

requires analytic or band-limited probes with bandwidth at least polynomial in R . The explicit formula then sees primes with

$$\log n \lesssim R^C/c.$$

Since $cQ \ll (\log Q)^{-C_0}$, this is generally longer than the ordinary packet-frequency range $\log n \asymp Q$. Therefore S1 is a separate long-prime actual-zeta nonrecurrence theorem, not a refinement of the ordinary Wronskian input.

- (iv) *Proxy source curvature is insufficient.* A zero-kernel proxy, flexible Dirichlet polynomial, B_2 -only diagnostic, or affine-subtracted complement curvature is not the proof object. Gate 1 and Gate 2 use

$$q''_{\text{can}} = B_2 + P_2 + D_2 + E_2.$$

- (v) *Generic determinant nonvanishing is insufficient.* A Wronskian or finite determinant can be non-identically zero while becoming too small on a one-parameter prime-log orbit. The missing theorem is an actual-zeta nonrecurrence/incidence lower bound, not formal rank.
- (vi) *Filtered-minor isolation is not locally available at current depth.* Isolating one subset-product frequency would require an m -space filter larger than the packet scale or derivative depth beyond the current budget.
- (vii) *Trace positivity is not coercivity.* A positive explicit-formula or trace identity gives nonnegativity, but Gate 1 requires a polynomial lower bound on the packet interval.
- (viii) *The scalar I -coordinate is not a faithful carrier quotient.* The order-nine scalar incidence wall appears after independent carrier directions are collapsed into the same scalar coordinate. Carrier or quotient row energy should be used unless scalar I -coordinate output is explicitly required.
- (ix) *Wall escape depends on row-output architecture.* Wall escape supplies a non-wall sample somewhere in the packet, not a row lower bound at a preassigned wall point. Exact-point architectures require a separate q_0 -adic wall-row audit.

Remark 18.15 (Numerical stress tests are not no-loss certificates). The Python and finite-prime stress tests are useful for locating candidate flattening walls, row degeneracies, and finite carrier singular values. They are not proof inputs unless converted into certified finite audits. A numerical experiment may be promoted to a finite audit only after specifying:

- (a) the exact finite basis or row table;
- (b) the coefficient ring or interval enclosure used for every entry;
- (c) the Gram matrices and their certified condition bounds;
- (d) the residual/tail estimates connecting the finite model to the analytic packet model;
- (e) the exact pass/fail inequality, with margin exceeding the downstream residual budget.

Until those data are recorded, numerical evidence remains diagnostic and should not be cited as a no-loss lemma.

Remark 18.16 (Numerical and surrogate evidence archive). The numerical and surrogate experiments in this draft are diagnostic only. They do not prove the canonical Hardy Gate–1 estimate

$$\text{Agg}L^2(q''_{\text{can}}),$$

the global branch-entry/no-loss theorem, the Fejér–Riesz persistence theorem, the raw $2Y$ row/routing certificate, or the packet-to- $K9$ transfer. Their value is to rule out obvious relaxed countermechanisms and to guide the finite audits.

The current evidence archive is:

branch	scope	
packet-frequency determinant stress	$R \leq 16$, $m_{\min} = 50$, reflected/actual-like dual and E_2	Y
local Z_α -amplitude v1	1152 rows	Y
v2 direct jet/cell-sum hardening	adversarial targets from v1/determinant runs	no Yellow/Or
multiblock $K = 8$ cell test	794 rows with log-uniform support	Yellow =
small Z_α adversarial rows	$ Z_\alpha \sim 10^{-15}\text{--}10^{-12}$	amplitude, jets

The strongest empirical message is:

$$Z_\alpha(m_0) \approx 0 \not\Rightarrow \text{amplitude / jet / cell collapse}$$

in the tested packet-frequency surrogates.

The numerical evidence also has clear limits:

- (1) It does not prove fixed-width cell nonrecurrence.
- (2) It does not prove cofactor-vector nonrecurrence.
- (3) It does not prove Wronskian angle nonalignment.
- (4) It does not prove singular anti-design exclusion.
- (5) It does not replace canonical transfer from model rows to the actual object $q''_{\text{can}} = B_2 + P_2 + D_2 + E_2$.

The first constraint not broken by the adversarial relaxed models remains full actual-zeta compatibility:

$$q''_{\text{can}} = B_2 + P_2 + D_2 + E_2$$

with fixed actual coefficients, completed dual/gamma/tail structure, and certified error control. Thus the numerical evidence supports prioritizing the actual-zeta incidence/nonrecurrence theorem; it is not a substitute for that theorem.

APPENDIX A. TOY MICROSCOPIC EXPANSION

Lemma A.1 (Uniform microscopic expansion of the toy local deformation). *Fix a compact interval*

$$D = [d_-, d_+] \subset (0, \infty),$$

and let

$$\Delta_{\text{toy}}(u, d; s)$$

denote the toy local deformation matrix in the symmetric microscopic regime, with off-line parameter u , separation parameter d , and microscopic node s .

Then there exist constants $\rho_{\text{toy}} > 0$ and $u_0(D) > 0$ such that for all

$$d \in D, \quad |s| < \rho_{\text{toy}}, \quad |u| < u_0(D),$$

one has the expansion

$$\Delta_{\text{toy}}(u, d; s) = u^2 M(d) + u^4 E(u, d; s),$$

where $M(d)$ is the toy anomaly matrix, $E(u, d; s)$ is analytic in s for $|s| < \rho_{\text{toy}}$, and

$$\sup_{\substack{d \in D \\ |s| < \rho_{\text{toy}} \\ |u| < u_0(D)}} \|E(u, d; s)\| \ll 1.$$

In particular, for the microscopic nodes

$$s_j = \frac{j}{Q^2}, \quad j = 1, \dots, N,$$

with $N \leq \rho_{\text{toy}} Q^2$, one has uniformly

$$\Delta_{\text{toy}}(u, d; s_j) = u^2 M(d) + u^4 E_j(u, d), \quad \sup_j \|E_j(u, d)\| \ll 1.$$

Proof. The toy local deformation is obtained from an explicit finite-dimensional off-line perturbation of the toy block. By construction it depends smoothly (in fact analytically) on the parameters (u, s, d) in a neighborhood of

$$(u, s, d) = (0, 0, d_0), \quad d_0 \in D.$$

At $u = 0$, the deformation vanishes, and by the even symmetry of the off-line split there is no linear term in u . Hence the first nontrivial term is quadratic:

$$\Delta_{\text{toy}}(u, d; s) = u^2 M(d, s) + O(u^4),$$

uniformly on compact d -ranges and for $|s|$ in a fixed microscopic disk.

By the jet-limit computation for the toy model, the leading quadratic coefficient is independent of the microscopic node s and equals the toy anomaly matrix $M(d)$. Thus

$$M(d, s) \equiv M(d),$$

and the expansion becomes

$$\Delta_{\text{toy}}(u, d; s) = u^2 M(d) + u^4 E(u, d; s),$$

with E analytic in s and uniformly bounded on compact parameter ranges. Restricting to the microscopic nodes $s_j = j/Q^2$ gives the stated uniform bound for $E_j(u, d)$. $\square$

Remark A.2 (Work in progress: toy microscopic expansion). The proof above is schematic: it uses smoothness in (u, s, d) , vanishing at $u = 0$, and even symmetry in the off-line split to obtain the quadratic leading term, and invokes the jet-limit computation to identify the coefficient as the toy anomaly matrix $M(d)$. A later pass should make the analyticity and uniform-bound steps quantitative by recording an explicit Taylor remainder with a constant depending only on D .

APPENDIX B. FINITE FRÉCHET COEFFICIENT CALCULUS FOR THE CORRECTED-WHITENING BLOCK

This section records the finite Fréchet/Sylvester coefficient recurrences and finite gates used by the corrected-whitening package frontier of Remark 12.146. These are table contracts, not completed source tables. Their role is to make the finite computations auditable by agents and scripts without re-opening convention choices already fixed in the main text.

Remark B.1 (Fréchet coefficient normalization). The finite tables in this section use homogeneous Taylor coefficients in the ordinary z -expansion of the inverse-square-root transfer. Thus Λ , H , and G denote the actual coefficient contributions to the expanded matrix series, not raw multilinear derivatives D, D^2, D^3 unless explicitly stated.

Equivalently, if a separate computation uses derivative-normalized tensors, then the dictionary is

$$H_{\text{Taylor}} = \frac{1}{2!} H_{\text{deriv}}, \quad G_{\text{Taylor}} = \frac{1}{3!} G_{\text{deriv}}.$$

All determinant tests in this paper use the Taylor-coefficient convention unless the contrary is explicitly stated.

Remark B.2 (Stage 1 source-table status). The finite Fréchet recurrences in this section become a proof tool only after the required source tables are supplied in the same ordinary- z , pre- Φ_K ,

tagged normalization. The current Stage 1 infrastructure reduces the needed source input to a finite midpoint-jet theorem:

$$\text{Jet}_{\text{Stage1}} = \left\{ q_0^{(0)}, \dots, q_0^{(9)}; (r_i^{[a]})^{(k)} : i = 1, 2, \quad a \in \mathcal{A}_{\text{BC}}, \quad 2 \leq k \leq 9 \right\}_m.$$

The lead local obstruction currently uses $a = 2$. Hence any Stage 1 table omitting the $a = 2$ source-linear primitive cannot evaluate the present Bottleneck-C frontier.

The table generator is infrastructure, not a zeta-source proof. A promotion requires a source theorem supplying these jets, or a reduction of them to named baseline objects, in the same normalization used for B_7^{f} . Scale estimates are not a substitute for these finite coefficient tables.

B.1. Source-linear same-point primitives. For $\varepsilon \in \{-, +\}$, identified with $\varepsilon = -1, +1$, write

$$X_{\varepsilon, n} := [z^n] \mathfrak{D}_Q(\delta G_{\varepsilon}^{\text{lin}}) = \sum_{a=2}^6 X_{\varepsilon, n}^a r_a.$$

For $n \leq 2$, the nonzero source-constrained coefficients are

$$\begin{aligned} X_{\varepsilon, 0}^2 &= \frac{1}{24\pi}(I - D), \\ X_{\varepsilon, 1}^2 &= \frac{\varepsilon}{4\pi Q} S, \quad X_{\varepsilon, 1}^3 = \frac{\varepsilon}{48\pi Q^2}(I - D), \\ X_{\varepsilon, 2}^2 &= \left(\frac{1}{8\pi Q^2} + \frac{q^2}{32\pi Q^4} \right) I + \left(\frac{1}{8\pi Q^2} - \frac{q^2}{32\pi Q^4} \right) D, \\ X_{\varepsilon, 2}^3 &= \frac{1}{16\pi Q^3} S, \quad X_{\varepsilon, 2}^4 = \frac{1}{192\pi Q^4}(I - D). \end{aligned}$$

All other $X_{\varepsilon, n}^a$ with $n \leq 2$, $2 \leq a \leq 6$, vanish.

Remark B.3. These coefficients are not free symbolic inputs. They are the residual-source normal form of the source-linear same-point perturbation.

B.2. Baseline mixed block coefficients. Let

$$\begin{aligned} s &= \frac{z}{Q^2}, \quad q_{\pm}(z) = q_0 \left(m \pm \frac{z}{2Q^2} \right), \\ \Delta_0(z) &= \Phi_0(t_-) - \Phi_0(t_+), \quad C(z) = \cos \Delta_0(z), \quad S(z) = \sin \Delta_0(z). \end{aligned}$$

The baseline mixed block is

$$N^{(0)}(z) = \frac{1}{\pi} \begin{pmatrix} -\frac{2S}{s} & \frac{S + q_+ s C}{s^2} \\ -\frac{S + q_- s C}{s^2} & \frac{(q_- + q_+) s C + (2 - q_- q_+ s^2) S}{2s^3} \end{pmatrix}.$$

For $j \geq 0$, write

$$N_j^{(0)} = [z^j] N^{(0)}(z).$$

Then $N_{0..4}^{(0)}$ are obtained entrywise from the coefficient formulas

$$\begin{aligned} (N_j^{(0)})_{11} &= -\frac{2Q^2}{\pi} [z^{j+1}] S(z), \\ (N_j^{(0)})_{12} &= \frac{Q^4}{\pi} \left([z^{j+2}] S(z) + \frac{1}{Q^2} [z^{j+1}] (q_+(z) C(z)) \right), \\ (N_j^{(0)})_{21} &= -\frac{Q^4}{\pi} \left([z^{j+2}] S(z) + \frac{1}{Q^2} [z^{j+1}] (q_-(z) C(z)) \right), \\ (N_j^{(0)})_{22} &= \frac{Q^6}{2\pi} \left(\frac{1}{Q^2} [z^{j+2}] ((q_- + q_+) C) + 2[z^{j+3}] S - \frac{1}{Q^4} [z^{j+1}] (q_- q_+ S) \right). \end{aligned}$$

B.3. First Fréchet coefficient table. Let $P_{\varepsilon,0}$ be the zeroth square-root coefficient of $G_{\varepsilon}^{(0)}$, and let $\Sigma_{\varepsilon}(C)$ denote the unique solution of

$$P_{\varepsilon,0}Z + ZP_{\varepsilon,0} = C.$$

For each source index a , define $S_{\varepsilon,n}^a$ recursively by

$$S_{\varepsilon,n}^a = \Sigma_{\varepsilon} \left(X_{\varepsilon,n}^a - \sum_{j=1}^n (P_{\varepsilon,j} S_{\varepsilon,n-j}^a + S_{\varepsilon,n-j}^a P_{\varepsilon,j}) \right).$$

Then

$$\Lambda_{\varepsilon,n}^a = - \sum_{i+j+k=n} W_{\varepsilon,i} S_{\varepsilon,j}^a W_{\varepsilon,k}.$$

The required ranges used below include

$$\Lambda_{2,0..4}^-, \quad \Lambda_{3,1..5}^-, \quad \Lambda_{4,2..6}^-.$$

B.4. Second Fréchet coefficient table. For an ordered source pair (a, b) , define $T_{\varepsilon,n}^{ab}$ by

$$T_{\varepsilon,n}^{ab} = \Sigma_{\varepsilon} \left(- \sum_{i+j=n} (S_{\varepsilon,i}^a S_{\varepsilon,j}^b + S_{\varepsilon,i}^b S_{\varepsilon,j}^a) - \sum_{j=1}^n (P_{\varepsilon,j} T_{\varepsilon,n-j}^{ab} + T_{\varepsilon,n-j}^{ab} P_{\varepsilon,j}) \right).$$

The corresponding second Fréchet coefficient is

$$H_{\varepsilon,n}^{ab} = \sum_{i+j+k+\ell+m=n} \left(W_{\varepsilon,i} S_{\varepsilon,j}^b W_{\varepsilon,k} S_{\varepsilon,\ell}^a W_{\varepsilon,m} + W_{\varepsilon,i} S_{\varepsilon,j}^a W_{\varepsilon,k} S_{\varepsilon,\ell}^b W_{\varepsilon,m} \right) - \sum_{i+j+k=n} W_{\varepsilon,i} T_{\varepsilon,j}^{ab} W_{\varepsilon,k}.$$

The five-family endpoint gate uses

$$H_{32,1..5}^-, \quad H_{34,3..7}^-, \quad H_{24,2..6}^-.$$

B.5. Third Fréchet coefficient table. For source triple (a, b, c) , define $U_{\varepsilon,n}^{abc}$ by the third Sylvester recursion

$$P_{\varepsilon,0} U_{\varepsilon,n}^{abc} + U_{\varepsilon,n}^{abc} P_{\varepsilon,0} = - \sum_{i+j=n} \left(S_{\varepsilon,i}^a T_{\varepsilon,j}^{bc} + S_{\varepsilon,i}^b T_{\varepsilon,j}^{ac} + S_{\varepsilon,i}^c T_{\varepsilon,j}^{ab} + T_{\varepsilon,i}^{bc} S_{\varepsilon,j}^a + T_{\varepsilon,i}^{ac} S_{\varepsilon,j}^b + T_{\varepsilon,i}^{ab} S_{\varepsilon,j}^c \right) - \sum_{j=1}^n (P_{\varepsilon,j} U_{\varepsilon,n-j}^{abc} + U_{\varepsilon,n-j}^{abc} P_{\varepsilon,j}).$$

The all-left third Fréchet coefficient $G_{abc,n}^-$ is then obtained from the coefficient of the cubic source monomial in

$$(G_{-}^{(0)} + \text{source})^{-1/2},$$

equivalently by the coefficient recursion determined by $U_{\varepsilon,n}^{abc}$, the lower S - and T -coefficients, and the inverse coefficients $W_{\varepsilon,n}$.

The seven-family audit for the first surviving residual requires

$$G_{324,3..7}^- \quad \text{and} \quad G_{324,3..7}^+.$$

Remark B.4 (Status of the third-Fréchet table). The recursion above defines the finite third-Fréchet coefficient table that a raw seven-family audit would require. It does not assert that the raw $L_3 N_0 R_0 / L_0 N_0 R_3$ completion is quotient-gauge. Under the current package architecture, this table is a diagnostic tool for raw local coefficients. The theorem object is the normalized corrected package $C_{\mathcal{W}}^{\text{corr}}$ of Definition C.8, not the raw seven-family Fréchet sum.

Remark B.5 (Convention: transfer degree versus source bidegree). We write d for homogeneous transfer degree in

$$\mathcal{T}(X) = \sum_{d \geq 1} \mathcal{T}_d(X),$$

and reserve (p, q) for source bidegree in the source-tagged expansion

$$C(\tau_1, \tau_2) = \sum_{p,q} \tau_1^p \tau_2^q C_{p,q}.$$

Remark B.6 (Fixed-sector projection in the moving fiber). When the midpoint is h , the fixed-sector projection is denoted $\pi_{\mathfrak{f}_h}$. The transpose-invariance identity

$$\pi_{\mathfrak{f}_h}(Z^T) = \pi_{\mathfrak{f}_h}(Z)$$

is applied in the moving midpoint fiber before midpoint trivialization to m .

Lemma B.7 (Universal first-collision formula by transfer degree). *Let*

$$\mathcal{T}(X) = \sum_{d \geq 1} \mathcal{T}_d(X)$$

be the homogeneous transfer expansion of the corrected-whitening map. On the collision chart

$$h_1 = h - \frac{\delta}{2}, \quad h_2 = h + \frac{\delta}{2}, \quad a_2 = -a_1(1 + \kappa\delta),$$

the first-collision contribution of homogeneous transfer degree d has the form

$$F_{1,d} = [z^7] \pi_{\mathfrak{f}_h} \left[(-1)^{d+1} d\kappa \mathcal{T}_d(X^d) + \frac{d}{2} (1 - (-1)^d) \mathcal{T}_d(X^{d-1}, X_h) \right].$$

Thus even d contributes only the amplitude term

$$[z^7] \pi_{\mathfrak{f}_h} \mathcal{T}_d(X^d),$$

whereas odd d contributes both the amplitude term and the source-motion term

$$[z^7] \pi_{\mathfrak{f}_h} \mathcal{T}_d(X^{d-1}, X_h).$$

Proof sketch. The factor $a_2 = -a_1(1 + \kappa\delta)$ produces the amplitude derivative term proportional to $d\kappa$. The endpoint displacement $h_1 = h - \delta/2$, $h_2 = h + \delta/2$ produces the midpoint-motion insertion. The parity of d controls whether the endpoint-motion contributions cancel or add: for even d the X_h -insertion cancels, while for odd d it appears with coefficient d . $\square$

B.6. Fourth-transfer coefficient table for Gate B. The corrected-whitening map is

$$\mathcal{W}(G_-, N, G_+) = G_-^{-1/2} N G_+^{-1/2}.$$

Let a labelled source input be

$$X_j = (A_j, Y_j, B_j),$$

where

$$A_j = \delta G_{-,j}^{\text{lin}}, \quad Y_j = \delta N_j^{\text{lin}}, \quad B_j = \delta G_{+,j}^{\text{lin}}.$$

Write

$$(G_-^{(0)} + A)^{-1/2} = L_0 + L_1[A] + L_2[A, A] + L_3[A, A, A] + L_4[A, A, A, A] + \cdots,$$

and

$$(G_+^{(0)} + B)^{-1/2} = R_0 + R_1[B] + R_2[B, B] + R_3[B, B, B] + R_4[B, B, B, B] + \cdots.$$

The middle block is source-linear:

$$N = N_0 + Y.$$

For a labelled four-source set $J = \{1, 2, 3, 4\}$, the fourth homogeneous corrected-whitening transfer is

$$\mathcal{T}_4(X_1, X_2, X_3, X_4) = \sum_{I \subseteq J} L_I^- N_0 R_{J \setminus I}^+ + \sum_{m \in J} \sum_{I \subseteq J \setminus \{m\}} L_I^- Y_m R_{J \setminus (I \cup \{m\})}^+.$$

The first sum consists of

$$L_4 N_0 R_0, \quad L_3 N_0 R_1, \quad L_2 N_0 R_2, \quad L_1 N_0 R_3, \quad L_0 N_0 R_4,$$

and the second sum consists of

$$L_3 Y R_0, \quad L_2 Y R_1, \quad L_1 Y R_2, \quad L_0 Y R_3.$$

The ordinary z^7 -coefficient is

$$[z^7]\mathcal{T}_4(J) = \sum_{I \subseteq J} \sum_{\alpha+\beta+\gamma=7} L_{I,\alpha}^- N_\beta^{(0)} R_{J \setminus I, \gamma}^+ + \sum_{m \in J} \sum_{I \subseteq J \setminus \{m\}} \sum_{\alpha+\beta+\gamma=7} L_{I,\alpha}^- Y_{m,\beta} R_{J \setminus (I \cup \{m\}), \gamma}^+.$$

For a source multiset, $\text{Sym } \mathcal{T}_4$ denotes the labelled sum divided by the automorphism factor of the multiset.

Lemma B.8 (All-subset midpoint parity). *Let K be any labelled subset of homogeneous source inputs and define*

$$|K| := \sum_{g \in K} g.$$

At leading homogeneous order, the left and right same-point inverse-square-root coefficients satisfy

$$R_K^+ = (-1)^{|K|} L_K^-.$$

Remark B.9 (Scope of all-subset midpoint parity). Lemma B.8 is a leading homogeneous coefficient statement. It uses the midpoint chart

$$t_\pm = h \pm \frac{z}{2Q^2}$$

and the Taylor-coefficient Fréchet convention. If one works with derivative-normalized Fréchet tensors, ordered unsymmetrized source slots, or non-leading ordinary coefficients, the parity factors and multiplicities must be translated back to this convention before applying the subset-complement cancellation.

Lemma B.10 (Symmetry of same-point subset coefficients). *For every labelled subset K ,*

$$(L_K^-)^T = L_K^-, \quad (R_K^+)^T = R_K^+.$$

Moreover $N_0[0]^T = N_0[0]$.

Lemma B.11 (Leading Y -middle parity). *The leading Y -middle input satisfies*

$$Y_{\text{lead}}^{[a]} = 0 \quad (a \text{ odd}),$$

and

$$(Y_{\text{lead}}^{[a]})^T = Y_{\text{lead}}^{[a]} \quad (a \text{ even}).$$

Remark B.12 (Scope of the leading Y -middle parity table). Lemma B.11 concerns the leading homogeneous ordinary-order coefficient of the source-linear mixed middle input in the active homogeneous scalar-grade convention. It is not a statement about every ordinary coefficient of a non-homogeneous matrix-output $M^{[5]}$ table.

In particular, older coefficient audits that found antisymmetric J -channel middle terms belong to the quarantined non-homogeneous matrix-output convention. They do not contradict the leading homogeneous Y -middle parity table used in Gate B.

Theorem B.13 (Odd total-grade subset-complement cancellation). *Let*

$$J = (g_1, \dots, g_d)$$

be a labelled homogeneous source multiset with total grade

$$|J| = \sum_i g_i = 7.$$

Then the leading ordinary-order amplitude contribution of the corrected-whitening transfer vanishes in fixed sector:

$$[z^7]\pi_{\mathbf{i}_h} \mathcal{T}_d(X^{[g_1]}, \dots, X^{[g_d]}) = 0.$$

Proof. For the N_0 -middle part, the leading contribution is

$$\mathcal{A}_{N_0}(J) = [z^7] \pi_{\mathfrak{f}_h} \sum_{I \subseteq J} L_I^- N_0[0] R_{J \setminus I}^+.$$

By Lemma B.8,

$$R_{J \setminus I}^+ = (-1)^{|J| - |I|} L_{J \setminus I}^-.$$

Since $|J| = 7$, this sign is $-(-1)^{|I|}$. Pair I with $J \setminus I$. The paired contribution is

$$(-1)^{|I|} \pi_{\mathfrak{f}_h} \left(-L_I^- N_0[0] L_{J \setminus I}^- + L_{J \setminus I}^- N_0[0] L_I^- \right).$$

The two matrix products are transposes, and $\pi_{\mathfrak{f}_h}$ is transpose-invariant. Hence the pair cancels.

For the Y -middle part, fix the source label $m \in J$ in the middle and write $a = g_m$, $K = J \setminus \{m\}$. If a is odd, then $Y_m = 0$ by Lemma B.11. If a is even, then Y_m is symmetric and $|K| = 7 - a$ is odd, so the same subset-complement pairing cancels the fixed-sector projection. $\square$

Lemma B.14 (Grade-preserving source-motion cancellation). *Let $J = (g_1, \dots, g_d)$ be a labelled homogeneous source multiset with*

$$|J| = \sum_i g_i = 7.$$

Replace one source slot by the grade-preserving part of its midpoint derivative,

$$(\partial_h \operatorname{Gr}_{g_i} r)_{\text{same}},$$

and keep all other source slots fixed. Then the corresponding fixed-sector source-motion contribution vanishes:

$$[z^7] \pi_{\mathfrak{f}_h} \mathcal{T}_d \left(X^{[g_1]}, \dots, X_{\text{same}, h}^{[g_i]}, \dots, X^{[g_d]} \right) = 0.$$

Proof sketch. The grade-preserving derivative has the same ordinary source parity and the same endpoint transpose parity as the undifferentiated homogeneous grade in that slot. Therefore the same subset-complement pairing used in Theorem B.13 applies: each labelled subset contribution is paired with its complement, the two matrix products are transposes, and the transpose-invariant projection $\pi_{\mathfrak{f}_h}$ kills the aggregate fixed-sector contribution. $\square$

Corollary B.15 (Amplitude closure of higher transfers). *The amplitude part of every Gate B transfer of total homogeneous grade 7 vanishes in $\mathfrak{f}_h$. In particular, the aggregate fourth-transfer positive-grade contribution*

$$\sum_{J \in \{(1,1,1,4), (1,1,2,3), (1,2,2,2)\}} [z^7] \pi_{\mathfrak{f}_h} \mathcal{T}_4(J)$$

vanishes.

The same argument also kills the first lower-shadow fourth-transfer aggregate

$$[z^7] \pi_{\mathfrak{f}_h} \mathcal{T}_4(0, 1, 1, 5) = 0.$$

More generally, the subset-complement argument applies to every labelled multiset of total homogeneous grade 7, including zero-padded lower-shadow multisets, provided the full Taylor source tower is differentiated before ordinary-order truncation.

Thus the amplitude parts also vanish for

$$d = 5 : \quad (1, 1, 1, 1, 3), (1, 1, 1, 2, 2),$$

$$d = 6 : \quad (1, 1, 1, 1, 1, 2),$$

and

$$d = 7 : \quad (1, 1, 1, 1, 1, 1, 1).$$

Lemma B.16 (Raw midpoint derivative of homogeneous scalar grades). *For*

$$\mathrm{Gr}_g r(h; t) = \frac{r^{(g+2)}(h)}{(g+2)!} (t-h)^{g+2},$$

one has

$$\partial_h \mathrm{Gr}_g r = (\partial_h \mathrm{Gr}_g r)_{\text{same}} + (\partial_h \mathrm{Gr}_g r)_{\text{lower}},$$

where

$$(\partial_h \mathrm{Gr}_g r)_{\text{same}} = \frac{r^{(g+3)}(h)}{(g+2)!} (t-h)^{g+2},$$

and

$$(\partial_h \mathrm{Gr}_g r)_{\text{lower}} = -\frac{r^{(g+2)}(h)}{(g+1)!} (t-h)^{g+1}.$$

Theorem B.17 (Moving homogeneous source-frame telescoping). *For every interior grade $g \geq 1$,*

$$\boxed{(\partial_h \mathrm{Gr}_{g-1} r)_{\text{same}} + (\partial_h \mathrm{Gr}_g r)_{\text{lower}} = 0.}$$

Consequently, after labelled symmetrization, the grade-lowering part of the raw source-motion insertion cancels source-slot by source-slot against the same-basis derivative of the adjacent lower grade.

Proof. By Lemma B.16,

$$(\partial_h \mathrm{Gr}_{g-1} r)_{\text{same}} = \frac{r^{(g+2)}(h)}{(g+1)!} (t-h)^{g+1},$$

whereas

$$(\partial_h \mathrm{Gr}_g r)_{\text{lower}} = -\frac{r^{(g+2)}(h)}{(g+1)!} (t-h)^{g+1}.$$

The signs and factorials agree exactly. The cancellation is labelled-slot local, so it survives the symmetrized multilinear transfer expansion. $\square$

Lemma B.18 (Affine boundary removal in the moving source frame). *In the $\mathfrak{D}_Q$ -scaled corrected source block, admissible source inputs are 2-jet normalized at the moving midpoint:*

$$f(h) = 0, \quad f'(h) = 0.$$

Equivalently, the source-frame projection is

$$\Pi_{\geq 2, h} f(t) = f(t) - f(h) - f'(h)(t-h).$$

Thus constant and affine modes are not independent raw source inputs.

This statement is a source-admissibility assertion for the holomorphic $\mathfrak{D}_Q$ -scaled corrected source variables. It is not a license to discard lower homogeneous grades. It uses the residual-source convention in which the moving midpoint affine jet has already been subtracted; without that convention an affine component would not be killed by a formal coefficient argument alone.

This is not a blanket assertion that all lower-shadow grades are discarded. The grade 0, 2, 4 source directions may appear in the moving Taylor tower. They are handled by the interior telescoping identity when they arise from source-motion, not by reclassifying them as baseline data. The only boundary mode removed here is the affine grade -1 term.

For the homogeneous scalar-grade basis, the remaining lower boundary

$$(\partial_h \mathrm{Gr}_0 r)_{\text{lower}} = -r^{(2)}(h)(t-h)$$

is affine and is killed by $\Pi_{\geq 2, h}$. Therefore no grade -1 source term enters

$$[z^7][\delta^1]\pi_{fh}(1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}.$$

The Taylor source tower is differentiated before ordinary-order truncation; in particular, grade 8 is included when proving ordinary order 7 cancellation and truncation is applied only after ∂_h .

Proof sketch. An affine perturbation $\psi(t) = t - h$ satisfies

$$\psi(t_{\pm}) = \pm \frac{z}{2Q^2}, \quad \psi'(t_{\pm}) = 1, \quad \psi''(t_{\pm}) = 0.$$

In the $\mathfrak{D}_Q$ -scaled source variables

$$d_{\pm}(z) = \frac{z^2}{Q^4} U_{\pm}(z), \quad e_{\pm}(z) = \frac{z}{Q^2} E_{\pm}(z),$$

an affine source would force $U_{\pm}$ and $E_{\pm}$ to have poles in z . Hence affine modes are outside the admissible holomorphic source sector. This is the source-frame reason the homogeneous scalar-grade tower starts at $g = 0$, i.e. at the second derivative. $\square$

Remark B.19 (No premature ordinary-order truncation). The cancellation in Theorem B.17 is a statement about the full moving Taylor source tower. For ordinary order 7, one must include the next homogeneous grade before differentiating: the lower part of $\partial_h \text{Gr}_8 r$ cancels the apparent upper boundary coming from the same-basis derivative of $\text{Gr}_7 r$. If the tower is truncated at grade 7 before ∂_h is applied, this cancellation is artificially destroyed.

Thus “through ordinary order 7” means: differentiate the full homogeneous source tower first, perform the adjacent-grade telescoping, and only then discard ordinary powers above 7.

Proposition B.20 (Decomposition of the Gate B first-collision coefficient). *Through ordinary order 7, the Gate B first-collision coefficient decomposes as*

$$F_1 = F_1^{\text{GateA}} + F_1^{\text{amp}} + F_1^{X_h, \text{same}} + F_1^{X_h, \text{lower}} + F_1^{\text{bdry}},$$

where:

- (1) F_1^{GateA} is the leading homogeneous cubic contribution;
- (2) F_1^{amp} is the amplitude contribution of higher transfer terms $\mathcal{T}_d(X^d)$ with total homogeneous grade 7;
- (3) $F_1^{X_h, \text{same}}$ is the grade-preserving part of the source-motion insertion;
- (4) $F_1^{X_h, \text{lower}}$ is the grade-lowering part of the source-motion insertion;
- (5) F_1^{bdry} consists of the lower affine boundary and the upper truncation boundary of the moving Taylor source frame.

Under the hypotheses of Gate A closure, all-subset midpoint parity, leading Y -middle parity, moving source-frame telescoping, and 2-jet source-frame normalization, every term on the right vanishes.

Proof sketch. The decomposition follows from Lemma B.7. Gate A handles the leading cubic term. Theorem B.13 handles the amplitude part. The grade-preserving component of X_h is handled by Lemma B.14. The grade-lowering part of X_h is handled by Theorem B.17. The lower affine boundary is removed by Lemma B.18, and the upper boundary is avoided by differentiating the full Taylor source tower before truncation. $\square$

Corollary B.21 (Conditional Gate B raw edge-divisibility through ordinary order 7). *Assume that the following source-frame hypotheses have been verified for the actual $\mathfrak{D}_Q$ -scaled corrected source model, not merely for a formal homogeneous Taylor model:*

- (1) Gate A aggregate closure;
- (2) all-subset midpoint parity for same-point whitening coefficients;
- (3) symmetry of same-point whitening subset coefficients and of $N_0[0]$;
- (4) transpose-invariance of $\pi_{\mathfrak{f}_h}$;
- (5) the leading Y -middle parity table;
- (6) moving homogeneous source-frame telescoping;
- (7) affine boundary removal by 2-jet source-frame normalization; and
- (8) differentiation of the full Taylor source tower before ordinary-order truncation.

Then

$$[z^7][\delta^1] \pi_{\mathfrak{f}_h} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} = 0.$$

Under these same hypotheses, equivalently,

$$(1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} = O(a_1 \delta^2)$$

through ordinary order 7 in fixed sector.

Remark B.22 (Dependency ledger for Gate B). Corollary B.21 is a raw edge-divisibility result, not a post-edge quotient compatibility result. Its inputs are logically independent and should not be silently replaced by quotient-level language.

The dependencies are:

- (1) Gate A aggregate closure, which removes the leading homogeneous cubic non-(1, 1) contribution.
- (2) All-subset midpoint parity for the same-point whitening coefficients, which pairs labelled subsets with their complements.
- (3) Symmetry of the relevant same-point subset coefficients and of $N_0[0]$.
- (4) Transpose-invariance of the moving fixed-sector projection $\pi_{\mathfrak{f}_h}$.
- (5) Leading Y -middle parity, which removes odd Y -middle entries and makes the even entries compatible with subset-complement cancellation.
- (6) Moving homogeneous source-frame telescoping, which cancels the grade-lowering source-motion term against the same-basis derivative of the adjacent lower grade.
- (7) 2-jet source-frame normalization, which removes the affine lower boundary produced by differentiating Gr_0 .
- (8) Differentiation of the full Taylor source tower before ordinary-order truncation, which prevents an artificial upper-boundary residual.

If any one of these inputs is removed, the statement should be demoted back to a conditional Gate B target rather than used as a proved package construction. Thus the corollary should be cited as a conditional raw edge-divisibility theorem: it closes Gate B for the corrected package only after the listed parity, source-frame, and full-tower-before-truncation hypotheses have been identified with the actual source model.

Remark B.23 (Gate B remains conditional on source-model hardening). Corollary B.21 is a proved implication from its stated source-frame hypotheses. It should not yet be read as an unconditional theorem of Definition 10.6 alone.

The tagged source block gives pre-whitening $\mathcal{K}$ -linear additivity. The remaining source-model obligation is to prove that the actual $\mathfrak{D}_Q$ -scaled corrected source variables feed the nonlinear corrected-whitening transfer

$$\mathcal{W}(G_-, N, G_+) = G_-^{-1/2} N G_+^{-1/2}$$

with the same homogeneous scalar grades, midpoint subset parities, source-motion telescoping, affine-boundary removal, and full-tower-before-truncation used in Gate B.

Thus the corollary is presently conditional in exactly this sense: the raw edge-divisibility conclusion follows once these source-frame hypotheses are verified for the actual corrected package, rather than only for the formal homogeneous Taylor tower.

Remark B.24 (Negative findings from the Gate A/Gate B audit). The Gate A/Gate B audit rules out several tempting shortcuts.

- (1) **Placement-level vanishing is false.** Individual transfer placements can have nonzero fixed-sector components. The cancellation is only after labelled subset-complement aggregation and transpose-invariant fixed-sector projection.
- (2) **Family-by-family gauge is too strong.** The package target is aggregate fixed-sector cancellation or quotient invisibility. It does not require each Fréchet family to be separately proportional to $A_5^{\mathfrak{f}}$.
- (3) **Post-edge quotient compatibility is too late.** A statement such as

$$Q_{7,m}^q(C) = Q_{7,m}^q(\Pi_{1,1}C)$$

cannot be used to prove raw edge-divisibility unless it is reformulated at the raw edge layer.

- (4) **Determinant scalarization is only a good-patch coordinate.** The quotient object is

$$\mathfrak{f}_m / \mathcal{C}A_5^{\mathfrak{f}}(m).$$

The scalar determinant test is a coordinate on a nondegenerate patch, not the definition of the quotient.

- (5) **The matrix-output $M^{[5]}$ convention is quarantined.** The active proof uses homogeneous scalar grades. Replacing the grade-five source by an ordinary matrix-output order-5 coefficient reopens the Y -slot families and requires a separate no-double-counting theorem against grades 0, 2, 4.
- (6) **The Z_j -table is post-Gate-B, not the Gate-B obstruction.** After raw edge-divisibility, $Z_j[M]$ is the next pre-trivialization table for the connection calculation. It is not a missing proof of the δ^2 -edge law itself.

Remark B.25 (Mechanism of the Gate B cancellation). The Gate B cancellation is not a brute-force finite coefficient accident. It is the combination of three structural symmetries: odd total-grade subset-complement cancellation for amplitude terms, moving Taylor-frame telescoping for source-motion lower terms, and 2-jet source-frame normalization for the affine boundary. These mechanisms act before Φ_K , before determinant scalarization, and before quotient extraction. The resulting raw edge-divisibility is exactly the input needed to construct the normalized order-7 corrected package.

Remark B.26 (What remains attackable in Gate B). After Corollary B.21, the leading cubic inventory, the $d \geq 4$ amplitude terms, and the raw source-motion lower-shadow terms should not be reopened as independent obstructions unless one of the stated hypotheses is rejected. A valid attack on Gate B must now target one of the hypotheses:

all-subset midpoint parity, leading Y -middle parity, moving source-frame telescoping, affine boundary r

In particular, a nonzero unsymmetrized transfer placement is not a Gate B counterexample unless it survives the labelled aggregate and the source-frame telescoping procedure.

Remark B.27 (Gate A is not Gate B). Gate A is the leading homogeneous cubic audit. Gate B is the raw edge-divisibility theorem for the projected corrected-whitening package. Gate A closure is a necessary local input, but it is not the same statement as edge-normalized package construction. Corollary B.21 records the additional transfer and source-frame mechanisms that promote the Gate A closure to the order-7 raw edge-divisibility statement.

Remark B.28 (Unsymmetrized subchannels versus aggregate cancellation). Individual transfer placements can have nonzero fixed-sector components. This does not contradict the aggregate cancellations above. The cancellations pair each labelled subset with its complement, use midpoint parity, and then project to the transpose-invariant fixed sector. No placement-by-placement vanishing is asserted.

Remark B.29 (Raw placement-level vanishing is false). The aggregate cancellations above must not be read as placement-level vanishing. A single fourth-transfer subchannel, for instance a diagonal same-point placement in the $(1, 1, 1, 4)$ class, can produce a nonzero fixed-sector I -component. The cancellation occurs only after summing the labelled subset-complement aggregate and applying the transpose-invariant projection $\pi_{\mathfrak{f}}$. Thus the correct theorem is aggregate fixed-sector cancellation, not termwise support-zero or family-by-family gauge.

Remark B.30 (Layering of the corrected package). The raw edge-divisibility statement is pre-normalization. Only after

$$(1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} = O(a_1 \delta^2)$$

has been established may one form the amplitude-normalized package

$$-a_1^{-1} \delta^{-2} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}.$$

Post-edge quotient compatibility statements involving $Q_{7,m}^q$ should not be used as the first proof of edge divisibility.

Remark B.31 (Post-edge quotient compatibility is not a proof of edge divisibility). The formal projection identity

$$\Pi_{1,1}\mathcal{T}_{d\geq 3} = 0$$

does not by itself prove Gate B. Gate B uses the retained complement

$$(1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}},$$

and must be proved before the normalized quotient package exists. Therefore a statement such as

$$Q_{7,m}^q(C) = Q_{7,m}^q(\Pi_{1,1}C)$$

is a post-edge compatibility statement unless it is reformulated at the raw edge level. It cannot be used as the first proof that

$$(1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}} = O(a_1\delta^2).$$

Definition B.32 (Pre-trivialization edge table after Gate B). After Corollary B.21, the edge-normalized package is holomorphic through ordinary order 7. For a source monomial M , define the pre-trivialization edge coefficient

$$Z_j[M](z) := -[\epsilon^j \delta^2 M] a_1^{-1} \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} [z^{\leq 7}] (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}},$$

where $\epsilon = h - m$. Equivalently,

$$Z_j[M](z) = [\epsilon^j M] \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} [z^{\leq 7}] \hat{C}_{\mathcal{W}}^{\text{amp}},$$

once the edge-normalized package $\hat{C}_{\mathcal{W}}^{\text{amp}}$ has been formed.

Remark B.33 (Polynomial versus vector notation for $Z_j[M]$). Definition B.32 first defines the pre-trivialization edge table as a polynomial

$$Z_j[M](z) \in \mathfrak{f}_m[z]_{\leq 7}.$$

When $Z_j[M]$ appears without the argument z in the moving-line connection formula, it denotes its ordinary z^7 -coefficient:

$$Z_j[M] := [z^7] Z_j[M](z) \in \mathfrak{f}_m.$$

Thus

$$q_{\text{conn},7}[M] = \sum_{i=1}^7 \omega_m(Z_{7-i}[M], A_i)$$

uses the vector coefficient $[z^7] Z_{7-i}[M](z)$, not the full polynomial.

Remark B.34 (Pre-trivialization versus post-trivialization coefficients). The coefficients $Z_j[M]$ are pre-trivialization coefficients. If $B_j[M]$ denotes coefficients after applying the representative-level midpoint trivialization, then $B_j[M]$ already includes the moving-line connection contribution. Therefore the connection sum

$$q_{\text{conn},7}[M] = \sum_{i=1}^7 \omega_m(Z_{7-i}[M], A_i)$$

must use $Z_j[M]$, not post-trivialization $B_j[M]$, or the connection is counted twice.

Remark B.35 (Next object after Gate B). After Gate B, the next finite table is not a raw seven-family coefficient table. It is the pre-trivialization δ^2 -edge table

$$Z_j[M] = -[\epsilon^j \delta^2 M] a_1^{-1} \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} [z^{\leq 7}] (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}.$$

This table is defined only after raw edge-divisibility makes the δ^{-2} -normalized package holomorphic. The finite Fréchet/Sylvester calculus supplies the raw coefficient infrastructure for computing $Z_j[M]$; it does not replace the edge-divisibility theorem.

B.7. Midpoint parity.

Lemma B.36 (Midpoint parity of Fréchet coefficients). *In the midpoint chart,*

$$t_{\pm} = m \pm \frac{z}{2Q^2},$$

the plus-side and minus-side coefficients satisfy

$$\Lambda_{a,n}^+ = (-1)^n \Lambda_{a,n}^-,$$

and

$$H_{ab,n}^+ = (-1)^n H_{ab,n}^-,$$

for the source indices used in the finite coefficient tables. The same parity convention applies to the third Fréchet coefficients once the corresponding $G_{abc,n}^{\pm}$ table is defined.

B.8. Fixed-sector projection rule.

Lemma B.37 (Fixed-sector projection of a J -middle product). *Let*

$$H = aI + bS + cD, \quad W = pI + qS + rD.$$

Then

$$\pi_{\mathfrak{f}}(HJW) = (cq - br)I + (cp - ar)S,$$

and

$$\pi_{\mathfrak{f}}(WJH) = -(cq - br)I - (cp - ar)S.$$

Consequently,

$$\pi_{\mathfrak{f}}(WJH) = -\pi_{\mathfrak{f}}(HJW).$$

B.9. Computed finite gates.

Remark B.38 (Scope of the computed finite gates subsection). The finite gates in this subsection record diagnostic computations from the ordinary matrix-output / raw-family audit. They are useful as coefficient checks and as warnings against termwise or raw-package shortcuts, but they are not the active Gate A/Gate B proof route.

In the active homogeneous scalar-grade proof, the Y -slot families are absent at B_7 , Gate A closes by aggregate fixed-sector cancellation, and Gate B closes through ordinary order 7 by odd total-grade subset-complement cancellation, moving source-frame telescoping, and affine boundary removal. Therefore the older Y^5/Y^6 , d_{324}^- , and five-family $N^{(0)}$ -weighted residual formulas below should be read as historical diagnostics for the raw package, not as remaining theorem obligations.

Remark B.39 (Source-coefficient parity in the Y -slot). For the grade-five mixed slot Y^a , the source-coefficient table gives

$$Y^5 = 0,$$

while

$$Y^6 = y_6 J, \quad y_6 = \frac{1}{53760\pi Q^9}.$$

Thus the Y^5 -slot residuals vanish individually, and the Y^6 -slot residuals cancel in left-right pairs by Lemma B.37.

Remark B.40 (First invisible endpoint residual). For the invisible endpoint pair $(3, 4)$ with middle source $q = 2$, the three-family C_2 gate reduces to one scalar

$$d_{324}^- = y_4 \Xi_{324}, \quad y_4 = -\frac{q^2}{640\pi Q^9},$$

where

$$\Xi_{324} = \mathcal{D}(H_{2,32}^- JW_0 - H_{1,32}^- JW_1).$$

The accompanying d_{324}^+ term vanishes by $Y^3 = 0$, and the d_{234}^+ term vanishes by source-degree support $H_{0..2;34}^- = 0$.

Remark B.41 (Five-family $N^{(0)}$ -weighted endpoint gate). In the older raw-family diagnostic audit, the N_0 -middle families did not collapse to the $r + t = 2$ range. They involved the full baseline middle factor and hence the convolution

$$n + s + t = 7.$$

For the first endpoint residual define

$$\begin{aligned} E_{324}^- &= \mathcal{D} \sum_{\substack{1 \leq n \leq 5 \\ 2 \leq t \leq 6 \\ n+t \leq 7}} (-1)^t H_{32,n}^- N_{7-n-t}^{(0)} \Lambda_{4,t}^-, \\ E_{234}^+ &= \mathcal{D} \sum_{\substack{0 \leq n \leq 4 \\ 3 \leq t \leq 7 \\ n+t \leq 7}} (-1)^t \Lambda_{2,n}^- N_{7-n-t}^{(0)} H_{34,t}^-, \\ E_{324}^+ &= \mathcal{D} \sum_{\substack{1 \leq n \leq 5 \\ 2 \leq t \leq 6 \\ n+t \leq 7}} (-1)^t \Lambda_{3,n}^- N_{7-n-t}^{(0)} H_{24,t}^-. \end{aligned}$$

The five-family residuals are

$$\begin{aligned} R_5^{(A)} &= 2y_4 \Xi_{324} + 2E_{324}^- + E_{234}^+, \\ R_5^{(C)} &= y_4 \Xi_{324} + E_{324}^- + 2E_{324}^+. \end{aligned}$$

These are finite coefficient identities, not consequences of support or generic Fréchet algebra.

In the active Gate B proof these raw-family residual formulas are superseded by Corollary B.21. They should not be listed as live post-Gate-B obstructions unless one rejects one of the Gate B hypotheses in Remark B.26.

B.10. Raw-package failure status.

Remark B.42 (Computational status of the raw package shortcut). Source-constrained non-constant baseline tests show that the raw five-family and raw seven-family Fréchet packages are not generally quotient-gauge under the provisional fixed-sector line. The constant baseline is quotient-inconclusive because $A_5^{\mathfrak{f}} = 0$. Nonconstant linear baselines give nonzero determinant residuals.

The possible Fréchet-normalization rescue has also been checked: both the derivative-normalized weights

$$(\lambda_2, \lambda_3) = (1, 1)$$

and the homogeneous Taylor weights

$$(\lambda_2, \lambda_3) = \left(\frac{1}{2}, \frac{1}{6} \right)$$

leave nonzero residuals in the tested nonconstant baselines.

These tests reject the raw UV-026 shortcut. They do not test UV-026 itself, which is the package quotient-gauge statement for

$$C_{\mathcal{W}}^{\text{corr}}.$$

The corrected package representative is the object defined in Definition C.8; the finite recurrences and raw gates are a coefficient toolkit, not a closure of the package theorem.

The word “inconclusive” for the constant baseline is meant in the precise sense of Remark C.10: when $A_5^{\mathfrak{f}} = 0$, the determinant coordinate does not define the fixed-target quotient. The nonconstant baseline failures occur on nondegenerate good patches and therefore reject the raw Fréchet shortcut under the tested convention. They do not replace the corrected package theorem, whose object is $C_{\mathcal{W}}^{\text{corr}}$.

In particular, these computations belong to the old non-homogeneous matrix-output $M^{[5]}$ / fixed-midpoint shortcut lane; they are not tests of the homogeneous scalar-grade Gate A/Gate B mechanism above.

APPENDIX C. CORRECTED FIXED-TARGET QUOTIENT PACKAGE

This section defines the corrected package object targeted by the package frontier of Remark 12.146. The sequence of definitions is: raw matrix cross-effect before Φ_K , source-bidegree projection, amplitude normalization, representative-level midpoint trivialization, the normalized package representative, the order-7 quotient extractor, and the lower moving package coefficients with the order-7 connection correction. The closing remark separates definition material from the theorem obligations that remain. Throughout this section, the shorthand raw labels the pre-projection cross-effect and amp labels the amplitude-normalized package; these are not new macros, only superscripts.

The definitions in this section are package definitions. Raw edge-divisibility through ordinary order 7 is supplied by Corollary B.21, conditional on the stated source-frame, parity, and boundary hypotheses. These definitions do not by themselves prove the later quotient-gauge theorem

$$B_7^f(C_{\mathcal{W}}^{\text{corr}}) \in \mathbf{CA}_5^f(m),$$

nor do they prove the post-Gate-B connection cancellation identities for M_A and M_C . Those obligations are listed in Remark C.25.

Remark C.1 (Naming convention for UV-026). In what follows, “UV-026” means the corrected package quotient-gauge target:

$$B_7^f(C_{\mathcal{W}}^{\text{corr}}) \in \mathbf{CA}_5^f(m),$$

where $C_{\mathcal{W}}^{\text{corr}}$ is the normalized, source-bidegree-projected, fixed-target corrected-whitening package representative.

The older statement that the raw seven-family cubic Fréchet inventory is already $A_5^f(m)$ -gauge is called the *raw UV-026 shortcut*. It is not the theorem object.

Thus, in the current draft, “UV-026” always refers to the corrected package quotient-gauge theorem. The live coefficient gates are merely local tests for that theorem, not separate theorem statements.

Theorem C.2 (Corrected UV-026 quotient-gauge theorem on the quintic-good patch). *Work on a good patch on which*

$$A_5^f(m) \neq 0.$$

Assume that Gate B raw edge-divisibility has supplied the holomorphic edge-normalized object

$$\widehat{C}_{\mathcal{W}}^{\text{amp}} = -a_1^{-1} \delta^{-2} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}$$

through ordinary order 7 on the collision chart. Let $C_{\mathcal{W}}^{\text{corr}}$ denote the fixed-target fixed-sector package representative obtained from this object by

$$C_{\mathcal{W}}^{\text{corr}} = B_{\text{pkg}}^f(h; z) = \sum_{n=0}^7 z^n \text{Triv}_{h \rightarrow m} Z_n(h),$$

where

$$\pi_{\mathfrak{f}_h}[z^{\leq 7}] \widehat{C}_{\mathcal{W}}^{\text{amp}}(h) = \sum_{n=0}^7 z^n Z_n(h).$$

Then the corrected UV-026 statement is

$$B_7^f(C_{\mathcal{W}}^{\text{corr}}) := [z^7] B_{\text{pkg}}^f(m; z) \in \mathbf{CA}_5^f(m),$$

equivalently

$$Q_{7,m}^q(C_{\mathcal{W}}^{\text{corr}}) = 0 \quad \text{in} \quad \mathfrak{f}_m / \mathbf{CA}_5^f(m).$$

Remark C.3 (UV-026 theorem object versus proof data). The theorem object in UV-026 is the quotient class

$$Q_{7,m}^q(C_{\mathcal{W}}^{\text{corr}}) \in \mathfrak{f}_m / \mathbf{CA}_5^f(m).$$

The source-bidegree projection, amplitude normalization, fixed-sector extraction, and midpoint trivialization are part of the definition of $C_{\mathcal{W}}^{\text{corr}}$. They should not be applied again to $C_{\mathcal{W}}^{\text{corr}}$.

The pre-trivialization edge table $Z_j[M]$ is not extra output data in the UV-026 statement. It is proof data needed to verify the connection cancellation for

$$M_A = \dot{r}_3 r_2 \dot{r}_4, \quad M_C = \dot{r}_3 \dot{r}_2 r_4.$$

The required proof identities are

$$R_7^{(A)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_A], A_i) = 0,$$

and

$$R_7^{(C)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_C], A_i) = 0.$$

Replacing $Z_j[M]$ by post-trivialization $B_j[M]$ inside this connection sum double-counts the moving-line correction.

Definition C.4 (Raw corrected-whitening matrix cross-effect before Φ_K). Let

$$\mathcal{B}_0 = (G_-^{(0)}, N^{(0)}, G_+^{(0)})$$

be the baseline corrected block, and let

$$\mathcal{B}_1(a_i, h_i)$$

denote the one-source tagged corrected block obtained by substituting only the i -th tagged source perturbation into the corrected same-point and mixed block formulae. Let

$$\mathcal{B}_2(a_1, h_1; a_2, h_2)$$

be the tagged two-source corrected block.

Let

$$\mathcal{W}(G_-, N, G_+) := G_-^{-1/2} N G_+^{-1/2}$$

be the corrected-whitening transfer, defined on the good analytic patch where the inverse square roots are holomorphic.

Define the matrix cross-effect before applying Φ_K , determinant scalarization, or quotient extraction by

$$\begin{aligned} C_{\mathcal{W}}^{\text{raw}}(a_1, h_1; a_2, h_2) &:= \mathcal{W}(\mathcal{B}_2(a_1, h_1; a_2, h_2)) - \mathcal{W}(\mathcal{B}_1(a_1, h_1)) \\ &\quad - \mathcal{W}(\mathcal{B}_1(a_2, h_2)) + \mathcal{W}(\mathcal{B}_0). \end{aligned}$$

This cross-effect is the correct interaction layer for one-amplitude collapse, but it does not by itself imply diagonal or collision vanishing of the order-7 quotient component; that remains the merger, renormalization, or same-reduced-image theorem.

Definition C.5 (Source-bidegree projection before quotient extraction). Work with formal source tags τ_1, τ_2 retained through the matrix cross-effect. For a tagged matrix series

$$F(\tau_1, \tau_2) = \sum_{p, q \geq 0} \tau_1^p \tau_2^q F_{p, q},$$

define

$$\Pi_{1,1} F := F_{1,1}.$$

The non-(1, 1) mixed source component is

$$(1 - \Pi_{1,1})_{\text{mix}} F := \sum_{\substack{p, q \geq 1 \\ (p, q) \neq (1, 1)}} F_{p, q}.$$

This projection is applied before fixed-sector extraction, before B_7^{f} , before determinant pairing, and before quotienting.

Definition C.6 (Amplitude-normalized non-(1, 1) package). On the collision chart

$$h_1 = h - \frac{\delta}{2}, \quad h_2 = h + \frac{\delta}{2}, \quad a_2 = -a_1(1 + \kappa\delta),$$

define the formally amplitude-normalized package by

$$\widehat{C}_{\mathcal{W}}^{\text{amp}} := -a_1^{-1}\delta^{-2}(1 - \Pi_{1,1})_{\text{mix}}C_{\mathcal{W}}^{\text{raw}}\left(a_1, h - \frac{\delta}{2}; -a_1(1 + \kappa\delta), h + \frac{\delta}{2}\right).$$

This is a matrix-valued object before Φ_K , before determinant scalarization, and before quotient extraction. Its use below is conditional on holomorphic extension to $\delta = 0$.

Definition C.7 (Representative-level midpoint trivialization). Let

$$A(h) := A_5^{\mathfrak{f}}(h) \in \mathfrak{f}_h$$

be the moving quintic fixed-sector line. Fix a good patch on which

$$A(m) \neq 0.$$

Fix an analytic identification

$$\tau_{h \rightarrow m} : \mathfrak{f}_h \longrightarrow \mathfrak{f}_m.$$

Write

$$A_h^m := \tau_{h \rightarrow m} A(h), \quad A_m := A(m).$$

Let

$$\omega_m : \mathfrak{f}_m \times \mathfrak{f}_m \rightarrow \mathbf{C}$$

be the fixed determinant form in the I, S -basis.

Choose a local transverse section $E_m \in \mathfrak{f}_m$ satisfying

$$\omega_m(E_m, A_m) = 1,$$

and an analytic linear functional

$$\ell_h : \mathfrak{f}_m \rightarrow \mathbf{C}$$

satisfying

$$\ell_h(A_h^m) = 1.$$

For $Z(h) \in \mathfrak{f}_h$, set

$$\widetilde{Z}(h) := \tau_{h \rightarrow m} Z(h),$$

and define

$$\text{Triv}_{h \rightarrow m} Z(h) := \ell_h(\widetilde{Z}(h))A_m + \omega_m(\widetilde{Z}(h), A_h^m)E_m.$$

Then

$$\omega_m(\text{Triv}_{h \rightarrow m} Z(h), A_m) = \omega_m(\widetilde{Z}(h), A_h^m).$$

Changing E_m or ℓ_h changes the representative only by a multiple of A_m , and hence does not change its class in

$$\mathfrak{f}_m / \mathbf{C}A_m.$$

Definition C.8 (Normalized corrected-whitening fixed-sector package representative). Assume the amplitude-normalized package

$$\widehat{C}_{\mathcal{W}}^{\text{amp}}$$

extends holomorphically to $\delta = 0$ through ordinary order 7.

First extract the moving fixed-sector z -polynomial

$$Z_{\mathcal{W}}^{\mathfrak{f}}(h; z) := \pi_{\mathfrak{f}_h}[z^{\leq 7}]\widehat{C}_{\mathcal{W}}^{\text{amp}}(h) \in \mathfrak{f}_h[z]_{\leq 7}.$$

Write

$$Z_{\mathcal{W}}^{\mathfrak{f}}(h; z) = \sum_{n=0}^7 z^n Z_n(h), \quad Z_n(h) \in \mathfrak{f}_h.$$

The midpoint trivialization is applied coefficientwise:

$$B_{\text{pkg}}^{\mathfrak{f}}(h; z) := \sum_{n=0}^7 z^n \text{Triv}_{h \rightarrow m} Z_n(h) \in \mathfrak{f}_m[z]_{\leq 7}.$$

Thus ordinary z -coefficients are extracted only after fixed-target trivialization.

Equivalently, expand the fixed-target z -polynomial in the moving midpoint parameter:

$$B_{\text{pkg}}^{\mathfrak{f}}(h; z) = \sum_{j=0}^7 (h - m)^j \mathcal{B}_j(z) + O((h - m)^8), \quad \mathcal{B}_j(z) \in \mathfrak{f}_m[z]_{\leq 7}.$$

Although the coefficients still depend on the moving parameter h , the representatives themselves have already been transported to the fixed midpoint fiber $\mathfrak{f}_m$.

When we write $C_{\mathcal{W}}^{\text{corr}}$ below, we mean this normalized fixed-target fixed-sector package representative $B_{\text{pkg}}^{\mathfrak{f}}(h; z)$, not the full unprojected matrix cross-effect.

Definition C.9 (Package-level order-7 quotient extractor). For the normalized fixed-target package representative, define

$$B_7^{\mathfrak{f}}(C_{\mathcal{W}}^{\text{corr}}) := [z^7] B_{\text{pkg}}^{\mathfrak{f}}(m; z) \in \mathfrak{f}_m.$$

The quotient-valued order-7 extractor is

$$Q_{7,m}^q(C_{\mathcal{W}}^{\text{corr}}) := \left[B_7^{\mathfrak{f}}(C_{\mathcal{W}}^{\text{corr}}) \right] \in \mathfrak{f}_m / \mathbf{C} A_5^{\mathfrak{f}}(m).$$

Remark C.10 (Good-patch convention for the quotient coordinate). The quotient

$$Q_{7,m}^q(C_{\mathcal{W}}^{\text{corr}}) \in \mathfrak{f}_m / \mathbf{C} A_5^{\mathfrak{f}}(m)$$

is the invariant object. On a good patch where

$$A_5^{\mathfrak{f}}(m) = u_5 I + v_5 S \neq 0,$$

the determinant coordinate

$$\mathcal{D}_m(Z) := (v_5 \pi_I - u_5 \pi_S) \pi_{\mathfrak{f}_m}(Z)$$

identifies this one-dimensional quotient with $\mathbf{C}$. This coordinate is not available on the rank-jump locus

$$A_5^{\mathfrak{f}}(m) = 0.$$

Consequently, baseline tests for which $A_5^{\mathfrak{f}}(m) = 0$ are quotient-inconclusive, not evidence for or against the quotient-gauge theorem.

Remark C.11 (Rank-jump locus is not covered by the good-patch quotient theorem). The corrected UV-026 theorem is a good-patch statement. The representative-level midpoint trivialization uses a nonzero quintic line

$$A_5^{\mathfrak{f}}(m) \neq 0$$

to identify the quotient

$$\mathfrak{f}_m / \mathbf{C} A_5^{\mathfrak{f}}(m).$$

On the rank-jump locus

$$A_5^{\mathfrak{f}}(m) = 0,$$

the determinant coordinate is unavailable. Moreover, the literal inclusion

$$B_7^{\mathfrak{f}}(C_{\mathcal{W}}^{\text{corr}}) \in \mathbf{C} A_5^{\mathfrak{f}}(m)$$

would reduce to the stronger assertion

$$B_7^{\mathfrak{f}}(C_{\mathcal{W}}^{\text{corr}}) = 0.$$

That zero statement is not supplied by the good-patch quotient theorem. Rank-jump cases therefore require a separate extension, blow-up, or limiting quotient-line theorem.

Remark C.12 (Two coefficient indices and two lower tables). There are two independent indexing conventions in the corrected package.

First, the subscript 7 in

$$B_7^f$$

denotes the ordinary z -order coefficient extracted from the fixed-target package polynomial.

Second, the subscript j in a lower moving table denotes the coefficient of $\epsilon^j = (h - m)^j$ after the ordinary z^7 coefficient has been selected.

There are also two lower tables. The pre-trivialization table is

$$Z_j[M] = -[z^7][\epsilon^j \delta^2 M] a_1^{-1} \tau_{h \rightarrow m} \pi_{f_h} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}},$$

and it is the table that enters the moving-line connection formula:

$$q_{\text{conn},7}[M] = \sum_{i=1}^7 \omega_m(Z_{7-i}[M], A_i).$$

The post-trivialization table is

$$B_j[M] = [z^7][\epsilon^j M] B_{\text{pkg}}^f(h; z),$$

after applying the representative-level midpoint trivialization. It records the chosen final representative, but it must not be inserted into the pre-trivialization connection formula.

Definition C.13 (Lower moving package coefficients and connection data). Let M be a residual-source monomial and set $\epsilon = h - m$. After Gate B raw edge-divisibility, define the pre-trivialization edge coefficients by

$$Z_j[M] := -[z^7][\epsilon^j \delta^2 M] a_1^{-1} \tau_{h \rightarrow m} \pi_{f_h} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}, \quad 0 \leq j \leq 6.$$

Equivalently,

$$Z_j[M] = [z^7][\epsilon^j M] \tau_{h \rightarrow m} \pi_{f_h} \hat{C}_{\mathcal{W}}^{\text{amp}}$$

once the edge-normalized package has been formed.

By contrast, let

$$B_{\text{pkg}}^f(h; z; M)$$

be the coefficient of M in the fixed-target package representative after representative-level midpoint trivialization, and write

$$B_{\text{pkg}}^f(h; z; M) = \sum_{j \geq 0} \epsilon^j \mathcal{B}_j[M](z), \quad \mathcal{B}_j[M](z) \in f_m[z]_{\leq 7}.$$

The post-trivialization lower representatives are

$$B_j[M] := [z^7] \mathcal{B}_j[M](z) \in f_m.$$

These coefficients are useful for recording the final representative, but they must not be inserted into the moving-line connection formula, because the trivialization has already transported along the moving quotient line.

For the connection formula, all moving-line coefficients are first transported to the fixed target f_m . Thus set

$$A_h^m := \tau_{h \rightarrow m} A_5^f(h), \quad A_h^m = \sum_{i \geq 0} \epsilon^i A_i, \quad A_i = u_i^A I + v_i^A S \in f_m.$$

Equivalently, every occurrence of

$$A_i = [\epsilon^i] A_5^f(h)$$

inside $\omega_m(\cdot, A_i)$ means

$$A_i = [\epsilon^i] \tau_{h \rightarrow m} A_5^f(h).$$

If

$$Z_j[M] = u_j^Z[M] I + v_j^Z[M] S,$$

then the order-7 quotient-line connection correction on M is

$$q_{\text{conn},7}[M] = \sum_{i=1}^7 \omega_m(Z_{7-i}[M], A_i).$$

In coordinates,

$$q_{\text{conn},7}[M] = \sum_{i=1}^7 (u_{7-i}^Z[M] v_i^A - u_i^A v_{7-i}^Z[M]).$$

If $A_i, Z_j[M]$ are derivative-normalized rather than coefficient-normalized, replace this formula by

$$q_{\text{conn},7}[M] = \sum_{i=1}^7 \binom{7}{i} \omega_m(Z_{7-i}[M], A_i).$$

Remark C.14 (Representative choices and lower coefficient tables). There are two different kinds of lower coefficients.

The pre-trivialization coefficients $Z_j[M]$ are extracted before applying $\text{Triv}_{h \rightarrow m}$. These are the coefficients that enter the moving-line connection formula

$$q_{\text{conn},7}[M] = \sum_{i=1}^7 \omega_m(Z_{7-i}[M], A_i).$$

The post-trivialization coefficients $B_j[M]$ record the chosen fixed-target representative after representative-level midpoint trivialization. They are useful for describing the final normalized representative, but they already contain the moving-line correction.

Consequently, one may not arbitrarily replace

$$Z_j[M] \mapsto Z_j[M] + c_j A_0$$

inside the connection sum unless the direct term and the moving-line expansion are transformed coherently as part of the same analytic representative choice. Likewise, post-trivialization shifts

$$B_j[M] \mapsto B_j[M] + c_j A_0$$

are harmless only when $B_j[M]$ is used solely as a quotient representative; they are not harmless if the shifted $B_j[M]$ is then inserted into a separate connection formula.

Thus coefficient tables may choose an analytic section, but they must keep the section fixed through the final quotient-coordinate calculation. Changing representatives in the middle of a finite table is not a valid simplification.

Remark C.15 (Pre- and post-trivialization lower coefficients). Definition C.13 uses pre-trivialization edge coefficients $Z_j[M]$ in the connection formula

$$q_{\text{conn},7}[M] = \sum_{i=1}^7 \omega_m(Z_{7-i}[M], A_i).$$

Post-trivialization coefficients $B_j[M]$ already include the moving-line connection contribution. Inserting $B_j[M]$ into the displayed pre-trivialization connection formula would double-count the moving-line correction.

Remark C.16 (Next package table required for the connection correction). The next computable package table is not a raw Fréchet-family table and not a post-trivialization representative table. The connection correction requires the transported moving-line table

$$A_i = [\epsilon^i] \tau_{h \rightarrow m} A_5^{\mathfrak{f}}(h), \quad 1 \leq i \leq 7, \quad \epsilon = h - m,$$

together with the pre-trivialization edge coefficients

$$Z_j[M_A] = -[z^7] [\epsilon^j \delta^2 M_A] a_1^{-1} \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}},$$

and

$$Z_j[M_C] = -[z^7] [\epsilon^j \delta^2 M_C] a_1^{-1} \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}, \quad 0 \leq j \leq 6,$$

where

$$M_A = \dot{r}_3 r_2 \dot{r}_4, \quad M_C = \dot{r}_3 \dot{r}_2 r_4.$$

The connection-cancellation identities to prove are

$$R_7^{(A)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_A], A_i) = 0,$$

and

$$R_7^{(C)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_C], A_i) = 0.$$

The post-trivialization coefficients $B_j[M]$ may be recorded after applying $\text{Triv}_{h \rightarrow m}$, but inserting them into the displayed connection formula double-counts the moving quotient-line correction.

Remark C.17 (Status of the M_A, M_C connection identities). The identities

$$R_7^{(A)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_A], A_i) = 0,$$

and

$$R_7^{(C)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_C], A_i) = 0$$

are not proved in the present draft by parity alone. The available antisymmetry and subset-complement mechanisms are Gate B edge mechanisms; they do not by themselves evaluate the post-Gate-B moving-line connection sum.

The viable closure mechanism is cancellation between the direct corrected order-7 residuals $R_7^{(A)}, R_7^{(C)}$ and the connection contributions computed from the lower pre-trivialization table $Z_j[M_A], Z_j[M_C]$. Therefore the current status is:

connection closure depends on the missing finite Z_j -tables and corrected R_7 -tables.

In particular, $R_7^{(A)}$ and $R_7^{(C)}$ must be corrected-package residuals in the same pre- Φ_K , pre-determinant convention. Old raw seven-family residuals must not be substituted for these quantities.

Lemma C.18 (Source-grade reflection). *Let S denote endpoint swap $h_- \leftrightarrow h_+$, equivalently $\delta \mapsto -\delta$. Under the homogeneous scalar-grade convention*

$$\text{Gr}_g r(t) = \frac{r^{(g+2)}(m)}{(g+2)!} (t-m)^{g+2},$$

the source-linear grade extractor P_g satisfies

$$P_g \circ S = (-1)^g S \circ P_g.$$

Proof. Reflection about the midpoint sends $t-m \mapsto -(t-m)$. Hence

$$(t-m)^{g+2} \mapsto (-1)^{g+2} (t-m)^{g+2} = (-1)^g (t-m)^{g+2}.$$

This is exactly the asserted commutation law. □

Lemma C.19 (Corrected mixed-block endpoint-swap identity). *The corrected mixed block satisfies*

$$N^{\text{corr}}(h_-, h_+)^T = N^{\text{corr}}(h_+, h_-).$$

Proof. Writing the corrected mixed block as

$$N^{\text{corr}}(q_-, q_+, \Delta; s) = \frac{1}{\pi} \begin{pmatrix} -\frac{2 \sin \Delta}{s} & \frac{\sin \Delta + q_+ s \cos \Delta}{s^2} \\ -\frac{\sin \Delta + q_- s \cos \Delta}{s^2} & \frac{(q_- + q_+) s \cos \Delta + (2 - q_- q_+ s^2) \sin \Delta}{2s^3} \end{pmatrix},$$

endpoint swap sends

$$(q_-, q_+, \Delta, s) \mapsto (q_+, q_-, -\Delta, -s).$$

Using $\sin(-\Delta) = -\sin \Delta$, $\cos(-\Delta) = \cos \Delta$, and $(-s)^2 = s^2$, $(-s)^3 = -s^3$, the swapped matrix is exactly the transpose of the original matrix. $\square$

Lemma C.20 (Edge-refined Y -middle parity). *Let Y_g be the source-linear grade- g middle block extracted from N^{corr} . Then*

$$Y_g(h_-, h_+)^T = (-1)^g Y_g(h_+, h_-).$$

Consequently, if

$$Y_g(h - \delta/2, h + \delta/2) = \sum_{e \geq 0} \delta^e Y_g^{(e)}(h),$$

then

$$\left(Y_{g,\beta}^{(e)}\right)^T = (-1)^{g+e} Y_{g,\beta}^{(e)}.$$

Proof. Apply Lemma C.19 and then extract the source-linear homogeneous grade using Lemma C.18. The coefficient of δ^e gains the additional factor $(-1)^e$ because endpoint swap sends $\delta \mapsto -\delta$. $\square$

Theorem C.21 (M_A, M_C pre-trivialization connection closure). *Let*

$$M_A = \dot{r}_3 r_2 \dot{r}_4, \quad M_C = \dot{r}_3 \dot{r}_2 r_4.$$

Assume all-subset midpoint parity, transpose-invariance of $\pi_{\mathfrak{h}}$, symmetry of $N^{(0)}$, and the source-grade reflection convention of Lemma C.18. Then the pre-trivialization table satisfies

$$Z_j[M_A] = Z_j[M_C] = 0, \quad 0 \leq j \leq 7.$$

Consequently,

$$R_7^{(A)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_A], A_i) = 0,$$

and

$$R_7^{(C)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_C], A_i) = 0.$$

Proof. For the N_0 -middle block, the labelled source multiset is $J = (3, 2, 4)$, hence $|J| = 9$. Complementing a placement $I \subseteq J$ gives a factor $(-1)^{|J|} = -1$, while the δ^2 -edge weights of M_A and M_C are complement-invariant because both monomials have two dotted endpoint sources. Hence every N_0 -middle complement pair cancels.

For the Y -middle block, fix a middle grade $m \in J$, set $K = J \setminus \{m\}$, and let $e \in \{0, 1\}$ be the number of edge derivatives carried by the middle source. The complement placement gains the matrix factor

$$(-1)^{|K|} (-1)^{m+e}.$$

The source-edge weight gains the factor $(-1)^e$. Since

$$|K| + m = |J| = 9,$$

the total complement factor is

$$(-1)^e (-1)^{|K|} (-1)^{m+e} = (-1)^{|J|} = -1.$$

Thus every Y -middle complement pair cancels. Therefore

$$Z_j[M_A] = Z_j[M_C] = 0$$

for $0 \leq j \leq 7$. The displayed connection identities follow immediately from

$$R_7^{(X)} = \omega_m(Z_7[M_X], A_0), \quad X \in \{A, C\}.$$

$\square$

Remark C.22 (Scope of the M_A, M_C connection closure). Theorem C.21 closes the M_A, M_C pre-trivialization connection branch under the stated source-grade reflection convention. It does not prove the full actual-source Gate B upgrade

$$\Theta_7^{\text{actual}} = 0,$$

and it does not address the promoted nonlinear mixed-block Bottleneck-C coefficient

$$E_{A,Y}^{\text{BC}} = \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \text{Sym}[z^7] D(G_-^{-1/2})[A_-] Y_{12}^{(2)} R_0, \quad Y_{12}^{(2)} = D^2 N_0[\xi_1, \xi_2],$$

which remains a separate actual-source/Bottleneck-C obstruction.

Remark C.23 (Connection table reduction for the residual monomials). The post-Gate-B connection identities for

$$M_A = \dot{r}_3 r_2 \dot{r}_4, \quad M_C = \dot{r}_3 \dot{r}_2 r_4$$

are finite pre-trivialization table identities. They are not raw seven-family identities and they must not be tested using the post-trivialization coefficients $B_j[M]$.

Write

$$\epsilon = h - m, \quad A_h^m := \tau_{h \rightarrow m} A_5^{\mathfrak{f}}(h) = \sum_{i \geq 0} \epsilon^i A_i,$$

so that

$$A_i = [\epsilon^i] \tau_{h \rightarrow m} A_5^{\mathfrak{f}}(h), \quad 1 \leq i \leq 7.$$

If

$$A_i = \alpha_i I + \beta_i S,$$

and

$$Z_j[M_X] = u_j^X I + v_j^X S, \quad X \in \{A, C\},$$

then the coefficient-normalized connection contribution is

$$q_{\text{conn},7}[M_X] = \sum_{i=1}^7 \omega_m(Z_{7-i}[M_X], A_i) = \sum_{i=1}^7 (u_{7-i}^X \beta_i - \alpha_i v_{7-i}^X).$$

Thus the two required scalar identities are

$$R_7^{(A)} + \sum_{i=1}^7 (u_{7-i}^A \beta_i - \alpha_i v_{7-i}^A) = 0,$$

and

$$R_7^{(C)} + \sum_{i=1}^7 (u_{7-i}^C \beta_i - \alpha_i v_{7-i}^C) = 0.$$

If instead the table uses derivative-normalized coefficients

$$A_i = \partial_\epsilon^i \left(\tau_{m+\epsilon \rightarrow m} A_5^{\mathfrak{f}}(m + \epsilon) \right) \big|_{\epsilon=0}, \quad Z_j[M] = \partial_\epsilon^j Z[M](m + \epsilon) \big|_{\epsilon=0},$$

then

$$A_i = \frac{A_i}{i!}, \quad Z_j[M] = \frac{Z_j[M]}{j!}.$$

Consequently the coefficient-normalized order-7 term is

$$q_{\text{conn},7}[M_X] = \frac{1}{7!} \sum_{i=1}^7 \binom{7}{i} \omega_m(Z_{7-i}[M_X], A_i).$$

Equivalently, after applying $\partial_\epsilon^7|_{\epsilon=0}$ to the whole order-7 identity, the derivative-normalized connection term is

$$\sum_{i=1}^7 \binom{7}{i} \omega_m(Z_{7-i}[M_X], A_i).$$

The coefficient-normalized and derivative-normalized conventions must not be mixed.

Definition C.24 (Finite Z_j -table target for M_A and M_C). For $X \in \{A, C\}$, set

$$M_A = \dot{r}_3 r_2 \dot{r}_4, \quad M_C = \dot{r}_3 \dot{r}_2 r_4.$$

The required pre-trivialization table entries are

$$Z_j[M_X] = -[z^7][\epsilon^j \delta^2 M_X] a_1^{-1} \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} (1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{raw}}, \quad 0 \leq j \leq 6.$$

Equivalently, after Gate B edge divisibility has supplied the holomorphic edge-normalized package,

$$Z_j[M_X] = [z^7][\epsilon^j M_X] \tau_{h \rightarrow m} \pi_{\mathfrak{f}_h} \hat{C}_{\mathcal{W}}^{\text{amp}}.$$

To compute these entries from the finite Fréchet/Sylvester calculus, one must keep the labelled source slots through the computation. For the labelled source triple $J = (3, 2, 4)$, the N_0 -middle contribution has the schematic contract

$$\sum_{I \subseteq J} \sum_{\alpha + \beta + \gamma = 7} \pi_{\mathfrak{f}_h} \left(L_{I, \alpha}^- N_{\beta}^{(0)} R_{J \setminus I, \gamma}^+ \right),$$

and the Y -middle contribution has the schematic contract

$$\sum_{m_0 \in J} \sum_{I \subseteq J \setminus \{m_0\}} \sum_{\alpha + \beta + \gamma = 7} \pi_{\mathfrak{f}_h} \left(L_{I, \alpha}^- Y_{m_0, \beta} R_{J \setminus (I \cup \{m_0\}), \gamma}^+ \right).$$

The operator

$$-[\epsilon^j \delta^2 M_X] a_1^{-1} \tau_{h \rightarrow m} (1 - \Pi_{1,1})_{\text{mix}}$$

is then applied to the labelled aggregate before determinant scalarization and before midpoint trivialization.

The current finite Fréchet recurrences supply the formal L - and R -coefficient infrastructure and the baseline $N^{(0)}$ -coefficient infrastructure. Before the edge-refined Y -middle parity theorem was isolated, this table was the direct finite computation needed for the M_A, M_C branch.

After Lemma C.20 and Theorem C.21, this table should be read as an audit fallback rather than as an open primary dependency. Under the stated actual source-grade reflection convention, the labelled $J = (3, 2, 4)$ N_0 -middle and Y -middle complement pairs cancel before midpoint trivialization:

$$Z_j[M_A] = Z_j[M_C] = 0, \quad 0 \leq j \leq 7.$$

Thus explicit evaluation of the finite Z_j -table is no longer required for the M_A, M_C connection branch unless the actual source-grade reflection convention or edge-refined Y -middle parity is later reopened.

Remark C.25 (Definition-level material and theorem obligations). Definitions C.4–C.13 specify the intended corrected package object. They do not prove its quotient-gauge property.

The following are theorem obligations.

- (1) **Source-weight-linear package input.** One must prove that the tagged pre-whitening source block induces the actual source-weight-linear input used by the corrected-whitening cross-effect. In particular, the construction must not reintroduce the rejected signed-lift shortcut in which multiplying an even whitened package by the source amplitude leaves higher $a^3, a^5, \dots$ terms. The package input must be invariant, functorial through the corrected-whitening transfer, and compatible with the quotient extractor $Q_{7,m}^q$. For the order-7 Gate B theorem, this requirement is realized concretely by the moving homogeneous source-frame hypotheses used in Corollary B.21; future attacks should therefore target those hypotheses directly rather than reintroducing the discarded signed-lift shortcut.
- (2) **Homogeneous scalar-grade source theorem.** One must prove that the residual scalar source convention

$$\text{Gr}_a r = \frac{r^{(a+2)}(m)}{(a+2)!} (t-m)^{a+2}$$

is the correct source-grade decomposition for the actual corrected package. In particular, the absence of the Y -slot families at $B_7^{\mathfrak{f}}$ relies on this homogeneous scalar-grade convention.

If a different convention defines $M^{[5]}$ as an ordinary matrix-output order-5 projection, then the Y -slot families

$$L_1 Y R_1, \quad L_2 Y R_0, \quad L_0 Y R_2$$

must be reopened and audited separately, as noted in Remark 12.155. The homogeneous scalar-grade convention is also the convention under which moving source-frame telescoping is stated; changing the source convention requires re-proving the telescoping and affine-boundary lemmas.

- (3) **Edge divisibility.** Corollary B.21 proves

$$[z^7][\delta^1]\pi_{\mathfrak{f}_h}(1 - \Pi_{1,1})_{\text{mix}}C_{\mathcal{W}}^{\text{raw}} = 0$$

under its stated source-frame, parity, and full-tower-before-truncation hypotheses. Equivalently,

$$(1 - \Pi_{1,1})_{\text{mix}}C_{\mathcal{W}}^{\text{raw}} = O(a_1\delta^2)$$

through ordinary order 7 in fixed sector.

The remaining obligation is not to re-prove the leading cubic or higher-transfer audits, but to ensure that the hypotheses of Corollary B.21 are part of the actual corrected source package, and to extend the edge law if a later argument requires orders beyond 7.

- (4) **Source-bidegree compatibility.** The source-bidegree projector must be applied before fixed-sector coefficient extraction, determinant pairing, or quotienting. At the raw edge level, Corollary B.21 proves the required first-collision divisibility through ordinary order 7 under its stated hypotheses:

$$[z^7][\delta^1]\pi_{\mathfrak{f}_h}(1 - \Pi_{1,1})_{\text{mix}}C_{\mathcal{W}}^{\text{raw}} = 0.$$

After this holomorphic edge normalization is available, the corresponding post-edge quotient formulation may be stated as

$$Q_{7,m}^q(C) = Q_{7,m}^q(\Pi_{1,1}C)$$

for the full corrected package, or equivalently as the quotient-gauge condition for the already-projected representative $C_{\mathcal{W}}^{\text{corr}}$. The post-edge statement should not be used as the proof of raw edge divisibility.

- (5) **Midpoint trivialization independence.** The quotient class in

$$\mathfrak{f}_m/\mathbf{C}A_5^{\mathfrak{f}}(m)$$

must be independent of the auxiliary representative choices E_m, ℓ_h .

- (6) **Connection cancellation.** For the residual monomials

$$M_A = \dot{r}_3 r_2 \dot{r}_4, \quad M_C = \dot{r}_3 \dot{r}_2 r_4,$$

one must prove the pre-trivialization connection identities

$$R_7^{(A)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_A], A_i) = 0,$$

and

$$R_7^{(C)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_C], A_i) = 0.$$

Here $Z_j[M]$ denotes the pre-trivialization edge coefficient of Definition C.13 (equivalently Definition B.32); it is not the post-trivialization coefficient $B_j[M]$.

- (7) **UV-026 package quotient gauge.** With the present definitions, $C_{\mathcal{W}}^{\text{corr}}$ already denotes the amplitude-normalized, non-(1, 1) mixed package after midpoint trivialization. Thus UV-026 is the quotient-gauge statement

$$B_7^{\mathfrak{f}}(C_{\mathcal{W}}^{\text{corr}}) \in \mathbf{C}A_5^{\mathfrak{f}}(m),$$

equivalently

$$Q_{7,m}^q(C_{\mathcal{W}}^{\text{corr}}) = 0.$$

If one instead works with an unprojected full corrected package $C_{\mathcal{W}}^{\text{corr,full}}$, then the equivalent full-package formulation is

$$B_7^f \left((1 - \Pi_{1,1})_{\text{mix}} C_{\mathcal{W}}^{\text{corr,full}} \right) \in \mathbf{CA}_5^f(m),$$

or

$$Q_{7,m}^q(C_{\mathcal{W}}^{\text{corr,full}}) = Q_{7,m}^q(\Pi_{1,1} C_{\mathcal{W}}^{\text{corr,full}}).$$

The source-bidegree projection must be applied before fixed-sector extraction, determinant pairing, or quotienting.

APPENDIX D. FINITE COEFFICIENT TABLE CONTRACTS FOR THE CORRECTED-WHITENING PACKAGE

Remark D.1 (Active convention for this appendix). This appendix is written under the homogeneous scalar-grade convention

$$\text{Gr}_a r = \frac{r^{(a+2)}(m)}{(a+2)!} (t-m)^{a+2}.$$

Under this convention the ordinary matrix-output $M^{[5]}$ route is quarantined. In particular, the Y -slot families

$$L_1 Y R_1, \quad L_2 Y R_0, \quad L_0 Y R_2$$

are absent at B_7 . If the project later changes to an ordinary matrix-output $M^{[5]}$ convention, then those families must be reopened and a no-double-counting theorem against grades 0, 2, 4 is required.

This appendix records the finite coefficient tables needed by the current corrected-whitening package frontier. These are table contracts, not completed source tables. Their role is to make the finite computations auditable by agents and scripts without re-opening convention choices already fixed in the main text.

All tables below are understood:

- (1) before Φ_K ;
- (2) before determinant scalarization;
- (3) in ordinary z ;
- (4) with source tags retained through the relevant 112/122 extraction;
- (5) in the same fixed-sector normalization used for B_7^f and $Q_{7,m}^q$.

The finite Fréchet recurrences in Section B become proof tools only after the source tables in this appendix are supplied. They are not, by themselves, quotient-gauge proofs.

D.1. Stage 1 source-table contract. The Stage 1 source-table generator requires the following scalar midpoint jets:

$$q_0^{(k)}(m), \quad 0 \leq k \leq 9,$$

and

$$(r_i^{[a]})^{(k)}(m), \quad i = 1, 2, \quad a \in \mathcal{A}_{\text{BC}}, \quad 2 \leq k \leq 9.$$

Thus the input table is

$$\text{Jet}_{\text{Stage1}} = \left\{ q_0^{(0)}, \dots, q_0^{(9)}; (r_1^{[1]})^{(2)}, \dots, (r_1^{[1]})^{(9)}; (r_1^{[5]})^{(2)}, \dots, (r_1^{[5]})^{(9)}; (r_2^{[1]})^{(2)}, \dots, (r_2^{[1]})^{(9)}; (r_2^{[5]})^{(2)}, \dots, (r_2^{[5]})^{(9)} \right\}_m$$

The current source supports ungraded one-pair source jets after affine removal, but it does not yet supply the graded source jets above as a proved theorem. In particular, the baseline q_0 -jet theorem

$$q_0^{(k)}(m), \quad 0 \leq k \leq 9,$$

is still a separate source obligation. Scale estimates do not substitute for this finite table.

D.2. Homogeneous scalar-grade status table. Under the homogeneous scalar-grade convention

$$\text{Gr}_a r = \frac{r^{(a+2)}(m)}{(a+2)!} (t-m)^{a+2},$$

the current B_7^f -status of the seven cubic families is:

Family	Status at B_7^f	Reason / hypothesis
$L_1 Y R_1$	absent	middle homogeneous grade-five in p
$L_2 Y R_0$	absent	second Fréchet terms cannot lower
$L_0 Y R_2$	absent	right-hand mirror of L_2
$L_2 N_0 R_1$	order-admissible placement, aggregate closed	cancels with $L_1 N_0 R_2$ by trans
$L_1 N_0 R_2$	order-admissible placement, aggregate closed	cancels with $L_2 N_0 R_1$ by trans
$L_3 N_0 R_0$	order-admissible placement, aggregate closed conditionally	cancels with $L_0 N_0 R_3$ under third-Fré
$L_0 N_0 R_3$	order-admissible placement, aggregate closed conditionally	right-hand mirror of L_3

The three Y -slot absences are order-count statements, not determinant proportionality statements. The four non- Y rows are not asserted to vanish placement by placement. They close only after the indicated aggregate pairing and fixed-sector projection. If a future theorem defines $M^{[5]}$ as an ordinary matrix-output order-5 projection, then the Y -slot families must be reopened. That route requires a no-double-counting theorem against the natural homogeneous scalar grades 0, 2, 4.

D.3. Mixed-input parity table. The full source-linear scaled mixed input satisfies

$$M(-z) = M(z)^T.$$

Consequently:

Component of $M(z)$	ordinary z -parity
diagonal entries	even
off-diagonal symmetric channel	even
off-diagonal antisymmetric channel	odd

Through ordinary order 7, the coefficient $[z^5]M$ lies in the off-diagonal antisymmetric channel and is source-supported by

$$r^{(2)}(m), \quad r^{(4)}(m), \quad r^{(6)}(m),$$

not by the homogeneous grade-five source $r^{(7)}(m)$. Therefore $[z^5]M$ should not be identified with homogeneous $\text{Gr}_5 r$ without a separate parity-corrected source-projection theorem.

D.4. Source primitive support table. For a homogeneous scalar source grade

$$\text{Gr}_a r = \frac{r^{(a+2)}(m)}{(a+2)!} (t-m)^{a+2},$$

the same-point input first appears at ordinary pre-whitening order a . The mixed input must also respect the transpose-parity table above, and need not have a nonzero coefficient at ordinary order a .

Source grade	same-point input	mixed input
1	$\delta G_{\pm}^{[1]}$ starts at z^1	use Lemma 12.148 for the corresponding mixed coefficient
5	$\delta G_{\pm}^{[5]}$ starts at z^5	$[z^5]\mathfrak{D}_Q \delta N^{[5]} = 0$ under the homogeneous convention

The grade-zero sector corresponding to $r^{(2)}(m)$ is not removed by affine normalization and should not be erased from the source table. Its correct status after the Gate B source-frame audit is more precise: grade 0 remains a live moving Taylor source direction, but its first-collision contribution is handled uniformly. Amplitude terms of total homogeneous grade 7, including zero-padded lower-shadow multisets, vanish by odd total-grade subset-complement cancellation; source-motion lower terms telescope against adjacent grades; and the sole lower boundary

$$(\partial_h \text{Gr}_0 r)_{\text{lower}} = -r^{(2)}(h)(t-h)$$

is affine and is removed by the 2-jet source-frame projection.

D.5. Baseline coefficient table contract. The baseline tables needed for the current cubic gates are:

$$G_{\pm}^{(0)}(z), \quad W_{\pm}(z) := G_{\pm}^{(0)}(z)^{-1/2}, \quad N_0(z) := N^{(0)}(z).$$

Write

$$W_{\pm}(z) = \sum_{n \geq 0} W_{\pm,n} z^n, \quad N_0(z) = \sum_{n \geq 0} N_0[n] z^n.$$

For the historical first non- Y coefficient audit $L_2 N_0 R_1 / L_1 N_0 R_2$, the leading aggregate calculation uses only the baseline mixed coefficient

$$N_0[0].$$

For the Stage 1 generator and later all-left/all-right audits, one needs the ordinary- z baseline tables through the ranges specified in Appendix D.6.

D.6. Fréchet table range contract. The finite Fréchet recurrences of Section B require the following coefficient ranges for the active UV-026 gates.

Object	Range	Use
$\Lambda_{\pm}^{[1]}$	$[z^1]$	grade-one first Fréchet input
$\Lambda_{\pm}^{[5]}$	$[z^5]$	grade-five first Fréchet input
$\mathcal{L}_{-}^{\{1,1\}}$	$[z^2]$	$L_2 N_0 R_1$ with (1, 1) on the left
$\mathcal{L}_{-}^{\{1,5\}}$	$[z^6]$	$L_2 N_0 R_1$ with (1, 5) on the left
$\mathcal{R}_{+}^{\{1,1\}}$	$[z^2]$	$L_1 N_0 R_2$ with (1, 1) on the right
$\mathcal{R}_{+}^{\{1,5\}}$	$[z^6]$	$L_1 N_0 R_2$ with (1, 5) on the right
G_{324}^{-}	$[z^3], \dots, [z^7]$	third-Fréchet parity audit / convention check
G_{324}^{+}	$[z^3], \dots, [z^7]$	right-side mirror in the same convention

Here $\mathcal{L}$ and $\mathcal{R}$ denote the left and right second Fréchet inverse-square-root outputs in the same fixed $B_7^{\dagger}$ normalization. The third-Fréchet rows are now a normalization check, not a pending Gate A determinant table. Under the symmetric multilinear third-Fréchet convention used in Lemma 12.142, the all-left/all-right pair closes in aggregate. If an ordered-triple coefficient convention is used instead, the ordered-permutation aggregate must be included consistently on both sides before invoking the transpose-pairing argument.

D.7. Diagnostic leading table for $L_2 N_0 R_1 / L_1 N_0 R_2$. Before the aggregate Gate A closure was isolated, the first homogeneous non- Y coefficient table to audit was

$$L_2 N_0 R_1 \quad \text{and} \quad L_1 N_0 R_2.$$

The table below is retained as a diagnostic coefficient contract. It is not a remaining live Gate A obstruction and it is not a family-by-family determinant-gauge requirement. The leading table is:

Tag vector	leading fixed-sector coefficient
$C_{112}^{L_2 N_0 R_1}$	$\pi_{\mathfrak{f}} \left([z^2] \mathcal{L}_{-}^{\{1,1\}} N_0[0] [z^5] \Lambda_{+}^{[5]} \right)$
$C_{122}^{L_2 N_0 R_1}$	$\pi_{\mathfrak{f}} \left([z^6] \mathcal{L}_{-}^{\{1,5\}} N_0[0] [z^1] \Lambda_{+}^{[1]} \right)$
$C_{112}^{L_1 N_0 R_2}$	$\pi_{\mathfrak{f}} \left([z^1] \Lambda_{-}^{[1]} N_0[0] [z^6] \mathcal{R}_{+}^{\{1,5\}} \right)$
$C_{122}^{L_1 N_0 R_2}$	$\pi_{\mathfrak{f}} \left([z^5] \Lambda_{-}^{[5]} N_0[0] [z^2] \mathcal{R}_{+}^{\{1,1\}} \right)$

Each entry is a fixed-sector vector in

$$\mathfrak{f} = \text{span}\{I, S\}.$$

These four vectors are retained as a coefficient-table contract, but they are no longer required to pass an A_5^f -line test family-by-family. Under the homogeneous scalar-grade convention, Lemma 12.141 proves the aggregate cancellation

$$C_{112}^{L_2 N_0 R_1} + C_{112}^{L_1 N_0 R_2} = 0, \quad C_{122}^{L_2 N_0 R_1} + C_{122}^{L_1 N_0 R_2} = 0$$

in $\mathfrak{f}$. This is stronger than quotient-gauge and does not use the A_5^f -line.

If this table is later used outside the homogeneous scalar-grade convention, then the old determinant coordinate

$$uv_5 - u_5v = 0$$

is only a good-patch audit coordinate after the fixed-sector vector has been computed; it is not the definition of the quotient object and not the current Gate A closure mechanism.

Remark D.2 (Aggregate closure versus coefficient-table diagnostics). The coefficient tables in this appendix are diagnostic contracts. They are designed to let agents reproduce the finite entries used by the aggregate proofs. They should not be read as replacing the aggregate statements in the main text. In particular, a nonzero placement-level vector is not a counterexample to Gate A unless the full aggregate specified in the relevant lemma also survives in $\mathfrak{f}$.

D.8. All-left/all-right cubic aggregate table. The all-left and all-right families

$$L_3 N_0 R_0, \quad L_0 N_0 R_3$$

are live placements by order count, but they are not remaining independent Gate A residuals under the third-Fréchet midpoint-parity convention. Let

$$G := G_{7;1,1,5}^-, \quad W := W_{-,0} = W_{+,0}, \quad N := N_0[0].$$

The two aggregate fixed-sector contributions are

$$\pi_{\mathfrak{f}}(GNW)$$

and

$$-\pi_{\mathfrak{f}}(WNG).$$

Since G, N, W are symmetric,

$$(GNW)^T = WNG,$$

and since $\pi_{\mathfrak{f}}$ is transpose-invariant,

$$\pi_{\mathfrak{f}}(GNW) = \pi_{\mathfrak{f}}(WNG).$$

Therefore

$$L_3 N_0 R_0 + L_0 N_0 R_3 = 0 \quad \text{in } \mathfrak{f}$$

at the aggregate fixed-sector level, conditional on the third-Fréchet midpoint-parity convention.

Thus the required table is not a pending quotient-gauge determinant table. The remaining obligation is to verify that the third-Fréchet coefficient normalization used by any coefficient agent is the same symmetric multilinear normalization used in Lemma 12.142. If an ordered-triple convention is used instead, the ordered-permutation aggregate must be included consistently on both sides before applying the transpose-pairing argument.

D.9. Validation checklist for coefficient agents. A valid coefficient-table submission must specify:

Item		Re
ordinary variable		
stage		
source tags		τ_1, τ_2 re
same-point input		raw vs
mixed input		M_i
baseline table		$N_0[$
projection		$\pi_{\mathfrak{f}}$ be
determinant coordinate		allowed only after the fixed-se
ordered vs symmetric source slots	state whether Fréchet coefficients are ordered-labelled or symmetric; if on	

For aggregate Gate A tables, determinant proportionality to $A_5^{\mathbf{f}}$ is not the closure mechanism unless explicitly stated. The primary object is the fixed-sector vector in $\mathbf{f}$, and the primary test is the aggregate identity in $\mathbf{f}$. The determinant coordinate

$$uv_5 - u_5v$$

is only a good-patch coordinate after the relevant fixed-sector vector or aggregate has already been formed.

A submission is not valid if it:

- (1) argues after Φ_K ;
- (2) uses determinant scalarization as the definition of the quotient;
- (3) replaces the active homogeneous grade by a matrix-output $M^{[5]}$ without the no-double-counting theorem;
- (4) treats recurrence formulas as quotient-gauge proofs without computing the fixed-sector vectors;
- (5) omits the baseline q_0 -jet convention or the grade-zero sector.

Remark D.3 (Final status of the Gate A/Gate B package layer). At this stage the corrected-package construction has the following status.

Gate A is closed at the aggregate fixed-sector level under the active homogeneous scalar-grade convention and the stated third-Fréchet midpoint parity convention. It should not be reopened as a family-by-family determinant-gauge problem.

Gate B is closed only in the conditional raw edge-divisibility sense of Corollary B.21: its use depends on the stated midpoint-parity, source-frame telescoping, 2-jet source-frame normalization, and full-Taylor-tower-before-truncation hypotheses. If one of those hypotheses is not verified in the final source model, Gate B reverts to an open raw edge-divisibility target, not to a post-edge quotient-compatibility statement.

After Gate B, the next local package table is the pre-trivialization $Z_j[M]$ -table, and the next local identities are

$$R_7^{(A)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_A], A_i) = 0, \quad R_7^{(C)} + \sum_{i=1}^7 \omega_m(Z_{7-i}[M_C], A_i) = 0.$$

The post-trivialization coefficients $B_j[M]$ must not be substituted into these connection sums.

Finally, none of Gate A, Gate B, or the $Z_j[M]$ -connection package by itself proves the global package-to- $\Xi_{\zeta}^{(N)}$ hidden-state bridge or the final finite-core contradiction. Those remain downstream theorem obligations.

APPENDIX E. UPSTREAM GATE-1 ARITHMETIC INPUTS AFTER THE ORDINARY-ROUTE AUDIT

The downstream local package of Section 14 reduces row-energy production to a single active upstream input:

$$\text{Agg}L^2(q_{\text{can}}'')$$

for the Hardy-normalized source $q_{\text{can}}'' = u_H = \partial_t^3 Z_H$, together with the canonical branch-complete source window. The live material in this section is the proof-level chain

$$\text{Hardy source API} \rightarrow \text{windowed finite AFE packets} \rightarrow W_{\text{branch}}^{\text{can}} \geq Q^{-C} W_{\text{ref}} \rightarrow \delta_{\text{src}} \geq Q^{-A}.$$

Older O1/O2/O3, S1/T1, pointwise Wronskian, numerical proxy, and raw branch-observable routes are archived diagnostics only.

E.1. Branch-complete local L^2 Gate-1. The active Gate-1 input is local scalar L^2 -source energy of

$$u = q_{\text{can}}''.$$

For a branch $b \in \{\text{ord}, \text{tr}, \text{sing}\}$, define

$$F_{b,m}(x) := \lambda_b^3 q_{\text{can}}''(m + \lambda_b x),$$

and

$$E_{1,b}^{L^2}(m) := \int_{I_b} |F_{b,m}(x)|^2 w_b(x) dx,$$

where $w_b \geq 0$ is a fixed smooth local branch window. Gate-1 on the packet m means

$$\boxed{E_{1,b}^{L^2}(m) \geq Q^{-A}.}$$

This definition is branch-independent after normalization and is the scalar source input consumed by the Gate-2 L^2 -real-rigidity theorem.

Definition E.1 (Aggregate scalar L^2 Gate-1 target). The proof-level Gate-1 target is the aggregate scalar source estimate

$$\text{Agg}L^2(q''_{\text{can}}).$$

For a branch $b \in \{\text{ord}, \text{tr}, \text{sing}\}$, define

$$F_{b,m}(x) := \lambda_b^3 q''_{\text{can}}(m + \lambda_b x),$$

and

$$E_{1,b}^{L^2}(m) := \int_{I_b} |F_{b,m}(x)|^2 w_b(x) dx.$$

In the active Hardy branch,

$$q''_{\text{can}} = u_H = \partial_t^3 Z_H,$$

and packet-normalized rows carry the explicit factor λ_b^3 .

Equivalently, using the proof-level canonical branch-complete window $W_{\text{branch}}^{\text{can}}$,

$$\text{Agg}L^2(q''_{\text{can}}) \iff \int |q''_{\text{can}}(t)|^2 W_{\text{branch}}^{\text{can}}(t) dt \geq Q^{-A} \int |q''_{\text{can}}(t)|^2 W_{\text{ref}}(t) dt,$$

up to the finite-AFE completion error already controlled by Corollary E.15.

This label is retained for compatibility with older references. It is not a separate hypothesis in the canonical Hardy branch; it is proved by Theorem E.71 and Corollary E.73.

E.2. Source hypothesis provenance.

Remark E.2 (Compatibility labels for old source hypotheses). The older source labels

`hyp:packet-core-micro-covering`, `hyp:packet-amplitude-decay-budget`, `hyp:ef-completion-b`

are retained only as compatibility labels for downstream references. In the active Hardy/canonical branch they are supplied as follows:

- packet-core coverage and branch/reference domination are supplied by the canonical branch-complete source window Definition E.59 and Theorem E.61;
- amplitude bounds and off-block decay are supplied by Lemma E.51, Lemma E.52, and Corollary E.53;
- EF completion is supplied by Corollary E.15;
- fixed-window reference noncollapse is supplied by Corollary E.19.

Thus these labels should not be read as open assumptions in the canonical branch. They remain open only for noncanonical branch windows, non-Hardy source choices, or drifting reference windows not covered by the stated nonconcentration theorem.

Remark E.3 (Standing zeta-side source hypotheses: actual Hardy-curvature version). The source package used below is no longer an abstract ordinary Dirichlet source with a distinguished coefficient $a_1 = 1$. It is the actual Hardy-normalized curvature source attached to ζ . Put

$$\chi(s) := \pi^{s-1/2} \frac{\Gamma((1-s)/2)}{\Gamma(s/2)}, \quad \zeta(s) = \chi(s)\zeta(1-s), \quad \chi(s)\chi(1-s) = 1.$$

On each ordinary source strip we fix the analytic square-root branch

$$B_H(s) := \chi(s)^{-1/2}$$

by the Hardy critical-line convention. Equivalently, on the critical line $s = 1/2 + it$, the boundary value

$$Z_H(t) := B_H(1/2 + it)\zeta(1/2 + it)$$

is the Hardy real scalar, up to the harmless global sign determined by the chosen branch. We write

$$\Phi_H^\#(s) := B_H(s)\zeta(s), \quad \mathcal{Z}_{\text{curv}}(s) := -i \frac{d^3}{ds^3} \Phi_H^\#(s)$$

The object $\mathcal{Z}_{\text{curv}}$ is a finite-order differential Hardy source: its half-plane expansion contains log-power coefficients and derivatives of the gamma/theta amplitude. Thus the old fixed-coefficient package $\mathcal{Z}(s) = \sum a_n n^{-s}$, $a_1 = 1$, is superseded in the active proof branch by the actual-curvature package below.

All source estimates in this subsection are made on ordinary zero-separated source strips where the chosen branch of B_H is analytic and where all routed zero/pole singularities have already been sent to the singular or transition branches. Branch changes multiply Z_H by a fixed sign on a connected ordinary strip and do not affect the quadratic estimates.

Lemma E.4 (Positive reference mass). *Assume W_{ref} is a nonnegative weight on Ω and is not identically zero. Then*

$$\int_{\Omega} W_{\text{ref}}(t) dt > 0.$$

Equivalently, under the Fourier convention used in this draft, $\widehat{W}_{\text{ref}}(0) > 0$.

Proof. Since $W_{\text{ref}} \geq 0$ and is not identically zero, it is positive on a set of positive measure. Therefore its integral over Ω is strictly positive. $\square$

Theorem E.5 (Actual Hardy-curvature source anchor). *Let*

$$Z_H(t) := B_H(1/2 + it)\zeta(1/2 + it), \quad \mathcal{Z}_{\text{curv}}(s) := -i(\Phi_H^\#)^{(3)}(s).$$

Then, on every ordinary source packet on which the Hardy branch is fixed,

$$\partial_t^3 Z_H(t) = \mathcal{Z}_{\text{curv}}\left(\frac{1}{2} + it\right).$$

More generally, if the packet-normalized derivative is $\mathcal{D}_p = \lambda_p \partial_t$, then

$$\mathcal{D}_p^3 \Phi_\zeta(t) = \lambda_p^3 \mathcal{Z}_{\text{curv}}\left(\frac{1}{2} + it\right),$$

where $\Phi_\zeta(t) = Z_H(t)$ in the active Hardy-normalized source branch. The factors λ_p^3 are carried in the packet row normalization and are polynomially harmless on admissible packets.

Proof. By definition $s = 1/2 + it$, hence $\partial_t = i\partial_s$. Therefore

$$\partial_t^3 Z_H(t) = i^3 (\Phi_H^\#)^{(3)}(1/2 + it) = -i (\Phi_H^\#)^{(3)}(1/2 + it) = \mathcal{Z}_{\text{curv}}(1/2 + it).$$

If $\mathcal{D}_p = \lambda_p \partial_t$, the displayed packet-normalized identity is obtained by multiplying by λ_p^3 . $\square$

Remark E.6 (The completed-source anchor is now an actual-source theorem). Theorem E.5 replaces the older abstract completed-source hypothesis. It proves the source trace for the actual Hardy-normalized scalar. What changes is the analytic source package: the source is not an ordinary fixed-coefficient Dirichlet series with $a_1 = 1$, but a finite differential Hardy source with gamma/theta amplitude derivatives and log-power coefficients. All downstream uses of

$$q_{\text{can}}''(t) = \mathcal{Z}(1/2 + it)$$

should now be read in the active branch as

$$q_{\text{can},p}''(t) = \lambda_p^3 \mathcal{Z}_{\text{curv}}(1/2 + it)$$

when packet-normalized derivatives are used, and as $q_{\text{can}}''(t) = \mathcal{Z}_{\text{curv}}(1/2 + it)$ in the physical derivative normalization. A proof branch that keeps a different real scalar or a logarithmic phase must provide its own corresponding AFE and residue package.

Remark E.7 (Label convention for the Hardy source anchor). The historical label `hyp:completed-source-anchor` is retained only for citation stability. In the active Hardy branch the source trace is the theorem

$$\partial_t^3 Z_H(t) = \mathcal{Z}_{\text{curv}}\left(\frac{1}{2} + it\right),$$

proved in Theorem E.5. Any future source branch using a different scalar or logarithmic phase must provide its own trace theorem, AFE package, and source-mass theorem.

Lemma E.8 (Odd functional equation for the actual Hardy-curvature source). *Let*

$$B_H(s) = \chi(s)^{-1/2}, \quad \Phi_H^\#(s) = B_H(s)\zeta(s), \quad \mathcal{Z}_{\text{curv}}(s) = -i(\Phi_H^\#)^{(3)}(s),$$

with the fixed Hardy branch on an ordinary source strip. Then

$$\Phi_H^\#(s) = \Phi_H^\#(1-s)$$

on that strip, and hence

$$\mathcal{Z}_{\text{curv}}(1-s) = -\mathcal{Z}_{\text{curv}}(s).$$

Proof. The functional equation is

$$\zeta(s) = \chi(s)\zeta(1-s), \quad \chi(s)\chi(1-s) = 1.$$

Since the Hardy branch is chosen so that

$$B_H(1-s) = \chi(1-s)^{-1/2} = \chi(s)^{1/2},$$

we have

$$\Phi_H^\#(s) = \chi(s)^{-1/2}\zeta(s) = \chi(s)^{1/2}\zeta(1-s) = B_H(1-s)\zeta(1-s) = \Phi_H^\#(1-s).$$

Differentiating the identity three times gives

$$(\Phi_H^\#)^{(3)}(s) = -(\Phi_H^\#)^{(3)}(1-s).$$

Multiplying by $-i$ gives

$$\mathcal{Z}_{\text{curv}}(1-s) = -i(\Phi_H^\#)^{(3)}(1-s) = i(\Phi_H^\#)^{(3)}(s) = -\mathcal{Z}_{\text{curv}}(s).$$

□

Remark E.9 (Why the old fixed-coefficient completed-source package is superseded). The actual Hardy-curvature source is not an ordinary fixed-coefficient Dirichlet source of the form

$$\sum_{n \geq 1} a_n n^{-s}, \quad a_1 = 1.$$

For $\Re s > 1$,

$$\Phi_H^\#(s) = B_H(s) \sum_{n \geq 1} n^{-s},$$

and therefore

$$(\Phi_H^\#)^{(3)}(s) = \sum_{r=0}^3 \binom{3}{r} B_H^{(3-r)}(s) \sum_{n \geq 1} (-\log n)^r n^{-s}.$$

The coefficients are log-power Dirichlet coefficients multiplied by analytic gamma/theta amplitudes depending on s . Thus the old $a_1 = 1$ noncollapse and ordinary fixed-coefficient AFE language should not be cited in the active Hardy-curvature proof branch. It is replaced by Theorem E.13 and Lemma E.17.

Remark E.10 (Alternative source choices are archived). Two natural alternatives are deliberately not used in the active branch. If one takes $\Phi_\zeta^\#(s) = \xi(s)$, then the trace identity

$$\partial_t^3 \xi(1/2 + it) = -i\xi^{(3)}(1/2 + it)$$

is clean, but the gamma factor in ξ introduces critical-line scaling that is incompatible with the present packet source normalization without a separate renormalization audit. If one takes Φ_ζ to be a logarithmic phase, the source becomes a logarithmic-derivative object with zero/pole

residues and principal-value singularities; that would require an explicit-formula source package rather than the smooth Hardy-curvature AFE used here. The active proof branch therefore uses the Hardy-normalized scalar $B_H(s)\zeta(s)$ and its third critical-line derivative.

Remark E.11 (Why the Hardy-normalized source is the active source). The choice

$$\Phi_H^\#(s) = \chi(s)^{-1/2}\zeta(s), \quad q_{\text{can}}''(t) = \partial_t^3 \Phi_H^\#(\tfrac{1}{2} + it),$$

is a source-level normalization, not a cosmetic rewrite. It is selected because it simultaneously satisfies the three requirements used by the finite-survival chain. First, its critical-line trace is an actual zeta object: with a fixed Hardy branch, $Z_H(t) = \Phi_H^\#(1/2 + it)$ is real on ordinary source strips and

$$\partial_t^3 Z_H(t) = \mathcal{Z}_{\text{curv}}(\tfrac{1}{2} + it), \quad \mathcal{Z}_{\text{curv}}(s) = -i(\Phi_H^\#)^{(3)}(s).$$

Second, the source admits a smooth finite AFE after replacing the old fixed-coefficient Dirichlet package by a finite differential Hardy package: the main terms have log-power coefficients and gamma/theta amplitude derivatives, and the residual satisfies the differentiated AFE tail estimates used in the packet layer. Third, Hardy normalization avoids the critical-line gamma-decay problem of using $\xi(s)$ directly while avoiding the poles, principal values, and zero-residue bookkeeping forced by logarithmic-phase sources. Thus the Hardy source is the least disruptive actual-zeta source compatible with the existing Gate-1/Gate-2, D4, and source-graded K9 architecture.

The tradeoff is explicit. The old ordinary source package $\mathcal{Z}(s) = \sum a_n n^{-s}$ with a fixed leading coefficient $a_1 = 1$ is not used in the active branch. It is superseded by the Hardy differential AFE and by the Hardy-curvature mean-square / mass-good packet selection theorem. If one uses packet-normalized derivatives $\mathcal{D}_p = \lambda_p \partial_t$, then the packet row is $\lambda_p^3 \mathcal{Z}_{\text{curv}}(1/2 + it)$; this factor is part of the source API and must be carried in the corresponding packet Grams and row normalizations.

Definition E.12 (Canonical actual Hardy-curvature source and Mellin cutoff). In the actual Hardy-curvature branch, the physical source is

$$q_{\text{curv}}(t) := \partial_t^3 Z_H(t) = \mathcal{Z}_{\text{curv}}(\tfrac{1}{2} + it).$$

On a packet with normalized derivative $\mathcal{D}_p = \lambda_p \partial_t$, the packet source is

$$q_{\text{can},p}''(t) := \mathcal{D}_p^3 \Phi_\zeta(t) = \lambda_p^3 q_{\text{curv}}(t).$$

Fix a smooth Mellin cutoff V whose Mellin transform $\widehat{V}$ is entire and rapidly decaying on vertical strips. The finite AFE below is obtained by applying $-i\partial_s^3$ to the smoothed AFE for $\Phi_H^\#(s) = B_H(s)\zeta(s)$.

Theorem E.13 (Smooth finite AFE package for the actual Hardy-curvature source). *Let $s = 1/2 + it$, with t in an ordinary source strip. There exist smooth weights $V_{1,r,k}(y;t)$, $V_{2,r,k}(y;t)$, a finite residue term $P_{\text{res},X}^{\text{curv}}(t)$, and a remainder $R_X^{\text{curv}}(t)$ such that*

$$q_{\text{curv}}(t) = F_X^{\text{curv}}(t) + R_X^{\text{curv}}(t),$$

where

$$\begin{aligned} F_X^{\text{curv}}(t) = & \sum_{0 \leq r+k \leq 3} \sum_{n \geq 1} c_{r,k}^+(t) (\log n)^r n^{-1/2-it} V_{1,r,k}(n/X; t) \\ & + \sum_{0 \leq r+k \leq 3} \sum_{n \geq 1} c_{r,k}^-(t) (\log n)^r n^{-1/2+it} V_{2,r,k}(nX; t) + P_{\text{res},X}^{\text{curv}}(t). \end{aligned}$$

Here the amplitudes $c_{r,k}^\pm(t)$ are finite linear combinations of the critical-line boundary values of $B_H^{(k)}(s)$, the differentiated functional-equation factors, and derivatives of the smooth AFE weights. They satisfy polynomial vertical-strip bounds on ordinary source strips. The critical-line coefficients obey, for every fixed r and every $\varepsilon > 0$,

$$(\log n)^r n^{-1/2} \ll_{r,\varepsilon} n^{-1/2+\varepsilon}.$$

For every fixed derivative order J and every $A > 0$, after choosing the Mellin cutoff depth sufficiently large,

$$\sum_{0 \leq j \leq J} \left(\|\partial_t^j R_X^{\text{curv}}\|_{L^2(W_{\text{ref}})} + \|\partial_t^j R_X^{\text{curv}}\|_{L^2(W_{\text{branch}}^{\text{can}})} \right) \ll_{A,J} X^{-A}.$$

On a packet with $\mathcal{D}_p = \lambda_p \partial_t$, the same estimate holds for $\lambda_p^3 R_X^{\text{curv}}$ with only a polynomial loss, since $Q^{-C} \leq |\lambda_p| \leq Q^C$ on admissible packets.

Proof. Start from the standard smoothed AFE for the Hardy-normalized analytic source $\Phi_H^\#(s) = B_H(s)\zeta(s)$ on the chosen ordinary strip:

$$\Phi_H^\#(s) = \sum_{n \geq 1} B_H(s) n^{-s} V_1(n/X) + \sum_{n \geq 1} B_H(1-s) n^{-(1-s)} V_2(nX; s) + P_{\text{res},X}^{(0)}(s) + R_X^{(0)}(s).$$

The residue term records the finitely many poles or branch-endpoint residues crossed in the Mellin contour shifts; on ordinary retained strips it is a finite smooth term. Apply $-i\partial_s^3$. Differentiating the first sum produces finite terms

$$-i \binom{3}{r} B_H^{(3-r)}(s) (-\log n)^r n^{-s} V_1(n/X), \quad 0 \leq r \leq 3,$$

and differentiating the second sum produces the dual terms, including finite derivatives of $B_H(1-s)$ and of the smooth weight $V_2(nX; s)$. These terms have exactly the displayed form. The residue term is differentiated and included in $P_{\text{res},X}^{\text{curv}}$. The remainder is

$$R_X^{\text{curv}}(t) := -i\partial_s^3 R_X^{(0)}(1/2 + it).$$

The tail estimates follow from the usual smoothed AFE contour bounds, the rapid decay of $\widehat{V}$, Stirling bounds for the gamma-ratio amplitude B_H and its derivatives, and the fact that only finitely many fixed s -derivatives are taken. The packet-normalized statement follows by the polynomial admissibility bounds for λ_p . $\square$

Corollary E.14 (Smooth AFE tail bound). *With $q_{\text{curv}} = F_X^{\text{curv}} + R_X^{\text{curv}}$ as in Theorem E.13, one has*

$$\|R_X^{\text{curv}}\|_{L^2(W_{\text{ref}})} + \|R_X^{\text{curv}}\|_{L^2(W_{\text{branch}}^{\text{can}})} \ll_A X^{-A}.$$

The finite main term includes all differentiated residue terms. A residue term is never placed in the error unless it is separately proved perturbative on the retained source window.

Proof. This is the case $J = 0$ of Theorem E.13. $\square$

Corollary E.15 (EF completion bound in reference-whitened coordinates). *Assume Theorem E.13. Assume also that the finite packet family used to form the source matrices is reference-regular, in the sense that*

$$\|G_{\text{ref}}^{-1/2}\|_{\text{op}} + \|G_{\text{ref}}^{1/2}\|_{\text{op}} \leq Q^{C_{\text{ref}}},$$

and that the packet count and the branch/reference weights are polynomially controlled. Let F_X denote the differentiated Hardy finite main term F_X^{curv} , with the packet factor λ_p^3 absorbed where normalized packet derivatives are used. Then the normalized residual correction

$$E = G_{\text{ref}}^{-1/2} (G_{\text{branch}}(q_{\text{can}}'') - G_{\text{branch}}(F_X)) G_{\text{ref}}^{-1/2}$$

satisfies, for every prescribed $A > 0$,

$$\|E\|_{\text{op}} \leq Q^{-A}$$

after choosing the smooth AFE cutoff depth sufficiently large.

Proof. Theorem E.13 gives an arbitrarily strong $L^2(W_{\text{ref}})$ and $L^2(W_{\text{branch}}^{\text{can}})$ bound for $q_{\text{can}}'' - F_X$, up to the polynomial packet-normalization factors already absorbed into the source rows. Cauchy-Schwarz and the polynomial packet count convert this into a branch-Gram perturbation with polynomial loss. The two reference-whitening factors cost at most $Q^{2C_{\text{ref}}}$. Choosing the AFE tail depth larger than the total polynomial loss gives the claim. $\square$

Remark E.16 (Differential Hardy AFE coefficient normalization). The active finite AFE coefficients are not fixed coefficients a_n with $a_1 = 1$. They are finite sums of critical-line amplitudes multiplying $(\log n)^r n^{-1/2 \pm it}$. For every fixed r ,

$$(\log n)^r n^{-1/2} \ll_{r,\varepsilon} n^{-1/2+\varepsilon}.$$

All gamma/theta derivatives are smooth packet amplitudes with polynomial vertical-strip bounds on ordinary source strips. Consequently the old $a_1 = 1$ noncollapse argument is superseded by the actual Hardy-curvature mean-square theorem below.

Lemma E.17 (Dyadic mean square for the actual Hardy curvature). *Let $W_T(t) = W(t/T)$ be a nonnegative smooth dyadic weight supported in $[T, 2T]$, with*

$$W^{(j)}(x) \ll_j 1, \quad \int W_T(t) dt \asymp T.$$

Then, for T sufficiently large,

$$\int |\partial_t^3 Z_H(t)|^2 W_T(t) dt \asymp T(\log T)^7.$$

In particular,

$$\int |q_{\text{curv}}(t)|^2 W_T(t) dt \geq c T(\log T)^7$$

for some absolute $c > 0$ depending only on the fixed dyadic weight.

Proof. The classical weighted mean-square theorem for Hardy derivatives gives, for each fixed k ,

$$\int |Z_H^{(k)}(t)|^2 W_T(t) dt = T \mathfrak{P}_{2k+1}(\log T) + O_k(T(\log T)^{2k}),$$

where $\mathfrak{P}_{2k+1}$ is a real polynomial of degree $2k+1$ with positive leading coefficient. For $k=3$, this is the displayed asymptotic with degree 7.

For completeness, we recall the standard proof mechanism. Differentiate the smoothed AFE for Z_H three times. The resulting Dirichlet polynomials have coefficients which are finite linear combinations of $(\log n)^r n^{-1/2}$, $0 \leq r \leq 3$, times smooth gamma/theta amplitudes. The weighted Montgomery–Vaughan mean-value theorem gives the diagonal main term, and the differentiated functional equation pairs the two halves into the Hardy real form. The diagonal contribution has leading size $T(\log T)^7$ and positive leading coefficient; off-diagonal and differentiated weight terms contribute to lower powers of $\log T$. Hence the lower and upper bounds follow. $\square$

Corollary E.18 (Finite-AFE main-term noncollapse). *Let W_{ref} be a canonical dyadic source reference weight on an ordinary source window $\Omega_T \subset [T, 2T]$, comparable to a smooth dyadic weight as in Lemma E.17. Then*

$$\int_{\Omega_T} |q_{\text{curv}}(t)|^2 W_{\text{ref}}(t) dt \geq Q^{-A} \int_{\Omega_T} W_{\text{ref}}(t) dt.$$

For the differentiated finite AFE $q_{\text{curv}} = F_X^{\text{curv}} + R_X^{\text{curv}}$, choose the cutoff depth so that

$$\|R_X^{\text{curv}}\|_{L^2(W_{\text{ref}})} \leq \frac{1}{2} \|q_{\text{curv}}\|_{L^2(W_{\text{ref}})}.$$

Then

$$\int_{\Omega_T} |F_X^{\text{curv}}(t)|^2 W_{\text{ref}}(t) dt \geq Q^{-A'} \int_{\Omega_T} W_{\text{ref}}(t) dt.$$

The same estimates hold for packet-normalized $q''_{\text{can},p} = \lambda_p^3 q_{\text{curv}}$ with only polynomial changes in the exponents.

Proof. The dyadic lower bound is Lemma E.17, after absorbing the fixed comparison between W_{ref} and a smooth dyadic weight. Since $(\log T)^7$ is polynomially bounded below in the packet scale, the lower bound has the stated Q^{-A} form. The finite-main estimate follows from the triangle inequality:

$$\|F_X^{\text{curv}}\|_{L^2(W_{\text{ref}})} \geq \|q_{\text{curv}}\|_{L^2(W_{\text{ref}})} - \|R_X^{\text{curv}}\|_{L^2(W_{\text{ref}})} \geq \frac{1}{2} \|q_{\text{curv}}\|_{L^2(W_{\text{ref}})}.$$

The packet-normalized version follows from $Q^{-C} \leq |\lambda_p| \leq Q^C$. $\square$

Corollary E.19 (Reference mean-square noncollapse). *On every canonical dyadic ordinary source window used by the active proof branch,*

$$\int |q''_{\text{can}}(t)|^2 W_{\text{ref}}(t) dt \geq Q^{-A} \int W_{\text{ref}}(t) dt,$$

where q''_{can} denotes either the physical source q_{curv} or the packet-normalized source $\lambda_p^3 q_{\text{curv}}$, according to the local packet normalization. Consequently the source selection has polynomially large reference mass before branch completion.

Proof. This is Corollary E.18, with the packet normalization convention stated in Theorem E.5. $\square$

Definition E.20 (Hardy-curvature mass-good packets). Let $\{K_p^+\}$ be the Whitney packet enlargement of a canonical dyadic source window. Define

$$M_p := \int_{K_p^+} |q''_{\text{can}}(t)|^2 W_{\text{ref}}(t) dt.$$

For a fixed exponent A_{mg} , call p mass-good if

$$M_p \geq Q^{-A_{\text{mg}}} \int_{K_p^+} W_{\text{ref}}(t) dt.$$

The active Gate-1 packet family is taken from the mass-good packets, together with the usual ordinary/admissible packet conditions.

Lemma E.21 (Mass-good packets carry polynomial Hardy-curvature mass). *For A_{mg} chosen larger than the fixed polynomial losses in the Whitney overlap and the dyadic upper bound, the union of mass-good packets carries a polynomial fraction of the total Hardy-curvature source mass:*

$$\sum_{p \in \mathcal{P}_{\text{mg}}} M_p \geq Q^{-A} \int_{\Omega_T} |q''_{\text{can}}(t)|^2 W_{\text{ref}}(t) dt.$$

Proof. The dyadic upper bound in Lemma E.17 and the bounded Whitney overlap give

$$\sum_p M_p \ll Q^C \int_{\Omega_T} |q''_{\text{can}}(t)|^2 W_{\text{ref}}(t) dt.$$

The total lower bound is Corollary E.19. If the mass-good packets carried less than a polynomial fraction of the total mass, then the bad packets, each satisfying

$$M_p < Q^{-A_{\text{mg}}} \int_{K_p^+} W_{\text{ref}},$$

would force the total mass to be smaller than the dyadic lower bound after choosing A_{mg} beyond the fixed overlap and scale losses. This contradiction proves the claim. $\square$

Remark E.22 (No arbitrary local Hardy-curvature lower bound). The Hardy-curvature lower bound is a dyadic/mass-good source theorem, not a pointwise lower bound on every arbitrary packet. A nonnegative weight can be supported in a tiny neighbourhood where $|\partial_t^3 Z_H|$ is very small; therefore no theorem of the form

$$\int_K |\partial_t^3 Z_H(t)|^2 W_K(t) dt \geq Q^{-A} \int_K W_K(t) dt$$

can hold for all adversarial local windows K . The active proof branch uses canonical dyadic reference windows and then selects mass-good Whitney packets. This is the source-side replacement for the old $a_1 = 1$ noncollapse argument.

Remark E.23 (Status after the actual Hardy-curvature source repair). The completed-source anchor is discharged in the active Hardy branch by Theorem E.5; the old abstract $a_1 = 1$ source package is superseded. The source noncollapse input is discharged on canonical dyadic ordinary source windows by Lemma E.17 and by mass-good packet selection. These changes do not alter the downstream canonical branch-complete window, the pre-extraction Fejér–Riesz Stack construction, or the maximal labelled $K9$ live/dead routing; those pieces use only the resulting finite AFE, polynomial coefficient/derivative bounds, and source mass in the canonical branch.

Definition E.24 (Active Hardy source API). In the active proof branch, the physical actual-zeta source is

$$u_H(t) := \partial_t^3 Z_H(t) = \mathcal{Z}_{\text{curv}}\left(\frac{1}{2} + it\right),$$

where

$$Z_H(t) = B_H\left(\frac{1}{2} + it\right) \zeta\left(\frac{1}{2} + it\right), \quad B_H(s) = \chi(s)^{-1/2}.$$

On a packet with normalized derivative

$$\mathcal{D}_p = \lambda_p \partial_t,$$

the packet-normalized source row is

$$u_{H,p}^{\text{norm}}(t) := \mathcal{D}_p^3 Z_H(t) = \lambda_p^3 u_H(t).$$

Thus a single global function $\mathcal{Z}_{\text{curv}}(1/2 + it)$ represents the physical source u_H , while packet-normalized rows carry the explicit factor λ_p^3 . All Gate–1 Grams, packet rows, and downstream source rows in this branch use this convention.

The finite AFE decomposition in this branch is

$$u_H(t) = F_X^{H,3}(t) + R_X^{H,3}(t),$$

where $F_X^{H,3}$ is the differentiated Hardy AFE main term of Theorem E.13. The packet-normalized finite AFE is

$$u_{H,p}^{\text{norm}}(t) = \lambda_p^3 F_X^{H,3}(t) + \lambda_p^3 R_X^{H,3}(t).$$

When no confusion is possible, we write

$$q_{\text{can}}'' = u_H, \quad F_X = F_X^{H,3}, \quad R_X = R_X^{H,3}$$

in physical coordinates, and insert the factor λ_p^3 whenever the normalized packet derivative $\mathcal{D}_p$ is used.

Definition E.25 (Hardy-differential source pieces). The source pieces appearing in the finite survival chain are the Hardy-differential pieces

$$B_2^H, \quad P_2^H, \quad D_2^H, \quad E_2^H,$$

defined by decomposing the differentiated Hardy AFE main term and residual as

$$q_{\text{can}}'' = u_H = B_2^H + P_2^H + D_2^H + E_2^H.$$

Here each term includes the relevant derivatives of $B_H(s)$, log-power coefficients $(\log n)^r$, differentiated cutoffs, and all residue terms produced by the contour shift. In the active Hardy branch, unsuperscripted symbols

$$B_2, P_2, D_2, E_2, F_X$$

mean these Hardy-differential objects unless an older archived source model is explicitly named. The old ordinary fixed-coefficient source pieces are not used in the proof branch leading to Gate–1.

Lemma E.26 (Hardy source and Gate–2 derivative compatibility). *In the active Hardy branch let*

$$\Phi_\zeta(t) = Z_H(t), \quad q_p := \mathcal{D}_p \Phi_\zeta(t).$$

Then

$$\mathcal{D}_p^2 q_p = \mathcal{D}_p^3 \Phi_\zeta(t) = \lambda_p^3 u_H(t).$$

Consequently, in the normalized packet convention,

$$q'' = q''_{\text{can},p} = u_{H,p}^{\text{norm}}.$$

The Gate-2 anisotropy identities remain exact:

$$\Gamma_G = q'' - 6q + 2q^3,$$

and

$$\Gamma_{G,2} = 2q(q')^2 - \frac{1}{6}\Gamma_G'',$$

where primes denote the same normalized derivative $\mathcal{D}_p$.

Proof. The first identity is the definition of q_p and the fact that $\mathcal{D}_p = \lambda_p \partial_t$ is constant on the packet. Thus

$$\mathcal{D}_p^2 q_p = \mathcal{D}_p^3 Z_H = \lambda_p^3 \partial_t^3 Z_H = \lambda_p^3 u_H.$$

The formula for Γ_G is the definition of the first Gate-2 anisotropy. The displayed form of $\Gamma_{G,2}$ follows from Lemma 2.5, with every prime interpreted as $\mathcal{D}_p$. No physical-coordinate derivative is mixed into these identities. $\square$

Remark E.27 (Downstream source API after the Hardy repair). The Hardy-normalized differential-source repair is not a drop-in proof of the old fixed-coefficient package. It replaces the source API. Downstream modules must receive

$$u_H, \quad \lambda_p^3 u_H, \quad F_X^{H,3}, \quad R_X^{H,3},$$

not the old ordinary Dirichlet source with a distinguished coefficient $a_1 = 1$. In particular:

- (1) every occurrence of B_2, P_2, D_2, E_2 in the source layer means the Hardy-differential piece of Definition E.25;
- (2) packet-normalized source Grams and packet rows carry λ_p^3 ;
- (3) the Gate-2 identities use the normalized derivative convention of Lemma E.26;
- (4) the old $a_1 = 1$ noncollapse proof is replaced by Lemma E.17 and the mass-good packet selection.

A branch using ξ , a logarithmic phase, or a different real completed scalar must provide its own trace, residue-aware AFE, source mass theorem, and Gate-2 compatibility audit.

Remark E.28 (Hardy source migration checklist). The active Hardy branch is a full source-API replacement. The following migration rules are part of the proof branch.

- (1) The old fixed-coefficient completed source and its $a_1 = 1$ noncollapse proof are archived; all active source estimates use the differential Hardy AFE.
- (2) Packet-normalized source rows carry the explicit factor λ_p^3 . A global function $\mathcal{Z}_{\text{curv}}(1/2 + it)$ represents the physical source only.
- (3) Gate-2 identities use the same derivative convention as the source row; in particular $q'' = q''_{\text{can}} = u_{H,p}^{\text{norm}}$ in normalized packet coordinates.
- (4) The pieces B_2, P_2, D_2, E_2 mean the Hardy-differential pieces unless an archived source model is explicitly named.
- (5) Residues, differentiated cutoffs, gamma/theta derivatives, and log-power coefficients are part of the Hardy finite main term or its explicit residual ledger; they are never silently dropped.

These rules are the downstream audit consequences of replacing the old completed-source package by the actual Hardy-curvature package.

Remark E.29 (Provenance of reference mean-square noncollapse). For a fixed reference weight, reference noncollapse is no longer a no-cancellation hypothesis for the Dirichlet polynomial. It follows from the analytic nonzero trace argument in Lemma E.17, and the finite-AFE main term inherits the lower bound from Corollary E.18. The previous proof route based on the $n = 1$ diagonal term and a negligible off-diagonal term should not be used.

If the reference window is allowed to depend on Q , the uniform polynomial lower bound remains a separate quantitative nonconcentration requirement, as recorded in Remark E.22.

Definition E.30 (Whitney packet cores). Let $\Omega \subset \mathbb{R}$ be the packet-admissible source region and

$$d(t) := \text{dist}(t, \Omega^c)$$

the associated distance-to-boundary function. Fix $0 < \varepsilon \leq \frac{1}{10}$. Choose packet centers $m_p \in \Omega$ so that the shrunken intervals

$$K_p^- := (m_p - \tfrac{1}{2}r_p, m_p + \tfrac{1}{2}r_p), \quad r_p := \varepsilon d(m_p),$$

are pairwise disjoint. Define the packet cores by

$$K_p := (m_p - r_p, m_p + r_p).$$

Theorem E.31 (Whitney selection / overlap comparability). *With the packet cores K_p from Definition E.30:*

(1) *if $K_p \cap K_q \neq \emptyset$, then*

$$r_p \asymp r_q;$$

(2) *the packet cores have bounded overlap:*

$$\sup_{t \in \Omega} \sum_p \mathbf{1}_{K_p}(t) \leq C_\varepsilon;$$

(3) *consequently, $\{K_p\}$ is a Whitney family for Ω , and the packet-core micro-covering needed for the source frame bound is automatic.*

Proof. Assume $K_p \cap K_q \neq \emptyset$. Pick $x \in K_p \cap K_q$. Then

$$|m_p - m_q| \leq r_p + r_q.$$

Since $d(\cdot)$ is 1-Lipschitz,

$$d(m_p) \leq d(m_q) + |m_p - m_q| \leq d(m_q) + r_p + r_q = d(m_q) + \varepsilon d(m_p) + \varepsilon d(m_q).$$

Thus

$$(1 - \varepsilon)d(m_p) \leq (1 + \varepsilon)d(m_q),$$

and symmetrically

$$(1 - \varepsilon)d(m_q) \leq (1 + \varepsilon)d(m_p).$$

Multiplying by ε gives

$$r_p = \varepsilon d(m_p) \asymp \varepsilon d(m_q) = r_q.$$

For bounded overlap, fix $t \in \Omega$, and let

$$\mathcal{P}(t) := \{p : t \in K_p\}.$$

For $p \in \mathcal{P}(t)$,

$$|t - m_p| \leq r_p = \varepsilon d(m_p).$$

By the same Lipschitz estimate,

$$(1 - \varepsilon)d(m_p) \leq d(t) \leq (1 + \varepsilon)d(m_p),$$

so all r_p for $p \in \mathcal{P}(t)$ are comparable to $d(t)$. Hence the pairwise disjoint shrunken intervals K_p^- all lie inside an interval of length $O(d(t))$, while each has length $\asymp d(t)$. Therefore only $O(1)$ many can fit, uniformly in t , and

$$\sup_{t \in \Omega} \sum_p \mathbf{1}_{K_p}(t) \leq C_\varepsilon.$$

This proves the Whitney property. □

Remark E.32 (Whitney family implies source thickness). If the packet construction chooses the packet cores by the maximal Whitney selection of Definition E.30, then the bounded-overlap and scale-comparability inputs in the source layer are unconditional consequences of the construction. This discharges property (1) of Theorem E.56 and clauses (1) and (3) of Theorem E.36's packet-structure assumptions.

Scope note. The theorem above is unconditional once the Whitney selection rule is adopted as part of the packet construction. The present draft adopts this selection rule by Definition E.30; compatibility of this rule with the existing packet usage in the source decomposition is a draft-level convention to be checked when the source construction is finalised. The remaining open packet-side input is the frequency-separation theorem for the resonant-block organization (item (4) of Theorem E.56).

Lemma E.33 (Finite-AFE packet geometry and branch-weight regularity). *Assume*

$$q''_{\text{can}} = F_X + R_X, \quad F_X := B_2 + P_{2,\leq X} + D_{2,\leq X} + P_{\text{res},X},$$

where the finite AFE uses the fixed smooth cutoff introduced in the source section. Let K_p be a Whitney packet core with enlargement K_p^+ and scale λ_p . Assume the routed branch/reference weights are scale-slow on good packet enlargements, namely for each fixed j ,

$$\sup_{u \in K_p^+} \lambda_p^j |\partial_u^j \log W_{\text{branch}}(u)| \leq Q^{C_j}.$$

Then on every good packet:

- (1) the packet cores have bounded overlap;
- (2) the branch weights satisfy the unscaled polynomial derivative bound

$$\sup_{u \in K_p^+} |\partial_u^j \log W_{\text{branch}}(u)| \leq Q^{C'_j}.$$

No relative derivative estimate for F_X is asserted in this lemma.

Proof. Bounded overlap is the Whitney geometry of Theorem E.31. The branch-weight regularity is a property of the routed branch/reference window construction, or equivalently of the explicit scale-slow cutoff model used in Lemma E.51; it is not a consequence of the AFE coefficients. On scale-admissible good packets $\lambda_p^{-1} \leq Q^C$, so the scale-slow bounds imply the displayed unscaled polynomial derivative bounds. The packet derivative estimates for F_X are supplied separately by Theorem E.36 through an absolute derivative budget plus mass-goodness. $\square$

Definition E.34 (Raw packet scale, scale-admissibility, and mass-good packets). For each Whitney packet core K_p , let λ_p^{raw} denote its geometric packet width from Definition E.30. A packet is called *scale-admissible* if

$$Q^{-C_\lambda} \leq \lambda_p^{\text{raw}} \leq Q^{C_\lambda}.$$

On the scale-admissible family we write $\lambda_p := \lambda_p^{\text{raw}}$.

Set

$$M_p := \int_{K_p^+} |F_X(t)|^2 W_{\text{ref}}(t) dt.$$

A scale-admissible packet is called *mass-good* if

$$M_p \geq Q^{-C_{\text{mass}}}.$$

No derivative estimate is included in the definition of scale-admissibility or mass-goodness.

Remark E.35 (Raw-scale admissibility versus collar no-loss). The upper bound $\lambda_p^{\text{raw}} \leq Q^C$ follows from the fact that the source-admissible region lies inside the finite-AFE window I_{ref} of polynomial diameter, so

$$d(m_p) \leq \text{diam}(I_{\text{ref}}) \ll Q^C.$$

The lower bound

$$\lambda_p^{\text{raw}} \geq Q^{-C_\lambda}$$

is part of the admissible-packet selection in Definition E.34. The role of Corollary E.49 is not to prove that every raw Whitney packet has this lower bound. Its role is to prove that discarding the sub-polynomial boundary-collar packets is no-loss for the source-frame mass. Thus the noncircular order is

raw Whitney packets $\rightarrow$ scale-admissible selection $\rightarrow$ packet envelope $\rightarrow$ boundary-collar no-loss.

Theorem E.36 (Finite-AFE derivative budget on raw packets). *Assume Theorem E.13. Let $F_X(t)$ be the differential Hardy finite-AFE main term $F_X^{\text{curv}}(t)$,*

$$\begin{aligned} F_X(t) = & \sum_{0 \leq r+k \leq 3} \sum_{n \geq 1} c_{r,k}^+(t) (\log n)^r n^{-1/2-it} V_{1,r,k}(n/X; t) \\ & + \sum_{0 \leq r+k \leq 3} \sum_{n \geq 1} c_{r,k}^-(t) (\log n)^r n^{-1/2+it} V_{2,r,k}(nX; t) + P_{\text{res},X}^{\text{curv}}(t), \end{aligned}$$

with all nonzero contour residues collected in $P_{\text{res},X}^{\text{curv}}$. Assume the source truncation range, residue term, dual AFE weights, and reference weight are polynomially controlled in the Q -scale used in the source section. Then for each raw Whitney packet core K_p and each fixed integer $j \geq 0$,

$$D_{p,j} := \int_{K_p^+} |\partial_t^j F_X(t)|^2 W_{\text{ref}}(t) dt \leq Q^{C_j}.$$

Consequently, on every mass-good packet (Definition E.34),

$$\int_{K_p^+} |\partial_t^j F_X(t)|^2 W_{\text{ref}}(t) dt \leq Q^{C_j + C_{\text{mass}}} \int_{K_p^+} |F_X(t)|^2 W_{\text{ref}}(t) dt.$$

Thus the packet derivative budget is polynomial on the mass-good packet family.

Proof. Differentiating each summand inserts factors $(-i \log n)^j$ (from the first sum) or $(i \log n)^j$ (from the second), together with finite Leibniz contributions from t -derivatives of the smooth amplitudes $c_{r,k}^\pm(t)$ and weights $V_{1,r,k}(n/X; t)$, $V_{2,r,k}(nX; t)$. Theorem E.13 gives polynomial bounds for these amplitudes and weight derivatives on the retained source window. The residue term $P_{\text{res},X}^{\text{curv}}$ has polynomially controlled derivatives by the same theorem.

The smooth cutoffs restrict both Dirichlet sums to polynomially controlled effective ranges, up to rapidly decaying tails. On those ranges,

$$|\log n|^{r+j} \leq Q^{C_{r,j}}, \quad 0 \leq r \leq 3.$$

Combined with $(\log n)^r n^{-1/2} \ll_{r,\varepsilon} n^{-1/2+\varepsilon}$ and the polynomial vertical-strip bounds $|c_{r,k}^\pm(t)| \leq Q^C$, the differentiated terms satisfy

$$|c_{r,k}^\pm(t)| n^{-1/2} |\log n|^{r+j} \leq Q^{C_{r,j}} n^{-1/2+\varepsilon}$$

on the effective supports. Since the number and length of the effective summation ranges are polynomial in Q , and since the derivatives of $V_{1,r,k}$, $V_{2,r,k}$ and $P_{\text{res},X}^{\text{curv}}$ are polynomially controlled, this gives the pointwise bound

$$\sup_{t \in \Omega} |\partial_t^j F_X(t)| \leq Q^{C_j}.$$

The reference weight has polynomial mass on every raw packet enlargement, so

$$D_{p,j} = \int_{K_p^+} |\partial_t^j F_X(t)|^2 W_{\text{ref}}(t) dt \leq Q^{C_j}.$$

This proves the absolute derivative budget without invoking any local no-cancellation estimate for F_X .

If p is mass-good, then

$$M_p = \int_{K_p^+} |F_X(t)|^2 W_{\text{ref}}(t) dt \geq Q^{-C_{\text{mass}}}.$$

Therefore

$$D_{p,j} \leq Q^{C_j} \leq Q^{C_j + C_{\text{mass}}} M_p,$$

which is the claimed polynomial relative derivative budget on the mass-good packet family. $\square$

Corollary E.37 (Packet derivative budget on mass-good packets). *The packet derivative budget used in the source-frame and envelope estimates holds on the mass-good packet family by Theorem E.36. It is a polynomial relative budget, not a $(\log X)^{2j}$ -relative budget.*

Remark E.38 (Status of the packet regularity theorem). Theorem E.36 provides an absolute derivative budget $D_{p,j} \leq Q^{C_j}$ from the residue-aware smooth AFE package and polynomial source-window control. On mass-good packets this gives a polynomial relative budget. This is not the false $(\log X)^{2j}$ -relative estimate. The word “relative” here always means “after mass-goodness.” In the active Hardy-curvature branch, the completed-source anchor is closed by Theorem E.5 and the smooth AFE package by Theorem E.13; the theorem above therefore inherits its inputs from those two source-side theorems. A different source branch must provide its own trace/AFE package.

Remark E.39 (Scale-normalized oscillatory-kernel bound). The oscillatory-kernel estimate must be stated in terms of a dimensionless separation parameter. The bound

$$\left| \int A_{pq}(t) e^{i(\phi_p(t) - \phi_q(t))} dt \right| \ll_N \Lambda_{pq}^{-N}$$

is a decay estimate only in the regime $\Lambda_{pq} \gtrsim 1$. If $\Lambda_{pq} \leq Q^{-\delta}$, then

$$\Lambda_{pq}^{-N} \geq Q^{\delta N},$$

so the right-hand side grows rather than decays.

Let

$$I_{pq} := \text{supp } A_{pq}, \quad \lambda_{pq} := |I_{pq}|, \quad \Delta\omega_{pq} := |\omega_p - \omega_q|.$$

The correct scale-normalized separation parameter is

$$\Theta_{pq} := 1 + \lambda_{pq} \Delta\omega_{pq}.$$

Equivalently, in the fully invariant form one may set

$$\Theta_{pq} := 1 + \lambda_{pq} \inf_{t \in I_{pq}} |\partial_t(\phi_p - \phi_q)(t)|.$$

On packet-normalized supports these two definitions are comparable, since

$$\inf_{t \in I_{pq}} |\partial_t(\phi_p - \phi_q)(t)| \asymp |\omega_p - \omega_q|.$$

Assume that the amplitude satisfies the scale-normalized derivative bounds

$$\sum_{k=0}^N \lambda_{pq}^k \left\| \partial_t^k A_{pq} \right\|_{L^1(I_{pq})} \leq Q^{C_N},$$

and that the phase is non-stationary on I_{pq} with polynomially controlled higher derivatives. Then repeated integration by parts gives

$$\left| \int A_{pq}(t) e^{i(\phi_p(t) - \phi_q(t))} dt \right| \ll_N Q^{C_N} (1 + \lambda_{pq} |\omega_p - \omega_q|)^{-N}.$$

Thus off-block decay requires the scale-normalized nonresonance condition

$$\lambda_{pq} |\omega_p - \omega_q| \geq Q^\kappa$$

for some $\kappa > 0$, not merely the unnormalized condition

$$|\omega_p - \omega_q| \geq Q^{-\delta}.$$

Indeed, under

$$\lambda_{pq} |\omega_p - \omega_q| \geq Q^\kappa,$$

the preceding estimate gives

$$\left| \int A_{pq}(t) e^{i(\phi_p(t) - \phi_q(t))} dt \right| \ll_N Q^{C_N - \kappa N}.$$

Consequently, if the integration-by-parts order N is chosen so that

$$\kappa N - C_N \geq A,$$

then

$$\left| \int A_{pq}(t) e^{i(\phi_p(t) - \phi_q(t))} dt \right| \ll_A Q^{-A}.$$

Lemma E.40 (Absolute packet-scale AFE envelope). *Assume the finite AFE truncation range, the packet window Ω , and the weights $W_{\text{ref}}, W_{\text{branch}}$ are polynomially controlled in Q . Then for every scale-admissible packet core K_p ,*

$$\sup_{t \in K_p} |F_X(t)|^2 W_{\text{ref}}(t) \leq Q^B.$$

Consequently, after the smooth AFE tail estimate,

$$\sup_{t \in K_p} |q_{\text{can}}''(t)|^2 W_{\text{ref}}(t) \leq Q^B$$

on the scale-admissible packet family.

If, in addition, W_{ref} is scale-slow on K_p^+ , then the sharper local Sobolev form also holds:

$$\sup_{t \in K_p} |F_X(t)|^2 W_{\text{ref}}(t) \ll Q^C \left((\lambda_p^{\text{raw}})^{-1} \int_{K_p^+} |F_X(u)|^2 W_{\text{ref}}(u) du + \lambda_p^{\text{raw}} \int_{K_p^+} |F_X'(u)|^2 W_{\text{ref}}(u) du \right).$$

Proof. For the absolute envelope, use the finite AFE expression

$$F_X(t) = \sum_{n \geq 1} a_n n^{-1/2-it} V_1(n/X) + \sum_{n \geq 1} \tilde{a}_n n^{-1/2+it} V_2(nX; \tfrac{1}{2} + it) + P_{\text{res},X}(t).$$

The smooth cutoffs restrict both sums to polynomially controlled effective ranges, and the coefficient bounds give

$$\sup_{t \in \Omega} |F_X(t)| \leq Q^C.$$

Since W_{ref} is polynomially bounded on the source window,

$$\sup_{t \in K_p} |F_X(t)|^2 W_{\text{ref}}(t) \leq Q^B.$$

The same bound for $q_{\text{can}}'' = F_X + R_X$ follows from Corollary E.14, after increasing B .

For the local Sobolev refinement, apply the one-dimensional Sobolev inequality at scale λ_p^{raw} to

$$g(t) := F_X(t) W_{\text{ref}}(t)^{1/2}.$$

Scale-slow variation of W_{ref} controls the derivative terms from $\partial_t W_{\text{ref}}^{1/2}$, giving the displayed estimate. The weighted derivative integral is controlled on mass-good packets by Theorem E.36; the absolute envelope above is the form used for collar exclusion. $\square$

Remark E.41 (Hardy form of the packet-scale AFE envelope). In Lemma E.40, the displayed proof line using

$$F_X(t) = \sum a_n n^{-1/2-it} V_1(n/X) + \sum \tilde{a}_n n^{-1/2+it} V_2(nX) + P_{\text{res},X}(t)$$

should be read in the active branch as the Hardy-differential finite AFE

$$\begin{aligned} F_X(t) = F_X^{H,3}(t) &= \sum_{0 \leq r+k \leq 3} \sum_{n \geq 1} c_{r,k}^+(t) (\log n)^r n^{-1/2-it} V_{1,r,k}(n/X; t) \\ &+ \sum_{0 \leq r+k \leq 3} \sum_{n \geq 1} c_{r,k}^-(t) (\log n)^r n^{-1/2+it} V_{2,r,k}(nX; t) + P_{\text{res},X}^{\text{curv}}(t). \end{aligned}$$

The conclusion of the lemma is unchanged. Indeed, the smooth cutoffs restrict the sums to polynomially controlled effective ranges, the log powers satisfy $(\log n)^r \ll_{r,\varepsilon} n^\varepsilon$, and the Hardy amplitude coefficients $c_{r,k}^\pm(t)$ and their fixed-order derivatives are polynomially bounded on ordinary source strips by Theorem E.13. Thus the same absolute envelope proof applies to $F_X^{H,3}$.

The old fixed-coefficient display is archived notation only and should not be used as a source-side input in the active branch.

Lemma E.42 (Two-sided packet-window mass). *Let J_p be the frequency window attached to a good packet p , and let $J_p^-, J_p^+ \subset J_p^M$ be two subintervals of the middle third separated by*

$$\text{dist}(J_p^-, J_p^+) \geq c_* |J_p|$$

and satisfying

$$|J_p^-|, |J_p^+| \geq c_* |J_p|.$$

Assume the packet amplitude factors as

$$A_p(\xi, t) = \chi_p(\xi) B_p(\xi, t),$$

where $\chi_p \equiv 1$ on J_p^M , $0 \leq \chi_p \leq 1$, and

$$e^{-C_0} \leq \frac{|B_p(\xi, t)|}{|B_p(\eta, t)|} \leq e^{C_0} \quad (\xi, \eta \in J_p).$$

Define

$$\rho_t(\xi) := \frac{|A_p(\xi, t)|^2}{\int_{J_p} |A_p(\eta, t)|^2 d\eta}.$$

Then there exists $\theta_0 > 0$ such that

$$\int_{J_p^-} \rho_t(\xi) d\xi \geq \theta_0, \quad \int_{J_p^+} \rho_t(\xi) d\xi \geq \theta_0.$$

Proof. The proof is the same on J_p^- and J_p^+ . Since $\chi_p \equiv 1$ on J_p^M and $\chi_p \leq 1$ on J_p ,

$$\int_{J_p^\pm} \rho_t(\xi) d\xi = \frac{\int_{J_p^\pm} |B_p(\xi, t)|^2 d\xi}{\int_{J_p} |\chi_p(\eta) B_p(\eta, t)|^2 d\eta}.$$

By uniform comparability of $|B_p|$,

$$\int_{J_p^\pm} |B_p(\xi, t)|^2 d\xi \geq e^{-2C_0} |J_p^\pm| \inf_{J_p} |B_p|^2,$$

whereas

$$\int_{J_p} |\chi_p(\eta) B_p(\eta, t)|^2 d\eta \leq e^{2C_0} |J_p| \inf_{J_p} |B_p|^2.$$

Since $|J_p^\pm| \geq c_* |J_p|$, the claim follows with

$$\theta_0 := c_* e^{-4C_0}.$$

□

Lemma E.43 (Packet-window variance lower bound from two-sided mass). *Under Lemma E.42,*

$$\text{Var}_{\rho_t}(\xi) \geq c |J_p|^2$$

for some absolute $c > 0$.

Proof. Let X, X' be independent random variables with law ρ_t . Then

$$2 \text{Var}_{\rho_t}(\xi) = \mathbb{E}|X - X'|^2.$$

On the event $X \in J_p^-$ and $X' \in J_p^+$, which has probability at least θ_0^2 , one has

$$|X - X'| \geq c_* |J_p|.$$

Therefore

$$2 \text{Var}_{\rho_t}(\xi) \geq \theta_0^2 c_*^2 |J_p|^2,$$

which proves the claim after changing constants. □

Remark E.44 (Phase-derivative transversality is diagnostic here). The two-sided packet-window variance estimate is useful when one wants to derive an explicit phase-transversality theorem from a spectral packet model. It is not a load-bearing input in the current source-frame proof. The active off-block estimate uses the scale-normalized resonant graph of Lemma E.46 and the integration-by-parts estimate of Lemma E.52. Thus no global or unnormalized lower bound for $|\omega_p - \omega_q|$ is assumed.

Remark E.45 (Identification of packet phase). For packets produced by the smooth-AFE construction in the form $\psi_p(t) = a_p(t)e^{i\omega_p t}$, the packet phase parameter ω_p agrees on the good packet set with the local phase derivative

$$-\partial_t \arg F_X(t)|_{t=m_p}.$$

Lemma E.46 (Scale-normalized frequency separation off resonant blocks). *For two packet atoms p, q , set*

$$I_{pq} := \text{supp}(a_p \overline{a_q} W_{\text{branch}}), \quad \lambda_{pq} := |I_{pq}|, \quad \Delta\omega_{pq} := |\omega_p - \omega_q|.$$

Fix $\kappa > 0$. Define the resonant graph by placing an edge between p and q when

$$\lambda_{pq} \Delta\omega_{pq} < Q^\kappa.$$

The resonant blocks are the connected components of this graph. If p and q lie in distinct resonant blocks, then

$$\lambda_{pq} |\omega_p - \omega_q| \geq Q^\kappa.$$

Proof. This is the definition of the connected components of the scale-normalized resonant graph. If p and q were in distinct blocks and nevertheless $\lambda_{pq} |\omega_p - \omega_q| < Q^\kappa$, the edge (p, q) would join the two components, a contradiction. $\square$

Remark E.47 (Frequency-gap closure status). For the source-frame proof, no unnormalized lower bound of the form $|\omega_p - \omega_q| \gg Q^{-\delta}$ is used. Resonant blocks are defined by the scale-normalized relation in Lemma E.46; off-block separation is therefore automatic in the dimensionless quantity $\lambda_{pq} |\omega_p - \omega_q|$. The earlier phase-derivative/variance chain may still be useful for constructing economically sized resonant blocks, but it is not needed for the validity of the off-block integration-by-parts estimate.

Definition E.48 (Finite-complexity source collar). The source window Ω is said to have polynomial collar complexity if, for all $0 < \eta \leq 1$,

$$\mu\{t \in \Omega : \text{dist}(t, \Omega^c) \leq \eta\} \leq Q^{C_\Omega} \eta.$$

This holds, for example, if Ω is a union of at most Q^{C_Ω} intervals. The good packet family is understood to consist of scale-admissible mass-good packets, with mass-goodness measured by

$$M_p = \int_{K_p^+} |F_X(t)|^2 W_{\text{ref}}(t) dt \geq Q^{-C_{\text{mass}}}.$$

Corollary E.49 (Boundary-collar exclusion). *Let*

$$C_\eta := \{t \in \Omega : d(t) \leq \eta\}, \quad d(t) = \text{dist}(t, \Omega^c).$$

Assume Lemma E.40, Corollary E.19, Corollary E.14, and the polynomial collar-complexity condition of Definition E.48. Then for each prescribed $A > 0$ there is C_{col} such that, with $\eta = Q^{-C_{\text{col}}}$,

$$\int_{C_\eta} |q''_{\text{can}}(t)|^2 W_{\text{ref}}(t) dt \leq Q^{-A} \int_{\Omega} |q''_{\text{can}}(t)|^2 W_{\text{ref}}(t) dt.$$

Moreover, after increasing C_{col} in terms of C_{mass} , no mass-good packet enlargement K_p^+ is contained in $C_{c\eta}$ for any fixed geometric constant c . Consequently every good packet satisfies

$$d(m_p) \geq Q^{-C}$$

for some fixed C .

Proof. By Lemma E.40,

$$\sup_{t \in \Omega} |q''_{\text{can}}(t)|^2 W_{\text{ref}}(t) \leq Q^B$$

on the scale-admissible packet region, after absorbing the smooth AFE tail. Using polynomial collar complexity,

$$\int_{C_\eta} |q''_{\text{can}}(t)|^2 W_{\text{ref}}(t) dt \leq Q^B \mu(C_\eta) \leq Q^{B+C_\Omega} \eta.$$

Choose C_{col} so that $Q^{B+C_\Omega} \eta \leq Q^{-A} c_0$, where

$$c_0 \leq \int_{\Omega} |q''_{\text{can}}(t)|^2 W_{\text{ref}}(t) dt$$

is supplied by Corollary E.19. This gives the collar-mass estimate.

Now suppose $K_p^+ \subset C_{c\eta}$. By $q''_{\text{can}} = F_X + R_X$,

$$\int_{K_p^+} |F_X|^2 W_{\text{ref}} \leq 2 \int_{C_{c\eta}} |q''_{\text{can}}|^2 W_{\text{ref}} + 2 \int_{\Omega} |R_X|^2 W_{\text{ref}}.$$

Increase C_{col} and choose the AFE cutoff depth so that the right side is $< Q^{-C_{\text{mass}}}$. This contradicts mass-goodness. Whitney geometry gives $K_p^+ \subset C_{c\eta}$ whenever $d(m_p) \leq c'\eta$, after adjusting constants. Hence every good packet has $d(m_p) \geq Q^{-C}$. $\square$

Proposition E.50 (Packet-scale normalization). *For every good packet p ,*

$$Q^{-C} \leq d(m_p) \leq Q^C, \quad \text{and hence} \quad Q^{-C} \leq \lambda_p \leq Q^C.$$

Proof. The upper bound $d(m_p) \leq Q^C$ is the finite-AFE localization bound already available in the draft (Remark E.35). The lower bound follows from Corollary E.49. Therefore

$$Q^{-C} \leq d(m_p) \leq Q^C, \quad \text{and hence} \quad Q^{-C} \leq \lambda_p \leq Q^C.$$

$\square$

Lemma E.51 (Windowed packet amplitude bounds). *Let the packet atoms have the windowed form*

$$\psi_p(t) = a_p(t) e^{i\omega_p t}, \quad a_p(t) = \nu_p^{-1} \lambda_p^{-1/2} \chi_p(t) \eta\left(\frac{t - m_p}{\lambda_p}\right),$$

where $\eta \in C_c^\infty(\mathbb{R})$, ν_p is the $L^2(W_{\text{ref}})$ -normalizing factor, and λ_p is polynomially controlled on the good packet family. Assume the source weights and the auxiliary cutoffs χ_p have polynomial scale-slow derivatives on overlapping packet cores. Then, for

$$A_{pq}(t) := a_p(t) \overline{a_q(t)} W_{\text{branch}}(t), \quad I_{pq} := \text{supp } A_{pq}, \quad \lambda_{pq} := |I_{pq}|,$$

one has, for each fixed N ,

$$\sum_{r=0}^N \lambda_{pq}^r \|\partial_t^r A_{pq}\|_{L^1(I_{pq})} \leq Q^{C_N}.$$

Proof. If $I_{pq} = \emptyset$ there is nothing to prove. Otherwise Whitney overlap comparability gives $\lambda_p \asymp \lambda_q \asymp \lambda_{pq}$. Repeated Leibniz differentiation of

$$\nu_p^{-1} \nu_q^{-1} \lambda_p^{-1/2} \lambda_q^{-1/2} \chi_p(t) \overline{\chi_q(t)} \eta\left(\frac{t - m_p}{\lambda_p}\right) \overline{\eta\left(\frac{t - m_q}{\lambda_q}\right)} W_{\text{branch}}(t)$$

shows that each derivative contributes at most the corresponding inverse packet scale, up to a polynomial power of Q . The reference-normalizing factors ν_p^{-1}, ν_q^{-1} are polynomially controlled on good packets, and the support has length $\asymp \lambda_{pq}$. Multiplying the r -th derivative estimate by λ_{pq}^r and integrating over I_{pq} gives the bound. $\square$

Lemma E.52 (Scale-normalized packet kernel decay). *Under Lemma E.51, for every fixed N ,*

$$|\langle \psi_p, \psi_q \rangle_{W_{\text{branch}}} | \leq Q^{C_N} (1 + \lambda_{pq} |\omega_p - \omega_q|)^{-N}.$$

Proof. Write

$$\langle \psi_p, \psi_q \rangle_{W_{\text{branch}}} = \int_{I_{pq}} A_{pq}(t) e^{i(\omega_p - \omega_q)t} dt.$$

The trivial estimate gives the claim when $\lambda_{pq}|\omega_p - \omega_q| \lesssim 1$. Otherwise integrate by parts N times:

$$\left| \int_{I_{pq}} A_{pq}(t) e^{i(\omega_p - \omega_q)t} dt \right| \leq |\omega_p - \omega_q|^{-N} \|\partial_t^N A_{pq}\|_{L^1(I_{pq})}.$$

Using Lemma E.51,

$$|\omega_p - \omega_q|^{-N} \|\partial_t^N A_{pq}\|_{L^1(I_{pq})} \leq Q^{C_N} (\lambda_{pq}|\omega_p - \omega_q|)^{-N}.$$

Combining this with the trivial bound yields the displayed $(1 + \lambda_{pq}|\omega_p - \omega_q|)^{-N}$ estimate. $\square$

Corollary E.53 (Scale-normalized off-block row-sum decay). *Assume the good packet family has cardinality $\#\mathcal{P}_{\text{good}} \leq Q^D$, resonant blocks are defined as in Lemma E.46, and the reference block Grams satisfy*

$$\|G_{\text{ref},B}^{-1/2}\|_{\text{op}} \leq Q^{C_{\text{ref}}}$$

on every good resonant block. Then, after separating resonant blocks, the off-block part of the reference-whitened branch Gram matrix satisfies

$$\sup_B \sum_{C \neq B} \left\| G_{\text{ref},B}^{-1/2} G_{\text{branch},BC} G_{\text{ref},C}^{-1/2} \right\|_{\text{op}} \leq Q^{-A}$$

for every prescribed $A > 0$, after choosing the integration-by-parts order N sufficiently large.

Proof. For q outside the block of p , Lemma E.46 gives

$$\lambda_{pq}|\omega_p - \omega_q| \geq Q^\kappa.$$

Lemma E.52 therefore gives

$$|\langle \psi_p, \psi_q \rangle_{W_{\text{branch}}}| \leq Q^{C_N - \kappa N}.$$

Thus every raw off-block matrix entry is $O(Q^{C_N - \kappa N})$. Multiplying on the left and right by the reference block inverse square roots costs at most $Q^{2C_{\text{ref}}}$, and summing over at most Q^D packet indices costs another polynomial factor. Choosing N so that $\kappa N - C_N \geq A + D + 2C_{\text{ref}} + 10$ gives the displayed whitened row-sum bound. $\square$

Remark E.54 (Use in the source frame lemma). The off-block control used in Lemma E.70 is supplied by the scale-normalized chain

windowed amplitude bounds $\Rightarrow$ scale-normalized kernel decay $\Rightarrow$ off-block row-sum control, namely Lemma E.51, Lemma E.52, and Corollary E.53. Resonant blocks are handled by branch/reference domination, not by oscillatory decay.

Remark E.55 (Packet-regularity bridge status). The packet-regularity bridge now has a scale-normalized form:

absolute derivative/envelope bounds and windowed packet amplitudes $\Rightarrow$ off-block oscillatory row-sum control

This closes the analytic off-diagonal estimate. It does not by itself prove branch/reference domination on good packet supports; that structural routing input is isolated in Hypothesis E.59 and used through Lemma E.69.

Theorem E.56 (Windowed packet construction regularity relative to construction data). *Adopt the Whitney packet cores of Definition E.30 and construct packet atoms by*

$$\psi_p(t) = a_p(t) e^{i\omega_p t}, \quad a_p(t) = \nu_p^{-1} \lambda_p^{-1/2} \chi_p(t) \eta\left(\frac{t - m_p}{\lambda_p}\right), \quad \eta \in C_c^\infty(\mathbb{R}),$$

on the good packet family. Assume the source weights and auxiliary cutoffs are polynomially scale-slow, the good packet family is scale-admissible, the packet normalizers are polynomially bounded, and every good resonant block has a polynomially conditioned reference Gram matrix:

$$\|G_{\text{ref},B}^{-1/2}\|_{\text{op}} \leq Q^{C_{\text{ref}}}.$$

Then:

- (1) the packet supports have bounded overlap;
- (2) $Q^{-C} \leq \lambda_p \leq Q^C$ on good packets;
- (3) the amplitudes satisfy

$$\sum_{r=0}^N \lambda_{pq}^r \|\partial_t^r (a_p \overline{a_q} W_{\text{branch}})\|_{L^1(I_{pq})} \leq Q^{C_N};$$

- (4) after defining resonant blocks by the scale-normalized relation

$$\lambda_{pq} |\omega_p - \omega_q| < Q^\kappa,$$

the off-block kernels satisfy

$$|\langle \psi_p, \psi_q \rangle_{W_{\text{branch}}}| \leq Q^{C_N} (1 + \lambda_{pq} |\omega_p - \omega_q|)^{-N};$$

- (5) the off-block branch Gram is polynomially small in the same reference-whitened block coordinates used by the block coercivity lemma.

Consequently, Corollary E.53 gives the whitened off-block row-sum decay used in the source-frame argument.

Proof. Bounded overlap follows from Theorem E.31. Polynomial scale bounds follow from scale-admissibility and Proposition E.50 on the good packet family. The amplitude derivative estimate is Lemma E.51. The off-block kernel estimate is Lemma E.52, with the scale-normalized nonresonance supplied by the definition of the resonant blocks in Lemma E.46. The passage to reference-whitened block coordinates is Corollary E.53, using the stated polynomial conditioning of $G_{\text{ref},B}$. This proves all claims. $\square$

Remark E.57 (What the windowed packet target closes after the canonical window). Theorem E.56 closes the source-side oscillatory-decay step once the windowed packet construction supplies scale-admissibility, polynomial normalizers, scale-slow cutoffs, and polynomial conditioning of the reference block Grams. Resonant blocks are defined by the scale-normalized relation, so no separate unnormalized frequency-gap theorem is used for off-block decay.

The AFE tail and fixed-window reference noncollapse are supplied by Theorem E.13, Corollary E.15, and Corollary E.19. In the canonical Gate-1 branch, branch/reference block domination is no longer an additional analytic input: it is supplied by the canonical branch-complete construction, Definition E.59, through Theorem E.61.

Honest closure state. In the active Hardy-curvature branch, Theorem E.5 proves the actual source anchor, and Lemma E.17 plus mass-good packet selection supplies reference source mass. Relative to the canonical branch-complete window and the windowed packet-construction data, Theorem E.71 gives $\delta_{\text{src}} \geq Q^{-A}$. The remaining source-side obligations are therefore constructional rather than analytic:

canonical branch-complete construction data, uniform reference nonconcentration if the reference window

The downstream raw row/routing certificate is handled in the canonical labelled branch by Definition 16.355 and Theorem 16.356; if that construction is not adopted, the legacy obstruction vector $\mathfrak{B}_{2Y,p}$ remains the precise finite audit item.

In the active canonical source-graded branch, δ_{coerc} , the positive-density extraction/static-line module, and the same-norm atomic comparison are handled by the certified downstream package recorded in Remark F.10. They should not be listed as open finite-survival gaps unless the proof leaves the canonical certified branch. The Graph/O2O3 gap δ_{graph} is closed by Theorem E.94 together with the canonical multiplier/anti-lock theorem Theorem E.106. It is not an active source-construction gap in the canonical Fejér–Riesz branch.

The label **thm:unconditional-packet-regularity** is retained for citation stability; the theorem is unconditional only relative to the stated windowed packet-construction data.

Remark E.58 (Source bridge proof register). The source bridge now separates analytic estimates from constructional routing estimates. The analytic estimates are supplied by the completed-source AFE, canonical reference noncollapse, absolute packet envelopes, and scale-normalized oscillatory integration by parts. The constructional routing estimate is supplied by the canonical branch-complete window:

$$W_{\text{branch}}^{\text{can}} \geq Q^{-C} W_{\text{ref}}, \quad G_{\text{branch},B}^{\text{can}} \geq Q^{-C} G_{\text{ref},B}.$$

Thus branch/reference domination is theorem-level in the canonical Gate-1 branch, by Theorem E.61. If some archived or raw branch window is used instead of $W_{\text{branch}}^{\text{can}}$, it must carry its own quadratic-form domination proof; the canonical theorem does not retrofit arbitrary older weights.

Definition E.59 (Canonical branch-complete source window). Let Ω_{good} be the good source region used by the Gate-1 packet family. Let

$$K_p = (m_p - r_p, m_p + r_p)$$

be the raw Whitney packet cover of Ω_{good} , with bounded overlap. Fix a nonnegative smooth bump w and a core parameter $0 < \kappa < 1$ such that

$$w(x) \geq c_w > 0 \quad (|x| \leq \kappa).$$

Assume the Whitney cores

$$K_p^\circ := \{t : |t - m_p| \leq \kappa r_p\}$$

cover Ω_{good} , and assume the reference weight is scale-slow on cores:

$$W_{\text{ref}}(m_p) \geq Q^{-C_{\text{sl}}} W_{\text{ref}}(t) \quad (t \in K_p^\circ).$$

The proof-level Gate-1 branch observation window is the canonical branch-complete window

$$W_{\text{branch}}^{\text{can}}(t) := \sum_p W_{\text{ref}}(m_p) w\left(\frac{t - m_p}{r_p}\right).$$

All Gate-1 branch Gram matrices in the canonical source branch are computed using $W_{\text{branch}}^{\text{can}}$. Older occurrences of W_{branch} in the Gate-1 source-frame argument are to be read as $W_{\text{branch}}^{\text{can}}$, unless explicitly marked as an archived or raw branch weight.

Lemma E.60 (Canonical pointwise branch/reference domination). *On Ω_{good} ,*

$$W_{\text{branch}}^{\text{can}}(t) \geq Q^{-C} W_{\text{ref}}(t)$$

for some absolute C .

Proof. Fix $t \in \Omega_{\text{good}}$. By the core-covering property there is a packet $p = p(t)$ with $t \in K_p^\circ$. Hence

$$w\left(\frac{t - m_p}{r_p}\right) \geq c_w.$$

The corresponding term in the defining sum gives

$$W_{\text{branch}}^{\text{can}}(t) \geq c_w W_{\text{ref}}(m_p) \geq c_w Q^{-C_{\text{sl}}} W_{\text{ref}}(t).$$

Absorbing c_w^{-1} into the polynomial exponent gives the claim. $\square$

Theorem E.61 (Canonical branch/reference block domination). *For the canonical branch-complete source window, every good resonant block B satisfies the quadratic-form domination*

$$G_{\text{branch},B}^{\text{can}} \geq Q^{-C} G_{\text{ref},B}.$$

Equivalently, for every coefficient vector $c = (c_p)_{p \in B}$,

$$\int \left| \sum_{p \in B} c_p \psi_p(t) \right|^2 W_{\text{branch}}^{\text{can}}(t) dt \geq Q^{-C} \int \left| \sum_{p \in B} c_p \psi_p(t) \right|^2 W_{\text{ref}}(t) dt.$$

Thus the branch-hole failure mode is absent in the canonical Gate-1 source branch.

Proof. Apply Lemma E.60 to

$$f_B(t) := \sum_{p \in B} c_p \psi_p(t).$$

Then

$$\int |f_B(t)|^2 W_{\text{branch}}^{\text{can}}(t) dt \geq Q^{-C} \int |f_B(t)|^2 W_{\text{ref}}(t) dt,$$

which is precisely the stated quadratic-form domination of the Gram matrices. $\square$

Remark E.62 (Canonical branch window is a construction choice). The preceding theorem is a theorem about the canonical branch-complete window $W_{\text{branch}}^{\text{can}}$. It is not a theorem about an arbitrary previously routed branch weight. If an older branch object is used elsewhere, then either it must be identified with $W_{\text{branch}}^{\text{can}}$ or it must carry its own proof of quadratic-form domination. In particular, AFE tail control, packet regularity, and positive reference mass do not by themselves imply $G_{\text{branch},B} \geq Q^{-C} G_{\text{ref},B}$ for an arbitrary branch window.

Remark E.63 (Scaling convention for the canonical branch window). The displayed definition of $W_{\text{branch}}^{\text{can}}$ is a pointwise t -weight. No extra powers of r_p are included in this pointwise convention. If a local-coordinate normalization uses factors such as r_p^k , those factors belong to the local energy normalization and must not be inserted again into the pointwise weight. This convention prevents a spurious scale loss in the branch/reference Gram comparison.

Remark E.64 (Whitney geometry versus admissible lower scale). The Whitney construction supplies bounded overlap and scale comparability on overlapping packets. A polynomial lower bound for the raw packet scale, $r_p \geq Q^{-C}$, is not a consequence of Whitney geometry alone. In the canonical source branch it is either an admissibility condition on the selected packet family or a consequence of a separately proved boundary-collar exclusion theorem. It is not used in the proof of Theorem E.61, which only needs the core covering and scale-slow reference weight hypotheses.

Remark E.65 (Legacy local-coordinate branch weights). Lemma E.77 records the local-coordinate Fubini form of older branch observables, where powers such as λ_b^5 come from the local energy normalization. This is not the proof-level pointwise branch window used for Gate-1 in the canonical branch. The proof-level Gate-1 Gram comparison uses $W_{\text{branch}}^{\text{can}}$ from Definition E.59. Any argument using the legacy Fubini weight must either identify its resulting pointwise weight with $W_{\text{branch}}^{\text{can}}$ or prove a separate quadratic-form branch/reference domination theorem for that weight.

Remark E.66 (Archived source hypothesis labels). The labels

`hyp:packet-amplitude-decay-budget`, `hyp:ef-completion-bound`, `hyp:reference-meansquare-n`

are archived compatibility labels. In the active Hardy/canonical branch:

`hyp:packet-amplitude-decay-budget`

is supplied by the windowed packet amplitude and scale-normalized oscillatory decay lemmas;

`hyp:ef-completion-bound`

is supplied by Corollary E.15; and

`hyp:reference-meansquare-noncollapse`

is supplied by Corollary E.19, subject to the uniformity caveat in Remark E.22 for moving Q -dependent reference windows. They should not be cited as active hypotheses unless the proof explicitly leaves the canonical Hardy branch.

Remark E.67 (Source-layer derivation register). In the active Hardy/canonical branch, the following source-side items are theorem-level outputs:

Hardy source trace, differential AFE tail, fixed-window source noncollapse, absolute packet envelope,

The branch/reference lower bound is also theorem-level in the canonical branch:

$$W_{\text{branch}}^{\text{can}} \geq Q^{-C} W_{\text{ref}}, \quad G_{\text{branch},B}^{\text{can}} \geq Q^{-C} G_{\text{ref},B}.$$

It is not a theorem about arbitrary archived branch windows. Therefore the only source-layer qualifications that remain are:

- (1) use the Hardy source API of Definition E.24;
- (2) use $W_{\text{branch}}^{\text{can}}$ as the proof-level Gate-1 observation window;
- (3) maintain the uniform reference nonconcentration condition if W_{ref} varies with Q .

Definition E.68 (Reference-whitened resonant block coordinates). For a good resonant block B , let

$$G_{\text{ref},B} := (\langle \psi_i, \psi_j \rangle_{W_{\text{ref}}})_{i,j \in B}, \quad G_{\text{branch},B} := (\langle \psi_i, \psi_j \rangle_{W_{\text{branch}}})_{i,j \in B}.$$

A good block is *reference-regular* if $G_{\text{ref},B}$ is positive definite and

$$\|G_{\text{ref},B}^{-1/2}\|_{\text{op}} + \|G_{\text{ref},B}^{1/2}\|_{\text{op}} \leq Q^{C_{\text{ref}}}.$$

On such a block the branch Gram in reference-whitened coordinates is

$$M_B := G_{\text{ref},B}^{-1/2} G_{\text{branch},B} G_{\text{ref},B}^{-1/2}.$$

All block lower bounds in the source frame are to be interpreted in these coordinates. Individual normalization of the atoms is not a substitute for this whitening unless the reference Gram is already the identity.

Lemma E.69 (Block coercivity from branch/reference domination). *Let B be a reference-regular good resonant block. If*

$$G_{\text{branch},B} \geq Q^{-C_{\text{br}}} G_{\text{ref},B}$$

as a quadratic form, then

$$M_B = G_{\text{ref},B}^{-1/2} G_{\text{branch},B} G_{\text{ref},B}^{-1/2} \geq Q^{-C_{\text{br}}} I_B.$$

Proof. For any vector $v \in \mathbb{C}^B$, set $u = G_{\text{ref},B}^{-1/2} v$. Then

$$\begin{aligned} v^* M_B v &= u^* G_{\text{branch},B} u \geq Q^{-C_{\text{br}}} u^* G_{\text{ref},B} u \\ &= Q^{-C_{\text{br}}} \|G_{\text{ref},B}^{1/2} G_{\text{ref},B}^{-1/2} v\|_2^2 = Q^{-C_{\text{br}}} \|v\|_2^2. \end{aligned}$$

This is the claimed matrix inequality in reference-whitened coordinates. □

Lemma E.70 (Source frame coercivity for Gate-1). *Let*

$$\mathcal{S}_X = \text{span}\{B_2, P_{2,\leq X}, D_{2,\leq X}, P_{\text{res},X}\}, \quad F_X = B_2 + P_{2,\leq X} + D_{2,\leq X} + P_{\text{res},X}.$$

Let $\{\psi_i\}_{i=1}^N$ be packet-adapted source atoms spanning $\mathcal{S}_X$, grouped into the scale-normalized resonant blocks of Lemma E.46. Assume:

- (1) *every good block is reference-regular in the sense of Definition E.68;*
- (2) *branch/reference block domination holds on every good block:*

$$G_{\text{branch},B} \geq Q^{-C_{\text{br}}} G_{\text{ref},B};$$

- (3) *the off-block branch Gram satisfies the reference-whitened row-sum bound of Corollary E.53;*
- (4) *the EF completion correction satisfies Corollary E.15.*

Then

$$\lambda_{\min}(M + E) \geq Q^{-A}$$

for some absolute $A > 0$, where M is the block reference-whitened branch Gram and E is the reference-whitened residual correction. Equivalently,

$$\|f\|_{L^2(W_{\text{branch}})}^2 \geq Q^{-A} \|f\|_{L^2(W_{\text{ref}})}^2 \quad (f \in \mathcal{S}_X).$$

Proof. By Lemma E.69, each resonant block satisfies

$$M_B \geq Q^{-C_{\text{br}}} I_B.$$

Thus the block-diagonal part M_{diag} of the reference-whitened matrix obeys

$$M_{\text{diag}} \geq Q^{-C_{\text{br}}} I.$$

By Corollary E.53, after choosing the integration-by-parts order sufficiently large,

$$\|M_{\text{off}}\|_{\text{op}} \leq Q^{-A_1}.$$

The EF completion estimate gives

$$\|E\|_{\text{op}} \leq Q^{-A_2}$$

after the AFE cutoff is chosen sufficiently deep. Taking A_1, A_2 larger than C_{br} by a fixed margin yields

$$M + E \geq (Q^{-C_{\text{br}}} - Q^{-A_1} - Q^{-A_2}) I \geq Q^{-A} I.$$

The final weighted L^2 inequality is the same statement pulled back from reference-whitened coordinates to the original finite source space. $\square$

Theorem E.71 (Source survival from completed source and canonical construction). *Assume:*

- (1) the actual Hardy source anchor and differential AFE package, Theorems E.5 and E.13;
- (2) the windowed packet construction regularity of Theorem E.56, including reference-regular resonant blocks;
- (3) the Gate-1 branch window is the canonical branch-complete window $W_{\text{branch}}^{\text{can}}$ of Definition E.59.

Then the Gate-1 source gap satisfies

$$\delta_{\text{src}} := \lambda_{\min}(M) - \|E\|_{\text{op}} \geq Q^{-A}$$

for some absolute $A > 0$. Equivalently,

$$\int |q''_{\text{can}}(t)|^2 W_{\text{branch}}^{\text{can}}(t) dt \geq Q^{-A} \int |q''_{\text{can}}(t)|^2 W_{\text{ref}}(t) dt,$$

up to the already quantified finite-AFE completion error.

Proof. The completed-source anchor identifies the actual canonical source curvature with the completed source trace to which the smooth AFE applies. Theorem E.13 and Corollary E.15 give the finite main term plus a polynomially small reference-whitened residual correction. The windowed packet construction gives bounded overlap, polynomial scales, amplitude derivative bounds, reference block conditioning, and reference-whitened off-block decay by Corollary E.53. The canonical branch-complete window gives the block-diagonal lower bound by Theorem E.61 and Lemma E.69. Lemma E.70 then yields $\lambda_{\min}(M + E) \geq Q^{-A}$, which is exactly $\delta_{\text{src}} \geq Q^{-A}$ after absorbing the residual correction. $\square$

Remark E.72 (Remaining source inputs after the canonical Hardy branch). In the active Hardy-curvature branch the source anchor is not an assumption: it is Theorem E.5. The analytic estimates after that anchor are explicit: the differential Hardy AFE tail bound, the dyadic Hardy-curvature mean-square lower bound, absolute packet derivative/envelope bounds, scale-normalized off-block oscillatory decay, and mass-good packet selection. The branch-hole obstruction is removed by the construction of $W_{\text{branch}}^{\text{can}}$ and Theorem E.61.

Thus the remaining source-side qualifications are:

- (1) the proof-level Gate-1 branch window must be $W_{\text{branch}}^{\text{can}}$;

- (2) if the reference window is allowed to vary with Q , the uniform polynomial reference nonconcentration in Remark E.22 must be maintained;
- (3) a different source branch, such as a ξ -source or logarithmic-phase source, must supply its own trace, AFE, residue, and mass package.

Branch/reference block domination is not a separate remaining input in the canonical branch, because it is Theorem E.61. The old completed-source model theorem is archived; it is superseded in the active branch by the actual Hardy-curvature source package.

Corollary E.73 (Gate–1 source survival in the canonical branch). *Assume the actual Hardy source anchor and differential AFE package, Theorems E.5 and E.13; the windowed packet construction regularity of Theorem E.56; and the canonical branch-complete source window of Definition E.59. Then Theorem E.71 gives*

$$\int |q''_{\text{can}}(t)|^2 W_{\text{branch}}^{\text{can}}(t) dt \geq Q^{-A} \int |q''_{\text{can}}(t)|^2 W_{\text{ref}}(t) dt,$$

equivalently $\text{Agg}L^2(q''_{\text{can}})$ holds with polynomial loss. The analytic inputs used here are the residue-aware AFE tail estimate Corollary E.15, fixed-window reference noncollapse Corollary E.19, packet derivative/envelope control, reference-whitened block coercivity, and scale-normalized off-block decay Corollary E.53.

Remark E.74 (Source-layer status). The source estimates are architecturally repaired in the following precise sense:

completed-source anchor+residue-aware smooth AFE+windowed packet regularity+canonical branch-complete

The packet derivative budget is absolute, with only a mass-good polynomial relative consequence; no unqualified $(\log X)^{2j}$ -relative estimate is used. In the active Hardy-curvature branch the completed-source anchor is closed by Theorem E.5; a different source branch must provide its own trace/AFE package. The source layer is otherwise conditional only on the decision to use the canonical branch-complete window as the proof-level Gate–1 observation window. If the reference window moves with Q , it also remains conditional on a uniform polynomial reference-noncollapse theorem.

Remark E.75 (Failure labels). After the corrections above, the off-diagonal packet-correlation failure is handled by the scale-normalized oscillatory estimate, and fixed-window reference mean-square collapse is excluded by Corollary E.19. In the canonical source branch, the branch-hole failure is removed by Theorem E.61. A branch-hole failure remains a valid label only for noncanonical or archived branch windows that are not identified with $W_{\text{branch}}^{\text{can}}$. For a Q -dependent reference window, a separate possible failure mode is lack of a uniform polynomial version of reference noncollapse, as noted in Remark E.22.

Remark E.76 (Resonant block conditioning). Resonant blocks are defined by the scale-normalized condition

$$\lambda_{pq} |\omega_p - \omega_q| < Q^\kappa.$$

Inside a resonant block no oscillatory decay is claimed. Block conditioning is instead supplied by branch/reference domination via Lemma E.69:

$$G_{\text{branch},B} \geq Q^{-A} G_{\text{ref},B}.$$

Remark E.77 (Archived Gate–1 proof tools). The older local-coordinate Fubini forms, branch-observable diagnostics, finite-AFE anti-lock heuristics, central-tube fold discussions, and degree-one determinant analogies are archived proof tools. They are not part of the proof-level Gate–1 chain in the canonical Hardy branch.

The live Gate–1 route is:

Hardy source API+ $W_{\text{branch}}^{\text{can}}$ +windowed packet regularity+reference-whitened block coercivity $\implies \text{Agg}L^2(q''_{\text{can}})$

Any future use of the archived diagnostics must explicitly prove a polynomial comparison with the canonical $W_{\text{branch}}^{\text{can}}$ -weighted source norm.

E.3. Archived ordinary packet-frequency route.

Remark E.78 (Archived O1/O2/O3 ordinary finite-row route). The O1/O2/O3 ordinary packet-frequency route is no longer an active Gate–1 definition. It is archived as a diagnostic finite-row route. Its old objects should not be used as substitutes for

$$\text{Agg}L^2(q''_{\text{can}})$$

unless one proves a transfer theorem from the corresponding finite-row quantity to the canonical scalar L^2 -source energy for the Hardy source.

The active upstream proof uses the canonical Hardy source, the canonical branch-complete window, and the aggregate source survival statement

$$\text{Agg}L^2(q''_{\text{can}}).$$

Any use of the old O1/O2/O3 route must be explicitly labelled as diagnostic unless a separate transfer theorem proves that it supplies this canonical Hardy L^2 -source estimate.

E.4. Finite/O2O3₂₃: analytic Riesz graph and exterior-pivot input. O2/O3 is now separated from Gate–2. It should be handled in an analytic R^C -conditioned Riesz basis, not in the downstream singular-discard carrier basis.

Let

$$A_{\leq X, m}$$

be the finite augmented matrix built from

$$F_X = B_2 + P_{2, \leq X} + D_{2, \leq X}$$

and the augmented column space

$$\mathbf{C} \cdot 1 + m_{\text{can}} \mathbf{C}_R[x].$$

The analytic normalization requirement is

$$\boxed{\|A_{\leq X, m}\|_{\text{op}} \leq R^C.}$$

Definition E.79 (Riesz graph model for O2/O3). The augmented column space has the graph form

$$\mathbf{C} \cdot 1 + m_{\text{can}} \mathbf{C}_R[x] = m_{\text{can}} (\mathbf{C} \cdot (1/m_{\text{can}}) + \mathbf{C}_R[x]).$$

Thus the pure branch obstruction is the non-polynomiality of $1/m_{\text{can}}$:

$$\text{dist}(1/m_{\text{can}, +}, \mathbf{C}_R[x]) \geq Q^{-A}$$

in the R^C -conditioned Riesz norm. In the favorable model

$$m_{\text{can}, +}(x) = \kappa x^2(1 + \varepsilon(x)),$$

this is an analytic divided-difference/Riesz problem for $1/m_{\text{can}, +}$.

Remark E.80 (Graph/O2/O3 route and near-pure Schur block). In the R^C -normalized analytic basis, the pure branch graph model is

$$\mathbf{C} \cdot 1 + m_{\text{can}} \mathbf{C}_R[x] = m_{\text{can}} (\mathbf{C} \cdot (1/m_{\text{can}}) + \mathbf{C}_R[x]).$$

Thus pure-branch O2/O3 reduces to non-polynomiality of $1/m_{\text{can}}$ in the finite Riesz norm:

$$\text{dist}_{\mathcal{R}_R}(1/m_{\text{can}}, \mathbf{C}_R[x]) \geq Q^{-A}.$$

The full determinant problem is not solved by the pure branch alone. Under a bounded branch-charge Stack weight, the Fejér–Riesz charge anti-lock theorem proves survival of the pure branch inside the full determinant once the packet model is expressed in smooth charge coordinates. Equivalently, after branch-charge decomposition,

$$D_{\text{full}} = \sum_{r=0}^n D_{-n+r}, \quad n = R + 2,$$

one needs the finite near-pure Schur complement

$$D_{-n} \notin_{Q^{-A}} \text{span}\{D_{-n+1}, \dots, D_{-n+d_0}\},$$

where d_0 is the bounded branch-charge degree of the smooth Stack weight. This statement is proved below by Theorem E.106 before capped extraction; hard extraction would destroy the branch-charge/Fourier structure needed for the proof.

Definition E.81 (Bounded-charge near-pure Schur block). Let $n = R + 2$, and decompose the full determinant into branch-charge sectors

$$D_{\text{full}} = \sum_r D_{-n+r}.$$

Let d_0 be the branch-charge bandwidth of the bounded-degree Stack weight used before capped extraction:

$$d_0 := d_N + 2d_H + 2d_{\Gamma, \text{eff}}.$$

Here $d_{\Gamma, \text{eff}}$ is computed from the final effective forbidden wall linear form after coherent amplitude projection. Then the pure plus sector D_{-n} can interact, under the pre-extraction Stack weight, only with the finite block

$$D_{-n+1}, \dots, D_{-n+d_0}.$$

The fixed- α graph-survival condition is therefore the finite Schur condition

$$D_{-n} \notin_{Q^{-A}} \text{span}\{D_{-n+1}, \dots, D_{-n+d_0}\}.$$

This is the finite Schur target proved by Theorem E.106 under the bounded-degree Fejér–Riesz charge model; an AlphaAdmissibility package remains only a fallback route.

Remark E.82 (Do not cap before determinant survival). The Fejér–Riesz charge anti-lock proof of finite/O2/O3 graph survival is carried out at the smooth aggregate level. Capped extraction destroys the branch-charge and Fourier structure used to prove

$$D_{-n} \notin_{Q^{-A}} \text{span}\{D_{-n+1}, \dots, D_{-n+d_0}\}.$$

Thus the proof order is

$$\boxed{\text{smooth aggregate} \rightarrow \text{branch survival/O2/O3} \rightarrow \text{capped extraction} \rightarrow \text{atomic D4 contradiction.}}$$

In particular, the Stack weight used for determinant survival should be a bounded-degree pre-extraction weight. Capped factors such as $F_{\text{wall}}^{\text{forb}}$ may be used only after determinant-survival and aggregate lower bounds are secured.

Hypothesis E.83 (Canonical quadratic multiplier theorem). On each ordinary branch subpacket used for finite/O2/O3, the canonical curvature multiplier has the form

$$m_{\text{can},+}(x) = \kappa(x - z_0)^2 u(x),$$

where

$$|\kappa|, |\kappa|^{-1} \leq Q^C, \quad |u(x)|, |u(x)|^{-1} \leq Q^C,$$

and u has R^C -controlled divided differences through order $R + 1$ in the active R^C -conditioned Riesz norm. Moreover, z_0 is polynomially close to the active branch scale, so that $1/m_{\text{can},+}$ has a controlled second-order pole visible to the finite polynomial space.

Consequently

$$\text{dist}_{\mathcal{R}_R}(1/m_{\text{can},+}, \mathbf{C}_R[x]) \geq e^{-O(R \log R)} \geq Q^{-C}.$$

Remark E.84 (Why the quadratic multiplier is the O2/O3 keystone). The pure-plus graph theorem

$$\text{dist}(1/m_{\text{can},+}, \mathbf{C}_R[x]) \geq Q^{-A}$$

is the analytic source of pure-plus determinant mass. In the model

$$m_{\text{can},+}(x) \sim \kappa x^2,$$

this is the elementary non-polynomiality of x^{-2} , with loss $e^{-O(R \log R)} = Q^{-O(1)}$.

The same multiplier also produces the branch-substitution ratio

$$r(x) = \frac{(1-x)^2}{x^2}$$

which enters the bounded-charge near-pure Schur block. Thus the same finite theorem supplies both:

$$\boxed{\text{pure-plus graph non-polynomiality}} \quad \text{and} \quad \boxed{\text{near-pure substitution-ratio separation.}}$$

Definition E.85 (Harmonic exterior O2/O3 quantity). For an $n \times n$ augmented matrix A , $n = R + 2$, define

$$\mathcal{H}_{23}(A) := \frac{\det(A^* A)}{e_{n-1}(A^* A)}.$$

Equivalently,

$$\mathcal{H}_{23}(A) = \frac{|\det A|^2}{\sum_{i,j} |\text{Cof}_{ij}(A)|^2},$$

with the value 0 when A is singular. If $\sigma_{\min}(A)^2$ is the least eigenvalue of $A^* A$, then

$$\frac{1}{n} \sigma_{\min}(A)^2 \leq \mathcal{H}_{23}(A) \leq \sigma_{\min}(A)^2.$$

Thus $\mathcal{H}_{23}$ is the scale-invariant aggregate substitute for raw determinant energy.

Hypothesis E.86 (Weighted exterior-pivot anti-trough for finite/O2O3). Let

$$d\nu := F_1 F_{\text{wall}}^{\text{forb}} N d\rho$$

be the Stack-C2 weight after dyadic normalization and removal of geometric bad sets. For

$$D(m) := \det A_{\leq X, m}, \quad C_{ij}(m) := \text{Cof}_{ij}(A_{\leq X, m}),$$

define the dominant-cofactor classes

$$\mathcal{C}_{ij} := \left\{ m : |C_{ij}(m)|^2 \geq n^{-2} \sum_{p,q} |C_{pq}(m)|^2 \right\}.$$

For every dominant cofactor class,

$$\int_{\mathcal{C}_{ij}} \min \left(Q^A \left| \frac{D(m)}{C_{ij}(m)} \right|^2, 1 \right) d\nu(m) \geq Q^{-B} \nu(\mathcal{C}_{ij}).$$

Equivalently, one may prove the logarithmic resultant form

$$\int_{\mathcal{C}_{ij}} \log_+ \frac{|C_{ij}(m)|^2}{|D(m)|^2} d\nu(m) \leq C(\log Q) \nu(\mathcal{C}_{ij}).$$

Definition E.87 (Full determinant survival under Stack weight). Let $A_{\text{full}}(m)$ denote the full scalar canonical O2/O3 augmented matrix built from

$$q''_{\text{can}} = B_2 + P_2 + D_2 + E_2.$$

Let

$$D_{\text{full}} := \det A_{\text{full}}.$$

If a branch-count decomposition is available, write

$$D_{\text{full}} = \sum_{\beta} D_{\beta}.$$

A phase-aligned branch determinant D_{dom} is a proof witness only if it survives inside the full determinant under the active Stack measure

$$d\nu = F_1 F_{\text{wall}}^{\text{forb}} N d\rho :$$

$$\int |D_{\text{full}}|^2 d\nu \geq Q^{-A} \int |D_{\text{dom}}|^2 d\nu.$$

A lower bound for D_{dom} alone is a projected branch certificate and does not prove scalar O2/O3 for the canonical source.

Remark E.88 (Cauchy–Binet determinant coefficient and divided differences). For a single branch with frequencies $x_n = \lambda \log n$, write the augmented columns as

$$h_0(x) = 1, \quad h_{j+1}(x) = m_{\text{can}}(x)x^j, \quad 0 \leq j \leq R.$$

If

$$A_m = (L_m(h_j x^k))_{0 \leq k, j \leq R+1},$$

then Cauchy–Binet gives

$$\det A_m = \sum_{\substack{S \\ |S|=R+2}} C_S e^{-im \log N_S}, \quad N_S := \prod_{n \in S} n.$$

For a pure branch one has

$$C_S = \left(\prod_{n \in S} a_n \right) (\det V_S)^2 \left(\prod_{n \in S} m_{\text{can}}(x_n) \right) [x_n : n \in S] \left(\frac{1}{m_{\text{can}}} \right),$$

where V_S is the Vandermonde matrix of the nodes x_n . Thus the analytic O3 content of the exterior determinant is the divided difference

$$[x_n : n \in S] \left(\frac{1}{m_{\text{can}}} \right).$$

In the favorable branch-normalized elliptic model

$$m_{\text{can},+}(x) = \kappa x^2(1 + \varepsilon(x)),$$

one compares $1/m_{\text{can},+}$ to $(\kappa x^2)^{-1}$. Since

$$(x^{-2})^{(R+1)} = (-1)^{R+1}(R+2)!x^{-R-3},$$

the model divided differences have fixed phase and non-small mass on $x > 0$. Therefore the analytic multiplier obstruction reduces to the finite audit that

$$\left(\frac{1}{m_{\text{can},+}} - \frac{1}{\kappa x^2} \right)^{(R+1)}$$

is perturbative relative to

$$\left(\frac{1}{\kappa x^2} \right)^{(R+1)}$$

on the chosen branch-routed subpacket.

Remark E.89 (Collision and branch-survival guardrail for determinant methods). A pure-branch determinant lower bound is not, by itself, a scalar finite/O2/O3 theorem for the full canonical matrix. The determinant expansion has product frequencies

$$\log N_S = \sum_{n \in S} \log n,$$

with $|S| = R + 2$, and therefore effective length $e^{O(RQ)}$. Exact and near multiplicative collisions can occur among the products N_S . Moreover, the full determinant also contains pure dual, mixed, background, and completion exterior terms.

Thus any determinant route must prove both:

collision/near-collision noncancellation for the relevant exterior coefficients

and

survival of the branch determinant inside the full scalar canonical determinant.

Prime-labelled or otherwise collision-safe subpackets are useful only as witnesses inside a sector-compatible component of the full determinant. A projected prime determinant alone certifies a projected map, not the scalar canonical augmented matrix.

Lemma E.90 (Riesz-normalized determinant controls $\mathcal{H}_{23}$). *Let A be an $n \times n$ matrix with $n = R + 2$, written in an R^C -conditioned domain/codomain basis so that*

$$\|A\| \leq R^C.$$

Then

$$\mathcal{H}_{23}(A) \geq Q^{-C} |\det A|^2.$$

Proof. If $s_1, \dots, s_n$ are the singular values of A , then

$$e_{n-1}(A^*A) = \sum_{i=1}^n \prod_{j \neq i} s_j^2 \leq n \|A\|^{2(n-1)}.$$

Thus

$$\mathcal{H}_{23}(A) \geq \frac{|\det A|^2}{n \|A\|^{2(n-1)}}.$$

Since $n = R + 2$, $\|A\| \leq R^C$, and

$$R^{O(R)} = Q^{O(1)},$$

the claimed polynomial loss follows. $\square$

Definition E.91 (Pre-extraction Fejér–Riesz Stack weight). On each resonant O2/O3 block B , before capped extraction, the active Stack measure is chosen inside the bounded-degree Fejér–Riesz charge algebra:

$$d\nu_{\text{FR},B} := |\Phi_B(z)|^2 d\theta d\sigma_B, \quad z = e^{i\theta},$$

where

$$\deg_z \Phi_B \leq D_W, \quad D_W = O(R),$$

and the coefficient norm and outer lower bound of Φ_B on the selected good set are polynomially controlled. The O2/O3 packet family, its masses, and the later caps are selected relative to this pre-extraction measure. No rough cap, sharp cutoff, or arbitrary smooth factor is inserted before the O2/O3 survival theorem unless it is itself represented inside the same bounded-degree Fejér–Riesz algebra.

Remark E.92 (No early capped extraction). The order of operations in the active O2/O3 branch is

pre-extraction Fejér–Riesz Stack weight $\longrightarrow$ Fejér–Riesz charge anti-lock $\longrightarrow$ capped extraction / downstream

Hard caps or arbitrary smooth Stack factors inserted before determinant survival are outside the active proof branch unless they are first encoded by bounded-degree Fejér–Riesz factors. This is why the construction closes model membership by design rather than by approximating an older Stack weight after the fact.

Lemma E.93 (Pre-extraction Fejér–Riesz model membership). *In the construction of Definition E.91, the O2/O3 packet weights belong to the bounded-degree Fejér–Riesz charge model with degree $D_W = O(R)$. If the contaminant charge bandwidth is d_0 , then the thickened finite charge model has degree at most $D_W + d_0$, with polynomially controlled coefficient and mass constants on the selected good set.*

Proof. The Stack density is exactly $|\Phi_B(z)|^2$ with $\deg_z \Phi_B \leq D_W$. Thus Fejér–Riesz factorization is not an additional lemma; it is part of the construction. Multiplication by the finite contaminant packet family increases the charge bandwidth by at most d_0 . The coefficient and outer bounds in Definition E.91 give the stated polynomial control on the selected good set. $\square$

Theorem E.94 (Canonical Fejér–Riesz construction closes O2/O3 model membership). *Assume the pre-extraction Stack measure is $d\nu_{\text{FR},B}$ from Definition E.91. Then the model-membership hypothesis in Theorem E.106 is satisfied. Consequently, whenever the canonical quadratic multiplier / near-pure anti-lock hypotheses of that theorem hold and*

$$D_W, d_0 = O(R), \quad R \log R = O(\log Q),$$

the O2/O3 graph gap satisfies

$$\delta_{\text{graph}} \geq Q^{-A}.$$

Proof. Lemma E.93 supplies the bounded-degree Fejér–Riesz charge model required by Theorem E.106. That theorem gives the finite anti-lock lower bound

$$\eta_0 \geq (D_W + d_0 + 1)^{-2d_0}.$$

Under $D_W, d_0 = O(R)$ and $R \log R = O(\log Q)$, this lower bound is polynomial in Q^{-1} . Renaming the exponent gives $\delta_{\text{graph}} \geq Q^{-A}$. $\square$

Remark E.95 (Fejér–Riesz construction is not a retrofit). The preceding theorem closes the bounded-degree model-membership problem only for the pre-extraction Fejér–Riesz construction. It does not prove that an arbitrary older Stack weight, or a Stack weight already modified by rough caps, has bounded Fejér–Riesz degree. If such a weight is retained, then its bounded-degree factorization and polynomial mass persistence must be proved separately.

Definition E.96 (Fejér–Riesz thickened O2/O3 Gram blocks). Work in the smooth pre-extraction Fejér–Riesz Hilbert space

$$\mathcal{H}_{\text{FR}} := L^2(d\nu_{\text{FR}}),$$

where the Stack weight has first been replaced on each resonant packet block by its Fejér–Riesz factor. Let

$$U = [u_i]_{i \in I_0}$$

be the finite column matrix whose columns are the thickened pure packet columns from the charge-zero determinant sector

$$D_{-n},$$

and let

$$V = [v_j]_{j \in I_S}$$

be the finite column matrix whose columns are the thickened contaminant packet columns from the bounded positive-charge sector

$$D_{-n+1}, \dots, D_{-n+d_0}.$$

The exact Gram blocks are

$$G_{00} := U^*U, \quad G_{0S} := U^*V, \quad G_{SS} := V^*V.$$

Equivalently,

$$(G_{00})_{ii'} = \langle u_i, u_{i'} \rangle_{\mathcal{H}_{\text{FR}}}, \quad (G_{0S})_{ij} = \langle u_i, v_j \rangle_{\mathcal{H}_{\text{FR}}}, \\ (G_{SS})_{jj'} = \langle v_j, v_{j'} \rangle_{\mathcal{H}_{\text{FR}}}.$$

After deleting redundant columns, equivalently after passing to the ranges of the two Gram matrices, we regard

$$G_{00} > 0, \quad G_{SS} > 0.$$

The normalized cross-Gram operator is

$$K := G_{00}^{-1/2} G_{0S} G_{SS}^{-1/2},$$

and the graph survival gap is

$$\delta_{\text{graph}} := 1 - \|K\|_{\text{op}}.$$

If redundant columns are retained, every occurrence of an inverse below is read as the corresponding inverse on the Gram range, or equivalently as the Moore–Penrose inverse on the full coordinate space.

Theorem E.97 (Thickened overlap Schur criterion). *With the notation of Definition E.96, the following quantitative statements are equivalent up to changing the polynomial exponent.*

(1) *The normalized cross-Gram has a polynomial gap:*

$$\|K\|_{\text{op}} \leq 1 - Q^{-A}.$$

(2) *The smallest principal angle between*

$$\mathcal{U} := \text{ran } U, \quad \mathcal{V} := \text{ran } V$$

has polynomial sine:

$$\sin \theta_{\min}(\mathcal{U}, \mathcal{V}) \geq Q^{-A}.$$

(3) *The Schur complement of the contaminant block satisfies*

$$G_{00} - G_{0S} G_{SS}^\dagger G_{S0} \geq Q^{-A} G_{00}.$$

Moreover, if

$$\left(\max_i \sum_j |K_{ij}| \right) \left(\max_j \sum_i |K_{ij}| \right) \leq (1 - Q^{-A})^2,$$

then the preceding three conditions hold, after again weakening the exponent if necessary.

Proof. Set

$$\widehat{U} := U G_{00}^{-1/2}, \quad \widehat{V} := V G_{SS}^{-1/2}.$$

The columns of $\widehat{U}$ and $\widehat{V}$ are orthonormal bases for the pure and contaminant ranges. Therefore

$$K = \widehat{U}^* \widehat{V},$$

and $\|K\|_{\text{op}}$ is the cosine of the smallest principal angle:

$$\|K\|_{\text{op}} = \cos \theta_{\min}(\mathcal{U}, \mathcal{V}).$$

Thus a bound $\|K\|_{\text{op}} \leq 1 - \varepsilon$ is equivalent, with polynomially comparable constants, to

$$\sin \theta_{\min}(\mathcal{U}, \mathcal{V}) = (1 - \|K\|_{\text{op}}^2)^{1/2} \gg \varepsilon^{1/2}.$$

Changing the exponent of Q gives the equivalence of the first two statements.

For the Schur complement, compute

$$G_{00} - G_{0S} G_{SS}^\dagger G_{S0} = G_{00}^{1/2} (I - K K^*) G_{00}^{1/2}.$$

Hence

$$I - K K^* \geq \eta I$$

is exactly the same as

$$G_{00} - G_{0S} G_{SS}^\dagger G_{S0} \geq \eta G_{00}.$$

Since $1 - \|K\|_{\text{op}}^2$ and $1 - \|K\|_{\text{op}}$ are polynomially comparable in the range under consideration, the Schur statement is equivalent to the norm statement after renaming the exponent.

Finally, Schur's test gives

$$\|K\|_{\text{op}}^2 \leq \left(\max_i \sum_j |K_{ij}| \right) \left(\max_j \sum_i |K_{ij}| \right),$$

which proves the sufficient row/column criterion. $\square$

Lemma E.98 (One-step shift lower bound). *Let F be a Hilbert-valued polynomial of degree at most N ,*

$$F(z) = \sum_{k=0}^N f_k z^k.$$

Then, for every $\lambda \in \mathbf{C}$,

$$\|(1 - \lambda z)F\|_{L^2(\mathbb{T})} \geq \frac{1}{N+1} \|F\|_{L^2(\mathbb{T})}.$$

Proof. Write

$$Y(z) := (1 - \lambda z)F(z) = \sum_{k=0}^{N+1} y_k z^k.$$

Then

$$y_0 = f_0, \quad y_k = f_k - \lambda f_{k-1} \quad (1 \leq k \leq N), \quad y_{N+1} = -\lambda f_N.$$

If $|\lambda| \leq 1$, the recurrence gives

$$f_k = \sum_{j=0}^k \lambda^j y_{k-j}.$$

The triangular convolution kernel has ℓ^1 -norm at most $N + 1$, and Young's inequality yields

$$\left(\sum_{k=0}^N \|f_k\|^2 \right)^{1/2} \leq (N + 1) \left(\sum_{k=0}^{N+1} \|y_k\|^2 \right)^{1/2}.$$

If $|\lambda| > 1$, solve backwards:

$$f_k = - \sum_{j=k+1}^{N+1} \lambda^{-(j-k)} y_j.$$

Again the triangular convolution kernel has ℓ^1 -norm at most $N + 1$, because $|\lambda|^{-1} < 1$. Thus the same estimate holds. Parseval's identity converts the coefficient estimate to the asserted $L^2(\mathbb{T})$ estimate. $\square$

Lemma E.99 (Finite predictor lower bound). *Let P be a scalar polynomial of degree at most D , and let*

$$b(z) = 1 + b_1 z + \cdots + b_d z^d$$

be any scalar polynomial with constant coefficient 1. Then

$$\|Pb\|_{L^2(\mathbb{T})} \geq \gamma_{D,d} \|P\|_{L^2(\mathbb{T})},$$

where

$$\gamma_{D,d} := \prod_{\ell=0}^{d-1} \frac{1}{D + \ell + 1}.$$

In particular,

$$\gamma_{D,d} \geq (D + d + 1)^{-d}.$$

Proof. Since $b(0) = 1$, the nonconstant part factors as

$$b(z) = \prod_{\ell=1}^s (1 - \lambda_\ell z), \quad s \leq d.$$

Apply Lemma E.98 successively. Before the $(\ell + 1)$ -st factor is applied, the degree is at most $D + \ell$. Hence

$$\|Pb\|_2 \geq \prod_{\ell=0}^{s-1} \frac{1}{D + \ell + 1} \|P\|_2.$$

Weakening the product to the first d factors gives the stated $\gamma_{D,d}$ bound. $\square$

Lemma E.100 (Fiberwise Fejér–Riesz charge anti-lock). *Let (Y, σ) be a measure space. For each $y \in Y$, let $\Phi_y(z)$ be a scalar polynomial in z of degree at most D , and let $\mathcal{E}_y$ be a Hilbert fiber. For measurable fields*

$$\xi(y) \in \mathcal{E}_y, \quad \eta_r(y) \in \mathcal{E}_y \quad (1 \leq r \leq d),$$

one has

$$\begin{aligned} \int_Y \int_{\mathbb{T}} \left\| \Phi_y(z) \left(\xi(y) - \sum_{r=1}^d z^r \eta_r(y) \right) \right\|^2 d\theta d\sigma(y) \\ \geq \gamma_{D,d}^2 \int_Y \int_{\mathbb{T}} \|\Phi_y(z) \xi(y)\|^2 d\theta d\sigma(y), \end{aligned}$$

where $\gamma_{D,d}$ is as in Lemma E.99.

Proof. Fix y . If $\xi(y) = 0$, there is nothing to prove. Otherwise set

$$e(y) := \frac{\xi(y)}{\|\xi(y)\|}.$$

Orthogonal projection onto the line $\mathbf{C}e(y)$ cannot increase norm, so

$$\begin{aligned} \left\| \Phi_y(z) \left(\xi(y) - \sum_{r=1}^d z^r \eta_r(y) \right) \right\|^2 \\ \geq \|\xi(y)\|^2 |\Phi_y(z)|^2 \left| 1 - \sum_{r=1}^d a_r(y) z^r \right|^2, \end{aligned}$$

where

$$a_r(y) := \frac{\langle \eta_r(y), e(y) \rangle}{\|\xi(y)\|}.$$

The scalar polynomial

$$b_y(z) := 1 - \sum_{r=1}^d a_r(y) z^r$$

has constant coefficient 1. Applying Lemma E.99 to $P = \Phi_y$ and $b = b_y$ gives

$$\int_{\mathbb{T}} |\Phi_y(z) b_y(z)|^2 d\theta \geq \gamma_{D,d}^2 \int_{\mathbb{T}} |\Phi_y(z)|^2 d\theta.$$

Multiplying by $\|\xi(y)\|^2$ and integrating in y proves the lemma. $\square$

Proposition E.101 (Block Fejér–Riesz charge anti-lock). *Fix a resonant $O2/O3$ packet block B . Let*

$$z = e^{i\theta}$$

be the branch-charge coordinate, and suppose that on B the smooth pre-extraction Stack weight has a Fejér–Riesz factorization

$$W_B(y, z) = |\Phi_B(y, z)|^2, \quad \Phi_B(y, z) = \sum_{\ell=0}^{D_W} \phi_{B,\ell}(y) z^\ell.$$

Let $\mathcal{E}_B(y)$ denote the non-charge packet coefficient fiber on this block. Assume that the pure block is contained in the charge-zero model

$$\mathcal{U}_B \subseteq \{ \Phi_B(y, z) \xi(y) : \xi(y) \in \mathcal{E}_B(y) \},$$

and that the contaminant block is contained in the positive-charge model

$$\mathcal{V}_B \subseteq \left\{ \Phi_B(y, z) \sum_{r=1}^{d_0} z^r \eta_r(y) : \eta_r(y) \in \mathcal{E}_B(y) \right\}.$$

Then every $u \in \mathcal{U}_B$ satisfies

$$\text{dist}(u, \mathcal{V}_B)^2 \geq \eta_0 \|u\|^2,$$

where

$$\eta_0 := \gamma_{D_W, d_0}^2 \geq (D_W + d_0 + 1)^{-2d_0}.$$

Consequently, for the block-normalized cross-Gram operator

$$K_B := G_{00,B}^{-1/2} G_{0S,B} G_{SS,B}^{-1/2},$$

one has

$$\|K_B\|_{\text{op}} \leq \sqrt{1 - \eta_0} \leq 1 - \frac{1}{2}\eta_0.$$

Proof. Let

$$\mathcal{V}_B^\sharp := \left\{ \Phi_B(y, z) \sum_{r=1}^{d_0} z^r \eta_r(y) : \eta_r(y) \in \mathcal{E}_B(y) \right\}.$$

Since $\mathcal{V}_B \subseteq \mathcal{V}_B^\sharp$,

$$\text{dist}(u, \mathcal{V}_B) \geq \text{dist}(u, \mathcal{V}_B^\sharp).$$

Write $u = \Phi_B \xi$. Lemma E.100, with $D = D_W$ and $d = d_0$, gives for every choice of $\eta_1, \dots, \eta_{d_0}$

$$\left\| \Phi_B \xi - \Phi_B \sum_{r=1}^{d_0} z^r \eta_r \right\|^2 \geq \gamma_{D_W, d_0}^2 \|\Phi_B \xi\|^2.$$

Taking the infimum over the η_r 's proves the distance estimate.

For unit $u \in \mathcal{U}_B$,

$$\text{dist}(u, \mathcal{V}_B)^2 = 1 - \|P_{\mathcal{V}_B} u\|^2.$$

Thus

$$\sup_{\|u\|=1, u \in \mathcal{U}_B} \|P_{\mathcal{V}_B} u\|^2 \leq 1 - \eta_0.$$

The left side is $\|K_B\|_{\text{op}}^2$, which gives

$$\|K_B\|_{\text{op}} \leq \sqrt{1 - \eta_0}.$$

Finally, $\sqrt{1 - x} \leq 1 - x/2$ for $0 \leq x \leq 1$, so

$$\|K_B\|_{\text{op}} \leq 1 - \frac{1}{2}\eta_0.$$

□

Lemma E.102 (Packetwise thickened overlap decay and off-block perturbation). *Let $\{u_i\}$ be the thickened pure packet family and $\{v_j\}$ the thickened contaminant packet family after Fejér–Riesz thickening. Partition the packet indices into the resonant blocks used in Proposition E.101, and let*

$$G_{00}^{\text{bd}} := \bigoplus_B G_{00,B}, \quad G_{SS}^{\text{bd}} := \bigoplus_B G_{SS,B}, \quad G_{0S}^{\text{bd}} := \bigoplus_B G_{0S,B}.$$

For every prescribed M , after choosing the packet integration-by-parts order sufficiently large, the off-block packet geometry gives

$$\left\| (G_{00}^{\text{bd}})^{-1/2} (G_{00} - G_{00}^{\text{bd}}) (G_{00}^{\text{bd}})^{-1/2} \right\|_{\text{op}} \leq Q^{-M},$$

$$\left\| (G_{SS}^{\text{bd}})^{-1/2} (G_{SS} - G_{SS}^{\text{bd}}) (G_{SS}^{\text{bd}})^{-1/2} \right\|_{\text{op}} \leq Q^{-M},$$

and

$$\left\| (G_{00}^{\text{bd}})^{-1/2} (G_{0S} - G_{0S}^{\text{bd}}) (G_{SS}^{\text{bd}})^{-1/2} \right\|_{\text{op}} \leq Q^{-M}.$$

Equivalently, if Δ_{ij} denotes the normalized packet separation, the available overlap bound has the form

$$|\langle \tilde{u}_i, \tilde{v}_j \rangle| \leq \mu_L(\Delta_{ij}), \quad \mu_L(\Delta) \ll_L (1 + \Delta)^{-L},$$

for block-whitened packets $\tilde{u}_i, \tilde{v}_j$, and the corresponding off-block row and column sums are $O(Q^{-M})$.

Proof. The Fejér–Riesz factor has bounded charge degree and polynomially controlled coefficients on every ordinary packet. Multiplication by this factor only increases the symbol seminorms by a polynomial amount. For two packets in different resonant blocks, the phase derivative of their overlap is bounded away from zero in the normalized packet scale. Repeated integration by parts therefore gives, for every L ,

$$|\langle \tilde{u}_i, \tilde{v}_j \rangle| \leq C_L (1 + \Delta_{ij})^{-L},$$

and the same estimate holds for pure–pure and contaminant–contaminant off-block entries.

The number of packet indices is polynomial in Q . Choosing L larger than this polynomial counting exponent plus M , the Schur test gives off-block row and column sums $O(Q^{-M})$. Applying Schur’s test separately to the pure–pure, contaminant–contaminant, and pure–contaminant off-block matrices yields the three displayed operator-norm estimates. This is purely the already-packaged smooth packet geometry; it does not use capped extraction or any downstream Y12, D4, or atomic information. $\square$

Remark E.103 (Provenance of O2/O3 off-block decay). The off-block estimates in Lemma E.102 are the Fejér–Riesz-thickened form of the packet oscillatory-decay estimates proved in the source packet-regularity package (Theorem E.36, Theorem E.56, Corollary E.53). Multiplication by the bounded-degree Fejér–Riesz factor $\Phi_B(y, z)$ changes symbol seminorms only by a polynomial factor, which is absorbed in the choice of integration-by-parts order; thickening therefore does not introduce an independent off-block coverage hypothesis beyond the source-side packet decay.

Lemma E.104 (Row/column coherence bound). *Let $K = (K_{ij})$ be any finite normalized overlap matrix. Define*

$$r_i := \sum_j |K_{ij}|, \quad c_j := \sum_i |K_{ij}|.$$

Then

$$\|K\|_{\text{op}}^2 \leq \left(\max_i r_i \right) \left(\max_j c_j \right).$$

In particular, if

$$\left(\max_i r_i \right) \left(\max_j c_j \right) < (1 - Q^{-A})^2,$$

then

$$\|K\|_{\text{op}} < 1 - Q^{-A}.$$

If each row has at most d_0 nonnegligible entries and each column at most d_1 , with $|K_{ij}| \leq \mu$, then

$$d_0 d_1 \mu^2 < (1 - Q^{-A})^2 \implies \|K\|_{\text{op}} < 1 - Q^{-A}.$$

Proof. For any vector x ,

$$|(Kx)_i| \leq \sum_j |K_{ij}| |x_j|.$$

Schur’s test gives

$$\|Kx\|_2^2 \leq \left(\max_i r_i \right) \left(\max_j c_j \right) \|x\|_2^2.$$

Taking the supremum over $x \neq 0$ proves the operator-norm bound. The bounded-incidence form follows from $r_i \leq d_0 \mu$ and $c_j \leq d_1 \mu$. $\square$

Lemma E.105 (Schur complement / principal-angle equivalence). *With*

$$K := G_{00}^{-1/2} G_{0S} G_{SS}^{-1/2},$$

the following are equivalent up to polynomial loss:

- (1) $\|K\|_{\text{op}} \leq 1 - Q^{-A}$;
- (2) $\sin \theta_{\min}(U, V) \geq Q^{-A}$;
- (3)

$$G_{00} - G_{0S} G_{SS}^\dagger G_{0S}^* \geq Q^{-A} G_{00}.$$

Proof. This is Theorem E.97 restated in the notation used by the downstream finite-gap ledger. The normalized columns $UG_{00}^{-1/2}$ and $VG_{SS}^{-1/2}$ give orthonormal bases for the two subspaces, so their cross-Gram has norm $\cos \theta_{\min}$. The Schur identity

$$G_{00} - G_{0S}G_{SS}^\dagger G_{0S}^* = G_{00}^{1/2}(I - KK^*)G_{00}^{1/2}$$

then gives the Schur equivalence. $\square$

Theorem E.106 (Fejér–Riesz charge anti-lock closes Graph/O2O3). *Assume the finite/O2O3 packet model of Definition E.96, the bounded positive-charge contaminant range*

$$D_{-n+1}, \dots, D_{-n+d_0},$$

the Fejér–Riesz block factorization of Proposition E.101, and the off-block packet geometry of Lemma E.102. Then

$$\delta_{\text{graph}} = 1 - \|G_{00}^{-1/2}G_{0S}G_{SS}^{-1/2}\|_{\text{op}} \geq \frac{1}{4}\eta_0,$$

where

$$\eta_0 := \gamma_{D_W, d_0}^2 \geq (D_W + d_0 + 1)^{-2d_0}.$$

In particular, if $D_W, d_0 = O(1)$, then $\delta_{\text{graph}} \gg 1$. If $D_W, d_0 = O(R)$, then

$$\delta_{\text{graph}} \geq e^{-O(R \log R)}.$$

Under the standing depth budget

$$R \log R = O(\log Q),$$

this gives

$$\delta_{\text{graph}} \geq Q^{-A}.$$

Equivalently,

$$G_{00} - G_{0S}G_{SS}^\dagger G_{0S} \geq Q^{-A}G_{00}.$$

Proof. First ignore off-block interactions. Let

$$K^{\text{bd}} := (G_{00}^{\text{bd}})^{-1/2}G_{0S}^{\text{bd}}(G_{SS}^{\text{bd}})^{-1/2}.$$

By Proposition E.101, each block satisfies

$$\|K_B\|_{\text{op}} \leq 1 - \frac{1}{2}\eta_0.$$

Since K^{bd} is the direct sum of the K_B 's,

$$\|K^{\text{bd}}\|_{\text{op}} \leq 1 - \frac{1}{2}\eta_0.$$

Now include off-block terms. Write

$$E_0 := (G_{00}^{\text{bd}})^{-1/2}(G_{00} - G_{00}^{\text{bd}})(G_{00}^{\text{bd}})^{-1/2},$$

$$E_S := (G_{SS}^{\text{bd}})^{-1/2}(G_{SS} - G_{SS}^{\text{bd}})(G_{SS}^{\text{bd}})^{-1/2},$$

and

$$E_{0S} := (G_{00}^{\text{bd}})^{-1/2}(G_{0S} - G_{0S}^{\text{bd}})(G_{SS}^{\text{bd}})^{-1/2}.$$

Then

$$G_{0S} = (G_{00}^{\text{bd}})^{1/2}(K^{\text{bd}} + E_{0S})(G_{SS}^{\text{bd}})^{1/2}.$$

Lemma E.102 gives, for any prescribed M ,

$$\|E_0\|_{\text{op}}, \|E_S\|_{\text{op}}, \|E_{0S}\|_{\text{op}} \leq Q^{-M}.$$

Set

$$A_0 := G_{00}^{-1/2}(G_{00}^{\text{bd}})^{1/2}, \quad A_S := G_{SS}^{-1/2}(G_{SS}^{\text{bd}})^{1/2}.$$

The relative identities

$$G_{00} = (G_{00}^{\text{bd}})^{1/2}(I + E_0)(G_{00}^{\text{bd}})^{1/2},$$

$$G_{SS} = (G_{SS}^{\text{bd}})^{1/2}(I + E_S)(G_{SS}^{\text{bd}})^{1/2}$$

imply

$$\|A_0\|_{\text{op}}, \|A_S\|_{\text{op}} \leq (1 - Q^{-M})^{-1/2}.$$

Moreover

$$K = A_0(K^{\text{bd}} + E_{0S})A_S^*.$$

Hence

$$\|K\|_{\text{op}} \leq (1 - Q^{-M})^{-1} \left(\|K^{\text{bd}}\|_{\text{op}} + Q^{-M} \right).$$

Choose M so large that the last multiplicative and additive losses together are at most $\eta_0/4$. Then

$$\|K\|_{\text{op}} \leq \|K^{\text{bd}}\|_{\text{op}} + \frac{1}{4}\eta_0 \leq 1 - \frac{1}{2}\eta_0 + \frac{1}{4}\eta_0 = 1 - \frac{1}{4}\eta_0.$$

Thus

$$\delta_{\text{graph}} = 1 - \|K\|_{\text{op}} \geq \frac{1}{4}\eta_0.$$

The lower bounds

$$\eta_0 \geq (D_W + d_0 + 1)^{-2d_0}$$

and, in the $O(R)$ -degree case,

$$\eta_0 \geq e^{-O(R \log R)}$$

are immediate from the definition of γ_{D_W, d_0} . The depth budget $R \log R = O(\log Q)$ gives $\eta_0 \geq Q^{-A}$. The Schur complement bound then follows from Lemma E.105. $\square$

Corollary E.107 (Graph/O2O3 is the immediate finite-gap consequence). *The Fejér–Riesz charge anti-lock theorem implies Graph/O2O3₂₃ with polynomial loss. In particular,*

$$\|G_{00}^{-1/2} G_{0S} G_{SS}^{-1/2}\|_{\text{op}} \leq 1 - Q^{-A},$$

and no additional graph hypothesis is required once the finite/O2O3 packet model has been placed in the smooth Fejér–Riesz charge coordinates.

Corollary E.108 (Exact target for Graph/O2O3₂₃). *The exact target is the principal-angle gap*

$$\delta_{\text{graph}} := 1 - \|K\|_{\text{op}} \geq Q^{-A}, \quad K = G_{00}^{-1/2} G_{0S} G_{SS}^{-1/2}.$$

Theorem E.106 proves this target from the bounded-charge Fejér–Riesz block model and the already packaged off-block packet geometry. The proof does not require capped extraction, the Y12 quotient, D4 row-normal information, or the final atomic contradiction.

Remark E.109 (Resolved graph/O2O3 bridge). The previous “thickened overlap too coherent” failure mode is removed by the charge anti-lock argument. Inside a resonant block the relevant mechanism is not entrywise smallness; the pure vector $\Phi_B \xi$ cannot be predicted by positive charges $\Phi_B \sum_{r=1}^{d_0} z^r \eta_r$ with loss better than $(D_W + d_0 + 1)^{-d_0}$. Off-block coherence is then perturbative by the packetwise oscillatory decay estimate. Thus the graph/O2O3 gate is closed at polynomial cost once the finite/O2O3 packet family is expressed in the Fejér–Riesz charge model above.

Remark E.110 (O2/O3 failure modes after charge anti-lock). After Theorem E.106, the remaining possible failure is not a Schur/principal-angle failure. It would have to be a failure of one of the inputs needed to form the finite charge model itself: lack of R^C -normalization, failure of the bounded-degree Fejér–Riesz Stack factorization, or failure to identify the pure sector as charge 0 and the contaminant sector as charges $1, \dots, d_0$ before capped extraction. Once those structural inputs are present, the normalized cross-Gram gap follows from the finite anti-lock proof.

Remark E.111 (What finite/O2O3 does and does not use). Finite/O2O3 uses the analytic Riesz basis for $A_{\leq X, m}$. It does not use the singular retained projection P_{sing} except through the weight $F_{\text{wall}}^{\text{forb}}$. Conversely, the singular-discard carrier projection is not required to make $A_{\leq X, m}$ functorial.

The following routes are no-gos as standalone proofs of finite/O2O3:

- (1) fixed-vector aggregate frames, since moving null directions may make every local $A_{\leq X, m}$ singular;

- (2) raw determinant energy without R^C -normalization, since the determinant-to- $\sigma_{\min}$ loss is super-polynomial;
- (3) generic Jensen/resultant/type arguments, since $\det A_{\leq X, m}$ has effective type QR while the zero density is only Q ;
- (4) pure branch determinant lower bounds without full determinant survival;
- (5) the Λ -wall factor alone, since it does not force $\mathcal{H}_{23}(A_{\leq X, m})$ to be non-small.

The actual remaining theorem is weighted exterior-pivot anti-trough for the full finite canonical matrix, not merely for a pure branch determinant.

Definition E.112 (α -admissibility for determinant survival). An α -twisted determinant survival argument is admissible only if the following package holds.

- (1) **Zero persistence.** Every fixed-target bad zero $Z_0(m_0) = 0$ in the regular stratum extends to a zero curve

$$Z_\alpha(m(\alpha)) = 0$$

on an interval I_α of polynomial length. Quantitatively,

$$|\partial_m Z_\alpha(m(\alpha))| \geq Q^{-C}, \quad \frac{dm}{d\alpha} = -\frac{\partial_\alpha Z_\alpha}{\partial_m Z_\alpha}.$$

- (2) **Badness persistence.** The Stack quantities

$$E_1, \quad E_{23}, \quad E_{\text{wall}}^{\text{forb}}, \quad D_{\text{full}}, \quad D_{\text{dom}}, \quad N$$

have polynomial log-Lipschitz variation along the zero curve, away from capped sublevel boundaries.

- (3) **Twist nondegeneracy.** If

$$D_{\text{full}}(m, \alpha) = \sum_{r=0}^{R+2} e^{ir\Theta(m, \alpha)} D_r(m, \alpha),$$

then along the zero curve

$$\left| \frac{d}{d\alpha} \Theta(m(\alpha), \alpha) \right| = \left| \partial_\alpha \Theta - \partial_m \Theta \frac{\partial_\alpha Z_\alpha}{\partial_m Z_\alpha} \right| \geq Q^{1-C}.$$

- (4) **Stack uniformity.** Gate-2, O2/O3, D4, and the forbidden functional Λ are defined with uniform polynomial constants on the α -tube.

Without these four properties, α -averaging is heuristic and cannot be used to prove the fixed $\alpha = 0$ target.

Remark E.113 (Regular-zero branch and near-multiple branch). The α -twist route applies only on the regular zero stratum

$$|\partial_m Z_\alpha| \geq Q^{-C}.$$

The complementary branch

$$|\partial_m Z_\alpha| < Q^{-C}$$

must not be ignored. It should be routed to a separate near-multiple-zero module. Such zeros may already seed Gate-1, since

$$Z_\alpha(m) \approx 0, \quad \partial_m Z_\alpha(m) \approx 0$$

forces local curvature unless the canonical source is flat to higher order. The required auxiliary theorem is

$$\text{NearMultipleGate1 : } |Z_\alpha| + |\partial_m Z_\alpha| \leq Q^{-A} \implies \int_I |\lambda^3 q_{\text{can}}''(m + \lambda x)|^2 dx \geq Q^{-A'}.$$

If this theorem fails, near-multiple zeros require a separate singular/flatness branch.

Definition E.114 (α -D4 compatibility). Let $\mathcal{K}_9$ be the covariant order-nine singular coherence residual, with residual codomain $\mathcal{Y}_9$. Let

$$\mathcal{T}_{\text{par}}^{(\alpha)} := \text{span}\{D_p \mathcal{K}_9(\partial_{p_a})\},$$

where p_a ranges over the allowed actual-source parameters, including α if α is admitted. Let $\mathcal{T}_7$ and $\mathcal{T}_{<7}$ be the order-seven and lower-order connection residual subspaces. Define

$$\mathcal{Q}_9^{\text{forb},\alpha} := \mathcal{Y}_9 / (\mathcal{T}_{\text{par}}^{(\alpha)} + \mathcal{T}_7 + \mathcal{T}_{<7}).$$

The α -twist is compatible with the Y12 D4 channel if

$$[\mathbf{C}_{Y12}] \neq 0 \quad \text{in} \quad \mathcal{Q}_9^{\text{forb},\alpha}.$$

Equivalently,

$$T_{\text{tr} \rightarrow \text{sing}}[\mathbf{C}_{Y12}] \notin \mathcal{T}_{\text{par}}^{(\alpha)} + \mathcal{T}_7 + \mathcal{T}_{<7}.$$

Definition E.115 (Y12-signature quotient). Let $\Sigma(Y12)$ denote the primitive order-nine mixed-block signature of the carrier

$$Y_{12}^{(2)} = D^2 N_0[\xi_1, \xi_2]$$

coupled to one same-point whitening slot. Let

$$\Pi_{\Sigma(Y12)} : \mathcal{Y}_9 \rightarrow \mathcal{Y}_{\Sigma(Y12)}$$

be the corresponding signature projection in the residual codomain of the covariant order-nine coherence residual $\mathcal{K}_9$.

Define the Y12-signature forbidden quotient by

$$\mathcal{Q}_{9,Y12}^{\text{forb},\alpha} := \mathcal{Y}_{\Sigma(Y12)} / \Pi_{\Sigma(Y12)}(\mathcal{T}_{\text{par}}^{(\alpha)} + \mathcal{T}_7 + \mathcal{T}_{<7} + \mathcal{II}_{\text{arith},9}).$$

Here $\mathcal{II}_{\text{arith},9}$ denotes the second-order actual-source curvature/tangent image at order nine. The Y12-channel survives iff

$$\Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing}}[\mathbf{C}_{Y12}] \neq 0 \quad \text{in} \quad \mathcal{Q}_{9,Y12}^{\text{forb},\alpha}.$$

Hypothesis E.116 (Second-order Y12-signature separation). The actual-source tangent, lower-connection, and actual-source second-curvature images have no primitive Y12-signature:

$$\Pi_{\Sigma(Y12)}(\mathcal{T}_{\text{par}}^{(\alpha)} + \mathcal{T}_7 + \mathcal{T}_{<7} + \mathcal{II}_{\text{arith},9}) = 0.$$

Moreover the transported D4 carrier has nonzero Y12-signature:

$$\Pi_{\Sigma(Y12)} T_{\text{tr} \rightarrow \text{sing}}[\mathbf{C}_{Y12}] \neq 0.$$

If α is admitted as an actual parameter, it is required to be source-linear in the sense that

$$\Pi_{\Sigma(Y12)} \partial_\alpha u = 0, \quad \Pi_{\Sigma(Y12)} \partial_\alpha^2 u = 0,$$

unless a separate source-nonlinear α -audit is supplied.

Remark E.117 (Why the Y12-signature test is sharper than full K_9). The full K_9 -upper may be stronger than needed. The D4 channel currently used by the rank-two route is the primitive Y12-signature component of the order-nine residual. Therefore it is enough to control the Y12-signature projection of actual EF residuals and to prove that the D4 carrier survives in $\mathcal{Q}_{9,Y12}^{\text{forb},\alpha}$. This avoids proving unnecessary completion estimates for unrelated order-nine residual sectors.

Remark E.118 (α can help O2/O3 while hurting D4). The α -twist is a possible determinant-survival tool because it may orthogonalize folded-dual branch counts. However, if α is admitted as an actual-source parameter, the actual singular tangent space expands by $\partial_\alpha u$. The D4 contradiction survives this enlargement only if

$$[\mathbf{C}_{Y12}] \neq 0 \quad \text{in} \quad \mathcal{Q}_9^{\text{forb},\alpha}.$$

Thus the proof-critical admissibility package is

$$\text{AlphaAdmissibility}^+ := \text{AlphaAdmissibility} + \text{AlphaD4Compatibility}.$$

There are two logically distinct uses of α .

First, α may be used as a proof-thickening or averaging parameter for determinant survival. In that use, one proves persistence back to the fixed-target problem at $\alpha = 0$, and $\partial_\alpha u$ need not be added to the downstream actual-source tangent space.

Second, α may be admitted as an actual source deformation. In that case $\partial_\alpha u$ belongs to the tangent space in $\mathcal{T}_{\text{par}}^{(\alpha)}$, and one must verify

$$[\text{C}_{Y12}] \neq 0 \quad \text{in} \quad \mathcal{Q}_9^{\text{forb}, \alpha}.$$

Therefore the proof should not automatically insert $\partial_\alpha u$ into the downstream tangent space merely because α is used as an auxiliary averaging parameter.

If α -admission kills $[\text{C}_{Y12}]$, then α may still serve as an auxiliary averaging parameter, but it must not be added to the downstream actual-source manifold without a replacement forbidden channel.

Remark E.119 (Two basis layers: analytic Riesz versus carrier projection). The analytic basis used for finite/O2/O3 and the carrier projection used for Λ have different purposes and should not be identified.

The finite/O2/O3 basis should be an R^C -conditioned analytic Riesz basis in which the augmented matrix satisfies

$$\|A_{\leq X, m}\| \leq R^C.$$

This is what makes

$$\mathcal{H}_{23}(A_{\leq X, m}) \geq Q^{-C} |\det A_{\leq X, m}|^2$$

a polynomial-loss implication.

The forbidden-channel projection P_{sing} , in contrast, should encode actual EF-coherent singular source retention. It must be non-functorial on at least one formal D4 carrier:

$$(I - P_{\text{sing}})T_{\text{tr} \rightarrow \text{sing}}[\text{C}_7] \neq 0 \quad \text{or} \quad (I - P_{\text{sing}})T_{\text{tr} \rightarrow \text{sing}}[\text{C}_{Y12}] \neq 0.$$

Using one fully functorial basis for both jobs risks making O2/O3 well conditioned while killing the forbidden D4 channel.

E.5. Archived transition and singular Gate–1 proof tools.

Remark E.120 (Transition and singular diagnostics are archived). The transition and singular point-jet/sine-moment objects are archived proof tools. They no longer define separate Gate–1 objects. In the active branch, ordinary, transition, and singular packets all feed the same canonical scalar source estimate

$$\int_{I_b} |\lambda_b^3 q_{\text{can}}''(m + \lambda_b x)|^2 dx \geq Q^{-A}.$$

Any future use of transition point jets or singular sine diagnostics must be accompanied by an explicit lower-frame theorem transferring them to this scalar L^2 -source energy. Without that transfer they remain diagnostics. The ordinary Wronskian and finite-row O1/O2/O3 discussions are also archived; they are not the active upstream input. The live upstream statement is $\text{Agg}L^2(q_{\text{can}}'')$ for the Hardy source.

E.6. Canonical audit for the named Gate–1 inputs.

Remark E.121 (Canonical audit for named Gate–1 inputs). Every named Gate–1 input must be stated for the same active Hardy source API:

$$q_{\text{can}}'' = u_H = B_2^H + P_2^H + D_2^H + E_2^H,$$

with packet-normalized rows carrying the explicit factor λ_p^3 , as in Definition E.24. The following substitutions are not allowed:

$$B_2\text{-only}, \quad P_2\text{-only}, \quad \text{zero-kernel-only}, \quad \text{flexible Dirichlet-polynomial proxies}.$$

The finalized gamma/background curvature B_2 is part of the canonical source. Diagnostics using B_2 alone are not Gate-1 candidates.

For the downstream rank-two analytic route, if B_2 is included in

$$p = q''_{\text{can}},$$

then the wall-completion module must include or approximate

$$B_2^H, \quad DB_2^H, \quad D^2B_2^H$$

in the scale-normalized H^2 -ledger.

Remark E.122 (Status after the canonical construction rules and Hardy-source repair). The proof-level source/O2O3 branch now uses three canonical choices. First, the actual source is the Hardy-curvature source $q_{\text{curv}} = \partial_t^3 Z_H$, with trace $\mathcal{Z}_{\text{curv}}(1/2 + it)$ supplied by Theorem E.5. Second, the Gate-1 observation window is $W_{\text{branch}}^{\text{can}}$, which gives

$$W_{\text{branch}}^{\text{can}} \geq Q^{-C} W_{\text{ref}}, \quad G_{\text{branch},B}^{\text{can}} \geq Q^{-C} G_{\text{ref},B}.$$

Third, the O2/O3 Stack weight is imposed before capped extraction inside the bounded-degree Fejér–Riesz charge algebra,

$$d\nu_{\text{FR},B} = |\Phi_B(z)|^2 d\theta d\sigma_B, \quad \deg \Phi_B = O(R).$$

The Hardy source theorem closes the completed-source anchor in the active Hardy branch. The canonical branch-complete window closes the branch-hole obstruction, and the pre-extraction Fejér–Riesz Stack closes the bounded-degree model-membership obstruction. The two construction rules themselves do not prove the source anchor; that role is played by the Hardy-curvature source package. These choices do not assert anything about archived noncanonical branch weights, scalar-collapsed Stack weights, or alternative source choices. They also do not remove the need to stay in the canonical source-graded $K9$ norm for the downstream same-norm comparison.

Remark E.123 (Canonical construction discipline). The active proof branch uses canonical choices only in the following no-loss sense. A canonical branch window, Stack weight, packet family, labelled lift, or quotient is proof-level only after one proves either polynomial domination of the previous object or super-polynomial loss outside the canonical object. Downstream arguments must use the same canonical objects and the same Hilbert norms. Archived arbitrary branch weights, post-extraction caps, scalar-collapsed observables, and unsplit nuisance quotients are diagnostic only unless separately compared to the canonical branch by a polynomial comparison theorem.

Concretely, the active proof branch is not the statement that every possible routing choice is harmless. It is the statement that a hypothetical obstruction is first transferred into the canonical branch below, and the contradiction is then proved entirely inside that branch.

Remark E.124 (Unsuperscripted notation in the canonical branch). In sections following the canonical construction ledger, unsuperscripted notation is used only under the active-branch convention

$$W_{\text{branch}} = W_{\text{branch}}^{\text{can}}, \quad q_p^Y = q_{p,\text{aff}}^Y, \quad P_{\text{sing},p} = P_{\text{sing},p}^{\text{cert}}, \quad \Lambda_p = \Lambda_p^{\text{gr},\text{full}},$$

unless explicitly stated otherwise. Any theorem using an archived branch weight, a raw quotient, a raw singular projection, or a scalar-collapsed observable must carry the corresponding superscript and must provide a polynomial comparison theorem before entering the proof-level chain.

Remark E.125 (Canonical branch dependency ledger). The order of construction in the active branch is fixed as follows:

Hardy source and ordinary source strip $\longrightarrow$ raw Whitney packets $\longrightarrow W_{\text{branch}}^{\text{can}} \longrightarrow d\nu_{\text{FR},B} \longrightarrow$ cap extraction

Thus the source lower bound is proved before cap extraction, the Fejér–Riesz charge model is imposed before hard caps, the live/dead split is performed before forming the exact killed quotient, and the final atomic comparison is taken in the same source-graded $K9$ norm as the lower bound. This ordering rules out adaptive definitions in which the quotient, cap, or Stack weight is chosen after seeing the live direction to be killed.

Remark E.126 (No-loss certificates for the active canonical choices). The canonical choices used later have the following no-loss certificates.

canonical choice	no-loss certificate	w
Hardy source $u_H = \partial_t^3 Z_H$	trace identity, differential AFE, dyadic mass-good selection	δ_{src}, A
$W_{\text{branch}}^{\text{can}}$	$W_{\text{branch}}^{\text{can}} \geq Q^{-C} W_{\text{ref}}$	$G_{\text{branch}}^{\text{can}},$
$d\nu_{\text{FR},B} = \Phi_B ^2 d\theta d\sigma_B$	$\deg \Phi_B = O(R)$ and polynomial outer/mass control	Gr
maximal labelled $K9$ live/dead split	$A_{a,p}^{\text{raw}} = A_{a,p}^{\text{dead}} + E_p L_p^{\text{row}} A_{a,p}^{\text{raw}}$	rank-two
canonical source-graded $K9$ lift	$\pi_{\text{coll}} \mathcal{L}_{K9,p}^{\text{gr}} = T_{\text{tr} \rightarrow \text{sing},p} J_{\text{tr},p}$ and same-norm upper/lower	AtomicY

If a later proof step uses an object outside this table, it must either identify that object with the listed canonical one or supply a separate no-loss theorem.

Remark E.127 (Fejér–Riesz Stack mass persistence). The pre-extraction Fejér–Riesz construction is a no-loss construction only on the selected good set where the factor Φ_B has polynomial coefficient bounds and a polynomial outer lower bound. In particular, model membership alone is not enough: if $|\Phi_B|^2$ were allowed to vanish on the active packet family, the anti-lock theorem would be applied to a measure that has already discarded the obstruction. The factor Φ_B is fixed by the pre-extraction Stack construction before the cap, wall, or live/dead quotient is chosen; it is not selected after seeing which live direction would otherwise survive. All O2/O3 masses in the active branch are therefore measured with $d\nu_{\text{FR},B}$ from Definition E.91, and any older Stack measure must be polynomially compared to it before it can be used in a proof.

Hypothesis E.128 (Fejér–Riesz Stack no-loss and no-early-cap certificate). For every active resonant block B , the pre-extraction Fejér–Riesz Stack measure

$$d\nu_{\text{FR},B} = |\Phi_B(z)|^2 d\theta d\sigma_B, \quad \deg \Phi_B = O(R),$$

is imposed before capped extraction and satisfies the no-loss estimate

$$\nu_{\text{FR},B}(\mathcal{P}_B^{\text{active,car}}) \geq Q^{-C_{\text{FR}}} \nu_{\text{pre},B}(\mathcal{P}_B^{\text{active}}) - Q^{-B} G_{\text{ref},B}.$$

Moreover every cap or extraction step applied after the FR Stack preserves the bounded-degree charge model up to declared residuals below Q^{-B} . No cap is allowed to select packets before the Stack construction in a way that destroys FR charge membership or carrier visibility.

Remark E.129 (Model membership is not FR no-loss). The statement that $d\nu_{\text{FR},B}$ puts the active packets into a bounded-degree charge model is not by itself a mass-persistence theorem. The O2/O3 anti-lock module may be applied only after Hypothesis E.128, or an equivalent verified persistence theorem, has shown that the FR Stack has not discarded a super-polynomial fraction of the source/graph/carrier mass. Thus

$$\text{FR model membership} \not\Rightarrow \text{FR no-loss persistence}$$

without an additional estimate.

Remark E.130 (Same-norm invariant). Every lower bound used in the final contradiction is measured in the same canonical source-graded $K9$ norm as the upper bound. More precisely, the active upper is the full-vector source-graded discard $\Lambda_p^{\text{gr,full}}$ and its orthogonal maximal-affine projection $\hat{\Lambda}_{\text{max},p}$. No scalar Y12-only norm, scalar I -coordinate, or primitive-line functional may replace this norm downstream unless a separate polynomial comparison lemma is proved in both directions on the selected cap family. This is the invariant that prevents the affine forcing component from being hidden by scalar cancellation.

Remark E.131 (Canonical packet-to-corrected-jet transfer discipline). The draft does not use an unqualified isomorphism between arbitrary packet energy and arbitrary corrected-jet energy. The active branch uses only the following named transfers:

- (1) Gate–1 source energy is transferred by the canonical branch/reference domination theorem $G_{\text{branch},B}^{\text{can}} \geq Q^{-C} G_{\text{ref},B}$.
- (2) Tail errors are transferred to source-graded $K9$ rows by the polynomial finite-jet upper bound for certified $K9$ rows.

- (3) Lower bounds are transferred only on the retained affine carrier plane, by the certified coercivity / finite-main Schur gap in the canonical source-graded Hilbert space.
- (4) Quotient transfer is performed only after exhaustive labelled live/dead splitting, so that the killed quotient contains row-dead remainders and not live $C_7/Y12$ components.

Thus the proof requires no hidden functorial comparison between every analytic packet and every finite jet. If a future argument needs such a comparison, it must be stated as a separate Gram-transfer theorem with polynomial constants.

Lemma E.132 (No full packet-to- $K9$ lower bound). *Let p be a good analytic packet and let*

$$T_p = (I - P_p^{\text{full}}) \mathcal{L}_{K9,p}^{\text{gr}} J_{\text{tr},p}$$

be any corrected $K9$ jet map using finitely many transition jets and finitely many labelled row projections. Then no constants C and Q can make

$$Q^{-C} \|v\|_{\text{pkt},J,p}^2 \leq \|T_p v\|_{\text{gr}}^2$$

hold for every $v \in H^J(K_p^+)$.

Consequently, all lower packet-to- $K9$ transfer statements in the proof must be restricted to a finite certified carrier or to a quotient of such a carrier.

Proof. The map T_p has finite rank because it depends on finitely many transition jets and finitely many finite row projections. The analytic packet space $H^J(K_p^+)$ is infinite-dimensional. Hence $\ker T_p$ is infinite-dimensional. Choose nonzero $v \in \ker T_p$. Then

$$\|T_p v\|_{\text{gr}} = 0, \quad \|v\|_{\text{pkt},J,p} > 0,$$

contradicting any lower bound of the displayed form. $\square$

Definition E.133 (Hardy-source finite carrier). On a good ordinary Hardy-normalized packet p , the Hardy-source finite carrier is a finite-dimensional subspace

$$\mathcal{V}_{\text{pkt},p}^{\text{car}} = \text{span}\{\psi_{p,1}, \dots, \psi_{p,N_p}\} \subset H^J(K_p^+)$$

containing the packet restrictions of the following proof-level objects:

- (i) the Hardy-differential AFE main pieces for $u_H = \partial_t^3 Z_H$;
- (ii) the gamma/theta derivative and log-power pieces appearing in the differentiated Hardy AFE;
- (iii) the pre-extraction Fejér–Riesz Stack charge pieces used by the finite/O2O3 branch;
- (iv) the transition/corrected-jet model pieces used by $\mathcal{L}_{K9,p}^{\text{gr}} J_{\text{tr},p}$;
- (v) the retained labelled affine $C_7/Y12$ directions.

The effective carrier

$$\mathcal{V}_{\text{pkt},p}^{\text{eff}} \subseteq \mathcal{V}_{\text{pkt},p}^{\text{car}}$$

is obtained by removing row-dead directions and declared Q^{-B} -residual directions.

Definition E.134 (Carrier Gram audit). For a certified carrier basis

$$\mathcal{B}_p = \{\psi_{p,1}, \dots, \psi_{p,N_p}\},$$

define the packet Gram matrix

$$G_{\text{pkt},p} = (\langle \psi_{p,i}, \psi_{p,j} \rangle_{\text{pkt},J,p})_{i,j},$$

the corrected $K9$ jet matrix

$$A_p = \begin{bmatrix} \Lambda_p^{\text{gr,full}} \psi_{p,1} & \dots & \Lambda_p^{\text{gr,full}} \psi_{p,N_p} \end{bmatrix},$$

and the source-graded target Gram matrix $G_{\text{jet},p}^{\text{gr}}$. The carrier Gram audit is the finite certificate

$$A_p^* G_{\text{jet},p}^{\text{gr}} A_p \geq Q^{-C_{\text{gram}}} G_{\text{pkt},p} \quad \text{on } \mathcal{V}_{\text{pkt},p}^{\text{eff}}.$$

Equivalently,

$$\sigma_{\min} \left((G_{\text{jet},p}^{\text{gr}})^{1/2} A_p G_{\text{pkt},p}^{-1/2} \big|_{\mathcal{V}_{\text{pkt},p}^{\text{eff}}} \right) \geq Q^{-C_{\text{gram}}}.$$

Theorem E.135 (Carrier-corrected packet-to- $K9$ transfer). *Let $\{p\}$ be a good ordinary Hardy-normalized canonical packet family. Let $J \geq J_{K9}$. Assume:*

- (a) *the packet Sobolev trace upper bound for $\Lambda_p^{\text{gr,full}}$;*
- (b) *the Hardy-source carrier decomposition*

$$u_H = u_{H,\text{car}} + u_{H,\text{tail}}, \quad u_{H,\text{car}} \in \bigoplus_p \mathcal{V}_{\text{pkt},p}^{\text{car}},$$

with

$$\|u_{H,\text{tail}}\|_{\text{pkt},J}^2 \leq Q^{-B} G_{\text{ref}};$$

- (c) *carrier capture:*

$$\sum_p \mu_p \|\Pi_{\text{eff},p} \Pi_{\text{car},p} u_H\|_{\text{pkt},J,p}^2 \geq Q^{-C_{\text{cap}}} G_{\text{ref}};$$

- (d) *the finite $K9$ carrier Gram audit*

$$A_p^* G_{\text{jet},p}^{\text{gr}} A_p \geq Q^{-C_{\text{gram}}} G_{\text{pkt},p}$$

on $\mathcal{V}_{\text{pkt},p}^{\text{eff}}$;

- (e) *label compatibility with $\mathfrak{R}_{2Y}(p)$ on the same transferred carrier columns.*

Then

$$\sum_p \mu_p \|\Lambda_p^{\text{gr,full}} \Pi_{\text{eff},p} \Pi_{\text{car},p} u_H\|_{\text{gr}}^2 \geq Q^{-C_{\text{cap}} - C_{\text{gram}} - C_{2Y}} G_{\text{ref}} - Q^{-B} G_{\text{ref}}.$$

In particular, choosing

$$B > C_{\text{cap}} + C_{\text{gram}} + C_{2Y} + C_{\text{atomic}} + 10$$

makes the tail contribution negligible at the final atomic scale.

Proof. The Sobolev trace theorem gives the upper estimate for all analytic packets and therefore controls the tail:

$$\sum_p \mu_p \|\Lambda_p^{\text{gr,full}} u_{H,\text{tail}}\|_{\text{gr}}^2 \leq Q^{C_{\text{tr}}} \|u_{H,\text{tail}}\|_{\text{pkt},J}^2 \leq Q^{-B+C_{\text{tr}}} G_{\text{ref}}.$$

On the effective carrier, the finite Gram audit gives

$$\sum_p \mu_p \|\Lambda_p^{\text{gr,full}} \Pi_{\text{eff},p} \Pi_{\text{car},p} u_H\|_{\text{gr}}^2 \geq Q^{-C_{\text{gram}}} \sum_p \mu_p \|\Pi_{\text{eff},p} \Pi_{\text{car},p} u_H\|_{\text{pkt},J,p}^2.$$

Carrier capture gives the displayed lower bound, and $\mathfrak{R}_{2Y}(p)$ ensures that the live $C_7/Y12$ part of the transferred columns is retained rather than quotient-killed. The residual/tail term is absorbed by the choice of B . $\square$

Remark E.136 (Carrier capture is not a consequence of aggregate source mass). The aggregate source bound

$$\text{Agg} L^2(u_H)$$

does not imply carrier capture. It is possible for an analytic function to have large packet Sobolev mass while all finite transition jets and corrected $K9$ rows vanish. The finite-rank no-go lemma E.132 is precisely this obstruction. Thus Theorem E.135 must always be cited with its carrier-capture and finite Gram audit hypotheses. Any proof step of the form

$$\|u_H\|_{\text{pkt},J}^2 \gg 1 \implies \|\Lambda_p^{\text{gr,full}} u_H\|_{\text{gr}}^2 \gg 1$$

without $\Pi_{\text{eff}} \Pi_{\text{car}}$ is invalid.

Lemma E.137 (Same-column finite-carrier requirement). *Let $\mathcal{B}_p = \{\psi_{p,1}, \dots, \psi_{p,N_p}\}$ be the packet carrier basis used in the finite $K9$ Gram audit. The raw $2Y$ certificate $\mathfrak{R}_{2Y}(p)$, the corrected Hardy-source package audit, and the carrier Gram audit must be evaluated on the same transferred columns*

$$\Lambda_p^{\text{gr,full}} \psi_{p,i}.$$

If any one of these modules is certified on a different column set, then the modules do not compose into a source-to-two/four-point bridge.

Proof. The finite Gram audit proves lower coercivity only for the image of the declared basis under $\Lambda_p^{\text{gr,full}}$. The row certificate proves live/dead routing only for the rows and columns it sees. The corrected package audit proves source identities only for its transferred source columns. If the column sets differ, a vector may be live in one module and invisible, dead, or unlabelled in another. Hence no common no-loss lower bound follows. $\square$

Definition E.138 (Corrected Hardy-source package audit). Let

$$\mathfrak{P}_2^\zeta(a_1, h_1; a_2, h_2)$$

denote the corrected two-atom source package produced from the actual Hardy-curvature source

$$u_H = \partial_t^3 Z_H$$

after canonical whitening, transition-jet correction, and source-graded $K9$ normalization. The corrected Hardy-source package audit consists of the following finite checks.

(a) **Swap symmetry:**

$$\mathfrak{P}_2^\zeta(a_1, h_1; a_2, h_2) = \mathfrak{P}_2^\zeta(a_2, h_2; a_1, h_1) + O(Q^{-B}).$$

(b) **One-amplitude collapse:**

$$\mathfrak{P}_2^\zeta(0, h_1; a_2, h_2) = a_2 F_{h_2} + O(Q^{-B}),$$

$$\mathfrak{P}_2^\zeta(a_1, h_1; 0, h_2) = a_1 F_{h_1} + O(Q^{-B}).$$

(c) **Diagonal merger:**

$$\mathfrak{P}_2^\zeta(a_1, h; a_2, h) = (a_1 + a_2) F_h + O(Q^{-B}).$$

(d) **Fixed-sector quotient nullity:** after non-(1, 1) source-bidegree removal,

$$B_7^\dagger(C_{\mathcal{W}}^{\text{corr}, \zeta}) \in \mathbf{C}A_5^\dagger(m) + O(Q^{-B}).$$

(e) **Same-column $K9/Y12$ compatibility:** all columns used in the package audit are the same transferred carrier columns used in $\mathfrak{R}_{2Y}(p)$ and in the finite $K9$ carrier Gram audit.

Remark E.139 (Corrected package residual scale). The residual term $O(Q^{-B})$ in the corrected Hardy-source package audit is not a qualitative placeholder. The exponent B must dominate the combined loss from Gate-2 rigidity, D4 wall escape, carrier capture, finite $K9$ Gram coercivity, raw $2Y$ routing, and the final atomic comparison. Equivalently, if

$$A_{\text{tot}} = A_{\text{src}} + C_{\text{cap}} + C_{\text{gram}} + C_{2Y} + C_{\text{D4}} + C_{\text{atomic}},$$

then the package residual must satisfy

$$B > A_{\text{tot}} + 10.$$

A corrected source identity proved only modulo an unspecified small error is not enough for the two-point/four-point bridge.

Remark E.140 (Two-point provenance no-go for post-whitened amplitudes). The two-point source identities must hold for the corrected source package used by the proof, not merely for a formal pre-whitened model. In particular, whitening or nonlinear normalization may turn linear amplitude dependence into even or nonlinear amplitude dependence. If the proof constructs $\mathfrak{P}_2^\zeta$ only after such a nonlinear operation, then the one-amplitude collapse and diagonal merger identities do not follow formally. They must be verified in the corrected package audit. Without those identities, the divisibility

$$\mathcal{I}_2^\zeta = a_1 a_2 (h_1 - h_2)^2 \mathcal{J}_2^\zeta + O(Q^{-B})$$

is not justified.

Remark E.141 (Mixed four-point no-new-carrier warning). The mixed four-point bridge uses only the quotient class of the corrected fixed-sector order-seven source term. If

$$\left[B_7^f(C_{\mathcal{W}}^{\text{corr}, \zeta}) \right] \neq 0 \quad \text{in} \quad \mathfrak{f}/\mathbf{CA}_5^f(m),$$

then the actual zeta source creates a new mixed four-point carrier and the scalar exits $T_v = 0$ and $\mathcal{R} = 0$ are no longer exhaustive. Thus the fixed-sector quotient nullity in Definition E.138 is a load-bearing no-loss certificate, not a cosmetic simplification.

Theorem E.142 (Actual zeta-source admissibility for the two/four-point package). *Let*

$$u_H(t) = \partial_t^3 Z_H(t)$$

be the active Hardy-curvature source on a good ordinary Hardy-normalized canonical packet family. Assume:

- (a) *ordinary packet no-loss routing, Hypothesis 15.19;*
- (b) *Gate-1 source survival for u_H ;*
- (c) *Gate-2 from L^2 -input real rigidity, Theorem 15.22;*
- (d) *gauge-locked wall escape and D4 row conversion, Theorem 15.29;*
- (e) *carrier-visible K9/Y12 transfer, Theorem 15.26;*
- (f) *the corrected Hardy-source package audit, Definition E.138.*

Then the actual zeta source supplies the finite source data required by the two-point and mixed four-point branches:

$$\text{SourceAdmiss}_{\zeta}^{2/4}.$$

More explicitly:

- (i) *the live source-graded K9/Y12 component is nonzero with polynomial lower bound:*

$$\sum_p \mu_p \left\| \Pi_{\text{live}} \Lambda_p^{\text{gr}, \text{full}}(u_H) \right\|_{\text{gr}}^2 \geq Q^{-A} G_{\text{ref}};$$

- (ii) *the corrected two-atom Hardy-source package satisfies swap symmetry, one-amplitude collapse, and diagonal merger modulo $O(Q^{-B})$ residuals;*
- (iii) *the mixed four-point corrected source cross-effect has no new fixed-sector quotient component:*

$$B_7^f(C_{\mathcal{W}}^{\text{corr}, \zeta}) \in \mathbf{CA}_5^f(m) + O(Q^{-B});$$

- (iv) *any remaining nonmixed four-point source-order clause is either bypassed by the final source-graded K9/Y12 bridge or is supplied as the finite determinant certificate recorded in the corrected package audit.*

Proof. Gate-1, ordinary packet no-loss routing, and the real-canonical Gate-2 rigidity theorem give analytic Gate-2 source anisotropy on the ordinary packet subfamily. Gauge-locked wall escape converts this source anisotropy into D4 row energy, and row mobility thickens pointwise row output into interval row energy. The carrier-visible transfer theorem then places a polynomial portion of this row energy in the retained source-graded K9/Y12 carrier.

The two-point and mixed four-point assertions are exactly the finite corrected Hardy-source package audit: swap symmetry, one-amplitude collapse, and diagonal merger imply the two-point quadratic factorization, while the fixed-sector quotient inclusion prevents a new mixed four-point carrier. The residuals are absorbed by choosing B larger than the total downstream loss exponent. $\square$

Hypothesis E.143 (Optional nonmixed four-point source-order certificate). If the nonmixed four-point branch is not bypassed by the final source-graded K9/Y12 bridge, the corrected Hardy-source package satisfies the following finite determinant sign certificate. On every good nonmixed patch, the nonmixed determinant

$$\Delta_{\text{nonmix}}^{\zeta}(h_1, h_2, h_3, h_4)$$

associated with the corrected zeta source has fixed orientation on ordered quadruples away from the declared scalar walls, up to $O(Q^{-B})$ residuals. Equivalently, all nonmixed endpoint-gap

recurrences are routed either to a declared scalar wall or to an already absorbed $K9/Y12$ carrier branch.

Remark E.144 (Nonmixed four-point fork). The corrected Hardy-source two-point/mixed-four-point bridge does not by itself settle any still-live nonmixed four-point endpoint-gap branch. There are two allowed statuses:

- (a) the final source-graded $K9/Y12$ atomic bridge bypasses the older nonmixed geometric endgame, in which case the nonmixed material is historical or diagnostic;
- (b) the nonmixed branch remains live, in which case a separate finite determinant/sign or endpoint-gap no-loss certificate must be supplied.

The proof may not use the two-point provenance identities or the mixed fixed-sector quotient nullity as a substitute for a nonmixed four-point theorem unless case (a) has been explicitly established.

Theorem E.145 (Complete actual-zeta bridge to the two/four-point package). *Assume:*

- (a) *ordinary Hardy branch entry and ordinary/singular packet no-loss routing;*
- (b) *Gate-1 for $u_H = \partial_t^3 Z_H$;*
- (c) *Gate-2 from L^2 -input real rigidity;*
- (d) *carrier-visible $K9/Y12$ transfer;*
- (e) *the corrected Hardy-source package audit;*
- (f) *the raw $2Y$ certificate $\mathfrak{R}_{2Y}(p)$ on the same transferred carrier columns.*

Then the actual zeta source supplies all source inputs required by the two-point and mixed four-point branches:

$$\text{SourceAdmiss}_\zeta^{2/4}.$$

Namely:

$$\mathcal{I}_2^\zeta = a_1 a_2 (h_1 - h_2)^2 \mathcal{J}_2^\zeta + O(Q^{-B}),$$

and

$$B_7^f(C_{\mathcal{W}}^{\text{corr}, \zeta}) \in \mathbf{CA}_5^f(m) + O(Q^{-B}).$$

Furthermore, the corresponding source mass is live in the retained source-graded $K9/Y12$ carrier:

$$\sum_p \mu_p \|\Pi_{\text{live}} \Lambda_p^{\text{gr}, \text{full}}(u_H)\|_{\text{gr}}^2 \geq Q^{-A} G_{\text{ref}}.$$

Proof. The live $K9/Y12$ lower bound is Theorem E.142. The two-point factorization is Theorem 12.125. The mixed four-point no-new-carrier assertion is Theorem 10.17. All three use the same transferred carrier columns, and $\mathfrak{R}_{2Y}(p)$ ensures that live $Y12$ contributions are retained rather than quotient-killed. Hence the actual zeta source has been bridged into the exact finite source package consumed by the two-point/four-point architecture. $\square$

Remark E.146 (No parallel Pick/de Branges proof is being invoked). The source bridge in this section uses the actual Hardy-curvature source

$$u_H = \partial_t^3 Z_H$$

and the corrected finite source package consumed by the two-point/four-point architecture. It does not assert Pick positivity for $-\Xi'/\Xi$, does not invoke a de Branges kernel positivity theorem, and does not attempt to prove RH by a parallel global criterion. Its only purpose is to certify that the actual zeta source satisfies the finite source provenance, no-new-carrier, and source-graded $K9/Y12$ -visibility conditions required downstream.

Remark E.147 (Source-substitution no-go). The active source is

$$u_H = \partial_t^3 Z_H.$$

A logarithmic, Laguerre, Pick, or de Branges kernel source may be useful as a diagnostic for how an off-line zero is seen by Ξ , but it is not the source used by the present two-point/four-point and $K9/Y12$ architecture. Replacing u_H by such a source invalidates the Hardy AFE theorem, Gate-2 rigidity, D4 row conversion, carrier Gram audit, and same-norm final comparison unless a

new transfer theorem maps that source into the active Hardy-curvature carrier with polynomial loss. No such substitution is made in the active proof.

Remark E.148 (Dependency diagram for the zeta-to-two/four-point bridge). The completed source bridge has the following dependency graph:

$$\begin{array}{c}
 u_H = \partial_t^3 Z_H \\
 \Downarrow \quad \text{Hardy AFE + ordinary packet no-loss} \\
 \text{Gate-1} \\
 \Downarrow \quad L^2\text{-input real rigidity} \\
 \text{Gate-2} \\
 \Downarrow \quad \text{gauge-locked wall escape + D4} \\
 C_7/Y12 \text{ row energy} \\
 \Downarrow \quad \text{carrier capture + finite } K9 \text{ Gram} + \mathfrak{R}_{2Y} \\
 \text{live source-graded } K9/Y12 \text{ carrier} \\
 \Downarrow \quad \text{corrected Hardy-source package audit} \\
 \text{two-point quadratic factorization} \quad + \quad \text{mixed four-point no-new-carrier.}
 \end{array}$$

Thus Gate–2 supplies live row energy, while the corrected package audit supplies the finite identities consumed by the two-point and four-point modules.

Theorem E.149 (Exhaustive labelled live/dead routing). *In the active maximal-labelled $K9$ branch, every raw nuisance column $A_{a,p}^{\text{raw}}$ entering the $Y12$ quotient admits a canonical decomposition*

$$A_{a,p}^{\text{raw}} = A_{a,p}^{\text{dead}} + E_p m_{a,p} + A_{a,p}^{\text{res}}, \quad m_{a,p} := L_p^{\text{row}} A_{a,p}^{\text{raw}},$$

where

$$L_p^{\text{row}} A_{a,p}^{\text{dead}} = 0.$$

Only the row-dead terms $A_{a,p}^{\text{dead}}$ are placed in the exact killed quotient. The live terms $E_p m_{a,p}$ are retained in the maximal labelled affine target, and the residual terms $A_{a,p}^{\text{res}}$ are controlled by the source-graded residual ledger. Consequently no atom-local live $C_7/Y12$ component is quotient-killed in the active branch.

Proof. This is the defining projection onto the retained labelled live plane $E_p = [C_{7,p}^{\text{tar}} \ C_{Y,p}^{\text{tar}}]$ using the certified rows $L_p^{\text{row}} E_p = I_2$. Subtracting $E_p L_p^{\text{row}} A_{a,p}^{\text{raw}}$ makes the dead remainder row-invisible. The maximal labelled construction of Definition 16.355 retains the live component and routes the residual component before the dead quotient is formed. $\square$

Remark E.150 (Boundary-collar route is no longer the Gate–1 proof mechanism). With the canonical branch-complete window, Gate–1 source domination follows from the pointwise inequality $W_{\text{branch}}^{\text{can}} \geq Q^{-C} W_{\text{ref}}$. Boundary-collar exclusion may still be used to justify admissible raw packet scales or to organize packet selection, but it is no longer the load-bearing proof of branch/reference domination in the canonical source branch. The draft should avoid running the collar-exclusion route and the canonical branch-complete route as competing proofs of the same Gate–1 lower bound. When collar exclusion is invoked later, it is used only as a no-loss certificate for scale-admissible packet selection, not as the proof of the Gate–1 branch/reference Gram domination.

Remark E.151 (Canonical construction ledger). In the final status tables, the following wording should be used.

module	canonical-branch status
δ_{src}	closed by the Hardy source mass theorem and $W_{\text{branch}}^{\text{can}}$
δ_{graph}	closed by pre-extraction Fejér–Riesz charge anti-lock
raw 2Y row/routing	closed after $\mathfrak{R}_{2Y}(p)$ is verified/imported and labelled routing is ex
actual source anchor	closed in the active Hardy-curvature branch
Gate–2 source compatibility	closed by Lemma E.26
RH-entry into the Hardy branch	separate entry theorem
carrier-corrected packet-to- $K9$ transfer	trace upper + carrier capture + finite $K9$ Gram audit + $\mathfrak{R}_{2Y}$ comp
zeta-to-two/four-point bridge	proved from actual zeta-source admissibility

Thus the canonical construction rules and the Hardy-curvature source repair close the source/O2O3/raw-2Y construction gaps after the proof has entered the ordinary Hardy-normalized canonical branch. They do not by themselves supply the global RH-entry theorem recorded in Remark E.155. A different source or routing choice requires its own audit.

The bridge from $u_H = \partial_t^3 Z_H$ to the two-point/four-point package is not a Pick/de Branges or logarithmic-source proof of RH. It is a finite-source admissibility theorem inside the ordinary Hardy-normalized canonical branch. Its role is only to provide the source identities and live carrier mass required by the existing two-point/four-point and $K9/Y12$ architecture.

Remark E.152 (Unified active no-go registry). The following routes are active no-go warnings for the present draft.

- (i) **No full packet-to-jet lower bound:** finite corrected $K9$ rows cannot be coercive on the full analytic packet Hilbert space.
- (ii) **No mean-square-to-entry shortcut:** dyadic Hardy mean-square mass does not localize the source to the packet attached to a specified off-line zero.
- (iii) **No hidden η^{-1} loss:** any RH-entry proof resolving the horizontal displacement $\eta = |\beta - 1/2|$ must either include η^{-1} in Q or prove η -insensitivity.
- (iv) **No source substitution:** log/Pick/Laguerre sources do not replace $u_H = \partial_t^3 Z_H$ inside the active bridge without a transfer theorem.
- (v) **No mismatched row certificates:** $\mathfrak{R}_{2Y}$, the finite $K9$ Gram audit, and the corrected source package must use the same transferred columns.
- (vi) **No unproved nonmixed closure:** two-point and mixed four-point source provenance does not close a still-live nonmixed branch unless the final $K9/Y12$ bridge has bypassed it or a separate nonmixed certificate is supplied.

Any proof step using one of these routes must cite the corresponding repair lemma, not the no-go route itself.

Remark E.153 (How to read the raw 2Y row in the canonical ledger). The maximal labelled live/dead construction is the correct quotient-normal-form repair, but it is not a replacement for the raw row calculation. It prevents live rows from being killed once the live coordinates and label set are known. The raw certificate $\mathfrak{R}_{2Y}(p)$ is the assertion that those coordinates and labels have in fact been constructed from the actual finite row algebra with polynomial conditioning and no unlabelled atom-local live remainder.

Remark E.154 (Referee-safe status of the canonical construction rules). Rules 1–2 are theorem-level only in the canonical construction branch. Rule 1 closes branch/reference block domination because the branch window is constructed as $W_{\text{branch}}^{\text{can}}$ and satisfies the pointwise lower bound against W_{ref} . Rule 2 closes bounded-degree Fejér–Riesz model membership because the O2/O3 Stack weight is chosen before capped extraction inside the Fejér–Riesz charge algebra. Neither rule is a retrofit theorem for arbitrary older branch or Stack weights.

The completed-source anchor is not proved by Rules 1–2; it is proved in the active branch by the separate Hardy source theorem, Theorem E.5, after replacing the old fixed-coefficient source package by the Hardy differential package. Likewise, raw 2Y row/routing is not proved

by Rules 1–2 or by the abstract maximal-labelled construction alone. The maximal-labelled construction supplies the correct live/dead quotient normal form once the detecting rows and live labels have been constructed; the raw coefficient construction is precisely the certificate $\mathfrak{R}_{2Y}(p)$, or an explicit import of that certificate.

Remark E.155 (Hardy source and the RH-entry reduction). The Hardy source repair ties the source layer to the actual zeta function, rather than to an abstract completed-source model. It does not by itself replace the separate global entry theorem which starts the finite-survival architecture from a hypothetical RH counterexample. The proof branch using $\Phi_\zeta = Z_H$ must cite or prove the implication

RH counterexample $\implies$ ordinary Hardy-normalized source branch satisfying Gate-1/Gate-2 entry hypotheses

Once this branch-entry theorem is invoked, all source estimates downstream use $u_H = \partial_t^3 Z_H$ and its packet-normalized form $\lambda_p^3 u_H$. A different initial scalar or phase requires a separate entry and source-package audit.

Remark E.156 (Canonical branch entry is not a formal WLOG step). The canonical branch is a proof-level normal form only after an entry theorem places the hypothetical obstruction into it with polynomial loss. Thus phrases such as “we work canonically” mean

obstruction $\implies$ canonical Hardy/Whitney/FR/ $K9$ branch

by a cited theorem, not by an informal change of notation. This distinction is essential for the Hardy source, the scale-admissible packet selection, and the maximal labelled $K9$ quotient.

APPENDIX F. CERTIFIED DOWNSTREAM STATUS AFTER THE CANONICAL SOURCE-GRADED CLOSURE

Remark F.1 (Current architecture ledger after canonical downstream closure). The current proof architecture is a finite-survival chain. On ordinary real-canonical routed blocks, the downstream certified branch now has the following status.

- (1) Aggregate L^2 -observability for q''_{can} , together with branch/reference domination, supplies Gate-1.
- (2) The Fejér–Riesz charge anti-lock theorem supplies δ_{graph} once the finite/O2O3 packet-model hypotheses are verified.
- (3) The certified live-row quotient supplies rank-two $Y12$ quotient survival on packets satisfying the raw row/routing certificate $\mathfrak{R}_{2Y}(p)$.
- (4) The certified singular retained space supplies the low-block $Y12$ -graph/coercivity gap δ_{coerc} .
- (5) The effective-carrier certificate supplies $\sigma_{\min}(L_{\text{eff},p}) \geq Q^{-C}$.
- (6) Actual-source D4 amplitude nonflatness supplies scalar amplitude mass downstream of Gate-1/Gate-2.
- (7) The canonical source-graded $K9$ lift closes the finite-main Schur complement and the affine-forcing same-norm upper/lower branch.
- (8) The extraction covariance gate is closed as a positive packet-density nonstatic/static fork: either $\delta_{\text{ext}}^{\text{cov}} \geq Q^{-A}$, or the static cap is sent to the canonical source-graded affine branch.

Thus the phrases “finite-main Schur obstruction”, “affine-forcing same-norm obstruction”, and “positive-density extraction selection” should no longer be listed as active downstream gaps in the canonical source-graded branch. Remaining global obligations, if any, are the explicit upstream packet/source inputs and the raw row/routing coefficient certificate recorded in their own sections.

Remark F.2 (Updated downstream module ledger after canonical closure). After the certified quotient/coercivity/effective-carrier repairs, the actual-source D4 amplitude repair, and the

canonical source-graded $K9$ same-norm repair, the downstream status table is:

module	status in the certified branch
Y12-quotient/rank-two carrier survival	proved after the raw $\mathfrak{R}_{2Y}(p)$ certificate supplies the rows and exhausts
Y12-graph/coercivity	proved on the certified low block
row-normal effective $D4$ carrier	proved by the carrier-quotient/effective-carrier certificate
$D4$ amplitude mass	discharged downstream of Gate-1/Gate-2
finite-main Schur / affine forcing	closed by the canonical source-graded $K9$ lift
extraction covariance/static line	closed as a positive-density nonstatic/static fork
same-norm atomic comparison	proved in the canonical source-graded norm

The entries marked closed are statements in the certified dead/live, source-graded branch. They are not statements about raw quotients, raw singular projections, or scalar-collapsed Y12-only norms. The only remaining downstream coefficient issue is the raw verification of $\mathfrak{R}_{2Y}(p)$, unless that verification has been imported elsewhere in the draft.

Remark F.3 (Structural takeaways from the cap/Y12 repairs). The downstream repairs leave four useful structural facts for later modules. First, the rank-two quotient condition has equivalent primal, Schur–Gram, distance, and bounded-dual-row forms; the coefficient work is to construct the rows, not to reprove the finite-dimensional equivalences. Second, exact nuisance must be row-dead: coherent or residual live-looking terms are never quotient-killed, and atom-local live terms are a genuine no-go. Third, $D4$ amplitude mass is a source theorem, not a quotient theorem. Fourth, the six-row carrier table is a carrier-quotient computation; scalar I -coordinate incidences are quarantined and do not affect the active carrier row-energy route.

Remark F.4 (Final bridge observable). The proof-level final comparison is not made with any legacy scalar-discard observable $(I - P_{\text{sing},p}^{\text{cert}})T_{\text{tr} \rightarrow \text{sing},p}J_{\text{tr},p}$. It is made with

$$\Lambda_p^{\text{gr,full}} = (I - P_p^{\text{full}})\mathcal{L}_{K9,p}^{\text{gr}}J_{\text{tr},p}$$

and, for the affine live contradiction, with its orthogonal maximal-affine projection $\hat{\Lambda}_{\text{max},p}$. This is the same norm in which the finite-main Schur complement is positive, the full-vector $K9$ upper is proved, and the affine $D4$ lower is measured.

Theorem F.5 (Canonical source-graded atomic contradiction). *Assume the selected routed cap family is in the canonical source-graded $K9$ branch. Assume:*

(1) *the finite-main Schur gap*

$$(E_p^{\text{aff}})^*(I - P_p^{\text{full}})E_p^{\text{aff}} \geq Q^{-C_E}I,$$

as supplied by Corollary 16.168;

(2) *clean carrier invertibility*

$$\sigma_{\min}(L_{\text{clean},p}) \geq Q^{-C_L};$$

(3) *source-weighted $D4$ amplitude mass*

$$\sum_{p \in C_j} \mu_p |H_p|^2 \|v_p\|^2 \geq Q^{-A_H} \sum_{p \in C_j} \mu_p \|v_p\|^2;$$

(4) *maximal affine row realization, Theorem 16.190;*

(5) *source-graded same-norm Type-I upper, Theorem 16.189.*

If the residual and upper exponents are chosen larger than the lower-bound losses, no nonzero routed cap family can occur. Equivalently, the final cap transversality lower bound and the Type-I upper give AtomicY12Contradiction in the canonical source-graded $K9$ norm.

Proof. Let

$$z_p = (H_p L_{\text{clean},p} v_p, (m_{\lambda,p})_{\lambda \in \mathfrak{A}_{K9,p}}).$$

The clean component alone gives

$$\|z_p\|^2 \geq |H_p|^2 \|L_{\text{clean},p} v_p\|^2 \geq Q^{-2C_L} |H_p|^2 \|v_p\|^2.$$

The finite-main Schur gap gives

$$\|(I - P_p^{\text{full}})E_p^{\text{aff}} z_p\|_{\text{gr}}^2 \geq Q^{-C_E} \|z_p\|^2.$$

Since $E_p^{\text{aff}} z_p$ lies in the maximal affine source-grade and P_p^{full} is source-grade saturated, the maximal-affine projection does not reduce this carrier contribution. Hence Theorem 16.190 and $\|x + y\|^2 \geq \frac{1}{2}\|x\|^2 - \|y\|^2$ imply

$$\begin{aligned} \sum_{p \in C_j} \mu_p \|\hat{\Lambda}_{\max, p}(q''_{\text{can}})\|_{\text{gr}}^2 &\geq Q^{-C_E - 2C_L - O(1)} \sum_{p \in C_j} \mu_p |H_p|^2 \|v_p\|^2 \\ &\quad - \sum_{p \in C_j} \mu_p \|\mathcal{R}_{p, \max}^{\text{id}}\|_{\text{gr}}^2. \end{aligned}$$

Choosing B_{id} larger than the polynomial lower losses and then using D4 amplitude mass gives

$$\sum_{p \in C_j} \mu_p \|\hat{\Lambda}_{\max, p}(q''_{\text{can}})\|_{\text{gr}}^2 \geq Q^{-A_{\text{low}}} \sum_{p \in C_j} \mu_p \|v_p\|^2$$

for $A_{\text{low}} = A_H + C_E + 2C_L + O(1)$. On the other hand, Theorem 16.189 and orthogonal projection give

$$\sum_{p \in C_j} \mu_p \|\hat{\Lambda}_{\max, p}(q''_{\text{can}})\|_{\text{gr}}^2 \leq Q^{-B_{\text{up}}} \sum_{p \in C_j} \mu_p \|v_p\|^2.$$

For a nonzero routed cap the mass on the right is positive. Taking $B_{\text{up}} > A_{\text{low}}$ gives a contradiction for sufficiently large Q . $\square$

Corollary F.6 (Static-line branch in the canonical source-graded norm). *On a quantitatively static cap, take $v_p = u_j$ to be the normalized static line direction. Under the same canonical source-graded inputs as Theorem F.5, the static-line branch also gives AtomicY12Contradiction. No scalar wall-escape condition for $H_p L_{\text{clean}, p} u_j + m_{\eta, p}$ is required, because the clean D4 component and every affine forcing component live in orthogonal source grades.*

Proof. The proof of Theorem F.5 uses only the clean component $H_p L_{\text{clean}, p} v_p$ for the lower bound. With $v_p = u_j$, clean carrier invertibility and D4 amplitude mass give the same lower estimate. The same-norm upper is unchanged. $\square$

Remark F.7 (Final bridge status in the canonical branch). The literal pair $(q_p^Y N_p, q_p^Y Y_p)$ remains a no-go because N_p is a killed nuisance column. The proof-level bridge is the retained affine $C_7/Y12$ carrier in the canonical source-graded $K9$ norm. In that branch the Jacobian/Wronskian/angle lower bound, the finite-main Schur gap, the maximal affine row realization, and the Type-I upper are all in the same source-graded Hilbert norm. Therefore the same-norm atomic compatibility obstruction is closed in the canonical source-graded branch. Scalar Y12-completion or scalar I -coordinate collapse is not a substitute for this branch.

Remark F.8 (Holistic consistency after the canonical $K9$ closure). The canonical source-graded insertion closes the finite-main/affine-forcing same-norm obstruction in the certified branch. The closure rests on four specific facts:

- (1) $\mathcal{L}_{K9, p}^{\text{gr}}$ is a fixed linear labelled lift before scalar collapse and is independent of the decomposition $q''_{\text{can}} = F_X + R_X$.
- (2) finite-main labels lie in $H_{p, \text{main}}$, affine live labels lie in $H_{p, \text{aff}}$, and these labelled subspaces are orthogonal.
- (3) the maximal affine target contains every live affine $K9$ source-grade label, including the odd- $\eta\eta$ / Bottleneck-C forcing row.
- (4) the upper and lower are both measured with $\Lambda_p^{\text{gr}, \text{full}}$ or its orthogonal maximal-affine projection; no scalar I -coordinate or primitive-only Y12 collapse occurs before the contradiction.

Consequently, older phrases such as “finite-main Schur obstruction”, “affine-forcing same-norm obstruction”, and “tail-only completion problem” are legacy warnings for noncanonical or scalar-collapsed routes. They are not active gaps in the canonical branch. The still-active downstream

audit, if not already discharged elsewhere, is the raw row/routing coefficient certificate $\mathfrak{R}_{2Y}(p)$ that constructs the certified live/dead quotient rows.

Remark F.9 (No-go registry is consolidated). The no-go registry has been consolidated in Remark 18.2. This label is retained only for citation stability.

Remark F.10 (Downstream status after source-graded final closure). The downstream status table is normalized as follows. The canonical source-graded final bridge is closed, but this does not erase upstream source, branch-entry, or raw row-certificate obligations.

module	current status
$\text{Agg}L^2(q''_{\text{can}})$	proved inside the ordinary Hardy canonical branch
Graph/O2O3_{23}	proved from pre-extraction Fejér–Riesz anti-lock
$Y12\text{-certified quotient and } L_{\text{eff},p}$	proved after $\mathfrak{R}_{2Y}(p)$ is verified/imported and exhaustive labelled routing is
δ_{coerc}	proved in the canonical source-graded branch
$D4$ amplitude mass	proved downstream of Gate–1/Gate–2
$\text{EFCompletion}_{K9}^{\text{gr,vec}}$	proved on ordinary routed blocks from the smooth AFE tail
δ_{ext} / static line	closed in positive packet-density formulation
AtomicY12Contradiction	proved in the canonical source-graded $K9$ branch

Thus the same-norm final bridge is no longer an active obstruction in the canonical source-graded branch. Remaining global obligations are upstream source, packet-model, branch-entry, or raw row-certificate obligations; they are not failures of the final atomic compatibility comparison.

Remark F.11 (Final status after reflecting the actual-source and labelled-routing repairs). In the active proof branch the following items should no longer be listed as open gaps:

- (1) the completed-source anchor: it is replaced by the actual Hardy-curvature source theorem, Theorem E.5;
- (2) the old $a_1 = 1$ reference noncollapse route: it is replaced by the dyadic Hardy-curvature mean-square theorem and mass-good packet selection;
- (3) branch/reference block domination: it is closed by the canonical branch-complete window;
- (4) Fejér–Riesz model membership: it is closed by the pre-extraction Fejér–Riesz Stack construction;
- (5) the primitive arithmetic vanishing target $c_{\text{arith},Y} = 0$: it is superseded by typed maximal labelled live/dead routing. A coherent live arithmetic row is retained as affine forcing rather than quotient-killed; an atom-local live row remains a failure of the raw packet certificate.

The remaining qualifications are branch-specific: the proof must use the canonical Hardy source, canonical branch window, pre-extraction Fejér–Riesz Stack, the verified/imported raw certificate $\mathfrak{R}_{2Y}(p)$, exhaustive maximal labelled $K9$ routing, and the canonical source-graded $K9$ norm. Departing from any of these canonical choices requires a separate audit for the departed branch.

Remark F.12 (Live upstream proof chain after pruning). After pruning, the active upstream chain is:

Hardy source anchor $\rightarrow$ differential Hardy AFE $\rightarrow$ mass-good Whitney packets $\rightarrow W_{\text{branch}}^{\text{can}} \geq Q^{-C} W_{\text{ref}} \rightarrow \delta_{\text{src}}$

The active graph branch is:

pre-extraction Fejér–Riesz Stack $\rightarrow$ finite charge anti-lock $\rightarrow \delta_{\text{graph}} \geq Q^{-A}$.

The active downstream branch is:

$\mathfrak{R}_{2Y}(p)$ verified/imported $\rightarrow$ maximal labelled $K9$ live/dead routing $\rightarrow$ source-graded same-norm comparison

Any route not fitting one of these chains is historical unless a transfer theorem explicitly maps it into the active chain with polynomial loss.

Remark F.13 (Archived numerical Gate–1 diagnostics). The numerical O1, zero-side proxy, same-scale quadrature, and packet-frequency determinant stress tests are archived diagnostics. They are not used as proof inputs for the canonical Hardy Gate–1 theorem, the Fejér–Riesz

O2/O3 theorem, the certified Y12/K9 quotient, or the final source-graded atomic contradiction. They may be moved to an appendix or supplementary numerical section.

Remark F.14 (Superseded Gate-1 warning). This warning is superseded in the active Hardy/canonical branch by Theorem E.71 and Corollary E.73. It remains valid only for archived noncanonical routes such as zero-kernel proxies, P_2 -only or B_2 -only diagnostics, scalar pointwise finite-row packages, and raw branch windows not identified with $W_{\text{branch}}^{\text{can}}$.

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